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(56) **References Cited**

U.S. PATENT DOCUMENTS

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2,788,341 A	4/1957	Schwyzter et al.
3,717,642 A	2/1973	Von Strandtmann
4,109,496 A	8/1978	Allemann et al.
4,650,796 A	3/1987	George et al.
4,868,103 A	9/1989	Stavrianopoulos et al.
5,223,409 A	6/1993	Ladner et al.
5,225,422 A	7/1993	Nagata et al.
5,283,317 A	2/1994	Saifer et al.
5,631,169 A	5/1997	Lakowicz et al.
5,677,158 A	10/1997	Zhou et al.
5,801,030 A	9/1998	McVey et al.
5,858,358 A	1/1999	June et al.
5,883,223 A	3/1999	Gray
6,180,612 B1	1/2001	Hockensmith et al.
6,309,634 B1	10/2001	Bankiewicz et al.
6,352,694 B1	3/2002	June et al.
6,534,055 B1	3/2003	June et al.
6,551,786 B2	4/2003	Manfredi
6,683,058 B1	1/2004	Tuszynski
6,692,964 B1	2/2004	June et al.
6,716,622 B2	4/2004	Curiel et al.
6,797,514 B2	9/2004	Berenson et al.
6,867,041 B2	3/2005	Berenson et al.
6,887,466 B2	5/2005	June et al.
6,905,680 B2	6/2005	June et al.
6,905,681 B1	6/2005	June et al.
6,905,874 B2	6/2005	Berenson et al.
6,995,011 B2	2/2006	Itoh et al.
7,056,883 B2	6/2006	Ito et al.
7,067,318 B2	6/2006	June et al.
7,144,575 B2	12/2006	June et al.
7,172,869 B2	2/2007	June et al.
7,175,843 B2	2/2007	June et al.
7,205,103 B2	4/2007	Emerson
7,232,566 B2	6/2007	June et al.
8,324,367 B2	12/2012	Kaemmerer et al.

(Continued)

FOREIGN PATENT DOCUMENTS

CN	104530013 B	6/2016
CN	108690020 A	10/2018

(Continued)

OTHER PUBLICATIONS

"FLI1 gene," MedlinePlus, published May 1, 2012, <<https://medlineplus.gov/genetics/>> (3 pages).

(Continued)

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ABSTRACT

The present invention relates to compositions and methods for the treatment of BAF-related disorders, such as cancers and viral infections.

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(56)

References Cited

U.S. PATENT DOCUMENTS

8,642,660	B2	2/2014	Goldfarb	2018/0213422	A1	7/2018	Kazmi et al.
8,697,359	B1	4/2014	Zhang	2018/0215766	A1	8/2018	Bair et al.
8,703,761	B2	4/2014	Forster et al.	2018/0258491	A1	9/2018	Jagani et al.
8,771,945	B1	7/2014	Zhang	2018/0327419	A1	11/2018	Bradner et al.
8,795,965	B2	8/2014	Zhang	2018/0328913	A1	11/2018	Kadoch et al.
8,945,861	B2	2/2015	Bomgarden et al.	2019/0071415	A1	3/2019	Bradner et al.
8,946,268	B2	2/2015	Lau et al.	2019/0076539	A1	3/2019	Phillips et al.
9,072,052	B2	6/2015	Griffin et al.	2019/0247509	A1	8/2019	Buckley et al.
9,126,985	B2	9/2015	Kley et al.	2020/0069669	A1	3/2020	Grassian et al.
9,271,978	B2	3/2016	Liu et al.	2020/0140456	A1	5/2020	Phillips et al.
9,353,051	B2	5/2016	Byrd et al.	2020/0206344	A1	7/2020	Kadoch et al.
9,410,943	B2	8/2016	Kadoch et al.	2021/0009568	A1	1/2021	Zhou et al.
9,546,206	B2	1/2017	Ring et al.	2021/0116454	A1	4/2021	Anton et al.
9,636,323	B2	5/2017	Lin et al.	2021/0171958	A1	6/2021	Chan et al.
9,708,338	B2	7/2017	Yukimasa et al.	2021/0198256	A1	7/2021	Nasveschuk et al.
9,708,348	B2	7/2017	Castro et al.	2021/0230190	A1	7/2021	Ruppel et al.
9,750,816	B2	9/2017	Bradner et al.	2021/0251988	A1	8/2021	Zhou et al.
9,821,068	B2	11/2017	Bradner et al.	2021/0260171	A1	8/2021	Zhou et al.
9,850,543	B2	12/2017	Jagani et al.	2022/0016083	A1	1/2022	Centore et al.
9,919,998	B2	3/2018	Ebright et al.	2022/0079940	A1	3/2022	Centore et al.
9,932,340	B2	4/2018	Dai et al.	2022/0098194	A1	3/2022	Nasveschuk et al.
10,105,420	B2	10/2018	Kadoch et al.	2022/0119378	A1	4/2022	Anthony et al.
10,131,637	B2	11/2018	Abdel-Meguid et al.	2022/0289711	A1	9/2022	Ruppel et al.
10,207,998	B2	2/2019	Derbyshire et al.	2023/0035235	A1	2/2023	Kadoch et al.
10,239,888	B2	3/2019	Bradner et al.	2023/0060334	A1	3/2023	Nasveschuk et al.
10,266,850	B2	4/2019	Doudna et al.	2023/0065463	A1	3/2023	Ruppel et al.
10,308,614	B2	6/2019	Albrecht et al.	2023/0066136	A1	3/2023	Ruppel et al.
10,464,925	B2	11/2019	Bradner et al.	2023/0072053	A1	3/2023	Ruppel et al.
10,646,575	B2	5/2020	Phillips et al.	2023/0077730	A1	3/2023	Ruppel et al.
10,660,968	B2	5/2020	Phillips et al.	2023/0079819	A1	3/2023	Vaswani et al.
10,669,253	B2	6/2020	Bradner et al.	2023/0121497	A1	4/2023	Vaswani et al.
10,849,982	B2	12/2020	Phillips et al.	2023/0138480	A1	5/2023	Anthony et al.
10,905,768	B1	2/2021	Phillips et al.	2023/0142883	A1	5/2023	Ruppel et al.
10,976,320	B2	4/2021	Dykhuizen et al.	2023/0331722	A1	10/2023	Ruppel et al.
11,059,801	B2	7/2021	Bradner et al.	2023/0416246	A1	12/2023	Ruppel et al.
11,149,254	B2	10/2021	Szalay et al.	2024/0002382	A1	1/2024	Ruppel et al.
11,185,592	B2	11/2021	Phillips et al.	2024/0067642	A1	2/2024	Ruppel et al.
11,285,218	B2 *	3/2022	Buckley A61P 35/00	2024/0101550	A1	3/2024	Vaswani et al.
11,306,105	B2	4/2022	Bradner et al.	2024/0150328	A1	5/2024	Zhou et al.
11,414,416	B1	8/2022	Ruppel et al.	2024/0150348	A1	5/2024	Ruppel et al.
11,485,732	B2	11/2022	Vaswani et al.	2024/0158387	A1	5/2024	Vaswani et al.
11,485,743	B2	11/2022	Mainolfi et al.	2024/0166668	A1	5/2024	Ruppel et al.
11,497,752	B2	11/2022	Anthony et al.	2024/0189318	A1	6/2024	Huang
11,524,949	B2	12/2022	Phillips et al.	2024/0190894	A1	6/2024	Gu et al.
11,560,381	B1	1/2023	Ruppel et al.				
11,583,586	B2	2/2023	Bradner et al.				
11,639,345	B2	5/2023	Bloch et al.				
11,691,972	B2	7/2023	Nasveschuk et al.				
11,773,085	B2	10/2023	Zhou et al.				
2002/0037281	A1	3/2002	Davidson et al.				
2003/0027335	A1	2/2003	Ruley et al.				
2004/0216178	A1	10/2004	Jones et al.				
2005/0079512	A1	4/2005	Emerson et al.				
2005/0130919	A1	6/2005	Xu et al.				
2006/0058255	A1	3/2006	Chen et al.				
2006/0121005	A1	6/2006	Berenson et al.				
2007/0105181	A1	5/2007	Pope et al.				
2008/0221157	A1	9/2008	Chakravarty et al.				
2010/0048565	A1	2/2010	Frenkel et al.				
2011/0061116	A1	3/2011	Haldar et al.				
2012/0035244	A1	2/2012	Chinnaiyan et al.				
2014/0287931	A1	9/2014	Bernards et al.				
2015/0057169	A1	2/2015	Siu et al.				
2016/0058872	A1	3/2016	Crew et al.				
2016/0130663	A1	5/2016	Kohn et al.				
2016/0176916	A1	6/2016	Bradner et al.				
2017/0014491	A1	1/2017	Kadoch et al.				
2017/0050968	A1	2/2017	Bennett et al.				
2017/0158709	A1	6/2017	Boloor				
2018/0044335	A1	2/2018	Martin et al.				
2018/0085465	A1	3/2018	Bradner et al.				
2018/0105500	A1	4/2018	Derbyshire et al.				
2018/0134684	A1	5/2018	Bradner et al.				
2018/0140722	A1	5/2018	Willis et al.				
2018/0187614	A1	7/2018	Dudar				

FOREIGN PATENT DOCUMENTS

EA	202192101	A1	12/2021
WO	WO-94/10300	A1	5/1994
WO	WO-95/30761	A2	11/1995
WO	WO-2000/024392	A1	5/2000
WO	WO-00/59888	A1	10/2000
WO	WO-00/59905	A1	10/2000
WO	WO-2005/039643	A2	5/2005
WO	WO-2005/112620	A2	12/2005
WO	WO-2006/005941	A1	1/2006
WO	WO-2006/051063	A1	5/2006
WO	WO-2008/022396	A1	2/2008
WO	WO-2009/086303	A2	7/2009
WO	WO-2010/007046	A2	1/2010
WO	WO-2011/115998	A2	9/2011
WO	WO-2012/003281	A2	1/2012
WO	WO-2012/085650	A1	6/2012
WO	WO-2013/106643	A2	7/2013
WO	WO-2013/116663	A1	8/2013
WO	WO-2014/150395	A1	9/2014
WO	WO-2015/103317	A1	7/2015
WO	WO-2015/120320	A1	8/2015
WO	WO-2015/121688	A1	8/2015
WO	WO-2016/054491	A1	4/2016
WO	WO-2016/105518	A1	6/2016
WO	WO-2016/138114	A1	9/2016
WO	WO-2016/139361	A1	9/2016
WO	WO-2016/160718	A1	10/2016
WO	WO-2016/207212	A1	12/2016
WO	WO-2017/007612	A1	1/2017
WO	WO-2017/024317	A2	2/2017
WO	WO-2017/024318	A1	2/2017
WO	WO-2017/087885	A1	5/2017

(56)

References Cited

FOREIGN PATENT DOCUMENTS

WO	WO-2017/118734	A1	7/2017
WO	WO-2017/158381	A1	9/2017
WO	WO-2017/197036	A1	11/2017
WO	WO-2017/197046	A1	11/2017
WO	WO-2017/197051	A1	11/2017
WO	WO-2017/197055	A1	11/2017
WO	WO-2017/197056	A1	11/2017
WO	WO-2017/223452	A1	12/2017
WO	WO-2018/064589	A1	4/2018
WO	WO-2018/148443	A1	8/2018
WO	WO-2018/160636	A1	9/2018
WO	WO-2018/177297	A1	10/2018
WO	WO-2019/040098	A1	2/2019
WO	WO-2019/099868	A2	5/2019
WO	WO-2019/152437	A1	8/2019
WO	WO-2019/152440	A1	8/2019
WO	WO-2019/165229	A1	8/2019
WO	WO-2019/226915	A1	11/2019
WO	WO-2020/010227	A1	1/2020
WO	WO-2020/035779	A1	2/2020
WO	WO-2020/051235	A1	3/2020
WO	WO-2020/081556	A2	4/2020
WO	WO-2020/081588	A1	4/2020
WO	WO-2020/106915	A1	5/2020
WO	WO-2020/127685	A1	6/2020
WO	WO-2020/132561	A1	6/2020
WO	WO-2020/160100	A1	8/2020
WO	WO-2020/160180	A1	8/2020
WO	WO-2020/160192	A1	8/2020
WO	WO-2020/160193	A2	8/2020
WO	WO-2020/160196	A1	8/2020
WO	WO-2020/160198	A1	8/2020
WO	WO-2020/264177	A1	12/2020
WO	WO-2021/022163	A2	2/2021
WO	WO-2021/055295	A1	3/2021
WO	WO-2021/081032	A1	4/2021
WO	WO-2021/155100	A1	8/2021
WO	WO-2021/155262	A1	8/2021
WO	WO-2021/155264	A1	8/2021
WO	WO-2021/155316	A1	8/2021
WO	WO-2021/155320	A1	8/2021
WO	WO-2021/155321	A2	8/2021
WO	WO-2021/178920	A1	9/2021
WO	WO-2021/183218	A1	9/2021
WO	WO-2021/236080	A1	11/2021
WO	WO-2022/198043	A1	9/2022
WO	WO-2023/009834	A2	2/2023
WO	WO-2023/039208	A1	3/2023
WO	WO-2023/109892	A1	6/2023
WO	WO-2023/196560	A1	10/2023
WO	WO-2023/196565	A1	10/2023
WO	WO-2023/196567	A2	10/2023

OTHER PUBLICATIONS

Adamo et al., "The oncogene ERG: a key factor in prostate cancer," *Oncogene* 35(4):403-14 (Jan. 28, 2016).

Advani et al., "A Phase I study of imatinib mesylate in combination with cytarabine and daunorubicin for c-kit positive relapsed acute myeloid leukemia," *Leuk Res.* 34(12):1622-6 (Dec. 2010).

Alazawi, "Foghorn Therapeutics," Blackseed Bio, last updated Mar. 4, 2022, retrieved Jul. 24, 2023, from <<https://blackseedbio.com/reports/fhtx#pipeline>> (26 pages).

Anders et al., "HTSeq—a Python framework to work with high-throughput sequencing data," *Bioinformatics* 31(2):166-9 (Jan. 15, 2015).

Asangani et al., "Therapeutic targeting of BET bromodomain proteins in castration-resistant prostate cancer," available in PMC Dec. 12, 2014. Published in final edited form as: *Nature*. 510(7504):278-82 (2014) (44 pages).

Attard et al., "Duplication of the fusion of TMPRSS2 to ERG sequences identifies fatal human prostate cancer," available in PMC Feb. 24, 2009. Published in final edited form as: *Oncogene*. 27(3):253-63 (2008) (19 pages).

Basuyaux et al., "The Ets transcription factors interact with each other and with the c-Fos/c-Jun complex via distinct protein domains in a DNA-dependent and -independent manner," *J Biol Chem.* 272(42):26188-95 (1997).

Bendall et al., "Prevention of amino acid conversion in SILAC experiments with embryonic stem cells," *Mol Cell Proteomics.* 7(9):1587-97 (2008).

Berger et al., "Androgen-induced differentiation and tumorigenicity of human prostate epithelial cells," *Cancer Res.* 64(24):8867-75 (2004).

Boulay et al., "Cancer-Specific Retargeting of BAF Complexes by a Prion-like Domain," *Cell* 171(1):163-78 (Sep. 21, 2017) (36 pages).

Börno et al., "Genome-wide DNA methylation events in TMPRSS2-ERG fusion-negative prostate cancers implicate an EZH2-dependent mechanism with miR-26a hypermethylation," *Cancer Discov.* 2(11):1024-35 (2012).

Camuzeaux et al., "Imaging Erg and Jun transcription factor interaction in living cells using fluorescence resonance energy transfer analyses," *Biochem Biophys Res Commun.* 332(4):1107-14 (2005).

Cancer Genome Atlas Research Network, "The Molecular Taxonomy of Primary Prostate Cancer," *Cell*. 163(4):1011-25 (2015) (16 pages).

Carell et al., "A novel procedure for the synthesis of libraries containing small organic molecules," *Angew Chem Int Ed Engl.* 33(20):2059-2061 (1994).

Carell et al., "A Solution-Phase Screening Procedure for the Isolation of Active Compounds from a Library of Molecules," *Angewandte Chemie International Edition in English.* 33(20): 2061-2064 (1994).

Centore et al., "Abstract 1224: Discovery of novel BAF inhibitors for the treatment of transcription factor-driven cancers," Poster Presentations—Proffered abstracts, *Cancer Research* 81(13_Supplement):1224 (Jul. 1, 2021) (2 pages).

Chandler et al., "ARID1a-DNA interactions are required for promoter occupancy by SWI/SNF," *Mol Cell Biol.* 33(2):265-80 (Jan. 2013).

Chen et al., "ETS factors reprogram the androgen receptor cistrome and prime prostate tumorigenesis in response to PTEN loss," available in PMC Feb. 1, 2014. Published in final edited form as: *Nat Med.* 19(8):1023-9 (2013) (21 pages).

Chng et al., "A transcriptional repressor co-regulatory network governing androgen response in prostate cancers," *EMBO J.* 31(12):2810-23 (2012).

Cho et al., "An unnatural biopolymer," *Science*. 261(5126):1303-1305 (1993).

Cull et al., "Screening for receptor ligands using large libraries of peptides linked to the C terminus of the lac repressor," *Proc Natl Acad Sci.* 89(5):1865-1869. (1992).

Cwirla et al., "Peptides on phage: A vast library of peptides for identifying ligands," *Proc Nati Acad Sci.* 87:6378-6382 (1990).

Database Registry, RN 1004932-80-2, entered Feb. 21, 2008 (1 page).

Database Registry, RN 1175782-23-6, entered Aug. 26, 2009 (1 page).

Database Registry, RN 1315743-98-6, entered Aug. 11, 2011 (1 page).

Database Registry, RN 878254-76-3, entered Mar. 28, 2006 (1 page).

Delattre et al., "Gene fusion with an ETS DNA-binding domain caused by chromosome translocation in human tumours," *Nature*. 359(6391):162-5 (1992).

Demichelis et al., "TMPRSS2:ERG gene fusion associated with lethal prostate cancer in a watchful waiting cohort," *Oncogene* 26:4596-4599 (2007).

Devlin et al., "Random Peptide Libraries: a Source of Specific Protein Binding Molecules," *Science*. 249(4967):404-406 (1990).

DeWitt et al., "'Diversomers': an approach to nonpeptide, nonoligomeric chemical diversity," *Proc Natl Acad Sci.* 90(15):6909-6913 (1993).

Dobin et al., "STAR: ultrafast universal RNA-seq aligner," *Bioinformatics.* 29(1):15-21 (2013).

Dominguez et al. "Beyond editing: Repurposing CRISPR-Cas9 for Precision Genome Regulation and Interrogation," available in PMC

(56)

References Cited**OTHER PUBLICATIONS**

- Jun. 27, 2016, published in final edited form as: *Nat Rev Mol Cell Biol.* 17(1):5-15 (Jan. 2016) (24 pages).
- Donaldson et al., "Solution structure of the ETS domain from murine Ets-1: a winged helix-turn-helix DNA binding motif," *EMBO J.* 15(1):125-34 (1996).
- Erb et al., "Recursive deconvolution of combinatorial chemical libraries," *Proc Natl Acad Sci.* 91(24):11422-11426 (1994).
- Extended European Search Report for European Application No. 19748410.8, dated Sep. 24, 2021 (10 pages).
- Extended European Search Report for European Application No. 19887386.1, dated Dec. 5, 2022 (18 pages).
- Extended European Search Report for European Application No. 20749261.2, dated Oct. 18, 2022 (8 pages).
- Extended European Search Report for European Application No. 21748261.1, dated Jan. 29, 2024 (17 pages).
- Fadul et al., "EWS/FLI utilizes NKX2-2 to repress mesenchymal features of Ewing sarcoma," *Genes Cancer* 6(3-4):129-43 (Mar. 2015).
- Fathi et al., "Differentiation syndrome with lower-intensity treatments for acute myeloid leukemia," *Am J Hematol.* 96(6):735-46 (Jun. 1, 2021) (13 Pages).
- Felici et al., "Selection of antibody ligands from a large library of oligopeptides expressed on a multivalent exposition vector," *J Mol Biol.* 222(2):301-10 (1991).
- Feng et al., "GFOLD: a generalized fold change for ranking differentially expressed genes from RNA-seq data," *Bioinformatics.* 28(21):2782-8 (2012).
- Fodor et al., "Multiplexed biochemical assays with biological chips," *Nature.* 364:555-556 (1993).
- Gaj et al., "ZFN, TALEN and CRISPR/Cas-based Methods for Genome Engineering," available in PMC Jul. 1, 2014, published in final edited form as: *Trends Biotechnol.* 31(7):397-405 (Jul. 2013) (20 pages).
- Gallop et al., "Applications of Combinatorial Technologies to Drug Discovery. 1. Background and Peptide Combinatorial Libraries," *J Med Chem* 37(9):1233-51 (1994).
- Gene Ontology Consortium, "Gene Ontology Consortium: going forward," *Nucleic Acids Res.* 43(Database issue):D1049-56 (2015).
- Godwin et al., "Gemtuzumab ozogamicin in acute myeloid leukemia," *Leukemia* 31(9):1855-68 (Sep. 2017).
- Grohar et al., "Ecteinascidin 743 interferes with the activity of EWS-FLI1 in Ewing sarcoma cells," *Neoplasia* 13(2):145-53 (Feb. 2011).
- Helgeson et al., "Characterization of TMPRSS2:ETV5 and SLC45A3:ETV5 gene fusions in prostate cancer," *Cancer Res.* 68(1):73-80 (2008).
- Herrero-Martin et al., "Stable interference of EWS-FLI1 in an Ewing sarcoma cell line impairs IGF-1/IGF-1R signalling and reveals TOPK as a new target," *Br J Cancer* 101(1):80-90 (Jul. 7, 2009).
- Ho et al., "An embryonic stem cell chromatin remodeling complex, esBAF, is essential for embryonic stem cell self-renewal and pluripotency," *Proc Natl Acad Sci U S A.* 106(13):5181-6 (2009).
- Houghten et al., "The use of synthetic peptide combinatorial libraries for the identification of bioactive peptides," *Biotechniques.* 13(3):412-21 (1992).
- Ichikawa et al., "An RNA-binding protein gene, TLS/FUS, is fused to ERG in human myeloid leukemia with t(16;21) chromosomal translocation," *Cancer Res.* 54(11):2865-8 (1994).
- International Preliminary Report on Patentability for International Application No. PCT/US2016/062911 issued May 22, 2018 (13 pages).
- International Preliminary Report on Patentability for International Application No. PCT/US2019/015722, issued Aug. 4, 2020 (6 pages).
- International Preliminary Report on Patentability for International Application No. PCT/US2019/056312, mailed Apr. 14, 2021 (10 pages).
- International Preliminary Report on Patentability for International Application No. PCT/US2020/033829, issued Nov. 17, 2022 (6 pages).
- International Preliminary Report on Patentability for International Application No. PCT/US2022/019506, issued Sep. 12, 2023 (6 pages).
- International Preliminary Report on Patentability for International Application No. PCT/US2022/020943, issued Sep. 12, 2023 (5 pages).
- International Search Report and Written Opinion for International Application No. PCT/US18/000339 dated Jan. 28, 2019 (13 pages).
- International Search Report and Written Opinion for International Application No. PCT/US19/15722, mailed May 31, 2019 (12 pages).
- International Search Report and Written Opinion for International Application No. PCT/US19/56312, dated Jan. 14, 2020 (14 pages).
- International Search Report and Written Opinion for International Application No. PCT/US19/56365 dated Jan. 30, 2020 (8 pages).
- International Search Report and Written Opinion for International Application No. PCT/US2016/062911 dated Mar. 3, 2017 (16 pages).
- International Search Report and Written Opinion for International Application No. PCT/US2020/015605, mailed Jun. 16, 2020 (15 pages).
- International Search Report and Written Opinion for International Application No. PCT/US2020/015723, mailed Jul. 2, 2020 (15 pages).
- International Search Report and Written Opinion for International Application No. PCT/US2020/033829, mailed Aug. 17, 2020 (14 pages).
- International Search Report and Written Opinion for International Application No. PCT/US2022/019506, dated Jun. 7, 2022 (9 pages).
- International Search Report and Written Opinion for International Application No. PCT/US2023/017839, mailed Sep. 6, 2023 (11 pages).
- International Search Report and Written Opinion for International Application No. PCT/US2023/077088, mailed Mar. 4, 2024 (17 pages).
- International Search Report and Written Opinion for International Application No. PCT/US21/15876, mailed on Apr. 7, 2021 (23 pages).
- International Search Report and Written Opinion for International Application No. PCT/US21/15878, mailed Jun. 4, 2021 (10 pages).
- International Search Report and Written Opinion for International Application No. PCT/US23/17821, mailed Jun. 30, 2023 (9 pages).
- International Search Report and Written Opinion for International Application No. PCT/US23/17829, mailed Aug. 23, 2023 (15 pages).
- International Search Report and Written Opinion for International Patent Application No. PCT/US2019/062525, mailed Feb. 18, 2020 (14 pages).
- International Search Report and Written Opinion for PCT/US2022/020943, mailed Jun. 14, 2022 (17 pages).
- Karim et al., "The ETS-domain: a new DNA-binding motif that recognizes a purine-rich core DNA sequence," *Genes Dev.* 4(9):1451-3 (1990).
- Kedage et al., "An Interaction with Ewing's Sarcoma Breakpoint Protein EWS Defines a Specific Oncogenic Mechanism of ETS Factors Rearranged in Prostate Cancer," *Cell Rep.* 17(5):1289-301 (Oct. 25, 2016) (14 pages).
- Klezovitch et al., "A causal role for ERG in neoplastic transformation of prostate epithelium," *Proc Natl Acad Sci U S A.* 105(6):2105-10 (2008).
- Kumar-Sinha et al., "Recurrent gene fusions in prostate cancer," available in PMC Jul. 16, 2009. Published in final edited form as: *Nat Rev Cancer.* 8(7):497-511 (2008) (29 pages).
- Kunderfranco et al., "ETS transcription factors control transcription of EZH2 and epigenetic silencing of the tumor suppressor gene Nkx3.1 in prostate cancer," *PLoS One.* 5(5):e10547 (2010) (17 pages).
- Lam et al., "A new type of synthetic peptide library for identifying ligand-binding activity," *Nature.* 354(6348):82-84 (1991).

(56)

References Cited

OTHER PUBLICATIONS

- Lam, "Application of combinatorial library methods in cancer research and drug discovery," *Anticancer Drug Des.* 12(3):145-67 (1997).
- Langmead et al., "Fast gapped-read alignment with Bowtie 2," available in PMC Apr. 1, 2013. Published in final edited form as: *Nat Methods.* 9(4):357-9 (2012) (8 pages).
- Link et al., "Targeting the BAF57 SWI/SNF subunit in prostate cancer: a novel platform to control androgen receptor activity," *Cancer Res.* 68(12):4551-8 (2008).
- Love et al., "Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2," *Genome Biol.* 15(12):550 (2014) (21 pages).
- Lupien et al., "FoxA1 translates epigenetic signatures into enhancer-driven lineage-specific transcription," *Cell.* 132(6):958-70 (2008).
- Machanic et al., "MEME-ChIP: motif analysis of large DNA datasets," *Bioinformatics.* 27(12):1696-7 (2011).
- Mackereth et al., "Diversity in structure and function of the Ets family PNT domains," *J Mol Biol.* 342(4):1249-64 (2004).
- Madura et al., "N-recognition/Ubc2 interactions in the N-end rule pathway," *J Biol Chem.* 268(16): 12046-54 (1993).
- Mel   et al., "The human transcriptome across tissues and individuals," available in PMC Aug. 24, 2015. Published in final edited form as: *Science.* 348(6235):660-5 (2015) (12 pages).
- Mill et al., "RUNX1-targeted therapy for AML expressing somatic or germline mutation in RUNX1," *Blood* 134(1):59-73 (Jul. 4, 2019).
- Mounir et al., "ERG signaling in prostate cancer is driven through PRMT5-dependent methylation of the Androgen Receptor," *Elife.* 5:e13964 (2016) (19 pages).
- Nagaich et al., "Rapid periodic binding and displacement of the glucocorticoid receptor during chromatin remodeling," *Mol Cell.* 14(2):163-74 (2004).
- Nam et al., "Expression of the TMPRSS2:ERG fusion gene predicts cancer recurrence after surgery for localised prostate cancer," *Br J Cancer.* 97(12):1690-5 (2007).
- Office Action for Chinese Patent Application No. 201980023925.9, dated Apr. 20, 2022 (13 pages).
- Ong et al., "A practical recipe for stable isotope labeling by amino acids in cell culture (SILAC)," *Nat Protoc.* 1(6):2650-60 (2006).
- Papillon et al., "Discovery of Orally Active Inhibitors of Brahma Homolog (BRM)/SMARCA2 ATPase Activity for the Treatment of Brahma Related Gene 1 (BRG1)/SMARCA4-Mutant Cancers," *J Med Chem.* 61(22): 10155-72 (Nov. 2018).
- Partial Supplementary European Search Report for European Patent Application No. 19887386.1, dated Jul. 20, 2022 (23 pages).
- Partial Supplementary European Search Report for European Patent Application No. 20936213.6, dated Feb. 8, 2024 (20 pages).
- Paulo et al., "FLI1 is a novel ETS transcription factor involved in gene fusions in prostate cancer," *Genes Chromosomes Cancer.* 51(3):240-9 (2012).
- Petrovics et al., "Frequent overexpression of ETS-related gene-1 (ERG1) in prostate cancer transcriptome," *Oncogene.* 24(23):3847-52 (2005).
- Pomerantz et al., "The androgen receptor cistrome is extensively reprogrammed in human prostate tumorigenesis," available in PMC May 1, 2016. Published in final edited form as: *Nat Genet.* 47(11):1346-51 (2015) (17 pages).
- Prensner et al., "The long noncoding RNA SchLAP1 promotes aggressive prostate cancer and antagonizes the SWI/SNF complex," available in PMC May 1, 2014. Published in final edited form as: *Nat Genet.* 45(11):1392-8 (2013) (26 pages).
- PubChem CID 117640569, "N-[2-[[4-(4-Methoxyphenyl)-1,3-thiazol-2-yl]amino]-2-oxoethyl]-1,3-thiazole-5-carboxamide," <https://pubchem.ncbi.nlm.nih.gov/compound/117640569>, created Feb. 23, 2016 (9 pages).
- PubChem CID 56442706, "1-(4-Methoxyphenyl)-N-[2-oxo-2-[4-(1,2,4-triazol-1-yl)anilino]ethyl]pyrazole-3-carboxamide," <https://pubchem.ncbi.nlm.nih.gov/compound/56442706>, created Jan. 25, 2012 (8 pages).
- PubChem CID 91946137, "N-[2-[(1-Ethylpyrazol-3-yl)amino]-2-oxoethyl]-1-methylpyrazole-3-carboxamide," <https://pubchem.ncbi.nlm.nih.gov/compound/91946137>, created Oct. 22, 2015 (8 pages).
- PubChem Compound Summary for CID 155037309, dated Dec. 19, 2020 (9 pages).
- PubChem Compound Summary for CID No. 136572628, "4-Chloro-N-[2-(cyclopentylamino)-2-oxoethyl]-5-nitro-1H-pyrazole-3-carboxamide," created Jan. 24, 2019, <<https://pubchem.ncbi.nlm.nih.gov/compound/136572628>>, (7 pages).
- PubChem Compound Summary for CID No. 49726797, "N-Methyl-N-(2-oxo-2-[(4-(pyridin-3-yl)thiazol-2-yl)amino]ethyl)-1H-indole-3-carboxamide," created Nov. 27, 2010, <<https://pubchem.ncbi.nlm.nih.gov/compound/49726797>>, (8 pages).
- PubChem Compound Summary for CID No. 91945707, "N-[2-[[4,5-Dimethyl-1,3-thiazol-2-yl]amino]-2-oxoethyl]-1-methylpyrazole-3-carboxamide," created Oct. 22, 2015 <<https://pubchem.ncbi.nlm.nih.gov/compound/91945707>>, (8 pages).
- PubChem Compound Summary for PubChem CID 49726798, "N-(2-[(4-(Furan-2-yl)thiazol-2-yl)amino]-2-oxoethyl)-N-methyl-1H-indole-3-carboxamide," created Nov. 27, 2010 <<https://pubchem.ncbi.nlm.nih.gov/compound/49726798>> (8 pages).
- PubChem Compound Summary for SID 172131678, dated Dec. 9, 2014 (8 pages).
- Pubchem, "Compound Summary for CID 108452511," <<https://pubchem.ncbi.nlm.nih.gov/compound/108452511>>, created Jan. 15, 2016, retrieved Jan. 4, 2021 (7 pages).
- Pubchem, "Compound Summary for CID 2955118," <<https://pubchem.ncbi.nlm.nih.gov/compound/2955118>>, created Jul. 29, 2005, retrieved Mar. 22, 2017 (13 pages).
- Pubchem, "Compound Summary for CID 7325930," <<https://pubchem.ncbi.nlm.nih.gov/compound/7325930>>, created Jul. 29, 2006, retrieved Mar. 22, 2017 (11 pages).
- Pubchem, "Compound Summary for CID 970466," <<https://pubchem.ncbi.nlm.nih.gov/compound/970466>>, created Jul. 9, 2005, retrieved Mar. 22, 2017 (11 pages).
- Quinlan et al., "BEDTools: a flexible suite of utilities for comparing genomic features," *Bioinformatics.* 26(6):841-2 (2010).
- Rajput et al., "Frequency of the TMPRSS2:ERG gene fusion is increased in moderate to poorly differentiated prostate cancers," *J Clin Pathol.* 60(11):1238-43 (2007).
- Ramos et al., "Current Approaches in the Treatment of Relapsed and Refractory Acute Myeloid Leukemia," *J Clin Med.* 4(4):665-95 (Apr. 2015).
- Rappsilber et al., "Protocol for micro-purification, enrichment, pre-fractionation and storage of peptides for proteomics using StageTips," *Nat Protoc.* 2(8):1896-906 (2007).
- Riggi et al., "EWS-FLI1 utilizes divergent chromatin remodeling mechanisms to directly activate or repress enhancer elements in Ewing sarcoma," *Cancer Cell* 26(5):668-81 (Nov. 10, 2014) (28 pages).
- Sankar et al., "Promiscuous partnerships in Ewing's sarcoma," *Cancer Genet.* 204(7):351-65 (Jul. 2011) (28 pages).
- Scott et al., "Searching for peptide ligands with an epitope library," *Science.* 249(4967):386-390 (1990).
- Shen et al., "The SWI/SNF ATPase Brm is a gatekeeper of proliferative control in prostate cancer," *Cancer Res.* 68(24):10154-62 (Dec. 2008).
- Shi et al., "Role of SWI/SNF in acute leukemia maintenance and enhancer-mediated Myc regulation," *Genes Dev.* 27(24):2648-62 (Dec. 2013).
- Siegel et al., "Cancer statistics, 2015," *CA Cancer J Clin.* 65(1):5-29 (2015).
- STN Registry Database, RN 1010893-05-6, entered Mar. 30, 2008 (2 pages).
- STN Registry Database, RN 1049271-26-2, entered Sep. 14, 2008 (2 pages).
- STN Registry Database, RN 1081662-32-9, entered Dec. 8, 2008 (2 pages).
- STN Registry Database, RN 1209112-42-4, entered Mar. 12, 2010 (2 pages).
- STN Registry Database, RN 1246047-75-5, entered Oct. 12, 2010 (2 pages).

(56)

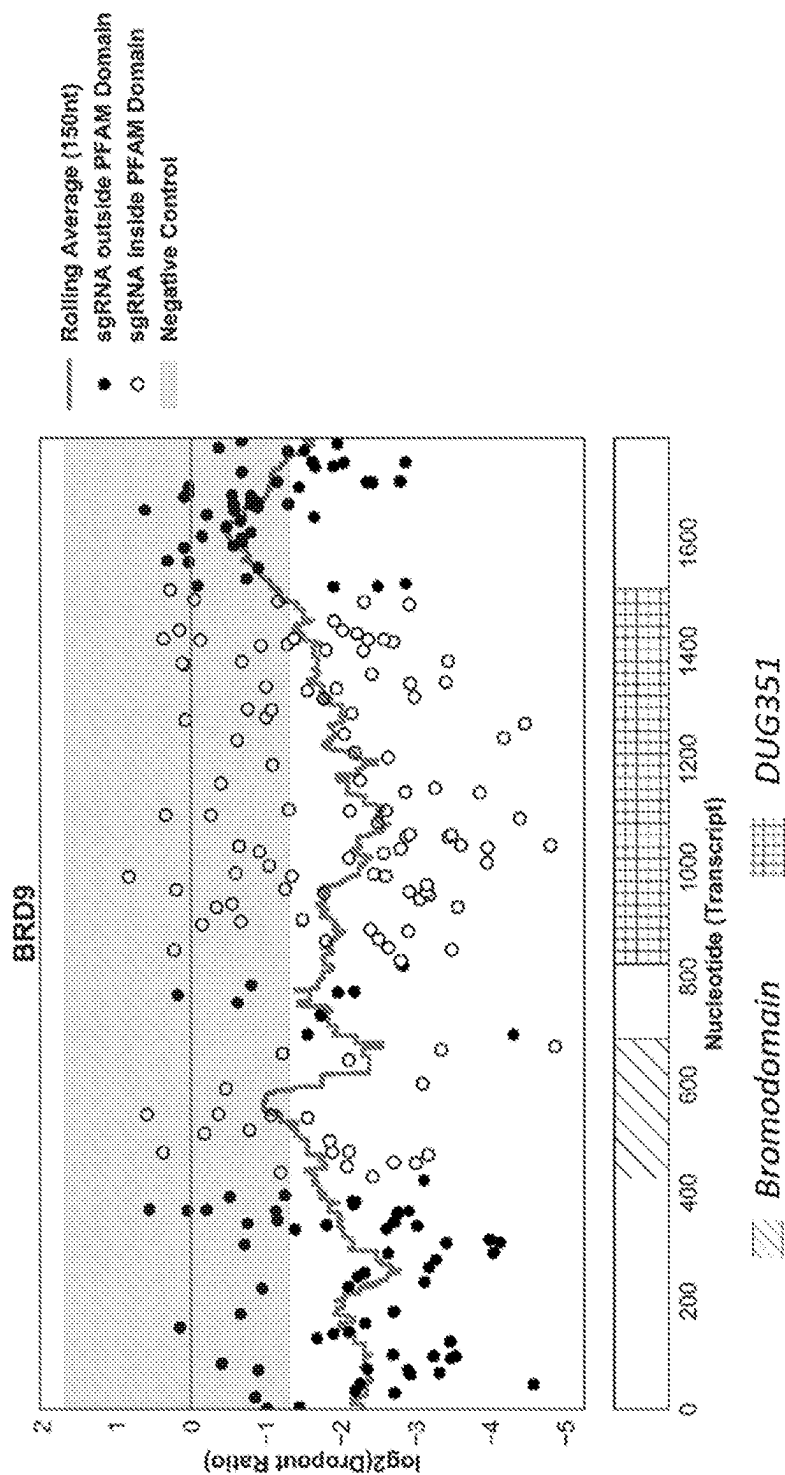
References Cited

OTHER PUBLICATIONS

- STN Registry Database, RN 1308280-67-2, entered Jun. 9, 2011 (2 pages).
- STN Registry Database, RN 1351682-19-3, entered Dec. 22, 2011 (2 pages).
- STN Registry Database, RN 1401558-47-1, entered Oct. 22, 2012 (2 pages).
- STN Registry Database, RN 1455783-72-8, entered Oct. 6, 2013 (2 pages).
- STN Registry Database, RN 1576383-94-2, entered Mar. 31, 2014 (2 pages).
- Takigami et al., "Synthetic siRNA targeting the breakpoint of EWS/Flt-1 inhibits growth of Ewing sarcoma xenografts in a mouse model," *Int J Cancer* 128(1):216-26 (Jan. 1, 2011).
- Wu et al., "Targeting the chromatin remodeling enzyme BRG1 increases the efficacy of chemotherapy drugs in breast cancer cells," *Oncotarget* 7(19):27158-75 (May 10, 2016).
- U.S. Appl. No. 18/292,426-A1, Chen et al.
- U.S. Appl. No. 18/292,508, Huang, Liyue.
- U.S. Appl. No. 62/798,374, Ruppel et al.
- U.S. Appl. No. 62/881,195, Ruppel et al.
- U.S. Appl. No. 62/985,774, Nasveschuk et al.
- U.S. Appl. No. 63/061,659, Nasveschuk et al.
- Brien et al., "Targeted degradation of BRD9 reverses oncogenic gene expression in synovial sarcoma," *eLife*. 7:e41305 (Nov. 15, 2018) (26 pages).
- Communication Pursuant to Rule 164(1) for European Patent Application 19746772.3, dated Oct. 7, 2021 (13 pages).
- Crawford et al., "Inhibition of bromodomain-containing protein 9 for the prevention of epigenetically-defined drug resistance," *Bioorg Med Chem Lett*. 27(15):3534-41(2017).
- Decision Denying Institution of Post-Grant Review of U.S. Pat. No. 11,414,416 dated Sep. 12, 2023 (4 Pages).
- Decision Denying Institution of Post-Grant Review of U.S. Pat. No. 11,560,381, dated Jan. 23, 2024 (4 Pages).
- Declaration of Christopher G. Nasveschuk, Ph.D. Under 37 C.F.R. §1.68 and §42.53(a), dated Oct. 19, 2023 (24 pages).
- Declaration of Kevin T. Sprott, Ph.D., Under 37 C.F.R. §1.68 and §42.53(a), dated Oct. 19, 2023 (74 pages).
- Declaration of Stevan W. Djuric, Ph.D. Under 37 C.F.R. §1.68 and §42.53(a), dated Oct. 19, 2023 (202 pages).
- Del Gaudio et al., "BRD9 binds cell type-specific chromatin regions regulating leukemic cell survival via STAT5 inhibition," *Cell Death Dis.* 10(5):338 (Apr. 2019) doi: 10.1038/s41419-019-1570-9 (14 pages).
- Extended European Search Report for European Application No. 19746772.3, dated Feb. 7, 2022 (15 pages).
- Extended European Search Report for European Application No. 20748462.7, dated Jan. 2, 2023 (10 pages).
- Hay et al., "Design and synthesis of potent and selective inhibitors of BRD7 and BRD9 bromodomains," *Med. Chem. Commun.* 6:1381-86 (2015).
- Hohmann et al., "Sensitivity and engineered resistance of myeloid leukemia cells to BRD9 inhibition," *Nat Chem Biol.* 12(9): 672-679 (Sep. 2016) (12 pages).
- International Preliminary Report on Patentability for International Application No. PCT/2020/015744, mailed Aug. 12, 2021 (8 pages).
- International Search Report and Written Opinion for International Application No. PCT/2020/015744, mailed Jun. 26, 2020 (16 pages).
- International Search Report and Written Opinion for International Patent Application No. PCT/US2019/015733, dated Jul. 1, 2019 (17 pages).
- International Search Report and Written Opinion for PCT/US2020/015746, mailed May 18, 2020 (11 pages).
- K. Luby-Phelps, "Cytoarchitecture and Physical Properties of Cytoplasm: Volume, Viscosity, Diffusion, Intracellular Surface Area," *Int Rev Cytol.* 192:189-221 (2000).
- Kadoch et al., "Mammalian SWI/SNF chromatin remodeling complexes and cancer: Mechanistic insights gained from human genomics," *Sci Adv.* 1(5):e1500447 (2015) (17 pages).
- Kadoch et al., "Proteomic and bioinformatic analysis of mammalian SWI/SNF complexes identifies extensive roles in human malignancy," *Nat Genet.* 45(6):592-601 (2013) (11 pages).
- Kadoch et al., "Reversible Disruption of mSWI/SNF (BAF) Complexes by the SS18-SSX Oncogenic Fusion in Synovial Sarcoma," *Cell.* 153(1):71-85 (2013).
- Martin et al., "Structure-Based Design of an in Vivo Active Selective BRD9 Inhibitor," *J Med Chem.* 59(10):4462-75 (2016).
- McBride et al., "Disruption of mammalian SWI/SNF and polycomb complexes in human sarcomas: mechanisms and therapeutic opportunities," *J Pathol.* 244(5): 638-649 (Apr. 2018).
- Michel et al., "Abstract PR15: BRD9 defines a novel mammalian SWI/SNF (BAF) complex configuration which supports proliferation in AML," *Clin Cancer Res.* 23(24_Suppl) Abstract PR15 (2017) (4 pages).
- Munoz et al. "CRISPR Screens Provide a Comprehensive Assessment of Cancer Vulnerabilities but Generate False-Positive Hits for Highly Amplified Genomic Regions," *Cancer Discov.* 6(8):900-13 (Aug. 2016) (14 pages).
- Nasir, M.S. et al., "Fluorescence Polarization: An analytical tool for Immunoassay and Drug Discovery," *Comb. Chem High Throughput Screen.* 2(4):177-190 (Aug. 1999).
- Pan et al., "A major chromatin regulator determines resistance of tumor cells to T cell-mediated killing," *Science.* 359(6377):770-75 (Jan. 2018) (11 pages).
- Petition for Post Grant Review of U.S. Pat. No. 11,414,416 filed May 1, 2023 (133 Pages).
- Petition for Post Grant Review of U.S. Pat. No. 11,560,381 filed Oct. 23, 2023 (163 Pages).
- Picaud et al., "9H-purine scaffold reveals induced-fit pocket plasticity of the BRD9 bromodomain," *J Med Chem.* 58(6):2718-36 (2015).
- PubChem CID 118028040, "6-Ethyl-4-(2-methoxy-5-methylsulfonylphenyl)-2-methyl-7,8-dihydro-5H-2,6-naphthyridin-1-one," created Feb. 23, 2020, retrieved Jun. 11, 2020 <<https://pubchem.ncbi.nlm.nih.gov/compound/118028040>> (10 pages).
- Remillard et al., "Degradation of the BAF Complex Factor BRD9 by Heterobifunctional Ligands," *Angew Chem Int Ed Engl.* 56(21):5738-43 (2017) (7 pages).
- Sender et al., "Revised Estimates for the No. of Human and Bacteria Cells in the Body," *PLoS Biol.* 14(8):e1002533 (Aug. 19, 2016) (14 pages).
- Supporting Information for Remillard et al., "Degradation of the BAF Complex Factor BRD9 by Heterobifunctional Ligands," *Angew Chem Int Ed Engl.* 56(21):5738-43 (2017) (43 pages).
- Teuscher et al., "A Versatile Method to Determine the Cellular Bioavailability of Small-Molecule Inhibitors," *J Med Chem.* 60(1): 157-169 (2017).
- Theodoulou et al., "Discovery of I-BRD9, a Selective Cell Active Chemical Probe for Bromodomain Containing Protein 9 Inhibition," *J Med Chem.* 59(4):1425-39 (2015).
- Tsherniak et al. "Defining a Cancer Dependency Map," available in PMC Jul. 27, 2018, published in final edited form as: *Cell.* 170(3):564-576 (Jul. 2017) (40 pages).
- Vangamudi et al., "The SMARCA2/4 ATPase Domain Surpasses the Bromodomain as a Drug Target in SWI/SNF-Mutant Cancers: Insights from cDNA Rescue and PFI-3 Inhibitor Studies," *Cancer Res.* 75(18):3865-78 (2015).
- Wang et al., "NMR Fragment Screening Hit Induces Plasticity of BRD7/9 Bromodomains," *Chembiochem.* 17(15):1456-63 (2016).
- Z.X. Wang, "An exact mathematical expression for describing competitive binding of two different ligands to a protein molecule," *FEBS Lett.* 360(2):111-114 (Feb. 1995).
- Zoppi et al., "Iterative Design and Optimization of Initially Inactive Proteolysis Targeting Chimeras (PROTACs) Identify VZ185 as a Potent, Fast, and Selective von Hippel-Lindau (VHL) Based Dual Degradator Probe of BRD9 and BRD7," *J Med Chem.* 62(2):699-726 (Jan. 2019).

* cited by examiner

FIG. 1



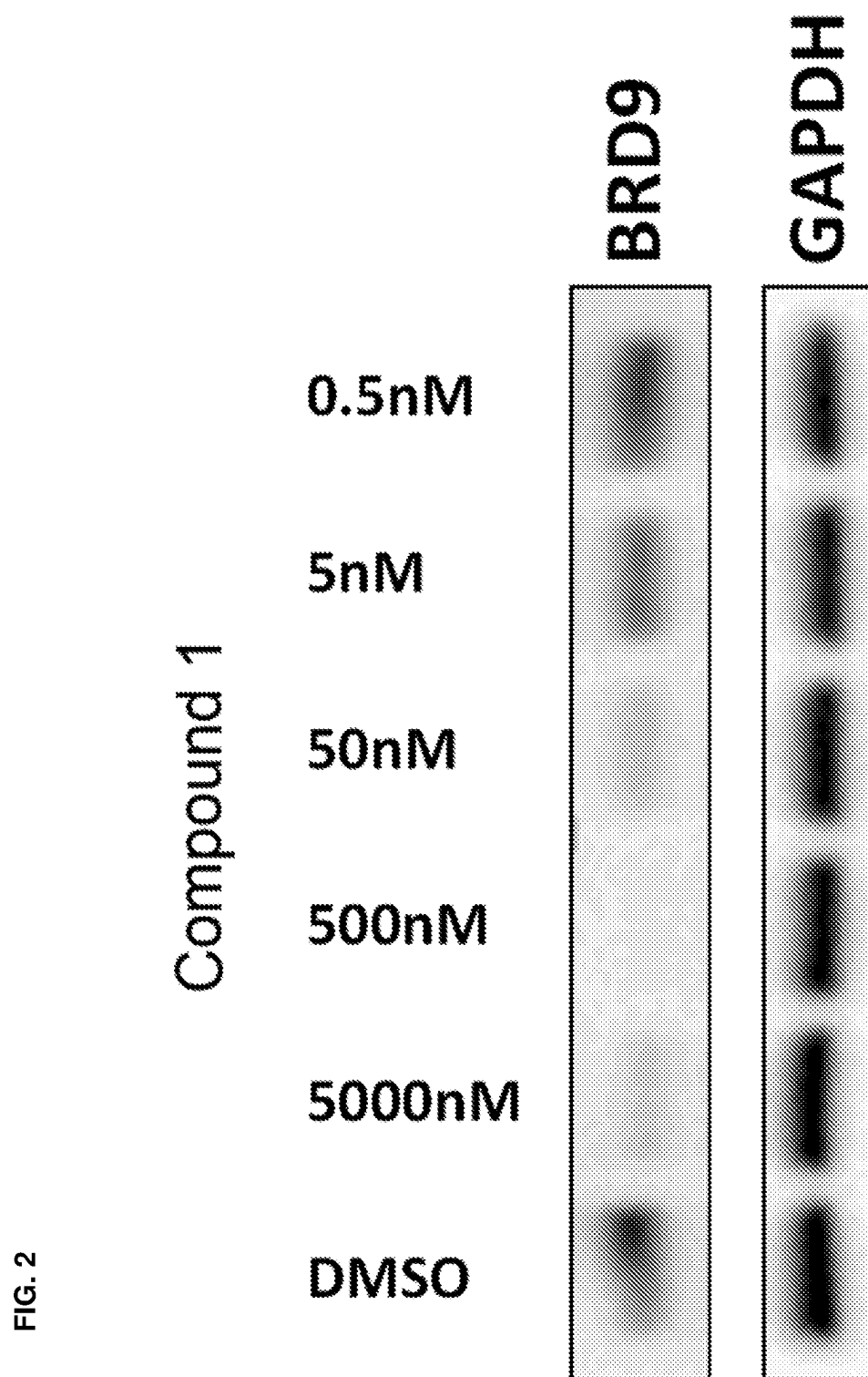
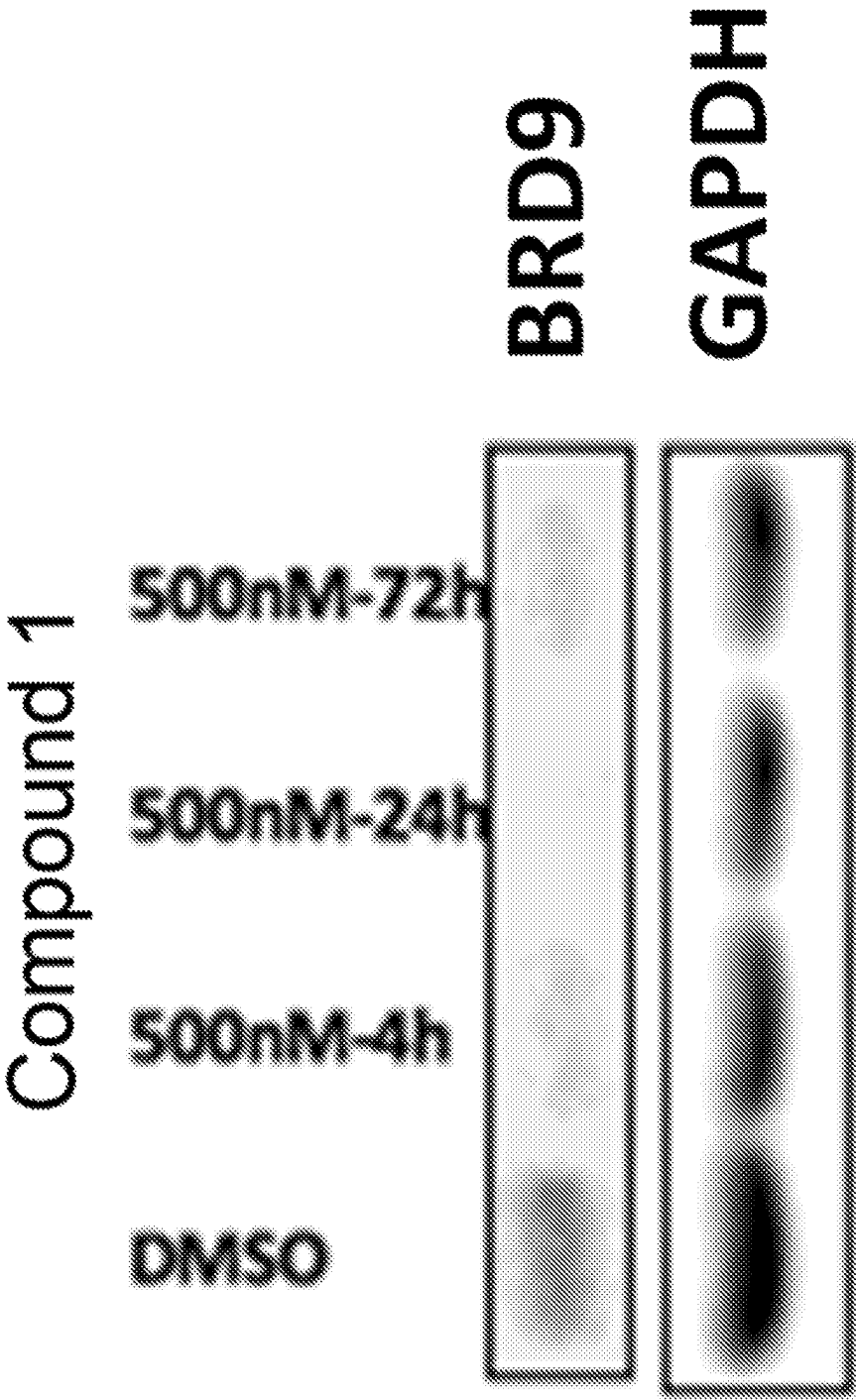


FIG. 3



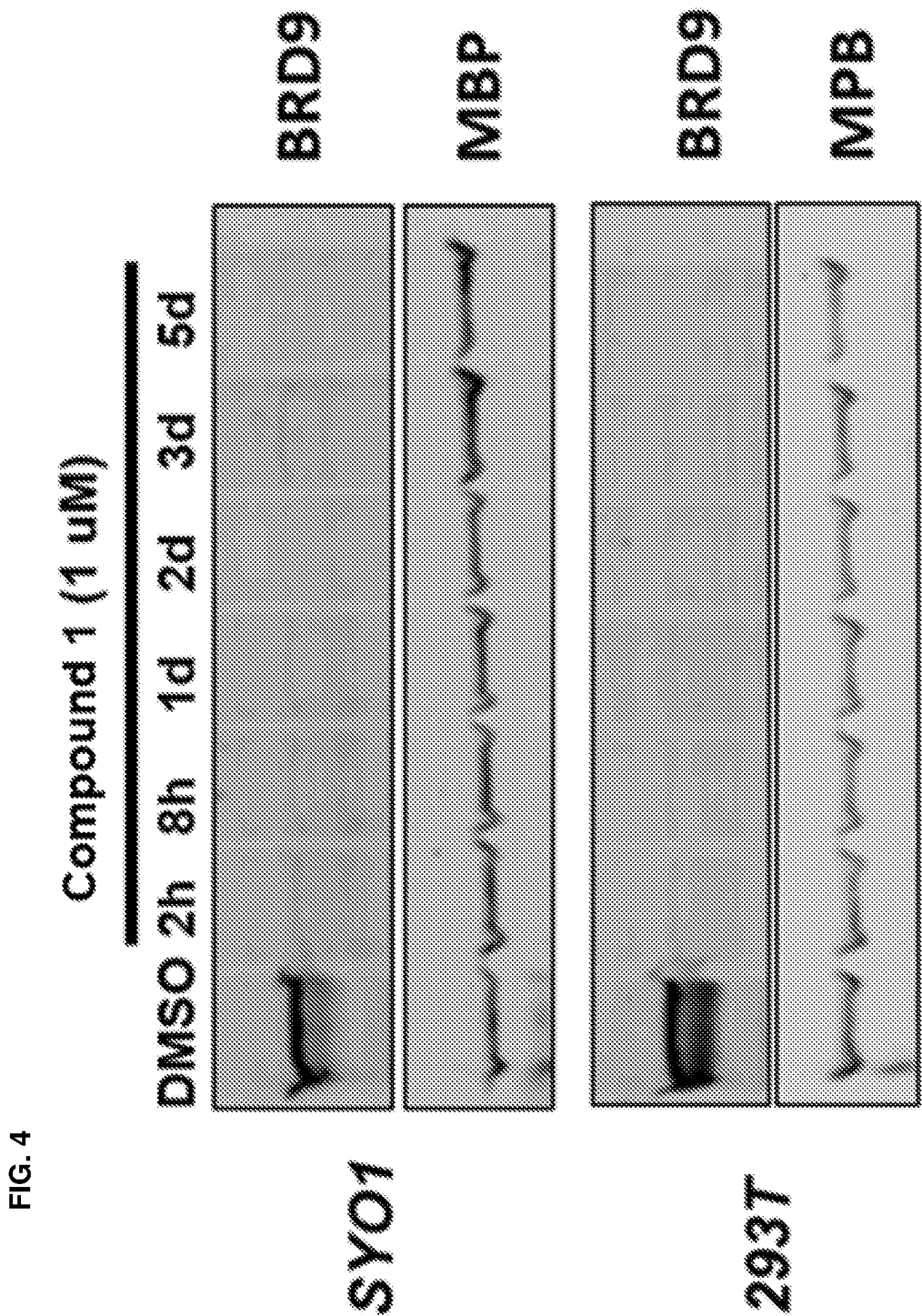
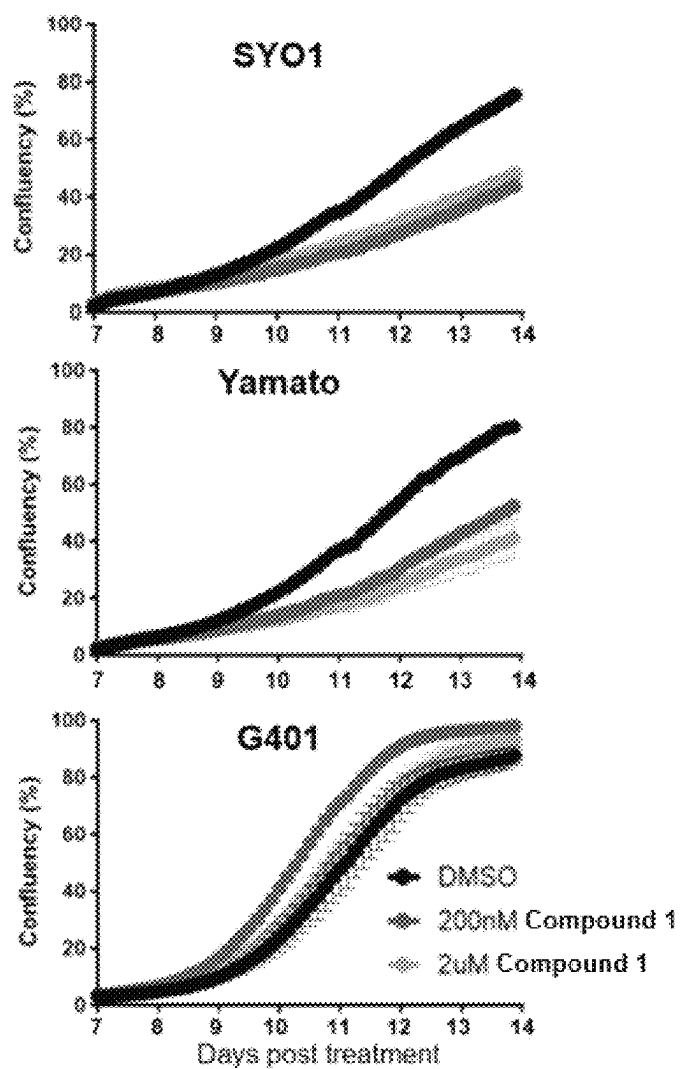


FIG. 5



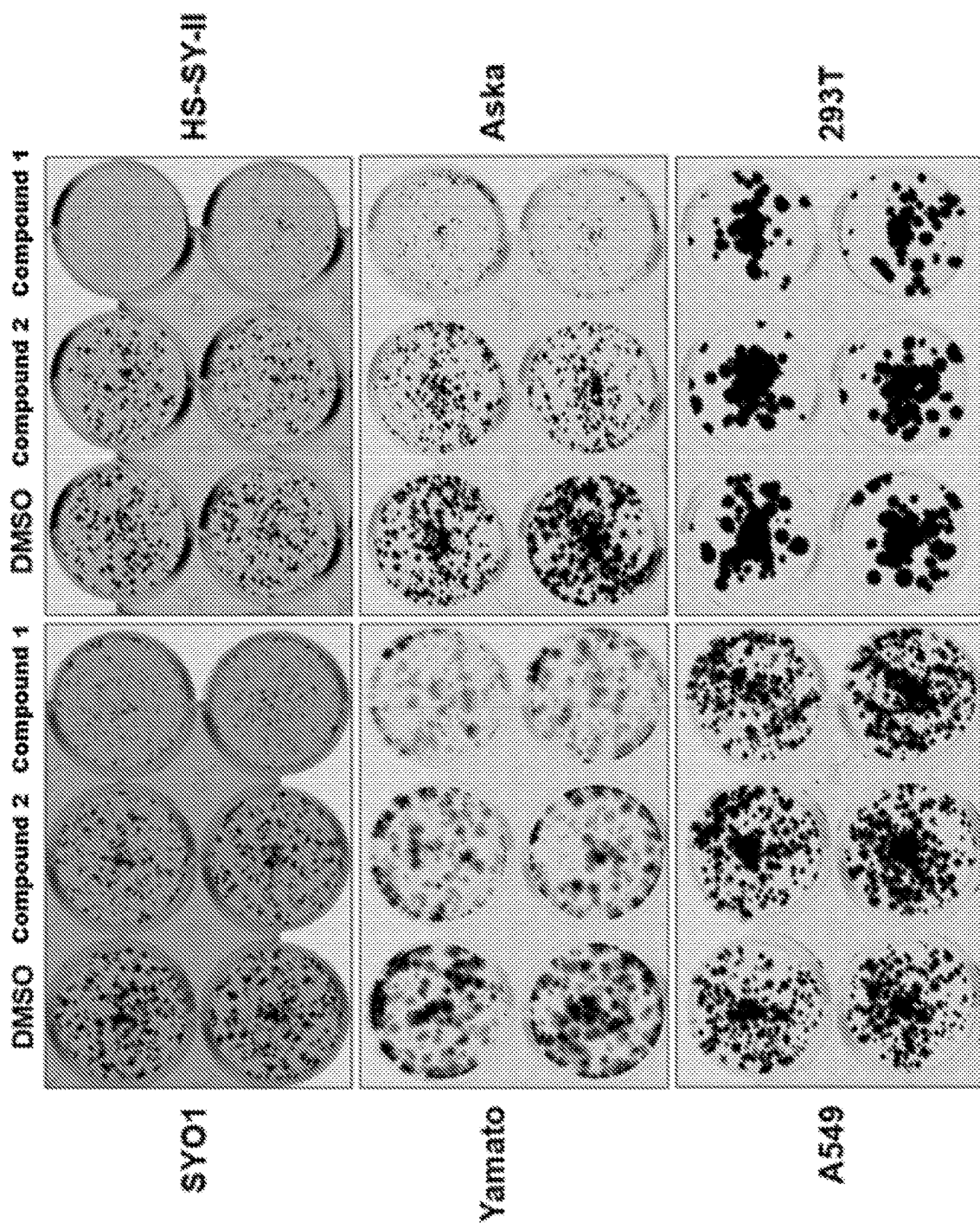


FIG. 6

FIG. 7

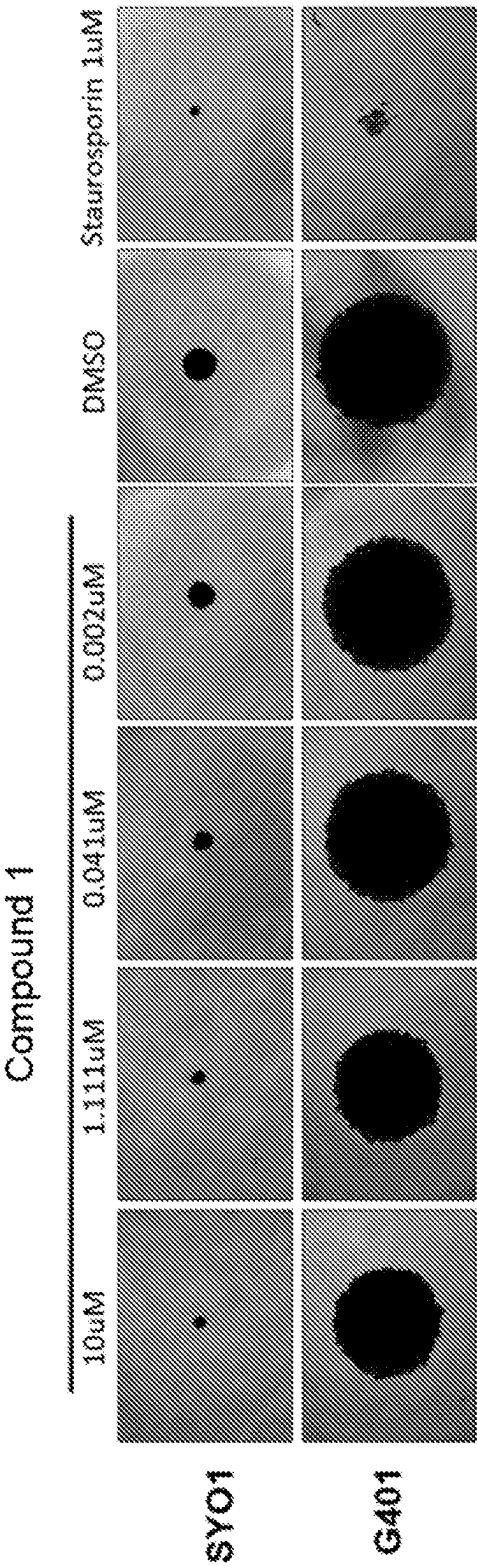
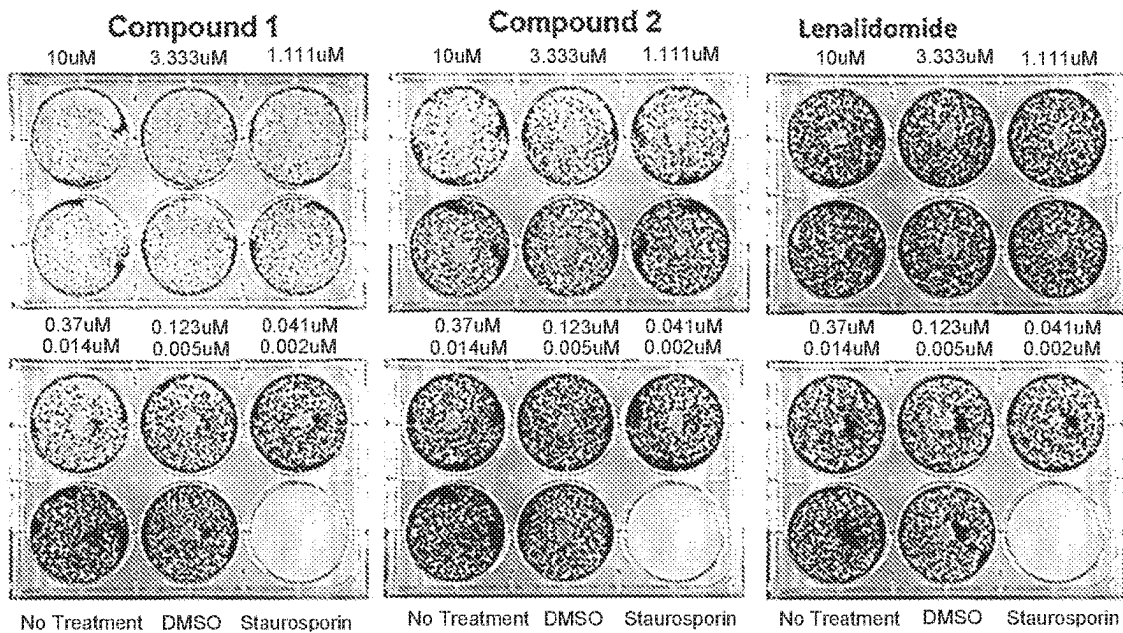
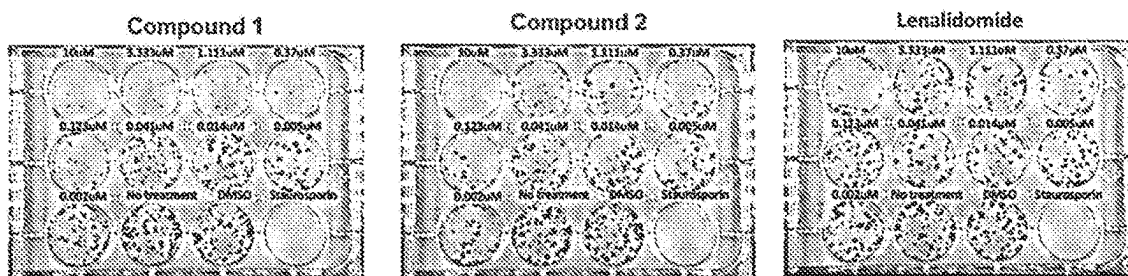


FIG. 8

SYO1



HS-SY-II



ASKA

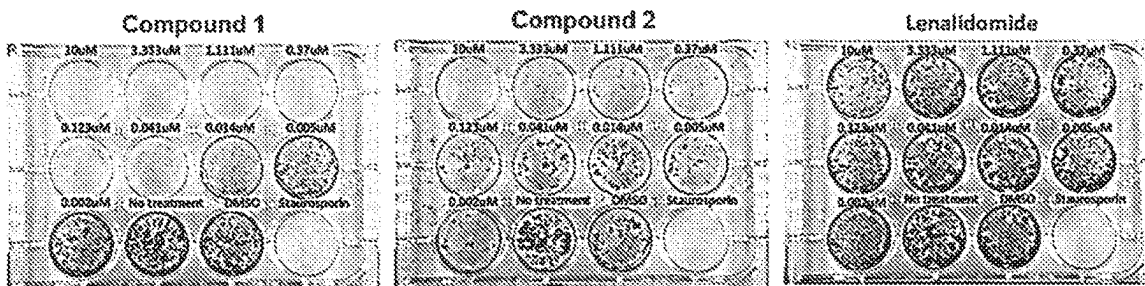
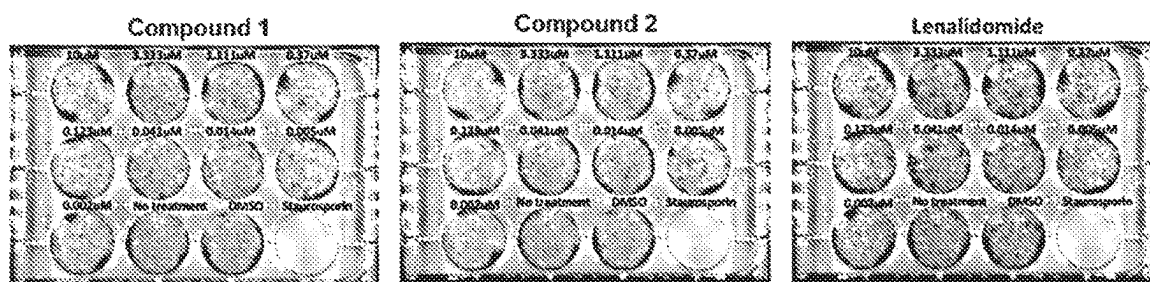
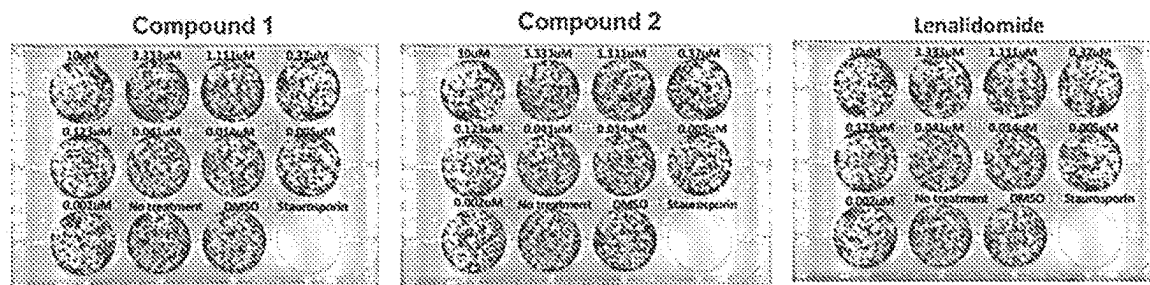


FIG. 9

RD



HCT116



Calu6

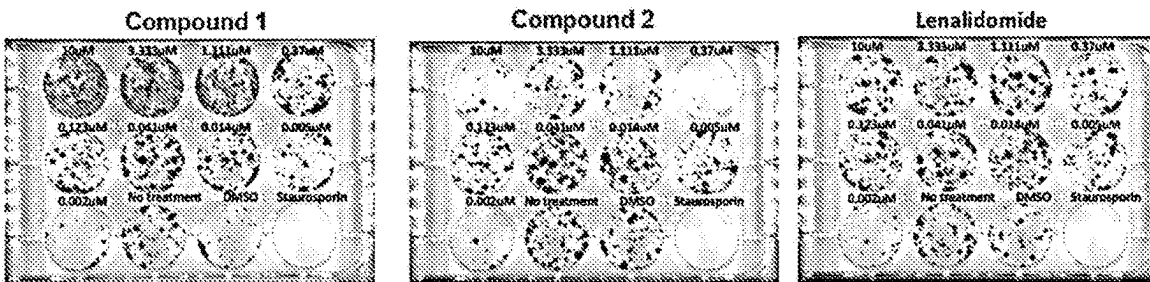


FIG. 10

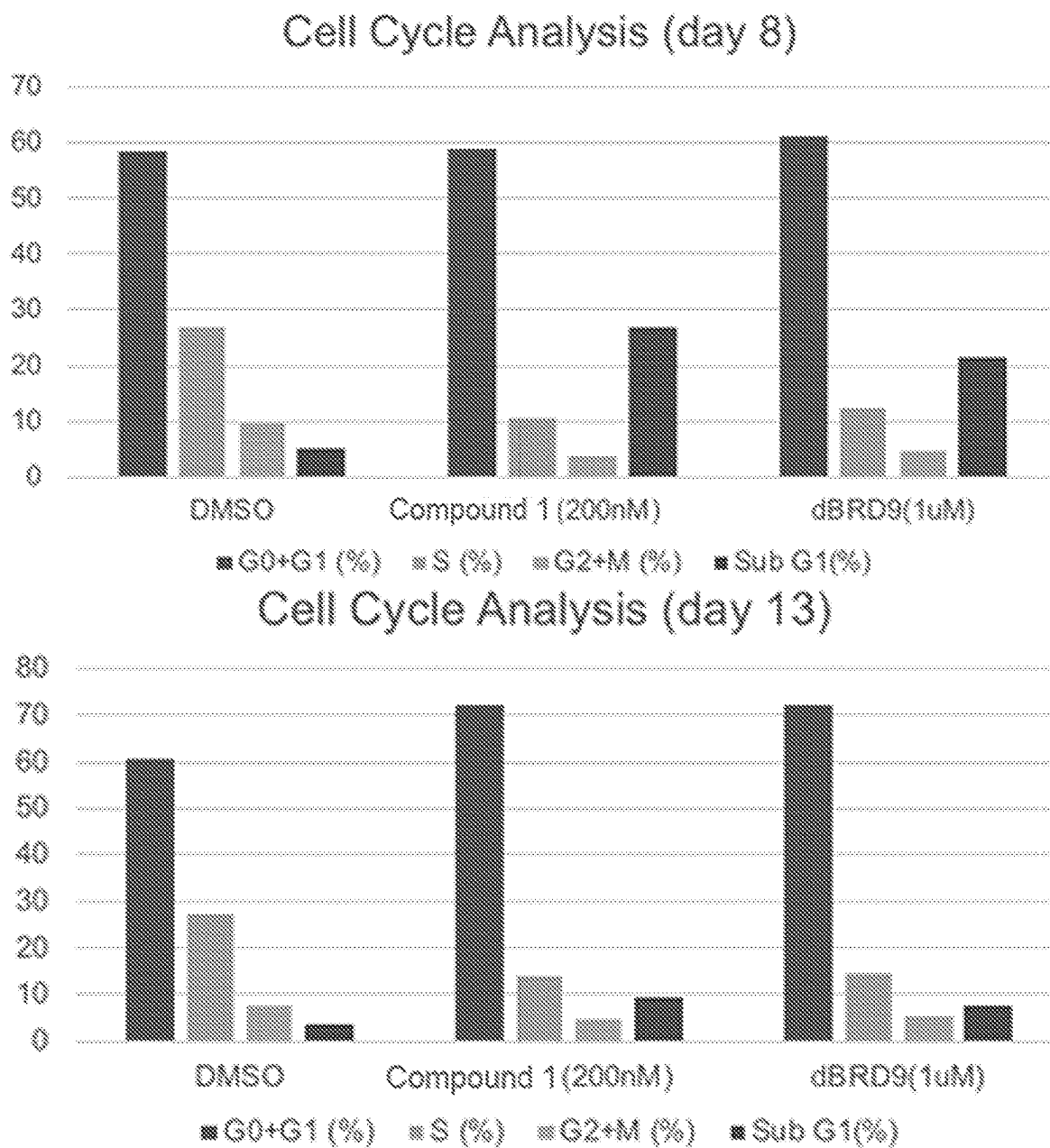


FIG. 11

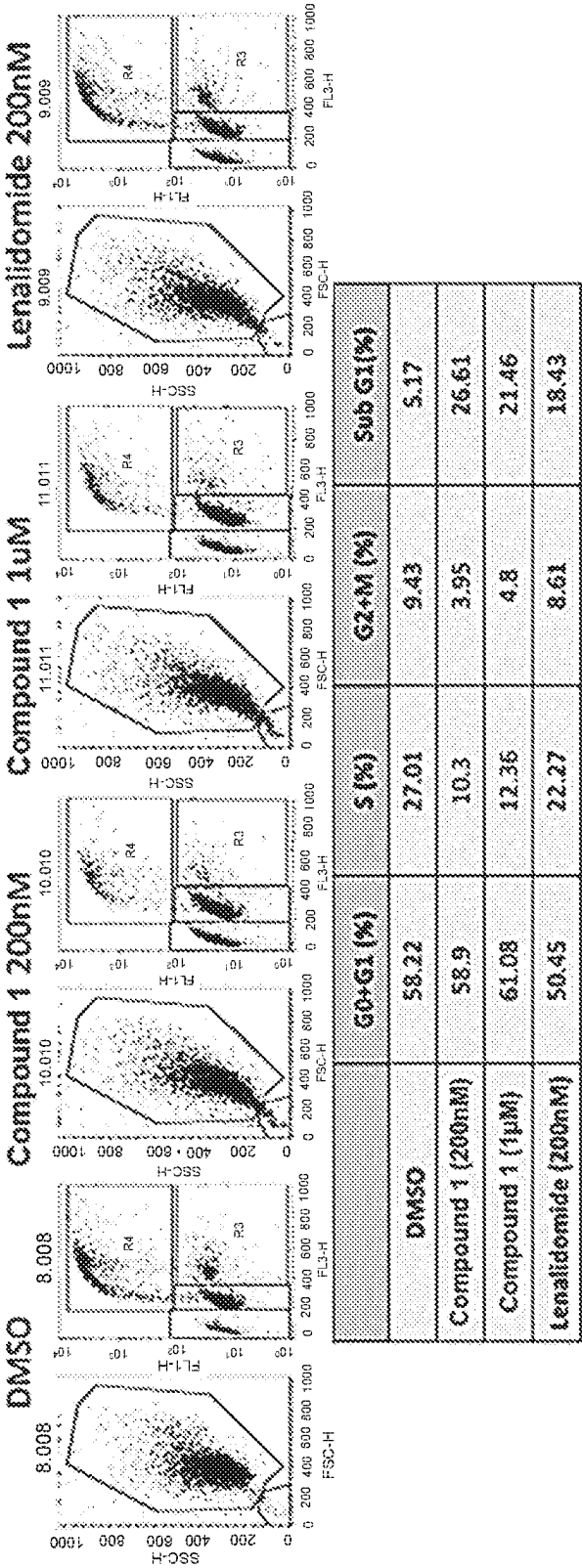
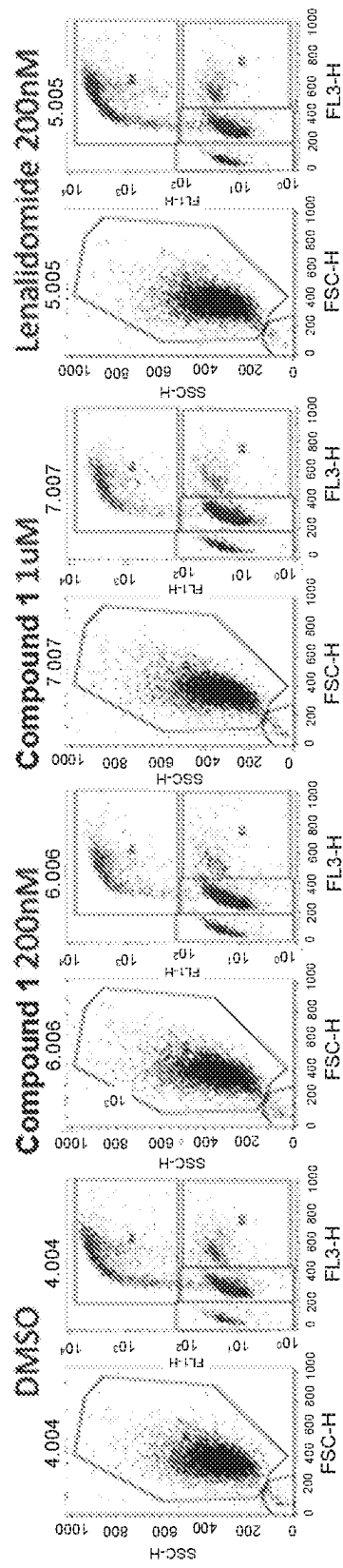
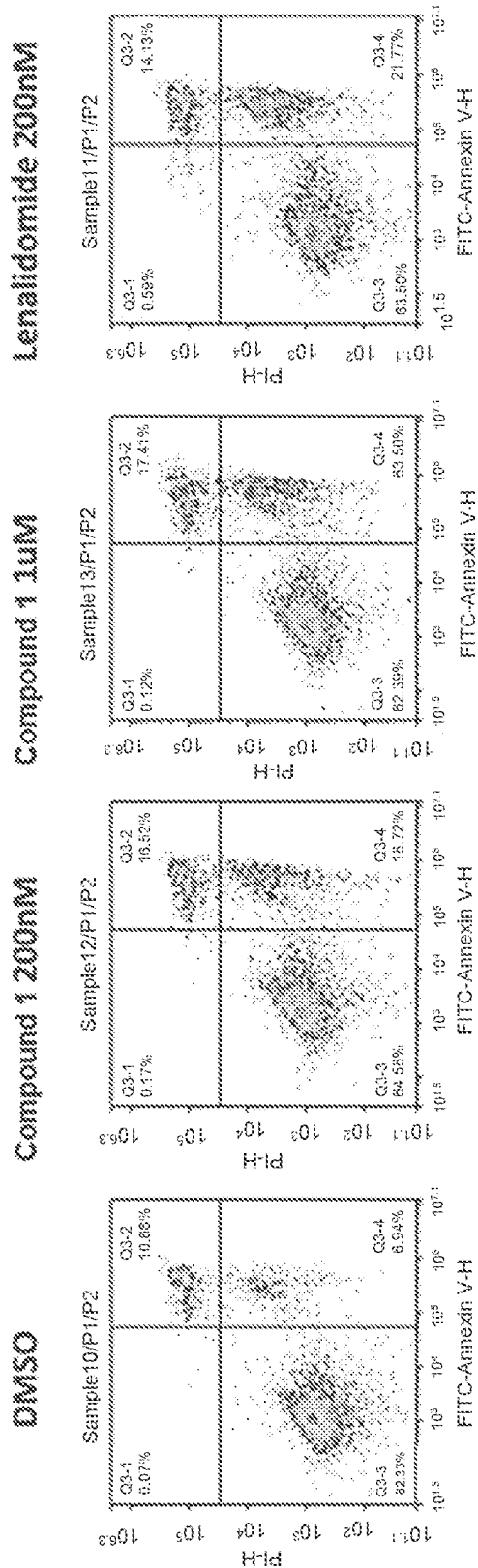


FIG. 12

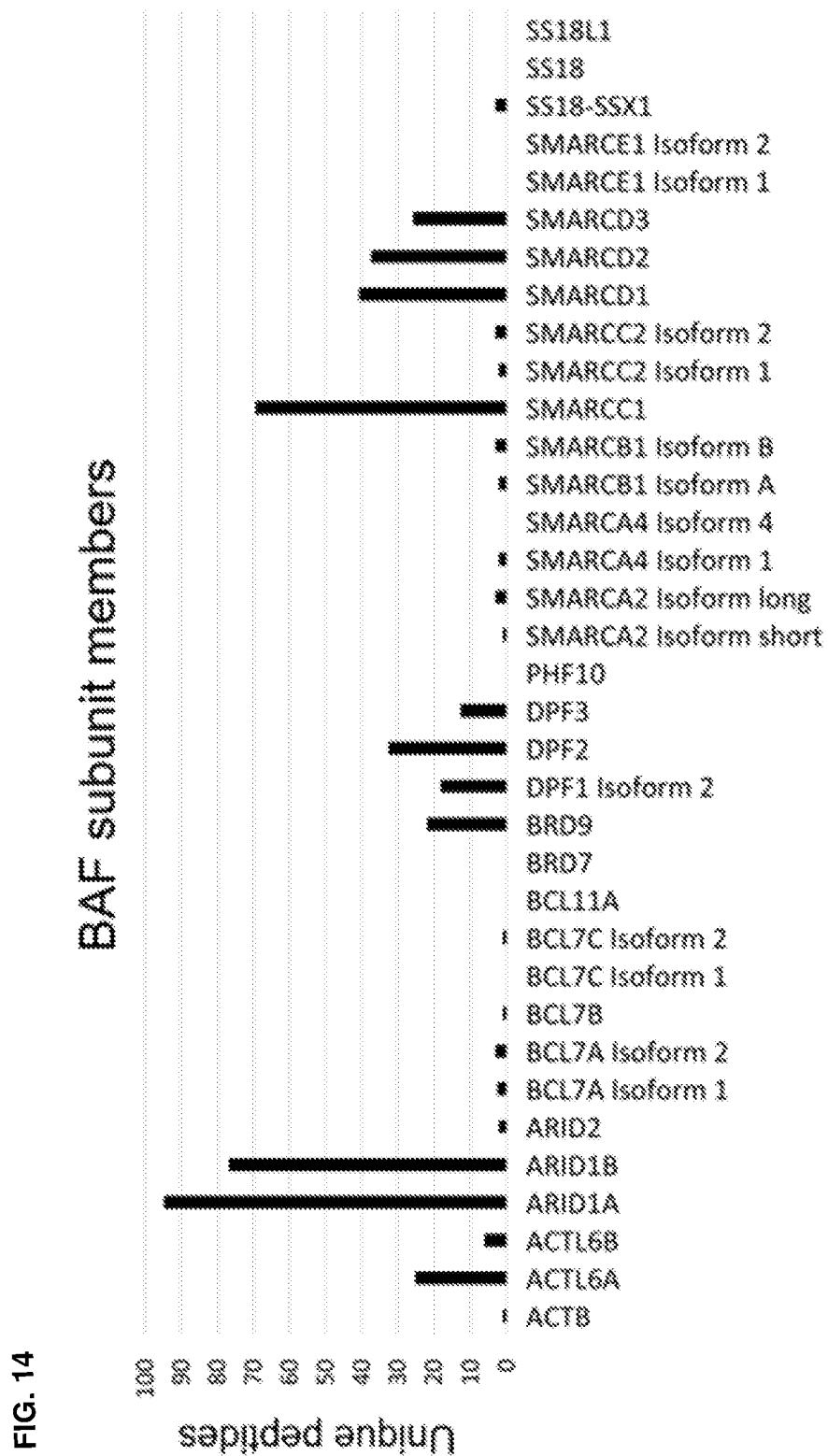


	G0+G1 (%)	S (%)	G2+M (%)	Sub G1 (%)
DMSO	60.83	27.29	7.53	3.88
Compound 1 (200nM)	71.9	14.06	4.76	9.1
Compound 1 (1uM)	72.21	14.53	5.08	7.91
Lenalidomide(200nM)	58.67	27.28	6.71	7.11

FIG. 13



	DMSO	Compound 1 (200nM)	Compound 1 (1uM)	Lenalidomide(200nM)
Early Apoptosis Cell	6.94	18.72	20.08	21.77
Late Apoptosis Cell	10.66	16.52	17.41	14.13



1

COMPOUNDS AND USES THEREOF

BACKGROUND

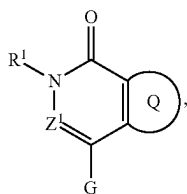
Disorders can be affected by the BAF complex. BRD9 is a component of the BAF complex. The present invention relates to useful compositions and methods for the treatment of BAF complex-related disorders, such as cancer and infection.

SUMMARY

Bromodomain-containing protein 9 (BRD9) is a protein encoded by the BRD9 gene on chromosome 5. BRD9 is a component of the BAF (BRG1- or BRM-associated factors) complex, a SWI/SNF ATPase chromatin remodeling complex, and belongs to family IV of the bromodomain-containing proteins. BRD9 is present in several SWI/SNF ATPase chromatin remodeling complexes and is upregulated in multiple cancer cell lines. Accordingly, agents that reduce the levels and/or activity of BRD9 may provide new methods for the treatment of disease and disorders, such as cancer and infection. The inventors have found that depleting BRD9 in cells results in the depletion of the SS18-SSX fusion protein in those cells. The SS18-SSX fusion protein has been detected in more than 95% of synovial sarcoma tumors and is often the only cytogenetic abnormality in synovial sarcoma. Additionally, evidence suggests that the BAF complex is involved in cellular antiviral activities. Thus, agents that degrade BRD9 (e.g., compounds) are useful in the treatment of disorders (e.g., cancers or infections) related to BAF, BRD9, and/or SS18-SSX.

The present disclosure features compounds and methods useful for treating BAF-related disorders (e.g., cancer or infection).

In an aspect, the disclosure features a compound having the structure of Formula I:



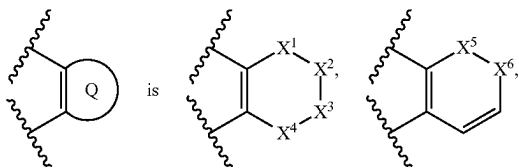
Formula I

where

R¹ is H, optionally substituted C₁-C₆ alkyl, optionally substituted C₂-C₆ alkenyl, optionally substituted C₁-C₆ heteroalkyl, or optionally substituted C₃-C₁₀ carbocyclyl;

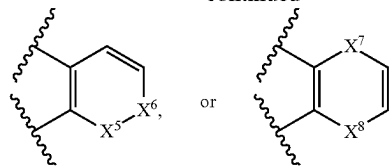
Z¹ is CR² or N;

R² is H, halogen, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted C₃-C₁₀ carbocyclyl, optionally substituted C₂-C₉ heterocyclyl, optionally substituted C₆-C₁₀ aryl, or optionally substituted C₂-C₉ heteroaryl;

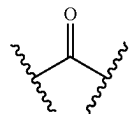


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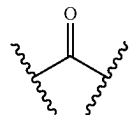
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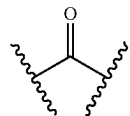
X¹ is a bond, O, NR^{3a},



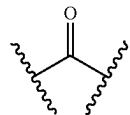
or CR^{4a}R^{5a},
X² is O, NR^{3b},



or CR^{4b}R^{5b},
X³ is O, NR^{3c},



or CR^{4c}R^{5c},
X⁴ is a bond, O, NR^{3d},



or CR^{4d}R^{5d},
X⁵ is O or NR^{3e} and X⁶ is CR^{4f}R^{5f}, or X⁵ is CR^{4e}R^{5e} and X⁶ is O or NR^{3f};

X⁷ is O, NR^{3g}, or CR^{4g}R^{5g};

X⁸ is O, NR^{3h}, or CR^{4h}R^{5h};

each of R^{3a}, R^{3b}, R^{3c}, and R^{3d} is, independently, H, halogen, hydroxyl, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted C₃-C₁₀ carbocyclyl, optionally substituted C₂-C₉ heterocyclyl, optionally substituted C₆-C₁₀ aryl, optionally substituted C₂-C₉ heteroaryl, optionally substituted C₂-C₆ alkenyl, optionally substituted C₂-C₆ heteroalkenyl, optionally substituted C₁-C₆ acyl, thiol, optionally substituted sulfone, optionally substituted sulfonamide, or optionally substituted amino, or R^{3a} and R^{4b}, R^{4a} and R^{3b}, R^{4a} and R^{4b}, R^{3b} and R^{4c}, R^{4b} and R^{4c}, R^{3c} and R^{4d}, R^{4c} and R^{4d}, and/or R^{3d} and R^{4c}, together with the atoms to which each is attached, combine to form optionally substituted C₂-C₉ heterocyclyl;

3

each of R^{4a} , R^{4b} , R^{4c} , and R^{4d} is, independently, H, halogen, hydroxyl, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, optionally substituted C_1 - C_6 acyl, thiol, optionally substituted sulfone, or optionally substituted amino, or R^{3a} and R^{4b} , R^{4a} and R^{3b} , R^{4b} and R^{4a} , R^{3b} and R^{4c} , R^{4b} and R^{4c} , R^{3c} and R^{4b} , R^{3c} and R^{4d} , R^{4c} and R^{4d} , and/or R^{3d} and R^{4c} , together with the atoms to which each is attached, combine to form optionally substituted C_2 - C_9 heterocyclyl;

each of R^{5a} , R^{5b} , R^{5c} , and R^{5d} is, independently, H, halogen, hydroxyl, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, hydroxyl, thiol, or optionally substituted amino;

each of R^{3e} , R^{3f} , R^{3g} , and R^{3h} is, independently, H, halogen, hydroxyl, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, optionally substituted C_1 - C_6 acyl, thiol, optionally substituted sulfone, optionally substituted sulfonamide, or optionally substituted amino, or R^{3e} and R^{4f} or R^{4e} and R^{3f} , together with the atoms to which each is attached, combine to form optionally substituted heterocyclyl;

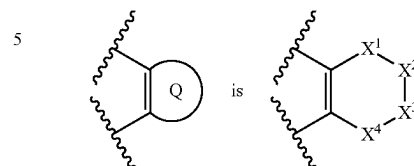
each of R^{4e} , R^{4f} , R^{4g} , and R^{4h} is, independently, H, halogen, hydroxyl, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, optionally substituted C_1 - C_6 acyl, thiol, optionally substituted sulfone, optionally substituted sulfonamide, or optionally substituted amino, or R^{3e} and R^{4f} or R^{4e} and R^{3f} , together with the atoms to which each is attached, combine to form optionally substituted heterocyclyl;

each of R^{5e} , R^{5f} , R^{5g} , and R^{5h} is, independently, H, halogen, hydroxyl, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, hydroxyl, thiol, or optionally substituted amino; and

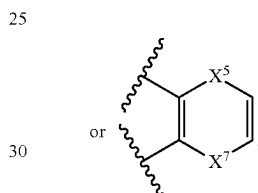
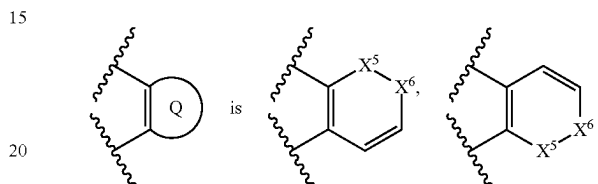
G is optionally substituted C_6 - C_{10} aryl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heteroaryl, or C_2 - C_9 heterocyclyl, or a pharmaceutically acceptable salt thereof.

4

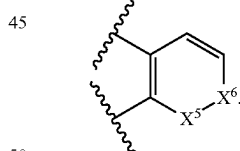
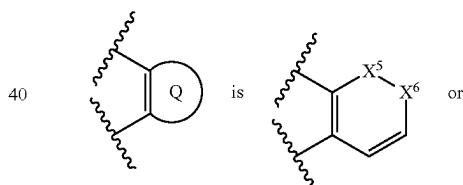
In some embodiments,



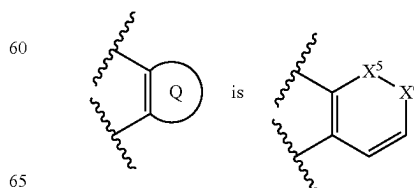
In some embodiments,



In some embodiments,

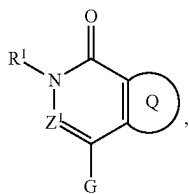


In some embodiments, is



5

In another aspect, the disclosure features a compound having the structure of Formula I:



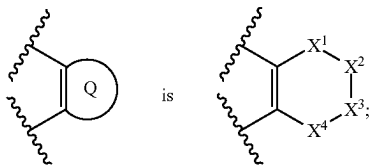
Formula I 5

where

R¹ is H, optionally substituted C₁-C₆ alkyl, optionally substituted C₂-C₆ alkenyl, optionally substituted C₁-C₆ heteroalkyl, or optionally substituted C₃-C₁₀ carbocyclyl;

Z¹ is CR² or N;

R² is H, halogen, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted C₃-C₁₀ carbocyclyl, optionally substituted C₂-C₉ heterocyclyl, optionally substituted C₆-C₁₀ aryl, or optionally substituted C₂-C₉ heteroaryl;



X¹ is a bond, O, NR^{3a}, or CR^{4a}R^{5a};

X² is O, NR^{3b}, or CR^{4b}R^{5b};

X³ is O, NR^{3c}, or CR^{4c}R^{5c};

X⁴ is a bond, O, NR^{3d}, or CR^{4d}R^{5d};

each of R^{3a}, R^{3b}, R^{3c}, and R^{3d} is, independently, H, halogen, hydroxyl, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted C₃-C₁₀ carbocyclyl, optionally substituted C₂-C₉ heterocyclyl, optionally substituted C₆-C₁₀ aryl, optionally substituted C₂-C₉ heteroaryl, optionally substituted C₂-C₆ alkenyl, optionally substituted C₂-C₆ heteroalkenyl, optionally substituted C₁-C₆ acyl, thiol, optionally substituted sulfone, optionally substituted sulfonamide, or optionally substituted amino, or R^{3a} and R^{4b}, R^{4a} and R^{3b}, R^{4b} and R^{4a}, R^{3b} and R^{4c}, R^{4b} and R^{4c}, R^{3c} and R^{4b}, R^{3c} and R^{4d}, R^{4c} and R^{4d}, and/or R^{3d} and R^{4c}, together with the atoms to which each is attached, combine to form optionally substituted C₂-C₉ heterocyclyl;

each of R^{4a}, R^{4b}, R^{4c}, and R^{4d} is, independently, H, halogen, hydroxyl, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted C₃-C₁₀ carbocyclyl, optionally substituted C₂-C₉ heterocyclyl, optionally substituted C₆-C₁₀ aryl, optionally substituted C₂-C₉ heteroaryl, optionally substituted C₂-C₆ alkenyl, optionally substituted C₂-C₆ heteroalkenyl, optionally substituted C₁-C₆ acyl, thiol, optionally substituted sulfone, or optionally substituted amino, or R^{3a} and R^{4b}, R^{4a} and R^{3b}, R^{4b} and R^{4a}, R^{3b} and R^{4c}, R^{4b} and R^{4c}, R^{3c} and R^{4b}, R^{3c} and R^{4d}, R^{4c} and R^{4d}, and/or R^{3d} and R^{4c}, together with the atoms to

6

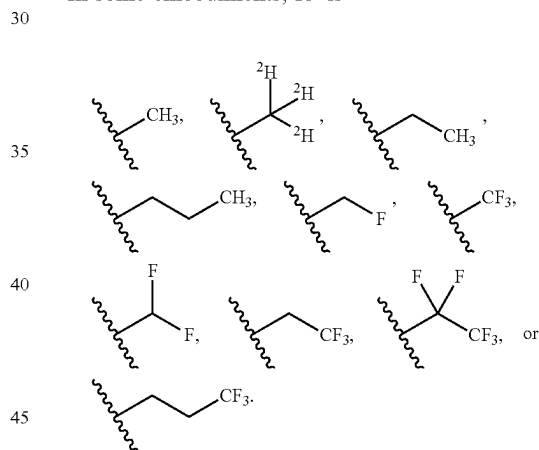
which each is attached, combine to form optionally substituted C₂-C₉ heterocyclyl; each of R^{5a}, R^{5b}, R^{5c}, and R^{5d} is, independently, H, halogen, hydroxyl, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted C₃-C₁₀ carbocyclyl, optionally substituted C₂-C₉ heterocyclyl, optionally substituted C₆-C₁₀ aryl, optionally substituted C₂-C₉ heteroaryl, optionally substituted C₂-C₆ alkenyl, optionally substituted C₂-C₆ heteroalkenyl, hydroxyl, thiol, or optionally substituted amino; and G is optionally substituted C₆-C₁₀ aryl, optionally substituted C₃-C₁₀ carbocyclyl, optionally substituted C₂-C₉ heteroaryl, or C₂-C₉ heterocyclyl, or a pharmaceutically acceptable salt thereof.

In some embodiments, R¹ is H, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, or optionally substituted C₃-C₁₀ carbocyclyl. In some embodiments, R¹ is H, optionally substituted C₁-C₆ alkyl, optionally substituted C₂-C₆ alkenyl, or optionally substituted C₃-C₁₀ carbocyclyl. In some embodiments, R¹ is H, optionally substituted C₁-C₆ alkyl, or optionally substituted C₃-C₁₀ carbocyclyl.

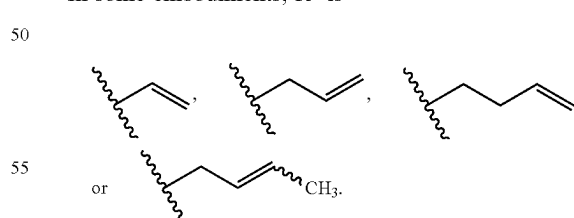
In some embodiments, R¹ is H. In some embodiments, R¹ is optionally substituted C₁-C₆ alkyl. In some embodiments, R¹ is optionally substituted C₂-C₆ alkenyl. In some embodiments, R¹ is optionally substituted C₃-C₁₀ carbocyclyl.

In some embodiments, optionally substituted C₁-C₆ alkyl is C₁-C₆ perfluoroalkyl.

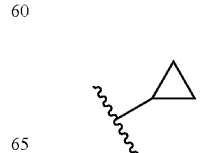
In some embodiments, R¹ is



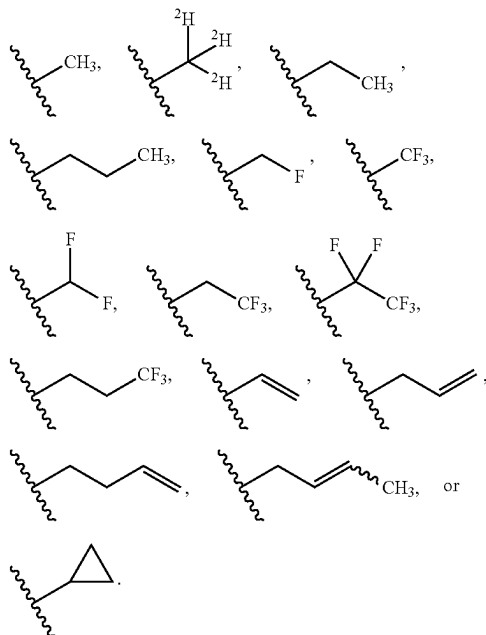
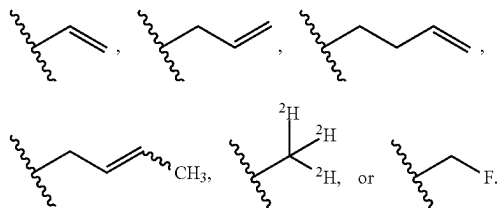
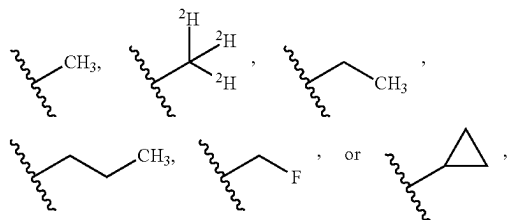
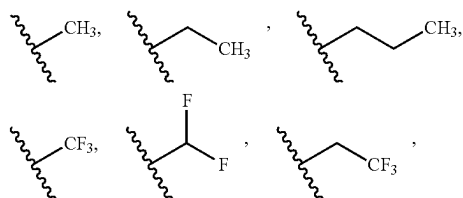
In some embodiments, R¹ is



In some embodiments, R¹ is

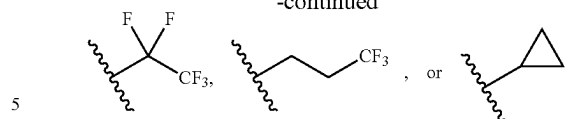
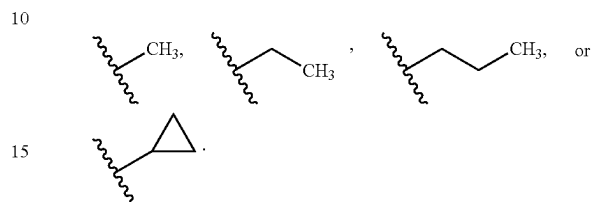
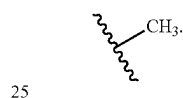


7

In some embodiments, R^1 is H,In some embodiments, R^1 isIn some embodiments, R^1 is H,In some embodiments, R^1 is H,

8

-continued

In some embodiments, R^1 is H,In some embodiments, R^1 is H orIn some embodiments, R^1 is H. In some embodiments, R^1 isIn some embodiments, Z^1 is CR^2 . In some embodiments, Z^1 is N.In some embodiments, R^2 is H, halogen, optionally substituted C_1 - C_6 alkyl, optionally substituted C_3 - C_{10} carbocycl, or optionally substituted C_6 - C_{10} aryl.In some embodiments, R^2 is H, halogen, or optionally substituted C_1 - C_6 alkyl.In some embodiments, R^2 is H, F, orIn some embodiments, R^2 is H. In some embodiments, R^2 is F. In some embodiments, R^2 isIn some embodiments, X^1 is a bond, O, NR^{3a} , or $CR^{4a}R^{5a}$; X^2 is O, NR^{3b} , or $CR^{4b}R^{5b}$; X^3 is O, NR^{3c} , or $CR^{4c}R^{5c}$; and X^4 is a bond, O, NR^{3d} , or $CR^{4d}R^{5d}$.In some embodiments, X^1 is a bond. In some embodiments, X^1 is O, NR^{3a} , or $CR^{4a}R^{5a}$. In some embodiments, X^1 is O or NR^{3a} . In some embodiments, X^1 is NR^{3a} or $CR^{4a}R^{5a}$. In some embodiments, X^2 is O or NR^{3b} . In some embodiments, X^2 is $CR^{4b}R^{5b}$. In some embodiments, X^2 is NR^{3b} or $CR^{4b}R^{5b}$.

9

In some embodiments, X^3 is O or NR^{3c} . In some embodiments, X^3 is $CR^{4c}R^{5c}$. In some embodiments, X^3 is NR^{3c} or $CR^{4c}R^{5c}$.

In some embodiments, X^4 is a bond. In some embodiments, X^4 is O, NR^{3d} , or $CR^{4d}R^{5d}$. In some embodiments, X^4 is O or NR^{3d} . In some embodiments, X^4 is NR^{3d} or $CR^{4d}R^{5d}$.

In some embodiments, X^1 is O, NR^{3a} , or $CR^{4a}R^{5a}$; X^2 is O, NR^{3b} , or $CR^{4b}R^{5b}$; X^3 is O, NR^{3c} , or $CR^{4c}R^{5c}$; and X^4 is O, NR^{3d} , or $CR^{4d}R^{5d}$.

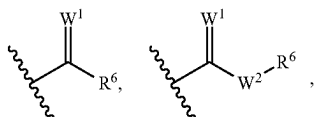
In some embodiments, X^1 is $CR^{4a}R^{5a}$; X^2 is NR^{3b} ; X^3 is $CR^{4c}R^{5c}$; and X^4 is $CR^{4d}R^{5d}$. In some embodiments, X^1 is $CR^{4a}R^{5a}$; X^2 is $CR^{4b}R^{5b}$; X^3 is NR^{3c} ; and X^4 is $CR^{4d}R^{5d}$. In some embodiments, X^1 is O or NR^{3a} ; X^2 is $CR^{4b}R^{5b}$; X^3 is $CR^{4c}R^{5c}$; and X^4 is O or NR^{3d} . In some embodiments, X^1 is a bond; X^2 is $CR^{4b}R^{5b}$; X^3 is O or NR^{3c} ; and X^4 is $CR^{4d}R^{5d}$. In some embodiments, X^1 is $CR^{4a}R^{5a}$; X^2 is $CR^{4b}R^{5b}$; X^3 is $CR^{4c}R^{5c}$; and X^4 is $CR^{4d}R^{5d}$. In some embodiments, X^5 is NR^{3e} and X^6 is $CR^{4f}R^{5f}$.

In some embodiments, each of R^{3a} , R^{3b} , R^{3c} , and R^{3d} is, independently, H, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_1 - C_6 acyl, optionally substituted sulfone, or optionally substituted sulfonamide. In some embodiments, each of R^{3a} , R^{3b} , R^{3c} , and R^{3d} is, independently, H, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted sulfone, or optionally substituted sulfonamide. In some embodiments, each of R^{3a} , R^{3b} , R^{3c} , and R^{3d} is, independently, H, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 acyl, optionally substituted sulfone, or optionally substituted sulfonamide. In some embodiments, each of R^{3a} , R^{3b} , R^{3c} , and R^{3d} is, independently, H, optionally substituted C_1 - C_6 alkyl, or optionally substituted C_1 - C_6 acyl.

In some embodiments, each of R^{3a} , R^{3b} , R^{3c} , and R^{3d} is, independently, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_1 - C_6 acyl, optionally substituted sulfone, or optionally substituted sulfonamide. In some embodiments, each of R^{3a} , R^{3b} , R^{3c} , and R^{3d} is, independently, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 acyl, optionally substituted sulfone, or optionally substituted sulfonamide.

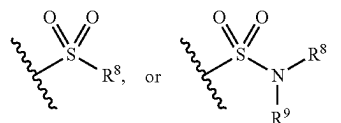
In some embodiments, each of R^{3a} , R^{3b} , R^{3c} , and R^{3d} is, independently, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, or optionally substituted C_1 - C_6 acyl. In some embodiments, each of R^{3a} , R^{3b} , R^{3c} , and R^{3d} is, independently, optionally substituted C_1 - C_6 alkyl or optionally substituted C_1 - C_6 acyl. In some embodiments, each of R^{3a} , R^{3b} , R^{3c} , and R^{3d} is, independently, optionally substituted C_1 - C_6 alkyl or optionally substituted C_1 - C_6 acyl. In some embodiments, each of R^{3a} , R^{3b} , R^{3c} , and R^{3d} is, independently, optionally substituted sulfone or optionally substituted sulfonamide.

In some embodiments, each of R^{3a} , R^{3b} , R^{3c} , and R^{3d} is, independently, H, optionally substituted C_1 - C_6 alkyl,



10

-continued



where

R^5 is H, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, or optionally substituted C_6 - C_{10} aryl;

W^1 is O or S;

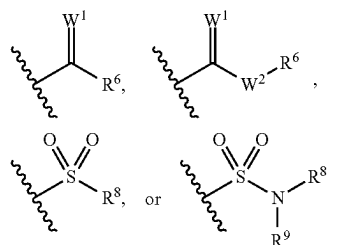
W^2 is NR^7 or O;

R^7 is H, optionally substituted C_1 - C_6 alkyl, or optionally substituted C_1 - C_6 heteroalkyl;

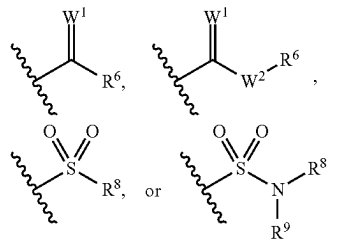
R^8 is optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, or optionally substituted C_6 - C_{10} aryl; and

R^9 is H, optionally substituted C_1 - C_6 alkyl, or optionally substituted C_1 - C_6 heteroalkyl.

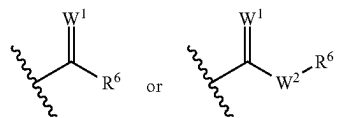
In some embodiments, each of R^{3a} , R^{3b} , R^{3c} , and R^{3d} is, independently, C_1 - C_6 alkyl,



In some embodiments, each of R^{3a} , R^{3b} , R^{3c} , and R^{3d} is, independently, C_1 - C_6 alkyl. In some embodiments, each of R^{3a} , R^{3b} , R^{3c} , and R^{3d} is, independently,

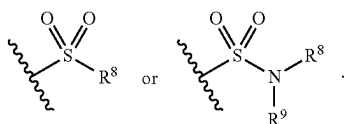


In some embodiments, each of R^{3a} , R^{3b} , R^{3c} , and R^{3d} is, independently,

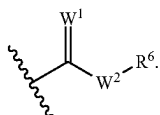


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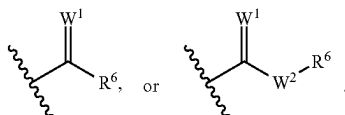
In some embodiments, each of R^{3a} , R^{3b} , R^{3c} , and R^{3d} is, independently,



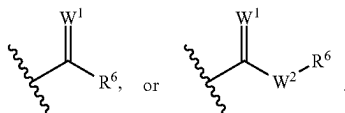
In some embodiments, each of R^{3a} , R^{3b} , R^{3c} , and R^{3d} is, independently,



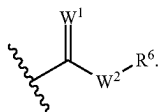
In some embodiments, each of R^{3a} , R^{3b} , R^{3c} , and R^{3d} is, independently, H, optionally substituted C_1 - C_6 alkyl,



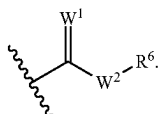
In some embodiments, each of R^{3a} , R^{3b} , R^{3c} , and R^{3d} is, independently, H, C_1 - C_6 alkyl,



In some embodiments, each of R^{3a} , R^{3b} , R^{3c} , and R^{3d} is, independently, H, C_1 - C_6 alkyl, or

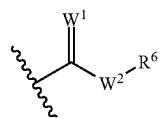


In some embodiments, each of R^{3a} , R^{3b} , R^{3c} , and R^{3d} is, independently, C_1 - C_6 alkyl



In some embodiments, each of R^{3a} , R^{3b} , R^{3c} , and R^{3d} is, independently, H, methyl, or

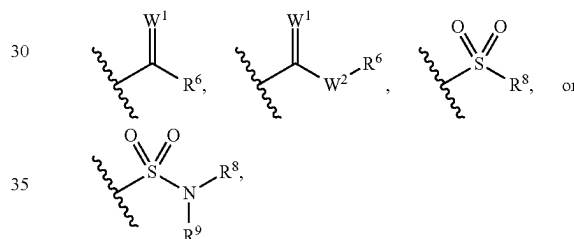
12



In some embodiments, each of R^{3a} , R^{3b} , R^{3c} , and R^{3d} is, independently, optionally substituted C_1 - C_6 alkyl.

In some embodiments, each of R^{4a} , R^{4b} , R^{4c} , and R^{4d} is, independently, H, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_1 - C_6 acyl, optionally substituted sulfone, or optionally substituted sulfonamide. In some embodiments, each of R^{4a} , R^{4b} , R^{4c} , and R^{4d} is, independently, H, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted sulfone, or optionally substituted sulfonamide. In some embodiments, each of R^{4a} , R^{4b} , R^{4c} , and R^{4d} is H. In some embodiments, each of R^{4a} , R^{4b} , R^{4c} , and R^{4d} is, independently, H, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 acyl, optionally substituted sulfone, or optionally substituted sulfonamide. In some embodiments, each of R^{4a} , R^{4b} , R^{4c} , and R^{4d} is, independently, H, optionally substituted C_1 - C_6 alkyl, or optionally substituted C_1 - C_6 acyl.

In some embodiments, each of R^{4a} , R^{4b} , R^{4c} , and R^{4d} is, independently, H, optionally substituted C_1 - C_6 alkyl,



where

R^6 is H, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, or optionally substituted C_6 - C_{10} aryl;

W^1 is O or S;

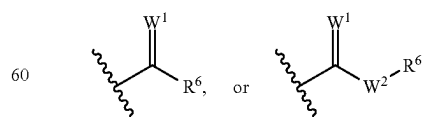
W^2 is NR^7 or O;

R^7 is H, optionally substituted C_1 - C_6 alkyl, or optionally substituted C_1 - C_6 heteroalkyl;

R^8 is optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, or optionally substituted C_6 - C_{10} aryl; and

R^9 is H, optionally substituted C_1 - C_6 alkyl, or optionally substituted C_1 - C_6 heteroalkyl.

In some embodiments, each of R^{4a} , R^{4b} , R^{4c} , and R^{4d} is, independently, H, optionally substituted C_1 - C_6 alkyl,



In some embodiments, each of R^{4a} , R^{4b} , R^{4c} , and R^{4d} is, independently, H or optionally substituted C_1 - C_6 alkyl. In some embodiments, each of R^{4a} , R^{4b} , R^{4c} , and R^{4d} is, independently, H.

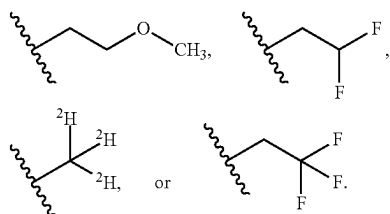
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In some embodiments, W^1 is O. In some embodiments, W^1 is S.

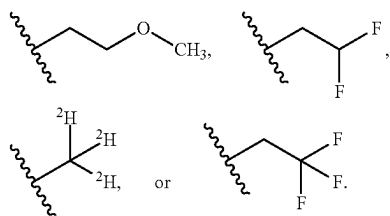
In some embodiments, W^2 is O. In some embodiments, W^2 is NR^7 .

In some embodiments, R^6 is H, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, or optionally substituted C_3 - C_{10} carbocyclyl. In some embodiments, R^6 is H, optionally substituted C_1 - C_6 alkyl, or optionally substituted C_1 - C_6 heteroalkyl. In some embodiments, R^6 is H, optionally substituted C_3 - C_{10} carbocyclyl, or optionally substituted C_6 - C_{10} aryl. In some embodiments, R^6 is optionally substituted C_1 - C_6 alkyl or optionally substituted C_1 - C_6 heteroalkyl. In some embodiments, R^6 is optionally substituted C_3 - C_{10} carbocyclyl or optionally substituted C_6 - C_{10} aryl

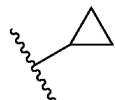
In some embodiments, R^6 is H, methyl, ethyl



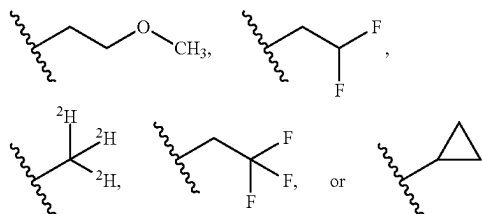
In some embodiments, R^6 is H. In some embodiments, R^6 is methyl, ethyl,



In some embodiments, R^6 is



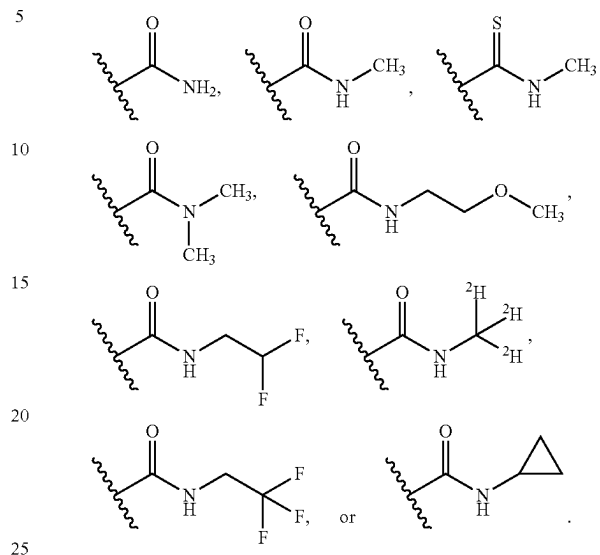
In some embodiments, R^6 is H, methyl, ethyl,



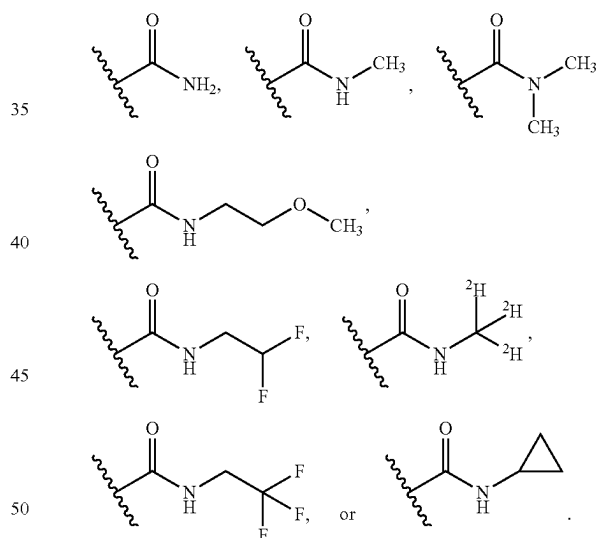
In some embodiments, R^7 is H or optionally substituted C_1 - C_6 alkyl. In some embodiments, R^7 is H or methyl.

14

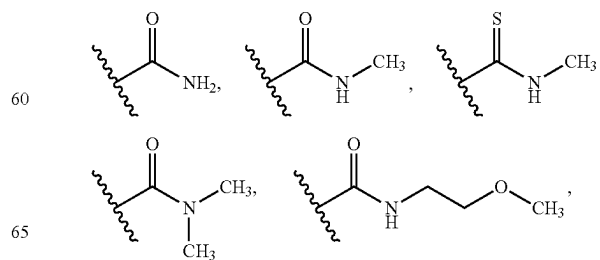
In some embodiments, each of R^{3a} , R^{3b} , R^{3c} , and R^{3d} is, independently, H, methyl,



In some embodiments, each of R^{3a} , R^{3b} , R^{3c} , and R^{3d} is, independently, H, methyl,

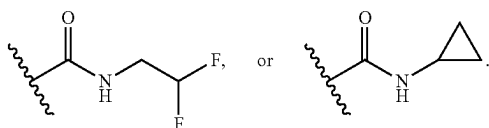


In some embodiments, R^{3a} is H, methyl,

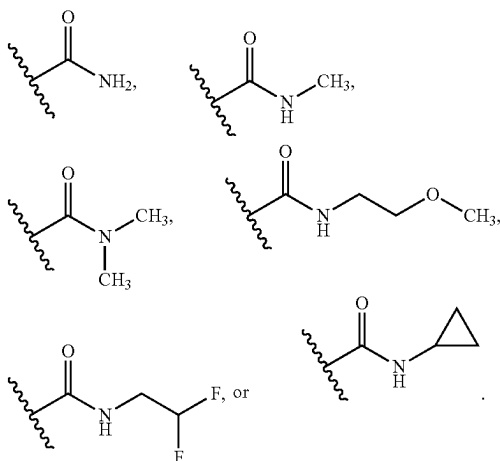


15

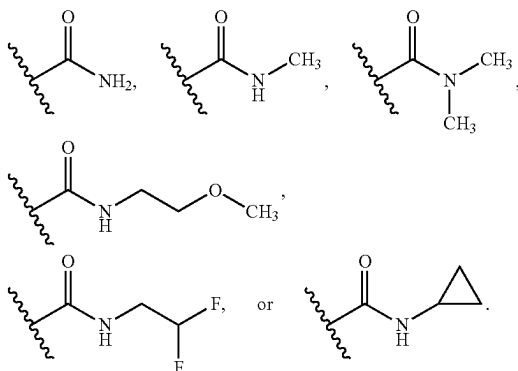
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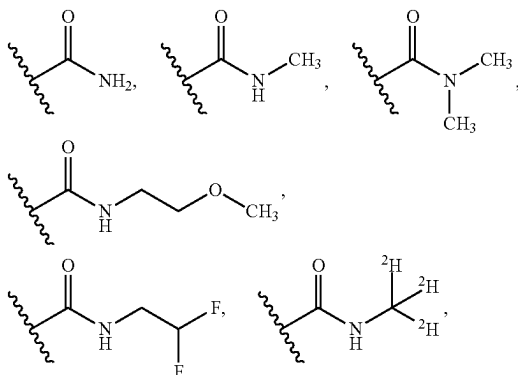
In some embodiments, R^{3b} is H, methyl,



In some embodiments, R^{3c} is H, methyl,

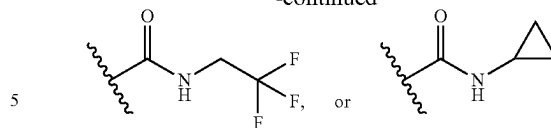


In some embodiments, R^{3d} is H, methyl,



16

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In some embodiments, R^{3a} and R^{4b} , R^{4a} and R^{3b} , R^{4b} and R^{4a} , R^{3b} and R^{4c} , R^{4b} and R^{4c} , R^{3c} and

R^{4b} , R^{3c} and R^{4d} , R^{4c} and R^{4d} , and/or R^{3d} and R^{4c} , together with the atoms to which each is attached, combine to form optionally substituted C_2 - C_9 heterocyclyl.

In some embodiments, R^{3a} and R^{4b} , R^{4b} and R^{4a} , R^{4b} and R^{4c} , R^{3c} and R^{4b} , R^{3c} and R^{4d} , and/or R^{3d} and R^{4c} , together with the atoms to which each is attached, combine to form optionally substituted C_2 - C_9 heterocyclyl.

In some embodiments, each of R^{5a} , R^{5b} , R^{5c} , and R^{5d} is, independently, H, optionally substituted C_1 - C_6 alkyl, or optionally substituted C_1 - C_6 heteroalkyl. In some embodiments, each of R^{5a} , R^{5b} , R^{5c} , and R^{5d} is H.

In some embodiments, each of R^{3e} , R^{3f} , R^{3g} , and R^{3h} is, independently, H, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_1 - C_6 acyl, optionally substituted sulfone, or optionally substituted sulfonamide.

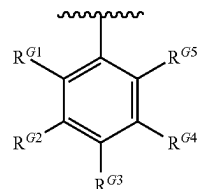
In some embodiments, each of R^{4e} , R^{4f} , R^{4g} , and R^{4h} is, independently, H, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_1 - C_6 acyl, optionally substituted sulfone, or optionally substituted sulfonamide.

In some embodiments, each of R^{5e} , R^{5f} , R^{5g} , and R^{5h} is, independently, H, optionally substituted C_1 - C_6 alkyl, or optionally substituted C_1 - C_6 heteroalkyl. In some embodiments, each of R^{5e} , R^{5f} , R^{5g} , and R^{5h} is H.

In some embodiments, G is optionally substituted C_3 - C_{10} carbocyclyl or optionally substituted C_2 - C_9 heterocyclyl. In some embodiments, G is optionally substituted C_6 - C_{10} aryl or optionally substituted C_2 - C_9 heteroaryl.

In some embodiments, G is optionally substituted C_3 - C_{10} carbocyclyl. In some embodiments, G is optionally substituted C_6 - C_{10} aryl. In some embodiments, G is optionally substituted C_2 - C_9 heterocyclyl. In some embodiments, G is optionally substituted C_2 - C_9 heteroaryl.

In some embodiments, G is



where

each of R^{G1} , R^{G2} , R^{G3} , R^{G4} , and R^{G5} is, independently, H, halogen, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, optionally substituted $-O-C_3-C_6$ carbocyclyl, optionally substituted $-C_1-C_3$ alkyl- C_3-C_6 carbocyclyl, optionally substituted $-C_1-C_3$ alkyl- C_2-C_5 heterocyclyl, hydroxyl, thiol, or optionally substituted

17

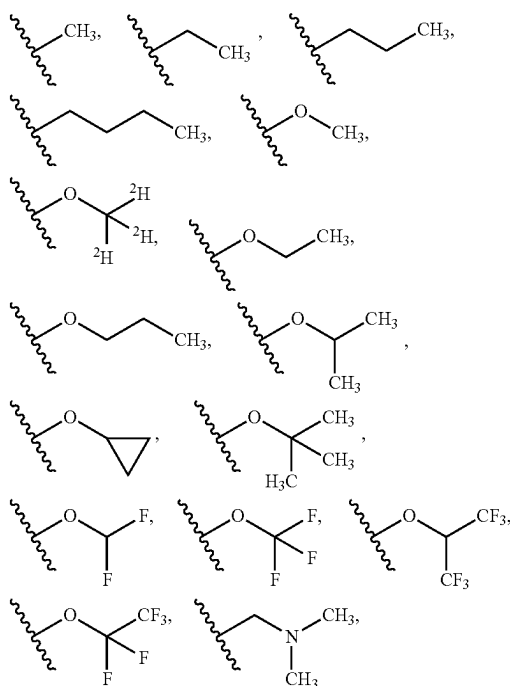
amino; or R^{G1} and R^{G2} , R^{G2} and R^{G3} , R^{G3} and R^{G4} , and/or R^{G4} and R^{G5} , together with the carbon atoms to which each is attached, combine to form optionally substituted C_6 - C_{10} aryl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heteroaryl, or optionally substituted C_2 - C_9 heterocyclyl.

In some embodiments, each of R^{G1} , R^{G2} , R^{G3} , R^{G4} , and R^{G5} is, independently, H, halogen, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, optionally substituted $-O-C_3-C_6$ carbocyclyl, optionally substituted $-C_1-C_3$ alkyl- C_3-C_6 carbocyclyl, optionally substituted $-C_1-C_3$ alkyl- C_2-C_5 heterocyclyl, hydroxyl, thiol, or optionally substituted amino; or R^{G1} and R^{G2} , R^{G2} and R^{G3} , R^{G3} and R^{G4} , and/or R^{G4} and R^{G5} , together with the carbon atoms to which each is attached, combine to form optionally substituted C_6 - C_{10} aryl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heteroaryl, or optionally substituted C_2 - C_9 heterocyclyl.

In some embodiments, each of R^{G1} , R^{G2} , R^{G3} , R^{G4} , and R^{G5} is, independently, H, halogen, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted $-O-C_3-C_6$ carbocyclyl, or optionally substituted $-C_1-C_3$ alkyl- C_2-C_5 heterocyclyl; or R^{G1} and R^{G2} , R^{G2} and R^{G3} , R^{G3} and R^{G4} , and/or R^{G4} and R^{G5} , together with the carbon atoms to which each is attached, combine to form optionally substituted C_2 - C_9 heteroaryl or optionally substituted C_2 - C_9 heterocyclyl.

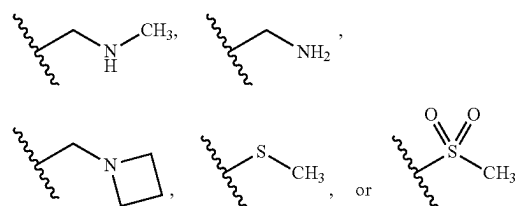
In some embodiments, each of R^{G1} , R^{G2} , R^{G3} , R^{G4} , and R^{G5} is, independently, H, halogen, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted $-O-C_3-C_6$ carbocyclyl, or optionally substituted $-C_1-C_3$ alkyl- C_2-C_5 heterocyclyl.

In some embodiments, each of R^{G1} , R^{G2} , R^{G3} , R^{G4} , and R^{G5} is, independently, H, F, Cl,

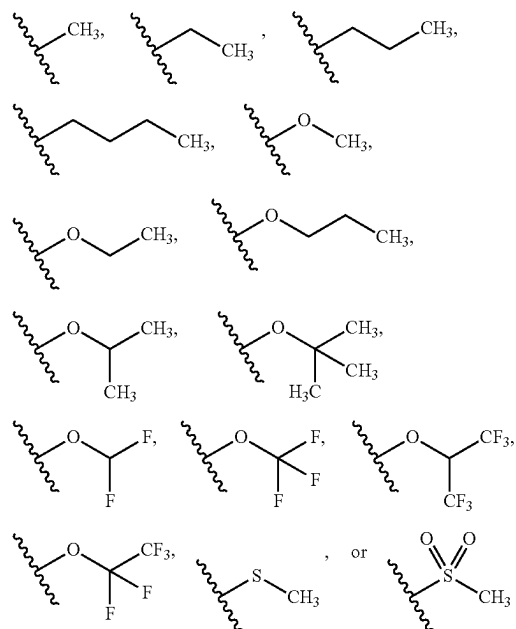


18

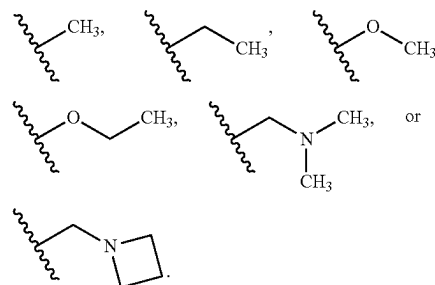
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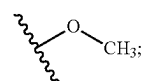
In some embodiments, each of R^{G1} , R^{G2} , R^{G3} , R^{G4} , and R^{G5} is, independently, H, F,



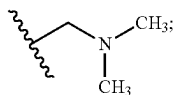
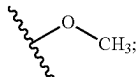
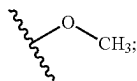
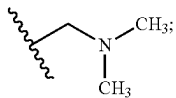
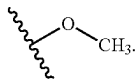
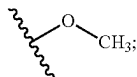
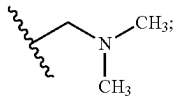
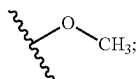
In some embodiments, each of R^{G1} , R^{G2} , R^{G3} , R^{G4} , and R^{G5} is, independently, H, F, Cl,



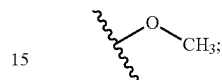
In some embodiments, R^{G1} is H; R^{G2} is



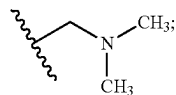
19

 R^{G3} is R^{G4} isand R^{G5} is H. In some embodiments, R^{G1} is H; R^{G2} is R^{G3} is R^{G4} is H; and R^{G5} isIn some embodiments, R^{G1} is H; R^{G2} is R^{G3} is R^{G4} is Cl or F; and R^{G5} is H. In some embodiments, R^{G1} is H; R^{G2} is

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 R^{G3} is10 R^{G4} is H; and R^{G5} is H. In some embodiments, R^{G1} is H; R^{G2} is R^{G3} is

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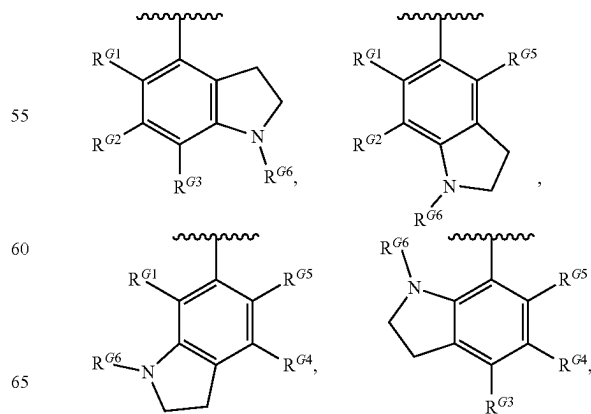
 R^{G4} is30 and R^{G5} is H.

35 In some embodiments, R^{G1} and R^{G2} , R^{G2} and R^{G3} , R^{G3} and R^{G4} , and/or R^{G4} and R^{G5} , together with the carbon atoms to which each is attached, combine to form optionally substituted C_2 - C_9 heteroaryl or optionally substituted C_2 - C_9 heterocyclyl.

40 In some embodiments, R^{G1} and R^{G2} , R^{G2} and R^{G3} , R^{G3} and R^{G4} , and/or R^{G4} and R^{G5} , together with the carbon atoms to which each is attached, combine to form optionally substituted C_2 - C_9 heterocyclyl. In some embodiments, R^{G1} and R^{G2} , R^{G2} and R^{G3} , R^{G3} and R^{G4} , and/or R^{G4} and R^{G5} , together with the carbon atoms to which each is attached, combine to form optionally substituted C_2 - C_9 heteroaryl.

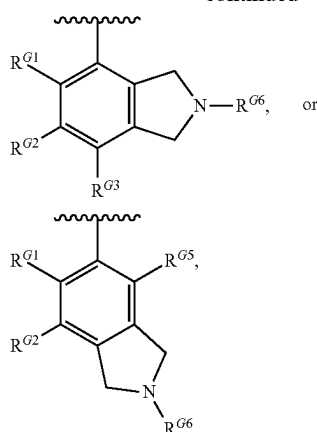
In some embodiments, G is

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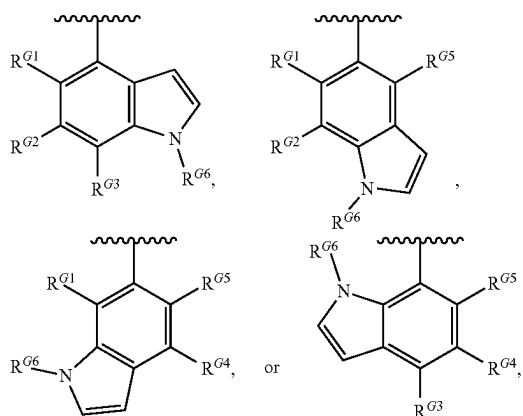


21

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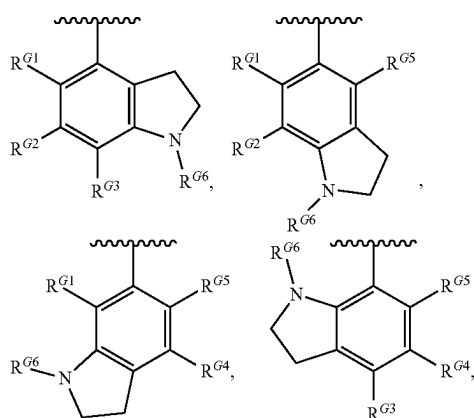
where R^{G6} is H or optionally substituted C₁-C₆ alkyl. In some embodiments, G is



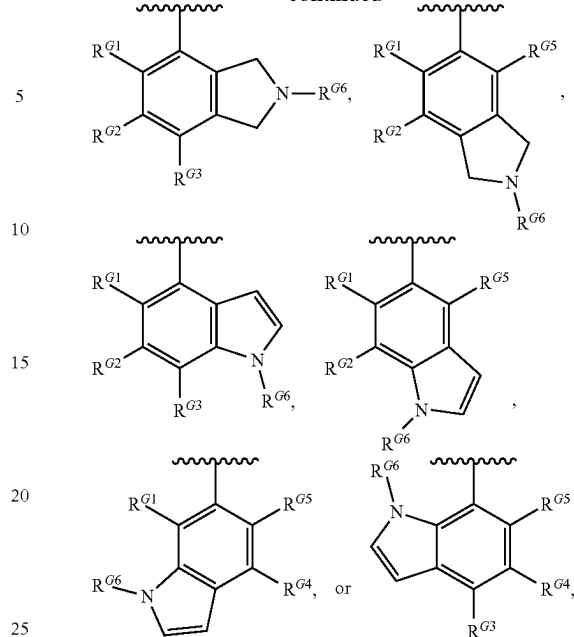
where R^{G6} is H or optionally substituted C₁-C₆ alkyl.

In some embodiments, R^{G1} and R^{G2}, R^{G2} and R^{G3}, R^{G3} and R^{G4}, and/or R^{G4} and R^{G5}, together with the carbon atoms to which each is attached, combine to form optionally substituted C₂-C₉ heterocycl or optionally substituted C₂-C₉ heteroaryl.

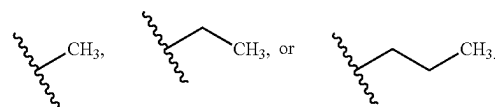
In some embodiments, G is

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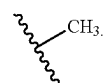
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where R^{G6} is H or optionally substituted C₁-C₆ alkyl. In some embodiments, R^{G6} is H,

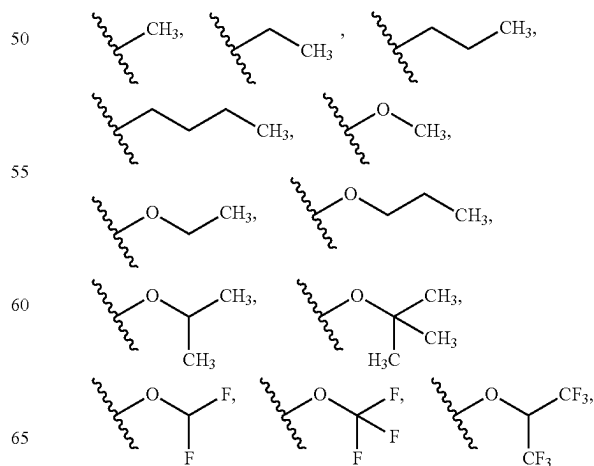


In some embodiments, R^{G6} is H or



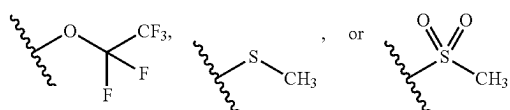
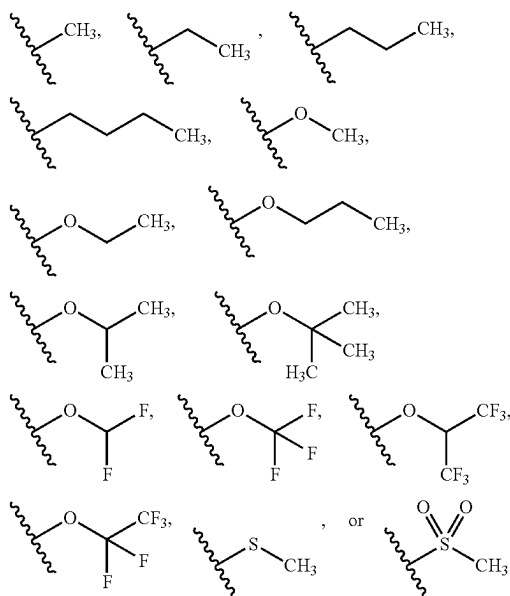
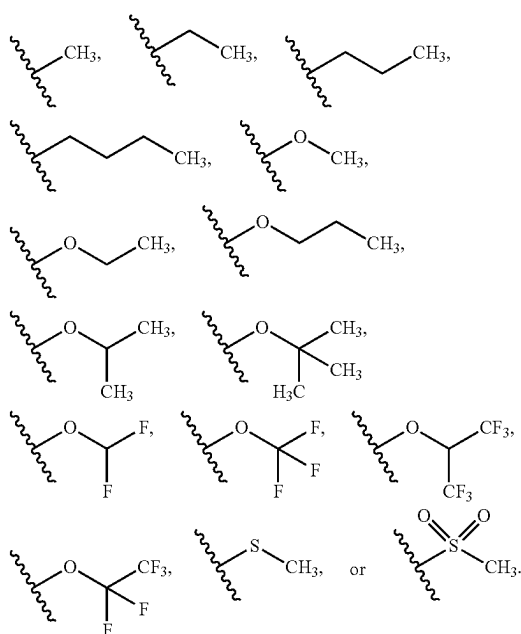
In some embodiments, R^{G6} is H.

In some embodiments, R^{G1} is H, F,

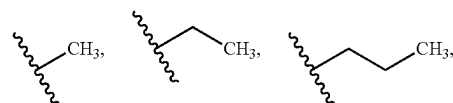


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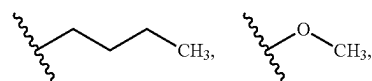
-continued

In some embodiments, R^{G1} is H.In some embodiments, R^{G2} is H, F,In some embodiments, R^{G2} is H.In some embodiments, R^{G3} is H, F,In some embodiments, R^{G3} is H.**24**In some embodiments, R^{G4} is H, F,

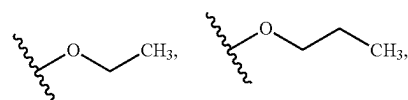
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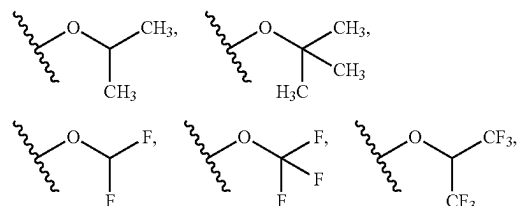
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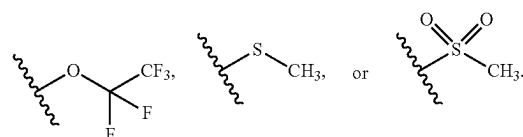
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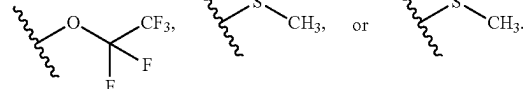
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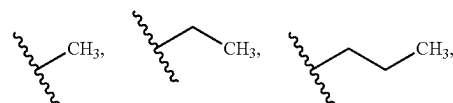
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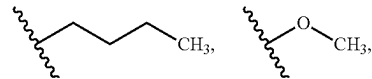
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In some embodiments, R^{G4} is H.In some embodiments, R^{G5} is H, F,

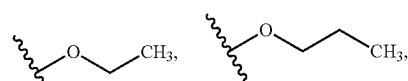
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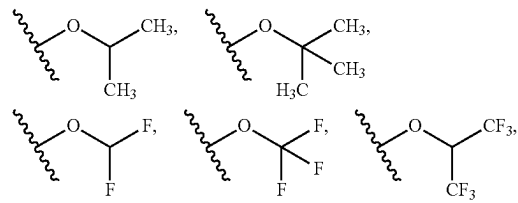
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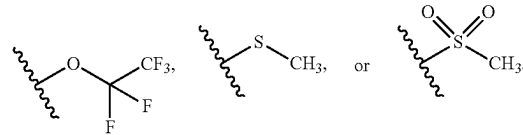
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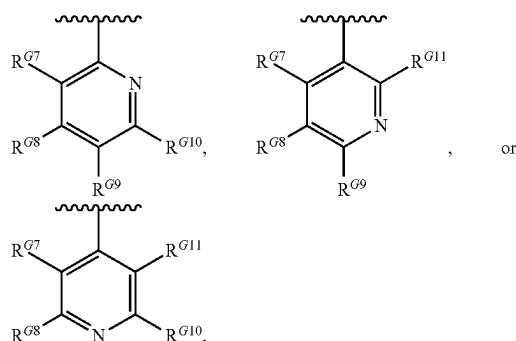
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In some embodiments, R^{G5} is H.

In some embodiments, one or more of R^{G1} , R^{G2} , R^{G3} , R^{G4} , and R^{G5} is H. In some embodiments, two or more of R^{G1} , R^{G2} , R^{G3} , R^{G4} , and R^{G5} is H. In some embodiments, three or more of R^{G1} , R^{G2} , R^{G3} , R^{G4} , and R^{G5} is H. In some embodiments, each of R^{G1} , R^{G2} , R^{G3} , R^{G4} , and R^{G5} is H.

65

In some embodiments, G is



where

each of R^{G7} , R^{G8} , R^{G9} , R^{G10} , and R^{G11} is, independently,

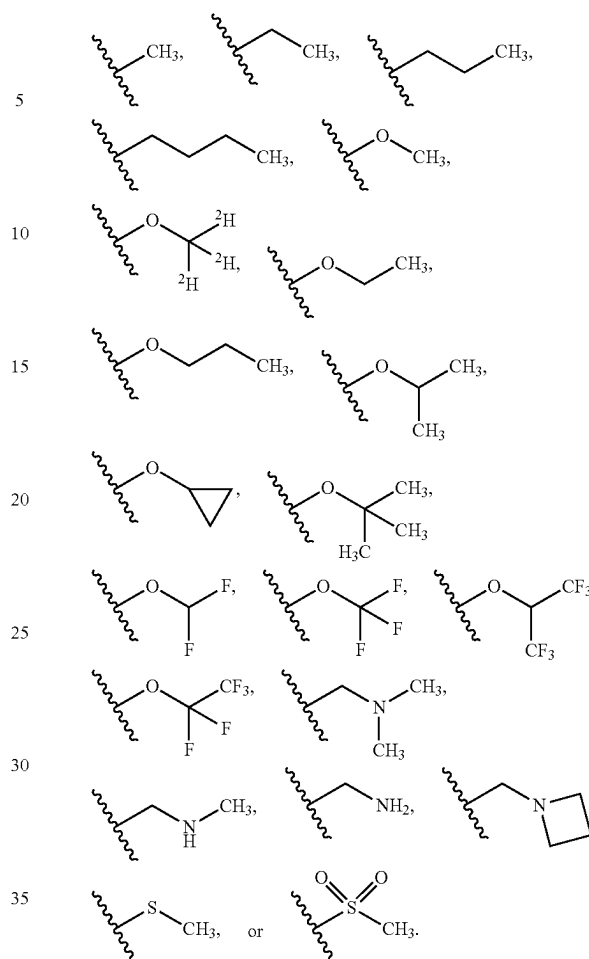
H, halogen, optionally substituted C₁-C₆ alkyl, optionally substituted C₃-C₁₀ carbocyclyl, optionally substituted C₂-C₉ heterocyclyl, optionally substituted C₆-C₁₀ aryl, optionally substituted C₂-C₉ heteroaryl, optionally substituted C₂-C₆ alkenyl, optionally substituted C₂-C₆ heteroalkenyl, optionally substituted —O—C₃-C₆ carbocyclyl, optionally substituted —C₁-C₃ alkyl-C₃-C₆ carbocyclyl, optionally substituted —C₁-C₃ alkyl-C₂-C₅ heterocyclyl, hydroxyl, thiol, or optionally substituted amino; or R^{G7} and R^{G8}, R^{G8} and R^{G9}, R^{G9} and R^{G10}, and/or R^{G10} and R^{G11}, together with the carbon atoms to which each is attached, combine to form optionally substituted C₆-C₁₀ aryl, optionally substituted C₃-C₁₀ carbocyclyl, optionally substituted C₂-C₉ heteroaryl, or C₇-C₉ heterocyclyl.

In some embodiments, each of R^{G7} , R^{G8} , R^{G9} , R^{G10} , and R^{G11} is, independently, H, halogen, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, hydroxyl, thiol, or optionally substituted amino; or R^{G7} and R^{G8} , R^{G8} and R^{G9} , R^{G9} and R^{G10} , and/or R^{G10} and R^{G11} , together with the carbon atoms to which each is attached, combine to form optionally substituted C_6 - C_{10} aryl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, or C_2 - C_9 heteroaryl.

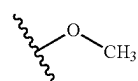
In some embodiments, each of R^{G7} , R^{G8} , R^{G9} , R^{G10} , and R^{G11} is, independently, H, halogen, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted $-O-C_3$ - C_6 carbocyclyl, or optionally substituted $-C_1$ - C_3 alkyl- C_2 - C_5 heterocyclyl; or R^{G7} and R^{G8} , R^{G8} and R^{G9} , R^{G9} and R^{G10} , and/or R^{G10} and R^{G11} , together with the carbon atoms to which each is attached, combine to form optionally substituted C_6 - C_{10} aryl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heteroaryl, or C_2 - C_9 heterocyclyl.

In some embodiments, each of R^{G7} , R^{G8} , R^{G9} , R^{G10} , and R^{G11} is, independently, H, halogen, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted $-O-C_3-C_6$ carbocyclyl, or optionally substituted $-C_1-C_3$ alkyl- C_2-C_5 heterocyclyl.

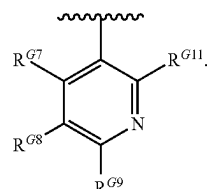
In some embodiments, each of R^{G7} , R^{G8} , R^{G9} , R^{G10} , and R^{G11} is, independently, H, F, Cl,



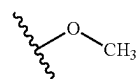
40 In some embodiments, R^{G^8} is



In some embodiments, G is



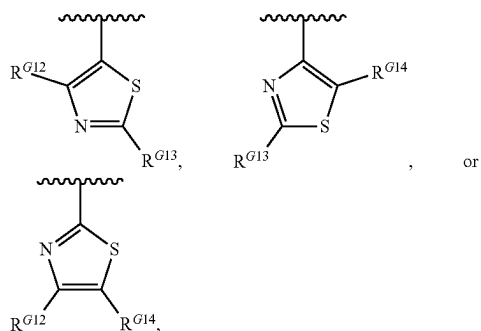
In some embodiments, R^{G7} is H; R^{G8} is



R^{G^9} is H ; and $R^{G^{11}}$ is H .

27

In some embodiments, G is

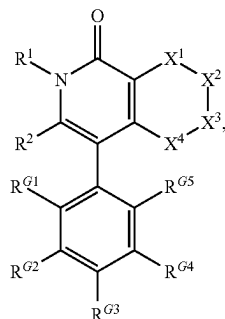


where

each of R^{G12} , R^{G13} , and R^{G14} is, independently, H, halogen, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, optionally substituted $-O-C_3-C_6$ carbocyclyl, optionally substituted $-C_1-C_3$ alkyl- C_3-C_6 carbocyclyl, optionally substituted $-C_1-C_3$ alkyl- C_2-C_5 heterocyclyl, hydroxyl, thiol, or optionally substituted amino; or R^{G12} and R^{G14} , together with the carbon atoms to which each is attached, combine to form optionally substituted C_6 - C_{10} aryl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heteroaryl, or optionally substituted C_2 - C_9 heterocyclyl.

In some embodiments, each of R^{G12} , R^{G13} , and R^{G14} is, independently, H, halogen, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, hydroxyl, thiol, or optionally substituted amino; or R^{G12} and R^{G14} , together with the carbon atoms to which each is attached, combine to form optionally substituted C_6 - C_{10} aryl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heteroaryl, or optionally substituted C_2 - C_9 heterocyclyl.

In some embodiments, the structure of Formula I has the structure of Formula Ia:

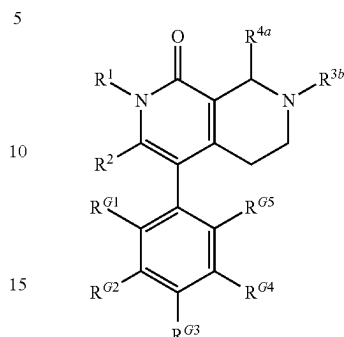


Formula Ia

or a pharmaceutically acceptable salt thereof.

28

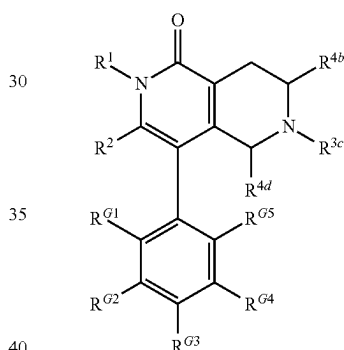
In some embodiments, the structure of Formula I has the structure of Formula Ib:



Formula Ib

or a pharmaceutically acceptable salt thereof.

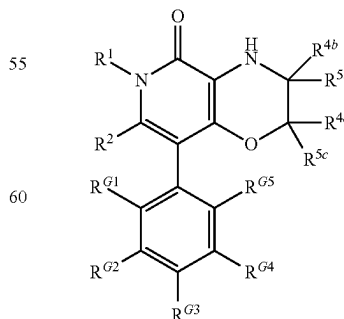
In some embodiments, the structure of Formula I has the structure of Formula Ic:



Formula Ic

or a pharmaceutically acceptable salt thereof.

In some embodiments, the structure of Formula I has the structure of Formula Id:

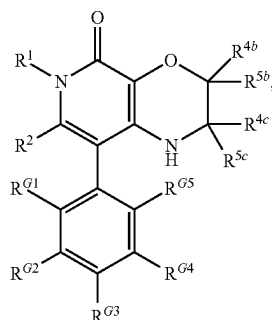


Formula Id

or a pharmaceutically acceptable salt thereof.

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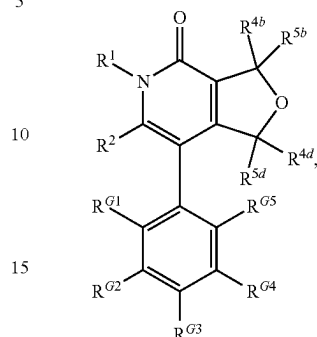
In some embodiments, the structure of Formula I has the structure of Formula Ie:



Formula Ie 5

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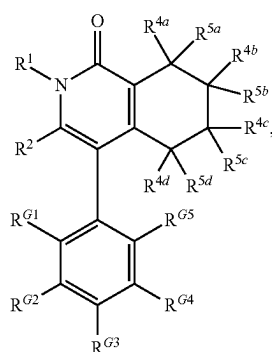
In some embodiments, the structure of Formula I has the structure of Formula Ih:



Formula Ih 10 15

or a pharmaceutically acceptable salt thereof.

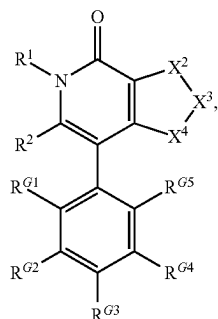
In some embodiments, the structure of Formula I has the structure of Formula If:



Formula If 25

or a pharmaceutically acceptable salt thereof.

In some embodiments, the structure of Formula I has the structure of Formula Ig:

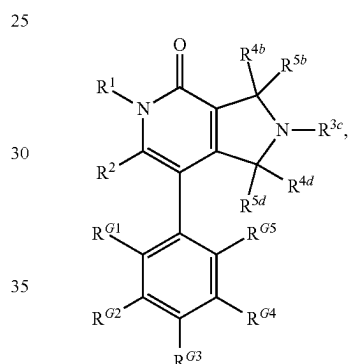


Formula Ig 30 35

or a pharmaceutically acceptable salt thereof.

or a pharmaceutically acceptable salt thereof.

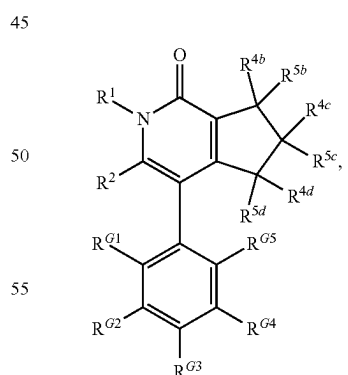
In some embodiments, the structure of Formula I has the structure of Formula Ii:



Formula Ii 40

or a pharmaceutically acceptable salt thereof.

In some embodiments, the structure of Formula I has the structure of Formula Ij:



Formula Ij 45 50 55

or a pharmaceutically acceptable salt thereof.

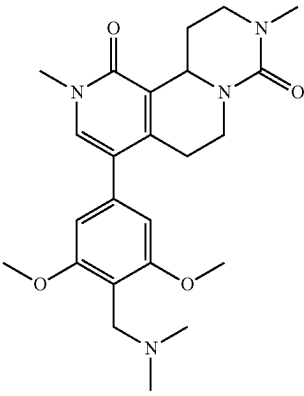
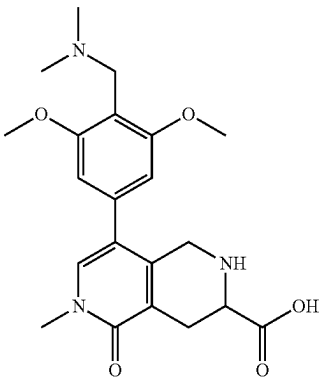
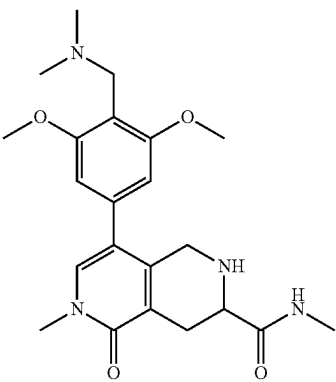
In some embodiments, the compound has the structure of any one of compounds B1-B21 in Table 1A, or a pharmaceutically acceptable salt thereof. In some embodiments, the compound has the structure of any one of compounds B22-B24 in Table 1B, or a pharmaceutically acceptable salt thereof.

31

In an aspect, the disclosure features a compound having the structure of any one of compounds B1-B21 in Table 1A, or a pharmaceutically acceptable salt thereof.

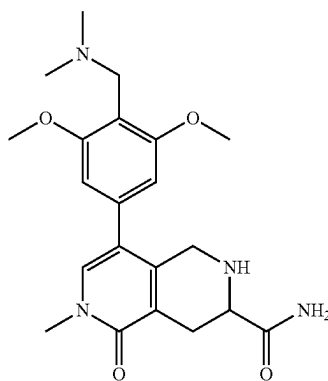
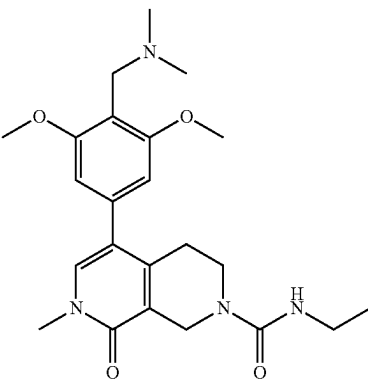
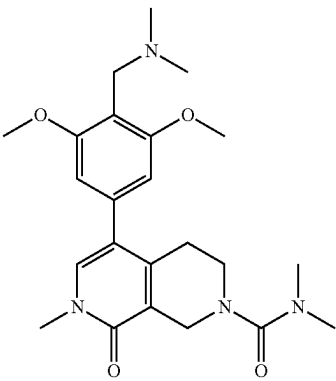
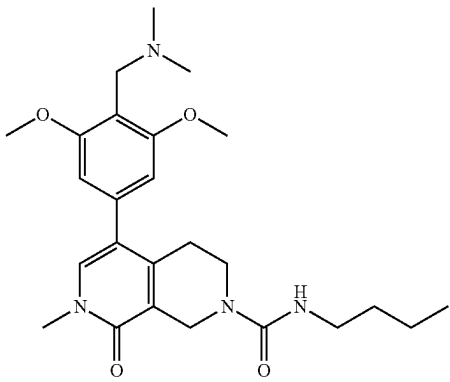
In another aspect, the disclosure features a compound having the structure of any one of compounds B22-B24 in Table 1B, or a pharmaceutically acceptable salt thereof.

TABLE 1A

Compounds B1-B21 of the Disclosure	
Compound No.	Structure
B1	
B2	
B3	

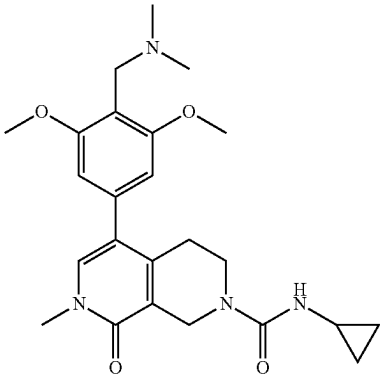
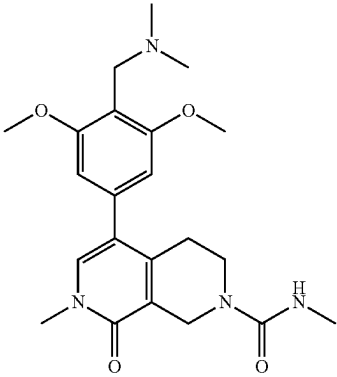
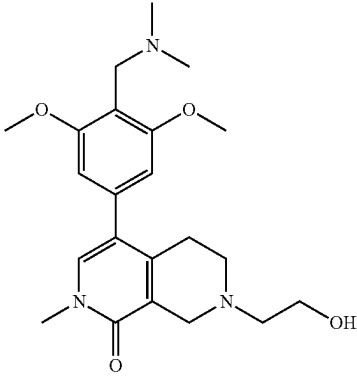
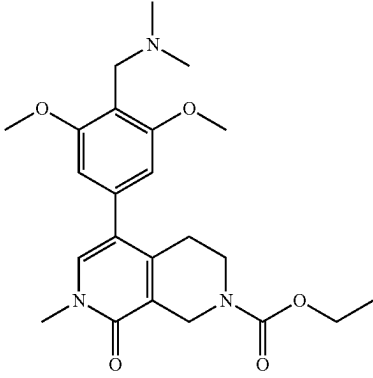
32

TABLE 1A-continued

Compounds B1-B21 of the Disclosure	
Compound No.	Structure
B4	
B5	
B6	
B7	

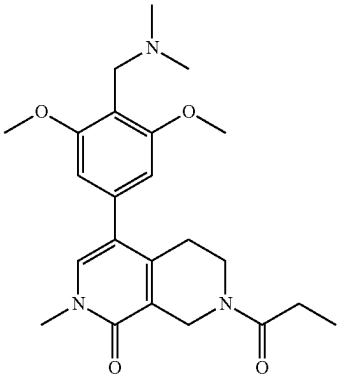
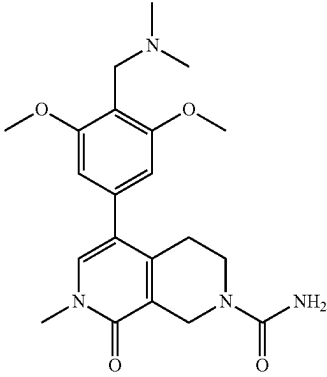
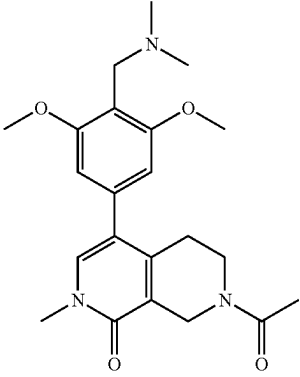
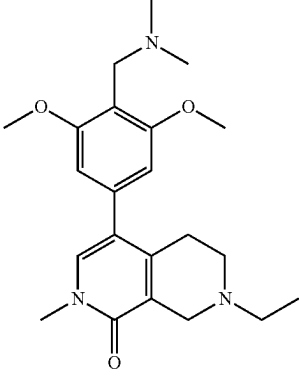
33

TABLE 1A-continued

Compounds B1-B21 of the Disclosure	
Compound No.	Structure
B8	
B9	
B10	
B11	

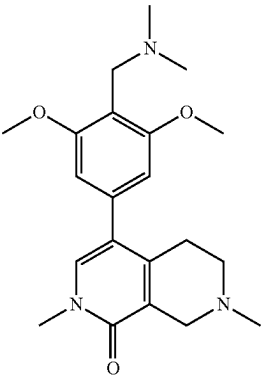
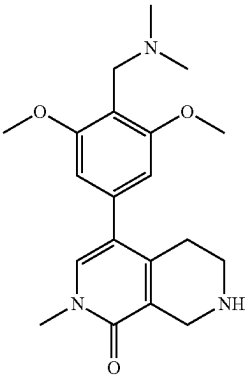
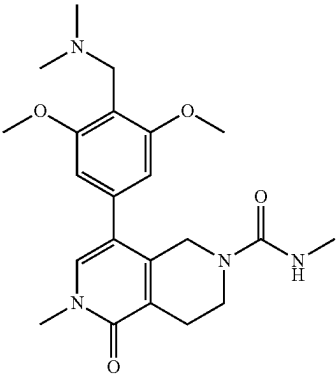
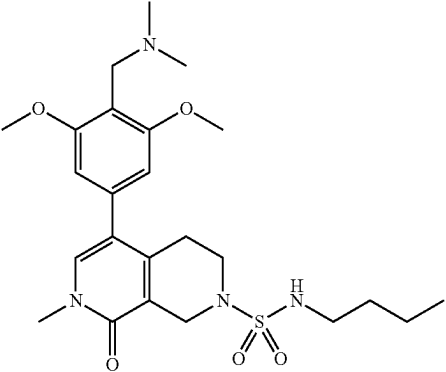
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TABLE 1A-continued

Compounds B1-B21 of the Disclosure	
Compound No.	Structure
B12	
B13	
B14	
B15	

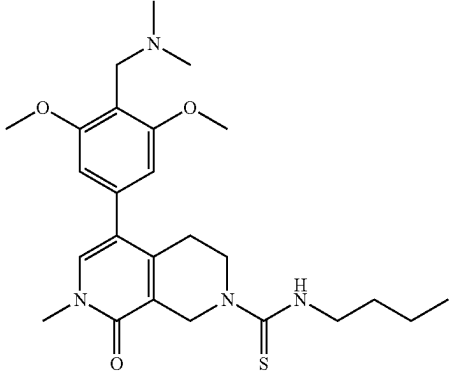
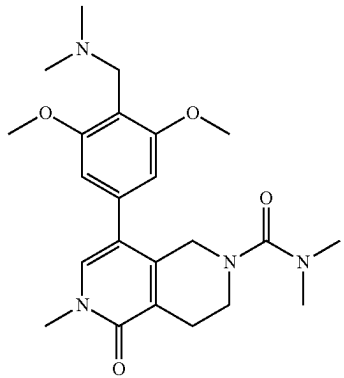
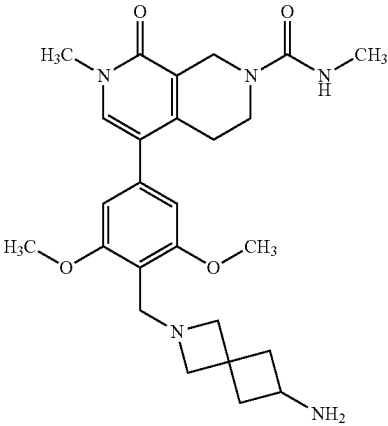
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TABLE 1A-continued

Compounds B1-B21 of the Disclosure	
Compound No.	Structure
B16	
B17	
B18	
B19	

36

TABLE 1A-continued

Compounds B1-B21 of the Disclosure	
Compound No.	Structure
B20	
B21	
TABLE 1B	
Compounds B22-B24 of the Disclosure	
Compound No.	Structure
B22	

37

TABLE 1B-continued

Compounds B22-B24 of the Disclosure	
Compound No.	Structure
B23	
B24	

In another aspect, the disclosure features a compound having the structure of Formula II:

A-L-B

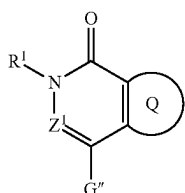
Formula II,

where

B is a degradation moiety,

L is a linker, and

A has the structure of Formula III:



where

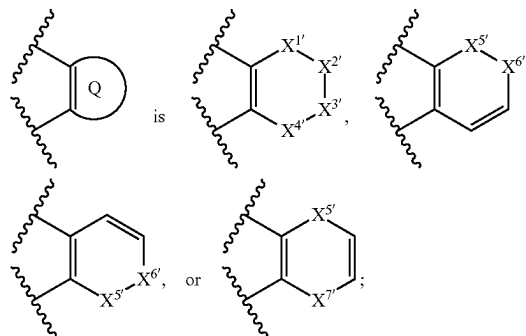
R¹ is H, optionally substituted C₁-C₆ alkyl, optionally substituted C₂-C₆ alkenyl, optionally substituted C₁-C₆ heteroalkyl, or optionally substituted C₃-C₁₀ carbocyclyl;

Z¹ is CR² or N;

R² is H, halogen, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally

38

substituted C₃-C₁₀ carbocyclyl, optionally substituted C₂-C₉ heterocyclyl, optionally substituted C₆-C₁₀ aryl, or optionally substituted C₂-C₉ heteroaryl;



X¹' is a bond, O, NR^{3a1}, or CR^{4a1}R^{5a1};

X²' is O, NR^{3b1}, or CR^{4b1}R^{5b1};

X³' is O, NR^{3c1}, or CR^{4c1}R^{5c1};

X⁴' is a bond, O, NR^{3d1}, or CR^{4d1}R^{5d1};

X⁵' is O, NR^{3e1}, or CR^{4e1}R^{5e1};

X⁶' is O, NR^{3f1}, or CR^{4f1}R^{5f1};

X⁷' is O, NR^{3g1}, or CR^{4g1}R^{5g1};

each of R^{3a1}, R^{3b1}, R^{3c1}, and R^{3d1} is, independently, H,



halogen, hydroxyl, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted C₃-C₁₀ carbocyclyl, optionally substituted C₂-C₉ heterocyclyl, optionally substituted C₆-C₁₀ aryl, optionally substituted C₂-C₉ heteroaryl, optionally substituted C₂-C₆ alkenyl, optionally substituted C₂-C₆ heteroalkenyl, optionally substituted C₁-C₆ acyl, thiol, optionally substituted sulfone, optionally substituted sulfonamide, or optionally substituted amino, or R^{3a1} and R^{4b1}, R^{4a1} and R^{3b1}, R^{4b1} and R^{4a1}, R^{3b1} and R^{4c1}, R^{4c1} and R^{3c1}, R^{4c1} and R^{4b1}, R^{3c1} and R^{4d1}, R^{4d1} and R^{4c1}, and/or R^{3d1} and R^{4c1}, together with the atoms to which each is attached, combine to form optionally substituted C₂-C₉ heterocyclyl;

R^{3a1} is absent, optionally substituted C₁-C₆ alkylene, optionally substituted C₁-C₆ heteroalkylene, optionally substituted C₃-C₁₀ carbocyclylene, optionally substituted C₂-C₉ heterocyclylene, optionally substituted C₆-C₁₀ arylene, optionally substituted C₂-C₉ heteroarylene, optionally substituted C₂-C₆ alkenylene, optionally substituted C₂-C₆ heteroalkenylene, optionally substituted sulfone, optionally substituted sulfonamide, or optionally substituted amino;

each of R^{4a1}, R^{4b1}, R^{4c1}, and R^{4d1} is, independently, H, halogen, hydroxyl, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted C₃-C₁₀ carbocyclyl, optionally substituted C₂-C₉ heterocyclyl, optionally substituted C₆-C₁₀ aryl, optionally substituted C₂-C₉ heteroaryl, optionally substituted C₂-C₆ alkenyl, optionally substituted C₂-C₆ heteroalkenyl, thiol, optionally substituted sulfone, or optionally substituted amino, or R^{3a1} and R^{4b1}, R^{4a1} and

39

R^{3b1} , R^{4b1} and R^{4a1} , R^{3b1} and R^{4c1} , R^{4b1} and R^{4c1} , R^{3c1} and R^{4b1} , R^{3c1} and R^{4d1} , R^{4c1} and R^{4d1} , and/or R^{3d1} and R^{4c1} , together with the atoms to which each is attached, combine to form optionally substituted C_2 - C_9 heterocyclyl;

each of R^{5a1} , R^{5b1} , R^{5i} , and R^{5d1} is, independently, H, halogen, hydroxyl, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, hydroxyl, thiol, or optionally substituted amino;

each of R^{3e1} , R^{3f1} , and R^{3g1} is, independently, H,



halogen, hydroxyl, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, optionally substituted C_1 - C_6 acyl, thiol, optionally substituted sulfone, optionally substituted sulfonamide, or optionally substituted amino, or R^{3e} and R^{4f} or R^{4e} and R^{3f} , together with the atoms to which each is attached, combine to form optionally substituted heterocyclyl;

each of R^{4e1} , R^{4f1} , and R^{4g1} is, independently, H, halogen, hydroxyl, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, optionally substituted C_1 - C_6 acyl, thiol, optionally substituted sulfone, optionally substituted sulfonamide, or optionally substituted amino, or R^{3e1} and R^{4f1} or R^{4e1} and R^{3f1} , together with the atoms to which each is attached, combine to form optionally substituted heterocyclyl;

each of R^{5e1} , R^{5f1} , and R^{5g1} is, independently, H, halogen, hydroxyl, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, hydroxyl, thiol, or optionally substituted amino;

G'' is



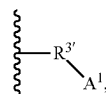
optionally substituted C_6 - C_{10} aryl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heteroaryl, or C_2 - C_9 heterocyclyl;

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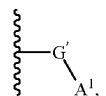
G' is optionally substituted C_3 - C_{10} carbocyclylene, C_2 - C_9 heterocyclylene, optionally substituted C_6 - C_{10} arylene, or optionally substituted C_2 - C_9 heteroarylene; and

A^1 is a bond between A and the linker,

where one of R^{3a1} , R^{3b1} , R^{3c1} , R^{3d1} , R^{3e1} , R^{3f1} , and R^{3g1} is

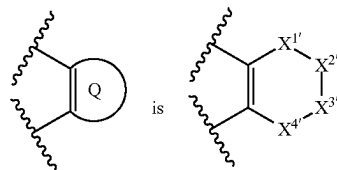


or G' is

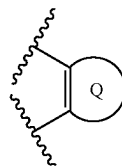


or a pharmaceutically acceptable salt thereof.

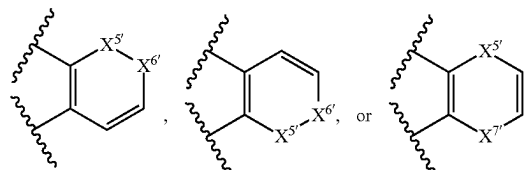
In some embodiments,



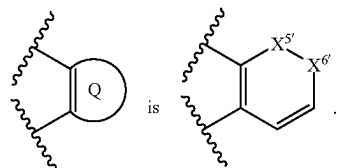
In some embodiments,



is



In some embodiments,



41

In another aspect, the disclosure features a compound having the structure of Formula II:

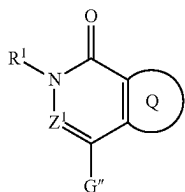
A-L-B

where

B is a degradation moiety,

L is a linker, and

A has the structure of Formula III:

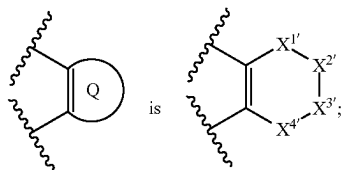


where

R¹ is, independently, H, optionally substituted C₁-C₆ alkyl, optionally substituted C₂-C₆ alkenyl, optionally substituted C₁-C₆ heteroalkyl, or optionally substituted C₃-C₁₀ carbocyclyl;

Z¹ is CR² or N;

R² is H, halogen, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted C₃-C₁₀ carbocyclyl, optionally substituted C₂-C₉ heterocyclyl, optionally substituted C₆-C₁₀ aryl, or optionally substituted C₂-C₉ heteroaryl;

X^{1'} is a bond, O, NR^{3a1}, or CR^{4a1}R^{5a1};X^{2'} is O, NR^{3b1}, or CR^{4b1}R^{5b1};X^{3'} is O, NR^{3c1}, or CR^{4c1}R^{5c1};X^{4'} is a bond, O, NR^{3d1}, or CR^{4d1}R^{5d1};each of R^{3a1}, R^{3b1}, R^{3c1}, and R^{3d1} is, independently, H,

halogen, hydroxyl, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted C₃-C₁₀ carbocyclyl, optionally substituted C₂-C₉ heterocyclyl, optionally substituted C₆-C₁₀ aryl, optionally substituted C₂-C₉ heteroaryl, optionally substituted C₂-C₆ alkenyl, optionally substituted C₂-C₆ heteroalkenyl, optionally substituted C₁-C₆ acyl, thiol, optionally substituted sulfone, optionally substituted sulfonamide, or optionally substituted amino, or R^{3a1} and R^{4b}, R^{4a1} and R^{3b1}, R^{4b1} and R^{4a1}, R^{3b1} and R^{4c1}, R^{4b1} and R^{4c1}, R^{3c1} and R^{4b1}, R^{3c1} and R^{4d1}, R^{4c1} and R^{4d1}, and/or R^{3d1} and R^{4c1}, together with the atoms to which each is attached, combine to form optionally substituted C₂-C₉ heterocyclyl;

42

R^{3a1} is absent, optionally substituted C₁-C₆ alkylene, optionally substituted C₁-C₆ heteroalkylene, optionally substituted C₃-C₁₀ carbocyclylene, optionally substituted C₂-C₉ heterocyclylene, optionally substituted C₆-C₁₀ arylen, optionally substituted C₂-C₉ heteroarylen, optionally substituted C₂-C₆ alkenylene, optionally substituted C₂-C₆ heteroalkenylene, optionally substituted sulfone, optionally substituted sulfonamide, or optionally substituted amino;

each of R^{4a1}, R^{4b1}, R^{4c1}, and R^{4d1} is, independently, H, halogen, hydroxyl, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted C₃-C₁₀ carbocyclyl, optionally substituted C₂-C₉ heterocyclyl, optionally substituted C₆-C₁₀ aryl, optionally substituted C₂-C₉ heteroaryl, optionally substituted C₂-C₆ alkenyl, optionally substituted C₂-C₆ heteroalkenyl, thiol, optionally substituted sulfone, or optionally substituted amino, or R^{3a1} and R^{4b}, R^{4a1} and R^{3b1}, R^{4b1} and R^{4a1}, R^{3b1} and R^{4c1}, R^{4b1} and R^{4c1}, R^{3c1} and R^{4b1}, R^{3c1} and R^{4d1}, R^{4c1} and R^{4d1}, and/or R^{3d1} and R^{4c1}, together with the atoms to which each is attached, combine to form optionally substituted C₂-C₉ heterocyclyl;

each of R^{5a1}, R^{5b1}, R^{5c1}, and R^{5d1} is, independently, H, halogen, hydroxyl, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted C₃-C₁₀ carbocyclyl, optionally substituted C₂-C₉ heterocyclyl, optionally substituted C₆-C₁₀ aryl, optionally substituted C₂-C₉ heteroaryl, optionally substituted C₂-C₆ alkenyl, optionally substituted C₂-C₆ heteroalkenyl, hydroxyl, thiol, or optionally substituted amino;

G'' is



optionally substituted C₆-C₁₀ aryl, optionally substituted C₃-C₁₀ carbocyclyl, optionally substituted C₂-C₉ heteroaryl, or C₂-C₉ heterocyclyl;

G' is optionally substituted C₃-C₁₀ carbocyclylene, C₂-C₉ heterocyclylene, optionally substituted C₆-C₁₀ arylen, or optionally substituted C₂-C₉ heteroarylen; and

A¹ is a bond between A and the linker, where one of R^{3a1}, R^{3b1}, R^{3c1}, and R^{3d1} is



or G is



or a pharmaceutically acceptable salt thereof.

In some embodiments, R¹ is H, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, or optionally substituted C₃-C₁₀ carbocyclyl. In some embodiments, R¹ is H, optionally substituted C₁-C₆ alkyl, optionally substituted C₂-C₆ alkenyl, or optionally substituted C₃-C₁₀

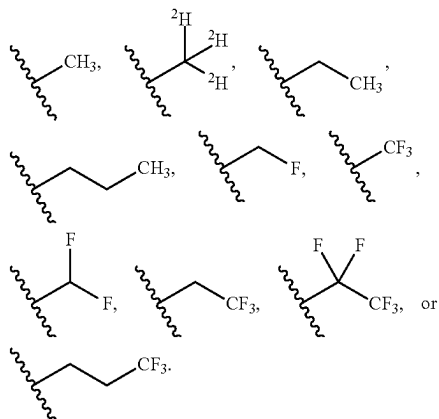
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carbocyclyl. In some embodiments, R^1 is H, optionally substituted C_1 - C_6 alkyl, or optionally substituted C_3 - C_{10} carbocyclyl.

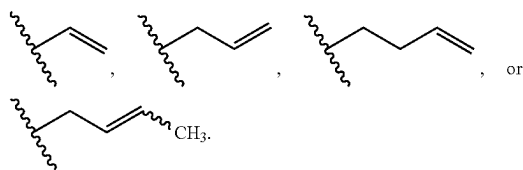
In some embodiments, R^1 is H. In some embodiments, R^1 is optionally substituted C_1 - C_6 alkyl. In some embodiments, R^1 is optionally substituted C_2 - C_6 alkenyl. In some embodiments, R^1 is optionally substituted C_3 - C_{10} carbocyclyl.

In some embodiments, optionally substituted C_1 - C_6 alkyl is C_1 - C_6 perfluoroalkyl.

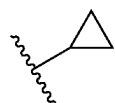
In some embodiments, R^1 is



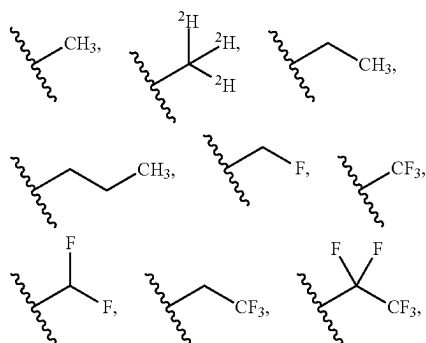
In some embodiments, R^1 is



In some embodiments, R^1 is

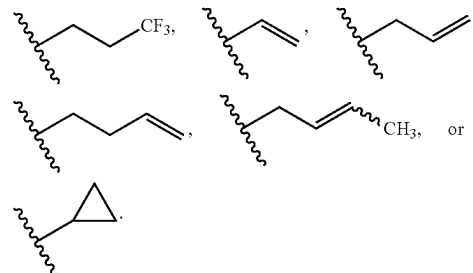


In some embodiments, R^1 is H,

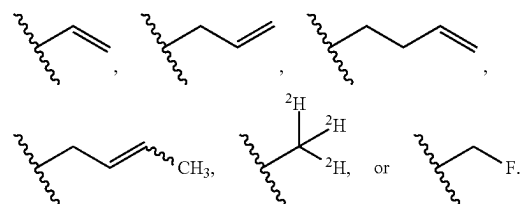


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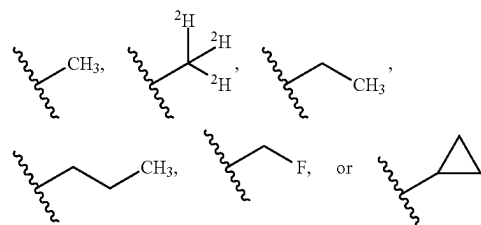
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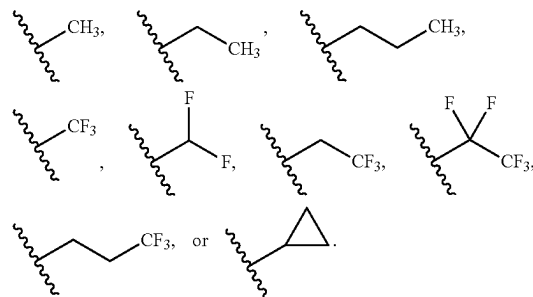
In some embodiments, R^1 is



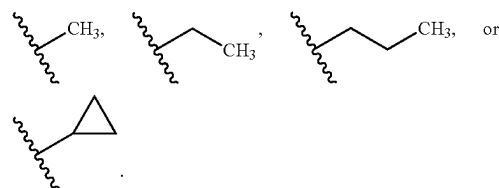
In some embodiments, R^1 is H,



In some embodiments, R^1 is H,



In some embodiments, R^1 is H,



45

In some embodiments, R^1 is H or



In some embodiments, R^1 is H. In some embodiments, R^1 is



In some embodiments, Z^1 is CR^2 . In some embodiments, Z^1 is N.

In some embodiments, R^2 is H, halogen, optionally substituted C_1 - C_6 alkyl, optionally substituted C_3 - C_{10} carbocycl, or optionally substituted C_6 - C_{10} aryl.

In some embodiments, R^2 is H, halogen, or optionally substituted C_1 - C_6 alkyl.

In some embodiments, R^2 is H, F, or



In some embodiments, R^2 is H. In some embodiments, R^2 is F. In some embodiments, R^2 is



In some embodiments, X^{1t} is a bond. In some embodiments, X^{1t} is O, NR^{3at} , or $CR^{4at}R^{5at}$. In some embodiments, X^{1t} is O or NR^{3at} . In some embodiments, X^{1t} is NR^{3at} or $CR^{4at}R^{5at}$.

In some embodiments, X^{2t} is O or NR^{3bt} . In some embodiments, X^{2t} is $CR^{4bt}R^{5bt}$. In some embodiments, X^{2t} is NR^{3bt} or $CR^{4bt}R^{5bt}$.

In some embodiments, X^{3t} is O or NR^{3ct} . In some embodiments, X^{3t} is $CR^{4ct}R^{5ct}$. In some embodiments, X^{3t} is NR^{3ct} or $CR^{4ct}R^{5ct}$.

In some embodiments, X^{4t} is a bond. In some embodiments, X^{4t} is O, NR^{3dt} , or $CR^{4dt}R^{5dt}$. In some embodiments, X^{4t} is O or NR^{3dt} . In some embodiments, X^{4t} is NR^{3dt} or $CR^{4dt}R^{5dt}$.

In some embodiments, X^{1t} is O, NR^{3at} , or $CR^{4at}R^{5at}$; X^{2t} is O, NR^{3bt} , or $CR^{4bt}R^{5bt}$; X^{3t} is O, NR^{3ct} , or $CR^{4ct}R^{5ct}$; and X^{4t} is O, NR^{3dt} , or $CR^{4dt}R^{5dt}$.

In some embodiments, X^{1t} is $CR^{4at}R^{5at}$; X^{2t} is NR^{3bt} ; X^{3t} is $CR^{4ct}R^{5ct}$; and X^{4t} is $CR^{4dt}R^{5dt}$.

In some embodiments, X^{1t} is $CR^{4at}R^{5at}$; X^{2t} is $CR^{4bt}R^{5bt}$; X^{3t} is NR^{3ct} ; and X^{4t} is $CR^{4dt}R^{5dt}$.

In some embodiments, X^{1t} is O or NR^{3at} ; X^{2t} is $CR^{4bt}R^{5bt}$; X^{3t} is $CR^{4ct}R^{5ct}$; and X^{4t} is O or NR^{3dt} .

In some embodiments, X^{1t} is a bond; X^{2t} is $CR^{4bt}R^{5bt}$; X^{3t} is O or NR^{3ct} ; and X^{4t} is $CR^{4dt}R^{5dt}$.

In some embodiments, X^{1t} is $CR^{4at}R^{5at}$; X^{2t} is $CR^{4bt}R^{5bt}$; X^{3t} is $CR^{4ct}R^{5ct}$; and X^{4t} is $CR^{4dt}R^{5dt}$.

46

In some embodiments, X^{5t} is $CR^{4et}R^{5et}$ and X^{6t} is NR^{3ft} .

In some embodiments, each of R^{3at} , R^{3bt} , R^{3ct} , and R^{3dt} is, independently, H,

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10

optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_1 - C_6 acyl, optionally substituted sulfone, or optionally substituted sulfonamide. In some embodiments, each of R^{3at} , R^{3bt} , R^{3ct} , and R^{3dt} is, independently, H,

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optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted sulfone, or optionally substituted sulfonamide. In some embodiments, each of R^{3at} , R^{3bt} , R^{3ct} , and R^{3dt} is, independently, H,

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optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 acyl, optionally substituted sulfone, or optionally substituted sulfonamide. In some embodiments, each of R^{3at} , R^{3bt} , R^{3ct} , and R^{3dt} is, independently, H,

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optionally substituted C_1 - C_6 alkyl, or optionally substituted C_1 - C_6 acyl.

In some embodiments, each of R^{3at} , R^{3bt} , R^{3ct} , and R^{3dt} is, independently,

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optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_1 - C_6 acyl, optionally substituted sulfone, or optionally substituted sulfonamide. In some embodiments, each of R^{3at} , R^{3bt} , R^{3ct} , and R^{3dt} is, independently,

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47

optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ acyl, optionally substituted sulfone, or optionally substituted sulfonamide.

In some embodiments, each of R^{3a1}, R^{3b1}, R^{3c1}, and R^{3d1} is, independently,



optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, or optionally substituted C₁-C₆ acyl. In some embodiments, each of R^{3a1}, R^{3b1}, R^{3c1}, and R^{3d1} is, independently,



optionally substituted C₁-C₆ alkyl or optionally substituted C₁-C₆ heteroalkyl. In some embodiments, each of R^{3a1}, R^{3b1}, R^{3c1}, and R^{3d1} is, independently,



optionally substituted C₁-C₆ alkyl or optionally substituted C₁-C₆ acyl. In some embodiments, each of R^{3a1}, R^{3b1}, R^{3c1}, and R^{3d1} is, independently,



or optionally substituted C₁-C₆ acyl. In some embodiments, each of R^{3a1}, R^{3c1}, R^{3b1}, and R^{3d1} is, independently,



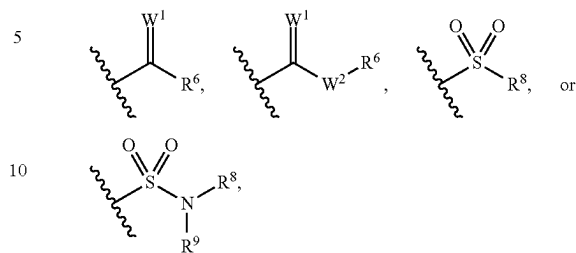
optionally substituted sulfone, or optionally substituted sulfonamide.

In some embodiments, each of R^{3a1}, R^{3b1}, R^{3c1}, and R^{3d1} is, independently, H,



48

optionally substituted C₁-C₆ alkyl,



where

R⁶ is H, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted C₃-C₁₀ carbocyclyl, or optionally substituted C₆-C₁₀ aryl;

W¹ is O or S;

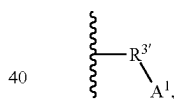
W² is NR⁷ or O;

R⁷ is H, optionally substituted C₁-C₆ alkyl, or optionally substituted C₁-C₆ heteroalkyl;

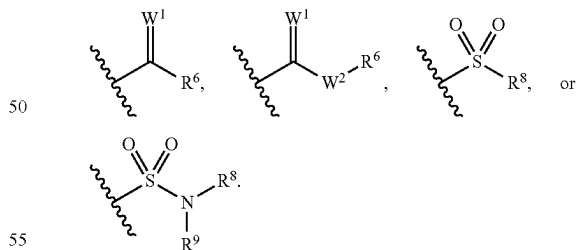
R⁸ is optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted C₃-C₁₀ carbocyclyl, or optionally substituted C₆-C₁₀ aryl; and

R⁹ is H, optionally substituted C₁-C₆ alkyl, or optionally substituted C₁-C₆ heteroalkyl.

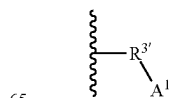
In some embodiments, each of R^{3a1}, R^{3b1}, R^{3c1}, and R^{3d1} is, independently,



C₁-C₆ alkyl,

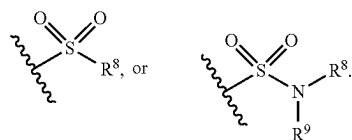
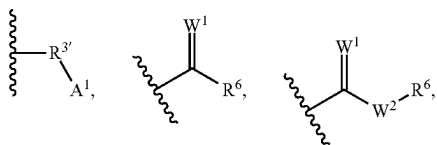


In some embodiments, each of R^{3a1}, R^{3b1}, R^{3c1}, and R^{3d1} is, independently,

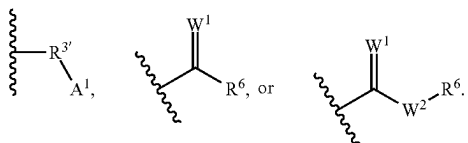


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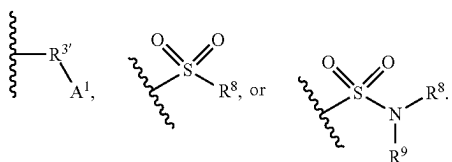
or C₁-C₆ alkyl. In some embodiments, each of R^{3a1}, R^{3b1}, R^{3c1}, and R^{3d1} is, independently



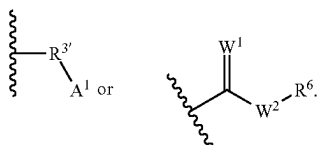
In some embodiments, each of R^{3a1}, R^{3b1}, R^{3c1}, and R^{3d1} is, independently,



In some embodiments, each of R^{3a1}, R^{3b1}, R^{3c1}, and R^{3d1} is, independently,



In some embodiments, each of R^{3a1}, R^{3b1}, R^{3c1}, and R^{3d1} is, independently,

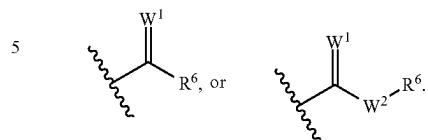


In some embodiments, each of R^{3a1}, R^{3b1}, R^{3c1}, and R^{3d1} is, independently, H,

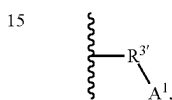


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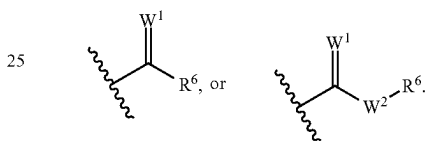
optionally substituted C₁-C₆ alkyl,



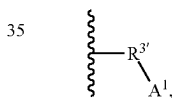
In some embodiments, each of R^{3a1}, R^{3b1}, R^{3c1}, and R^{3d1} is, independently, H,



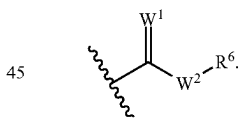
C₁-C₆ alkyl,



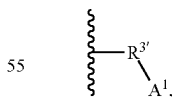
In some embodiments, each of R^{3a1}, R^{3b1}, R^{3c1}, and R^{3d1} is, independently, H,



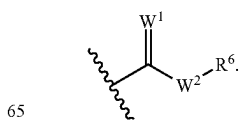
C₁-C₆ alkyl, or



In some embodiments, each of R^{3a1}, R^{3b1}, R^{3c1}, and R^{3d1} is, independently,



C₁-C₆ alkyl, or

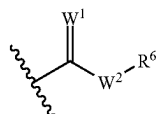


51

In some embodiments, each of R^{3a1} , R^{3b1} , R^{3c1} , and R^{3d1} is, independently, H,



methyl, or



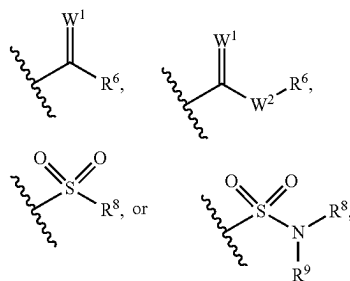
In some embodiments, each of R^{3a1} , R^{3b1} , R^{3c1} , and R^{3d1} is, independently



or optionally substituted C_1 - C_6 alkyl.

In some embodiments, each of R^{4a1} , R^{4b1} , R^{4c1} , and R^{4d1} is, independently, H, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_1 - C_6 acyl, optionally substituted sulfone, or optionally substituted sulfonamide. In some embodiments, each of R^{4a1} , R^{4b1} , R^{4c1} , and R^{4d1} is, independently, H, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted sulfone, or optionally substituted sulfonamide. In some embodiments, each of R^{4a1} , R^{4b1} , R^{4c1} , and R^{4d1} is H. In some embodiments, each of R^{4a1} , R^{4b1} , R^{4c1} , and R^{4d1} is, independently, H, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 acyl, optionally substituted sulfone, or optionally substituted sulfonamide. In some embodiments, each of R^{4a1} , R^{4b1} , R^{4c1} , and R^{4d1} is, independently, H, optionally substituted C_1 - C_6 alkyl, or optionally substituted C_1 - C_6 acyl.

In some embodiments, each of R^{4a1} , R^{4b1} , R^{4c1} , and R^{4d1} is, independently, H, optionally substituted C_1 - C_6 alkyl,



where

R^6 is H, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, or optionally substituted C_6 - C_{10} aryl;

W^1 is O or S;

W^2 is NR^7 or O;

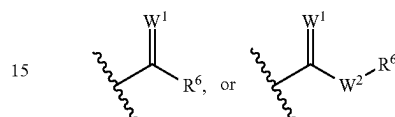
52

R^7 is H, optionally substituted C_1 - C_6 alkyl, or optionally substituted C_1 - C_6 heteroalkyl;

R^8 is optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, or optionally substituted C_6 - C_{10} aryl; and

R^9 is H, optionally substituted C_1 - C_6 alkyl, or optionally substituted C_1 - C_6 heteroalkyl.

In some embodiments, each of R^{4a1} , R^{4b1} , R^{4c1} , and R^{4d1} is, independently, H, optionally substituted C_1 - C_6 alkyl,



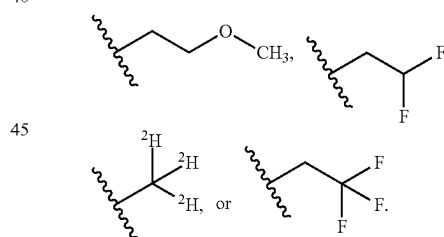
In some embodiments, each of R^{4a1} , R^{4b1} , R^{4c1} , and R^{4d1} is, independently, H or optionally substituted C_1 - C_6 alkyl. In some embodiments, each of R^{4a1} , R^{4b1} , R^{4c1} , and R^{4d1} is, independently, H.

In some embodiments, W^1 is O. In some embodiments, W^1 is S.

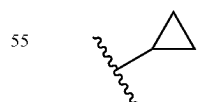
In some embodiments, W^2 is O. In some embodiments, W^2 is NR^7 .

In some embodiments, R^6 is H, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, or optionally substituted C_3 - C_{10} carbocyclyl. In some embodiments, R^6 is H, optionally substituted C_1 - C_6 alkyl, or optionally substituted C_1 - C_6 heteroalkyl. In some embodiments, R^6 is H, optionally substituted C_3 - C_{10} carbocyclyl, or optionally substituted C_6 - C_{10} aryl. In some embodiments, R^6 is optionally substituted C_1 - C_6 alkyl or optionally substituted C_1 - C_6 heteroalkyl. In some embodiments, R^6 is optionally substituted C_3 - C_{10} carbocyclyl or optionally substituted C_6 - C_{10} aryl.

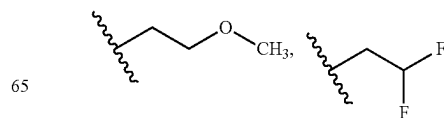
In some embodiments, R^6 is H, methyl, ethyl,



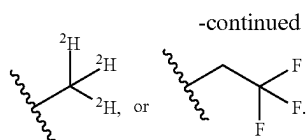
In some embodiments, R^6 is H or



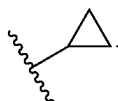
In some embodiments, R^6 is methyl, ethyl,



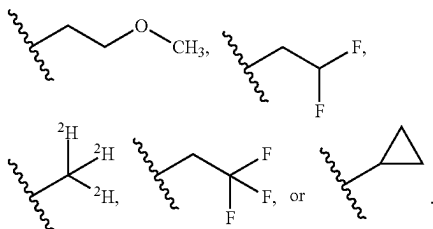
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In some embodiments, R^6 is or



In some embodiments, R^6 is H. In some embodiments, R^6 is H, methyl, ethyl,

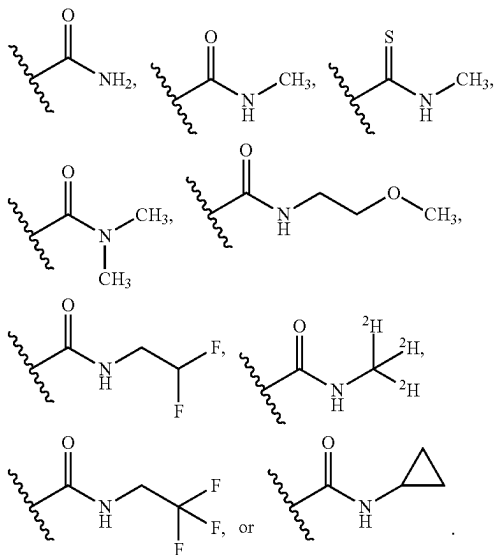


In some embodiments, R^7 is H or optionally substituted C_1 - C_6 alkyl. In some embodiments, R^7 is H or methyl.

In some embodiments, each of R^{3a1} , R^{3b1} , R^{3c1} , and R^{3d1} is, independently, H,

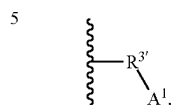


methyl,

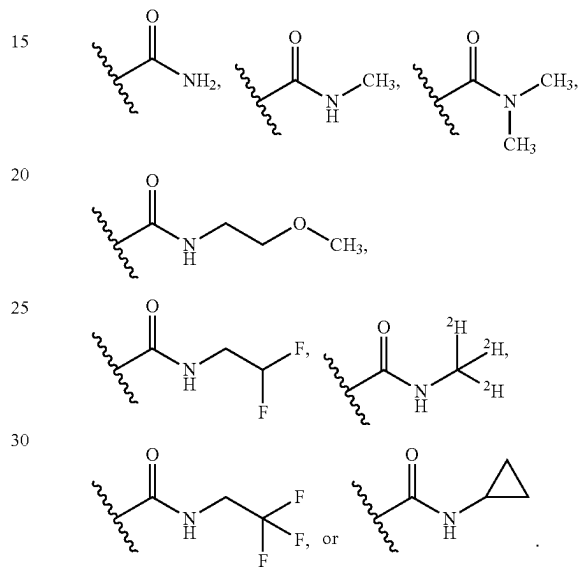


54

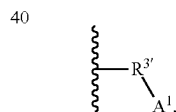
In some embodiments, each of R^{3a1} , R^{3b1} , R^{3c1} , and R^{3d1} is, independently, H,



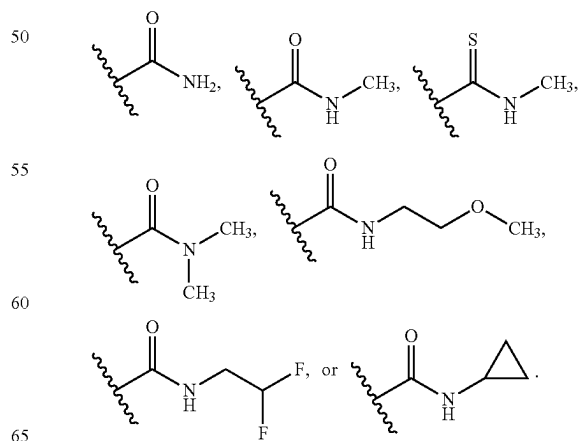
methyl,



In some embodiments, R^{3a1} is H,



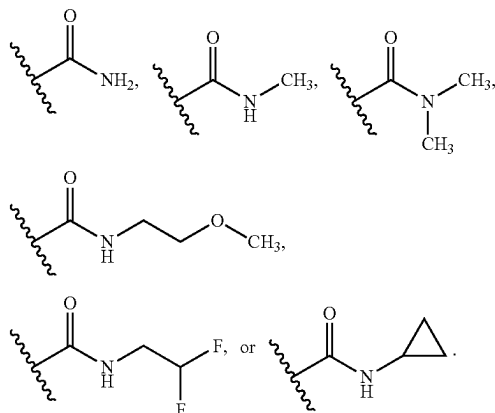
methyl



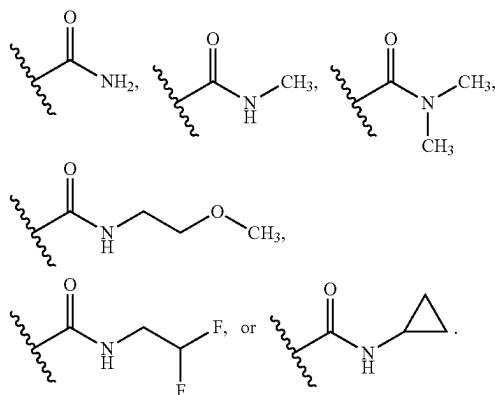
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In some embodiments, R^{3bt} is H,

methyl,

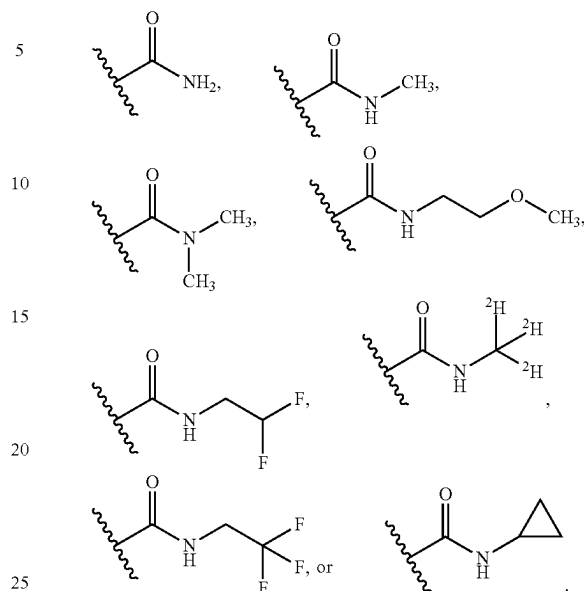
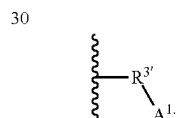
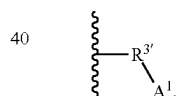
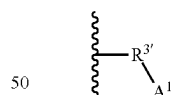
In some embodiments, R^{3ct} is H

methyl,

In some embodiments, R^{3dt} is H,

56

methyl,

In some embodiments, one of R^{3at} , R^{3bt} , R^{3ct} , and R^{3dt} isIn some embodiments, R^{3bt} isIn some embodiments, R^{3ct} isIn some embodiments, R^{3t} is absent.

In some embodiments, R^{3at} and R^{4bt} , R^{4at} and R^{3bt} , R^{4bt} and R^{4at} , R^{3bt} and R^{4ct} , R^{4bt} and R^{4ct} , R^{3ct} and R^{4bt} , R^{3ct} and R^{4dt} , R^{4ct} and R^{4dt} , and/or R^{3dt} and R^{4ct} , together with the atoms to which each is attached, combine to form optionally substituted C_2 - C_9 heterocyclyl.

In some embodiments, R^{3at} and R^{4bt} , R^{4at} and R^{3bt} , R^{3bt} and R^{4ct} , R^{4bt} and R^{3ct} , R^{3ct} and R^{4dt} , and/or R^{4ct} and R^{4dt} , together with the atoms to which each is attached, combine to form optionally substituted C_2 - C_9 heterocyclyl.

In some embodiments, each of R^{5a} , R^{5b} , R^{5c} , and R^{5d} is, independently, H, optionally substituted C_1 - C_6 alkyl, or optionally substituted C_1 - C_6 heteroalkyl. In some embodiments, each of R^{5a} , R^{5b} , R^{5c} , and R^{5d} is H.

57

In some embodiments, each of R^{3ei} , R^{3fi} , and R^{3gi} is, independently H,



optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_1 - C_6 acyl, optionally substituted sulfone, or optionally substituted sulfonamide.

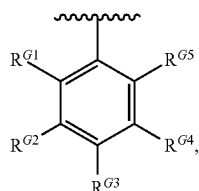
In some embodiments, each of R^{4ei} , R^{4fi} , and R^{4gi} is, independently, H, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_1 - C_6 acyl, optionally substituted sulfone, or optionally substituted sulfonamide.

In some embodiments, each of R^{5ei} , R^{5fi} , and R^{5gi} is, independently, H, optionally substituted C_1 - C_6 alkyl, or optionally substituted C_1 - C_6 heteroalkyl. In some embodiments, each of R^{5ei} , R^{5fi} , and R^{5gi} is H.

In some embodiments, G" is optionally substituted C_3 - C_{10} carbocyclyl or optionally substituted C_2 - C_9 heterocyclyl. In some embodiments, G" is optionally substituted C_6 - C_{10} aryl or optionally substituted C_2 - C_9 heteroaryl.

In some embodiments, G" is optionally substituted C_3 - C_{10} carbocyclyl. In some embodiments, G is optionally substituted C_6 - C_{10} aryl. In some embodiments, G is optionally substituted C_2 - C_9 heterocyclyl. In some embodiments, G" is optionally substituted C_2 - C_9 heteroaryl.

In some embodiments, G" is



where

each of R^{G1} , R^{G2} , R^{G3} , R^{G4} , and R^{G5} is, independently, H, halogen, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, optionally substituted $-O-C_3-C_6$ carbocyclyl, optionally substituted $-C_1-C_3$ alkyl- C_3-C_6 carbocyclyl, optionally substituted $-C_1-C_3$ alkyl- C_2-C_5 heterocyclyl, hydroxyl, thiol, or optionally substituted amino; or R^{G1} and R^{G2} , R^{G2} and R^{G3} , R^{G3} and R^{G4} , and/or R^{G4} and R^{G5} , together with the carbon atoms to which each is attached, combine to form optionally substituted C_6 - C_{10} aryl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heteroaryl, or optionally substituted C_2 - C_9 heterocyclyl.

In some embodiments, each of R^{G1} , R^{G2} , R^{G3} , R^{G4} , and R^{G5} is, independently, H, halogen, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted

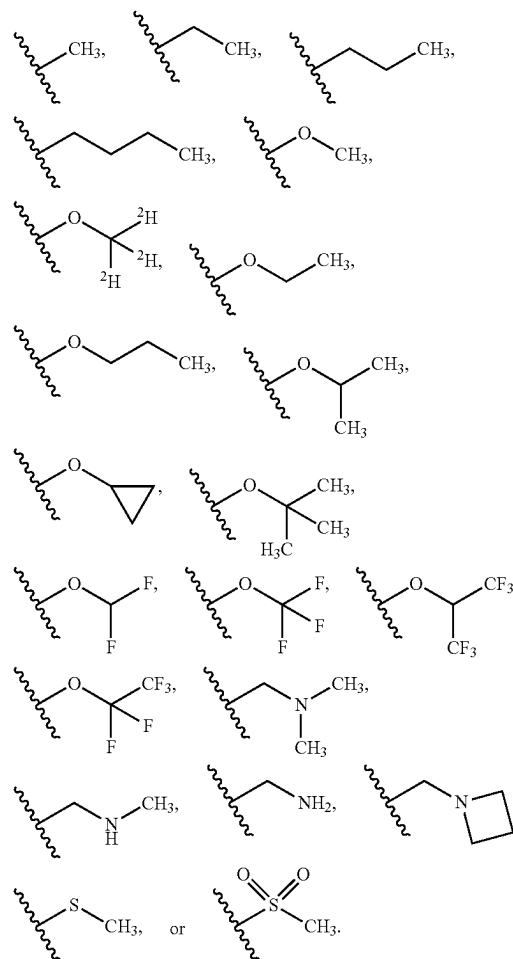
58

tuted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, optionally substituted $-O-C_3-C_6$ carbocyclyl, optionally substituted $-C_1-C_3$ alkyl- C_3-C_6 carbocyclyl, optionally substituted $-C_1-C_3$ alkyl- C_2-C_5 heterocyclyl, hydroxyl, thiol, or optionally substituted amino; or R^{G1} and R^{G2} , R^{G2} and R^{G3} , R^{G3} and R^{G4} , and/or R^{G4} and R^{G5} , together with the carbon atoms to which each is attached, combine to form optionally substituted C_6 - C_{10} aryl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heteroaryl, or optionally substituted C_2 - C_9 heterocyclyl.

In some embodiments, each of R^{G1} , R^{G2} , R^{G3} , R^{G4} , and R^{G5} is, independently, H, halogen, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted $-O-C_3-C_6$ carbocyclyl, or optionally substituted $-C_1-C_3$ alkyl- C_2-C_5 heterocyclyl; or R^{G1} and R^{G2} , R^{G2} and R^{G3} , R^{G3} and R^{G4} , and/or R^{G4} and R^{G5} , together with the carbon atoms to which each is attached, combine to form optionally substituted C_2 - C_9 heteroaryl or optionally substituted C_2 - C_9 heterocyclyl.

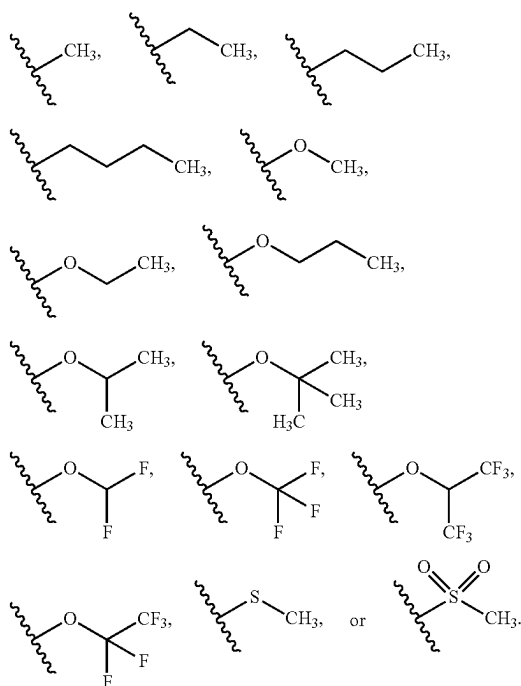
In some embodiments, each of R^{G1} , R^{G2} , R^{G3} , R^{G4} , and R^{G5} is, independently, H, halogen, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted $-O-C_3-C_6$ carbocyclyl, or optionally substituted $-C_1-C_3$ alkyl- C_2-C_5 heterocyclyl.

In some embodiments, each of R^{G1} , R^{G2} , R^{G3} , R^{G4} , and R^{G5} is, independently, H, F, Cl,

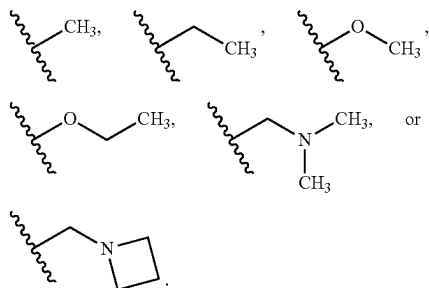


59

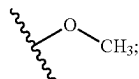
In some embodiments, each of R^{G1} , R^{G2} , R^{G3} , R^{G4} and R^{G5} is, independently, H, F



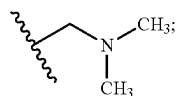
In some embodiments, each of R^{G1} , R^{G2} , R^{G3} , R^{G4} , and R^{G5} is, independently, H, F, Cl,



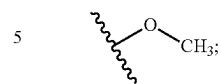
In some embodiments, R^{G1} is H; R^{G2} is



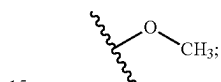
R^{G3} is



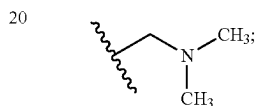
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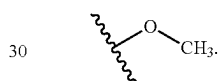
and R^{G5} is H. In some embodiments, R^{G1} is H; R^{G2} is



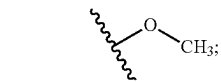
R^{G3} is



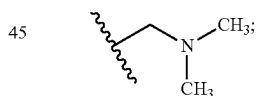
R^{G4} is H; and R^{G5} is



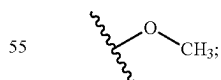
In some embodiments, R^{G1} is H; R^{G2} is



R^{G3} is



R^{G4} is Cl or F; and R^{G5} is H. In some embodiments, R^{G1} is H; R^{G2} is

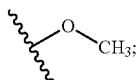


R^{G3} is

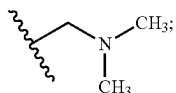


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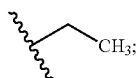
R^{G4} is H; and R^{G5} is H. In some embodiments, R^{G1} is H; R^{G2} is



R^{G3} is



R^{G4} is



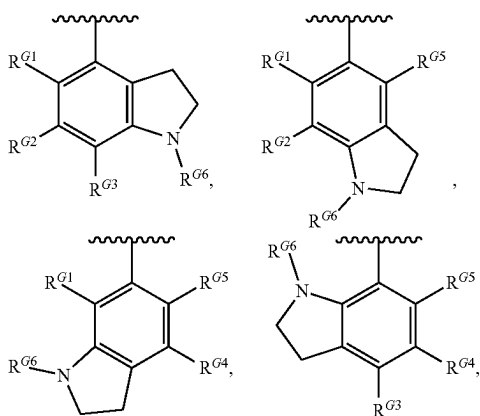
and R^{G5} is H.

In some embodiments, R^{G1} and R^{G2} , R^{G2} and R^{G3} , R^{G3} and R^{G4} , and/or R^{G4} and R^{G5} , together with the carbon atoms to which each is attached, combine to form optionally substituted C_2 - C_9 heteroaryl or optionally substituted C_2 - C_9 heterocyclyl.

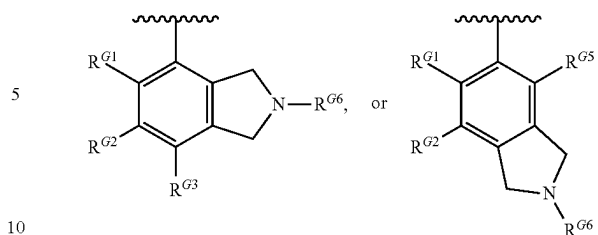
In some embodiments, R^{G1} and R^{G2} , R^{G2} and R^{G3} , R^{G3} and R^{G4} , and/or R^{G4} and R^{G5} , together with the carbon atoms to which each is attached, combine to form optionally substituted C_2 - C_9 heterocyclyl. In some embodiments, R^{G1} and R^{G2} , R^{G2} and R^{G3} , R^{G3} and R^{G4} , and/or R^{G4} and R^{G5} , together with the carbon atoms to which each is attached, combine to form optionally substituted C_2 - C_9 heteroaryl.

In some embodiments, R^{G1} and R^{G2} , R^{G2} and R^{G3} , R^{G3} and R^{G4} , and/or R^{G4} and R^{G5} , together with the carbon atoms to which each is attached, combine to form optionally substituted C_2 - C_9 heterocyclyl. In some embodiments, R^{G1} and R^{G2} , R^{G2} and R^{G3} , R^{G3} and R^{G4} , and/or R^{G4} and R^{G5} , together with the carbon atoms to which each is attached, combine to form optionally substituted C_2 - C_9 heteroaryl.

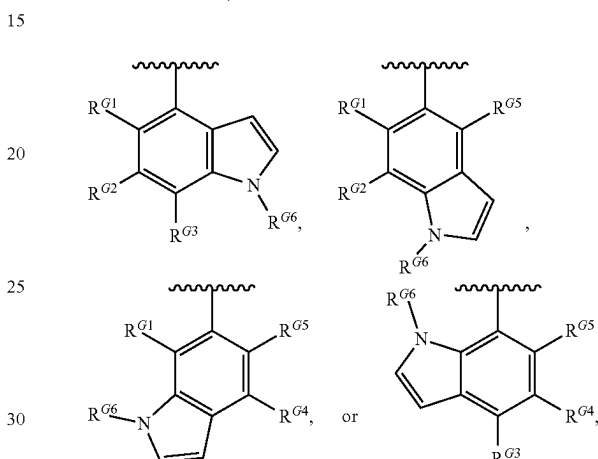
In some embodiments, G'' is

**62**

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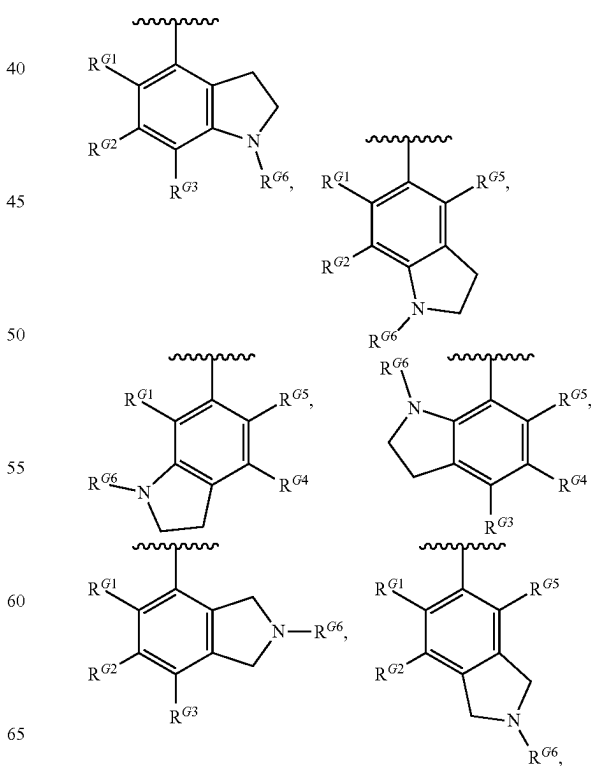


where R^{G6} is H or optionally substituted C_1 - C_6 alkyl. In some embodiments, G'' is



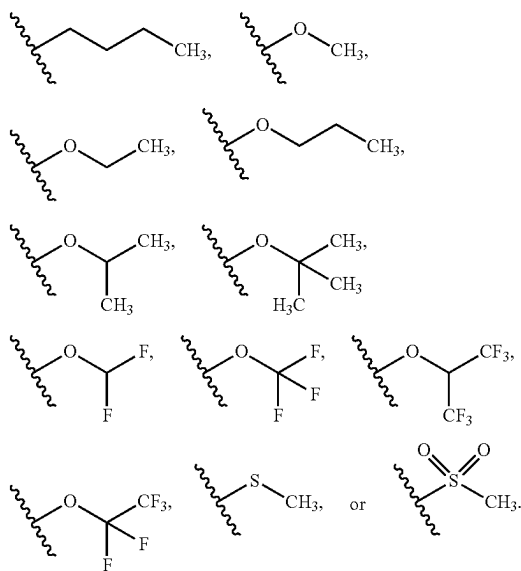
where R^{G6} is H or optionally substituted C_1 - C_6 alkyl.

In some embodiments, G'' is



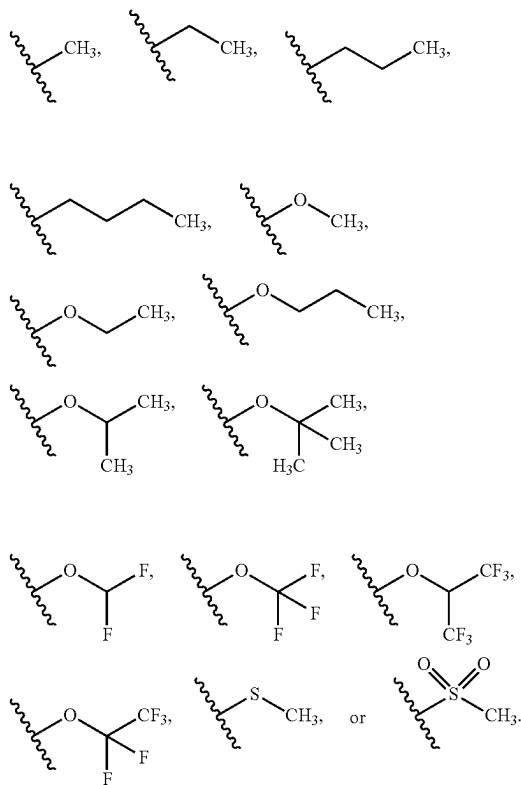
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In some embodiments, R^{G4} is H.

In some embodiments, R^{G5} is H, F,

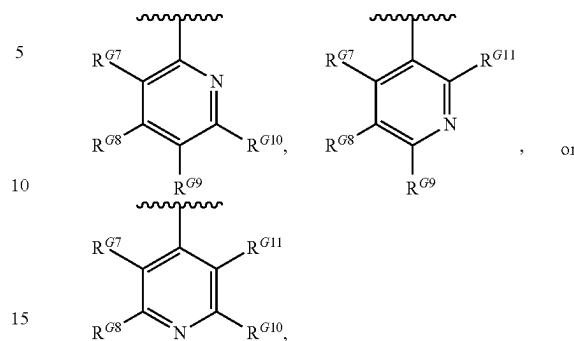


In some embodiments, R^{G5} is H.

In some embodiments, one or more of R^{G1} , R^{G2} , R^{G3} , R^{G4} , and R^{G5} is H. In some embodiments, two or more of R^{G1} , R^{G2} , R^{G3} , R^{G4} , and R^{G5} is H. In some embodiments, three or more of R^{G1} , R^{G2} , R^{G3} , R^{G4} , and R^{G5} is H. In some embodiments, each of R^{G1} , R^{G2} , R^{G3} , R^{G4} , and R^{G5} is H.

66

In some embodiments, G'' is



where

each of R^{G7} , R^{G8} , R^{G9} , R^{G10} , and R^{G11} is, independently, H, halogen, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, optionally substituted —O—C_3 - C_6 carbocyclyl, optionally substituted —C_1 - C_3 alkyl- C_3 - C_6 carbocyclyl, optionally substituted —C_1 - C_3 alkyl- C_2 - C_5 heterocyclyl, hydroxyl, thiol, or optionally substituted amino; or R^{G7} and R^{G8} , R^{G8} and R^{G9} , R^{G9} and R^{G10} , and/or R^{G10} and R^{G11} , together with the carbon atoms to which each is attached, combine to form optionally substituted C_6 - C_{10} aryl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heteroaryl, or C_2 - C_9 heterocyclyl.

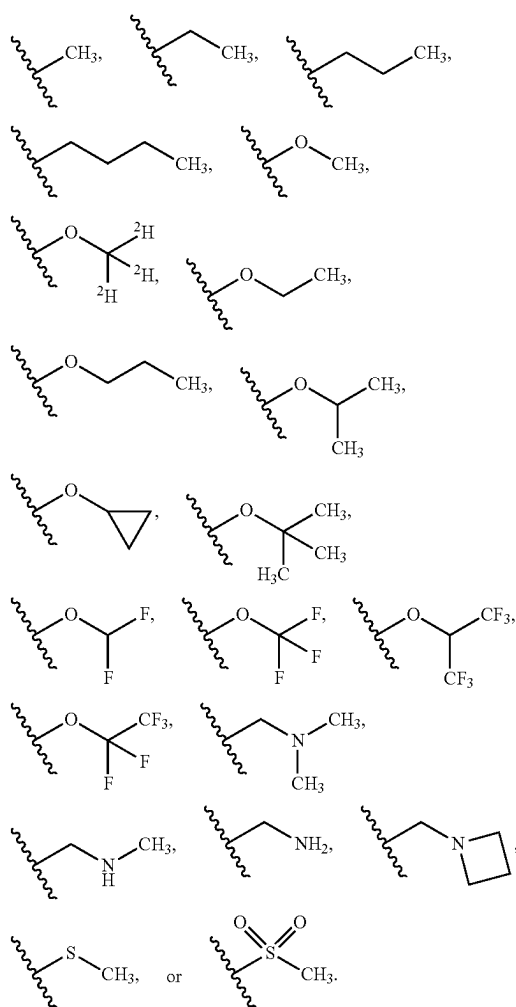
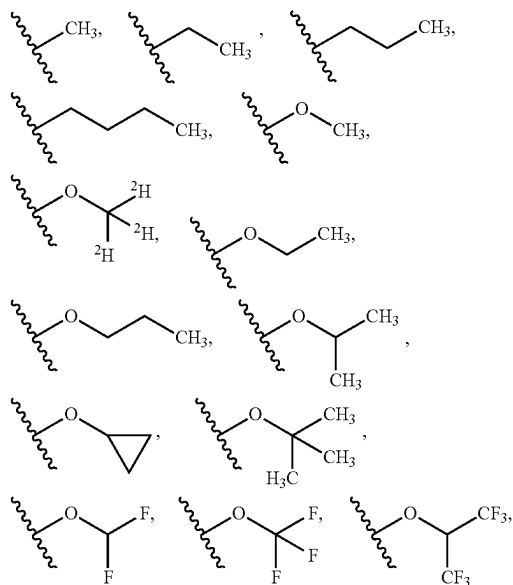
In some embodiments, each of R^{G7} , R^{G8} , R^{G9} , R^{G10} , and R^{G11} is, independently, H, halogen, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, hydroxyl, thiol, or optionally substituted amino; or R^{G7} and R^{G8} , R^{G8} and R^{G9} , R^{G9} and R^{G10} , and/or R^{G10} and R^{G11} , together with the carbon atoms to which each is attached, combine to form optionally substituted C_6 - C_{10} aryl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heteroaryl, or C_2 - C_9 heterocyclyl.

In some embodiments, each of R^{G7} , R^{G8} , R^{G9} , R^{G10} , and R^{G11} is, independently, H, halogen, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted —O—C_3 - C_6 carbocyclyl, or optionally substituted —C_1 - C_3 alkyl- C_2 - C_5 heterocyclyl; or R^{G7} and R^{G8} , R^{G8} and R^{G9} , R^{G9} and R^{G10} , and/or R^{G10} and R^{G11} , together with the carbon atoms to which each is attached, combine to form optionally substituted C_6 - C_{10} aryl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heteroaryl, or C_2 - C_9 heterocyclyl.

In some embodiments, each of R^{G7} , R^{G8} , R^{G9} , R^{G10} , and R^{G11} is, independently, H, halogen, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted —O—C_3 - C_6 carbocyclyl, or optionally substituted —C_1 - C_3 alkyl- C_2 - C_5 heterocyclyl.

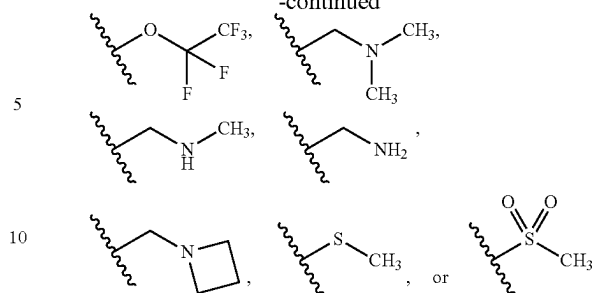
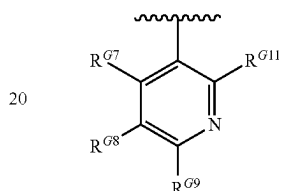
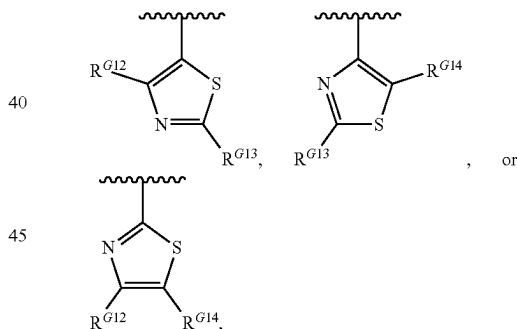
In some embodiments, each of R^{G7} , R^{G8} , R^{G9} , R^{G10} , and R^{G11} is, independently, H, F, Cl,

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In some embodiments, R^{G8} is

68

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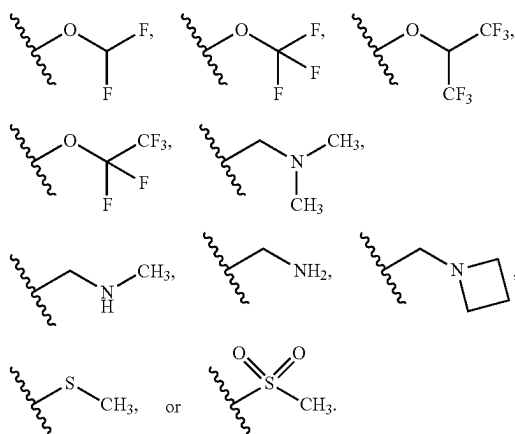
In some embodiments, G'' isIn some embodiments, R^{G7} is H; R^{G8} is R^{G9} is H; and R^{G11} is H.In some embodiments, G'' is

where

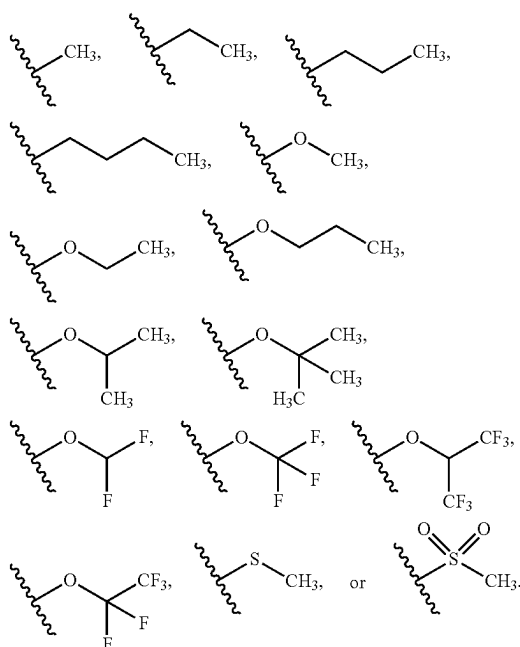
each of R^{G12} , R^{G13} , and R^{G14} is, independently, H, halogen, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocycl, optionally substituted C_2 - C_9 heterocycl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, optionally substituted —O—C_3 - C_6 carbocycl, optionally substituted —C_1 - C_3 alkyl- C_3 - C_6 carbocycl, optionally substituted —C_1 - C_3 alkyl- C_2 - C_5 heterocycl, hydroxyl, thiol, or optionally substituted amino; or R^{G12} and R^{G14} , together with the carbon atoms to which each is attached, combine to form optionally substituted C_6 - C_{10} aryl, optionally substituted C_3 - C_{10} carbocycl, optionally substituted C_2 - C_9 heteroaryl, or optionally substituted C_2 - C_9 heterocycl.

71

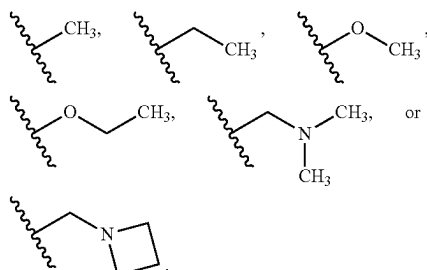
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In some embodiments, each of R^{G1t} , R^{G2t} , R^{G3t} , R^{G4t} , and R^{G5t} is, independently, H, A¹, F,



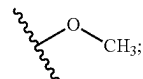
In some embodiments, each of R^{G1t} , R^{G2t} , R^{G3t} , R^{G4t} , and R^{G5t} is, independently, H, A¹, F, Cl,



In some embodiments, R^{G3t} is A¹.

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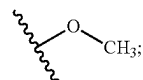
In some embodiments, R^{G1t} is H; R^{G2t} is



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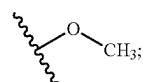
R^{G3t} is A¹; R^{G4t} is

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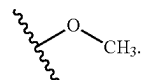
15 and R^{G5t} is H. In some embodiments, R^{G1t} is H; R^{G2t} is

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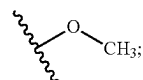


R^{G3t} is A; R^{G4t} is H; and R^{G5t} is

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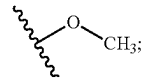
30 In some embodiments, R^{G1t} is H; R^{G2t} is



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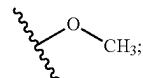
R^{G3t} is A¹; R^{G4t} is Cl or F; and R^{G5t} is H. In some embodiments, R^{G1t} is H; R^{G2t} is

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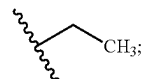
45 R^{G3t} is A¹; R^{G4t} is H; and R^{G5t} is H. In some embodiments, R^{G1t} is H; R^{G2t} is

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R^{G3t} is A¹; R^{G4t} is

55



60 and R^{G5t} is H.

In some embodiments, R^{G1t} and R^{G2t} , R^{G2t} and R^{G3t} , R^{G3t} and R^{G4t} , and/or R^{G4t} and R^{G5t} , together with the carbon

65 atoms to which each is attached, combine to form $\textcircled{\text{M}}$; and

$\textcircled{\text{M}}$ is optionally substituted C₂-C₉ heterocyclyl, which is

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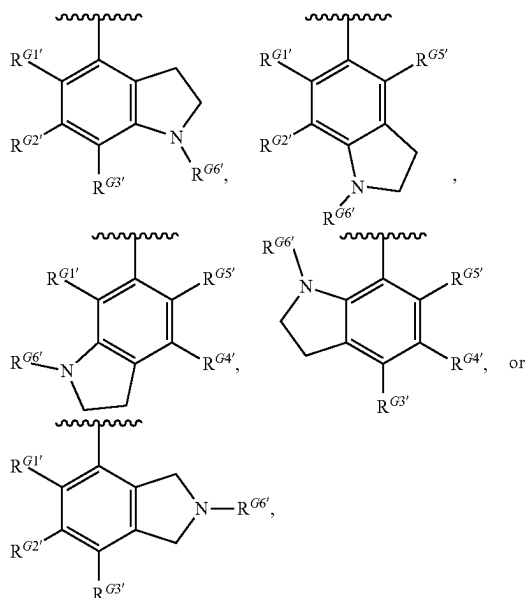
optionally substituted with A^1 , where one of $R^{G1'}$, $R^{G2'}$,

$R^{G3'}$, $R^{G4'}$ and $R^{G5'}$ is A^1 , or $\textcircled{M}$ is substituted with A^1 . In some embodiments, $R^{G1'}$, and $R^{G2'}$, $R^{G3'}$, and $R^{G4'}$, and/or $R^{G4'}$ and $R^{G5'}$, together with the carbon atoms

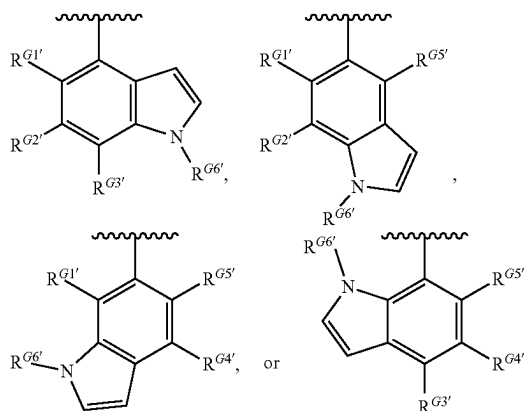
to which each is attached, combine to form $\textcircled{M}$; and

$\textcircled{M}$ is optionally substituted C_2 - C_9 heteroaryl, which is optionally substituted with A^1 , where one of $R^{G1'}$, $R^{G2'}$,

$R^{G3'}$, $R^{G4'}$, and $R^{G5'}$ is A^1 , or $\textcircled{M}$ is substituted with A^1 . In some embodiments, G' is



where $R^{G6'}$ is H, A^1 , or optionally substituted C_1 - C_6 alkyl. In some embodiments, G' is



where $R^{G6'}$ is H, A^1 , or optionally substituted C_1 - C_6 alkyl. In some embodiments, $R^{G1'}$, and $R^{G2'}$, $R^{G3'}$, and $R^{G4'}$, and/or $R^{G4'}$ and $R^{G5'}$, together with the carbon

atoms to which each is attached, combine to form $\textcircled{M}$; and

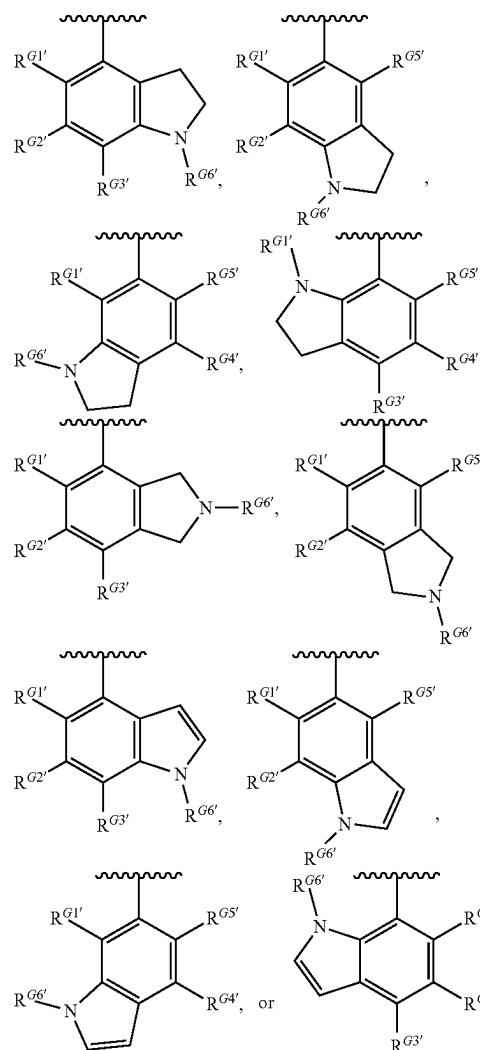
$\textcircled{M}$ is optionally substituted C_2 - C_9 heterocyclyl or option-

74

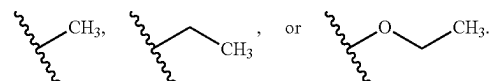
ally substituted C_2 - C_9 heteroaryl, any of which is optionally substituted with A^1 , where one of $R^{G1'}$, $R^{G2'}$, $R^{G3'}$, $R^{G4'}$, and

$R^{G5'}$ is A^1 , or $\textcircled{M}$ is substituted with A^1 .

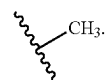
In some embodiments, G' is



where $R^{G6'}$ is H, A^1 , or optionally substituted C_1 - C_6 alkyl. In some embodiments, $R^{G6'}$ is H, A^1 ,

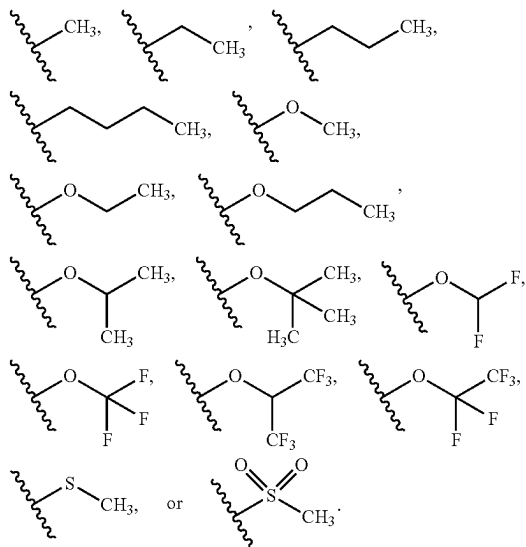
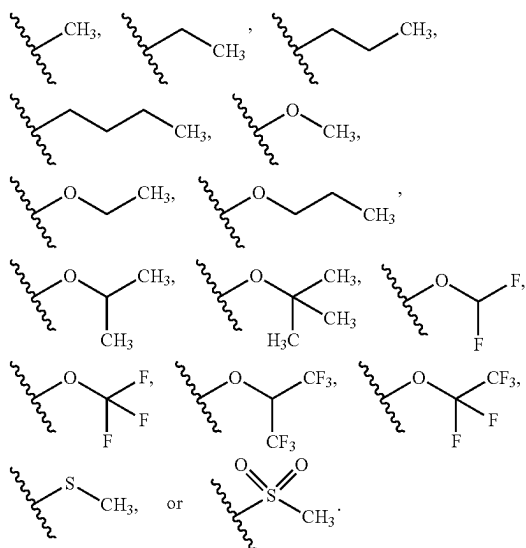
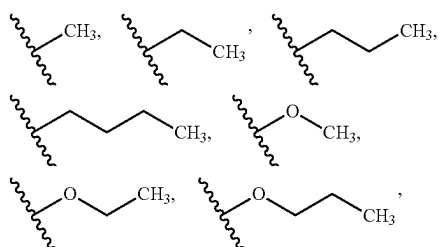


In some embodiments, $R^{G6'}$ is H, A^1 , or

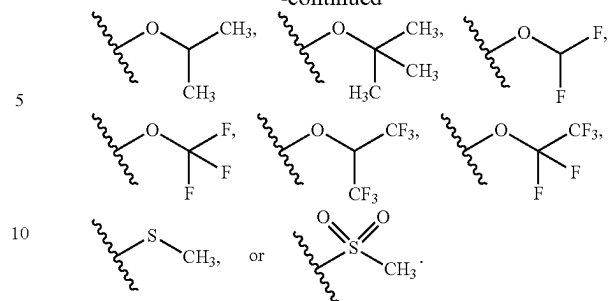


In some embodiments, $R^{G6'}$ is H or A^1 .

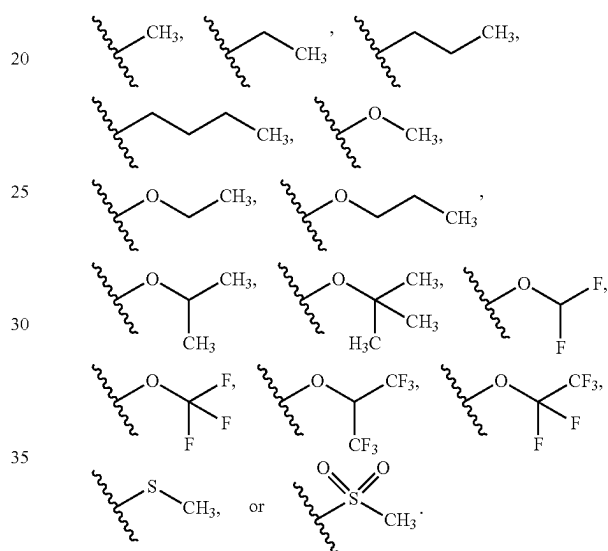
In some embodiments, $R^{G6'}$ is H. In some embodiments, $R^{G6'}$ is A^1 .

75In some embodiments, R^{G1t} is H, A¹, F,In some embodiments, R^{G1t} is H.In some embodiments, R^{G2t} is H, A¹, F,In some embodiments, R^{G2t} is H.In some embodiments, R^{G3t} is H, A¹, F,**76**

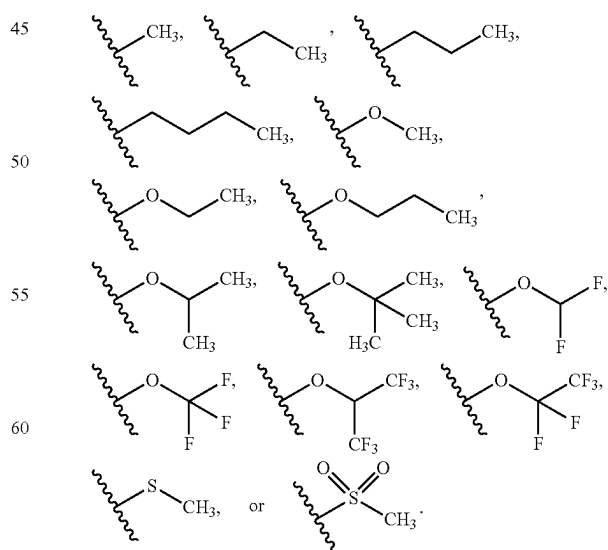
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In some embodiments, R^{G3t} is H.
 In some embodiments, R^{G4t} is H, A¹, F,



In some embodiments, R^{G4t} is H.
 In some embodiments, R^{G5t} is H, A¹, F,

In some embodiments, R^{G5t} is H.

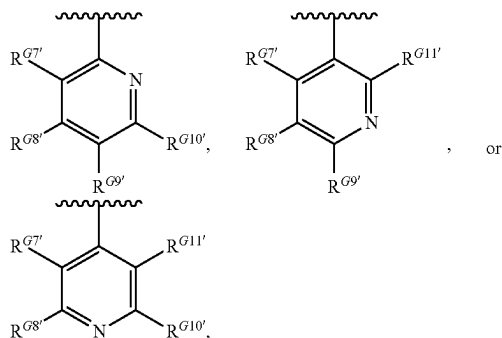
77

In some embodiments, one or more of R^{G1} , R^{G2} , R^{G3} , R^{G4} , and R^{G5} is H. In some embodiments, two or more of R^{G1} , R^{G2} , R^{G3} , R^{G4} , and R^{G5} is H. In some embodiments, three or more of R^{G1} , R^{G2} , R^{G3} , R^{G4} , and R^{G5} is H.

In some embodiments, R^{G1} is A^1 . In some embodiments, R^{G2} is A^1 . In some embodiments, R^{G3} is A^1 . In some embodiments, R^{G4} is A^1 . In some embodiments, R^{G5} is A^1 .

In some embodiments, $\textcircled{M}$ is substituted with A^1 .

In some embodiments, G^1 is



where

each of R^{G7} , R^{G8} , R^{G9} , R^{G10} , and R^{G11} is, independently, H, A^1 , halogen, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, optionally substituted $-O-C_3-C_6$ carbocyclyl, optionally substituted $-C_1-C_3$ alkyl- C_3-C_6 carbocyclyl, optionally substituted $-C_1-C_3$ alkyl- C_2-C_5 heterocyclyl, hydroxyl, thiol, or optionally substituted amino; or R^{G7} and R^{G8} , R^{G8} and R^{G9} , R^{G9} and R^{G10} , and/or R^{G10} and R^{G11} , together with the carbon atoms to which each is attached, combine to form

$\textcircled{M}$; and $\textcircled{M}$ is optionally substituted C_6 - C_{10} aryl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heteroaryl, or C_2 - C_9 heterocyclyl, any of which is optionally substituted with A^1 , where one of R^{G7} , R^{G8} , R^{G9} , R^{G10} , and R^{G11} is A^1 ; or

$\textcircled{M}$ is substituted with A^1 .

In some embodiments, each of R^{G7} , R^{G8} , R^{G9} , R^{G10} , and R^{G11} is, independently, H, A^1 , halogen, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, optionally substituted $-O-C_3-C_6$ carbocyclyl, optionally substituted $-C_1-C_3$ alkyl- C_3-C_6 carbocyclyl, optionally substituted $-C_1-C_3$ alkyl- C_2-C_5 heterocyclyl, hydroxyl, thiol, or optionally substituted amino; or R^{G7} and R^{G8} , R^{G8} and R^{G9} , R^{G9} and R^{G10} , and/or R^{G10} and R^{G11} , together with the carbon atoms to which each is

attached, combine to form $\textcircled{M}$; and $\textcircled{M}$ is optionally substituted C_6 - C_{10} aryl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heteroaryl, or C_2 - C_9

78

heterocyclyl, any of which is optionally substituted with A^1 , where one of R^{G7} , R^{G8} , R^{G9} , R^{G10} , and R^{G11} is A^1 ; or

$\textcircled{M}$ is substituted with A^1 .

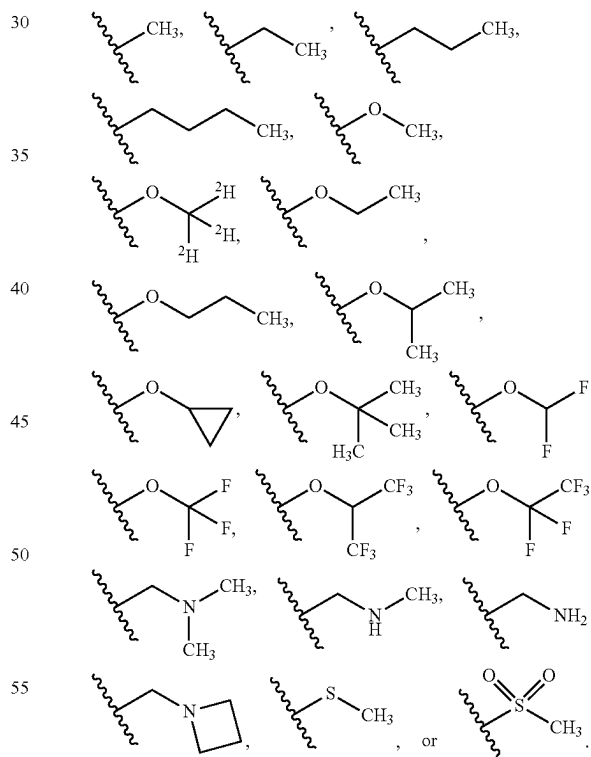
In some embodiments, each of R^{G7} , R^{G8} , R^{G9} , R^{G10} , and R^{G11} is, independently, H, A^1 , halogen, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted $-O-C_3-C_6$ carbocyclyl, or optionally substituted $-C_1-C_3$ alkyl- C_2-C_5 heterocyclyl; or R^{G7} and R^{G8} , R^{G8} and R^{G9} , R^{G9} and R^{G10} , and/or R^{G10} and R^{G11} , together with the carbon atoms to which each is

attached, combine to form $\textcircled{M}$; and $\textcircled{M}$ is optionally substituted C_6 - C_{10} aryl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heteroaryl, or C_2 - C_9 heterocyclyl, any of which is optionally substituted with A^1 , where one of R^{G7} , R^{G8} , R^{G9} , R^{G10} , and R^{G11} is A^1 ; or

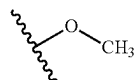
$\textcircled{M}$ is substituted with A^1 .

In some embodiments, each of R^{G7} , R^{G8} , R^{G9} , R^{G10} , and R^{G11} is, independently, H, A^1 , halogen, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted $-O-C_3-C_6$ carbocyclyl, or optionally substituted $-C_1-C_3$ alkyl- C_2-C_5 heterocyclyl.

In some embodiments, each of R^{G7} , R^{G8} , R^{G9} , R^{G10} , and R^{G11} is, independently, H, A^1 , F, Cl,

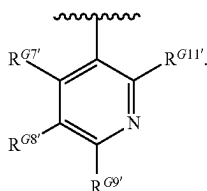


In some embodiments, R^{G8} is

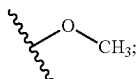


79

In some embodiments, G' is

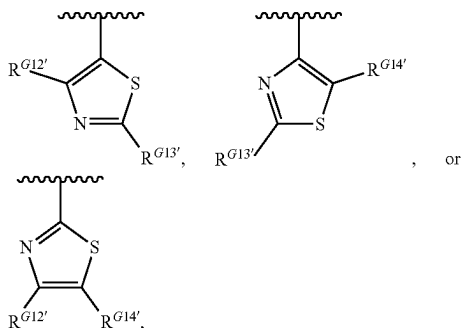


In some embodiments, $R^{G7'}$ is H; $R^{G8'}$ is



$R^{G9'}$ is A^1 ; and $R^{G11'}$ is H.

In some embodiments, G' is



where

each of $R^{G12'}$, $R^{G13'}$, and $R^{G14'}$ is, independently, H, A^1 , halogen, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, optionally substituted $-O-C_3-C_6$ carbocyclyl, optionally substituted $-C_1-C_3$ alkyl- C_3-C_6 carbocyclyl, optionally substituted $-C_1-C_3$ alkyl- C_2-C_5 heterocyclyl, hydroxyl, thiol, or optionally substituted amino; or $R^{G12'}$ and $R^{G14'}$ together with the carbon atoms to which each is attached, combine to form

$\textcircled{M}$; and $\textcircled{M}$ is optionally substituted C_6 - C_{10} aryl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heteroaryl, or optionally substituted C_2 - C_9 heterocyclyl, any of which is optionally substituted with A^1 , where one of $R^{G12'}$, $R^{G13'}$, and $R^{G14'}$ is

A^1 ; or $\textcircled{M}$ is substituted with A^1 .

In some embodiments, each of $R^{G12'}$, $R^{G13'}$, and $R^{G14'}$ is, independently, H, A^1 , halogen, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, option-

80

ally substituted $-O-C_3-C_6$ carbocyclyl, optionally substituted $-C_1-C_3$ alkyl- C_3-C_6 carbocyclyl, optionally substituted $-C_1-C_3$ alkyl- C_2-C_5 heterocyclyl, hydroxyl, thiol, or optionally substituted amino; or $R^{G12'}$ and $R^{G14'}$, together with the carbon atoms to which each is attached, combine to

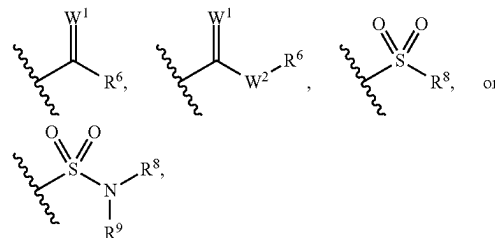
form $\textcircled{M}$; and $\textcircled{M}$ is optionally substituted C_6 - C_{10} aryl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heteroaryl, or optionally substituted C_2 - C_9 heterocyclyl, any of which is optionally substituted with A^1 ,

where one of $R^{G12'}$, $R^{G13'}$, and $R^{G14'}$ is A^1 ; or $\textcircled{M}$ is substituted with A^1 .

In some embodiments, each of $R^{3a'}$, $R^{3b'}$, $R^{3c'}$, and $R^{3d'}$ is, independently, H, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_1 - C_6 acyl, optionally substituted sulfone, or optionally substituted sulfonamide. In some embodiments, each of $R^{3a'}$, $R^{3b'}$, $R^{3c'}$, and $R^{3d'}$ is, independently, H, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted sulfone, or optionally substituted sulfonamide. In some embodiments, each of $R^{3a'}$, $R^{3b'}$, $R^{3c'}$, and $R^{3d'}$ is, independently, H, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 acyl, optionally substituted sulfone, or optionally substituted sulfonamide. In some embodiments, each of $R^{3a'}$, $R^{3b'}$, $R^{3c'}$, and $R^{3d'}$ is, independently, H, optionally substituted C_1 - C_6 alkyl, or optionally substituted C_1 - C_6 acyl.

In some embodiments, each of $R^{3a'}$, $R^{3b'}$, $R^{3c'}$, and $R^{3d'}$ is, independently, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_1 - C_6 acyl, optionally substituted sulfone, or optionally substituted sulfonamide. In some embodiments, each of $R^{3a'}$, $R^{3b'}$, $R^{3c'}$, and $R^{3d'}$ is, independently, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_1 - C_6 acyl, optionally substituted sulfone, or optionally substituted sulfonamide. In some embodiments, each of $R^{3a'}$, $R^{3b'}$, $R^{3c'}$, and $R^{3d'}$ is, independently, optionally substituted C_1 - C_6 alkyl or optionally substituted C_1 - C_6 acyl. In some embodiments, each of $R^{3a'}$, $R^{3b'}$, $R^{3c'}$, and $R^{3d'}$ is, independently, or optionally substituted C_1 - C_6 acyl. In some embodiments, each of $R^{3a'}$, $R^{3b'}$, $R^{3c'}$, and $R^{3d'}$ is, independently, optionally substituted sulfone, or optionally substituted sulfonamide.

In some embodiments, each of $R^{3a'}$, $R^{3b'}$, $R^{3c'}$, and $R^{3d'}$ is, independently, H, optionally substituted C_1 - C_6 alkyl,



81

where

R⁶ is H, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted C₃-C₁₀ carbocyclyl, or optionally substituted C₆-C₁₀ aryl;

W¹ is O or S;

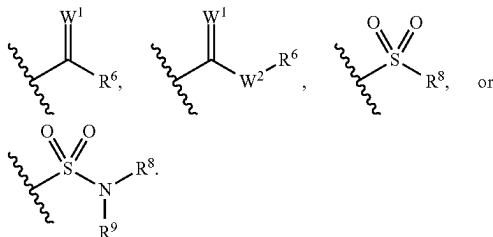
W² is NR⁷ or O;

R⁷ is H, optionally substituted C₁-C₆ alkyl, or optionally substituted C₁-C₆ heteroalkyl;

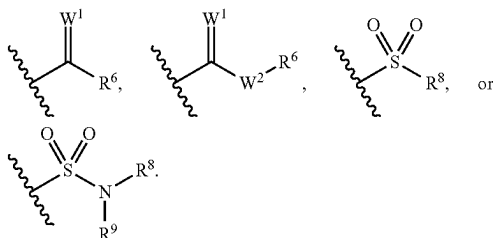
R⁸ is optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted C₃-C₁₀ carbocyclyl, or optionally substituted C₆-C₁₀ aryl; and

R⁹ is H, optionally substituted C₁-C₆ alkyl, or optionally substituted C₁-C₆ heteroalkyl.

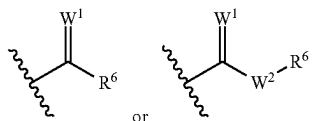
In some embodiments, each of R^{3a1}, R^{3b1}, R^{3c1}, and R^{3d1} is, independently, C₁-C₆ alkyl,



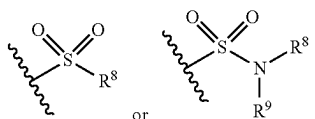
In some embodiments, each of R^{3a1}, R^{3b1}, R^{3c1}, and R^{3d1} is, independently, or C₁-C₆ alkyl. In some embodiments, each of R^{3a1}, R^{3b1}, R^{3c1}, and R^{3d1} is, independently,



In some embodiments, each of R^{3a1}, R^{3b1}, R^{3c1}, and R^{3d1} is, independently,

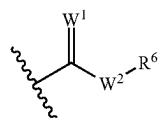


In some embodiments, each of R^{3a1}, R^{3b1}, R^{3c1}, and R^{3d1} is, independently,

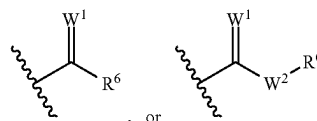


82

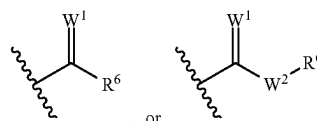
In some embodiments, each of R^{3a1}, R^{3b1}, R^{3c1}, and R^{3d1} is, independently,



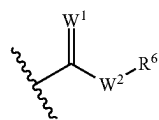
In some embodiments, each of R^{3a1}, R^{3b1}, R^{3c1}, and R^{3d1} is, independently, H, optionally substituted C₁-C₆ alkyl,



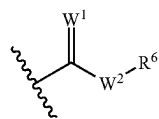
In some embodiments, each of R^{3a1}, R^{3b1}, R^{3c1}, and R^{3d1} is, independently, H, C₁-C₆ alkyl,



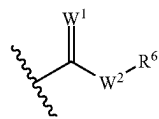
In some embodiments, each of R^{3a1}, R^{3b1}, R^{3c1}, and R^{3d1} is, independently, H, C₁-C₆ alkyl, or



In some embodiments, each of R^{3a1}, R^{3b1}, R^{3c1}, and R^{3d1} is, independently, C₁-C₆ alkyl or



In some embodiments, each of R^{3a1}, R^{3b1}, R^{3c1}, and R^{3d1} is, independently, H, methyl, or



In some embodiments, each of R^{3a1}, R^{3b1}, R^{3c1}, and R^{3d1} is, independently, optionally substituted C₁-C₆ alkyl.

In some embodiments, W¹ is O. In some embodiments, W¹ is S.

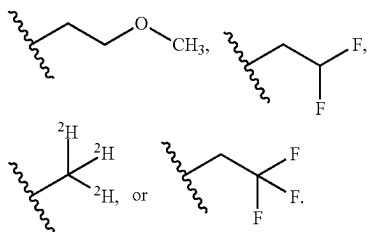
In some embodiments, W² is O. In some embodiments, W² is NR⁷.

In some embodiments, R⁶ is H, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, or

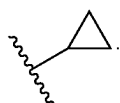
83

optionally substituted C₃-C₁₀ carbocyclyl. In some embodiments, R⁶ is H, optionally substituted C₁-C₆ alkyl, or optionally substituted C₁-C₆ heteroalkyl. In some embodiments, R⁶ is H, optionally substituted C₃-C₁₀ carbocyclyl, or optionally substituted C₆-C₁₀ aryl. In some embodiments, R⁶ is optionally substituted C₁-C₆ alkyl or optionally substituted C₁-C₆ heteroalkyl. In some embodiments, R⁶ is optionally substituted C₃-C₁₀ carbocyclyl or optionally substituted C₆-C₁₀ aryl

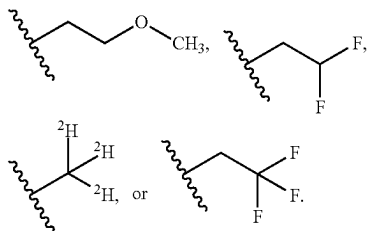
In some embodiments, R⁶ is H, methyl, ethyl,



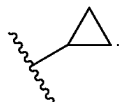
In some embodiments, R⁶ is H or



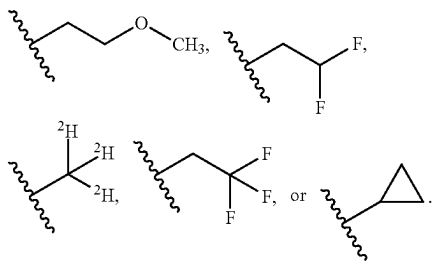
In some embodiments, R⁶ is methyl, ethyl,



In some embodiments, R⁶ is or



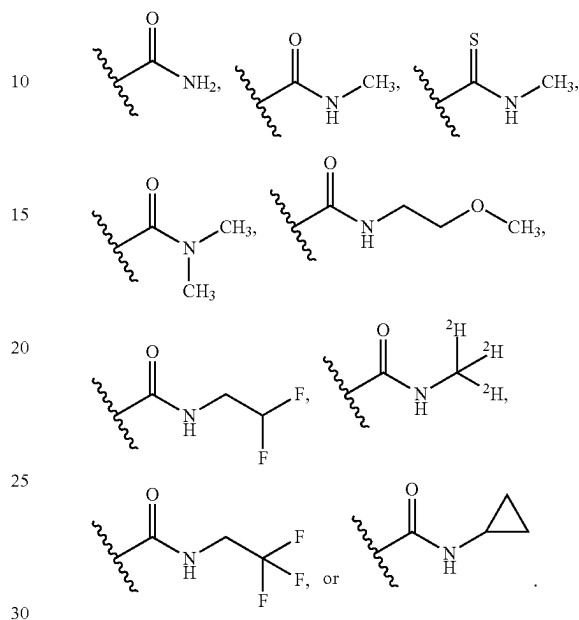
In some embodiments, R⁶ is H. In some embodiments, R⁶ is H, methyl, ethyl,



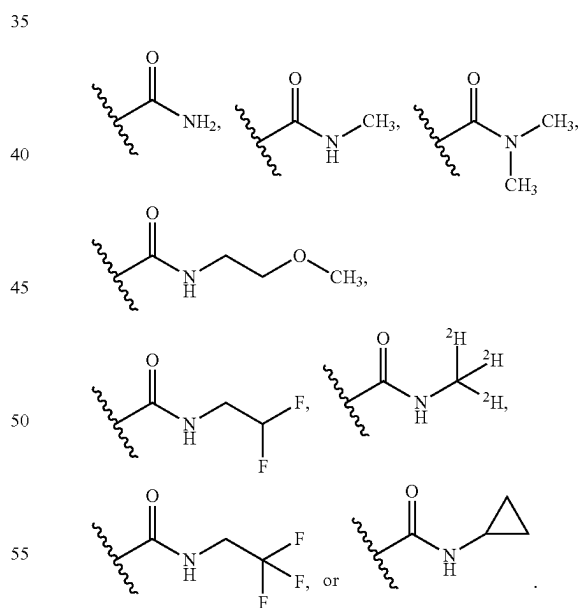
84

In some embodiments, R⁷ is H or optionally substituted C₁-C₆ alkyl. In some embodiments, R⁷ is H or methyl.

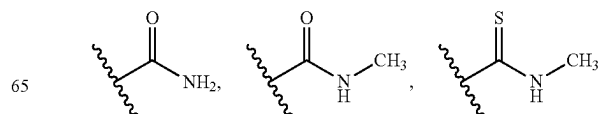
In some embodiments, each of R^{3a1}, R^{3b1}, R^{3c1}, and R^{3d1} is, independently, H, methyl,



In some embodiments, each of R^{3a1}, R^{3b1}, R^{3c1}, and R^{3d1} is, independently, H, methyl,

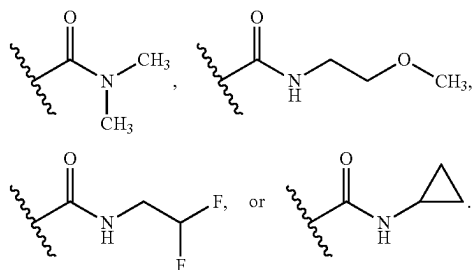
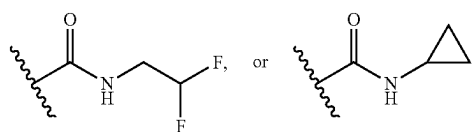
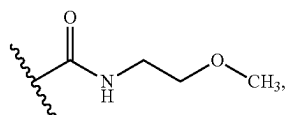
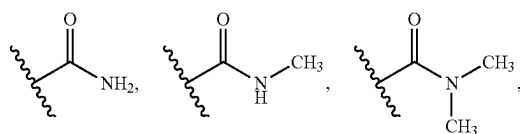
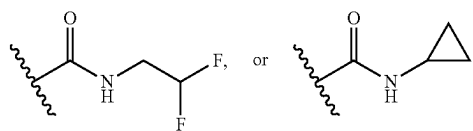
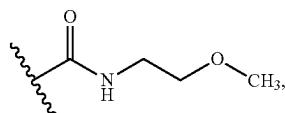
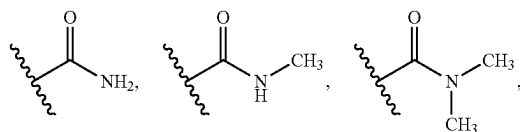
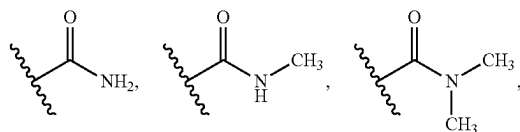


In some embodiments, R^{3a1} is H, methyl,

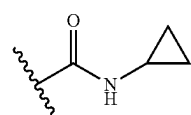
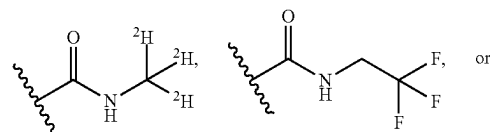
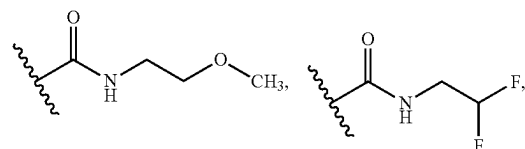


85

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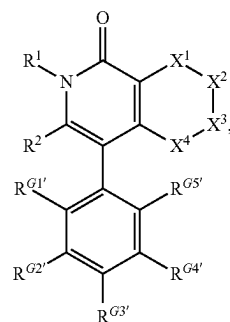
In some embodiments, R^{3b1} is H, methyl,In some embodiments, R^{3c1} is H, methyl,In some embodiments, R^{3d1} is H, methyl,**86**

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In some embodiments, A has the structure of Formula IIIa:

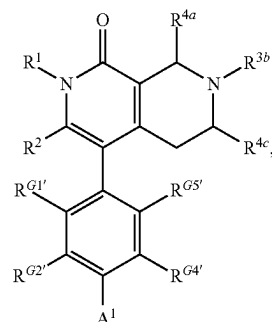
Formula IIIa



or a pharmaceutically acceptable salt thereof.

In some embodiments, A has the structure of Formula IIIb:

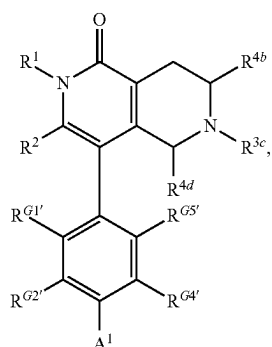
Formula IIIb



or a pharmaceutically acceptable salt thereof.

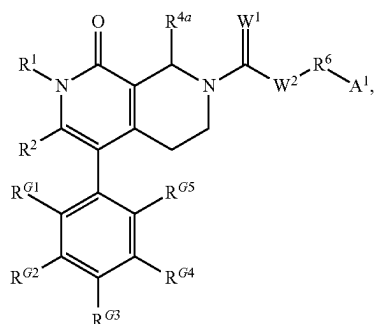
87

In some embodiments, A has the structure of Formula IIIc:



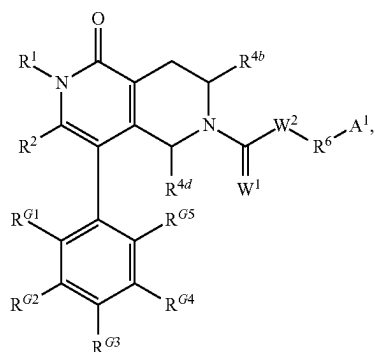
or a pharmaceutically acceptable salt thereof.

In some embodiments, A has the structure of Formula IIId:



or a pharmaceutically acceptable salt thereof.

In some embodiments, A has the structure of Formula IIIe:



or a pharmaceutically acceptable salt thereof.

88

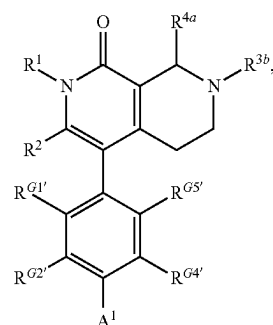
In some embodiments, A has the structure of Formula IIIf:

Formula IIIc

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or a pharmaceutically acceptable salt thereof.

In some embodiments, A has the structure of Formula IIIg:

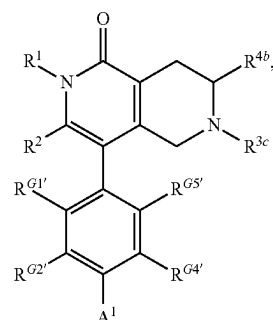
25

Formula IIId

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or a pharmaceutically acceptable salt thereof.

In some embodiments, A has the structure of Formula IIIh:

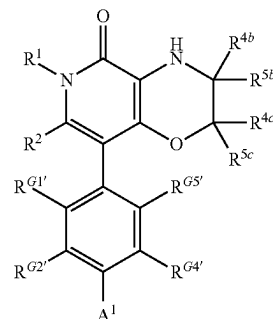
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Formula IIIe

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or a pharmaceutically acceptable salt thereof.

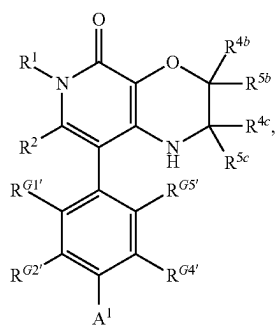
Formula IIIf

Formula IIIg

Formula IIIh

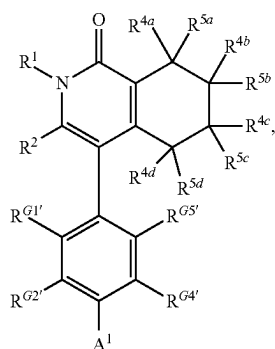
89

In some embodiments, A has the structure of Formula IIIi:



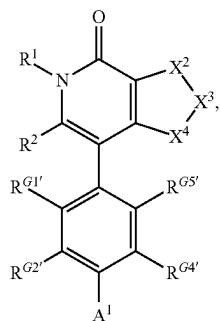
or a pharmaceutically acceptable salt thereof.

In some embodiments, A has the structure of Formula IIIj:



or a pharmaceutically acceptable salt thereof.

In some embodiments, A has the structure of Formula IIIk:



or a pharmaceutically acceptable salt thereof.

90

In some embodiments, A has the structure of Formula IIIIm:

Formula IIIi

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Formula IIIj

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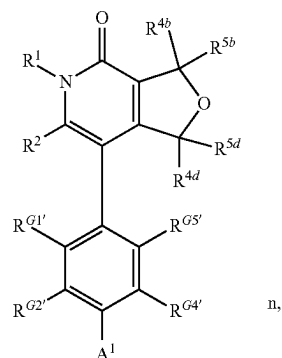
Formula IIIk

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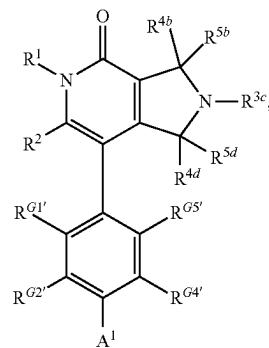
Formula IIIIm



or a pharmaceutically acceptable salt thereof.

In some embodiments, A has the structure of Formula IIIIn:

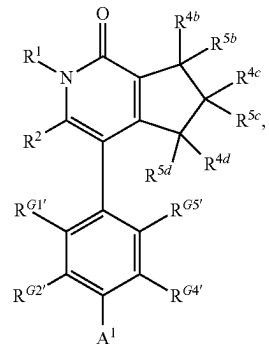
Formula IIIIn



or a pharmaceutically acceptable salt thereof.

In some embodiments, A has the structure of Formula IIIo:

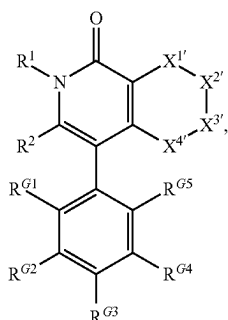
Formula IIIo



or a pharmaceutically acceptable salt thereof.

91

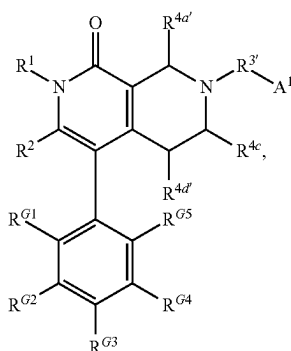
In some embodiments, A has the structure of Formula IIIp:



Formula IIIp 5

or a pharmaceutically acceptable salt thereof.

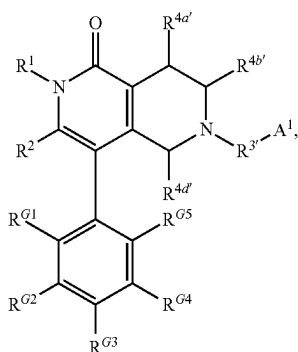
In some embodiments, A has the structure of Formula IIIq:



Formula IIIq

or a pharmaceutically acceptable salt thereof.

In some embodiments, A has the structure of Formula IIIr:



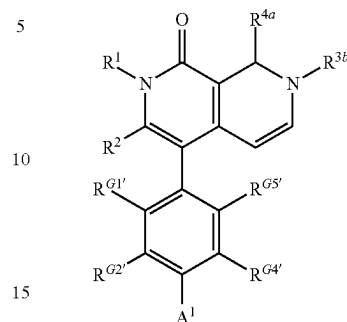
Formula IIIr

or a pharmaceutically acceptable salt thereof.

92

In some embodiments, A has the structure of Formula IIIs:

Formula IIIs



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or a pharmaceutically acceptable salt thereof.

In some embodiments, the degradation moiety is a ubiquitin ligase binding moiety.

In some embodiments, the ubiquitin ligase binding moiety comprises Cereblon ligands, IAP (Inhibitors of Apoptosis) ligands, mouse double minute 2 homolog (MDM2), or von Hippel-Lindau (VHL) ligands, or derivatives or analogs thereof.

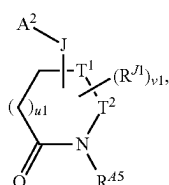
In some embodiments, the degradation moiety is a ubiquitin ligase binding moiety.

In some embodiments, the ubiquitin ligase binding moiety comprises Cereblon ligands, IAP (Inhibitors of Apoptosis) ligands, mouse double minute 2 homolog (MDM2), or von Hippel-Lindau (VHL) ligands, or derivatives or analogs thereof.

In some embodiments, the degradation moiety includes the structure of Formula Y:

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Formula Y

where

A² is a bond between the degradation moiety and the linker;

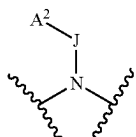
v₁ is 0, 1, 2, 3, 4, or 5;

u₁ is 1, 2, or 3;

T¹ is a bond or

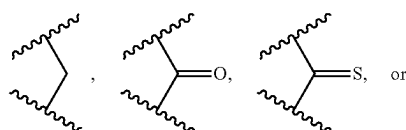
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T² is

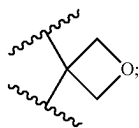
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93

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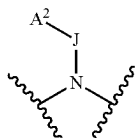


$R^{5.4}$ is H, optionally substituted C_1 - C_6 alkyl, or optionally substituted C_1 - C_6 heteroalkyl;

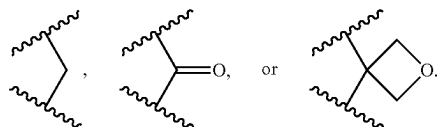
each R^{J1} is, independently, halogen, optionally substituted C_1 - C_6 alkyl, or optionally substituted C_1 - C_6 heteroalkyl; and

J is absent, optionally substituted C_3 - C_{10} carbocyclylene, optionally substituted C_6 - C_{10} arylene, optionally substituted C_2 - C_9 heterocyclylene, or optionally substituted C_2 - C_9 heteroarylene, or a pharmaceutically acceptable salt thereof.

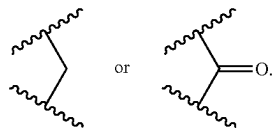
In some embodiments, T^1 is a bond. In some embodiments, T^1 is



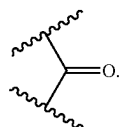
In some embodiments, T^2 is



In some embodiments, T^2 is or



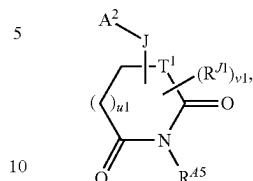
In some embodiments, T^2 is



94

In some embodiments, the structure of Formula Y has the structure of Formula Y1:

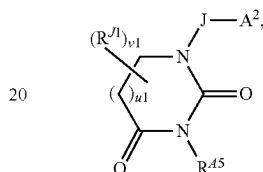
Formula Y1



or a pharmaceutically acceptable salt thereof.

In some embodiments, the structure of Formula Y has the structure of Formula Y2:

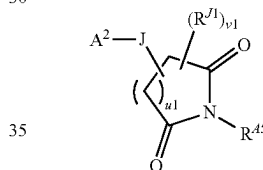
Formula Y2



or a pharmaceutically acceptable salt thereof.

In some embodiments, the structure of Formula Y has the structure of Formula Z:

Formula Z

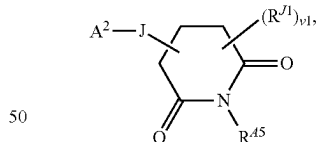


or a pharmaceutically acceptable salt thereof.

In some embodiments, $u1$ is 1. In some embodiments, $u1$ is 2. In some embodiments $u1$ is 3.

In some embodiments, the structure of Formula Z has the structure of Formula AA:

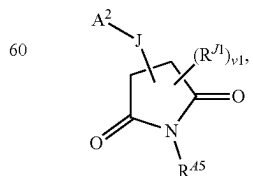
Formula AA



or a pharmaceutically acceptable salt thereof.

In some embodiments, the structure of Formula Z has the structure of Formula AB:

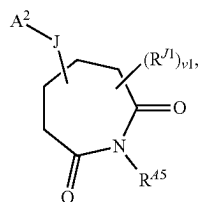
Formula AB



or a pharmaceutically acceptable salt thereof.

95

In some embodiments, the structure of Formula Z has the structure of Formula AC:

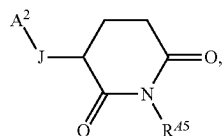


Formula AC 5

or a pharmaceutically acceptable salt thereof.

In some embodiments, v1 is 0, 1, 2, or 3. In some embodiments, v1 is 0. In some embodiments, v1 is 1. In some embodiments, v1 is 2. In some embodiments, v1 is 3.

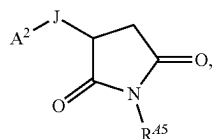
In some embodiments, the structure of Formula AA has the structure of Formula AA1:



Formula AA1 25

or a pharmaceutically acceptable salt thereof.

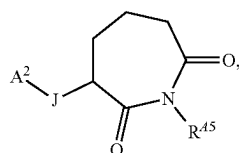
In some embodiments, the structure of Formula AB has the structure of Formula AB1:



Formula AB1 35

or a pharmaceutically acceptable salt thereof.

In some embodiments, the structure of Formula AC has the structure of Formula AC1:



Formula AC1 50

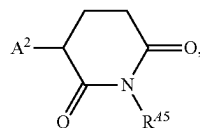
or a pharmaceutically acceptable salt thereof.

In some embodiments, J is absent. In some embodiments, J is optionally substituted C₃-C₁₀ carbocyclylene or optionally substituted C₆-C₁₀ arylene. In some embodiments, J is optionally substituted C₂-C₉ heterocyclylene or optionally substituted C₂-C₉ heteroarylene.

In some embodiments, J is optionally substituted heterocyclylene. In some embodiments, J is optionally substituted C₆-C₁₀ arylene.

96

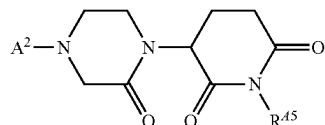
In some embodiments, the structure of Formula AA has the structure of Formula AA2:



Formula AA1

or a pharmaceutically acceptable salt thereof.

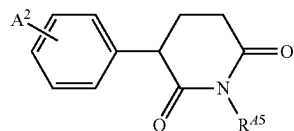
In some embodiments, the structure of Formula AA has the structure of Formula AA3:



Formula AA3 20

or a pharmaceutically acceptable salt thereof.

In some embodiments, the structure of Formula AA has the structure of Formula AA4:

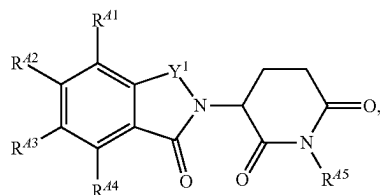


Formula AA4 30

or a pharmaceutically acceptable salt thereof.

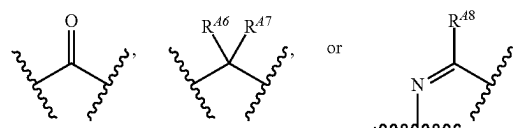
In some embodiments, R⁴⁵ is H or optionally substituted C₁-C₆ alkyl. In some embodiments, R⁴⁵ is H or methyl. In some embodiments, R⁴⁵ is H. In some embodiments, R⁴⁵ is methyl.

In some embodiments, the structure of Formula AA has the structure of Formula A:



Formula A 45

where
Y¹ is



R⁴⁵ is H, optionally substituted C₁-C₆ alkyl, or optionally substituted C₁-C₆ heteroalkyl;
R⁴⁶ is H or optionally substituted C₁-C₆ alkyl; and R⁴⁷ is H or optionally substituted C₁-C₆ alkyl; or R⁴⁶ and R⁴⁷,

97

together with the carbon atom to which each is bound, combine to form optionally substituted C₃-C₆ carbocyclyl or optionally substituted C₂-C₅ heterocyclyl; or R⁴⁶ and R⁴⁷, together with the carbon atom to which each is bound, combine to form optionally substituted C₃-C₆ carbocyclyl or optionally substituted C₂-C₅ heterocyclyl;

R⁴⁸ is H, optionally substituted C₁-C₆ alkyl, or optionally substituted C₁-C₆ heteroalkyl;

each of R⁴¹, R⁴², R⁴³, and R⁴⁴ is, independently, H, A², halogen, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted C₃-C₁₀ carbocyclyl, optionally substituted C₂-C₉ heterocyclyl, optionally substituted C₆-C₁₀ aryl, optionally substituted C₂-C₉ heteroaryl, optionally substituted C₂-C₆ alkenyl, optionally substituted C₂-C₆ heteroalkenyl, optionally substituted —O—C₃-C₆ carbocyclyl, hydroxyl, thiol, or optionally substituted amino; or R⁴¹ and R⁴², R⁴² and R⁴³, and/or R⁴³ and R⁴⁴, together with the carbon atoms to which each is attached,

combine to form $\textcircled{\text{N}}$; and $\textcircled{\text{N}}$ is optionally substituted C₆-C₁₀ aryl, optionally substituted C₃-C₁₀ carbocyclyl, optionally substituted C₂-C₉ heteroaryl, or C₂-C₉ heterocyclyl, any of which is optionally substituted with A², where one of R⁴¹, R⁴², R⁴³, and R⁴⁴ is

A², or $\textcircled{\text{N}}$ is substituted with A², or a pharmaceutically acceptable salt thereof.

In some embodiments, each of R⁴¹, R⁴², R⁴³, and R⁴⁴ is, independently, H, A², halogen, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted C₃-C₁₀ carbocyclyl, optionally substituted C₂-C₉ heterocyclyl, optionally substituted C₆-C₁₀ aryl, optionally substituted C₂-C₉ heteroaryl, optionally substituted C₂-C₆ alkenyl, optionally substituted C₂-C₆ heteroalkenyl, hydroxyl, thiol, or optionally substituted amino; or R⁴¹ and R⁴², R⁴² and R⁴³, and/or R⁴³ and R⁴⁴, together with the carbon atoms to which each is attached, combine to form

$\textcircled{\text{N}}$; and $\textcircled{\text{N}}$ is optionally substituted C₆-C₁₀ aryl, optionally substituted C₃-C₁₀ carbocyclyl, optionally substituted C₂-C₉ heteroaryl, or C₂-C₉ heterocyclyl, any of which is optionally substituted with A², where one of R⁴¹, R⁴²,

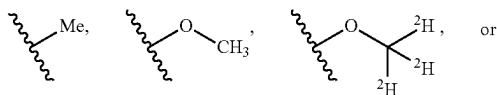
R⁴³, and R⁴⁴ is A², or $\textcircled{\text{N}}$ is substituted with A², or a pharmaceutically acceptable salt thereof.

In some embodiments, each of R⁴¹, R⁴², R⁴³, and R⁴⁴ is, H, A², halogen, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted —O—C₃-C₆ carbocyclyl, hydroxyl, optionally substituted amino; or R⁴¹ and R⁴², R⁴² and R⁴³, or R⁴³ and R⁴⁴, together with the carbon atoms to which each is attached,

combine to form $\textcircled{\text{N}}$; and $\textcircled{\text{N}}$ is optionally substituted C₂-C₉ heterocyclyl, which is optionally substituted with A²,

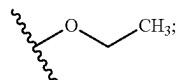
where one of R⁴¹, R⁴², R⁴³, and R⁴⁴ is A², or $\textcircled{\text{N}}$ is substituted with A².

In some embodiments, each of R⁴¹, R⁴², R⁴³, and R⁴⁴ is, independently, H, A², F,



98

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or R⁴¹ and R⁴², R⁴² and R⁴³, or R⁴³ and R⁴⁴, together with the carbon atoms to which each is attached,

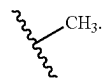
combine to form $\textcircled{\text{N}}$; and $\textcircled{\text{N}}$ is optionally substituted C₂-C₉ heterocyclyl, which is optionally substituted with A², where one of R⁴¹, R⁴², R⁴³, and R⁴⁴ is

A², or $\textcircled{\text{N}}$ is substituted with A².

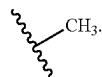
In some embodiments, R⁴¹ is A². In some embodiments, R⁴² is A². In some embodiments, R⁴³ is A². In some embodiments, R⁴⁴ is A². In some embodiments, R⁴⁵ is A².

In some embodiments, R⁴⁵ is H or optionally substituted C₁-C₆ alkyl.

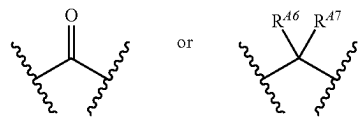
In some embodiments, R⁴⁵ is H or



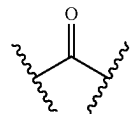
In some embodiments, R⁴⁵ is H. In some embodiments, R⁴⁵ is



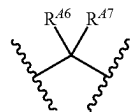
In some embodiments, Y¹ is



In some embodiments, Y¹ is

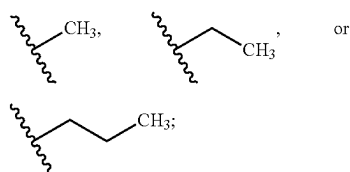


In some embodiments, Y¹ is

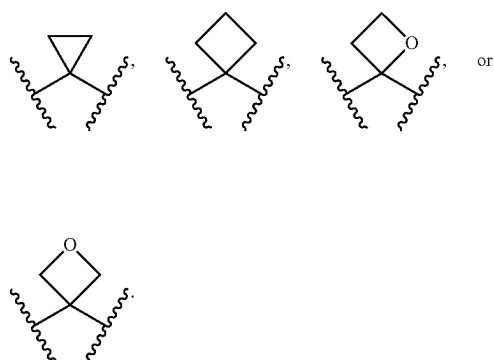


99

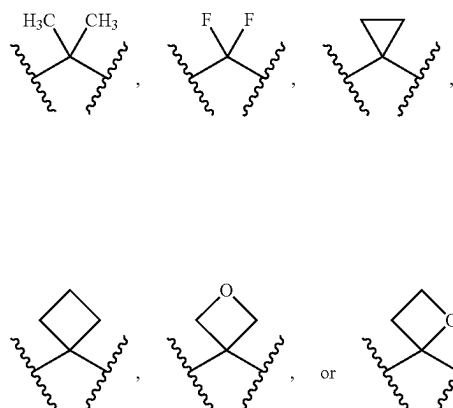
In some embodiments, each of R^{46} and R^{47} is, independently, H, F,



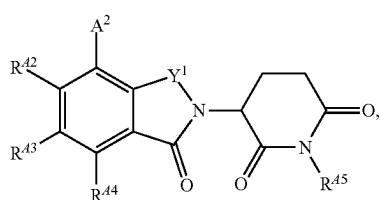
or R^{46} and R^{47} , together with the carbon atom to which each is bound, combine to form



In some embodiments, Y^1 is



In some embodiments, the structure of Formula A has the structure of Formula A1:

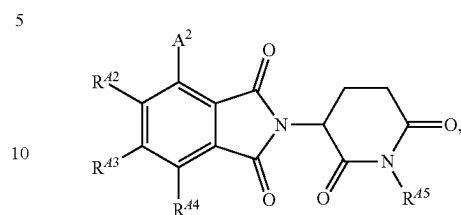


Formula A1

or a pharmaceutically acceptable salt thereof.

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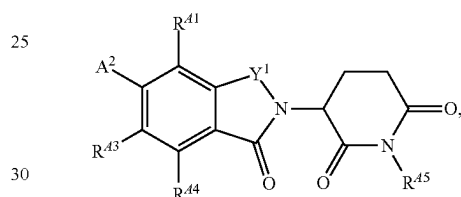
In some embodiments, the structure of Formula A has the structure of Formula A2:



Formula A2

or a pharmaceutically acceptable salt thereof.

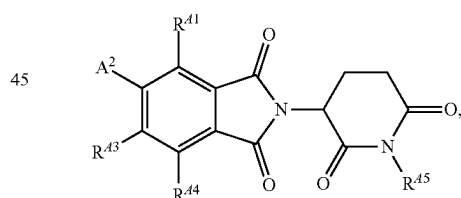
In some embodiments, the structure of Formula A has the structure of Formula A3:



Formula A3

or a pharmaceutically acceptable salt thereof.

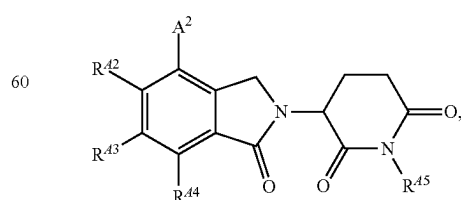
In some embodiments, the structure of Formula A has the structure of Formula A4:



Formula A4

or a pharmaceutically acceptable salt thereof.

In some embodiments, the structure of Formula A has the structure of Formula A5:

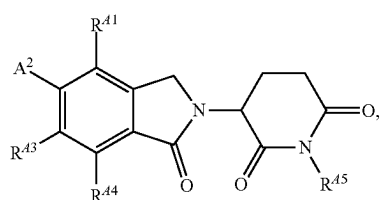


Formula A5

or a pharmaceutically acceptable salt thereof.

101

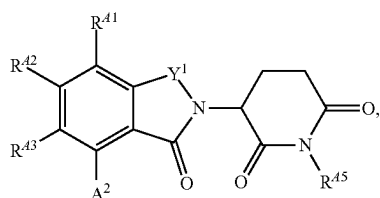
In some embodiments, the structure of Formula A has the structure of Formula A6:



Formula A6

or a pharmaceutically acceptable salt thereof.

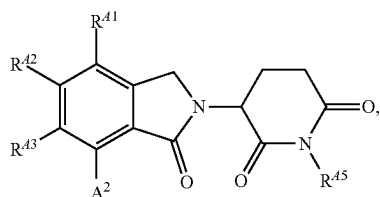
In some embodiments, the structure of Formula A has the structure of Formula A7:



Formula A7

or a pharmaceutically acceptable salt thereof.

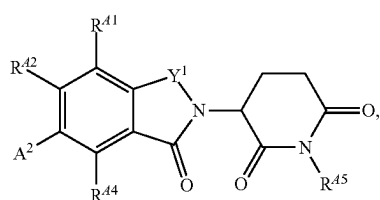
In some embodiments, the structure of Formula A has the structure of Formula A8:



Formula A8

or a pharmaceutically acceptable salt thereof.

In some embodiments, the structure of Formula A has the structure of Formula A9:



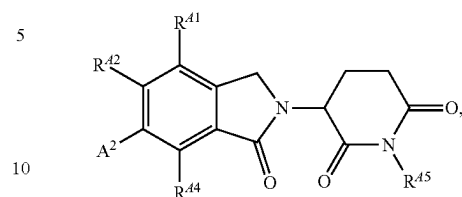
Formula A9

or a pharmaceutically acceptable salt thereof.

102

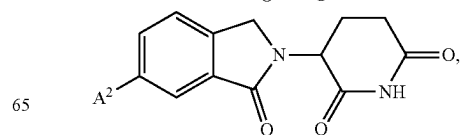
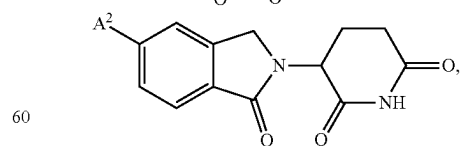
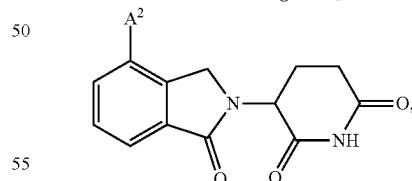
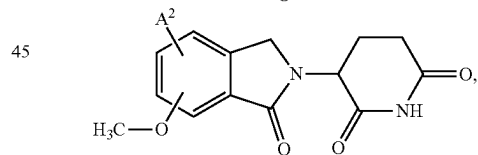
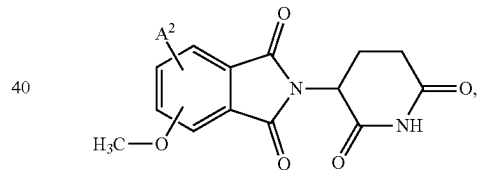
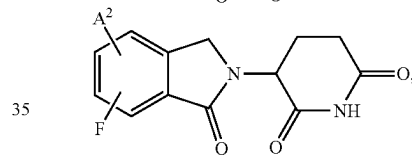
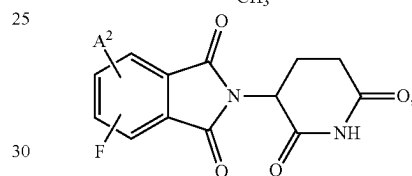
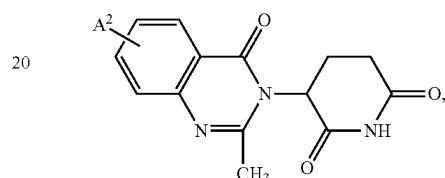
In some embodiments, the structure of Formula A has the structure of Formula A10:

Formula A10



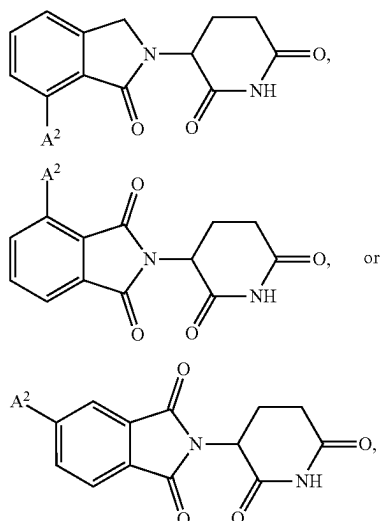
or a pharmaceutically acceptable salt thereof.

In some embodiments, wherein the structure of Formula A is



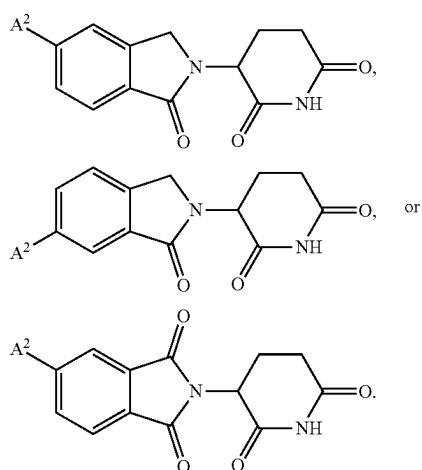
103

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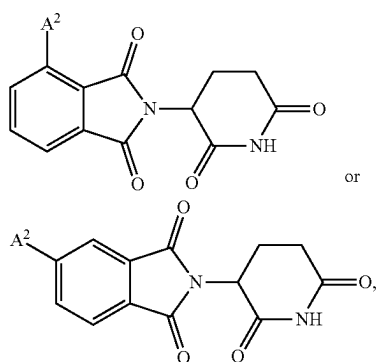


or derivative or analog thereof.

In some embodiments, the structure of Formula A is



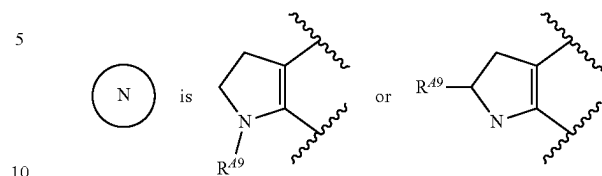
In some embodiments, the structure of Formula A is



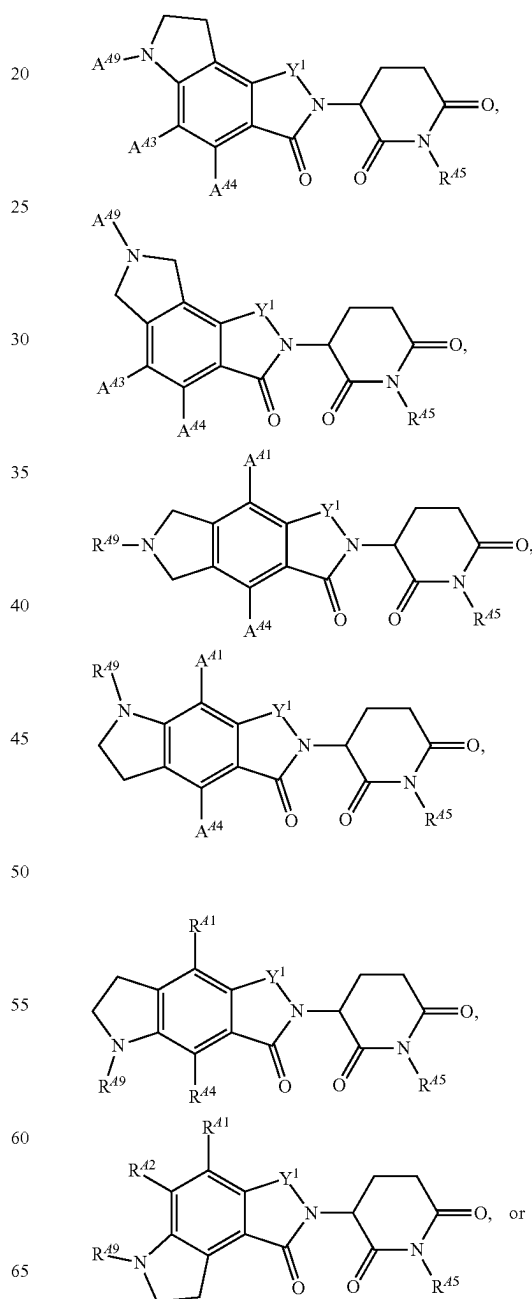
or derivative or analog thereof.

104

In some embodiments,

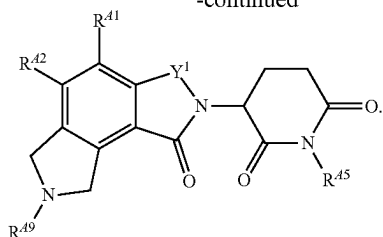
where R^{49} is H, A^2 , optionally substituted C_1 - C_6 alkyl, or optionally substituted C_1 - C_6 heteroalkyl.

In some embodiments, the structure of Formula A is



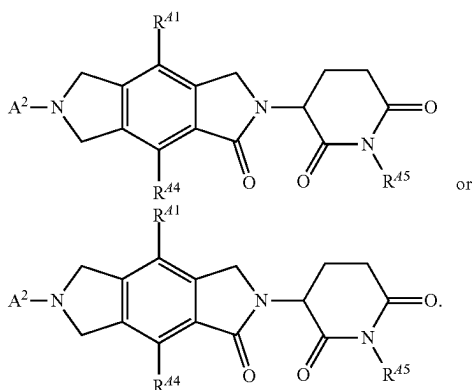
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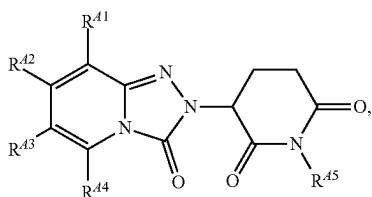


In some embodiments, R^{49} is H, A^2 , or optionally substituted C_1 - C_6 alkyl. In some embodiments, R^{49} is H, A^2 , or methyl. In some embodiments, R^{9A} is H. In some embodiments, R^{9A} is methyl. In some embodiments, R^{49} is A^2 .

In some embodiments, the structure of Formula A is



In some embodiments, the structure of Formula AA has the structure of Formula B:



Formula B

where

R^{45} is H, optionally substituted C_1 - C_6 alkyl, or optionally substituted C_1 - C_6 heteroalkyl;

each of R^{41} , R^{42} , R^{43} , and R^{44} is, independently, H, A^2 , halogen, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, optionally substituted $-O-C_3-C_6$ carbocyclyl, hydroxyl, thiol, or optionally substituted amino; or R^{41} and R^{42} , R^{42} and R^{43} , and/or R^{43} and R^{44} , together with the carbon atoms to which each is attached,

combine to form N ; and N is optionally substituted C_6 - C_{10} aryl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heteroaryl, or

106

C_2 - C_9 heterocyclyl, any of which is optionally substituted with A^2 , where one of R^{41} , R^{42} , R^{43} , and R^{44} is

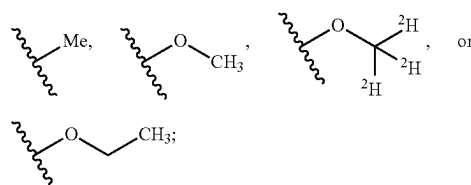
A^2 , or N is substituted with A^2 , or a pharmaceutically acceptable salt thereof.

In some embodiments, each of R^{41} , R^{42} , R^{43} , and R^{44} is, H, A^2 , halogen, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted $-O-C_3-C_6$ carbocyclyl, hydroxyl, optionally substituted amino; or R^{41} and R^{42} , R^{42} and R^{43} , or R^{43} and R^{44} , together with the carbon atoms to which each is attached,

combine to form N ; and N is optionally substituted C_2 - C_9 heterocyclyl, which is optionally substituted with A^2 ,

where one of R^{41} , R^{42} , R^{43} , and R^{44} is A^2 , or N is substituted with A^2 .

In some embodiments, each of R^{41} , R^{42} , R^{43} , and R^{44} is, independently, H, A^2 , F,



or R^{41} and R^{42} , R^{42} and R^{43} , or R^{43} and R^{44} , together with the carbon atoms to which each is attached,

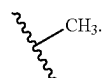
combine to form N ; and N is optionally substituted C_2 - C_9 heterocyclyl, which is optionally substituted with A^2 , where one of R^{41} , R^{42} , R^{43} , and R^{44} is

A^2 , or N is substituted with A^2 .

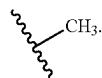
In some embodiments, R^{41} is A^2 . In some embodiments, R^{42} is A^2 . In some embodiments, R^{43} is A^2 . In some embodiments, R^{44} is A^2 . In some embodiments, R^{45} is A^2 .

In some embodiments, R^{45} is H or optionally substituted C_1 - C_6 alkyl.

In some embodiments, R^{45} is H or

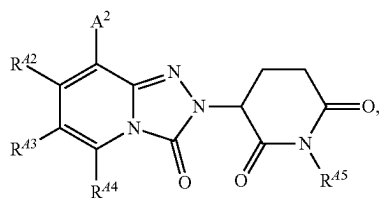


In some embodiments, R^{45} is H. In some embodiments, R^{45} is



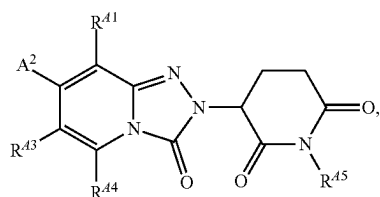
107

In some embodiments, the structure of Formula B has the structure of Formula B1:



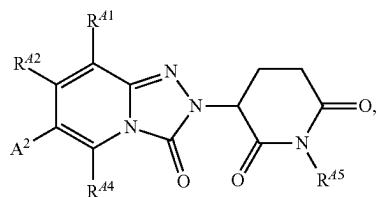
or a pharmaceutically acceptable salt thereof.

In some embodiments, the structure of Formula B has the structure of Formula B2:



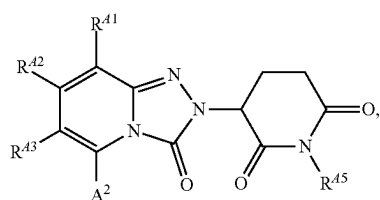
or a pharmaceutically acceptable salt thereof.

In some embodiments, the structure of Formula B has the structure of Formula B3:



or a pharmaceutically acceptable salt thereof.

In some embodiments, the structure of Formula B has the structure of Formula B4:



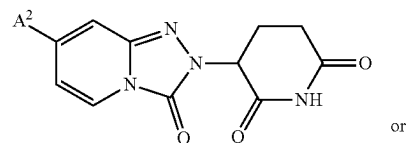
or a pharmaceutically acceptable salt thereof.

108

In some embodiments, the structure of Formula B is

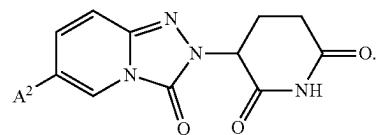
Formula B1

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or

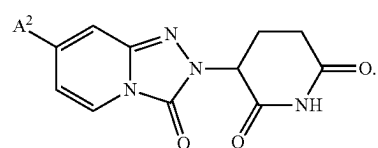
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Formula B2

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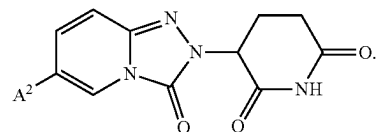
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In some embodiments, the structure of Formula B is

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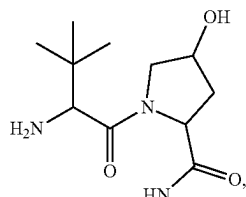
Formula B3

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In some embodiments, the ubiquitin ligase binding moiety comprises a von Hippel-Lindau ligand.

In some embodiments, the von Hippel-Lindau ligand has the structure of

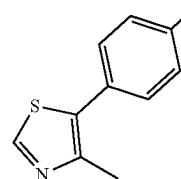
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Formula B4

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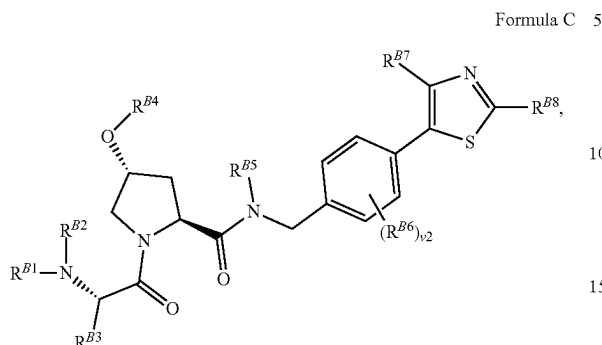


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or derivative or analog thereof.

109

In some embodiments, the degradation moiety includes the structure of Formula C:



where

R^{B1} is H, A^2 , optionally substituted C_1 - C_6 alkyl, or optionally substituted C_1 - C_6 heteroalkyl;

R^{B2} is H, optionally substituted C_1 - C_6 alkyl, or optionally substituted C_1 - C_6 heteroalkyl;

R^{B3} is A^2 , optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_1 - C_6 alkyl C_3 - C_{10} carbocyclyl, or optionally substituted C_1 - C_6 alkyl C_6 - C_{10} aryl;

R^{B4} is H, optionally substituted C_1 - C_6 alkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_1 - C_6 alkyl C_3 - C_{10} carbocyclyl, or optionally substituted C_1 - C_6 alkyl C_6 - C_{10} aryl;

R^{B5} is H, optionally substituted C_1 - C_6 alkyl, or optionally substituted C_1 - C_6 heteroalkyl;

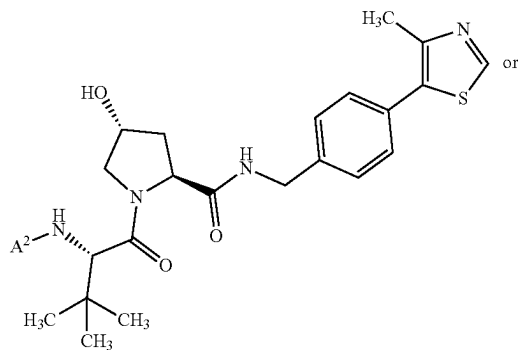
$v2$ is 0, 1, 2, 3, or 4;

each R^{B6} is, independently, halogen, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, hydroxy, thiol, or optionally substituted amino; and

each of R^{B7} and R^{B8} is, independently, H, halogen, optionally substituted C_1 - C_6 alkyl, or optionally substituted C_6 - C_{10} aryl,

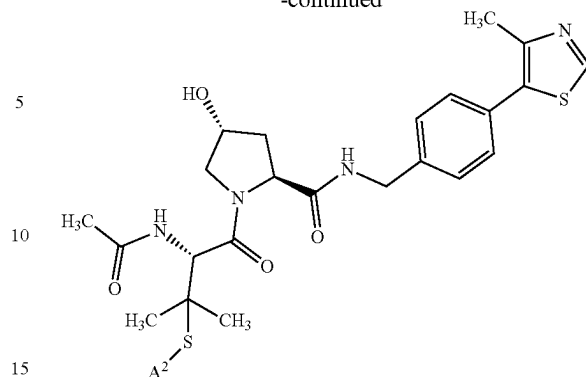
where one of R^{B1} and R^{B3} is A^2 , or a pharmaceutically acceptable salt thereof.

In some embodiments, the structure of Formula C is



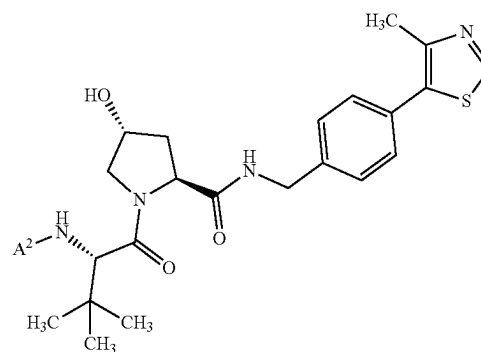
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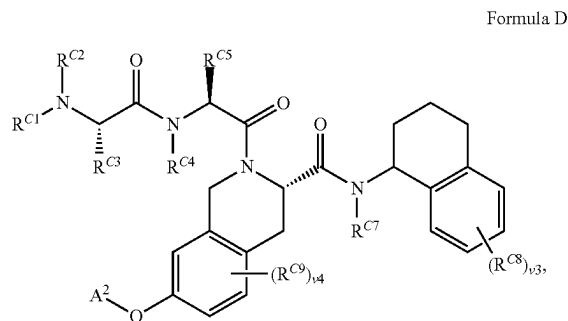


or derivative or analog thereof.

In some embodiments, the structure of Formula C is



In some embodiments, the degrader moiety includes the structure of Formula D:



where

A^2 is a bond between B and the linker;

each of R^{C1} , R^{C2} , and R^{C7} is, independently, H, optionally substituted C_1 - C_6 alkyl, or optionally substituted C_1 - C_6 heteroalkyl;

R^{C3} is optionally substituted C_1 - C_6 alkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_1 - C_6 alkyl C_3 - C_{10} carbocyclyl, or optionally substituted C_1 - C_6 alkyl C_6 - C_{10} aryl;

R^{C5} is optionally substituted C_1 - C_6 alkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_1 - C_6 alkyl C_3 - C_{10} carbocyclyl, or optionally substituted C_1 - C_6 alkyl C_6 - C_{10} aryl;

111

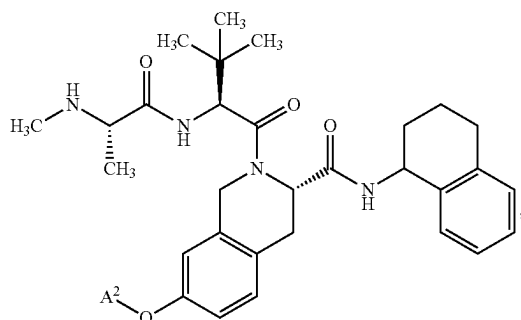
v3 is 0, 1, 2, 3, or 4;

each R^{C8} is, independently, halogen, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, hydroxy, thiol, or optionally substituted amino;

v4 is 0, 1, 2, 3, or 4; and

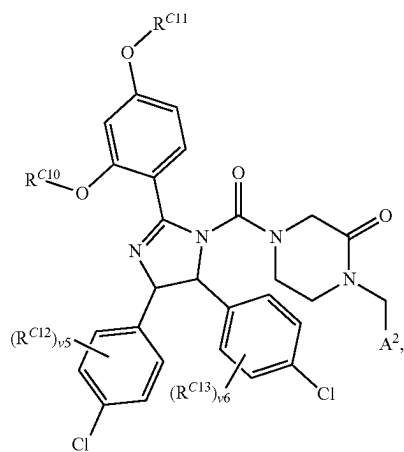
each R^{C9} is, independently, halogen, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, hydroxy, thiol, or optionally substituted amino, or a pharmaceutically acceptable salt thereof.

In some embodiments, the structure of Formula D is



or derivative or analog thereof.

In some embodiments, the degrader moiety includes the structure of Formula E:



where

A^2 is a bond between B and the linker;

each of R^{C10} and R^{C11} is, independently, H, optionally substituted C_1 - C_6 alkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_1 - C_6 alkyl C_3 - C_{10} carbocyclyl, or optionally substituted C_1 - C_6 alkyl C_6 - C_{10} aryl;

112

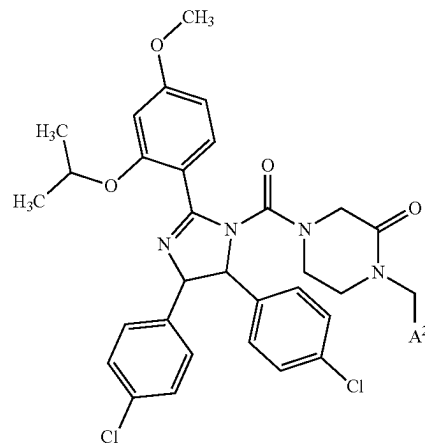
v5 is 0, 1, 2, 3, or 4;

each R^{C12} is, independently, halogen, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, hydroxy, thiol, or optionally substituted amino;

v6 is 0, 1, 2, 3, or 4; and

each R^{C21} is, independently, halogen, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, hydroxy, thiol, or optionally substituted amino, or a pharmaceutically acceptable salt thereof.

In some embodiments, the structure of Formula E is

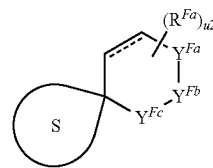


Formula E

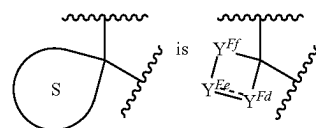
or derivative or analog thereof.

In some embodiments, the degradation moiety includes the structure of Formula FA:

Formula FA



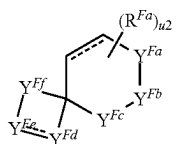
where



or a bicyclic moiety which is substituted with A^2 and substituted with one or more groups independently selected from H, R^{F1} , and oxo;

113

--- is a single bond or a double bond;
 u_2 is 0, 1, 2, or 3;
 A^2 is a bond between the degrader and the linker;
 Y^{Fa} is $CR^{Fb}R^{Fc}$, $C=O$, $C=S$, $C=CH_2$, SO_2 , $S(O)$, $P(O)Oalkyl$, $P(O)NHalkyl$, $P(O)N(alkyl)_2$, $P(O)alkyl$, $P(O)OH$, $P(O)NH_2$;
 Y^{Fb} is NH , NR^{FF1} , CH_2 , CHR^{FF1} , $C(R^{FF1})_2$, O , or S ;
 Y^{Fc} is $CR^{Fd}R^{Fe}$, $C=O$, $C=S$, $C=CH_2$, SO_2 , $S(O)$, $P(O)Oalkyl$, $P(O)NHalkyl$, $P(O)N(alkyl)_2$, $P(O)alkyl$, $P(O)OH$, $P(O)NH_2$;
each of R^{Fb} , R^{Fc} , R^{Fd} , and R^{Fe} is, independently, H , $alkyl$, $aliphatic$, $heteroaliphatic$, $aryl$, $heteroaryl$, $carbocyclyl$, $hydroxyl$, $alkoxy$, $amino$, $-NHalkyl$, or $-Nalkyl_2$;
or R^{Fb} and R^{Fc} , together with the carbon atom to which each is attached, combine to form a 3-, 4-, 5-, or 6-membered spirocarbocyclene, or a 4-, 5-, or 6-membered spiroheterocyclene comprising 1 or 2 heteroatoms selected from N and O ;
or R^{Fd} and R^{Fe} , together with the carbon atom to which each is attached, combine to form a 3-, 4-, 5-, or 6-membered spirocarbocyclene, or a 4-, 5-, or 6-membered spiroheterocyclene comprising 1 or 2 heteroatoms selected from N and O ; and
or R^{Fd} and R^{Fb} , together with the carbon atoms to which each is attached, combine to form a 1, 2, 3, or 4 carbon bridged ring;
each of Y^{Fd} and Y^{Ff} is, independently, CH_2 , CHR^{FF2} , $C(R^{FF2})_2$, $C(O)$, N , NH , NR^{FF3} , O , S , or $S(O)$;
 Y^{Fe} is a bond or a divalent moiety attached to Y^{Fd} and Y^{Ff} that contains 1 to 5 contiguous carbon atoms that form a 3 to 8-membered ring,
wherein 1, 2, or 3 carbon atoms can be replaced with a nitrogen, oxygen, or sulfur atom;
wherein one of the ring atoms is substituted with A^2 and the others are substituted with one or more groups independently selected from H and R^{FF1} ; and
wherein the contiguous atoms of Y^{Fe} can be attached through a single or double bond;
each R^{FF1} is, independently, H , $alkyl$, $alkenyl$, $alkynyl$, $aliphatic$, $heteroaliphatic$, $carbocyclyl$, $halogen$, $hydroxyl$, $amino$, $cyano$, $alkoxy$, $aryl$, $heteroaryl$, $heterocyclyl$, $alkylamino$, $alkylhydroxyl$, or $haloalkyl$;
each R^{FF2} is, independently, $alkyl$, $alkene$, $alkyne$, $halogen$, $hydroxyl$, $alkoxy$, $azide$, $amino$, $-C(O)H$, $-C(O)OH$, $-C(O)(aliphatic, including alkyl)$, $-C(O)O(aliphatic, including alkyl)$, $-NH(aliphatic, including alkyl)$, $-N(aliphatic including alkyl)(aliphatic including alkyl)$, $-NHSO_2alkyl$, $-N(alkyl)SO_2alkyl$, $-NHSO_2aryl$, $-N(alkyl)SO_2aryl$, $-NHSO_2alkenyl$, $-N(alkyl)SO_2alkenyl$, $-NHSO_2alkynyl$, $-N(alkyl)SO_2alkynyl$, $aliphatic$, $heteroaliphatic$, $aryl$, $heteroaryl$, $heterocyclic$, $carbocyclic$, $cyano$, $nitro$, $nitroso$, $-SH$, $-Salkyl$, or $haloalkyl$; and
 R^{FF3} is $alkyl$, $alkenyl$, $alkynyl$, $-C(O)H$, $-C(O)OH$, $-C(O)alkyl$, or $-C(O)Oalkyl$,
wherein if Y^{Fd} or Y^{Ff} is substituted with A^2 , then Y^{Fe} is a bond, or a pharmaceutically acceptable salt thereof.
In some embodiments, the compound of Formula FA has the structure of Formula FA1:

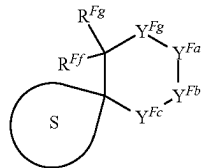


Formula FA1

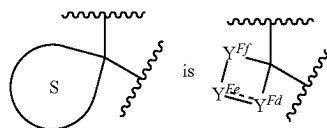
114

In some embodiments, the degradation moiety includes the structure of Formula FB:

Formula FB



where



or a bicyclic moiety which is substituted with A^2 and substituted with one or more groups independently selected from H , R^{FF1} , and oxo ;

A^2 is a bond between the degrader and the linker;

Y^{Fa} is $CR^{Fb}R^{Fc}$, $C=O$, $C=S$, $C=CH_2$, SO_2 , $S(O)$, $P(O)Oalkyl$, $P(O)NHalkyl$, $P(O)N(alkyl)_2$, $P(O)alkyl$, $P(O)OH$, $P(O)NH_2$;

each of Y^{Fb} and Y^{Fg} is, independently, NH , NR^{FF1} , CH_2 , CHR^{FF1} , $C(R^{FF1})_2$, O , or S ;

Y^{Fc} is $CR^{Fd}R^{Fe}$, $C=O$, $C=S$, $C=CH_2$, SO_2 , $S(O)$, $P(O)Oalkyl$, $P(O)NHalkyl$, $P(O)N(alkyl)_2$, $P(O)alkyl$, $P(O)OH$, $P(O)NH_2$;

each of R^{Fb} , R^{Fc} , R^{Fd} , R^{Fe} , R^{Ff} , and R^{Fg} is, independently, H , $alkyl$, $aliphatic$, $heteroaliphatic$, $aryl$, $heteroaryl$, $carbocyclyl$, $hydroxyl$, $alkoxy$, $amino$, $-NHalkyl$, or $-Nalkyl_2$;

or R^{Fb} and R^{Fc} , together with the carbon atom to which each is attached, combine to form a 3-, 4-, 5-, or 6-membered spirocarbocyclene, or a 4-, 5-, or 6-membered spiroheterocyclene comprising 1 or 2 heteroatoms selected from N and O ;

or R^{Fd} and R^{Fe} , together with the carbon atom to which each is attached, combine to form a 3-, 4-, 5-, or 6-membered spirocarbocyclene, or a 4-, 5-, or 6-membered spiroheterocyclene comprising 1 or 2 heteroatoms selected from N and O ;

or R^{Ff} and R^{Fg} , together with the carbon atom to which each is attached, combine to form a 3-, 4-, 5-, or 6-membered spirocarbocyclene, or a 4-, 5-, or 6-membered spiroheterocyclene comprising 1 or 2 heteroatoms selected from N and O ;

or R^{Fd} and R^{Fb} , together with the carbon atoms to which each is attached, combine to form a 1, 2, 3, or 4 carbon bridged ring;

or R^{Fd} and R^{Ff} , together with the carbon atoms to which each is attached, combine to form a 1, 2, 3, or 4 carbon bridged ring;

or R^{Fb} and R^{Fg} , together with the carbon atoms to which each is attached, combine to form a 1, 2, 3, or 4 carbon bridged ring;

each of Y^{Fd} and Y^{Ff} is, independently, CH_2 , CHR^{FF2} , $C(R^{FF2})_2$, $C(O)$, N , NH , NR^{FF3} , O , S , or $S(O)$;

or a pharmaceutically acceptable salt thereof.

115

Y^{Fe} is a bond or a divalent moiety attached to Y^{Fd} and Y^{Ff} that contains 1 to 5 contiguous carbon atoms that form a 3 to 3-membered ring,

wherein 1, 2, or 3 carbon atoms can be replaced with a nitrogen, oxygen, or sulfur atom;

wherein one of the ring atoms is substituted with A^2 and the others are substituted with one or more groups independently selected from H and R^{FF1} ; and

wherein the contiguous atoms of Y^{Fe} can be attached through a single or double bond;

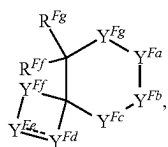
each R^{FF1} is, independently, H, alkyl, alkenyl, alkynyl, aliphatic, heteroaliphatic, carbocyclyl, halogen, hydroxyl, amino, cyano, alkoxy, aryl, heteroaryl, heterocyclyl, alkylamino, alkylhydroxyl, or haloalkyl;

each R^{FF2} is, independently, alkyl, alkene, alkyne, halogen, hydroxyl, alkoxy, azide, amino, $-C(O)H$, $-C(O)OH$, $-C(O)(\text{aliphatic, including alkyl})$, $-C(O)O(\text{aliphatic, including alkyl})$, $-NH(\text{aliphatic, including alkyl})$, $-N(\text{aliphatic including alkyl})(\text{aliphatic including alkyl})$, $-NHSO_2\text{alkyl}$, $-N(\text{alkyl})SO_2\text{alkyl}$, $-NHSO_2\text{aryl}$, $-N(\text{alkyl})SO_2\text{aryl}$, $-NHSO_2\text{alkenyl}$, $-N(\text{alkyl})SO_2\text{alkynyl}$, $-NHSO_2\text{alkynyl}$, $-N(\text{alkyl})SO_2\text{alkynyl}$, aliphatic, heteroaliphatic, aryl, heteroaryl, heterocyclic, carbocyclic, cyano, nitro, nitroso, $-SH$, $-Salkyl$, or haloalkyl; and

R^{FF3} is alkyl, alkenyl, alkynyl, $-C(O)H$, $-C(O)OH$, $-C(O)alkyl$, or $-C(O)Oalkyl$,

wherein if Y^{Fd} or is substituted with A^2 , then Y^{Fe} is a bond, or a pharmaceutically acceptable salt thereof.

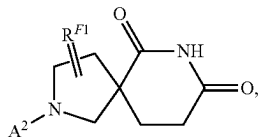
In some embodiments, the compound of Formula FB has the structure of Formula FB1:



Formula FB1

or a pharmaceutically acceptable salt thereof.

In some embodiments, the degradation moiety includes the structure of Formula F1:

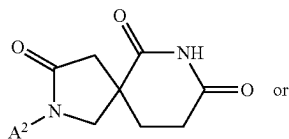


Formula F1

where A^2 is a bond between the degrader and the linker; and R^{F1} is absent or O, or a pharmaceutically acceptable salt thereof.

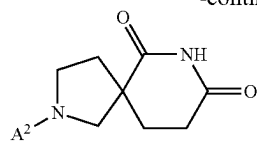
In some embodiments, R^{F1} is absent. In some embodiments, R^{F1} is O.

In some embodiments, the structure of Formula F1 is



116

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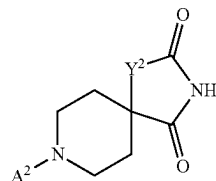


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In some embodiments, the degradation moiety includes the structure Formula F2:

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Formula F2



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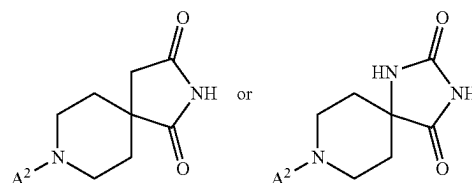
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where A^2 is a bond between the degrader and the linker; and Y^2 is CH_2 or NH, or a pharmaceutically acceptable salt thereof.

In some embodiments, Y^2 is NH. In some embodiments, Y^2 is CH_2 .

In some embodiments, structure of Formula F2 is



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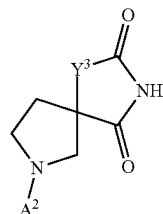
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In some embodiments, the degradation moiety includes the structure Formula G:

Formula F1

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Formula G



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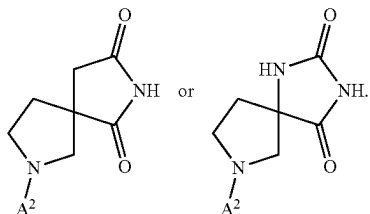
where A^2 is a bond between the degrader and the linker; and Y^3 is CH_2 or NH, or a pharmaceutically acceptable salt thereof.

In some embodiments, Y^3 is NH. In some embodiments, Y^3 is CH_2 .

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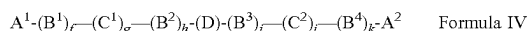
117

In some embodiments, structure of Formula G is



The degradation moiety may also include structures found in, e.g., WO2017/197036; WO2019/204354, WO2019/236483, WO2020/010177; and WO2020/010227, the structures of which are herein incorporated by reference.

In some embodiments, the linker has the structure of Formula IV:



where

A¹ is a bond between the linker and A;

A² is a bond between B and the linker;

each of B¹, B², B³, and B⁴ is, independently, optionally substituted C₁-C₂ alkylene, optionally substituted C₁-C₃ heteroalkylene, O, S, S(O)₂, or NR^N;

each R^N is, independently, H, optionally substituted C₁₋₄ alkyl, optionally substituted C₂₋₄ alkenyl, optionally substituted C₂₋₄ alkynyl, optionally substituted C₂₋₆ heterocyclyl, optionally substituted C₆₋₁₂ aryl, or optionally substituted C₁₋₇ heteroalkyl;

each of C¹ and C² is, independently, carbonyl, thiocarbonyl, sulphonyl, or phosphoryl;

each of f, g, h, i, j, and k is, independently, 0 or 1; and

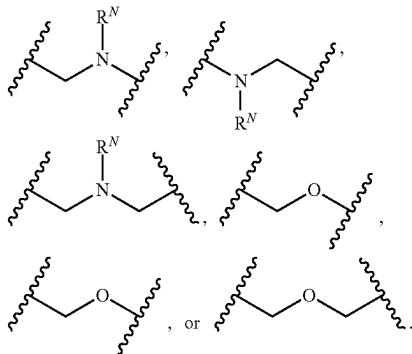
D is optionally substituted C₁₋₁₀ alkylene, optionally substituted C₂₋₁₀ alkenylene, optionally substituted C₂₋₁₀ alkynylene, optionally substituted C₂₋₆ heterocyclylene, optionally substituted C₆₋₁₂ arylene, optionally substituted C₂-C₁₀ polyethylene glycol, or optionally substituted C₁₋₁₀ heteroalkylene, or a chemical bond linking A¹-(B¹)_f-(C¹)_g-(B²)_h to -(B³)_i-(C²)_j-(B⁴)_k-A².

In some embodiments, each of B¹, B², B³, and B⁴ is, independently, optionally substituted C₁-C₄ alkylene, optionally substituted C₁-C₄ heteroalkylene, or NR^N.

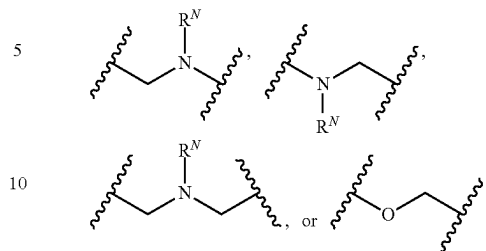
In some embodiments, each R^N is, independently, H or optionally substituted C₁-C₄ alkylene.

In some embodiments, each R^N is, independently, H or methyl.

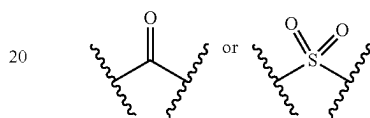
In some embodiments, each of B¹ and B⁴ is, independently,



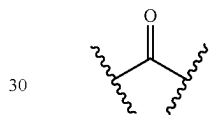
118

In some embodiments, B¹ is

In some embodiments, each of C¹ and C² is, independently



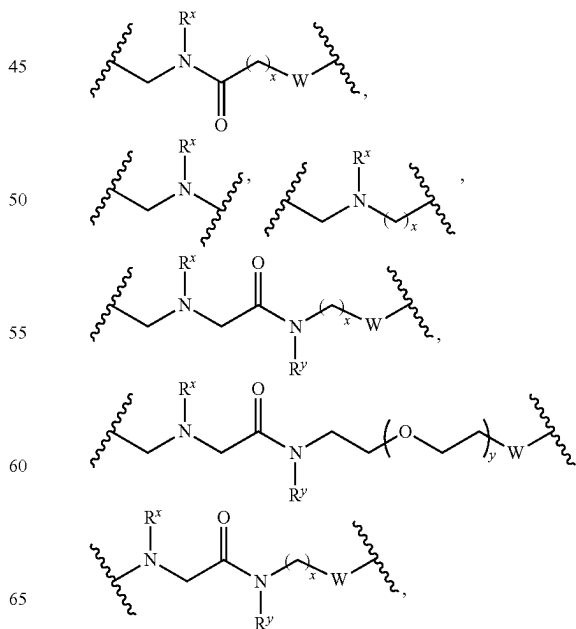
In some embodiments, C¹ is



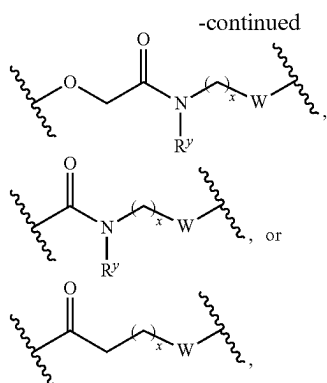
In some embodiments, B² is NR^N. In some embodiments, B² is optionally substituted C₁-C₄ alkylene.

In some embodiments, f is 0. In some embodiments, f is 1. In some embodiments, g is 1. In some embodiments, h is 0. In some embodiments, h is 1. In some embodiments, i is 0. In some embodiments, j is 0. In some embodiments, k is 0.

In some embodiments, the linker has the structure of



119



wherein

x is 1, 2, 3, 4, 5, 6, 7, or 8;

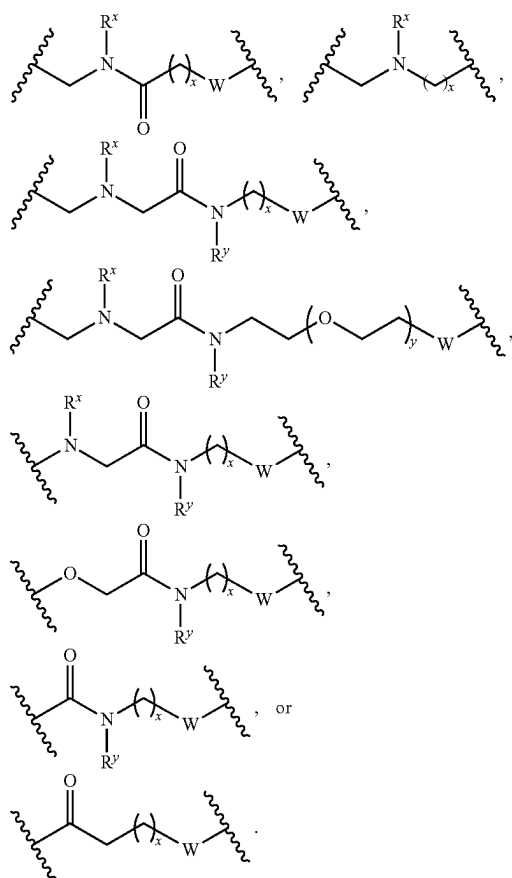
y is 1, 2, 3, or 4;

R^x is H, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, or optionally substituted C₃-C₆ carbocyclyl;

R^y is H, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, or optionally substituted C₃-C₆ carbocyclyl; and

W is O or NR^w, wherein R^w is H, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, or optionally substituted C₃-C₆ carbocyclyl.

In some embodiments, the linker has the structure of



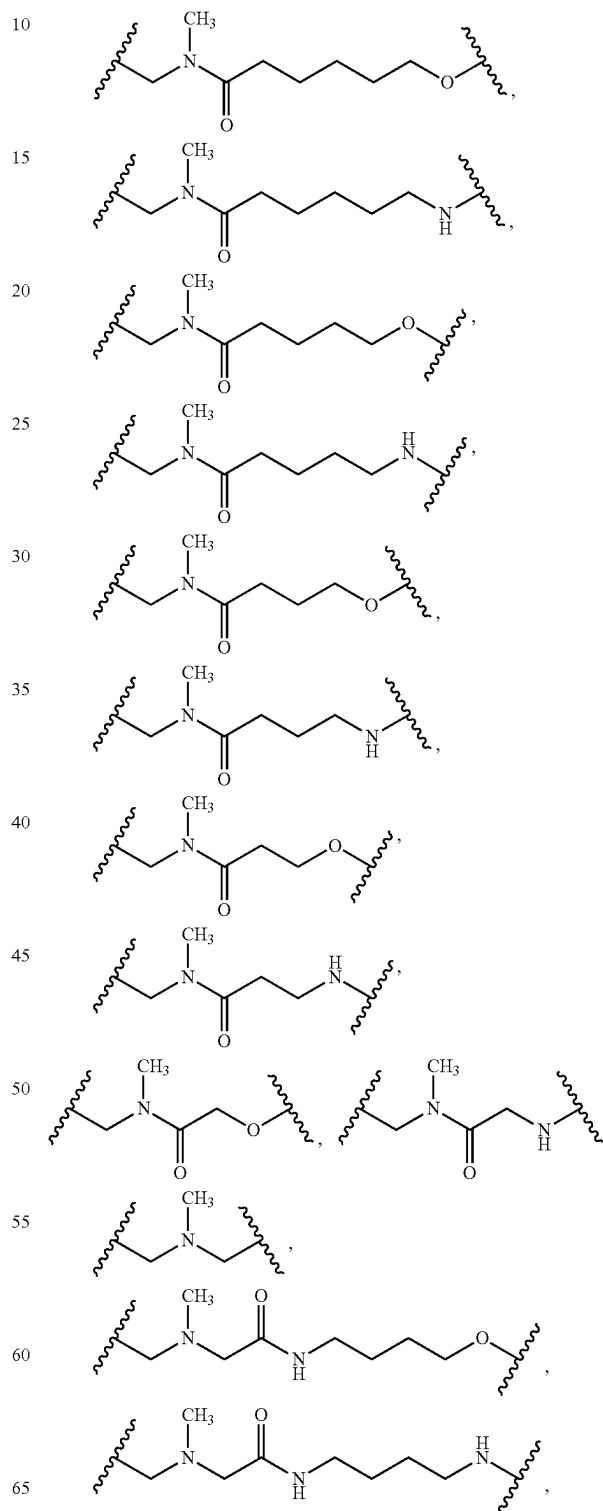
In some embodiments, R^x is H or me optionally substituted C₁-C₆ alkyl. In some embodiments, R^y is H or option-

120

ally substituted C₁-C₆ alkyl. In some embodiments, R^w is H or optionally substituted C₁-C₆ alkyl.

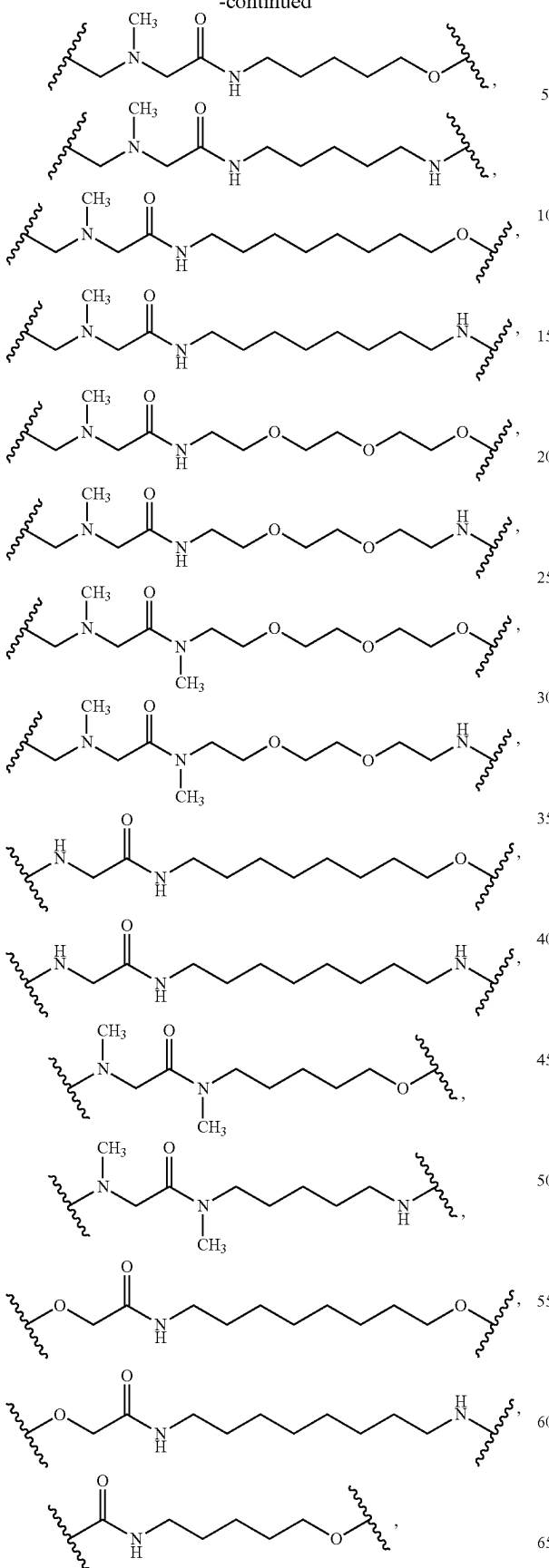
In some embodiments, R^x is H or methyl. In some embodiments, RY is H or methyl. In some embodiments, R^w is H or methyl.

In some embodiments, the linker has the structure of

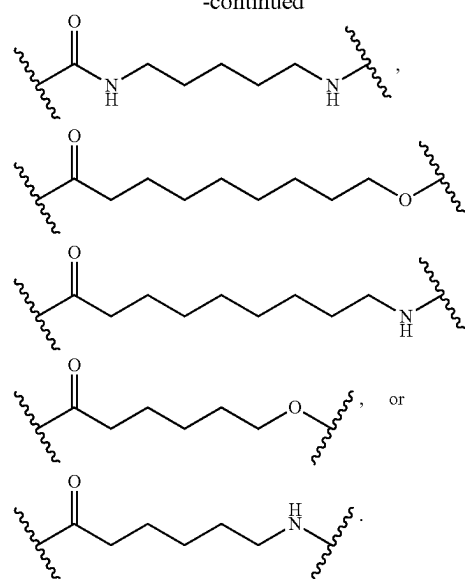


121

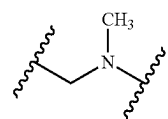
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**122**

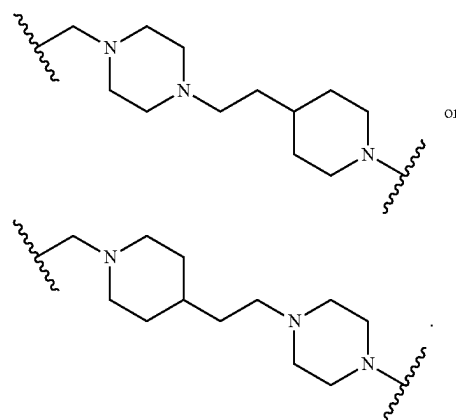
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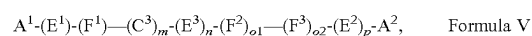
In some embodiments, the linker has the structure of



In some embodiments, the linker has the structure of



In some embodiments, the linker has the structure of Formula V:



where

A^1 is a bond between the linker and A;

A^2 is a bond between B and the linker;

each of m, n, o1, o2, and p is, independently, 0 or 1;

each of E^1 and E^2 is, independently, O, S, NR^N , optionally substituted C_{1-10} alkylene, optionally substituted C_{2-10} alkenylene, optionally substituted C_{2-10} alkynylene, optionally substituted C_2-C_{10} polyethylene glycol, or optionally substituted C_{1-10} heteroalkylene;

123

E^3 is optionally substituted C_1 - C_6 alkylene, optionally substituted C_1 - C_6 heteroalkylene, O, S, or NR^N ; each R^N is, independently, H, optionally substituted C_{1-4} alkyl, optionally substituted C_{2-4} alkenyl, optionally substituted C_{2-4} alkynyl, optionally substituted C_{2-6} heterocyclyl, optionally substituted C_{6-12} aryl, or optionally substituted C_{1-7} heteroalkyl;

C^3 is carbonyl, thiocarbonyl, sulphonyl, or phosphoryl; and

each of F^1 , F^2 , and F^3 is, independently, optionally substituted C_3 - C_{10} carbocyclylene, optionally substituted C_{2-10} heterocyclylene, optionally substituted C_6 - C_{10} arylene, or optionally substituted C_2 - C_9 heteroarylene.

In some embodiments, the linker has the structure of Formula Va:



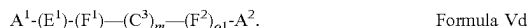
In some embodiments, the linker has the structure of Formula Vb:



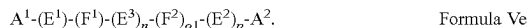
In some embodiments, the linker has the structure of Formula Vc:



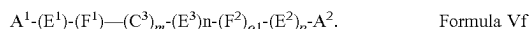
In some embodiments, the linker has the structure of Formula Vd:



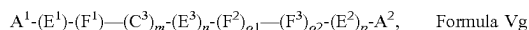
In some embodiments, the linker has the structure of Formula Ve:



In some embodiments, the linker has the structure of Formula Vf:



In some embodiments, the linker has the structure of Formula Vg:



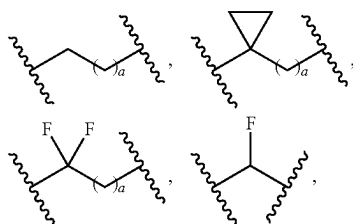
In some embodiments, each of E^1 and E^2 is, independently, NR^N , optionally substituted C_{1-10} alkyl, optionally substituted C_{2-10} polyethylene glycol, or optionally substituted C_{1-10} heteroalkyl.

In some embodiments, E^3 is optionally substituted C_1 - C_6 alkylene, O, S, or NR^N ;

In some embodiments, E^3 is optionally substituted C_1 - C_6 alkylene. In some embodiments, E^3 is optionally substituted C_1 - C_3 alkylene. In some embodiments, E^3 is O, S, or NR^N .

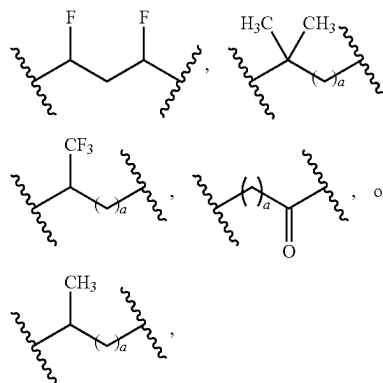
In some embodiments, E^3 is C_1 - C_6 alkylene. In some embodiments, E^3 is C_1 - C_3 alkylene. In some embodiments, E^3 is O.

In some embodiments, E^3 is



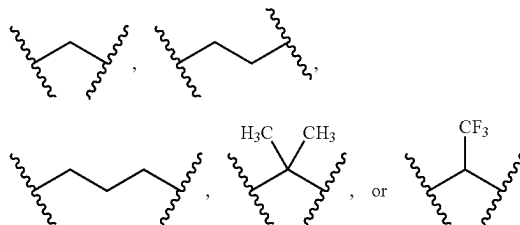
124

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where a is 0, 1, 2, 3, 4, or 5.

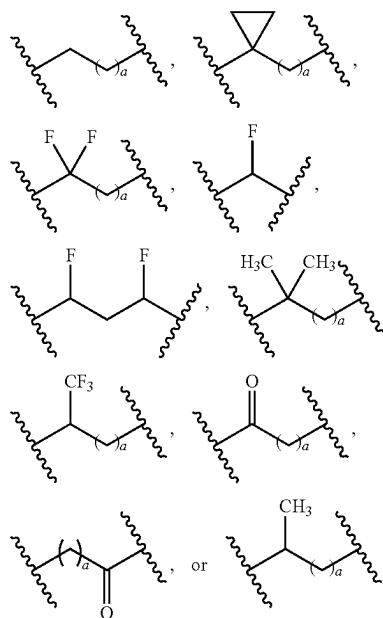
In some embodiments, E^3 is



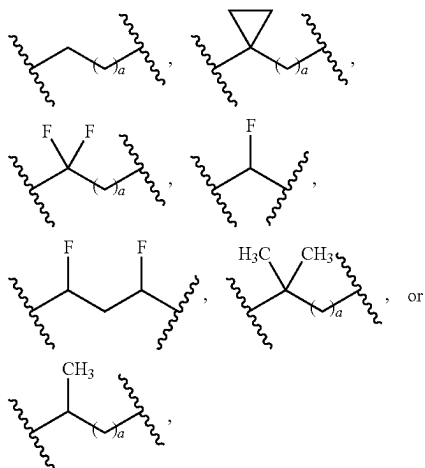
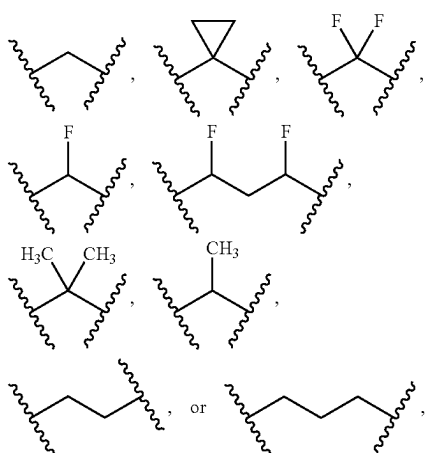
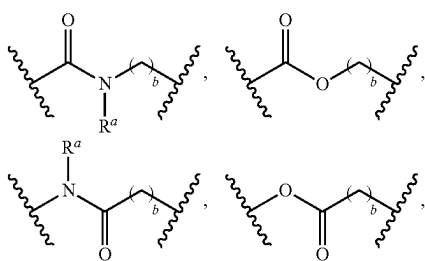
In some embodiments, each R^N is, independently, H or optionally substituted C_{1-4} alkyl.

In some embodiments, each R^N is, independently, H or methyl.

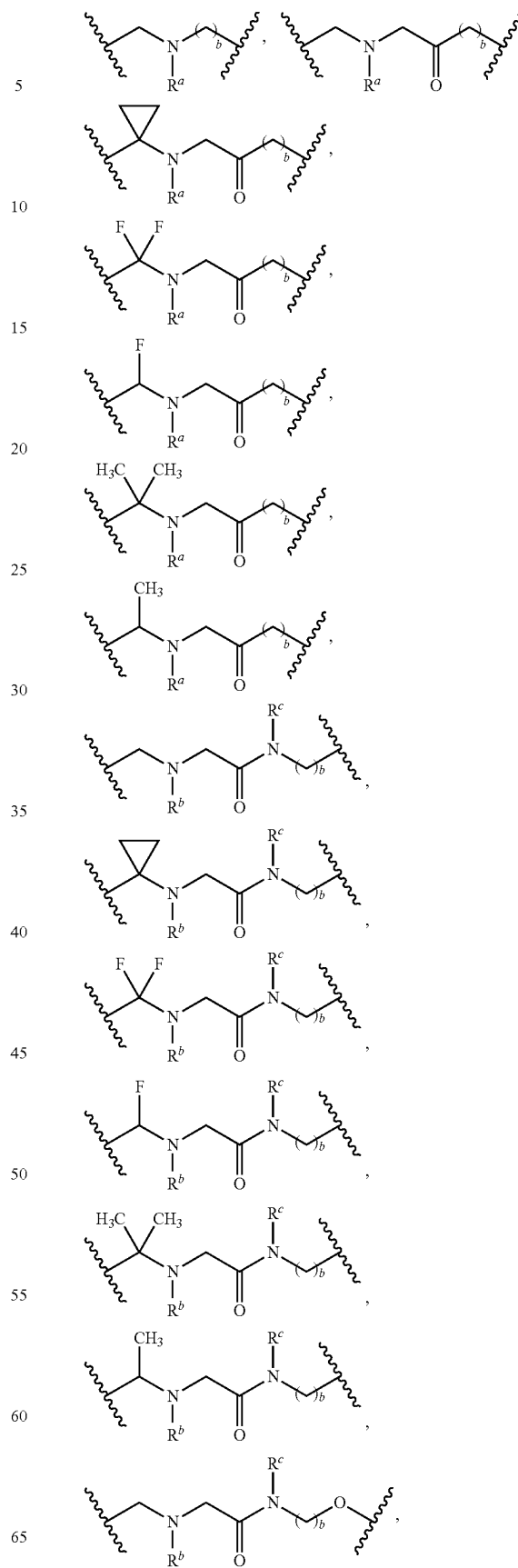
In some embodiments, E^1 is



where a is 0, 1, 2, 3, 4, or 5.

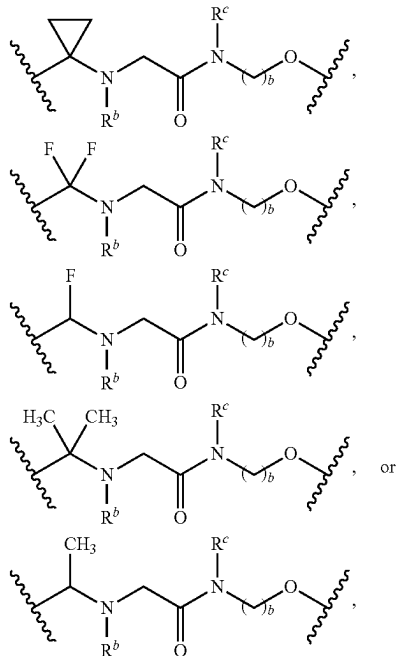
125In some embodiments, E^1 iswhere a is 0, 1, 2, 3, 4, or 5.In some embodiments, E^1 isIn some embodiments, E^1 isIn some embodiments, E^1 is**126**

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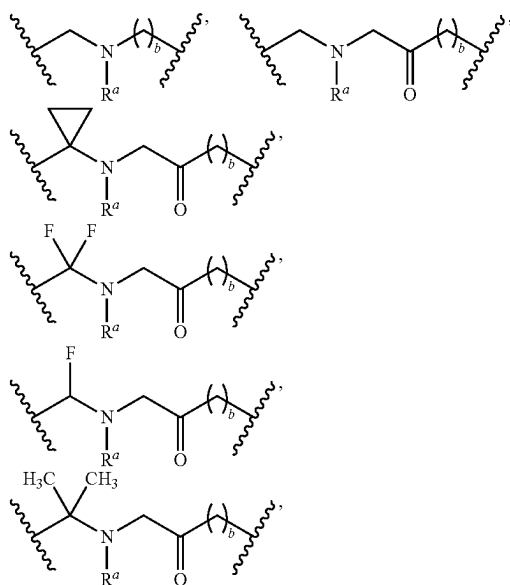


127

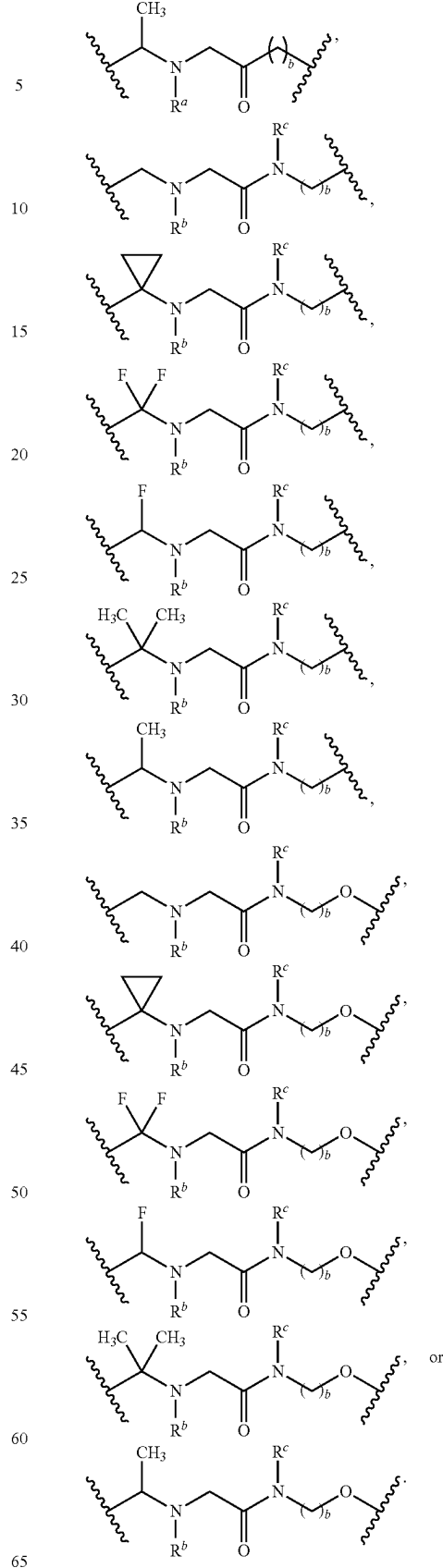
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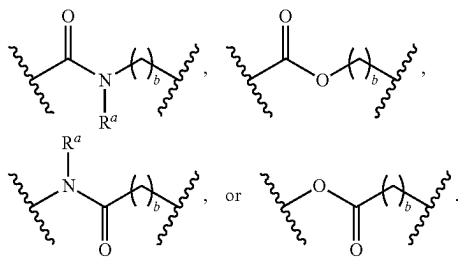
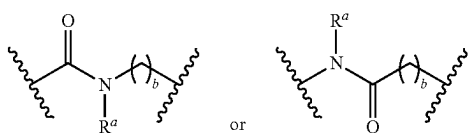


where

 b is 0, 1, 2, 3, 4, 5, or 6; R^a is H, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, or optionally substituted C_3 - C_6 carbocyclyl; R^b is H, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, or optionally substituted C_3 - C_6 carbocyclyl; and R^c is H, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, or optionally substituted C_3 - C_6 carbocyclyl.In some embodiments, E^1 is**128**

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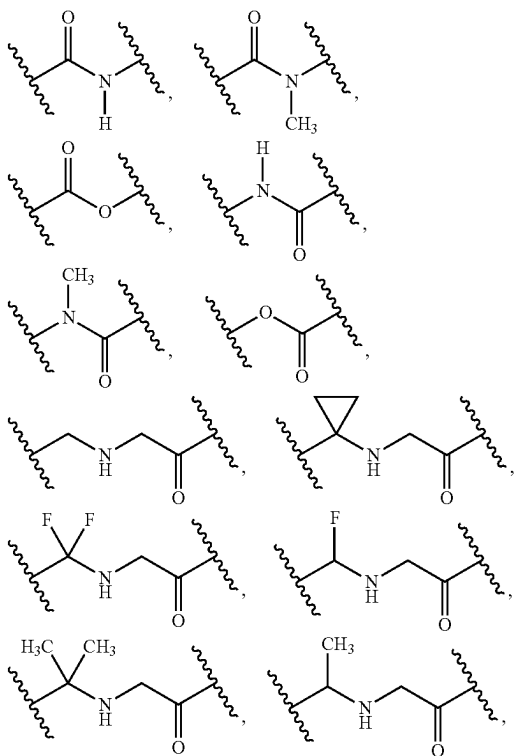


129In some embodiments, E¹ isIn some embodiments, E¹ is

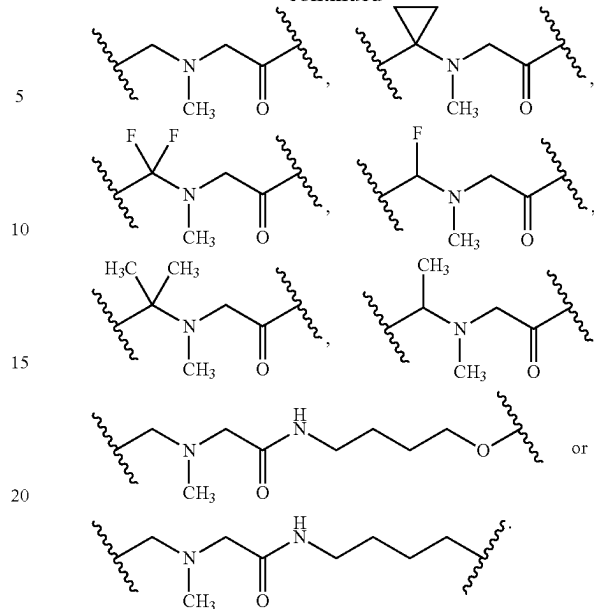
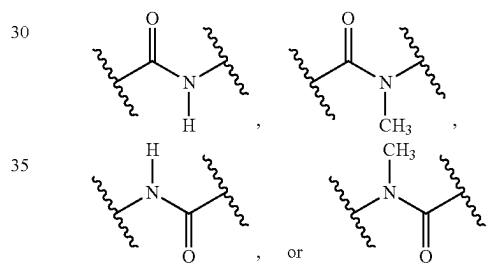
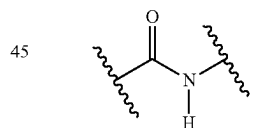
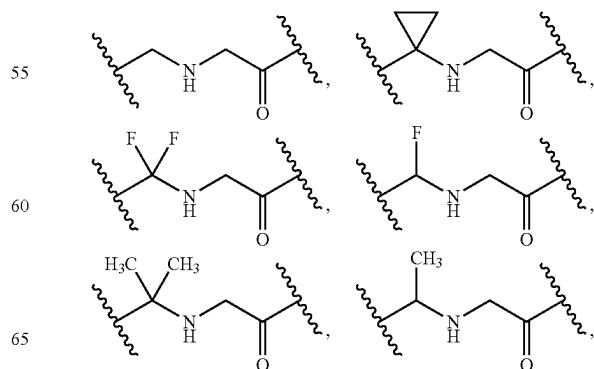
In some embodiments, R^a is H or optionally substituted C₁-C₆ alkyl. In some embodiments, R^b is H or optionally substituted C₁-C₆ alkyl. In some embodiments, R^c is H or optionally substituted C₁-C₆ alkyl.

In some embodiments, R^a is H or methyl. In some embodiments, R^b is H or methyl. In some embodiments, R^c is H or methyl.

In some embodiments, b is 0, 1, 2, or 3. In some embodiments, b is 0. In some embodiments, b is 1. In some embodiments, b is 2. In some embodiments, b is 3.

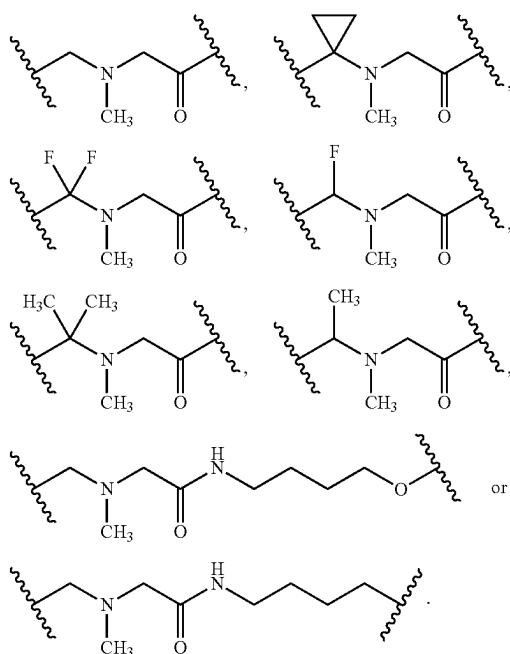
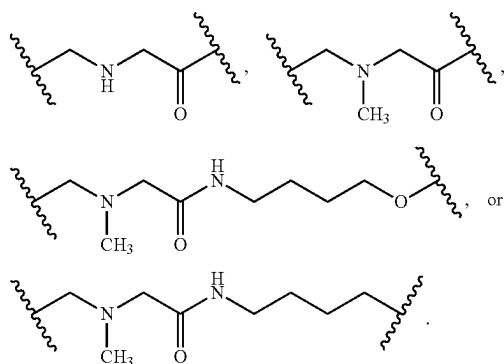
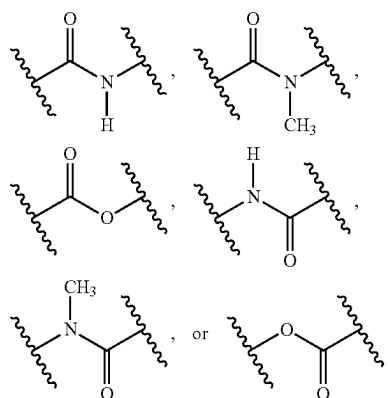
In some embodiments, E¹ is**130**

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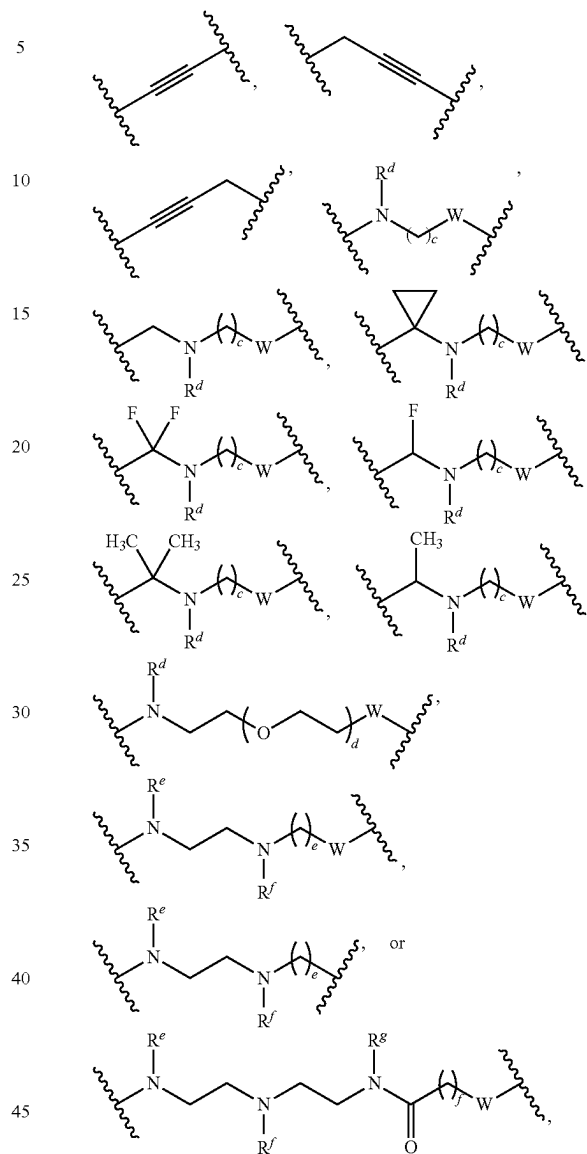
In some embodiments, E¹ isIn some embodiments, E¹ isIn some embodiments, E¹ is

131

-continued

In some embodiments, E¹ isIn some embodiments, E¹ is

132

In some embodiments, E² is O, NR^w,

wherein

c is 0, 1, 2, 3, 4, 5, 6, 7, or 8;

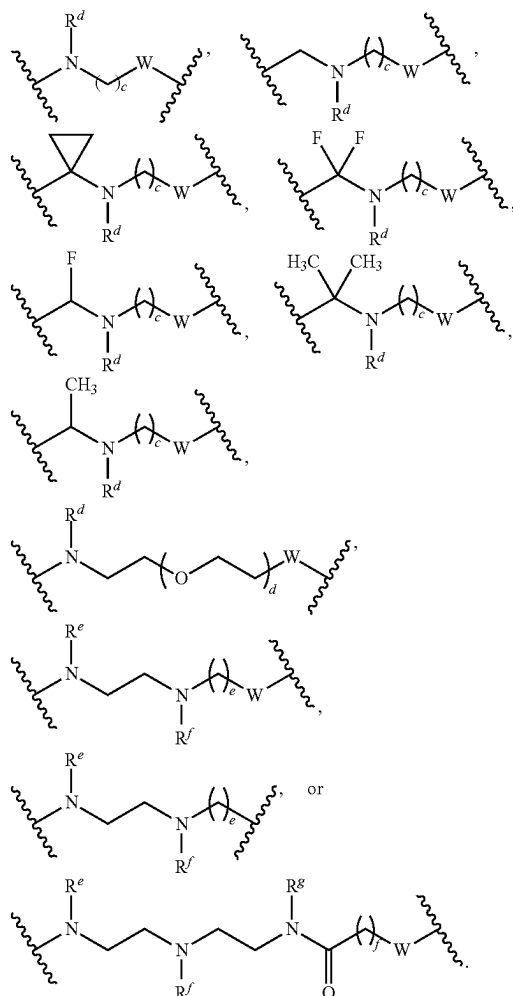
d is 0, 1, 2, or 3;

e is 0, 1, 2, 3, 4, 5, or 6;

f is 0, 1, 2, 3, or 4;

R^d is H, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, or optionally substituted C₃-C₆ carbocyclyl;R^e is H, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, or optionally substituted C₃-C₆ carbocyclyl;R^f is H, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, or optionally substituted C₃-C₆ carbocyclyl;R^g is H, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, or optionally substituted C₃-C₆ carbocyclyl; andW is O or NR^w, wherein R^w is H or optionally substituted C₁-C₆ alkyl.

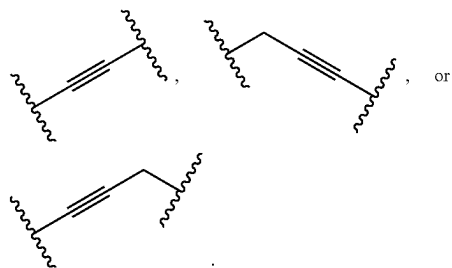
133

In some embodiments, E² is O, NR^w,

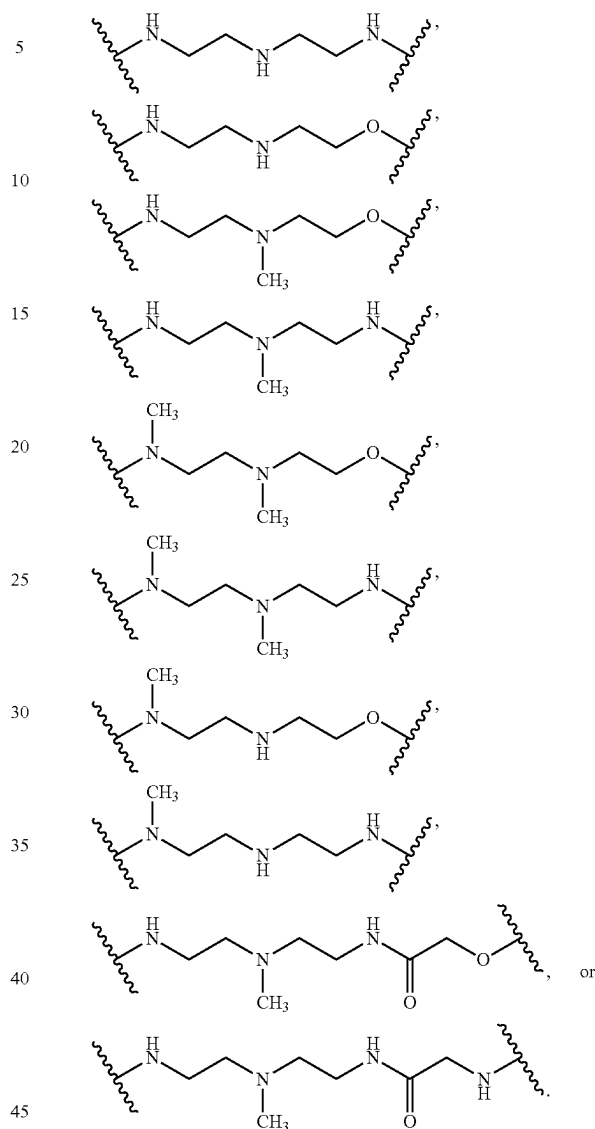
In some embodiments, R^d is H or optionally substituted C₁-C₆ alkyl. In some embodiments, R^e is H or optionally substituted C₁-C₆ alkyl. In some embodiments, R^f is H or optionally substituted C₁-C₆ alkyl. In some embodiments, R^g is H or optionally substituted C₁-C₆ alkyl. In some

embodiments, R^w is H or optionally substituted C₁-C₆ alkyl. In some embodiments, R^d is H or methyl. In some embodiments, R^e is H or methyl. In some embodiments, R^f is H or methyl. In some embodiments, R^g is H or methyl. In some embodiments, R^w is H or methyl.

In some embodiments, E² is



134

In some embodiments, E² is O,

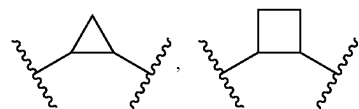
In some embodiments, each of F¹, F², or F³ is, independently, optionally substituted C₃-C₁₀ carbocyclylene.

In some embodiments, the C₃-C₁₀ carbocyclylene is monocyclic. In some embodiments, the C₃-C₁₀ carbocyclylene is polycyclic.

In some embodiments, the C₃-C₁₀ carbocyclylene is bicyclic.

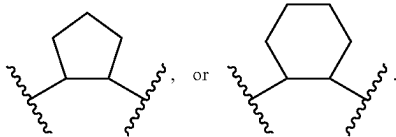
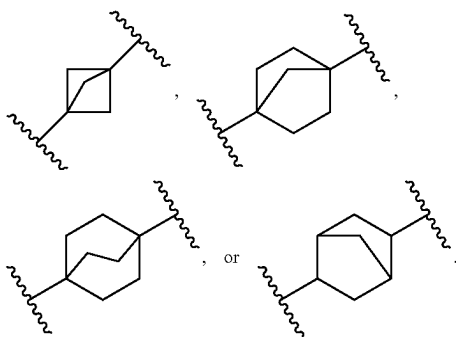
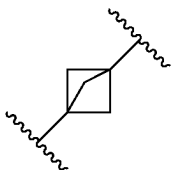
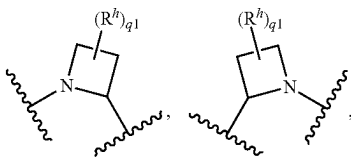
In some embodiments, the C₃-C₁₀ carbocyclylene is bridged. In some embodiments, the C₃-C₁₀ carbocyclylene is fused. In some embodiments, the C₃-C₁₀ carbocyclylene is spirocyclic.

In some embodiments, the C₃-C₁₀ carbocyclylene is

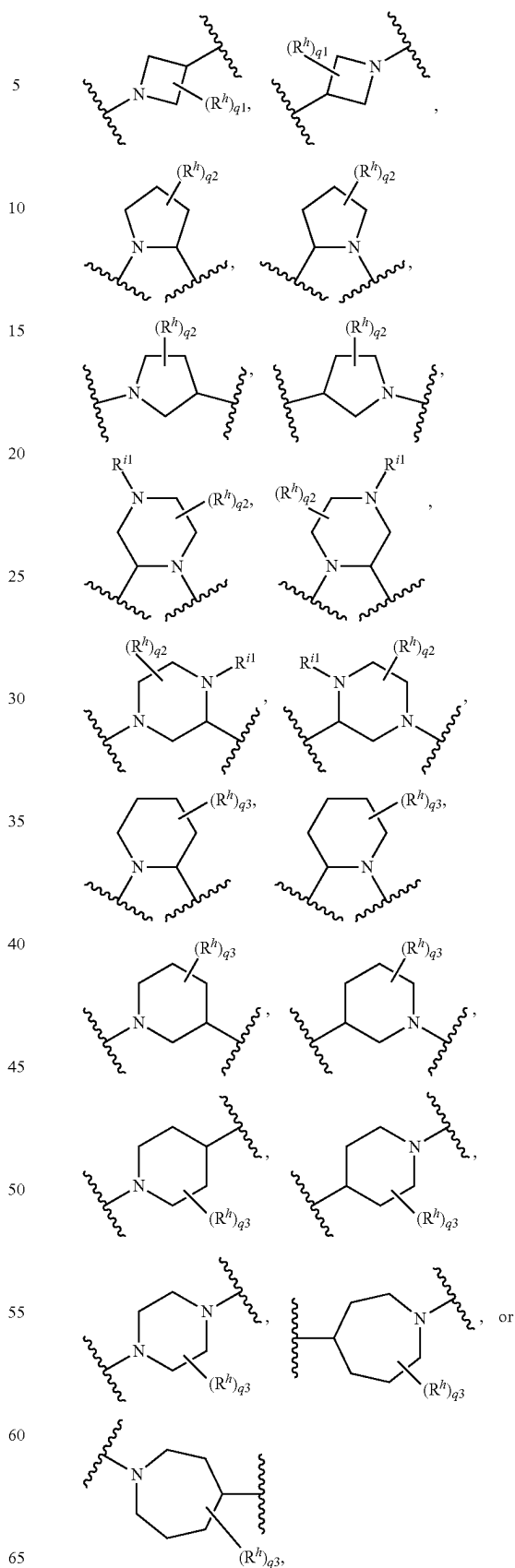


135

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In some embodiments, F^2 isIn some embodiments, the C_3 - C_{10} carbocyclylene isIn some embodiments, F^1 isIn some embodiments, each of F^1 , F^2 , or F^3 is, independently, optionally substituted C_2 - C_9 heterocyclylene.In some embodiments, the C_2 - C_9 heterocyclylene is monocyclic. In some embodiments, the C_2 - C_9 heterocyclylene is polycyclic.In some embodiments, the C_2 - C_9 heterocyclylene is bicyclic.In some embodiments, the C_2 - C_9 heterocyclylene is bridged. In some embodiments, the C_2 - C_9 heterocyclylene is fused. In some embodiments, the C_2 - C_9 heterocyclylene is spirocyclic.In some embodiments, the C_2 - C_9 heterocyclylene includes a quaternary amine.In some embodiments, the C_2 - C_9 heterocyclylene is**136**

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137

where

q1 is 0, 1, 2, 3, or 4;

q2 is 0, 1, 2, 3, 4, 5, or 6;

q3 is 0, 1, 2, 3, 4, 5, 6, 7, or 8;

each R^h is, independently, ^2H , halogen, optionally substituted $\text{C}_1\text{-C}_6$ alkyl, OR^{i2} , or $\text{NR}^{i3}\text{R}^{i4}$; or two R^h groups, together with the carbon atom to which each is attached, combine to form optionally substituted $\text{C}_3\text{-C}_{10}$ carbocyclyl or optionally substituted $\text{C}_2\text{-C}_9$ heterocyclyl; or two R^h groups, together with the carbon atoms to which each is attached, combine to form optionally substituted $\text{C}_3\text{-C}_{10}$ carbocyclyl or optionally substituted $\text{C}_2\text{-C}_9$ heterocyclyl;

R^{i1} is H or optionally substituted $\text{C}_1\text{-C}_6$ alkyl;

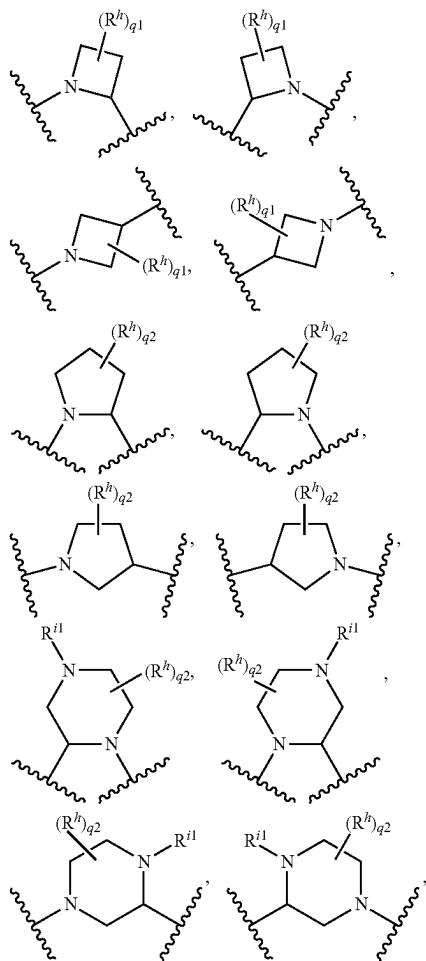
R^{i2} is H, optionally substituted $\text{C}_1\text{-C}_6$ alkyl, optionally substituted $\text{C}_1\text{-C}_6$ heteroalkyl, or optionally substituted $\text{C}_3\text{-C}_6$ carbocyclyl;

R^{i3} is H or optionally substituted $\text{C}_1\text{-C}_6$ alkyl; and

R^{i4} is H or optionally substituted $\text{C}_1\text{-C}_6$ alkyl.

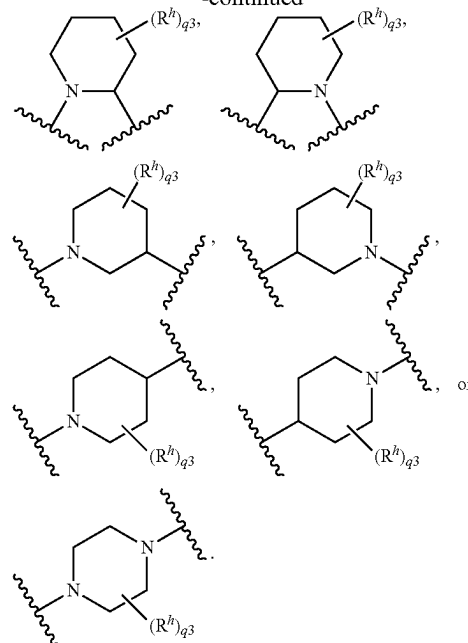
In some embodiments, each R^h is, independently, halogen, optionally substituted $\text{C}_1\text{-C}_6$ alkyl, OR^{i2} , or $\text{NR}^{i3}\text{R}^{i4}$. In some embodiments, R^{i1} is H or optionally substituted $\text{C}_1\text{-C}_6$ alkyl. In some embodiments, R^{i2} is H or optionally substituted $\text{C}_1\text{-C}_6$ alkyl. In some embodiments, R^{i3} is H or optionally substituted $\text{C}_1\text{-C}_6$ alkyl. In some embodiments, R^{i4} is H or optionally substituted $\text{C}_1\text{-C}_6$ alkyl.

In some embodiments, the $\text{C}_2\text{-C}_9$ heterocyclylene is



138

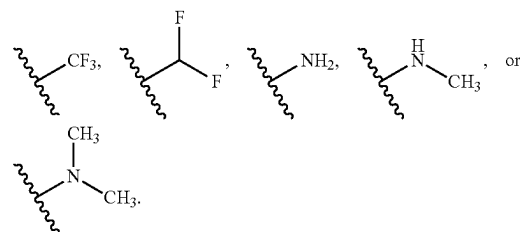
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In some embodiments, each R^h is, independently, halogen, optionally substituted $\text{C}_1\text{-C}_6$ alkyl, OR^{i2} , or $\text{NR}^{i3}\text{R}^{i4}$. In some embodiments, each R^h is, independently, halogen, optionally substituted $\text{C}_1\text{-C}_6$ alkyl, or $\text{NR}^{i3}\text{R}^{i4}$.

In some embodiments, each R^h is, independently, ^2H , halogen, cyano, optionally substituted $\text{C}_1\text{-C}_6$ alkyl, OR^{i2} , or $\text{NR}^{i3}\text{R}^{i4}$. In some embodiments, two R^h groups, together with the carbon atom to which each is attached, combine to form optionally substituted $\text{C}_3\text{-C}_{10}$ carbocyclyl or optionally substituted $\text{C}_2\text{-C}_9$ heterocyclyl. In some embodiments, two R^h groups, together with the carbon atoms to which each is attached, combine to form optionally substituted $\text{C}_3\text{-C}_{10}$ carbocyclyl or optionally substituted $\text{C}_2\text{-C}_9$ heterocyclyl.

In some embodiments, each R^h is, independently, H, ^2H , F, methyl,



In some embodiments, each R^h is, independently, F, methyl, or $\text{NR}^{i3}\text{R}^{i4}$.

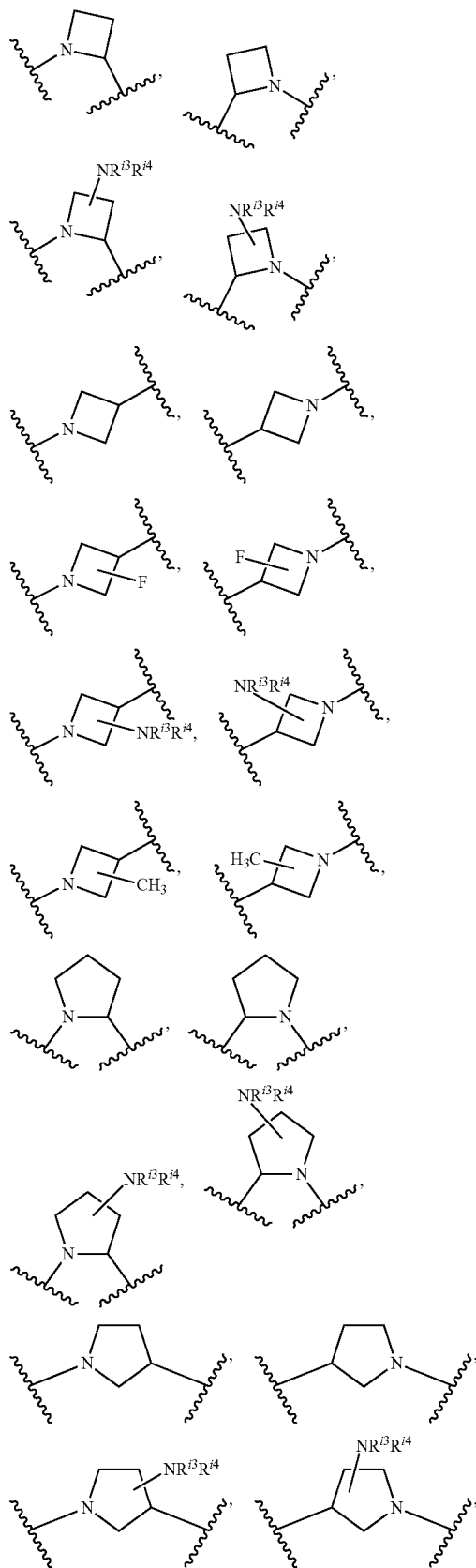
In some embodiments, q1 is 0, 1, or 2. In some embodiments, q1 is 0. In some embodiments, q1 is 1. In some embodiments, q1 is 2.

In some embodiments, q2 is 0, 1, or 2. In some embodiments, q2 is 0. In some embodiments, q2 is 1. In some embodiments, q2 is 2.

In some embodiments, q3 is 0, 1, or 2. In some embodiments, q3 is 0. In some embodiments, q3 is 1. In some embodiments, q3 is 2.

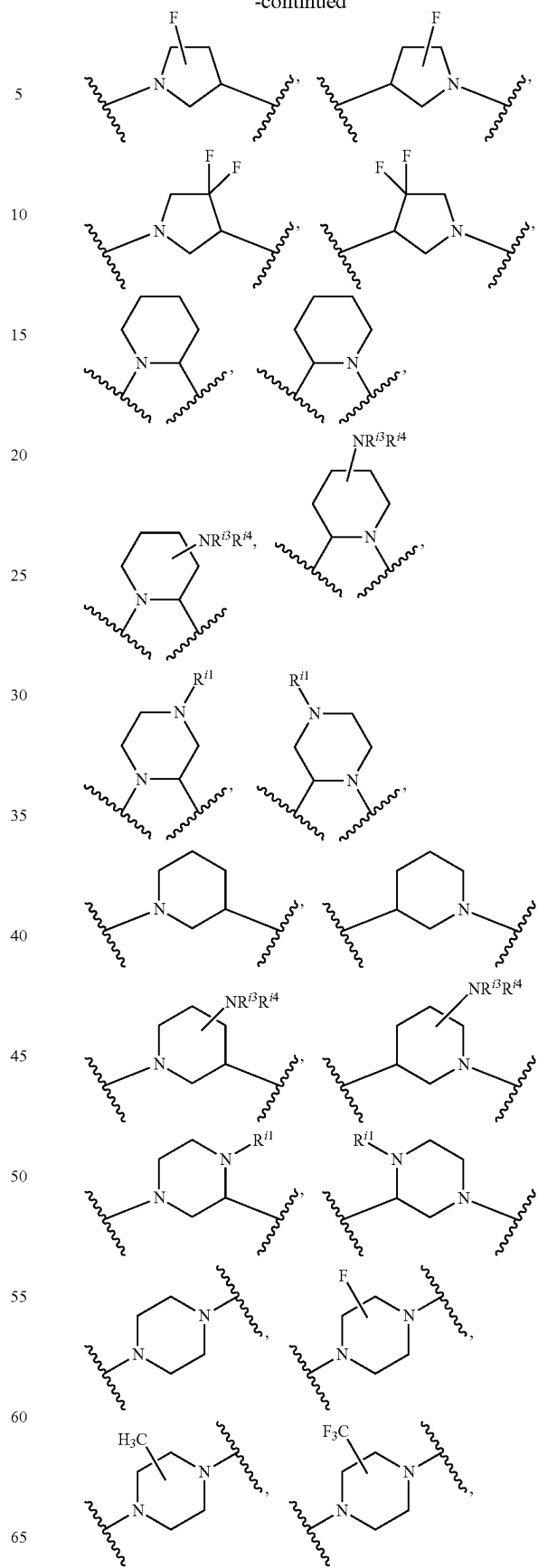
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In some embodiments, the C₂-C₉ heterocyclylene is



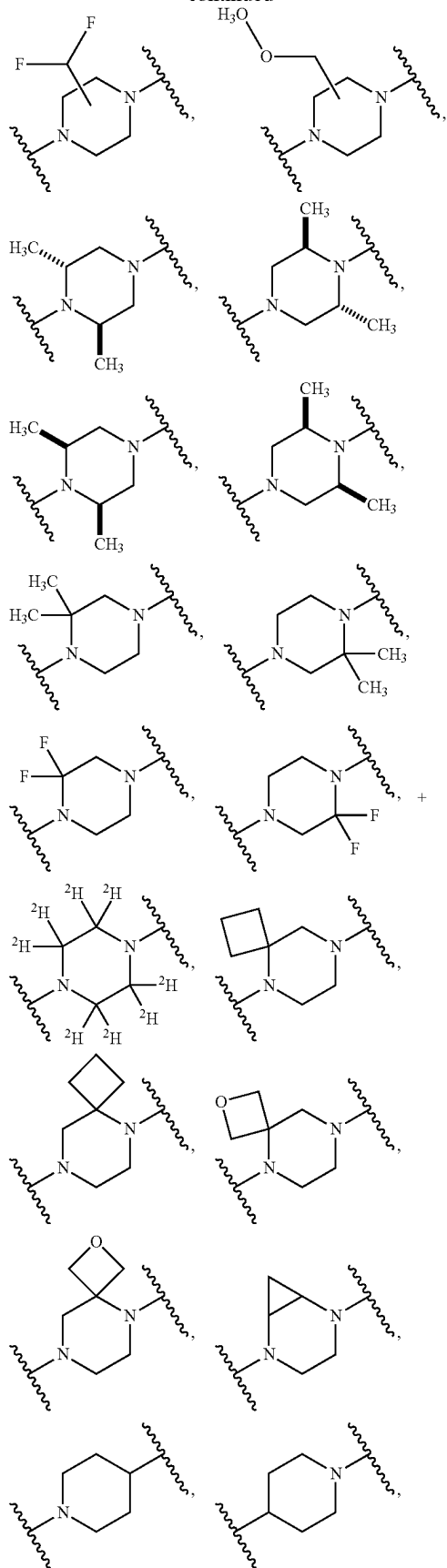
140

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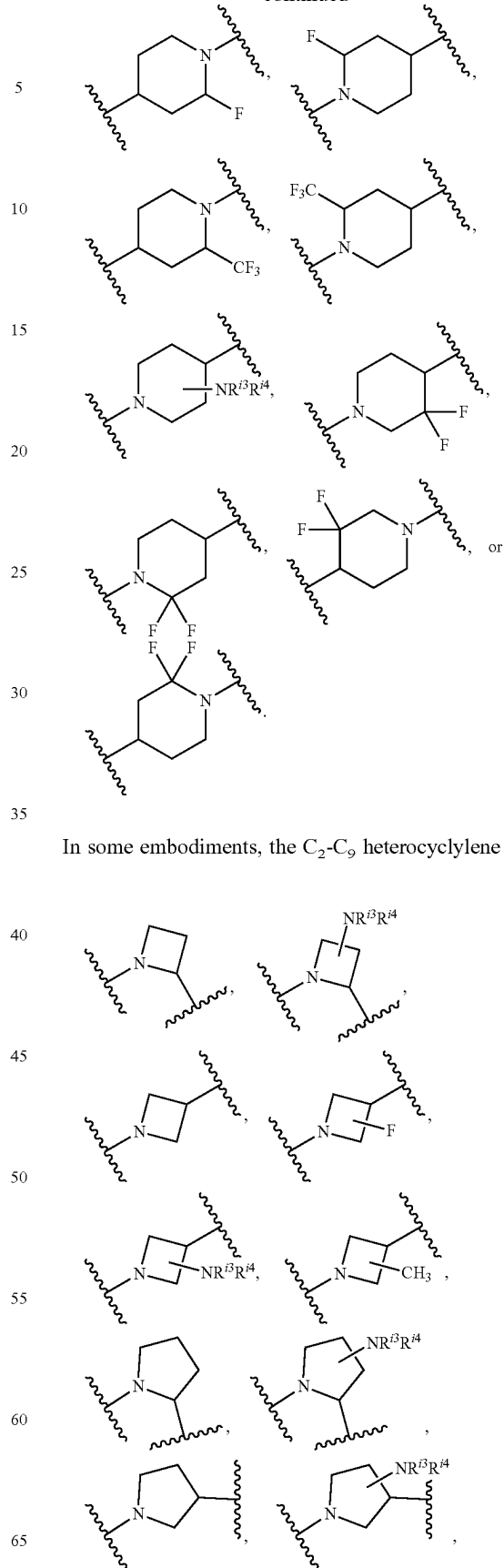
141

-continued



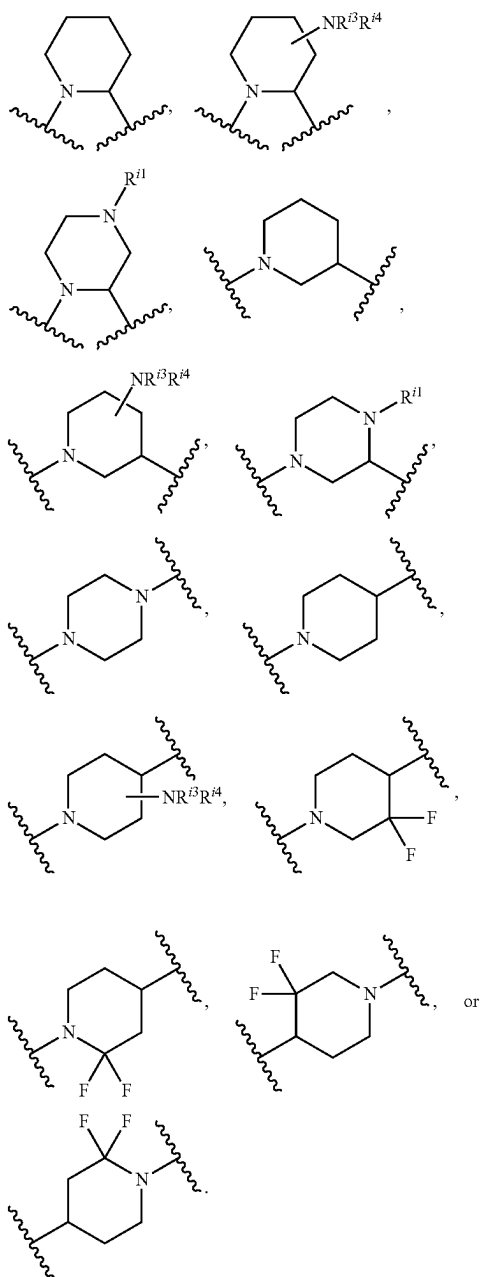
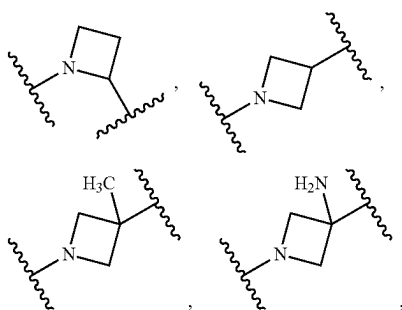
142

-continued

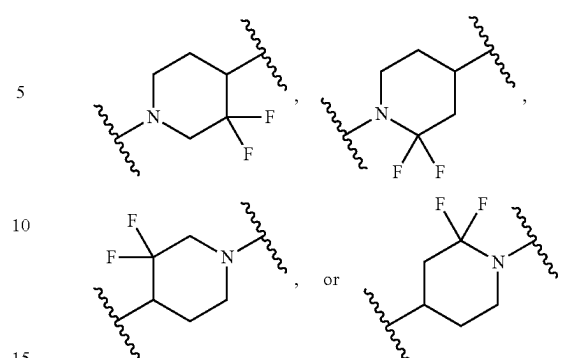
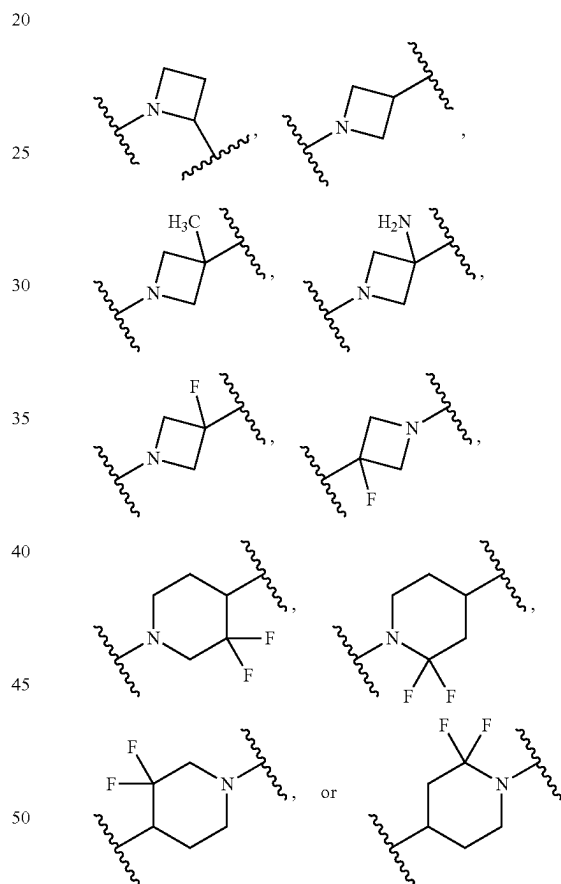
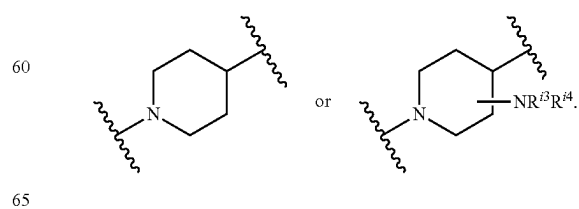


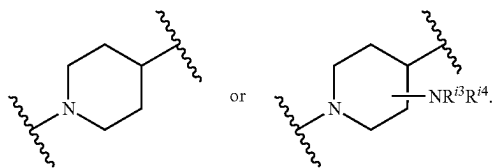
143

-continued

In some embodiments, F¹ is**144**

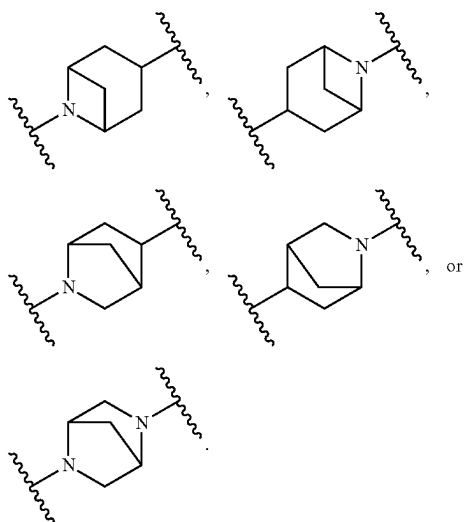
-continued

In some embodiments, F¹ isIn some embodiments, F² is

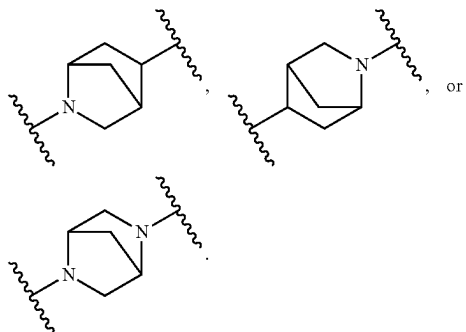
145In some embodiments, R^3 is

In some embodiments, R^{i1} is H or methyl. In some embodiments, R^{i2} is H or methyl. In some embodiments, R^{i3} is H or methyl. In some embodiments, R^{i4} is H or methyl.

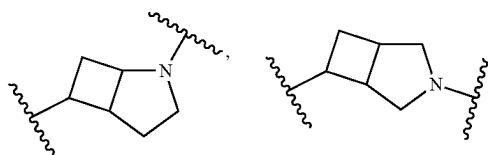
In some embodiments, the C_2 - C_9 heterocyclene is



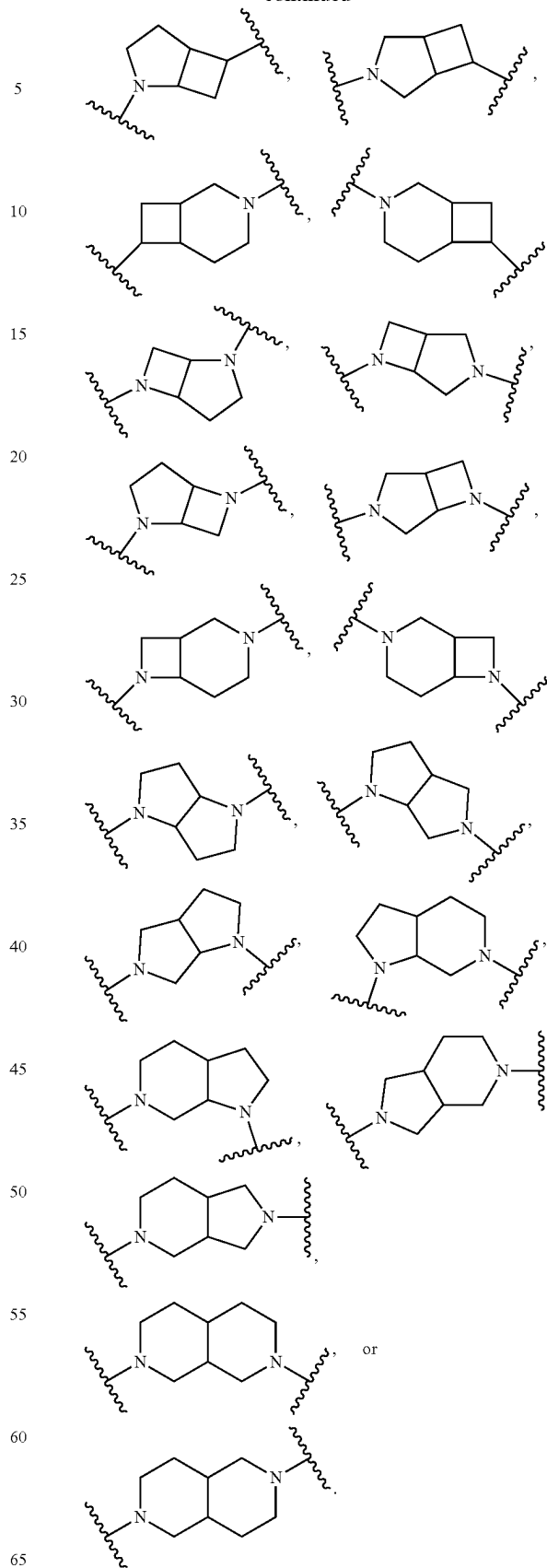
In some embodiments, the C_2 - C_9 heterocyclene is



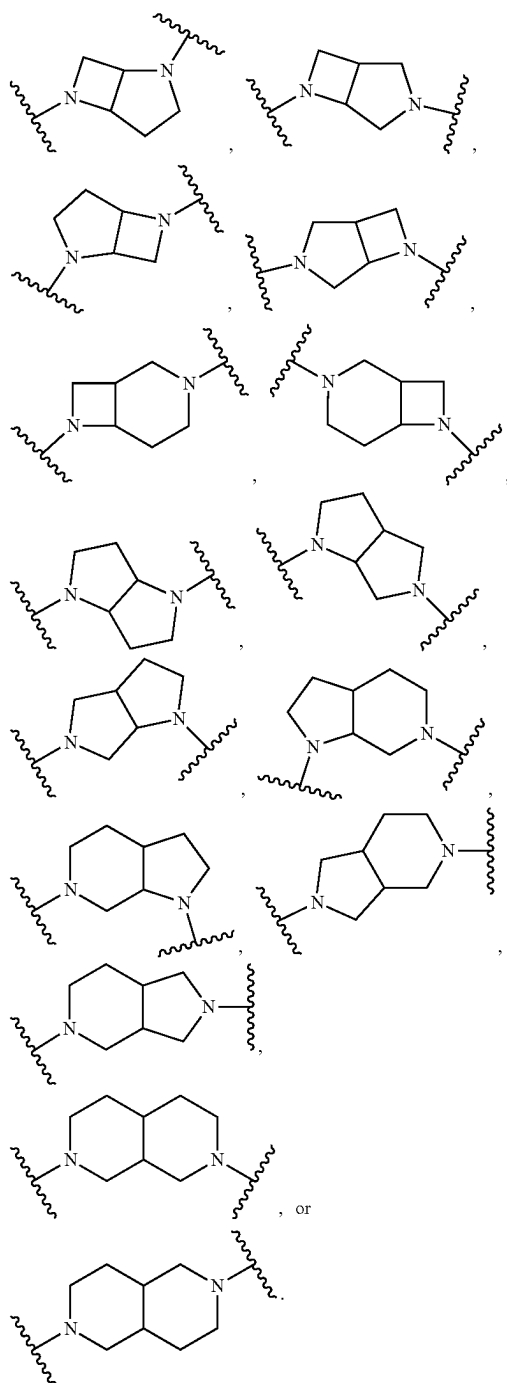
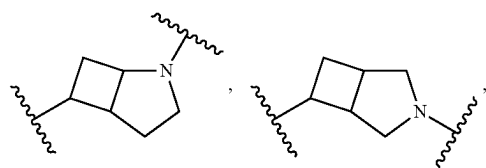
In some embodiments, the C_2 - C_9 heterocyclene is

**146**

-continued

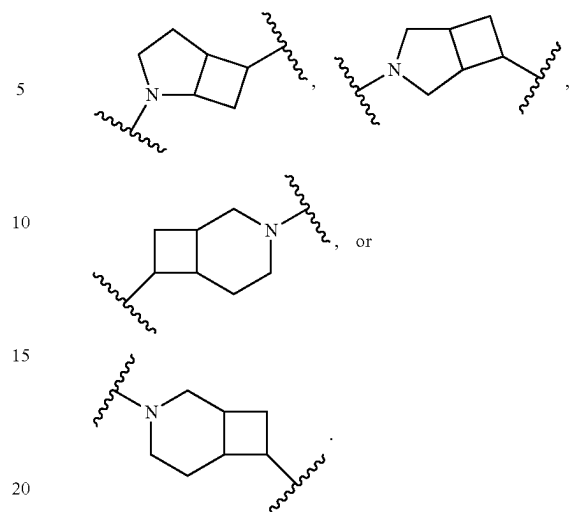
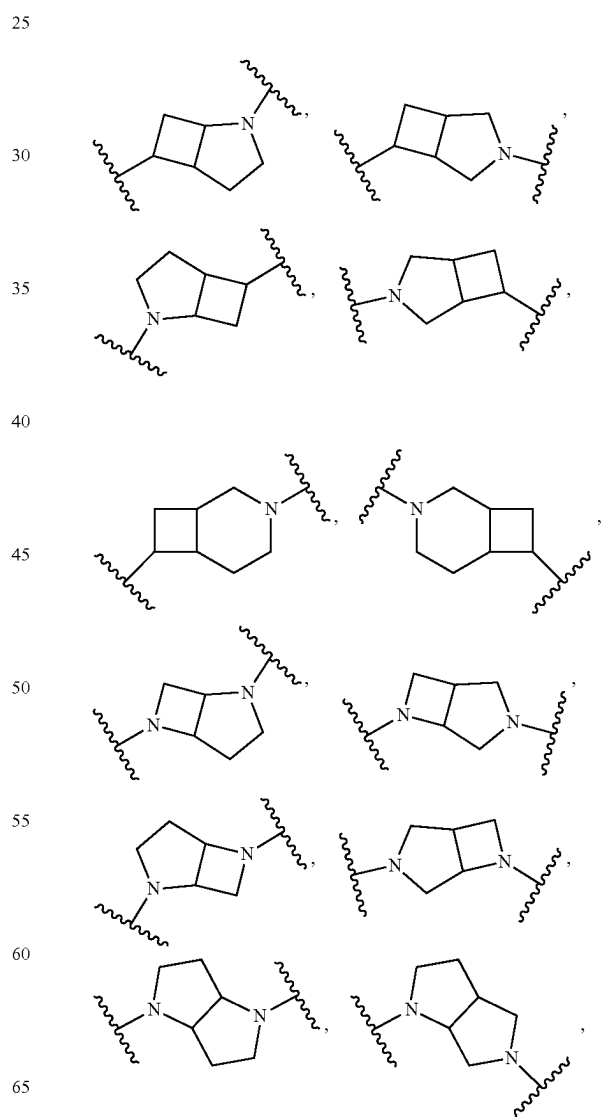


147

In some embodiments, the C₂-C₉ heterocyclylene isIn some embodiments, the C₂-C₉ heterocyclylene is

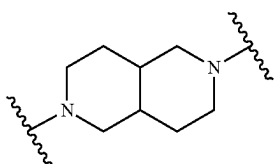
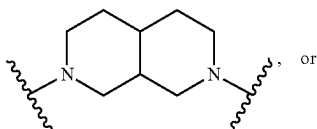
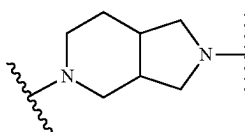
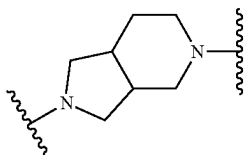
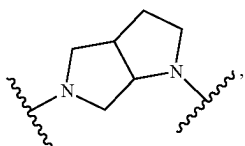
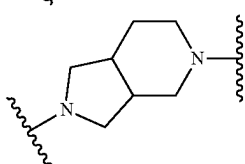
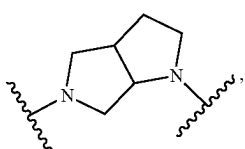
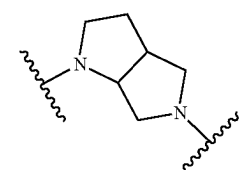
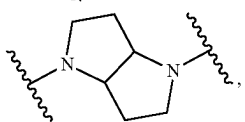
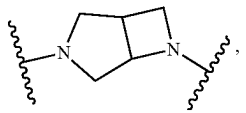
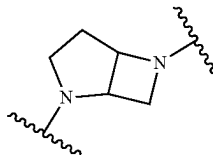
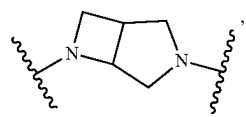
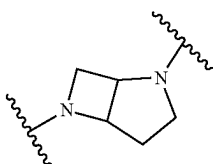
148

-continued

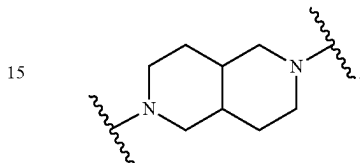
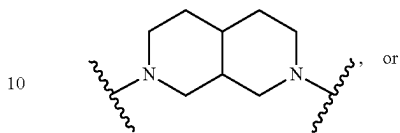
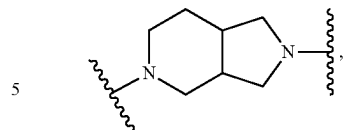
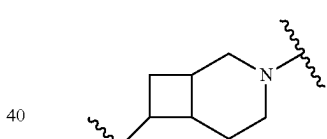
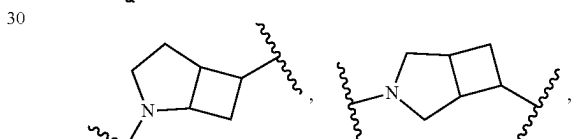
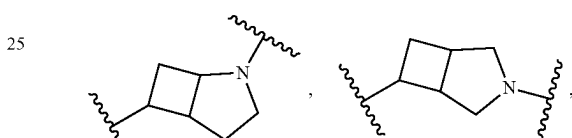
In some embodiments, F¹ is

149

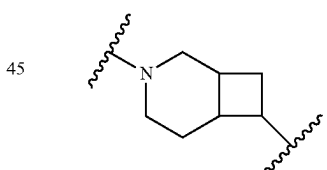
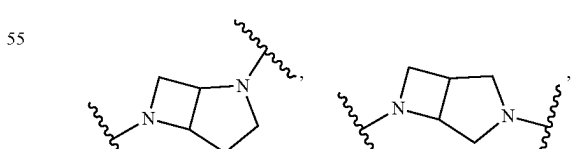
-continued

In some embodiments, F¹ is**150**

-continued

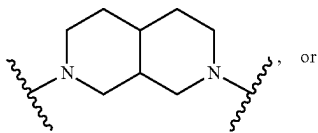
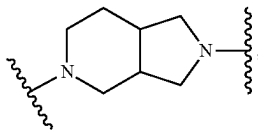
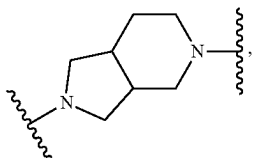
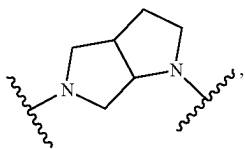
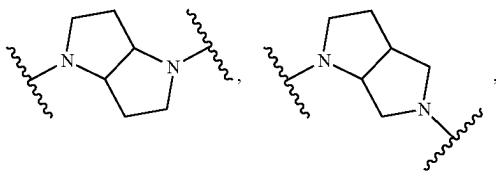
In some embodiments, F¹ is

, or

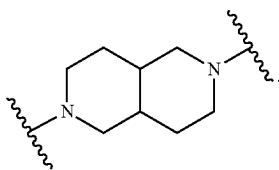
In some embodiments, F² is

151

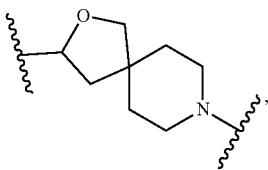
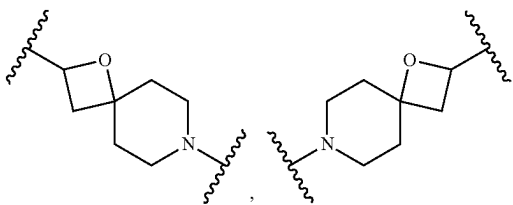
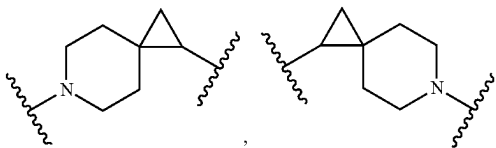
-continued



or

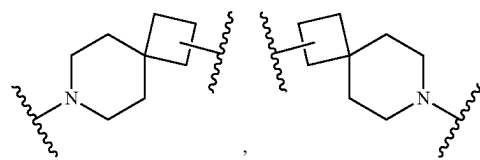
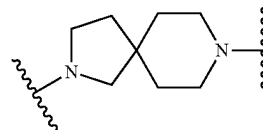
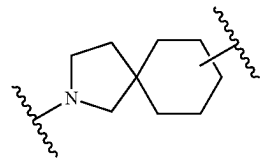
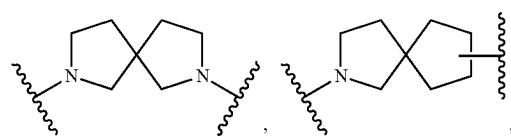
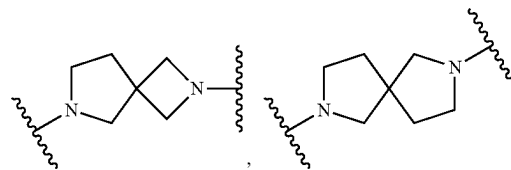
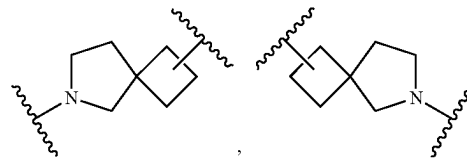
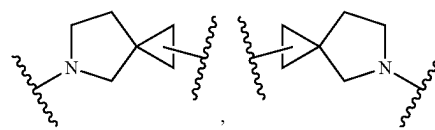
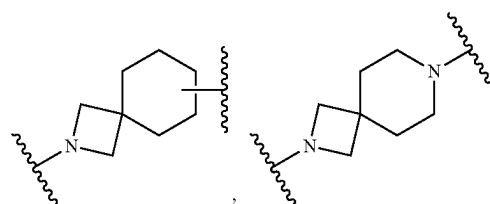
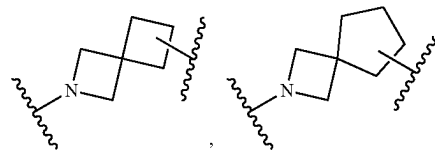
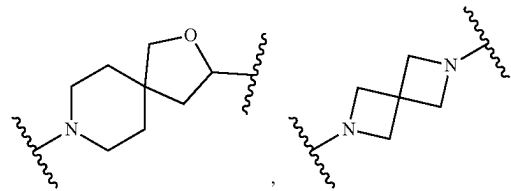


In some embodiments, the C₂-C₉ heterocyclylene is



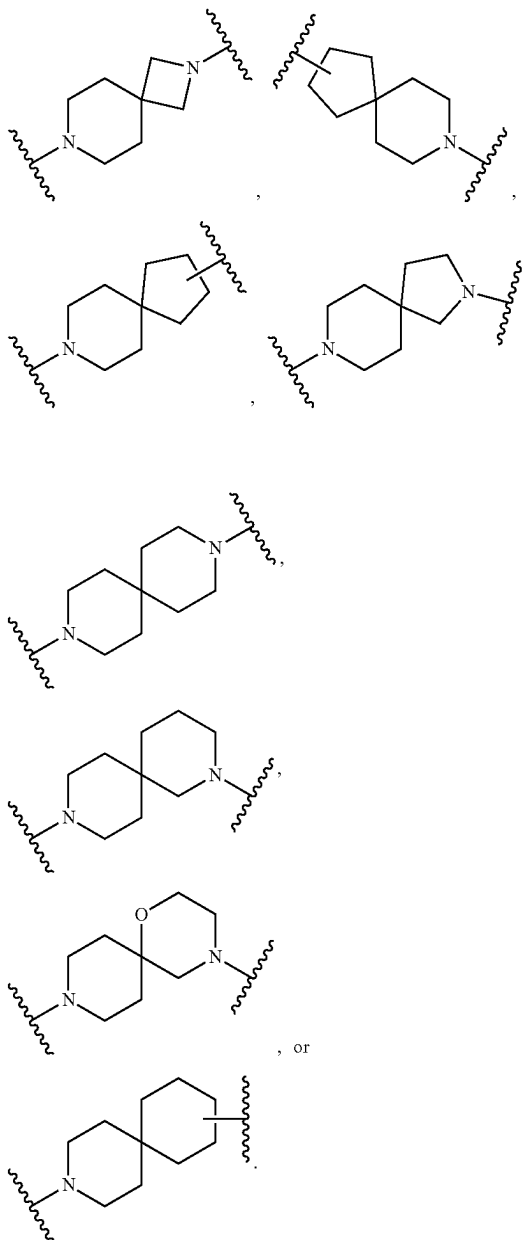
152

-continued

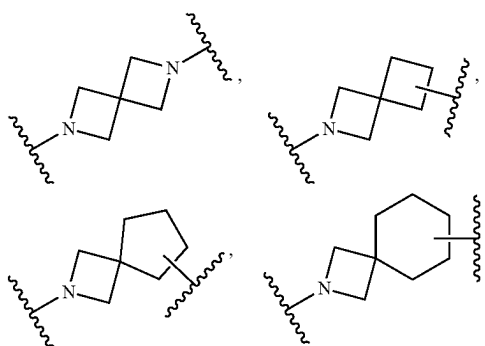


153

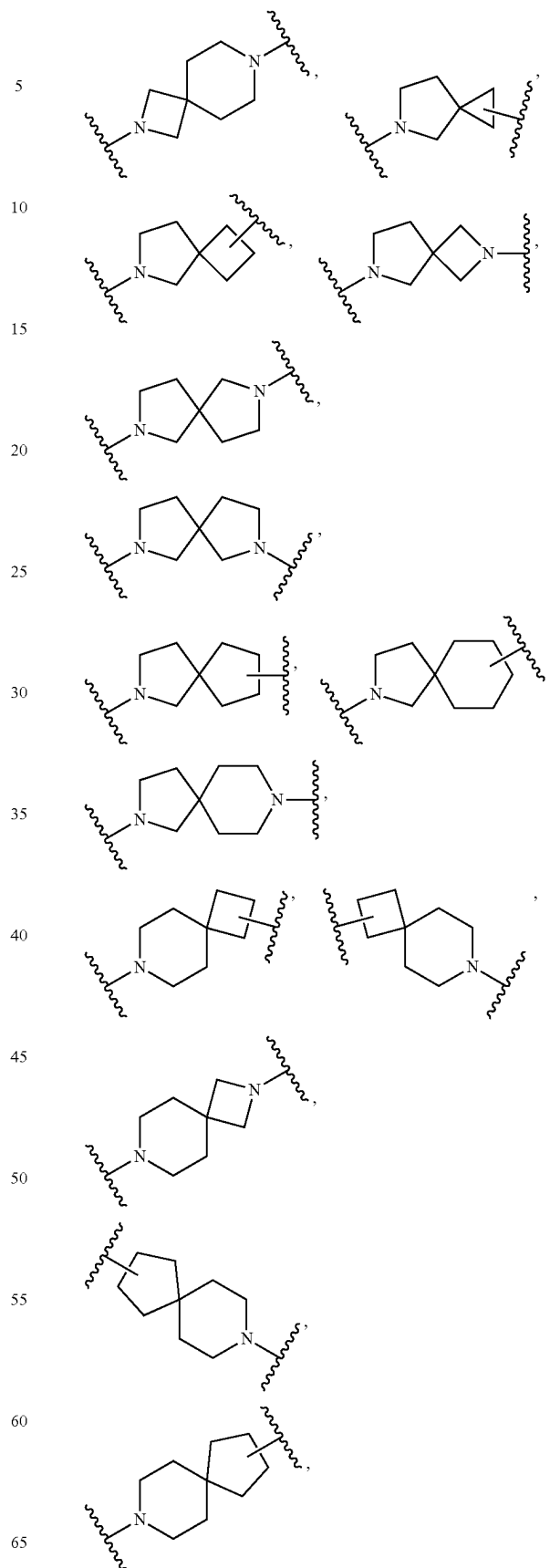
-continued



In some embodiments, the C₂-C₉ heterocyclene is

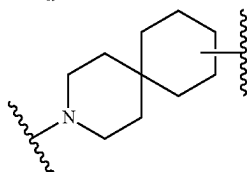
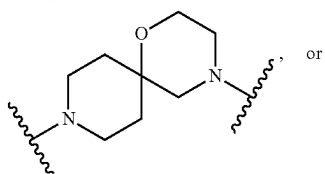
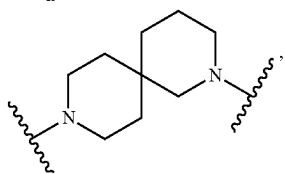
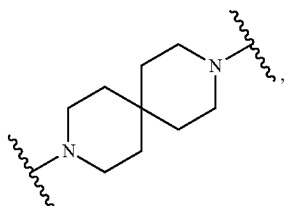
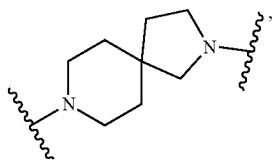
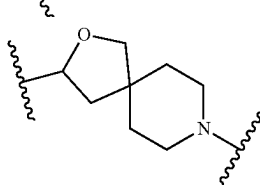
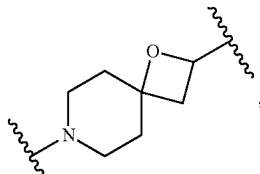
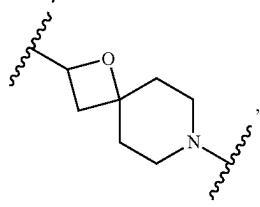
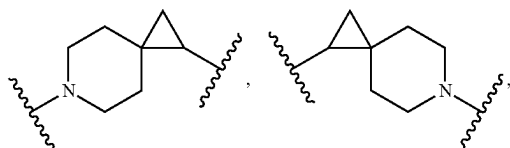
**154**

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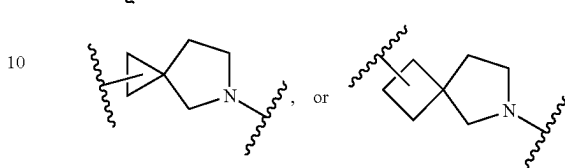
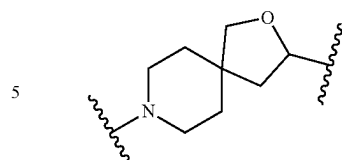
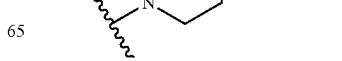
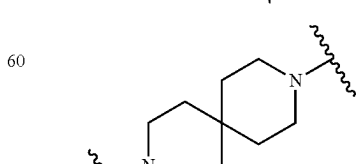
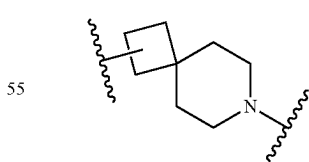
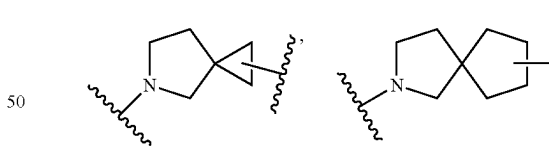
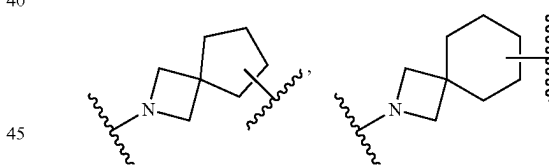
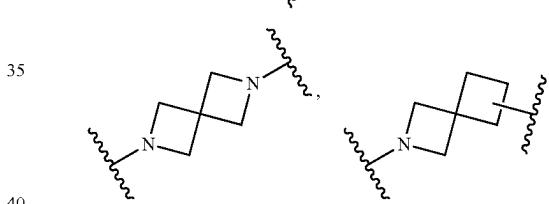
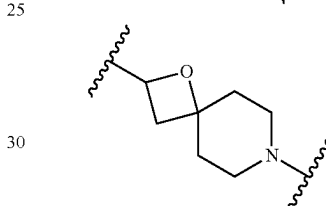
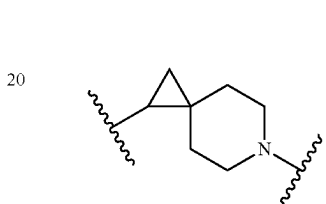


155

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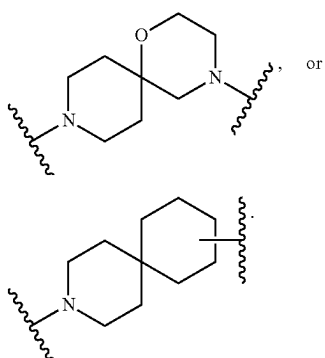
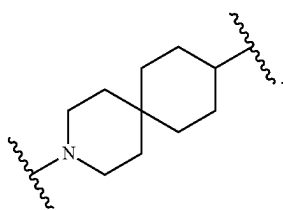
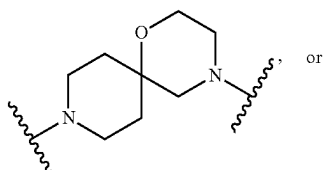
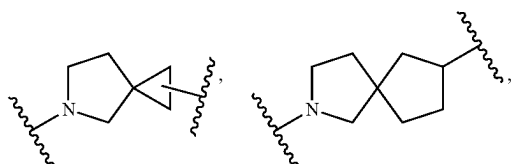
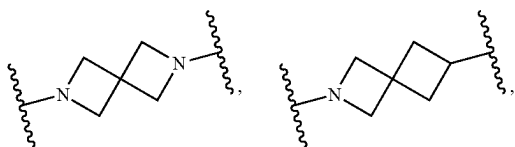
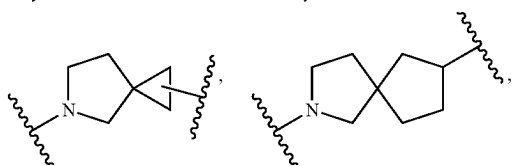
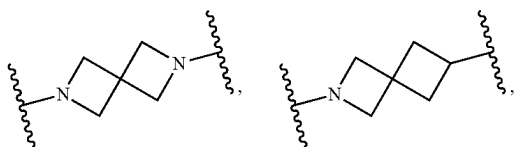
In some embodiments, the C₂-C₉ heterocyclylene is**156**

-continued

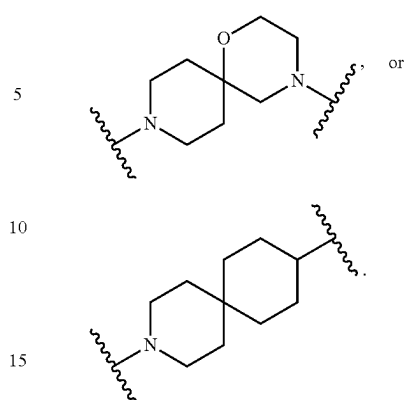
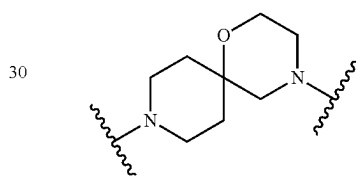
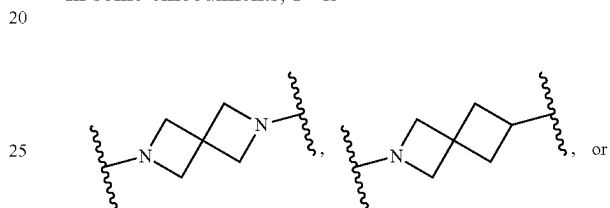
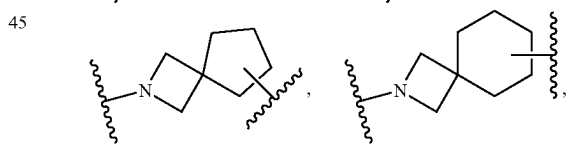
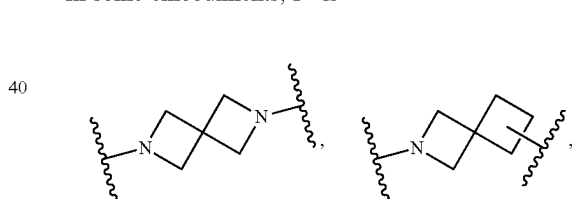
In some embodiments, F¹ is

157

-continued

In some embodiments, F¹ isIn some embodiments, F¹ is**158**

-continued

In some embodiments, F¹ isIn some embodiments, F² is

50

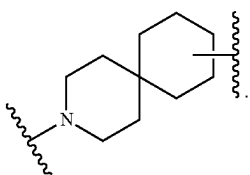
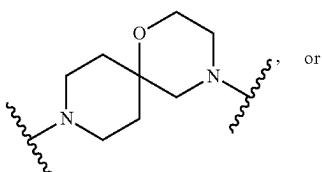
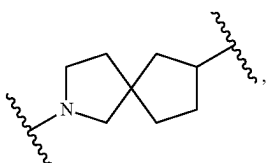
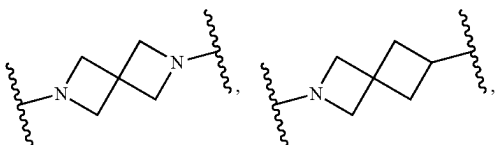
55

60

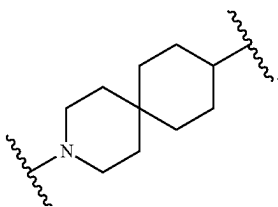
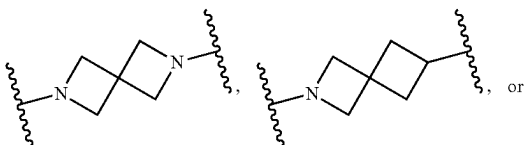
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159

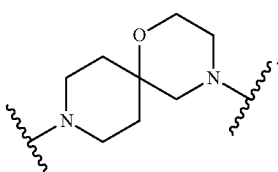
-continued

In some embodiments, F² is

or

In some embodiments, F² is

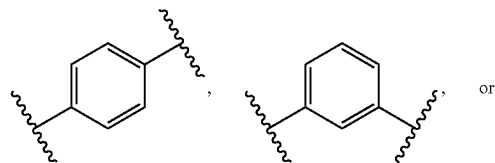
or



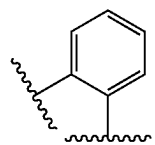
In some embodiments, each of F¹, F², or F³ is, independently, optionally substituted C₆-C₁₀ arylene.

160In some embodiments, the C₆-C₁₀ arylene is

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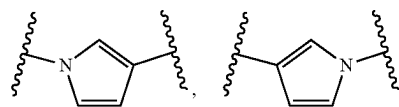
15

In some embodiments, each of F¹, F², or F³ is, independently, optionally substituted C₂-C₉ heteroarylene.

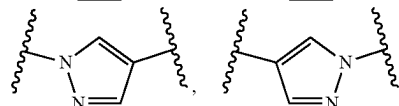
In some embodiments, the C₂-C₉ heteroarylene is

20

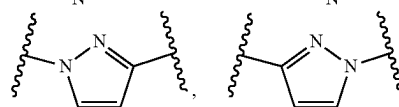
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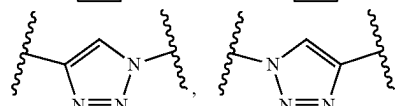
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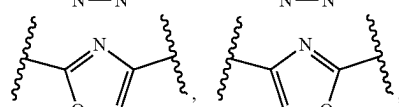
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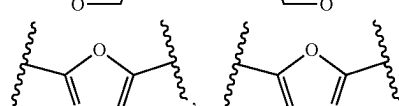
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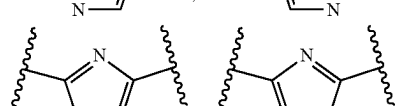
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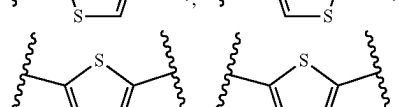
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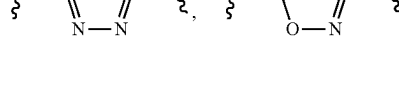
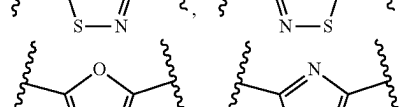
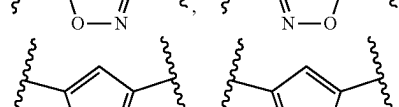
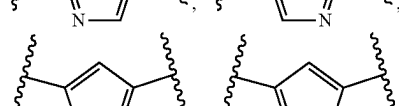
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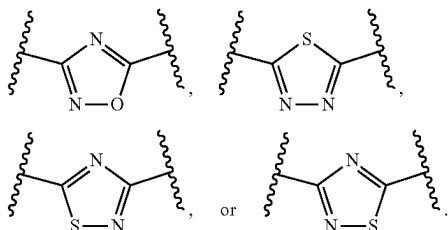
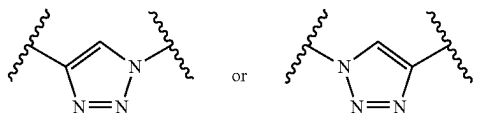
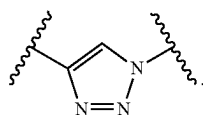
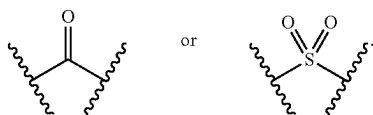
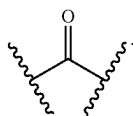


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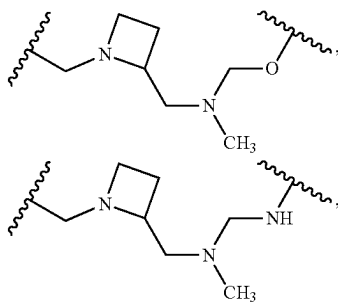
161

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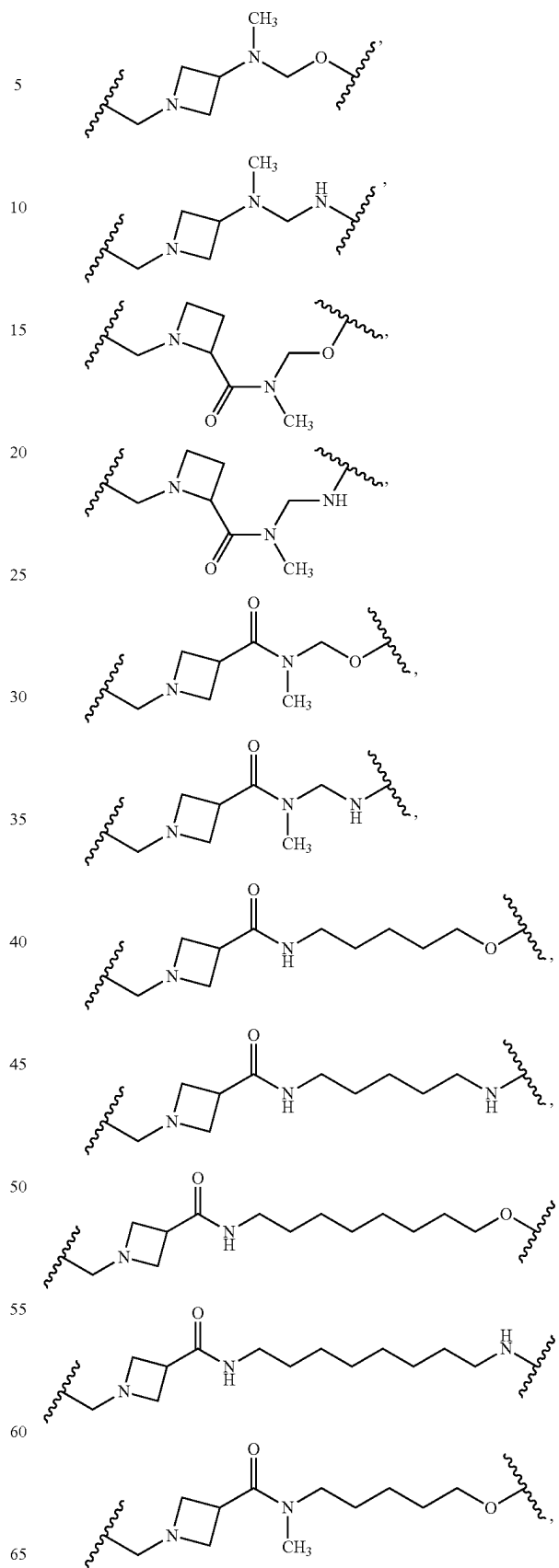
In some embodiments, F^2 isIn some embodiments, F^2 isIn some embodiments, C^3 isIn some embodiments, C^3 is

In some embodiments, m is 1. In some embodiments, p is 1.

In some embodiments, the linker has the structure of

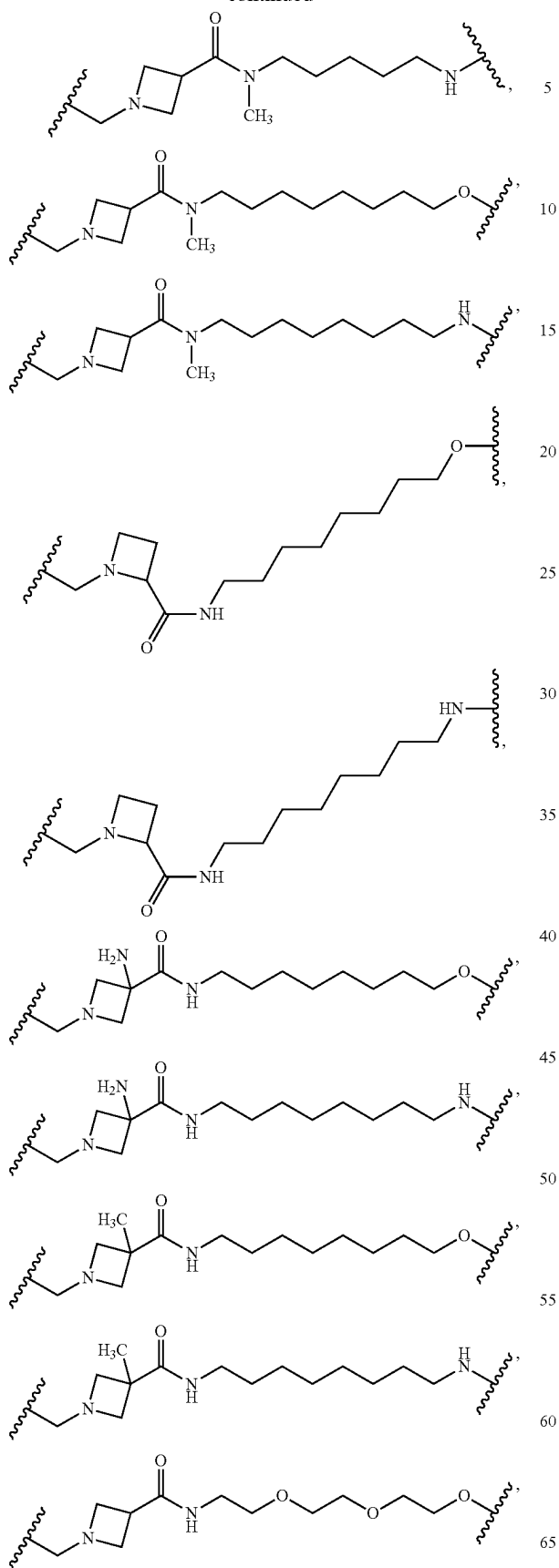
**162**

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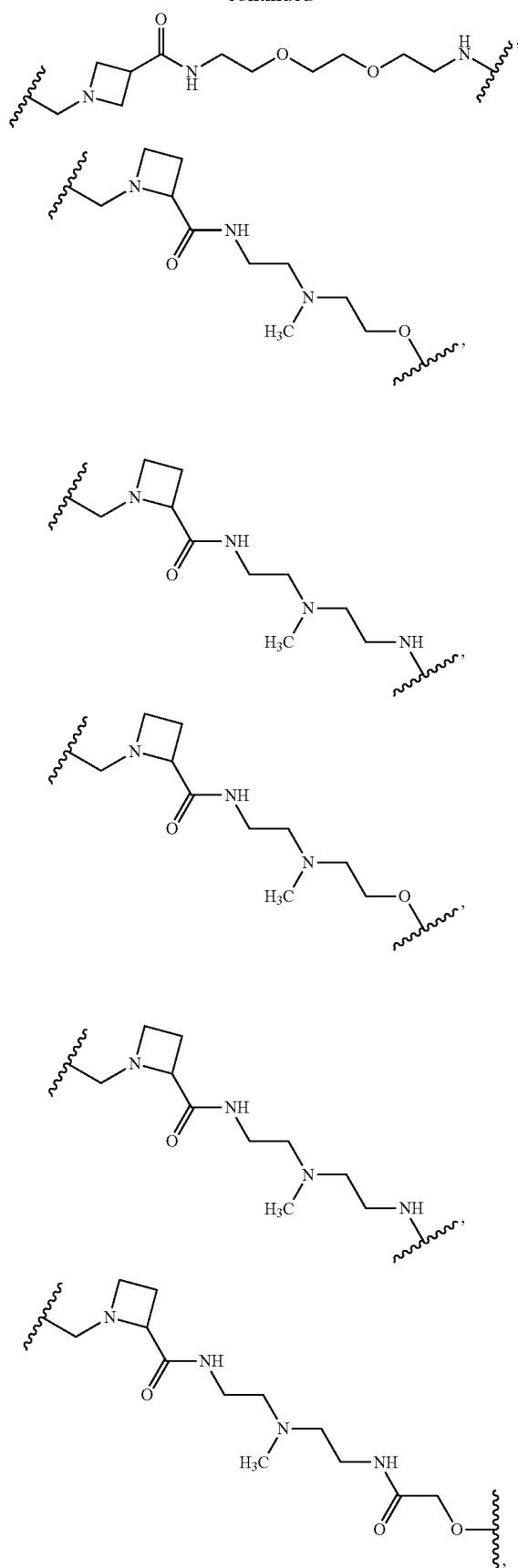


163

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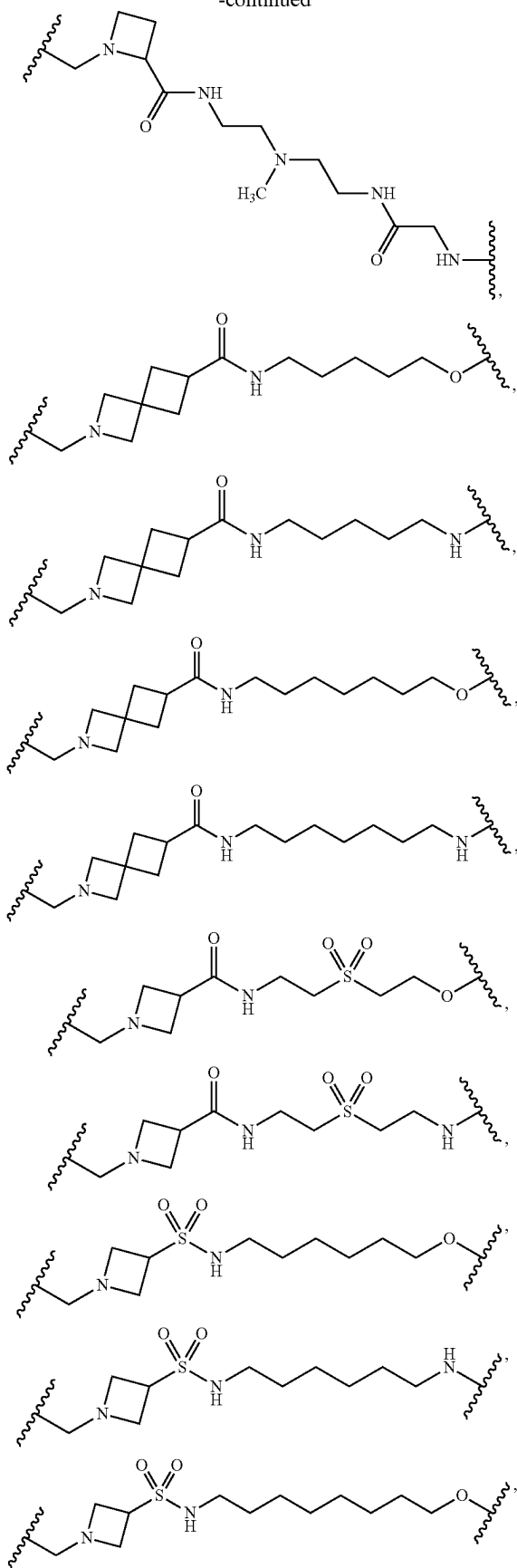
**164**

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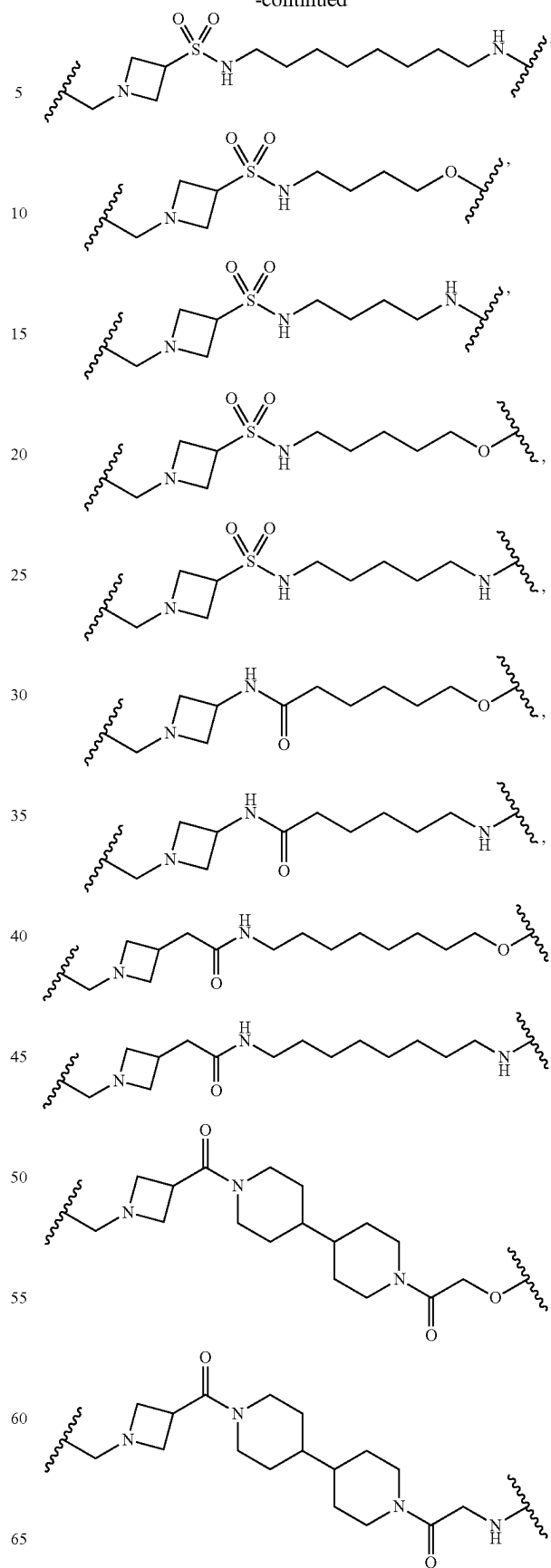


165

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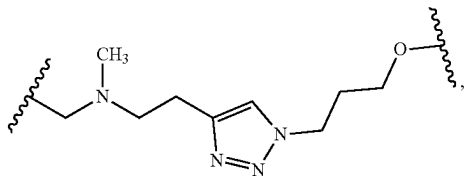
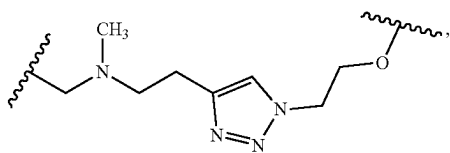
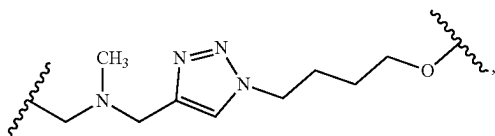
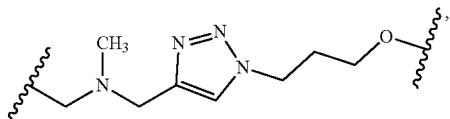
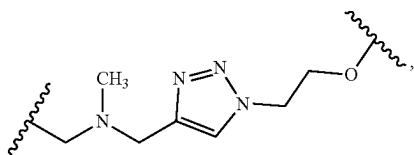
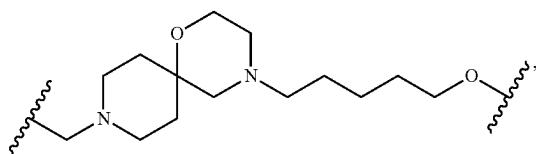
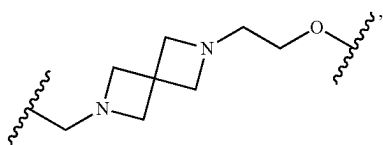
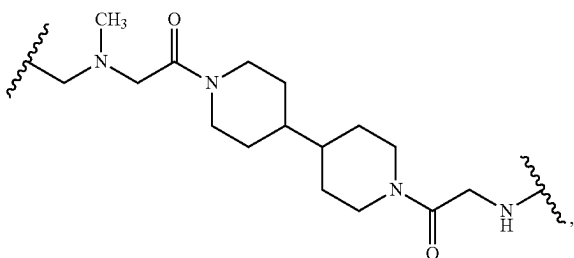
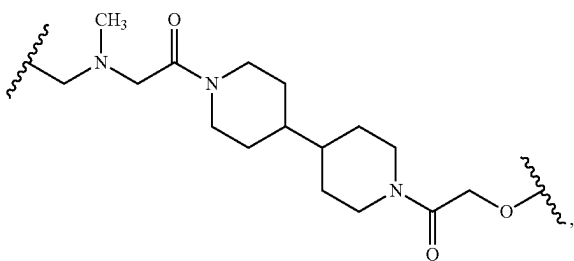
**166**

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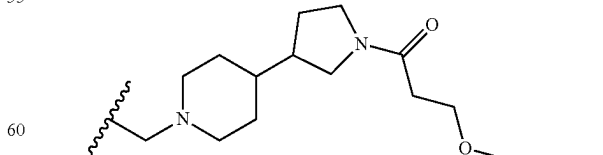
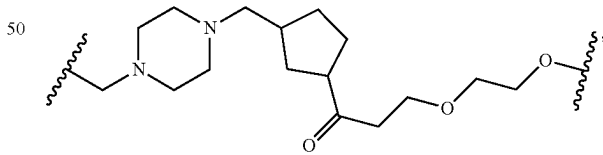
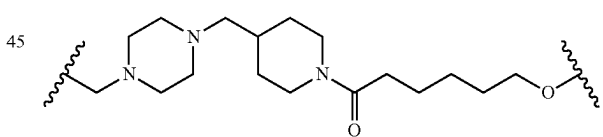
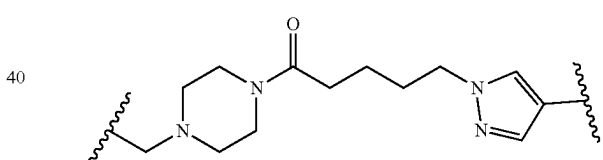
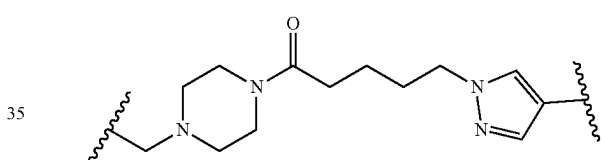
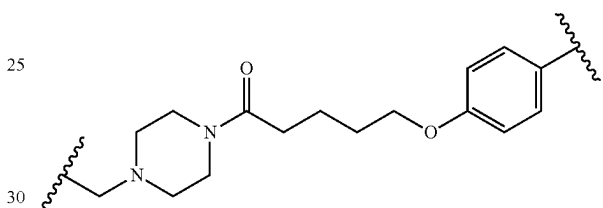
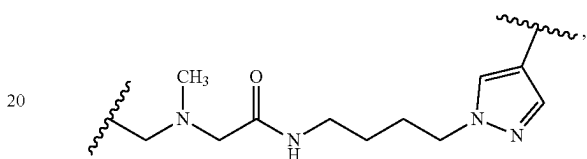
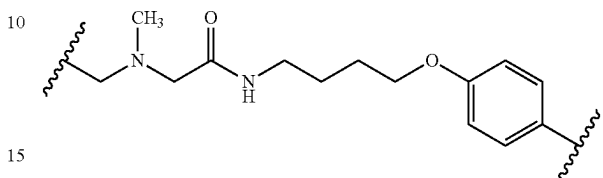
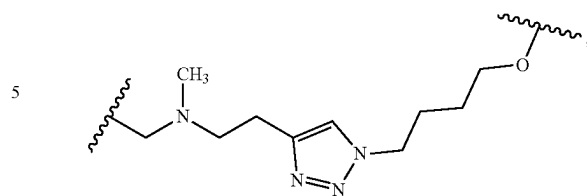


167

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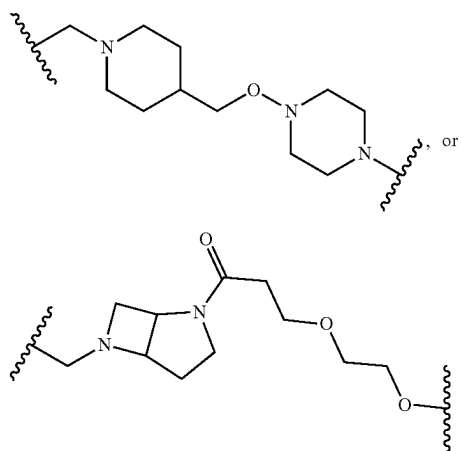
**168**

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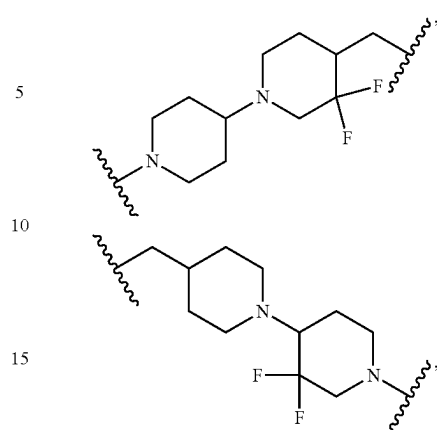


169

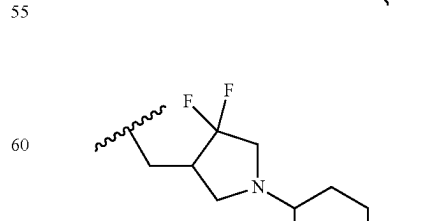
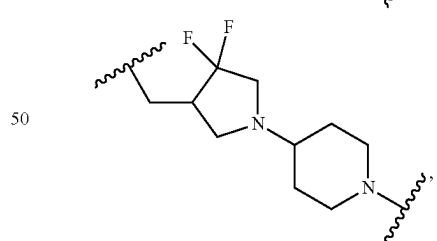
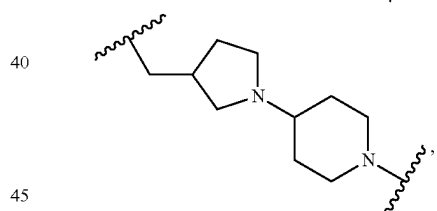
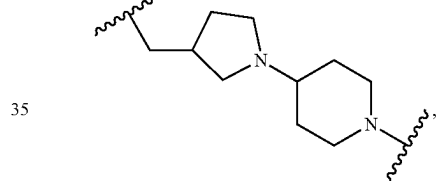
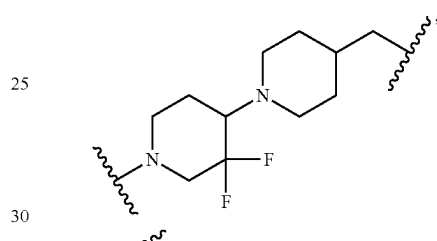
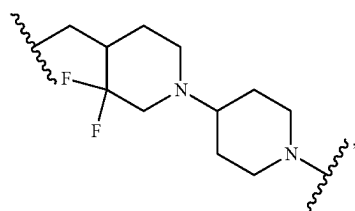
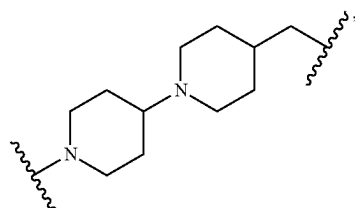
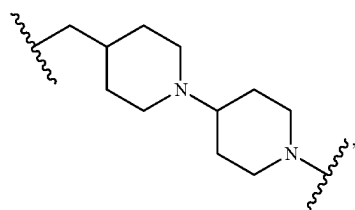
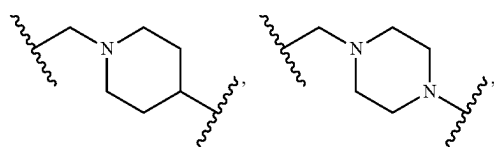
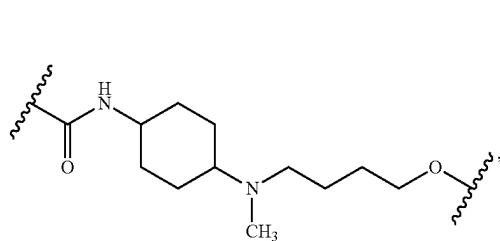
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**170**

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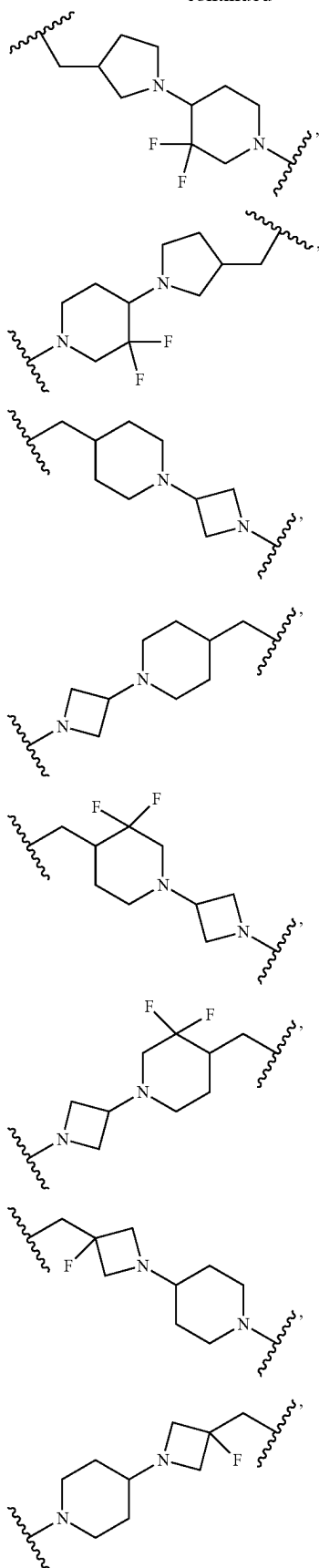


In some embodiments, the linker has the structure of

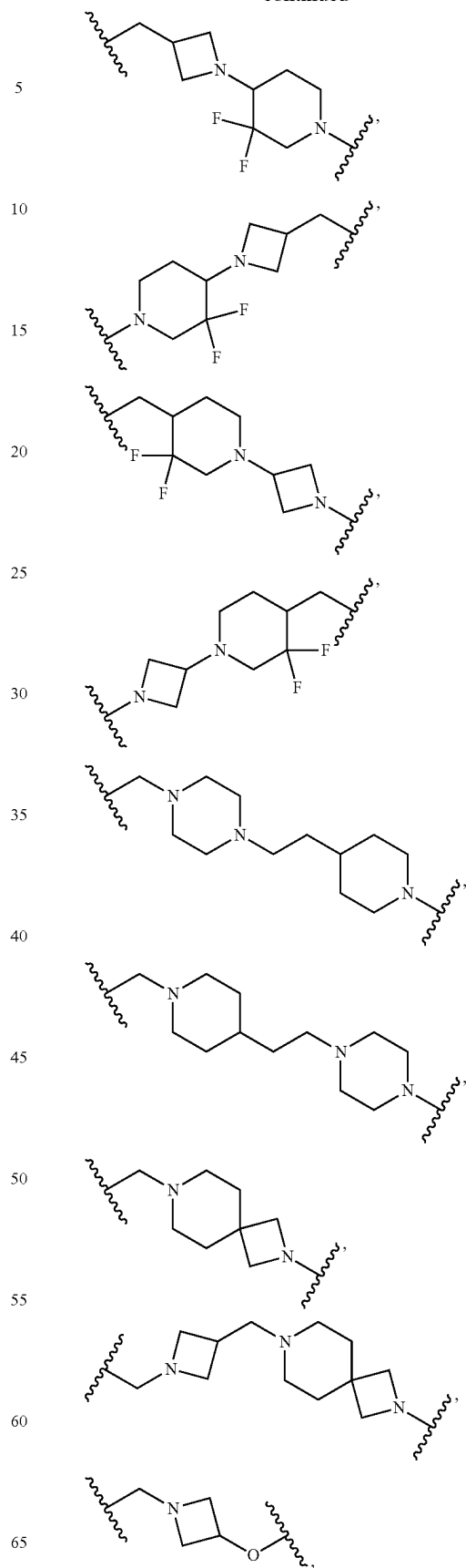


171

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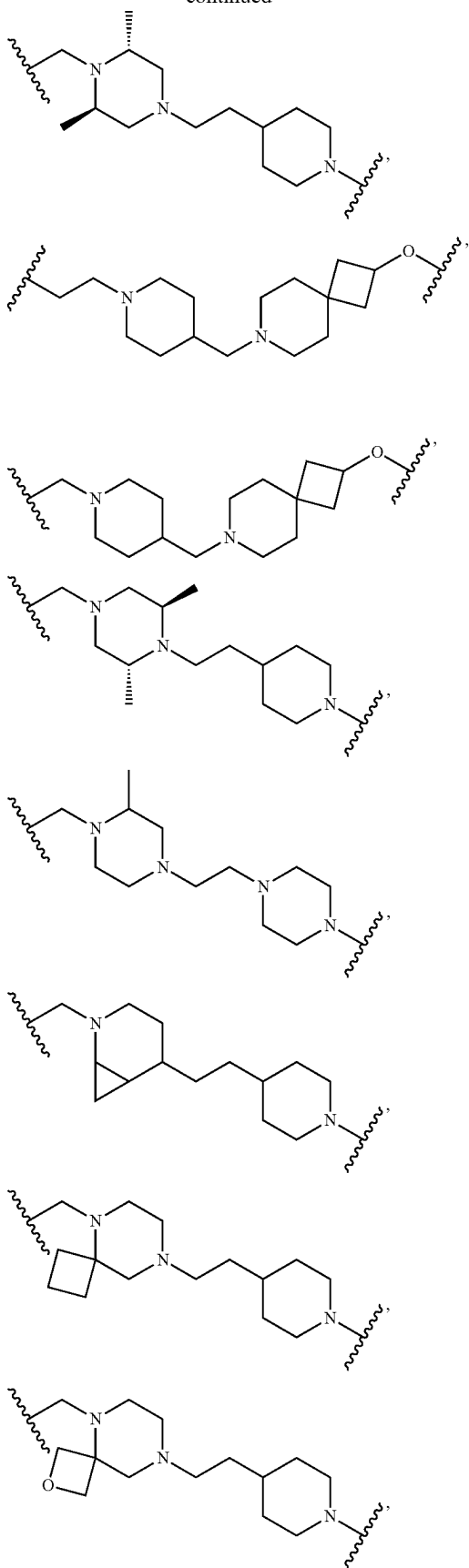
**172**

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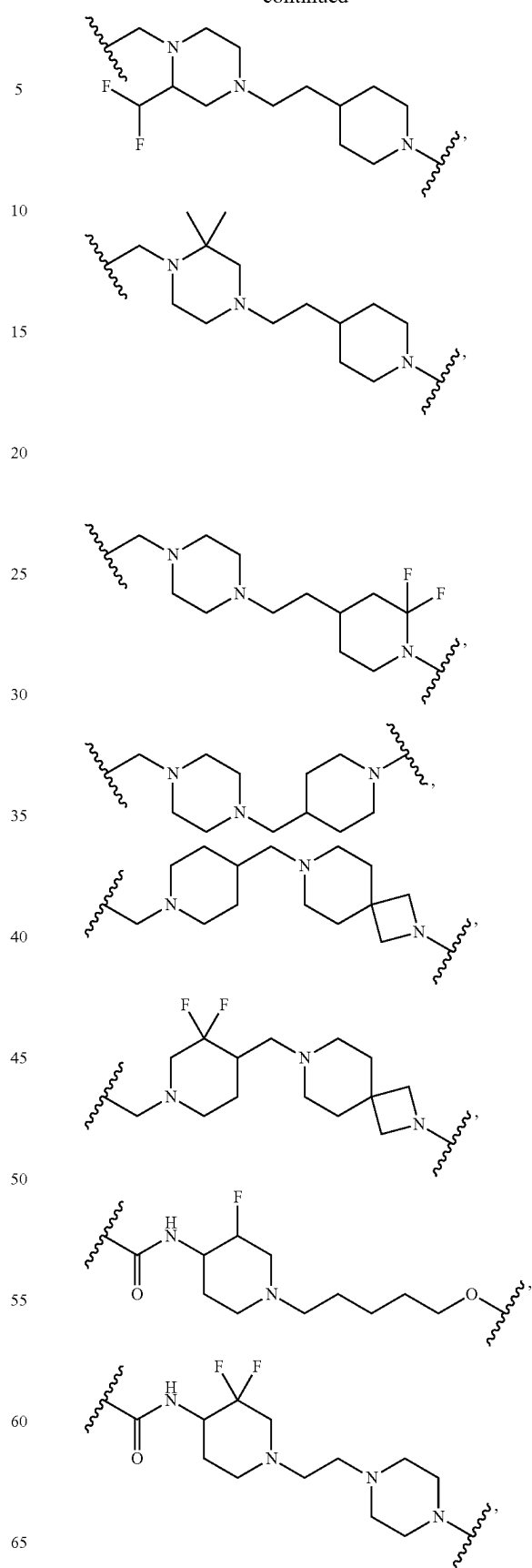


173

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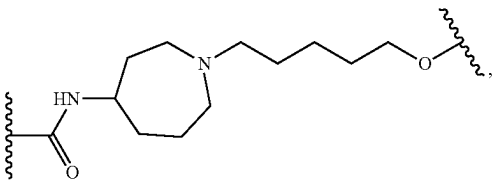
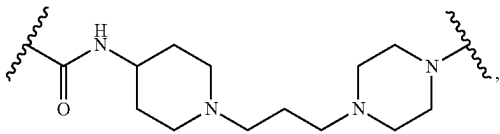
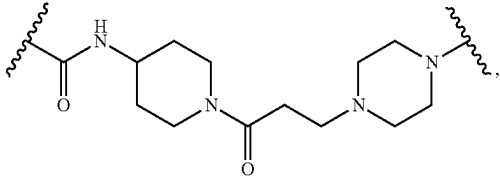
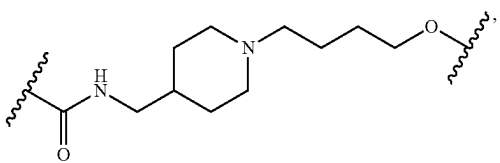
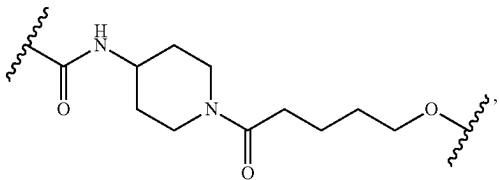
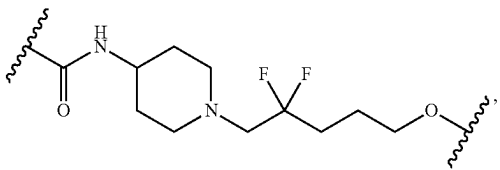
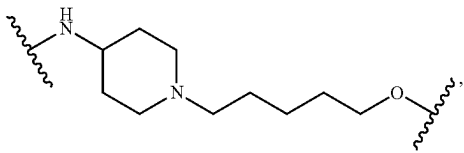
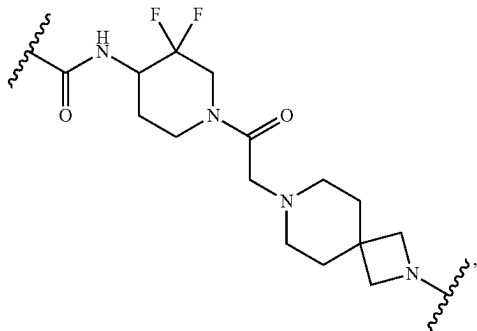
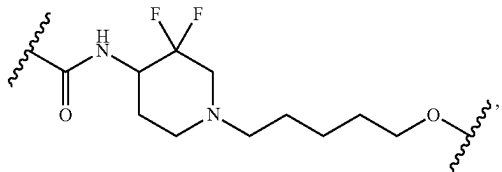
**174**

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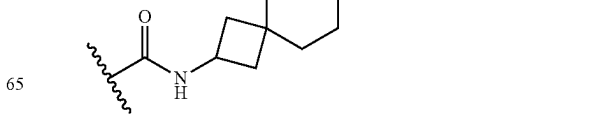
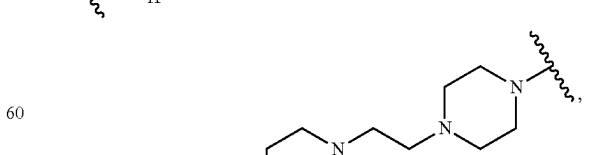
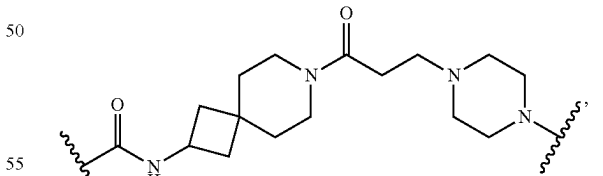
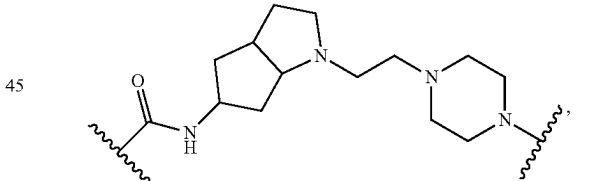
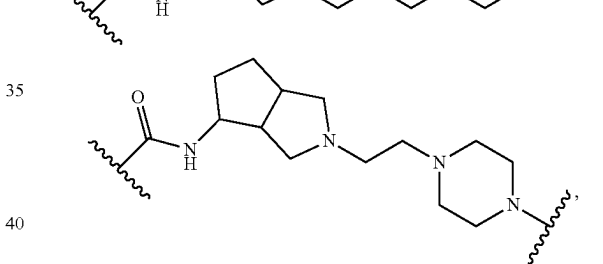
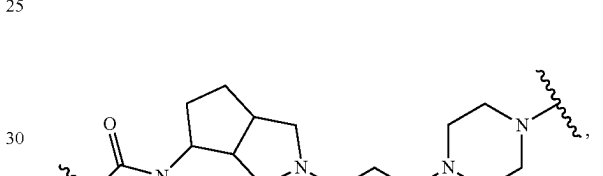
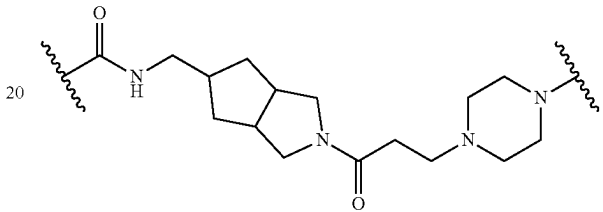
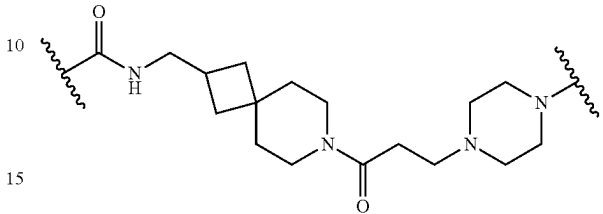
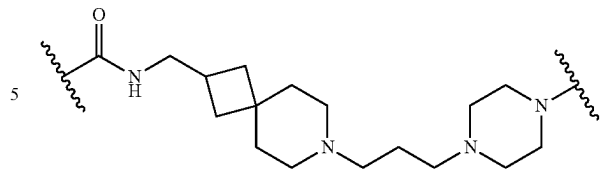
175

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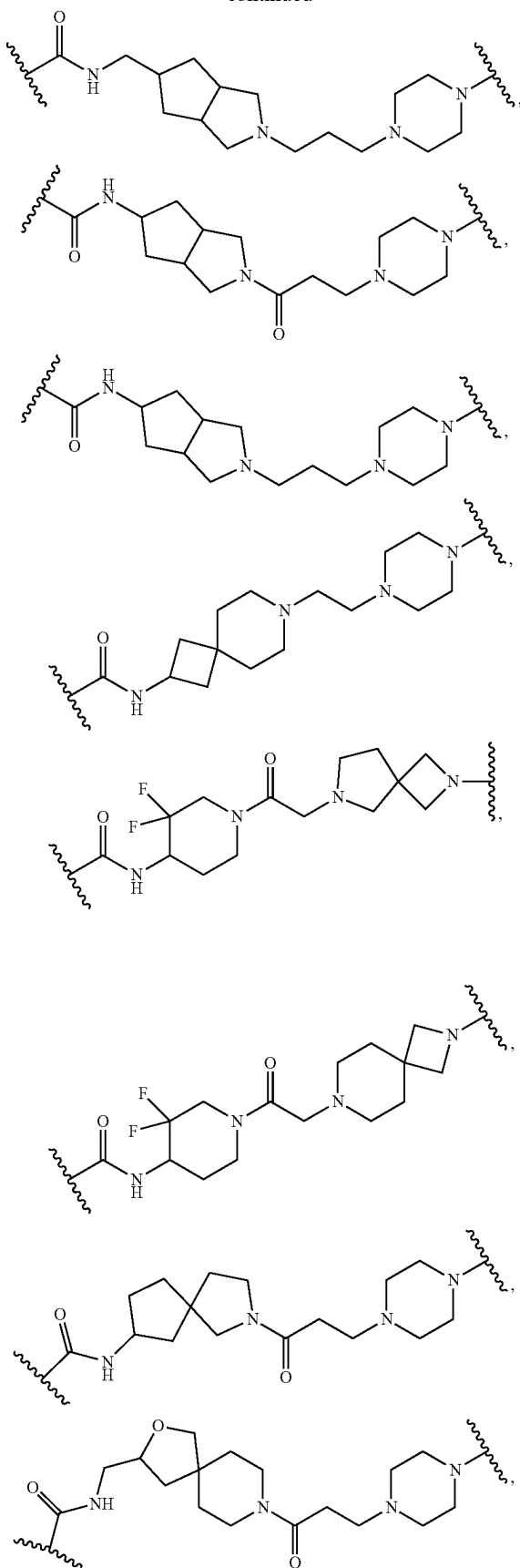
176

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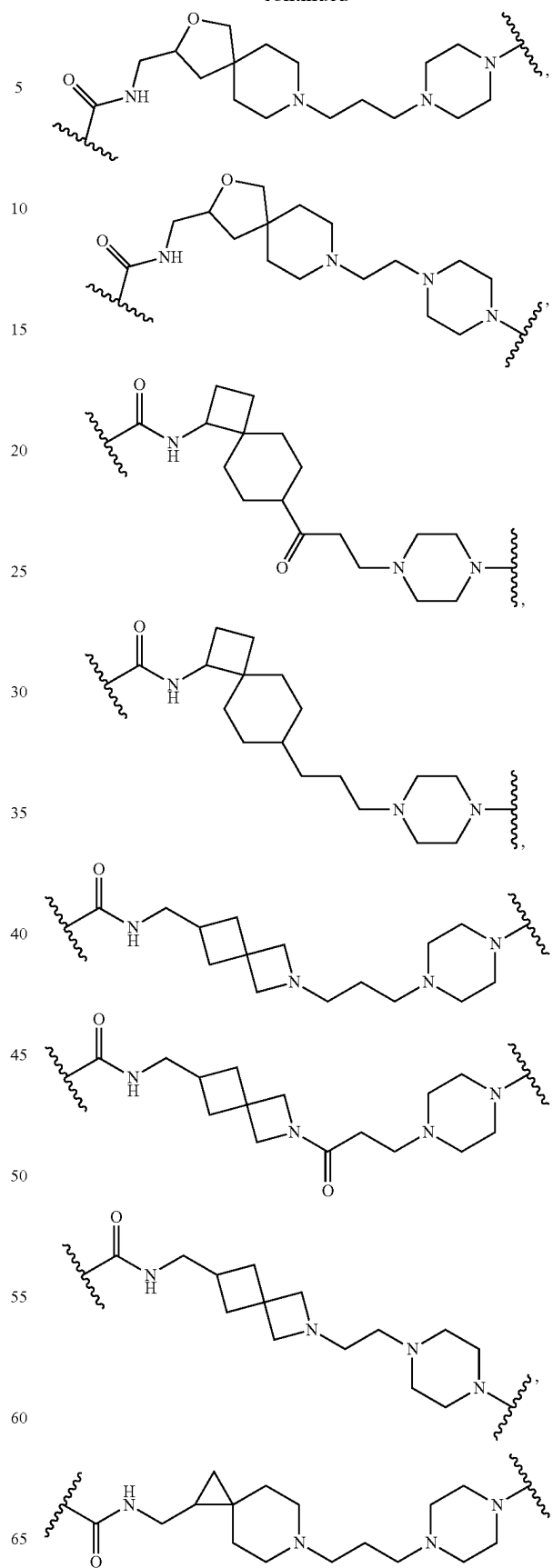


177

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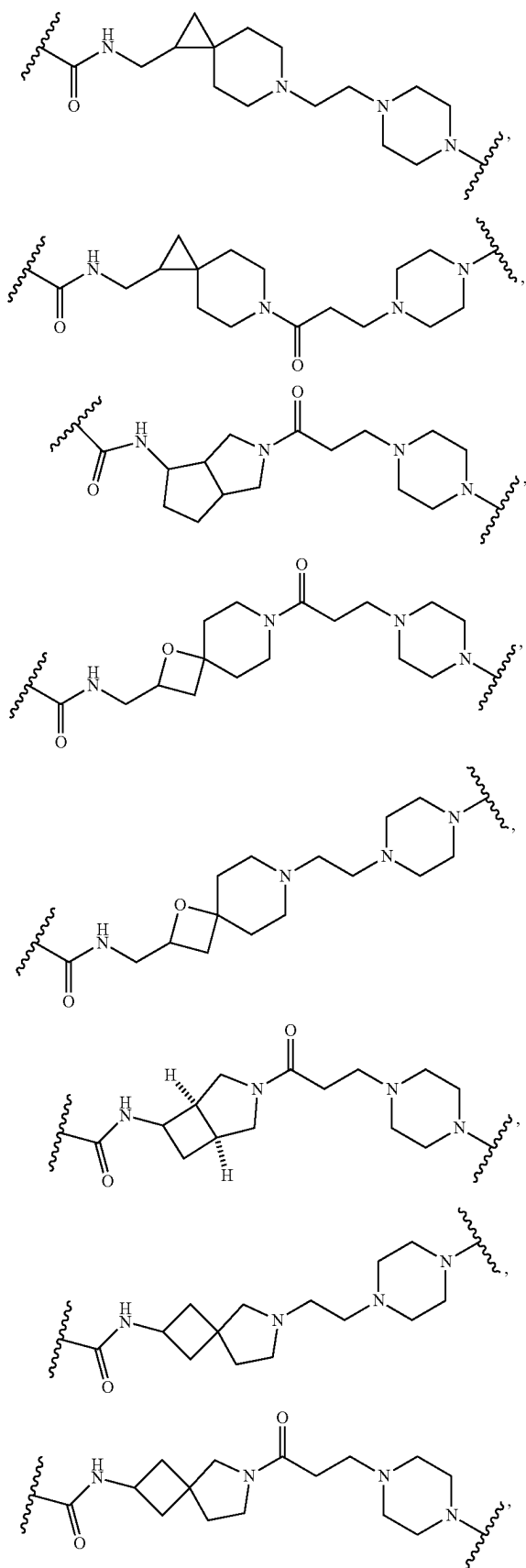
**178**

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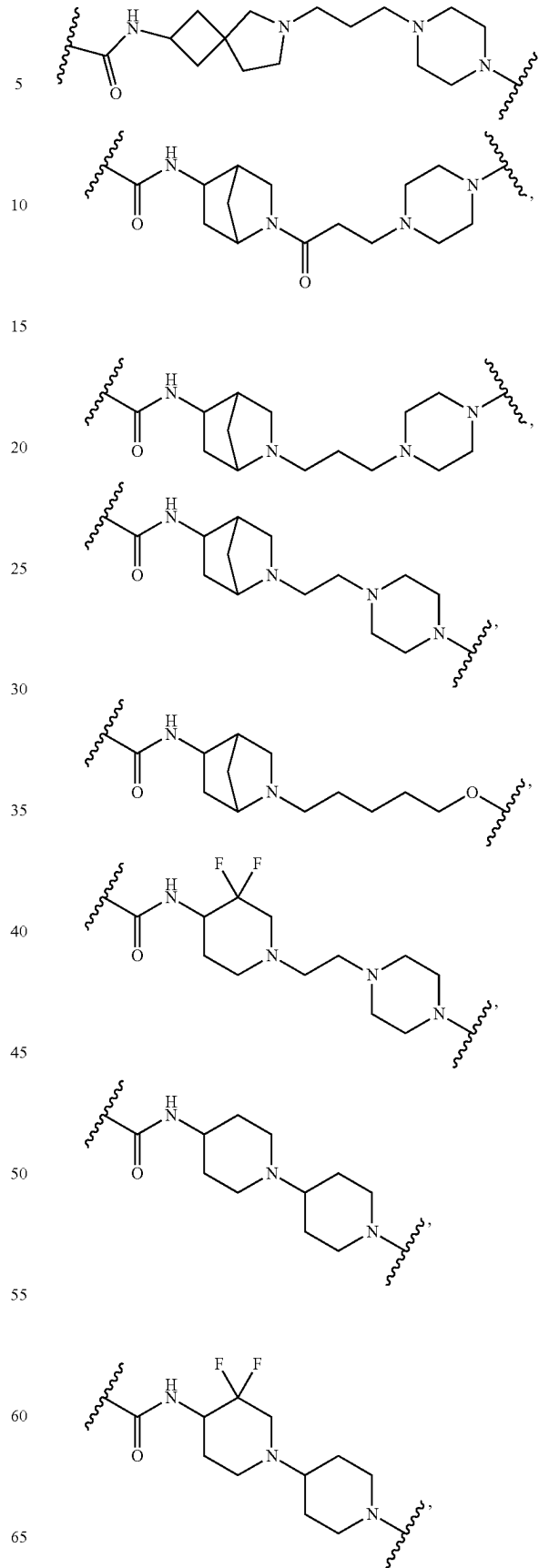


179

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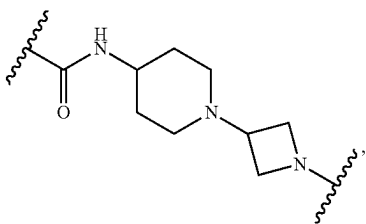
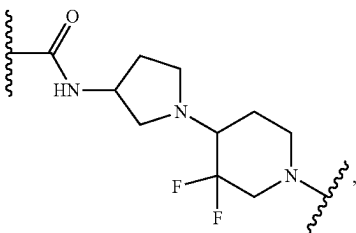
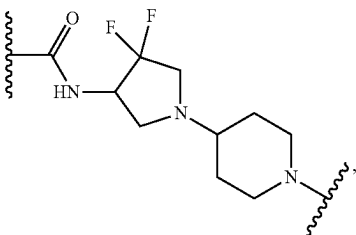
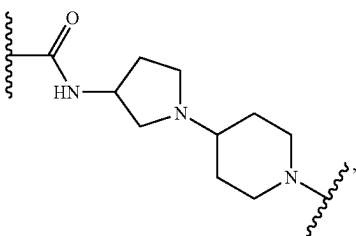
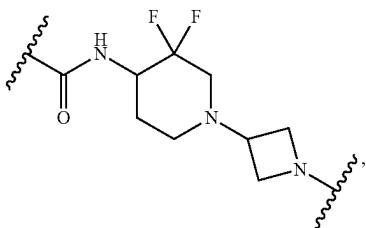
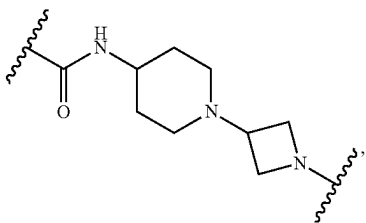
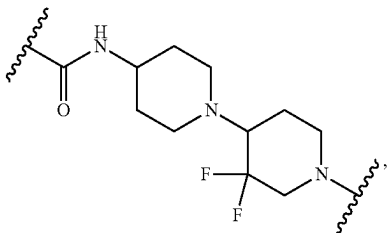
**180**

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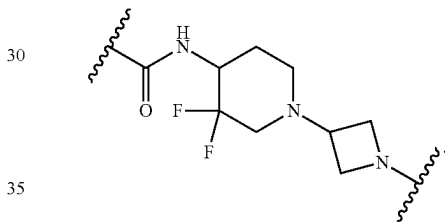
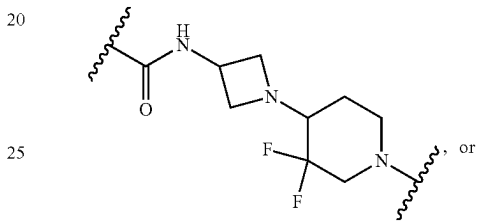
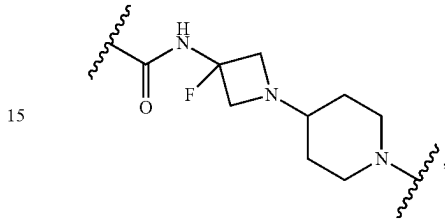
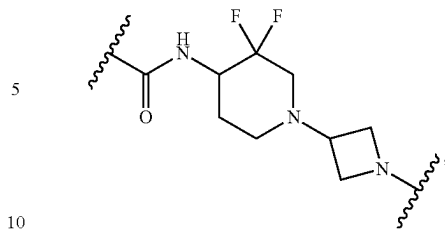
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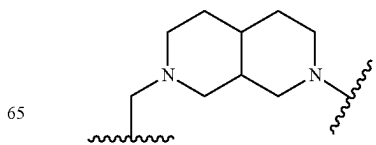
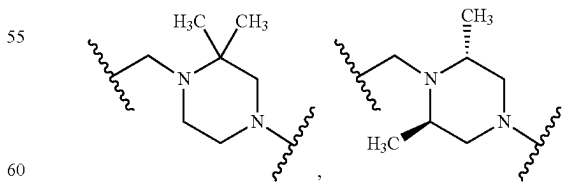
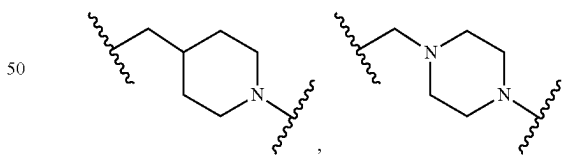
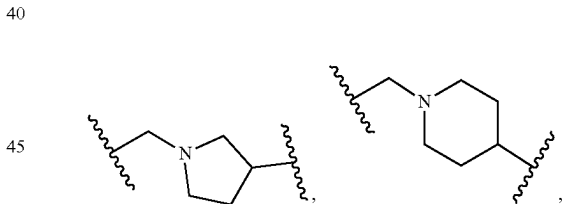


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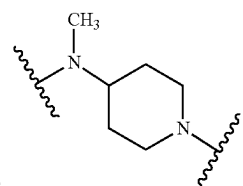
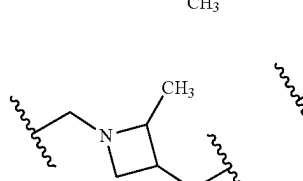
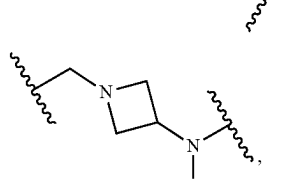
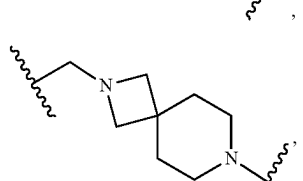
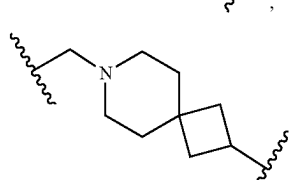
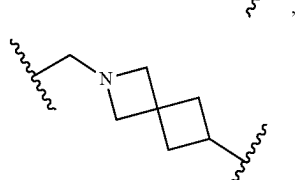
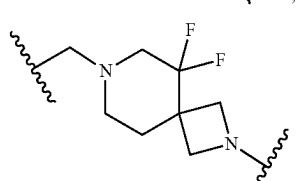
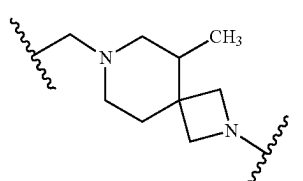
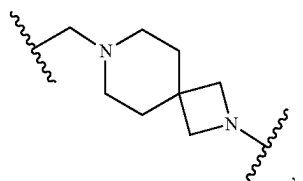
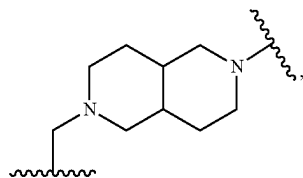


In some embodiments, the linker has the structure of:

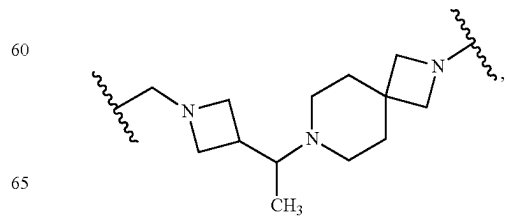
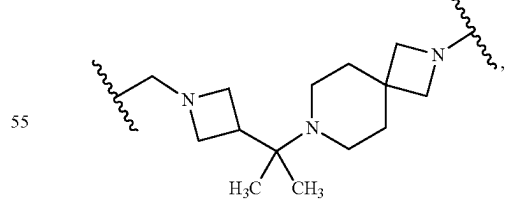
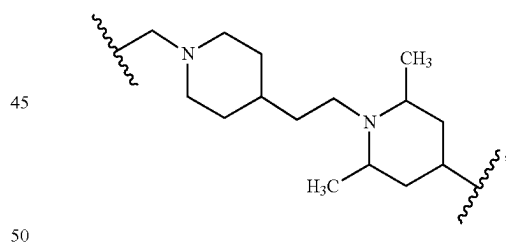
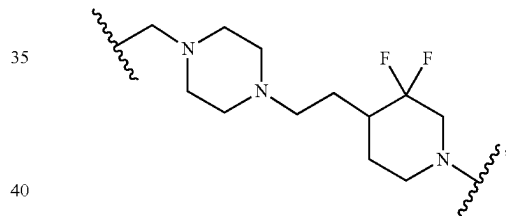
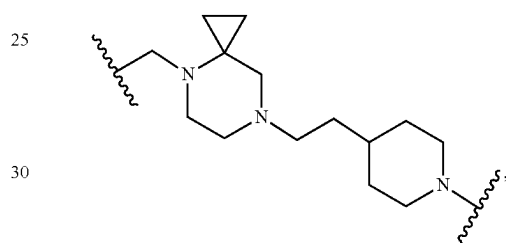
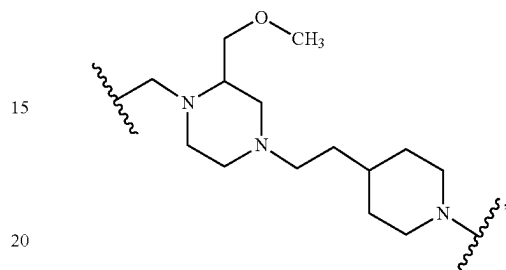
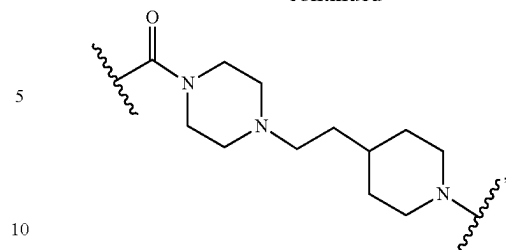


183

-continued

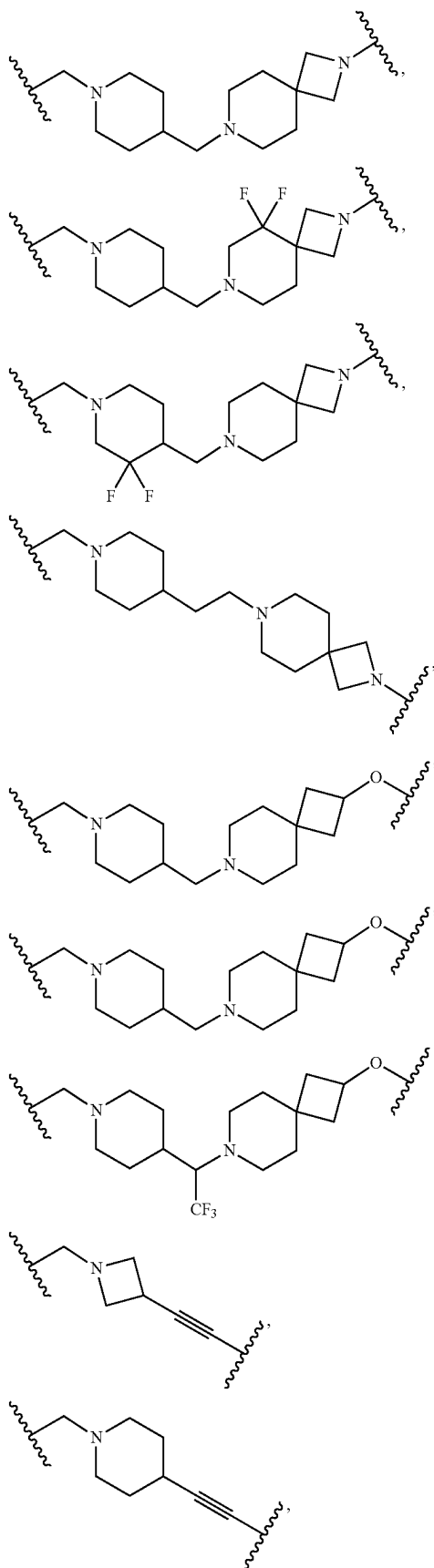
**184**

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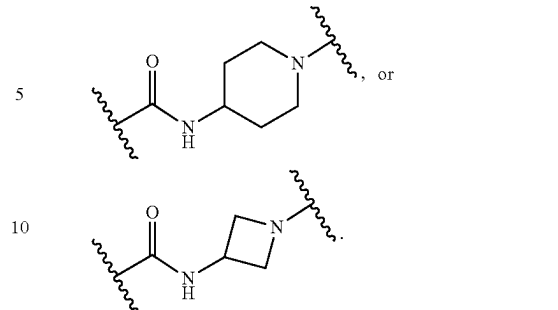


185

-continued

**186**

-continued



15 In some embodiments, the linker is a bond.

In some embodiments, the linker is optionally substituted C_3 - C_{10} carbocyclylene, optionally substituted C_{2-10} heterocyclylene, optionally substituted C_6 - C_{10} arylene, or optionally substituted C_2 - C_9 heteroarylene.

20 In some embodiments, the linker is optionally substituted C_3 - C_{10} carbocyclylene or optionally substituted C_{2-10} heterocyclylene. In some embodiments, the linker is optionally substituted C_6 - C_{10} arylene or optionally substituted C_2 - C_9 heteroarylene.

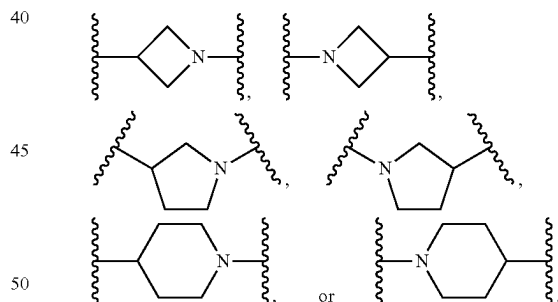
25 In some embodiments, the linker is optionally substituted C_{2-10} heterocyclylene.

In some embodiments, the C_2 - C_9 heterocyclylene is monocyclic. In some embodiments, the C_2 - C_9 heterocyclylene is polycyclic.

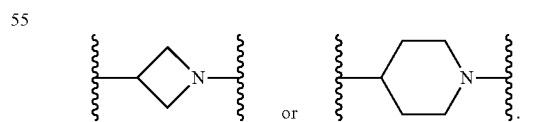
30 In some embodiments, the C_2 - C_9 heterocyclylene is bicyclic.

In some embodiments, the C_2 - C_9 heterocyclylene is bridged. In some embodiments, the C_2 - C_9 heterocyclylene is fused. In some embodiments, the C_2 - C_9 heterocyclylene is spirocyclic.

In some embodiments, the linker has the structure of



In some embodiments, the linker has the structure of



60 In some embodiments, the compound has the structure of any one of compounds D1-D38 in Table 2A, or a pharmaceutically acceptable salt thereof. In some embodiments, the compound has the structure of any one of compounds D1-D33 in Table 2A, or a pharmaceutically acceptable salt thereof. In some embodiments, the compound has the structure of any one of compounds D34-D38 in Table 2A, or a

187

pharmaceutically acceptable salt thereof. In some embodiments, the compound has the structure of any one of compounds D39-D302 in Table 2B, or a pharmaceutically acceptable salt thereof. In some embodiments, the compound has the structure of any one of compounds D303-D375 in Table 2C, or a pharmaceutically acceptable salt thereof.

In some embodiments, the compound has the structure of any one of compounds D9, D22, D25, D28, or D29 in Table 2A, or a pharmaceutically acceptable salt thereof. In some embodiments, the compound has the structure of any one of compounds D39-D49, D51, D52, D55, D56, D58, D58, D63, D64, D68, D69, D71-D73, D76, D78, D80-89, D91, D93, D95-107, D109, D110, D112-119, D121, D123-D1256, D127, D128, D130-D136, D138-D143, D145-D147, D149, D151, D152, D154-D166, D169-D174, D178-D183, D195-D201, D222, D224, D227, D231, D246, D251, D253-

188

256, D283, or D299-302 in Table 2B, or a pharmaceutically acceptable salt thereof. In some embodiments, the compound has the structure of any one of compounds D303, D304, D306-D312, D314, D315, D317-D327, D329, D330, D332-336, D338-D342, D344-D348, D350, OR D353-D373, or a pharmaceutically acceptable salt thereof.

In an aspect, the disclosure features compounds D1-D38 in Table 2A, or a pharmaceutically acceptable salt thereof.

In another aspect, the disclosure features compounds D1-D33 in Table 2A, or a pharmaceutically acceptable salt thereof.

In another aspect, the disclosure features compounds D39-D302 in Table 2B, or a pharmaceutically acceptable salt thereof.

In yet another aspect, the disclosure features compounds D303-D375 in Table 2C, or a pharmaceutically acceptable salt thereof.

TABLE 2A

Compounds D1-D38 of the Disclosure	
Compound No.	Structure
D1	
D2	

TABLE 2A-continued

D5

TABLE 2A-continued

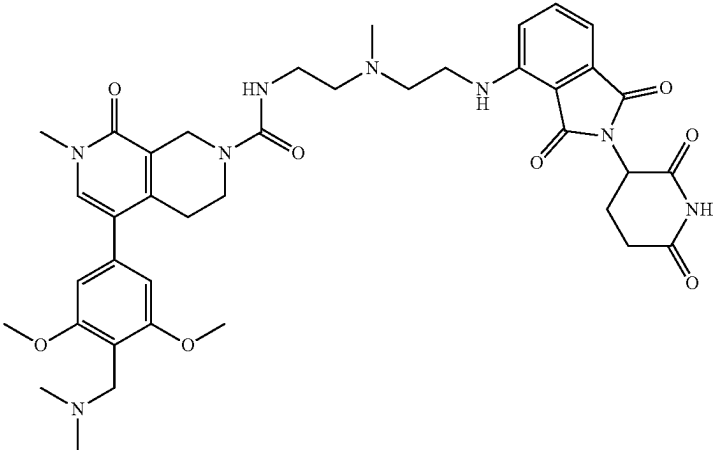
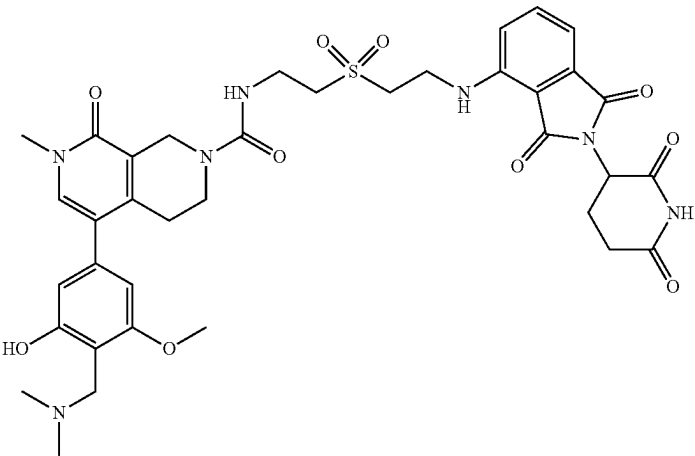
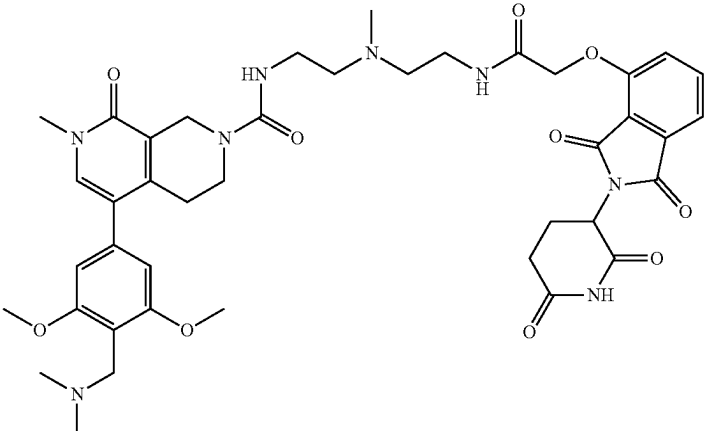
Compounds D1-D38 of the Disclosure	
Compound No.	Structure
D6	 Chemical structure of compound D6: A 6-methoxy-2-(dimethylamino)-3,4-dimethoxyphenyl group is attached to a pyridine ring. The pyridine ring is linked via a methylene bridge to a piperidine ring. The piperidine ring is connected to a carbonyl group, which is further linked to a dimethylamino group. This dimethylamino group is connected to a carbonyl group, which is further linked to a benzimidazole ring system. The benzimidazole ring system is connected to a piperidine ring, which is further connected to a carbonyl group.
D7	 Chemical structure of compound D7: A 6-methoxy-2-(dimethylamino)-3,4-dimethoxyphenyl group is attached to a pyridine ring. The pyridine ring is linked via a methylene bridge to a piperidine ring. The piperidine ring is connected to a carbonyl group, which is further linked to a dimethylamino group. This dimethylamino group is connected to a carbonyl group, which is further linked to a benzimidazole ring system. The benzimidazole ring system is connected to a piperidine ring, which is further connected to a carbonyl group.
D8	 Chemical structure of compound D8: A 6-methoxy-2-(dimethylamino)-3,4-dimethoxyphenyl group is attached to a pyridine ring. The pyridine ring is linked via a methylene bridge to a piperidine ring. The piperidine ring is connected to a carbonyl group, which is further linked to a dimethylamino group. This dimethylamino group is connected to a carbonyl group, which is further linked to a benzimidazole ring system. The benzimidazole ring system is connected to a piperidine ring, which is further connected to a carbonyl group.

TABLE 2A-continued

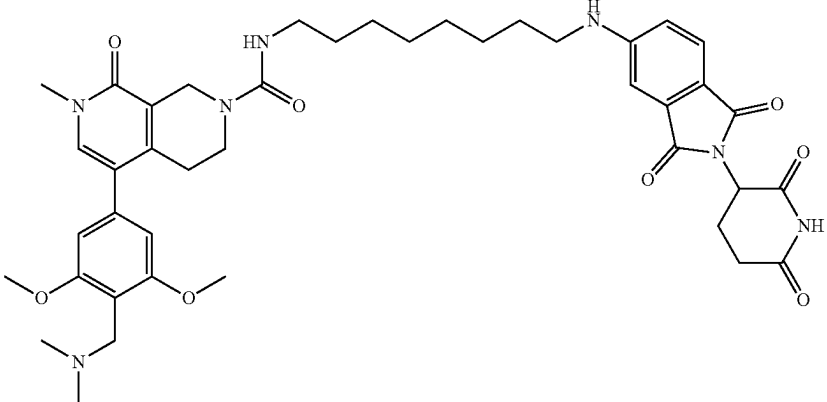
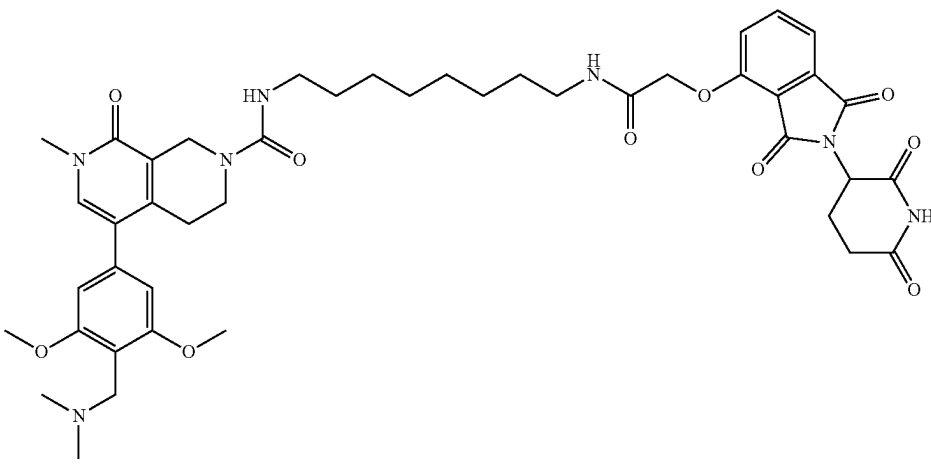
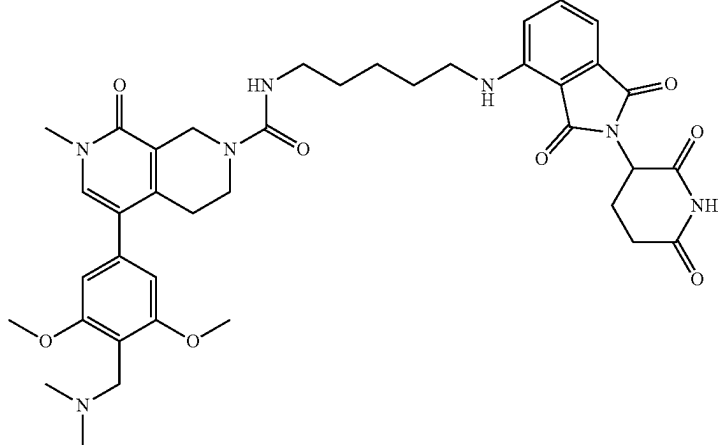
Compounds D1-D38 of the Disclosure	
Compound No.	Structure
D9	 Chemical structure of compound D9: A 6-methoxy-2-(dimethylamino)phenyl group is attached to a 4-methyl-2,3,4,5-tetrahydroquinolin-6(1H)-one. This is linked via a 10-carbon aliphatic chain to a 4-((2,3,4,5-tetrahydro-1H-benzotriazin-4-ylidene)amino)benzamide derivative.
D10	 Chemical structure of compound D10: A 6-methoxy-2-(dimethylamino)phenyl group is attached to a 4-methyl-2,3,4,5-tetrahydroquinolin-6(1H)-one. This is linked via a 10-carbon aliphatic chain to a 4-((2,3,4,5-tetrahydro-1H-benzotriazin-4-ylidene)amino)benzamide derivative.
D11	 Chemical structure of compound D11: A 6-methoxy-2-(dimethylamino)phenyl group is attached to a 4-methyl-2,3,4,5-tetrahydroquinolin-6(1H)-one. This is linked via a 10-carbon aliphatic chain to a 4-((2,3,4,5-tetrahydro-1H-benzotriazin-4-ylidene)amino)benzamide derivative.

TABLE 2A-continued

Compounds D1-D38 of the Disclosure	
Compound No.	Structure
D12	
D13	
D14	

TABLE 2A-continued

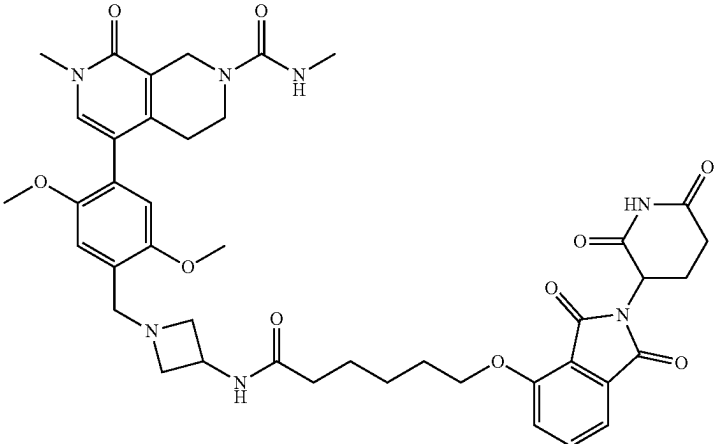
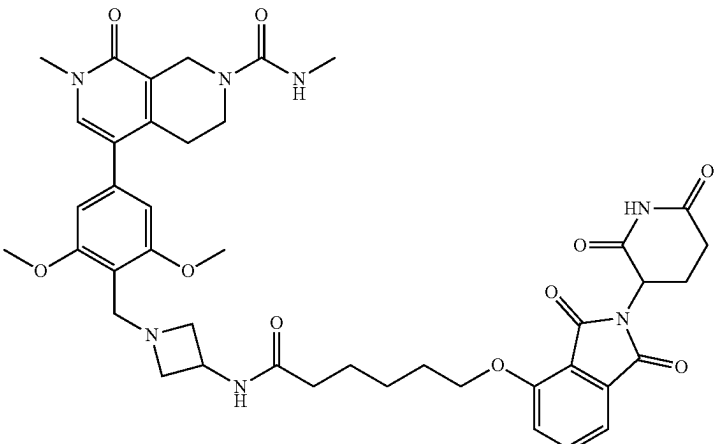
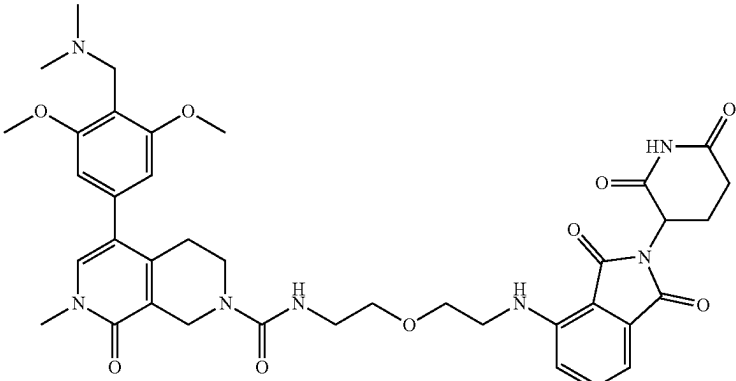
Compounds D1-D38 of the Disclosure	
Compound No.	Structure
D15	 Chemical structure of compound D15: A complex molecule featuring a 6,7-dimethoxy-2-methyl-1,2,3,4-tetrahydroquinolin-4(1H)-one moiety connected via a methylene bridge to a 4-methyl-1,3-dioxolane ring. This ring is further connected via an amide linkage to a 7-((6-methoxy-2-methyl-1,2,3,4-tetrahydroquinolin-4(1H)-one)methoxy)heptan-1-one moiety, which is then connected via an ether linkage to a 1-((6-methoxy-2-methyl-1,2,3,4-tetrahydroquinolin-4(1H)-one)methoxy)pyrrolidine-2-one moiety.
D16	 Chemical structure of compound D16: A complex molecule featuring a 6,7-dimethoxy-2-methyl-1,2,3,4-tetrahydroquinolin-4(1H)-one moiety connected via a methylene bridge to a 4-methyl-1,3-dioxolane ring. This ring is further connected via an amide linkage to a 7-((6-methoxy-2-methyl-1,2,3,4-tetrahydroquinolin-4(1H)-one)methoxy)heptan-1-one moiety, which is then connected via an ether linkage to a 1-((6-methoxy-2-methyl-1,2,3,4-tetrahydroquinolin-4(1H)-one)methoxy)pyrrolidine-2-one moiety.
D17	 Chemical structure of compound D17: A complex molecule featuring a 6,7-dimethoxy-2-methyl-1,2,3,4-tetrahydroquinolin-4(1H)-one moiety connected via a methylene bridge to a 4-methyl-1,3-dioxolane ring. This ring is further connected via an amide linkage to a 7-((6-methoxy-2-methyl-1,2,3,4-tetrahydroquinolin-4(1H)-one)methoxy)heptan-1-one moiety, which is then connected via an ether linkage to a 1-((6-methoxy-2-methyl-1,2,3,4-tetrahydroquinolin-4(1H)-one)methoxy)pyrrolidine-2-one moiety.

TABLE 2A-continued

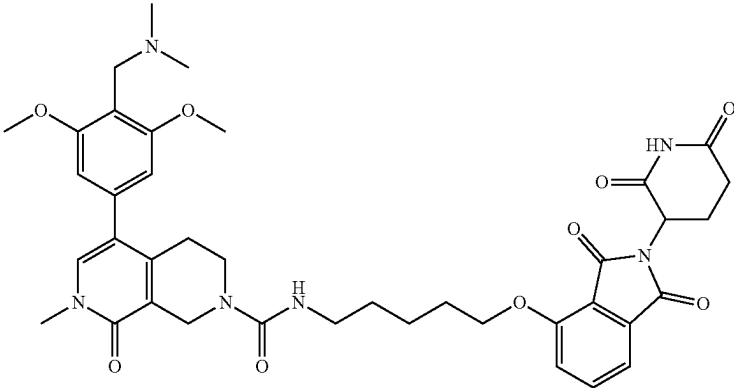
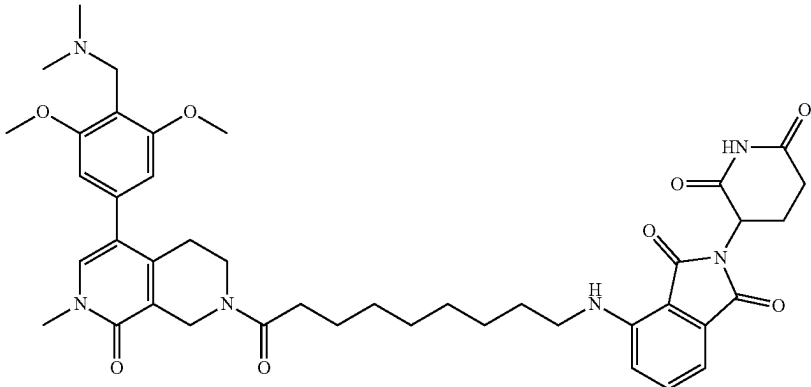
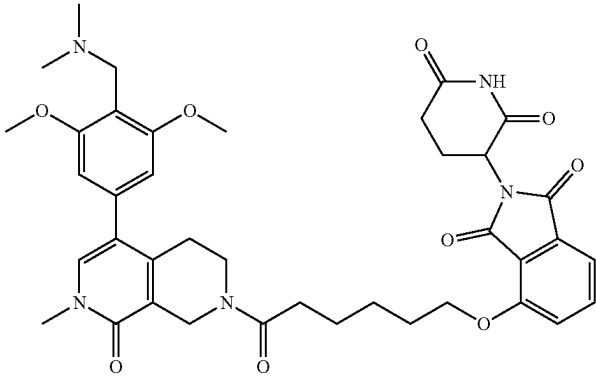
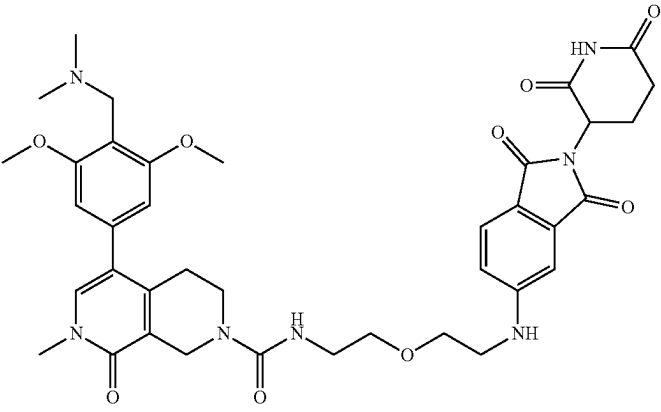
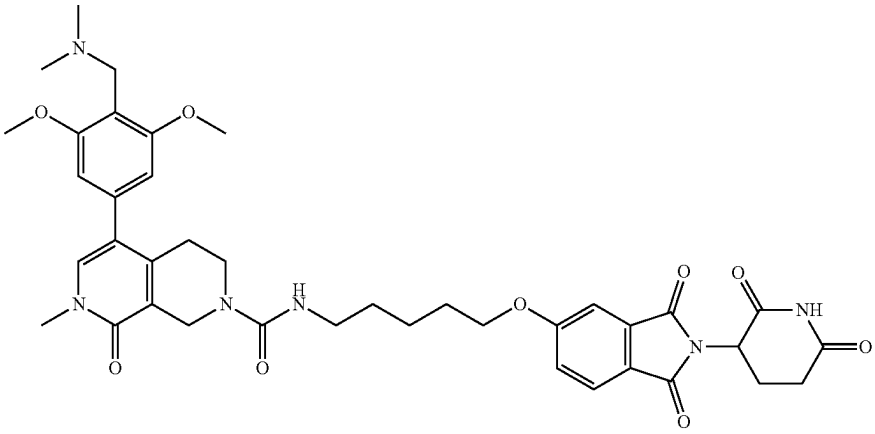
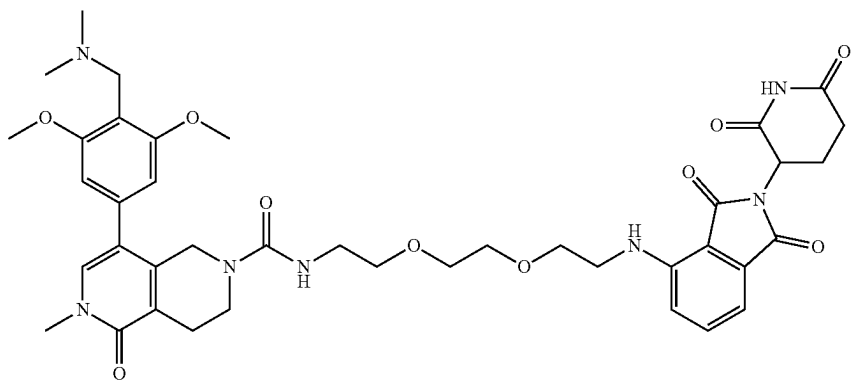
Compounds D1-D38 of the Disclosure	
Compound No.	Structure
D18	 Chemical structure of compound D18: A 6,7-dimethoxy-2-(dimethylaminomethyl)phenyl group is attached to the 4-position of a 1,2,3,4-tetrahydropyridine ring. The nitrogen at position 1 of the tetrahydropyridine is methylated. The nitrogen at position 3 is part of an amide linkage to a 6-heptoxy-1H-indolizino[1,2-b]pyridine-2,3-dione system. The heptyloxy chain is connected to the 6-position of the indolizino system via an ether linkage.
D19	 Chemical structure of compound D19: A 6,7-dimethoxy-2-(dimethylaminomethyl)phenyl group is attached to the 4-position of a 1,2,3,4-tetrahydropyridine ring. The nitrogen at position 1 of the tetrahydropyridine is methylated. The nitrogen at position 3 is part of an amide linkage to a 1H-indolizino[1,2-b]pyridine-2,3-dione system. The heptyl chain is connected to the 6-position of the indolizino system via an amide linkage.
D20	 Chemical structure of compound D20: A 6,7-dimethoxy-2-(dimethylaminomethyl)phenyl group is attached to the 4-position of a 1,2,3,4-tetrahydropyridine ring. The nitrogen at position 1 of the tetrahydropyridine is methylated. The nitrogen at position 3 is part of an amide linkage to a 1H-indolizino[1,2-b]pyridine-2,3-dione system. The heptyloxy chain is connected to the 6-position of the indolizino system via an ether linkage.

TABLE 2A-continued

Compounds D1-D38 of the Disclosure	
Compound No.	Structure
D21	 Chemical structure of compound D21: A 6,7,8-trimethyl-1,2,3,4-tetrahydroquinoline-2-carboxamide derivative. The quinoline ring is substituted at the 6-position with a 3,4,5-trimethoxybenzyl group. The amide group is linked via a 3-ethoxypropyl chain to a 1,2,3,4-tetrahydroquinolin-2(1H)-one derivative, which is further substituted with a 3,4,5-trimethoxybenzyl group.
D22	 Chemical structure of compound D22: A 6,7,8-trimethyl-1,2,3,4-tetrahydroquinoline-2-carboxamide derivative. The quinoline ring is substituted at the 6-position with a 3,4,5-trimethoxybenzyl group. The amide group is linked via a 6-ethoxyhexyl chain to a 1,2,3,4-tetrahydroquinolin-2(1H)-one derivative, which is further substituted with a 3,4,5-trimethoxybenzyl group.
D23	 Chemical structure of compound D23: A 6,7,8-trimethyl-1,2,3,4-tetrahydroquinoline-2-carboxamide derivative. The quinoline ring is substituted at the 6-position with a 3,4,5-trimethoxybenzyl group. The amide group is linked via a 3-ethoxypropyl chain to a 1,2,3,4-tetrahydroquinolin-2(1H)-one derivative, which is further substituted with a 3,4,5-trimethoxybenzyl group.

Compounds D1-D38 of the Disclosure

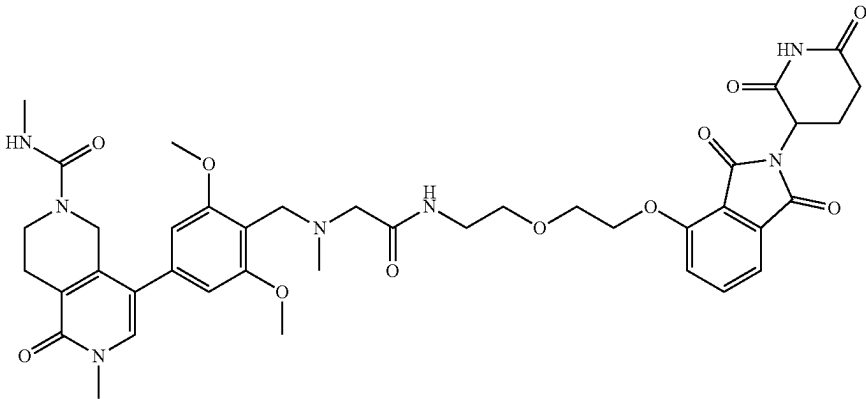
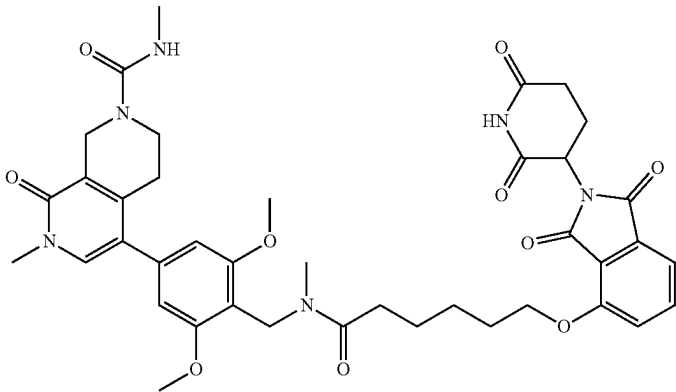
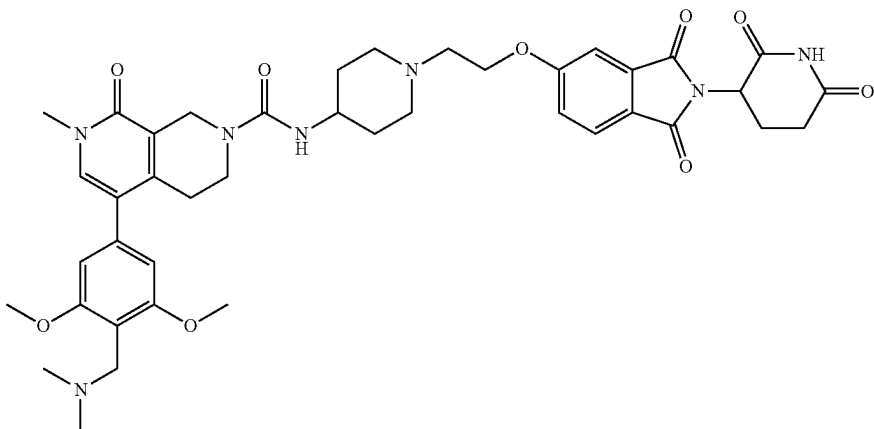
Compound No.	Structure
D24	
D25	
D26	

TABLE 2A-continued

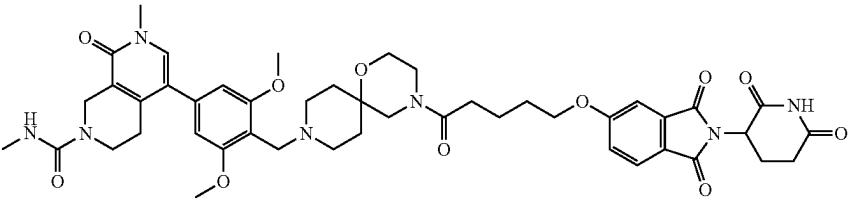
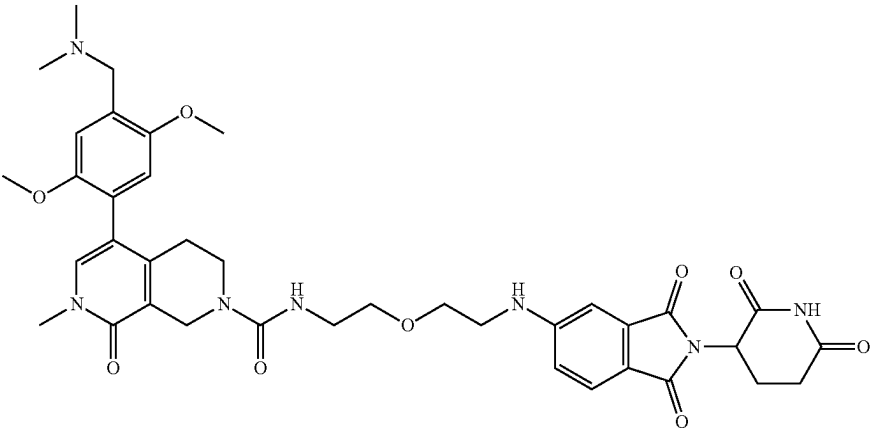
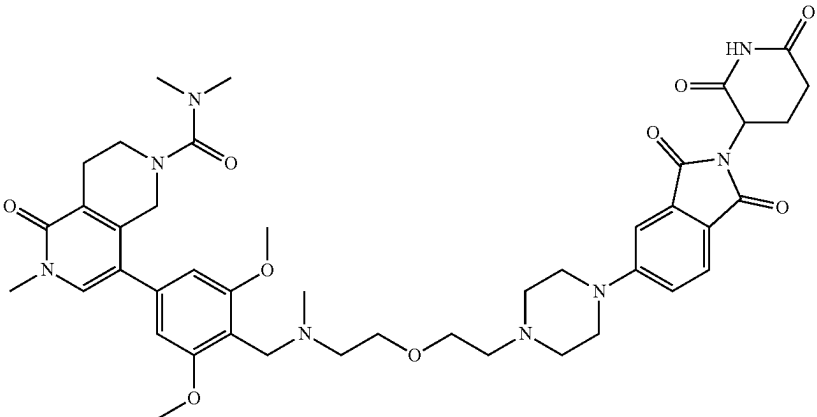
Compounds D1-D38 of the Disclosure	
Compound No.	Structure
D31	 Chemical structure of compound D31: A complex molecule featuring a central benzene ring substituted with a 4-methoxyphenyl group, a 2-methoxyphenyl group, and a 4-methoxyphenyl group. The central benzene ring is also substituted with a 4-methoxyphenyl group. The structure includes a piperidine ring, a morpholine ring, and a succinimide ring, connected by various linkers including a carbonyl group and an ether group.
D32	 Chemical structure of compound D32: A complex molecule featuring a central benzene ring substituted with a 4-methoxyphenyl group, a 2-methoxyphenyl group, and a 4-methoxyphenyl group. The central benzene ring is also substituted with a 4-methoxyphenyl group. The structure includes a piperidine ring, a morpholine ring, and a succinimide ring, connected by various linkers including a carbonyl group and an ether group.
D33	 Chemical structure of compound D33: A complex molecule featuring a central benzene ring substituted with a 4-methoxyphenyl group, a 2-methoxyphenyl group, and a 4-methoxyphenyl group. The central benzene ring is also substituted with a 4-methoxyphenyl group. The structure includes a piperidine ring, a morpholine ring, and a succinimide ring, connected by various linkers including a carbonyl group and an ether group.

TABLE 2A-continued

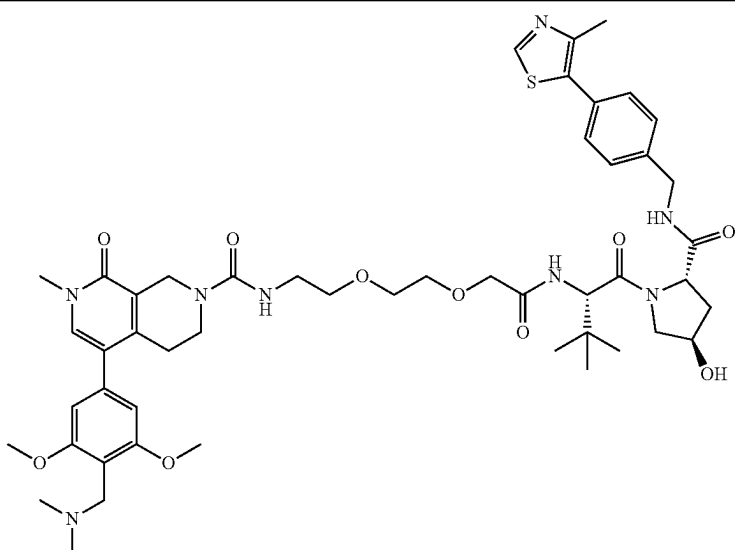
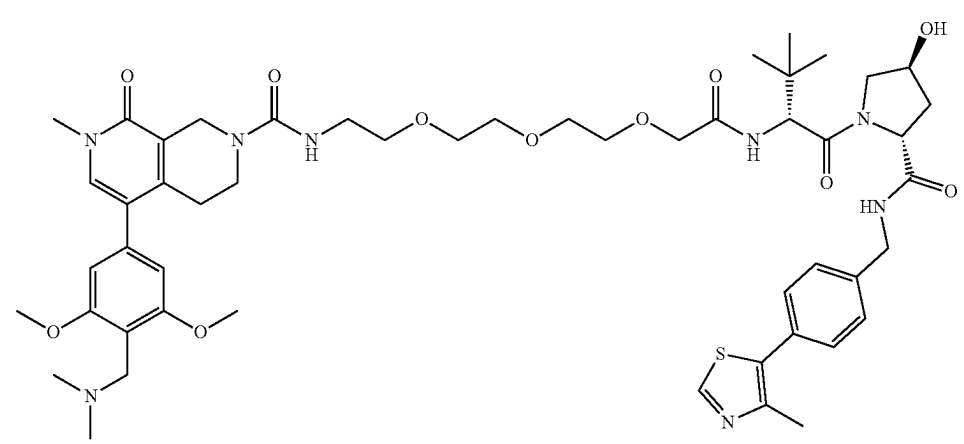
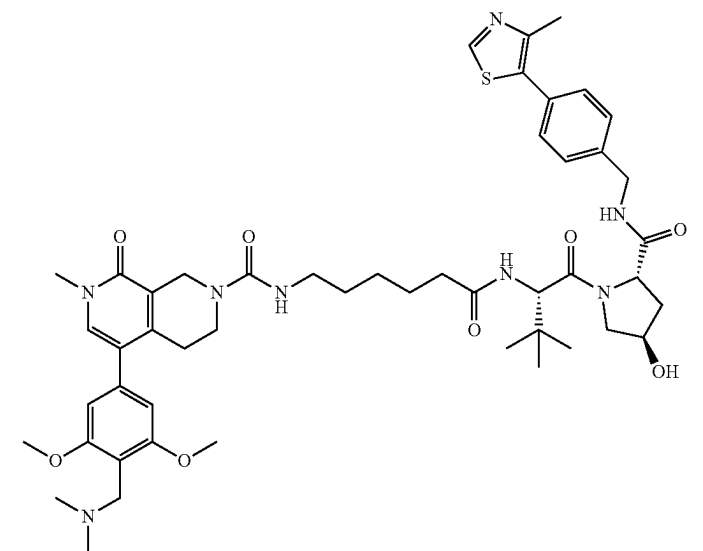
Compounds D1-D38 of the Disclosure	
Compound No.	Structure
D34	
D35	
D36	

TABLE 2A-continued

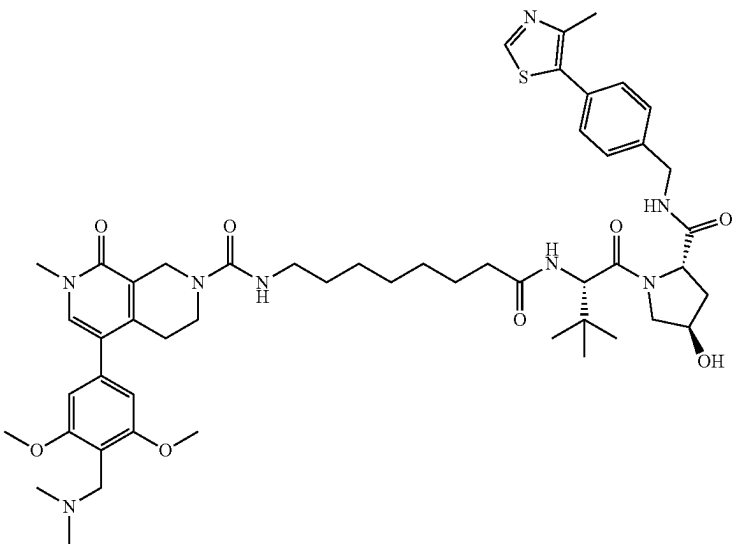
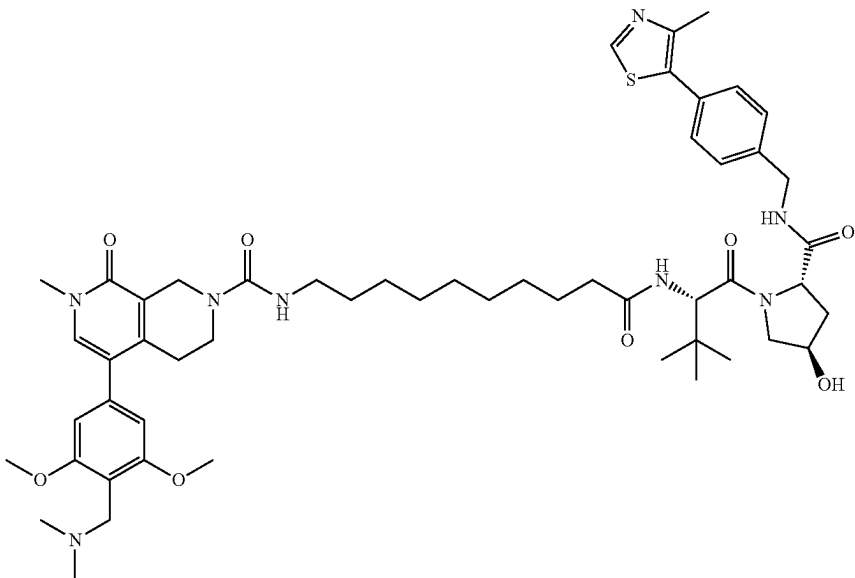
Compounds D1-D38 of the Disclosure	
Compound No.	Structure
D37	 Chemical structure of Compound D37. It features a 6-methoxy-2-(dimethylamino)pyridine-3-carboxamide moiety linked via a long alkyl chain (10 carbons) to a 4-(4-methyl-1,3,4-thiazol-2-yl)phenyl group. The 4-methyl-1,3,4-thiazol-2-yl group is further linked via a chiral center (C1) to a 4-hydroxy-2-methyl-5-oxo-1,3-dihydro-1H-imidazole-3-carboxamide moiety.
D38	 Chemical structure of Compound D38. It features a 6-methoxy-2-(dimethylamino)pyridine-3-carboxamide moiety linked via a long alkyl chain (10 carbons) to a 4-(4-methyl-1,3,4-thiazol-2-yl)phenyl group. The 4-methyl-1,3,4-thiazol-2-yl group is further linked via a chiral center (C1) to a 4-hydroxy-2-methyl-5-oxo-1,3-dihydro-1H-imidazole-3-carboxamide moiety.

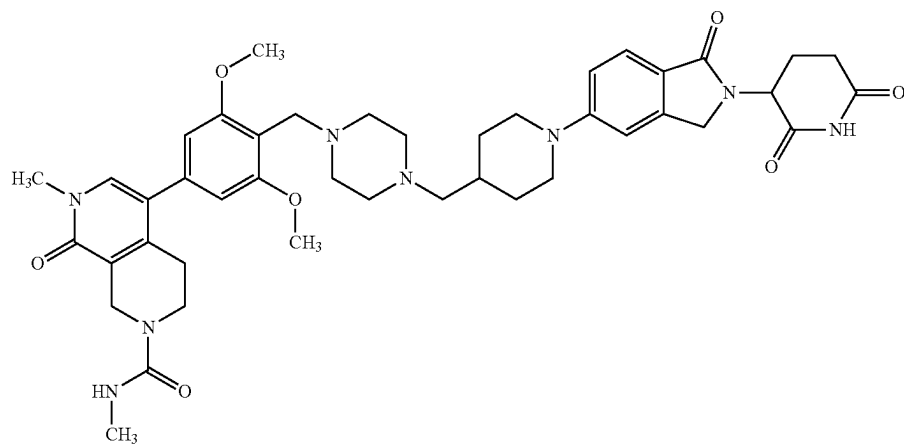
TABLE 2B

Compounds D39-D302 of the Disclosure

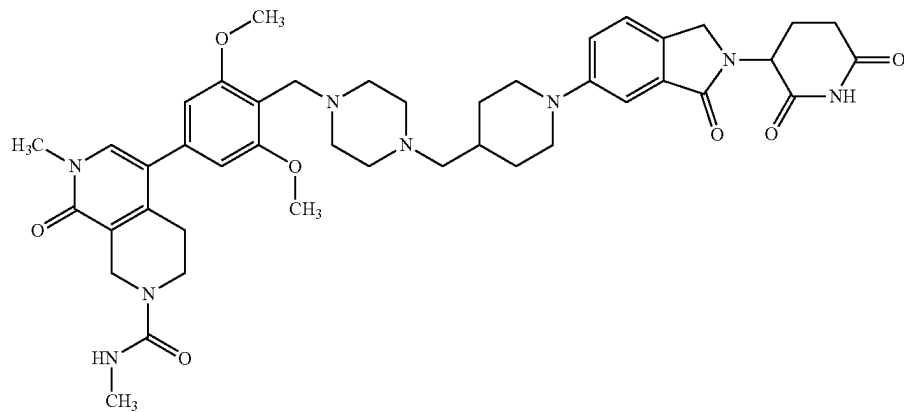
Compound
No.

Structure

D39



D40



D41

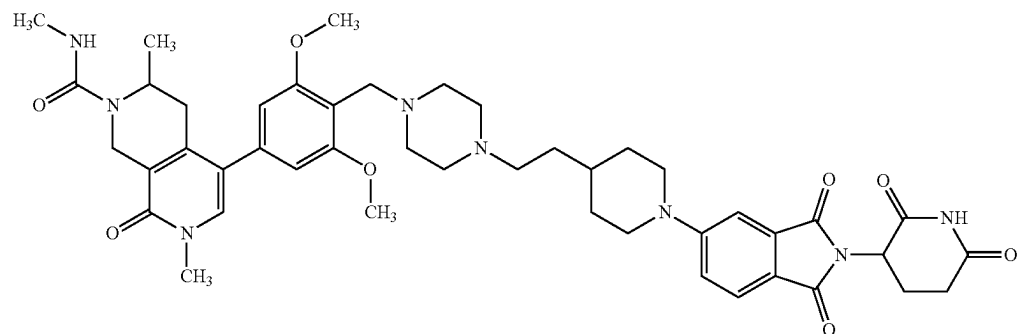


TABLE 2B-continued

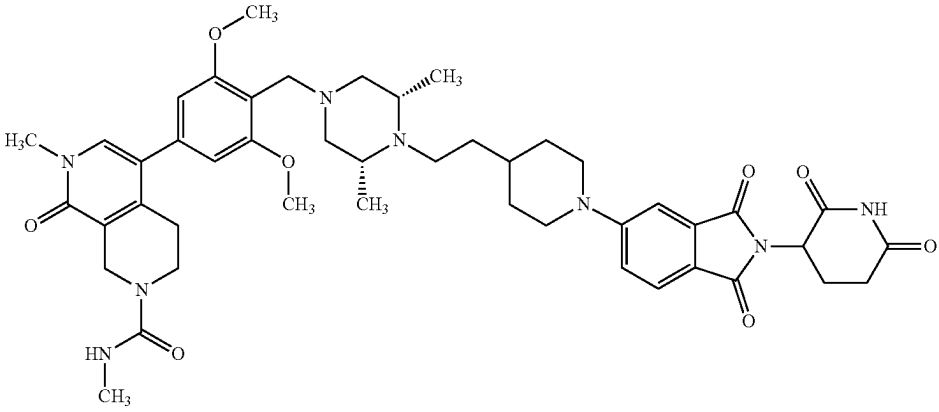
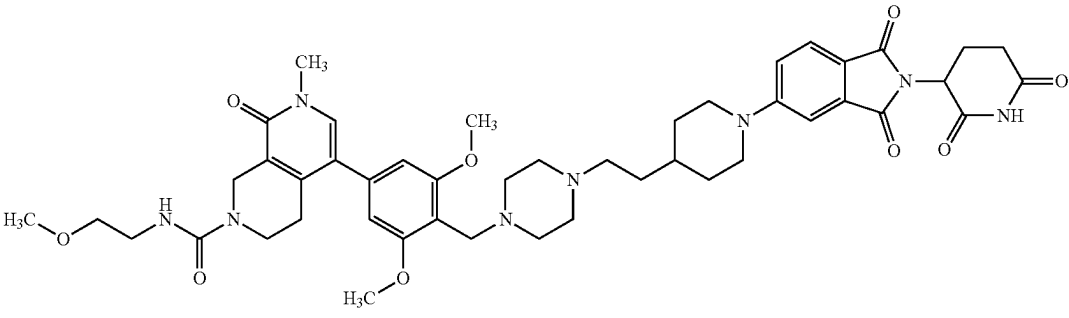
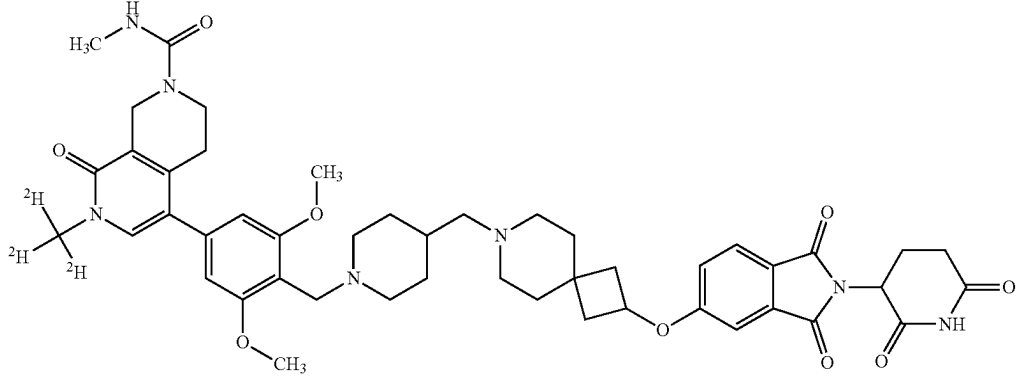
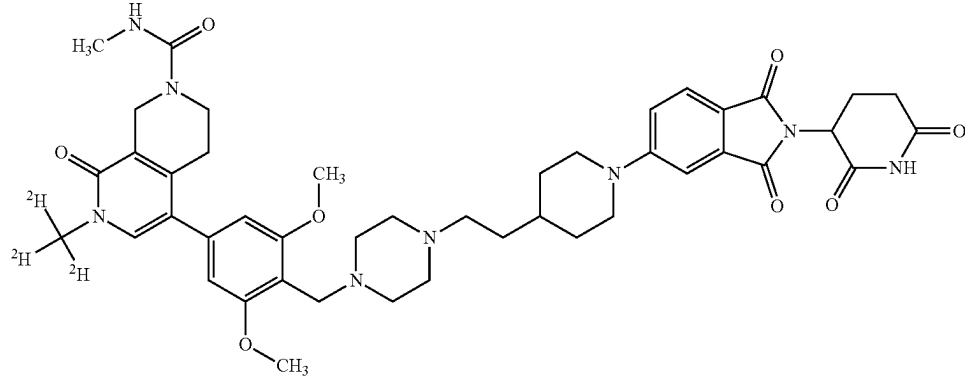
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D42	
D43	
D44	
D45	

TABLE 2B-continued

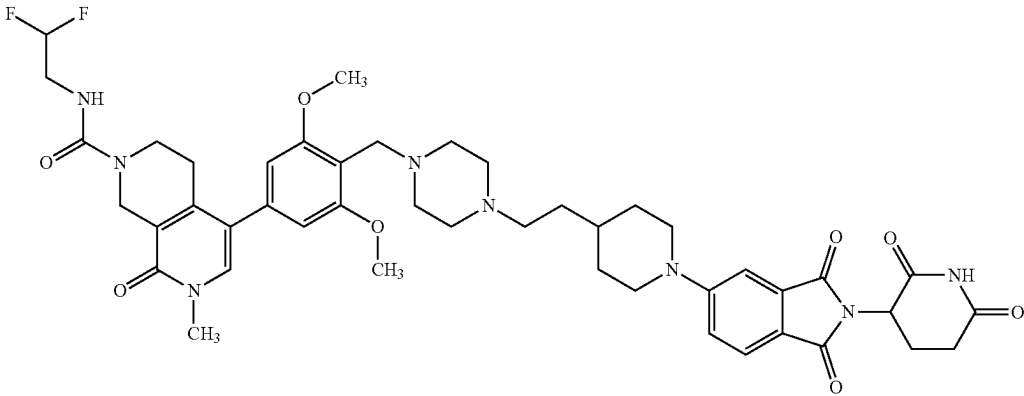
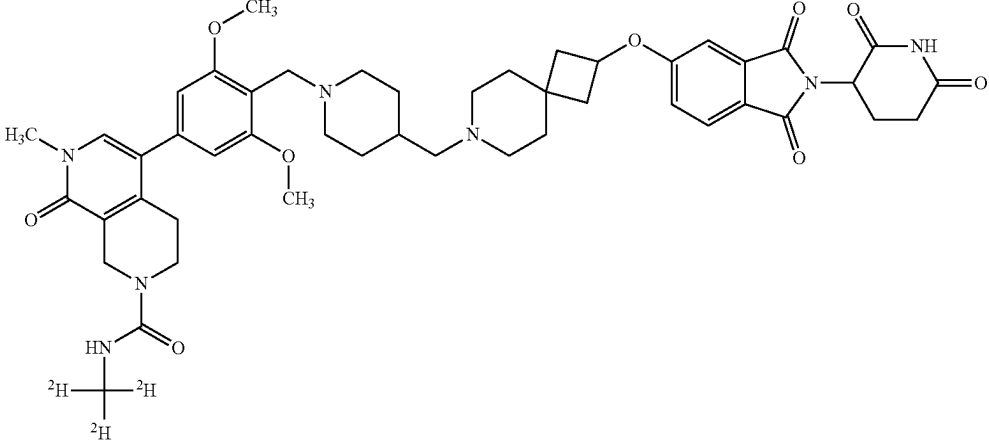
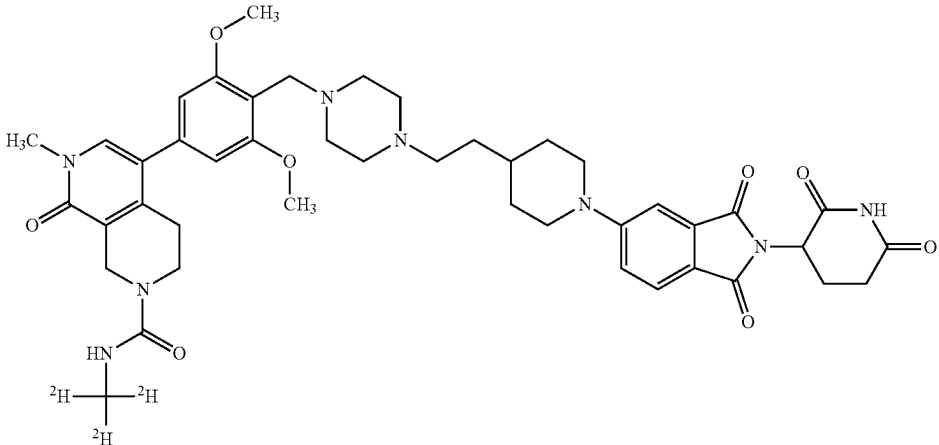
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D46	
D47	
D48	

TABLE 2B-continued

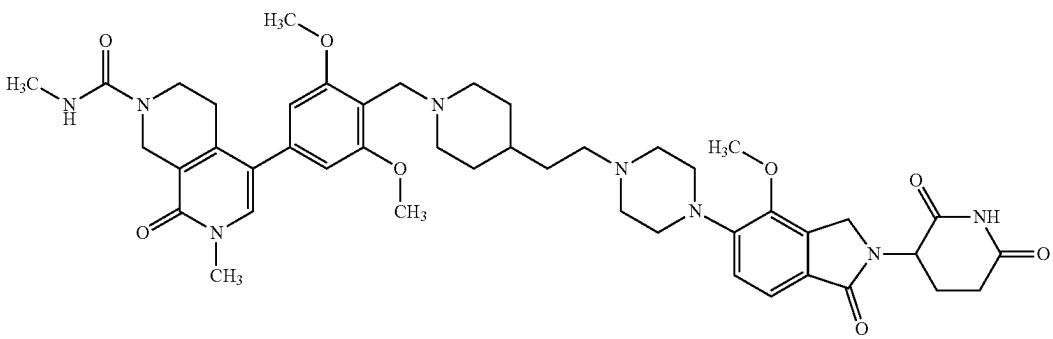
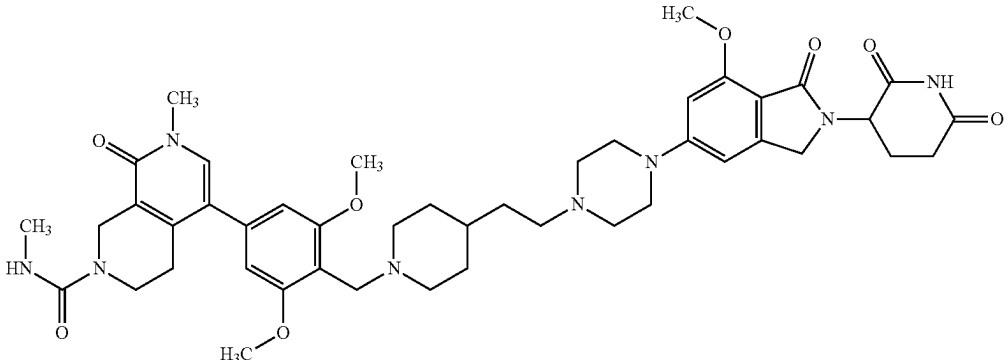
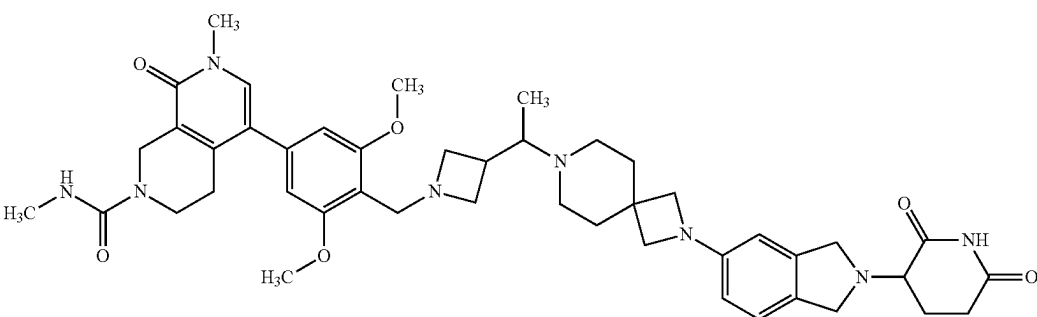
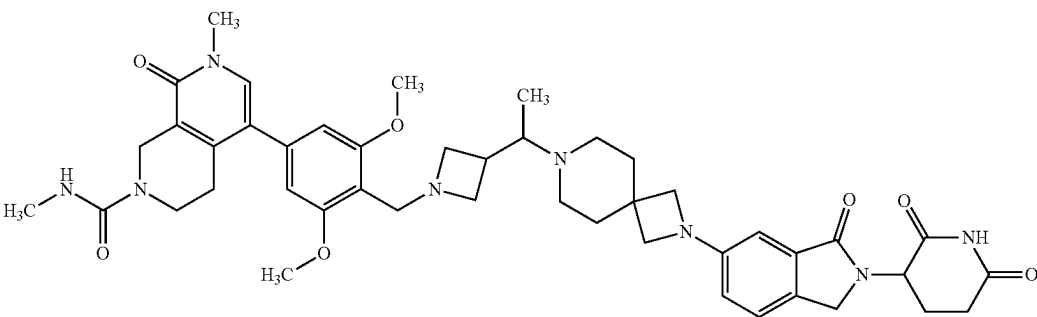
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D49	
D50	
D51	
D52	

TABLE 2B-continued

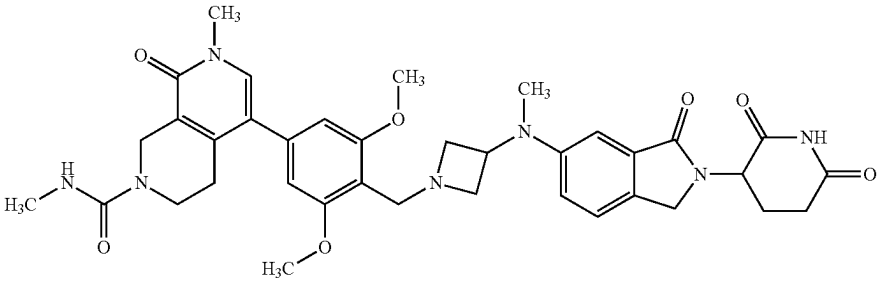
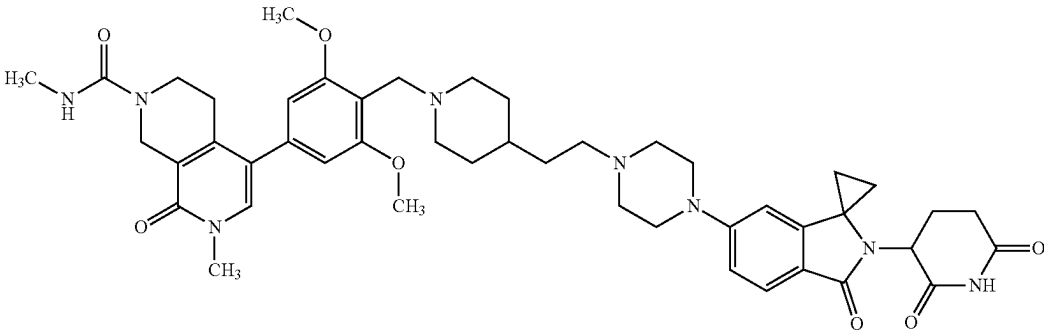
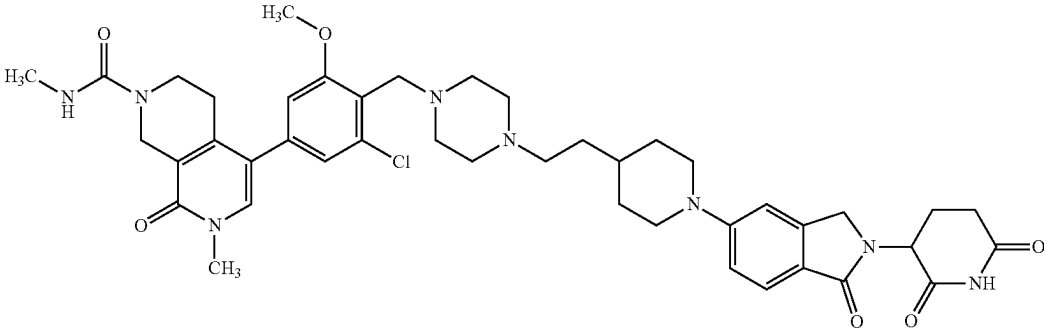
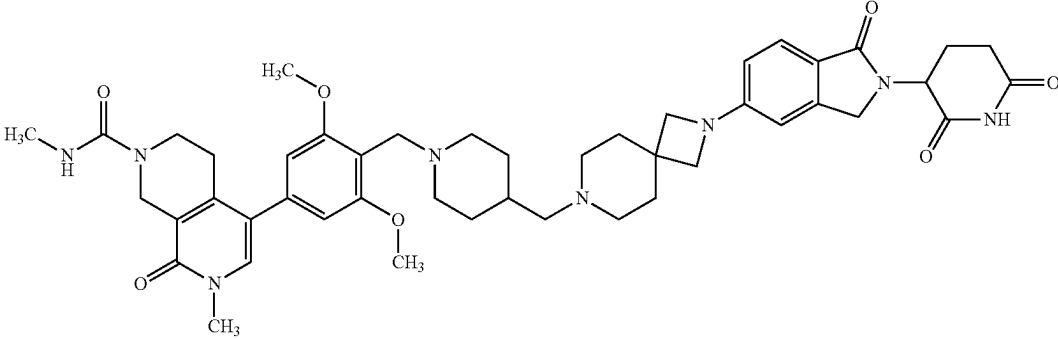
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D53	
D54	
D55	
D56	

TABLE 2B-continued

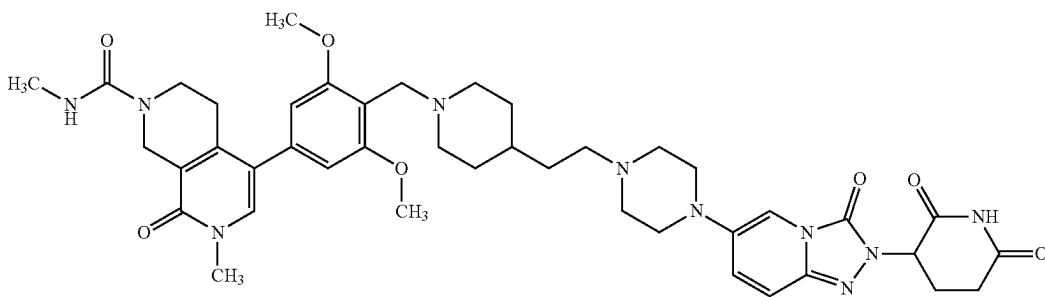
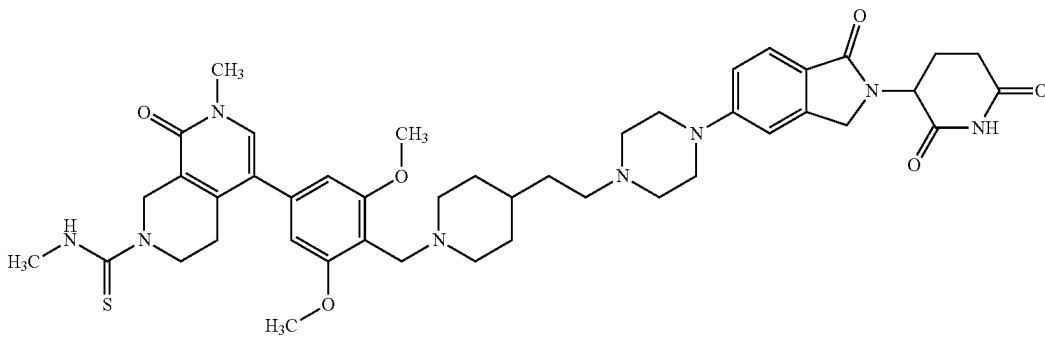
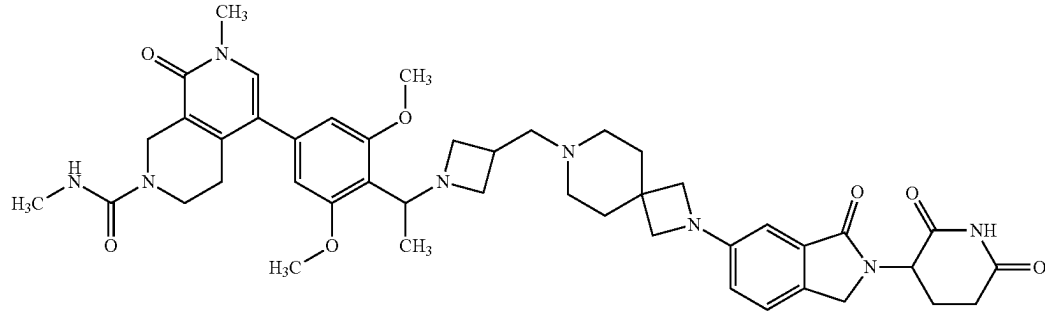
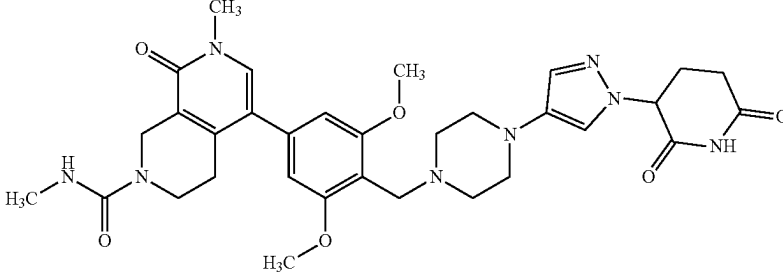
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D57	
D58	
D59	
D60	

TABLE 2B-continued

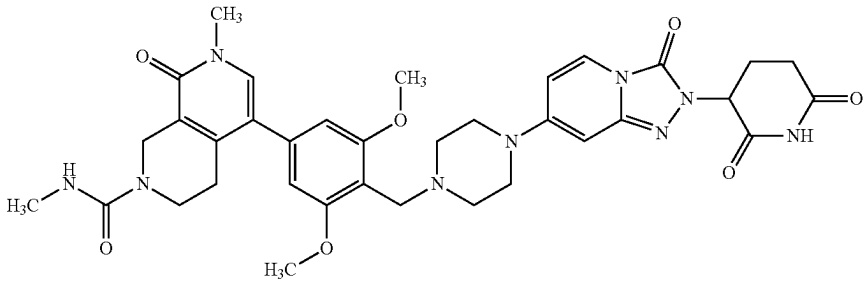
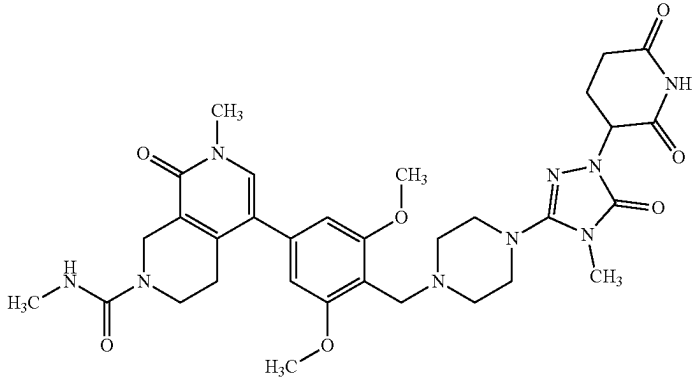
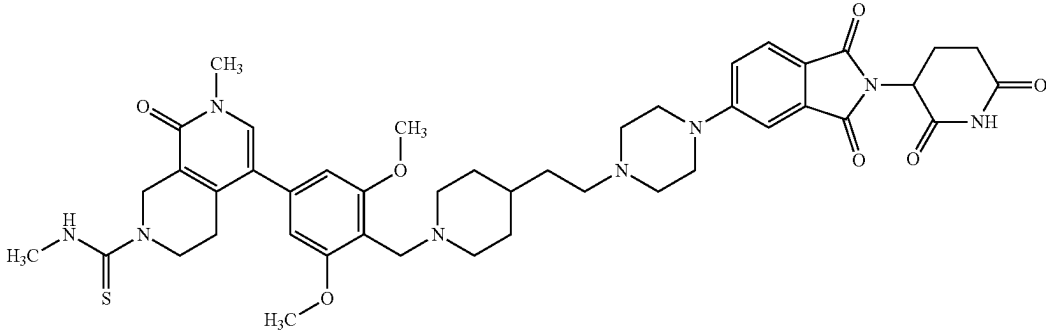
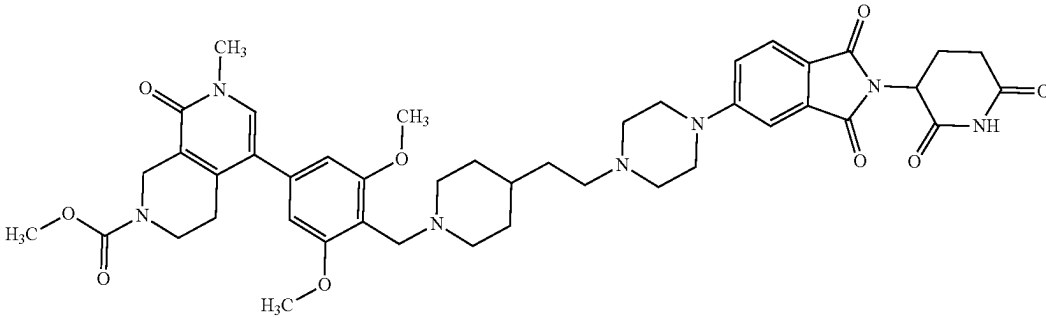
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D61	
D62	
D63	
D64	

TABLE 2B-continued

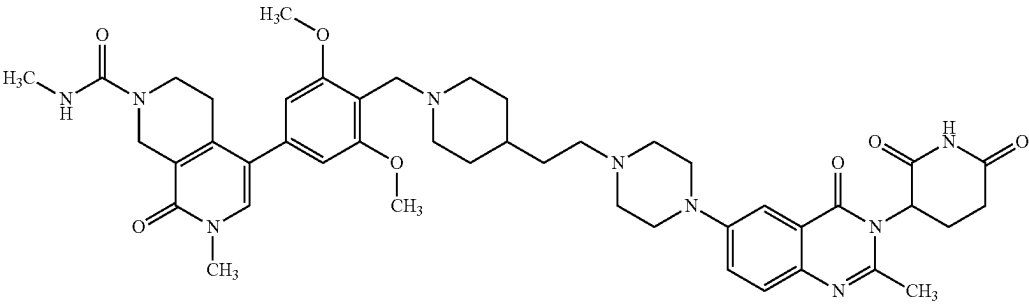
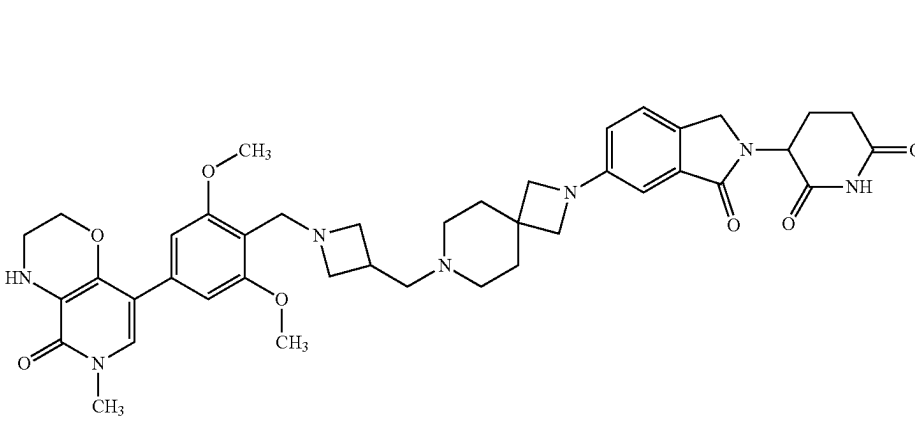
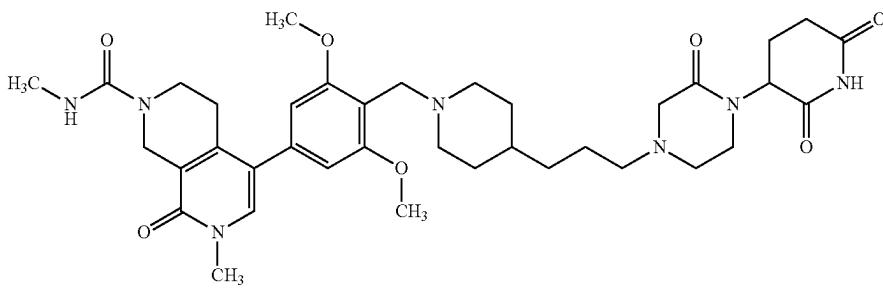
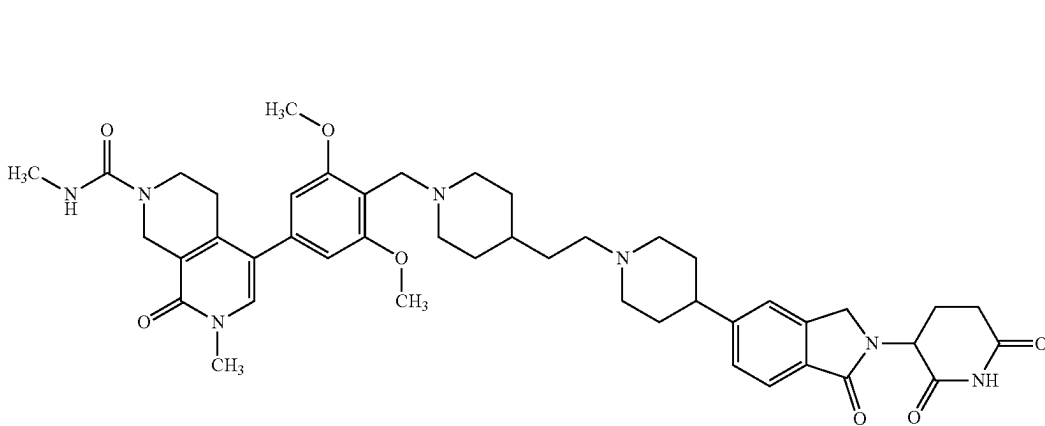
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D65	 Chemical structure of compound D65: A complex molecule featuring a central benzene ring substituted with two methoxy groups and two side chains. The left side chain consists of a 6-methyl-4-(methylamino)pyrimidin-2(1H)-one ring connected via a methylene bridge to a piperidine ring, which is further connected to a methylamino group. The right side chain consists of a piperidine ring connected via a methylene bridge to a 4-methyl-6-oxo-1,2,3,4-tetrahydropyrimidin-2-yl group.
D66	 Chemical structure of compound D66: A complex molecule featuring a central benzene ring substituted with two methoxy groups and two side chains. The left side chain consists of a 6-methyl-4-(methylamino)pyrimidin-2(1H)-one ring connected via a methylene bridge to a piperidine ring, which is further connected to a methylamino group. The right side chain consists of a piperidine ring connected via a methylene bridge to a 4-methyl-6-oxo-1,2,3,4-tetrahydropyrimidin-2-yl group.
D67	 Chemical structure of compound D67: A complex molecule featuring a central benzene ring substituted with two methoxy groups and two side chains. The left side chain consists of a 6-methyl-4-(methylamino)pyrimidin-2(1H)-one ring connected via a methylene bridge to a piperidine ring, which is further connected to a methylamino group. The right side chain consists of a piperidine ring connected via a methylene bridge to a 4-methyl-6-oxo-1,2,3,4-tetrahydropyrimidin-2-yl group.
D68	 Chemical structure of compound D68: A complex molecule featuring a central benzene ring substituted with two methoxy groups and two side chains. The left side chain consists of a 6-methyl-4-(methylamino)pyrimidin-2(1H)-one ring connected via a methylene bridge to a piperidine ring, which is further connected to a methylamino group. The right side chain consists of a piperidine ring connected via a methylene bridge to a 4-methyl-6-oxo-1,2,3,4-tetrahydropyrimidin-2-yl group.

TABLE 2B-continued

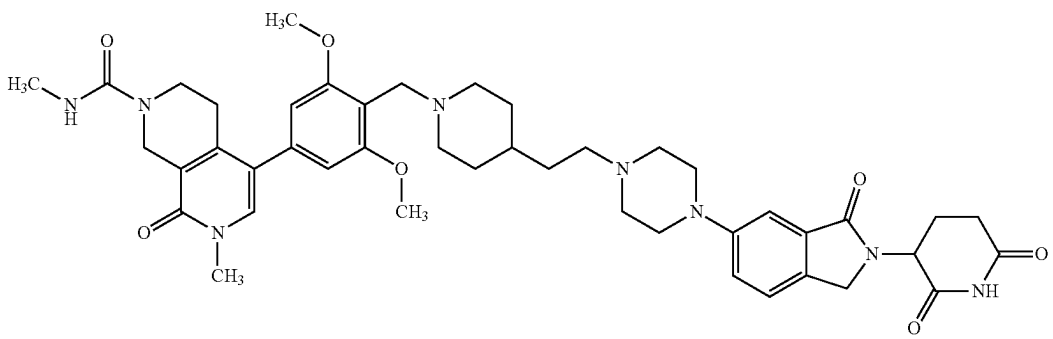
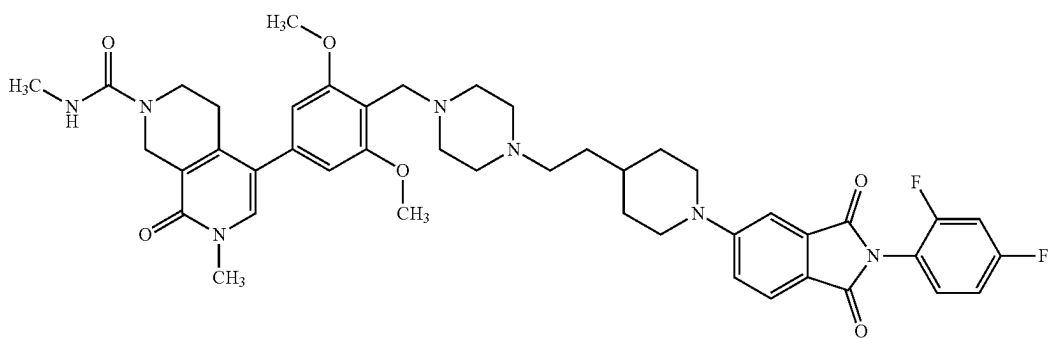
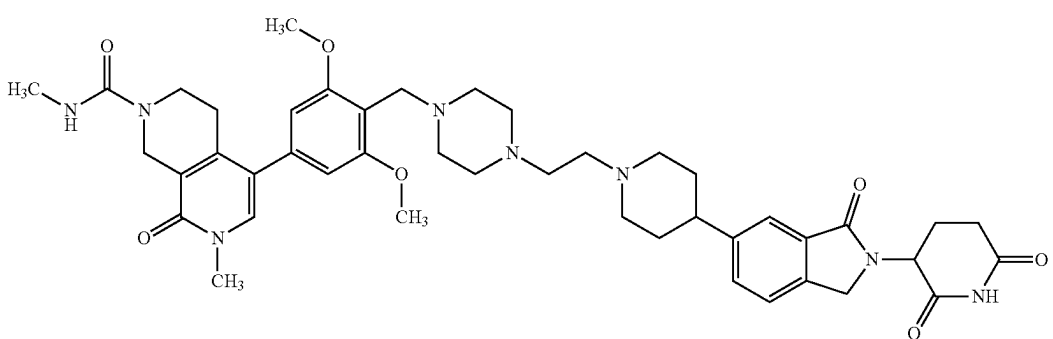
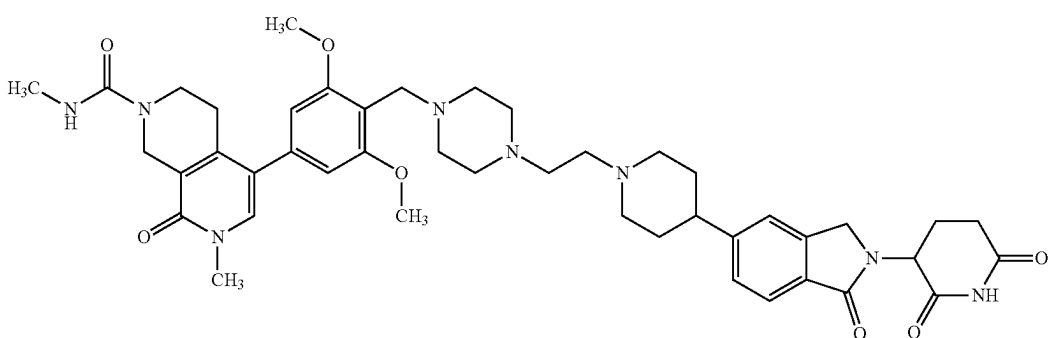
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D69	
D70	
D71	
D72	

TABLE 2B-continued

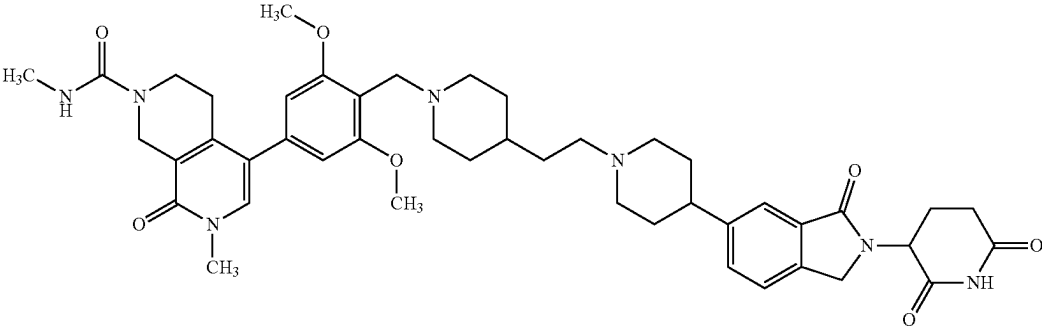
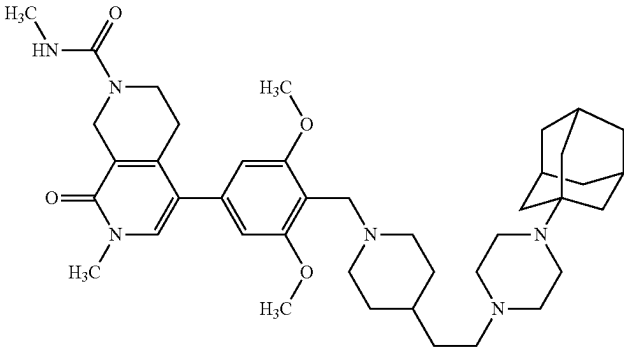
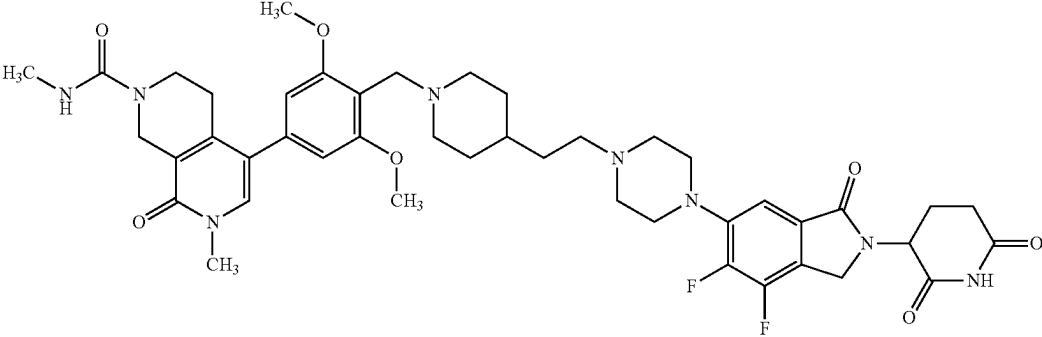
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D73	
D74	
D75	

TABLE 2B-continued

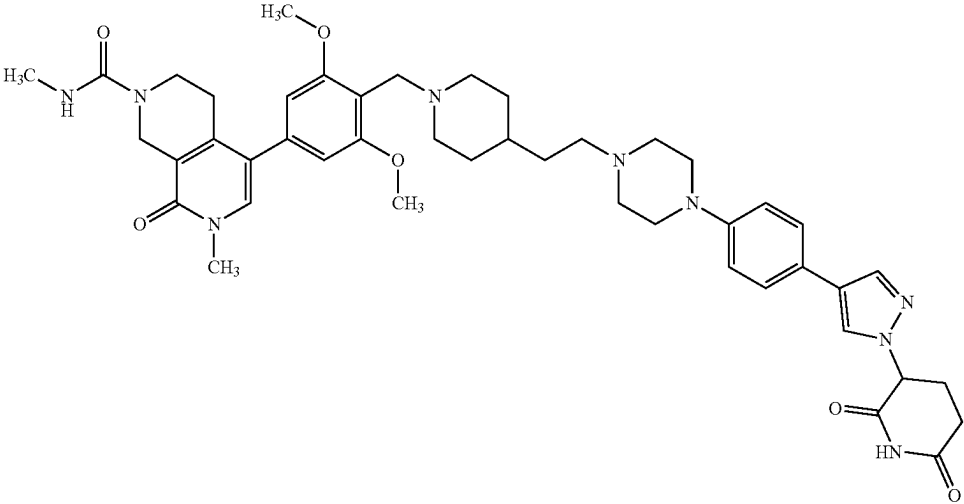
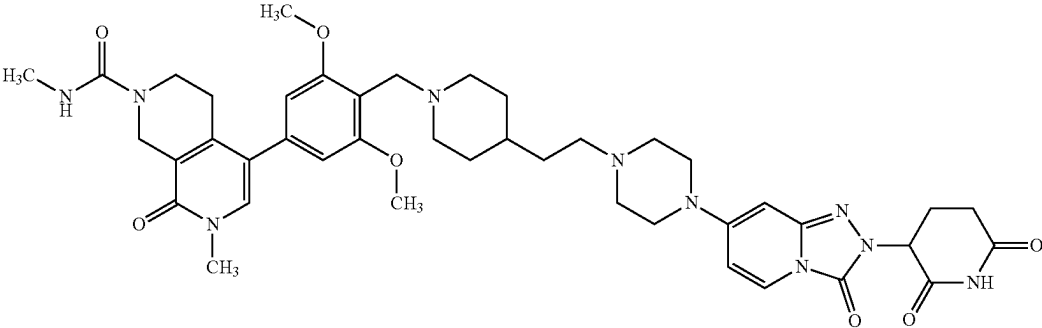
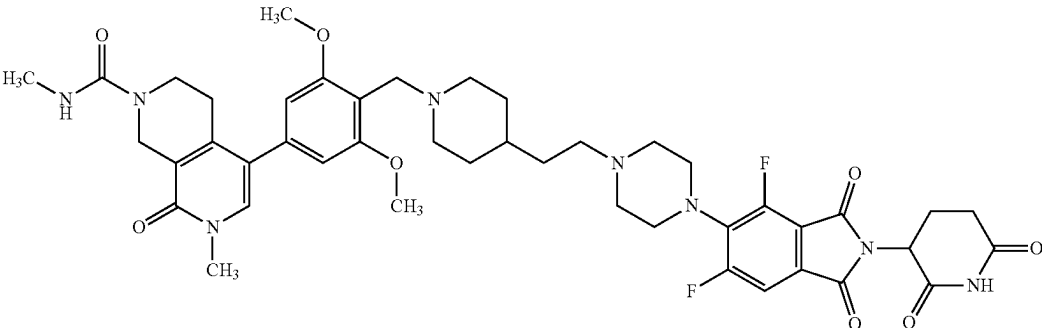
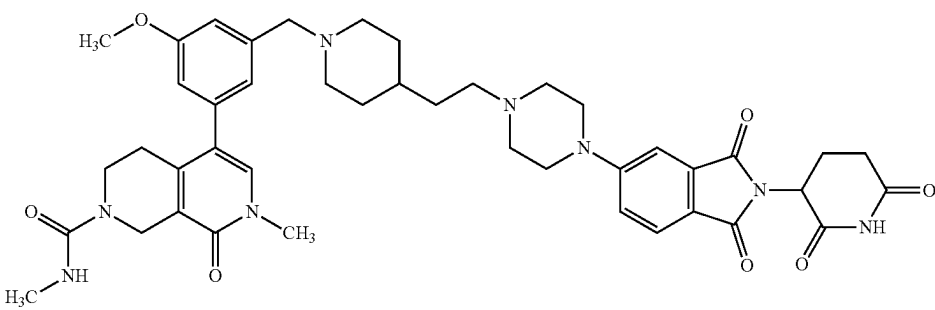
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D76	
D77	
D78	
D79	

TABLE 2B-continued

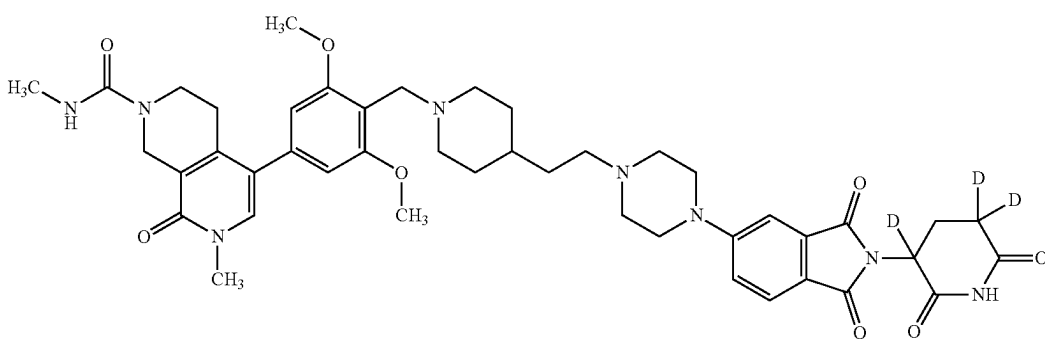
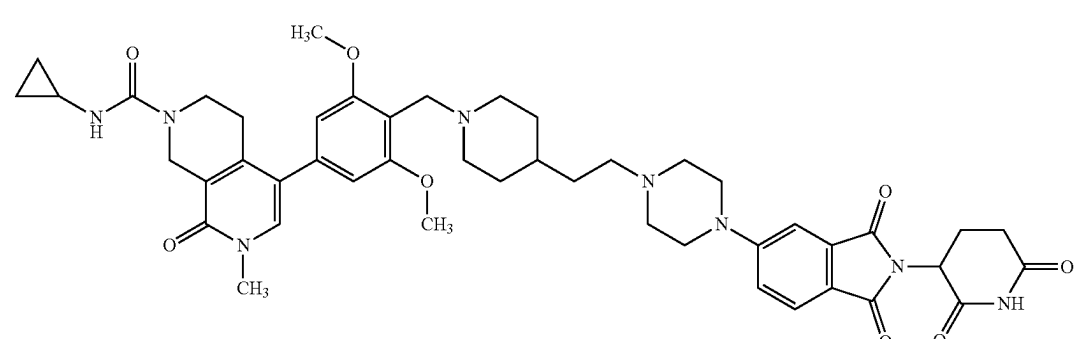
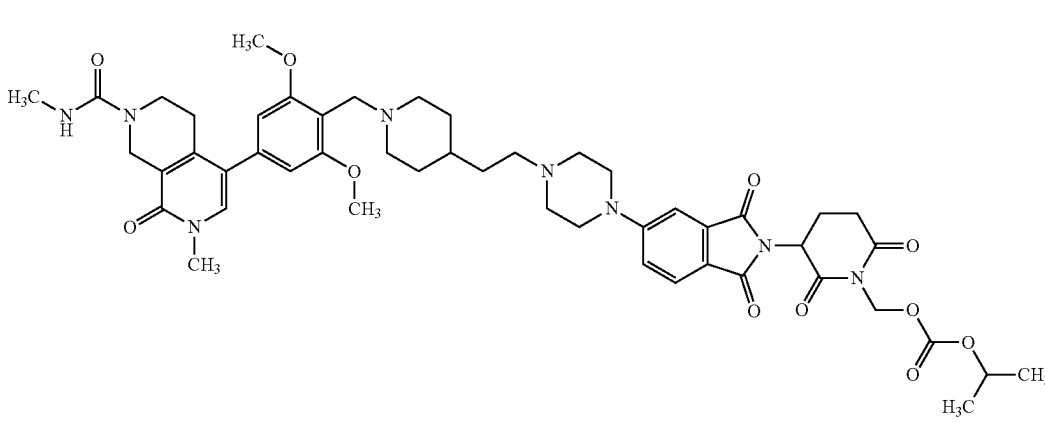
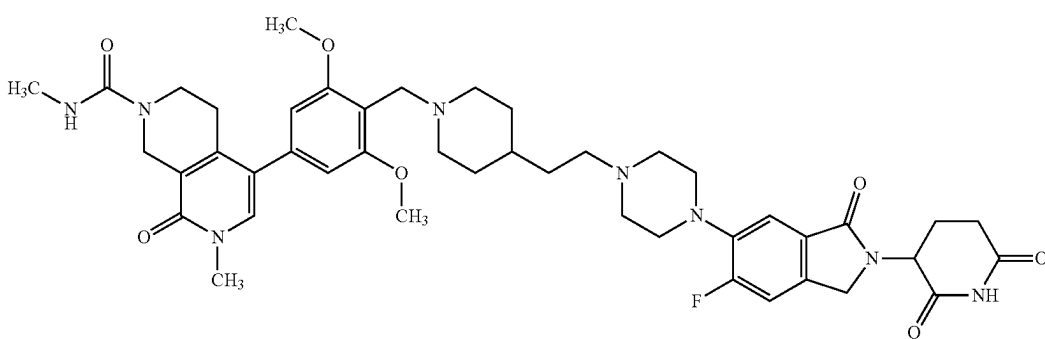
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D80	
D81	
D82	
D83	

TABLE 2B-continued

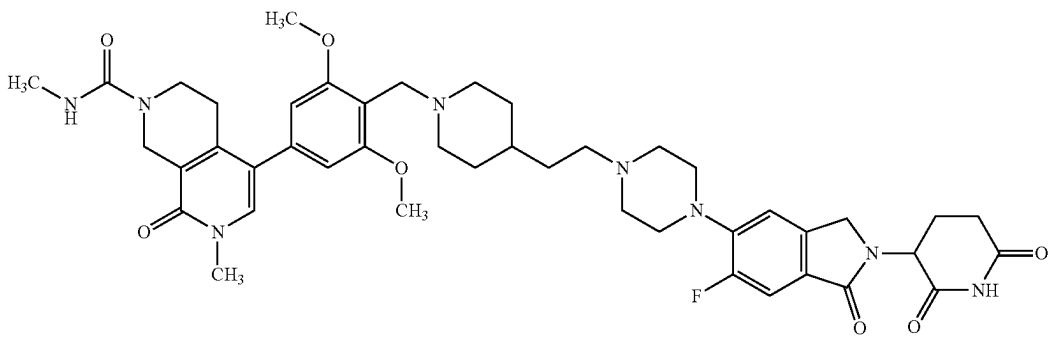
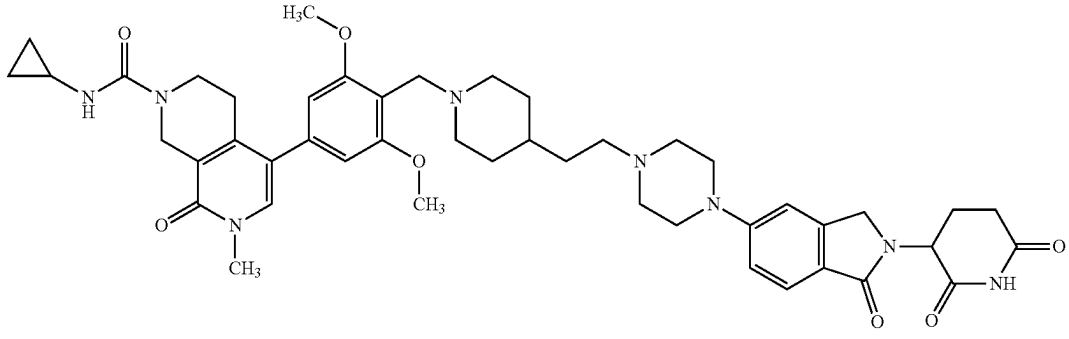
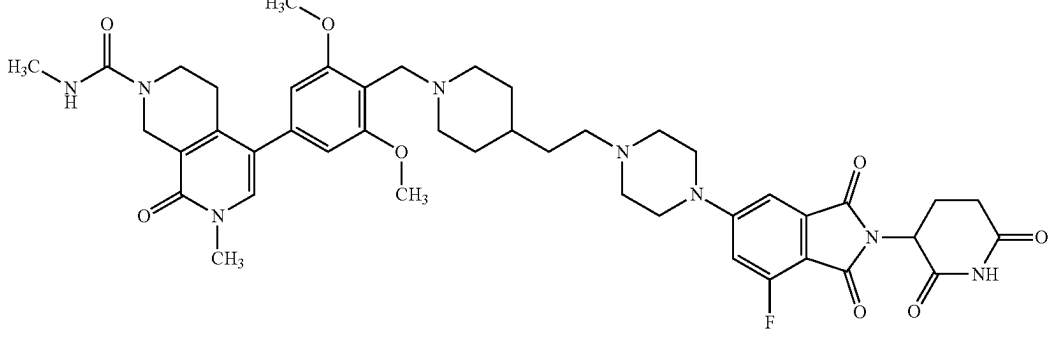
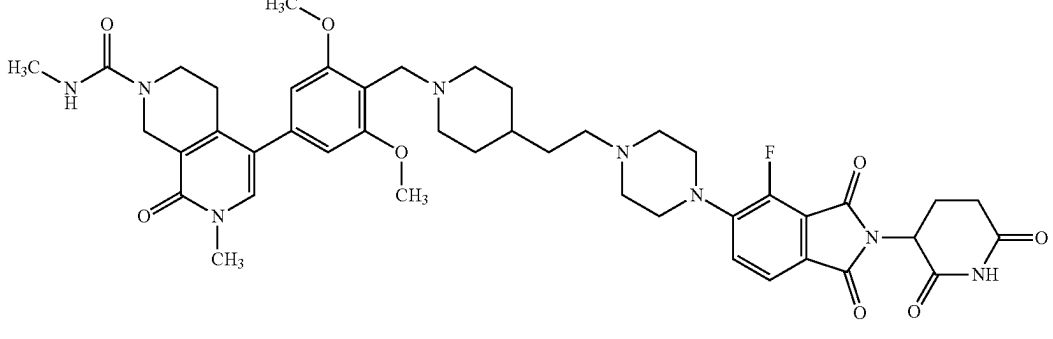
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D84	
D85	
D86	
D87	

TABLE 2B-continued

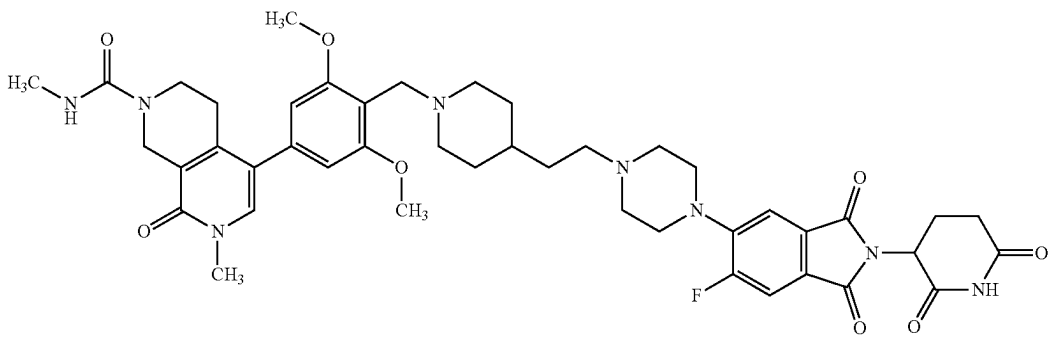
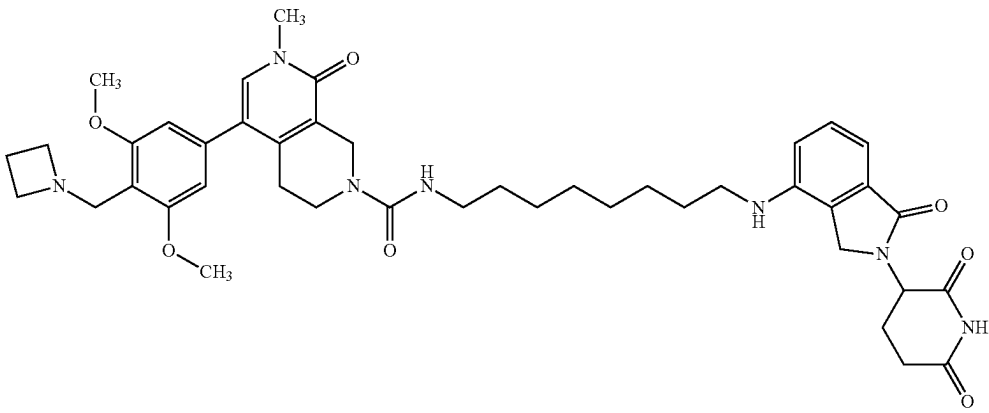
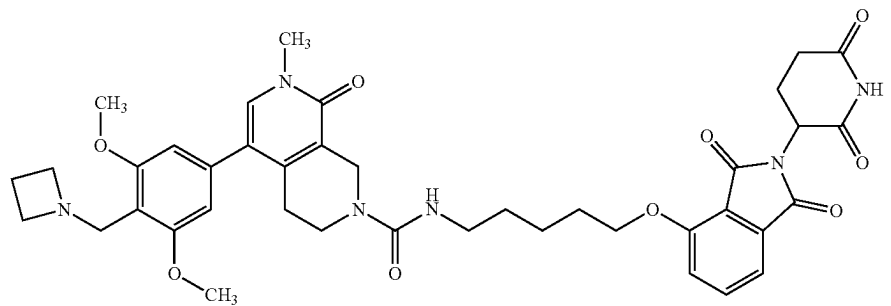
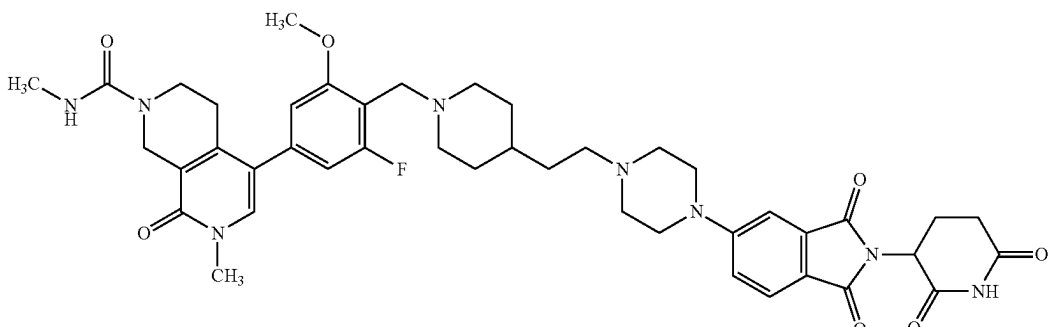
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D88	
D89	
D90	
D91	

TABLE 2B-continued

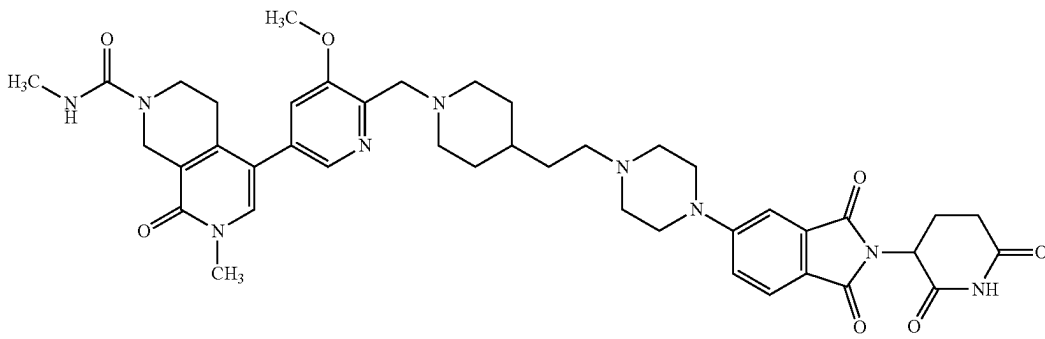
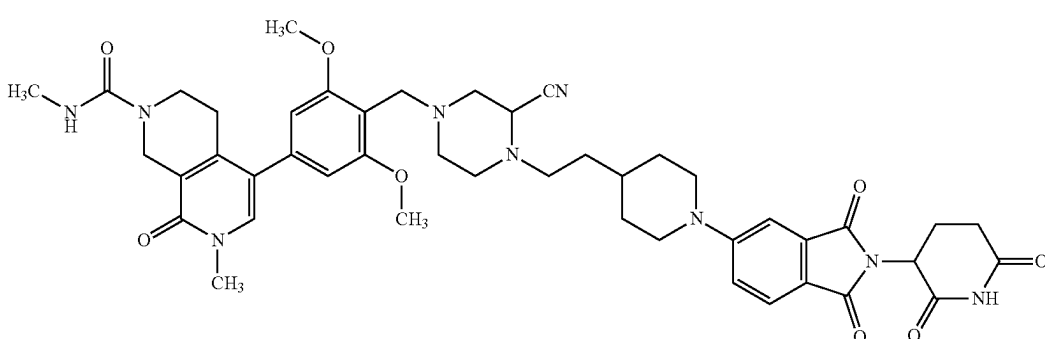
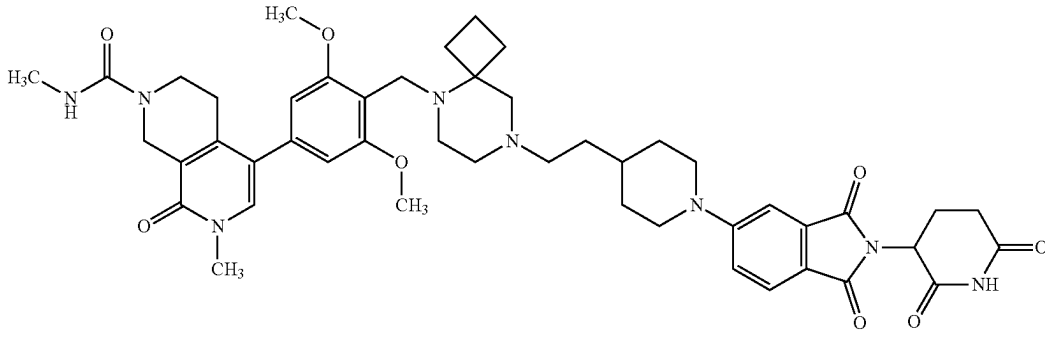
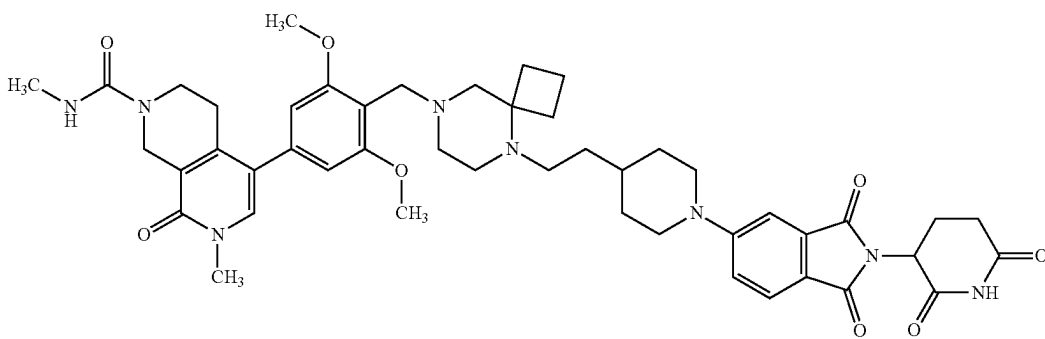
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D92	
D93	
D94	
D95	

TABLE 2B-continued

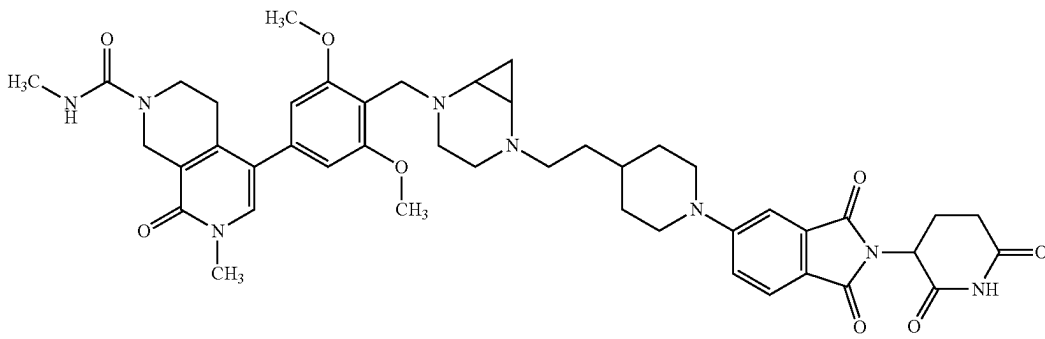
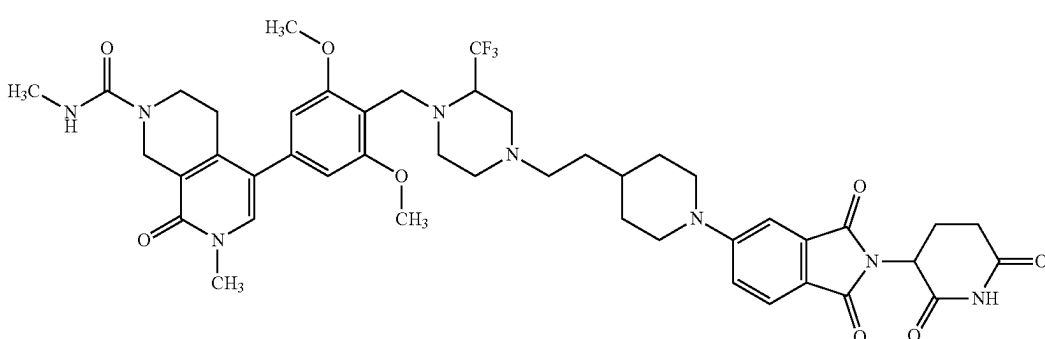
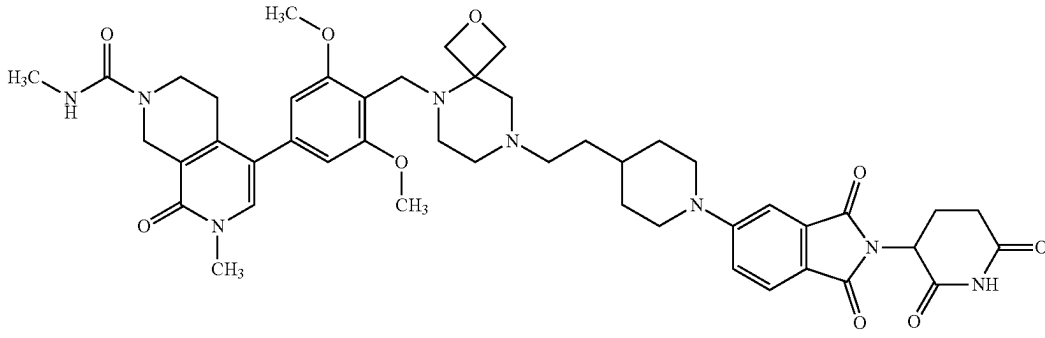
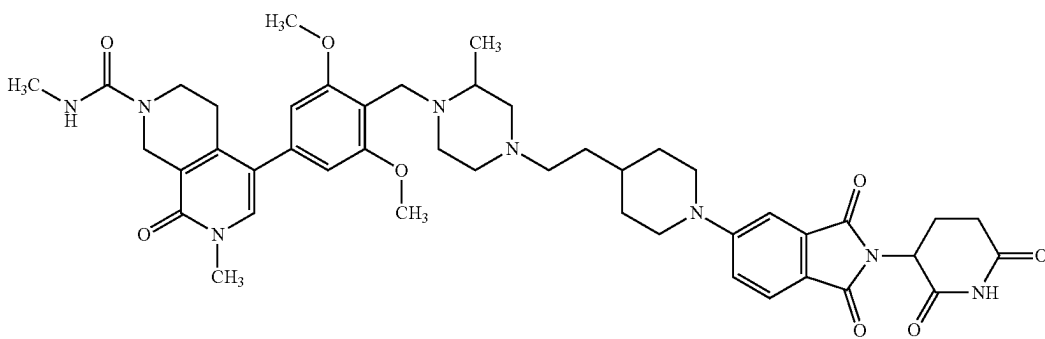
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D96	
D97	
D98	
D99	

TABLE 2B-continued

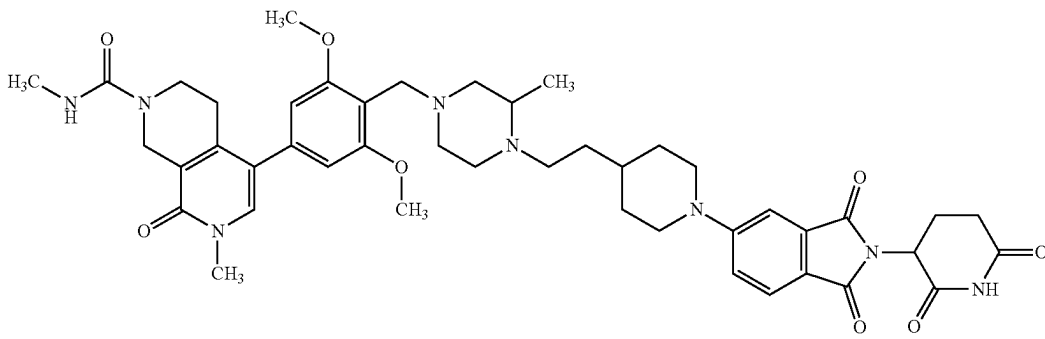
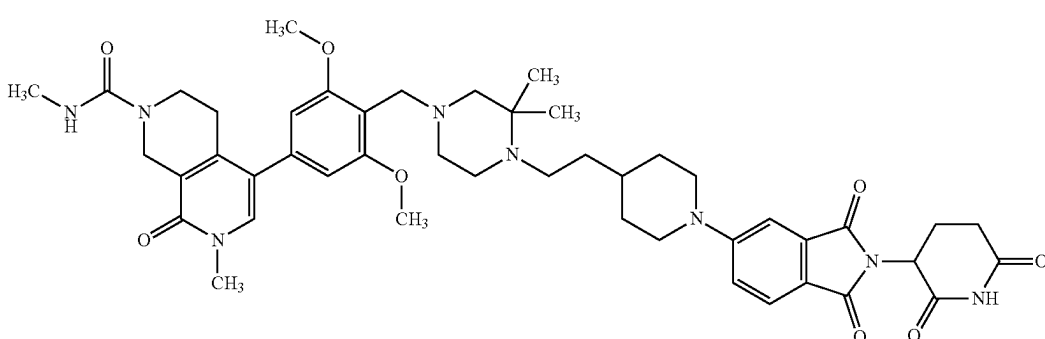
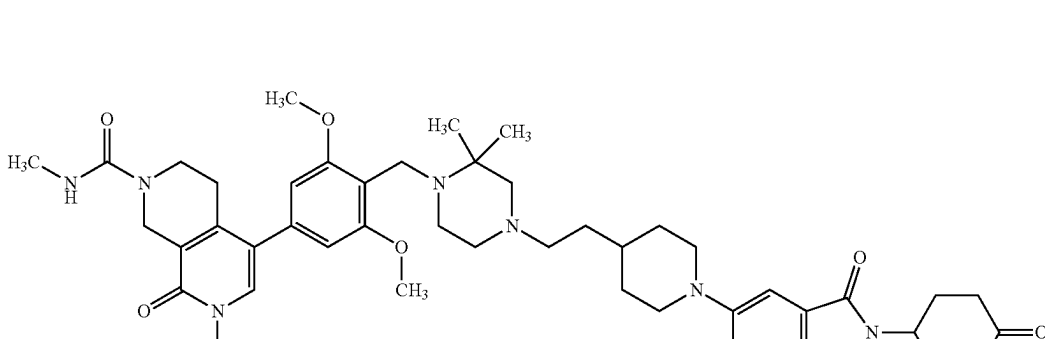
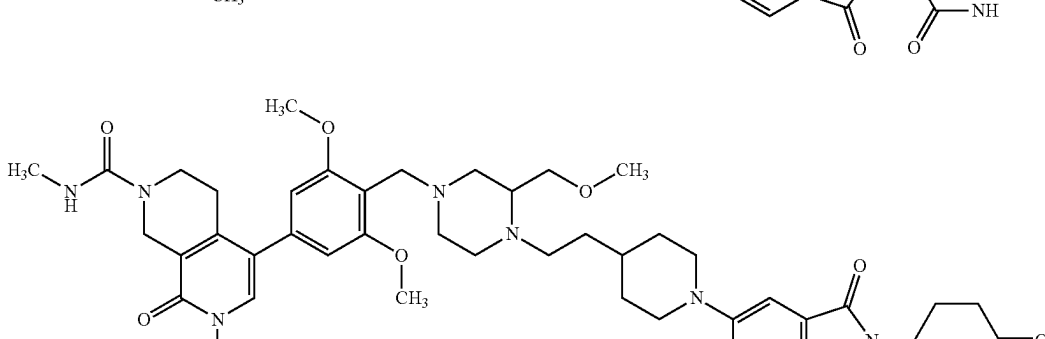
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D100	
D101	
D102	
D103	

TABLE 2B-continued

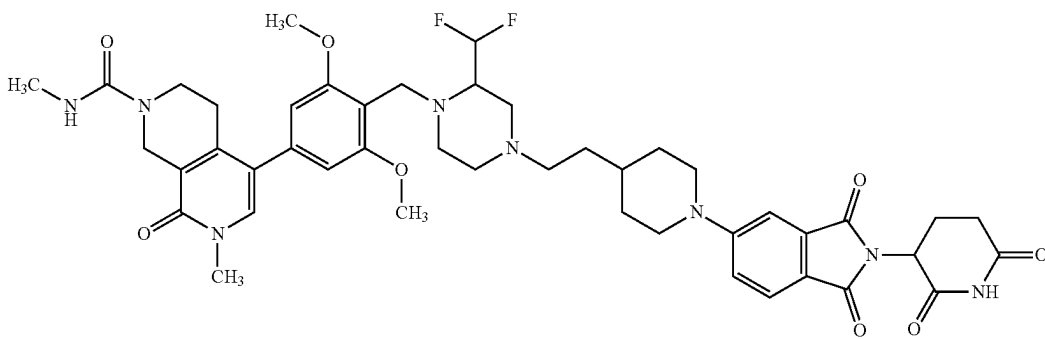
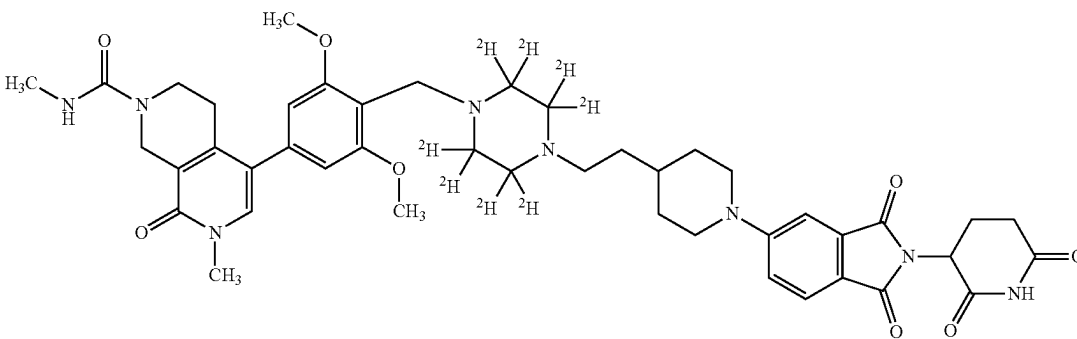
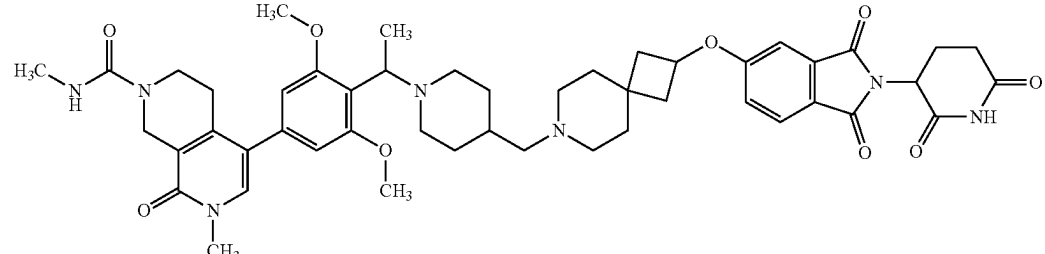
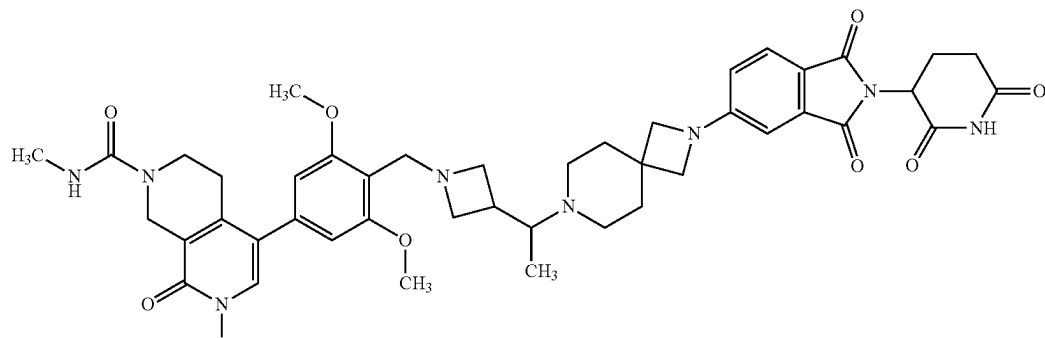
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D104	
D105	
D106	
D107	

TABLE 2B-continued

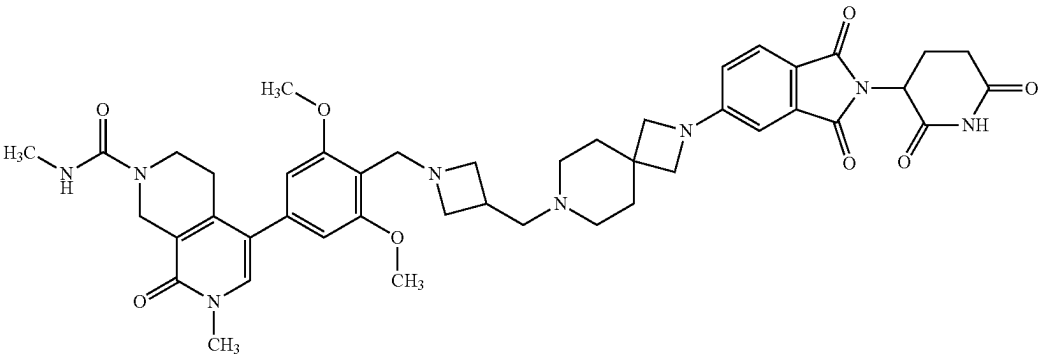
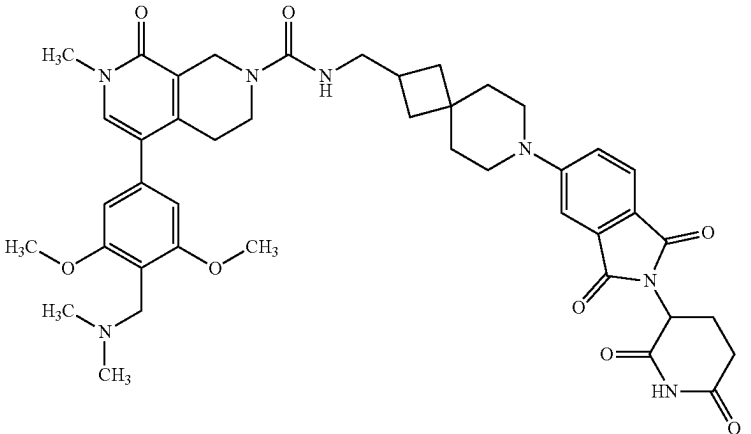
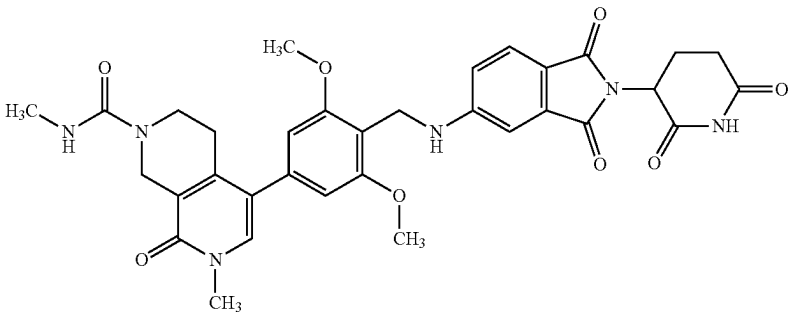
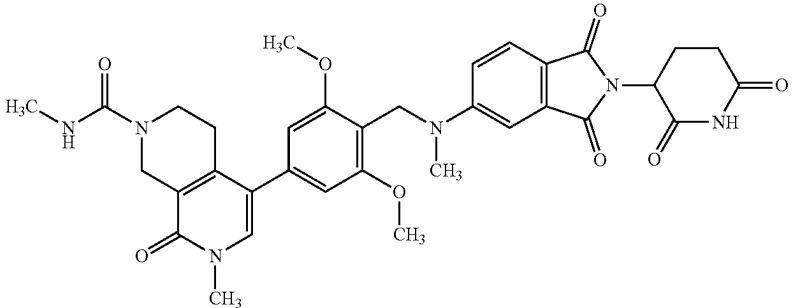
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D109	 Chemical structure of compound D109: A complex molecule featuring a central benzene ring substituted with a methoxy group and a methyl 2-(4-methoxyphenyl)-6-methyl-1,2,3,4-tetrahydropyridin-5(1H)-ylidene group. This central moiety is linked via a methylene bridge to a 4-(1,2,3,4-tetrahydropyridin-5-yl)-1,2,3,4-tetrahydropyridine-5-carboxamide group, which is further connected to a 1,2,3,4-tetrahydropyridine-5-carboxamide moiety.
D110	 Chemical structure of compound D110: A complex molecule featuring a central benzene ring substituted with two methoxy groups and a dimethylamino group. This central moiety is linked via a methylene bridge to a 1,2,3,4-tetrahydropyridine-5-carboxamide moiety, which is further connected to a 1,2,3,4-tetrahydropyridine-5-carboxamide moiety.
D111	 Chemical structure of compound D111: A complex molecule featuring a central benzene ring substituted with a methoxy group and a methyl 2-(4-methoxyphenyl)-6-methyl-1,2,3,4-tetrahydropyridin-5(1H)-ylidene group. This central moiety is linked via a methylene bridge to a 4-(1,2,3,4-tetrahydropyridin-5-yl)-1,2,3,4-tetrahydropyridine-5-carboxamide group, which is further connected to a 1,2,3,4-tetrahydropyridine-5-carboxamide moiety.
D112	 Chemical structure of compound D112: A complex molecule featuring a central benzene ring substituted with a methoxy group and a methyl 2-(4-methoxyphenyl)-6-methyl-1,2,3,4-tetrahydropyridin-5(1H)-ylidene group. This central moiety is linked via a methylene bridge to a 4-(1,2,3,4-tetrahydropyridin-5-yl)-1,2,3,4-tetrahydropyridine-5-carboxamide group, which is further connected to a 1,2,3,4-tetrahydropyridine-5-carboxamide moiety.

TABLE 2B-continued

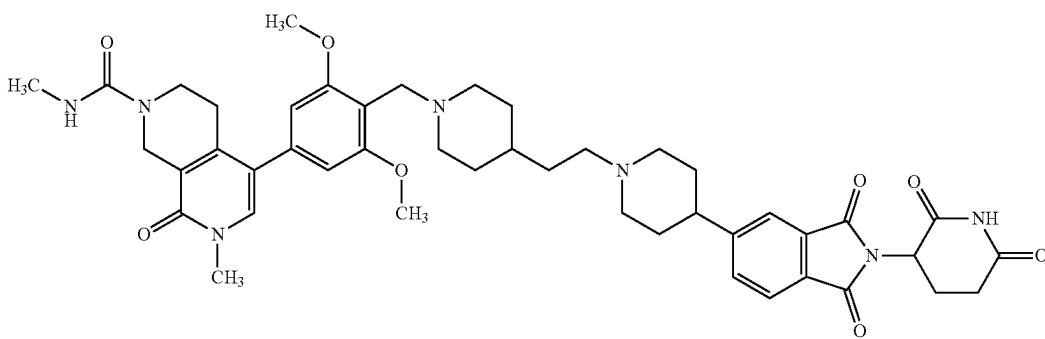
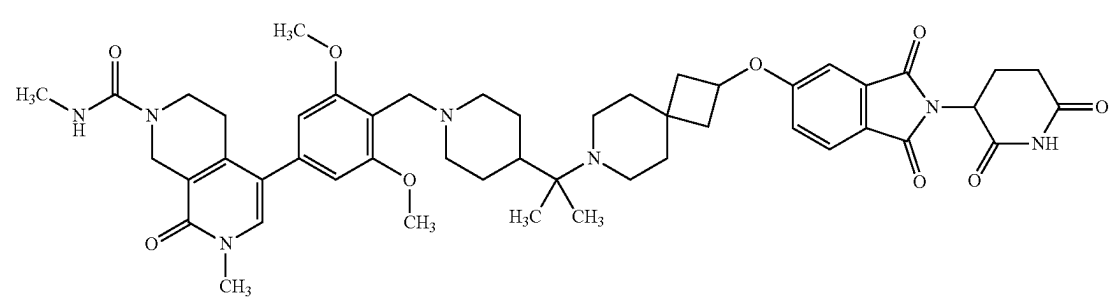
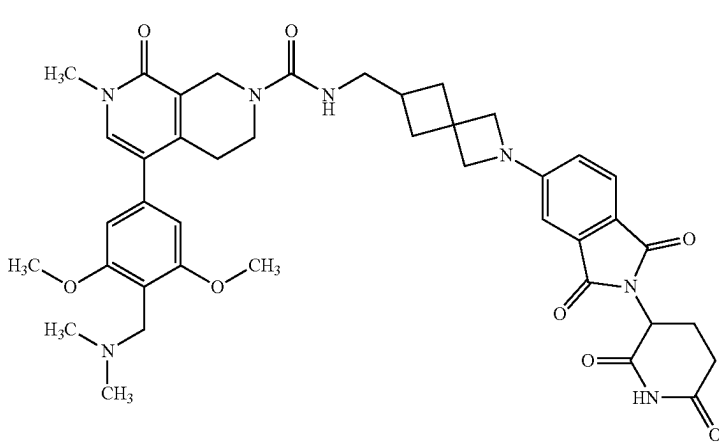
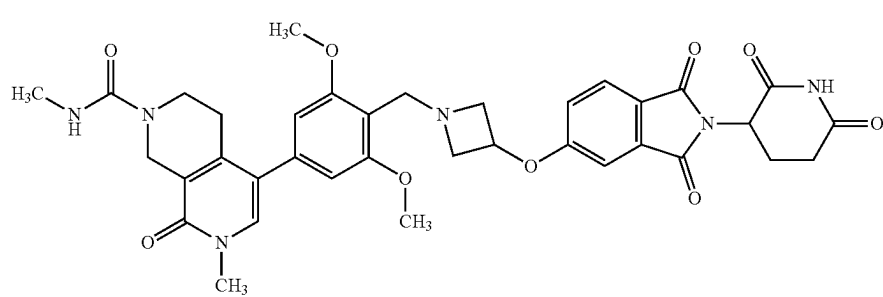
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D113	
D114	
D115	
D116	

TABLE 2B-continued

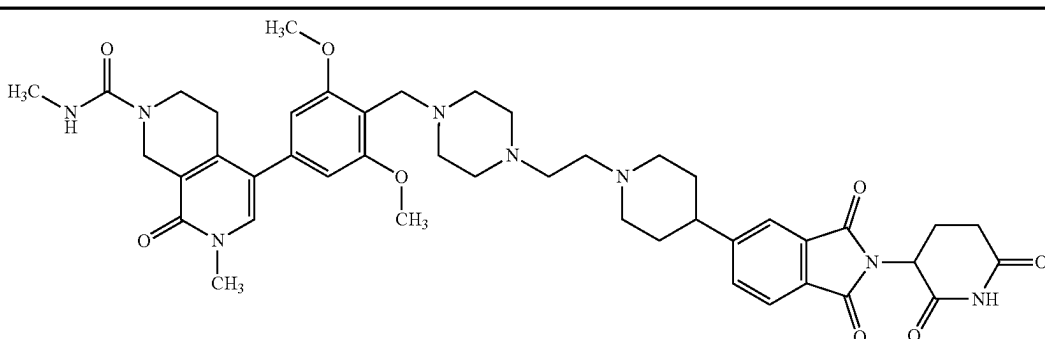
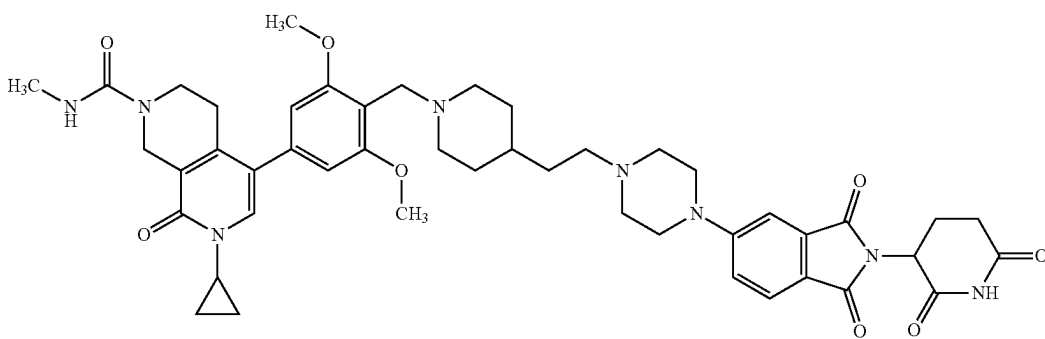
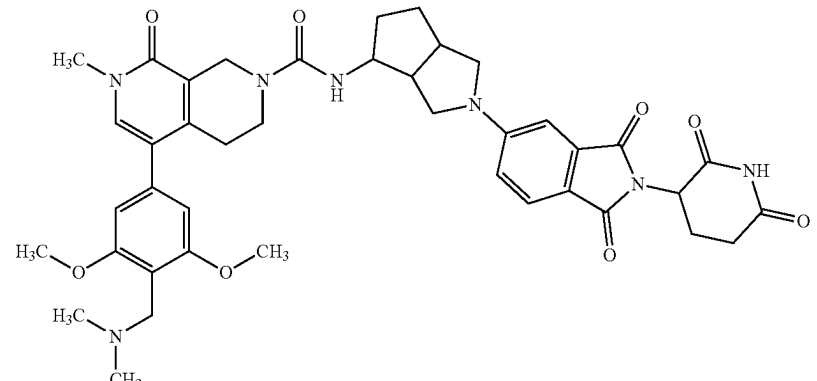
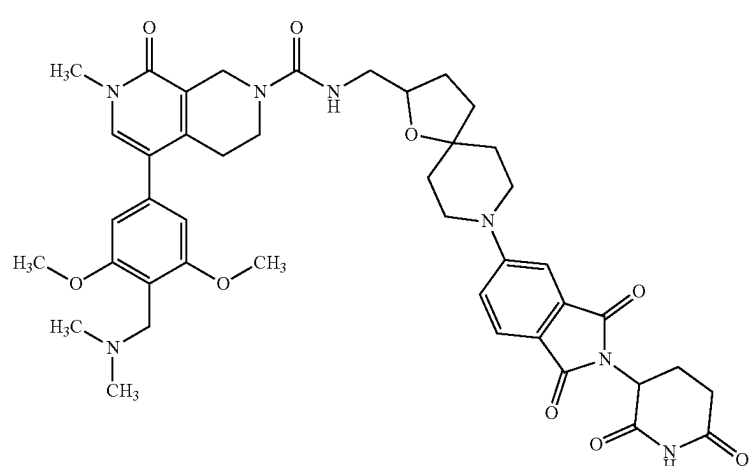
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D117	
D118	
D119	
D120	

TABLE 2B-continued

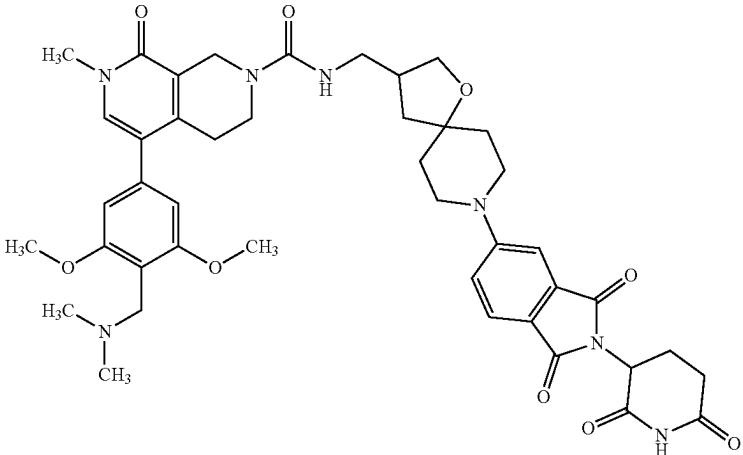
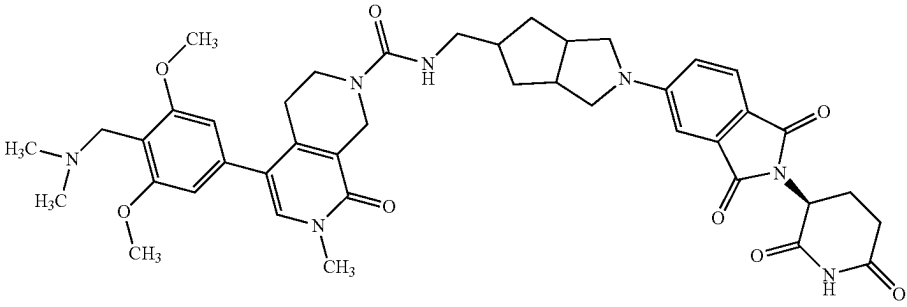
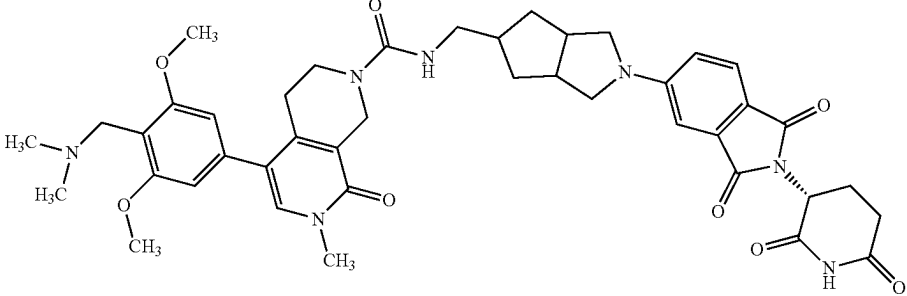
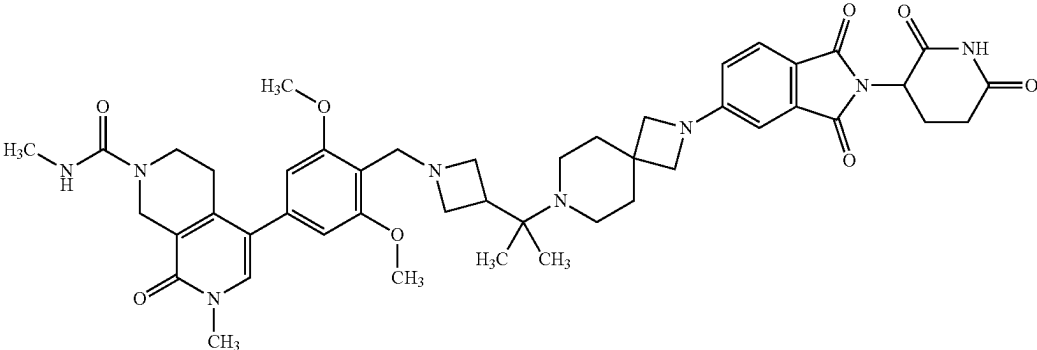
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D121	
D122	
D123	
D124	

TABLE 2B-continued

Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D125	
D126	
D127	
D128	
D129	

TABLE 2B-continued

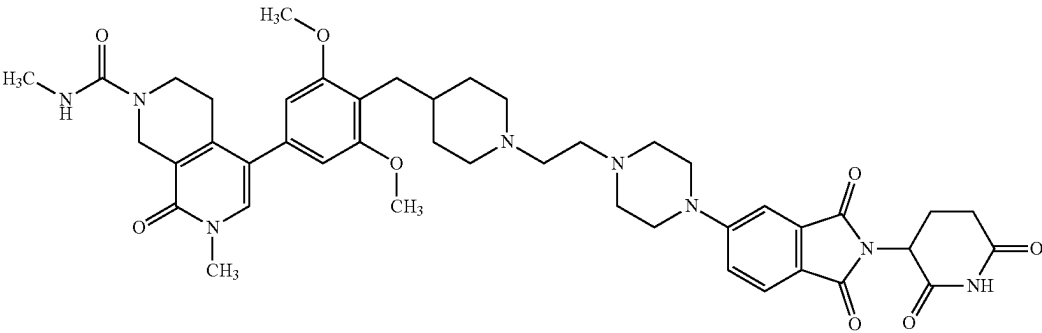
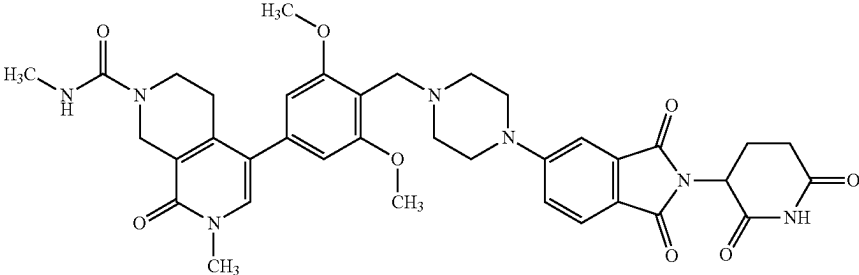
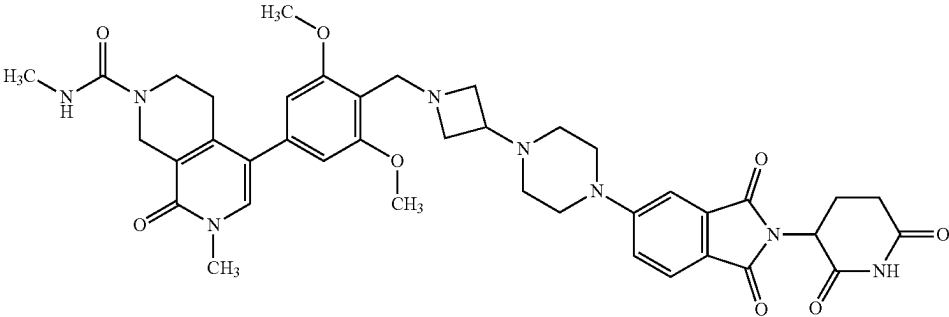
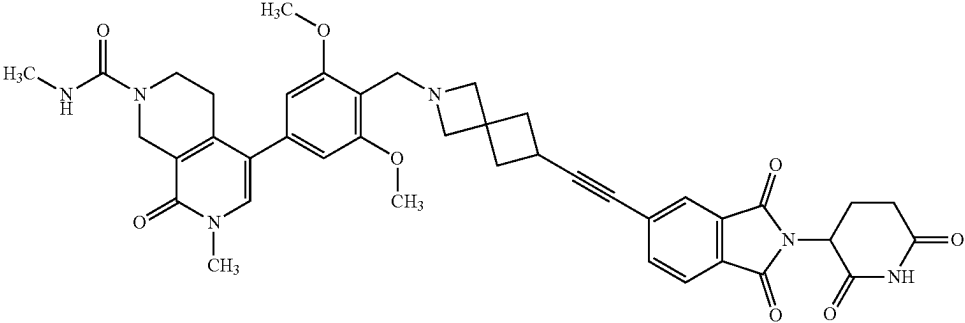
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D130	 Chemical structure of compound D130: A complex molecule featuring a 6-methoxy-2-(methylamino)pyridine-3-carboxamide moiety linked via a methylene bridge to a 3,4-dimethoxyphenyl ring. This phenyl ring is further connected through a methylene bridge to a piperidine ring, which is linked via a methylene bridge to a piperazine ring. The piperazine ring is connected via a methylene bridge to a benzimidazole-1-one system, which is finally linked via a methylene bridge to a 2,6-pyridinedione moiety.
D131	 Chemical structure of compound D131: A complex molecule featuring a 6-methoxy-2-(methylamino)pyridine-3-carboxamide moiety linked via a methylene bridge to a 3,4-dimethoxyphenyl ring. This phenyl ring is further connected through a methylene bridge to a piperazine ring, which is linked via a methylene bridge to a benzimidazole-1-one system, which is finally linked via a methylene bridge to a 2,6-pyridinedione moiety.
D132	 Chemical structure of compound D132: A complex molecule featuring a 6-methoxy-2-(methylamino)pyridine-3-carboxamide moiety linked via a methylene bridge to a 3,4-dimethoxyphenyl ring. This phenyl ring is further connected through a methylene bridge to a pyrrolidine ring, which is linked via a methylene bridge to a piperazine ring. The piperazine ring is connected via a methylene bridge to a benzimidazole-1-one system, which is finally linked via a methylene bridge to a 2,6-pyridinedione moiety.
D133	 Chemical structure of compound D133: A complex molecule featuring a 6-methoxy-2-(methylamino)pyridine-3-carboxamide moiety linked via a methylene bridge to a 3,4-dimethoxyphenyl ring. This phenyl ring is further connected through a methylene bridge to a bicyclo[3.1.0]hexane system, which is linked via an ethynyl group to a benzimidazole-1-one system, which is finally linked via a methylene bridge to a 2,6-pyridinedione moiety.

TABLE 2B-continued

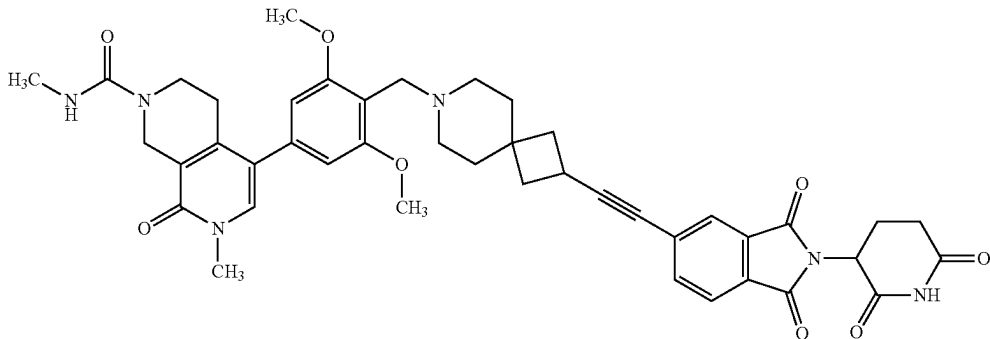
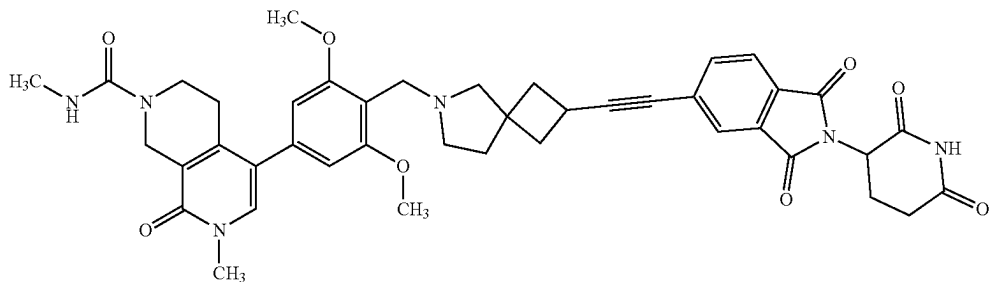
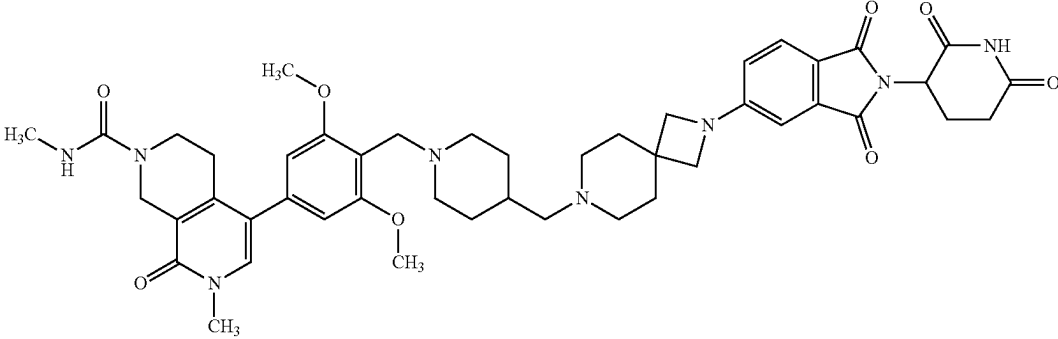
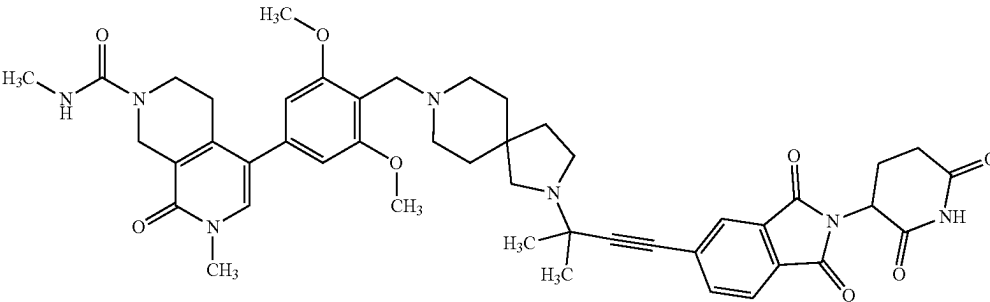
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D134	
D135	
D136	
D137	

TABLE 2B-continued

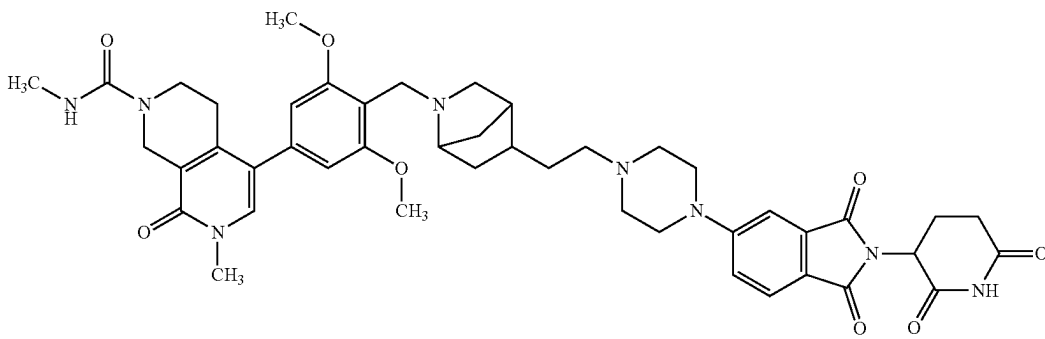
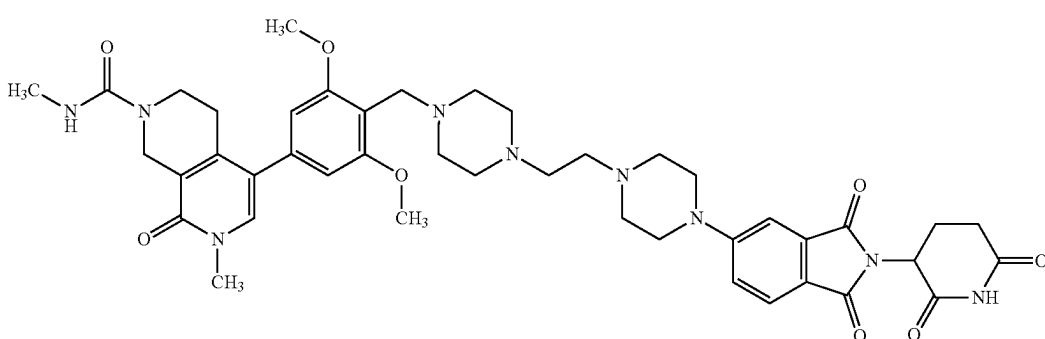
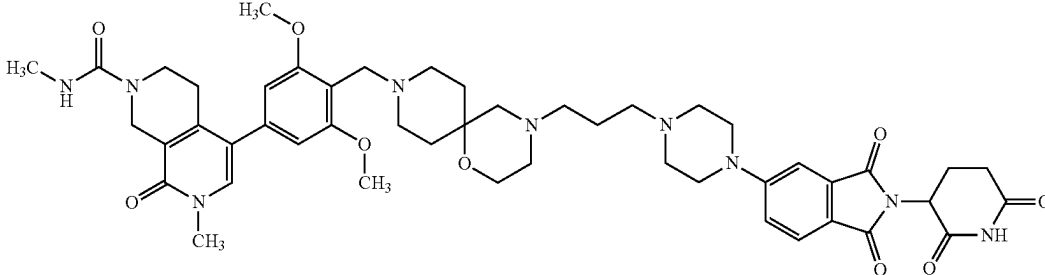
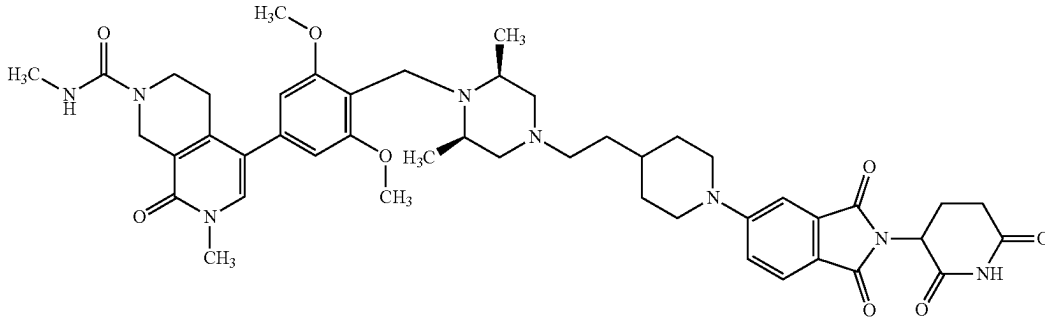
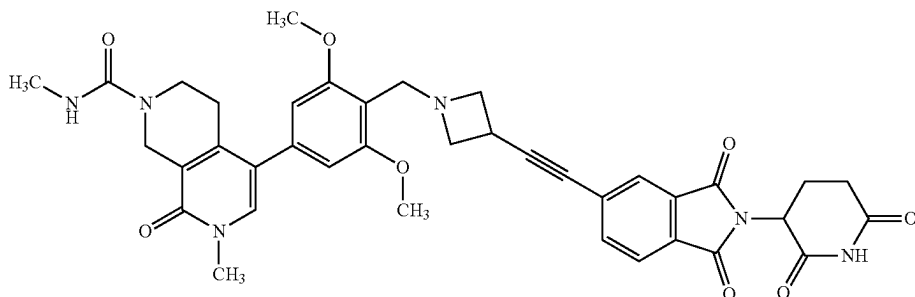
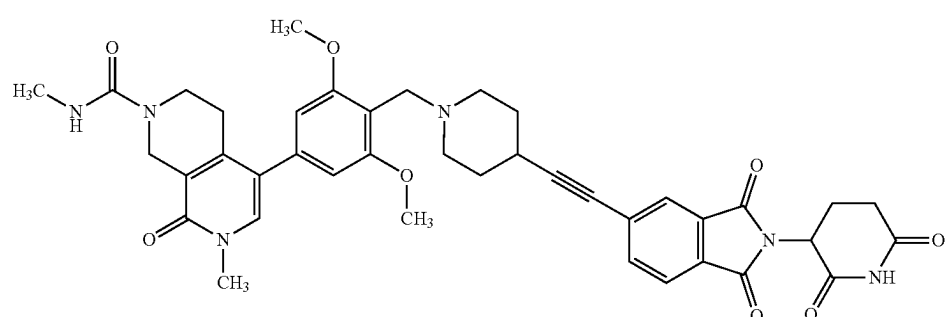
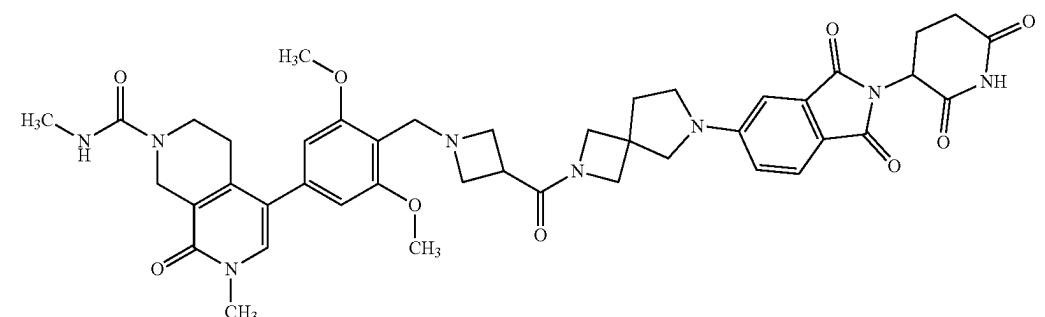
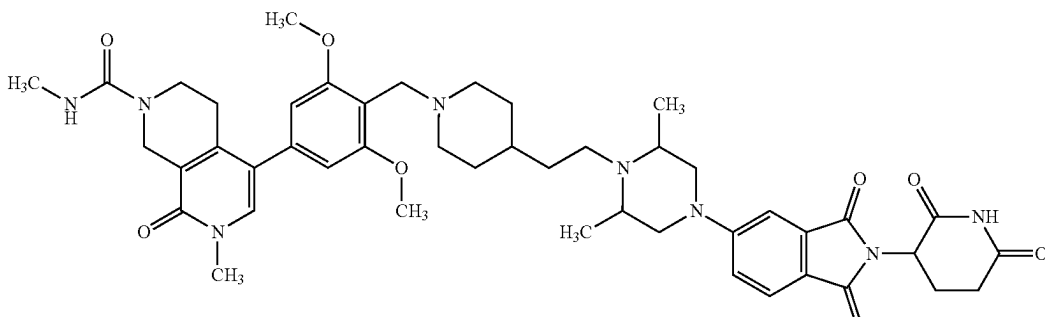
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D138	
D139	
D140	
D141	

TABLE 2B-continued

Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D142	
D143	
D144	
D145	

Compounds D39-D302 of the Disclosure

Compound No.	Structure
D146	
D147	
D148	
D149	

Compounds D39-D302 of the Disclosure

Compound No.	Structure
D150	
D151	
D152	
D153	

TABLE 2B-continued

Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D154	 <chem>CN(C)C(=O)N1CCc2cc(CN(C)C(=O)N1)cc(C1=CC=C(C=C1)C2=CC(OC)=C(OC)C2CN1CCC(F)(F)CN1CCC2=CC3=C(C=C1)C(=O)N4CC(=O)NCC4=O)C3</chem>
D155	 <chem>CN(C)C(=O)N1CCc2cc(CN(C)C(=O)N1)cc(C1=CC=C(C=C1)C2=CC(OC)=C(OC)C2CN1CCC(F)(F)CN1CCC2=CC3=C(C=C1)C(=O)N4CC(=O)NCC4=O)C3</chem>
D156	 <chem>CN(C)C(=O)N1CCc2cc(CN(C)C(=O)N1)cc(C1=CC=C(C=C1)C2=CC(OC)=C(OC)C2CN1CCC(F)(F)CN1CCC2=CC3=C(C=C1)C(=O)N4CC(=O)NCC4=O)C3</chem>
D157	 <chem>CN(C)C(=O)N1CCc2cc(CN(C)C(=O)N1)cc(C1=CC=C(C=C1)C2=CC(OC)=C(OC)C2CN1CCC(F)(F)CN1CCC2=CC3=C(C=C1)C(=O)N4CC(=O)NCC4=O)C3</chem>

TABLE 2B-continued

[illegible]

TABLE 2B-continued

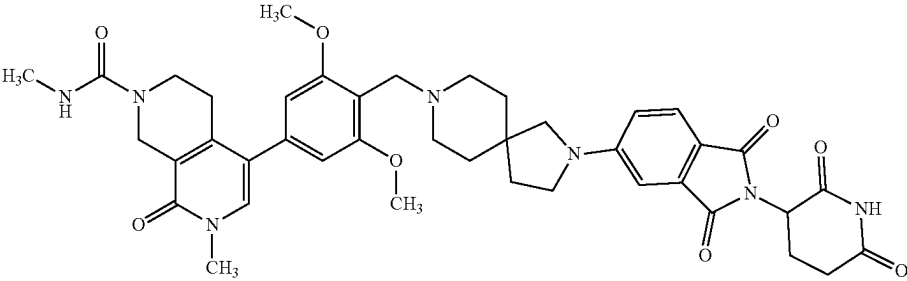
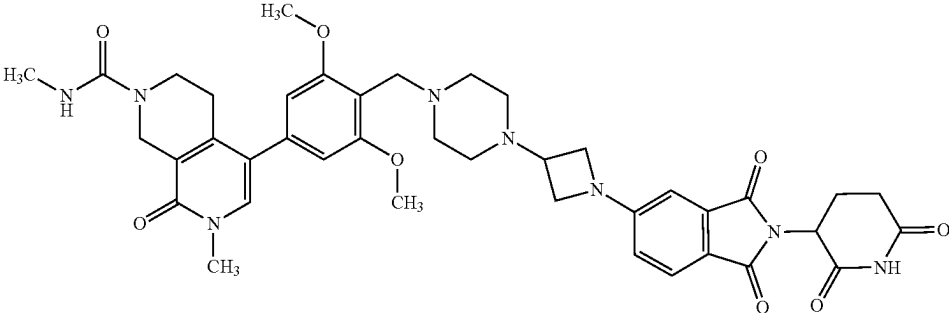
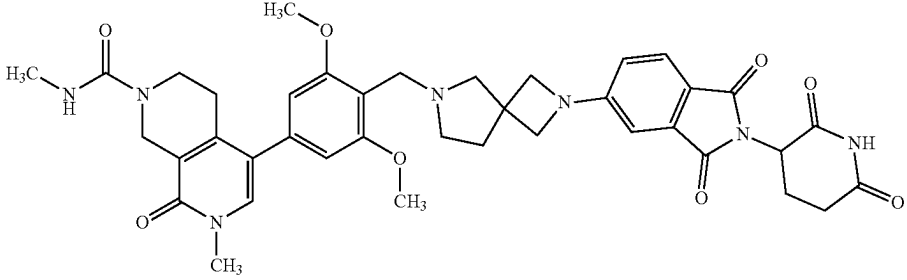
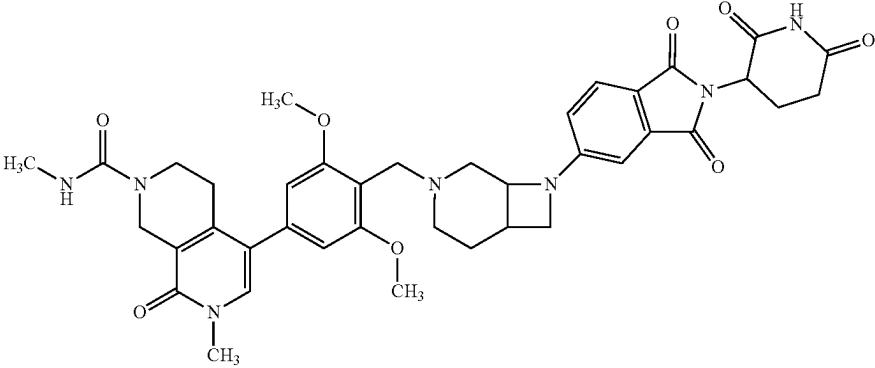
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D163	 Chemical structure of compound D163: A complex molecule featuring a 4-methyl-6-((methylamino)carbonyl)piperidine-2-carboxamide moiety linked via a biphenyl system to a 3,4-dimethoxyphenyl group. This is further connected to a piperidine ring, which is linked to a pyrrolidine ring, and finally to a benzimidazole moiety substituted with a 2,6-pyridinedione ring.
D164	 Chemical structure of compound D164: Similar to D163, but the piperidine ring is replaced by a pyrrolidine ring, and the benzimidazole moiety is substituted with a 2,6-pyridinedione ring.
D165	 Chemical structure of compound D165: Similar to D163, but the piperidine ring is replaced by a pyrrolidine ring, and the benzimidazole moiety is substituted with a 2,6-pyridinedione ring.
D166	 Chemical structure of compound D166: Similar to D163, but the piperidine ring is replaced by a pyrrolidine ring, and the benzimidazole moiety is substituted with a 2,6-pyridinedione ring.

TABLE 2B-continued

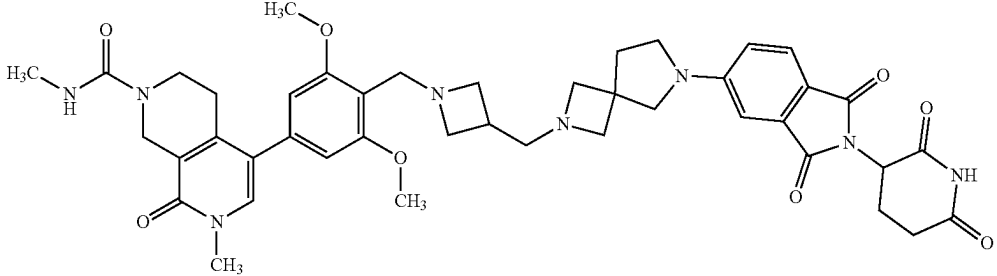
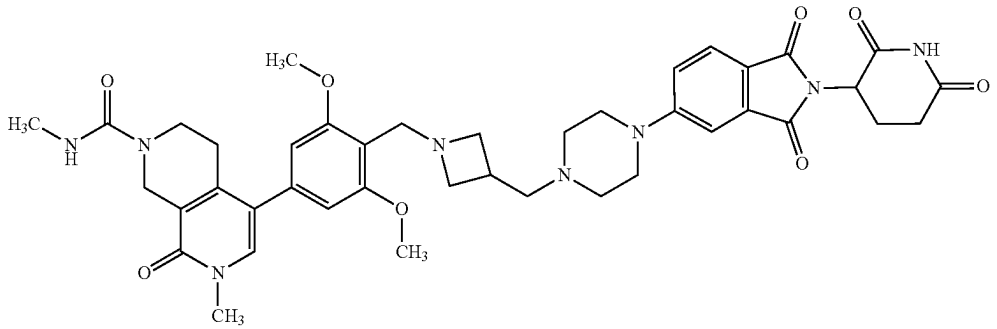
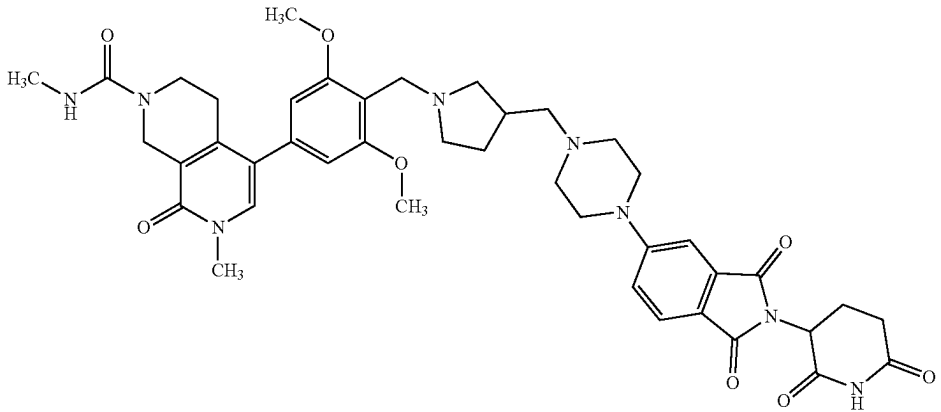
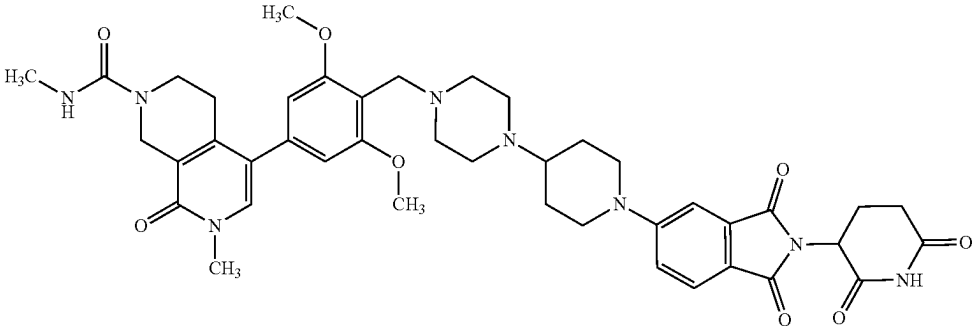
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D167	
D168	
D169	
D170	

TABLE 2B-continued

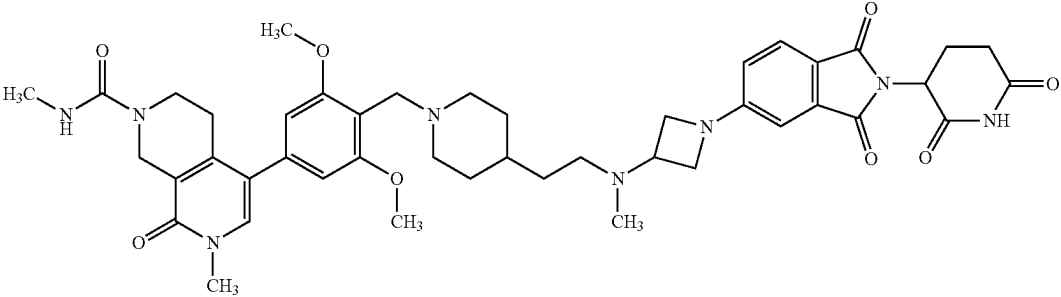
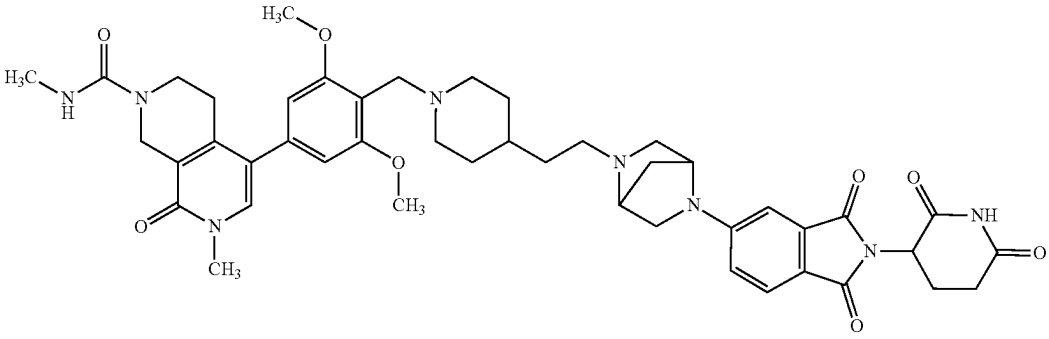
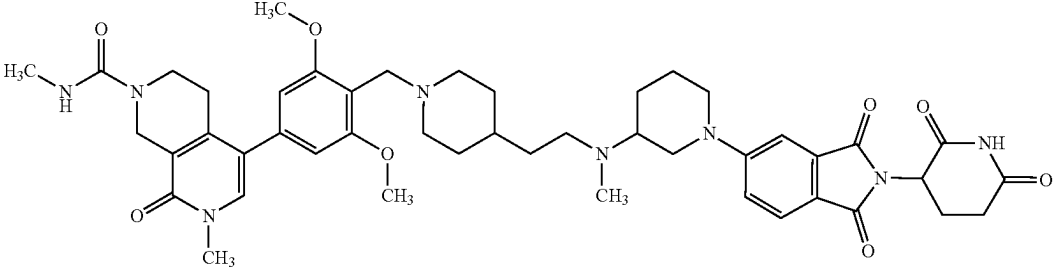
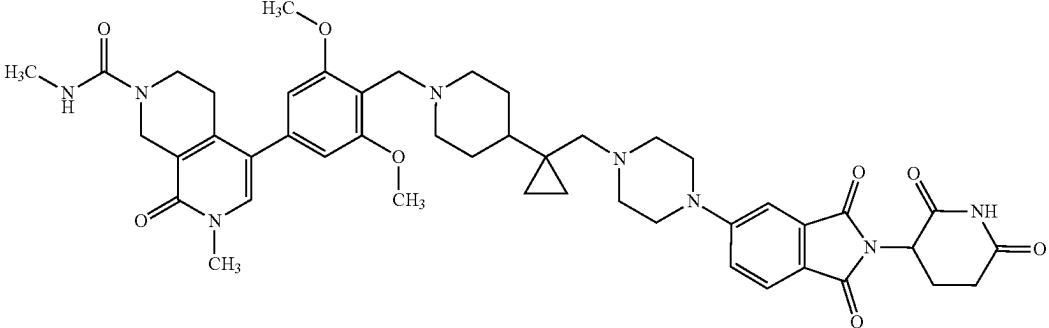
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D171	
D172	
D173	
D174	

TABLE 2B-continued

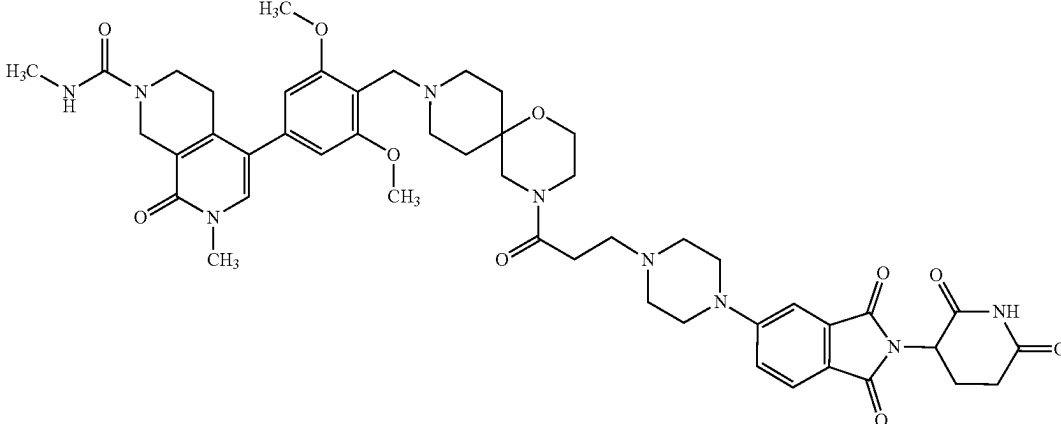
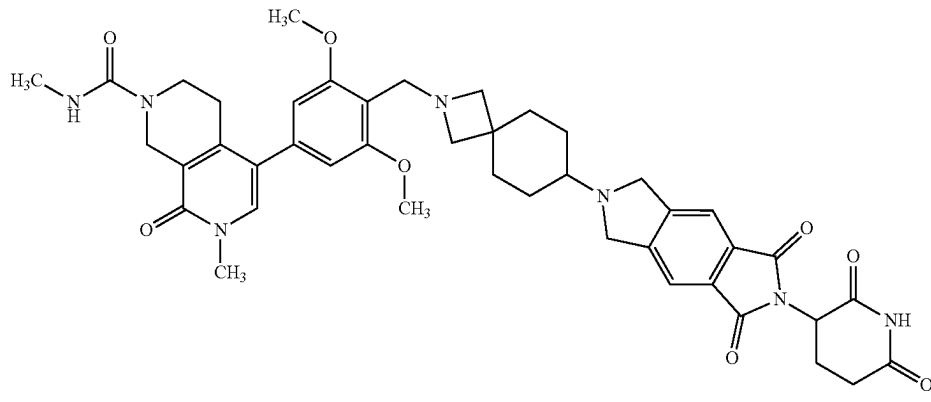
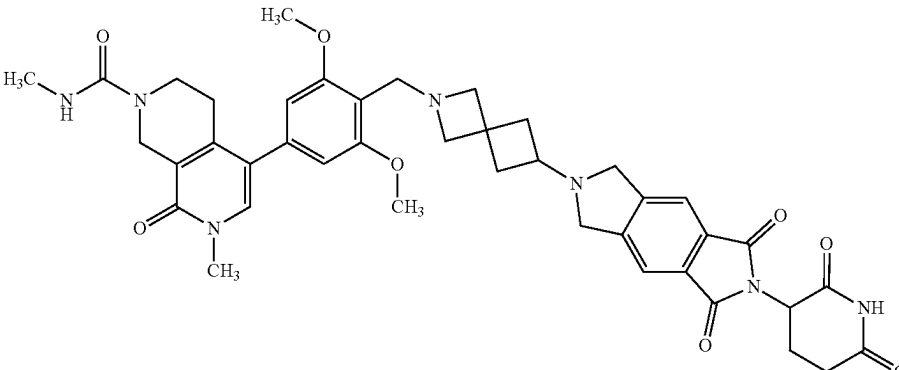
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D175	 Chemical structure of compound D175: A complex molecule featuring a 6-methoxy-2-methyl-4-((methylamino)carbonyl)pyridine ring system. This is linked via a methylene bridge to a 3,4-dimethoxyphenyl ring. The 3,4-dimethoxyphenyl ring is further connected to a 4-morpholinylmethyl group. The morpholine ring is linked to a 2-((methylamino)carbonyl)ethyl group, which is connected to a 4-piperidinylmethyl group. The piperidine ring is linked to a 1,2,3,4-tetrahydroquinolin-5(1H)-one system, which is further connected to a 2,6-pyridinedione ring.
D176	 Chemical structure of compound D176: A complex molecule featuring a 6-methoxy-2-methyl-4-((methylamino)carbonyl)pyridine ring system. This is linked via a methylene bridge to a 3,4-dimethoxyphenyl ring. The 3,4-dimethoxyphenyl ring is further connected to a 1,2,3,4-tetrahydroquinolin-5(1H)-one system, which is further connected to a 2,6-pyridinedione ring.
D177	 Chemical structure of compound D177: A complex molecule featuring a 6-methoxy-2-methyl-4-((methylamino)carbonyl)pyridine ring system. This is linked via a methylene bridge to a 3,4-dimethoxyphenyl ring. The 3,4-dimethoxyphenyl ring is further connected to a 1,2,3,4-tetrahydroquinolin-5(1H)-one system, which is further connected to a 2,6-pyridinedione ring.

TABLE 2B-continued

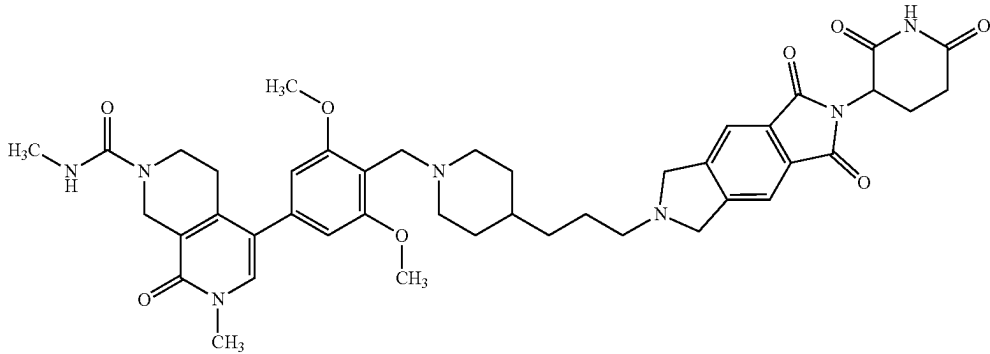
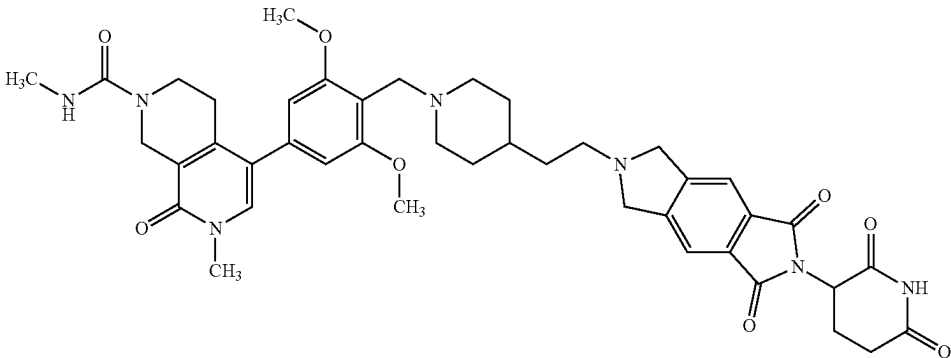
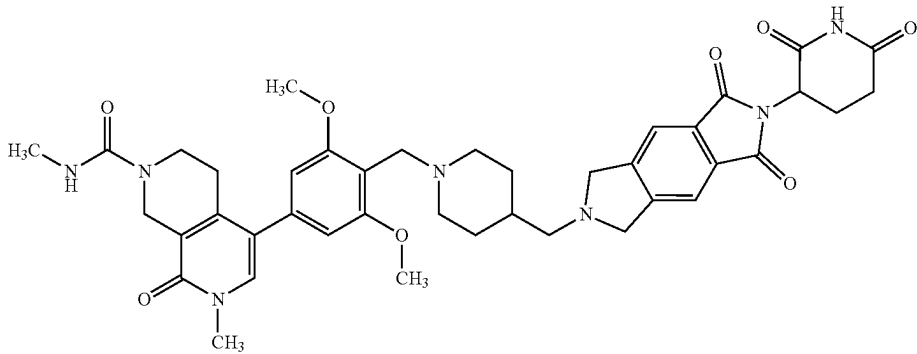
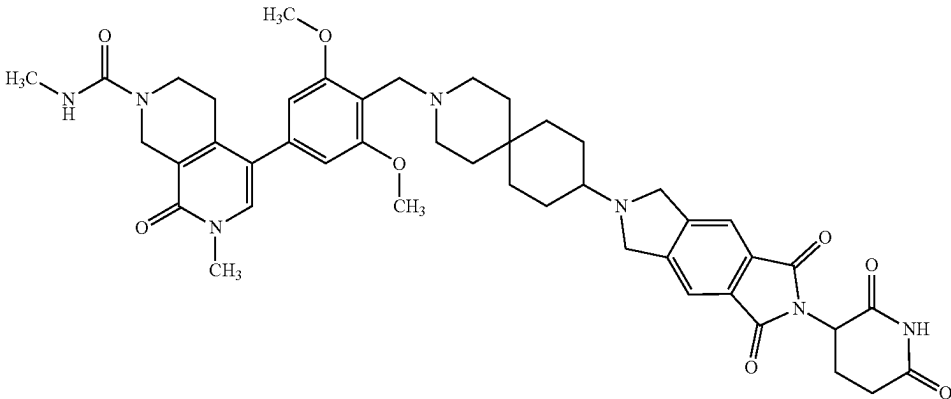
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D178	 Chemical structure of compound D178: A complex molecule featuring a central benzene ring substituted with two methoxy groups and two methylene groups. One methylene group is connected to a piperidine ring, which is further linked via a propyl chain to a bicyclic system (indole-fused). The other methylene group is connected to a pyridine ring, which is substituted with a methyl group and a methylamino group. The pyridine ring is also linked to a piperidine ring, which is connected to a bicyclic system (indole-fused).
D179	 Chemical structure of compound D179: A complex molecule featuring a central benzene ring substituted with two methoxy groups and two methylene groups. One methylene group is connected to a piperidine ring, which is further linked via a propyl chain to a bicyclic system (indole-fused). The other methylene group is connected to a pyridine ring, which is substituted with a methyl group and a methylamino group. The pyridine ring is also linked to a piperidine ring, which is connected to a bicyclic system (indole-fused).
D180	 Chemical structure of compound D180: A complex molecule featuring a central benzene ring substituted with two methoxy groups and two methylene groups. One methylene group is connected to a piperidine ring, which is further linked via a propyl chain to a bicyclic system (indole-fused). The other methylene group is connected to a pyridine ring, which is substituted with a methyl group and a methylamino group. The pyridine ring is also linked to a piperidine ring, which is connected to a bicyclic system (indole-fused).
D181	 Chemical structure of compound D181: A complex molecule featuring a central benzene ring substituted with two methoxy groups and two methylene groups. One methylene group is connected to a piperidine ring, which is further linked via a propyl chain to a bicyclic system (indole-fused). The other methylene group is connected to a pyridine ring, which is substituted with a methyl group and a methylamino group. The pyridine ring is also linked to a piperidine ring, which is connected to a bicyclic system (indole-fused).

TABLE 2B-continued

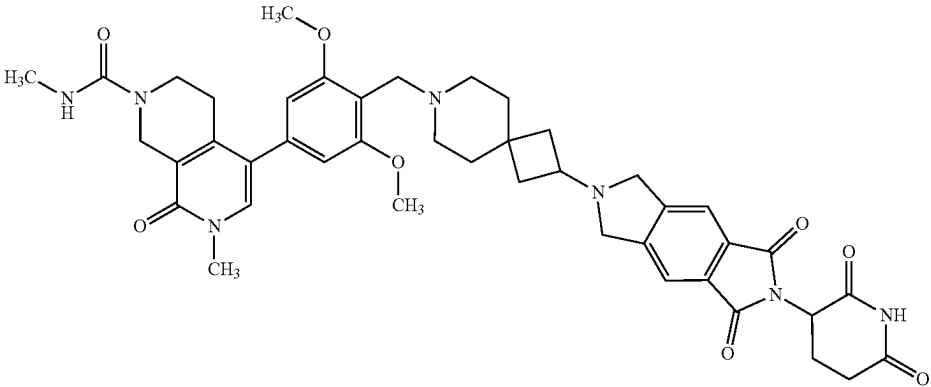
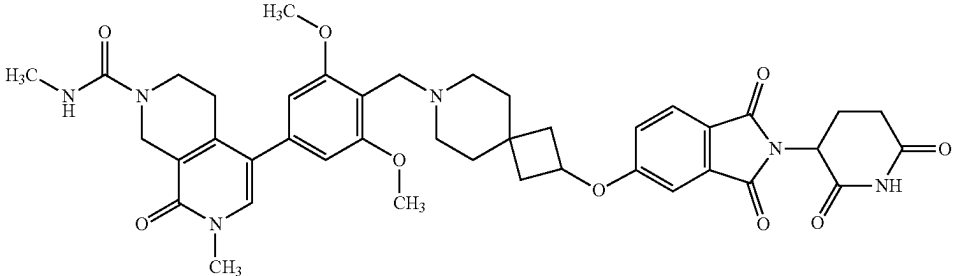
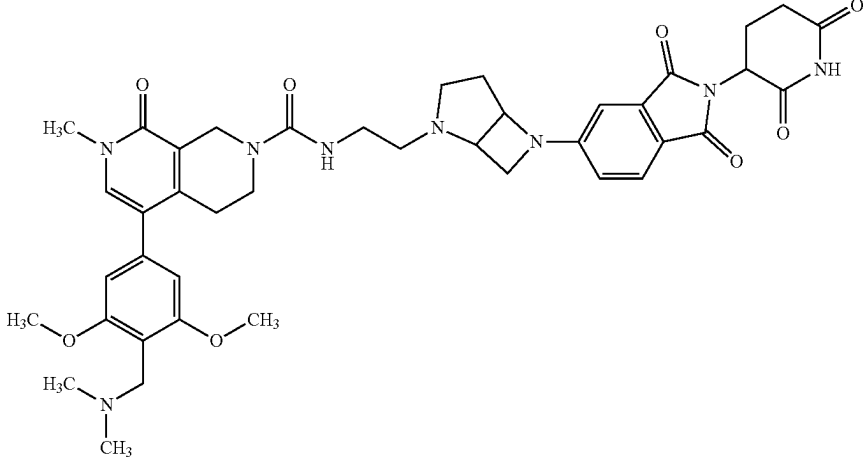
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D182	 Chemical structure of compound D182: A complex molecule featuring a 6-methoxy-2-(4-methoxy-2-methyl-1,2,3,4-tetrahydropyridin-6-yl)-4-methoxyphenyl group connected via a piperidine ring to a cyclobutane ring, which is further connected to a 1,2,3,4-tetrahydroquinoline-5-carboxamide moiety, and finally to a 2,6-pyridinedicarboxamide moiety.
D183	 Chemical structure of compound D183: A complex molecule featuring a 6-methoxy-2-(4-methoxy-2-methyl-1,2,3,4-tetrahydropyridin-6-yl)-4-methoxyphenyl group connected via a piperidine ring to a cyclobutane ring, which is further connected to a 1,2,3,4-tetrahydroquinoline-5-carboxamide moiety, and finally to a 2,6-pyridinedicarboxamide moiety.
D184	 Chemical structure of compound D184: A complex molecule featuring a 6-methoxy-2-(4-methoxy-2-methyl-1,2,3,4-tetrahydropyridin-6-yl)-4-methoxyphenyl group connected via a piperidine ring to a cyclobutane ring, which is further connected to a 1,2,3,4-tetrahydroquinoline-5-carboxamide moiety, and finally to a 2,6-pyridinedicarboxamide moiety.

TABLE 2B-continued

Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D185	
D186	
D187	

TABLE 2B-continued

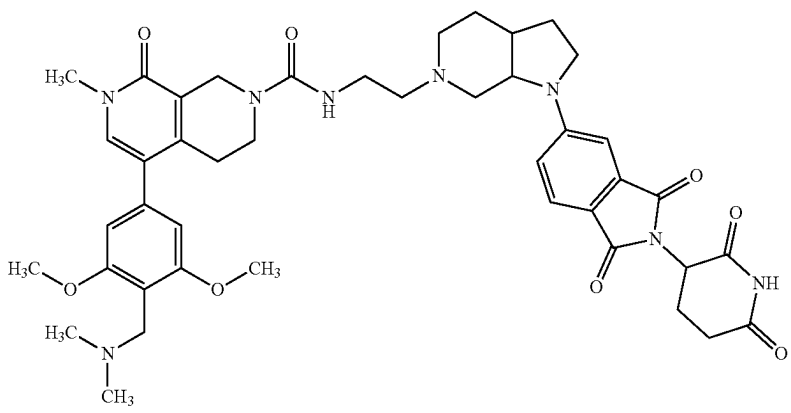
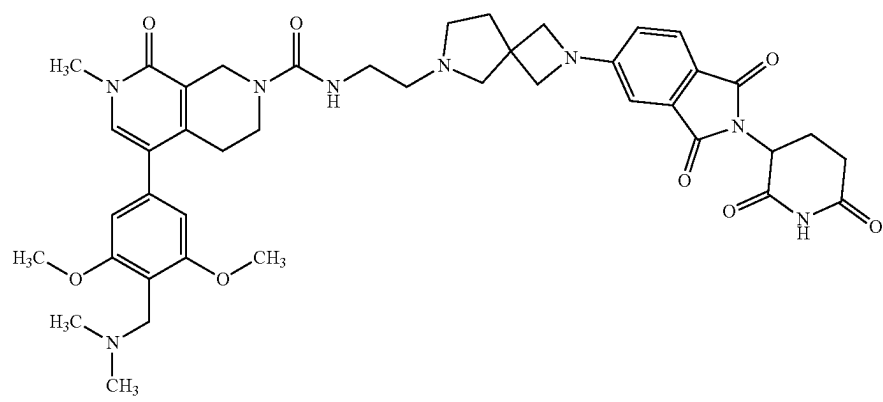
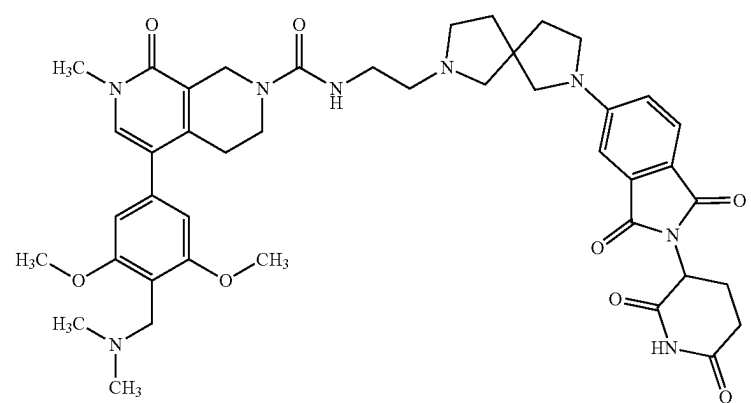
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D188	 Chemical structure of compound D188: A complex molecule featuring a 6-methyl-4-methylpiperidin-2(1H)-one ring system. The piperidine ring is substituted with a methyl group at the 6-position and a methylpiperidin-2(1H)-one group at the 4-position. The piperidine ring is further substituted with a 2,4-dimethoxyphenyl group at the 2-position. The 2,4-dimethoxyphenyl group is substituted with a dimethylamino group at the 3-position. The piperidine ring is further substituted with a 2,4-dimethoxyphenyl group at the 2-position. The 2,4-dimethoxyphenyl group is substituted with a dimethylamino group at the 3-position. The piperidine ring is further substituted with a 2,4-dimethoxyphenyl group at the 2-position. The 2,4-dimethoxyphenyl group is substituted with a dimethylamino group at the 3-position.
D189	 Chemical structure of compound D189: A complex molecule featuring a 6-methyl-4-methylpiperidin-2(1H)-one ring system. The piperidine ring is substituted with a methyl group at the 6-position and a methylpiperidin-2(1H)-one group at the 4-position. The piperidine ring is further substituted with a 2,4-dimethoxyphenyl group at the 2-position. The 2,4-dimethoxyphenyl group is substituted with a dimethylamino group at the 3-position. The piperidine ring is further substituted with a 2,4-dimethoxyphenyl group at the 2-position. The 2,4-dimethoxyphenyl group is substituted with a dimethylamino group at the 3-position. The piperidine ring is further substituted with a 2,4-dimethoxyphenyl group at the 2-position. The 2,4-dimethoxyphenyl group is substituted with a dimethylamino group at the 3-position.
D190	 Chemical structure of compound D190: A complex molecule featuring a 6-methyl-4-methylpiperidin-2(1H)-one ring system. The piperidine ring is substituted with a methyl group at the 6-position and a methylpiperidin-2(1H)-one group at the 4-position. The piperidine ring is further substituted with a 2,4-dimethoxyphenyl group at the 2-position. The 2,4-dimethoxyphenyl group is substituted with a dimethylamino group at the 3-position. The piperidine ring is further substituted with a 2,4-dimethoxyphenyl group at the 2-position. The 2,4-dimethoxyphenyl group is substituted with a dimethylamino group at the 3-position. The piperidine ring is further substituted with a 2,4-dimethoxyphenyl group at the 2-position. The 2,4-dimethoxyphenyl group is substituted with a dimethylamino group at the 3-position.

TABLE 2B-continued

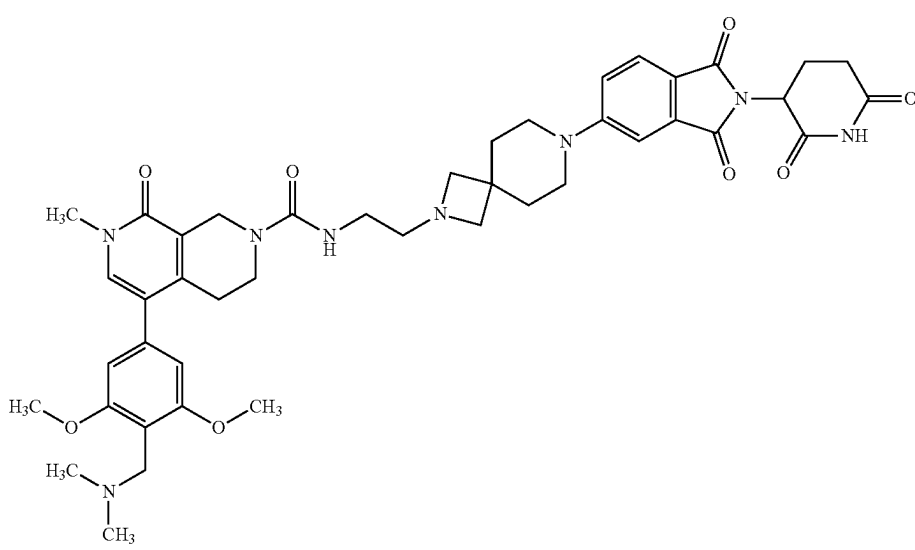
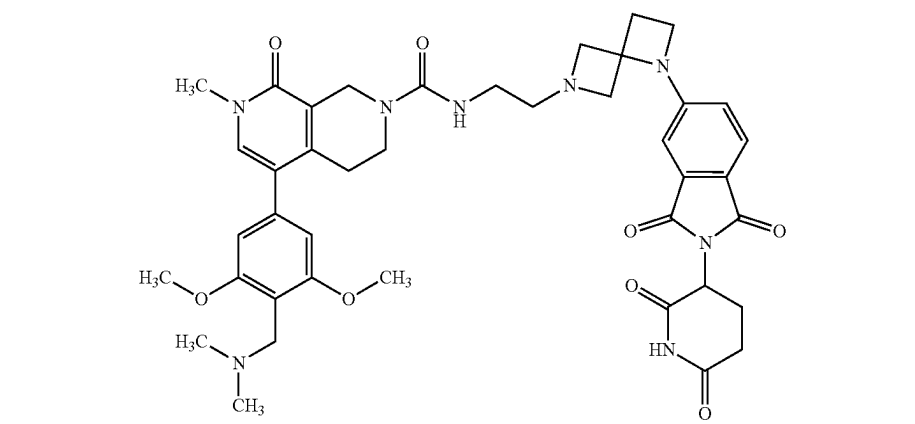
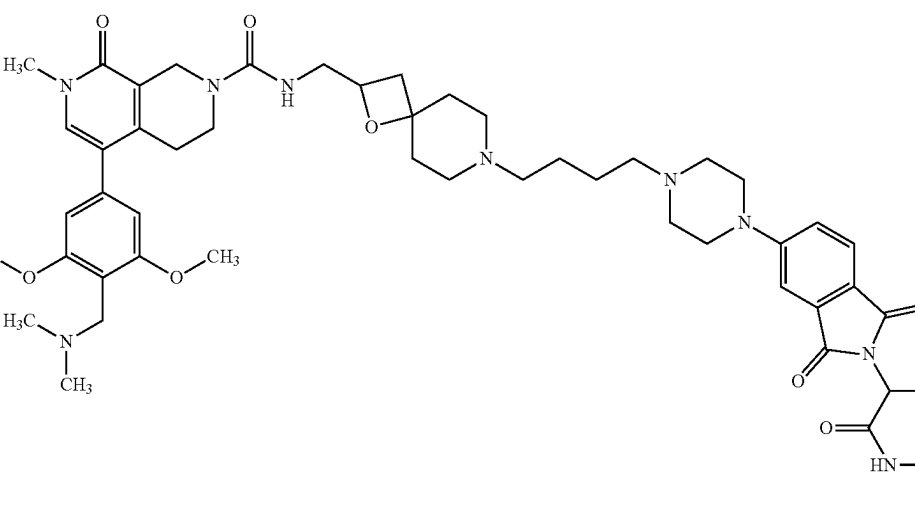
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D191	
D192	
D193	

TABLE 2B-continued

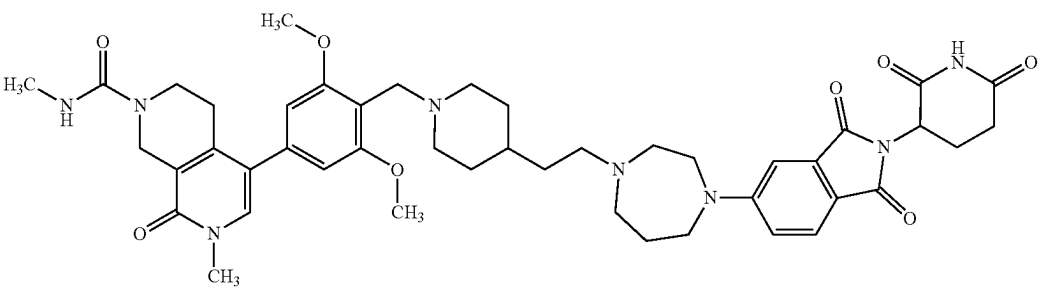
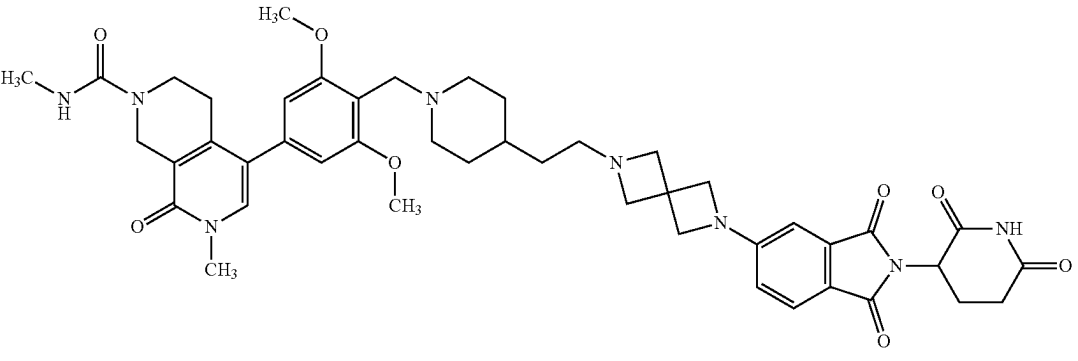
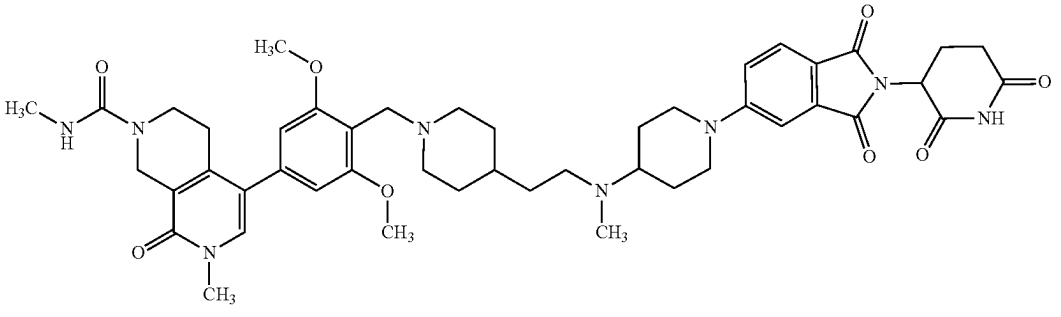
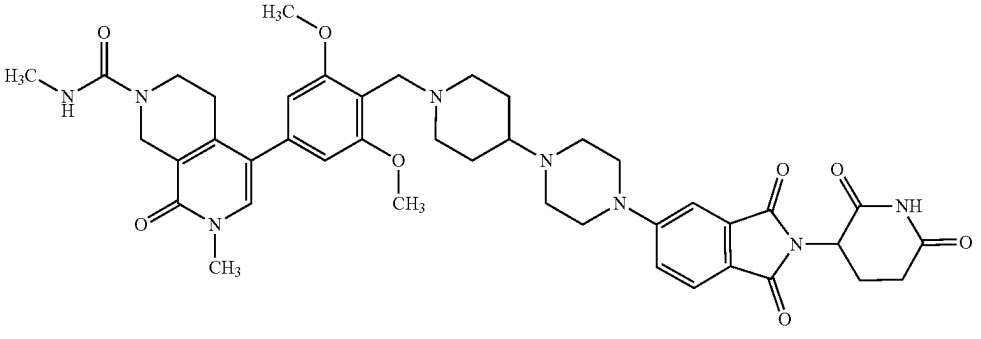
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D198	 Chemical structure of compound D198: A complex molecule featuring a pyridine ring substituted with a methylamino group, a methyl group, and a methoxy-substituted benzene ring. This benzene ring is further substituted with a piperidine ring, which is connected via a propyl chain to an azepane ring. The azepane ring is connected to a benzimidazole system, which is substituted with a piperidine ring and a methylamino group.
D199	 Chemical structure of compound D199: A complex molecule featuring a pyridine ring substituted with a methylamino group, a methyl group, and a methoxy-substituted benzene ring. This benzene ring is further substituted with a piperidine ring, which is connected via a propyl chain to a bicyclic azetidine system. The bicyclic azetidine system is connected to a benzimidazole system, which is substituted with a piperidine ring and a methylamino group.
D200	 Chemical structure of compound D200: A complex molecule featuring a pyridine ring substituted with a methylamino group, a methyl group, and a methoxy-substituted benzene ring. This benzene ring is further substituted with a piperidine ring, which is connected via a propyl chain to a piperidine ring. The piperidine ring is connected to a benzimidazole system, which is substituted with a piperidine ring and a methylamino group.
D201	 Chemical structure of compound D201: A complex molecule featuring a pyridine ring substituted with a methylamino group, a methyl group, and a methoxy-substituted benzene ring. This benzene ring is further substituted with a piperidine ring, which is connected via a propyl chain to a piperazine ring. The piperazine ring is connected to a benzimidazole system, which is substituted with a piperidine ring and a methylamino group.

TABLE 2B-continued

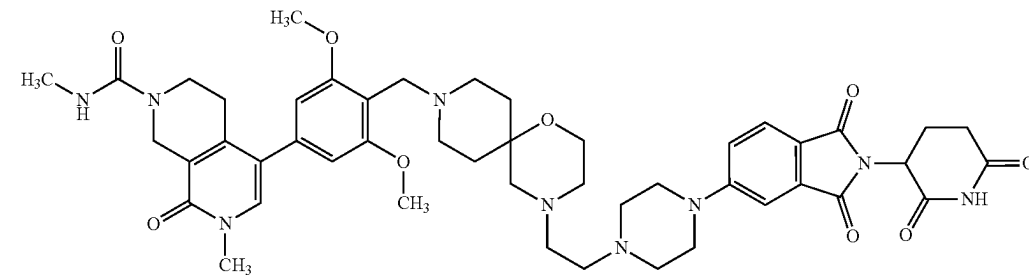
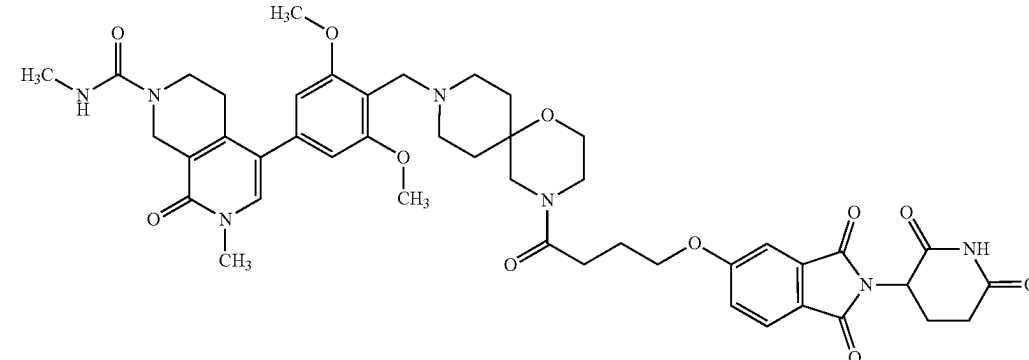
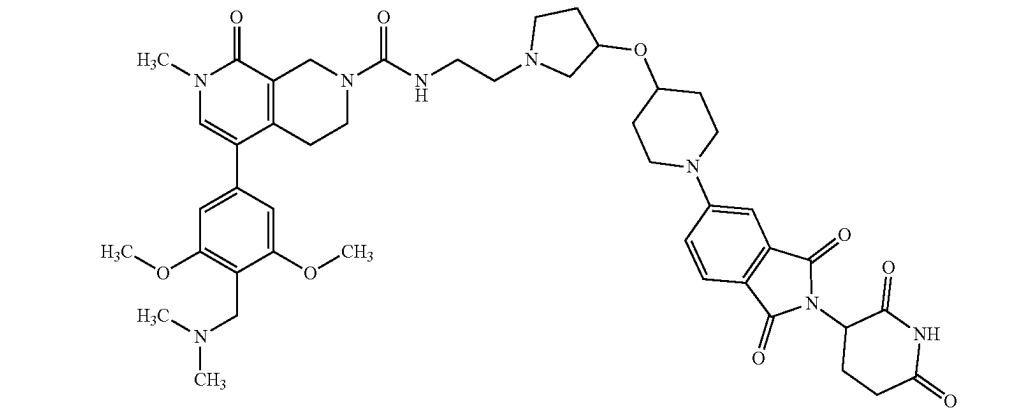
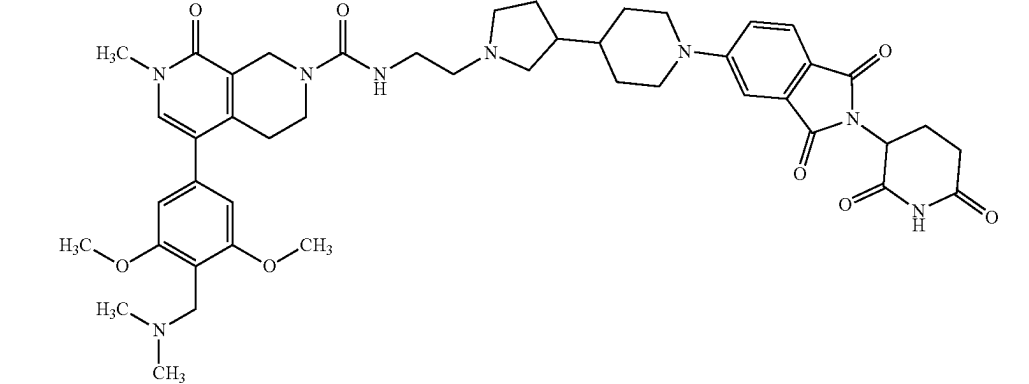
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D202	
D203	
D204	
D205	

TABLE 2B-continued

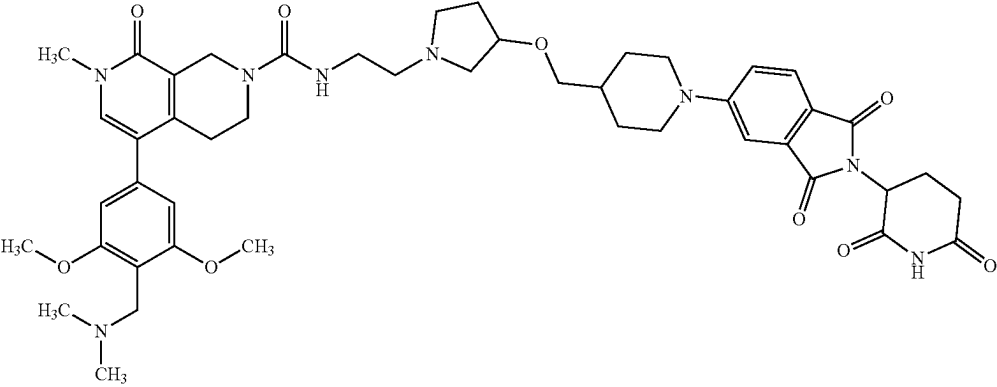
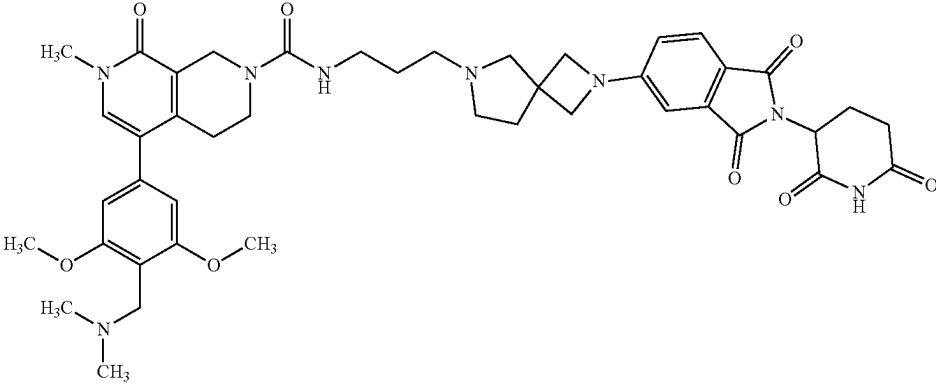
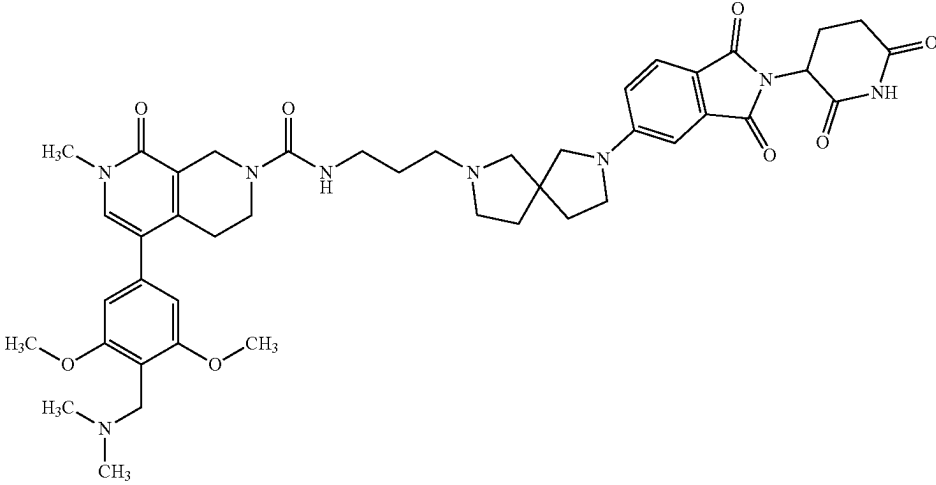
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D206	 Chemical structure of compound D206: A complex molecule featuring a 6-methyl-4-(dimethylamino)quinoline-2,3-dione moiety linked via an amide bond to a 3-(4-methoxyphenyl)-2-methyl-5-oxo-1,4-dihydropyridine-6-carboxamide moiety. The pyridine ring is substituted with a methoxy group and a dimethylamino group. The amide linkage connects to a 3-(4-methoxyphenyl)-2-methyl-5-oxo-1,4-dihydropyridine-6-carboxamide moiety.
D207	 Chemical structure of compound D207: A complex molecule featuring a 6-methyl-4-(dimethylamino)quinoline-2,3-dione moiety linked via an amide bond to a 3-(4-methoxyphenyl)-2-methyl-5-oxo-1,4-dihydropyridine-6-carboxamide moiety. The pyridine ring is substituted with a methoxy group and a dimethylamino group. The amide linkage connects to a 3-(4-methoxyphenyl)-2-methyl-5-oxo-1,4-dihydropyridine-6-carboxamide moiety.
D208	 Chemical structure of compound D208: A complex molecule featuring a 6-methyl-4-(dimethylamino)quinoline-2,3-dione moiety linked via an amide bond to a 3-(4-methoxyphenyl)-2-methyl-5-oxo-1,4-dihydropyridine-6-carboxamide moiety. The pyridine ring is substituted with a methoxy group and a dimethylamino group. The amide linkage connects to a 3-(4-methoxyphenyl)-2-methyl-5-oxo-1,4-dihydropyridine-6-carboxamide moiety.

TABLE 2B-continued

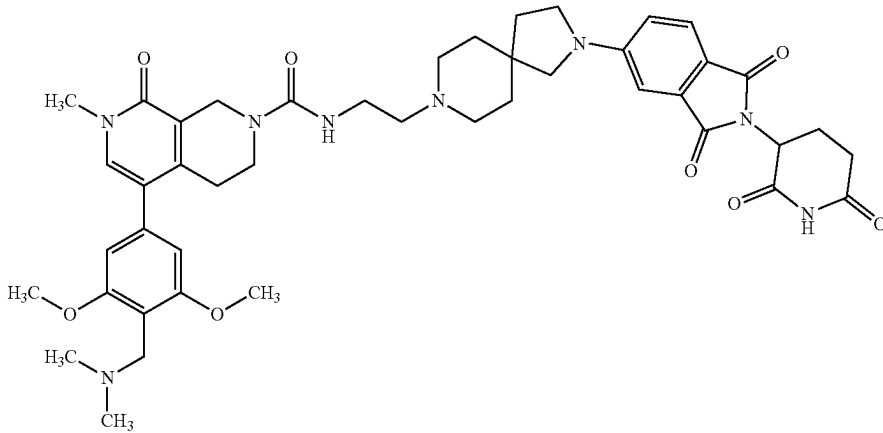
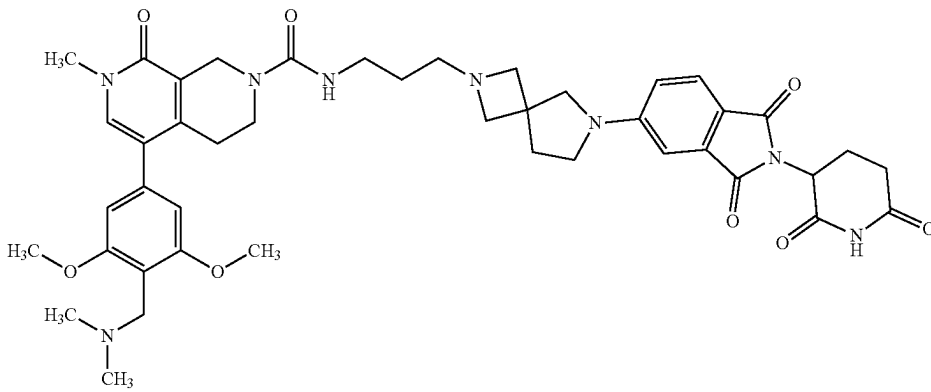
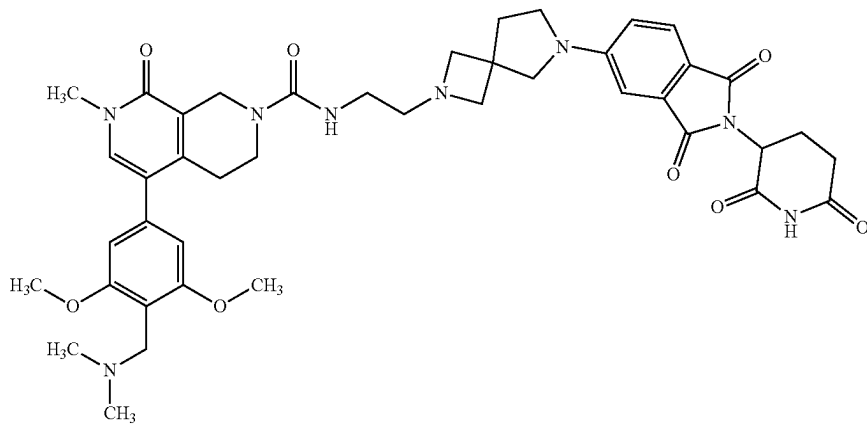
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D209	 Chemical structure of compound D209: A complex molecule featuring a 6-methyl-4-(dimethylamino)quinoline-2-carboxamide group linked via an amide bond to a 1,4-bis(piperidin-2-yl)butane chain. This chain is further linked via an amide bond to a 1,4-bis(4-methoxyphenyl)piperazine chain, which is finally linked via an amide bond to a 1,4-bis(4-methoxyphenyl)piperazine chain. The structure also includes a 1,4-bis(4-methoxyphenyl)piperazine chain and a 1,4-bis(4-methoxyphenyl)piperazine chain.
D210	 Chemical structure of compound D210: A complex molecule featuring a 6-methyl-4-(dimethylamino)quinoline-2-carboxamide group linked via an amide bond to a 1,4-bis(piperidin-2-yl)butane chain. This chain is further linked via an amide bond to a 1,4-bis(4-methoxyphenyl)piperazine chain, which is finally linked via an amide bond to a 1,4-bis(4-methoxyphenyl)piperazine chain. The structure also includes a 1,4-bis(4-methoxyphenyl)piperazine chain and a 1,4-bis(4-methoxyphenyl)piperazine chain.
D211	 Chemical structure of compound D211: A complex molecule featuring a 6-methyl-4-(dimethylamino)quinoline-2-carboxamide group linked via an amide bond to a 1,4-bis(piperidin-2-yl)butane chain. This chain is further linked via an amide bond to a 1,4-bis(4-methoxyphenyl)piperazine chain, which is finally linked via an amide bond to a 1,4-bis(4-methoxyphenyl)piperazine chain. The structure also includes a 1,4-bis(4-methoxyphenyl)piperazine chain and a 1,4-bis(4-methoxyphenyl)piperazine chain.

TABLE 2B-continued

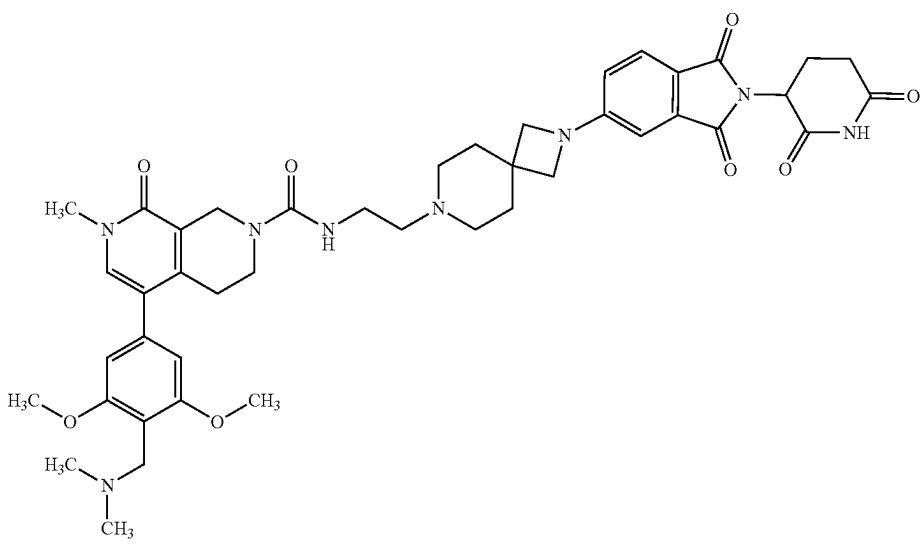
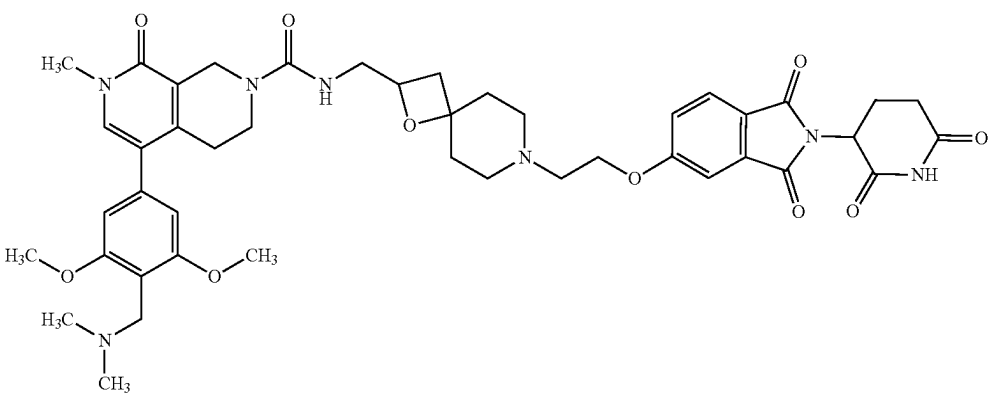
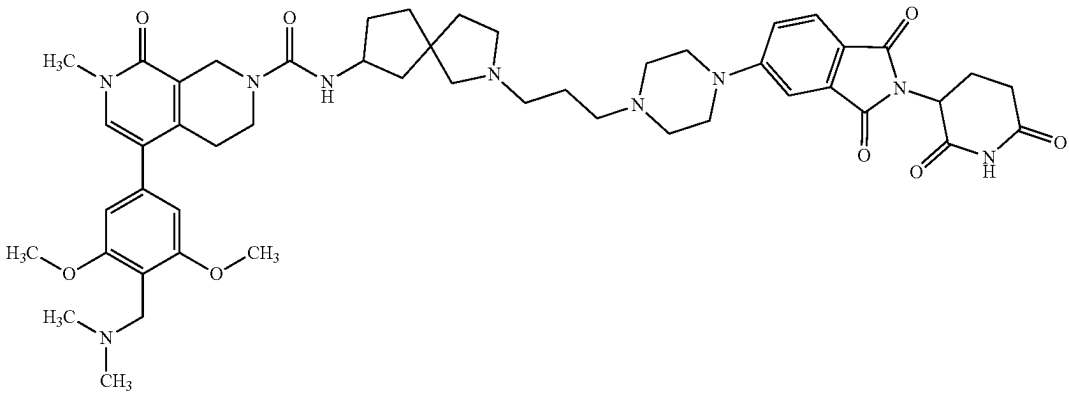
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D212	 Chemical structure of compound D212: A 6-methoxy-2-(dimethylaminomethyl)-3,4-dihydroquinolin-2(1H)-one derivative. The quinoline ring is substituted with a 4-methoxyphenyl group at position 6 and a dimethylaminomethyl group at position 3. The quinoline ring is connected via an amide linkage to a 4-(2,2,6,6-tetramethyl-1,3-dioxane-5-yl)butanamide derivative. The butanamide chain is further substituted with a 4-(2,2,6,6-tetramethyl-1,3-dioxane-5-yl)butanamide derivative.
D213	 Chemical structure of compound D213: A 6-methoxy-2-(dimethylaminomethyl)-3,4-dihydroquinolin-2(1H)-one derivative. The quinoline ring is substituted with a 4-methoxyphenyl group at position 6 and a dimethylaminomethyl group at position 3. The quinoline ring is connected via an amide linkage to a 4-(2,2,6,6-tetramethyl-1,3-dioxane-5-yl)butanamide derivative. The butanamide chain is further substituted with a 4-(2,2,6,6-tetramethyl-1,3-dioxane-5-yl)butanamide derivative.
D214	 Chemical structure of compound D214: A 6-methoxy-2-(dimethylaminomethyl)-3,4-dihydroquinolin-2(1H)-one derivative. The quinoline ring is substituted with a 4-methoxyphenyl group at position 6 and a dimethylaminomethyl group at position 3. The quinoline ring is connected via an amide linkage to a 4-(2,2,6,6-tetramethyl-1,3-dioxane-5-yl)butanamide derivative. The butanamide chain is further substituted with a 4-(2,2,6,6-tetramethyl-1,3-dioxane-5-yl)butanamide derivative.

TABLE 2B-continued

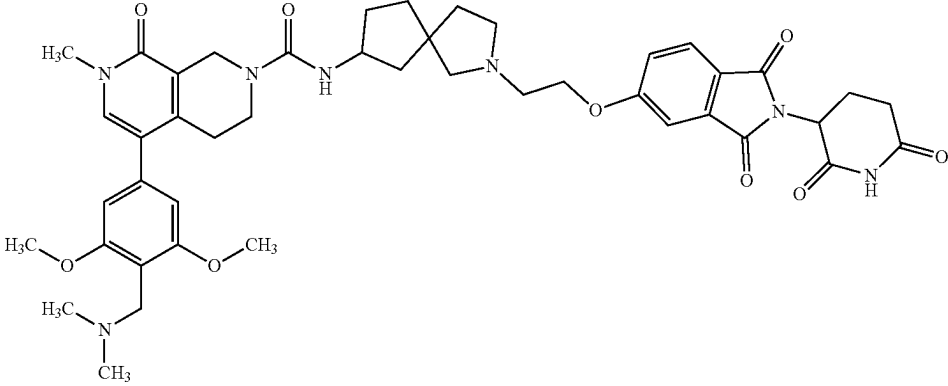
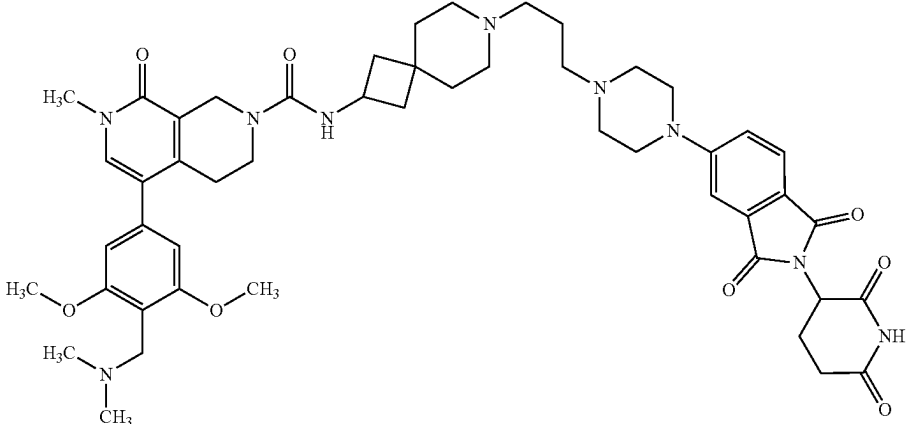
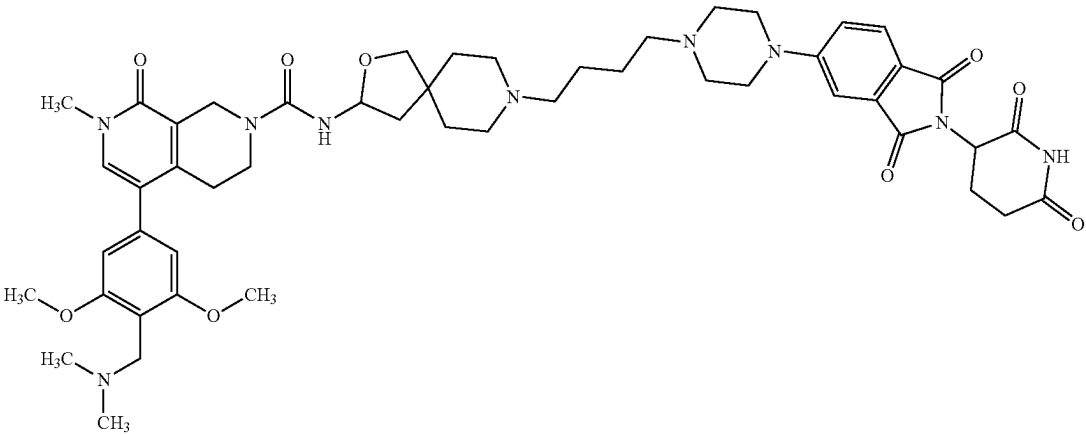
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D215	 Chemical structure of compound D215: A 6-methyl-2,3,4,5-tetrahydroquinolin-6(1H)-one core substituted at the 8-position with a 4-(dimethylamino)-3,5-dimethoxyphenyl group. The 4-position of the quinolinone ring is substituted with a carbamate group linked to a 1,4-diazabicyclo[2.2.2]octane moiety, which is further connected via an ethyl chain to an ether linkage. The ether is connected to a benzimidazole ring system, which is substituted at the 2-position with a 2,6-pyridinedione derivative.
D216	 Chemical structure of compound D216: A 6-methyl-2,3,4,5-tetrahydroquinolin-6(1H)-one core substituted at the 8-position with a 4-(dimethylamino)-3,5-dimethoxyphenyl group. The 4-position of the quinolinone ring is substituted with a carbamate group linked to a 1,4-diazabicyclo[2.2.2]octane moiety, which is further connected via an ethyl chain to a piperazine ring. The piperazine ring is connected to a benzimidazole ring system, which is substituted at the 2-position with a 2,6-pyridinedione derivative.
D217	 Chemical structure of compound D217: A 6-methyl-2,3,4,5-tetrahydroquinolin-6(1H)-one core substituted at the 8-position with a 4-(dimethylamino)-3,5-dimethoxyphenyl group. The 4-position of the quinolinone ring is substituted with a carbamate group linked to a 1,4-diazabicyclo[2.2.2]octane moiety, which is further connected via an ethyl chain to a piperazine ring. The piperazine ring is connected to a benzimidazole ring system, which is substituted at the 2-position with a 2,6-pyridinedione derivative.

TABLE 2B-continued

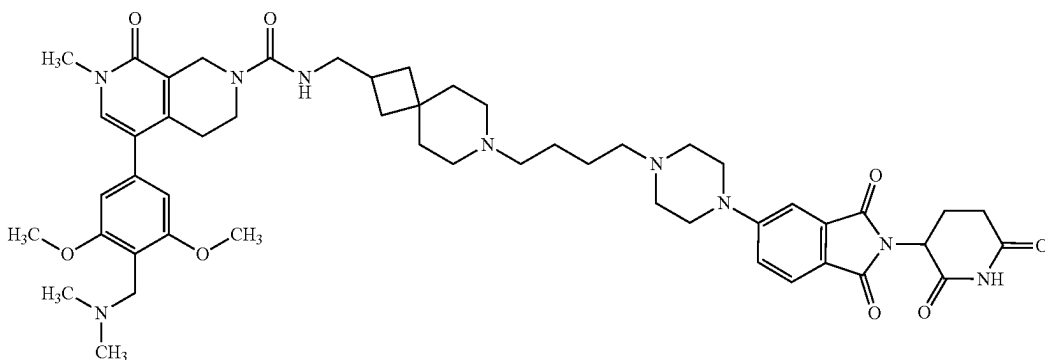
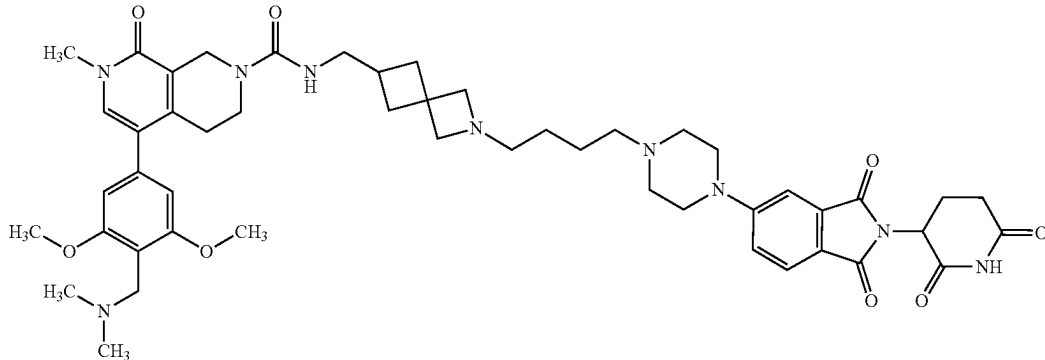
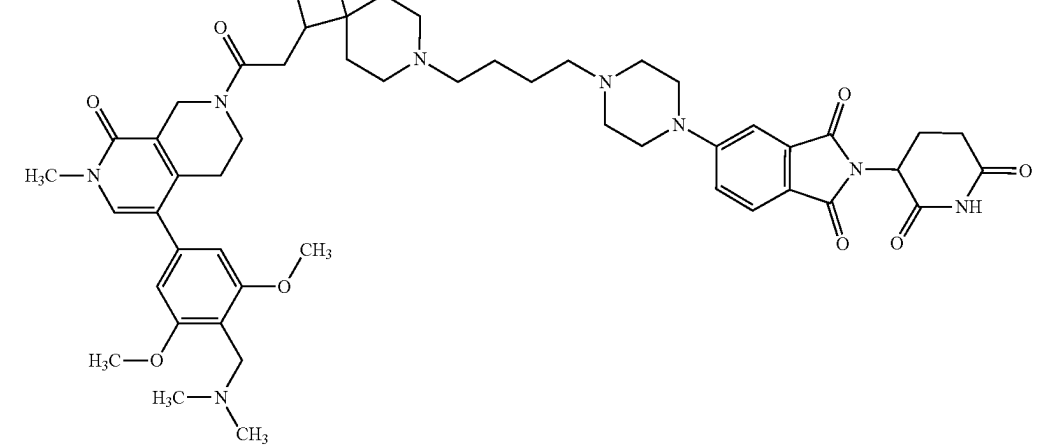
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D218	 Chemical structure of compound D218: A complex molecule featuring a 6-methyl-4-(dimethylamino)pyridine-2-carboxamide moiety linked via an amide bond to a 1,4-bis(piperidin-1-yl)-1,4-bis(methyl)benzene moiety, which is further linked via an amide bond to a 1,4-bis(piperidin-1-yl)-1,4-bis(methyl)benzene moiety, and finally linked via an amide bond to a 1,4-bis(piperidin-1-yl)-1,4-bis(methyl)benzene moiety.
D219	 Chemical structure of compound D219: A complex molecule featuring a 6-methyl-4-(dimethylamino)pyridine-2-carboxamide moiety linked via an amide bond to a 1,4-bis(piperidin-1-yl)-1,4-bis(methyl)benzene moiety, which is further linked via an amide bond to a 1,4-bis(piperidin-1-yl)-1,4-bis(methyl)benzene moiety, and finally linked via an amide bond to a 1,4-bis(piperidin-1-yl)-1,4-bis(methyl)benzene moiety.
D220	 Chemical structure of compound D220: A complex molecule featuring a 6-methyl-4-(dimethylamino)pyridine-2-carboxamide moiety linked via an amide bond to a 1,4-bis(piperidin-1-yl)-1,4-bis(methyl)benzene moiety, which is further linked via an amide bond to a 1,4-bis(piperidin-1-yl)-1,4-bis(methyl)benzene moiety, and finally linked via an amide bond to a 1,4-bis(piperidin-1-yl)-1,4-bis(methyl)benzene moiety.

TABLE 2B-continued

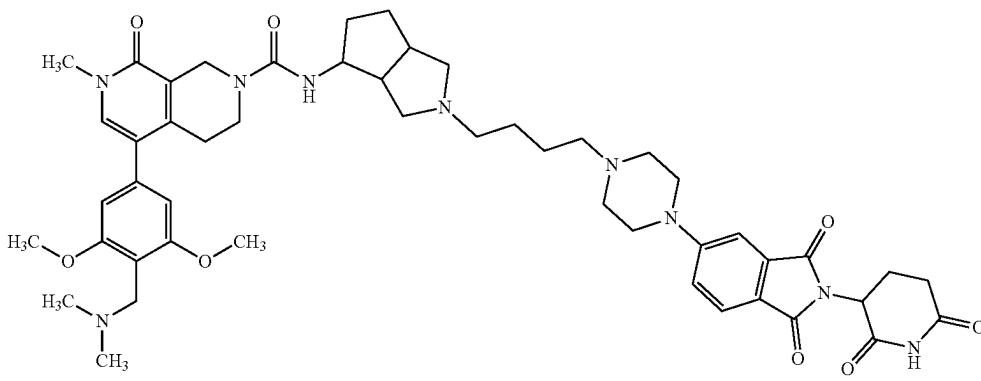
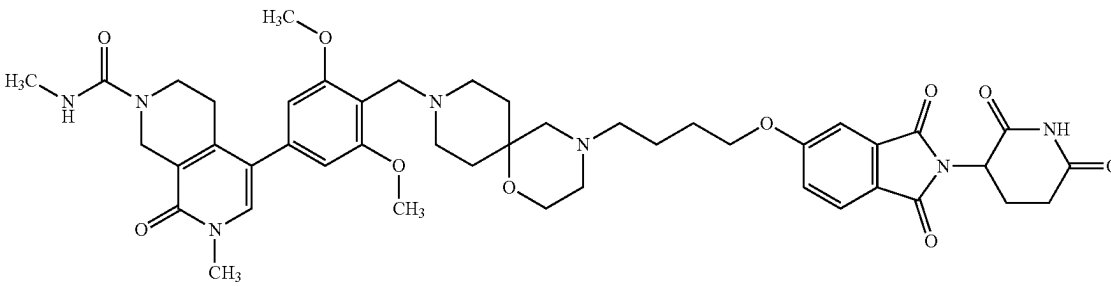
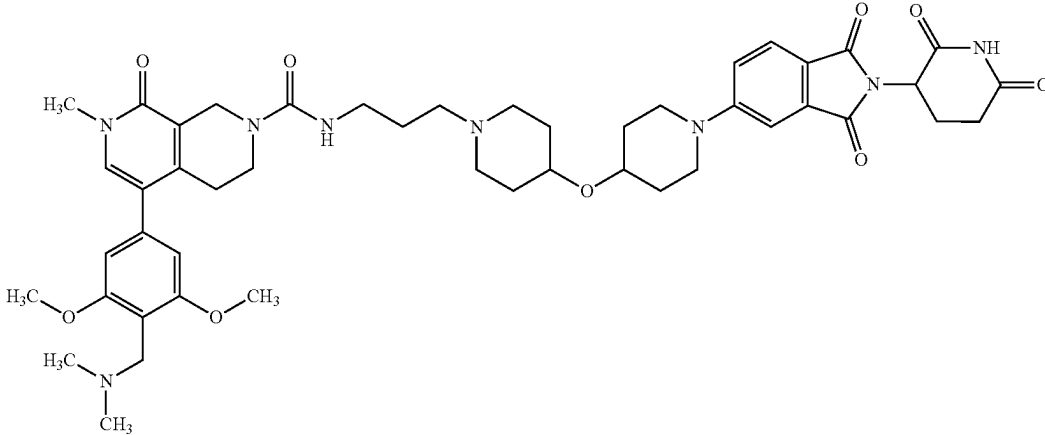
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D221	 Chemical structure of compound D221: A complex molecule featuring a 6-methyl-4-(dimethylamino)pyridine-2-carboxamide moiety linked via a piperidine ring to a 1,2,3,4-tetrahydro-1H-benzod[e][1,2-b:4,5-b']diazepine system. This is further connected via a long chain containing a piperazine ring to a 1,2,3,4-tetrahydro-1H-benzod[e][1,2-b:4,5-b']diazepine system, which is finally linked to a 1,2,3,4-tetrahydro-1H-benzod[e][1,2-b:4,5-b']diazepine system.
D222	 Chemical structure of compound D222: A complex molecule featuring a 6-methoxy-4-(dimethylamino)pyridine-2-carboxamide moiety linked via a piperidine ring to a 1,2,3,4-tetrahydro-1H-benzod[e][1,2-b:4,5-b']diazepine system. This is further connected via a long chain containing a piperazine ring to a 1,2,3,4-tetrahydro-1H-benzod[e][1,2-b:4,5-b']diazepine system, which is finally linked to a 1,2,3,4-tetrahydro-1H-benzod[e][1,2-b:4,5-b']diazepine system.
D223	 Chemical structure of compound D223: A complex molecule featuring a 6-methoxy-4-(dimethylamino)pyridine-2-carboxamide moiety linked via a piperidine ring to a 1,2,3,4-tetrahydro-1H-benzod[e][1,2-b:4,5-b']diazepine system. This is further connected via a long chain containing a piperazine ring to a 1,2,3,4-tetrahydro-1H-benzod[e][1,2-b:4,5-b']diazepine system, which is finally linked to a 1,2,3,4-tetrahydro-1H-benzod[e][1,2-b:4,5-b']diazepine system.

TABLE 2B-continued

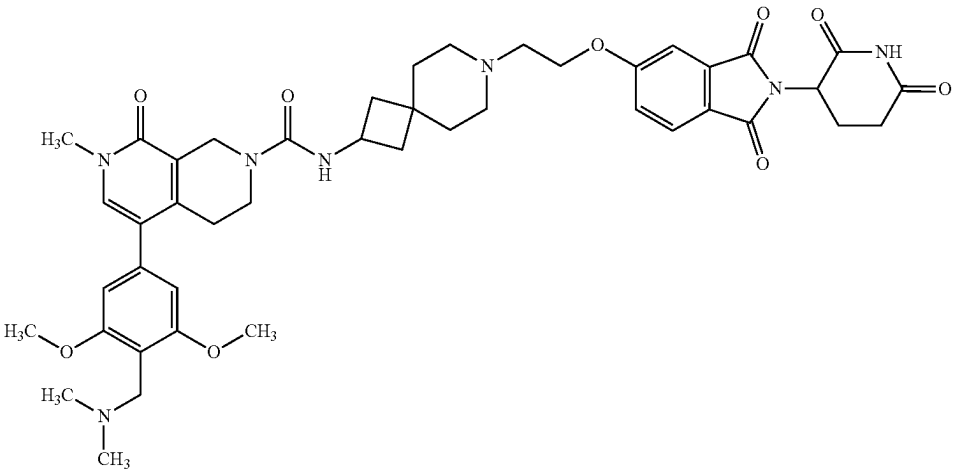
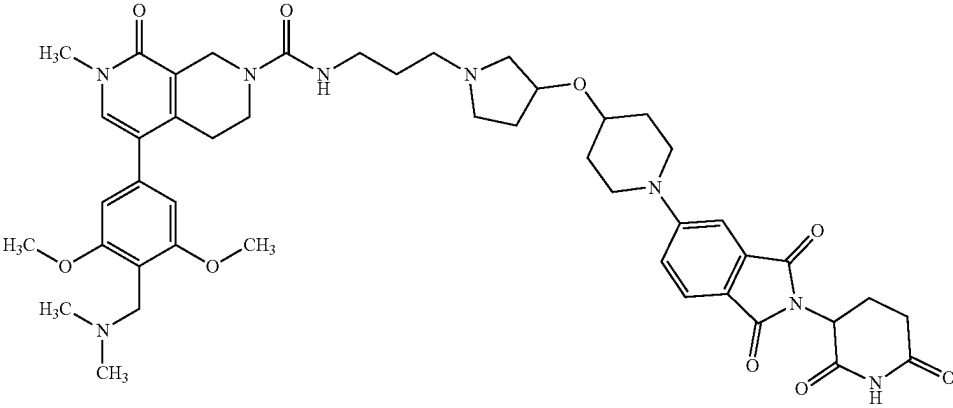
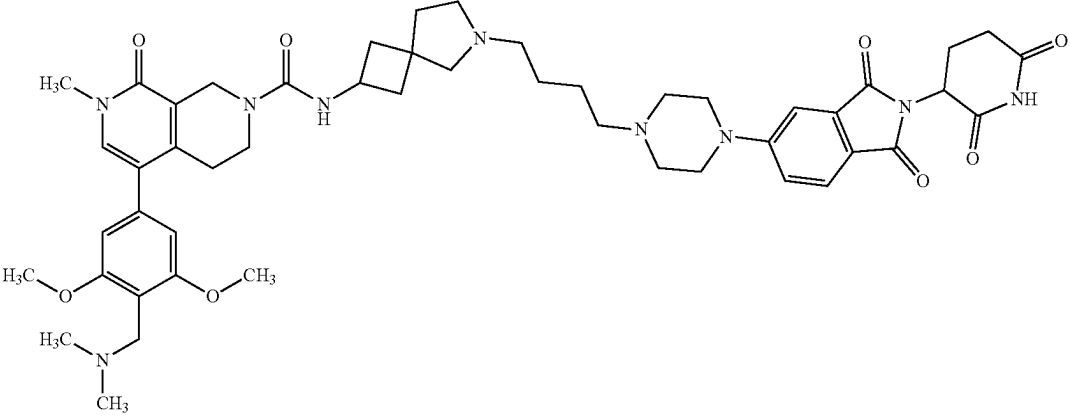
Compound No.	Structure
D224	 Chemical structure of compound D224: A complex molecule featuring a 6-methyl-2,3,4,5-tetrahydroquinolin-6(1H)-one core. The quinoline ring is substituted with a 4-methoxyphenyl group at position 8 and a 2-(dimethylamino)ethyl group at position 7. The quinoline nitrogen is methylated. The quinoline ring is connected via an amide linkage to a 1,4-dioxane ring, which is further connected via an ether linkage to a 1,2,3,4-tetrahydroquinolin-2(1H)-one moiety. The quinoline nitrogen is methylated.
D225	 Chemical structure of compound D225: A complex molecule featuring a 6-methyl-2,3,4,5-tetrahydroquinolin-6(1H)-one core. The quinoline ring is substituted with a 4-methoxyphenyl group at position 8 and a 2-(dimethylamino)ethyl group at position 7. The quinoline nitrogen is methylated. The quinoline ring is connected via an amide linkage to a 1,4-dioxane ring, which is further connected via an ether linkage to a 1,2,3,4-tetrahydroquinolin-2(1H)-one moiety. The quinoline nitrogen is methylated.
D226	 Chemical structure of compound D226: A complex molecule featuring a 6-methyl-2,3,4,5-tetrahydroquinolin-6(1H)-one core. The quinoline ring is substituted with a 4-methoxyphenyl group at position 8 and a 2-(dimethylamino)ethyl group at position 7. The quinoline nitrogen is methylated. The quinoline ring is connected via an amide linkage to a 1,4-dioxane ring, which is further connected via an ether linkage to a 1,2,3,4-tetrahydroquinolin-2(1H)-one moiety. The quinoline nitrogen is methylated.

TABLE 2B-continued

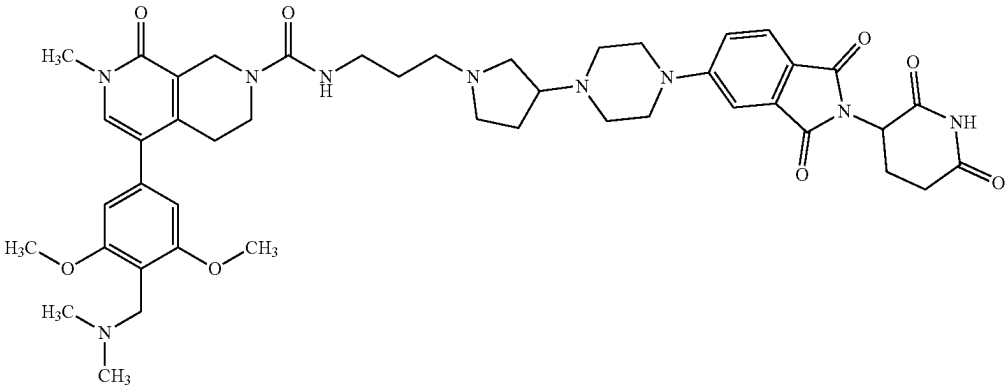
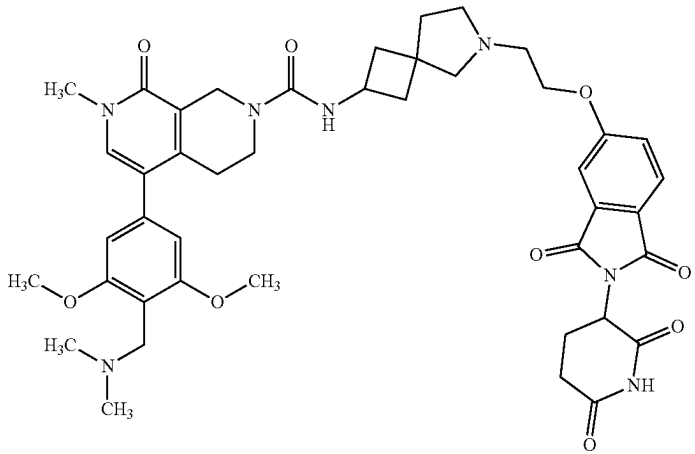
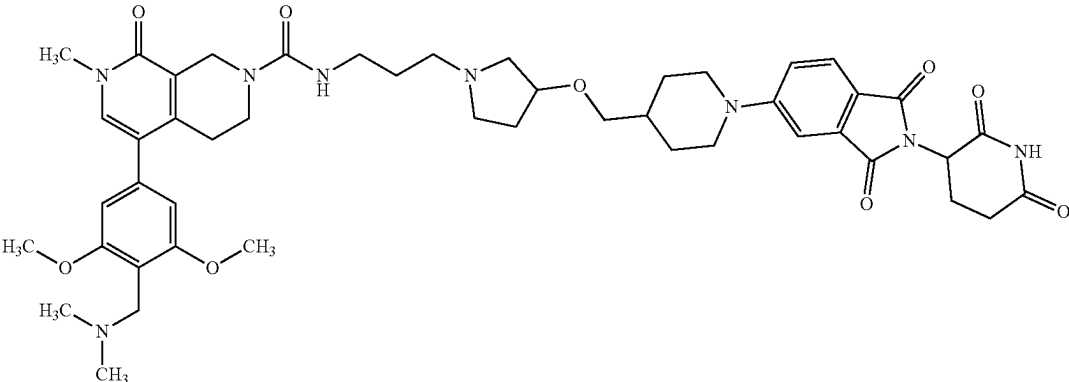
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D227	 Chemical structure of compound D227: A complex molecule featuring a 6-methyl-4-(dimethylamino)quinoline-2,3-dione moiety linked via an amide bond to a 4-(3-(4-methoxy-2-(dimethylamino)ethoxy)phenyl)-2,3-dioxo-1,2,3,4-tetrahydroquinoline moiety. The linker consists of a 4-((4-methoxy-2-(dimethylamino)ethoxy)phenyl)-2,3-dioxo-1,2,3,4-tetrahydroquinoline moiety connected via an amide bond to a 4-(3-(4-methoxy-2-(dimethylamino)ethoxy)phenyl)-2,3-dioxo-1,2,3,4-tetrahydroquinoline moiety.
D228	 Chemical structure of compound D228: A complex molecule featuring a 6-methyl-4-(dimethylamino)quinoline-2,3-dione moiety linked via an amide bond to a 4-(3-(4-methoxy-2-(dimethylamino)ethoxy)phenyl)-2,3-dioxo-1,2,3,4-tetrahydroquinoline moiety. The linker consists of a 4-(3-(4-methoxy-2-(dimethylamino)ethoxy)phenyl)-2,3-dioxo-1,2,3,4-tetrahydroquinoline moiety connected via an amide bond to a 4-(3-(4-methoxy-2-(dimethylamino)ethoxy)phenyl)-2,3-dioxo-1,2,3,4-tetrahydroquinoline moiety.
D229	 Chemical structure of compound D229: A complex molecule featuring a 6-methyl-4-(dimethylamino)quinoline-2,3-dione moiety linked via an amide bond to a 4-(3-(4-methoxy-2-(dimethylamino)ethoxy)phenyl)-2,3-dioxo-1,2,3,4-tetrahydroquinoline moiety. The linker consists of a 4-(3-(4-methoxy-2-(dimethylamino)ethoxy)phenyl)-2,3-dioxo-1,2,3,4-tetrahydroquinoline moiety connected via an amide bond to a 4-(3-(4-methoxy-2-(dimethylamino)ethoxy)phenyl)-2,3-dioxo-1,2,3,4-tetrahydroquinoline moiety.

TABLE 2B-continued

Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D230	
D231	
D232	

TABLE 2B-continued

Compound No.	Structure
D233	
D234	
D235	

TABLE 2B-continued

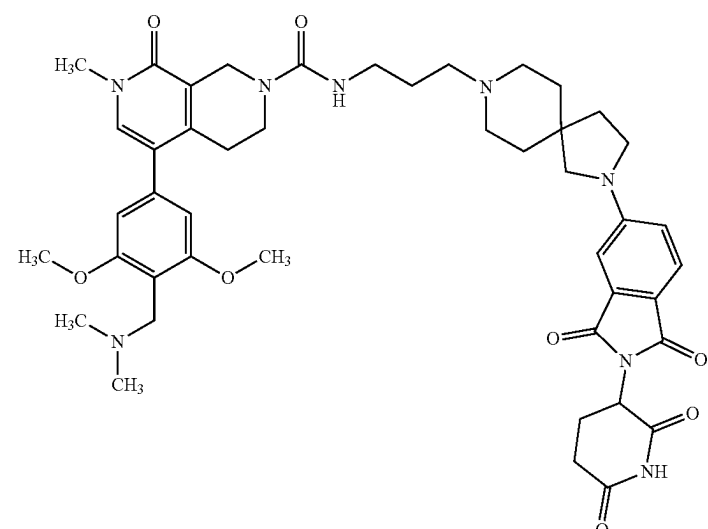
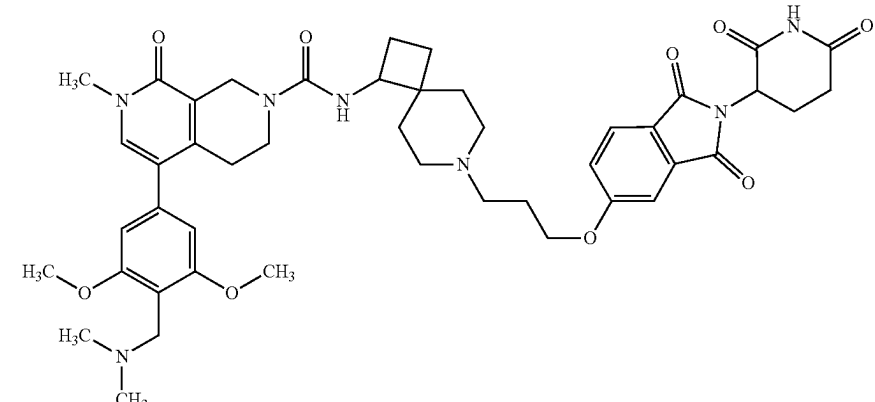
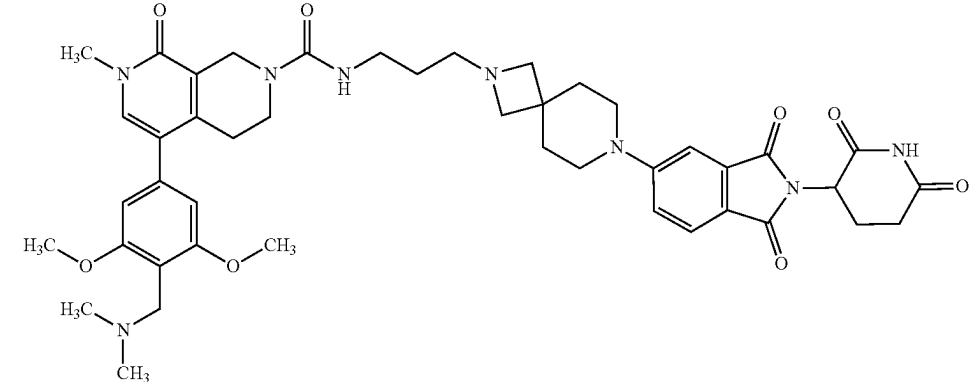
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D236	
D237	
D238	

TABLE 2B-continued

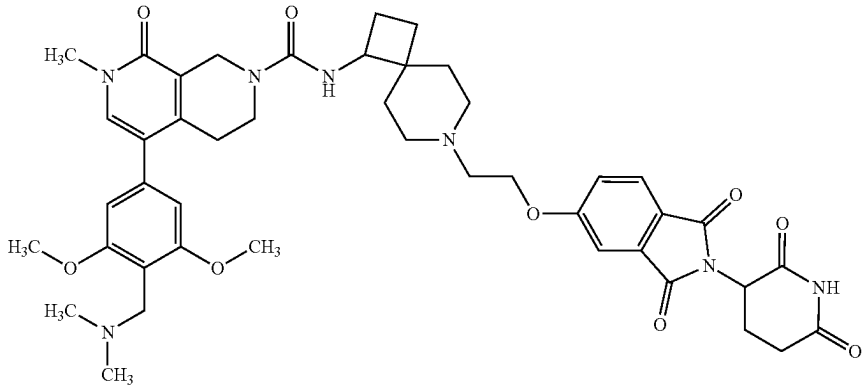
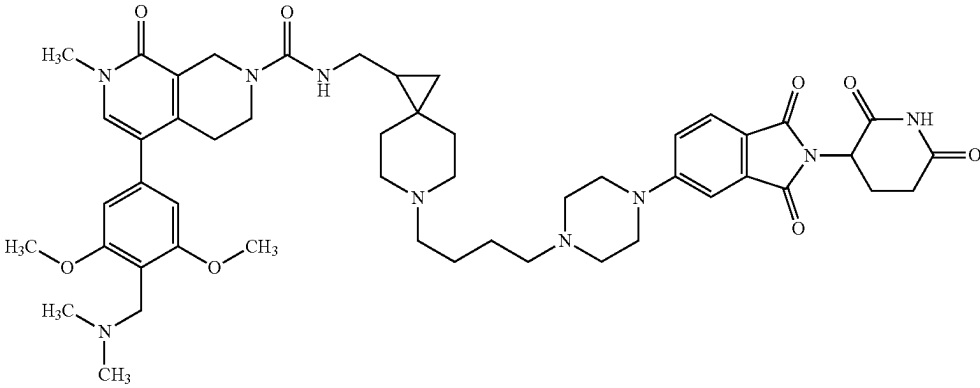
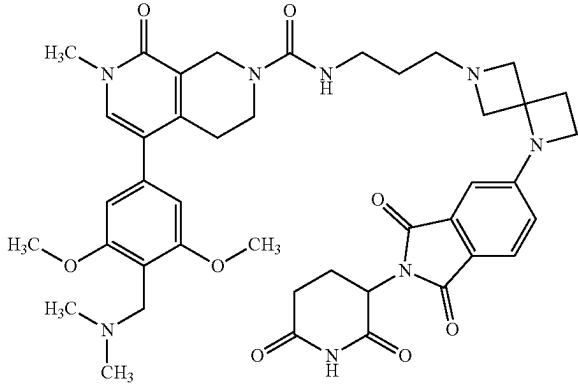
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D239	
D240	
D241	

TABLE 2B-continued

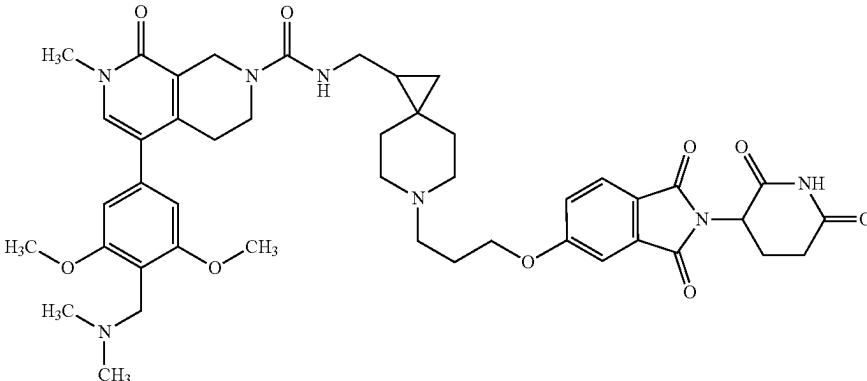
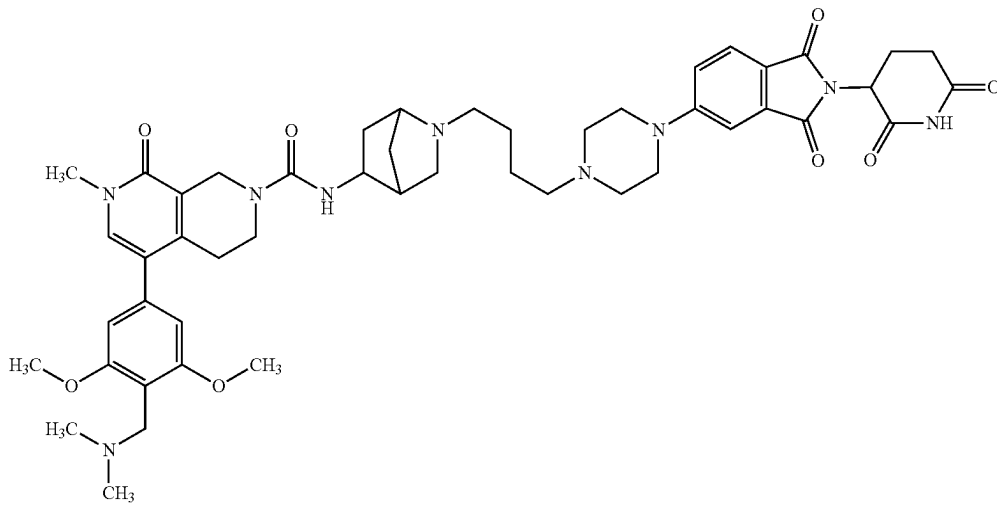
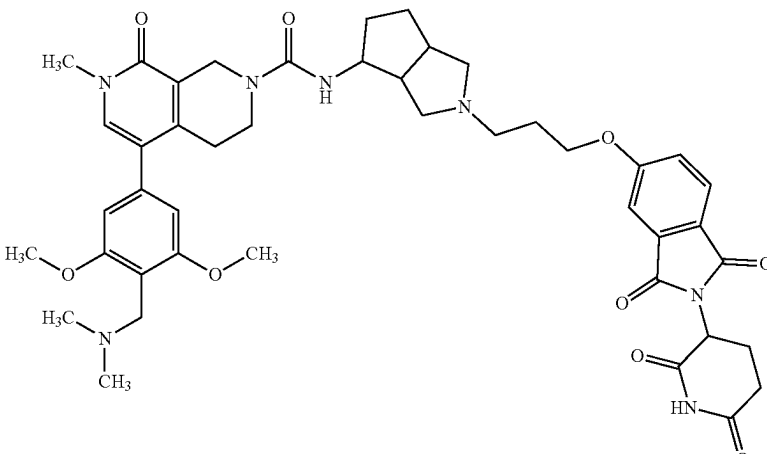
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D242	
D243	
D244	

TABLE 2B-continued

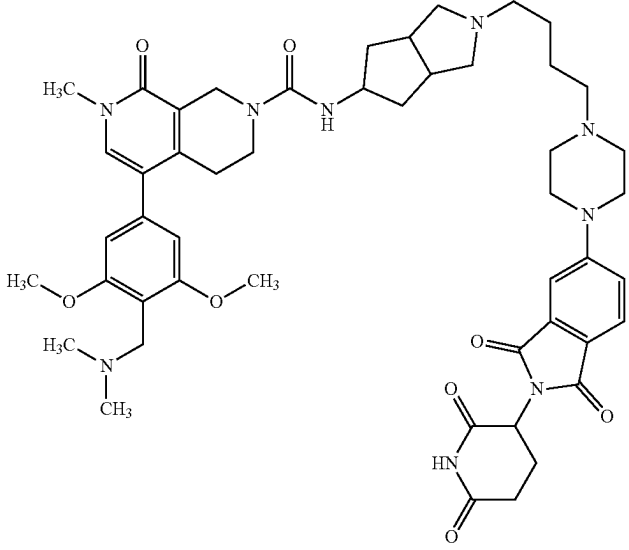
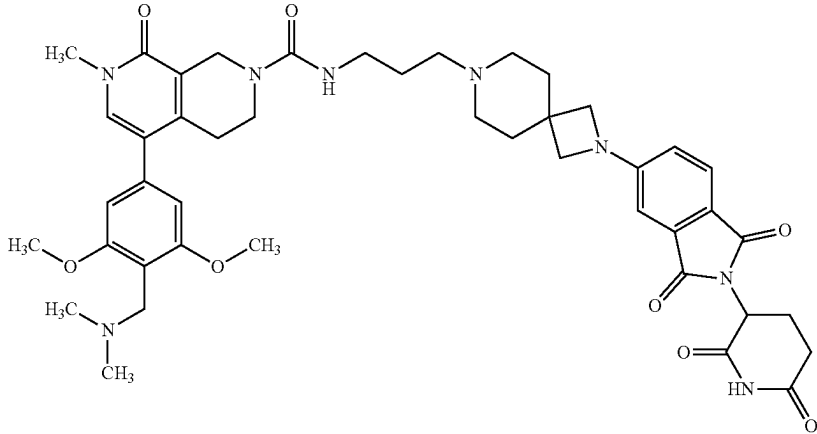
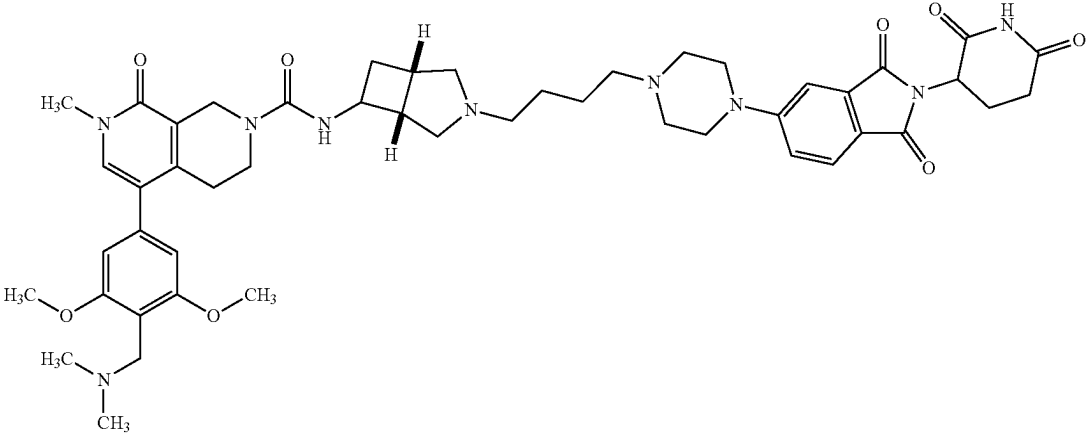
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D245	
D246	
D247	

TABLE 2B-continued

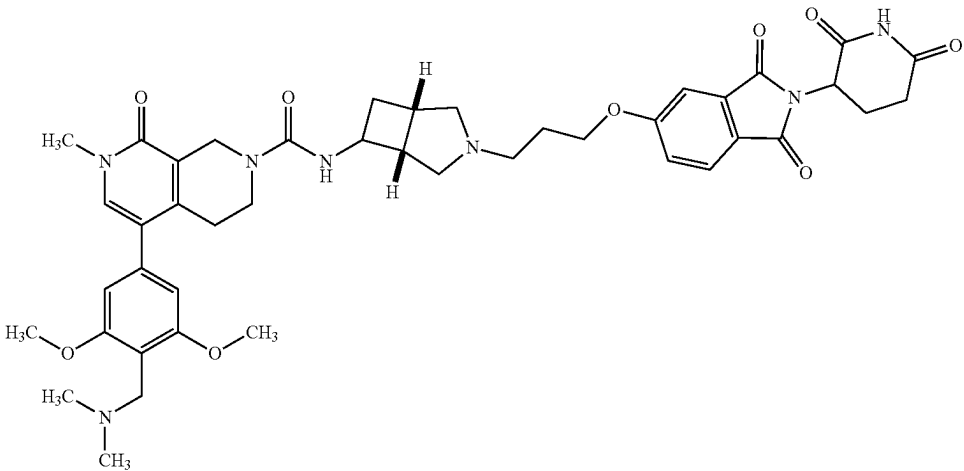
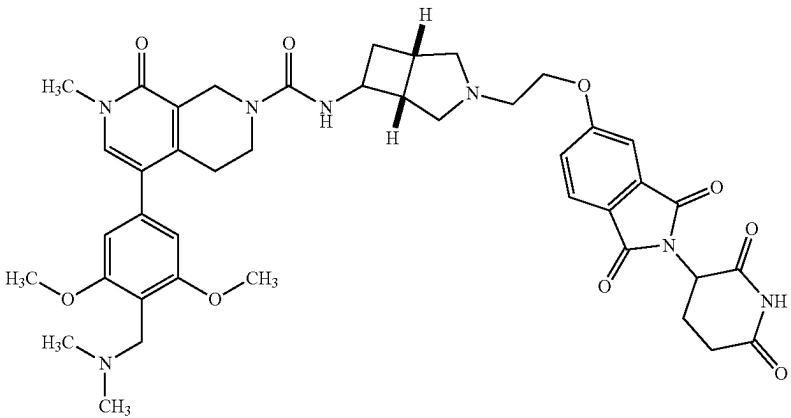
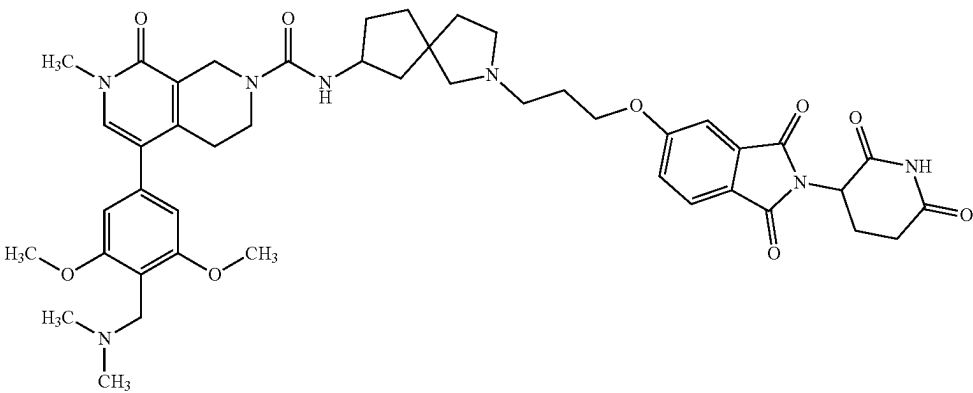
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D248	 Chemical structure of compound D248: A complex molecule featuring a 6-methoxy-2-(dimethylamino)phenyl group connected to a 4-methyl-6-oxo-1,2,3,4-tetrahydropyrimidin-5-yl group. This is linked via an amide bond to a bicyclic amine (8-azabicyclo[3.2.1]octane), which is further connected via a propyl chain to a 4-(2-oxo-2,3,4,5-tetrahydropyridin-5-yloxy)phenyl group, and finally to a 2-oxo-2,3,4,5-tetrahydropyridine ring.
D249	 Chemical structure of compound D249: Similar to D248, but the bicyclic amine is replaced by a 1,4-diazepane ring, and the amide linkage is reversed, connecting the 4-methyl-6-oxo-1,2,3,4-tetrahydropyrimidin-5-yl group to the 1,4-diazepane ring.
D250	 Chemical structure of compound D250: Similar to D248, but the bicyclic amine is replaced by a 1,4-diazepane ring, and the amide linkage is reversed, connecting the 4-methyl-6-oxo-1,2,3,4-tetrahydropyrimidin-5-yl group to the 1,4-diazepane ring.

TABLE 2B-continued

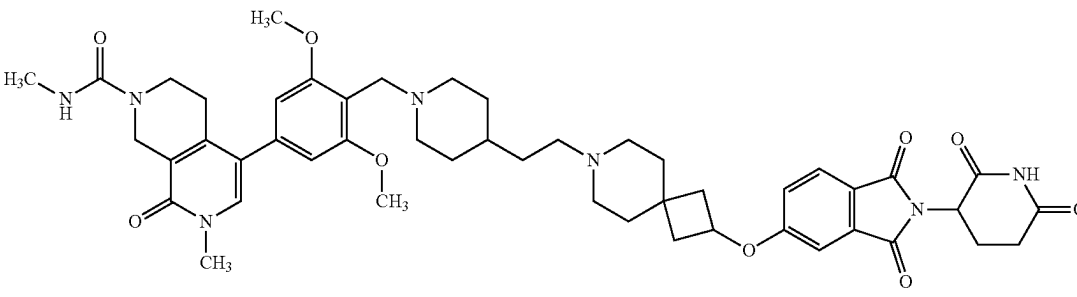
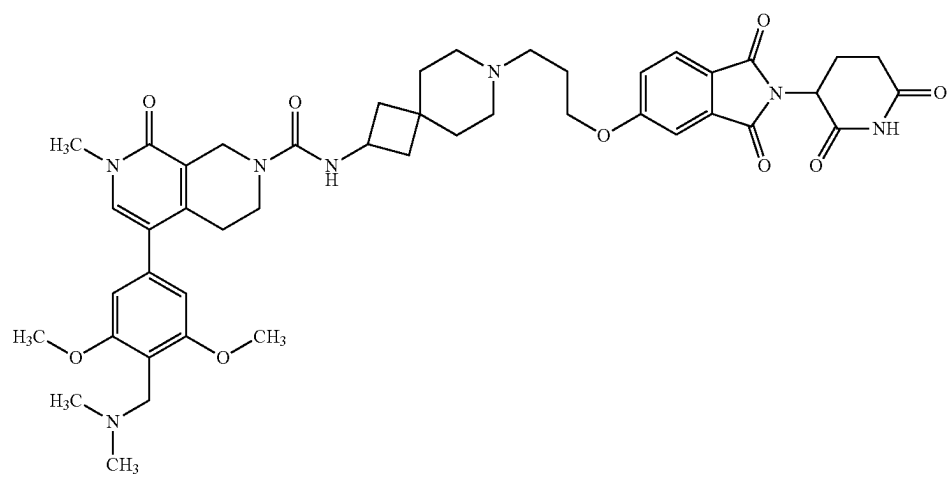
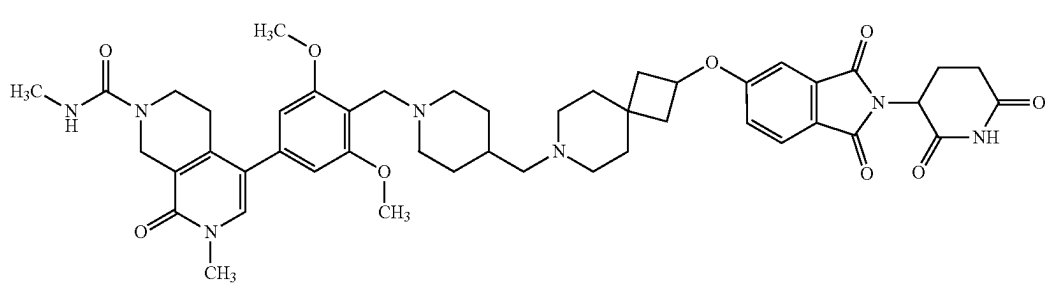
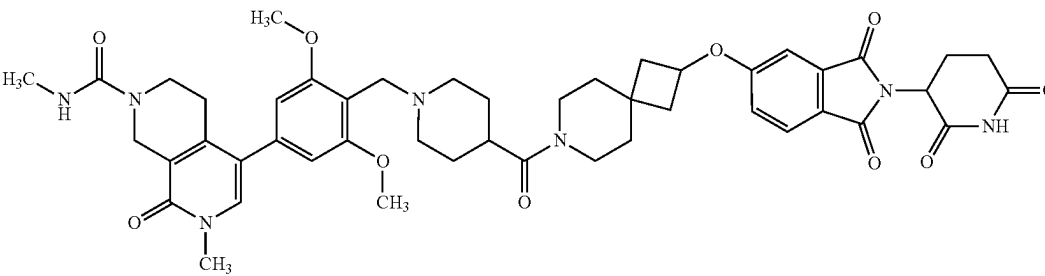
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D251	 Chemical structure of compound D251: A complex molecule featuring a pyridine ring substituted with a methylamino group, a methoxy group, and a side chain containing a piperidine ring, a cyclobutane ring, and a phthalimide moiety.
D252	 Chemical structure of compound D252: A complex molecule featuring a pyridine ring substituted with a methylamino group, a methoxy group, and a side chain containing a piperidine ring, a cyclobutane ring, and a phthalimide moiety.
D253	 Chemical structure of compound D253: A complex molecule featuring a pyridine ring substituted with a methylamino group, a methoxy group, and a side chain containing a piperidine ring, a cyclobutane ring, and a phthalimide moiety.
D254	 Chemical structure of compound D254: A complex molecule featuring a pyridine ring substituted with a methylamino group, a methoxy group, and a side chain containing a piperidine ring, a cyclobutane ring, and a phthalimide moiety.

TABLE 2B-continued

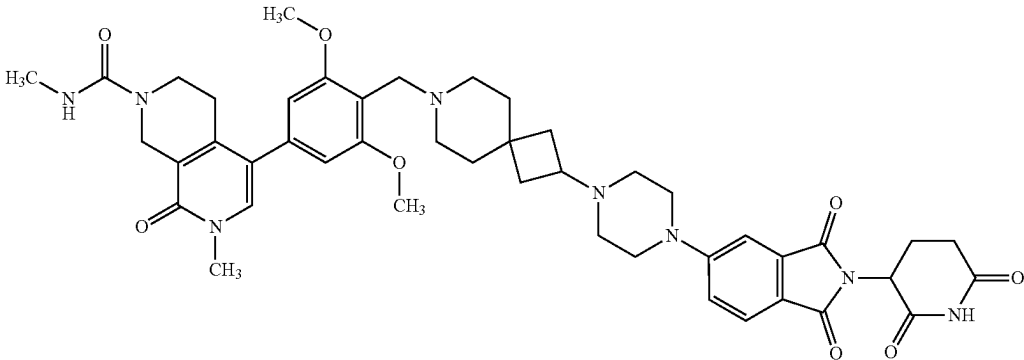
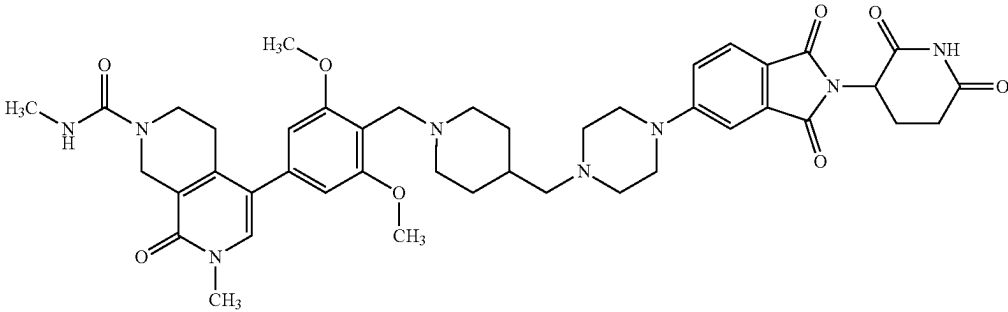
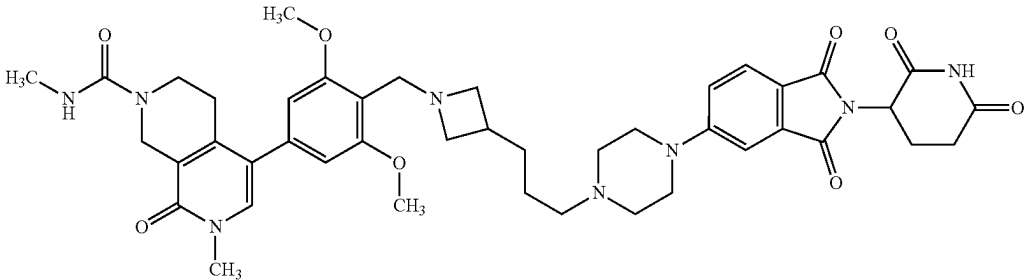
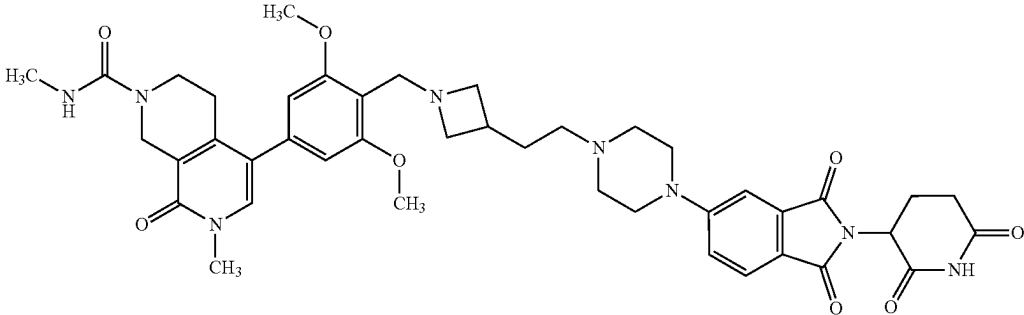
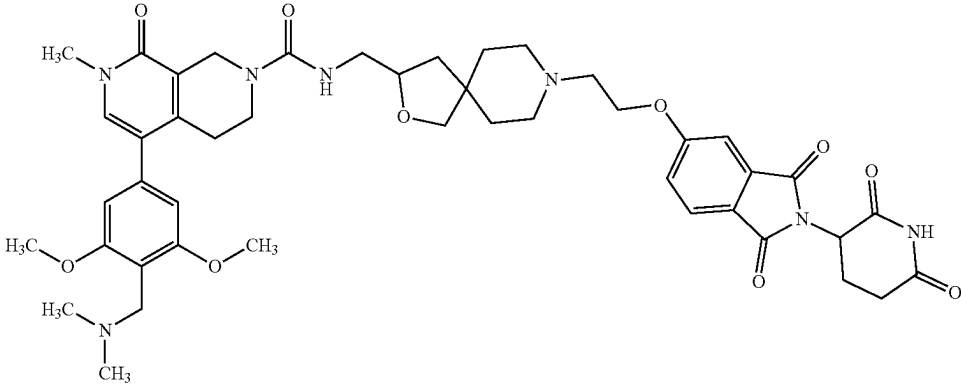
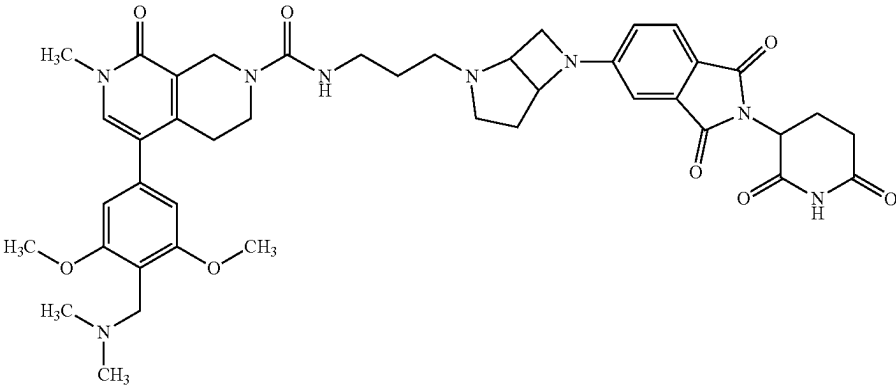
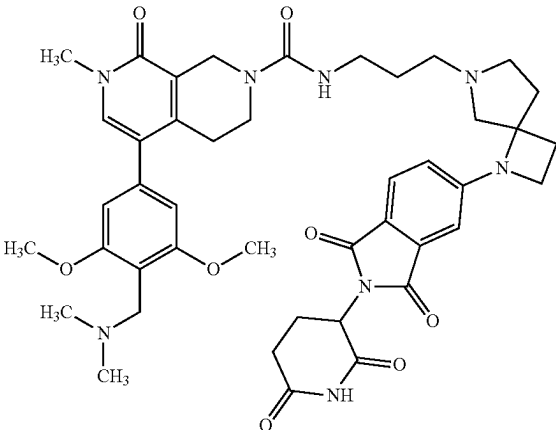
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D255	
D256	
D257	
D258	

TABLE 2B-continued

Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D259	 Chemical structure of compound D259: A complex molecule featuring a 6-methyl-4-(dimethylamino)quinoline-2-carboxamide group linked via an amide bond to a 1,4-dioxane-2-ylmethyl group. This is further connected to a 4-(2-methoxy-5-(dimethylamino)phenyl)-6-methylquinoline-2-carboxamide moiety. The structure also includes a 1,4-bis(2-methoxy-5-(dimethylamino)phenyl)-2,5-dimethylquinoline-2-carboxamide moiety.
D260	 Chemical structure of compound D260: A complex molecule featuring a 6-methyl-4-(dimethylamino)quinoline-2-carboxamide group linked via an amide bond to a 1,4-dioxane-2-ylmethyl group. This is further connected to a 4-(2-methoxy-5-(dimethylamino)phenyl)-6-methylquinoline-2-carboxamide moiety. The structure also includes a 1,4-bis(2-methoxy-5-(dimethylamino)phenyl)-2,5-dimethylquinoline-2-carboxamide moiety.
D261	 Chemical structure of compound D261: A complex molecule featuring a 6-methyl-4-(dimethylamino)quinoline-2-carboxamide group linked via an amide bond to a 1,4-dioxane-2-ylmethyl group. This is further connected to a 4-(2-methoxy-5-(dimethylamino)phenyl)-6-methylquinoline-2-carboxamide moiety. The structure also includes a 1,4-bis(2-methoxy-5-(dimethylamino)phenyl)-2,5-dimethylquinoline-2-carboxamide moiety.

Compounds D39-D302 of the Disclosure

Structure

CN1CCCC1C2C(=O)N3C(=O)c4ccc(OCCCCN5C6CC7CC8CC(C7)CC8N5C6C(=O)N9C(=O)c10ccc(C1(C)C)cc10C2)c1cc(OC)c(CN(C)C)c1CN1CCCC1C(=O)N2C(=O)c3ccc(OCCN4C5CC6CC7CC8CC(C)N(C)CC8C7C6CC5C4)cc3C2=OCN1C=CC2=C1C(=O)N(C)CC2C3=CC=C(C=C3)C(OC)C(OC)C4=CC=C5C(=O)N(C)CC(=O)N5C6=CC=C7C(=O)N(C)CC(=O)N7C6OCCN8C9CC1CC2CC(C1)CC2N8

TABLE 2B-continued

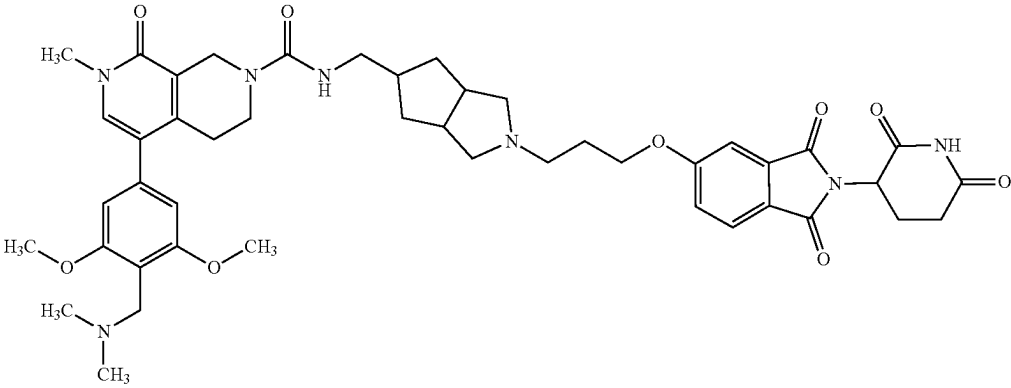
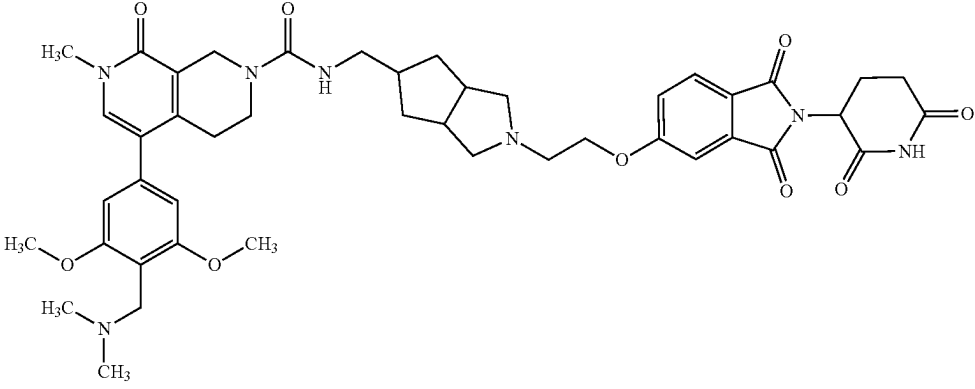
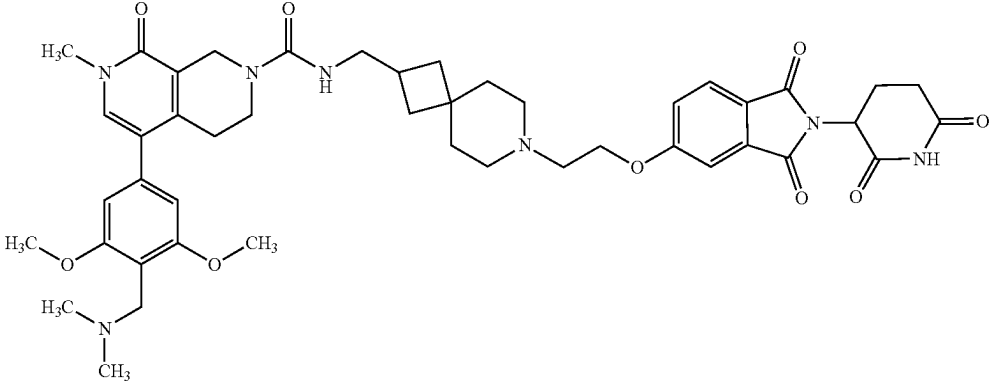
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D265	 Chemical structure of compound D265: A complex molecule featuring a 6-methoxy-2-(dimethylamino)phenyl group attached to a 4-methoxy-2-(dimethylamino)phenyl ring. This ring is further substituted with a 1-methyl-4-oxo-1,2,3,4-tetrahydropyridine-5-carboxamide group. The amide nitrogen is connected to a 1,2,3,4-tetrahydronaphthalene system, which is linked via a 3-ethoxy-1H-indole-2,3-dione moiety to a 2,6-pyridinedione ring.
D266	 Chemical structure of compound D266: A complex molecule featuring a 6-methoxy-2-(dimethylamino)phenyl group attached to a 4-methoxy-2-(dimethylamino)phenyl ring. This ring is further substituted with a 1-methyl-4-oxo-1,2,3,4-tetrahydropyridine-5-carboxamide group. The amide nitrogen is connected to a 1,2,3,4-tetrahydronaphthalene system, which is linked via a 3-ethoxy-1H-indole-2,3-dione moiety to a 2,6-pyridinedione ring.
D267	 Chemical structure of compound D267: A complex molecule featuring a 6-methoxy-2-(dimethylamino)phenyl group attached to a 4-methoxy-2-(dimethylamino)phenyl ring. This ring is further substituted with a 1-methyl-4-oxo-1,2,3,4-tetrahydropyridine-5-carboxamide group. The amide nitrogen is connected to a 1,2,3,4-tetrahydronaphthalene system, which is linked via a 3-ethoxy-1H-indole-2,3-dione moiety to a 2,6-pyridinedione ring.

TABLE 2B-continued

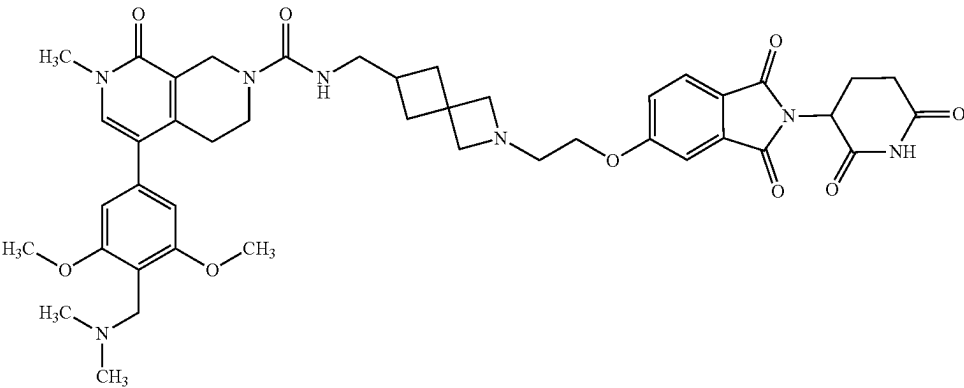
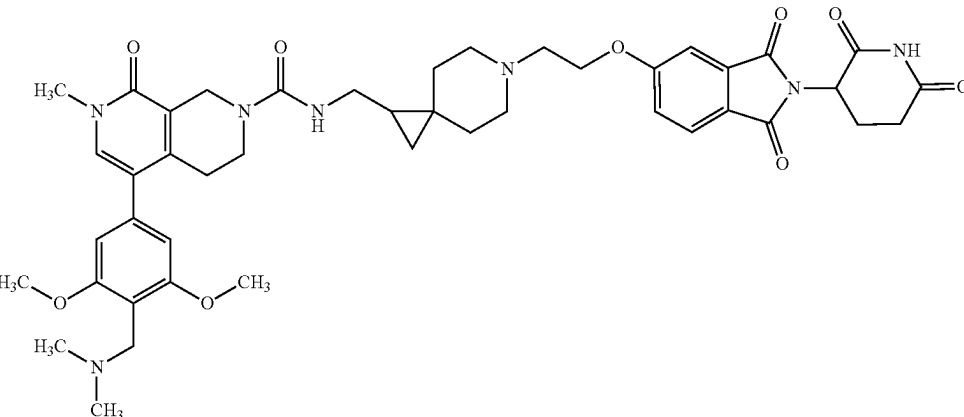
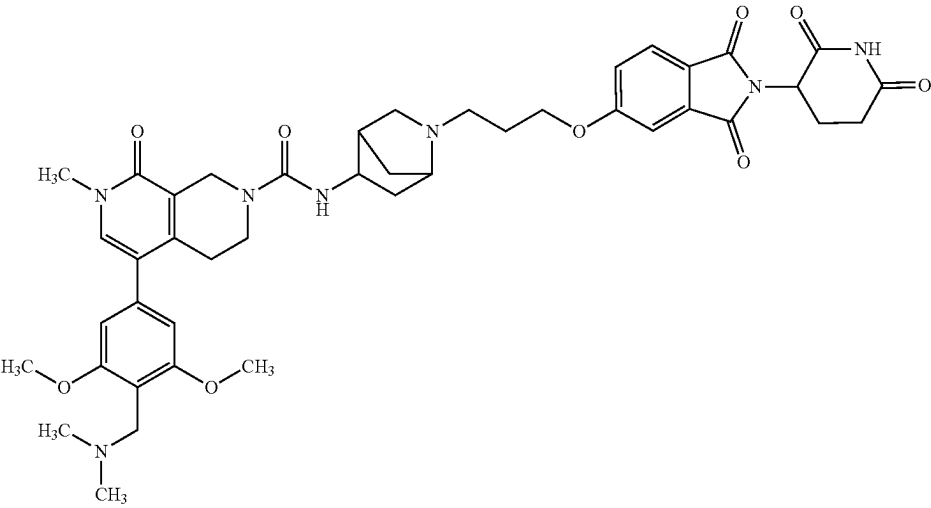
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D268	
D269	
D270	

TABLE 2B-continued

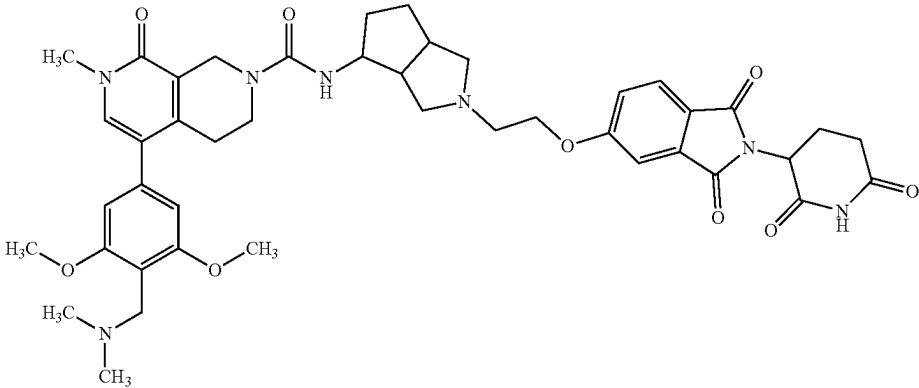
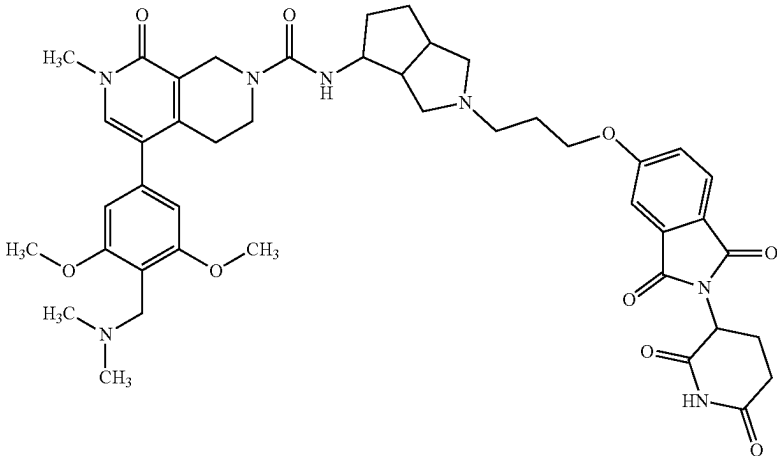
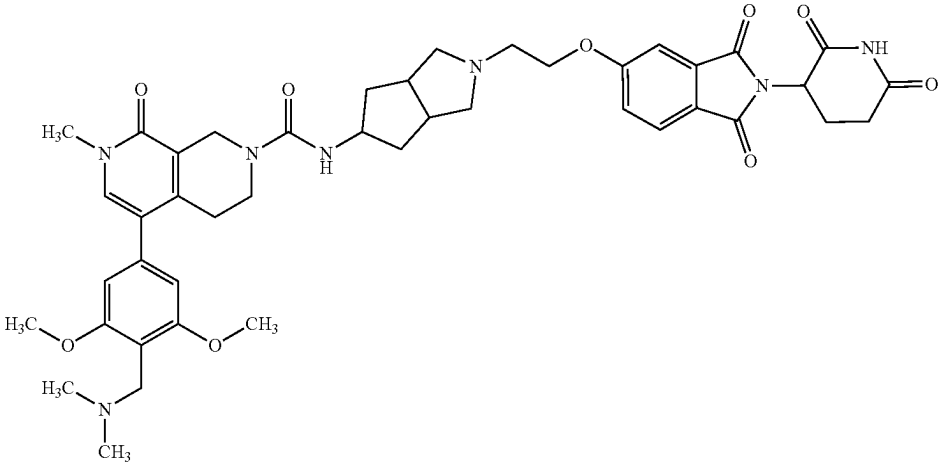
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D271	 Chemical structure of compound D271: A complex molecule featuring a 6-methyl-2,3,4,5-tetrahydroquinolin-2(1H)-one core. The quinoline ring is substituted with a 4-methoxy-2-(dimethylamino)phenyl group at position 6 and a 4-((2-oxo-1,2,3,4-tetrahydropyridin-5-yl)oxy)butyl group at position 3. The quinoline ring is also substituted with a 4-methoxy-2-(dimethylamino)phenyl group at position 6 and a 4-((2-oxo-1,2,3,4-tetrahydropyridin-5-yl)oxy)butyl group at position 3.
D272	 Chemical structure of compound D272: A complex molecule featuring a 6-methyl-2,3,4,5-tetrahydroquinolin-2(1H)-one core. The quinoline ring is substituted with a 4-methoxy-2-(dimethylamino)phenyl group at position 6 and a 4-((2-oxo-1,2,3,4-tetrahydropyridin-5-yl)oxy)butyl group at position 3. The quinoline ring is also substituted with a 4-methoxy-2-(dimethylamino)phenyl group at position 6 and a 4-((2-oxo-1,2,3,4-tetrahydropyridin-5-yl)oxy)butyl group at position 3.
D273	 Chemical structure of compound D273: A complex molecule featuring a 6-methyl-2,3,4,5-tetrahydroquinolin-2(1H)-one core. The quinoline ring is substituted with a 4-methoxy-2-(dimethylamino)phenyl group at position 6 and a 4-((2-oxo-1,2,3,4-tetrahydropyridin-5-yl)oxy)butyl group at position 3. The quinoline ring is also substituted with a 4-methoxy-2-(dimethylamino)phenyl group at position 6 and a 4-((2-oxo-1,2,3,4-tetrahydropyridin-5-yl)oxy)butyl group at position 3.

TABLE 2B-continued

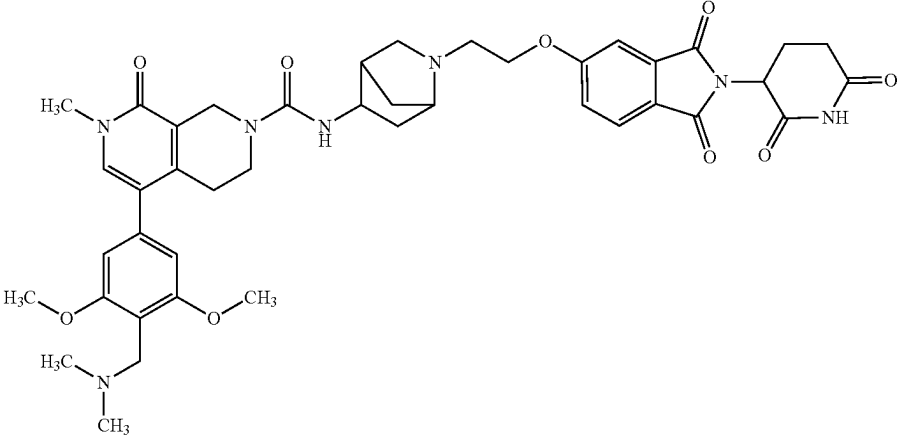
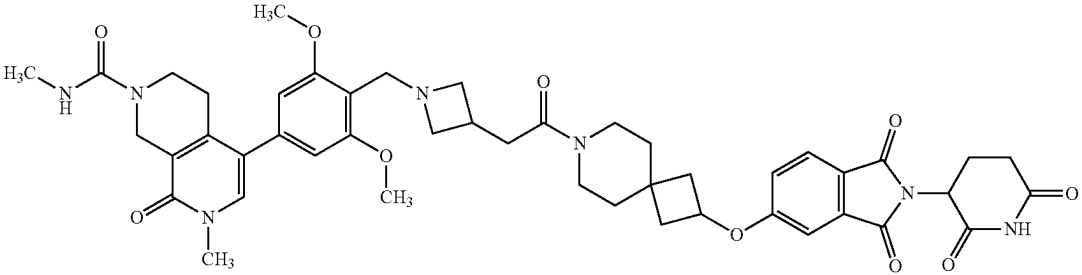
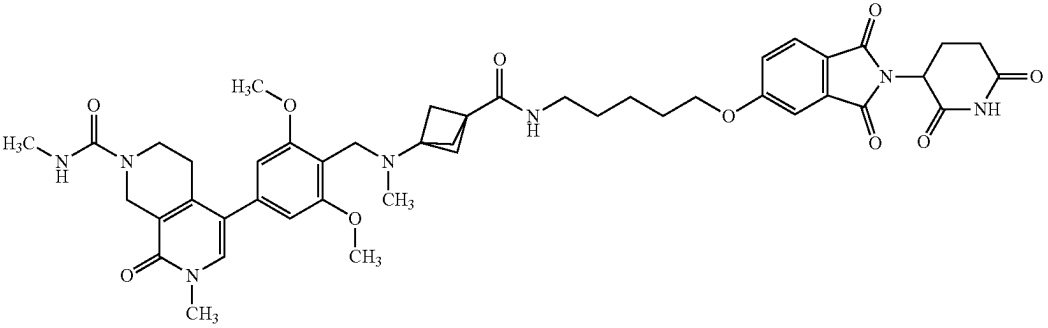
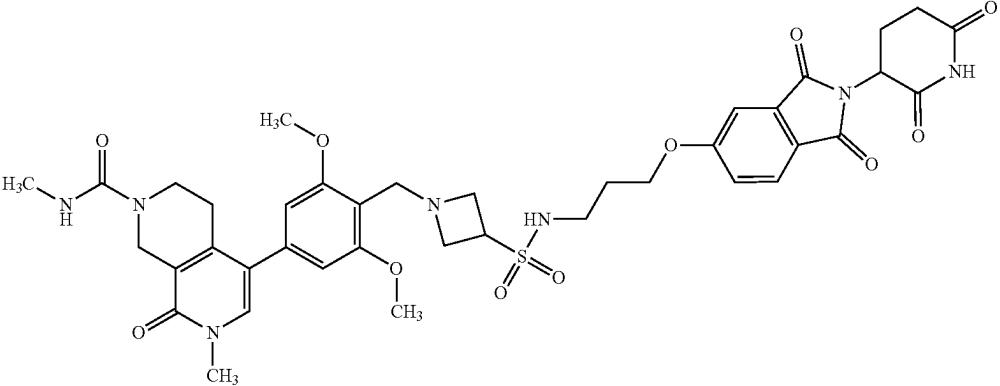
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D274	
D275	
D276	
D277	

TABLE 2B-continued

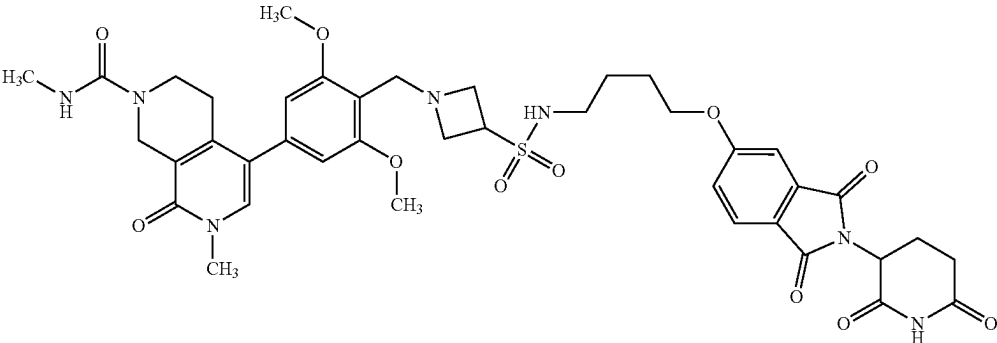
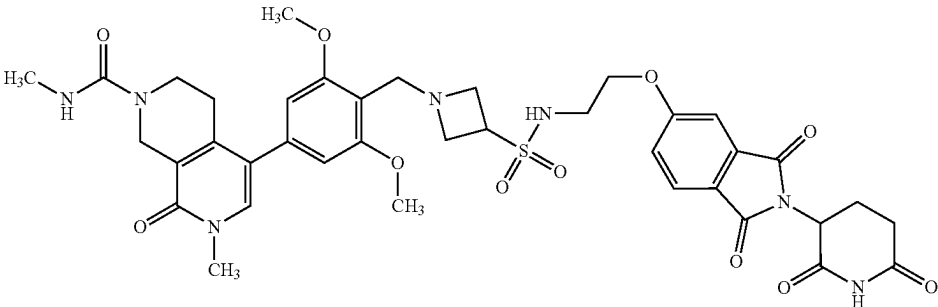
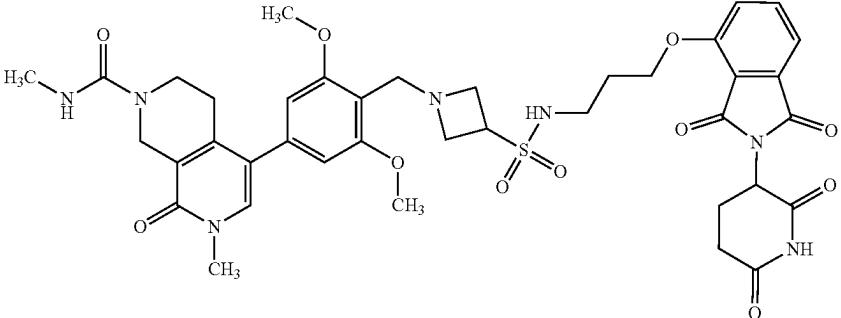
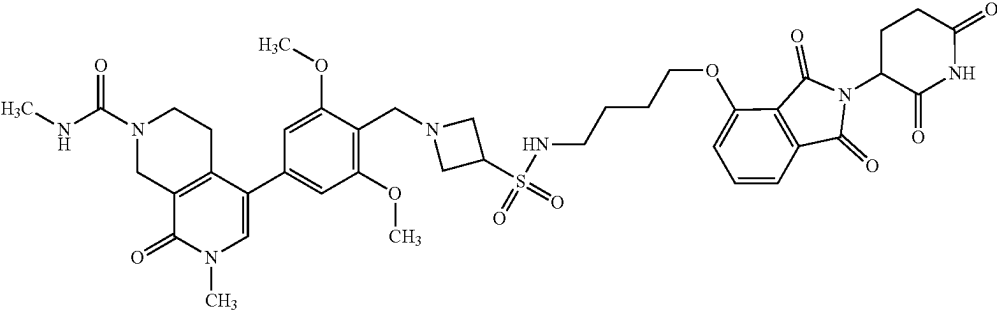
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D278	
D279	
D280	
D281	

TABLE 2B-continued

Compound No.	Structure
D282	
D283	
D284	
D285	

TABLE 2B-continued

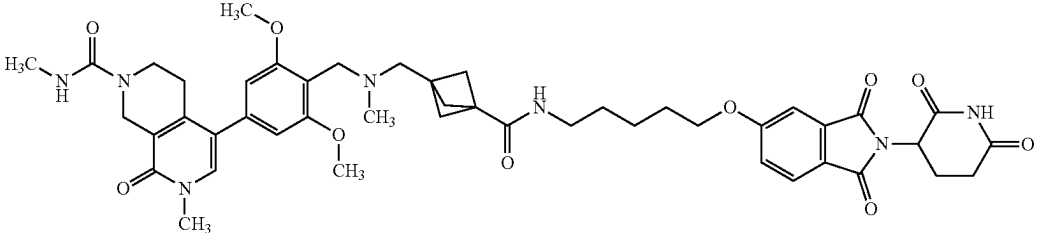
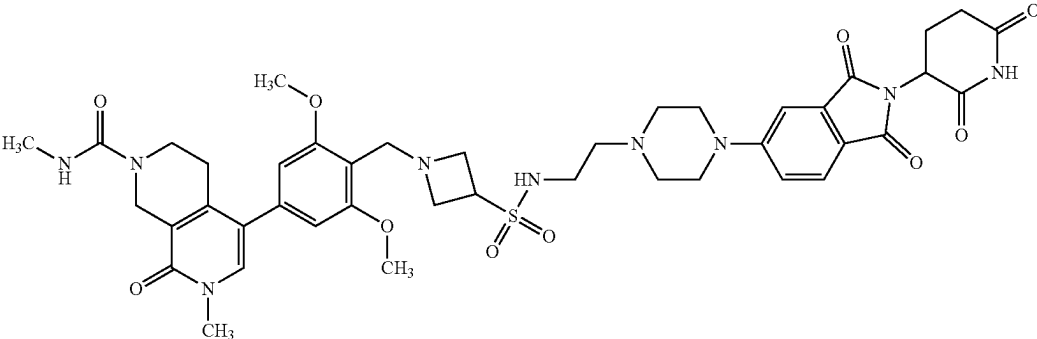
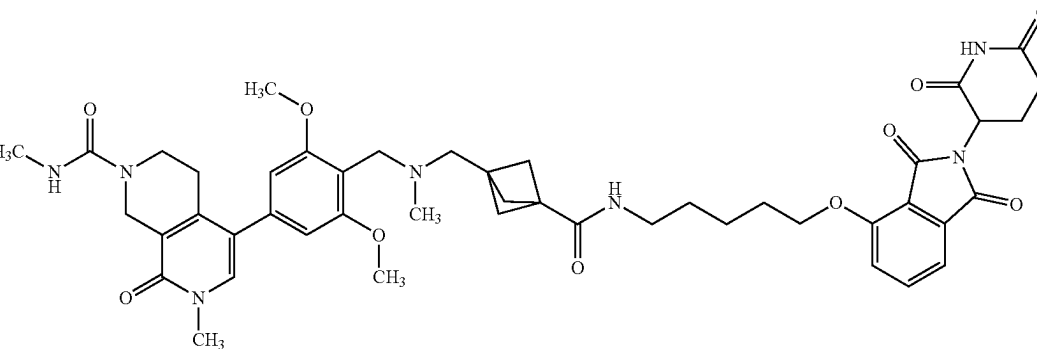
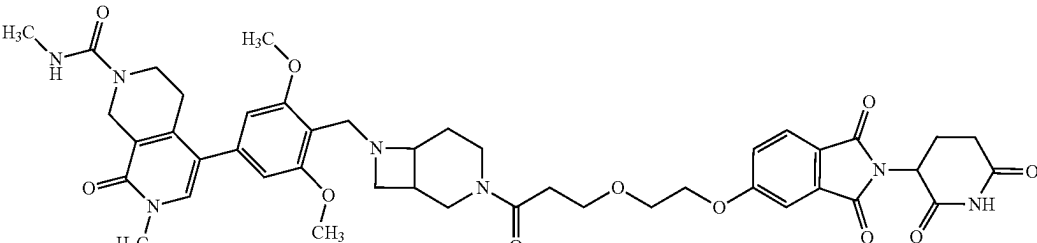
Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D286	
D287	
D288	
D289	

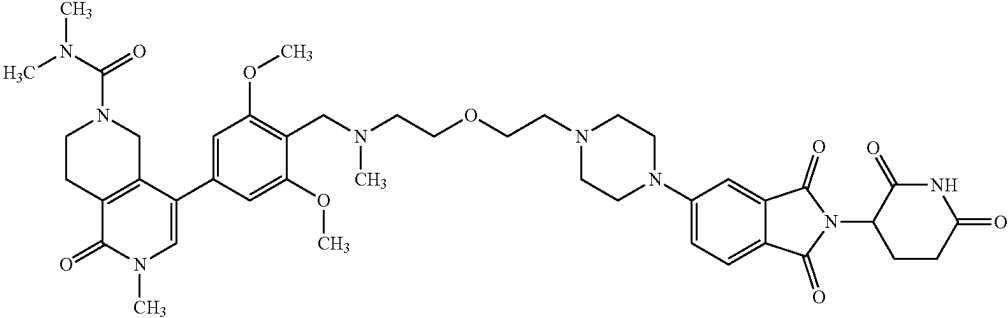
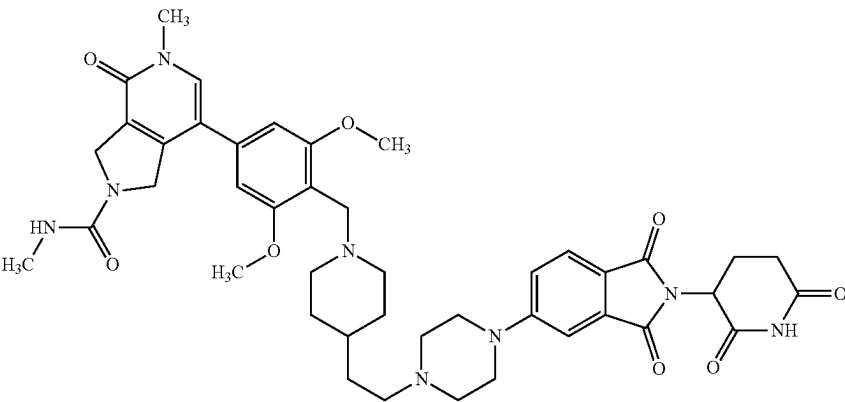
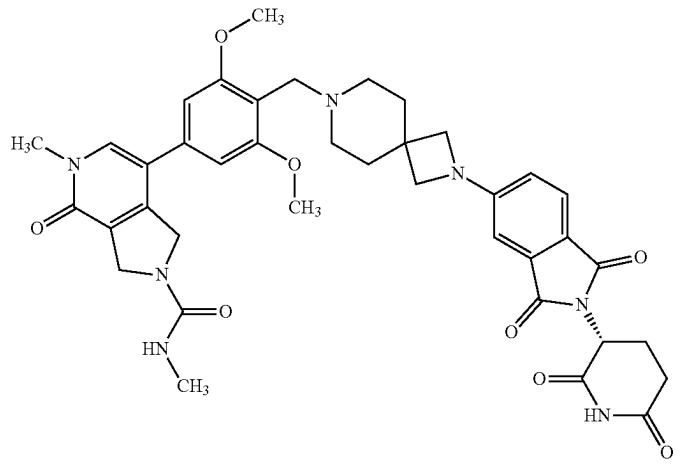
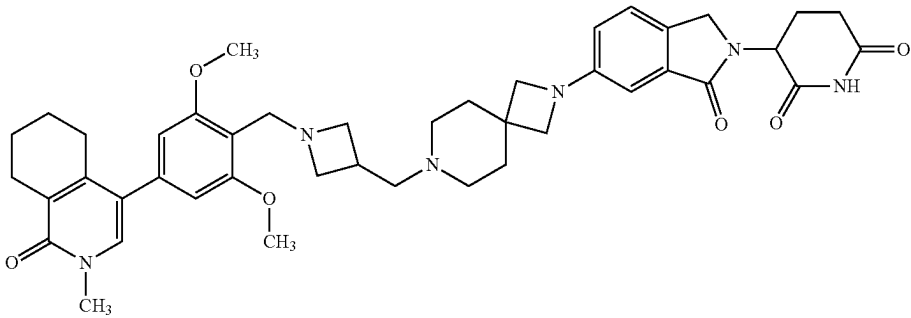
TABLE 2B-continued

Compound No.	Structure
D290	
D291	
D292	
D293	

TABLE 2B-continued

Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D294	
D295	
D296	
D297	

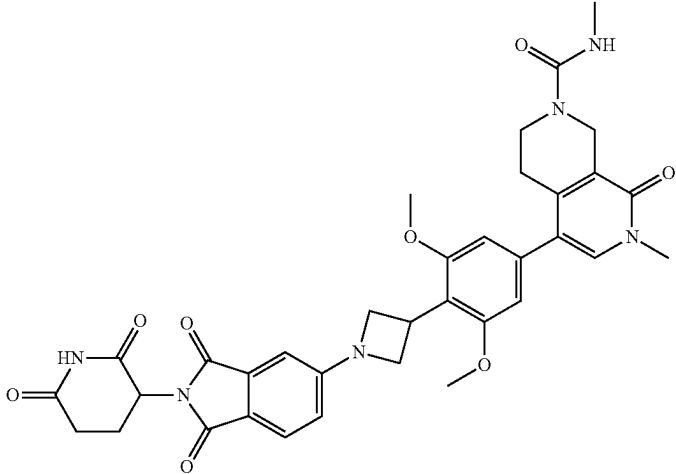
TABLE 2B-continued

Compounds D39-D302 of the Disclosure	
Compound No.	Structure
D298	
D299	
D300	
D301	

Compounds D39-D302 of the Disclosure

Compounds D303-D375 of the Disclosure	
Compound No.	Structure
D303	

D303



D304

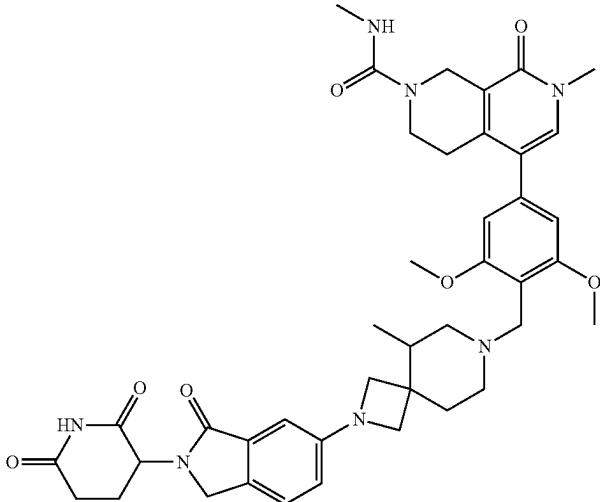


TABLE 2C-continued

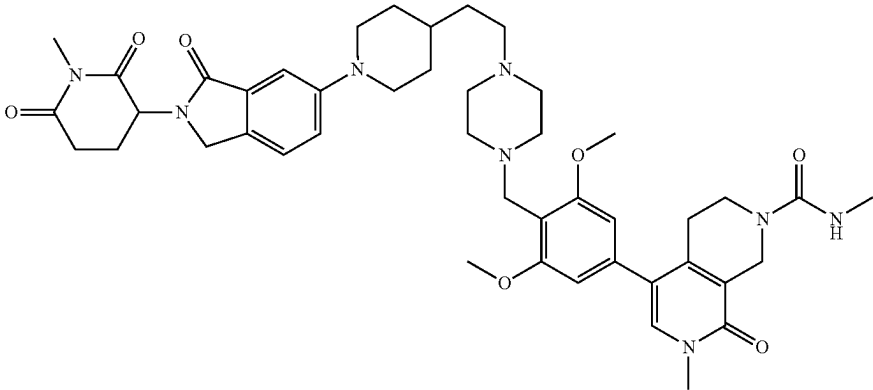
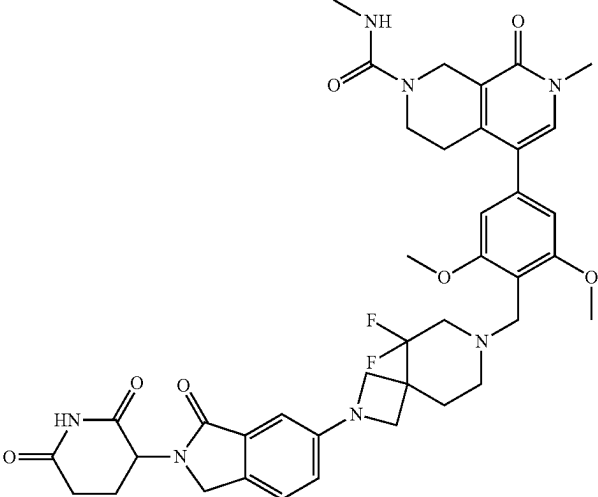
Compounds D303-D375 of the Disclosure	
Compound No.	Structure
D305	 Chemical structure of compound D305: A complex molecule featuring a 6-membered lactam ring (piperidin-2-one) substituted with a methyl group and a 1,2,3,4-tetrahydro-1H-benzodioxepin-1-one moiety. This is connected via a piperidine ring to a 1,4-bis(methoxy)phenyl group, which is further linked to a 1,2,3,4-tetrahydroquinoline-6-carboxamide moiety.
D306	 Chemical structure of compound D306: A complex molecule featuring a 6-membered lactam ring (piperidin-2-one) substituted with a methyl group and a 1,2,3,4-tetrahydro-1H-benzodioxepin-1-one moiety. This is connected via a piperidine ring to a 1,4-bis(methoxy)phenyl group, which is further linked to a 1,2,3,4-tetrahydroquinoline-6-carboxamide moiety.

TABLE 2C-continued

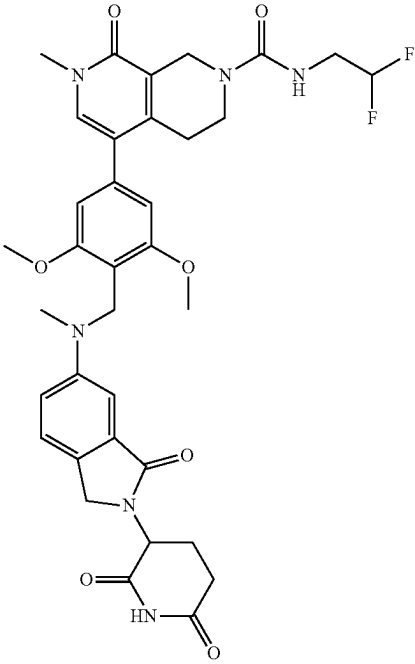
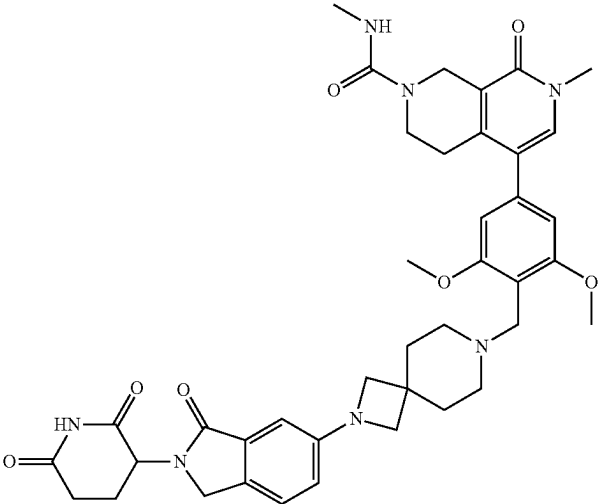
Compounds D303-D375 of the Disclosure	
Compound No.	Structure
D307	 Chemical structure of compound D307: A complex molecule featuring a 6-methyl-4-oxo-1,2,3,4-tetrahydroquinoline-5-carboxamide group linked via a methylene bridge to a 3,5-dimethoxyphenyl ring. This phenyl ring is further linked via a methylene bridge to a 4-methyl-1H-pyrrolo[2,3-b]pyridine-2-carboxamide group. The pyrrole ring is substituted with a 2,6-pyridinedicarboxamide group.
D308	 Chemical structure of compound D308: A complex molecule featuring a 6-methyl-4-oxo-1,2,3,4-tetrahydroquinoline-5-carboxamide group linked via a methylene bridge to a 3,5-dimethoxyphenyl ring. This phenyl ring is further linked via a methylene bridge to a 4-methyl-1H-pyrrolo[2,3-b]pyridine-2-carboxamide group. The pyrrole ring is substituted with a 2,6-pyridinedicarboxamide group.

TABLE 2C-continued

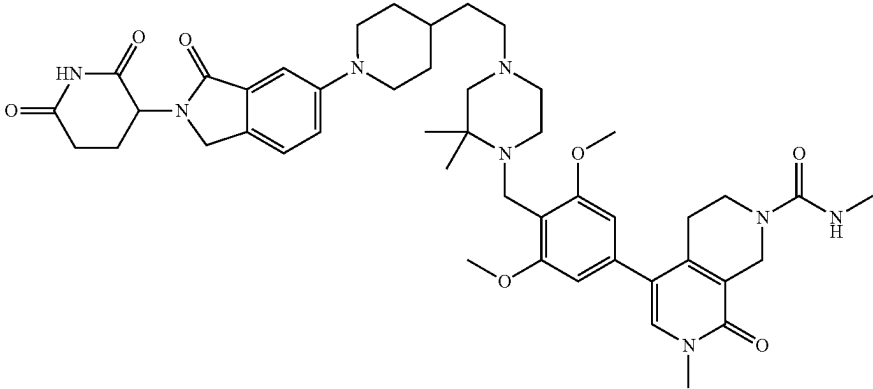
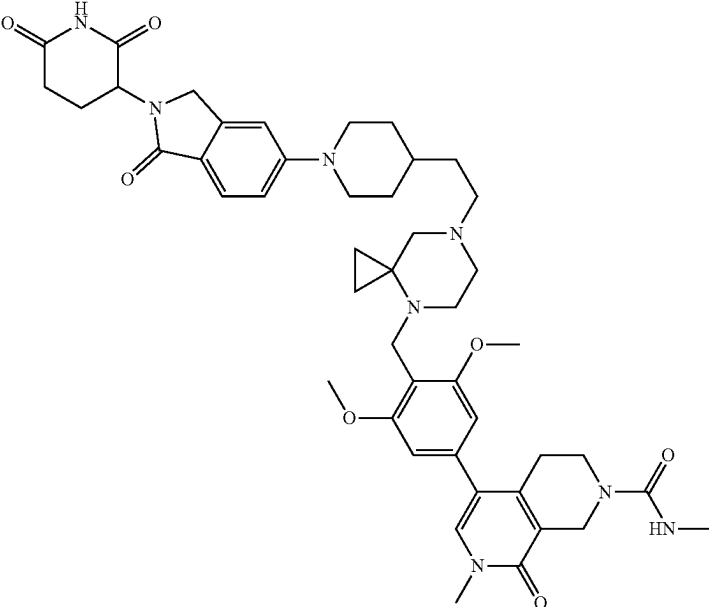
Compounds D303-D375 of the Disclosure	
Compound No.	Structure
D309	 Chemical structure of Compound D309: A complex molecule featuring a 6,7-dihydro-5H-benzocyclohepta[b]pyridine-5-carboxamide moiety linked via a piperidine ring to a 4-methoxy-2-(methylamino)pyridine-3-carboxamide moiety. The structure includes a piperidine ring, a methoxy group, and a methylamino group.
D310	 Chemical structure of Compound D310: A complex molecule featuring a 6,7-dihydro-5H-benzocyclohepta[b]pyridine-5-carboxamide moiety linked via a piperidine ring to a 4-methoxy-2-(methylamino)pyridine-3-carboxamide moiety. The structure includes a piperidine ring, a methoxy group, and a methylamino group.

TABLE 2C-continued

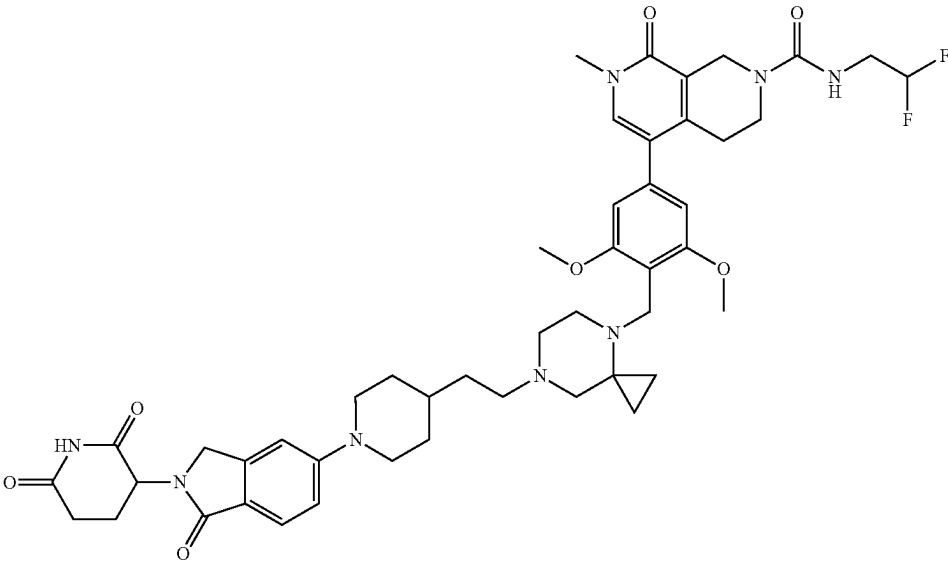
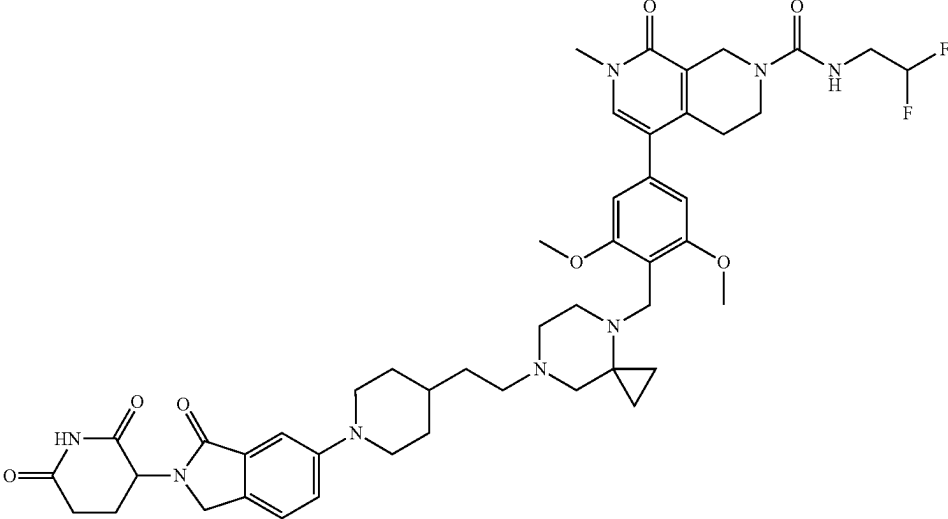
Compounds D303-D375 of the Disclosure	
Compound No.	Structure
D311	 The chemical structure of Compound D311 is a complex molecule. It features a 6,7-dihydroquinolin-2(1H)-one core. At the 4-position of the quinoline ring, there is a piperidine ring. This piperidine ring is connected via a (2E)-2-((2,4-dimethoxyphenyl)methyl)azetidine-1-carboxamide linker to a 2,4-dimethoxyphenyl group. The 2-position of this phenyl group is substituted with a 6-methyl-7-((2E)-2-((2,4-dimethoxyphenyl)methyl)azetidine-1-carboxamido)-1,2,3,4-tetrahydroquinolin-2-one moiety. The 4-position of the quinoline ring is also substituted with a 2-((2E)-2-((2,4-dimethoxyphenyl)methyl)azetidine-1-carboxamido)-6-methyl-1,2,3,4-tetrahydroquinolin-2-one moiety. The 2-position of this quinoline ring is substituted with a 2-((2E)-2-((2,4-dimethoxyphenyl)methyl)azetidine-1-carboxamido)-6-methyl-1,2,3,4-tetrahydroquinolin-2-one moiety. The 4-position of the quinoline ring is also substituted with a 2-((2E)-2-((2,4-dimethoxyphenyl)methyl)azetidine-1-carboxamido)-6-methyl-1,2,3,4-tetrahydroquinolin-2-one moiety.
D312	 The chemical structure of Compound D312 is identical to the structure of Compound D311, as described above.

TABLE 2C-continued

D314

TABLE 2C-continued

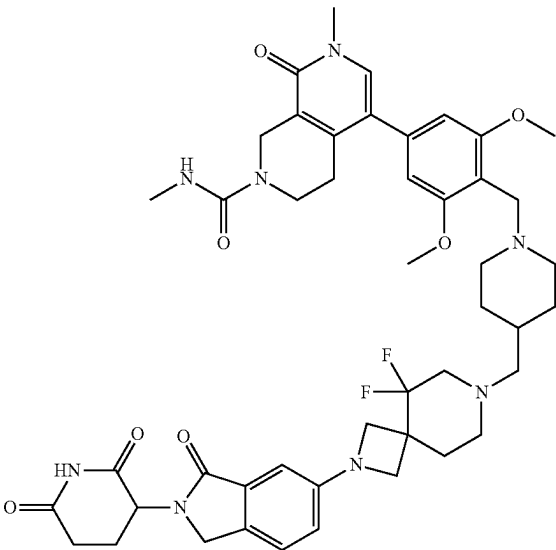
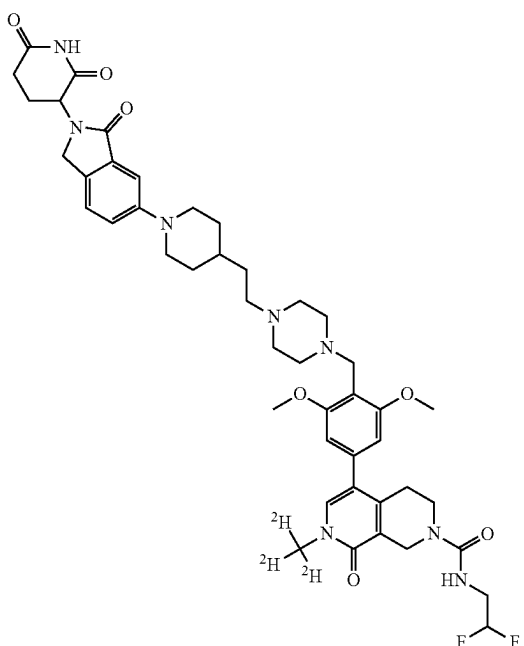
Compounds D303-D375 of the Disclosure	
Compound No.	Structure
D315	 Chemical structure of Compound D315: A complex molecule featuring a central benzene ring substituted with a 1-methyl-4-oxo-1,2,3,4-tetrahydropyridine-5-yl group, a 3,4-dimethoxybenzyl group, and a 1-(2-oxo-2,3,4,5-tetrahydropyridin-5-yl) group. The 1-methyl-4-oxo-1,2,3,4-tetrahydropyridine-5-yl group is further substituted with a methylamino group. The 1-(2-oxo-2,3,4,5-tetrahydropyridin-5-yl) group is further substituted with a 1,1-difluoro-2-methyl-2,3,4,5-tetrahydropyridin-5-yl group.
D316	 Chemical structure of Compound D316: A complex molecule featuring a central benzene ring substituted with a 1-(2-oxo-2,3,4,5-tetrahydropyridin-5-yl) group, a 1-(2-oxo-2,3,4,5-tetrahydropyridin-5-yl) group, and a 1-(2-oxo-2,3,4,5-tetrahydropyridin-5-yl) group. The 1-(2-oxo-2,3,4,5-tetrahydropyridin-5-yl) group is further substituted with a 1,1-difluoro-2-methyl-2,3,4,5-tetrahydropyridin-5-yl group. The 1-(2-oxo-2,3,4,5-tetrahydropyridin-5-yl) group is further substituted with a 1,1-difluoro-2-methyl-2,3,4,5-tetrahydropyridin-5-yl group.

TABLE 2C-continued

Chemical structure of compound 10, showing a complex molecule with a piperidine ring, a benzimidazole core, and a pyridine ring, with various substituents including a morpholine ring and a methoxy group.

TABLE 2C-continued

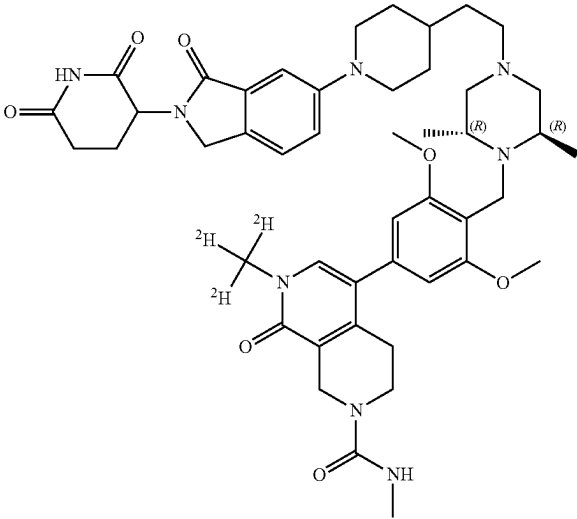
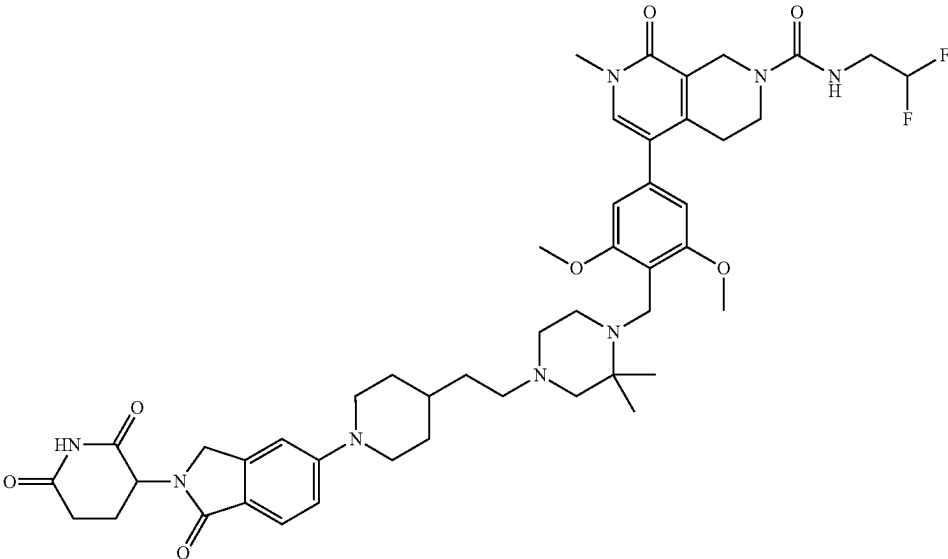
Compounds D303-D375 of the Disclosure	
Compound No.	Structure
D319	 Chemical structure of Compound D319: A complex molecule featuring a central pyridine ring substituted with a 2,6-dimethoxyphenyl group, a 4-(2-methyl-2H-tetrazol-5-yl) group, and a 4-(2-methyl-2H-tetrazol-5-yl) group. The pyridine ring is also substituted with a 2,6-dimethoxyphenyl group and a 4-(2-methyl-2H-tetrazol-5-yl) group. The structure includes a 2,6-dimethoxyphenyl group, a 4-(2-methyl-2H-tetrazol-5-yl) group, and a 4-(2-methyl-2H-tetrazol-5-yl) group. The structure is a complex molecule with multiple rings and functional groups, including a 2,6-dimethoxyphenyl group, a 4-(2-methyl-2H-tetrazol-5-yl) group, and a 4-(2-methyl-2H-tetrazol-5-yl) group.
D320	 Chemical structure of Compound D320: A complex molecule featuring a central pyridine ring substituted with a 2,6-dimethoxyphenyl group, a 4-(2-methyl-2H-tetrazol-5-yl) group, and a 4-(2-methyl-2H-tetrazol-5-yl) group. The pyridine ring is also substituted with a 2,6-dimethoxyphenyl group and a 4-(2-methyl-2H-tetrazol-5-yl) group. The structure includes a 2,6-dimethoxyphenyl group, a 4-(2-methyl-2H-tetrazol-5-yl) group, and a 4-(2-methyl-2H-tetrazol-5-yl) group. The structure is a complex molecule with multiple rings and functional groups, including a 2,6-dimethoxyphenyl group, a 4-(2-methyl-2H-tetrazol-5-yl) group, and a 4-(2-methyl-2H-tetrazol-5-yl) group.

TABLE 2C-continued

The chemical structure is a complex molecule with several fused and linked rings. It includes a piperidine ring, a benzimidazole system, a morpholine ring, a 3,4-dimethoxyphenyl group, and a 1,2,3,4-tetrahydropyridine ring. A 2,2,3,3-tetradeuterioacetamide group is attached to the tetrahydropyridine ring. The structure is labeled with (R) for stereochemistry and 2H for deuterium atoms.

TABLE 2C-continued

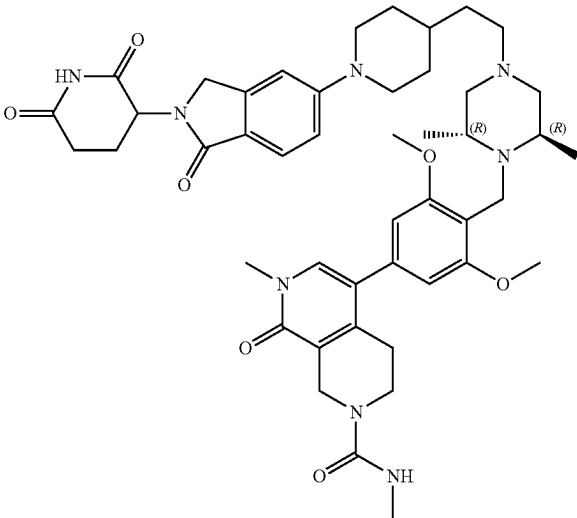
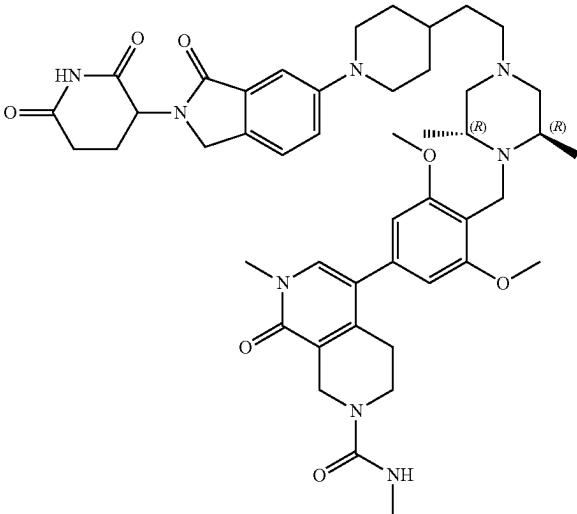
Compounds D303-D375 of the Disclosure	
Compound No.	Structure
D325	 Chemical structure of compound D325. It features a central benzene ring substituted with a 6-oxo-1,2,3,4-tetrahydropyridine-3-carbonyl group, a 1-methyl-2-oxo-1,2,3,4-tetrahydropyridine-3-carbonyl group, and a 1-methyl-2-oxo-1,2,3,4-tetrahydropyridine-3-carbonyl group. The structure also includes a 1-methyl-2-oxo-1,2,3,4-tetrahydropyridine-3-carbonyl group and a 1-methyl-2-oxo-1,2,3,4-tetrahydropyridine-3-carbonyl group.
D326	 Chemical structure of compound D326. It features a central benzene ring substituted with a 6-oxo-1,2,3,4-tetrahydropyridine-3-carbonyl group, a 1-methyl-2-oxo-1,2,3,4-tetrahydropyridine-3-carbonyl group, and a 1-methyl-2-oxo-1,2,3,4-tetrahydropyridine-3-carbonyl group. The structure also includes a 1-methyl-2-oxo-1,2,3,4-tetrahydropyridine-3-carbonyl group and a 1-methyl-2-oxo-1,2,3,4-tetrahydropyridine-3-carbonyl group.

TABLE 2C-continued

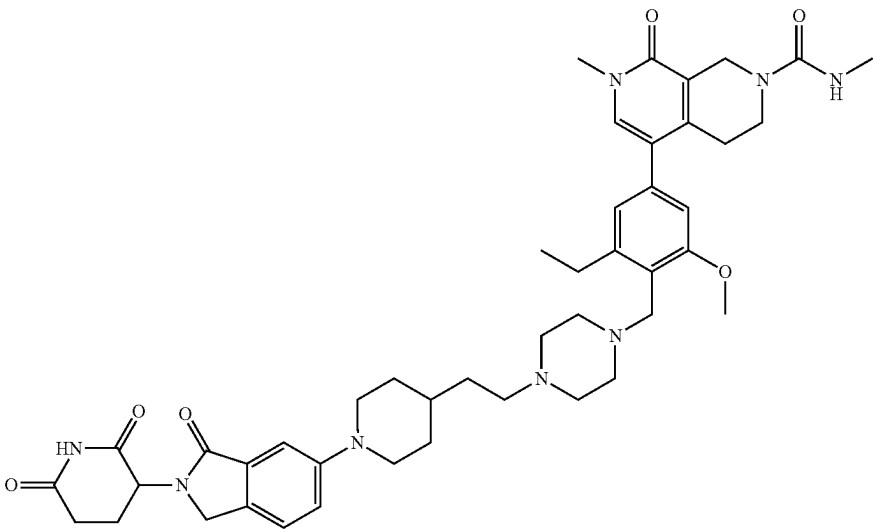
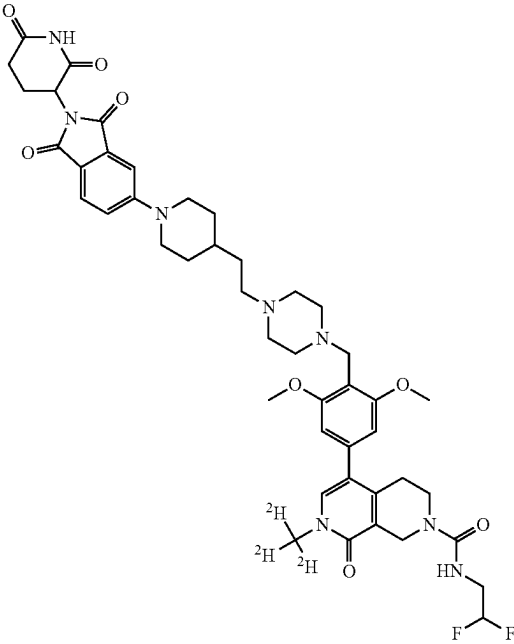
Compounds D303-D375 of the Disclosure	
Compound No.	Structure
D327	 Chemical structure of Compound D327: A complex molecule featuring a 6,7-dihydro-5H-benzocyclohepta[b]pyridine-5-carboxamide core. The core is substituted with a piperidine ring at the 2-position, which is further linked via a 2,2'-bis(piperidin-1-yl)ethane bridge to a 4-ethoxy-2-methylphenyl group. This phenyl group is also substituted with a 4-methyl-6-(N-methylacetamido)pyridine-2-carboxamide moiety.
D328	 Chemical structure of Compound D328: A complex molecule featuring a 6,7-dihydro-5H-benzocyclohepta[b]pyridine-5-carboxamide core. The core is substituted with a piperidine ring at the 2-position, which is further linked via a 2,2'-bis(piperidin-1-yl)ethane bridge to a 4-methoxy-2-methylphenyl group. This phenyl group is also substituted with a 4-(2,2,2-trifluoroethyl)amino-6-(2,2,2-trifluoroethyl)pyridine-2-carboxamide moiety.

TABLE 2C-continued

Compounds D303-D375 of the Disclosure

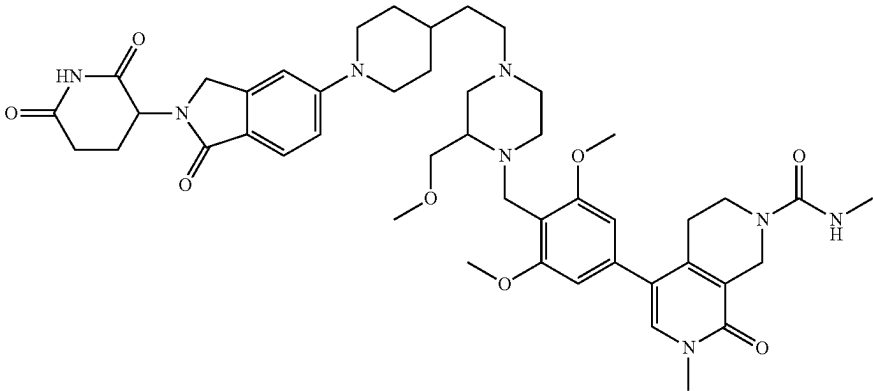
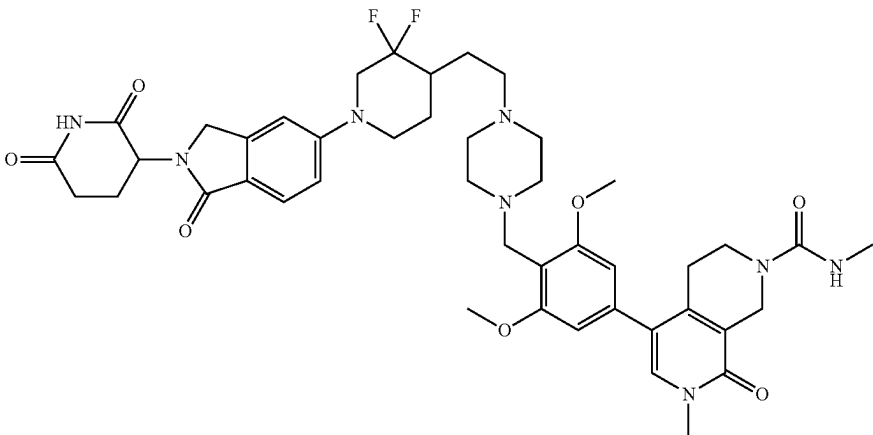
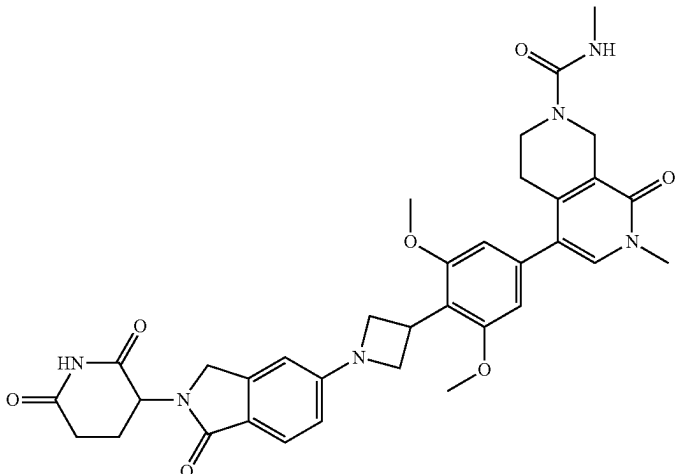
Compound No.	Structure
D329	
D330	
D331	

TABLE 2C-continued

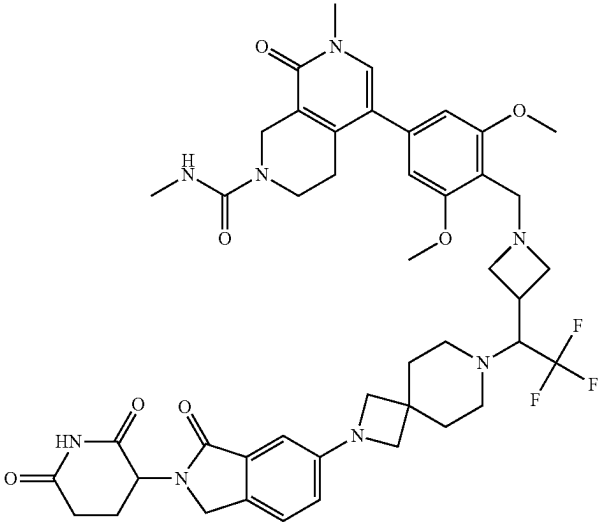
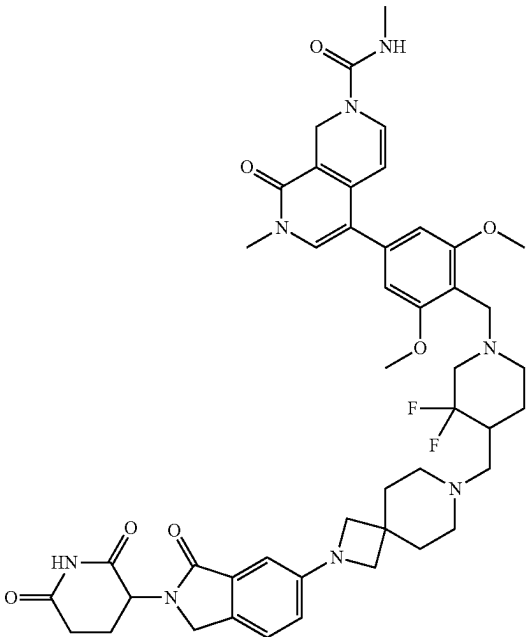
Compounds D303-D375 of the Disclosure	
Compound No.	Structure
D334	 Chemical structure of Compound D334: A complex molecule featuring a central benzene ring substituted with a 1-methyl-4-oxo-1,2,3,4-tetrahydropyridine-5-yl group, a 4-methoxy-2-((2-methoxy-4-((2,2,2-trifluoroethyl)amino)ethyl)azetidin-1-yl)phenyl group, and a 1-((2,2,2-trifluoroethyl)amino)ethyl group. The molecule also includes a 1-methyl-4-oxo-1,2,3,4-tetrahydropyridine-5-yl group and a 1-((2,2,2-trifluoroethyl)amino)ethyl group.
D335	 Chemical structure of Compound D335: A complex molecule featuring a central benzene ring substituted with a 1-methyl-4-oxo-1,2,3,4-tetrahydropyridine-5-yl group, a 4-methoxy-2-((2-methoxy-4-((2,2,2-trifluoroethyl)amino)ethyl)azetidin-1-yl)phenyl group, and a 1-((2,2,2-trifluoroethyl)amino)ethyl group. The molecule also includes a 1-methyl-4-oxo-1,2,3,4-tetrahydropyridine-5-yl group and a 1-((2,2,2-trifluoroethyl)amino)ethyl group.

TABLE 2C-continued

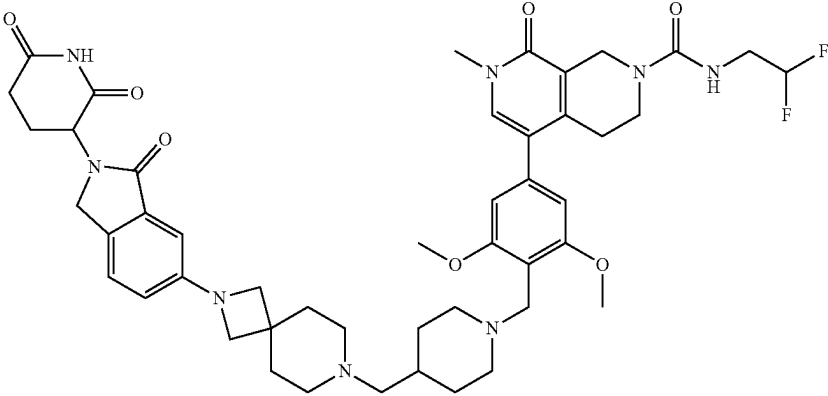
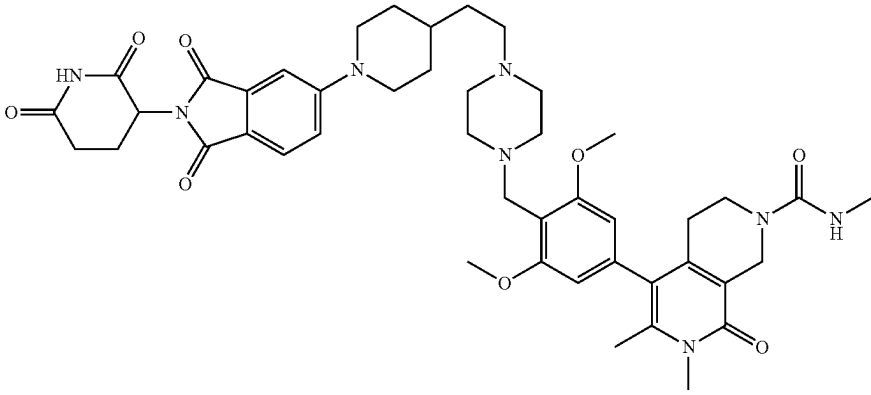
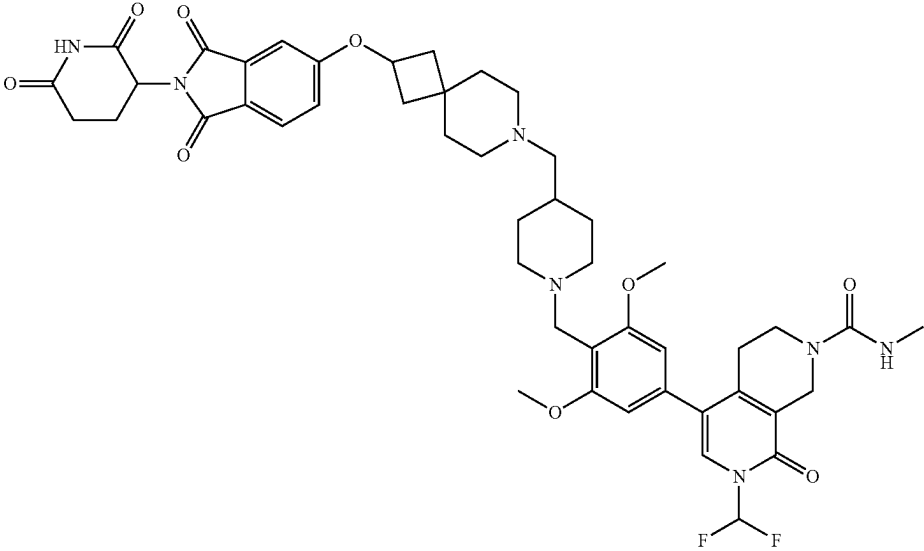
Compounds D303-D375 of the Disclosure	
Compound No.	Structure
D336	 Chemical structure of Compound D336: A complex molecule featuring a central piperidine ring connected via a methylene bridge to a 4-methoxyphenyl group. The 4-methoxyphenyl group is further connected to a 6-methoxy-2-methyl-4-(dimethylamino)pyrimidin-5-yl group. The piperidine ring is also connected to a 4-methoxyphenyl group, which is linked to a 6-methoxy-2-methyl-4-(dimethylamino)pyrimidin-5-yl group. The piperidine ring is further connected to a 4-methoxyphenyl group, which is linked to a 6-methoxy-2-methyl-4-(dimethylamino)pyrimidin-5-yl group.
D337	 Chemical structure of Compound D337: A complex molecule featuring a central piperidine ring connected via a methylene bridge to a 4-methoxyphenyl group. The 4-methoxyphenyl group is further connected to a 6-methoxy-2-methyl-4-(dimethylamino)pyrimidin-5-yl group. The piperidine ring is also connected to a 4-methoxyphenyl group, which is linked to a 6-methoxy-2-methyl-4-(dimethylamino)pyrimidin-5-yl group. The piperidine ring is further connected to a 4-methoxyphenyl group, which is linked to a 6-methoxy-2-methyl-4-(dimethylamino)pyrimidin-5-yl group.
D338	 Chemical structure of Compound D338: A complex molecule featuring a central piperidine ring connected via a methylene bridge to a 4-methoxyphenyl group. The 4-methoxyphenyl group is further connected to a 6-methoxy-2-methyl-4-(dimethylamino)pyrimidin-5-yl group. The piperidine ring is also connected to a 4-methoxyphenyl group, which is linked to a 6-methoxy-2-methyl-4-(dimethylamino)pyrimidin-5-yl group. The piperidine ring is further connected to a 4-methoxyphenyl group, which is linked to a 6-methoxy-2-methyl-4-(dimethylamino)pyrimidin-5-yl group.

TABLE 2C-continued

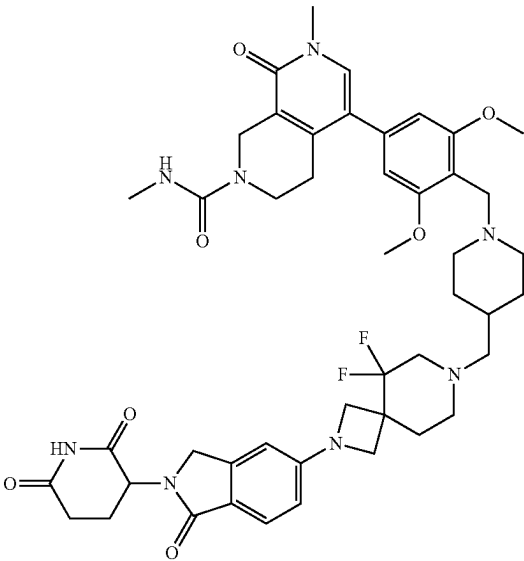
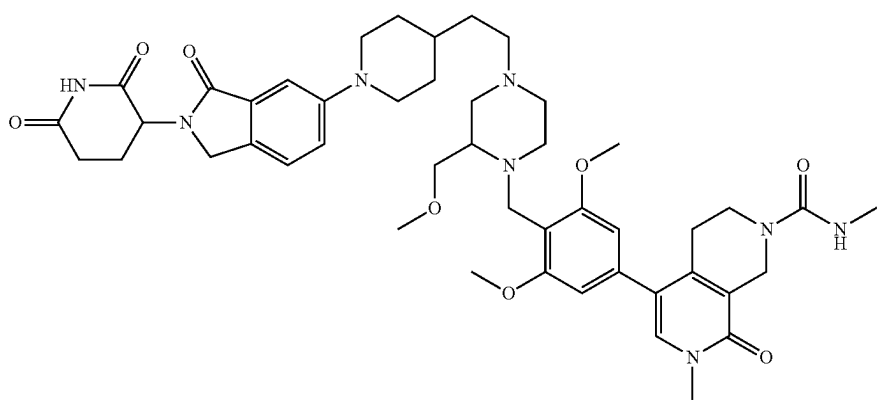
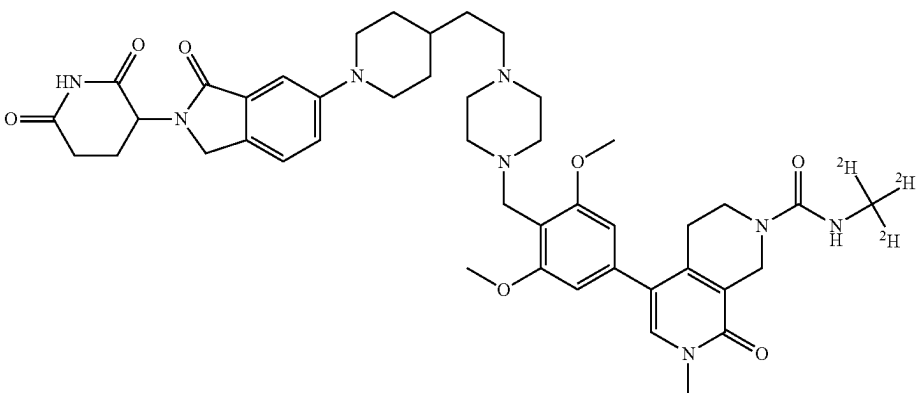
Compounds D303-D375 of the Disclosure	
Compound No.	Structure
D339	 Chemical structure of Compound D339: A complex molecule featuring a central benzimidazole core. The benzimidazole is substituted with a 6-methoxy-4-(2-methyl-2-oxo-1,2,3,4-tetrahydropyridin-6-yl)phenyl group and a 2-(2,2-difluoro-1-(4-methoxy-2-(2-methoxy-4-(2-methyl-2-oxo-1,2,3,4-tetrahydropyridin-6-yl)phenyl)pyrimidin-5-yl)ethyl)ethyl group. The pyrimidine ring is also substituted with a methyl group and a carbonyl group.
D340	 Chemical structure of Compound D340: A complex molecule featuring a central benzimidazole core. The benzimidazole is substituted with a 6-methoxy-4-(2-methyl-2-oxo-1,2,3,4-tetrahydropyridin-6-yl)phenyl group and a 2-(2,2-difluoro-1-(4-methoxy-2-(2-methoxy-4-(2-methyl-2-oxo-1,2,3,4-tetrahydropyridin-6-yl)phenyl)pyrimidin-5-yl)ethyl)ethyl group. The pyrimidine ring is also substituted with a methyl group and a carbonyl group.
D341	 Chemical structure of Compound D341: A complex molecule featuring a central benzimidazole core. The benzimidazole is substituted with a 6-methoxy-4-(2-methyl-2-oxo-1,2,3,4-tetrahydropyridin-6-yl)phenyl group and a 2-(2,2-difluoro-1-(4-methoxy-2-(2-methoxy-4-(2-methyl-2-oxo-1,2,3,4-tetrahydropyridin-6-yl)phenyl)pyrimidin-5-yl)ethyl)ethyl group. The pyrimidine ring is also substituted with a methyl group and a carbonyl group. The structure includes deuterium labels (^2H) on the terminal methyl group of the side chain.

TABLE 2C-continued

Compounds D303-D375 of the Disclosure	
Compound No.	Structure
D342	
D343	
D344	

TABLE 2C-continued

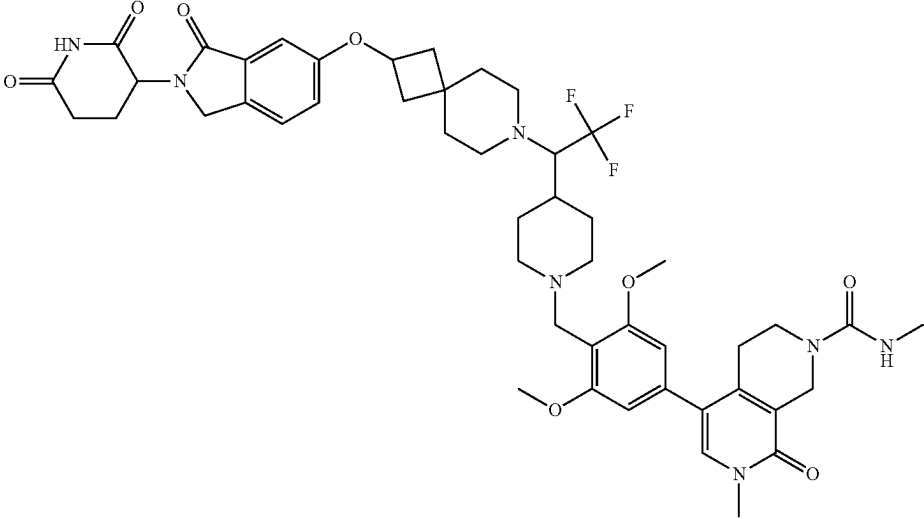
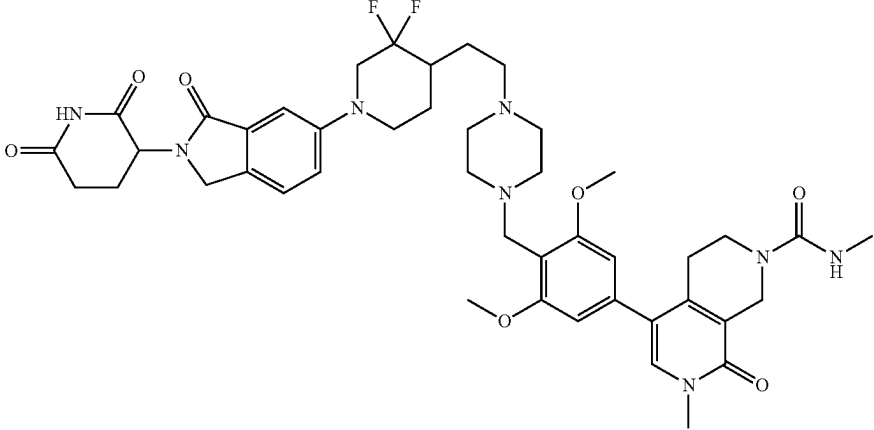
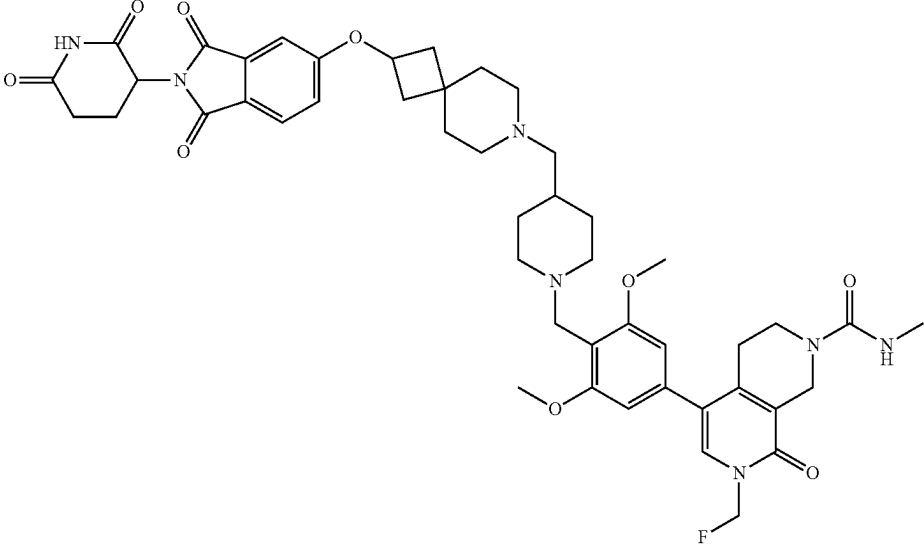
Compounds D303-D375 of the Disclosure	
Compound No.	Structure
D345	
D346	
D347	

TABLE 2C-continued

Compounds D303-D375 of the Disclosure	
Compound No.	Structure
D348	
D349	
D350	
D351	

TABLE 2C-continued

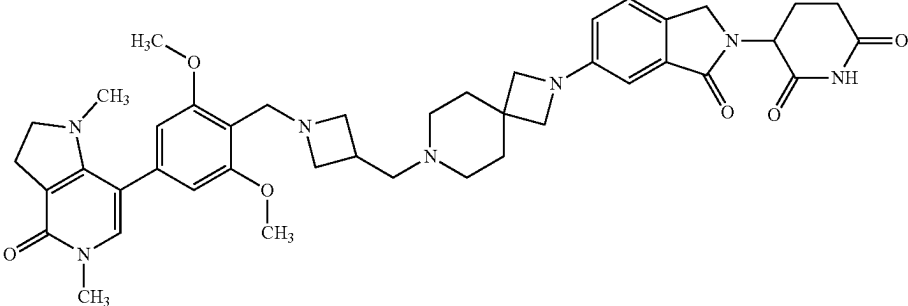
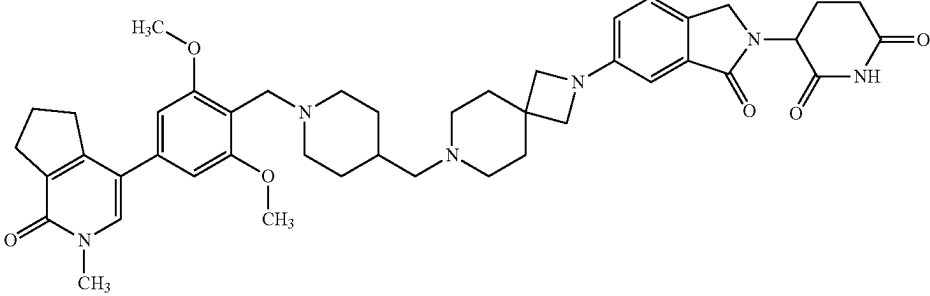
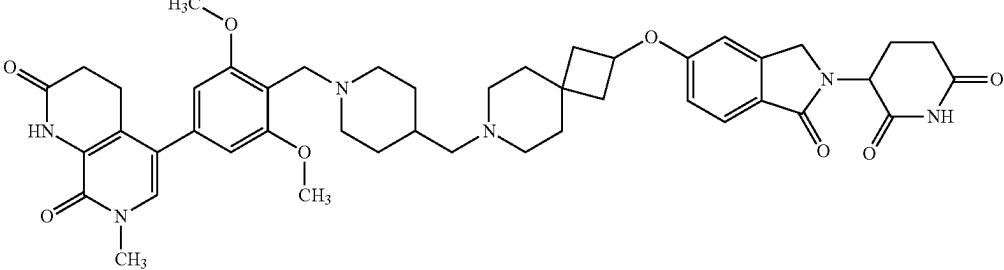
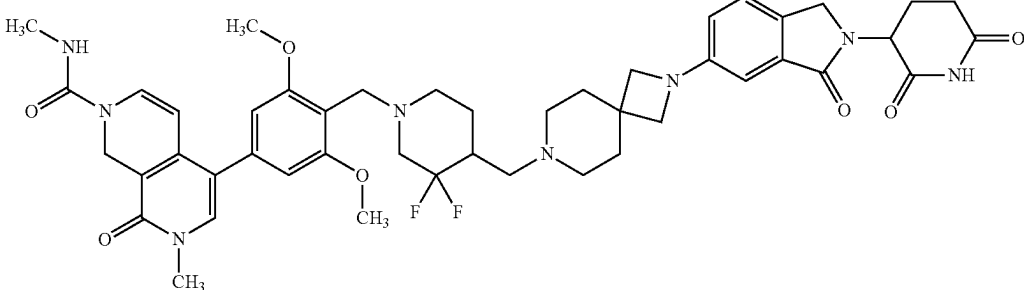
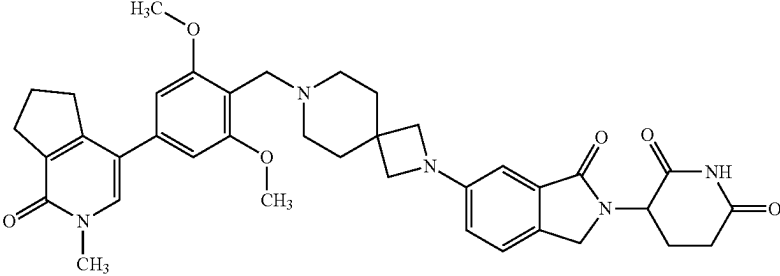
Compounds D303-D375 of the Disclosure	
Compound No.	Structure
D352	
D353	
D354	
D355	
D356	

TABLE 2C-continued

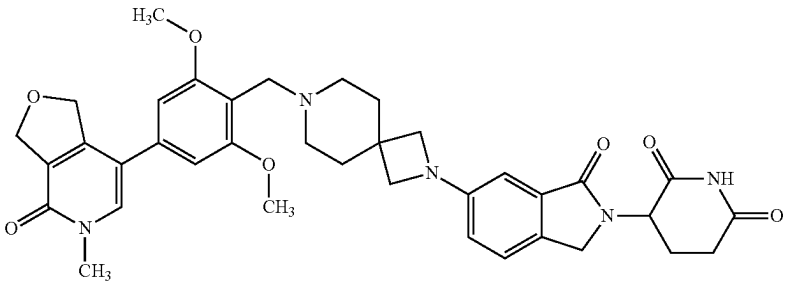
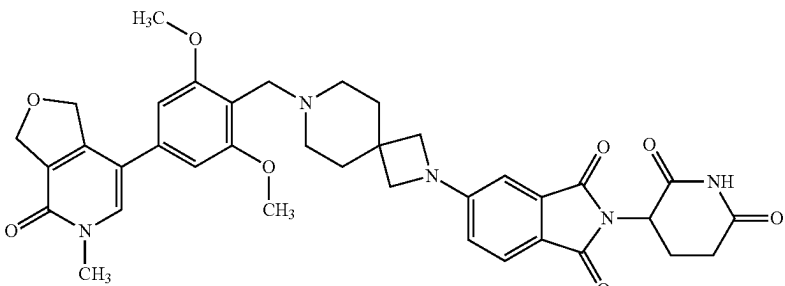
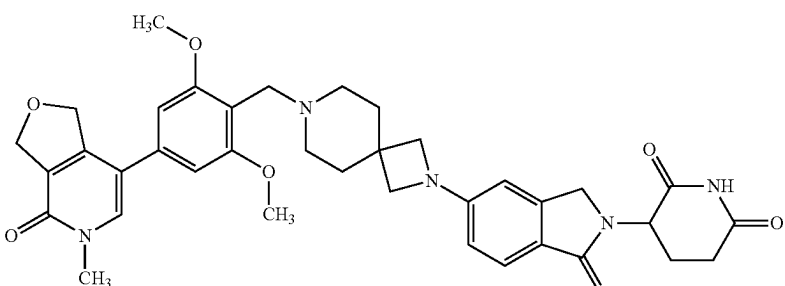
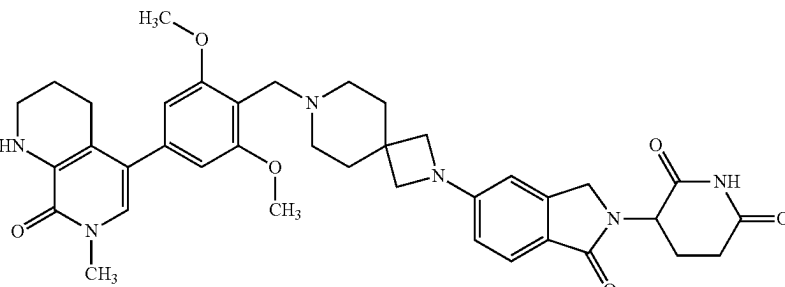
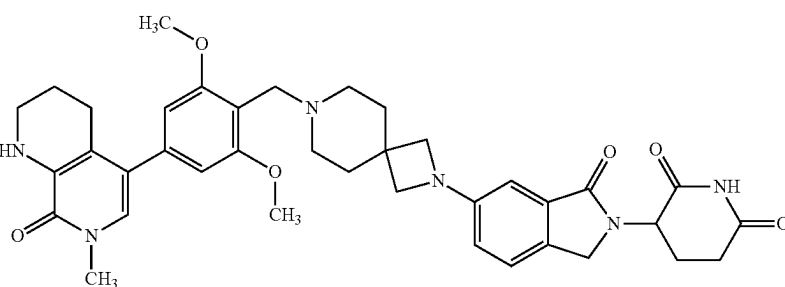
Compounds D303-D375 of the Disclosure	
Compound No.	Structure
D357	 Chemical structure of compound D357: A 4-methyl-2,3-dihydro-4H-pyridine-5(1H)-one ring is connected at the 5-position to a 3,4-dimethoxyphenyl group. The 3-position of the phenyl ring is connected via a methylene bridge to a piperidine ring, which is further connected to a bicyclo[3.1.0]hexane system. This bicyclo system is connected via its nitrogen atom to a 5,6-dihydro-1H-benzodioxole-2,3-dione ring, which is finally connected to a 2,6-pyridinedione ring.
D358	 Chemical structure of compound D358: A 4-methyl-2,3-dihydro-4H-pyridine-5(1H)-one ring is connected at the 5-position to a 3,4-dimethoxyphenyl group. The 3-position of the phenyl ring is connected via a methylene bridge to a piperidine ring, which is further connected to a bicyclo[3.1.0]hexane system. This bicyclo system is connected via its nitrogen atom to a 5,6-dihydro-1H-benzodioxole-2,3-dione ring, which is finally connected to a 2,6-pyridinedione ring.
D359	 Chemical structure of compound D359: A 4-methyl-2,3-dihydro-4H-pyridine-5(1H)-one ring is connected at the 5-position to a 3,4-dimethoxyphenyl group. The 3-position of the phenyl ring is connected via a methylene bridge to a piperidine ring, which is further connected to a bicyclo[3.1.0]hexane system. This bicyclo system is connected via its nitrogen atom to a 5,6-dihydro-1H-benzodioxole-2,3-dione ring, which is finally connected to a 2,6-pyridinedione ring.
D360	 Chemical structure of compound D360: A 4-methyl-2,3-dihydro-4H-pyridine-5(1H)-one ring is connected at the 5-position to a 3,4-dimethoxyphenyl group. The 3-position of the phenyl ring is connected via a methylene bridge to a piperidine ring, which is further connected to a bicyclo[3.1.0]hexane system. This bicyclo system is connected via its nitrogen atom to a 5,6-dihydro-1H-benzodioxole-2,3-dione ring, which is finally connected to a 2,6-pyridinedione ring.
D361	 Chemical structure of compound D361: A 4-methyl-2,3-dihydro-4H-pyridine-5(1H)-one ring is connected at the 5-position to a 3,4-dimethoxyphenyl group. The 3-position of the phenyl ring is connected via a methylene bridge to a piperidine ring, which is further connected to a bicyclo[3.1.0]hexane system. This bicyclo system is connected via its nitrogen atom to a 5,6-dihydro-1H-benzodioxole-2,3-dione ring, which is finally connected to a 2,6-pyridinedione ring.

TABLE 2C-continued

Compounds D303-D375 of the Disclosure	
Compound No.	Structure
D362	 <chem>CN1C=CC2=C(C1)C3=CC(OC)=C(C3CN4C5=CC(=C(C=C5)C(=O)N4C6=CC(=O)NC(=O)C6)OC4)C2</chem>
D363	 <chem>CN1C=CC2=C(C1)C3=CC(OC)=C(C3CN4C5=CC(=C(C=C5)C(=O)N4C6=CC(=O)NC(=O)C6)OC4)C2</chem>
D364	 <chem>CN1C=CC2=C(C1)C3=CC(OC)=C(C3CN4C5=CC(=C(C=C5)C(=O)N4C6=CC(=O)NC(=O)C6)OC4)C2</chem>
D365	 <chem>CN1C=CC2=C(C1)C3=CC(OC)=C(C3CN4C5=CC(=C(C=C5)C(=O)N4C6=CC(=O)NC(=O)C6)OC4)C2</chem>
D366	 <chem>CN1C=CC2=C(C1)C3=CC(OC)=C(C3CN4C5=CC(=C(C=C5)C(=O)N4C6=CC(=O)NC(=O)C6)OC4)C2</chem>

TABLE 2C-continued

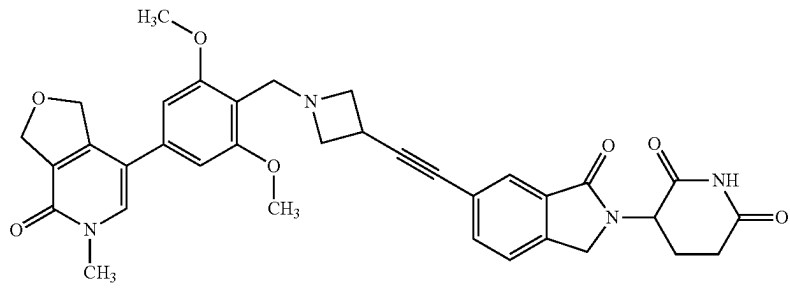
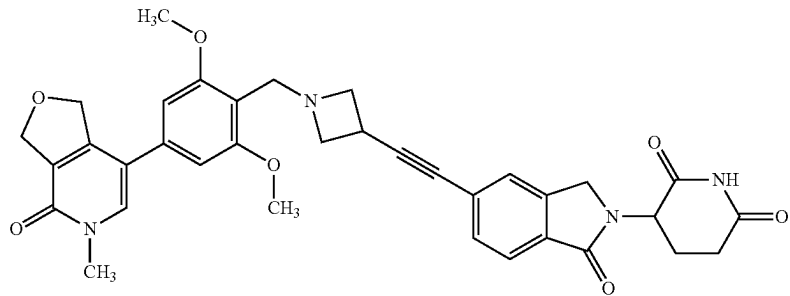
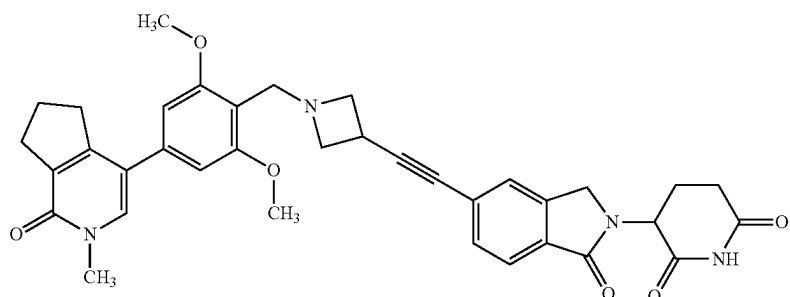
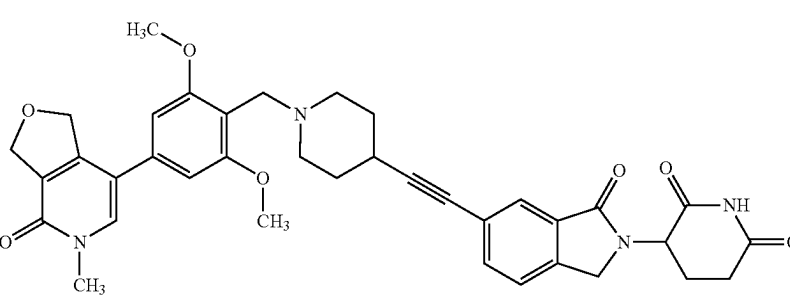
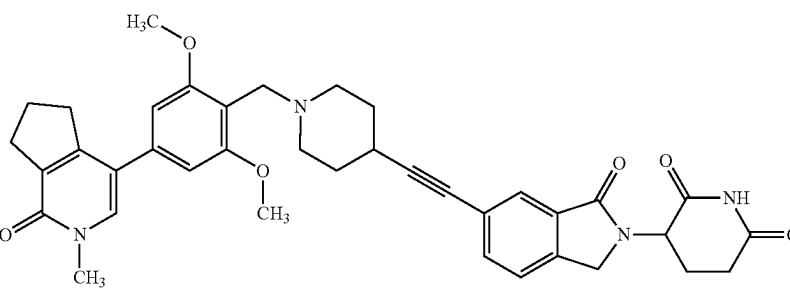
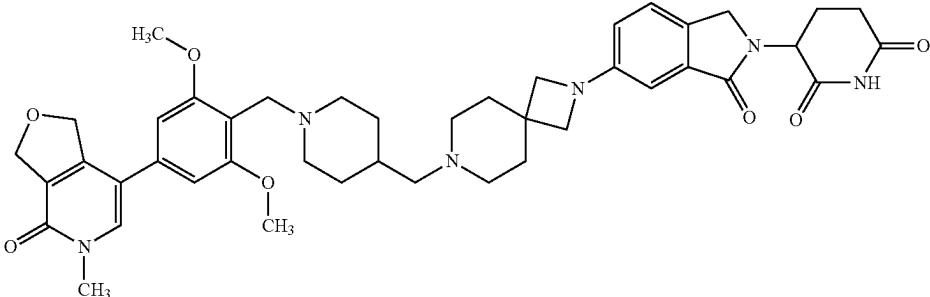
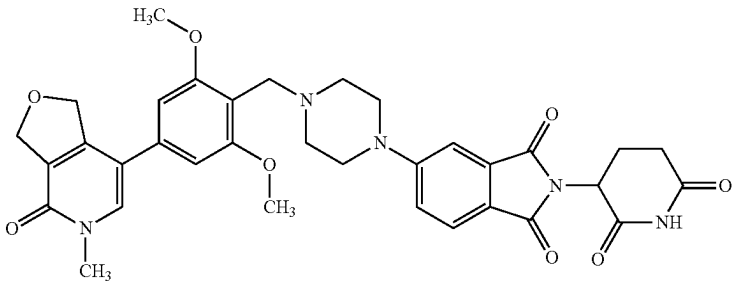
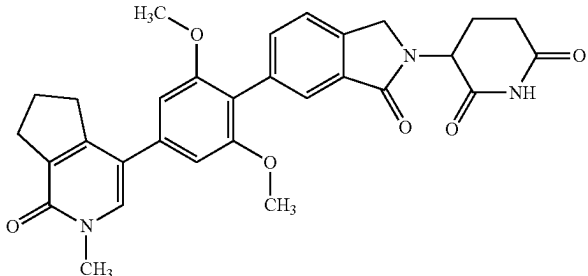
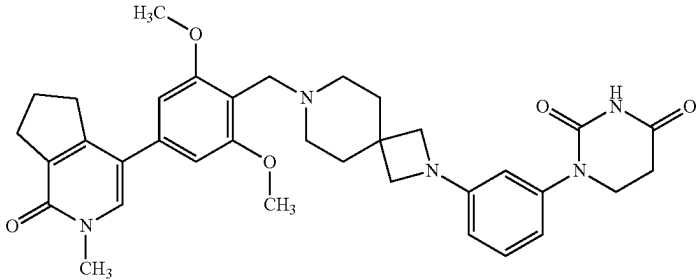
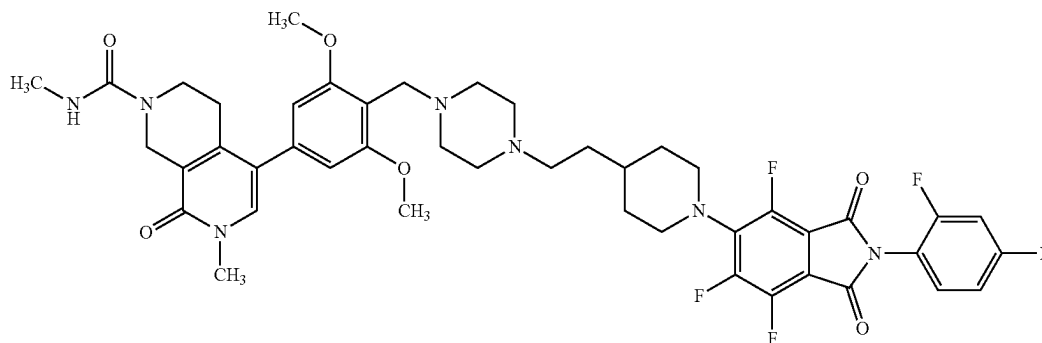
Compounds D303-D375 of the Disclosure	
Compound No.	Structure
D367	
D368	
D369	
D370	
D371	

TABLE 2C-continued

Compounds D303-D375 of the Disclosure	
Compound No.	Structure
D372	 Chemical structure of compound D372: A complex molecule featuring a central benzene ring substituted with two methoxy groups (H₃C-O- and -O-CH₃) and a 4-methyl-5,6-dihydro-2H-pyran-2-yl group. This central ring is connected via a methylene bridge to a piperidine ring, which is further linked to a 1,4-bis(piperidin-1-yl)butane chain. The other end of this chain is connected to a 1,4-bis(piperidin-1-yl)butane chain, which is then linked to a 1,4-bis(piperidin-1-yl)butane chain. The final part of the molecule is a 1,4-bis(piperidin-1-yl)butane chain connected to a 1,4-bis(piperidin-1-yl)butane chain, which is then linked to a 1,4-bis(piperidin-1-yl)butane chain. The final part of the molecule is a 1,4-bis(piperidin-1-yl)butane chain connected to a 1,4-bis(piperidin-1-yl)butane chain, which is then linked to a 1,4-bis(piperidin-1-yl)butane chain.
D373	 Chemical structure of compound D373: A complex molecule featuring a central benzene ring substituted with two methoxy groups (H₃C-O- and -O-CH₃) and a 4-methyl-5,6-dihydro-2H-pyran-2-yl group. This central ring is connected via a methylene bridge to a piperidine ring, which is further linked to a 1,4-bis(piperidin-1-yl)butane chain. The other end of this chain is connected to a 1,4-bis(piperidin-1-yl)butane chain, which is then linked to a 1,4-bis(piperidin-1-yl)butane chain. The final part of the molecule is a 1,4-bis(piperidin-1-yl)butane chain connected to a 1,4-bis(piperidin-1-yl)butane chain, which is then linked to a 1,4-bis(piperidin-1-yl)butane chain.
D374	 Chemical structure of compound D374: A complex molecule featuring a central benzene ring substituted with two methoxy groups (H₃C-O- and -O-CH₃) and a 4-methyl-5,6-dihydro-2H-pyran-2-yl group. This central ring is connected via a methylene bridge to a piperidine ring, which is further linked to a 1,4-bis(piperidin-1-yl)butane chain. The other end of this chain is connected to a 1,4-bis(piperidin-1-yl)butane chain, which is then linked to a 1,4-bis(piperidin-1-yl)butane chain. The final part of the molecule is a 1,4-bis(piperidin-1-yl)butane chain connected to a 1,4-bis(piperidin-1-yl)butane chain, which is then linked to a 1,4-bis(piperidin-1-yl)butane chain.
D375	 Chemical structure of compound D375: A complex molecule featuring a central benzene ring substituted with two methoxy groups (H₃C-O- and -O-CH₃) and a 4-methyl-5,6-dihydro-2H-pyran-2-yl group. This central ring is connected via a methylene bridge to a piperidine ring, which is further linked to a 1,4-bis(piperidin-1-yl)butane chain. The other end of this chain is connected to a 1,4-bis(piperidin-1-yl)butane chain, which is then linked to a 1,4-bis(piperidin-1-yl)butane chain. The final part of the molecule is a 1,4-bis(piperidin-1-yl)butane chain connected to a 1,4-bis(piperidin-1-yl)butane chain, which is then linked to a 1,4-bis(piperidin-1-yl)butane chain.

In an aspect, the disclosure features a compound having the structure of DD1, or a pharmaceutically acceptable salt thereof.



DD1

In another aspect, the disclosure features a pharmaceutical composition including any of the foregoing compounds, or pharmaceutically acceptable salts thereof, and a pharmaceutically acceptable excipient.

In an aspect, the disclosure features a method of inhibiting the level and/or activity of BRD9 in a cell, the method involving contacting the cell with an effective amount of any of the foregoing compounds, or pharmaceutically acceptable salts thereof, or a pharmaceutical composition thereof.

In another aspect, the disclosure features a method of reducing the level and/or activity of BRD9 in a cell, the method involving contacting the cell with an effective amount of any of the foregoing compounds, or pharmaceutically acceptable salts thereof, or a pharmaceutical composition thereof.

In some embodiments, the cell is a cancer cell.

In some embodiments, the cancer is a malignant, rhabdoid tumor, a CD8+ T-cell lymphoma, endometrial carcinoma, ovarian carcinoma, bladder cancer, stomach cancer, pancreatic cancer, esophageal cancer, prostate cancer, renal cell carcinoma, melanoma, colorectal cancer, a sarcoma (e.g., a soft tissue sarcoma, synovial sarcoma, Ewing's sarcoma, osteosarcoma, rhabdomyosarcoma, adult fibrosarcoma, alveolar soft-part sarcoma, angiosarcoma, clear cell sarcoma, desmoplastic small round cell tumor, epithelioid sarcoma, fibromyxoid sarcoma, gastrointestinal stromal tumor, Kaposi sarcoma, liposarcoma, leiomyosarcoma, malignant mesenchymoma malignant peripheral nerve sheath tumors, myxofibrosarcoma, low-grade rhabdomyosarcoma), non-small cell lung cancer (e.g., squamous or adenocarcinoma), stomach cancer, or breast cancer. In some embodiments, the cancer is a malignant, rhabdoid tumor, a CD8+ T-cell lymphoma, endometrial carcinoma, ovarian carcinoma, bladder cancer, stomach cancer, pancreatic cancer, esophageal cancer, prostate cancer, renal cell carcinoma, melanoma, or colorectal cancer. In some embodiments, the cancer is a sarcoma (e.g., synovial sarcoma or Ewing's sarcoma), non-small cell lung cancer (e.g., squamous or adenocarcinoma), stomach cancer, or breast cancer. In some embodiments, the cancer is sarcoma (e.g., synovial sarcoma or Ewing's sarcoma). In some embodiments, the sarcoma is synovial sarcoma.

In an aspect, the disclosure features a method of treating a BAF complex-related disorder in a subject in need thereof, the method involving administering to the subject an effective amount of any of the foregoing compounds, or phar-

maceutically acceptable salts thereof, or a pharmaceutical composition thereof. In some embodiments, the BAF complex-related disorder is cancer. In some embodiments, the BAF complex-related disorder is infection.

In another aspect, the disclosure features a method of treating an SS18-SSX fusion protein-related disorder in a subject in need thereof, the method involving administering to the subject an effective amount of any of the foregoing compounds, or pharmaceutically acceptable salts thereof, or a pharmaceutical composition thereof. In some embodiments, the SS18-SSX fusion protein-related disorder is cancer. In some embodiments, the SS18-SSX fusion protein-related disorder is infection. In some embodiments of any of the foregoing methods, the SS18-SSX fusion protein is a SS18-SSX1 fusion protein, a SS18-SSX2 fusion protein, or a SS18-SSX4 fusion protein.

In yet another aspect, the disclosure features a method of treating a BRD9-related disorder in a subject in need thereof, the method involving administering to the subject an effective amount of any of the foregoing compounds, or pharmaceutically acceptable salts thereof, or a pharmaceutical composition thereof. In some embodiments, the BRD9-related disorder is cancer. In some embodiments, the BRD9-related disorder is infection.

In some embodiments, the cancer is squamous cell carcinoma, basal cell carcinoma, adenocarcinoma, hepatocellular carcinomas, and renal cell carcinomas, cancer of the bladder, bowel, breast, cervix, colon, esophagus, head, kidney, liver, lung, neck, ovary, pancreas, prostate, and stomach; leukemias; benign and malignant lymphomas, particularly Burkitt's lymphoma and Non-Hodgkin's lymphoma; benign and malignant melanomas; myeloproliferative diseases; sarcomas, including Ewing's sarcoma, hemangiosarcoma, Kaposi's sarcoma, liposarcoma, myosarcomas, peripheral neuroepithelioma, synovial sarcoma, gliomas, astrocytomas, oligodendrogliomas, ependymomas, glioblastomas, neuroblastomas, ganglioneuromas, gangliogliomas, medulloblastomas, pineal cell tumors, meningiomas, meningeal sarcomas, neurofibromas, and Schwannomas; bowel cancer, breast cancer, prostate cancer, cervical cancer, uterine cancer, lung cancer, ovarian cancer, testicular cancer, thyroid cancer, astrocytoma, esophageal cancer, pancreatic cancer, stomach cancer, liver cancer, colon cancer, melanoma; carcinosarcoma, Hodgkin's disease, Wilms' tumor and teratocarcinomas. Additional cancers which may be treated using the disclosed compounds according to the

present invention include, for example, acute granulocytic leukemia, acute lymphocytic leukemia (ALL), acute myelogenous leukemia (AML), adenocarcinoma, adenocarcinoma, adrenal cancer, adrenocortical carcinoma, anal cancer, anaplastic astrocytoma, angiosarcoma, appendix cancer, astrocytoma, Basal cell carcinoma, B-Cell lymphoma, bile duct cancer, bladder cancer, bone cancer, bone marrow cancer, bowel cancer, brain cancer, brain stem glioma, breast cancer, triple (estrogen, progesterone and HER-2) negative breast cancer, double negative breast cancer (two of estrogen, progesterone and HER-2 are negative), single negative (one of estrogen, progesterone and HER-2 is negative), estrogen-receptor positive, HER2-negative breast cancer, estrogen receptor-negative breast cancer, estrogen receptor positive breast cancer, metastatic breast cancer, luminal A breast cancer, luminal B breast cancer, Her2-negative breast cancer, HER2-positive or negative breast cancer, progesterone receptor-negative breast cancer, progesterone receptor-positive breast cancer, recurrent breast cancer, carcinoid tumors, cervical cancer, cholangiocarcinoma, chondrosarcoma, chronic lymphocytic leukemia (CLL), chronic myelogenous leukemia (CML), colon cancer, colorectal cancer, craniopharyngioma, cutaneous lymphoma, cutaneous melanoma, diffuse astrocytoma, ductal carcinoma in situ (DCIS), endometrial cancer, ependymoma, epithelioid sarcoma, esophageal cancer, ewing sarcoma, extrahepatic bile duct cancer, eye cancer, fallopian tube cancer, fibrosarcoma, gallbladder cancer, gastric cancer, gastrointestinal cancer, gastrointestinal carcinoid cancer, gastrointestinal stromal tumors (GIST), germ cell tumor glioblastoma multiforme (GBM), glioma, hairy cell leukemia, head and neck cancer, hemangioendothelioma, Hodgkin lymphoma, hypopharyngeal cancer, infiltrating ductal carcinoma (IDC), infiltrating lobular carcinoma (ILC), inflammatory breast cancer (IBC), intestinal Cancer, intrahepatic bile duct cancer, invasive/infiltrating breast cancer, Islet cell cancer, jaw cancer, Kaposi sarcoma, kidney cancer, laryngeal cancer, leiomyosarcoma, leptomeningeal metastases, leukemia, lip cancer, liposarcoma, liver cancer, lobular carcinoma in situ, low-grade astrocytoma, lung cancer, lymph node cancer, lymphoma, male breast cancer, medullary carcinoma, medulloblastoma, melanoma, meningioma, Merkel cell carcinoma, mesenchymal chondrosarcoma, mesenchymous, mesothelioma metastatic breast cancer, metastatic melanoma metastatic squamous neck cancer, mixed gliomas, monodermal teratoma, mouth cancer mucinous carcinoma, mucosal melanoma, multiple myeloma, Mycosis Fungoides, myelodysplastic syndrome, nasal cavity cancer, nasopharyngeal cancer, neck cancer, neuroblastoma, neuroendocrine tumors (NETs), non-Hodgkin's lymphoma, non-small cell lung cancer (NSCLC), oat cell cancer, ocular cancer, ocular melanoma, oligodendroglioma, oral cancer, oral cavity cancer, oropharyngeal cancer, osteogenic sarcoma, osteosarcoma, ovarian cancer, ovarian epithelial cancer ovarian germ cell tumor, ovarian primary peritoneal carcinoma, ovarian sex cord stromal tumor, Paget's disease, pancreatic cancer, papillary carcinoma, paranasal sinus cancer, parathyroid cancer, pelvic cancer, penile cancer, peripheral nerve cancer, peritoneal cancer, pharyngeal cancer, pheochromocytoma, pilocytic astrocytoma, pineal region tumor, pineoblastoma, pituitary gland cancer, primary central nervous system (CNS) lymphoma, prostate cancer, rectal cancer, renal cell carcinoma, renal pelvis cancer, rhabdomyosarcoma, salivary gland cancer, soft tissue sarcoma, bone sarcoma, sarcoma, sinus cancer, skin cancer, small cell lung cancer (SCLC), small intestine cancer, spinal cancer, spinal column cancer, spinal cord cancer, squamous cell carcinoma, stomach can-

cer, synovial sarcoma, T-cell lymphoma, testicular cancer, throat cancer, thymoma/thymic carcinoma, thyroid cancer, tongue cancer, tonsil cancer, transitional cell cancer, tubal cancer, tubular carcinoma, undiagnosed cancer, ureteral cancer, urethral cancer, uterine adenocarcinoma, uterine cancer, uterine sarcoma, vaginal cancer, vulvar cancer, T-cell lineage acute lymphoblastic leukemia (T-ALL), T-cell lineage lymphoblastic lymphoma (T-LL), peripheral T-cell lymphoma, Adult T-cell leukemia, Pre-B ALL, Pre-B lymphomas, large B-cell lymphoma, Burkitts lymphoma, B-cell ALL, Philadelphia chromosome positive ALL, Philadelphia chromosome positive CML, juvenile myelomonocytic leukemia (JMML), acute promyelocytic leukemia (a subtype of AML), large granular lymphocytic leukemia, Adult T-cell chronic leukemia, diffuse large B cell lymphoma, follicular lymphoma; Mucosa-Associated Lymphatic Tissue lymphoma (MALT), small cell lymphocytic lymphoma, mediastinal large B cell lymphoma, nodal marginal zone B cell lymphoma (NMZL); splenic marginal zone lymphoma (SMZL); intravascular large B-cell lymphoma; primary effusion lymphoma; or lymphomatoid granulomatosis; B-cell prolymphocytic leukemia; splenic lymphoma/leukemia, unclassifiable, splenic diffuse red pulp small B-cell lymphoma; lymphoplasmacytic lymphoma; heavy chain diseases, for example, Alpha heavy chain disease, Gamma heavy chain disease, Mu heavy chain disease, plasma cell myeloma, solitary plasmacytoma of bone; extraosseous plasmacytoma; primary cutaneous follicle center lymphoma, T cell/histocyte rich large B-cell lymphoma, DLBCL associated with chronic inflammation; Epstein-Barr virus (EBV)+ DLBCL of the elderly; primary mediastinal (thymic) large B-cell lymphoma, primary cutaneous DLBCL, leg type, ALK+ large B-cell lymphoma, plasmablastic lymphoma; large B-cell lymphoma arising in HHV8-associated multicentric, Castleman disease; B-cell lymphoma, unclassifiable, with features intermediate between diffuse large B-cell lymphoma, or B-cell lymphoma, unclassifiable, with features intermediate between diffuse large B-cell lymphoma and classical Hodgkin lymphoma.

In some embodiments, the cancer is a malignant, rhabdoid tumor, a CD8+ T-cell lymphoma, endometrial carcinoma, ovarian carcinoma, bladder cancer, stomach cancer, pancreatic cancer, esophageal cancer, prostate cancer, renal cell carcinoma, melanoma, colorectal cancer, a sarcoma (e.g., a soft tissue sarcoma, synovial sarcoma, Ewing's sarcoma, osteosarcoma, rhabdomyosarcoma, adult fibrosarcoma, alveolar soft-part sarcoma, angiosarcoma, clear cell sarcoma, desmoplastic small round cell tumor, epithelioid sarcoma, fibromyxoid sarcoma, gastrointestinal stromal tumor, Kaposi sarcoma, liposarcoma, leiomyosarcoma, malignant mesenchymoma malignant peripheral nerve sheath tumors, myxofibrosarcoma, low-grade rhabdomyosarcoma), non-small cell lung cancer (e.g., squamous or adenocarcinoma), stomach cancer, or breast cancer. In some embodiments, the cancer is a malignant, rhabdoid tumor, a CD8+ T-cell lymphoma, endometrial carcinoma, ovarian carcinoma, bladder cancer, stomach cancer, pancreatic cancer, esophageal cancer, prostate cancer, renal cell carcinoma, melanoma, or colorectal cancer. In some embodiments, the cancer is a sarcoma (e.g., synovial sarcoma or Ewing's sarcoma), non-small cell lung cancer (e.g., squamous or adenocarcinoma), stomach cancer, or breast cancer. In some embodiments, the cancer is sarcoma (e.g., synovial sarcoma or Ewing's sarcoma). In some embodiments, the sarcoma is synovial sarcoma.

In some embodiments, the infection is viral infection (e.g., an infection with a virus of the Retroviridae family such as the lentiviruses (e.g. Human immunodeficiency virus (HIV) and deltaretroviruses (e.g., human T cell leukemia virus I (HTLV-I), human T cell leukemia virus II (HTLV-II)); Hepadnaviridae family (e.g. hepatitis B virus (HBV)); Flaviviridae family (e.g. hepatitis C virus (HCV)); Adenoviridae family (e.g. Human Adenovirus); Herpesviridae family (e.g. Human cytomegalovirus (HCMV), Epstein-Barr virus, herpes simplex virus 1 (HSV-1), herpes simplex virus 2 (HSV-2), human herpesvirus 6 (HHV-6), Herpesvirus K*, CMV, varicella-zoster virus); Papillomaviridae family (e.g. Human Papillomavirus (HPV, HPV E1)); Parvoviridae family (e.g. Parvovirus B19); Polyomaviridae family (e.g. JC virus and BK virus); Paramyxoviridae family (e.g. Measles virus); or Togaviridae family (e.g. Rubella virus)). In some embodiments, the disorder is Coffin Siris, Neurofibromatosis (e.g., NF-1, NF-2, or Schwannomatosis), or Multiple Meningioma. In an aspect, the disclosure features a method of treating a cancer in a subject in need thereof, the method including administering to the subject an effective amount of any of the foregoing compounds, or pharmaceutically acceptable salts thereof, or any of the foregoing pharmaceutical compositions.

In some embodiments, the cancer is squamous cell carcinoma, basal cell carcinoma, adenocarcinoma, hepatocellular carcinomas, and renal cell carcinomas, cancer of the bladder, bowel, breast, cervix, colon, esophagus, head, kidney, liver, lung, neck, ovary, pancreas, prostate, and stomach; leukemias; benign and malignant lymphomas, particularly Burkitt's lymphoma and Non-Hodgkin's lymphoma; benign and malignant melanomas; myeloproliferative diseases; sarcomas, including Ewing's sarcoma, hemangiosarcoma, Kaposi's sarcoma, liposarcoma, myosarcomas, peripheral neuroepithelioma, synovial sarcoma, gliomas, astrocytomas, oligodendrogliomas, ependymomas, glioblastomas, neuroblastomas, ganglioneuromas, gangliogliomas, medulloblastomas, pineal cell tumors, meningiomas, meningeal sarcomas, neurofibromas, and Schwannomas; bowel cancer, breast cancer, prostate cancer, cervical cancer, uterine cancer, lung cancer, ovarian cancer, testicular cancer, thyroid cancer, astrocytoma, esophageal cancer, pancreatic cancer, stomach cancer, liver cancer, colon cancer, melanoma; carcinosarcoma, Hodgkin's disease, Wilms' tumor and teratocarcinomas. Additional cancers which may be treated using the disclosed compounds according to the present invention include, for example, acute granulocytic leukemia, acute lymphocytic leukemia (ALL), acute myelogenous leukemia (AML), adenocarcinoma, adenocarcinoma, adrenal cancer, adrenocortical carcinoma, anal cancer, anaplastic astrocytoma, angiosarcoma, appendix cancer, astrocytoma, Basal cell carcinoma, B-Cell lymphoma, bile duct cancer, bladder cancer, bone cancer, bone marrow cancer, bowel cancer, brain cancer, brain stem glioma, breast cancer, triple (estrogen, progesterone and HER-2) negative breast cancer, double negative breast cancer (two of estrogen, progesterone and HER-2 are negative), single negative (one of estrogen, progesterone and HER-2 is negative), estrogen-receptor positive, HER2-negative breast cancer, estrogen receptor-negative breast cancer, estrogen receptor positive breast cancer, metastatic breast cancer, luminal A breast cancer, luminal B breast cancer, Her2-negative breast cancer, HER2-positive or negative breast cancer, progesterone receptor-negative breast cancer, progesterone receptor-positive breast cancer, recurrent breast cancer, carcinoid tumors, cervical cancer, cholangiocarcinoma, chondrosarcoma, chronic lymphocytic leukemia (CLL), chronic myel-

ogenous leukemia (CML), colon cancer, colorectal cancer, craniopharyngioma, cutaneous lymphoma, cutaneous melanoma, diffuse astrocytoma, ductal carcinoma in situ (DCIS), endometrial cancer, ependymoma, epithelioid sarcoma, esophageal cancer, ewing sarcoma, extrahepatic bile duct cancer, eye cancer, fallopian tube cancer, fibrosarcoma, gallbladder cancer, gastric cancer, gastrointestinal cancer, gastrointestinal carcinoid cancer, gastrointestinal stromal tumors (GIST), germ cell tumor glioblastoma multiforme (GBM), glioma, hairy cell leukemia, head and neck cancer, hemangioendothelioma, Hodgkin lymphoma, hypopharyngeal cancer, infiltrating ductal carcinoma (IDC), infiltrating lobular carcinoma (ILC), inflammatory breast cancer (IBC), intestinal Cancer, intrahepatic bile duct cancer, invasive/infiltrating breast cancer, Islet cell cancer, jaw cancer, Kaposi sarcoma, kidney cancer, laryngeal cancer, leiomyosarcoma, leptomeningeal metastases, leukemia, lip cancer, liposarcoma, liver cancer, lobular carcinoma in situ, low-grade astrocytoma, lung cancer, lymph node cancer, lymphoma, male breast cancer, medullary carcinoma, medulloblastoma, melanoma, meningioma, Merkel cell carcinoma, mesenchymal chondrosarcoma, mesenchymous, mesothelioma metastatic breast cancer, metastatic melanoma metastatic squamous neck cancer, mixed gliomas, monodermal teratoma, mouth cancer mucinous carcinoma, mucosal melanoma, multiple myeloma, Mycosis Fungoides, myelodysplastic syndrome, nasal cavity cancer, nasopharyngeal cancer, neck cancer, neuroblastoma, neuroendocrine tumors (NETs), non-Hodgkin's lymphoma, non-small cell lung cancer (NSCLC), oat cell cancer, ocular cancer, ocular melanoma, oligodendroglioma, oral cancer, oral cavity cancer, oropharyngeal cancer, osteogenic sarcoma, osteosarcoma, ovarian cancer, ovarian epithelial cancer ovarian germ cell tumor, ovarian primary peritoneal carcinoma, ovarian sex cord stromal tumor, Paget's disease, pancreatic cancer, papillary carcinoma, paranasal sinus cancer, parathyroid cancer, pelvic cancer, penile cancer, peripheral nerve cancer, peritoneal cancer, pharyngeal cancer, pheochromocytoma, pilocytic astrocytoma, pineal region tumor, pineoblastoma, pituitary gland cancer, primary central nervous system (CNS) lymphoma, prostate cancer, rectal cancer, renal cell carcinoma, renal pelvis cancer, rhabdomyosarcoma, salivary gland cancer, soft tissue sarcoma, bone sarcoma, sarcoma, sinus cancer, skin cancer, small cell lung cancer (SCLC), small intestine cancer, spinal cancer, spinal column cancer, spinal cord cancer, squamous cell carcinoma, stomach cancer, synovial sarcoma, T-cell lymphoma, testicular cancer, throat cancer, thymoma/thymic carcinoma, thyroid cancer, tongue cancer, tonsil cancer, transitional cell cancer, tubal cancer, tubular carcinoma, undiagnosed cancer, ureteral cancer, urethral cancer, uterine adenocarcinoma, uterine cancer, uterine sarcoma, vaginal cancer, vulvar cancer, T-cell lineage acute lymphoblastic leukemia (T-ALL), T-cell lineage lymphoblastic lymphoma (T-LL), peripheral T-cell lymphoma, Adult T-cell leukemia, Pre-B ALL, Pre-B lymphomas, large B-cell lymphoma, Burkitts lymphoma, B-cell ALL, Philadelphia chromosome positive ALL, Philadelphia chromosome positive CML, juvenile myelomonocytic leukemia (JMML), acute promyelocytic leukemia (a subtype of AML), large granular lymphocytic leukemia, Adult T-cell chronic leukemia, diffuse large B cell lymphoma, follicular lymphoma; Mucosa-Associated Lymphatic Tissue lymphoma (MALT), small cell lymphocytic lymphoma, mediastinal large B cell lymphoma, nodal marginal zone B cell lymphoma (NMZL); splenic marginal zone lymphoma (SMZL); intravascular large B-cell lymphoma; primary effusion lymphoma; or lymphomatoid granulomatosis;

B-cell prolymphocytic leukemia; splenic lymphoma/leukemia, unclassifiable, splenic diffuse red pulp small B-cell lymphoma; lymphoplasmacytic lymphoma; heavy chain diseases, for example, Alpha heavy chain disease, Gamma heavy chain disease, Mu heavy chain disease, plasma cell myeloma, solitary plasmacytoma of bone; extraosseous plasmacytoma; primary cutaneous follicle center lymphoma, T cell/histiocyte rich large B-cell lymphoma, DLBCL associated with chronic inflammation; Epstein-Barr virus (EBV)+ DLBCL of the elderly; primary mediastinal (thymic) large B-cell lymphoma, primary cutaneous DLBCL, leg type, ALK+ large B-cell lymphoma, plasmablastic lymphoma; large B-cell lymphoma arising in HHV8-associated multicentric, Castleman disease; B-cell lymphoma, unclassifiable, with features intermediate between diffuse large B-cell lymphoma, or B-cell lymphoma, unclassifiable, with features intermediate between diffuse large B-cell lymphoma and classical Hodgkin lymphoma.

In some embodiments, the cancer is a malignant, rhabdoid tumor, a CD8+ T-cell lymphoma, endometrial carcinoma, ovarian carcinoma, bladder cancer, stomach cancer, pancreatic cancer, esophageal cancer, prostate cancer, renal cell carcinoma, melanoma, colorectal cancer, a sarcoma (e.g., a soft tissue sarcoma, synovial sarcoma, Ewing's sarcoma, osteosarcoma, rhabdomyosarcoma, adult fibrosarcoma, alveolar soft-part sarcoma, angiosarcoma, clear cell sarcoma, desmoplastic small round cell tumor, epithelioid sarcoma, fibromyxoid sarcoma, gastrointestinal stromal tumor, Kaposi sarcoma, liposarcoma, leiomyosarcoma, malignant mesenchymoma malignant peripheral nerve sheath tumors, myxofibrosarcoma, low-grade rhabdomyosarcoma), non-small cell lung cancer (e.g., squamous or adenocarcinoma), stomach cancer, or breast cancer. In some embodiments, the cancer is a malignant, rhabdoid tumor, a CD8+ T-cell lymphoma, endometrial carcinoma, ovarian carcinoma, bladder cancer, stomach cancer, pancreatic cancer, esophageal cancer, prostate cancer, renal cell carcinoma, melanoma, or colorectal cancer. In some embodiments, the cancer is a sarcoma (e.g., synovial sarcoma or Ewing's sarcoma), non-small cell lung cancer (e.g., squamous or adenocarcinoma), stomach cancer, or breast cancer. In some embodiments, the cancer is sarcoma (e.g., synovial sarcoma or Ewing's sarcoma). In some embodiments, the sarcoma is synovial sarcoma.

In another aspect, the disclosure features a method for treating a viral infection in a subject in need thereof. This method includes administering to the subject an effective amount of any of the foregoing compounds, or pharmaceutically acceptable salts thereof, or any of the foregoing pharmaceutical compositions. In some embodiments, the viral infection is an infection with a virus of the Retroviridae family such as the lentiviruses (e.g. Human immunodeficiency virus (HIV) and deltaretroviruses (e.g., human T cell leukemia virus I (HTLV-I), human T cell leukemia virus II (HTLV-II)); Hepadnaviridae family (e.g. hepatitis B virus (HBV)), Flaviviridae family (e.g. hepatitis C virus (HCV)), Adenoviridae family (e.g. Human Adenovirus), Herpesviridae family (e.g. Human cytomegalovirus (HCMV), Epstein-Barr virus, herpes simplex virus 1 (HSV-1), herpes simplex virus 2 (HSV-2), human herpesvirus 6 (HHV-6), Herpesvirus K*, CMV, varicella-zoster virus), Papillomaviridae family (e.g. Human Papillomavirus (HPV, HPV E1)), Parvoviridae family (e.g. Parvovirus B19), Polyomaviridae family (e.g. JC virus and BK virus), Paramyxoviridae family (e.g. Measles virus), Togaviridae family (e.g. Rubella virus).

In another embodiment of any of the foregoing methods, the method further includes administering to the subject an additional anticancer therapy (e.g., chemotherapeutic or cytotoxic agent or radiotherapy).

In particular embodiments, the additional anticancer therapy is: a chemotherapeutic or cytotoxic agent (e.g., doxorubicin or ifosfamide), a differentiation-inducing agent (e.g., retinoic acid, vitamin D, cytokines), a hormonal agent, an immunological agent, or an anti-angiogenic agent. Chemotherapeutic and cytotoxic agents include, but are not limited to, alkylating agents, cytotoxic antibiotics, antineoplastic agents, vinca alkaloids, etoposides, and others (e.g., paclitaxel, taxol, docetaxel, taxotere, cis-platinum). A list of additional compounds having anticancer activity can be found in L. Brunton, B. Chabner and B. Knollman (eds). Goodman and Gilman's The Pharmacological Basis of Therapeutics, Twelfth Edition, 2011, McGraw Hill Companies, New York, NY.

In particular embodiments, the compound of the invention and the additional anticancer therapy and any of the foregoing compounds or pharmaceutical compositions are administered within 28 days of each other (e.g., within 21, 14, 10, 7, 5, 4, 3, 2, or 1 days) or within 24 hours (e.g., 12, 6, 3, 2, or 1 hours; or concomitantly) each in an amount that together are effective to treat the subject.

Chemical Terms

The terminology employed herein is for the purpose of describing particular embodiments and is not intended to be limiting.

For any of the following chemical definitions, a number following an atomic symbol indicates that total number of atoms of that element that are present in a particular chemical moiety. As will be understood, other atoms, such as hydrogen atoms, or substituent groups, as described herein, may be present, as necessary, to satisfy the valences of the atoms. For example, an unsubstituted C₂ alkyl group has the formula —CH₂CH₃. When used with the groups defined herein, a reference to the number of carbon atoms includes the divalent carbon in acetal and ketal groups but does not include the carbonyl carbon in acyl, ester, carbonate, or carbamate groups. A reference to the number of oxygen, nitrogen, or sulfur atoms in a heteroaryl group only includes those atoms that form a part of a heterocyclic ring.

Herein a phrase of the form "optionally substituted X" (e.g., optionally substituted alkyl) is intended to be equivalent to "X, wherein X is optionally substituted" (e.g., "alkyl, wherein said alkyl is optionally substituted"). It is not intended to mean that the feature "X" (e.g., alkyl) per se is optional. As described herein, certain compounds of interest may contain one or more "optionally substituted" moieties. In general, the term "substituted", whether preceded by the term "optionally" or not, means that one or more hydrogens of the designated moiety are replaced with a suitable substituent, e.g., any of the substituents or groups described herein. Unless otherwise indicated, an "optionally substituted" group may have a suitable substituent at each substitutable position of the group, and when more than one position in any given structure may be substituted with more than one substituent selected from a specified group, the substituent may be either the same or different at every position. Combinations of substituents envisioned by the present disclosure are preferably those that result in the formation of stable or chemically feasible compounds. The term "stable", as used herein, refers to compounds that are not substantially altered when subjected to conditions to

allow for their production, detection, and, in certain embodiments, their recovery, purification, and use for one or more of the purposes disclosed herein.

The term “aliphatic,” as used herein, refers to a saturated or unsaturated, straight, branched, or cyclic hydrocarbon. “Aliphatic” is intended herein to include, but is not limited to, alkyl, alkenyl, alkynyl, cycloalkyl, cycloalkenyl, and cycloalkynyl moieties, and thus incorporates each of these definitions. In one embodiment, “aliphatic” is used to indicate those aliphatic groups having 1-20 carbon atoms. The aliphatic chain can be, for example, mono-unsaturated, di-unsaturated, tri-unsaturated, or polyunsaturated, or alkynyl. Unsaturated aliphatic groups can be in a cis or trans configuration. In one embodiment, the aliphatic group contains from 1 to about 12 carbon atoms, more generally from 1 to about 6 carbon atoms or from 1 to about 4 carbon atoms. In one embodiment, the aliphatic group contains from 1 to about 8 carbon atoms. In certain embodiments, the aliphatic group is C₁-C₂, C₁-C₃, C₁-C₄, C₁-C₅, or C₁-C₆. The specified ranges as used herein indicate an aliphatic group having each member of the range described as an independent species. For example, the term C₁-C₆ aliphatic as used herein indicates a straight or branched alkyl, alkenyl, or alkynyl group having from 1, 2, 3, 4, 5, or 6 carbon atoms and is intended to mean that each of these is described as an independent species. For example, the term C₁-C₄ aliphatic as used herein indicates a straight or branched alkyl, alkenyl, or alkynyl group having from 1, 2, 3, or 4 carbon atoms and is intended to mean that each of these is described as an independent species. In one embodiment, the aliphatic group is substituted with one or more functional groups that results in the formation of a stable moiety.

The term “heteroaliphatic,” as used herein, refers to an aliphatic moiety that contains at least one heteroatom in the chain, for example, an amine, carbonyl, carboxy, oxo, thio, phosphate, phosphonate, nitrogen, phosphorus, silicon, or boron atoms in place of a carbon atom. In one embodiment, the only heteroatom is nitrogen. In one embodiment, the only heteroatom is oxygen. In one embodiment, the only heteroatom is sulfur. “Heteroaliphatic” is intended herein to include, but is not limited to, heteroalkyl, heteroalkenyl, heteroalkynyl, heterocycloalkyl, heterocycloalkenyl, and heterocycloalkynyl moieties. In one embodiment, “heteroaliphatic” is used to indicate a heteroaliphatic group (cyclic, acyclic, substituted, unsubstituted, branched or unbranched) having 1-20 carbon atoms. In one embodiment, the heteroaliphatic group is optionally substituted in a manner that results in the formation of a stable moiety. Nonlimiting examples of heteroaliphatic moieties are polyethylene glycol, polyalkylene glycol, amide, polyamide, polylactide, polyglycolide, thioether, ether, alkyl-heterocycle-alkyl, —O-alkyl-O-alkyl, and alkyl-O-haloalkyl.

The term “acyl,” as used herein, represents a hydrogen or an alkyl group that is attached to a parent molecular group through a carbonyl group, as defined herein, and is exemplified by formyl (i.e., a carboxyaldehyde group), acetyl, trifluoroacetyl, propionyl, and butanoyl. Exemplary unsubstituted acyl groups include from 1 to 6, from 1 to 11, or from 1 to 21 carbons.

The term “alkyl,” as used herein, refers to a branched or straight-chain monovalent saturated aliphatic hydrocarbon radical of 1 to 20 carbon atoms (e.g., 1 to 16 carbon atoms, 1 to 10 carbon atoms, 1 to 6 carbon atoms, or 1 to 3 carbon atoms). An “alkylene” is a divalent alkyl group.

The term “alkenyl,” as used herein, alone or in combination with other groups, refers to a straight chain or branched hydrocarbon residue having a carbon-carbon double bond

and having 2 to 20 carbon atoms (e.g., 2 to 16 carbon atoms, 2 to 10 carbon atoms, 2 to 6, or 2 carbon atoms). An “alkenylene” is a divalent alkenyl group.

The term “alkynyl,” as used herein, alone or in combination with other groups, refers to a straight chain or branched hydrocarbon residue having a carbon-carbon triple bond and having 2 to 20 carbon atoms (e.g., 2 to 16 carbon atoms, 2 to 10 carbon atoms, 2 to 6, or 2 carbon atoms). An “alkynylene” is a divalent alkynyl group.

The term “amino,” as used herein, represents —N(R^{N1})₂, wherein each R^{N1} is, independently, H, OH, NO₂, N(R^{N2})₂, SO₂OR^{N2}, SO₂R^{N2}, SOR^{N2}, an N-protecting group, alkyl, alkoxy, aryl, arylalkyl, cycloalkyl, acyl (e.g., acetyl, trifluoroacetyl, or others described herein), wherein each of these recited R^{N1} groups can be optionally substituted; or two R^{N1} combine to form an alkylene or heteroalkylene, and wherein each R^{N2} is, independently, H, alkyl, or aryl. The amino groups of the compounds described herein can be an unsubstituted amino (i.e., —NH₂) or a substituted amino (i.e., —N(R^{N1})₂).

The term “aryl,” as used herein, refers to an aromatic mono- or polycarbocyclic radical of, e.g., 6 to 12, carbon atoms having at least one aromatic ring. Examples of such groups include, but are not limited to, phenyl, naphthyl, 1,2,3,4-tetrahydronaphthyl, 1,2-dihydronaphthyl, indanyl, and 1H-indenyl.

The term “arylalkyl,” as used herein, represents an alkyl group substituted with an aryl group. Exemplary unsubstituted arylalkyl groups are from 7 to 30 carbons (e.g., from 7 to 16 or from 7 to 20 carbons, such as C₁-C₆ alkyl C₆-C₁₀ aryl, C₁-C₁₀ alkyl C₆-C₁₀ aryl, or C₁-C₂₀ alkyl C₆-C₁₀ aryl), such as, benzyl and phenethyl. In some embodiments, the alkyl and the aryl each can be further substituted with 1, 2, 3, or 4 substituent groups as defined herein for the respective groups.

The term “azido,” as used herein, represents a —N₃ group.

The term “bridged cyclyl,” as used herein, refers to a bridged polycyclic group of 5 to 20 atoms, containing from 1 to 3 bridges. Bridged cyclyl includes bridged carbocyclyl (e.g., norbornyl) and bridged heterocyclyl (e.g., 1,4-diazabicyclo[2.2.2]octane).

The term “cyano,” as used herein, represents a —CN group.

The term “carbocyclyl,” as used herein, refers to a non-aromatic C₃-C₁₂, monocyclic or polycyclic (e.g., bicyclic or tricyclic) structure in which the rings are formed by carbon atoms. Carbocyclyl structures include cycloalkyl groups (e.g., cyclohexyl) and unsaturated carbocyclyl radicals (e.g., cyclohexenyl). Polycyclic carbocyclyl includes spirocyclic carbocyclyl, bridged carbocyclyl, and fused carbocyclyl. A “carbocyclylene” is a divalent carbocyclyl group.

The term “cycloalkyl,” as used herein, refers to a saturated, non-aromatic, monovalent mono- or polycarbocyclic radical of 3 to 10, preferably 3 to 6 carbon atoms. This term is further exemplified by radicals such as cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl, cycloheptyl, norbornyl, and adamantyl.

The terms “halo” or “halogen,” as used herein, mean a fluorine (fluoro), chlorine (chloro), bromine (bromo), or iodine (iodo) radical.

The term “heteroalkyl,” as used herein, refers to an alkyl group, as defined herein, in which one or more of the constituent carbon atoms have been replaced by nitrogen, oxygen, or sulfur. In some embodiments, the heteroalkyl group can be further substituted with 1, 2, 3, or 4 substituent groups as described herein for alkyl groups. Examples of

heteroalkyl groups are an "alkoxy" which, as used herein, refers to alkyl-O— (e.g., methoxy and ethoxy), and an "alkylamino" which, as used herein, refers to —N(alkyl)R^{Na}, where R^{Na} is H or alkyl (e.g., methylamino). A "heteroalkylene" is a divalent heteroalkyl group.

The term "heteroalkenyl," as used herein, refers to an alkenyl group, as defined herein, in which one or more of the constituent carbon atoms have been replaced by nitrogen, oxygen, or sulfur. In some embodiments, the heteroalkenyl group can be further substituted with 1, 2, 3, or 4 substituent groups as described herein for alkenyl groups. Examples of heteroalkenyl groups are an "alkenoxy" which, as used herein, refers to alkenyl-O—. A "heteroalkenylene" is a divalent heteroalkenyl group.

The term "heteroalkynyl," as used herein, refers to an alkynyl group, as defined herein, in which one or more of the constituent carbon atoms have been replaced by nitrogen, oxygen, or sulfur. In some embodiments, the heteroalkynyl group can be further substituted with 1, 2, 3, or 4 substituent groups as described herein for alkynyl groups. Examples of heteroalkynyl groups are an "alkynoxy" which, as used herein, refers to alkynyl-O—. A "heteroalkynylene" is a divalent heteroalkynyl group.

The term "heteroaryl," as used herein, refers to an aromatic monocyclic or polycyclic structure of 5 to 12 atoms having at least one aromatic ring containing 1, 2, or 3 ring atoms selected from nitrogen, oxygen, and sulfur, with the remaining ring atoms being carbon. One or two ring carbon atoms of the heteroaryl group may be replaced with a carbonyl group. Examples of heteroaryl groups are pyridyl, pyrazoyl, benzoxazolyl, benzoimidazolyl, benzothiazolyl, imidazolyl, oxazolyl, and thiazolyl. A "heteroarylene" is a divalent heteroaryl group.

The term "heteroarylalkyl," as used herein, represents an alkyl group substituted with a heteroaryl group. Exemplary unsubstituted heteroarylalkyl groups are from 7 to 30 carbons (e.g., from 7 to 16 or from 7 to 20 carbons, such as C₁-C₆ alkyl C₂-C₉ heteroaryl, C₁-C₁₀ alkyl C₂-C₉ heteroaryl, or C₁-C₂₀ alkyl C₂-C₉ heteroaryl). In some embodiments, the alkyl and the heteroaryl each can be further substituted with 1, 2, 3, or 4 substituent groups as defined herein for the respective groups.

The term "heterocyclyl," as used herein, refers to a monocyclic or polycyclic radical (e.g., bicyclic or tricyclic) having 3 to 12 atoms having at least one non-aromatic ring containing 1, 2, 3, or 4 ring atoms selected from N, O, or S, and no aromatic ring containing any N, O, or S atoms. Polycyclic heterocyclyl includes spirocyclic heterocyclyl, bridged heterocyclyl, and fused heterocyclyl. Examples of heterocyclyl groups include, but are not limited to, morpholinyl, thiomorpholinyl, furyl, piperazinyl, piperidinyl, pyranyl, pyrrolidinyl, tetrahydropyranyl, tetrahydrofuranyl, and 1,3-dioxanyl. A "heterocyclylene" is a divalent heterocyclyl group.

The term "heterocyclylalkyl," as used herein, represents an alkyl group substituted with a heterocyclyl group. Exemplary unsubstituted heterocyclylalkyl groups are from 7 to 30 carbons (e.g., from 7 to 16 or from 7 to 20 carbons, such as C₁-C₆ alkyl C₂-C₉ heterocyclyl, C₁-C₁₀ alkyl C₂-C₉ heterocyclyl, or C₁-C₂₀ alkyl C₂-C₉ heterocyclyl). In some embodiments, the alkyl and the heterocyclyl each can be further substituted with 1, 2, 3, or 4 substituent groups as defined herein for the respective groups.

The term "hydroxyalkyl," as used herein, represents alkyl group substituted with an —OH group.

The term "hydroxyl," as used herein, represents an —OH group.

The term "imine," as used herein, represents =NR^N group, where R^N is, e.g., H or alkyl.

The term "N-protecting group," as used herein, represents those groups intended to protect an amino group against undesirable reactions during synthetic procedures. Commonly used N-protecting groups are disclosed in Greene, "Protective Groups in Organic Synthesis," 3rd Edition (John Wiley & Sons, New York, 1999). N-protecting groups include, but are not limited to, acyl, aryloyl, or carbamyl groups such as formyl, acetyl, propionyl, pivaloyl, t-butylacetyl, 2-chloroacetyl, 2-bromoacetyl, trifluoroacetyl, trichloroacetyl, phthalyl, o-nitrophenoxycarbonyl, α-chlorobutyryl, benzoyl, 4-chlorobenzoyl, 4-bromobenzoyl, 4-nitrobenzoyl, and chiral auxiliaries such as protected or unprotected D, L, or D, L-amino acids such as alanine, leucine, and phenylalanine; sulfonyl-containing groups such as benzenesulfonyl, and p-toluenesulfonyl; carbamate forming groups such as benzyloxycarbonyl, p-chlorobenzyloxycarbonyl, p-methoxybenzyloxycarbonyl, p-nitrobenzyloxycarbonyl, 2-nitrobenzyloxycarbonyl, p-bromobenzyloxycarbonyl, 3,4-dimethoxybenzyloxycarbonyl, 3,5-dimethoxybenzyloxycarbonyl, 2,4-dimethoxybenzyloxycarbonyl, 4-methoxybenzyloxycarbonyl, 2-nitro-4,5-dimethoxybenzyloxycarbonyl, 3,4,5-trimethoxybenzyloxycarbonyl, 1-(p-biphenyl)-1-methylethoxycarbonyl, α,α-dimethyl-3,5-dimethoxybenzyloxycarbonyl, benzhydryloxy carbonyl, t-butyloxycarbonyl, diisopropylmethoxycarbonyl, isopropylloxycarbonyl, ethoxycarbonyl, methoxycarbonyl, allyloxycarbonyl, 2,2,2-trichloroethoxycarbonyl, phenoxycarbonyl, 4-nitrophenoxy carbonyl, fluorenyl-9-methoxycarbonyl, cyclopentylloxycarbonyl, adamantylloxycarbonyl, cyclohexylloxycarbonyl, and phenylthiocarbonyl, arylalkyl groups such as benzyl, triphenylmethyl, and benzyloxymethyl, and silyl groups, such as trimethylsilyl. Preferred N-protecting groups are alloc, formyl, acetyl, benzoyl, pivaloyl, t-butylacetyl, alanil, phenylsulfonyl, benzyl, t-butyloxycarbonyl (Boc), and benzyloxycarbonyl (Cbz).

The term "nitro," as used herein, represents an —NO₂ group.

The term "oxo," as used herein, represents an =O group.

The term "thiol," as used herein, represents an —SH group.

The alkyl, alkenyl, alkynyl, heteroalkyl, heteroalkenyl, heteroalkynyl, carbocyclyl (e.g., cycloalkyl), aryl, heteroaryl, and heterocyclyl groups may be substituted or unsubstituted. When substituted, there will generally be 1 to 4 substituents present, unless otherwise specified. Substituents include, for example: alkyl (e.g., unsubstituted and substituted, where the substituents include any group described herein, e.g., aryl, halo, hydroxy), aryl (e.g., substituted and unsubstituted phenyl), carbocyclyl (e.g., substituted and unsubstituted cycloalkyl), halogen (e.g., fluoro), hydroxyl, heteroalkyl (e.g., substituted and unsubstituted methoxy, ethoxy, or thioalkoxy), heteroaryl, heterocyclyl, amino (e.g., NH₂ or mono- or dialkyl amino), azido, cyano, nitro, oxo, sulfonyl, or thiol. Aryl, carbocyclyl (e.g., cycloalkyl), heteroaryl, and heterocyclyl groups may also be substituted with alkyl (unsubstituted and substituted such as arylalkyl (e.g., substituted and unsubstituted benzyl)).

Compounds described herein (e.g., compounds of the invention) can have one or more asymmetric carbon atoms and can exist in the form of optically pure enantiomers, mixtures of enantiomers such as, for example, racemates, optically pure diastereoisomers, mixtures of diastereoisomers, diastereoisomeric racemates, or mixtures of diastereoisomeric racemates. The optically active forms can be

obtained for example by resolution of the racemates, by asymmetric synthesis or asymmetric chromatography (chromatography with a chiral adsorbent or eluant). That is, certain of the disclosed compounds may exist in various stereoisomeric forms. Stereoisomers are compounds that differ only in their spatial arrangement. Enantiomers are pairs of stereoisomers whose mirror images are not superimposable, most commonly because they contain an asymmetrically substituted carbon atom that acts as a chiral center. "Enantiomer" means one of a pair of molecules that are mirror images of each other and are not superimposable. Diastereomers are stereoisomers that are not related as mirror images, most commonly because they contain two or more asymmetrically substituted carbon atoms and represent the configuration of substituents around one or more chiral carbon atoms. Enantiomers of a compound can be prepared, for example, by separating an enantiomer from a racemate using one or more well-known techniques and methods, such as, for example, chiral chromatography and separation methods based thereon. The appropriate technique and/or method for separating an enantiomer of a compound described herein from a racemic mixture can be readily determined by those of skill in the art. "Racemate" or "racemic mixture" means a compound containing two enantiomers, wherein such mixtures exhibit no optical activity; i.e., they do not rotate the plane of polarized light. "Geometric isomer" means isomers that differ in the orientation of substituent atoms in relationship to a carbon-carbon double bond, to a cycloalkyl ring, or to a bridged bicyclic system. Atoms (other than H) on each side of a carbon-carbon double bond may be in an E (substituents are on opposite sides of the carbon-carbon double bond) or Z (substituents are oriented on the same side) configuration. "R," "S," "S*," "R*," "E," "Z," "cis," and "trans," indicate configurations relative to the core molecule. Certain of the disclosed compounds may exist in atropisomeric forms. Atropisomers are stereoisomers resulting from hindered rotation about single bonds where the steric strain barrier to rotation is high enough to allow for the isolation of the conformers. The compounds described herein (e.g., the compounds of the invention) may be prepared as individual isomers by either isomer-specific synthesis or resolved from an isomeric mixture. Conventional resolution techniques include forming the salt of a free base of each isomer of an isomeric pair using an optically active acid (followed by fractional crystallization and regeneration of the free base), forming the salt of the acid form of each isomer of an isomeric pair using an optically active amine (followed by fractional crystallization and regeneration of the free acid), forming an ester or amide of each of the isomers of an isomeric pair using an optically pure acid, amine or alcohol (followed by chromatographic separation and removal of the chiral auxiliary), or resolving an isomeric mixture of either a starting material or a final product using various well known chromatographic methods. When the stereochemistry of a disclosed compound is named or depicted by structure, the named or depicted stereoisomer is at least 60%, 70%, 80%, 90%, 99%, or 99.9% by weight relative to the other stereoisomers. When a single enantiomer is named or depicted by structure, the depicted or named enantiomer is at least 60%, 70%, 80%, 90%, 99%, or 99.9% by weight optically pure. When a single diastereomer is named or depicted by structure, the depicted or named diastereomer is at least 60%, 70%, 80%, 90%, 99%, or 99.9% by weight pure. Percent optical purity is the ratio of the weight of the enantiomer or over the weight of the enantiomer plus the weight of its optical isomer. Diastereomeric purity by weight

is the ratio of the weight of one diastereomer or over the weight of all the diastereomers. When the stereochemistry of a disclosed compound is named or depicted by structure, the named or depicted stereoisomer is at least 60%, 70%, 80%, 90%, 99%, or 99.9% by mole fraction pure relative to the other stereoisomers. When a single enantiomer is named or depicted by structure, the depicted or named enantiomer is at least 60%, 70%, 80%, 90%, 99%, or 99.9% by mole fraction pure. When a single diastereomer is named or depicted by structure, the depicted or named diastereomer is at least 60%, 70%, 80%, 90%, 99%, or 99.9% by mole fraction pure. Percent purity by mole fraction is the ratio of the moles of the enantiomer or over the moles of the enantiomer plus the moles of its optical isomer. Similarly, percent purity by moles fraction is the ratio of the moles of the diastereomer or over the moles of the diastereomer plus the moles of its isomer. When a disclosed compound is named or depicted by structure without indicating the stereochemistry, and the compound has at least one chiral center, it is to be understood that the name or structure encompasses either enantiomer of the compound free from the corresponding optical isomer, a racemic mixture of the compound, or mixtures enriched in one enantiomer relative to its corresponding optical isomer. When a disclosed compound is named or depicted by structure without indicating the stereochemistry and has two or more chiral centers, it is to be understood that the name or structure encompasses a diastereomer free of other diastereomers, a number of diastereomers free from other diastereomeric pairs, mixtures of diastereomers, mixtures of diastereomeric pairs, mixtures of diastereomers in which one diastereomer is enriched relative to the other diastereomer(s), or mixtures of diastereomers in which one or more diastereomer is enriched relative to the other diastereomers. The invention embraces all of these forms.

Compounds of the present disclosure also include all of the isotopes of the atoms occurring in the intermediate or final compounds. "Isotopes" refers to atoms having the same atomic number but different mass numbers resulting from a different number of neutrons in the nuclei. For example, isotopes of hydrogen include tritium and deuterium.

Unless otherwise stated, structures depicted herein are also meant to include compounds that differ only in the presence of one or more isotopically enriched atoms. Exemplary isotopes that can be incorporated into compounds of the present invention include isotopes of hydrogen, carbon, nitrogen, oxygen, phosphorus, sulfur, fluorine, chlorine, and iodine, such as ^2H , ^3H , ^{11}C , ^{13}C , ^{14}C , ^{15}N , ^{15}O , ^{17}O , ^{18}O , ^{32}P , ^{33}P , ^{35}S , ^{18}F , ^{36}Cl , ^{123}I and ^{125}I . Isotopically-labeled compounds (e.g., those labeled with ^3H and ^{14}C) can be useful in compound or substrate tissue distribution assays. Tritiated (i.e., ^3H) and carbon-14 (i.e., ^{14}C) isotopes can be useful for their ease of preparation and detectability. Further, substitution with heavier isotopes such as deuterium (i.e., ^2H) may afford certain therapeutic advantages resulting from greater metabolic stability (e.g., increased in vivo half-life or reduced dosage requirements). In some embodiments, one or more hydrogen atoms are replaced by ^2H or ^3H , or one or more carbon atoms are replaced by ^{13}C - or ^{14}C -enriched carbon. Positron emitting isotopes such as ^{15}O , ^{13}N , ^{11}C , and ^{18}F are useful for positron emission tomography (PET) studies to examine substrate receptor occupancy. Preparations of isotopically labelled compounds are known to those of skill in the art. For example, isotopically labeled compounds can generally be prepared by following procedures analogous to those disclosed for compounds of

the present invention described herein, by substituting an isotopically labeled reagent for a non-isotopically labeled reagent.

As is known in the art, many chemical entities can adopt a variety of different solid forms such as, for example, amorphous forms or crystalline forms (e.g., polymorphs, hydrates, solvate). In some embodiments, compounds of the present invention may be utilized in any such form, including in any solid form. In some embodiments, compounds described or depicted herein may be provided or utilized in hydrate or solvate form.

Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. Methods and materials are described herein for use in the present disclosure; other, suitable methods and materials known in the art can also be used. The materials, methods, and examples are illustrative only and not intended to be limiting. All publications, patent applications, patents, sequences, database entries, and other references mentioned herein are incorporated by reference in their entirety. In case of conflict, the present specification, including definitions, will control.

Definitions

In this application, unless otherwise clear from context, (i) the term “a” may be understood to mean “at least one”; (ii) the term “or” may be understood to mean “and/or”; and (iii) the terms “including” and “including” may be understood to encompass itemized components or steps whether presented by themselves or together with one or more additional components or steps.

As used herein, the terms “about” and “approximately” refer to a value that is within 10% above or below the value being described. For example, the term “about 5 nM” indicates a range of from 4.5 to 5.5 nM.

As used herein, the term “administration” refers to the administration of a composition (e.g., a compound or a preparation that includes a compound as described herein) to a subject or system. Administration to an animal subject (e.g., to a human) may be by any appropriate route. For example, in some embodiments, administration may be bronchial (including by bronchial instillation), buccal, enteral, interdermal, intra-arterial, intradermal, intragastric, intramedullary, intramuscular, intranasal, intraperitoneal, intrathecal, intratumoral, intravenous, intraventricular, mucosal, nasal, oral, rectal, subcutaneous, sublingual, topical, tracheal (including by intratracheal instillation), transdermal, vaginal, and vitreal.

As used herein, the term “adult soft tissue sarcoma” refers to a sarcoma that develops in the soft tissues of the body, typically in adolescent and adult subjects (e.g., subjects who are at least 10 years old, 11 years old, 12 years old, 13 years old, 14 years old, 15 years old, 16 years old, 17 years old, 18 years old, or 19 years old). Non-limiting examples of adult soft tissue sarcoma include, but are not limited to, synovial sarcoma, fibrosarcoma, malignant fibrous histiocytoma, dermatofibrosarcoma, liposarcoma, leiomyosarcoma, hemangiosarcoma, Kaposi’s sarcoma, lymphangiosarcoma, malignant peripheral nerve sheath tumor/neurofibrosarcoma, extraskeletal chondrosarcoma, extraskeletal osteosarcoma, extraskeletal myxoid chondrosarcoma, and extraskeletal mesenchymal.

The term “antisense,” as used herein, refers to a nucleic acid comprising a polynucleotide that is sufficiently complementary to all or a portion of a gene, primary transcript, or

processed mRNA, so as to interfere with expression of the endogenous gene (e.g., BRD9). “Complementary” polynucleotides are those that are capable of base pairing according to the standard Watson-Crick complementarity rules.

Specifically, purines will base pair with pyrimidines to form a combination of guanine paired with cytosine (G:C) and adenine paired with either thymine (A:T) in the case of DNA, or adenine paired with uracil (A:U) in the case of RNA. It is understood that two polynucleotides may hybridize to each other even if they are not completely complementary to each other, provided that each has at least one region that is substantially complementary to the other.

The term “antisense nucleic acid” includes single-stranded RNA as well as double-stranded DNA expression cassettes that can be transcribed to produce an antisense RNA. “Active” antisense nucleic acids are antisense RNA molecules that are capable of selectively hybridizing with a primary transcript or mRNA encoding a polypeptide having at least 80% sequence identity (e.g., 80%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.9% identity, or more) with the targeted polypeptide sequence (e.g., a BRD9 polypeptide sequence). The antisense nucleic acid can be complementary to an entire coding strand, or to only a portion thereof. In some embodiments, an antisense nucleic acid molecule is antisense to a “coding region” of the coding strand of a nucleotide sequence. The term “coding region” refers to the region of the nucleotide sequence comprising codons that are translated into amino acid residues. In some embodiments, the antisense nucleic acid molecule is antisense to a “noncoding region” of the coding strand of a nucleotide sequence. The term “noncoding region” refers to 5' and 3' sequences that flank the coding region that are not translated into amino acids (i.e., also referred to as 5' and 3' untranslated regions). The antisense nucleic acid molecule can be complementary to the entire coding region of mRNA, or can be antisense to only a portion of the coding or noncoding region of an mRNA. For example, the antisense oligonucleotide can be complementary to the region surrounding the translation start site. An antisense oligonucleotide can be, for example, about 5, 10, 15, 20, 25, 30, 35, 40, 45, or 50 nucleotides in length.

As used herein, the term “BAF complex” refers to the BRG1- or HRBM-associated factors complex in a human cell.

As used herein, the term “BAF complex-related disorder” refers to a disorder that is caused or affected by the level and/or activity of a BAF complex.

As used herein, the terms “GBAF complex” and “GBAF” refer to a SWI/SNF ATPase chromatin remodeling complex in a human cell. GBAF complex subunits may include, but are not limited to, ACTB, ACTL6A, ACTL6B, BICRA, BICRAL, BRD9, SMARCA2, SMARCA4, SMARCC1, SMARCD1, SMARCD2, SMARCD3, and SS18. The term “cancer” refers to a condition caused by the proliferation of malignant neoplastic cells, such as tumors, neoplasms, carcinomas, sarcomas, leukemias, and lymphomas.

As used herein, the term “BRD9” refers to bromodomain-containing protein 9, a component of the BAF (BRG1- or BRM-associated factors) complex, a SWI/SNF ATPase chromatin remodeling complex, and belongs to family IV of the bromodomain-containing proteins. BRD9 is encoded by the BRD9 gene, the nucleic acid sequence of which is set forth in SEQ ID NO: 1. The term “BRD9” also refers to natural variants of the wild-type BRD9 protein, such as proteins having at least 85% identity (e.g., 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%,

98%, 99%, 99.9% identity, or more) to the amino acid sequence of wild-type BRD9, which is set forth in SEQ ID NO: 2.

As used herein, the term “BRD9-related disorder” refers to a disorder that is caused or affected by the level and/or activity of BRD9. The term “cancer” refers to a condition caused by the proliferation of malignant neoplastic cells, such as tumors, neoplasms, carcinomas, sarcomas, leukemias, and lymphomas.

As used herein, a “combination therapy” or “administered in combination” means that two (or more) different agents or treatments are administered to a subject as part of a defined treatment regimen for a particular disease or condition. The treatment regimen defines the doses and periodicity of administration of each agent such that the effects of the separate agents on the subject overlap. In some embodiments, the delivery of the two or more agents is simultaneous or concurrent and the agents may be co-formulated. In some embodiments, the two or more agents are not co-formulated and are administered in a sequential manner as part of a prescribed regimen. In some embodiments, administration of two or more agents or treatments in combination is such that the reduction in a symptom, or other parameter related to the disorder is greater than what would be observed with one agent or treatment delivered alone or in the absence of the other. The effect of the two treatments can be partially additive, wholly additive, or greater than additive (e.g., synergistic). Sequential or substantially simultaneous administration of each therapeutic agent can be effected by any appropriate route including, but not limited to, oral routes, intravenous routes, intramuscular routes, and direct absorption through mucous membrane tissues. The therapeutic agents can be administered by the same route or by different routes. For example, a first therapeutic agent of the combination may be administered by intravenous injection while a second therapeutic agent of the combination may be administered orally.

A “compound of the present invention” and similar terms as used herein, whether explicitly noted or not, refers to compounds useful for treating BAF-related disorders (e.g., cancer or infection) described herein, including, e.g., compounds of Formula I or Formula II (e.g., compounds of Table 2A, Table 2B, and Table 2C), as well as salts (e.g., pharmaceutically acceptable salts), solvates, hydrates, stereoisomers (including atropisomers), and tautomers thereof. Those skilled in the art will appreciate that certain compounds described herein can exist in one or more different isomeric (e.g., stereoisomers, geometric isomers, atropisomers, and tautomers) or isotopic (e.g., in which one or more atoms has been substituted with a different isotope of the atom, such as hydrogen substituted for deuterium) forms. Unless otherwise indicated or clear from context, a depicted structure can be understood to represent any such isomeric or isotopic form, individually or in combination. Compounds described herein can be asymmetric (e.g., having one or more stereocenters). All stereoisomers, such as enantiomers and diastereomers, are intended unless otherwise indicated. Compounds of the present disclosure that contain asymmetrically substituted carbon atoms can be isolated in optically active or racemic forms. Methods on how to prepare optically active forms from optically active starting materials are known in the art, such as by resolution of racemic mixtures or by stereoselective synthesis. Many geometric isomers of olefins, C=N double bonds, and the like can also be present in the compounds described herein, and all such stable isomers are contemplated in the present disclosure. Cis and trans geometric isomers of the compounds of the present

disclosure are described and may be isolated as a mixture of isomers or as separated isomeric forms. In some embodiments, one or more compounds depicted herein may exist in different tautomeric forms. As will be clear from context, unless explicitly excluded, references to such compounds encompass all such tautomeric forms. In some embodiments, tautomeric forms result from the swapping of a single bond with an adjacent double bond and the concomitant migration of a proton. In certain embodiments, a tautomeric form may be a prototropic tautomer, which is an isomeric protonation states having the same empirical formula and total charge as a reference form. Examples of moieties with prototropic tautomeric forms are ketone-enol pairs, amide-imidic acid pairs, lactam-lactim pairs, amide-imidic acid pairs, enamine-imine pairs, and annular forms where a proton can occupy two or more positions of a heterocyclic system, such as, 1H- and 3H-imidazole, 1H-, 2H- and 4H-1,2,4-triazole, 1H- and 2H-isindole, and 1H- and 2H-pyrazole. In some embodiments, tautomeric forms can be in equilibrium or sterically locked into one form by appropriate substitution. In certain embodiments, tautomeric forms result from acetal interconversion.

As used herein, the term “degrader” refers to a small molecule compound including a degradation moiety, wherein the compound interacts with a protein (e.g., BRD9) in a way which results in degradation of the protein, e.g., binding of the compound results in at least 5% reduction of the level of the protein, e.g., in a cell or subject.

As used herein, the term “degradation moiety” refers to a moiety whose binding results in degradation of a protein, e.g., BRD9. In one example, the moiety binds to a protease or a ubiquitin ligase that metabolizes the protein, e.g., BRD9.

By “determining the level of a protein” is meant the detection of a protein, or an mRNA encoding the protein, by methods known in the art either directly or indirectly. “Directly determining” means performing a process (e.g., performing an assay or test on a sample or “analyzing a sample” as that term is defined herein) to obtain the physical entity or value. “Indirectly determining” refers to receiving the physical entity or value from another party or source (e.g., a third-party laboratory that directly acquired the physical entity or value). Methods to measure protein level generally include, but are not limited to, western blotting, immunoblotting, enzyme-linked immunosorbent assay (ELISA), radioimmunoassay (RIA), immunoprecipitation, immunofluorescence, surface plasmon resonance, chemiluminescence, fluorescent polarization, phosphorescence, immunohistochemical analysis, matrix-assisted laser desorption/ionization time-of-flight (MALDI-TOF) mass spectrometry, liquid chromatography (LC)-mass spectrometry, microcytometry, microscopy, fluorescence activated cell sorting (FACS), and flow cytometry, as well as assays based on a property of a protein including, but not limited to, enzymatic activity or interaction with other protein partners. Methods to measure mRNA levels are known in the art.

As used herein, the terms “effective amount,” “therapeutically effective amount,” and “a sufficient amount” of an agent that reduces the level and/or activity of BRD9 (e.g., in a cell or a subject) described herein refer to a quantity sufficient to, when administered to the subject, including a human, effect beneficial or desired results, including clinical results, and, as such, an “effective amount” or synonym thereto depends on the context in which it is being applied. For example, in the context of treating cancer, it is an amount of the agent that reduces the level and/or activity of BRD9 sufficient to achieve a treatment response as com-

pared to the response obtained without administration of the agent that reduces the level and/or activity of BRD9. The amount of a given agent that reduces the level and/or activity of BRD9 described herein that will correspond to such an amount will vary depending upon various factors, such as the given agent, the pharmaceutical formulation, the route of administration, the type of disease or disorder, the identity of the subject (e.g., age, sex, and/or weight) or host being treated, and the like, but can nevertheless be routinely determined by one of skill in the art. Also, as used herein, a “therapeutically effective amount” of an agent that reduces the level and/or activity of BRD9 of the present disclosure is an amount which results in a beneficial or desired result in a subject as compared to a control. As defined herein, a therapeutically effective amount of an agent that reduces the level and/or activity of BRD9 of the present disclosure may be readily determined by one of ordinary skill by routine methods known in the art. Dosage regimen may be adjusted to provide the optimum therapeutic response.

As used herein, the term “inhibitor” refers to any agent which reduces the level and/or activity of a protein (e.g., BRD9). Non-limiting examples of inhibitors include small molecule inhibitors, degraders, antibodies, enzymes, or polynucleotides (e.g., siRNA).

The term “inhibitory RNA agent” refers to an RNA, or analog thereof, having sufficient sequence complementarity to a target RNA to direct RNA interference. Examples also include a DNA that can be used to make the RNA. RNA interference (RNAi) refers to a sequence-specific or selective process by which a target molecule (e.g., a target gene, protein, or RNA) is down-regulated. Generally, an interfering RNA (“iRNA”) is a double-stranded short-interfering RNA (siRNA), short hairpin RNA (shRNA), or single-stranded micro-RNA (miRNA) that results in catalytic degradation of specific mRNAs, and also can be used to lower or inhibit gene expression.

By “level” is meant a level of a protein, or mRNA encoding the protein, as compared to a reference. The reference can be any useful reference, as defined herein. By a “decreased level” or an “increased level” of a protein is meant a decrease or increase in protein level, as compared to a reference (e.g., a decrease or an increase by about 5%, about 10%, about 15%, about 20%, about 25%, about 30%, about 35%, about 40%, about 45%, about 50%, about 55%, about 60%, about 65%, about 70%, about 75%, about 80%, about 85%, about 90%, about 95%, about 100%, about 150%, about 200%, about 300%, about 400%, about 500%, or more; a decrease or an increase of more than about 10%, about 15%, about 20%, about 50%, about 75%, about 100%, or about 200%, as compared to a reference; a decrease or an increase by less than about 0.01-fold, about 0.02-fold, about 0.1-fold, about 0.3-fold, about 0.5-fold, about 0.8-fold, or less; or an increase by more than about 1.2-fold, about 1.4-fold, about 1.5-fold, about 1.8-fold, about 2.0-fold, about 3.0-fold, about 3.5-fold, about 4.5-fold, about 5.0-fold, about 10-fold, about 15-fold, about 20-fold, about 30-fold, about 40-fold, about 50-fold, about 100-fold, about 1000-fold, or more). A level of a protein may be expressed in mass/vol (e.g., g/dL, mg/mL, µg/mL, ng/mL) or percentage relative to total protein or mRNA in a sample.

The terms “miRNA” and “microRNA” refer to an RNA agent, preferably a single-stranded agent, of about 10-50 nucleotides in length, preferably between about 15-25 nucleotides in length, which is capable of directing or mediating RNA interference. Naturally-occurring miRNAs are generated from stem-loop precursor RNAs (i.e., pre-miRNAs) by Dicer. The term “Dicer” as used herein,

includes Dicer as well as any Dicer ortholog or homolog capable of processing dsRNA structures into siRNAs, miRNAs, siRNA-like or miRNA-like molecules. The term microRNA (“miRNA”) is used interchangeably with the term “small temporal RNA” (“stRNA”) based on the fact that naturally-occurring miRNAs have been found to be expressed in a temporal fashion (e.g., during development).

By “modulating the activity of a BAF complex,” is meant altering the level of an activity related to a BAF complex (e.g., GBAF), or a related downstream effect. The activity level of a BAF complex may be measured using any method known in the art, e.g., the methods described in Kadoch et al. Cell 153:71-85 (2013), the methods of which are herein incorporated by reference.

“Percent (%) sequence identity” with respect to a reference polynucleotide or polypeptide sequence is defined as the percentage of nucleic acids or amino acids in a candidate sequence that are identical to the nucleic acids or amino acids in the reference polynucleotide or polypeptide sequence, after aligning the sequences and introducing gaps, if necessary, to achieve the maximum percent sequence identity. Alignment for purposes of determining percent nucleic acid or amino acid sequence identity can be achieved in various ways that are within the capabilities of one of skill in the art, for example, using publicly available computer software such as BLAST, BLAST-2, or Megalign software. Those skilled in the art can determine appropriate parameters for aligning sequences, including any algorithms needed to achieve maximal alignment over the full length of the sequences being compared. For example, percent sequence identity values may be generated using the sequence comparison computer program BLAST. As an illustration, the percent sequence identity of a given nucleic acid or amino acid sequence, A, to, with, or against a given nucleic acid or amino acid sequence, B, (which can alternatively be phrased as a given nucleic acid or amino acid sequence, A that has a certain percent sequence identity to, with, or against a given nucleic acid or amino acid sequence, B) is calculated as follows:

100 multiplied by (the fraction X/Y)

where X is the number of nucleotides or amino acids scored as identical matches by a sequence alignment program (e.g., BLAST) in that program’s alignment of A and B, and where Y is the total number of nucleic acids in B. It will be appreciated that where the length of nucleic acid or amino acid sequence A is not equal to the length of nucleic acid or amino acid sequence B, the percent sequence identity of A to B will not equal the percent sequence identity of B to A.

A “pharmaceutically acceptable excipient,” as used herein, refers any ingredient other than the compounds described herein (for example, a vehicle capable of suspending or dissolving the active compound) and having the properties of being substantially nontoxic and non-inflammatory in a patient. Excipients may include, for example: antiadherents, antioxidants, binders, coatings, compression aids, disintegrants, dyes (colors), emollients, emulsifiers, fillers (diluents), film formers or coatings, flavors, fragrances, glidants (flow enhancers), lubricants, preservatives, printing inks, sorbents, suspending or dispersing agents, sweeteners, and waters of hydration. Exemplary excipients include, but are not limited to: butylated hydroxytoluene (BHT), calcium carbonate, calcium phosphate (dibasic), calcium stearate, croscarmellose, crosslinked polyvinyl pyrrolidone, citric acid, crospovidone, cysteine, ethylcellulose, gelatin, hydroxypropyl cellulose, hydroxypropyl methylcellulose, lactose, magnesium stearate, maltitol, mannitol,

methionine, methylcellulose, methyl paraben, microcrystalline cellulose, polyethylene glycol, polyvinyl pyrrolidone, povidone, pregelatinized starch, propyl paraben, retinyl palmitate, shellac, silicon dioxide, sodium carboxymethyl cellulose, sodium citrate, sodium starch glycolate, sorbitol, starch (corn), stearic acid, sucrose, talc, titanium dioxide, vitamin A, vitamin E, vitamin C, and xylitol.

As used herein, the term “pharmaceutically acceptable salt” means any pharmaceutically acceptable salt of the compound of any of the compounds described herein. For example, pharmaceutically acceptable salts of any of the compounds described herein include those that are within the scope of sound medical judgment, suitable for use in contact with the tissues of humans and animals without undue toxicity, irritation, allergic response and are commensurate with a reasonable benefit/risk ratio. Pharmaceutically acceptable salts are well known in the art. For example, pharmaceutically acceptable salts are described in: Berge et al., *J. Pharmaceutical Sciences* 66:1-19, 1977 and in *Pharmaceutical Salts: Properties, Selection, and Use*, (Eds. P. H. Stahl and C. G. Wermuth), Wiley-VCH, 2008. The salts can be prepared in situ during the final isolation and purification of the compounds described herein or separately by reacting a free base group with a suitable organic acid.

The compounds described herein may have ionizable groups so as to be capable of preparation as pharmaceutically acceptable salts. These salts may be acid addition salts involving inorganic or organic acids or the salts may, in the case of acidic forms of the compounds described herein, be prepared from inorganic or organic bases. Frequently, the compounds are prepared or used as pharmaceutically acceptable salts prepared as addition products of pharmaceutically acceptable acids and bases and methods for preparation of the appropriate salts are well-known in the art. Salts may be prepared from pharmaceutically acceptable non-toxic acids and bases including inorganic and organic acids and bases. Representative acid addition salts include acetate, adipate, alginate, ascorbate, aspartate, benzenesulfonate, benzoate, bisulfate, borate, butyrate, camphorate, camphorsulfonate, citrate, cyclopentanepropionate, digluconate, dodecylsulfate, ethanesulfonate, fumarate, glucoheptonate, glycerophosphate, hemisulfate, heptonate, hexanoate, hydrobromide, hydrochloride, hydroiodide, 2-hydroxyethanesulfonate, lactobionate, lactate, laurate, lauryl sulfate, malate, maleate, malonate, methanesulfonate, 2-naphthalenesulfonate, nicotinate, nitrate, oleate, oxalate, palmitate, pamoate, pectinate, persulfate, 3-phenylpropionate, phosphate, picrate, pivalate, propionate, stearate, succinate, sulfate, tartrate, thiocyanate, toluenesulfonate, undecanoate, and valerate salts. Representative alkali or alkaline earth metal salts include sodium, lithium, potassium, calcium, and magnesium, as well as nontoxic ammonium, quaternary ammonium, and amine cations, including, but not limited to ammonium, tetramethylammonium, tetraethylammonium, methylamine, dimethylamine, trimethylamine, triethylamine, and ethylamine.

The term “pharmaceutical composition,” as used herein, represents a composition containing a compound described herein formulated with a pharmaceutically acceptable excipient, and manufactured or sold with the approval of a governmental regulatory agency as part of a therapeutic regimen for the treatment of disease in a mammal. Pharmaceutical compositions can be formulated, for example, for oral administration in unit dosage form (e.g., a tablet, capsule, caplet, gelcap, or syrup); for topical administration (e.g., as a cream, gel, lotion, or ointment); for intravenous

administration (e.g., as a sterile solution free of particulate emboli and in a solvent system suitable for intravenous use); or in any other pharmaceutically acceptable formulation.

By “reducing the activity of BRD9,” is meant decreasing the level of an activity related to an BRD9, or a related downstream effect. A non-limiting example of inhibition of an activity of BRD9 is decreasing the level of a BAF complex (e.g., GBAF) in a cell. The activity level of BRD9 may be measured using any method known in the art. In some embodiments, an agent which reduces the activity of BRD9 is a small molecule BRD9 inhibitor. In some embodiments, an agent which reduces the activity of BRD9 is a small molecule BRD9 degrader.

By “reducing the level of BRD9,” is meant decreasing the level of BRD9 in a cell or subject. The level of BRD9 may be measured using any method known in the art.

By a “reference” is meant any useful reference used to compare protein or mRNA levels. The reference can be any sample, standard, standard curve, or level that is used for comparison purposes. The reference can be a normal reference sample or a reference standard or level. A “reference sample” can be, for example, a control, e.g., a predetermined negative control value such as a “normal control” or a prior sample taken from the same subject; a sample from a normal healthy subject, such as a normal cell or normal tissue; a sample (e.g., a cell or tissue) from a subject not having a disease; a sample from a subject that is diagnosed with a disease, but not yet treated with a compound described herein; a sample from a subject that has been treated by a compound described herein; or a sample of a purified protein (e.g., any described herein) at a known normal concentration. By “reference standard or level” is meant a value or number derived from a reference sample. A “normal control value” is a pre-determined value indicative of non-disease state, e.g., a value expected in a healthy control subject. Typically, a normal control value is expressed as a range (“between X and Y”), a high threshold (“no higher than X”), or a low threshold (“no lower than X”). A subject having a measured value within the normal control value for a particular biomarker is typically referred to as “within normal limits” for that biomarker. A normal reference standard or level can be a value or number derived from a normal subject not having a disease or disorder (e.g., cancer); a subject that has been treated with a compound described herein. In preferred embodiments, the reference sample, standard, or level is matched to the sample subject sample by at least one of the following criteria: age, weight, sex, disease stage, and overall health. A standard curve of levels of a purified protein, e.g., any described herein, within the normal reference range can also be used as a reference.

The terms “short interfering RNA” and “siRNA” (also known as “small interfering RNAs”) refer to an RNA agent, preferably a double-stranded agent, of about 10-50 nucleotides in length, the strands optionally having overhanging ends comprising, for example 1, 2 or 3 overhanging nucleotides (or nucleotide analogs), which is capable of directing or mediating RNA interference. Naturally-occurring siRNAs are generated from longer dsRNA molecules (e.g., >25 nucleotides in length) by a cell’s RNAi machinery (e.g., Dicer or a homolog thereof).

The term “shRNA”, as used herein, refers to an RNA agent having a stem-loop structure, comprising a first and second region of complementary sequence, the degree of complementarity and orientation of the regions being sufficient such that base pairing occurs between the regions, the first and second regions being joined by a loop region, the

loop resulting from a lack of base pairing between nucleotides (or nucleotide analogs) within the loop region.

As used herein, the term “subject” refers to any organism to which a composition in accordance with the invention may be administered, e.g., for experimental, diagnostic, prophylactic, and/or therapeutic purposes. Typical subjects include any animal (e.g., mammals such as mice, rats, rabbits, non-human primates, and humans). A subject may seek or be in need of treatment, require treatment, be receiving treatment, be receiving treatment in the future, or be a human or animal who is under care by a trained professional for a particular disease or condition.

As used herein, the term “SS18-SSX fusion protein-related disorder” refers to a disorder that is caused or affected by the level and/or activity of SS18-SSX fusion protein.

As used herein, the terms “treat,” “treated,” or “treating” mean both therapeutic treatment and prophylactic or preventative measures wherein the object is to prevent or slow down (lessen) an undesired physiological condition, disorder, or disease, or obtain beneficial or desired clinical results. Beneficial or desired clinical results include, but are not limited to, alleviation of symptoms; diminishment of the extent of a condition, disorder, or disease; stabilized (i.e., not worsening) state of condition, disorder, or disease; delay in onset or slowing of condition, disorder, or disease progression; amelioration of the condition, disorder, or disease state or remission (whether partial or total), whether detectable or undetectable; an amelioration of at least one measurable physical parameter, not necessarily discernible by the patient; or enhancement or improvement of condition, disorder, or disease. Treatment includes eliciting a clinically significant response without excessive levels of side effects. Treatment also includes prolonging survival as compared to expected survival if not receiving treatment.

As used herein, the terms “variant” and “derivative” are used interchangeably and refer to naturally-occurring, synthetic, and semi-synthetic analogues of a compound, peptide, protein, or other substance described herein. A variant or derivative of a compound, peptide, protein, or other substance described herein may retain or improve upon the biological activity of the original material.

The details of one or more embodiments of the invention are set forth in the description below. Other features, objects, and advantages of the invention will be apparent from the description and from the claims.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a series of graphs illustrating the effect of specific guide RNA (sgRNA) targeting of the BRD9 BAF complex subunit on synovial sarcoma cell growth. The Y-axis indicated the dropout ratio. The X-axis indicates the nucleotide position of the BRD9 gene. The grey box indicates the range of the negative control sgRNAs in the screen. The SYO1 cell line carries SS18-SSX2 fusion protein. The breakpoint joining the N-terminal region of SS18 to the C-terminal region of SSX2 are indicated by the black lines in their respective panel. The linear protein sequence is shown with BRD9 PFAM domains annotated from the PFAM database.

FIG. 2 is an image illustrating dose dependent depletion of BRD9 levels in a synovial sarcoma cell line (SYO1) in the presence of a BRD9 degrader.

FIG. 3 is an image illustrating sustained suppression of BRD9 levels in a synovial sarcoma cell line (SYO1) in the presence of a BRD9 degrader over 72 hours.

FIG. 4 is an image illustrating sustained suppression of BRD9 levels in two cell lines (293T and SYO1) in the presence of a BRD9 degrader over 5 days.

FIG. 5 is an image illustrating sustained suppression of BRD9 levels in synovial sarcoma cell lines (SYO1 and Yamato) in the presence of a BRD9 degrader over 7 days compared to the levels in cells treated with CRISPR reagents.

FIG. 6 is an image illustrating the effect on cell growth of six cell lines (SYO1, Yamato, A549, HS-SY-II, ASKA, and 293T) in the presence of a BRD9 degrader and a BRD9 inhibitor.

FIG. 7 is an image illustrating the effect on cell growth of two cell lines (SYO1 and G401) in the presence of a BRD9 degrader.

FIG. 8 is an image illustrating the effect on cell growth of three synovial sarcoma cell lines (SYO1, HS-SY-II, and ASKA) in the presence of a BRD9 degrader, BRD9 binder and E3 ligase binder.

FIG. 9 is an image illustrating the effect on cell growth of three non-synovial sarcoma cell lines (RD, HCT116, and Calu6) in the presence of a BRD9 degrader, BRD9 binder and E3 ligase binder.

FIG. 10 is a graph illustrating the percentage of SYO1 in various cell cycle phases following treatment with DMSO, Compound 1 at 200 nM, or Compound 1 at 1 μM for 8 or 13 days.

FIG. 11 is a series of contour plots illustrating the percentage of SYO1 cells in various cell cycle phases following treatment with DMSO, Compound 1 at 200 nM, Compound 1 at 1 μM, or lenalidomide at 200 nM for 8 days. Numerical values corresponding to each contour plot are found in the table below.

FIG. 12 is a series of contour plots illustrating the percentage of SYO1 cells in various cell cycle phases following treatment with DMSO, Compound 1 at 200 nM, Compound 1 at 1 μM, or lenalidomide at 200 nM for 13 days. Numerical values corresponding to each contour plot are found in the table below.

FIG. 13 is a series of contour plots illustrating the percentage of early- and late-apoptotic SYO1 cells following treatment with DMSO, Compound 1 at 200 nM, Compound 1 at 1 μM, or lenalidomide at 200 nM for 8 days. Numerical values corresponding to each contour plot are found in the table below.

FIG. 14 is a graph illustrating the proteins present in BAF complexes including the SS18-SSX fusion protein.

DETAILED DESCRIPTION

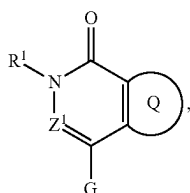
The present disclosure features compositions and methods useful for the treatment of BAF-related disorders (e.g., cancer and infection). The disclosure further features compositions and methods useful for inhibition of the level and/or activity of BRD9, e.g., for the treatment of disorders such as cancer (e.g., sarcoma) and infection (e.g., viral infection), e.g., in a subject in need thereof.

Compounds

Compounds described herein reduce the level of an activity related to BRD9, or a related downstream effect, or reduce the level of BRD9 in a cell or subject. Exemplary compounds described herein have the structure according to Formula I or Formula II.

433

Formula I is:

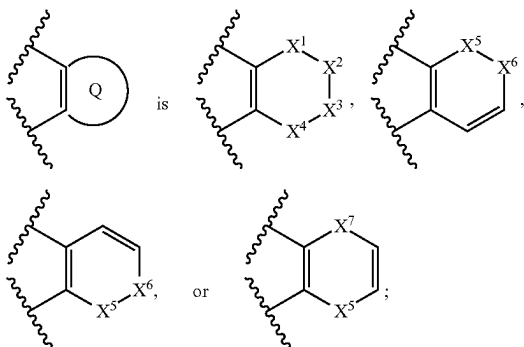


where

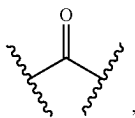
R¹ is H, optionally substituted C₁-C₆ alkyl, optionally substituted C₂-C₆ alkenyl, optionally substituted C₁-C₆ heteroalkyl, or optionally substituted C₃-C₁₀ carbocyclyl;

Z¹ is CR² or N;

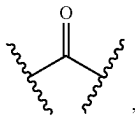
R² is H, halogen, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted C₃-C₁₀ carbocyclyl, optionally substituted C₂-C₉ heterocyclyl, optionally substituted C₆-C₁₀ aryl, or optionally substituted C₂-C₉ heteroaryl;



X¹ is a bond, O, NR^{3a},



or CR^{4a}R^{5a};
X² is O, NR^{3b},

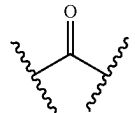


434

or CR^{4b}R^{5b};
X³ is O, NR^{3c},

Formula I

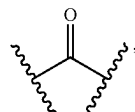
5



10

or CR^{4c}R^{5c};
X⁴ is a bond, O, NR^{3d},

15



20

or CR^{4d}R^{5d};
X⁵ is O or NR^{3e} and X⁶ is CR^{4f}R^{5f}, or X⁵ is CR^{4e}R^{5e} and X⁶ is O or NR^{3f};
X⁷ is O, NR^{3g}, or CR^{4g}R^{5g};
X⁸ is O, NR^{3h}, or CR^{4h}R^{5h};

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each of R^{3a}, R^{3b}, R^{3c}, and R^{3d} is, independently, H, halogen, hydroxyl, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted C₃-C₁₀ carbocyclyl, optionally substituted C₂-C₉ heterocyclyl, optionally substituted C₆-C₁₀ aryl, optionally substituted C₂-C₉ heteroaryl, optionally substituted C₂-C₆ alkenyl, optionally substituted C₂-C₆ heteroalkenyl, optionally substituted C₁-C₆ acyl, thiol, optionally substituted sulfone, optionally substituted sulfonamide, or optionally substituted amino, or R^{3a} and R^{4b}, R^{4a} and R^{3b}, R^{4b} and R^{4a}, R^{3b} and R^{4c}, R^{4b} and R^{4c}, R^{3c} and R^{4b}, R^{3c} and R^{4d}, R^{4c} and R^{4d}, and/or R^{3d} and R^{4c}, together with the atoms to which each is attached, combine to form optionally substituted C₂-C₉ heterocyclyl;

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each of R^{4a}, R^{4b}, R^{4c}, and R^{4d} is, independently, H, halogen, hydroxyl, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted C₃-C₁₀ carbocyclyl, optionally substituted C₂-C₉ heterocyclyl, optionally substituted C₆-C₁₀ aryl, optionally substituted C₂-C₉ heteroaryl, optionally substituted C₂-C₆ alkenyl, optionally substituted C₂-C₆ heteroalkenyl, optionally substituted C₁-C₆ acyl, thiol, optionally substituted sulfone, or optionally substituted amino, or R^{3a} and R^{4b}, R^{4a} and R^{3b}, R^{4b} and R^{4a}, R^{3b} and R^{4c}, R^{4b} and R^{4c}, R^{3c} and R^{4b}, R^{3c} and R^{4d}, R^{4c} and R^{4d}, and/or R^{3d} and R^{4c}, together with the atoms to which each is attached, combine to form optionally substituted C₂-C₉ heterocyclyl;

each of R^{5a}, R^{5b}, R^{5c}, and R^{5d} is, independently, H, halogen, hydroxyl, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted C₃-C₁₀ carbocyclyl, optionally substituted C₂-C₉ heterocyclyl, optionally substituted C₆-C₁₀ aryl, optionally substituted C₂-C₉ heteroaryl, optionally substituted C₂-C₆ alkenyl, optionally substituted C₂-C₆ heteroalkenyl, hydroxyl, thiol, or optionally substituted amino;

each of R^{3e}, R^{3f}, R^{3g}, and R^{3h} is, independently, H, halogen, hydroxyl, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted C₃-C₁₀ carbocyclyl, optionally substituted C₂-C₉ heterocyclyl, optionally substituted C₆-C₁₀ aryl, optionally substituted C₂-C₉ heteroaryl, optionally substituted C₂-C₆ alkenyl, optionally substituted C₂-C₆

435

heteroalkenyl, optionally substituted C_1-C_6 acyl, thiol, optionally substituted sulfone, optionally substituted sulfonamide, or optionally substituted amino, or R^{3e} and R^{4f} or R^{4e} and R^{3f} , together with the atoms to which each is attached, combine to form optionally substituted heterocyclyl;

each of R^{4e} , R^{4f} , R^{4g} , and R^{4h} is, independently, H, halogen, hydroxyl, optionally substituted C_1-C_6 alkyl, optionally substituted C_1-C_6 heteroalkyl, optionally substituted C_3-C_{10} carbocyclyl, optionally substituted C_2-C_9 heterocyclyl, optionally substituted C_6-C_{10} aryl, optionally substituted C_2-C_9 heteroaryl, optionally substituted C_2-C_6 alkenyl, optionally substituted C_2-C_6 heteroalkenyl, optionally substituted C_1-C_6 acyl, thiol, optionally substituted sulfone, optionally substituted sulfonamide, or optionally substituted amino, or R^{3e} and R^{4f} or R^{4e} and R^{3f} , together with the atoms to which each is attached, combine to form optionally substituted heterocyclyl;

each of R^{5e} , R^{5f} , R^{5g} , and R^{5h} is, independently, H, halogen, hydroxyl, optionally substituted C_1-C_6 alkyl, optionally substituted C_1-C_6 heteroalkyl, optionally substituted C_3-C_{10} carbocyclyl, optionally substituted C_2-C_9 heterocyclyl, optionally substituted C_6-C_{10} aryl, optionally substituted C_2-C_9 heteroaryl, optionally substituted C_2-C_6 alkenyl, optionally substituted C_2-C_6 heteroalkenyl, hydroxyl, thiol, or optionally substituted amino; and

G is optionally substituted C_6-C_{10} aryl, optionally substituted C_3-C_{10} carbocyclyl, optionally substituted C_2-C_9 heteroaryl, or C_2-C_9 heterocyclyl, or a pharmaceutically acceptable salt thereof.

Formula II is:

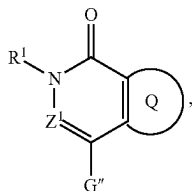
A-L-B

where

B is a degradation moiety,

L is a linker, and

A has the structure of Formula III:



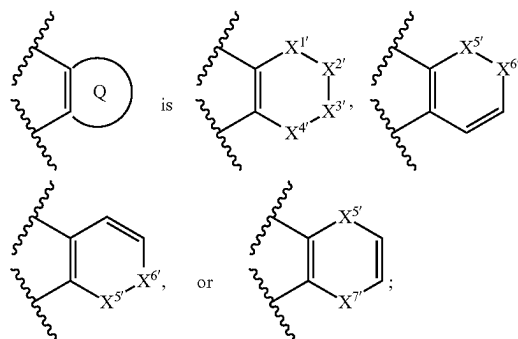
where

R^1 is H, optionally substituted C_1-C_6 alkyl, optionally substituted C_2-C_6 alkenyl, optionally substituted C_1-C_6 heteroalkyl, or optionally substituted C_3-C_{10} carbocyclyl;

Z^1 is CR^2 or N;

R^2 is H, halogen, optionally substituted C_1-C_6 alkyl, optionally substituted C_1-C_6 heteroalkyl, optionally substituted C_3-C_{10} carbocyclyl, optionally substituted C_2-C_9 heterocyclyl, optionally substituted C_6-C_{10} aryl, or optionally substituted C_2-C_9 heteroaryl;

436



$X^{1'}$ is a bond, O, $NR^{3a'}$, or $CR^{4a'}R^{5a'}$,
 $X^{2'}$ is O, $NR^{3b'}$, or $CR^{4b'}R^{5b'}$,
 $X^{3'}$ is O, $NR^{3c'}$, or $CR^{4c'}R^{5c'}$,
 $X^{4'}$ is a bond, O, $NR^{3d'}$, or $CR^{4d'}R^{5d'}$,
 $X^{5'}$ is O, $NR^{3e'}$, or $CR^{4e'}R^{5e'}$,
 $X^{6'}$ is O, $NR^{3f'}$, or $CR^{4f'}R^{5f'}$,
 $X^{7'}$ is O, $NR^{3g'}$, or $CR^{4g'}R^{5g'}$,
each of $R^{3a'}$, $R^{3b'}$, $R^{3c'}$, and $R^{3d'}$ is, independently, H,



halogen, hydroxyl, optionally substituted C_1-C_6 alkyl, optionally substituted C_1-C_6 heteroalkyl, optionally substituted C_3-C_{10} carbocyclyl, optionally substituted C_2-C_9 heterocyclyl, optionally substituted C_6-C_{10} aryl, optionally substituted C_2-C_9 heteroaryl, optionally substituted C_2-C_6 alkenyl, optionally substituted C_2-C_6 heteroalkenyl, optionally substituted C_1-C_6 acyl, thiol, optionally substituted sulfone, optionally substituted sulfonamide, or optionally substituted amino, or $R^{3a'}$ and $R^{4b'}$, $R^{4a'}$ and $R^{3b'}$, $R^{4b'}$ and $R^{4a'}$, $R^{3b'}$ and $R^{4a'}$, $R^{4c'}$ and $R^{3c'}$, $R^{4c'}$ and $R^{4b'}$, $R^{3c'}$ and $R^{4b'}$, $R^{4d'}$ and $R^{4c'}$, and/or $R^{3d'}$ and $R^{4c'}$, together with the atoms to which each is attached, combine to form optionally substituted C_2-C_9 heterocyclyl;

$R^{3'}$ is absent, optionally substituted C_1-C_6 alkylene, optionally substituted C_1-C_6 heteroalkylene, optionally substituted C_3-C_{10} carbocyclylene, optionally substituted C_2-C_9 heterocyclylene, optionally substituted C_6-C_{10} arylene, optionally substituted C_2-C_9 heteroarylene, optionally substituted C_2-C_6 alkenylene, optionally substituted C_2-C_6 heteroalkenylene, optionally substituted sulfone, optionally substituted sulfonamide, or optionally substituted amino;

each of $R^{4a'}$, $R^{4b'}$, $R^{4c'}$, and $R^{4d'}$ is, independently, H, halogen, hydroxyl, optionally substituted C_1-C_6 alkyl, optionally substituted C_1-C_6 heteroalkyl, optionally substituted C_3-C_{10} carbocyclyl, optionally substituted C_2-C_9 heterocyclyl, optionally substituted C_6-C_{10} aryl, optionally substituted C_2-C_9 heteroaryl, optionally substituted C_2-C_6 alkenyl, optionally substituted C_2-C_6 heteroalkenyl, thiol, optionally substituted sulfone, or optionally substituted amino, or $R^{3a'}$ and $R^{4b'}$, $R^{4a'}$ and $R^{3b'}$, $R^{4b'}$ and $R^{4a'}$, $R^{3b'}$ and $R^{4a'}$, $R^{4c'}$ and $R^{3c'}$, $R^{4c'}$ and $R^{4b'}$, $R^{3c'}$ and $R^{4b'}$, $R^{4d'}$ and $R^{4c'}$, and/or $R^{3d'}$ and $R^{4c'}$, together with the atoms to which each is attached, combine to form optionally substituted C_2-C_9 heterocyclyl;

437

each of R^{5a} , R^{5b} , R^{5c} , and R^{5d} is, independently, H, halogen, hydroxyl, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, hydroxyl, thiol, or optionally substituted amino;

each of R^{3e} , R^{3f} , and R^{3g} is, independently, H,



halogen, hydroxyl, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, optionally substituted C_1 - C_6 acyl, thiol, optionally substituted sulfone, optionally substituted sulfonamide, or optionally substituted amino, or R^{3e} and R^{4f} or R^{4e} and R^{3f} , together with the atoms to which each is attached, combine to form optionally substituted heterocyclyl;

each of R^{4e} , R^{4f} , and R^{4g} is, independently, H, halogen, hydroxyl, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, optionally substituted C_1 - C_6 acyl, thiol, optionally substituted sulfone, optionally substituted sulfonamide, or optionally substituted amino, or R^{3e} and R^{4f} or R^{4e} and R^{3f} , together with the atoms to which each is attached, combine to form optionally substituted heterocyclyl;

each of R^{5e} , R^{5f} , and R^{5g} is, independently, H, halogen, hydroxyl, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, hydroxyl, thiol, or optionally substituted amino;

G'' is



optionally substituted C_6 - C_{10} aryl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heteroaryl, or C_2 - C_9 heterocyclyl;

G' is optionally substituted C_3 - C_{10} carbocyclylene, C_2 - C_9 heterocyclylene, optionally substituted C_6 - C_{10} arylene, or optionally substituted C_2 - C_9 heteroarylene; and

438

A^1 is a bond between A and the linker, where one of R^{3a} , R^{3b} , R^{3c} , R^{3d} , R^{3e} , R^{3f} , and R^{3g} is



or G is



or a pharmaceutically acceptable salt thereof.

In some embodiments, the compound has the structure of any one of compounds D1-D38 in Table 2A, or a pharmaceutically acceptable salt thereof. In some embodiments, the compound has the structure of any one of compounds D39-D302 in Table 2B, or a pharmaceutically acceptable salt thereof. In some embodiments, the compound has the structure of any one of compounds D303-D375 in Table 2C, or a pharmaceutically acceptable salt thereof.

Other embodiments, as well as exemplary methods for the synthesis or production of these compounds, are described herein.

Pharmaceutical Uses

The compounds described herein are useful in the methods of the invention and, while not bound by theory, are believed to exert their desirable effects through their ability to modulate the level, status, and/or activity of a BAF complex, e.g., by inhibiting the activity or level of the BRD9 protein in a cell within the BAF complex in a mammal.

An aspect of the present invention relates to methods of treating disorders related to BRD9 such as cancer in a subject in need thereof. In some embodiments, the compound is administered in an amount and for a time effective to result in one of (or more, e.g., two or more, three or more, four or more of): (a) reduced tumor size, (b) reduced rate of tumor growth, (c) increased tumor cell death (d) reduced tumor progression, (e) reduced number of metastases, (f) reduced rate of metastasis, (g) decreased tumor recurrence (h) increased survival of subject, and (i) increased progression free survival of a subject.

Treating cancer can result in a reduction in size or volume of a tumor. For example, after treatment, tumor size is reduced by 5% or greater (e.g., 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or greater) relative to its size prior to treatment. Size of a tumor may be measured by any reproducible means of measurement. For example, the size of a tumor may be measured as a diameter of the tumor.

Treating cancer may further result in a decrease in number of tumors. For example, after treatment, tumor number is reduced by 5% or greater (e.g., 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or greater) relative to number prior to treatment. Number of tumors may be measured by any reproducible means of measurement, e.g., the number of tumors may be measured by counting tumors visible to the naked eye or at a specified magnification (e.g., 2x, 3x, 4x, 5x, 10x, or 50x).

Treating cancer can result in a decrease in number of metastatic nodules in other tissues or organs distant from the primary tumor site. For example, after treatment, the number of metastatic nodules is reduced by 5% or greater (e.g., 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90% or greater) relative to number prior to treatment. The number of meta-

static nodules may be measured by any reproducible means of measurement. For example, the number of metastatic nodules may be measured by counting metastatic nodules visible to the naked eye or at a specified magnification (e.g., 2x, 10x, or 50x).

Treating cancer can result in an increase in average survival time of a population of subjects treated according to the present invention in comparison to a population of untreated subjects. For example, the average survival time is increased by more than 30 days (more than 60 days, 90 days, or 120 days). An increase in average survival time of a population may be measured by any reproducible means. An increase in average survival time of a population may be measured, for example, by calculating for a population the average length of survival following initiation of treatment with the compound described herein. An increase in average survival time of a population may also be measured, for example, by calculating for a population the average length of survival following completion of a first round of treatment with a pharmaceutically acceptable salt of a compound described herein.

Treating cancer can also result in a decrease in the mortality rate of a population of treated subjects in comparison to an untreated population. For example, the mortality rate is decreased by more than 2% (e.g., more than 5%, 10%, or 25%). A decrease in the mortality rate of a population of treated subjects may be measured by any reproducible means, for example, by calculating for a population the average number of disease-related deaths per unit time following initiation of treatment with a pharmaceutically acceptable salt of a compound described herein. A decrease in the mortality rate of a population may also be measured, for example, by calculating for a population the average number of disease-related deaths per unit time following completion of a first round of treatment with a pharmaceutically acceptable salt of a compound described herein.

Combination Therapies

A method of the invention can be used alone or in combination with an additional therapeutic agent, e.g., other agents that treat cancer or symptoms associated therewith, or in combination with other types of therapies to treat cancer. In combination treatments, the dosages of one or more of the therapeutic compounds may be reduced from standard dosages when administered alone. For example, doses may be determined empirically from drug combinations and permutations or may be deduced by isobolographic analysis (e.g., Black et al., *Neurology* 65:S3-S6 (2005)). In this case, dosages of the compounds when combined should provide a therapeutic effect.

In some embodiments, the second therapeutic agent is a chemotherapeutic agent (e.g., a cytotoxic agent or other chemical compound useful in the treatment of cancer). These include alkylating agents, antimetabolites, folic acid analogs, pyrimidine analogs, purine analogs and related inhibitors, vinca alkaloids, epipodophyllotoxins, antibiotics, L-Asparaginase, topoisomerase inhibitors, interferons, platinum coordination complexes, anthracenedione substituted urea, methyl hydrazine derivatives, adrenocortical suppressant, adrenocorticosteroides, progestins, estrogens, antiestrogen, androgens, antiandrogen, and gonadotropin-releasing hormone analog. Also included is 5-fluorouracil (5-FU), leucovorin (LV), irinotecan, oxaliplatin, capecitabine, paclitaxel, and doxorubicin. Non-limiting examples of chemotherapeutic agents include alkylating agents such as thiopeta and cyclophosphamide; alkyl sulfonates such as busulfan, improsulfan and piposulfan; aziridines such as benzodopa, carboquone, meturedopa, and uredopa; ethylenimines and

methylamelamines including altretamine, triethylenemelamine, triethylenephosphoramide, triethylenethiophosphoramide and trimethylolmelamine; acetogenins (especially bullatacin and bullatacinone); a camptothecin (including the synthetic analogue topotecan); bryostatins; calystatins; CC-1065 (including its adozelesin, carzelesin and bizelesin synthetic analogues); cryptophycins (particularly cryptophycin 1 and cryptophycin 8); dolastatin; duocarmycin (including the synthetic analogues, KW-2189 and CB1-TM1); eleutherobin; pancratistatin; a sarcodictyin; spongistatin; nitrogen mustards such as chlorambucil, chlornaphazine, cholophosphamide, estramustine, ifosfamide, mechlorethamine, mechlorethamine oxide hydrochloride, melphalan, novembichin, phenesterine, prednimustine, trofosfamide, uracil mustard; nitrosureas such as carmustine, chlorozotocin, fotemustine, lomustine, nimustine, and ranimustine; antibiotics such as the enediyne antibiotics (e.g., calicheamicin, especially calicheamicin gammaII and calicheamicin omegaII (see, e.g., *Agnew, Chem. Intl. Ed Engl.* 33:183-186 (1994)); dynemicin, including dynemicin A; bisphosphonates, such as clodronate; an esperamicin; as well as neocarzinostatin chromophore and related chromoprotein enediyne antiobiotic chromophores), aclacinomysins, actinomycin, authramycin, azaserine, bleomycins, cactinomycin, carabacin, caminomycin, carzinophilin, chromomycinis, dactinomycin, daunorubicin, detorubicin, 6-diazo-5-oxo-L-norleucine, ADRIAMYCIN® (doxorubicin, including morpholino-doxorubicin, cyanomorpholino-doxorubicin, 2-pyrrolino-doxorubicin and deoxydoxorubicin), epirubicin, esorubicin, idarubicin, marcellomycin, mitomycins such as mitomycin C, mycophenolic acid, nogalamycin, olivomycins, peplomycin, potfiromycin, puromycin, quelamycin, rodorubicin, streptonigrin, streptozocin, tubercidin, ubenimex, zinostatin, zorubicin; anti-metabolites such as methotrexate and 5-fluorouracil (5-FU); folic acid analogues such as denopterin, methotrexate, pteropterin, trimetrexate; purine analogs such as fludarabine, 6-mercaptopurine, thiamiprine, thioguanine; pyrimidine analogs such as ancitabine, azacitidine, 6-azauridine, carmofur, cytarabine, dideoxyuridine, doxifluridine, enocitabine, flouxuridine; androgens such as calusterone, dromostanolone propionate, epitostanol, mepitiostane, testolactone; anti-adrenals such as aminoglutethimide, mitotane, trilostane; folic acid replenisher such as frolinic acid; aceglutone; aldophosphamide glycoside; aminolevulinic acid; eniluracil; amsacrine; bestabucil; bisantrene; edatraxate; defofamine; demecolcine; diaziquone; elfomithine; elliptinium acetate; an epothilone; etoglucid; gallium nitrate; hydroxyurea; lentinan; lonidainine; maytansinoids such as maytansine and ansamitocins; mitoguazone; mitoxantrone; mopidanmol; nitraerine; pentostatin; phenamet; pirarubicin; losoxantrone; podophyllinic acid; 2-ethylhydrazide; procarbazine; PSK® polysaccharide complex (JHS Natural Products, Eugene, OR); razoxane; rhizoxin; sizofuran; spirogermanium; tenuazonic acid; triaziquone; 2,2',2"-trichlorotriethylamine; trichothecenes (especially T-2 toxin, verracurin A, roridin A and anguidine); urethan; vindesine; dacarbazine; mannomustine; mitobronitol; mitolactol; pipobroman; gacytosine; arabinoside ("Ara-C"); cyclophosphamide; thiopeta; taxoids, e.g., TAXOL® (paclitaxel; Bristol-Myers Squibb Oncology, Princeton, NJ), ABRAXANE®, cremophor-free, albumin-engineered nanoparticle formulation of paclitaxel (American Pharmaceutical Partners, Schaumburg, IL), and TAXOTERE® doxorubicin (Rhône-Poulenc Rorer, Antony, France); chlorambucil; GEMZAR® gemcitabine; 6-thioguanine; mercaptopurine; methotrexate; platinum coordination complexes such as cisplatin, oxaliplatin and carboplatin; vin-

blastine; platinum; etoposide (VP-16); ifosfamide; mitoxantrone; vincristine; NAVELBINE® vinorelbine; novantrone; teniposide; edatrexate; daunomycin; aminopterin; xeloda; ibandronate; irinotecan (e.g., CPT-11); topoisomerase inhibitor RFS 2000; difluoromethylornithine (DMFO); retinoids such as retinoic acid; capecitabine; and pharmaceutically acceptable salts, acids or derivatives of any of the above. Two or more chemotherapeutic agents can be used in a cocktail to be administered in combination with the first therapeutic agent described herein. Suitable dosing regimens of combination chemotherapies are known in the art and described in, for example, Saltz et al., *Proc. Am. Soc. Clin. Oncol.* 18:233a (1999), and Douillard et al., *Lancet* 355(9209):1041-1047 (2000).

In some embodiments, the second therapeutic agent is a therapeutic agent which is a biologic such as a cytokine (e.g., interferon or an interleukin (e.g., IL-2)) used in cancer treatment. In some embodiments the biologic is an anti-angiogenic agent, such as an anti-VEGF agent, e.g., bevacizumab (AVASTIN®). In some embodiments the biologic is an immunoglobulin-based biologic, e.g., a monoclonal antibody (e.g., a humanized antibody, a fully human antibody, an Fc fusion protein or a functional fragment thereof) that agonizes a target to stimulate an anti-cancer response, or antagonizes an antigen important for cancer. Such agents include RITUXAN® (rituximab); ZENAPAX® (daclizumab); SIMULECT® (basiliximab); SYNAGIS® (palivizumab); REMICADE® (infliximab); HERCEPTIN® (trastuzumab); MYLOTARG® (gemtuzumab ozogamicin); CAMPATH® (alemtuzumab); ZEVALIN® (ibritumomab tiuxetan); HUMIRA® (adalimumab); XOLAIR® (omalizumab); BEXXAR® (tositumomab-I-131); RAPTIVA® (efalizumab); ERBITUX® (cetuximab); AVASTIN® (bevacizumab); TYSABRI® (natalizumab); ACTEMRA® (tocilizumab); VECTIBIX® (panitumumab); LUCENTIS® (ranibizumab); SOLIRIS® (eculizumab); CIMZIA® (certolizumab pegol); SIMPONI® (golimumab); ILARIS® (canakinumab); STELARA® (ustekinumab); ARZERRA® (ofatumumab); PROLIA® (denosumab); NUMAX® (motavizumab); ABTHRAX® (raxibacumab); BENLYSTA® (belimumab); YERVOY® (ipilimumab); ADCETRIS® (brentuximab vedotin); PERJETA® (pertuzumab); KADCYLA® (ado-trastuzumab emtansine); and GAZYVA® (obinituzumab). Also included are antibody-drug conjugates.

The second agent may be a therapeutic agent which is a non-drug treatment. For example, the second therapeutic agent is radiation therapy, cryotherapy, hyperthermia, and/or surgical excision of tumor tissue.

The second agent may be a checkpoint inhibitor. In one embodiment, the inhibitor of checkpoint is an inhibitory antibody (e.g., a monospecific antibody such as a monoclonal antibody). The antibody may be, e.g., humanized or fully human. In some embodiments, the inhibitor of checkpoint is a fusion protein, e.g., an Fc-receptor fusion protein. In some embodiments, the inhibitor of checkpoint is an agent, such as an antibody, that interacts with a checkpoint protein. In some embodiments, the inhibitor of checkpoint is an inhibitor (e.g., an inhibitory antibody or small molecule inhibitor) of CTLA-4 (e.g., an anti-CTLA4 antibody or fusion a protein such as ipilimumab/YERVOY® or tremelimumab). In some embodiments, the inhibitor of checkpoint is an inhibitor (e.g., an inhibitory antibody or small molecule inhibitor) of PD-1 (e.g., nivolumab/OPDIVO®; pembrolizumab/KEYTRUDA®; pidilizumab/CT-

011). In some embodiments, the inhibitor of checkpoint is an inhibitor (e.g., an inhibitory antibody or small molecule inhibitor) of PDL1 (e.g., MPDL3280A/RG7446; MEDI4736; MSB0010718C; BMS 936559). In some embodiments, the inhibitor of checkpoint is an inhibitor (e.g., an inhibitory antibody or Fc fusion or small molecule inhibitor) of PDL2 (e.g., a PDL2/Ig fusion protein such as AMP 224). In some embodiments, the inhibitor of checkpoint is an inhibitor (e.g., an inhibitory antibody or small molecule inhibitor) of B7-H3 (e.g., MGA271), B7-H4, BTLA, HVEM, TIM3, GAL9, LAG3, VISTA, KIR, 2B4, CD160, CGEN-15049, CHK 1, CHK2, A2aR, B-7 family ligands, or a combination thereof.

In some embodiments, the anti-cancer therapy is a T cell adoptive transfer (ACT) therapy. In some embodiments, the T cell is an activated T cell. The T cell may be modified to express a chimeric antigen receptor (CAR). CAR modified T (CAR-T) cells can be generated by any method known in the art. For example, the CAR-T cells can be generated by introducing a suitable expression vector encoding the CAR to a T cell. Prior to expansion and genetic modification of the T cells, a source of T cells is obtained from a subject. T cells can be obtained from a number of sources, including peripheral blood mononuclear cells, bone marrow, lymph node tissue, cord blood, thymus tissue, tissue from a site of infection, ascites, pleural effusion, spleen tissue, and tumors. In certain embodiments of the present invention, any number of T cell lines available in the art, may be used. In some embodiments, the T cell is an autologous T cell. Whether prior to or after genetic modification of the T cells to express a desirable protein (e.g., a CAR), the T cells can be activated and expanded generally using methods as described, for example, in U.S. Pat. Nos. 6,352,694; 6,534,055; 6,905,680; 6,692,964; 5,858,358; 6,887,466; 6,905,681; 7,144,575; 7,067,318; 7,172,869; 7,232,566; 7,175,843; 5,883,223; 6,905,874; 6,797,514; 6,867,041; and U.S. Patent Application Publication No. 20060121005.

In any of the combination embodiments described herein, the first and second therapeutic agents are administered simultaneously or sequentially, in either order. The first therapeutic agent may be administered immediately, up to 1 hour, up to 2 hours, up to 3 hours, up to 4 hours, up to 5 hours, up to 6 hours, up to 7 hours, up to, 8 hours, up to 9 hours, up to 10 hours, up to 11 hours, up to 12 hours, up to 13 hours, 14 hours, up to hours 16, up to 17 hours, up to 18 hours, up to 19 hours up to 20 hours, up to 21 hours, up to 22 hours, up to 23 hours up to 24 hours or up to 1-7, 1-14, 1-21 or 1-30 days before or after the second therapeutic agent.

50 Pharmaceutical Compositions

The pharmaceutical compositions described herein are preferably formulated into pharmaceutical compositions for administration to human subjects in a biologically compatible form suitable for administration in vivo.

The compounds described herein may be used in the form of the free base, in the form of salts, solvates, and as prodrugs. All forms are within the methods described herein. In accordance with the methods of the invention, the described compounds or salts, solvates, or prodrugs thereof may be administered to a patient in a variety of forms depending on the selected route of administration, as will be understood by those skilled in the art. The compounds described herein may be administered, for example, by oral, parenteral, buccal, sublingual, nasal, rectal, patch, pump, intratumoral, or transdermal administration and the pharmaceutical compositions formulated accordingly. Parenteral administration includes intravenous, intraperitoneal, subcu-

taneous, intramuscular, transepithelial, nasal, intrapulmonary, intrathecal, rectal, and topical modes of administration. Parenteral administration may be by continuous infusion over a selected period of time.

A compound described herein may be orally administered, for example, with an inert diluent or with an assimilable edible carrier, or it may be enclosed in hard or soft shell gelatin capsules, or it may be compressed into tablets, or it may be incorporated directly with the food of the diet. For oral therapeutic administration, a compound described herein may be incorporated with an excipient and used in the form of ingestible tablets, buccal tablets, troches, capsules, elixirs, suspensions, syrups, and wafers. A compound described herein may also be administered parenterally. Solutions of a compound described herein can be prepared in water suitably mixed with a surfactant, such as hydroxypropylcellulose. Dispersions can also be prepared in glycerol, liquid polyethylene glycols, DMSO, and mixtures thereof with or without alcohol, and in oils. Under ordinary conditions of storage and use, these preparations may contain a preservative to prevent the growth of microorganisms. Conventional procedures and ingredients for the selection and preparation of suitable formulations are described, for example, in Remington's Pharmaceutical Sciences (2012, 22nd ed.) and in The United States Pharmacopeia: The National Formulary (USP 41 NF36), published in 2018. The pharmaceutical forms suitable for injectable use include sterile aqueous solutions or dispersions and sterile powders for the extemporaneous preparation of sterile injectable solutions or dispersions. In all cases the form must be sterile and must be fluid to the extent that may be easily administered via syringe. Compositions for nasal administration may conveniently be formulated as aerosols, drops, gels, and powders. Aerosol formulations typically include a solution or fine suspension of the active substance in a physiologically acceptable aqueous or non-aqueous solvent and are usually presented in single or multidose quantities in sterile form in a sealed container, which can take the form of a cartridge or refill for use with an atomizing device. Alternatively, the sealed container may be a unitary dispensing device, such as a single dose nasal inhaler or an aerosol dispenser fitted with a metering valve which is intended for disposal after use. Where the dosage form includes an aerosol dispenser, it will contain a propellant, which can be a compressed gas, such as compressed air or an organic propellant, such as fluorochlorohydrocarbon. The aerosol dosage forms can also take the form of a pump-atomizer. Compositions suitable for buccal or sublingual administration include tablets, lozenges, and pastilles, where the active ingredient is formulated with a carrier, such as sugar, acacia, tragacanth, gelatin, and glycerine. Compositions for rectal administration are conveniently in the form of suppositories containing a conventional suppository base, such as cocoa butter. A compound described herein may be administered intratumorally, for example, as an intratumoral injection. Intratumoral injection is injection directly into the tumor vasculature and is specifically contemplated for discrete, solid, accessible tumors. Local, regional, or systemic administration also may be appropriate. A compound described herein may advantageously be contacted by administering an injection or multiple injections to the tumor, spaced for example, at approximately, 1 cm intervals. In the case of surgical intervention, the present invention may be used preoperatively, such as to render an inoperable tumor subject to resection. Continuous administration also may be applied where appropriate, for example, by implanting a catheter into a tumor or into tumor vasculature.

The compounds described herein may be administered to an animal, e.g., a human, alone or in combination with pharmaceutically acceptable carriers, as noted herein, the proportion of which is determined by the solubility and chemical nature of the compound, chosen route of administration, and standard pharmaceutical practice.

Dosages

The dosage of the compounds described herein, and/or compositions including a compound described herein, can vary depending on many factors, such as the pharmacodynamic properties of the compound; the mode of administration; the age, health, and weight of the recipient; the nature and extent of the symptoms; the frequency of the treatment, and the type of concurrent treatment, if any; and the clearance rate of the compound in the animal to be treated. One of skill in the art can determine the appropriate dosage based on the above factors. The compounds described herein may be administered initially in a suitable dosage that may be adjusted as required, depending on the clinical response. In general, satisfactory results may be obtained when the compounds described herein are administered to a human at a daily dosage of, for example, between 0.05 mg and 3000 mg (measured as the solid form). Dose ranges include, for example, between 10-1000 mg (e.g., 50-800 mg). In some embodiments, 50, 100, 150, 200, 250, 300, 350, 400, 450, 500, 550, 600, 650, 700, 750, 800, 850, 900, 950, or 1000 mg of the compound is administered.

Alternatively, the dosage amount can be calculated using the body weight of the patient. For example, the dose of a compound, or pharmaceutical composition thereof, administered to a patient may range from 0.1-100 mg/kg (e.g., 0.1-50 mg/kg (e.g., 0.25-25 mg/kg)). In exemplary, non-limiting embodiments, the dose may range from 0.5-5.0 mg/kg (e.g., 0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, or 5.0 mg/kg) or from 5.0-20 mg/kg (e.g., 5.5, 6.0, 6.5, 7.0, 7.5, 8.0, 8.5, 9.0, 9.5, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 mg/kg).

Kits

The invention also features kits including (a) a pharmaceutical composition including an agent that reduces the level and/or activity of BRD9 in a cell or subject described herein, and (b) a package insert with instructions to perform any of the methods described herein. In some embodiments, the kit includes (a) a pharmaceutical composition including an agent that reduces the level and/or activity of BRD9 in a cell or subject described herein, (b) an additional therapeutic agent (e.g., an anti-cancer agent), and (c) a package insert with instructions to perform any of the methods described herein.

EXAMPLES

Example 1—High Density Tiling sgRNA Screen Against Human BAF Complex Subunits in Synovial Sarcoma Cell Line SYO1

The following example shows that BRD9 sgRNA inhibits cell growth in synovial sarcoma cells.

Procedure: To perform high density sgRNA tiling screen, an sgRNA library against BAF complex subunits was custom synthesized at Cellecta (Mountain View, CA). Sequences of DNA encoding the BRD9-targeting sgRNAs used in this screen are listed in Table 3. Negative and positive control sgRNA were included in the library. Negative controls consisted of 200 sgRNAs that do not target human genome. The positive controls are sgRNAs targeting essential genes (CDC16, GTF2B, HSPA5, HSPA9,

445

PAFAH1B1, PCNA, POLR2L, RPL9, and SF3A3). DNA sequences encoding all positive and negative control sgRNAs are listed in Table 4. Procedures for virus production, cell infection, and performing the sgRNA screen were previously described (Tsherniak et al, *Cell* 170:564-576 (2017); Munoz et al, *Cancer Discovery* 6:900-913 (2016)). For each sgRNA, 50 counts were added to the sequencing counts and for each time point the resulting counts were normalized to the total number of counts. The log 2 of the ratio between the counts (defined as dropout ratio) at day 24 and day 1 post-infection was calculated. For negative control sgRNAs, the 2.5 and 97.5 percentile of the log 2 dropout ratio of all non-targeting sgRNAs was calculated and considered as background (grey box in the graph). Protein domains were obtained from PFAM regions defined for the UNIPROT identifier: Q9H8M2.

Results: As shown in FIG. 1, targeted inhibition of the GBAF complex component BRD9 by sgRNA resulted in growth inhibition of the SYO1 synovial sarcoma cell line. sgRNAs against other components of the BAF complexes resulted in increased proliferation of cells, inhibition of cell growth, or had no effect on SYO1 cells. These data show that targeting various subunits of the GBAF complex represents a therapeutic strategy for the treatment of synovial sarcoma.

TABLE 3

BRD9 sgRNA Library	
SEQ ID NO.	Nucleic Acid Sequence
203	CAAGAAGCACAAGAAGCACA
204	CTTGTGCTTCTTGCCCATGG
205	CTTCTTGTGCTTCTTGCCCA
206	ACAAGAAGCACAAGGCCGAG
207	CTCGTAGGACGAGCGCCACT
208	CGAGTGGCGCTCGTCTACG
209	GAGTGGCGCTCGTCTACGA
210	AGGCTTCTCCAGGGGCTTGT
211	AGATTATGCCGACAAGCCCC
212	ACCTTCAGGACTAGCTTTAG
213	AGCTTTAGAGGCTTCTCCAG
214	CTAGCTTTAGAGGCTTCTCC
215	TAGCTTTAGAGGCTTCTCCA
216	CTAAAGCTAGTCCTGAAGGT
217	GCCTCTAAAGCTAGTCCTGA
218	CTTCACTTCTCCGACCTTC
219	AAGCTAGTCCTGAAGGTCGG
220	AGTGAAGTGAAGTGAAGTCTC
221	GTGACTGAACTCTCAGGATC
222	ATAGTAACTGGAGTCGTGGC
223	CATCATAGTAACTGGAGTCG
224	TGACCTGTATCATAGTAAC

446

TABLE 3-continued

BRD9 sgRNA Library	
SEQ ID NO.	Nucleic Acid Sequence
225	ACTCCAGTTACTATGATGAC
226	CTTTGTGCCTCTCTCGCTCA
227	GGTCAGACCATTGAGCGAGAG
228	GAAGAAGAAGAAGTCCGAGA
229	GTCCAGATGCTTCTCTTCT
230	GTCCGAGAAGGAGAAGCATC
231	GGAGAAGCATCTGGACGATG
232	TGAGGAAAGAAGGAAGCGAA
233	ATCTGGACGATGAGGAAAGA
234	AGAAGAAGCGGAAGCGAGAG
235	GAAGAAGCGGAAGCGAGAGA
236	CCGCCAGGAAGAGAAGAAG
237	AGAGAGGGAGCACTGTGACA
238	AGGGAGCACTGTGACACGGA
239	GAGGGAGCACTGTGACACGG
240	GCACTGTGACACGGAGGGAG
241	GAGGCTGACGACTTTGATCC
242	AGGCTGACGACTTTGATCCT
243	TCCACCTCCACCTTCTTCCC
244	CGACTTTGATCCTGGGAAGA
245	CTTTGATCCTGGGAAGAAGG
246	TGATCCTGGGAAGAAGGTGG
247	TCCTGGGAAGAAGGTGGAGG
248	CGGACTGGCCGATCTGGGGG
249	ACGCTCGGACTGGCCGATCT
250	AGGTGGAGCCGCCCCAGAT
251	CGCTCGGACTGGCCGATCTG
252	GCTCGGACTGGCCGATCTGG
253	CACGCTCGGACTGGCCGATC
254	TGTGTCCGGCACGCTCGGAC
255	CTGGCTGTGTCCGGCACGCT
256	ATCGGCCAGTCCGAGCGTGC
257	CACCCTTGCCCTGGCTGTGTC
258	CGAGCGTGCCGGACACAGCC
259	TGTTCAGGAGTTGCTGAAT
260	CACACCTATTAGCAACTCC
261	GCTGGCGGAGGAAGTGTTC
262	TTTACCTCTGAAGCTGGCGG

447

TABLE 3-continued

BRD9 sgRNA Library		
SEQ ID NO.	Nucleic Acid Sequence	
263	CCCCGGTTTACCTCTGAAGC	5
264	ACTTCCTCCGCCAGCTTCAG	
265	CAGGAAAAGCAAAAATCCA	
266	GCTTTCAGAAAAGATCCCCA	
267	AGGAAAAGCAAAAATCCAT	10
268	GGAAAAGCAAAAATCCATG	
269	GGAGCAATTGCATCCGTGAC	
270	GTCACGGATGCAATTGCTCC	
271	TTTATTATCATTGAATATCC	15
272	AATGATAATAAAACATCCCCA	
273	ATAAAACATCCCATGGATTT	
274	TTCATGGTGCCAAAATCCAT	
275	TTTCATGGTGCCAAAATCCA	20
276	TAATGAATACAAGTCAGTTA	
277	CAAGTCAGTTACGGAATTTA	
278	ATAATGCAATGACATACAAT	
279	AACTTGTAGTACACGGTATC	25
280	CTTCGCCAACTTGTAGTACA	
281	AGATACCGTGTACTACAAGT	
282	GCGAAGAAGATCCTTCACGC	
283	TCATCTTAAAGCCTGCGTGA	30
284	TTCTCAGCAGGCAGCTCTTT	
285	CAATGAAGATACAGCTGTTG	
286	ACTGGTACAACCTCAGGGAC	
287	CTTGTAAGTACAACTTCA	35
288	ACTTGTAAGTACAACTTC	
289	TTGGCAGTTTCTACTTGATC	
290	TACCTGATAAATCTCTACT	
291	AGCCGAGTAGAGAAGTTATC	40
292	AGCTGCATGTTTGAGCCTGA	
293	GCTGCATGTTTGAGCCTGAA	
294	AAGCTGCAGGCATTCCCTTC	
295	GGTACTGTCCGTCAAGCTGC	45
296	AGGGAATGCCTGCAGCTTGA	
297	CTTGACGGACAGTACCGCAG	
298	CGCCAGCACGTGCTCCTCTG	
299	TACCGCAGAGGAGCACGTGC	50
		55
		60
		65

448

TABLE 3-continued

BRD9 sgRNA Library	
SEQ ID NO.	Nucleic Acid Sequence
300	AGAGGAGCACGTGCTGGCGC
301	GGAGCACGTGCTGGCGCTGG
302	AGCACGCAGCTGACGAAGCT
303	GCACGCAGCTGACGAAGCTC
304	CAGCTGACGAAGCTCGGGAC
305	AAGCTCGGGACAGGATCAAC
306	CCTTGCCGCCTGGGAGGAAC
307	AGGATCAACCGGTTCTCTCC
308	ATCAACCGGTTCTCTCCAGG
309	GCACTACCTTGCCGCCTGGG
310	AGAGCACTACCTTGCCGCCT
311	CCGGTTCTCTCCAGGCGGCA
312	TCCTCTTCTAGATAGCCATC
313	ATGGGCTATCTGAAGAGGAA
314	GGGCTATCTGAAGAGGAACG
315	TGGGCTATCTGAAGAGGAAC
316	TATCTGAAGAGGAACGGGGA
317	ATCTGAAGAGGAACGGGGAC
318	TGTTGACCACGCTGTAGAGC
319	GCTCTACAGCGTGGTCAACA
320	CGGGAGCCTGCTCTACAGCG
321	CGTGGTCAACACGGCCGAGC
322	CCCACCATCAGCGTCCGGCT
323	ACGGCCGAGCCGGACGCTGA
324	GGGCACCCACCATCAGCGTC
325	GCCGAGCCGGACGCTGATGG
326	CCATGTCCGTGTTGCAGAGG
327	CCGAGCCGGACGCTGATGGT
328	CGAGCTCAAGTCCACCGGCT
329	GCGAGCTCAAGTCCACCGGG
330	AGAGCGAGCTCAAGTCCACC
331	GAGAGCGAGCTCAAGTCCAC
332	GAAGCCTGGGAGTAGCTTAC
333	CTCTCCAGTAAGCTACTCCC
334	AGCCCAGCGTGGTGAAGCCT
335	AAGCCCAGCGTGGTGAAGCC
336	ACTCCAGGCTTCAACACGC
337	CTCCAGGCTTCAACACGCT

449

TABLE 3-continued

BRD9 sgRNA Library	
SEQ ID NO.	Nucleic Acid Sequence
338	CTCGTCTTTGAAGCCCAGCG
339	CACTGGAGAGAAAGGTGACT
340	GCACTGGAGAGAAAGGTGAC
341	AGTAGTGGCACTGGAGAGAA
342	CGAAAGCGCAGTAGTGGCAC
343	CTGCATCGAAAGCGCAGTAG
344	ATGCAGAATAATTGAGTATT
345	AGTATTTGGCGACTTGAAGT
346	CGACTTGAAGTCGGACGAGA
347	GAGCTGCTCTACTCAGCCTA
348	CACGCCTGTCTCATCTCCGT
349	TCAGCCTACGGAGATGAGAC
350	CAGGCGTGCAGTGTGCGCTG
351	CCGCGGCCCTCTAGCCTGC
352	CATCCTTCACAACTCCTGC
353	TAGCCTGCAGGAGTTGTGA
354	CAGGAGTTTGTGAAGGATGC
355	AGGAGTTTGTGAAGGATGCT
356	TGGGAGCTACAGCAAGAAAG
357	GAGCTACAGCAAGAAAGTGG
358	GAAAGTGGTGGACGACCTCC
359	CGCCTGTGATCTGGTCCAGG
360	CTCCGCCTGTGATCTGGTCC
361	GACCTCCTGGACGATCAC
362	CTCCTGGACGATCACAGG
363	GCTGGAAGAGCGTCTAGAG
364	TGCAGCCCACCTGCTTCAGC
365	GACGCTCTTCCAGCTGAAGC
366	CTCTTCCAGCTGAAGCAGGT
367	GCTCTTCCAGCTGAAGCAGG
368	CCTCCAGATGAAGCCAAGGT
369	GCTTCATCTGGAGGCTTCAT
370	GGCTTCATCTGGAGGCTTCA
371	CTTACCTTGGCTTCATCTGG
372	AAACTTACCTTGGCTTCATC
373	GAAGCCTCCAGATGAAGCCA
374	TCCTAGGGTGTCCCCAACCT

450

TABLE 3-continued

BRD9 sgRNA Library	
SEQ ID NO.	Nucleic Acid Sequence
375	CCTAGGGTGTCCCCAACCTG
376	GTGTCTGTCTCCACAGGTTG
377	TGTGTCTGTCTCCACAGGTT
378	CCACAGGTTGGGGACACCCCT
379	AGAGCTGTGTGTCTCTCCTA
380	CAGAGCTGCTGCTGTCTCCT
381	AGACAGCAGCAGCTCTGTTC
382	ATCCACAGAAACGTCGGGAT
383	GAGATATCCACAGAAACGTC
384	GGAGATATCCACAGAAACGT
385	GTCCTATCCCAGCTTTCTG
386	TCTCCATGCTCAGCTCTCTG
387	CTCACCAGAGAGCTGAGCA
388	ATCTCCATGCTCAGCTCTCT
389	TATCTCCATGCTCAGCTCTC
390	ATGTCCTGTTTACACAGGGA
391	TTACACAGGGAAGGTGAAGA
392	AGTTCAAATGGCTGTCGTCA
393	TGACGACAGCCATTTGAACT
394	AAGTTCAAATGGCTGTCGTC
395	TCGTCTCATCCAAGTTCAAA
396	TGAGACGACGAAGCTCCTGC
397	GTGCTTCGTGCAGGTCCTGC
398	GCAGGACCTGCACGAAGCAC
399	GCTCCGCCTGTGCTTCGTGC
400	GGACCTGCACGAAGCACAGG
401	CACGAAGCACAGGCGGAGCG
402	AGGCGGAGCGCGGCGCTCT
403	AGGGAGCTGAGGTTGGACGA
404	GTTGGACAGGGAGCTGAGGT
405	AGGCGTTGGACAGGGAGCTG
406	CCCTCTCGGAGGCGTTGGAC
407	CCTCTCGGAGGCGTTGGACA
408	CTGGTCCCTCTCGGAGGCGT
409	CCCTGTCCAACGCCTCCGAG
410	CCTGTCCAACGCCTCCGAGA
411	GTGGTGTGCTGCTCCCTCTCGG
412	CAGGTGGTGTGCTCCCTCT

451

TABLE 3-continued

BRD9 sgRNA Library	
SEQ ID NO.	Nucleic Acid Sequence
413	GCATCTCACCAGGTGGTGC
414	CGAGAGGGACCAGCACCACC
415	GAGAGGGACCAGCACCACCT
416	GTGGGGGCATCTCACCAGG
417	CCCCGACACTCAGGCGAGAA
418	TCCCCGACACTCAGGCGAGA
419	AGCCCTTCTCGCCTGAGTGT
420	CTGGCTGCTCCCCGACACTC
421	CCCTTCTCGCCTGAGTGTG
422	GCCCTTCTCGCCTGAGTGTG
423	TAGGGGTCGTGGGTGACGTC

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452

TABLE 3-continued

BRD9 sgRNA Library	
SEQ ID NO.	Nucleic Acid Sequence
424	AAGAAACTCATAGGGGTCGT
425	GAAGAAACTCATAGGGGTCG
426	GAGACTGAAGAAACTCATAG
427	GGAGACTGAAGAAACTCATA
428	TGGAGACTGAAGAAACTCAT
429	TCTTCAGTCTCCAGAGCCTG
430	TTGGCAGAGGCCGAGGCTC
431	TAGGTCTTGGCAGAGGCCGC
432	CTAGAGTTAGGTCTTGGCAG
433	GGTGGTCTAGAGTTAGGTCT

TABLE 4

Control sgRNA Library		
SEQ ID NO.	gRNA Label	Nucleic Acid Sequence
4341	sg_Non_Targeting_Human_0001 Non_Targeting_Human	Non_Targeting_Human GTAGCGAACGTGTCGGCGCT
4351	sg_Non_Targeting_Human_0002 Non_Targeting_Human	Non_Targeting_Human GACCGGAACGATCTCGCGTA
4361	sg_Non_Targeting_Human_0003 Non_Targeting_Human	Non_Targeting_Human GGCAGTCGTTCCGTTGATAT
4371	sg_Non_Targeting_Human_0004 Non_Targeting_Human	Non_Targeting_Human GCTTGAGCACATACGCGAAT
4381	sg_Non_Targeting_Human_0005 Non_Targeting_Human	Non_Targeting_Human GTGGTAGAATAACGTATTAC
4391	sg_Non_Targeting_Human_0006 Non_Targeting_Human	Non_Targeting_Human GTCATACATGGATAAGGCTA
4401	sg_Non_Targeting_Human_0007 Non_Targeting_Human	Non_Targeting_Human GATACACGAAGCATCACTAG
4411	sg_Non_Targeting_Human_0008 Non_Targeting_Human	Non_Targeting_Human GAACGTTGGCACTACTTCAC
4421	sg_Non_Targeting_Human_0009 Non_Targeting_Human	Non_Targeting_Human GATCCATGTAATGCGTTCGA
4431	sg_Non_Targeting_Human_0010 Non_Targeting_Human	Non_Targeting_Human GTCGTGAAGTGCATTGATC
4441	sg_Non_Targeting_Human_0011 Non_Targeting_Human	Non_Targeting_Human GTTCGACTCGCGTGACCGTA
4451	sg_Non_Targeting_Human_0012 Non_Targeting_Human	Non_Targeting_Human GAATCTACCGCAGCGGTTCCG
4461	sg_Non_Targeting_Human_0013 Non_Targeting_Human	Non_Targeting_Human GAAGTGACGTGCGATTGATA
4471	sg_Non_Targeting_Human_0014 Non_Targeting_Human	Non_Targeting_Human GCGGTGTATGACAACCGCCG

TABLE 4-continued

Control sgRNA Library		
SEQ ID NO. sgRNA Label	Gene	Nucleic Acid Sequence
4481 sg_Non_Targeting_Human_0015 Non_Targeting_Human	Non_Targeting_Human	GTACCGCGCCTGAAGTTCGC
4491 sg_Non_Targeting_Human_0016 Non_Targeting_Human	Non_Targeting_Human	GCAGCTCGTGTGCTGCTACTC
4501 sg_Non_Targeting_Human_0017 Non_Targeting_Human	Non_Targeting_Human	GCGCCTTAAGAGTACTCATC
4511 sg_Non_Targeting_Human_0018 Non_Targeting_Human	Non_Targeting_Human	GAGTGTGTCGTTGCTCCTA
4521 sg_Non_Targeting_Human_0019 Non_Targeting_Human	Non_Targeting_Human	GCAGCTCGACCTCAAGCCGT
4531 sg_Non_Targeting_Human_0020 Non_Targeting_Human	Non_Targeting_Human	GTATCCTGACCTACGCGCTG
4541 sg_Non_Targeting_Human_0021 Non_Targeting_Human	Non_Targeting_Human	GTGTATCTCAGCACGCTAAC
4551 sg_Non_Targeting_Human_0022 Non_Targeting_Human	Non_Targeting_Human	GTCGTCATACAACGGCAACG
4561 sg_Non_Targeting_Human_0023 Non_Targeting_Human	Non_Targeting_Human	GTCGTGCGCTTCCGGCGGTA
4571 sg_Non_Targeting_Human_0024 Non_Targeting_Human	Non_Targeting_Human	GCGGTCTCAGTAAGCGCGT
4581 sg_Non_Targeting_Human_0025 Non_Targeting_Human	Non_Targeting_Human	GCTCTGCTGCGGAAGGATTC
4591 sg_Non_Targeting_Human_0026 Non_Targeting_Human	Non_Targeting_Human	GCATGGAGGAGCGTCGCAGA
4601 sg_Non_Targeting_Human_0027 Non_Targeting_Human	Non_Targeting_Human	GTAGCGCGCGTAGGAGTGGC
4611 sg_Non_Targeting_Human_0028 Non_Targeting_Human	Non_Targeting_Human	GATCACCTGCATTCTGACAC
4621 sg_Non_Targeting_Human_0029 Non_Targeting_Human	Non_Targeting_Human	GCACACCTAGATATCGAATG
4631 sg_Non_Targeting_Human_0030 Non_Targeting_Human	Non_Targeting_Human	GTTGATCAACGCGCTTCGCG
4641 sg_Non_Targeting_Human_0031 Non_Targeting_Human	Non_Targeting_Human	GCGTCTCACTCACTCCATCG
4651 sg_Non_Targeting_Human_0032 Non_Targeting_Human	Non_Targeting_Human	GCCGACCAACGTCAAGCGGTA
4661 sg_Non_Targeting_Human_0033 Non_Targeting_Human	Non_Targeting_Human	GGATACGGTGGTCAATCTA
4671 sg_Non_Targeting_Human_0034 Non_Targeting_Human	Non_Targeting_Human	GAATCCAGTGGCGCGACAA
4681 sg_Non_Targeting_Human_0035 Non_Targeting_Human	Non_Targeting_Human	GCACTGTGAGTGAACGATA
4691 sg_Non_Targeting_Human_0036 Non_Targeting_Human	Non_Targeting_Human	GCGATCCTCAAGTATGCTCA
4701 sg_Non_Targeting_Human_0037 Non_Targeting_Human	Non_Targeting_Human	GCTAATATCGACACGGCCGC
4711 sg_Non_Targeting_Human_0038 Non_Targeting_Human	Non_Targeting_Human	GGAGATGCATCGAAGTCGAT
4721 sg_Non_Targeting_Human_0039 Non_Targeting_Human	Non_Targeting_Human	GGATGCACTCCATCTCGTCT

TABLE 4-continued

Control sgRNA Library		
SEQ ID NO. sgRNA Label	Gene	Nucleic Acid Sequence
4731 sg_Non_Targeting_Human_0040 Non_Targeting_Human	Non_Targeting_Human	GTGCCGAGTAATAACGCGAG
4741 sg_Non_Targeting_Human_0041 Non_Targeting_Human	Non_Targeting_Human	GAGATTCCGATGTAACGTAC
4751 sg_Non_Targeting_Human_0042 Non_Targeting_Human	Non_Targeting_Human	GTCGTACAGCAGGATTGC
4761 sg_Non_Targeting_Human_0043 Non_Targeting_Human	Non_Targeting_Human	GCGTTAGTCACTTAGCTCGA
4771 sg_Non_Targeting_Human_0044 Non_Targeting_Human	Non_Targeting_Human	GTTACACGGTGTCCGATAG
4781 sg_Non_Targeting_Human_0045 Non_Targeting_Human	Non_Targeting_Human	GGATAGGTGACCTTAGTACG
4791 sg_Non_Targeting_Human_0046 Non_Targeting_Human	Non_Targeting_Human	GTATGAGTCAAGCTAATGCG
4801 sg_Non_Targeting_Human_0047 Non_Targeting_Human	Non_Targeting_Human	GCAACTATTGGAATACGTGA
4811 sg_Non_Targeting_Human_0048 Non_Targeting_Human	Non_Targeting_Human	GTTACCTTCGCTCGTCTATA
4821 sg_Non_Targeting_Human_0049 Non_Targeting_Human	Non_Targeting_Human	GTACCGAGCACCACAGGCCG
4831 sg_Non_Targeting_Human_0050 Non_Targeting_Human	Non_Targeting_Human	GTCAGCCATCGGATAGAGAT
4841 sg_Non_Targeting_Human_0051 Non_Targeting_Human	Non_Targeting_Human	GTACGGCACTCCTAGCCGCT
4851 sg_Non_Targeting_Human_0052 Non_Targeting_Human	Non_Targeting_Human	GTCCTGTCGTATGCTTGCA
4861 sg_Non_Targeting_Human_0053 Non_Targeting_Human	Non_Targeting_Human	GCCGCAATATATGCGGTAAG
4871 sg_Non_Targeting_Human_0054 Non_Targeting_Human	Non_Targeting_Human	GCGCACGTATAATCCTGCGT
4881 sg_Non_Targeting_Human_0055 Non_Targeting_Human	Non_Targeting_Human	GTGCACAACACGATCCACGA
4891 sg_Non_Targeting_Human_0056 Non_Targeting_Human	Non_Targeting_Human	GCACAATGTTGACGTAAGTG
4901 sg_Non_Targeting_Human_0057 Non_Targeting_Human	Non_Targeting_Human	GTAAGATGCTGCTCACCGTG
4911 sg_Non_Targeting_Human_0058 Non_Targeting_Human	Non_Targeting_Human	GTCGGTGATCCAACGTATCG
4921 sg_Non_Targeting_Human_0059 Non_Targeting_Human	Non_Targeting_Human	GAGCTAGTAGGACGCAAGAC
4931 sg_Non_Targeting_Human_0060 Non_Targeting_Human	Non_Targeting_Human	GTACGTGGAAGCTTGTGGCC
4941 sg_Non_Targeting_Human_0061 Non_Targeting_Human	Non_Targeting_Human	GAGAACTGCCAGTTCTCGAT
4951 sg_Non_Targeting_Human_0062 Non_Targeting_Human	Non_Targeting_Human	GCCATTGCGCGCGCACTTC
4961 sg_Non_Targeting_Human_0063 Non_Targeting_Human	Non_Targeting_Human	GCACACGACCAATCCGCTTC

TABLE 4-continued

Control sgRNA Library		
SEQ ID NO. sgRNA Label	Gene	Nucleic Acid Sequence
4971 sg_Non_Targeting_Human_0064 Non_Targeting_Human	Non_Targeting_Human	GAGGTGATCGATTAAGTACA
4981 sg_Non_Targeting_Human_0065 Non_Targeting_Human	Non_Targeting_Human	GTCACCTCGCAGACGCCTAAC
4991 sg_Non_Targeting_Human_0066 Non_Targeting_Human	Non_Targeting_Human	GCGCTACGGAATCATACTGTT
5001 sg_Non_Targeting_Human_0067 Non_Targeting_Human	Non_Targeting_Human	GGTAGGACCTCAGGCGCGC
5011 sg_Non_Targeting_Human_0068 Non_Targeting_Human	Non_Targeting_Human	GAAGTGCATCTTGTGTAGT
5021 sg_Non_Targeting_Human_0069 Non_Targeting_Human	Non_Targeting_Human	GATCCTGATCCGCGCGCGC
5031 sg_Non_Targeting_Human_0070 Non_Targeting_Human	Non_Targeting_Human	GGTATGCGCGATCCTGAGTT
5041 sg_Non_Targeting_Human_0071 Non_Targeting_Human	Non_Targeting_Human	GCGGAGCTAGAGACGGTCA
5051 sg_Non_Targeting_Human_0072 Non_Targeting_Human	Non_Targeting_Human	GAATGGCAATTACGGCTGAT
5061 sg_Non_Targeting_Human_0073 Non_Targeting_Human	Non_Targeting_Human	GTATGGTGAGTAGTCGCTTG
5071 sg_Non_Targeting_Human_0074 Non_Targeting_Human	Non_Targeting_Human	GTGTAATTGCGTCTAGTCGG
5081 sg_Non_Targeting_Human_0075 Non_Targeting_Human	Non_Targeting_Human	GGTCCTGGCGAGGAGCCTTG
5091 sg_Non_Targeting_Human_0076 Non_Targeting_Human	Non_Targeting_Human	GAAGATAAGTCGCTGTCTCG
5101 sg_Non_Targeting_Human_0077 Non_Targeting_Human	Non_Targeting_Human	GTCGGCGTTCTGTTGTGACT
5111 sg_Non_Targeting_Human_0078 Non_Targeting_Human	Non_Targeting_Human	GAGGCAAGCCGTTAGGTGTA
5121 sg_Non_Targeting_Human_0079 Non_Targeting_Human	Non_Targeting_Human	GCGGATCCAGATCTCATTCG
5131 sg_Non_Targeting_Human_0080 Non_Targeting_Human	Non_Targeting_Human	GGAACATAGGAGCACGTAGT
5141 sg_Non_Targeting_Human_0081 Non_Targeting_Human	Non_Targeting_Human	GTCATCATTATGGCGTAAGG
5151 sg_Non_Targeting_Human_0082 Non_Targeting_Human	Non_Targeting_Human	GCGACTAGCGCCATGAGCGG
5161 sg_Non_Targeting_Human_0083 Non_Targeting_Human	Non_Targeting_Human	GGCGAAGTTCGACATGACAC
5171 sg_Non_Targeting_Human_0084 Non_Targeting_Human	Non_Targeting_Human	GCTGTCGTGTGGAGGCTATG
5181 sg_Non_Targeting_Human_0085 Non_Targeting_Human	Non_Targeting_Human	GCGGAGAGCATTGACCTCAT
5191 sg_Non_Targeting_Human_0086 Non_Targeting_Human	Non_Targeting_Human	GACTAATGGACCAAGTCAGT
5201 sg_Non_Targeting_Human_0087 Non_Targeting_Human	Non_Targeting_Human	GCGGATTAGAGGTAATGCGG

TABLE 4-continued

Control sgRNA Library		
SEQ ID NO. sgRNA Label	Gene	Nucleic Acid Sequence
5211 sg_Non_Targeting_Human_0088 Non_Targeting_Human	Non_Targeting_Human	GCCGACGGCAATCAGTACGC
5221 sg_Non_Targeting_Human_0089 Non_Targeting_Human	Non_Targeting_Human	GTAACCTCTCGAGCGATAGA
5231 sg_Non_Targeting_Human_0090 Non_Targeting_Human	Non_Targeting_Human	GACTTGTATGTGGCTTACGG
5241 sg_Non_Targeting_Human_0091 Non_Targeting_Human	Non_Targeting_Human	GTCACTGTGGTCGAACATGT
5251 sg_Non_Targeting_Human_0092 Non_Targeting_Human	Non_Targeting_Human	GTACTCCAATCCGCGATGAC
5261 sg_Non_Targeting_Human_0093 Non_Targeting_Human	Non_Targeting_Human	GCGTTGGCAGCATGTTACGG
5271 sg_Non_Targeting_Human_0094 Non_Targeting_Human	Non_Targeting_Human	GAACCAGCCGGCTAGTATGA
5281 sg_Non_Targeting_Human_0095 Non_Targeting_Human	Non_Targeting_Human	GTATACTAGCTAACCACACG
5291 sg_Non_Targeting_Human_0096 Non_Targeting_Human	Non_Targeting_Human	GAATCGGAATAGTTGATTTCG
5301 sg_Non_Targeting_Human_0097 Non_Targeting_Human	Non_Targeting_Human	GAGCACTTGCATGAGGCGGT
5311 sg_Non_Targeting_Human_0098 Non_Targeting_Human	Non_Targeting_Human	GAACGGCGATGAAGCCAGCC
5321 sg_Non_Targeting_Human_0099 Non_Targeting_Human	Non_Targeting_Human	GCAACCGAGATGAGAGGTTTC
5331 sg_Non_Targeting_Human_0100 Non_Targeting_Human	Non_Targeting_Human	GCAAGATCAATATGCGTGAT
5341 sg_Non_Targeting_Human_GA_0101 Non_Targeting_Human	Non_Targeting_Human	ACGGAGGCTAAGCGTCGCAA
5351 sg_Non_Targeting_Human_GA_0102 Non_Targeting_Human	Non_Targeting_Human	CGCTTCCGCGCCCGTTCAA
5361 sg_Non_Targeting_Human_GA_0103 Non_Targeting_Human	Non_Targeting_Human	ATCGTTTCCGCTTAACGGCG
5371 sg_Non_Targeting_Human_GA_0104 Non_Targeting_Human	Non_Targeting_Human	GTAGGCGCGCCGCTCTCTAC
5381 sg_Non_Targeting_Human_GA_0105 Non_Targeting_Human	Non_Targeting_Human	CCATATCGGGGCGAGACATG
5391 sg_Non_Targeting_Human_GA_0106 Non_Targeting_Human	Non_Targeting_Human	TACTAACGCCGCTCCTACAG
5401 sg_Non_Targeting_Human_GA_0107 Non_Targeting_Human	Non_Targeting_Human	TGAGGATCATGTCGAGCGCC
5411 sg_Non_Targeting_Human_GA_0108 Non_Targeting_Human	Non_Targeting_Human	GGGCCCCGATAGGATATCGC
5421 sg_Non_Targeting_Human_GA_0109 Non_Targeting_Human	Non_Targeting_Human	TAGACAACCGCGGAGAATGC
5431 sg_Non_Targeting_Human_GA_0110 Non_Targeting_Human	Non_Targeting_Human	ACGGGCGGCTATCGCTGACT
5441 sg_Non_Targeting_Human_GA_0111 Non_Targeting_Human	Non_Targeting_Human	CGCGGAAATTTTACCGACGA

TABLE 4-continued

Control sgRNA Library			
SEQ ID			Nucleic Acid Sequence
NO. sgRNA Label	Gene		
5451 sg_Non_Targeting_Human_GA_0112 Non_Targeting_Human	Non_Targeting_Human		CTTACAATCGTCGGTCCAAT
5461 sg_Non_Targeting_Human_GA_0113 Non_Targeting_Human	Non_Targeting_Human		GCGTGCCTCCCGGTTACCC
5471 sg_Non_Targeting_Human_GA_0114 Non_Targeting_Human	Non_Targeting_Human		CGGAGTAACAAGCGACGGA
5481 sg_Non_Targeting_Human_GA_0115 Non_Targeting_Human	Non_Targeting_Human		CGAGTGTATACGCACCGTT
5491 sg_Non_Targeting_Human_GA_0116 Non_Targeting_Human	Non_Targeting_Human		CGACTAACCGAACTTTTT
5501 sg_Non_Targeting_Human_GA_0117 Non_Targeting_Human	Non_Targeting_Human		CAACGGTTCTCCCGCTAC
5511 sg_Non_Targeting_Human_GA_0118 Non_Targeting_Human	Non_Targeting_Human		CAGGAGTCGCCGATACGCGT
5521 sg_Non_Targeting_Human_GA_0119 Non_Targeting_Human	Non_Targeting_Human		TTCACGTCGTCTCGCGACCA
5531 sg_Non_Targeting_Human_GA_0120 Non_Targeting_Human	Non_Targeting_Human		GTGTCGGATTCCCGCGTTA
5541 sg_Non_Targeting_Human_GA_0121 Non_Targeting_Human	Non_Targeting_Human		CACGAACCTCACCCGCGCA
5551 sg_Non_Targeting_Human_GA_0122 Non_Targeting_Human	Non_Targeting_Human		CGCTAGTACGCTCCTCTATA
5561 sg_Non_Targeting_Human_GA_0123 Non_Targeting_Human	Non_Targeting_Human		TCGCGCTTGGGTATACGCT
5571 sg_Non_Targeting_Human_GA_0124 Non_Targeting_Human	Non_Targeting_Human		CTATCTCGAGTGGTAATGCG
5581 sg_Non_Targeting_Human_GA_0125 Non_Targeting_Human	Non_Targeting_Human		AATCGACTCGAACTTCGTGT
5591 sg_Non_Targeting_Human_GA_0126 Non_Targeting_Human	Non_Targeting_Human		CCCGATGGACTATACCGAAC
5601 sg_Non_Targeting_Human_GA_0127 Non_Targeting_Human	Non_Targeting_Human		ACGTTCGAGTACGACCAGCT
5611 sg_Non_Targeting_Human_GA_0128 Non_Targeting_Human	Non_Targeting_Human		CGCGACGACTCAACCTAGTC
5621 sg_Non_Targeting_Human_GA_0129 Non_Targeting_Human	Non_Targeting_Human		GGTCACCGATCGAGAGCTAG
5631 sg_Non_Targeting_Human_GA_0130 Non_Targeting_Human	Non_Targeting_Human		CTCAACCGACCGTATGGTCA
5641 sg_Non_Targeting_Human_GA_0131 Non_Targeting_Human	Non_Targeting_Human		CGTATTCGACTCTCAACGCG
5651 sg_Non_Targeting_Human_GA_0132 Non_Targeting_Human	Non_Targeting_Human		CTAGCCGCCGAGATCGAGCC
5661 sg_Non_Targeting_Human_GA_0133 Non_Targeting_Human	Non_Targeting_Human		GAATCGACCGACACTAATGT
5671 sg_Non_Targeting_Human_GA_0134 Non_Targeting_Human	Non_Targeting_Human		ACTTCAGTTCGGCGTAGTCA
5681 sg_Non_Targeting_Human_GA_0135 Non_Targeting_Human	Non_Targeting_Human		GTGCGATGTCGCTTCAACGT

TABLE 4-continued

Control sgRNA Library		
SEQ ID NO. gRNA Label	Gene	Nucleic Acid Sequence
5691 sg_Non_Targeting_Human_GA_0136 Non_Targeting_Human	Non_Targeting_Human	CGCCTAATTCCGGATCAAT
5701 sg_Non_Targeting_Human_GA_0137 Non_Targeting_Human	Non_Targeting_Human	CGTGGCCGGAACCGTCATAG
5711 sg_Non_Targeting_Human_GA_0138 Non_Targeting_Human	Non_Targeting_Human	ACCCTCCGAATCGTACGGA
5721 sg_Non_Targeting_Human_GA_0139 Non_Targeting_Human	Non_Targeting_Human	AAACGGTACGACAGCGTGTG
5731 sg_Non_Targeting_Human_GA_0140 Non_Targeting_Human	Non_Targeting_Human	ACATAGTCGACGGCTCGATT
5741 sg_Non_Targeting_Human_GA_0141 Non_Targeting_Human	Non_Targeting_Human	GATGGCGCTTCAGTCGTCGG
5751 sg_Non_Targeting_Human_GA_0142 Non_Targeting_Human	Non_Targeting_Human	ATAATCCGGAACGCTCGAC
5761 sg_Non_Targeting_Human_GA_0143 Non_Targeting_Human	Non_Targeting_Human	CGCCGGGCTGACAATTAACG
5771 sg_Non_Targeting_Human_GA_0144 Non_Targeting_Human	Non_Targeting_Human	CGTCGCCATATGCCGGTGGC
5781 sg_Non_Targeting_Human_GA_0145 Non_Targeting_Human	Non_Targeting_Human	CGGGCCTATAACACCATCGA
5791 sg_Non_Targeting_Human_GA_0146 Non_Targeting_Human	Non_Targeting_Human	CGCCGTTCGAGATACTTGA
5801 sg_Non_Targeting_Human_GA_0147 Non_Targeting_Human	Non_Targeting_Human	CGGGACGTCGCGAAAATGTA
5811 sg_Non_Targeting_Human_GA_0148 Non_Targeting_Human	Non_Targeting_Human	TCGGCATACGGGACACACGC
5821 sg_Non_Targeting_Human_GA_0149 Non_Targeting_Human	Non_Targeting_Human	AGCTCCATCGCCGCGATAAT
5831 sg_Non_Targeting_Human_GA_0150 Non_Targeting_Human	Non_Targeting_Human	ATCGTATCATCAGCTAGCGC
5841 sg_Non_Targeting_Human_GA_0151 Non_Targeting_Human	Non_Targeting_Human	TCGATCGAGGTGCATTCGG
5851 sg_Non_Targeting_Human_GA_0152 Non_Targeting_Human	Non_Targeting_Human	CTCGACAGTTCGTCCCAGC
5861 sg_Non_Targeting_Human_GA_0153 Non_Targeting_Human	Non_Targeting_Human	CGGTAGTATTAATCGCTGAC
5871 sg_Non_Targeting_Human_GA_0154 Non_Targeting_Human	Non_Targeting_Human	TGAACGCGTGTTCCTTGCA
5881 sg_Non_Targeting_Human_GA_0155 Non_Targeting_Human	Non_Targeting_Human	CGACGCTAGGTAACGTAGAG
5891 sg_Non_Targeting_Human_GA_0156 Non_Targeting_Human	Non_Targeting_Human	CATTGTTGAGCGGCGCGCT
5901 sg_Non_Targeting_Human_GA_0157 Non_Targeting_Human	Non_Targeting_Human	CCGCTATTGAAACGCCAC
5911 sg_Non_Targeting_Human_GA_0158 Non_Targeting_Human	Non_Targeting_Human	AGACACGTCACCGTCAAAA
5921 sg_Non_Targeting_Human_GA_0159 Non_Targeting_Human	Non_Targeting_Human	TTTACGATCTAGCGCGTAG

TABLE 4-continued

Control sgRNA Library		
SEQ ID NO. sgRNA Label	Gene	Nucleic Acid Sequence
5931 sg_Non_Targeting_Human_GA_0160 Non_Targeting_Human	Non_Targeting_Human	TTCGCACGATTGCACCTTGG
5941 sg_Non_Targeting_Human_GA_0161 Non_Targeting_Human	Non_Targeting_Human	GGTTAGAGACTAGGCGCGCG
5951 sg_Non_Targeting_Human_GA_0162 Non_Targeting_Human	Non_Targeting_Human	CCTCCGTGCTAACGCGGACG
5961 sg_Non_Targeting_Human_GA_0163 Non_Targeting_Human	Non_Targeting_Human	TTATCGCGTAGTGCTGACGT
5971 sg_Non_Targeting_Human_GA_0164 Non_Targeting_Human	Non_Targeting_Human	TACGCTTGCCTTTAGCGTCC
5981 sg_Non_Targeting_Human_GA_0165 Non_Targeting_Human	Non_Targeting_Human	CGCGGCCCCACGCGTCATCGC
5991 sg_Non_Targeting_Human_GA_0166 Non_Targeting_Human	Non_Targeting_Human	AGCTCGCCATGTCGGTTCTC
6001 sg_Non_Targeting_Human_GA_0167 Non_Targeting_Human	Non_Targeting_Human	AACTAGCCCCGAGCAGCTTCG
6011 sg_Non_Targeting_Human_GA_0168 Non_Targeting_Human	Non_Targeting_Human	CGCAAGGTGTCGGTAACCCCT
6021 sg_Non_Targeting_Human_GA_0169 Non_Targeting_Human	Non_Targeting_Human	CTTCGACGCCATCGTGCTCA
6031 sg_Non_Targeting_Human_GA_0170 Non_Targeting_Human	Non_Targeting_Human	TCCTGGATACCGGTGGTTA
6041 sg_Non_Targeting_Human_GA_0171 Non_Targeting_Human	Non_Targeting_Human	ATAGCCCGCGCTCATTACTT
6051 sg_Non_Targeting_Human_GA_0172 Non_Targeting_Human	Non_Targeting_Human	GTCGTCCGGATTACAAAAT
6061 sg_Non_Targeting_Human_GA_0173 Non_Targeting_Human	Non_Targeting_Human	TAATGCTGCACACGCCGAAT
6071 sg_Non_Targeting_Human_GA_0174 Non_Targeting_Human	Non_Targeting_Human	TATCGCTTCCGATTAGTCCG
6081 sg_Non_Targeting_Human_GA_0175 Non_Targeting_Human	Non_Targeting_Human	GTACCATACCGCGTACCCTT
6091 sg_Non_Targeting_Human_GA_0176 Non_Targeting_Human	Non_Targeting_Human	TAAGATCCGCGGTGGCAAC
6101 sg_Non_Targeting_Human_GA_0177 Non_Targeting_Human	Non_Targeting_Human	GTAGACGTCGTGAGCTTCAC
6111 sg_Non_Targeting_Human_GA_0178 Non_Targeting_Human	Non_Targeting_Human	TCGCGGACATAGGGCTCTAA
6121 sg_Non_Targeting_Human_GA_0179 Non_Targeting_Human	Non_Targeting_Human	AGCGCAGATAGCGCGTATCA
6131 sg_Non_Targeting_Human_GA_0180 Non_Targeting_Human	Non_Targeting_Human	GTTTCGCTTCGTAACGAGGAA
6141 sg_Non_Targeting_Human_GA_0181 Non_Targeting_Human	Non_Targeting_Human	GACCCCCGATAACTTTTGAC
6151 sg_Non_Targeting_Human_GA_0182 Non_Targeting_Human	Non_Targeting_Human	ACGTCCATACTGTCGGCTAC
6161 sg_Non_Targeting_Human_GA_0183 Non_Targeting_Human	Non_Targeting_Human	GTACCATTGCCGGCTCCCTA

TABLE 4-continued

Control sgRNA Library		
SEQ ID NO. sgRNA Label	Gene	Nucleic Acid Sequence
6171 sg_Non_Targeting_Human_GA_0184 Non_Targeting_Human Non_Targeting_Human		TGTTCCGTAGGTCGGTATA
6181 sg_Non_Targeting_Human_GA_0185 Non_Targeting_Human Non_Targeting_Human		TCTGGCTTGACACGACCGTT
6191 sg_Non_Targeting_Human_GA_0186 Non_Targeting_Human Non_Targeting_Human		CGTAGGTCCGGTAAGTGCG
6201 sg_Non_Targeting_Human_GA_0187 Non_Targeting_Human Non_Targeting_Human		AGCACGTAATGTCCGTGGAT
6211 sg_Non_Targeting_Human_GA_0188 Non_Targeting_Human Non_Targeting_Human		AAGGCGCGCAATGTGGCAG
6221 sg_Non_Targeting_Human_GA_0189 Non_Targeting_Human Non_Targeting_Human		ACTGCGGAGCGCCCAATATC
6231 sg_Non_Targeting_Human_GA_0190 Non_Targeting_Human Non_Targeting_Human		CGTCGAGTGCTCGAACTCCA
6241 sg_Non_Targeting_Human_GA_0191 Non_Targeting_Human Non_Targeting_Human		TCGCAGCGCGTGGGATCGG
6251 sg_Non_Targeting_Human_GA_0192 Non_Targeting_Human Non_Targeting_Human		ATCTGTCTAATTCGGATCG
6261 sg_Non_Targeting_Human_GA_0193 Non_Targeting_Human Non_Targeting_Human		TGCGGCGTAATGCTTGAAG
6271 sg_Non_Targeting_Human_GA_0194 Non_Targeting_Human Non_Targeting_Human		CGAACTTAATCCCGTGGCAA
6281 sg_Non_Targeting_Human_GA_0195 Non_Targeting_Human Non_Targeting_Human		GCCGTGTTGCTGGATACGCC
6291 sg_Non_Targeting_Human_GA_0196 Non_Targeting_Human Non_Targeting_Human		TACCTCCGGATACGGACTG
6301 sg_Non_Targeting_Human_GA_0197 Non_Targeting_Human Non_Targeting_Human		CCGTTGGACTATGGCGGGTC
6311 sg_Non_Targeting_Human_GA_0198 Non_Targeting_Human Non_Targeting_Human		GTACGGGGCGATCATCCACA
6321 sg_Non_Targeting_Human_GA_0199 Non_Targeting_Human Non_Targeting_Human		AAGAGTAGTAGACGCCCGGG
6331 sg_Non_Targeting_Human_GA_0200 Non_Targeting_Human Non_Targeting_Human		AAGAGCGAATCGATTTCTGTG
6343 sg_hCDC16_CC_1 CDC16	CDC16	TCAACACCAGTGCCTGACGG
6353 sg_hCDC16_CC_2 CDC16	CDC16	AAAGTAGCTTCACTCTCTCG
6363 sg_hCDC16_CC_3 CDC16	CDC16	GAGCCAACCAATAGATGTCC
6373 sg_hCDC16_CC_4 CDC16	CDC16	GCGCCGCCATGAACCTAGAG
6383 sg_hGTF2B_CC_1 GTF2B	GTF2B	ACAAAGGTTGGAACAGAACC
6393 sg_hGTF2B_CC_2 GTF2B	GTF2B	GGTGACCGGGTTATTGATGT
6403 sg_hGTF2B_CC_3 GTF2B	GTF2B	TTAGTGGAGGACTACAGAGC
6413 sg_hGTF2B_CC_4 GTF2B	GTF2B	ACATATAGCCCGTAAAGCTG
6423 sg_hHSPA5_CC_1 HSPA5	HSPA5	CGTTGGCGATGATCTCCACG
6433 sg_hHSPA5_CC_2 HSPA5	HSPA5	TGGCCTTTTCTACCTCGCGC
6443 sg_hHSPA5_CC_3 HSPA5	HSPA5	AATGGAGATACTCATCTGGG

TABLE 4-continued

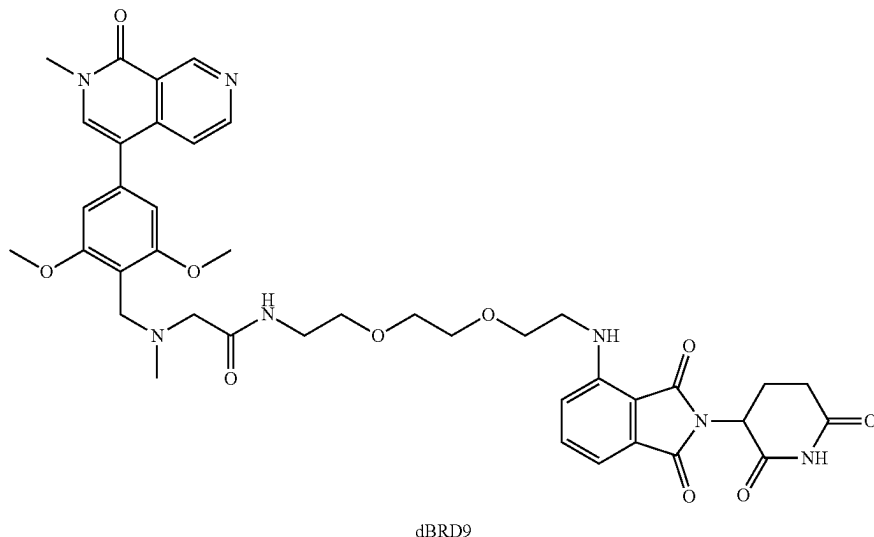
Control sgRNA Library			
SEQ ID NO. gRNA Label	Gene	Nucleic Acid Sequence	
6453 sg_hHSPA5_CC_4 HSPA5	HSPA5	GAAGCCCGTCCAGAAAGTGT	
6463 sg_hHSPA9_CC_1 HSPA9	HSPA9	CAATCTGAGGAAGTCCACGA	
6473 sg_hHSPA9_CC_2 HSPA9	HSPA9	AGGCTGCGGCGCCACGAGA	
6483 sg_hHSPA9_CC_3 HSPA9	HSPA9	ACTTTGACCAGGCCTTGCTA	
6493 sg_hHSPA9_CC_4 HSPA9	HSPA9	ACCTTCCATAACTGCCACGC	
6503 sg_hPAFAH1B1_CC_1 PAFAH1B1	PAFAH1B1	CGAGGCGTACATACCCAAGG	
6513 sg_hPAFAH1B1_CC_2 PAFAH1B1	PAFAH1B1	ATGGTACGGCCAAATCAAGA	
6523 sg_hPAFAH1B1_CC_3 PAFAH1B1	PAFAH1B1	TCTTGTAATCCCATACGCGT	
6533 sg_hPAFAH1B1_CC_4 PAFAH1B1	PAFAH1B1	ATTACAGGACACAGAGAAT	
6543 sg_hPCNA_CC_1 PCNA	PCNA	CCAGGGCTCCATCCTCAAGA	
6553 sg_hPCNA_CC_2 PCNA	PCNA	TGAGCTGCACCAAGAGACG	
6563 sg_hPCNA_CC_3 PCNA	PCNA	ATGTCTGCAGATGTACCCCT	
6573 sg_hPCNA_CC_4 PCNA	PCNA	CGAAGATAACGCGGATACCT	
6583 sg_hPOLR2L_CC_1 POLR2L	POLR2L	GCTGCAGGCCGAGTACACCG	
6593 sg_hPOLR2L_CC_2 POLR2L	POLR2L	ACAAGTGGGAGGCTTACCTG	
6603 sg_hPOLR2L_CC_3 POLR2L	POLR2L	GCAGCGTACAGGGATGATCA	
6613 sg_hPOLR2L_CC_4 POLR2L	POLR2L	GCAGTAGCGCTTCAGGCCCA	
6623 sg_hRPL9_CC_1 RPL9	RPL9	CAAATGGTGGGGTAACAGAA	
6633 sg_hRPL9_CC_2 RPL9	RPL9	GAAAGGAAGTGGCTACCGTT	
6643 sg_hRPL9_CC_3 RPL9	RPL9	AGGGCTTCCGTTACAAGATG	
6653 sg_hRPL9_CC_4 RPL9	RPL9	GAACAAGCAACACCTAAAAG	
6663 sg_hSF3A3_CC_1 SF3A3	SF3A3	TGAGGAGAAGGAACGGCTCA	
6673 sg_hSF3A3_CC_2 SF3A3	SF3A3	GGAAGAATGCAGAGTATAAG	
6683 sg_hSF3A3_CC_3 SF3A3	SF3A3	GGAATTTGAGGAAGTCTCTGA	
6693 sg_hSF3A3_CC_4 SF3A3	SF3A3	GCTCACCGGCCATCCAGGAA	
6703 sg_hSF3B3_CC_1 SF3B3	SF3B3	ACTGGCCAGGAACGATGCGA	
6713 sg_hSF3B3_CC_2 SF3B3	SF3B3	GCAGCTCCAAGATCTTCCCA	
6723 sg_hSF3B3_CC_3 SF3B3	SF3B3	GAATGAGTACACAGAACGGA	
6733 sg_hSF3B3_CC_4 SF3B3	SF3B3	GGAGCAGGACAAGGTCGGGG	

471

Example 2—BRD9 Degradator Depletes BRD9 Protein

The following example demonstrates the depletion of the BRD9 protein in synovial sarcoma cells treated with a BRD9 degrader.

Procedure: Cells were treated with DMSO or the BRD9 degrader, Compound 1 (also known as dBRD9, see Remillard et al, *Angew. Chem. Int. Ed. Engl.* 56(21):5738-5743 (2017); see structure of compound 1 below), for indicated doses and timepoints.



472

Procedures:

Cells were treated with DMSO or the BRD9 degrader, Compound 1, at indicated concentrations, and proliferation was monitored from day 7 to day 14 by measuring confluency over time using an IncuCyte live cell analysis system (FIG. 5). Growth medium and compounds were refreshed every 3-4 days.

Cells were seeded into 12-well plates and treated with DMSO, 1 μ M BRD9 inhibitor, Compound 2 (also known as BI-7273, see Martin et al, *J Med Chem.* 59(10):4462-4475 (2016); see structure of compound 2 below), or 1 μ M BRD9 degrader, Compound 1.

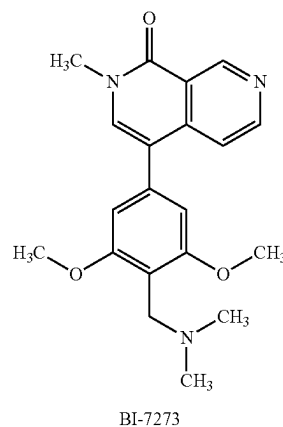
(Compound 1)

Whole cell extracts were fractionated by SDS-PAGE and transferred to a polyvinylidene difluoride membrane using a transfer apparatus according to the manufacturer's protocols (Bio-Rad). After incubation with 5% nonfat milk in TBST (10 mM Tris, pH 8.0, 150 mM NaCl, 0.5% Tween 20) for 60 min, the membrane was incubated with antibodies against BRD9 (1:1,000, Bethyl laboratory A303-781A), GAPDH (1:5,000, Cell Signaling Technology), and/or MBP (1:1,000, BioRad) overnight at 4° C. Membranes were washed three times for 10 min and incubated with anti-mouse or anti-rabbit antibodies conjugated with either horseradish peroxidase (HRP, FIGS. 2-3) or IRDye (FIG. 4, 1:20,000, LI-COR) for at least 1 h. Blots were washed with TBST three times and developed with either the ECL system according to the manufacturer's protocols (FIGS. 2-3) or scanned on an Odyssey CLx Imaging system (FIG. 4).

Results: Treatment of SYO1 synovial sarcoma cells with the BRD9 degrader Compound 1 results in dose dependent (FIG. 2) and time dependent (FIG. 3) depletion of BRD9 in the cells. Further, as shown in FIG. 4, the depletion of BRD9 by Compound 1 is replicated in a non-synovial sarcoma cell line (293T) and may be sustained for at least 5 days.

Example 3—Inhibition of Growth of Synovial Cell Lines by BRD9 Inhibitors and BRD9 Degradators

The following example demonstrates that BRD9 degraders and inhibitors selectively inhibit growth of synovial sarcoma cells.



(Compound 2)

The number of cells was optimized for each cell line. Growth medium and compounds were refreshed every 3-5 days. SYO1, Yamato, A549, 293T and HS-SY-II cells were fixed and stained at day 11. ASKA cells were fixed and stained at day 23. Staining was done by incubation with crystal violet solution (0.5 g Crystal Violet, 27 ml 37% Formaldehyde, 100 mL 10xPBS, 10 mL Methanol, 863 dH2O to 1 L) for 30 min followed by 3x washes with water and drying the plates for at least 24 h at room temperature. Subsequently plates were scanned on an Odyssey CLx Imaging system (FIG. 6).

473

Cells were seeded into 96-well ultra low cluster plate (Costar, #7007) in 200 μ L complete media and treated at day 2 with DMSO, Staurosporin, or BRD9 degrader, Compound 1, at indicated doses (FIG. 3C). Media and compounds were changed every 5 d and cell colonies were imaged at day 14.

Results: As shown in FIGS. 5, 6, and 7, treatment of synovial sarcoma cell lines (SYO1, Yamato, HS-SY-II, and ASKA) with a BRD9 inhibitor, Compound 2, or a BRD9 degrader, Compound 1, results in inhibition of the growth of the cells, but does not result in inhibition of the growth of non-synovial control cancer cell lines (293T, A549, G401).

Example 4—Selective Inhibition of Growth of Synovial Cell Lines by BRD9 Degraders and BRD9 Binders

The following example demonstrates that BRD9 degraders and binders selectively inhibit growth of synovial sarcoma cells.

Procedure: Cells were seeded into 6-well or 12-well plates and were treated daily with a BRD9 degrader (Compound 1), a bromo-domain BRD9 binder (Compound 2), E3 ligase binder (lenalidomide), DMSO, or staurosporin (positive control for cell killing), at indicated concentrations. The number of cells was optimized for each cell line. Growth media was refreshed every 5 days. By day 14, medium was removed, cells were washed with PBS, and stained using 500 μ L of 0.005% (w/v) crystal violet solution in 25% (v/v) methanol for at least 1 hour at room temperature. Subsequently plates were scanned on an Odyssey CLx Imaging system.

Results: As shown in FIGS. 8 and 9, treatment of synovial sarcoma cell lines (SYO1, HS-SY-II, and ASKA) with Compound 1 or Compound 2 resulted in inhibition of the growth of the cells, but did not result in inhibition of the growth of non-synovial control cancer cell lines (RD, HCT116, and Calu6). Overall, Compound 1 showed most significant growth inhibition in all synovial cell lines.

Example 5—Inhibition of Cell Growth in Synovial Sarcoma Cells

The following example shows that BRD9 degraders inhibit cell growth and induce apoptosis in synovial sarcoma cells.

Procedure: SYO1 cells were treated for 8 or 13 days with DMSO, a BRD9 degrader (Compound 1) at 200 nM or 1 μ M, or an E3 ligase binder (lenalidomide) at 200 nM. Compounds were refreshed every 5 days. Cell cycle analysis was performed using the Click-iT™ Plus EdU Flow Cytometry Assay (Invitrogen). The apoptosis assay was performed using the Annexin V-FITC Apoptosis Detection Kit (Sigma A9210). Assays were performed according to the manufacturer's protocol.

Results: As shown in FIGS. 10-13, treatment with Compound 1 for 8 or 13 days resulted in reduced numbers of cells in the S-phase of the cell cycle as compared to DMSO and lenalidomide. Treatment with Compound 1 for 8 days also resulted in increased numbers of early- and late-apoptotic cells as compared to DMSO controls.

Example 6—Composition for SS18-SSX1-BAF

The following example shows the identification of BRD9 as a component of SS18-SSX containing BAF complexes.

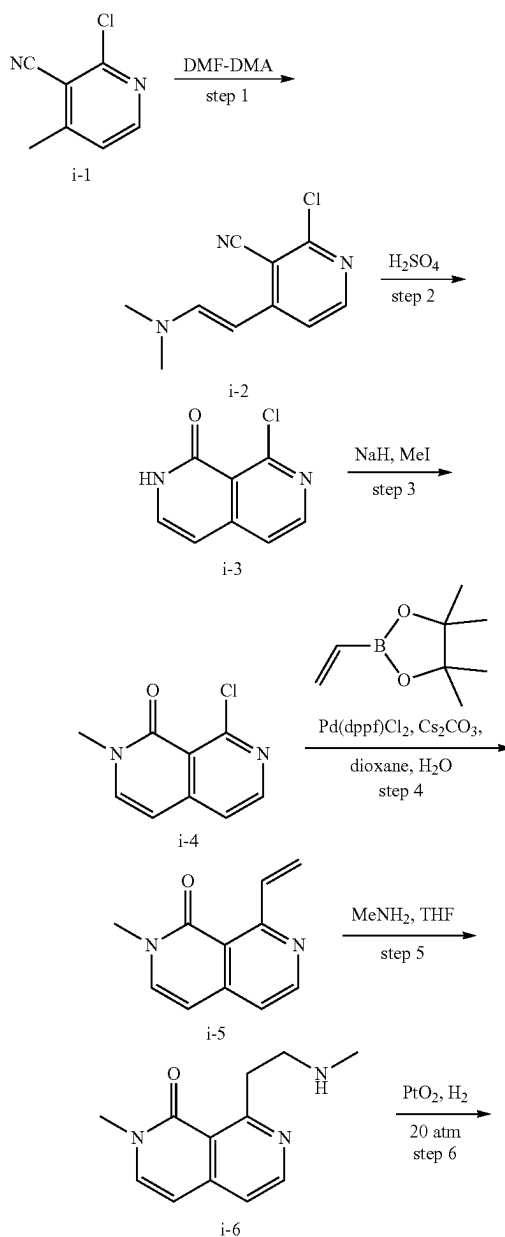
Procedure: A stable 293T cell line expressing HA-SS18SSX1 was generated using lentiviral integration.

474

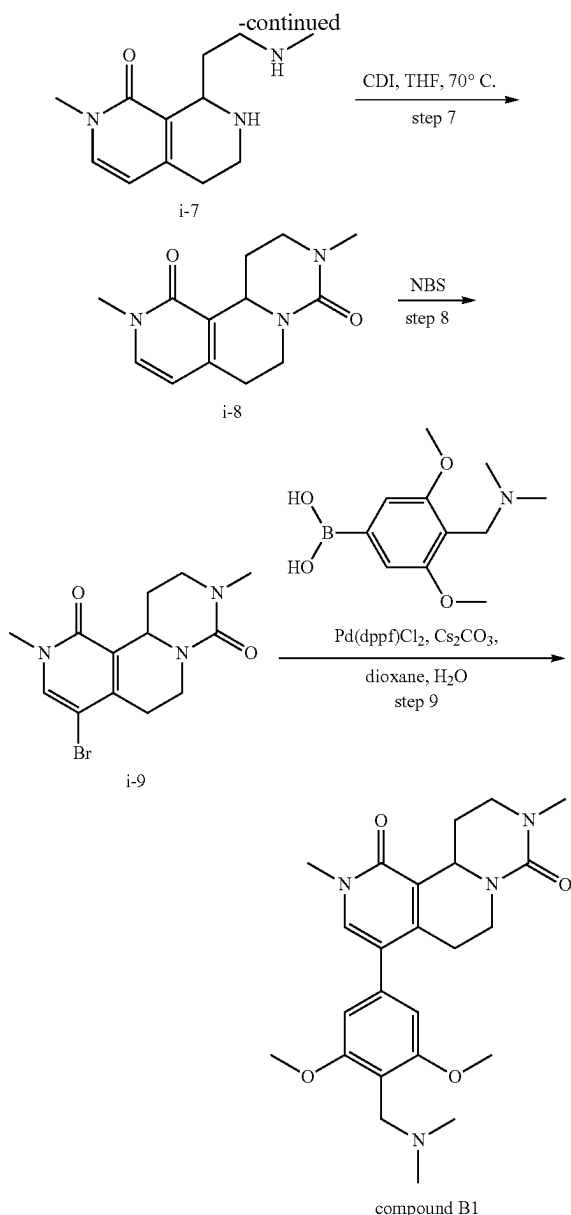
SS18-SSX1 containing BAF complexes were subject to affinity purification and subsequent mass spectrometry analysis revealed SS18-SSX1 interacting proteins.

Results: As shown in FIG. 14, BAF complexes including the SS18-SSX fusion protein also included BRD9. More than 5 unique peptides were identified for ARID1A (95 peptides), ARID1B (77 peptides), SMARCC1 (69 peptides), SMARCD1 (41 peptides), SMARCD2 (37 peptides), DPF2 (32 peptides), SMARCD3 (26 peptides), ACTL6A (25 peptides), BRD9 (22 peptides), DPF1 Isoform 2 (18 peptides), DPF3 (13 peptides), and ACTL6B (6 peptides).

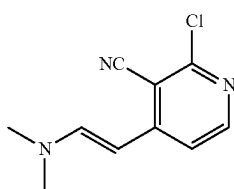
Example 7—Preparation of 4-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-2,9-dimethyl-5H,6H,10H,11H, 11aH-pyrimido[4,3-a] 2,7-naphthyridine-1,8-dione (Compound B1)



475



Step 1: Preparation of 6-bromo-3-methyl-[1,2,4]triazolo[4,3-a]pyridin-8-amine (i-2)

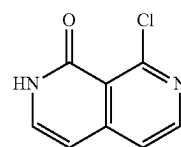


A solution of 2-chloro-4-methylpyridine-3-carbonitrile (3.00 g, 19.662 mmol, 1.00 equiv) in DMF-DMA (120.00 mL, 896.257 mmol, 45.58 equiv) was stirred for 6 hours at 110° C. The resulting mixture was concentrated. The residue was applied onto a silica gel column with ethyl acetate/

476

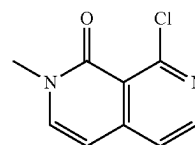
petroleum ether (1:1). This resulted in 3.4 g (83.27%) of 2-chloro-4-[(E)-2-(dimethylamino)ethenyl]pyridine-3-carbonitrile as a light yellow solid. LCMS (ESI) m/z: [M+H]⁺=208.

Step 2: Preparation of 8-chloro-1,2-dihydro-2,7-naphthyridin-1-one (i-3)



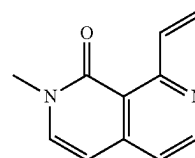
Into a 250-mL round-bottom flask, was placed 2-chloro-4-[(E)-2-(dimethylamino)ethenyl]pyridine-3-carbonitrile (3.40 g, 16.373 mmol, 1.00 equiv), H₂SO₄ (100.00 mL). The resulting solution was stirred for 2 hours at 110° C. The reaction was then quenched by the addition of 100 mL of water/ice. The pH of the solution was adjusted to pH>7 by addition of aqueous (aq.) K₂CO₃. The resulting solution was extracted with ethyl acetate (3×300 mL) and the organic layers combined and concentrated. This resulted in 2.7 g (91.32%) of 8-chloro-1,2-dihydro-2,7-naphthyridin-1-one as a light yellow solid. LCMS (ESI) m/z: [M+H]⁺=181.

Step 3: Preparation of 8-chloro-1,2-dihydro-2,7-naphthyridin-1-one (i-4)



To a solution of 8-chloro-1,2-dihydro-2,7-naphthyridin-1-one (2.20 g, 12.182 mmol, 1.00 equiv) in THE (120.00 mL) was added NaH (321.00 mg, 13.376 mmol, 1.10 equiv) and MeI (3.45 g, 24.306 mmol, 2.00 equiv). The resulting solution was stirred for 2 hours at room temperature. The reaction was then quenched by the addition of 100 mL of water and extracted with ethyl acetate (3×250 mL), and the organic layers were combined and concentrated. This resulted in 2 g (84.36%) of 8-chloro-2-methyl-1,2-dihydro-2,7-naphthyridin-1-one as a light yellow solid. LCMS (ESI) m/z: [M+H]⁺=195.

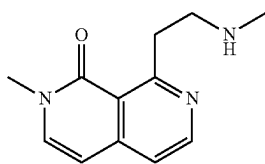
Step 4: Preparation of 8-ethenyl-2-methyl-1,2-dihydro-2,7-naphthyridin-1-one (i-5)



477

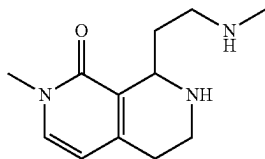
To a solution of 8-chloro-2-methyl-1,2-dihydro-2,7-naphthyridin-1-one (2.0 g, 10.276 mmol, 1.00 equiv) in 1,4-dioxane (72.00 mL) and H₂O (24.00 mL), was added Cs₂CO₃ (10.0 g, 30.727 mmol, 2.99 equiv) and 2-ethenyl-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (3.2 g, 20.758 mmol, 2.02 equiv), Pd(dppf)Cl₂·CH₂Cl₂ (1.68 g, 2.055 mmol, 0.20 equiv) at 25° C. The resulting solution was stirred for 2 hours at 80° C. The resulting mixture was concentrated. The residue was applied onto a silica gel column with ethyl acetate/petroleum ether (1:1). This resulted in 980 mg (51.21%) of 8-ethenyl-2-methyl-1,2-dihydro-2,7-naphthyridin-1-one as a light yellow solid. LCMS (ESI) m/z: [M+H]⁺=187.

Step 5: Preparation of 2-methyl-8-[2-(methylamino)ethyl]-1,2-dihydro-2,7-naphthyridin-1-one (i-6)



To a solution of 8-ethenyl-2-methyl-1,2-dihydro-2,7-naphthyridin-1-one (580 mg, 2.67 mmol, 1.00 equiv) in THE (2.00 mL) was added 4 M MeNH₂ THF solution (20 mL) dropwise at 25° C. The resulting solution was stirred for 5 hours at 90° C. The reaction was cooled and concentrated, the crude product was purified by Flash-Prep-HPLC (Conditions (IntelFlash-1): Column, C18 silica gel; mobile phase, acetonitrile (MeCN or ACN) in water, 10% to 50% gradient in 10 minutes; detector, UV 254 nm). This resulted in 480 mg (71.0%) of 2-methyl-8-[2-(methylamino)ethyl]-1,2-dihydro-2,7-naphthyridin-1-one as light yellow oil. LCMS (ESI) m/z: [M+H]⁺=218.

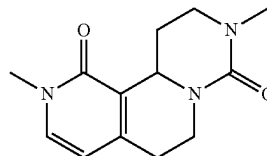
Step 6: Preparation of 2-methyl-8-[2-(methylamino)ethyl]-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one (i-7)



To a solution of 2-methyl-8-[2-(methylamino)ethyl]-1,2-dihydro-2,7-naphthyridin-1-one (480.00 mg, 2.201 mmol, 1.00 equiv) in MeOH (10.00 mL) was added PtO₂ (386.71 mg, 1.703 mmol, 0.74 equiv) under high pressure of H₂ (22.0 atm) atmosphere at 25° C. The resulting solution was stirred for 12 hours at room temperature. The solids were filtered out. The filtrate was concentrated. This resulted in 350 mg crude of 2-methyl-8-[2-(methylamino)ethyl]-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one as light yellow oil, that was used directly without further purification. LCMS (ESI) m/z: [M+H]⁺=222.

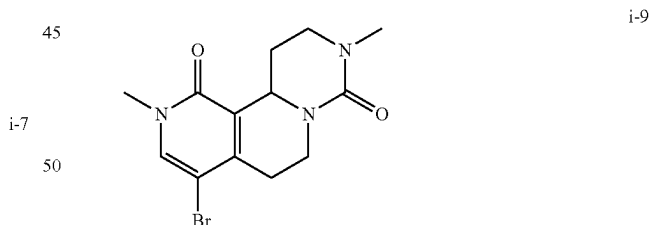
478

Step 7: Preparation of 2,9-dimethyl-1H,2H,5H,6H,8H,9H,10H,11H,11aH-pyrimido[4,3-a]2,7-naphthyridine-1,8-dione (i-8)



To a solution of 2-methyl-8-[2-(methylamino)ethyl]-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one (350.00 mg, 1.576 mmol, 1.00 equiv) in THE (3.00 mL) was added CDI (1,1'-carbonyldiimidazole) (511.0 mg, 3.153 mmol, 2.00 equiv) at room temperature. The resulting solution was stirred for 6 hours at 70° C. The mixture was concentrated. This resulted in 340 mg crude of 2,9-dimethyl-1H,2H,5H,6H,8H,9H,10H,11H,11aH-pyrimido[4,3-a]2,7-naphthyridine-1,8-dione as light yellow oil, that was used directly without further purification. LCMS (ESI) m/z: [M+H]⁺=248.

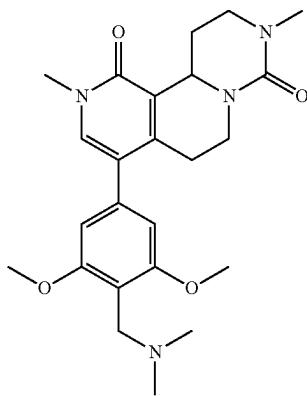
Step 8: Preparation of 4-bromo-2,9-dimethyl-1H,2H,5H,6H,8H,9H,10H,11H,11aH-pyrimido[4,3-a]2,7-naphthyridine-1,8-dione (i-9)



To a solution of 2,9-dimethyl-1H,2H,5H,6H,8H,9H,10H,11H,11aH-pyrimido[4,3-a]2,7-naphthyridine-1,8-dione (340.00 mg, 1.371 mmol, 1.00 equiv) in HOAc (20.00 mL) was added NBS (732.1 mg, 4.113 mmol, 2.99 equiv) at room temperature. The resulting solution was stirred for 12 hours at room temperature. The mixture was concentrated. The crude product was purified by Flash-Prep-HPLC (conditions (IntelFlash-1): Column, C18 silica gel; mobile phase, MeCN/H₂O=0 increasing to MeCN/H₂O=100 within 30 minutes; Detector, 254 nm. This resulted in 385 mg of 4-bromo-2,9-dimethyl-1H,2H,5H,6H,8H,9H,10H,11H,11aH-pyrimido[4,3-a]2,7-naphthyridine-1,8-dione as a light yellow solid. LCMS (ESI) m/z: [M+H]⁺=326.

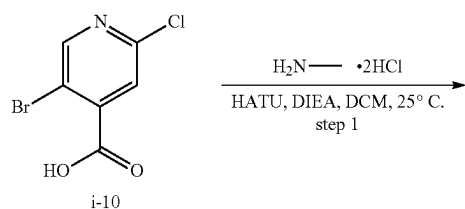
479

Step 9: Preparation of 4-((dimethylamino)methyl)-3,5-dimethoxyphenyl)-2,9-dimethyl-5,6,9,10,11,11a-hexahydro-1H-pyrimido[6,1-a][2,7]naphthyridine-1,8 (2H)-dione (Compound B1)



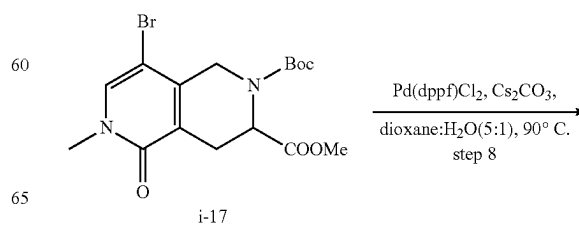
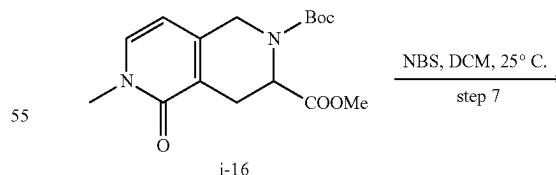
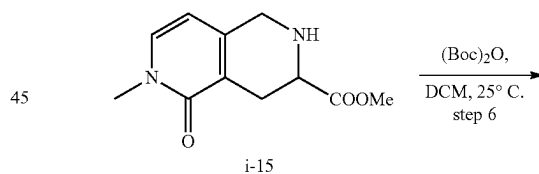
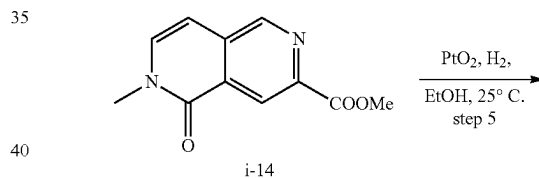
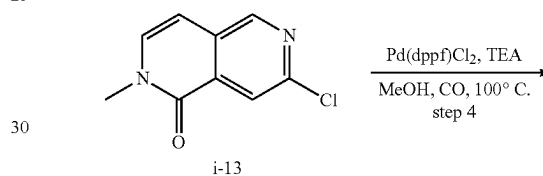
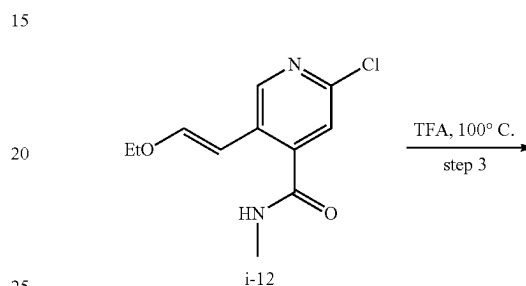
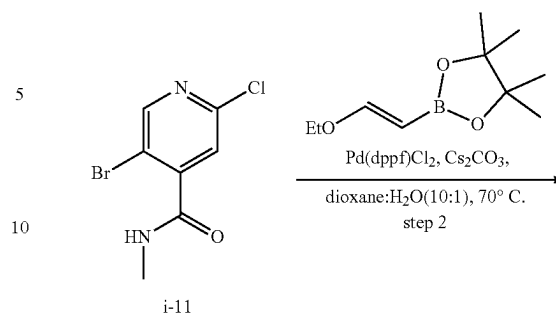
To a solution of 4-bromo-2,9-dimethyl-5H,6H,10H,11H,11aH-pyrimido[4,3-a]2,7-naphthyridine-1,8-dione (110.00 mg, 0.337 mmol, 1.00 equiv) in Dioxane (4.00 mL) and H₂O (1.00 mL) was added 4-[(dimethylamino)methyl]-3,5-dimethoxyphenylboronic acid (161.25 mg, 0.674 mmol, 2.00 equiv), Cs₂CO₃ (329.62 mg, 1.012 mmol, 3.00 equiv) and Pd(dppf)Cl₂ (49.35 mg, 0.067 mmol, 0.20 equiv) at 25° C. The resulting solution was stirred for 12 hours at 80° C. The solids were filtered out. The filtrate was concentrated. The crude product was purified by Prep-HPLC with the following conditions: Column, SunFire Prep C18 OBD Column, 19, 150 mm 5 μm 10 nm; mobile phase, Water (0.1% FA) and ACN (3% Phase B up to 10% in 35 minutes); Detector, UV. This resulted in 12 mg (7.31%) of 4-4-((dimethylamino)methyl)-3,5-dimethoxyphenyl)-2,9-dimethyl-5,6,9,10,11,11a-hexahydro-1H-pyrimido[6,1-a][2,7]naphthyridine-1,8 (2H)-dione formate as a light yellow semi-solid. ¹H NMR (300 MHz, Methanol-d₄) δ 8.51 (s, 1.35H, FA), 7.66 (s, 1H), 6.74 (s, 2H), 4.75-4.46 (m, 2H), 4.38 (s, 2H), 3.96 (s, 6H), 3.64 (s, 3H), 3.61-3.50 (m, 1H), 3.29 (dd, J=5.3, 1.8 Hz, 1H), 2.99 (s, 4H), 2.89 (s, 6H), 2.85-2.73 (m, 1H), 2.70-2.54 (m, 1H), 2.39 (d, J=16.7 Hz, 1H), 1.73-1.49 (m, 1H). LCMS (ESI) m/z: [M+H]⁺=441.30.

Example 8—Preparation 8-[4-((dimethylamino)methyl)-3,5-dimethoxyphenyl]-6-methyl-5-oxo-1,2,3,4,5,6-hexahydro-2,6-naphthyridine-3-carboxylic acid (Compound B2)



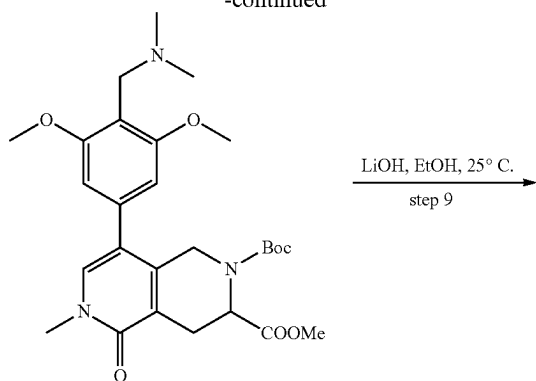
480

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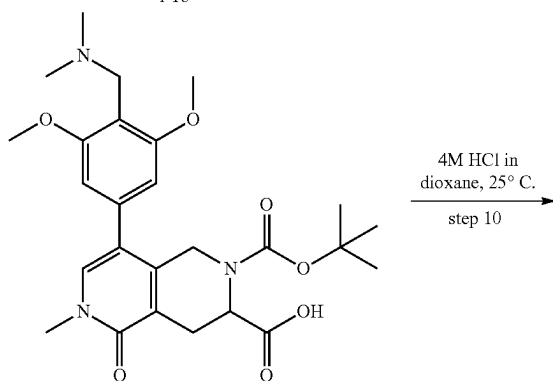


481

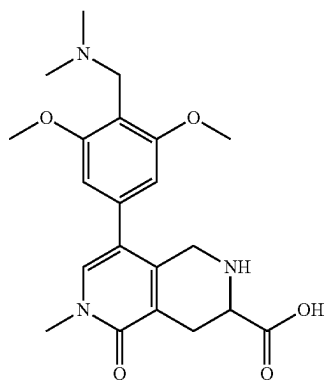
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i-18

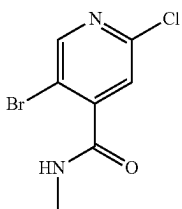


i-19



compound B2

Step 1: Preparation of
5-bromo-2-chloro-N-methylpyridine-4-carboxamide
(i-11)

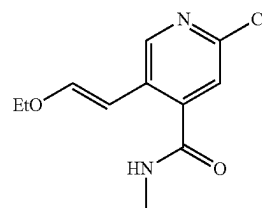


i-11

482

To a solution of 5-bromo-2-chloropyridine-4-carboxylic acid (10 g, 42.292 mmol, 1 equiv) and methanamine (1.58 g, 50.751 mmol, 1.2 equiv) in solvent DCM (100 mL) was added HATU (24.12 g, 63.438 mmol, 1.5 equiv) and DIEA (27.33 g, 211.461 mmol, 5 equiv). The resulting solution was stirred at 25° C. for 2 hours. The resulting mixture was diluted with H₂O (50 mL) and extracted with DCM (3×50 mL). The organic layers were dried over anhydrous sodium sulfate. After filtration, the filtrate was concentrated under reduced pressure. The residue was purified by silica gel column chromatography, eluted with PE/EA (5:1) to afford 5-bromo-2-chloro-N-methylpyridine-4-carboxamide (8.3 g, 78.66%) as a white solid. LCMS (ESI) m/z: [M+H]⁺=248.9, 250.9.

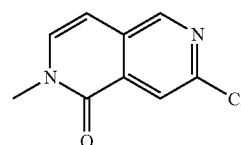
Step 2: Preparation of 2-chloro-5-[(E)-2-ethoxyethenyl]-N-methylpyridine-4-carboxamide (i-12)



i-12

To a solution of 5-bromo-2-chloro-N-methylpyridine-4-carboxamide (8.00 g, 32.065 mmol, 1.00 equiv) and 2-[(E)-2-ethoxyethenyl]-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (6.35 g, 32.065 mmol, 1.0 equiv) in dioxane (50.00 mL) and H₂O (5.00 mL) was added Pd(dppf)Cl₂ (2.35 g, 3.207 mmol, 0.1 equiv) and Cs₂CO₃ (20.90 g, 64.131 mmol, 2.0 equiv). The resulting solution was stirred at 90° C. for 2 hours under N₂ atmosphere. The resulting mixture was concentrated under reduced pressure. The residue was purified by silica gel column chromatography, eluted with PE/EA (4:1) to afford 2-chloro-5-[(E)-2-ethoxyethenyl]-N-methylpyridine-4-carboxamide (5.4 g, 69.97%) as a yellow solid. LCMS (ESI) m/z: [M+H]⁺=241.1.

Step 3: Preparation of 7-chloro-2-methyl-1,2-dihydro-2,6-naphthyridin-1-one (i-13)



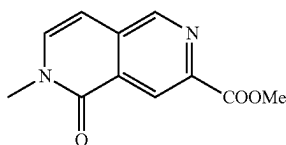
i-13

To a solution of 2-chloro-5-[(E)-2-ethoxyethenyl]-N-methylpyridine-4-carboxamide (5.40 g, 22.435 mmol, 1.00 equiv) in solvent TFA (20.00 mL) was refluxed at 100° C. for 3 hours. The mixture was basified with sodium bicarbonate saturated solution to pH 9. The resulting mixture was extracted with DCM (3×30 mL). The combined organic layers were washed with saturated sodium chloride solution and dried over anhydrous sodium sulfate. After filtration, the filtrate was concentrated under reduced pressure. The resi-

483

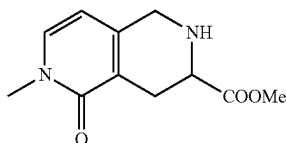
due was purified by silica gel column chromatography, eluted with PE/EA (5:1) to afford 7-chloro-2-methyl-1,2-dihydro-2,6-naphthyridin-1-one (3.2 g, 73.29%) as a yellow solid. LCMS (ESI) m/z: [M+H]⁺=195.0.

Step 4: Preparation of methyl 6-methyl-5-oxo-5,6-dihydro-2,6-naphthyridine-3-carboxylate (i-14)



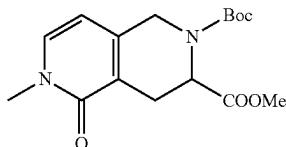
To a solution of 7-chloro-2-methyl-1,2-dihydro-2,6-naphthyridin-1-one (3.20 g, 16.442 mmol, 1.00 equiv) and Pd(dppf)Cl₂ (2.41 g, 3.288 mmol, 0.2 equiv) in MeOH (30.00 mL) and TEA (5 mL) was refluxed at 100° C. for 15 hours under 20 atm CO atmosphere. The resulting mixture was concentrated under reduced pressure. The residue was purified by silica gel column chromatography, eluted with PE/EA (4:1) to afford methyl 6-methyl-5-oxo-5,6-dihydro-2,6-naphthyridine-3-carboxylate (1.8 g, 50.17%) as a yellow solid. LCMS (ESI) m/z: [M+H]⁺=219.1.

Step 5: Preparation of methyl 6-methyl-5-oxo-1,2,3,4,5,6-hexahydro-2,6-naphthyridine-3-carboxylate (i-15)



A solution of methyl 6-methyl-5-oxo-5,6-dihydro-2,6-naphthyridine-3-carboxylate (1.80 g, 8.249 mmol, 1.00 equiv) and PtO₂ (0.94 g, 0.004 mmol, 0.5 equiv) in EtOH (20 mL) was stirred at 25° C. for 15 hours under 20 atm H₂ atmosphere. The resulting mixture was concentrated under reduced pressure then purified by silica gel column chromatography, eluted with DCM/MeOH (10/1) to afford methyl 6-methyl-5-oxo-1,2,3,4,5,6-hexahydro-2,6-naphthyridine-3-carboxylate (960 mg, 52.37%) as a yellow solid. LCMS (ESI) m/z: [M+H]⁺=223.1.

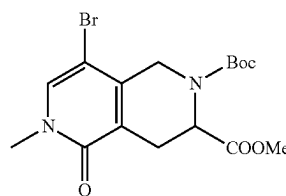
Step 6: Preparation of 2-tert-butyl 3-methyl 6-methyl-5-oxo-1,2,3,4,5,6-hexahydro-2,6-naphthyridine-2,3-dicarboxylate (i-16)



484

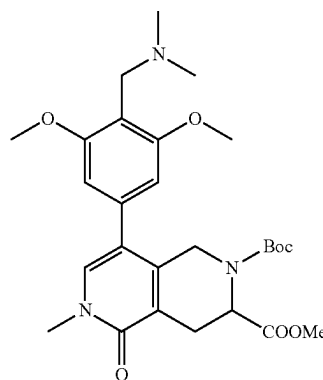
A solution of methyl 6-methyl-5-oxo-1,2,3,4,5,6-hexahydro-2,6-naphthyridine-3-carboxylate (900.00 mg, 4.050 mmol, 1.00 equiv) and (Boc)₂O (2.65 g, 12.149 mmol, 3.0 equiv) in DCM (30.00 mL) was stirred at 25° C. for 1 hour. The resulting mixture was concentrated under reduced pressure then purified by silica gel column chromatography, eluted with PE/EA (5/1) to afford 2-tert-butyl 3-methyl 6-methyl-5-oxo-1,2,3,4,5,6-hexahydro-2,6-naphthyridine-2,3-dicarboxylate (760 mg, 58.22%) as a yellow solid. LCMS (ESI) m/z: [M+H]⁺=323.2.

Step 7: Preparation of 2-tert-butyl 3-methyl 8-bromo-6-methyl-5-oxo-1,2,3,4,5,6-hexahydro-2,6-naphthyridine-2,3-dicarboxylate (i-17)



A solution of 2-tert-butyl 3-methyl 6-methyl-5-oxo-1,2,3,4,5,6-hexahydro-2,6-naphthyridine-2,3-dicarboxylate (760.00 mg, 2.358 mmol, 1.00 equiv) and NBS (503.54 mg, 2.829 mmol, 1.20 equiv) in DCM (10.00 mL) was stirred at 25° C. for 2 hours. The resulting mixture was extracted with DCM (20 mL×3). The combined organic layers were washed with H₂O and then dried over anhydrous sodium sulfate. After filtration, the filtrate was concentrated under reduced pressure and purified by silica gel column chromatography, eluted with PE/EA (4/1) to afford 2-tert-butyl 3-methyl 8-bromo-6-methyl-5-oxo-1,2,3,4,5,6-hexahydro-2,6-naphthyridine-2,3-dicarboxylate (680 mg, 71.88%) as a yellow solid. LCMS (ESI) m/z: [M+H]⁺=401.1, 403.1.

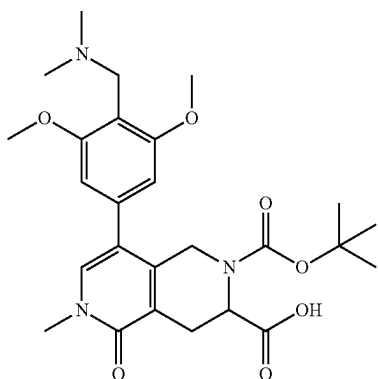
Step 8: Preparation of 2-tert-butyl 3-methyl 8-[[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-6-methyl-5-oxo-1,2,3,4,5,6-hexahydro-2,6-naphthyridine-2,3-dicarboxylate (i-18)



485

To a solution of 2-tert-butyl 3-methyl 8-bromo-6-methyl-5-oxo-1,2,3,4,5,6-hexahydro-2,6-naphthyridine-2,3-dicarboxylate (680.00 mg, 1.695 mmol, 1.00 equiv) and [4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl] boronic acid (607.74 mg, 2.542 mmol, 1.50 equiv) in dioxane (10.00 mL) and H₂O (2.00 mL) was added Pd(dppf)Cl₂ (248.00 mg, 0.339 mmol, 0.20 equiv) and Cs₂CO₃ (1656.48 mg, 5.084 mmol, 3.00 equiv). The resulting solution was stirred at 90° C. for 2 hours. The resulting mixture was concentrated under reduced pressure then purified by silica gel column chromatography, eluted with DCM/MeOH (10/1) to afford 2-tert-butyl 3-methyl 8-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-6-methyl-5-oxo-1,2,3,4,5,6-hexahydro-2,6-naphthyridine-2,3-dicarboxylate (480 mg, 54.93%) as a yellow solid. LCMS (ESI) m/z: [M+H]⁺=516.3.

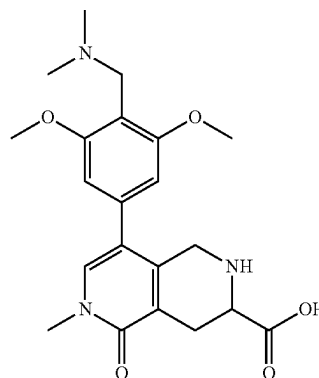
Step 9: Preparation of 2-[(tert-butoxy)carbonyl]-8-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-6-methyl-5-oxo-1,2,3,4,5,6-hexahydro-2,6-naphthyridine-3-carboxylic acid (i-19)



A solution of 2-tert-butyl 3-methyl 8-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-6-methyl-5-oxo-1,2,3,4,5,6-hexahydro-2,6-naphthyridine-2,3-dicarboxylate (480.00 mg, 0.931 mmol, 1.00 equiv) and LiOH (44.59 mg, 1.862 mmol, 2.00 equiv) in solvent EtOH (20.00 mL) was stirred at 25° C. for 2 hours. The residue was purified by reverse flash chromatography with the following conditions: column, C18 silica gel; mobile phase, MeCN in water, 10% to 20% gradient in 8 minutes; detector, UV 254 nm to afford 2-[(tert-butoxy)carbonyl]-8-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-6-methyl-5-oxo-1,2,3,4,5,6-hexahydro-2,6-naphthyridine-3-carboxylic acid (210 mg, 44.97%) as a yellow solid. LCMS (ESI) m/z: [M+H]⁺=502.2.

486

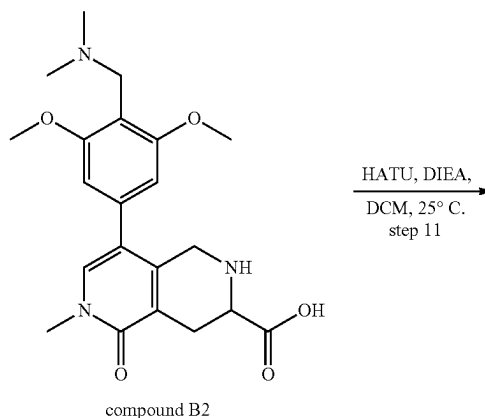
Step 10: Preparation of 8-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-6-methyl-5-oxo-1,2,3,4,5,6-hexahydro-2,6-naphthyridine-3-carboxylic acid (Compound B2)



compound B2

2-[(tert-butoxy)carbonyl]-8-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-6-methyl-5-oxo-1,2,3,4,5,6-hexahydro-2,6-naphthyridine-3-carboxylic acid (30.00 mg, 0.060 mmol, 1.00 equiv) was added to 4 M HCl 1,4-dioxane solution (5.00 mL), and the resulting solution was stirred at 25° C. for 1 hour. The residue was purified by reverse flash chromatography with the following conditions: column, C18 silica gel; mobile phase, MeCN in water, 10% to 50% gradient in 30 minutes; detector, UV 254 nm to afford 8-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-6-methyl-5-oxo-1,2,3,4,5,6-hexahydro-2,6-naphthyridine-3-carboxylic acid (18 mg, 74.96%) as a yellow solid. ¹H NMR (400 MHz, Methanol-d₄) δ 7.71 (s, 1H), 6.74 (s, 2H), 4.55-4.26 (m, 4H), 4.07 (d, J=16.9 Hz, 1H), 3.96 (s, 6H), 3.66 (s, 3H), 3.47-3.34 (m, 1H), 2.90 (m, 7H). LCMS (ESI) m/z: [M+H]⁺=402.25.

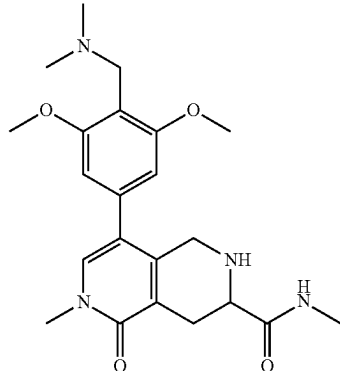
Example 9—Preparation of and 8-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-N,6-dimethyl-5-oxo-1,2,3,4,5,6-hexahydro-2,6-naphthyridine-3-carboxamide and 8-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-6-methyl-5-oxo-1,2,3,4,5,6-hexahydro-2,6-naphthyridine-3-carboxamide (Compounds B3 and B4)



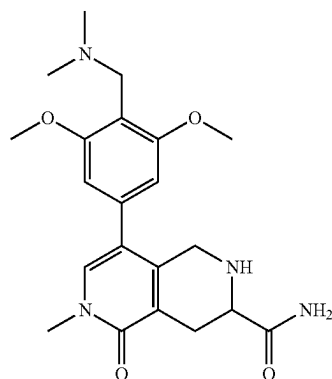
compound B2

487

-continued

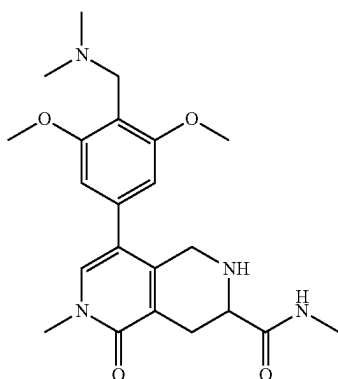


compound B3



compound B4

Preparation of 8-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-N,6-dimethyl-5-oxo-1,2,3,4,5,6-hexahydro-2,6-naphthyridine-3-carboxamide (Compound B3)

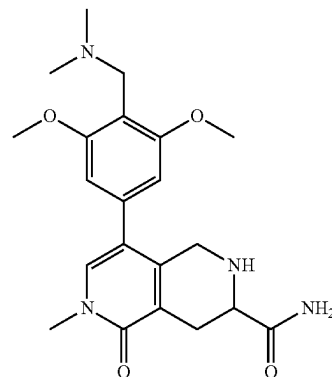


compound B3

488

8-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-6-methyl-5-oxo-1,2,3,4,5,6-hexahydro-2,6-naphthyridine-3-carboxylic acid (30.00 mg, 0.075 mmol, 1.00 equiv), HATU (42.62 mg, 0.112 mmol, 1.50 equiv), and methyl amine (4.64 mg, 0.149 mmol, 2.00 equiv), and diisopropylethylamine (57.9 mg, 0.448 mmol, 6.00 equiv) were dissolved in DCM (5.0 mL). The resulting solution was stirred at 25° C. for 3 hours. The crude product was purified by Prep-HPLC (conditions: C18 silica gel; mobile phase, MeCN in water, 10% to 50% gradient in 30 minutes; detector, UV 254 nm) to afford 8-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-N,6-dimethyl-5-oxo-1,2,3,4,5,6-hexahydro-2,6-naphthyridine-3-carboxamide (6.8 mg, 22%) as a yellow solid. ¹H NMR (400 MHz, Methanol-d₄) δ 7.50 (s, 1H), 6.62 (s, 2H), 3.88 (s, 6H), 3.85-3.69 (m, 4H), 3.63 (s, 3H), 3.52 (dd, J=10.4, 4.8 Hz, 1H), 2.96 (dd, J=17.5, 4.7 Hz, 1H), 2.81 (s, 3H), 2.55 (dd, J=17.7, 10.4 Hz, 1H), 2.42 (s, 6H). LCMS (ESI) m/z: [M+H]⁺=415.30.

Preparation of 8-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-6-methyl-5-oxo-1,2,3,4,5,6-hexahydro-2,6-naphthyridine-3-carboxamide (Compound B4)

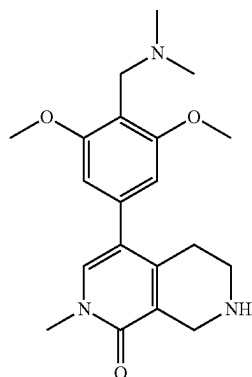


compound B4

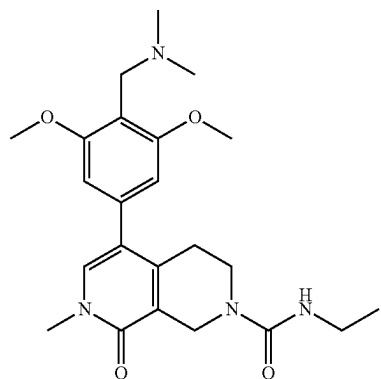
8-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-6-methyl-5-oxo-1,2,3,4,5,6-hexahydro-2,6-naphthyridine-3-carboxylic acid (30.00 mg, 0.075 mmol, 1.00 equiv), HATU (42.62 mg, 0.112 mmol and 1.50 equiv), DIEA (57.95 mg, 0.448 mmol, 6.00 equiv) were dissolved in DCM (5.00 mL). Then NH₃ (2.55 mg, 0.149 mmol, 2.00 equiv) in DCM was added to the reaction, and the resulting solution was stirred at 25° C. for 3 hours. The crude product (30 mg) was purified by Prep-HPLC (conditions: C18 silica gel; mobile phase, MeCN in water, 10% to 50% gradient in 30 minutes; detector, UV 254 nm) to afford 8-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-6-methyl-5-oxo-1,2,3,4,5,6-hexahydro-2,6-naphthyridine-3-carboxamide (4.7 mg, 15.71%) as a yellow solid. ¹H NMR (400 MHz, Methanol-d₄) δ 7.50 (s, 1H), 6.59 (s, 2H), 3.86 (s, 6H), 3.83-3.66 (m, 4H), 3.63 (s, 3H), 3.57 (dd, J=10.4, 4.8 Hz, 1H), 3.05-2.94 (m, 1H), 2.58 (dd, J=17.8, 10.3 Hz, 1H), 2.32 (s, 6H). LCMS (ESI) m/z: [M+H]⁺=401.30.

489

Example 10—Preparation of 5-(4-((dimethylamino)methyl)-3,5-dimethoxyphenyl)-N-ethyl-7-methyl-8-oxo-3,4,7,8-tetrahydro-2,7-naphthyridine-2 (1H)-carboxamide (Compound B5)



compound B17



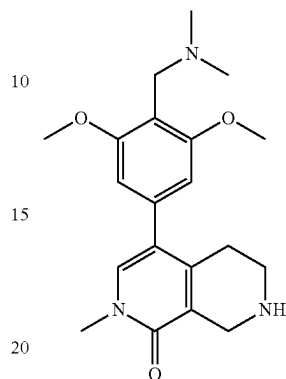
compound B5

Using the same procedure as for the synthesis of Compound B9 and substituting with ethyl isocyanate afforded 5-(4-((dimethylamino)methyl)-3,5-dimethoxyphenyl)-N-ethyl-7-methyl-8-oxo-3,4,7,8-tetrahydro-2,7-naphthyridine-2 (1H)-carboxamide (18.8 mg, 521% yield) as a white solid. ¹H NMR (400 MHz, DMSO-d₆) δ 7.61 (s, 1H), 6.61 (t, J=5.4 Hz, 1H), 6.55 (s, 2H), 4.19 (s, 2H), 3.75 (s, 5H), 3.47 (s, 3H), 3.43-3.34 (m, 4H), 3.11-2.97 (m, 2H), 2.09 (s, 6H), 1.00 (t, J=7.1 Hz, 3H). LCMS (ESI) m/z: [M+H]⁺=429.2.

490

Example 11—Preparation of 5-(4-((dimethylamino)methyl)-3,5-dimethoxyphenyl)-N,N,7-trimethyl-8-oxo-3,4,7,8-tetrahydro-2,7-naphthyridine-2 (1H)-carboxamide (Compound B6)

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compound B17

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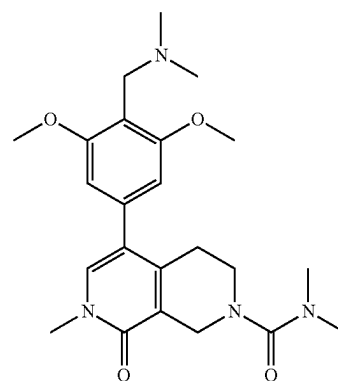
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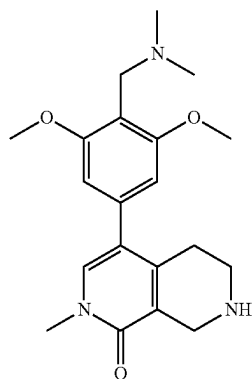


compound B6

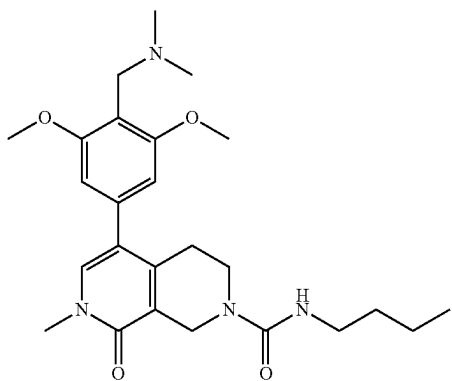
To a solution of 4-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-2-methyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one (30 mg, 83.9 μmol, 1 equiv) in DCM (1 mL) was added triethylamine (15.1 μL, 109 μmol, 1.3 equiv) and N,N-dimethylcarbonyl chloride (8.48 μL, 92.2 μmol, 1.1 equiv) at RT. The reaction was stirred at RT for 2 hours. The reaction mixture was concentrated in vacuo. The crude was purified by flash chromatography eluting with 0-15% MeOH with 0.1% NH₄OH in DCM to afford 5-(4-((dimethylamino)methyl)-3,5-dimethoxyphenyl)-N,N,7-trimethyl-8-oxo-3,4,7,8-tetrahydro-2,7-naphthyridine-2 (1H)-carboxamide (5.2 mg, 14.5% yield) as a white solid. ¹H NMR (400 MHz, Chloroform-d) δ 7.14 (s, 1H), 6.40 (s, 2H), 4.33 (s, 2H), 3.83 (s, 6H), 3.58 (s, 3H), 3.58 (m, 2H), 3.37 (t, J=5.6 Hz, 2H), 2.89 (s, 5H), 2.63-2.52 (m, 2H), 1.25 (s, 6H). LCMS (ESI) m/z: [M+H]⁺=429.2.

491

Example 12—Preparation of N-butyl-5-(4-((dimethylamino)methyl)-3,5-dimethoxyphenyl)-7-methyl-8-oxo-3,4,7,8-tetrahydro-2,7-naphthyridine-2 (1H)-carboxamide (Compound B7)



compound B17



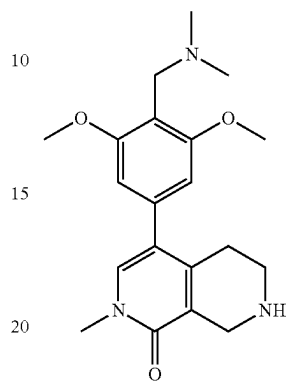
compound B7

Using the same procedure as for the synthesis of Compound B9 and substituting with 1-isocyanatobutane afforded N-butyl-5-(4-((dimethylamino)methyl)-3,5-dimethoxyphenyl)-7-methyl-8-oxo-3,4,7,8-tetrahydro-2,7-naphthyridine-2 (1H)-carboxamide (18.8 mg, 49% yield) as a white solid. ¹H NMR (400 MHz, Chloroform-d) δ 7.18 (s, 1H), 6.37 (s, 2H), 4.68 (t, J=5.5 Hz, 1H), 4.28 (s, 2H), 3.82 (s, 6H), 3.61 (s, 3H), 3.61 (m, 2H), 3.53 (s, 1H), 3.28 (td, J=7.1, 5.3 Hz, 2H), 2.57 (t, J=5.6 Hz, 2H), 2.30 (s, 6H), 1.59-1.45 (m, 9H), 1.43-1.30 (m, 2H), 0.93 (t, J=7.3 Hz, 3H). LCMS (ESI) m/z: [M+H]⁺=457.2.

492

Example 13—Preparation of N-cyclopropyl-5-(4-((dimethylamino)methyl)-3,5-dimethoxyphenyl)-7-methyl-8-oxo-3,4,7,8-tetrahydro-2,7-naphthyridine-2 (1H)-carboxamide (Compound B8)

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compound B17

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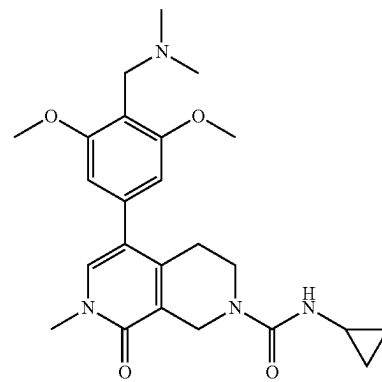
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compound B8

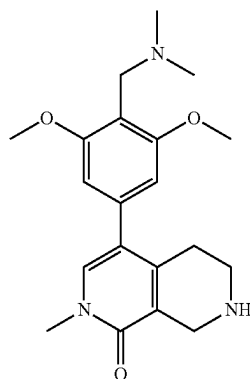
Using the same procedure as for the synthesis of Compound B9 and substituting with cyclopropyl isocyanate afforded N-cyclopropyl-5-(4-((dimethylamino)methyl)-3,5-dimethoxyphenyl)-7-methyl-8-oxo-3,4,7,8-tetrahydro-2,7-naphthyridine-2 (1H)-carboxamide (7.8 mg, 21% yield) as a white solid. ¹H NMR (400 MHz, Chloroform-d) δ 7.18 (s, 1H), 6.37 (s, 2H), 4.93 (d, J=1.9 Hz, 1H), 4.23 (s, 2H), 3.82 (s, 6H), 3.60 (d, J=2.2 Hz, 5H), 3.54 (s, 2H), 2.70 (ddd, J=6.9, 4.6, 2.7 Hz, 1H), 2.56 (t, J=5.7 Hz, 2H), 2.31 (s, 6H), 1.25 (s, 6H), 0.79-0.69 (m, 2H), 0.56-0.38 (m, 2H). LCMS (ESI) m/z: [M+H]⁺=441.2.

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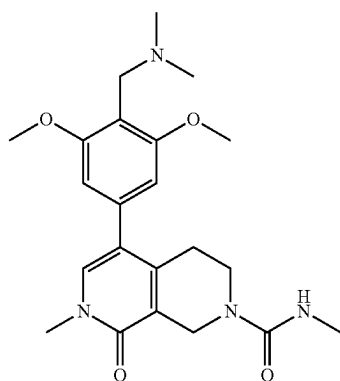
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Example 14—Preparation of 5-(4-((dimethylamino)methyl)-3,5-dimethoxyphenyl)-N,7-dimethyl-8-oxo-3,4,7,8-tetrahydro-2,7-naphthyridine-2 (1H)-carboxamide (Compound B9)



compound B17



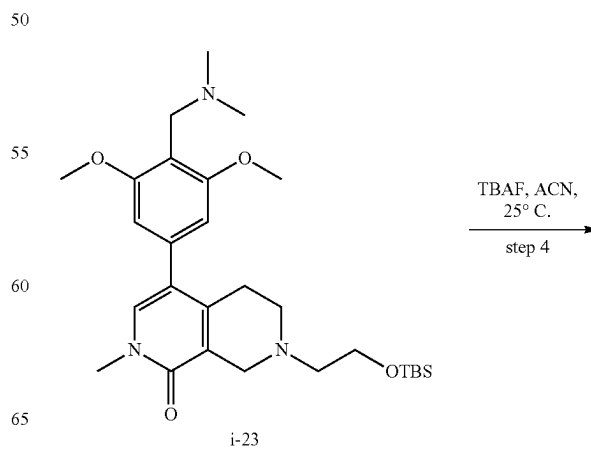
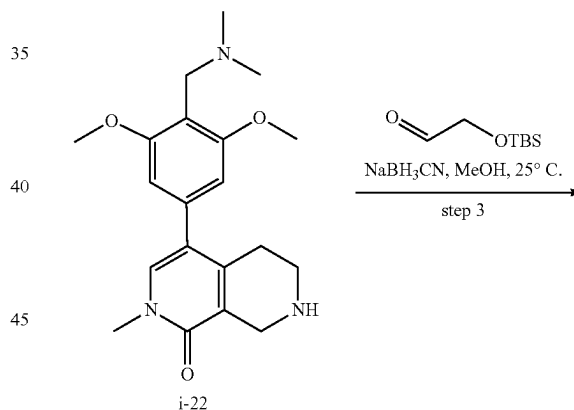
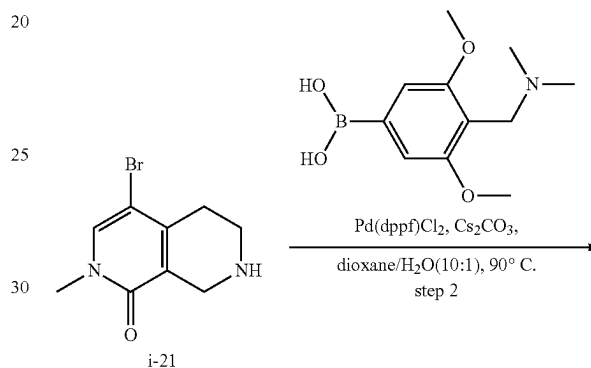
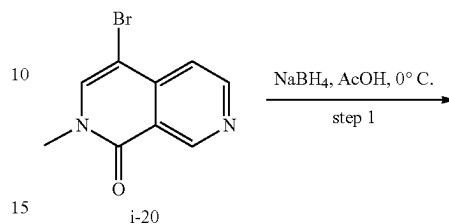
compound B9

To a solution of 4-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-2-methyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one (20 mg, 55.9 μ mol, 1 equiv) in anhydrous DCM (1 mL) was added N-methyl-1H-imidazole-1-carboxamide (7.7 mg, 61.4 μ mol, 1.1 equiv) and triethylamine (10 μ L, 72.6 μ mol, 1.3 equiv). The reaction mixture was stirred at room temperature for 3 hours and concentrated in vacuo. The crude was purified by flash chromatography eluting with 0-15% MeOH w 0.1% NH_4OH in DCM to afford 5-(4-((dimethylamino)methyl)-3,5-dimethoxyphenyl)-N,7-dimethyl-8-oxo-3,4,7,8-tetrahydro-2,7-naphthyridine-2 (1H)-carboxamide (4.0 mg, 18% yield) as a white solid. ^1H NMR (400 MHz, DMSO-d_6) δ 7.60 (s, 1H), 6.55 (s, 2H), 6.54 (bs, 1H), 4.19 (s, 2H), 3.75 (s, 6H), 3.47 (s, 3H), 3.39 (dd, $J=12.4, 7.0$ Hz, 4H), 2.57 (d, $J=4.3$ Hz, 3H), 2.48 (m, 2H), 2.10 (s, 6H). LCMS (ESI) m/z : $[\text{M}+\text{H}]^+=415.3$.

494

Example 15—Preparation of N-(6-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-3-methyl-[1,2,4]triazolo[4,3-a]pyridin-8-yl)acetamide (Compound B10)

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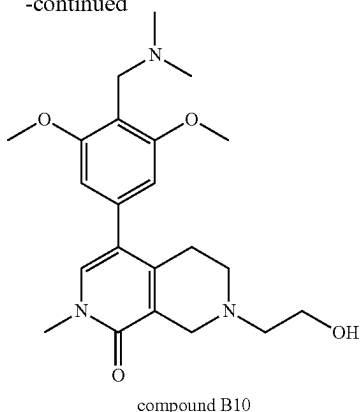
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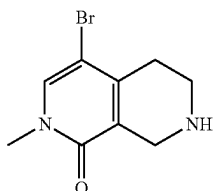
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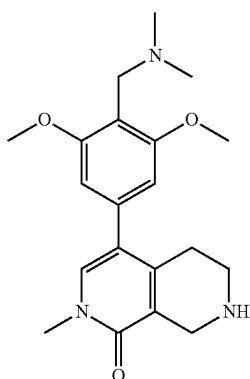


Step 1: Preparation of 4-bromo-2-methyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one (i-21)



To a solution of 4-bromo-2-methyl-1,2-dihydro-2,7-naphthyridin-1-one (238 mg, 0.996 mmol, 1 equiv) in AcOH (2.0 mL) was added NaBH₄ (263.64 mg, 6.969 mmol, 7 equiv). The mixture was stirred at 0° C. for 1 hour. Ammonium hydroxide was added to the resulting mixture until pH above 7, and then the resulting mixture was extracted with DCM (30 mL×3). The combined organic layers were dried over sodium sulfate. After filtration, the filtrate was concentrated under reduced pressure. The residue was purified by silica gel column chromatography, eluted with DCM/MeOH (10:1) to afford 4-bromo-2-methyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one (180 mg, 74.38%) as a white solid. LCMS (ESI) m/z: [M+H]⁺=243.

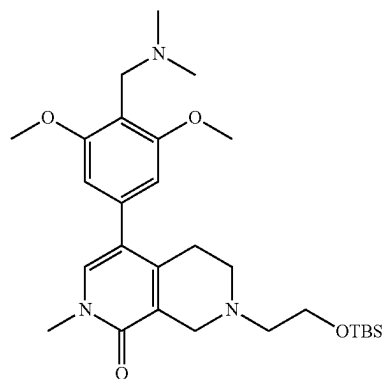
Step 2: Preparation of 4-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-2-methyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one (i-22)



496

To a solution of 4-bromo-2-methyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one (242 mg, 0.995 mmol, 1 equiv) and [4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]boronic acid (237.99 mg, 0.995 mmol, 1 equiv) in dioxane (5 mL) and H₂O (0.5 mL) was added Pd(dppf)Cl₂ (72.84 mg, 0.100 mmol, 0.1 equiv) and Cs₂CO₃ (973.02 mg, 2.986 mmol, 3.0 equiv). The resulting solution was stirred at 90° C. for 2 hours under N₂. The mixture was diluted with 50 mL of H₂O, and the resulting mixture was extracted with DCM (30 mL×3). The combined organic layers were dried over sodium sulfate, and then concentrated under reduced pressure. The residue was purified by silica gel column chromatography, eluted with DCM/MeOH (10:1) to afford 4-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-2-methyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one (200 mg, 56.21%) as a yellow solid. LCMS (ESI) m/z: [M+H]⁺=358.

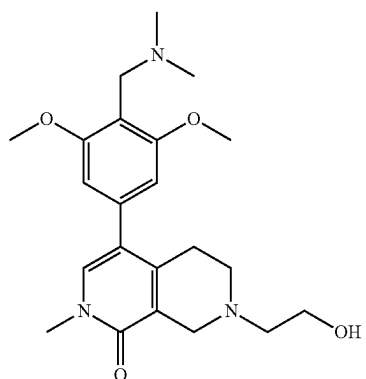
Step 3: Preparation of 7-[2-[(tert-butyldimethylsilyl)oxy]ethyl]-4-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-2-methyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one (i-23)



To a solution of 4-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-2-methyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one (200 mg, 0.560 mmol, 1 equiv) and 2-[(tert-butyldimethylsilyl)oxy]acetaldehyde (146.30 mg, 0.839 mmol, 1.5 equiv) in MeOH (5 mL) was added NaBH₃CN (105.48 mg, 1.679 mmol, 3.0 equiv). The mixture was stirred at 25° C. for 1 hour. Then the mixture was concentrated under reduced pressure. The residue was purified by silica gel column chromatography, eluted with PE/EtOAc (1:1) to afford 7-[2-[(tert-butyldimethylsilyl)oxy]ethyl]-4-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-2-methyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one (224 mg, 77.5%) as a brown oil. LCMS (ESI) m/z: [M+H]⁺=516.

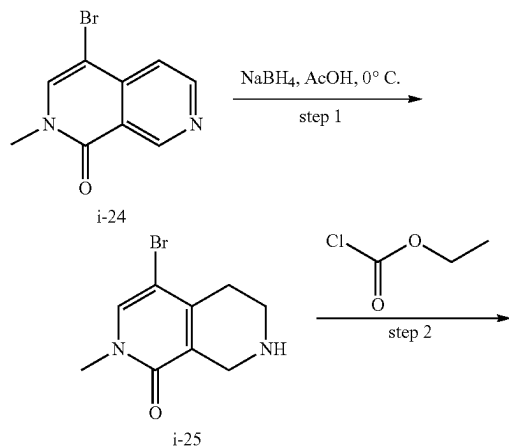
497

Step 4: Preparation of 4-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-7-(2-hydroxyethyl)-2-methyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one (Compound B10)



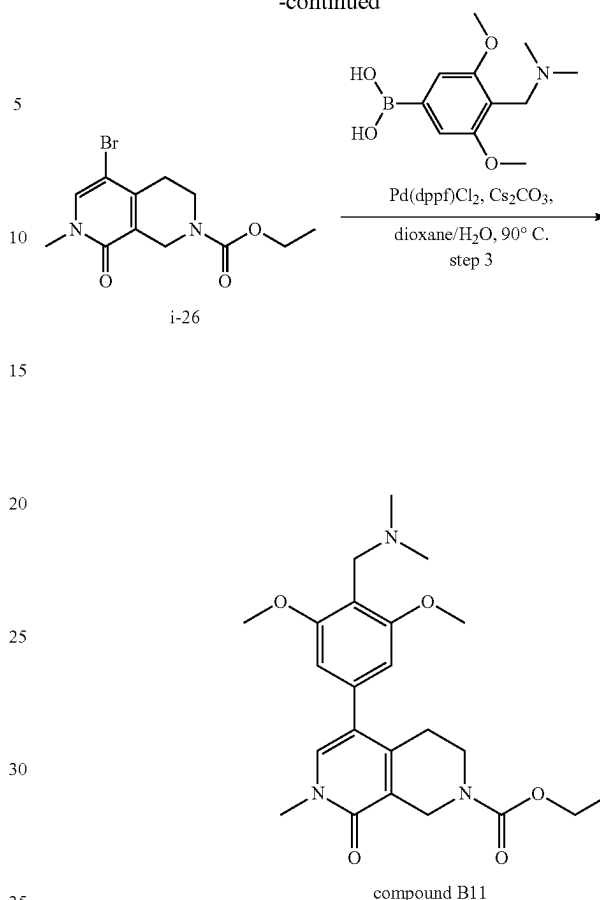
To a solution of 7-[2-[(tert-butyldimethylsilyl)oxy]ethyl]-4-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-2-methyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one (90 mg, 0.174 mmol, 1 equiv) was dissolved in ACN (2.0 mL) was added TBAF (91.25 mg, 0.349 mmol, 2.0 equiv). The resulting solution was stirred at 25° C. for 1 hour. The mixture was concentrated under reduced pressure. The residue was purified by Prep-(conditions: column, Xselect Peptide CSH 19*150 mm 5 μm; Mobile Phase A: Water (0.1% FA), Mobile Phase B: MeOH-HPLC; Flow rate: 25 mL/minute; Gradient: 15% B to 15% B in 12 minutes; 220 nm; R_t: 9.51 minutes; detector, UV 254 nm) to afford 4-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-7-(2-hydroxyethyl)-2-methyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one (10 mg, 14.27%) as a yellow solid. ¹H NMR (300 MHz, Methanol-d₄) δ 7.51 (s, 1H), 6.61 (s, 2H), 3.87 (s, 6H), 3.83-3.72 (m, 4H), 3.60 (d, J=14.7 Hz, 5H), 2.79-2.68 (m, 6H), 2.37 (s, 6H). LCMS (ESI) m/z: [M+H]⁺=402.40.

Example 16—Preparation of Ethyl 5-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-7-methyl-8-oxo-1,2,3,4,7,8-hexahydro-2,7-naphthyridine-2-carboxylate (Compound B11)

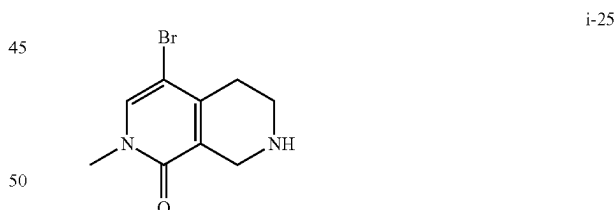


498

-continued



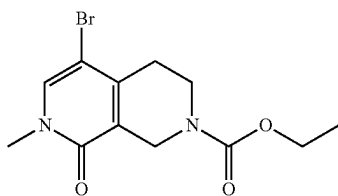
Step 1: Preparation of 4-Bromo-2-methyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one (i-25)



To a solution of 4-bromo-2-methyl-1,2-dihydro-2,7-naphthyridin-1-one (500 mg, 2.091 mmol, 1 equiv) in AcOH (4.20 mL) was added NaBH₄ (553.87 mg, 14.640 mmol, 7 equiv) at 0° C. The resulting solution was stirred at 0° C. for 1 hour. Ammonia was added to the resulting mixture until pH above 7. Then the mixture was diluted with water (10 mL) and extracted with DCM (30 mL×3). The combined organic layers were dried over saturated sodium sulfate. After filtration, the filtrate was concentrated under reduced pressure. The residue was purified by silica gel column chromatography, eluted with DCM/MeOH (10:1) to afford 4-bromo-2-methyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one (520 mg, 74.38%) as a white solid. LCMS (ESI) m/z: [M+H]⁺=243.

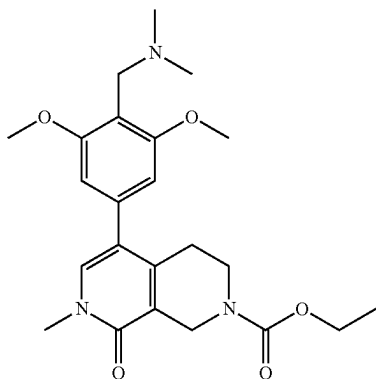
499

Step 2: Preparation of ethyl 5-bromo-7-methyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxylate (i-26)



To a solution of 4-bromo-2-methyl-5,6,7,8-tetrahydro-2,7-naphthyridin-1-one (100.00 mg, 0.411 mmol, 1.00 equiv) in (5.00 mL) was added NaH (19.74 mg, 0.494 mmol, 1.20 equiv, 60%). Then ethyl chloroformate (66.96 mg, 0.617 mmol, 1.50 equiv) was added at 0° C. The resulting mixture was stirred for 2 hours at room temperature. The resulting mixture was concentrated under reduced pressure. The residue was purified by Prep-TLC (PE/EtOAc 5:1) to afford ethyl 5-bromo-7-methyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxylate (120 mg, 92.56%) as a yellow solid. LCMS (ESI) m/z: [M+H]⁺=315.

Step 3: Preparation of ethyl 5-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-7-methyl-8-oxo-1,2,3,4,7,8-hexahydro-2,7-naphthyridine-2-carboxylate (Compound B11)

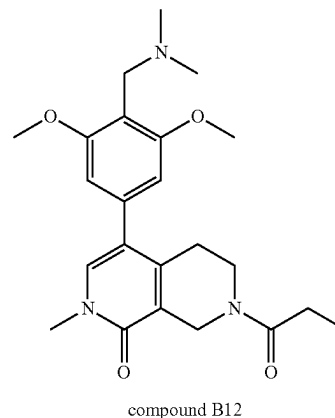
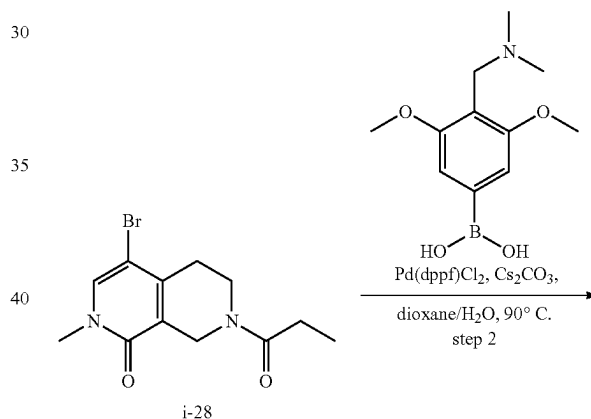
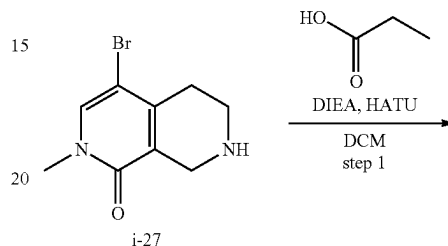


To a solution of ethyl 5-bromo-7-methyl-8-oxo-1,2,3,4,7,8-hexahydro-2,7-naphthyridine-2-carboxylate (96 mg, 0.305 mmol, 1 equiv) and [4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]boronic acid (72.82 mg, 0.305 mmol, 1 equiv) in dioxane (2 mL) and H₂O (0.5 mL) was added Cs₂CO₃ (297.73 mg, 0.914 mmol, 3 equiv) and Pd(dppf)Cl₂ (33.43 mg, 0.046 mmol, 0.15 equiv). The resulting solution was stirred at 90° C. for 2 hours (under N₂ atmosphere). The resulting mixture was concentrated under reduced pressure. The crude product was purified by Prep-HPLC (conditions: XBridge Shield RP18 OBD Column, 5 μm, 19*150 mm; Mobile Phase A: Water (0.05% NH₃/H₂O), Mobile Phase B: ACN; Flow rate: 25 mL/minute; Gradient: 24% B to 37% B in 8 minutes; 220 nm; Rt: 7.9 minutes) to afford ethyl 5-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-7-methyl-8-oxo-1,2,3,4,7,8-hexahydro-2,7-naphthyridine-2-

500

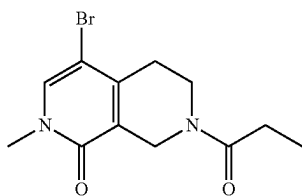
carboxylate (8.2 mg, 6.09%) as a light brown solid. ¹H NMR (300 MHz, Methanol-d₄) δ 7.56 (s, 1H), 6.61 (s, 2H), 4.47 (s, 2H), 4.20 (q, J=7.1 Hz, 2H), 3.87 (s, 6H), 3.71 (s, 2H), 3.62 (d, J=8.9 Hz, 5H), 2.65 (t, J=5.8 Hz, 2H), 2.34 (s, 6H), 1.31 (t, J=7.1 Hz, 3H). LCMS (ESI) m/z: [M+H]⁺=430.20.

Example 17—Preparation of 4-(3,5-dimethoxy-4-methylphenyl)-2-methyl-7-propanoyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one (Compound B12)



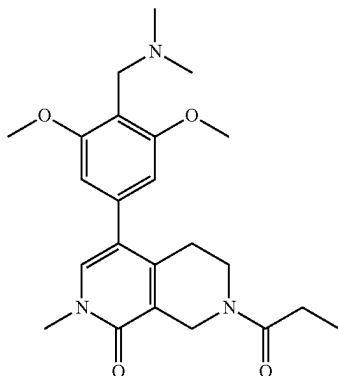
501

Step 1: Preparation of 4-bromo-2-methyl-7-propanoyl-6,8-dihydro-5H-2,7-naphthyridin-1-one (i-28)



To a stirred mixture of 4-bromo-2-methyl-5,6,7,8-tetrahydro-2,7-naphthyridin-1-one (500.00 mg, 2.057 mmol, 1.00 equiv) and propanoic acid (182.83 mg, 2.468 mmol, 1.20 equiv) in DCM (25.00 mL) was added DIEA (79.75 mg, 0.617 mmol, 3.00 equiv). The mixture was stirred at room temperature for 5 minutes, then HATU (938.44 mg, 2.468 mmol, 1.20 equiv) was added. The mixture was stirred for 2 hours at room temperature, and the resulting mixture was concentrated under reduced pressure. The residue was purified by silica gel column chromatography, eluted with CH₂Cl₂/MeOH (20:1) to afford 4-bromo-2-methyl-7-propanoyl-6,8-dihydro-5H-2,7-naphthyridin-1-one (502 mg, 78.88%) as a white solid.

Step 2: Preparation of 4-(3,5-dimethoxy-4-methylphenyl)-2-methyl-7-propanoyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one (Compound B12)

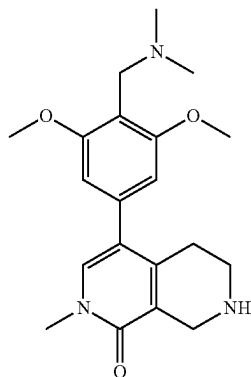


To a solution of 4-bromo-2-methyl-7-propanoyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one (200 mg, 0.669 mmol, 1 equiv) and [4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]boronic acid (191.80 mg, 0.802 mmol, 1.20 equiv) in dioxane (10 mL) and H₂O (1 mL) was added Cs₂CO₃ (653.45 mg, 2.006 mmol, 3.00 equiv) and Pd(dppf)Cl₂·CH₂Cl₂ (49.13 mg, 0.060 mmol, 0.09 equiv). After stirring for 2 hours at 90° C. under a nitrogen atmosphere, the resulting mixture was concentrated under reduced pressure. The residue was purified by silica gel column chromatography, eluted with DCM/MeOH (10/1) to afford 4-(3,5-dimethoxy-4-methylphenyl)-2-methyl-7-propanoyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one (130 mg, 51.44%). ¹H NMR (400 MHz, Methanol-d₄) δ 7.61 (d, J=10.5 Hz, 1H), 6.75 (d, J=6.3 Hz, 2H), 4.56 (d, J=14.2 Hz, 1H), 4.39

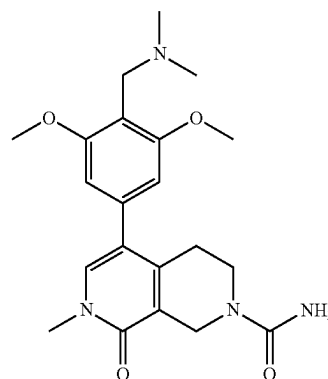
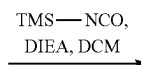
502

(s, 2H), 3.96 (d, J=2.6 Hz, 6H), 3.76-3.57 (m, 5H), 2.90 (s, 6H), 2.72 (d, J=6.1 Hz, 1H), 2.64 (s, 1H), 2.53 (dq, J=15.0, 7.5 Hz, 1H), 1.31 (s, 1H), 1.17 (td, J=7.5, 3.8 Hz, 3H). LCMS (ESI) m/z: [M+H]⁺=414.30.

Example 18—Preparation of 7-acetyl-4-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-2-methyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one (Compound B13)



compound B17

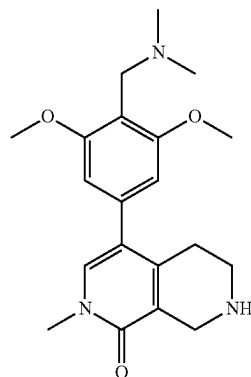


compound B13

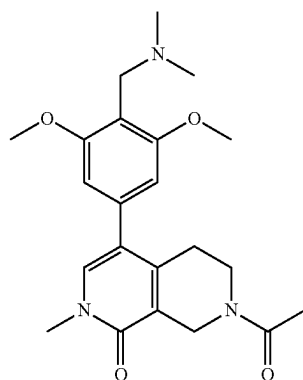
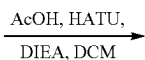
To a stirred solution of 4-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-2-methyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one (45.4 mg, 0.127 mmol, 1 equiv) in DCM (3 mL) was added isocyanatotrimethylsilane (29.26 mg, 0.254 mmol, 2 equiv) and TEA (38.56 mg, 0.381 mmol, 3 equiv). The resulting mixture was stirred for 2 hours at room temperature. The crude product was purified by Prep-HPLC (conditions: XBridge Prep Phenyl OBD Column 5 μm, 19*250 mm; Mobile Phase A: Water (10 mmol/L NH₄HCO₃), Mobile Phase B: ACN; Flow rate: 25 mL/minute; Gradient: 14% B to 20% B in 8 minutes; 254 nm; R_t: 7.18 minutes) to afford 5-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-7-methyl-8-oxo-1,2,3,4,7,8-hexahydro-2,7-naphthyridine-2-carboxamide (6.6 mg, 12.35%) as a white solid. ¹H NMR (400 MHz, MeOD) δ 7.57 (s, 1H), 6.61 (s, 2H), 4.41 (s, 2H), 3.87 (s, 6H), 3.68 (s, 2H), 3.64 (s, 3H), 3.55 (t, 2H), 2.68-2.65 (m, 2H), 2.31 (s, 6H). LCMS (ESI) m/z: [M+H]⁺=401.4.

503

Example 19—Preparation of 7-acetyl-4-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-2-methyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one (Compound B14)



compound B17



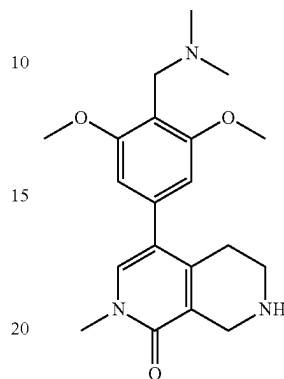
compound B14

To the solution of 4-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-2-methyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one (35.7 mg, 0.100 mmol, 1 equiv) in DCM (3 mL) was added acetic acid (7.20 mg, 0.120 mmol, 1.2 equiv), HATU (56.96 mg, 0.150 mmol, 1.5 equiv), and DIEA (38.72 mg, 0.300 mmol, 3 equiv). The resulting solution was stirred at room temperature for 1 hour. The resulting solution was concentrated. The crude product was purified by Prep-HPLC (conditions: XBridge Shield RP18 OBD Column, 5 μ m, 19*150 mm; Mobile Phase A: Water (0.1% FA), Mobile Phase B: ACN; Flow rate: 25 mL/minute; Gradient: 0% B to 15% B in 8 minutes; 254/220 nm; R_f : 7.03 minutes) to afford 7-acetyl-4-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-2-methyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one (7.3 mg, 17.64%) as a white solid. ^1H NMR (400 MHz, MeOD) δ 7.58 (d, 1H), 6.61 (d, 2H), 4.55 (d, 2H), 3.87 (d, 6H), 3.75-3.67 (m, 3H), 3.64 (d, 4H), 2.74 (t, 1H), 2.64 (t, 1H), 2.33 (d, 6H), 2.22 (d, 3H). LCMS (ESI) m/z : $[\text{M}+\text{H}]^+=400.25$.

504

Example 20—Preparation of 4-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-7-ethyl-2-methyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one formic acid (Compound B15 formic acid)

5



compound B17



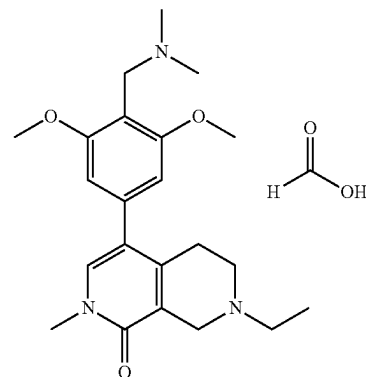
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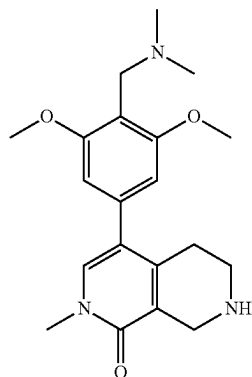


compound B15 formic acid

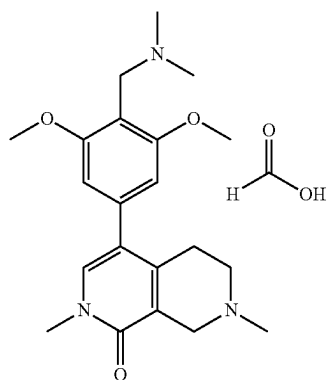
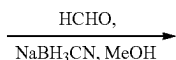
To the solution of 4-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-2-methyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one (20 mg, 0.056 mmol, 1 equiv) in MeOH (2 mL) was added acetaldehyde (24.65 mg, 0.560 mmol, 10 equiv) and NaBH_3CN (10.55 mg, 0.168 mmol, 3 equiv). The resulting solution was stirred at room temperature for 1 hour. The resulting solution was concentrated. The crude product was purified by Prep-HPLC (conditions: XBridge Shield RP18 OBD Column, 5 μ m, 19*150 mm; Mobile Phase A: Water (0.1% FA), Mobile Phase B: ACN; Flow rate: 25 mL/minute; Gradient: 0% B to 15% B in 8 minutes; 254/220 nm; R_f : 7.03 minutes) to afford 4-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-7-ethyl-2-methyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one formic acid (10.1 mg) as a white oil. ^1H NMR (400 MHz, Methanol- d_4) δ 8.47 (s, 1H), 7.59 (s, 1H), 6.73 (s, 2H), 4.38 (s, 2H), 3.96 (s, 6H), 3.74 (s, 2H), 3.64 (s, 3H), 2.89 (s, 1 OH), 2.76 (t, $J=5.8$ Hz, 2H), 1.31 (t, $J=7.2$ Hz, 3H). LCMS (ESI) m/z : $[\text{M}+\text{H}]^+=386.30$.

505

Example 21—Preparation of 4-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-2,7-dimethyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one formic acid (Compound B16 formic acid)



compound B17



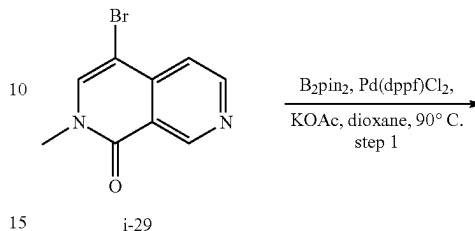
compound B16 formic acid

To the solution of 4-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-2-methyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one (20 mg, 0.056 mmol, 1 equiv) in MeOH (2 mL) was added formaldehyde (16.80 mg, 0.560 mmol, 10 equiv), NaBH₃CN (10.55 mg, 0.168 mmol, 3 equiv). The resulting solution was stirred at room temperature for 1 hour. The resulting solution was concentrated. The crude product was purified by Prep-HPLC (conditions: XBridge Shield RP18 OBD Column, 5 μm, 19*150 mm; Mobile Phase A: Water (0.1% FA), Mobile Phase B: ACN; Flow rate: 25 mL/minute; Gradient: 0% B to 15% B in 8 minutes; 254/220 nm; R_f: 7.12 minutes) to afford 4-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-2,7-dimethyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one formic acid (10.8 mg) as a white oil. ¹H NMR (400 MHz, Methanol-d₄) δ 8.48 (s, 1H), 7.58 (s, 1H), 6.73 (s, 2H), 4.38 (s, 2H), 3.96 (s, 6H), 3.67 (s, 2H), 3.63 (s, 3H), 2.89 (s, 6H), 2.84 (t, J=5.7 Hz, 2H), 2.75 (d, J=5.8 Hz, 2H), 2.66 (s, 3H). LCMS (ESI) m/z: [M+H]⁺=372.25.

506

Example 22—Preparation of 4-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-2-methyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one (Compound B17)

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i-29

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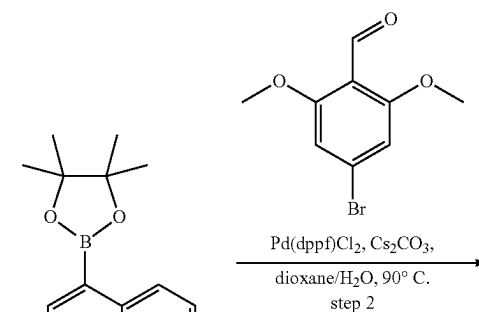
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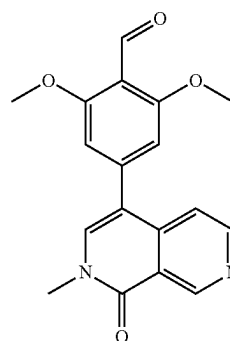
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160

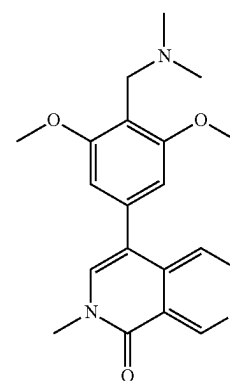
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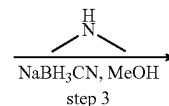
i-30



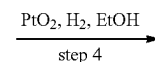
i-31



i-32



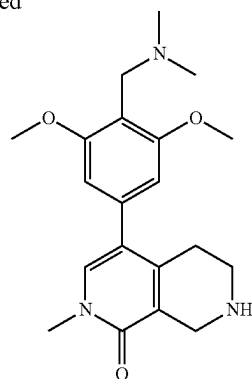
step 3



step 4

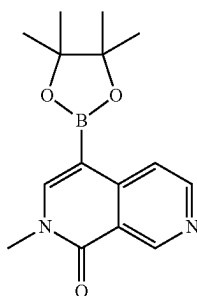
507

-continued



compound B17

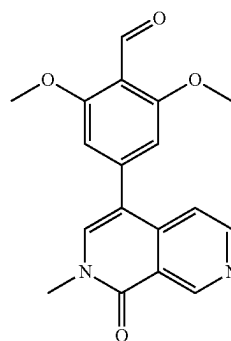
Step 1: Preparation of 2-methyl-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2-dihydro-2,7-naphthyridin-1-one (i-30)



To the solution of 4-bromo-2-methyl-1,2-dihydro-2,7-naphthyridin-1-one (2.7 g, 11.294 mmol, 1 equiv) in dioxane (15 mL) was added 4,4,5,5-tetramethyl-2-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,3,2-dioxaborolane (3.44 g, 13.552 mmol, 1.2 equiv), Pd(dppf)Cl₂ (0.83 g, 1.129 mmol, 0.1 equiv), and AcOK (3.33 g, 33.881 mmol, 3 equiv). The resulting solution was stirred at 90° C. for 2 hours under nitrogen atmosphere. The resulting solution was concentrated. The residue was purified by Flash column chromatography with EtOAc/PE (0-100%) to afford 2-methyl-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2-dihydro-2,7-naphthyridin-1-one (1.62 g, 50.13%) as light yellow solid. LCMS (ESI) m/z: [M+H]⁺=287.

508

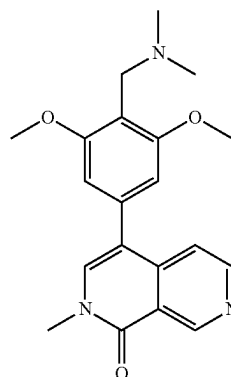
Step 2: Preparation of 2,6-dimethoxy-4-(2-methyl-1-oxo-1,2-dihydro-2,7-naphthyridin-4-yl)benzaldehyde (i-31)



i-31

To the solution of 2-methyl-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2-dihydro-2,7-naphthyridin-1-one (1.62 g, 5.662 mmol, 1 equiv) in dioxane (30 mL) was added 4-bromo-2,6-dimethoxybenzaldehyde (1.39 g, 5.662 mmol, 1 equiv), Pd(dppf)Cl₂ (414.26 mg, 0.566 mmol, 0.1 equiv), Cs₂CO₃ (5.53 g, 16.985 mmol, 3 equiv), H₂O (3 mL). The resulting solution was stirred at 90° C. for 2 hours under nitrogen atmosphere. The solution was concentrated. The residue was purified by Flash column chromatography with EtOAc/PE (0-100%) to give compound 2,6-dimethoxy-4-(2-methyl-1-oxo-1,2-dihydro-2,7-naphthyridin-4-yl)benzaldehyde (1.02 g, 55.55%) as yellow solid. LCMS (ESI) m/z: [M+H]⁺=325.

Step 3: Preparation of 4-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-2-methyl-1,2-dihydro-2,7-naphthyridin-1-one (i-32)



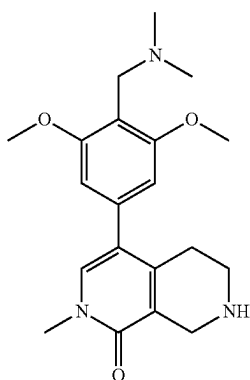
i-32

To the solution of 2,6-dimethoxy-4-(2-methyl-1-oxo-1,2-dihydro-2,7-naphthyridin-4-yl)benzaldehyde (1.20 g, 3.700 mmol, 1.00 equiv) in MeOH (10.00 mL) was added dimethylamine (362.02 mg, 4.440 mmol, 1.20 equiv) and NaBH₃CN (697.52 mg, 11.100 mmol, 3.00 equiv). The resulting solution was stirred at room temperature for 2 hours. The residue was purified by silica gel column chromatography, eluted with CH₂Cl₂/MeOH (20:1) to afford 4-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-2-methyl-1,2-dihydro-2,7-naphthyridin-1-one (1.02 g, 55.55%) as yellow solid. LCMS (ESI) m/z: [M+H]⁺=325.

509

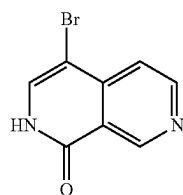
methyl-1,2-dihydro-2,7-naphthyridin-1-one (1 g, 76.48%) as a yellow solid. LCMS (ESI) m/z: [M+H]⁺=354.

Step 4: Preparation of 4-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-2-methyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one (Compound B17)

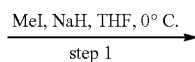


To a stirred solution of 4-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-2-methyl-1,2-dihydro-2,7-naphthyridin-1-one (1 g, 2.829 mmol, 1 equiv) in MeOH (15 mL) was added PtO₂ (1 g, 4.404 mmol, 1.56 equiv). The resulting mixture was stirred for 6 hours at room temperature under hydrogen atmosphere. The resulting mixture was filtered, the filter cake was washed with methanol (3×100 mL). The filtrate was concentrated under reduced pressure. The crude product was purified by Prep-HPLC (conditions: XBridge Prep C18 OBD Column, 5 μm, 19×150 mm; Mobile Phase A: Water (0.1% FA), Mobile Phase B: ACN; Flow rate: 25 mL/minute; Gradient: 15% B to 35% B in 8 minutes; 254/220 nm; R_t: 5.35 minutes) to afford 4-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-2-methyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one (27.2 mg, 2.59%) as a light yellow solid. ¹H NMR (400 MHz, DMSO-d₆) δ 7.54 (s, 1H), 6.55 (s, 2H), 3.77 (s, 6H), 3.60 (s, 2H), 3.46 (s, 3H), 3.36 (s, 3H), 2.80 (s, 2H), 2.40 (s, 2H), 2.11 (s, 6H). LCMS (ESI) m/z: [M+H]⁺=358.30.

Example 23—Preparation of 4-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-2-methyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one (Compound B17)

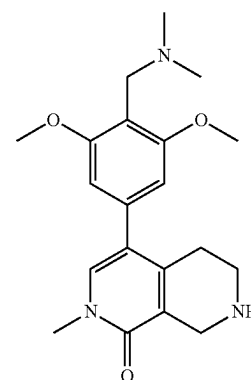
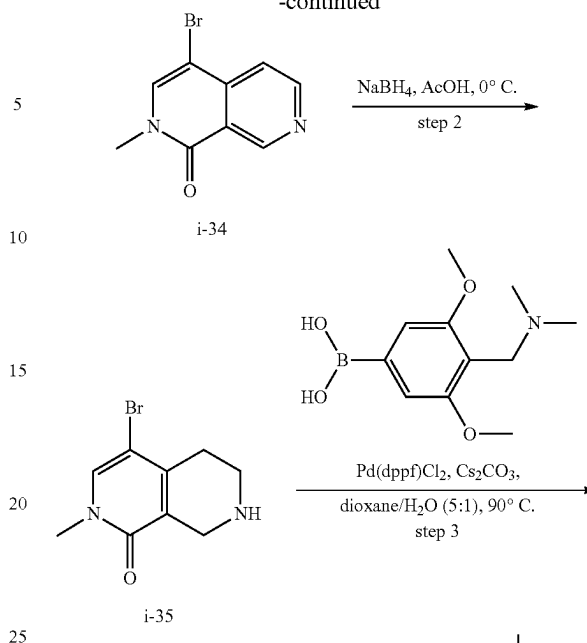


i-33



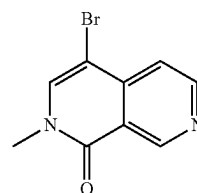
510

-continued



compound B17

Step 1: Preparation of 4-bromo-2-methyl-1,2-dihydro-2,7-naphthyridin-1-one (i-34)

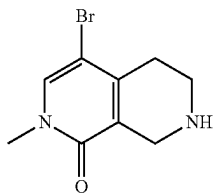


i-34

To a solution of 4-bromo-1,2-dihydro-2,7-naphthyridin-1-one (4.96 g, 22.040 mmol, 1 equiv) and NaH (0.74 g, 30.856 mmol, 1.40 equiv) in DMF (30 mL, 387.653 mmol, 17.59 equiv) was added iodomethane (8.95 g, 63.035 mmol, 2.86 equiv). The resulting solution was stirred at 0° C. for 2 hours under N₂ atmosphere. The resulting mixture was filtered, and the filter cake was washed with ice water to afford 4-bromo-2-methyl-1,2-dihydro-2,7-naphthyridin-1-one (4.5 g, 76.86%), which was used directly without further purification. LCMS (ESI) m/z: [M+H]⁺=239.0, 241.0.

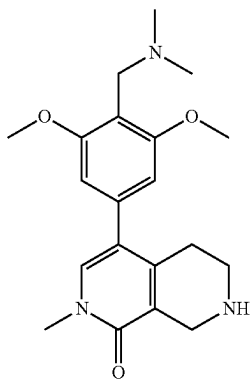
511

Step 2: Preparation of 4-bromo-2-methyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one (i-35)



NaBH_4 (4.43 g, 117.120 mmol, 7 equiv) was slowly added to a solution of 4-bromo-2-methyl-1,2-dihydro-2,7-naphthyridin-1-one (4.00 g, 16.731 mmol, 1.00 equiv) in AcOH (20.00 mL). The resulting solution was stirred at 0° C. for 1 hour. Ammonia was added to the resulting mixture until pH above 7. Then the resulting mixture was extracted with DCM (3×30 mL). The combined organic layers were dried over by saturated sodium sulfate. After filtration, the filtrate was concentrated under reduced pressure. The residue was purified by silica gel column chromatography, eluted with DCM/MeOH (10:1) to afford 4-bromo-2-methyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one (2.2 g, 52.64%) as a yellow solid. LCMS (ESI) m/z : $[\text{M}+\text{H}]^+=243.0$, 245.0.

Step 3: Preparation of 4-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-2-methyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one (Compound B17)

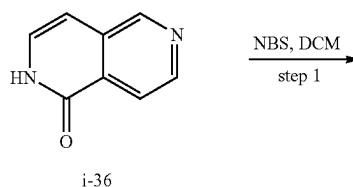


To a solution of 4-bromo-2-methyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one (2.00 g, 8.227 mmol, 1.00 equiv) and [4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]boronic acid (1.97 g, 8.227 mmol, 1.00 equiv) in dioxane (30 mL) and H_2O (6 mL) was added $\text{Pd}(\text{dppf})\text{Cl}_2$ (0.60 g, 0.823 mmol, 0.1 equiv) and Cs_2CO_3 (8.04 g, 24.681 mmol, 3.0 equiv). The resulting solution was stirred at 90° C. for 2 hours under N_2 atmosphere. The residue was purified by reverse flash chromatography (conditions: column, C18 silica gel; mobile phase, MeOH in water, 10% to 50% gradient in 30 minutes; detector, UV 254 nm) to afford 4-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-2-methyl-1,2,5,6,7,8-hexahydro-2,7-naphthyridin-1-one (1.54 g, 52.21%) as a yellow solid. ^1H NMR (300 MHz, Methanol- d_4) δ 7.50 (s, 1H), 6.59 (s, 2H), 3.88 (s, 1H), 3.86 (s, 6H), 3.81 (s, 2H), 3.64 (d, $J=14.9$ Hz, 5H), 3.63 (s, 1H), 3.55

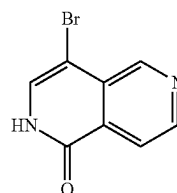
512

(s, 1H), 2.99 (dt, $J=25.5$, 5.8 Hz, 2H), 2.60 (dt, $J=27.3$, 5.7 Hz, 2H), 2.30 (s, 6H). LCMS (ESI) m/z : $[\text{M}+\text{H}]^+=358.25$.

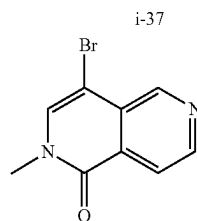
Example 24—Preparation of 8-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-N,6-dimethyl-5-oxo-3,4-dihydro-1H-2,6-naphthyridine-2-carboxamide (Compound B18)



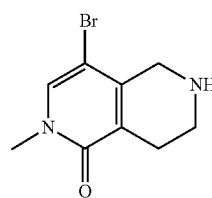
NBS, DCM
step 1



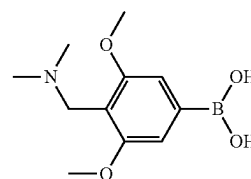
MeI, DMF
step 2



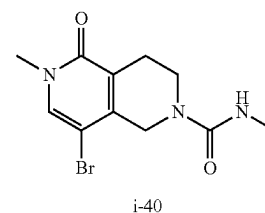
NaBH_4 , AcOH
step 3



TEA, DCM
step 4

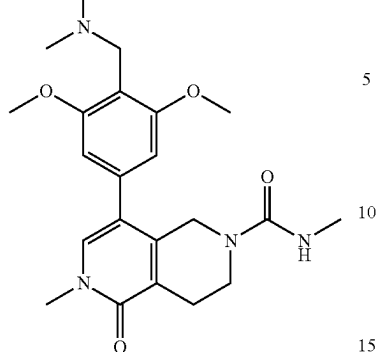


$\text{Pd}(\text{dppf})\text{Cl}_2$, Cs_2CO_3 ,
dioxane/ H_2O (5:1), 90° C.
step 5



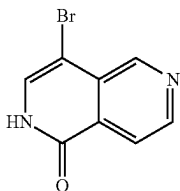
513

-continued



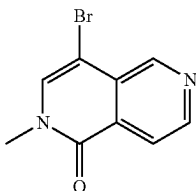
compound B18

Step 1: Preparation of
4-bromo-2H-2,6-naphthyridin-1-one (i-37)



To a stirred solution of 2H-2,6-naphthyridin-1-one (584.00 mg, 3.996 mmol, 1.00 equiv) in DCM (10.00 mL) was added NBS (640.09 mg, 3.596 mmol, 0.9 equiv) in portions at room temperature under air atmosphere. The mixture was stirred for another 1 hour. The reaction mixture was concentrated and purified by silica gel column chromatography, eluted with $\text{CH}_2\text{Cl}_2/\text{MeOH}$ (10:1) to afford 4-bromo-2H-2,6-naphthyridin-1-one (1.2 g, 86.74%) as a light yellow solid. LCMS (ESI) m/z: $[\text{M}+\text{H}]^+=225$, 227.

Step 2: Preparation of
4-bromo-2-methyl-2,6-naphthyridin-1-one (i-38)

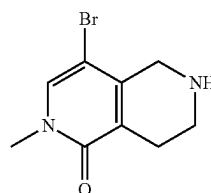


To a stirred solution of 4-bromo-2H-2,6-naphthyridin-1-one (600.00 mg, 2.666 mmol, 1.00 equiv) in DMF (15.00 mL) was added NaH (127.96 mg, 5.332 mmol, 2 equiv) in portions at 0°C . under nitrogen atmosphere. Then MeI (1513.71 mg, 10.665 mmol, 4 equiv) was added drop-wise. The mixture was stirred for another 1 hour at room temperature and quenched with water at 0°C . The product was precipitated by the addition of water. The precipitated solids were collected by filtration and washed with water (2×20

514

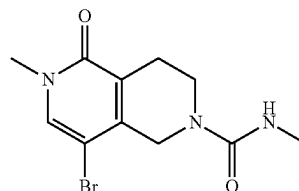
mL). The crude product 4-bromo-2-methyl-2,6-naphthyridin-1-one (369 mg, 57.89%) was used in the next step directly without further purification. LCMS (ESI) m/z: $[\text{M}+\text{H}]^+=239$, 241.

Step 3: Preparation of 4-bromo-2-methyl-5,6,7,8-tetrahydro-2,6-naphthyridin-1-one (i-39)



To a stirred solution of 4-bromo-2-methyl-2,6-naphthyridin-1-one (119.50 mg, 0.500 mmol, 1.00 equiv) in AcOH (5.00 mL) was added NaBH_4 (132.38 mg, 3.499 mmol, 7.00 equiv) in portions at 0°C . under nitrogen atmosphere. The resulting mixture was stirred for 5 minutes at room temperature. Then, the mixture was poured into ice water, basified with ammonium hydroxide, and extracted with CH_2Cl_2 (3×50 mL). The combined organic layers were concentrated under reduced pressure and the resulting crude product 4-bromo-2-methyl-5,6,7,8-tetrahydro-2,6-naphthyridin-1-one (128 mg, 87.43%) was used directly in the next step. LCMS (ESI) m/z: $[\text{M}+\text{H}]^+=243$, 245.

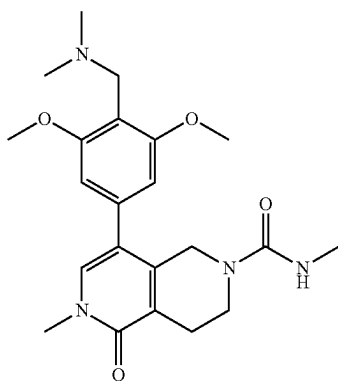
Step 4: Preparation of 8-bromo-N,6-dimethyl-5-oxo-3,4-dihydro-1H-2,6-naphthyridine-2-carboxamide (i-40)



To a stirred solution of 4-bromo-2-methyl-5,6,7,8-tetrahydro-2,6-naphthyridin-1-one (128.00 mg, 0.527 mmol, 1.00 equiv) and N-methylimidazole-1-carboxamide (79.06 mg, 0.632 mmol, 1.20 equiv) in DCM (2.00 mL) was added Et_3N (532.79 mg, 5.265 mmol, 10 equiv). The resulting mixture was stirred for 2 hours at room temperature under air atmosphere. The mixture was concentrated and purified by silica gel column chromatography, eluted with $\text{CH}_2\text{Cl}_2/\text{MeOH}$ (12:1) to afford 8-bromo-N,6-dimethyl-5-oxo-3,4-dihydro-1H-2,6-naphthyridine-2-carboxamide (153 mg, 80.35%). LCMS (ESI) m/z: $[\text{M}+\text{H}]^+=300$, 302.

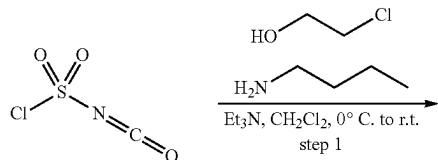
515

Step 5: Preparation of 8-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-N,6-dimethyl-5-oxo-3,4-dihydro-1H-2,6-naphthyridine-2-carboxamide (Compound B18)



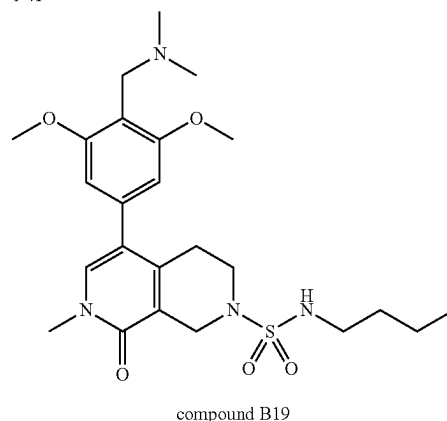
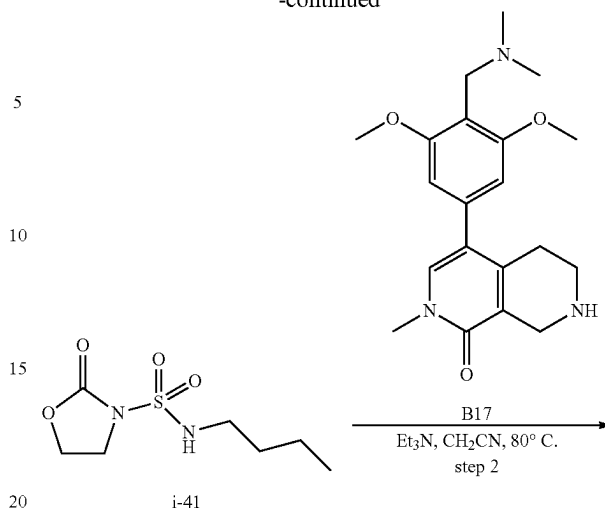
To a solution of 8-bromo-N,6-dimethyl-5-oxo-3,4-dihydro-1H-2,6-naphthyridine-2-carboxamide (153.00 mg, 0.510 mmol, 1.00 equiv) and [[2,6-dimethoxy-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)phenyl]methyl]dimethylamine (163.74 mg, 0.510 mmol, 1 equiv) in dioxane (5.00 mL) and H₂O (1.00 mL) was added Cs₂CO₃ (498.25 mg, 1.529 mmol, 3 equiv) and Pd(dppf)Cl₂·CH₂Cl₂ (41.63 mg, 0.051 mmol, 0.1 equiv). After stirring for 1.5 hours at 100° C. under a nitrogen atmosphere, the resulting mixture was concentrated under reduced pressure. The residue was purified by silica gel column chromatography, eluted with CH₂Cl₂/MeOH (12:1) to afford a crude product, and the crude was further purified by Prep-HPLC (conditions: Xselect CSH F-Phenyl OBD column, 19*250, 5 μm; Mobile Phase A: Water (0.1% FA), Mobile Phase B: ACN; Flow rate: 25 mL/minute; Gradient: 4 B to 22 B in 8 minutes; 254/220 nm; R_T: 6.32 minutes) to afford 8-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-N,6-dimethyl-5-oxo-3,4-dihydro-1H-2,6-naphthyridine-2-carboxamide (49 mg, 23.19%) as a white solid. ¹H NMR (300 MHz, Methanol-d₄) δ 8.56 (brs, 0.5H, FA), 7.52 (s, 1H), 6.74 (s, 2H), 4.39 (s, 2H), 4.28 (s, 2H), 3.96 (s, 6H), 3.67-3.59 (m, 5H), 2.81 (s, 6H), 2.75-2.67 (m, 5H). LCMS (ESI) m/z: [M+H]⁺=415.35.

Example 25—Preparation of N-butyl-5-{4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl}-7-methyl-8-oxo-1,2,3,4,7,8-hexahydro-2,7-naphthyridine-2-sulfonamide (compound B19)

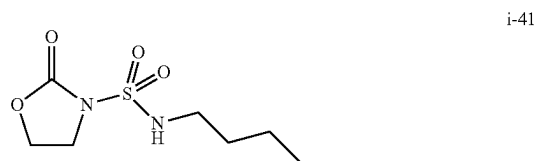


516

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Step 1: Preparation of N-butyl-2-oxo-1,3-oxazolidine-3-sulfonamide (i-41)

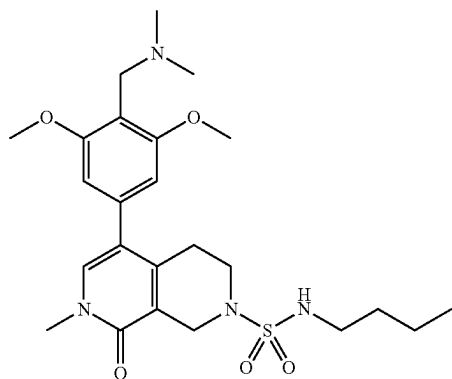


To a solution of chlorosulfonyl isocyanate (539 μL, 6.21 mmol, 1.00 equiv) in dry dichloromethane (8.8 mL) at 0° C. under nitrogen atmosphere was added a solution of 2-chloroethanol (416 μL, 6.21 mmol, 1.00 equiv) in dry dichloromethane (2.6 mL) dropwise over 30 minutes. The reaction mixture was then stirred at 0° C. for an additional 30 minutes. A solution of butylamine (674 μL, 6.83 mmol, 1.10 equiv) and triethylamine (1.88 mL, 13.6 mmol, 2.20 equiv) in dry dichloromethane (5.2 mL) was then added dropwise and the reaction mixture was warmed to room temperature and stirred for 2 hours. Then 1 N aqueous hydrochloric acid was added to adjust the pH to 2. The organic layer was separated and washed with 1 N aqueous hydrochloric acid (1×7 mL) then water (1×7 mL), dried over sodium sulfate, filtered, and concentrated under reduced pressure to afford N-butyl-2-oxo-1,3-oxazolidine-3-sulfonamide (1.49 g,

517

100%) as a white solid. The crude product was used in the next step without further purification.

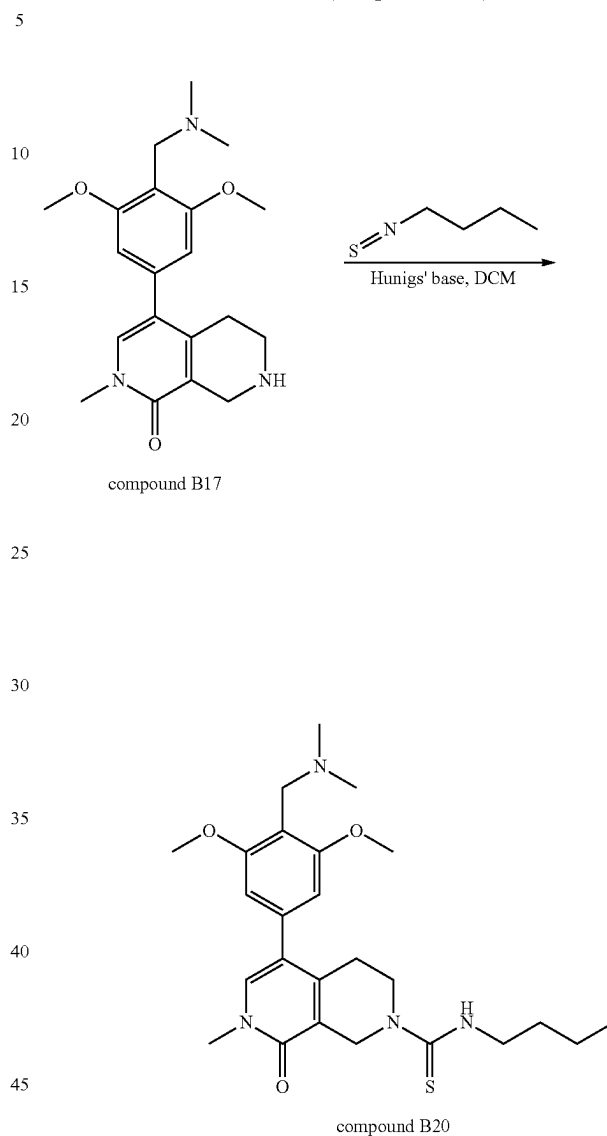
Step 2: Preparation of N-butyl-5-{4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl}-7-methyl-8-oxo-1,2,3,4,7,8-hexahydro-2,7-naphthyridine-2-sulfonamide (compound B19)



To a mixture of 4-4-((dimethylamino)methyl)-3,5-dimethoxyphenyl)-2-methyl-5,6,7,8-tetrahydro-2,7-naphthyridin-1 (2H)-one (48.0 mg, 134.6 μ mol, 1.50 equiv) and N-butyl-2-oxo-1,3-oxazolidine-3-sulfonamide (20 mg, 89.8 μ mol, 1.00 equiv) in dry acetonitrile (0.44 mL) at room temperature was added triethylamine (33.6 μ L, 242 μ mol, 2.70 equiv). The reaction mixture was stirred at 80° C. for 6 hours. The reaction mixture was then cooled down to room temperature and concentrated under reduced pressure. The residue was purified by flash silica gel chromatography, elution gradient 0 to 100% dichloromethane/methanol/ammonium hydroxide (90:10:1) in dichloromethane. Fractions containing the expected product were evaporated to dryness to afford 15.9 mg of impure product. Purification by Prep-HPLC (conditions: waters Xterra C18 Column, 19*100 mm, 10 μ m particles; mobile phase A=0.1% ammonium hydroxide in water, mobile phase B=acetonitrile; flow Rate=40 mL/minute; gradient: 40-82% B in 6 minutes, then a 2 minutes hold at 98% B; wavelength=215 and 254 nm) afforded N-butyl-5-{4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl}-7-methyl-8-oxo-1,2,3,4,7,8-hexahydro-2,7-naphthyridine-2-sulfonamide (3.5 mg, 7.9%) as a white solid. ¹H NMR (400 MHz, DMSO-d₆) δ 7.64 (s, 1H), 7.36 (s, 1H), 6.57 (s, 2H), 4.00 (s, 2H), 3.77 (s, 6H), 3.49 (s, 3H), 3.40 (s, 2H), 3.23 (t, J=5.6 Hz, 2H), 2.89 (t, J=7.0 Hz, 2H), 2.63 (t, J=5.4 Hz, 2H), 2.10 (s, 6H), 1.46-1.37 (m, 2H), 1.34-1.25 (m, 2H), 0.85 (t, J=7.3 Hz, 3H). LCMS (ESI) m/z: [M+H]⁺=493.6.

518

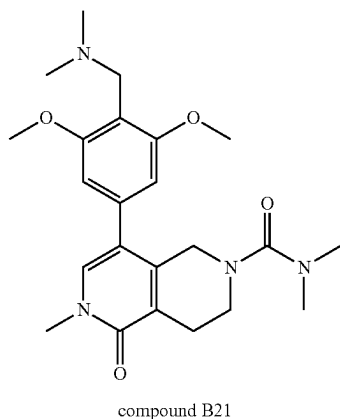
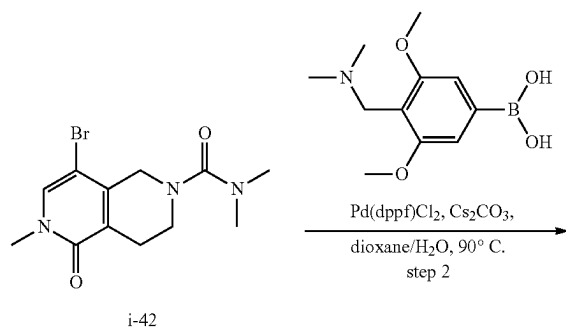
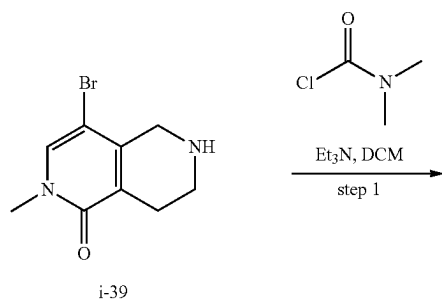
Example 26—Preparation of N-butyl-5-(4-((dimethylamino)methyl)-3,5-dimethoxyphenyl)-7-methyl-8-oxo-3,4,7,8-tetrahydro-2,7-naphthyridine-2 (1H)-carbothioamide (compound B20)



4-4-((dimethylamino)methyl)-3,5-dimethoxyphenyl)-2-methyl-5,6,7,8-tetrahydro-2,7-naphthyridin-1 (2H)-one (25 mg, 0.069 mmol, 1.00 equiv) was dissolved in dichloromethane. Diisopropylethylamine (0.0126 mL, 0.104 mmol, 1.50 equiv) was then added followed by 1-isothiocyanatobutane (0.0133 mL, 0.0768 mmol, 1.10 equiv). The reaction was allowed to stir at room temperature for 1 hour. The solvent was removed under reduced pressure and the resulting oil was purified by prep-HPLC to obtain N-butyl-5-(4-((dimethylamino)methyl)-3,5-dimethoxyphenyl)-7-methyl-8-oxo-3,4,7,8-tetrahydro-2,7-naphthyridine-2 (1H)-carbothioamide (30 mg, 47%). ¹H NMR (400 MHz, DMSO-d₆) δ 8.16 (s, 1H), 7.80 (t, J=5.3 Hz, 1H), 7.65 (s, 1H), 6.59 (s, 2H), 4.59 (s, 2H), 3.87 (q, J=5.1, 4.7 Hz, 2H), 3.77 (s, 6H), 3.51 (d, J=15.7 Hz, 7H), 3.15 (s, 1H), 2.57 (t, J=5.6 Hz, 2H), 2.19 (s, 6H), 1.52 (tt, J=8.0, 6.6 Hz, 2H), 1.27 (h, J=7.3 Hz, 2H), 0.87 (t, J=7.3 Hz, 3H). LCMS (ESI) m/z: [M+H]⁺=473.4.

519

Example 27—Preparation of 8-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-N,N,6-trimethyl-5-oxo-3,4-dihydro-1H-2,6-naphthyridine-2-carboxamide (compound B21)

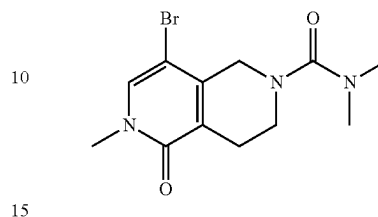


520

Step 1: preparation of 8-bromo-N,N,6-trimethyl-5-oxo-3,4-dihydro-1H-2,6-naphthyridine-2-carboxamide (i-42)

5

i-42

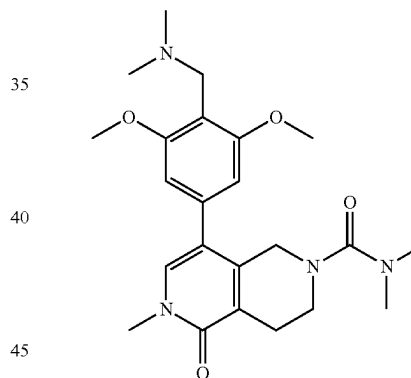


Using the same procedure as described in Example 24, step 4 and substituting with dimethylcarbamyl chloride (25.8 mg, 0.240 mmol, 1.20 equiv) afforded 8-bromo-N,N,6-trimethyl-5-oxo-3,4-dihydro-1H-2,6-naphthyridine-2-carboxamide (73 mg, 94%) as an off-white solid. LCMS (ESI) m/z: [M+H]⁺=314.

Step 2: Preparation of 8-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-N,N,6-trimethyl-5-oxo-3,4-dihydro-1H-2,6-naphthyridine-2-carboxamide (compound B21)

30

compound B21



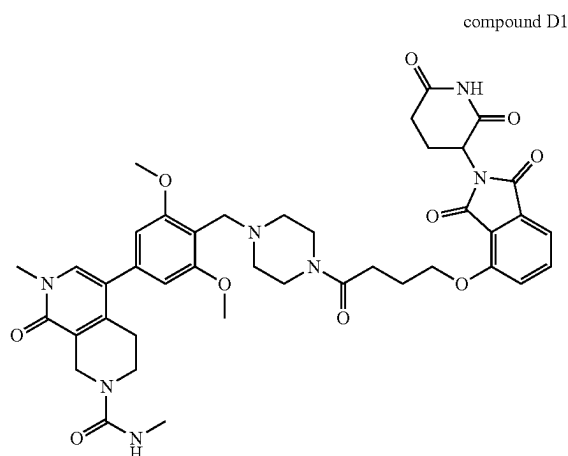
Using the same procedure as described in Example 23, step 2 and substituting with 8-bromo-N,N,6-trimethyl-5-oxo-3,4-dihydro-1H-2,6-naphthyridine-2-carboxamide (64.2 mg, 0.204 mmol, 1.00 equiv) afforded 8-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-N,N,6-trimethyl-5-oxo-3,4-dihydro-1H-2,6-naphthyridine-2-carboxamide (26.1 mg, 29%) as an off-white solid. ¹H NMR (400 MHz, DMSO-d₆) δ 7.57 (s, 1H), 6.60 (s, 2H), 4.00 (s, 2H), 3.78 (s, 6H), 3.48 (s, 3H), 3.46 (s, 2H), 3.37 (t, J=5.9 Hz, 2H), 2.70 (s, 6H), 2.57 (t, J=6.0 Hz, 2H), 2.14 (s, 6H). LCMS (ESI) m/z: [M+H]⁺=429.35

Compound B22: LCMS 482.2.

Compound B23: LCMS 511.2; ¹H NMR (400 MHz, DMSO-d₆) δ 7.60 (s, 1H), 6.55 (s, 3H), 4.19 (s, 2H), 4.06 (d, J=5.3 Hz, 1H), 3.75 (s, 7H), 3.48 (d, J=8.1 Hz, 6H), 3.38 (t, J=5.5 Hz, 2H), 3.24 (s, 1H), 3.15 (d, J=4.6 Hz, 2H), 2.96 (s, 0H), 2.57 (d, J=4.2 Hz, 3H), 2.37 (s, 4H), 2.13 (s, 4H).

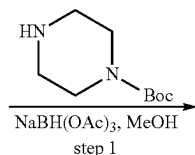
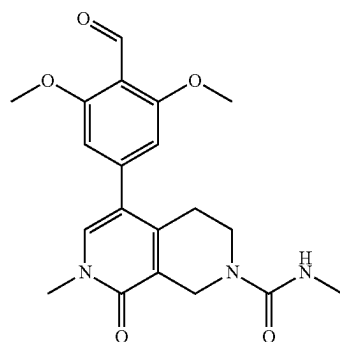
521

Example 28—Preparation of 5-(4-[[4-(4-[[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]oxy]butanoyl)piperazin-1-yl]methyl]-3,5-dimethoxyphenyl)-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (Compound D1)



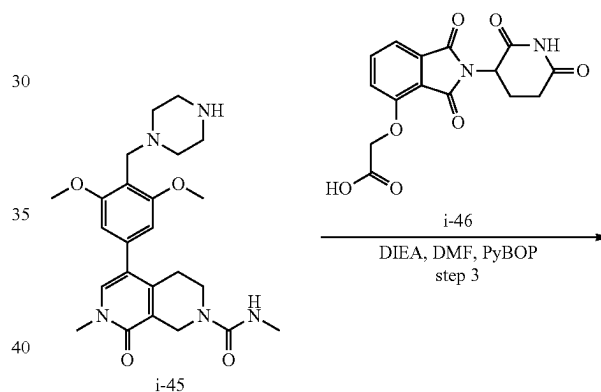
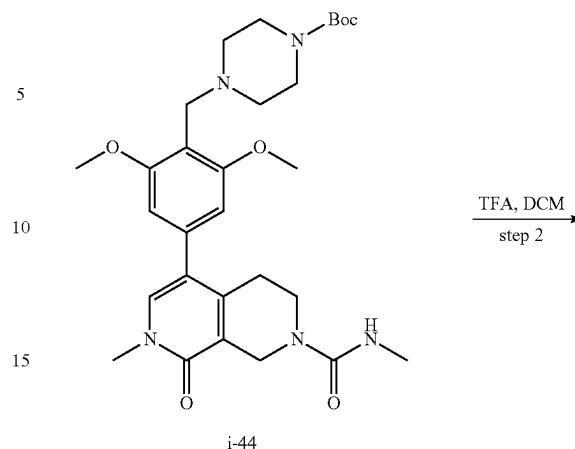
Compound D1 was prepared in a similar manner to the preparation of compound D2. PyBOP in step 3 was substituted with HATU. 5-(4-[[4-(4-[[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]oxy]butanoyl)piperazin-1-yl]methyl]-3,5-dimethoxyphenyl)-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (21.2 mg) was obtained as a white solid. ¹H NMR (300 MHz, Methanol-d₄) δ 7.80 (t, J=7.9 Hz, 1H), 7.57 (s, 1H), 7.48 (d, J=7.8 Hz, 2H), 6.64 (s, 2H), 5.11 (dd, J=12.1, 5.4 Hz, 1H), 4.36 (s, 2H), 4.30 (t, J=5.7 Hz, 2H), 4.00 (s, 2H), 3.88 (s, 6H), 3.80-3.69 (m, 4H), 3.64 (s, 3H), 3.53 (d, J=5.7 Hz, 2H), 2.94-2.81 (m, 5H), 2.78 (s, 4H), 2.72 (t, J=7.1 Hz, 3H), 2.67-2.59 (m, 2H), 2.22-2.09 (m, 3H). LCMS (ESI) m/z: [M+H]⁺=798.40.

Example 29—Preparation of 5-(4-[[4-(2-[[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]oxy]acetyl)piperazin-1-yl]methyl]-3,5-dimethoxyphenyl)-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (Compound D2)



522

-continued



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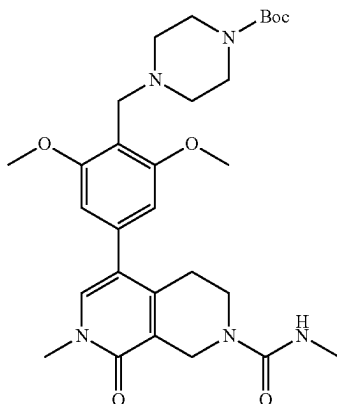
60

65

compound D2

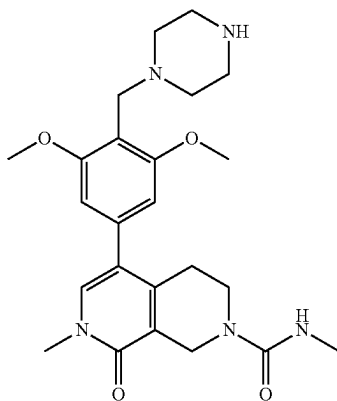
523

Step 1: Preparation of tert-butyl 4-([2,6-dimethoxy-4-[2-methyl-7-(methylcarbamoyl)-1-oxo-6,8-dihydro-5H-2,7-naphthyridin-4-yl]phenyl]methyl)piperazine-1-carboxylate (i-44)



A mixture of 5-(4-formyl-3,5-dimethoxyphenyl)-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (400.00 mg, 1.038 mmol, 1.00 equiv) and tert-butyl piperazine-1-carboxylate (193.30 mg, 1.038 mmol, 1.00 equiv) in MeOH (2 mL) was stirred for 30 minutes at room temperature under air atmosphere. To the above mixture was added NaBH(AcO)₃ (439.92 mg, 2.076 mmol, 2.00 equiv) in portions for 2 hours at room temperature. The resulting mixture was concentrated under vacuum. The residue was purified by reverse flash chromatography (conditions: column, C18 silica gel; mobile phase, MeOH in water, 10% to 50% gradient in 50 minutes; detector, UV 254 nm). This resulted in tert-butyl 4-([2,6-dimethoxy-4-[2-methyl-7-(methylcarbamoyl)-1-oxo-6,8-dihydro-5H-2,7-naphthyridin-4-yl]phenyl]methyl)piperazine-1-carboxylate (300 mg, 52.02%) as a yellow oil. LCMS (ESI) m/z: [M+H]⁺=556.

Step 2: Preparation of 5-[3,5-dimethoxy-4-(piperazin-1-ylmethyl)phenyl]-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (i-45)



524

A solution of TFA (1.00 mL) and tert-butyl 4-([2,6-dimethoxy-4-[2-methyl-7-(methylcarbamoyl)-1-oxo-6,8-dihydro-5H-2,7-naphthyridin-4-yl]phenyl]methyl)piperazine-1-carboxylate (10.00 mg, 0.018 mmol, 1.00 equiv) in DCM (2.00 mL) was stirred for 1 hour at room temperature under air atmosphere. The reaction mixture was concentrated under vacuum. The crude product mixture was used in the next step directly without further purification. LCMS (ESI) m/z: [M+H]⁺=456.

Step 3: 5-(4-([4-(2-([2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]oxy)acetyl)piperazin-1-yl]methyl)-3,5-dimethoxyphenyl)-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (Compound D2)

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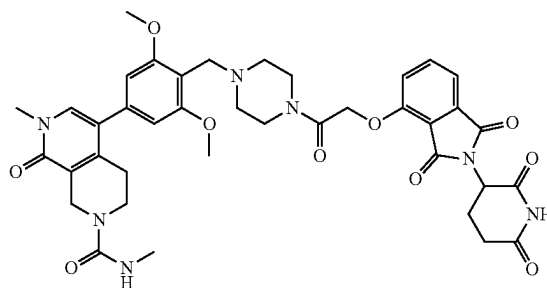
50

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65

compound D2



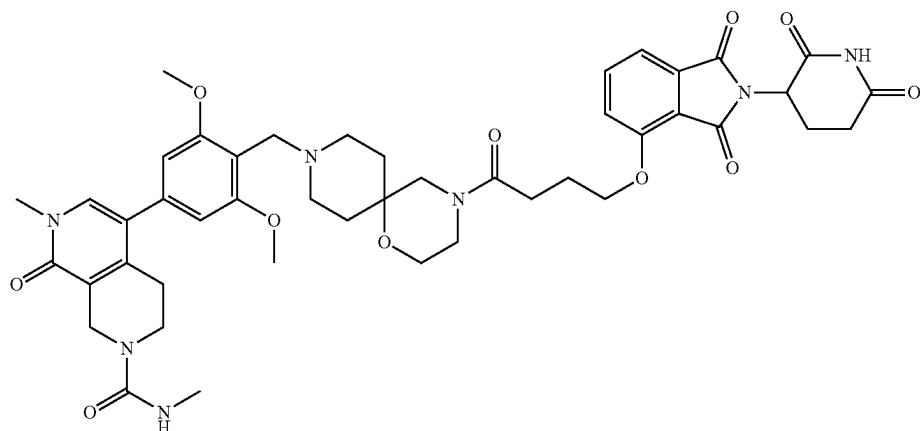
A mixture of [[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]oxy]acetic acid (36.47 mg, 0.110 mmol, 1.00 equiv), DIEA (70.93 mg, 0.549 mmol, 5.00 equiv), PyBOP (114.23 mg, 0.220 mmol, 2.00 equiv), and [[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]oxy]acetic acid (36.47 mg, 0.110 mmol, 1.00 equiv) in DMF (2 mL) was stirred for 2 hours at room temperature under air atmosphere. The crude product was purified by Prep-HPLC (conditions: SunFire C18 OBD Prep Column, 100 Å, 5 µm, 19 mm×250 mm; Mobile Phase A: water (0.1% FA), Mobile Phase B: ACN; Flow rate: 25 mL/minute; Gradient: 7 B to 20 B in 12 minutes; 254 nm; R_t: 10.95 minutes) to afford 5-(4-([4-(2-([2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]oxy)acetyl)piperazin-1-yl]methyl)-3,5-dimethoxyphenyl)-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (21.2 mg) as a white solid. ¹H NMR (300 MHz, Methanol-d₄) δ 7.85-7.72 (m, 1H), 7.61-7.49 (m, 2H), 7.48-7.35 (m, 1H), 6.74 (s, 2H), 5.13 (dd, J=12.2, 5.4 Hz, 3H), 5.04 (d, J=4.2 Hz, 1H), 4.66-4.42 (m, 3H), 4.40-4.18 (m, 3H), 3.96 (s, 6H), 3.72-3.62 (m, 4H), 3.61-3.45 (m, 5H), 2.96-2.70 (m, 7H), 2.69-2.57 (m, 2H), 2.22-2.09 (m, 1H). LCMS (ESI) m/z: [M+H]⁺=770.55.

525

526

Example 30—Preparation of 5-(4-[[4-(4-[[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]oxy]butanoyl)-1-oxa-4,9-diazaspiro[5.5]undecan-9-yl]methyl]-3,5-dimethoxyphenyl)-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (Compound D3)

5



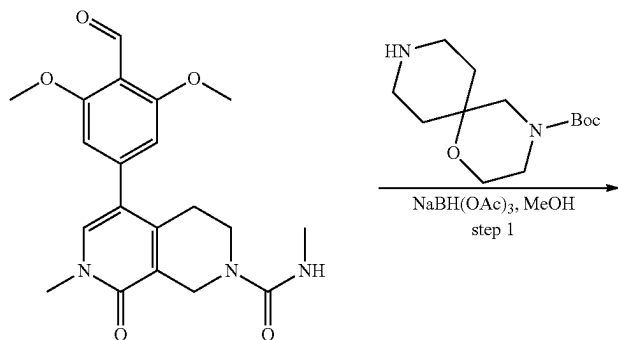
compound D3

Compound D3 was prepared in a similar manner to the preparation of compound D4. Compound D3 (19 mg, 19.2%) was obtained as an off-white solid. ¹H NMR (300 MHz, Methanol-d₄) δ 7.84-7.76 (m, 1H), 7.60 (d, J=7.8 Hz, 1H), 7.51-7.43 (m, 2H), 6.71 (d, J=5.0 Hz, 2H), 5.17-5.06 (m, 1H), 4.41-4.20 (m, 6H), 3.99-3.86 (m, 6H), 3.83-3.67 (m, 4H), 3.66-3.62 (m, 3H), 3.59-3.46 (m, 4H), 2.96-2.69 (m, 8H), 2.69-2.51 (m, 3H), 2.43-1.91 (m, 6H), 1.88-1.61 (m, 2H). LCMS (ESI) m/z: [M+H]⁺=868.80.

40

Example 31—Preparation of 5-(4-[[4-(5-[[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]oxy]pentanoyl)-1-oxa-4,9-diazaspiro[5.5]undecan-9-yl]methyl]-3,5-dimethoxyphenyl)-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide formic acid (Compound D4)

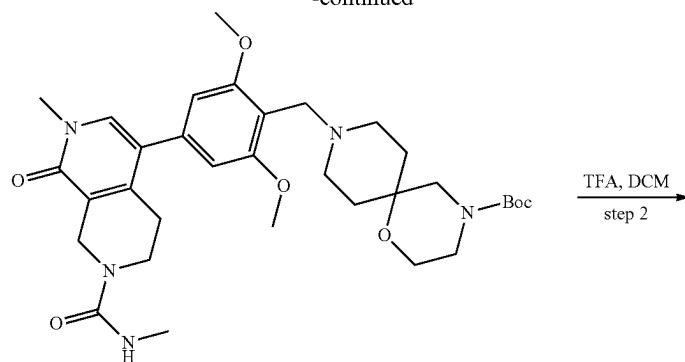
45



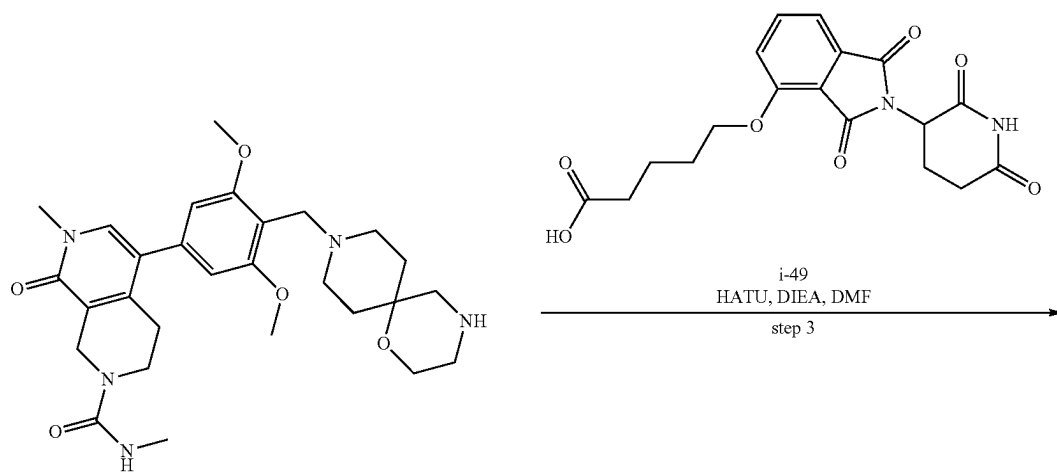
527

528

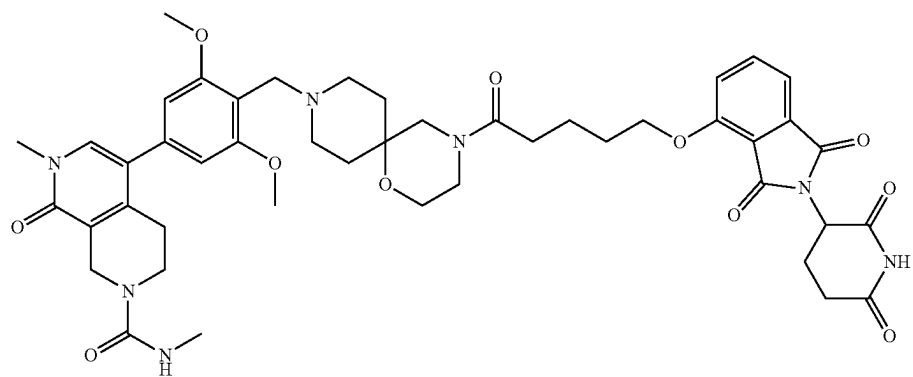
-continued



i-47



i-48

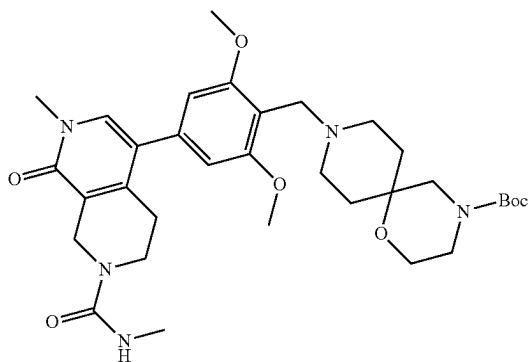


compound D4

529

Step 1: Preparation of tert-butyl 9-([2,6-dimethoxy-4-[2-methyl-7-(methylcarbamoyl)-1-oxo-6,8-dihydro-5H-2,7-naphthyridin-4-yl]phenyl]methyl)-1-oxa-4,9-diazaspiro[5.5]undecane-4-carboxylate (i-47)

5

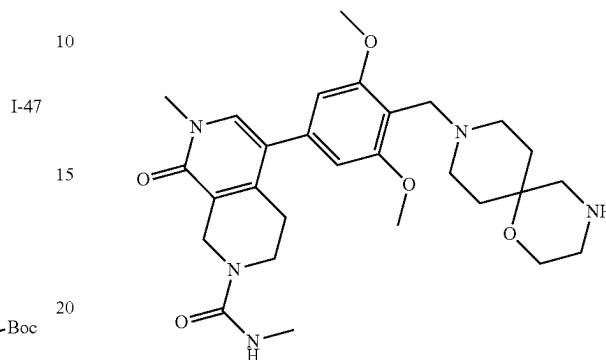


Using a similar procedure as described in Example 29, step 1 and substituting with tert-butyl 1-oxa-4,9-diazaspiro[5.5]undecane-4-carboxylate (93.1 mg, 0.363 mmol, 1 equiv) afforded tert-butyl 9-([2,6-dimethoxy-4-[2-methyl-7-(methylcarbamoyl)-1-oxo-6,8-dihydro-5H-2,7-naphthyridin-4-yl]phenyl]methyl)-1-oxa-4,9-diazaspiro[5.5]undecane-4-carboxylate (290 mg, 93.1%) as a light brown oil. LCMS (ESI) m/z: [M+H]⁺=626.

530

Step 2: Preparation of 5-(3,5-dimethoxy-4-[1-oxa-4,9-diazaspiro[5.5]undecan-9-ylmethyl]phenyl)-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (i-48)

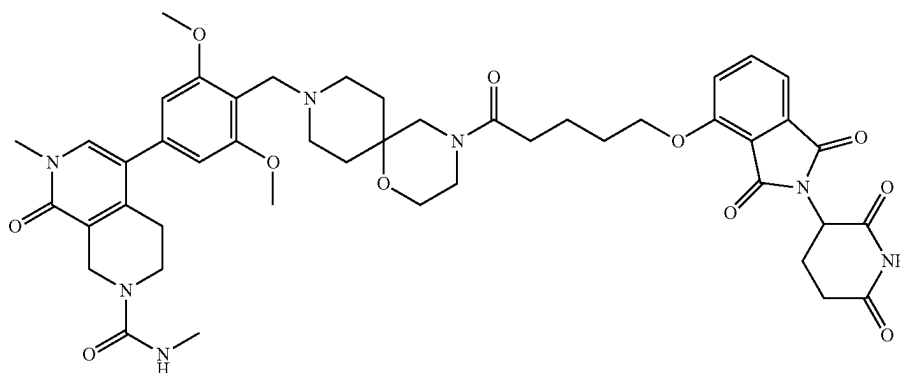
i-48



Using a similar procedure as described in Example 29, step 2 afforded 5-(3,5-dimethoxy-4-[1-oxa-4,9-diazaspiro[5.5]undecan-9-ylmethyl]phenyl)-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (114 mg, 46.8%) as a light brown solid. LCMS (ESI) m/z: [M+H]⁺=526.

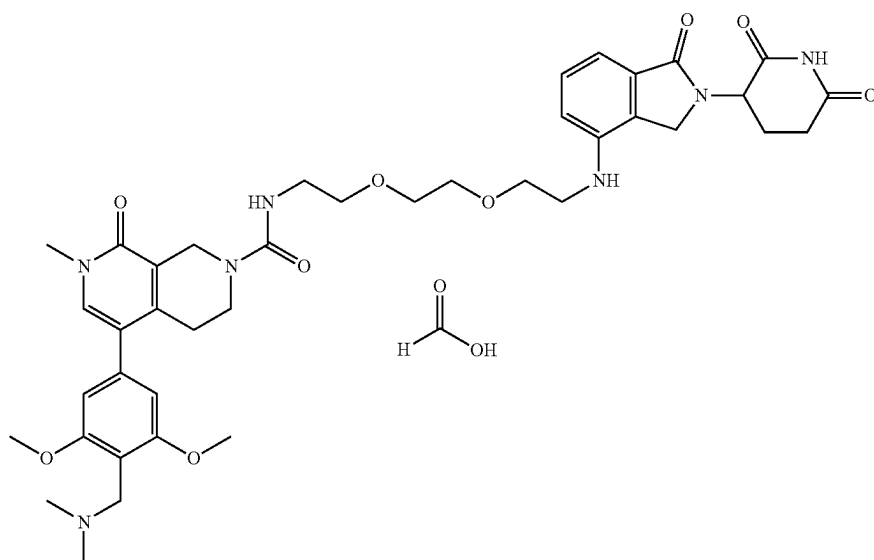
Step 3: Preparation of 5-(4-[[4-(5-[[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]oxy]pentanoyl)-1-oxa-4,9-diazaspiro[5.5]undecan-9-yl]methyl]-3,5-dimethoxyphenyl)-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide formic acid (Compound D4)

compound D4



Using a similar procedure as described in Example 29, step 3 and substituting with 5-[[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]oxy]pentanoic acid (28.5 mg, 0.076 mmol, 1.00 equiv) and 5-(3,5-dimethoxy-4-[1-oxa-4,9-diazaspiro[5.5]undecan-9-ylmethyl]phenyl)-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (40 mg, 0.076 mmol, 1.00 equiv) afforded 5-(4-[[4-(5-[[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]oxy]pentanoyl)-1-oxa-4,9-diazaspiro[5.5]undecan-9-yl]methyl]-3,5-dimethoxyphenyl)-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide formic acid (23 mg, 32.6%) as an off-white solid. ¹H NMR (300 MHz, Methanol-d₄) δ 8.55 (brs, 0.6H, formic acid), 7.79 (t, J=7.9 Hz, 1H), 7.58 (s, 1H), 7.46 (d, J=8.7 Hz, 2H), 6.70 (d, J=5.7 Hz, 2H), 5.15-5.07 (m, 1H), 4.39-4.25 (m, 6H), 3.93 (d, J=5.5 Hz, 6H), 3.82-3.72 (m, 2H), 3.64 (s, 5H), 3.59-3.50 (m, 4H), 3.31-3.05 (m, 4H), 2.93-2.83 (m, 1H), 2.78 (s, 3H), 2.77-2.56 (m, 6H), 2.21-2.03 (m, 3H), 2.01-1.68 (m, 6H). LCMS (ESI) m/z: [M+H]⁺=882.60.

Example 32—Preparation of 5-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-N-[2-[2-(2-[[2-(2,6-dioxopiperidin-3-yl)-1-oxo-3H-isindol-4-yl]amino]ethoxy)ethoxy]ethyl]-7-methyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide formic acid (Compound D5 formic acid)

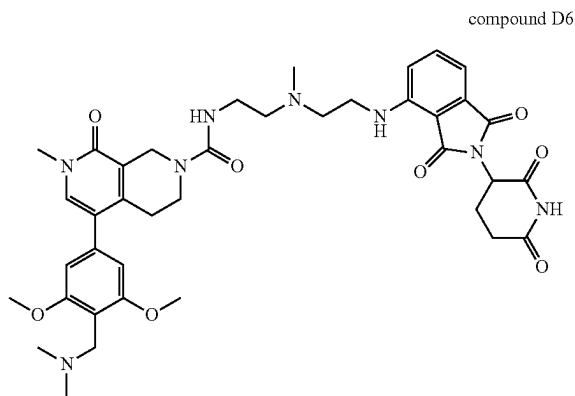


compound D5 formic acid

533

Compound D5 was prepared in a similar manner to the preparation of compound D21. Compound D5 formic acid (12.8 mg, 21.5%) was obtained as a green solid. ¹H NMR (300 MHz, Methanol-d₄) δ 8.56 (brs, 1.7H, FA), 7.55 (s, 1H), 7.24 (t, J=7.7 Hz, 1H), 7.09 (d, J=7.6 Hz, 1H), 6.91 (d, J=7.9 Hz, 1H), 6.71 (s, 2H), 5.19 (dd, J=13.4, 5.1 Hz, 1H), 4.39-4.27 (m, 5H), 4.09-3.98 (m, 2H), 3.94 (s, 6H), 3.67-3.50 (m, 14H), 3.40 (t, J=5.4 Hz, 2H), 3.04-2.91 (m, 2H), 2.85 (s, 6H), 2.61 (s, 2H), 2.53-2.39 (m, 1H), 2.24-2.12 (m, 1H). LCMS (ESI) m/z: [M-H]⁺=774.37.

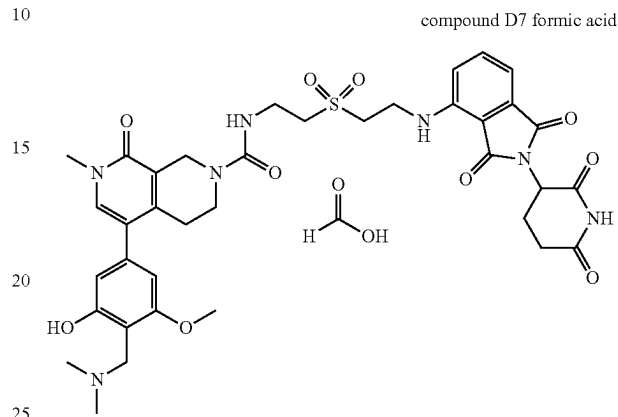
Example 33—Preparation of 5-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-N-[2-[[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]amino]ethyl](methylamino)ethyl]-7-methyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (Compound D6)



Compound D6 was prepared in a similar manner to the preparation of compound D21. Compound D6 (2.1 mg, 3.45%) was obtained as a green solid. ¹H NMR (300 MHz, Acetonitrile-d₃) δ 9.17 (s, 1H), 7.58 (t, J=7.7 Hz, 1H), 7.43 (s, 1H), 7.15 (d, J=8.4 Hz, 1H), 7.06 (d, J=7.0 Hz, 1H), 6.67-6.47 (m, 4H), 4.95 (dd, J=12.2, 5.2 Hz, 1H), 4.35-4.17 (m, 4H), 3.87 (s, 6H), 3.77 (d, J=5.8 Hz, 2H), 3.55 (s, 4H), 3.44-3.24 (m, 5H), 2.92 (s, 3H), 2.77 (s, 6H), 2.73-2.60 (m, 3H), 2.54 (s, 2H), 2.30-2.22 (m, 1H), 2.14-2.02 (m, 1H). LCMS (ESI) m/z: [M-H]⁺=757.36.

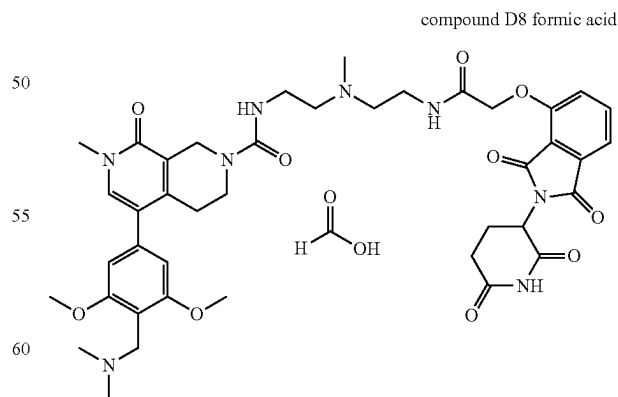
534

Example 34—Preparation of 5-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-N-[2-[[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]amino]ethanesulfonyl]ethyl]-7-methyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide formic acid (Compound D7 formic acid)



Compound D7 was prepared in a similar manner to the preparation of compound D21. Compound D7 formic acid (13.1 mg, 22.5%) was obtained as a green solid. ¹H NMR (300 MHz, Methanol-d₄) δ 8.56 (brs, 1H, FA), 7.66-7.54 (m, 2H), 7.12 (dd, J=17.2, 7.8 Hz, 2H), 6.69 (s, 2H), 5.05 (dd, J=12.6, 5.4 Hz, 1H), 4.35 (s, 2H), 4.24 (s, 2H), 3.94 (s, 6H), 3.89 (t, J=6.4 Hz, 2H), 3.70 (t, J=6.3 Hz, 2H), 3.63 (s, 3H), 3.49 (t, J=6.3 Hz, 4H), 3.42-3.37 (m, 2H), 2.86-2.68 (m, 9H), 2.67-2.60 (m, 3H), 2.13-2.01 (m, 1H). LCMS (ESI) m/z: [M+H]⁺=792.45.

Example 35—Preparation of N-(2-[[2-(5-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-7-methyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxylamino)ethyl](methylamino)ethyl]-2-[[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]oxy]acetamide formic acid (Compound D8 formic acid)



Compound D7 was prepared in a similar manner to the preparation of compound D21. Compound D8 formic acid (10.9 mg, 19.24%) was obtained as a white solid. ¹H NMR (300 MHz, Methanol-d₄) δ 8.42 (brs, 2H, FA), 7.72-7.63 (m,

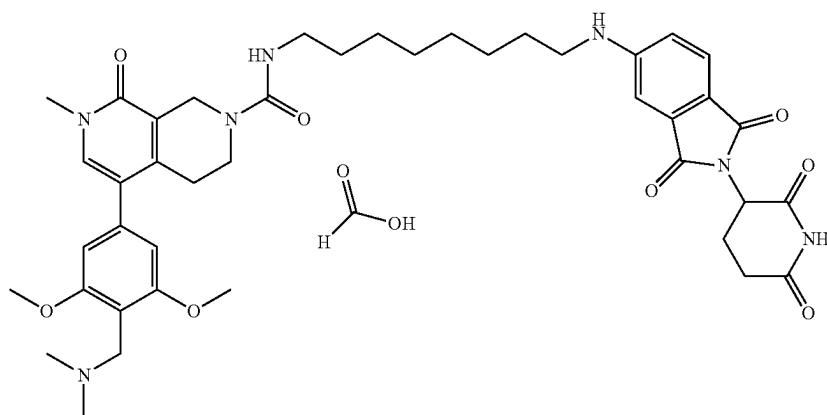
535

1H), 7.51 (s, 1H), 7.43-7.37 (m, 2H), 6.67 (s, 2H), 5.12 (dd, J=12.7, 5.4 Hz, 1H), 4.81 (s, 2H), 4.39 (s, 2H), 4.15 (s, 2H), 3.98 (s, 6H), 3.63-3.55 (m, 5H), 3.42 (s, 3H), 2.97-2.85 (m, 11H), 2.82-2.63 (m, 3H), 2.59 (s, 3H), 2.54-2.39 (m, 2H), 2.21-2.09 (m, 1H). LCMS (ESI) m/z: [M+H]⁺=815.36.

5

Example 36—Preparation of 5-(4-((dimethylamino) methyl)-3,5-dimethoxyphenyl)-N-(8-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino) octyl)-7-methyl-8-oxo-3,4,7,8-tetrahydro-2,7-naphthyridine-2 (1H)-carboxamide formic acid (Compound D9 formic acid)

10



compound D9 formic acid

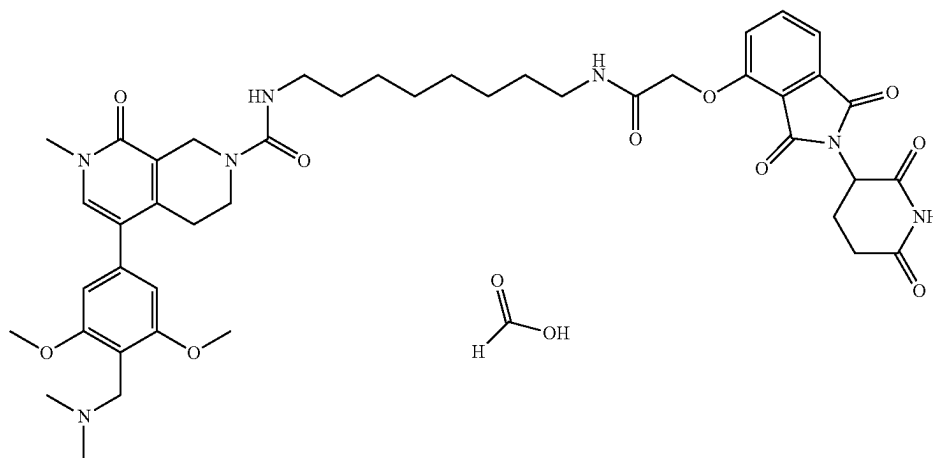
Compound D9 was prepared in a similar manner to the preparation of compound D21. Compound D9 formic acid (9.7 mg, 23.93%) was obtained as a light yellow solid. LCMS (ESI) m/z: [M+H]⁺=784.60. ¹H NMR (300 MHz, Methanol-d₄) δ 8.56 (br s, 1H, FA), 7.61-7.50 (m, 2H), 6.96 (d, J=2.1 Hz, 1H), 6.82 (dd, J=8.4, 2.1 Hz, 1H), 6.69 (s, 2H), 5.05 (dd, J=12.4, 5.5 Hz, 1H), 4.36 (s, 2H), 4.21 (s, 2H), 3.93 (s, 6H), 3.63 (s, 3H), 3.54 (t, J=5.6 Hz, 2H), 3.25-3.15 (m, 4H), 2.96-2.81 (m, 1H), 2.75 (s, 8H), 2.63 (t, J=5.2 Hz, 2H),

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2.15-2.03 (m, 1H), 1.73-1.61 (m, 2H), 1.60-1.49 (m, 2H), 1.48-1.33 (m, 8H).

Example 37—Preparation of 5-(4-((dimethylamino) methyl)-3,5-dimethoxyphenyl)-N-(8-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy) acetamido)octyl)-7-methyl-8-oxo-3,4,7,8-tetrahydro-2,7-naphthyridine-2 (1H)-carboxamide formic acid (Compound D10 formic acid)

40



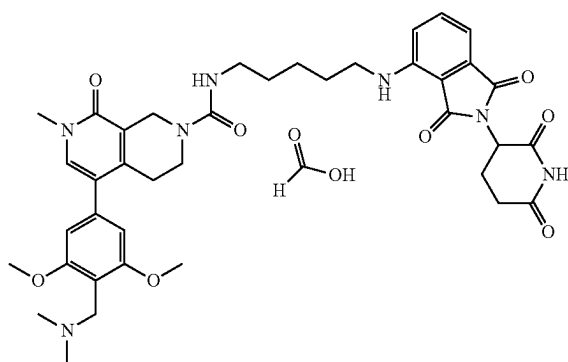
compound D10 formic acid

537

Compound D10 was prepared in a similar manner to the preparation of compound D21. Compound D10 formic acid (8.5 mg, 21.8%) was obtained as a white solid. LCMS (ESI) m/z : $[M+H]^+=842.65$. 1H NMR (300 MHz, Methanol- d_4) δ 8.56 (br s, 0.6H, FA), 7.82 (dd, $J=8.4$, 7.4 Hz, 1H), 7.61-7.52 (m, 2H), 7.44 (d, $J=8.4$ Hz, 1H), 6.71 (s, 2H), 5.15 (dd, $J=12.4$, 5.5 Hz, 1H), 4.77 (s, 2H), 4.36 (s, 2H), 4.27 (s, 2H), 3.94 (s, 6H), 3.64 (s, 3H), 3.54 (t, $J=5.6$ Hz, 2H), 3.30 (s, 2H), 3.19 (t, $J=7.0$ Hz, 2H), 2.98-2.83 (m, 1H), 2.83-2.70 (m, 8H), 2.64 (t, $J=5.4$ Hz, 2H), 2.22-2.10 (m, 1H), 1.65-1.47 (m, 4H), 1.36 (s, 8H).

Example 38—Preparation of 5-(4-((dimethylamino)methyl)-3,5-dimethoxyphenyl)-N-(5-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl)amino)pentyl)-7-methyl-8-oxo-3,4,7,8-tetrahydro-2,7-naphthyridine-2 (1H)-carboxamide formic acid (Compound D11 formic acid)

compound D11 formic acid

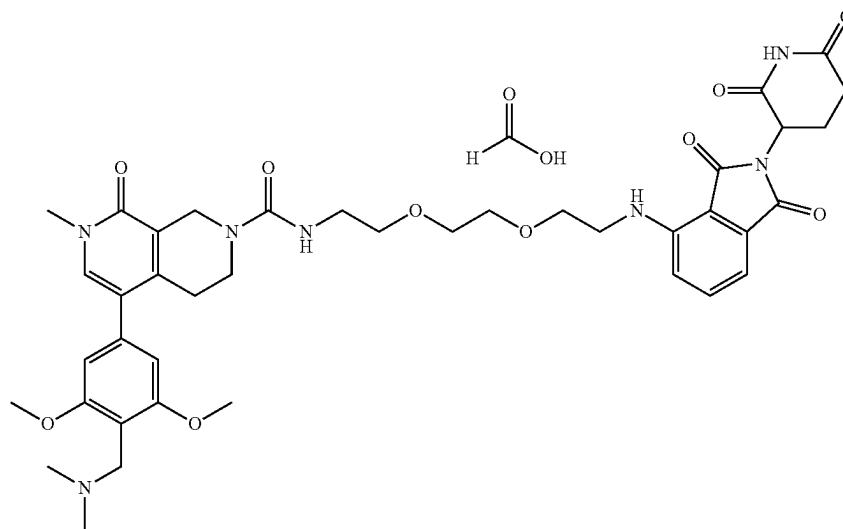


538

Compound D11 was prepared in a similar manner to the preparation of compound D21. Compound D11 formic acid (8.7 mg, 20.7%) was obtained as a light yellow solid. 1H NMR (300 MHz, Methanol- d_4) δ 8.56 (br s, 1H, FA), 7.60-7.52 (m, 2H), 7.04 (dd, $J=9.4$, 7.8 Hz, 2H), 6.71 (s, 2H), 5.04 (dd, $J=12.2$, 5.4 Hz, 1H), 4.36 (s, 2H), 4.26 (s, 2H), 3.94 (s, 6H), 3.64 (s, 3H), 3.53 (t, $J=5.6$ Hz, 2H), 3.36 (t, $J=6.8$ Hz, 2H), 3.25 (t, $J=6.8$ Hz, 2H), 2.87-2.72 (m, 8H), 2.70-2.59 (m, 3H), 2.15-2.03 (m, 1H), 1.78-1.66 (m, 2H), 1.66-1.56 (m, 2H), 1.56-1.45 (m, 2H). LCMS (ESI) m/z : $[M+H]^+=742.55$.

Example 39—Preparation of 5-[4-((dimethylamino)methyl)-3,5-dimethoxyphenyl]-N-[2-[2-([2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]amino)ethoxy]ethoxy]ethyl]-7-methyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide formic acid (Compound D12 formic acid)

compound D12 formic acid



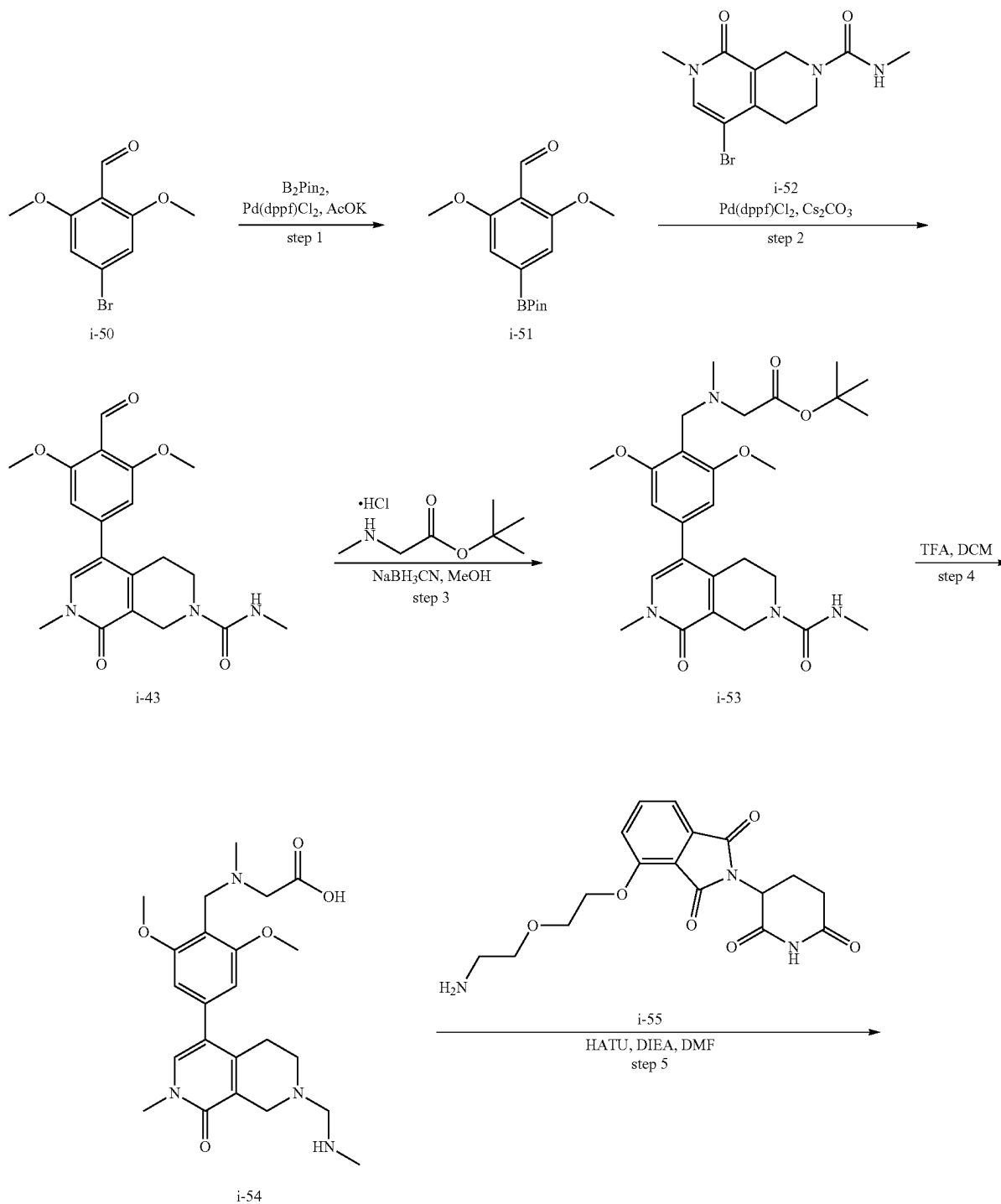
539

Compound D12 was prepared in a similar manner to the preparation of compound D21. Compound D12 formic acid (100 mg, 17.1%) was obtained as a yellow solid. ¹H NMR (300 MHz, Methanol-d₄) δ 8.56 (br s, 0.9H, FA), 7.54 (s, 1H), 7.48 (dd, J=8.6, 7.1 Hz, 1H), 7.06 (d, J=8.5 Hz, 1H), 6.99 (d, J=7.1 Hz, 1H), 6.69 (s, 2H), 5.03 (dd, J=12.5, 5.4 Hz, 1H), 4.34 (s, 2H), 4.30 (s, 2H), 3.95 (s, 6H), 3.79 (t, J=5.2 Hz, 2H), 3.72-3.65 (m, 4H), 3.64-3.58 (m, 5H), 3.56-3.46 (m, 4H), 3.41 (t, J=5.4 Hz, 2H), 2.89-2.79 (m,

540

7H), 2.77-2.64 (m, 2H), 2.63-2.52 (m, 2H), 2.16-2.05 (m, 1H). LCMS (ESI) m/z: [M+H]⁺=812.45.

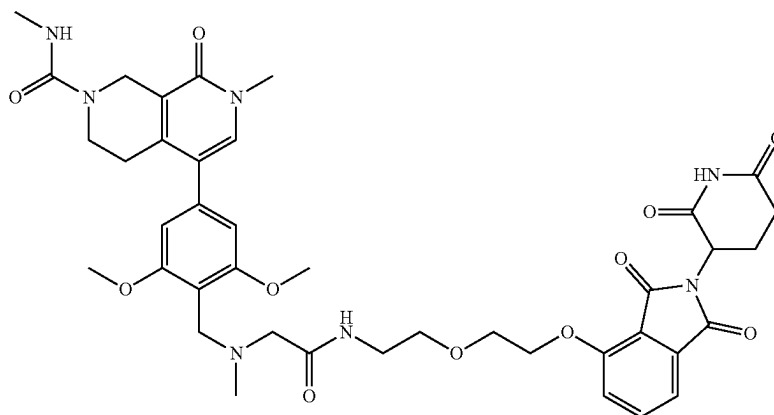
Example 40—Preparation of 5-(4-[[[2-(2-[[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]oxy]ethoxy)ethyl] carbamoyl]methyl)(methylamino)methyl]-3,5-dimethoxyphenyl)-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (Compound D13)



541

542

-continued



compound D13

Step 1: Preparation of 2,6-dimethoxy-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)benzaldehyde (i-51)

Step 2: Preparation of 5-(4-formyl-3,5-dimethoxyphenyl)-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (i-43)

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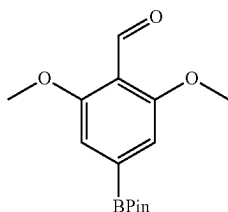
i-43

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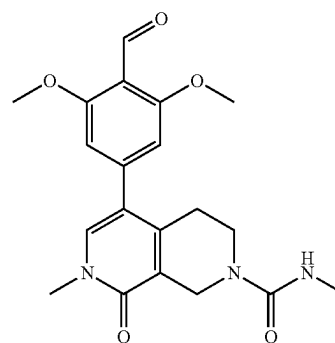
i-51

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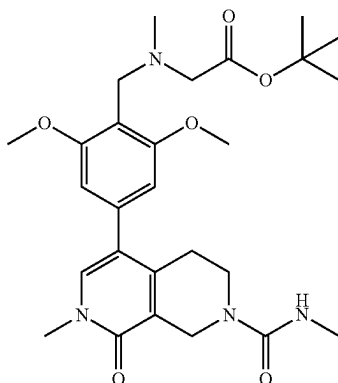
To a stirred solution of bis(pinacolato)diboron (7.62 g, 29.991 mmol, 1.50 equiv) and bis(pinacolato)diboron (7.62 g, 29.991 mmol, 1.50 equiv) in dioxane (70.00 mL) was added Pd(dppf)Cl₂·CH₂Cl₂ (1.63 g, 1.999 mmol, 0.10 equiv) and AcOK (5.89 g, 59.982 mmol, 3.00 equiv). The resulting mixture was stirred for 1 hour at 90° C. under nitrogen atmosphere. Then the reaction was concentrated and purified by silica gel column chromatography, eluted with PE/EtOAc (10:1) to afford 2,6-dimethoxy-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)benzaldehyde (6.3 g, 92.76%) as an orange solid. LCMS (ESI) m/z: [M+H]⁺=293.



To a stirred solution of 2,6-dimethoxy-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)benzaldehyde (291.99 mg, 0.999 mmol, 1.00 equiv) and 5-bromo-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (300.00 mg, 0.999 mmol, 1.00 equiv) in dioxane (5.00 mL) and H₂O (0.60 mL) was added Cs₂CO₃ (976.95 mg, 2.998 mmol, 3.00 equiv) and Pd(dppf)Cl₂·CH₂Cl₂ (81.62 mg, 0.100 mmol, 0.1 equiv). After stirring for 3 hours at 100° C. under a nitrogen atmosphere, the resulting mixture was concentrated under reduced pressure. The residue was purified by silica gel column chromatography, eluted with CH₂Cl₂/MeOH (19:1) to afford 5-(4-formyl-3,5-dimethoxyphenyl)-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (257 mg, 66.72%) as a brown semi-solid. LCMS (ESI) m/z: [M+H]⁺=386.

543

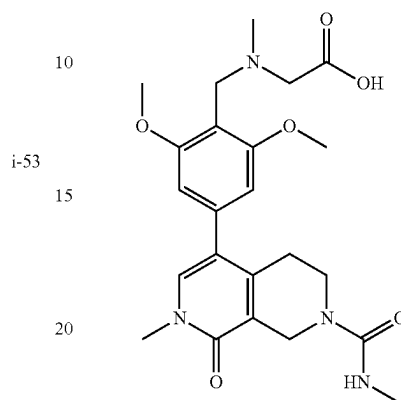
Step 3: Preparation of tert-butyl 2-([(2,6-dimethoxy-4-[2-methyl-7-(methylcarbamoyl)-1-oxo-6,8-dihydro-5H-2,7-naphthyridin-4-yl]phenyl)methyl](methyl)amino]acetate (i-53)



To a stirred solution of 5-(4-formyl-3,5-dimethoxyphenyl)-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (222.00 mg, 0.576 mmol, 1.00 equiv) and tert-butyl 2-(methylamino)acetate hydrochloride (104.64 mg, 0.576 mmol, 1 equiv) in MeOH (5.00 mL) was added NaBH_3CN (72.39 mg, 1.152 mmol, 2 equiv) in portions. The resulting mixture was stirred for 1 hour at room temperature under air atmosphere. Then the reaction mixture was concentrated, and the resulting residue was purified by silica gel column chromatography, eluted with $\text{CH}_2\text{Cl}_2/\text{MeOH}$ (19:1) to afford tert-butyl 2-([(2,6-dimethoxy-4-[2-methyl-7-(methylcarbamoyl)-1-oxo-6,8-dihydro-5H-2,7-naphthyridin-4-yl]phenyl)methyl](methyl)amino]acetate (268 mg, 90.41%) as a light yellow solid. LCMS (ESI) m/z : $[\text{M}+\text{H}]^+=515$.

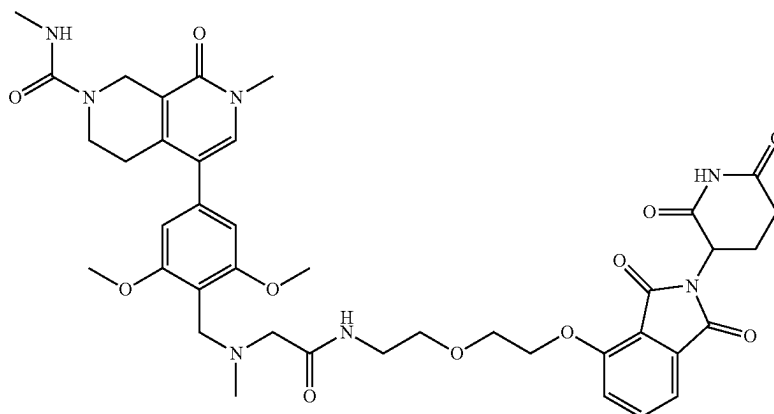
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Step 4: Preparation of [(2,6-dimethoxy-4-[2-methyl-7-(methylcarbamoyl)-1-oxo-6,8-dihydro-5H-2,7-naphthyridin-4-yl]phenyl)methyl](methyl)amino]acetic acid (i-54)



A solution of tert-butyl 2-([(2,6-dimethoxy-4-[2-methyl-7-(methylcarbamoyl)-1-oxo-6,8-dihydro-5H-2,7-naphthyridin-4-yl]phenyl)methyl](methyl)amino]acetate (268.00 mg, 0.521 mmol, 1.00 equiv) and TFA (2.00 mL, 26.926 mmol, 51.70 equiv) in DCM (3 mL) was stirred for 1 hour at room temperature under air atmosphere. The reaction mixture was concentrated, and the resulting residue was purified by reverse flash chromatography (conditions: C18 silica gel column; mobile phase, MeCN in water, 10% to 50% gradient in 10 minutes; detector, UV 254 nm) to afford [(2,6-dimethoxy-4-[2-methyl-7-(methylcarbamoyl)-1-oxo-6,8-dihydro-5H-2,7-naphthyridin-4-yl]phenyl)methyl](methyl)amino]acetic acid (135 mg, 56.54%) as a light yellow oil. LCMS (ESI) m/z : $[\text{M}+\text{H}]^+=459$.

Step 5: Preparation of 5-(4-([(2-(2-([(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]oxy)ethoxy)ethyl]carbamoyl)methyl](methyl)amino)methyl]-3,5-dimethoxyphenyl)-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (Compound D13)

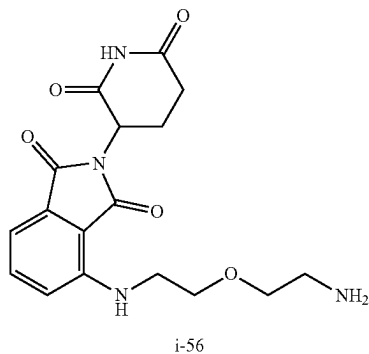


compound D13

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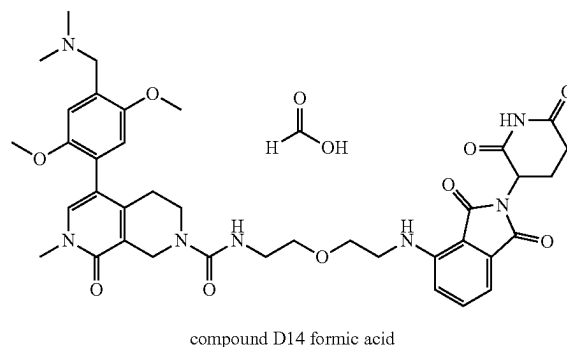
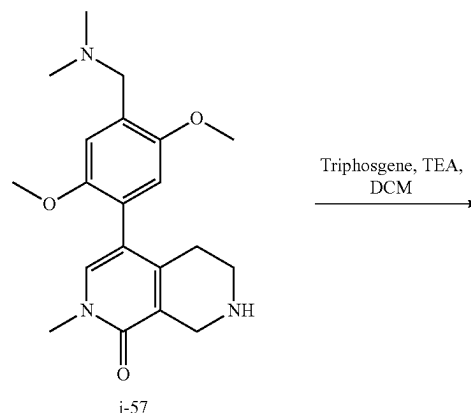
To a stirred solution of [(2,6-dimethoxy-4-[2-methyl-7-(methylcarbamoyl)-1-oxo-6,8-dihydro-5H-2,7-naphthyridin-4-yl]phenyl)methyl](methyl)amino]acetic acid (70.00 mg, 0.153 mmol, 1.00 equiv) and 4-[2-(2-aminoethoxy)ethoxy]-2-(2,6-dioxopiperidin-3-yl)isoindole-1,3-dione (55.17 mg, 0.153 mmol, 1.00 equiv) in DMF (1.00 mL) was added DIEA (98.66 mg, 0.763 mmol, 5.00 equiv) and HATU (116.10 mg, 0.305 mmol, 2.00 equiv). The resulting mixture was stirred for 1 hour at room temperature. Then the solution was directly purified by Prep-HPLC (conditions: SunFire C18 OBD Prep Column, 100 Å, 5 µm, 19 mm×250 mm; Mobile Phase A: water (0.1% FA), Mobile Phase B: ACN; Flow rate: 25 mL/minute; Gradient: 9% B to 25% B in 12 minutes; 254 nm; R_t: 10.82 minutes) to afford 5-(4-[[[2-(2-[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]oxy)ethoxy]ethyl]carbamoyl]methyl](methyl)amino]methyl]-3,5-dimethoxyphenyl)-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (23 mg, 18.79%) as a yellow solid. ¹H NMR (300 MHz, Methanol-d₄) δ 8.52 (br s, 0.4H, FA), 7.66-7.57 (m, 1H), 7.54 (s, 1H), 7.30 (d, J=7.2 Hz, 1H), 7.16 (d, J=8.4 Hz, 1H), 6.63 (s, 2H), 5.11 (dd, J=12.8, 5.4 Hz, 1H), 4.32 (s, 2H), 4.10-3.93 (m, 4H), 3.90 (s, 6H), 3.65-3.48 (m, 1 OH), 3.42-3.37 (m, 3H), 2.88 (dd, J=12.6, 4.6 Hz, 2H), 2.77 (s, 3H), 2.65 (d, J=18.1 Hz, 3H), 2.55-2.48 (m, 3H), 2.07-2.12 (s, 1H). LCMS (ESI) m/z: [M+H]⁺=802.55.

Example 41—Preparation of 5-[4-[(dimethylamino)methyl]-2,5-dimethoxyphenyl]-N-[2-(2-[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]amino)ethoxy]ethyl]-7-methyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide formic acid (Compound D14 formic acid)



546

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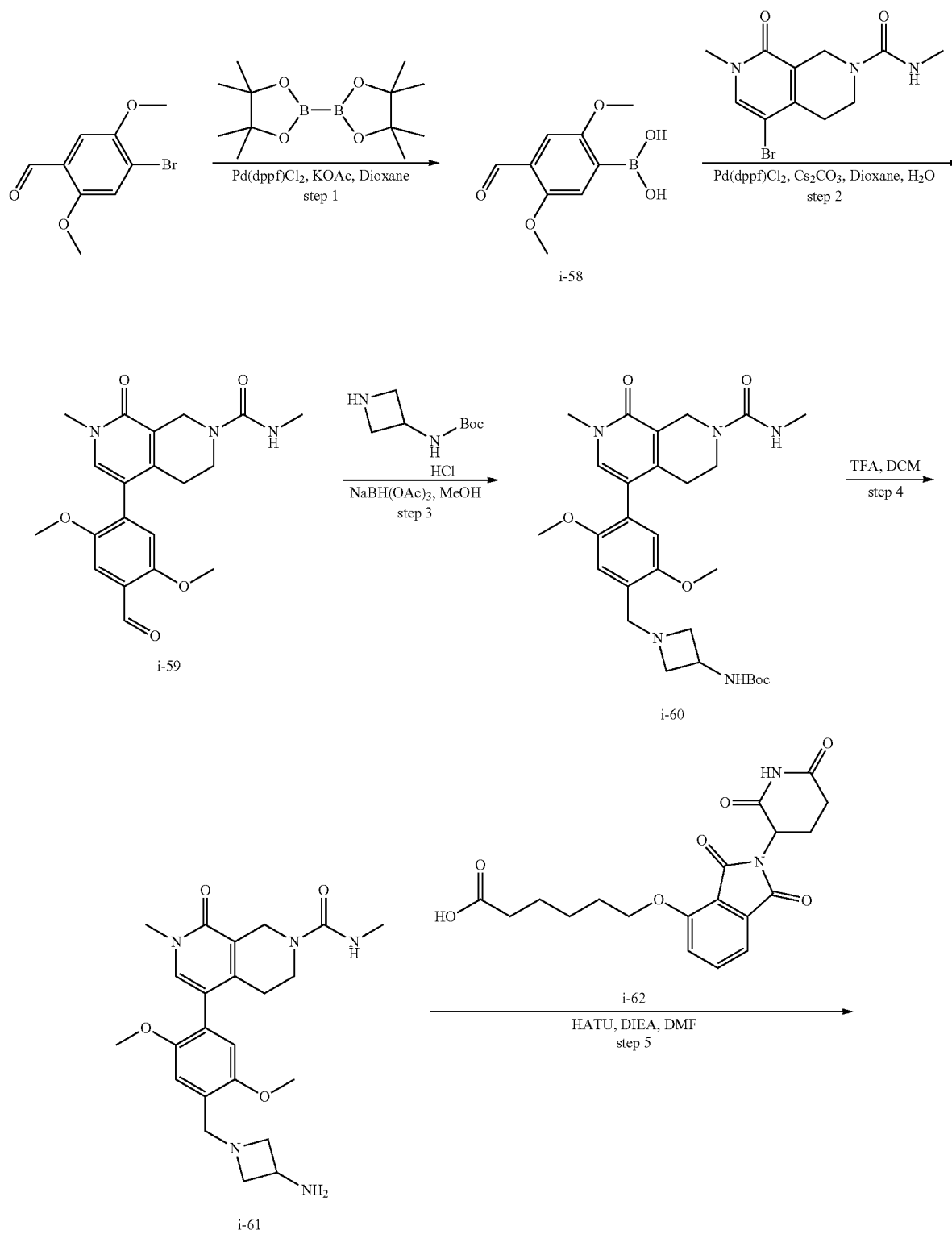


To a stirred solution of triphosgene (13.2 mg, 0.044 mmol, 0.40 equiv) in DCM (1 mL) was added a solution of 4-[[2-(2-aminoethoxy)ethyl]amino]-2-(2,6-dioxopiperidin-3-yl)isoindole-1,3-dione (40.0 mg, 0.111 mmol, 1.00 equiv) and TEA (50 µL) in DCM (0.6 mL) dropwise at 0° C. After 2 minutes, additional TEA (30 µL) was added dropwise. The resulting mixture was stirred for additional 10 minutes at 0° C. 4-[4-[(dimethylamino)methyl]-2,5-dimethoxyphenyl]-2-methyl-5,6,7,8-tetrahydro-2,7-naphthyridin-1-one (39.7 mg, 0.111 mmol, 1.00 equiv) was then added in one portion. The reaction was stirred for additional 5 minutes at 0° C. and then warmed to room temperature for 25 minutes. The reaction solution was concentrated under vacuum. The crude product was purified by Prep-HPLC (conditions: SunFire C18 OBD Prep Column, 100 Å, 5 µm, 19 mm×250 mm; Mobile Phase A: water (0.1% FA), Mobile Phase B: ACN; Flow rate: 25 mL/minute; Gradient: 9 B to 19 B in 14 minutes; 254 nm; RT: 15.53 minutes) to afford 5-[4-[(dimethylamino)methyl]-2,5-dimethoxyphenyl]-N-[2-(2-[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]amino)ethoxy]ethyl]-7-methyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide formic acid (5.9 mg, 6%) as a yellow solid. ¹H NMR (400 MHz, Methanol-d₄) δ 8.56 (s, 0.4H, FA), 7.56 (dd, J=8.6, 7.1 Hz, 1H), 7.44 (s, 1H), 7.11 (d, J=9.3 Hz, 2H), 7.03 (d, J=7.0 Hz, 1H), 6.95 (s, 1H), 5.04 (dd, J=12.4, 5.4 Hz, 1H), 4.36 (br s, 2H), 4.09 (s, 2H), 3.89 (s, 3H), 3.81-3.70 (m, 6H), 3.66-3.56 (m, 6H), 3.51 (t, J=5.2 Hz, 2H), 3.43 (t, J=5.3 Hz, 2H), 2.89-2.62 (m, 9H), 2.53 (br s, 1H), 2.32 (br s, 1H), 2.11-2.02 (m, 1H). LCMS (ESI) m/z: [M+H]⁺=744.50.

547

Example 42—Preparation of 5-(4-[[3-(6-[[2-(2,6-Dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]oxy]hexanamido)azetidin-1-yl]methyl]-2,5-dimethoxyphenyl)-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide formic acid
(Compound D15 formic acid)

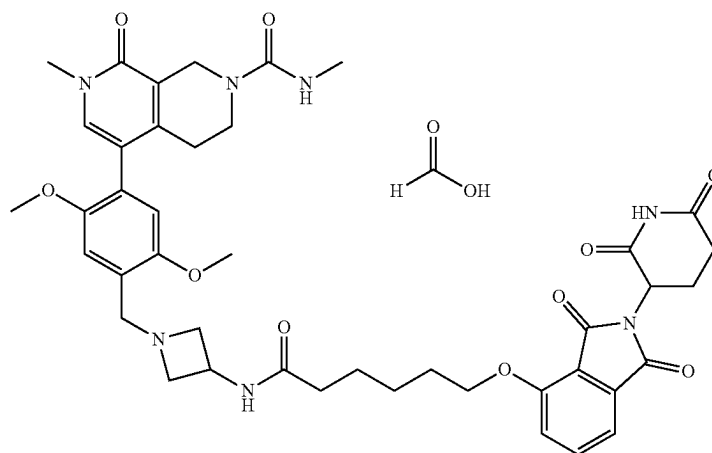
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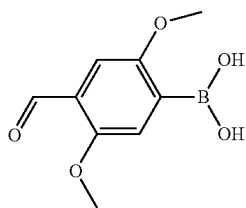
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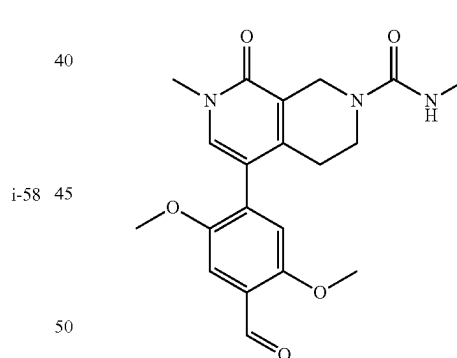
compound D15 formic acid

Step 1: Preparation of
4-Formyl-2,5-dimethoxyphenylboronic acid (i-58)

Step 2: Preparation of 5-(4-Formyl-2,5-dimethoxy-
phenyl)-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-
naphthyridine-2-carboxamide (i-59)



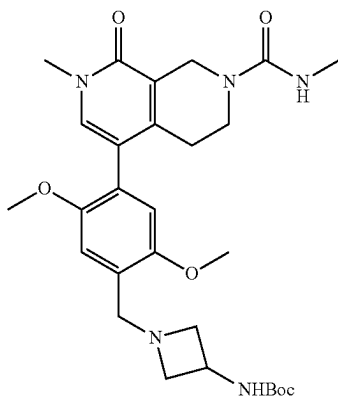
To a stirred solution of 4-bromo-2,5-dimethoxybenzaldehyde (200.00 mg, 0.816 mmol, 1.00 equiv) and bis(pinacolato)diboron (248.68 mg, 0.979 mmol, 1.2 equiv) in dioxane (2 mL) was added KOAc (160.19 mg, 1.632 mmol, 2 equiv) and Pd(dppf)Cl₂ (59.71 mg, 0.082 mmol, 0.1 equiv). The mixture was stirred at 90° C. for 1 hour (under N₂ atmosphere). The resulting mixture was concentrated under reduced pressure to afford 4-formyl-2,5-dimethoxyphenylboronic acid (400 mg, crude) as a brown solid. The crude product was used in the next step directly without further purification. LCMS (ESI) m/z: [M+H]⁺=211.3.



To a stirred solution of 4-formyl-2,5-dimethoxyphenylboronic acid (107.00 mg, 0.510 mmol, 1.00 equiv) and 5-bromo-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (152.94 mg, 0.510 mmol, 1.00 equiv) in dioxane (1.00 mL) and H₂O (5.00 mL) was added CS₂CO₃ (332.04 mg, 1.019 mmol, 2 equiv) and Pd(dppf)Cl₂ (37.28 mg, 0.051 mmol, 0.1 equiv). The mixture was stirred at 90° C. for 1 hour (under N₂ atmosphere). The resulting mixture was concentrated under reduced pressure. The residue was purified by silica gel column chromatography, eluted with CH₂Cl₂/MeOH (10:1) to afford 5-(4-formyl-2,5-dimethoxyphenyl)-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (247 mg, crude) as a brown solid. LCMS (ESI) m/z: [M+H]⁺=386.2.

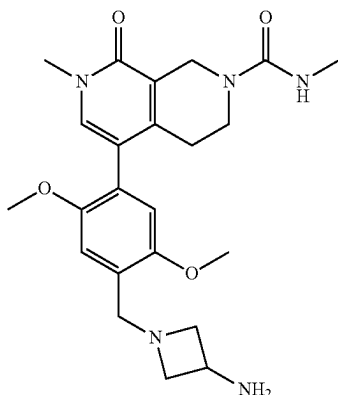
551

Step 3: Preparation of tert-Butyl N-[1-([2,5-dimethoxy-4-[2-methyl-7-(methylcarbamoyl)-1-oxo-6,8-dihydro-5H-2,7-naphthyridin-4-yl]phenyl)methyl]azetidin-3-yl]carbamate (i-60)



To a stirred solution of 5-(4-formyl-2,5-dimethoxyphenyl)-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (100.00 mg, 0.259 mmol, 1.00 equiv) and tert-butyl N-(azetidin-3-yl)carbamate (44.69 mg, 0.259 mmol, 1 equiv) in MeOH (2.00 mL) was added NaBH (OAc)₃ (109.98 mg, 0.519 mmol, 2 equiv). The mixture was stirred at room temperature for 1 h. The resulting mixture was concentrated under vacuum. The residue was purified by Prep-TLC (CH₂Cl₂/MeOH 10:1) to afford tert-butyl N-[1-([2,5-dimethoxy-4-[2-methyl-7-(methylcarbamoyl)-1-oxo-6,8-dihydro-5H-2,7-naphthyridin-4-yl]phenyl)methyl]azetidin-3-yl]carbamate (100 mg, 71.16%) as a brown solid. LCMS (ESI) m/z: [M+H]⁺=542.2

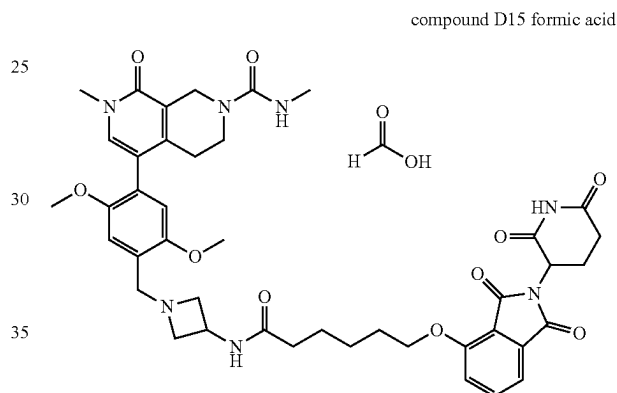
Step 4: Preparation of afford 5-[4-[(3-Aminoazetidin-1-yl)methyl]-2,5-dimethoxyphenyl]-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (i-61)



552

To a stirred solution of tert-butyl N-[1-([2,5-dimethoxy-4-[2-methyl-7-(methylcarbamoyl)-1-oxo-6,8-dihydro-5H-2,7-naphthyridin-4-yl]phenyl)methyl]azetidin-3-yl]carbamate (100.00 mg, 0.185 mmol, 1.00 equiv) in DCM (2.00 mL) was added TFA (0.40 mL). The resulting mixture was concentrated under reduced pressure to afford 5-[4-[(3-aminoazetidin-1-yl)methyl]-2,5-dimethoxyphenyl]-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (80 mg, 98.14%) as a yellow solid. The crude product was used in the next step directly without further purification. LCMS (ESI) m/z: [M+H]⁺=442.2.

Step 5: Preparation of afford 5-(4-[3-(6-[2-(2,6-Dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]oxy)hexanamido]azetidin-1-yl)methyl]-2,5-dimethoxyphenyl)-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide formic acid (Compound D15 formic acid)

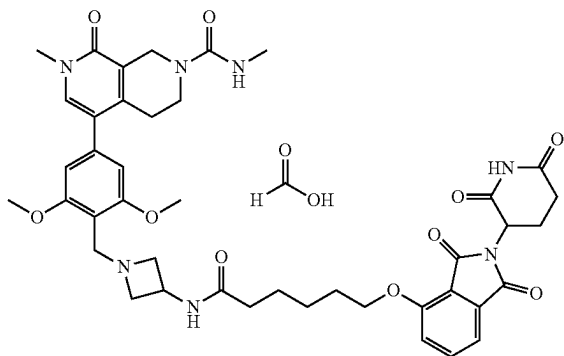


To a solution of 5-[4-[(3-aminoazetidin-1-yl)methyl]-2,5-dimethoxyphenyl]-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (70.00 mg, 0.159 mmol, 1.00 equiv) and 6-[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]oxy]hexanoic acid (61.57 mg, 0.159 mmol, 1.00 equiv) in DMF (1 mL) was added DIEA (102.45 mg, 0.793 mmol, 5.00 equiv) and HATU (90.42 mg, 0.238 mmol, 1.50 equiv). The resulting solution was stirred at room temperature for 1 hour. Without any additional work-up, the mixture was purified by prep-HPLC (conditions: SunFire C18 OBD Prep Column, 100 Å, 5 µm, 19 mm×250 mm; Mobile Phase A: water (0.1% FA), Mobile Phase B:ACN; Flow rate: 25 mL/minute; Gradient: 9 B to 23 B in 14 minutes; 254 nm; R_t: 14.33 minutes) to give 5-(4-[3-(6-[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]oxy)hexanamido]azetidin-1-yl)methyl]-2,5-dimethoxyphenyl)-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide formic acid (9.7 mg, 6.84%) as a white solid. ¹H NMR (400 MHz, Methanol-d₄) δ 8.49 (br s, 1H, FA), 7.78 (dd, J=8.4, 7.4 Hz, 1H), 7.49-7.41 (m, 3H), 7.06 (s, 1H), 6.88 (s, 1H), 5.10 (dd, J=12.5, 5.5 Hz, 1H), 4.55-4.44 (m, 1H), 4.34 (s, 2H), 4.25 (t, J=6.1 Hz, 2H), 4.11 (s, 2H), 4.06 (t, J=8.5 Hz, 2H), 3.86 (s, 3H), 3.76 (s, 3H), 3.74-3.64 (m, 3H), 3.61 (s, 3H), 2.95-2.65 (m, 6H), 2.52 (s, 2H), 2.29 (t, J=7.3 Hz, 2H), 2.18-2.07 (m, 1H), 1.96-1.85 (m, 2H), 1.81-1.69 (m, 2H), 1.66-1.54 (m, 2H). LCMS (ESI) m/z: [M+H]⁺=812.45.

553

Example 43—Preparation of 5-(4-[[3-(6-[[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]oxy]hexanamido)azetidin-1-yl]methyl]-3,5-dimethoxyphenyl)-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide formic acid
(Compound D16 formic acid)

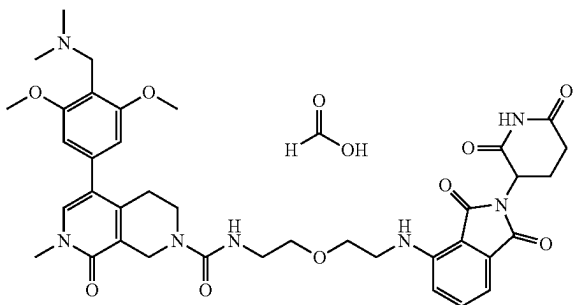
compound D16 formic acid



Compound D16 was prepared in a similar manner to the preparation of compound D21. 15 Compound D16 formic acid (15 mg, 40.8%) was obtained as a white solid. ¹H NMR (400 MHz, Methanol-d₄) δ 8.52 (br s, 1H, FA), 7.78 (dd, J=8.4, 7.3 Hz, 1H), 7.56 (s, 1H), 7.45 (dd, J=7.8, 2.7 Hz, 2H), 6.68 (s, 2H), 5.10 (dd, J=12.4, 5.5 Hz, 1H), 4.55-4.46 (m, 1H), 4.41-4.32 (m, 4H), 4.25 (t, J=6.1 Hz, 2H), 4.19 (s, 2H), 3.92 (s, 7H), 3.64 (s, 3H), 3.54 (t, J=5.6 Hz, 2H), 2.89-2.66 (m, 6H), 2.62 (t, J=5.6 Hz, 2H), 2.29 (t, J=7.3 Hz, 2H), 2.17-2.08 (m, 1H), 1.94-1.85 (m, 2H), 1.79-1.71 (m, 2H), 1.66-1.55 (m, 2H). LCMS (ESI) m/z: [M+H]⁺=812.45.

Example 44—Preparation of 5-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-N-[2-(2-[[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]amino]ethoxy)ethyl]-7-methyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide formic acid
(Compound D17 Formic Acid)

compound D17 formic acid

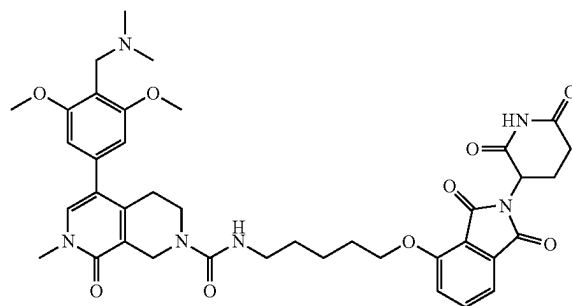


554

Compound D17 formic acid was prepared in a similar manner to the preparation of compound D21. Compound D17 formic acid (75.8 mg, 22%) was obtained as a yellow solid. ¹H NMR (300 MHz, Methanol-d₄) δ 8.57 (s, 0.1H, FA), 7.62-7.50 (m, 2H), 7.11 (d, J=8.5 Hz, 1H), 7.03 (d, J=7.1 Hz, 1H), 6.72 (s, 2H), 5.03 (dd, J=12.3, 5.3 Hz, 1H), 4.38 (s, 2H), 4.27 (s, 2H), 3.94 (s, 6H), 3.74 (t, J=5.0 Hz, 2H), 3.67-3.61 (m, 5H), 3.51 (q, J=4.9 Hz, 4H), 3.44 (t, J=5.3 Hz, 2H), 2.80 (s, 7H), 2.78-2.56 (m, 4H), 2.13-2.00 (m, 1H). LCMS (ESI) m/z: [M+H]⁺=744.45.

Example 45—Preparation of 5-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-N-(5-[[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]oxy]pentyl)-7-methyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (Compound D18)

compound D18

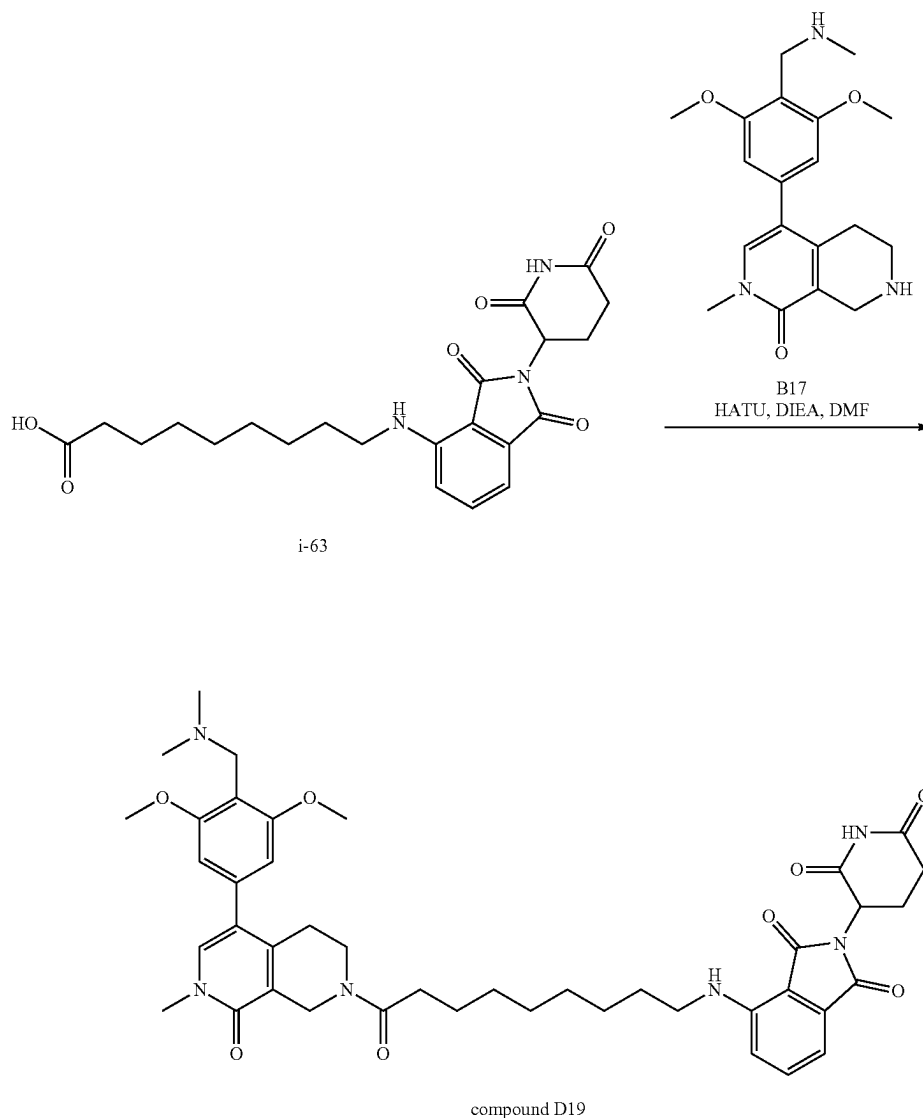


Compound D18 was prepared in a similar manner to the preparation of compound D21. Compound D18 (5 mg, 12.10%) was obtained as a yellow solid. ¹H NMR (300 MHz, Methanol-d₄) δ 7.78 (dd, J=8.5, 7.2 Hz, 1H), 7.59 (s, 1H), 7.46 (d, J=1.7 Hz, 1H), 7.43 (s, 1H), 6.74 (s, 2H), 5.12 (dd, J=12.3, 5.4 Hz, 1H), 4.38 (s, 4H), 4.26 (t, J=6.1 Hz, 2H), 3.96 (s, 6H), 3.63 (s, 3H), 3.54 (d, J=5.6 Hz, 2H), 3.27 (t, J=6.6 Hz, 1H), 2.89 (s, 7H), 2.90-2.79 (m, 1H), 2.79-2.59 (m, 3H), 2.18-2.05 (m, 1H), 1.97-1.86 (m, 2H), 1.65 (s, 5H). LCMS (ESI) m/z: [M+H]⁺=743.65.

555

Example 46—Preparation of 4-[[9-(5-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-7-methyl-8-oxo-1,2,3,4,7,8-hexahydro-2,7-naphthyridin-2-yl)-9-oxononyl]amino]-2-(2,6-dioxopiperidin-3-yl)-2,3-dihydro-1H-isoindole-1,3-dione (Compound D19)

556

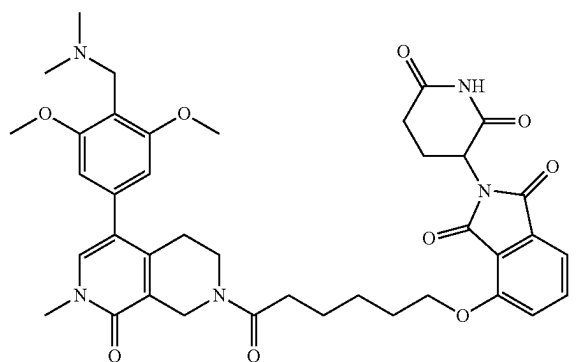


To a stirred solution of 9-[[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxo-2,3-dihydro-1H-isoindol-4-yl]amino]nonanoic acid (100 mg, 0.233 mmol, 1 equiv) in DMF (2 mL) was added HATU (115.02 mg, 0.303 mmol, 1.3 equiv) and DIEA (150.46 mg, 1.164 mmol, 5.0 equiv). The mixture was stirred at 25° C. for 30 minutes, and then 4-[[9-(5-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-7-methyl-8-oxo-1,2,3,4,7,8-hexahydro-2,7-naphthyridin-2-yl)-9-oxononyl]amino]-2-(2,6-dioxopiperidin-3-yl)-2,3-dihydro-1H-isoindole-1,3-dione (25 mg, 32.5 μmol, 13.96%) as a yellow solid. ¹H NMR (400 MHz, Methanol-d₄) δ 7.64-7.51 (m, 2H), 7.04 (dd, J=7.9, 6.2 Hz, 2H), 6.72 (d, J=5.9 Hz, 2H), 5.12-5.01 (m, 1H), 4.62 (s, 2H), 4.55 (d, J=5.5 Hz, 2H), 4.33 (d, J=11.1 Hz, 2H), 3.95 (d, J=1.9 Hz, 6H), 3.75-3.63 (m, 5H), 2.86 (d, J=2.3 Hz, 7H), 2.80-2.68 (m, 3H), 2.67-2.59 (m, 1H), 2.50 (dt, J=17.8, 7.6 Hz, 2H), 2.17-2.08 (m, 1H), 1.67 (dd, J=13.6, 6.8 Hz, 4H), 1.45-1.31 (m, 8H). LCMS (ESI) m/z: [M+H]⁺=769.70.

OBD Column, 5 μm, 19*150 mm; Mobile Phase A: Water (0.1% FA), Mobile Phase B: ACN; Flow rate: 25 mL/minute; Gradient: 20% B to 40% B in 8 minutes; 254/220 nm; R_t: 7.08 minutes; Detector, 254 nm) to give 4-[[9-(5-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-7-methyl-8-oxo-1,2,3,4,7,8-hexahydro-2,7-naphthyridin-2-yl)-9-oxononyl]amino]-2-(2,6-dioxopiperidin-3-yl)-2,3-dihydro-1H-isoindole-1,3-dione (25 mg, 32.5 μmol, 13.96%) as a yellow solid. ¹H NMR (400 MHz, Methanol-d₄) δ 7.64-7.51 (m, 2H), 7.04 (dd, J=7.9, 6.2 Hz, 2H), 6.72 (d, J=5.9 Hz, 2H), 5.12-5.01 (m, 1H), 4.62 (s, 2H), 4.55 (d, J=5.5 Hz, 2H), 4.33 (d, J=11.1 Hz, 2H), 3.95 (d, J=1.9 Hz, 6H), 3.75-3.63 (m, 5H), 2.86 (d, J=2.3 Hz, 7H), 2.80-2.68 (m, 3H), 2.67-2.59 (m, 1H), 2.50 (dt, J=17.8, 7.6 Hz, 2H), 2.17-2.08 (m, 1H), 1.67 (dd, J=13.6, 6.8 Hz, 4H), 1.45-1.31 (m, 8H). LCMS (ESI) m/z: [M+H]⁺=769.70.

557

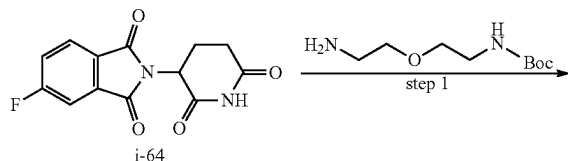
Example 47—Preparation of ([4-[7-(6-[[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxo-2,3-dihydro-1H-isoin-
dol-4-yl]oxy]hexanoyl)-2-methyl-1-oxo-1,2,5,6,7,8-
hexahydro-2,7-naphthyridin-4-yl]-2,6-
dimethoxyphenyl]methyl)(methyl)aminy
(Compound D20)



compound D20

Compound D20 was prepared in a similar manner to the preparation of compound D19 and by substituting the carboxylic acid i-36 with 6-[[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxo-2,3-dihydro-1H-isoin-
dol-4-yl]oxy]hexanoic acid (130 mg, 0.336 mmol, 1.20 equiv) in dimethylformamide (3.0 mL). Compound D20 (7.8 mg, 3.74%) was obtained as a white solid. ¹H NMR (400 MHz, Methanol-d₄) δ 8.56 (s, 1H), 7.76 (td, J=7.9, 3.5 Hz, 1H), 7.59 (d, J=12.7 Hz, 1H), 7.43 (q, J=5.8 Hz, 2H), 6.71 (s, 2H), 5.08 (dd, J=12.6, 5.5 Hz, 1H), 4.55 (s, 2H), 4.25 (dt, J=6.5, 3.2 Hz, 4H), 3.93 (d, J=4.5 Hz, 6H), 3.70 (dt, J=10.4, 5.6 Hz, 2H), 3.63 (d, J=2.9 Hz, 3H), 2.78 (d, J=6.0 Hz, 7H), 2.66 (s, 2H), 2.77-2.51 (m, 4H), 2.11 (tdd, J=10.7, 5.9, 3.1 Hz, 1H), 1.91 (h, J=6.4 Hz, 2H), 1.76 (p, J=7.5 Hz, 2H), 1.70-1.57 (m, 2H). LCMS (ESI) m/z: [M+H]⁺=728.50.

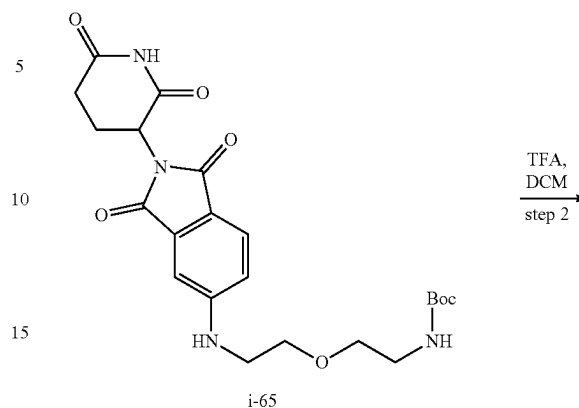
Example 48—Preparation of 5-[4-[(dimethylamino) methyl]-3,5-dimethoxyphenyl]-N-[2-(2-[[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoin-
dol-5-yl]amino]ethoxy)ethyl]-7-methyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide formic acid
(Compound D21 formic acid)



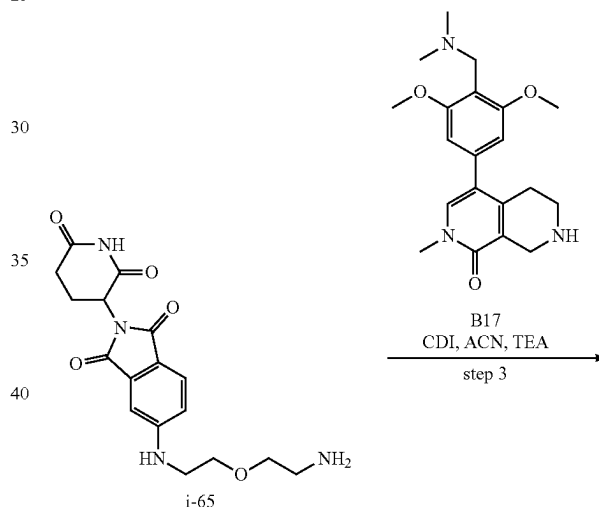
i-64

558

-continued



i-65

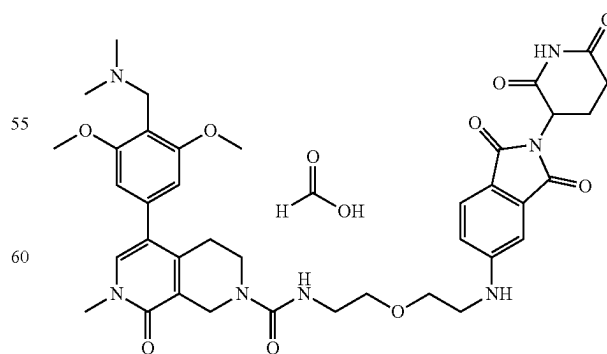


B17

CDI, ACN, TEA

step 3

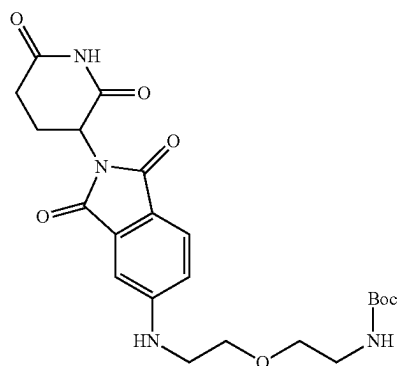
i-65



compound D21 formic acid

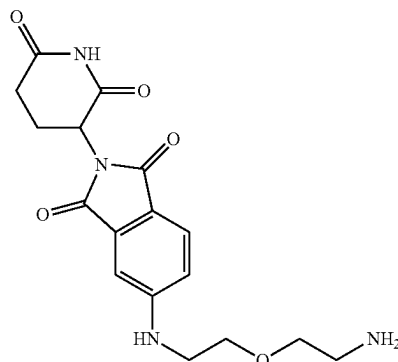
559

Step 1: Preparation of tert-butyl N-[2-(2-[[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-5-yl]amino]ethoxy)ethyl]carbamate (i-65)



To a stirred solution of 2-(2,6-dioxopiperidin-3-yl)-5-fluoroisindole-1,3-dione (1.00 g, 3.620 mmol, 1.00 equiv) and tert-butyl N-[2-(2-aminoethoxy)ethyl]carbamate (1.48 g, 7.241 mmol, 2.00 equiv) in NMP (10 mL) was added DIEA (935.79 mg, 7.241 mmol, 2.00 equiv) at room temperature. The resulting mixture was stirred for 4 hours at 90° C. under nitrogen atmosphere. The residue was purified by reverse flash chromatography (conditions: column, C18 silica gel; mobile phase, ACN in water, 0% to 50% gradient in 20 minutes; detector, UV 254 nm). This resulted in tert-butyl N-[2-(2-[[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-5-yl]amino]ethoxy)ethyl]carbamate (430 mg, 25.79%) as a yellow oil. LCMS (ESI) m/z: [M+H]⁺=461.20.

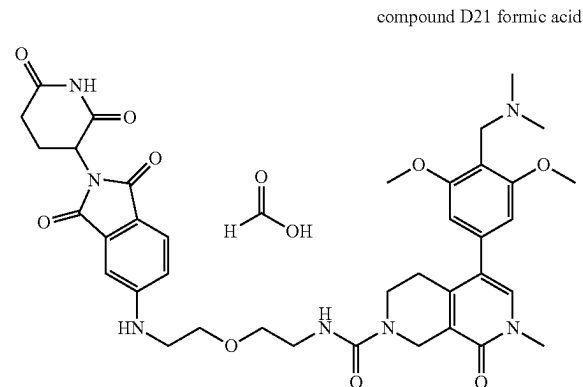
Step 2: Preparation of 5-[[2-(2-aminoethoxy)ethyl]amino]-2-(2,6-dioxopiperidin-3-yl)isindole-1,3-dione (i-66)



560

A solution of tert-butyl N-[2-(2-[[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-5-yl]amino]ethoxy)ethyl] carbamate (400.00 mg, 0.869 mmol, 1.00 equiv) and TFA (1.00 mL) in DCM was stirred for 2 hours at room temperature. The resulting mixture was concentrated under vacuum. This resulted in 5-[[2-(2-aminoethoxy)ethyl]amino]-2-(2,6-dioxopiperidin-3-yl)isindole-1,3-dione (220 mg, 70.28%) as a brown solid. LCMS (ESI) m/z: [M+H]⁺=361.14.

Step 3: Preparation of 5-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-N-[2-(2-[[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-5-yl]amino]ethoxy)ethyl]-7-methyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide formic acid (Compound D21 formic acid)



compound D21 formic acid

A solution of 5-[[2-(2-aminoethoxy)ethyl]amino]-2-(2,6-dioxopiperidin-3-yl)isindole-1,3-dione (60.00 mg, 0.166 mmol, 1.00 equiv) and CDI (29.70 mg, 0.183 mmol, 1.10 equiv) in acetonitrile (1.5 mL) and DMF (0.3 mL) was stirred for 2 hours at room temperature. Then 4-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-2-methyl-5,6,7,8-tetrahydro-2,7-naphthyridin-1-one (119.03 mg, 0.333 mmol, 2.00 equiv) and TEA (33.70 mg, 0.333 mmol, 2.00 equiv) was added to the reaction mixture. The resulting mixture was stirred for overnight at room temperature. The crude product was purified by Prep-HPLC (conditions: SunFire C18 OBD Prep Column, 100 Å, 5 µm, 19 mm×250 mm; mobile phase, Water (0.05% TFA) and ACN (12% Phase B up to 26% in 15 minutes); Detector, UV). This resulted in 5-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-N-[2-(2-[[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-5-yl]amino]ethoxy)ethyl]-7-methyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (4 mg, 3.23%) as a green semi-solid. ¹H NMR (400 MHz, Acetonitrile-d₃) δ 9.31 (s, 1H), 9.03 (brs, 1.0H, FA), 7.51 (d, J=8.3 Hz, 1H), 7.40 (s, 1H), 7.00 (d, J=2.1 Hz, 1H), 6.91 (dd, J=8.3, 2.2 Hz, 1H), 6.60 (s, 2H), 6.18 (s, 1H), 5.67 (t, J=5.7 Hz, 1H), 4.92 (dd, J=12.5, 5.4 Hz, 1H), 4.28 (d, J=4.8 Hz, 4H), 3.85 (s, 6H), 3.71 (t, J=5.0 Hz, 2H), 3.58-3.49 (m, 7H), 3.39 (q, J=5.3 Hz, 2H), 3.34 (t, J=5.0 Hz, 2H), 2.77 (s, 6H), 2.75-2.65 (m, 2H), 2.57 (t, J=5.6 Hz, 2H), 1.35-1.24 (m, 1H). LCMS (ESI) m/z: [M+H]⁺=744.50.

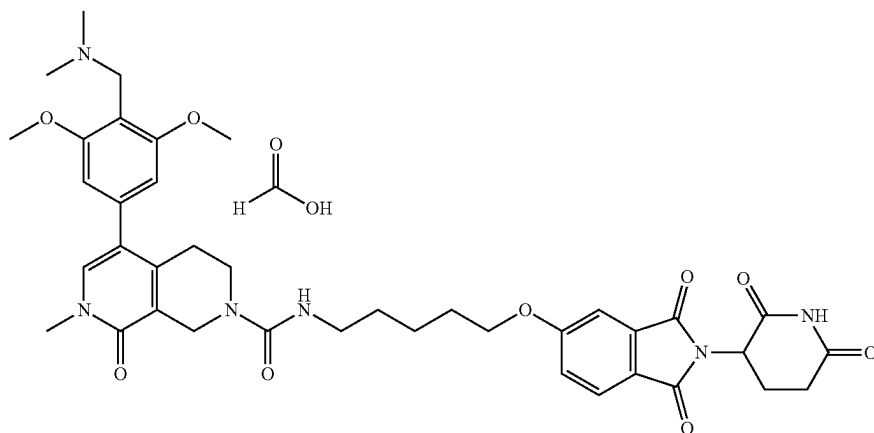
561

562

Example 49—Preparation of 5-[4-[(dimethylamino) methyl]-3,5-dimethoxyphenyl]-N-(5-[[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]amino] penty]-7-methyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide formic acid (Compound D22 formic acid)

5

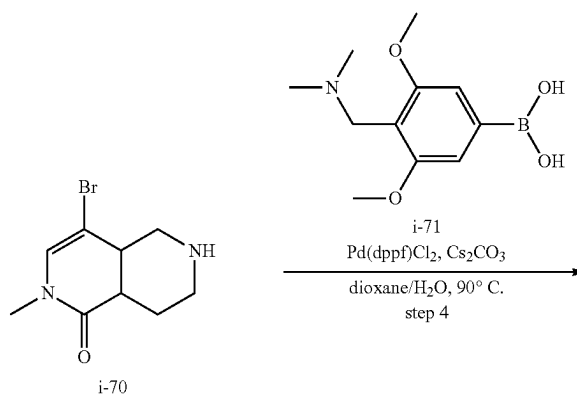
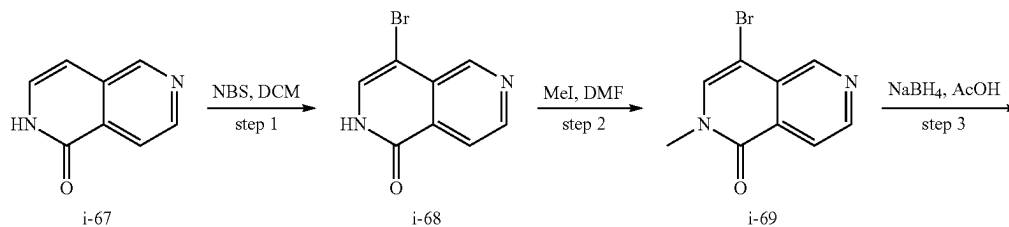
compound D22 formic acid



Compound D22 formic acid was prepared in a similar manner to the preparation of compound D21. Compound D22 formic acid (8.7 mg, 20.8%) was obtained as a light yellow solid. ¹H NMR (400 MHz, Methanol-d₄) δ 8.57 (brs, 0.7H, FA), 7.79 (d, J=8.3 Hz, 1H), 7.58 (s, 1H), 7.37 (d, J=2.2 Hz, 1H), 7.31 (dd, J=8.3, 2.3 Hz, 1H), 6.66 (s, 2H), 5.10 (dd, J=12.6, 5.4 Hz, 1H), 4.36 (s, 2H), 4.17 (t, J=6.3 Hz, 2H), 4.10 (s, 2H), 3.91 (s, 6H), 3.64 (s, 3H), 3.54 (t, J=5.6 Hz, 2H), 3.27 (t, J=6.6 Hz, 2H), 2.93-2.70 (m, 3H), 2.70-

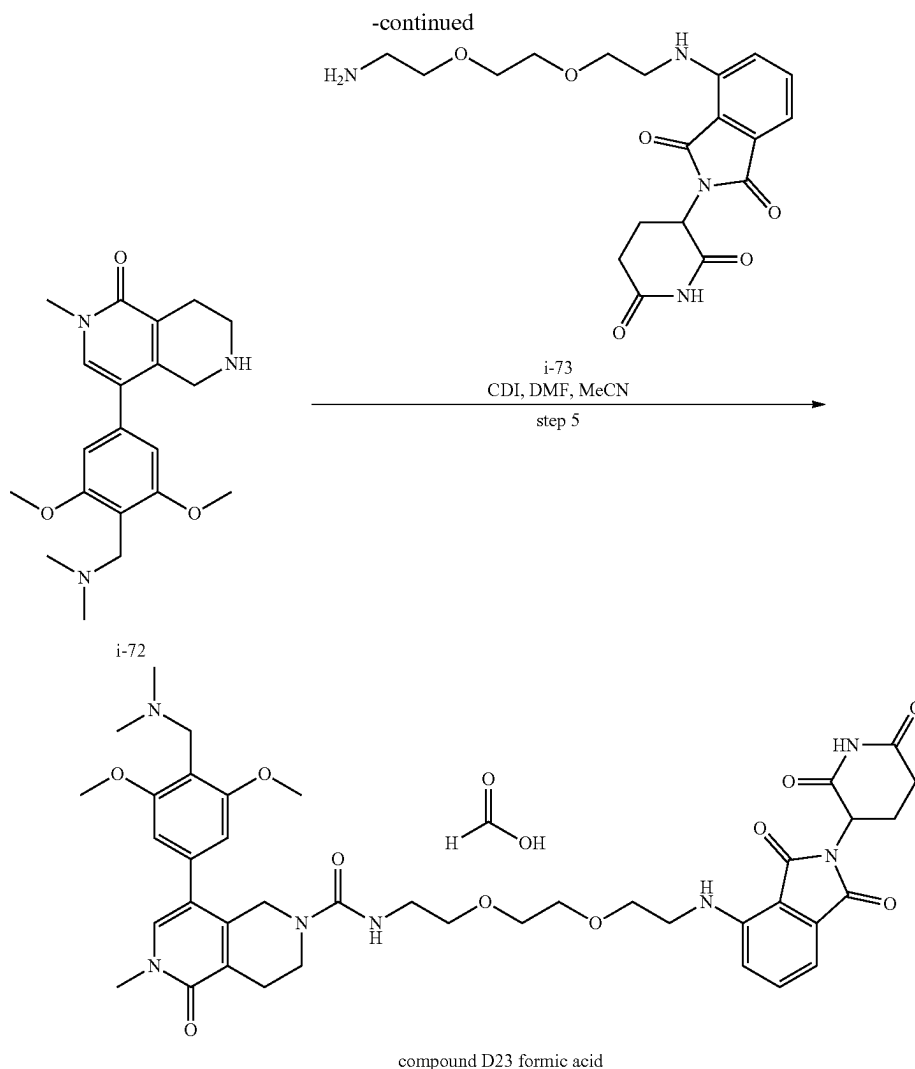
2.58 (m, 8H), 2.17-2.07 (m, 1H), 1.89 (p, J=6.5 Hz, 2H), 1.70-1.52 (m, 4H). LCMS (ESI) m/z: [M+H]⁺=743.35.

Example 50—Preparation of 8-[4-[(dimethylamino) methyl]-3,5-dimethoxyphenyl]-N-[2-[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]amino] ethoxy]ethyl]-6-methyl-5-oxo-3,4-dihydro-1H-2,6-naphthyridine-2-carboxamide formic acid (Compound D23 formic acid)



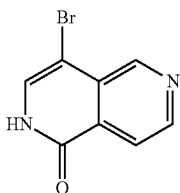
563

564



Step 1: Preparation of
4-bromo-2H-2,6-naphthyridin-1-one (i-68)

45



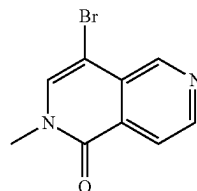
i-68

50

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Step 2: Preparation of
4-bromo-2-methyl-2,6-naphthyridin-1-one (i-69)

i-69



To a stirred solution of 2H-2,6-naphthyridin-1-one (584.00 mg, 3.996 mmol, 1.00 equiv) in DCM (10.00 mL) was added NBS (640.09 mg, 3.596 mmol, 0.9 equiv) in portions at room temperature under air atmosphere. The mixture was stirred for another 1 hour. The reaction mixture was concentrated and purified by silica gel column chromatography, eluted with $\text{CH}_2\text{Cl}_2/\text{MeOH}$ (12:1) to afford 4-bromo-2H-2,6-naphthyridin-1-one (1.18 g, 85.29%) as a light yellow solid. LCMS (ESI) m/z: $[\text{M}+\text{H}]^+=225, 227$.

60

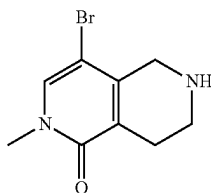
65

To a stirred solution of 4-bromo-2H-2,6-naphthyridin-1-one (1.18 g, 5.243 mmol, 1.00 equiv) in MeI (2.98 g, 20.974 mmol, 4.00 equiv) was added NaH (0.25 g, 10.487 mmol, 2.00 equiv) in portions at 0°C under nitrogen atmosphere. Then MeI (1513.71 mg, 10.665 mmol, 4 equiv) was added drop-wise. The mixture was stirred for another 1 hour at room temperature and quenched with water at 0°C . The product was precipitated by the addition of water. The precipitated solids were collected by filtration and washed with water (2×20 mL). The crude product 4-bromo-2-

565

methyl-2, 6-naphthyridin-1-one (568 mg, 45.31%) was used in the next step directly without further purification. LCMS (ESI) m/z: [M+H]⁺=239, 241.

Step 3: Preparation of 4-bromo-2-methyl-5,6,7,8-tetrahydro-2,6-naphthyridin-1-one (i-70)



To a stirred solution of 4-bromo-2-methyl-2, 6-naphthyridin-1-one (239.10 mg, 1.000 mmol, 1.00 equiv) in AcOH (5.00 mL) was added NaBH₄ (264.86 mg, 7.001 mmol, 7.00 equiv) in portions at 0° C. under nitrogen atmosphere. The resulting mixture was stirred for 5 minutes at room temperature. Then, the mixture was poured into ice water, basified with ammonium hydroxide, and extracted with CH₂Cl₂ (3×50 mL). The combined organic layers were concentrated under reduced pressure and the resulting crude product 4-bromo-2-methyl-5,6,7,8-tetrahydro-2,6-naphthyridin-1-one (237 mg, 80.91%) was used directly in the next step. LCMS (ESI) m/z: [M+H]⁺=243, 245.

566

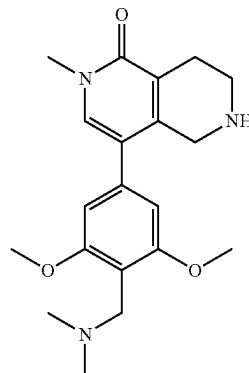
Step 4: Preparation of 4-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-2-methyl-5,6,7,8-tetrahydro-2,6-naphthyridin-1-one (i-72)

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i-72

i-70

25

30

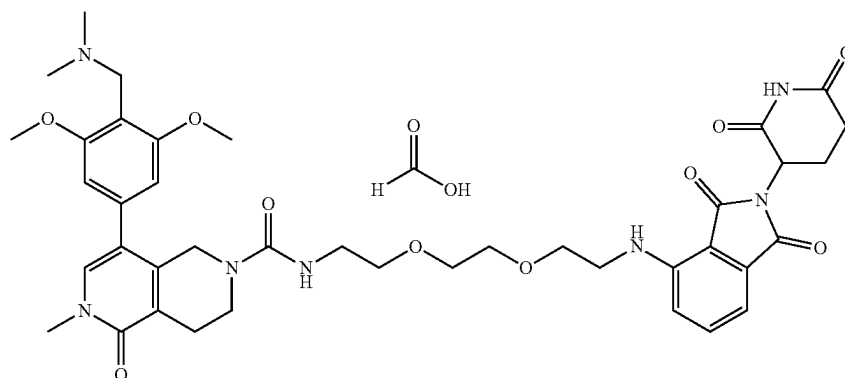
35

40

To a solution of 4-bromo-2-methyl-5,6,7,8-tetrahydro-2,6-naphthyridin-1-one (237.00 mg, 0.975 mmol, 1.00 equiv) and [[2,6-dimethoxy-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)phenyl]methyl]dimethylamine (313.15 mg, 0.975 mmol, 1 equiv) in dioxane (5.00 mL) and H₂O (1.00 mL) was added Cs₂CO₃ (952.92 mg, 2.925 mmol, 3 equiv) and Pd(dppf)Cl₂·CH₂Cl₂ (79.61 mg, 0.097 mmol, 0.1 equiv). After stirring for 1.5 h at 100° C. under a nitrogen atmosphere, the resulting mixture was concentrated under reduced pressure. The residue was purified by silica gel column chromatography, eluted with CH₂Cl₂/MeOH (12:1) to afford the crude product, 4-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-2-methyl-5,6,7,8-tetrahydro-2,6-naphthyridin-1-one (524 mg, 79.69%) as a dark brown solid. LCMS (ESI) m/z: [M+H]⁺=358.

Step 5: Preparation of 8-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-N-[2-[2-(2-[[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]amino]ethoxy)ethoxy]ethyl]-6-methyl-5-oxo-3,4-dihydro-1H-2,6-naphthyridine-2-carboxamide formic acid (Compound D23 formic acid)

compound D23 formic acid



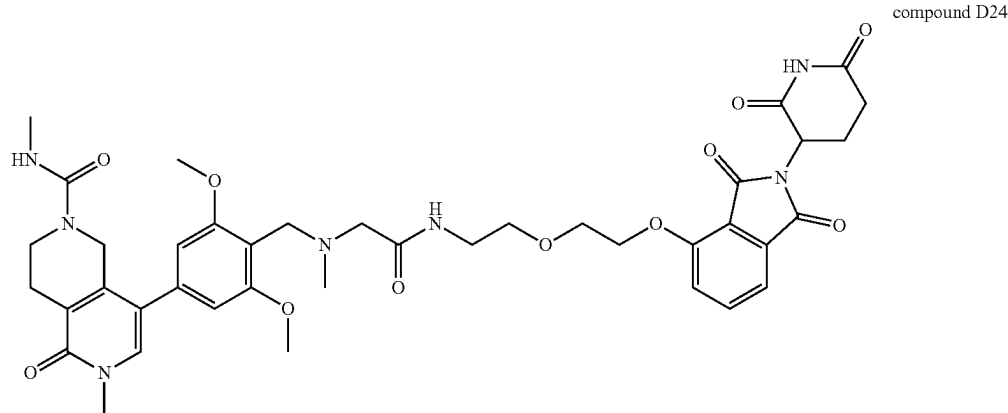
567

To a stirred solution of 4-([2-(2-(2-aminoethoxy)ethoxy)ethyl]amino)-2-(2,6-dioxopiperidin-3-yl)isoindole-1,3-dione trifluoroacetic acid salt (40.40 mg, 0.078 mmol, 1.00 equiv) in DMF (0.30 mL) and MeCN (0.90 mL) was added CDI (13.90 mg, 0.086 mmol, 1.10 equiv). The mixture was stirred for 2 hours at room temperature under air atmosphere. Then 4-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-2-methyl-5,6,7,8-tetrahydro-2,6-naphthyridin-1-one (41.78 mg, 0.117 mmol, 1.50 equiv) and Et₃N (23.66 mg, 0.234 mmol, 3.00 equiv) was added, and the final reaction mixture was stirred for overnight at room temperature under air atmosphere. Without any additional work-up, the resulting mixture was purified by Prep-HPLC (conditions: Xselect CSH F-Phenyl OBD Column 19*150 mm 5 μm; Mobile Phase A: Water (0.1% FA), Mobile Phase B: ACN; Flow rate: 25 mL/minute; Gradient: 9 B to 20 B in 10 minutes; 254 nm; R_t: 9.75 minutes) to afford 8-[4-[(dimeth-

568

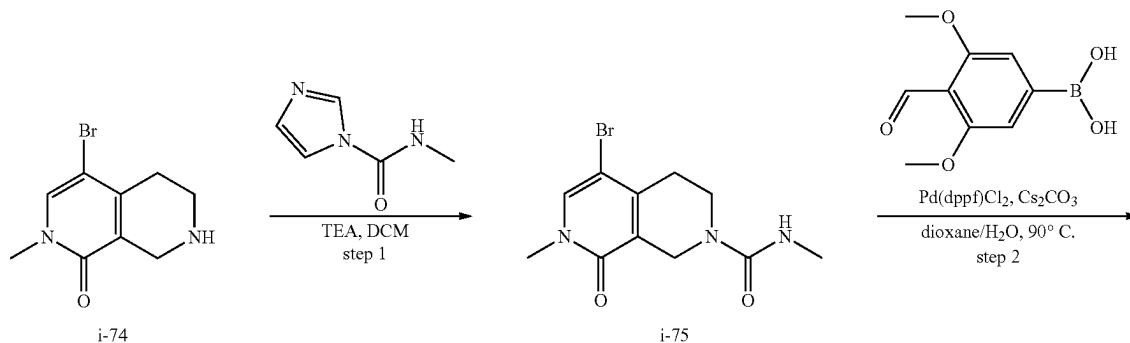
ylamino)methyl]-3,5-dimethoxyphenyl]-N-[2-(2-([2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]amino)ethoxy)ethoxy]ethyl]-6-methyl-5-oxo-3,4-dihydro-1H-2,6-naphthyridine-2-carboxamide (21 mg, 33.49%) as a yellow solid. ¹H NMR (300 MHz, Methanol-d₄) δ 8.57 (brs, 0.7H, FA), 7.58-7.48 (m, 2H), 7.07 (dd, J=14.0, 7.8 Hz, 2H), 6.71 (s, 2H), 5.03 (dd, J=12.4, 5.4 Hz, 1H), 4.36 (s, 2H), 4.19 (s, 2H), 3.94 (s, 6H), 3.73 (t, J=5.2 Hz, 2H), 3.68-3.57 (m, 9H), 3.57-3.47 (m, 4H), 3.35 (s, 1H), 2.92-2.75 (m, 2H), 2.74 (s, 6H), 2.71-2.58 (m, 4H), 2.14-2.04 (m, 1H). LCMS (ESI) m/z: [M+H]⁺=788.50.

Example 51—Preparation of 8-(((2-((2-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl)oxy)ethoxy)ethyl)amino)-2-oxoethyl)(methyl)amino)methyl)-3,5-dimethoxyphenyl]-N,6-dimethyl-5-oxo-3,4,5,6-tetrahydro-2,6-naphthyridine-2 (1H)-carboxamide (Compound D24)



Compound D24 was prepared in a similar manner to the preparation of compound D13 and compound D23.

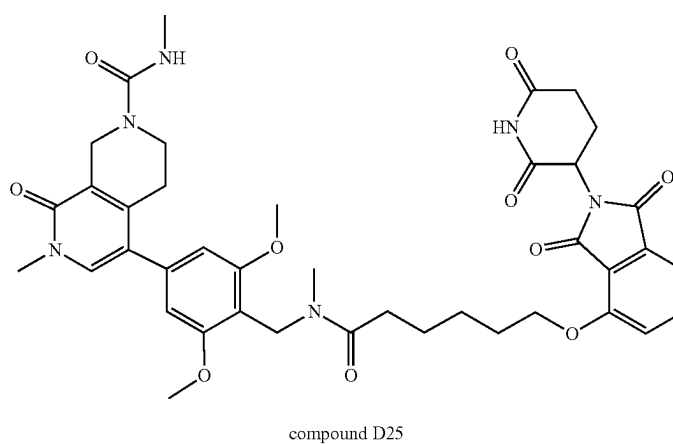
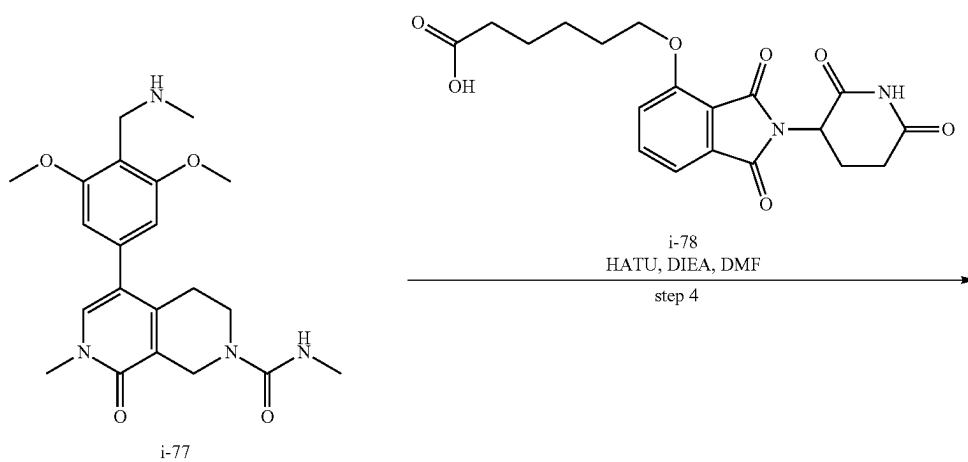
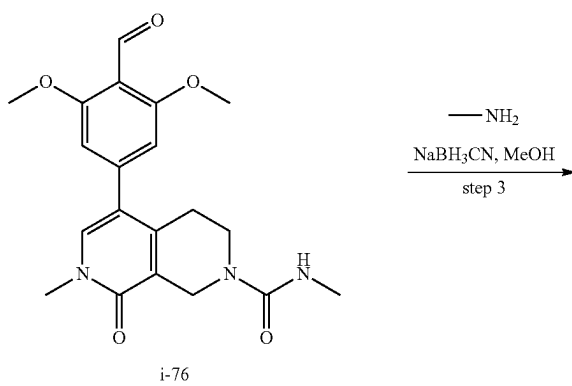
Example 52—Preparation of 5-[4-[(6-([2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]oxy)-N-methylhexanamido)methyl]-3,5-dimethoxyphenyl]-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (Compound D25)



569

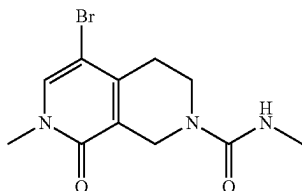
570

-continued



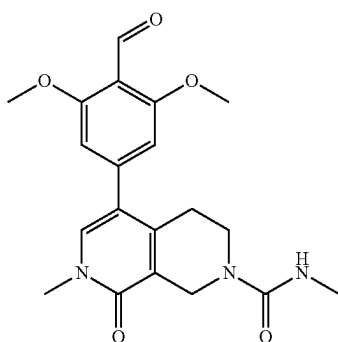
571

Step 1: Preparation of 5-bromo-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (1-75)



Using a similar procedure as described in Example 24, step 4 and substituting with 4-bromo-2-methyl-5,6,7,8-tetrahydro-2,7-naphthyridin-1-one (243 mg, 1.000 mmol) afforded 5-bromo-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (348 mg, 98.6%) as a light yellow solid. LCMS (ESI) m/z: [M+H]⁺=300, 302.

Step 2: Preparation of 5-(4-formyl-3,5-dimethoxyphenyl)-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (i-76)



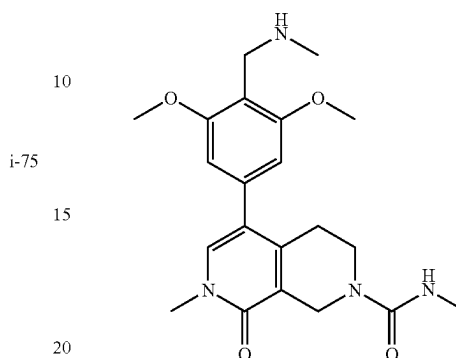
Using a similar procedure as described in Example 23, step 3 and substituting with 5-bromo-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (100 mg, 0.333 mmol, 1.00 equiv) and 2,6-dimethoxy-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)benzaldehyde (97.3 mg, 0.333 mmol, 1.00 equiv) afforded 5-(4-formyl-3,5-dimethoxyphenyl)-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (132 mg, 91.5%) as a light yellow solid. LCMS (ESI) m/z: [M+H]⁺=386.

572

Step 3: Preparation of 5-[3,5-dimethoxy-4-[(methylamino)methyl]phenyl]-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (i-77)

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i-77

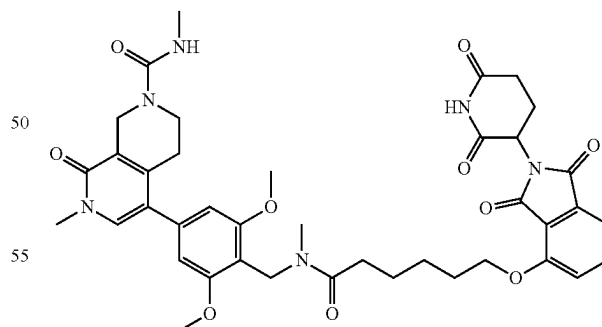


To a stirred solution of 5-(4-formyl-3,5-dimethoxyphenyl)-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (100 mg, 0.259 mmol, 1.00 equiv) and triethylamine (78.76 mg, 0.778 mmol, 3.00 equiv) in methanol (1.00 mL) was added sodium cyanoborohydride (32.6 mg, 0.519 mmol, 2.00 equiv) in portions at room temperature. Solvent was then evaporated under reduced pressure and the residue was purified by silica gel column chromatography, eluted with dichloromethane/methanol (12:1) to afford 5-[3,5-dimethoxy-4-[(methylamino)methyl]phenyl]-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (71 mg, 68.3%) as a yellow solid. LCMS (ESI) m/z: [M+H]⁺=400.

Step 4: Preparation of 5-[4-[(6-[[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]oxy]-N-methylhexanamido)methyl]-3,5-dimethoxyphenyl]-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (Compound D25)

i-76

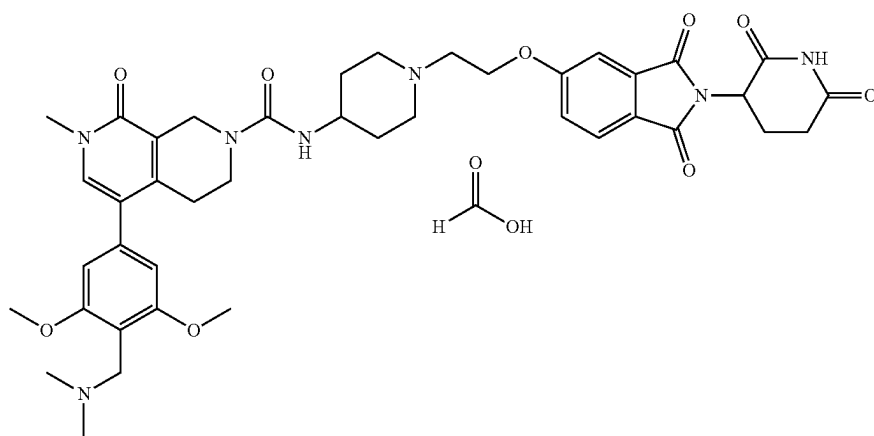
compound D25



Using a similar procedure as described in Example 46 and substituting with 5-[3,5-dimethoxy-4-[(methylamino)methyl]phenyl]-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (60 mg, 0.150 mmol, 1.00 equiv) and 6-[[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]oxy]hexanoic acid (58.2 mg, 0.150 mmol, 1.00 equiv) afforded 5-[4-[(6-[[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]oxy]-N-methylhexanamido)methyl]-3,

5-dimethoxyphenyl]-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (30 mg, 25.5%) as an off-white solid. ¹H NMR (300 MHz, Methanol-d₄) δ 7.76 (dd, J=8.6, 7.2 Hz, 1H), 7.55 (d, J=2.8 Hz, 1H), 7.49-7.37 (m, 2H), 6.60 (d, J=14.9 Hz, 2H), 5.08 (dd, J=12.3, 5.5 Hz, 1H), 4.76-4.57 (m, 2H), 4.35 (s, 2H), 4.25 (t, J=6.1 Hz, 2H), 3.85 (d, J=11.5 Hz, 6H), 3.63 (d, J=3.6 Hz, 3H), 3.52 (q, J=5.2 Hz, 2H), 2.83 (d, J=2.0 Hz, 4H), 2.77 (d, J=6.8 Hz, 4H), 2.72-2.55 (m, 4H), 2.51-2.39 (m, 1H), 2.20-2.00 (m, 1H), 1.87 (d, J=7.8 Hz, 2H), 1.66 (dd, J=21.4, 2.6 Hz, 4H). LCMS (ESI) m/z: [M+H]⁺=771.40.

Example 53—Preparation of 5-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-N-[1-(2-[[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-5-yl]oxy]ethyl)piperidin-4-yl]-7-methyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide formic acid (Compound D26 formic acid)



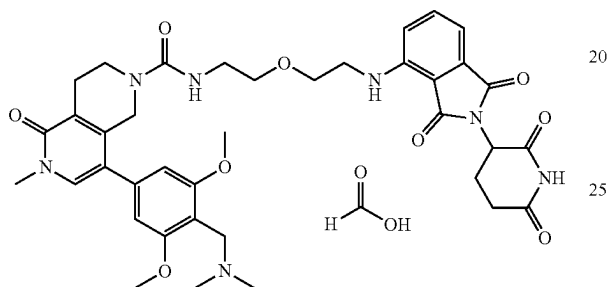
compound D26 formic acid

575

Compound D26 was prepared in a similar manner to the preparation of compound D21. Compound D26 formic acid (4.5 mg, 4.68%) was obtained as a white solid. ¹H NMR (300 MHz, Methanol-d₄) δ 8.48 (brs, 1.6H, FA), 7.85 (d, J=8.3 Hz, 1H), 7.60 (s, 1H), 7.49 (d, J=2.2 Hz, 1H), 7.39 (dd, J=8.3, 2.3 Hz, 1H), 6.74 (s, 2H), 5.13 (dd, J=12.3, 5.4 Hz, 1H), 4.39 (d, J=3.8 Hz, 6H), 3.95 (s, 6H), 3.78-3.68 (m, 1H), 3.64 (s, 3H), 3.56 (t, J=5.5 Hz, 2H), 3.29-3.22 (m, 2H), 3.13 (s, 2H), 2.89 (s, 6H), 2.86-2.71 (m, 3H), 2.69-2.54 (m, 4H), 2.22-2.09 (m, 1H), 2.01 (d, J=13.0 Hz, 2H), 1.81-1.64 (m, 2H). LCMS (ESI) m/z: [M+H]⁺=784.45.

Example 54—Preparation of 8-[4-[(dimethylamino) methyl]-3,5-dimethoxyphenyl]-N-[2-(2-[[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]amino] ethoxy)ethyl]-6-methyl-5-oxo-3,4-dihydro-1H-2,6-naphthyridine-2-carboxamide formic acid
(Compound D27 formic acid)

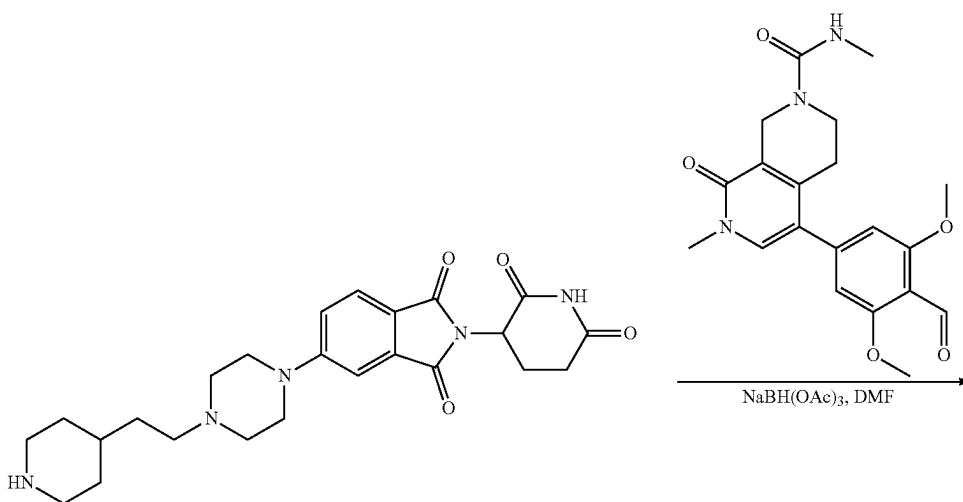
compound D27 formic acid



576

Compound D27 was prepared in a similar manner to the preparation of compound D23. Compound D27 formic acid (21 mg, 35.8%) was obtained as a yellow solid. ¹H NMR (300 MHz, DMSO-d₆) δ 11.09 (s, 1H), 8.19 (brs, 0.4H, FA), 7.63-7.52 (m, 2H), 7.13 (d, J=8.6 Hz, 1H), 7.04 (d, J=7.1 Hz, 1H), 6.73-6.54 (m, 4H), 5.06 (dd, J=12.8, 5.3 Hz, 1H), 4.26 (s, 2H), 3.80 (s, 6H), 3.64-3.51 (m, 7H), 3.47 (s, 3H), 3.45-3.41 (m, 5H), 3.20-3.12 (m, 3H), 2.96-2.83 (m, 1H), 2.61 (s, 1H), 2.27 (s, 6H), 2.07-1.97 (m, 1H). LCMS (ESI) m/z: [M+H]⁺=744.35

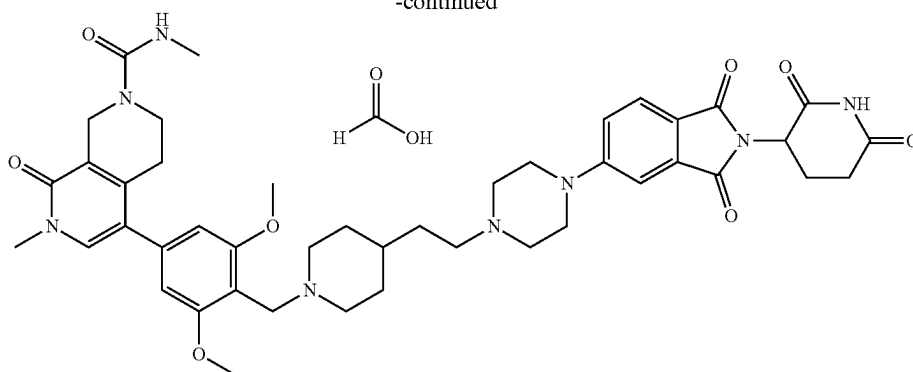
Example 55—Preparation of 5-(4-[[4-(2-[4-[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-5-yl]piperazin-1-yl]ethyl)piperidin-1-yl]methyl]-3,5-dimethoxyphenyl)-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide formic acid
(Compound D28 formic acid)



577

578

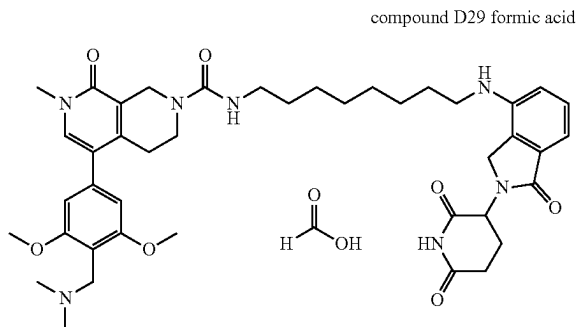
-continued



compound D28 formic acid

Using a similar procedure as described in Example 29, step 1 and substituting with 2-(2,6-dioxopiperidin-3-yl)-5-[4-[2-(piperidin-4-yl)ethyl]piperazin-1-yl]isoindole-1,3-dione (40 mg, 0.088 mmol, 1.00 equiv) in DMF (2 ml) and 5-(4-formyl-3,5-dimethoxyphenyl)-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide (34 mg, 0.088 mmol, 1.00 equiv) afforded 5-(4-[4-(2-[4-[2-(2,6-dioxo piperidin-3-yl)-1,3-dioxoisindol-5-yl]piperazin-1-yl]ethyl)piperidin-1-yl]methyl)-3,5-dimethoxyphenyl)-N,7-dimethyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide formic acid (11.1 mg, 15.3%) as a yellow solid. ¹H NMR (400 MHz, Methanol-d₄) δ 8.34 (brs, 2.4H, FA), 7.72 (d, J=8.5 Hz, 1H), 7.58 (s, 1H), 7.40 (d, J=1.9 Hz, 1H), 7.27 (d, J=10.4 Hz, 1H), 6.73 (s, 2H), 5.09 (dd, J=12.5, 5.4 Hz, 1H), 4.36 (s, 4H), 3.96 (s, 7H), 3.64 (s, 3H), 3.57-3.49 (m, 8H), 3.20-3.04 (m, 2H), 2.96-2.82 (m, 2H), 2.78 (s, 4H), 2.76-2.67 (m, 6H), 2.67-2.60 (m, 2H), 2.59-2.52 (m, 2H), 2.14 (s, 1H), 2.01 (s, 2H), 1.77-1.48 (m, 5H). LCMS (ESI) m/z: [M+H]⁺=823.45.

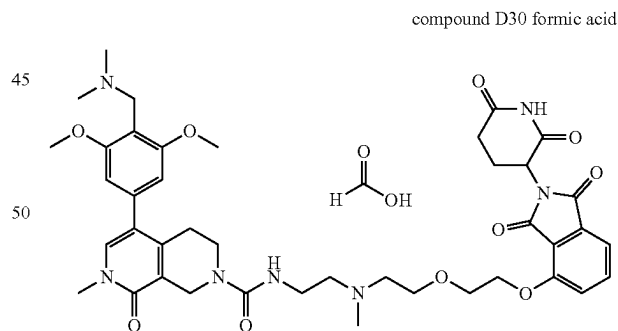
Example 56—Preparation of 5-(4-((dimethylamino)methyl)-3,5-dimethoxyphenyl)-N-(8-((2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-4-yl)amino)octyl)-7-methyl-8-oxo-3,4,7,8-tetrahydro-2,7-naphthyridine-2 (1H)-carboxamide formic acid (Compound D29 formic acid)



compound D29 formic acid

Compound D29 was prepared in a similar manner to the preparation of compound D21. Compound D29 formic acid (2.4 mg, 4.9%) was obtained as a white solid. ¹H NMR (300 MHz, Methanol-d₄) δ 8.57 (brs, 0.8H, FA), 7.57 (s, 1H), 7.32 (t, J=7.8 Hz, 1H), 7.07 (d, J=7.4 Hz, 1H), 6.82 (d, J=7.9 Hz, 1H), 6.69 (s, 2H), 5.17 (dd, J=13.3, 5.2 Hz, 1H), 4.36 (s, 2H), 4.29 (d, J=3.2 Hz, 2H), 4.18 (s, 2H), 3.92 (s, 6H), 3.64 (s, 3H), 3.53 (t, J=5.8 Hz, 2H), 3.21 (q, J=7.3 Hz, 4H), 2.97-2.80 (m, 2H), 2.73 (s, 6H), 2.67-2.60 (m, 2H), 2.49 (dd, J=13.1, 4.8 Hz, 1H), 2.26-2.15 (m, 1H), 1.74-1.62 (m, 2H), 1.60-1.50 (m, 2H), 1.49-1.35 (m, 8H). LCMS (ESI) m/z: [M+H]⁺=770.25.

Example 57—Preparation of 5-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-N-(2-[[2-(2-[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-4-yl]oxy]ethoxy)ethyl](methyl)amino)ethyl]-7-methyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide formic acid (Compound D30 formic acid)



compound D30 formic acid

Compound D30 was prepared in a similar manner to the preparation of compound D21. Compound D30 formic acid (3 mg, 3.1%) was obtained as an off-white solid. ¹H NMR (400 MHz, Methanol-d₄) δ 8.55 (brs, 2.6H, FA), 7.70 (dd, J=8.5, 7.2 Hz, 1H), 7.57 (s, 1H), 7.44 (d, J=8.5 Hz, 1H), 7.36 (d, J=7.3 Hz, 1H), 5.10 (dd, J=12.7, 5.5 Hz, 1H), 4.42 (t, J=4.1 Hz, 2H), 4.37 (s, 2H), 4.21 (s, 2H), 4.04-3.98 (m, 4H), 3.97 (s, 6H), 3.63 (s, 3H), 3.55-3.45 (m, 3H), 3.44-3.35 (m, 3H), 3.26 (s, 2H), 2.93-2.82 (m, 1 OH), 2.79-2.65 (m, 2H), 2.63-2.48 (m, 2H), 2.19-2.10 (m, 1H). LCMS (ESI) m/z: [M+H]⁺=802.30.

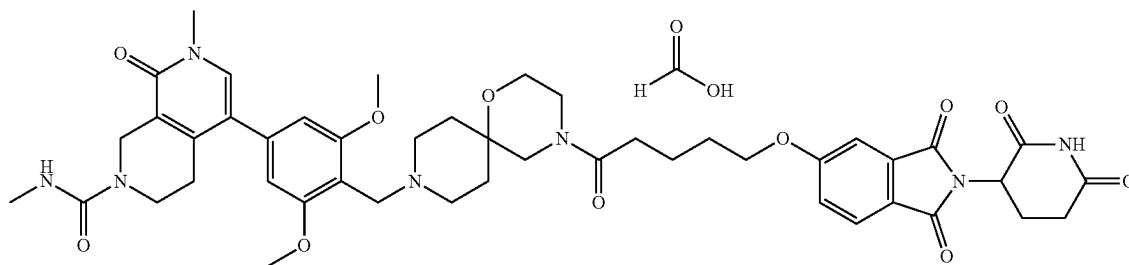
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Example 58—Preparation of 5-((4-((5-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)oxy)pentanoyl)-1-oxa-4,9-diazaspiro[5.5]undecan-9-yl)methyl)-3,5-dimethoxyphenyl)-N,7-dimethyl-8-oxo-3,4,7,8-tetrahydro-2,7-naphthyridine-2 (1H)-carboxamide formic acid (Compound D31 Formic Acid)

5

compound D31 formic acid



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Compound D31 was prepared in a similar manner to the preparation of compound D4. Compound D31 formic acid (3.6 mg, 9.4%) was obtained as a white solid. ¹H NMR (400 MHz, Methanol-d₄) δ 8.55 (brs, 0.8H, FA), 7.81 (dd, J=8.3, 5.2 Hz, 1H), 7.59 (s, 1H), 7.41 (d, J=2.3 Hz, 1H), 7.33 (dd, J=8.4, 2.2 Hz, 1H), 6.70 (d, J=4.2 Hz, 2H), 5.12 (dd, J=12.6, 5.4 Hz, 1H), 4.36 (s, 2H), 4.34-4.26 (m, 2H), 4.24-4.18 (m, 2H), 3.93 (d, J=3.0 Hz, 6H), 3.81-3.71 (m, 2H), 3.67-3.59 (m, 5H), 3.57-3.47 (m, 4H), 3.30-3.12 (m, 4H), 2.96-2.82 (m, 2H), 2.78 (s, 3H), 2.76-2.72 (m, 1H), 2.66-2.47 (m, 4H), 2.19-2.02 (m, 3H), 1.97-1.74 (m, 6H). LCMS (ESI) m/z: [M+H]⁺=882.25.

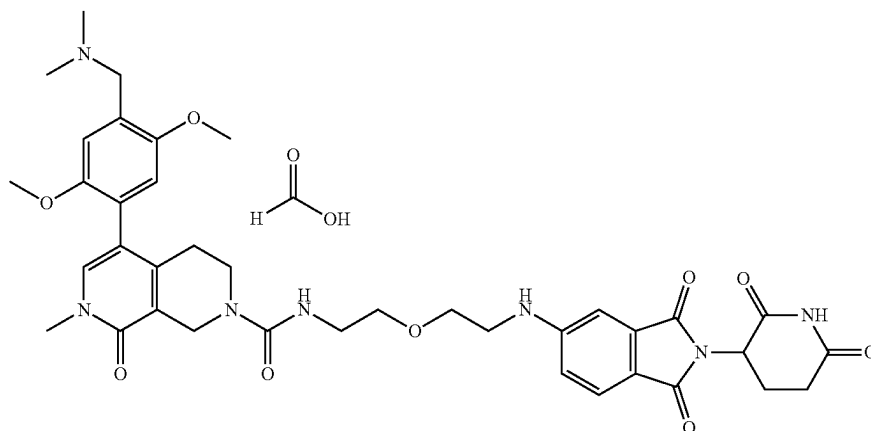
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Example 59—Preparation of 5-[4-[(dimethylamino)methyl]-2,5-dimethoxyphenyl]-N-[2-(2-[[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-5-yl]amino]ethoxy)ethyl]-7-methyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carboxamide formic acid (Compound D32 Formic Acid)

40

compound D32 formic acid

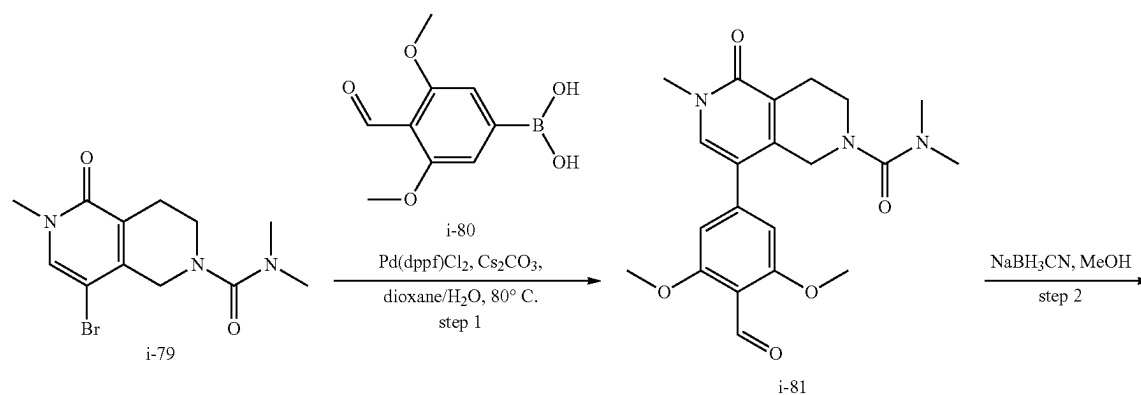


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Compound D32 was prepared in a similar manner to the preparation of compound D21. Compound D32 formic acid (24.2 mg, 19.5%) was obtained as a green solid. ¹H NMR (400 MHz, Methanol-d₄) δ 8.55 (brs, 0.7H, FA), 7.51 (d, J=8.3 Hz, 1H), 7.43 (s, 1H), 7.07 (s, 1H), 7.03 (d, J=2.2 Hz, 1H), 6.91-6.87 (m, 2H), 5.03 (dd, J=12.6, 5.4 Hz, 1H), 4.33 (s, 2H), 4.06-3.93 (m, 2H), 3.86 (s, 3H), 3.73 (s, 6H), 3.61 (s, 6H), 3.45-3.38 (m, 4H), 2.90-2.84 (m, 1H), 2.78-2.69 (m, 2H), 2.69-2.47 (m, 8H), 2.11-2.04 (m, 1H). LCMS (ESI) m/z: [M+H]⁺=744.33.

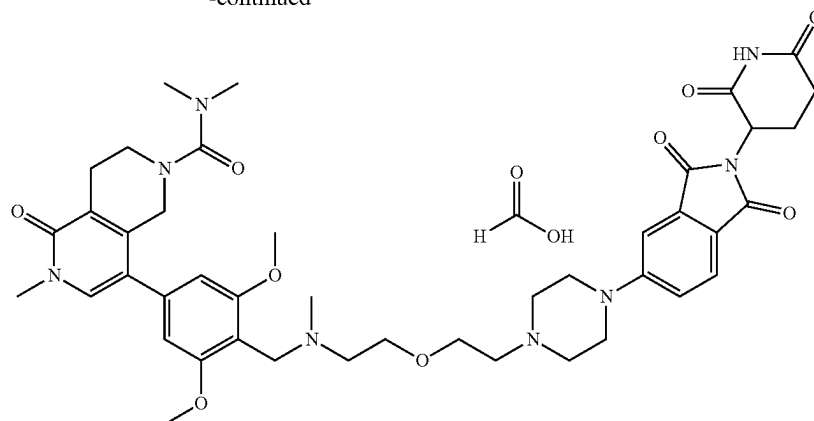
Example 60—Preparation of 8-4-(((2-(2-(4-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)piperazin-1-yl)ethoxy)ethyl)(methyl)amino)methyl)-3,5-dimethoxyphenyl)-N,N,6-trimethyl-5-oxo-3,4,5,6-tetrahydro-2,6-naphthyridine-2 (1H)-carboxamide formic acid (Compound D33 formic acid)



583

584

-continued



compound D33 formic acid

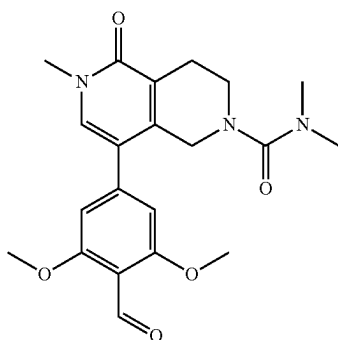
Step 1: Preparation of 8-(4-formyl-3,5-dimethoxyphenyl)-N,N,6-trimethyl-5-oxo-3,4,5,6-tetrahydro-2,6-naphthyridine-2 (1H)-carboxamide (i-81)

Step 2: Preparation of 8-(4-(((2-(2-(4-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)piperazin-1-yl)ethoxy)ethyl)(methylamino)methyl)-3,5-dimethoxyphenyl)-N,N,6-trimethyl-5-oxo-3,4,5,6-tetrahydro-2,6-naphthyridine-2 (1H)-carboxamide formic acid (Compound D33 formic acid)

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compound D33



i-81

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45

formic acid

Using a similar procedure as described in Example 23, step 3 and substituting with 8-bromo-N,N,6-trimethyl-5-oxo-3,4-dihydro-1H-2,6-naphthyridine-2-carboxamide (200 mg, 0.637 mmol, 1.00 equiv) and 4-formyl-3,5-dimethoxyphenylboronic acid (200.5 mg, 0.955 mmol, 1.50 equiv) afforded 8-(4-formyl-3,5-dimethoxyphenyl)-N,N,6-trimethyl-5-oxo-3,4,5,6-tetrahydro-2,6-naphthyridine-2 (1H)-carboxamide (200.0 mg, 78.7%) as a yellow oil. LCMS (ESI) m/z: [M+H]⁺=400.

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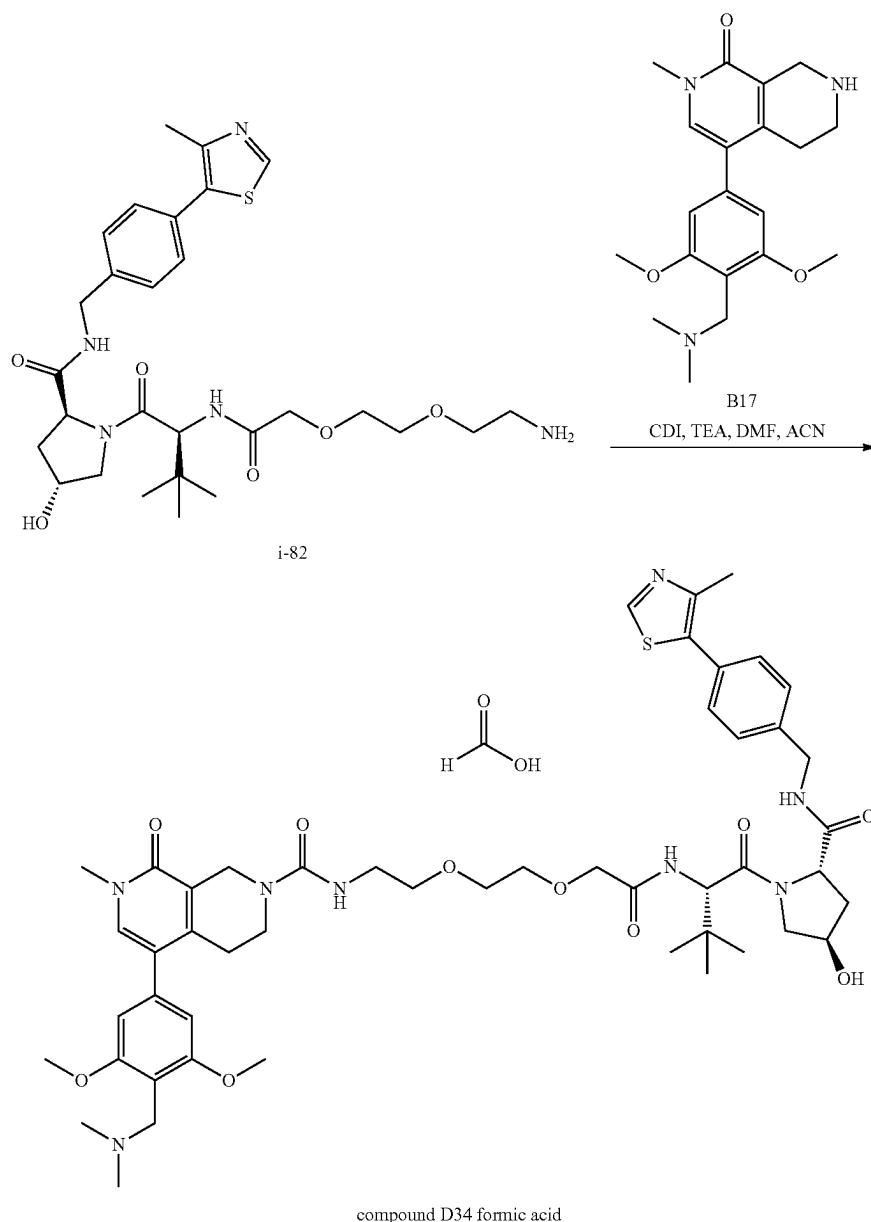
65

Using a similar procedure as described in Example 52, step 3 and substituting with 8-(4-formyl-3,5-dimethoxyphenyl)-N,N,6-trimethyl-5-oxo-3,4-dihydro-1H-2,6-naphthyridine-2-carboxamide (30.0 mg, 0.075 mmol, 1.00 equiv) and 2-(2,6-dioxopiperidin-3-yl)-5-(4-[2-[2-(methylamino)ethoxy]ethyl]piperazin-1-yl)isindole-1,3-dione (33.3 mg, 0.075 mmol, 1.00 equiv) afforded 8-[4-[[[2-(2-[4-[2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindol-5-yl]piperazin-1-yl]ethoxy]ethyl](methylamino)methyl)-3,5-dimethoxyphenyl]-N,N,6-trimethyl-5-oxo-3,4-dihydro-1H-2,6-naphthyridine-2-carboxamide formic acid (7.6 mg, 11.2%) as a yellow green solid. ¹H-NMR (400 MHz, Methanol-d₄) δ 8.47 (brs, 1.4H, FA), 7.69 (d, J=8.5 Hz, 1H), 7.50 (s, 1H), 7.34 (d, J=2.3 Hz, 1H), 7.24 (dd, J=8.6, 2.3 Hz, 1H), 6.77 (s, 2H), 5.08 (dd, J=12.4, 5.5 Hz, 1H), 4.50 (s, 2H), 4.09 (s, 2H), 3.97 (s, 6H), 3.92-3.87 (m, 2H), 3.76 (t, J=5.2 Hz, 2H), 3.61 (s, 3H), 3.51-3.45 (m, 8H), 2.90-2.86 (m, 4H), 2.83 (s, 6H), 2.79-2.71 (m, 1 OH), 2.17-2.07 (m, 1H). LCMS (ESI) m/z: [M+H]⁺=827.50.

585

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Example 61—Preparation of (2S,4R)-1-[(2S)-2-(2-[2-[2-(5-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-7-methyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carbonylamino)ethoxy]ethoxy]acetamido)-3,3-dimethylbutanoyl]-4-hydroxy-N-[[4-(4-methyl-1,3-thiazol-5-yl)phenyl]methyl]pyrrolidine-2-carboxamide formic acid (Compound D34 formic acid)



To a stirred mixture of CDI (9.29 mg, 0.057 mmol, 1.10 equiv) in ACN (0.50 mL) and DMF (0.10 mL) was added (2S,4R)-1-[(2S)-2-[2-[2-(2-aminoethoxy)ethoxy]acetamido]-3,3-dimethylbutanoyl]-4-hydroxy-N-[[4-(4-methyl-1,3-thiazol-5-yl)phenyl]methyl]pyrrolidine-2-carboxamide (30 mg, 0.052 mmol, 1.00 equiv) and ACN (0.30 mL) dropwise at room temperature under nitrogen atmosphere. After 3 hours, 4-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-2-methyl-5,6,7,8-tetrahydro-2,7-naphthyri-

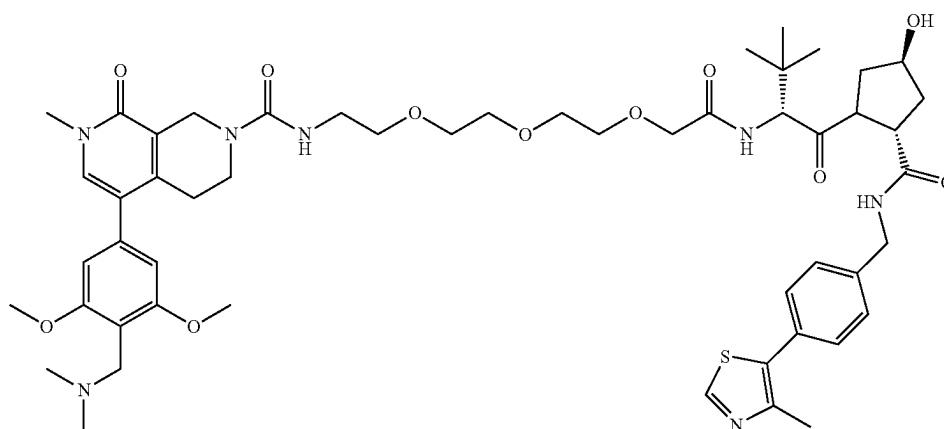
din-1-one (22.35 mg, 0.063 mmol, 1.20 equiv) and TEA (15.82 mg, 0.156 mmol, 3.00 equiv) were added. The resulting mixture was stirred at room temperature for 12 hours under nitrogen atmosphere. Without any additional work-up, the mixture was purified by Prep-HPLC (conditions: Gemini-NX C18 AXAI Packed, 21.2*150 mm, 5 μ m; Mobile Phase A: Water (0.1% FA), Mobile Phase B: ACN; Flow rate: 25 mL/minute; Gradient: 10% B to 25% B in 12 minutes; 254/220 nm; RT: 12.30 minutes). This resulted in

587

588

(2S,4R)-1-[(2S)-2-(2-[2-(2-(5-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-7-methyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carbonylamino)ethoxy]ethoxy]acetamido)-3,3-dimethylbutanoyl]-4-hydroxy-N-[[4-(4-methyl-1,3-thiazol-5-yl)phenyl]methyl]pyrrolidine-2-carboxamide formic acid (13.9 mg, 26.23%) as a white solid. ¹H NMR (400 MHz, Methanol-d₄) δ 8.88 (s, 1H), 8.56 (brs, 1.0H, FA), 7.56 (s, 1H), 7.49-7.38 (m, 4H), 6.67 (s, 2H), 4.84-4.75 (m, 2H), 4.67-4.59 (m, 1H), 4.52 (s, 1H), 4.48-4.26 (m, 5H), 4.03-3.83 (m, 1 OH), 3.77-3.61 (m, 7H), 3.59 (s, 3H), 3.55-3.48 (m, 1H), 3.43-3.34 (m, 2H), 2.84 (s, 6H), 2.69-2.52 (m, 2H), 2.48 (s, 3H), 2.31 (dd, J=13.1, 7.7 Hz, 1H), 2.18-2.07 (m, 1H), 1.08 (s, 9H). LCMS (ESI) m/z: [M+H]⁺= 959.55.

Example 62—Preparation of (2R,4S)-1-[(2R)-2-(2-[2-(2-[2-(5-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-7-methyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carbonylamino)ethoxy]ethoxy]ethoxy]acetamido)-3,3-dimethylbutanoyl]-4-hydroxy-N-[[4-(4-methyl-1,3-thiazol-5-yl)phenyl]methyl]pyrrolidine-2-carboxamide (Compound D35)

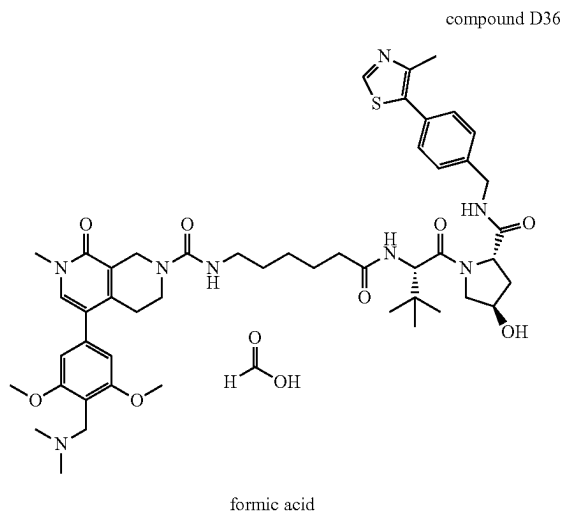


compound D35

589

Compound D35 was prepared in a similar manner to the preparation of compound D34. Compound D35 (24.3 mg, 29.8%) was obtained as an off-white solid. ¹H NMR (300 MHz, DMSO-d₆) δ 8.98 (s, 1H), 8.59 (s, 1H), 8.17 (brs, 0.4H, FA), 7.64 (s, 1H), 7.40 (s, 5H), 6.66 (d, J=18.5 Hz, 3H), 5.15 (s, 1H), 4.63-4.53 (m, 1H), 4.51-4.32 (m, 3H), 4.31-4.17 (m, 3H), 3.97 (s, 2H), 3.81 (s, 7H), 3.72 (s, 2H), 3.63-3.47 (m, 12H), 3.40 (s, 4H), 3.20 (s, 2H), 2.44 (s, 6H), 2.35 (s, 5H), 2.13-2.00 (m, 1H), 1.98-1.82 (m, 1H), 0.95 (s, 9H). LCMS (ESI) m/z: [M+H]⁺=1003.60.

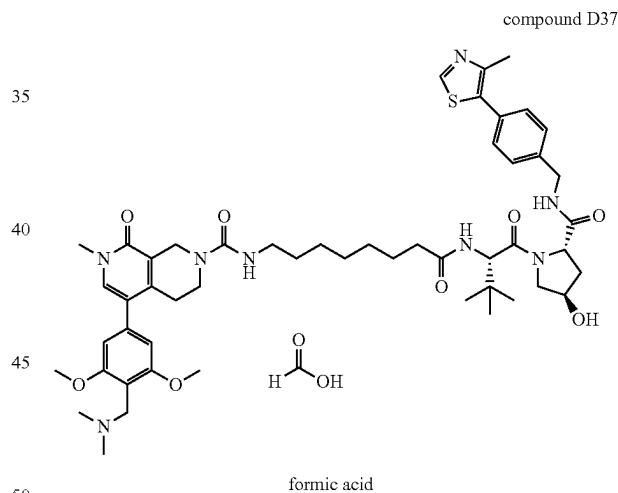
Example 63—Preparation of (2S,4R)-1-[(2S)-2-[6-(5-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-7-methyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carbonylamino)hexanamido]-3,3-dimethylbutanoyl]-4-hydroxy-N-[[4-(4-methyl-1,3-thiazol-5-yl)phenyl]methyl]pyrrolidine-2-carboxamide (Compound D36)



Compound D36 was prepared in a similar manner to the preparation of compound D34. Compound D36 formic acid (13.3 mg, 24.6%) was obtained as a white solid. ¹H NMR (400 MHz, DMSO-d₆) δ 9.05 (brs, 0.7H, FA), 8.99 (s, 1H), 8.56 (t, J=6.0 Hz, 1H), 7.83 (d, J=9.3 Hz, 1H), 7.66 (s, 1H), 7.41 (q, J=8.2 Hz, 4H), 6.73 (s, 2H), 6.65 (t, J=5.4 Hz, 1H), 6.51 (s, 0.3H, FA), 5.12 (d, J=3.6 Hz, 1H), 4.55 (d, J=9.4 Hz, 1H), 4.47-4.39 (m, 2H), 4.35 (s, 1H), 4.23 (s, 5H), 3.87 (s, 6H), 3.71-3.62 (m, 2H), 3.51 (s, 3H), 3.41 (t, J=5.5 Hz, 2H), 3.30-3.26 (m, 2H), 3.08-2.98 (m, 2H), 2.74 (s, 6H), 2.45 (s, 3H), 2.31-2.22 (m, 1H), 2.17-2.00 (m, 2H), 1.95-1.87 (m, 1H), 1.55-1.38 (m, 4H), 1.29-1.20 (m, 2H), 0.93 (s, 9H). LCMS (ESI) m/z: [M+H]⁺=927.55.

590

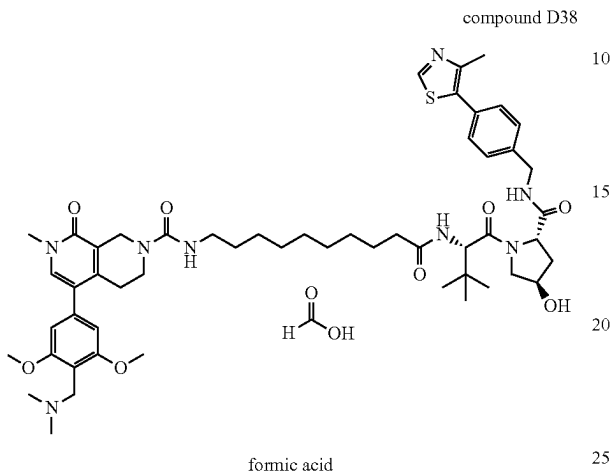
Example 64—Preparation of (2S,4R)-1-[(2S)-2-[8-(5-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-7-methyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carbonylamino)octanamido]-3,3-dimethylbutanoyl]-4-hydroxy-N-[[4-(4-methyl-1,3-thiazol-5-yl)phenyl]methyl]pyrrolidine-2-carboxamide formic acid (Compound D37)



Compound D37 was prepared in a similar manner to the preparation of compound D34. Compound D37 formic acid (19.9 mg, 37.9%) was obtained as a white solid. ¹H NMR (400 MHz, Methanol-d₄) δ 8.89 (s, 1H), 8.57 (brs, 0.6H, FA), 7.58 (s, 1H), 7.51-7.40 (m, 4H), 6.69 (s, 2H), 4.66 (s, 1H), 4.62-4.49 (m, 3H), 4.38 (d, J=12.6 Hz, 3H), 4.16 (s, 2H), 3.92 (s, 6H), 3.90 (s, 1H), 3.82 (dd, J=11.0, 3.9 Hz, 1H), 3.64 (s, 3H), 3.53 (t, J=5.5 Hz, 2H), 3.20 (t, J=7.1 Hz, 2H), 2.71 (s, 6H), 2.63 (t, J=5.2 Hz, 2H), 2.49 (s, 3H), 2.36-2.19 (m, 3H), 2.15-2.05 (m, 1H), 1.63 (t, J=6.8 Hz, 2H), 1.54 (t, J=7.0 Hz, 2H), 1.36 (s, 6H), 1.05 (s, 9H). LCMS (ESI) m/z: [M+H]⁺=955.55.

591

Example 65—Preparation of (2S,4R)-1-[(2S)-2-[10-(5-[4-[(dimethylamino)methyl]-3,5-dimethoxyphenyl]-7-methyl-8-oxo-3,4-dihydro-1H-2,7-naphthyridine-2-carbonylamino)decanamido]-3,3-dimethylbutanoyl]-4-hydroxy-N-[[4-(4-methyl-1,3-thiazol-5-yl)phenyl]methyl]pyrrolidine-2-carboxamide (Compound D38)



592

Compound D35 was prepared in a similar manner to the preparation of compound D34. Compound D35 formic acid (15.3 mg, 29.6%) was obtained as a white solid. ¹H NMR (400 MHz, Methanol-d₄) δ 8.90 (s, 1H), 8.57 (brs, 0.5H, FA), 7.59 (s, 1H), 7.52-7.41 (m, 4H), 6.69 (s, 2H), 4.65 (s, 1H), 4.62-4.49 (m, 3H), 4.40-4.34 (m, 3H), 4.14 (s, 2H), 3.92 (s, 7H), 3.82 (dd, J=10.9, 3.9 Hz, 1H), 3.64 (s, 3H), 3.54 (t, J=5.6 Hz, 2H), 3.20 (t, J=7.2 Hz, 2H), 2.69 (s, 6H), 2.63 (t, J=5.1 Hz, 2H), 2.49 (s, 3H), 2.37-2.19 (m, 3H), 2.14-2.05 (m, 1H), 1.68-1.48 (m, 4H), 1.34 (s, 1 OH), 1.05 (s, 9H). LCMS (ESI) m/z: [M+H]⁺=983.65.

Example 66—Preparation of Compounds D39-D302 and DD1

In analogy to the procedures described in the examples above, compounds D39-D302 and DD1 were prepared using the appropriate starting materials

Compound No.	LCMS	¹ H NMR
D39	LCMS (ESI) m/z: [M + H] ⁺ = 795.6	¹ H NMR (400 MHz, Methanol-d ₄) δ 8.39 (s, 2H, FA), 7.66-7.61 (m, 1H), 7.59 (s, 1H), 7.09 (d, J = 7.9 Hz, 2H), 6.72 (s, 2H), 5.11 (dd, J = 13.3, 5.2 Hz, 1H), 4.40 (d, J = 6.8 Hz, 2H), 4.35 (d, J = 7.5 Hz, 4H), 3.95 (s, 8H), 3.64 (s, 3H), 3.55 (t, J = 5.6 Hz, 2H), 3.32-3.17 (m, 4H), 2.99-2.69 (m, 11H), 2.69-2.60 (m, 2H), 2.49 (td, J = 13.1, 4.8 Hz, 1H), 2.41 (d, J = 7.0 Hz, 2H), 2.16 (dtd, J = 12.8, 5.3, 2.4 Hz, 1H), 1.91 (d, J = 13.4 Hz, 2H), 1.84 (d, J = 3.6 Hz, 1H), 1.40-1.26 (m, 2H).
D40	LCMS (ESI) m/z: [M + H] ⁺ = 795.6	¹ H NMR (400 MHz, Methanol-d ₄) δ 8.52 (s, 1H, FA), 7.58 (s, 1H), 7.45 (d, J = 8.4 Hz, 1H), 7.38-7.29 (m, 2H), 6.70 (s, 2H), 5.15 (dd, J = 13.3, 5.1 Hz, 1H), 4.48-4.39 (m, 2H), 4.36 (s, 2H), 4.23 (s, 2H), 3.93 (s, 6H), 3.82-3.74 (m, 2H), 3.64 (s, 3H), 3.55 (t, J = 5.6 Hz, 2H), 3.18 (s, 4H), 2.92 (ddd, J = 18.3, 13.5, 5.4 Hz, 2H), 2.86-2.59 (m, 11H), 2.49 (td, J = 13.2, 4.7 Hz, 1H), 2.40 (d, J = 7.1 Hz, 2H), 2.19 (ddd, J = 10.0, 5.2, 2.6 Hz, 1H), 1.93 (d, J = 13.0 Hz, 2H), 1.77 (s, 1H), 1.45-1.33 (m, 2H).
D41	LCMS (ESI) m/z: [M + H] ⁺ = 837.5	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.09 (s, 1H), 7.70-7.64 (m, 2H), 7.33 (d, J = 2.3 Hz, 1H), 7.29-7.22 (m, 1H), 6.69 (s, 2H), 6.58 (d, J = 5.0 Hz, 1H), 5.07 (dd, J = 12.9, 5.4 Hz, 1H), 4.62-4.46 (m, 2H), 4.07 (d, J = 13.0 Hz, 3H), 3.89-3.80 (m, 1H), 3.85 (s, 6H), 3.60-3.30 (m, 9H), 3.14-2.82 (m, 8H), 2.78 (d, J = 4.8 Hz, 1H), 2.63-2.52 (m, 5H), 2.17 (d, J = 17.0 Hz, 1H), 2.06-1.98 (m, 1H), 1.76 (d, J = 12.7 Hz, 2H), 1.60-1.52 (m, 3H), 1.27-1.13 (m, 2H), 0.86 (d, J = 6.7 Hz, 3H).
D42	LCMS (ESI) m/z: [M + H] ⁺ = 851.75	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.09 (s, 1H), 8.14 (s, 0H), 7.64 (d, J = 9.1 Hz, 2H), 7.31 (d, J = 2.2 Hz, 1H), 7.27-7.18 (m, 1H), 6.60 (d, J = 5.7 Hz, 3H), 5.07 (dd, J = 12.9, 5.4 Hz, 1H), 4.21 (s, 2H), 4.05 (d, J = 12.6 Hz, 2H), 3.79 (s, 6H), 3.66-3.46 (m, 5H), 3.41 (dd, J = 8.1, 2.8 Hz, 2H), 3.31 (s, 4H), 3.00-2.90 (m, 4H), 2.92-2.81 (m, 3H), 2.63-2.55 (m, 4H), 2.09-1.94 (m, 3H), 1.76 (d, J = 12.5 Hz, 2H), 1.63-1.33 (m, 3H), 1.30-1.13 (m, 3H), 1.10-0.95 (m, 6H).
D43	LCMS (ESI) m/z: [M + H] ⁺ = 867.35	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.09 (s, 1H), 8.16 (s, FA, 1H), 7.67-7.62 (m, 2H), 7.30 (d, J = 2.3 Hz, 1H), 7.23 (dd, J = 8.8, 2.3 Hz, 1H), 6.71 (t, J = 5.5 Hz, 1H), 6.59 (s, 2H), 5.07 (dd, J = 12.9, 5.4 Hz, 1H), 4.22 (s, 2H), 4.03 (d, J = 13.0 Hz, 2H), 3.78 (s, 6H), 3.53 (s, 2H), 3.50 (s, 3H), 3.45-3.31 (m, 8H), 3.24 (s, 3H), 3.24-3.16 (m, 2H), 2.98-2.82 (m, 3H), 2.63-2.53 (m, 2H), 2.43 (s, 5H), 2.35-2.30 (m, 3H), 2.04-1.97 (m, 1H), 1.74 (d, J = 12.7 Hz, 2H), 1.57 (s, 1H), 1.40-1.37 (m, 2H), 1.23-1.10 (m, 2H).
D44	LCMS (ESI) m/z: [M + H] ⁺ = 867.75	¹ H NMR (400 MHz, MeOD) δ 8.49 (s, 2FA, 2H), 7.82 (d, J = 8.3 Hz, 1H), 7.58 (s, 1H), 7.29 (d, J = 2.2 Hz, 1H), 7.25 (dd, J = 8.3, 2.3 Hz, 1H), 6.72 (s, 2H), 5.13 (dd, J = 12.6, 5.5 Hz, 1H), 4.38-4.30 (m, 3H), 3.95 (s, 5H), 3.90-3.81 (m, 1H), 3.80-3.62 (m, 1H), 3.58-3.50 (m,

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Compound No.	LCMS	¹ H NMR
D45	LCMS (ESI) m/z: [M + H] ⁺ = 826.35	3H), 3.52-3.45 (m, 1H), 3.15-3.04 (m, 2H), 2.78 (s, 4H), 2.77-2.74 (m, 1H), 2.73-2.68 (m, 1H), 2.67-2.57 (m, 3H), 2.56-2.50 (m, 4H), 2.25-2.09 (m, 1H), 2.09-1.97 (m, 6H), 1.88-1.79 (m, 4H), 1.58-1.53 (m, 2H), 1.38-1.29 (m, 3H). ¹ H NMR (400 MHz, MeOD) δ 8.49 (s, 2FA, 2H), 7.82 (d, J = 8.3 Hz, 1H), 7.58 (s, 1H), 7.29 (d, J = 2.2 Hz, 1H), 7.25 (dd, J = 8.3, 2.3 Hz, 1H), 6.72 (s, 2H), 5.13 (dd, J = 12.6, 5.5 Hz, 1H), 4.38-4.30 (m, 3H), 3.95 (s, 5H), 3.90-3.81 (m, 1H), 3.80-3.62 (m, 1H), 3.58-3.50 (m, 3H), 3.52-3.45 (m, 1H), 3.15-3.04 (m, 2H), 2.78 (s, 4H), 2.77-2.74 (m, 1H), 2.73-2.68 (m, 1H), 2.67-2.57 (m, 3H), 2.56-2.50 (m, 4H), 2.25-2.09 (m, 1H), 2.09-1.97 (m, 6H), 1.88-1.79 (m, 4H), 1.58-1.53 (m, 2H), 1.38-1.29 (m, 3H).
D46	LCMS (ESI) m/z: [M + H] ⁺ = 873.75	¹ H NMR (400 MHz, DMSO) δ 11.09 (s, 1H), 8.20 (s, FA, 1H), 7.68-7.61 (m, 2H), 7.30 (d, J = 2.2 Hz, 1H), 7.22 (dd, J = 8.7, 2.3 Hz, 1H), 7.13 (t, J = 5.7, 5.7 Hz, 1H), 6.59 (s, 2H), 6.15-5.81 (m, 1H), 5.07 (dd, J = 12.9, 5.4 Hz, 1H), 4.26 (s, 2H), 4.03 (d, J = 12.9 Hz, 2H), 3.78 (s, 6H), 3.55-3.48 (m, 5H), 3.47-3.36 (m, 5H), 2.99-2.83 (m, 3H), 2.63-2.52 (m, 4H), 2.44-2.38 (m, 5H), 2.35-2.26 (m, 4H), 2.04-1.97 (m, 1H), 1.78-1.70 (m, 2H), 1.65-1.47 (m, 1H), 1.40-1.32 (m, 2H), 1.26-1.11 (m, 2H).
D47	LCMS (ESI) m/z: [M + H] ⁺ = 867.55	¹ H NMR (400 MHz, MeOD) δ 8.51 (s, 1H), 7.81 (d, J = 8.2 Hz, 1H), 7.58 (s, 1H), 7.32-7.20 (m, 2H), 6.72 (s, 2H), 5.12 (dd, J = 12.6, 5.4 Hz, 1H), 4.41-4.31 (m, 4H), 3.95 (s, 6H), 3.64 (s, 3H), 3.59-3.45 (m, 5H), 3.11-3.07 (m, 2H), 2.92-2.82 (m, 1H), 2.81-2.69 (m, 5H), 2.67-2.59 (m, 3H), 2.53-2.57 (m, 4H), 2.20-2.09 (m, 1H), 2.08-1.96 (m, 5H), 1.87-1.78 (m, 4H), 1.68-1.43 (m, 2H).
D48	LCMS (ESI) m/z: [M + H] ⁺ = 826.55	¹ H NMR (400 MHz, MeOD) δ 7.68 (d, J = 8.5 Hz, 1H), 7.57 (s, 1H), 7.36 (d, J = 2.3 Hz, 1H), 7.23 (dd, J = 8.6, 2.3 Hz, 1H), 6.73 (s, 2H), 5.08 (dd, J = 12.5, 5.5 Hz, 1H), 4.48-4.43 (m, 2H), 4.36 (s, 2H), 4.07 (d, J = 13.1 Hz, 2H), 3.94 (s, 6H), 3.64 (s, 4H), 3.58-3.50 (m, 9H), 3.22-3.17 (m, 2H), 3.01 (t, J = 12.4, 12.4 Hz, 2H), 2.93-2.83 (m, 1H), 2.80-2.70 (m, 2H), 2.69-2.59 (m, 2H), 2.18-2.06 (m, 1), 1.88 (d, J = 12.9 Hz, 2H), 1.74-1.68 (m, 3H), 1.44-1.34 (m, 2H).
D49	LCMS (ESI) m/z: [M + H] ⁺ = 839.31	¹ H NMR (400 MHz, Methanol-d4) δ 7.61-7.50 (m, 2H), 7.18 (d, J = 8.3 Hz, 1H), 6.73 (s, 2H), 5.13 (dd, J = 13.3, 5.2 Hz, 1H), 4.54-4.45 (m, 2H), 4.38-4.34 (m, 4H), 3.95 (s, 9H), 3.65-3.61 (m, 4H), 3.59-3.51 (m, 4H), 3.37-3.32 (m, 2H), 3.18-3.08 (m, 2H), 3.04-2.98 (m, 4H), 2.96-2.86 (m, 1H), 2.86-2.80 (m, 2H), 2.79-2.74 (m, 5H), 2.64 (t, J = 5.7 Hz, 2H), 2.59-2.44 (m, 1H), 2.25-2.11 (m, 1H), 2.07-1.95 (m, 2H), 1.80-1.47 (m, 5H).
D50	LCMS (ESI) m/z: [M + H] ⁺ = 839.35	¹ H NMR (400 MHz, Methanol-d4) δ 7.56 (s, 1H), 6.77-6.64 (m, 3H), 6.57 (s, 1H), 5.04 (dd, J = 13.2, 5.4 Hz, 1H), 4.48-4.28 (m, 6H), 4.17-3.98 (m, 2H), 3.98-3.86 (m, 10H), 3.76-3.45 (m, 10H), 3.28-3.20 (m, 3H), 3.16-3.01 (m, 2H), 2.94-2.81 (m, 1H), 2.81-2.71 (m, 4H), 2.66-2.56 (m, 2H), 2.43 (qd, J = 13.0, 4.7 Hz, 1H), 2.19-2.07 (m, 1H), 2.06-1.87 (m, 3H), 1.86-1.69 (m, 3H), 1.67-1.48 (m, 2H).
D51	LCMS (ESI) m/z: [M + H] ⁺ = 821.40	¹ H NMR (400 MHz, Methanol-d4) δ 7.66-7.52 (m, 2H), 6.72 (s, 2H), 6.57 (d, J = 8.1 Hz, 2H), 5.11 (dd, J = 13.3, 5.1 Hz, 1H), 4.70-4.09 (m, 10H), 3.96 (s, 6H), 3.91-3.73 (m, 4H), 3.64 (s, 3H), 3.55 (t, J = 5.6 Hz, 2H), 3.34 (s, 4H), 3.25-3.10 (m, 1H), 3.08-3.00 (m, 1H), 2.98-2.84 (m, 1H), 2.84-2.72 (m, 4H), 2.62 (t, J = 5.3 Hz, 2H), 2.58-2.40 (m, 1H), 2.35-2.06 (m, 5H), 1.38 (d, J = 6.4 Hz, 3H).
D52	LCMS (ESI) m/z: [M + H] ⁺ = 821.55	¹ H NMR (400 MHz, Methanol-d4) δ 7.57 (s, 1H), 7.42 (d, J = 8.2 Hz, 1H), 6.88 (d, J = 2.3 Hz, 1H), 6.80 (dd, J = 8.2, 2.3 Hz, 1H), 6.73 (s, 2H), 5.14 (dd, J = 13.3, 5.2 Hz, 1H), 4.60-4.45 (m, 2H), 4.43-4.31 (m, 5H), 4.31-4.10 (m, 3H), 3.97 (s, 6H), 3.89-3.70 (m, 5H), 3.64 (s, 3H), 3.55 (t, J = 5.6 Hz, 2H), 3.32 (s, 4H), 3.17-3.00 (m, 1H), 2.99-2.85 (m, 1H), 2.85-2.80 (m, 1H), 2.78 (s, 3H), 2.62 (t, J = 5.5 Hz, 2H), 2.57-2.42 (m, 1H), 2.39-2.05 (m, 5H), 1.39 (d, J = 6.6 Hz, 3H).
D53	LCMS (ESI) m/z: [M + H] ⁺ = 698.50	¹ H NMR (400 MHz, Methanol-d4) δ 7.58 (s, 1H), 7.53 (dd, J = 8.4, 2.0 Hz, 1H), 7.24 (dd, J = 8.4, 2.4 Hz, 1H), 7.13 (dd, J = 8.1, 2.3 Hz, 1H), 6.73 (d, J = 6.8 Hz, 2H), 5.15 (dd, J = 13.3, 5.1 Hz, 1H), 4.61 (s, 1H), 4.57-4.46 (m, 3H), 4.46-4.33 (m, 5H), 4.33-4.26 (m, 1H), 4.23-4.14 (m, 1H), 3.97 (s, 2H), 3.91 (s, 4H), 3.64 (s, 3H), 3.54 (t, J = 5.5 Hz, 2H), 2.98 (d, J = 2.8 Hz, 3H), 2.98-2.86 (m, 1H), 2.82 (dd, J = 4.8, 2.5 Hz, 1H), 2.78 (s, 3H), 2.62 (s, 2H), 2.51 (qd, J = 13.2, 4.7 Hz, 1H), 2.19 (ddd, J = 10.6, 5.3, 2.9 Hz, 1H).
D54	LCMS (ESI) m/z: [M + H] ⁺ = 835.25	¹ H NMR (300 MHz, Methanol-d4) δ 8.46 (s, 2H, FA), 7.68-7.56 (m, 2H), 7.07 (d, J = 9.3 Hz, 1H), 6.75-6.65 (m, 3H), 4.37 (s, 4H), 3.96 (s, 7H), 3.65 (s, 3H), 3.60-3.50 (m, 4H), 3.50-3.40 (m, 4H), 3.20-3.10 (m, 2H), 2.80-2.50 (m, 13H), 2.10-1.90 (m, 3H), 1.80-1.40 (m, 10H).
D55	LCMS (ESI) m/z: [M + H] ⁺ = 813.35	¹ H NMR (300 MHz, DMSO-d6) δ 10.96 (s, 1H), 7.66 (s, 1H), 7.52 (d, J = 8.7 Hz, 1H), 7.15-6.99 (m, 4H), 6.62 (s, 1H), 5.15-4.98 (m, 1H), 4.40-4.13 (m, 4H), 3.99-3.72 (m, 7H), 3.61-3.31 (m, 7H), 3.23-

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Compound No.	LCMS	¹ H NMR
D56	LCMS (ESI) m/z: [M + H] ⁺ = 835.85	2.94 (m, 5H), 2.95-2.69 (m, 3H), 2.66-2.56 (m, 4H), 2.45-2.22 (m, 6H), 2.05-1.88 (m, 1H), 1.75 (d, J = 12.3 Hz, 2H), 1.67-1.46 (m, 3H), 1.37-1.11 (m, 2H). ¹ H NMR (300 MHz, MeOD) δ 8.45 (s, 2FA, 2H), 7.60 (d, J = 9.4 Hz, 2H), 6.73 (s, 2H), 6.55 (d, J = 7.4 Hz, 2H), 5.10 (dd, J = 13.2, 5.1 Hz, 1H), 4.37 (d, J = 4.4 Hz, 6H), 3.95 (s, 6H), 3.74 (s, 4H), 3.64 (s, 3H), 3.55 (t, J = 5.2, 5.2 Hz, 4H), 3.20-3.06 (m, 2H), 2.97-2.77 (m, 1H), 2.81-2.75 (m, 5H), 2.73-2.59 (m, 5H), 2.51-2.45 (m, 3H), 2.23-1.88 (m, 8H), 1.56 (s, 2H).
D57	LCMS (ESI) m/z: [M + H] ⁺ = 811.30	¹ H NMR (400 MHz, Methanol-d4) δ 7.58 (s, 1H), 7.35 (dd, J = 10.1, 2.2 Hz, 1H), 7.30 (d, J = 7.6 Hz, 1H), 7.22 (dd, J = 10.1, 4.0 Hz, 1H), 6.77-6.71 (m, 2H), 5.36 (dd, J = 12.6, 5.3 Hz, 1H), 4.37 (d, J = 6.9 Hz, 4H), 4.04-3.87 (m, 6H), 3.65 (d, J = 6.4 Hz, 5H), 3.60-3.52 (m, 4H), 3.36-3.43 (m, 2H), 3.28 (d, J = 10.1 Hz, 3H), 3.18-3.05 (m, 2H), 2.93 (d, J = 5.0 Hz, 1H), 2.90 (d, J = 5.2 Hz, 1H), 2.86 (d, J = 4.0 Hz, 1H), 2.80-2.84 (m, 1H), 2.79 (s, 3H), 2.70 (dd, J = 13.0, 4.7 Hz, 1H), 2.63 (s, 3H), 2.33 (t, J = 5.0 Hz, 1H), 2.02 (t, J = 15.9 Hz, 3H), 1.80 (t, J = 8.2 Hz, 2H), 1.65-1.48 (m, 2H).
D58	825.5	¹ H NMR (400 MHz, DMSO-d6) δ 10.96 (s, 1H), 8.14 (s, 1H, FA), 7.89 (q, J = 4.2 Hz, 1H), 7.69 (s, 1H), 7.54 (d, J = 8.4 Hz, 1H), 7.13-7.03 (m, 2H), 6.73 (s, 2H), 5.06 (dd, J = 13.2, 5.1 Hz, 1H), 4.59 (s, 2H), 4.34 (d, J = 17.0 Hz, 1H), 4.21 (d, J = 17.0 Hz, 1H), 4.16 (s, 2H), 3.93 (t, J = 5.5 Hz, 2H), 3.87 (s, 6H), 3.53 (s, 3H), 3.48-3.40 (m, 6H), 3.01-2.84 (m, 6H), 2.69-2.55 (m, 7H), 2.47-2.30 (m, 3H), 2.01-1.92 (m, 1H), 1.84 (d, J = 13.0 Hz, 2H), 1.68-1.29 (m, 5H).
D59	LCMS (ESI) m/z: [M + H] ⁺ = 821.55	¹ H NMR (300 MHz, Methanol-d4) δ 8.44 (s, 3H, FA), 7.59 (s, 1H), 7.40 (d, J = 8.2 Hz, 1H), 6.88-6.69 (m, 4H), 5.21-5.04 (m, 2H), 4.43-4.34 (m, 4H), 4.10-3.87 (m, 9H), 3.76-3.61 (m, 8H), 3.56 (s, 2H), 3.19-3.05 (m, 1H), 2.96-2.83 (m, 1H), 2.79 (s, 3H), 2.77-2.71 (m, 3H), 2.68-2.46 (m, 7H), 2.20 (s, 1H), 1.91 (s, 4H), 1.55 (d, J = 6.6 Hz, 3H).
D60	LCMS (ESI) m/z: [M + H] ⁺ = 633.30	¹ H NMR (300 MHz, DMSO-d6) δ 11.01 (s, 1H), 8.14 (s, 1H, FA), 7.65 (s, 1H), 7.30 (s, 1H), 7.22 (s, 1H), 6.65-6.57 (m, 3H), 5.19 (dd, J = 11.6, 4.5 Hz, 1H), 4.22 (s, 2H), 3.81 (s, 6H), 3.67 (s, 2H), 3.50 (s, 3H), 3.43-3.31 (m, 6H), 2.96-2.70 (m, 6H), 2.67-2.56 (m, 6H), 2.19-2.11 (m, 1H).
D61	LCMS (ESI) m/z: [M + H] ⁺ = 700.25	¹ H NMR (400 MHz, DMSO-d6) δ 7.79 (d, J = 7.6 Hz, 1H), 7.62 (s, 1H), 6.75-6.63 (m, 3H), 6.28 (d, J = 1.6 Hz, 1H), 5.23 (dd, J = 12.5, 5.3 Hz, 1H), 4.31 (s, 2H), 4.22 (s, 2H), 3.94-3.87 (m, 2H), 3.87 (s, 6H), 3.50 (s, 3H), 3.4-3.36 (m, 4H), 3.32-3.10 (m, 4H), 2.94-2.81 (m, 1H), 2.70-2.60 (m, 1H), 2.59 (s, 3H), 2.58-2.50 (m, 2H), 2.49-2.42 (m, 1H), 2.22-2.08 (m, 1H).
D62	LCMS (ESI) m/z: [M + H] ⁺ = 664.40	¹ H NMR (400 MHz, Methanol-d4) δ 8.54 (s, 0.3H, FA), 7.56 (s, 1H), 6.62 (s, 2H), 5.05 (dd, J = 12.6, 5.3 Hz, 1H), 4.37 (s, 2H), 3.88 (s, 6H), 3.84 (s, 2H), 3.64 (s, 3H), 3.54 (t, J = 5.6 Hz, 2H), 3.23 (s, 3H), 3.18 (d, J = 5.3 Hz, 4H), 2.93-2.72 (m, 9H), 2.65 (t, J = 5.6 Hz, 2H), 2.62-2.49 (m, 1H), 2.25-2.12 (m, 1H).
D63	839.7	¹ H NMR (300 MHz, DMSO-d6) δ 11.09 (s, 1H), 8.21 (s, 1H, FA), 7.86 (d, J = 4.4 Hz, 1H), 7.68 (t, J = 4.3 Hz, 2H), 7.33 (d, J = 2.2 Hz, 1H), 7.25 (dd, J = 8.7, 2.3 Hz, 1H), 6.61 (s, 2H), 5.07 (dd, J = 12.7, 5.4 Hz, 1H), 4.60 (s, 2H), 3.98-3.85 (m, 2H), 3.79 (s, 6H), 3.78 (s, 1H), 3.58 (s, 3H), 3.51 (s, 3H), 3.47-3.38 (m, 4H), 2.98-2.82 (m, 6H), 2.65-2.54 (m, 3H), 2.49-2.44 (m, 3H), 2.33 (t, J = 7.3 Hz, 2H), 2.19-1.95 (m, 3H), 1.63 (d, J = 12.0 Hz, 2H), 1.48-1.02 (m, 5H).
D64	824.7	¹ H NMR (300 MHz, DMSO-d6) δ 11.09 (s, 1H), 8.15 (s, 0.4H, FA), 7.74-7.63 (m, 2H), 7.36 (d, J = 2.2 Hz, 1H), 7.28 (dd, J = 8.6, 2.2 Hz, 1H), 6.73 (s, 2H), 5.08 (dd, J = 12.7, 5.4 Hz, 1H), 4.29 (s, 2H), 4.17 (s, 2H), 3.87 (s, 6H), 3.66 (s, 3H), 3.54-3.44 (m, 11H), 3.07-2.79 (m, 4H), 2.64-2.53 (m, 7H), 2.49-2.40 (m, 2H), 2.09-1.97 (m, 1H), 1.85 (d, J = 13.0 Hz, 2H), 1.66-1.53 (m, 1H), 1.45 (s, 4H).
D65	LCMS (ESI) m/z: [M + H] ⁺ = 836.65	¹ H NMR (300 MHz, DMSO-d6) δ 7.73-7.52 (m, 3H), 7.40 (d, J = 2.6 Hz, 1H), 6.71 (d, J = 2.0 Hz, 2H), 5.23 (dd, J = 11.4, 5.7 Hz, 1H), 4.20 (d, J = 8.2 Hz, 4H), 3.94 (s, 2H), 3.86 (s, 6H), 3.67-3.62 (m, 1H), 3.51 (s, 4H), 3.45-3.30 (m, 4H), 3.28-3.08 (m, 6H), 3.07-2.92 (m, 3H), 2.93-2.75 (m, 1H), 2.68-2.55 (m, 9H), 2.23-2.08 (m, 1H), 1.92-1.70 (m, 3H), 1.68-1.52 (m, 2H), 1.52-1.28 (m, 2H).
D66	752.3	¹ H NMR (300 MHz, Methanol-d4) δ 7.41 (d, J = 8.2 Hz, 1H), 7.23 (s, 1H), 6.88 (d, J = 2.2 Hz, 1H), 6.86 (s, 2H), 6.80 (dd, J = 8.2, 2.3 Hz, 1H), 5.14 (dd, J = 13.3, 5.1 Hz, 1H), 4.48 (s, 2H), 4.35 (dd, J = 27.7, 5.0 Hz, 6H), 4.13 (d, J = 10.3 Hz, 2H), 3.96 (s, 6H), 3.77 (d, J = 17.5 Hz, 4H), 3.63 (s, 3H), 3.49 (dd, J = 10.8, 6.4 Hz, 7H), 3.10 (d, J = 1.7 Hz, 2H), 2.99-2.84 (m, 1H), 2.79 (dd, J = 13.0, 2.7 Hz, 1H), 2.60-2.41 (m, 1H), 2.32-2.02 (m, 4H).
D67	706.52	¹ H NMR (400 MHz, DMSO-d6) δ 10.81 (s, 1H), 8.17 (s, 2H), 7.60 (s, 1H), 6.56 (s, 3H), 5.73 (s, 1H), 4.19 (s, 2H), 3.76 (s, 6H), 3.51 (s, 2H), 3.47 (s, 3H), 3.38 (t, J = 5.5 Hz, 2H), 3.26-3.08 (m, 1H), 3.07-2.92

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Compound No.	LCMS	¹ H NMR
D68	LCMS (ESI) m/z: [M + H] ⁺ = 808.43	(m, 2H), 2.81 (d, J = 11.5 Hz, 2H), 2.57 (d, J = 4.2 Hz, 3H), 2.28 (t, J = 7.4 Hz, 3H), 2.04 (s, 1H), 1.57 (d, J = 12.0 Hz, 2H), 1.15 (s, 3H), 1.05 (d, J = 11.4 Hz, 2H). ¹ H NMR (300 MHz, DMSO-d ₆) δ 11.00 (s, 1H), 8.23 (s, 2H, FA), 7.69-7.62 (m, 2H), 7.49 (s, 1H), 7.40 (d, J = 7.9 Hz, 1H), 6.61 (s, 3H), 5.11 (dd, J = 13.2, 5.1 Hz, 1H), 4.46-4.26 (m, 3H), 4.22 (s, 2H), 3.80 (s, 6H), 3.64 (s, 2H), 3.50 (s, 3H), 3.44-3.36 (m, 2H), 3.05 (d, J = 10.6 Hz, 2H), 2.92 (d, J = 10.4 Hz, 3H), 2.67-2.57 (m, 5H), 2.49-2.30 (m, 3H), 2.29-2.06 (m, 4H), 2.04-1.94 (m, 1H), 1.87-1.57 (m, 6H), 1.40 (s, 2H), 1.34-1.07 (m, 3H).
D69	LCMS (ESI) m/z: [M + H] ⁺ = 809.40	¹ H NMR (400 MHz, Methanol-d ₄) δ 7.67-7.48 (m, 2H), 7.45-7.32 (m, 2H), 6.73 (d, J = 6.3 Hz, 2H), 5.25-5.09 (m, 1H), 4.52-4.42 (m, 2H), 4.37 (d, J = 3.6 Hz, 3H), 3.96 (s, 2H), 3.95 (s, 6H), 3.75 (d, J = 13.9 Hz, 2H), 3.64 (s, 3H), 3.60 (d, J = 12.7 Hz, 2H), 3.54 (d, J = 5.6 Hz, 2H), 3.32-3.25 (m, 3H), 3.19 (s, 1H), 3.18-3.05 (m, 2H), 2.98-2.86 (m, 1H), 2.82 (dd, J = 4.8, 2.5 Hz, 1H), 2.78 (s, 3H), 2.64 (t, J = 5.6 Hz, 2H), 2.51 (qd, J = 13.2, 4.7 Hz, 1H), 2.24-2.14 (m, 1H), 2.04 (d, J = 13.9 Hz, 2H), 2.00-1.86 (m, 1H), 1.86-1.70 (m, 3H), 1.68-1.51 (m, 2H).
D70	LCMS (ESI) m/z: [M + H] ⁺ = 824.35	¹ H NMR (300 MHz, DMSO-d ₆) δ 8.15 (s, 1H, FA), 7.73 (d, J = 8.5 Hz, 1H), 7.66-7.49 (m, 3H), 7.38 (d, J = 2.2 Hz, 1H), 7.33-7.23 (m, 2H), 6.62-6.54 (m, 3H), 4.22 (s, 2H), 4.07 (d, J = 13.1 Hz, 2H), 3.78 (s, 6H), 3.55-3.49 (m, 5H), 3.44-3.39 (m, 4H), 2.97 (t, J = 12.5 Hz, 3H), 2.60 (d, J = 4.2 Hz, 4H), 2.44-2.31 (m, 8H), 1.76 (d, J = 12.9 Hz, 2H), 1.65-1.52 (m, 1H), 1.42-1.30 (m, 2H), 1.25-1.13 (m, 2H).
D71	LCMS (ESI) m/z: [M + H] ⁺ = 809.43	¹ H NMR (300 MHz, DMSO-d ₆) δ 7.86-7.52 (m, 4H), 6.73 (s, 2H), 5.10 (dd, J = 13.2, 5.2 Hz, 1H), 4.51-4.28 (m, 2H), 4.24 (d, J = 9.8 Hz, 4H), 3.88 (s, 6H), 3.67-3.58 (m, 2H), 3.51 (s, 3H), 3.43-3.28 (m, 6H), 3.22-2.99 (m, 6H), 2.96-2.72 (m, 4H), 2.67-2.53 (m, 8H), 2.48-2.33 (m, 2H), 2.10-1.95 (m, 4H).
D72	LCMS (ESI) m/z: [M + H] ⁺ = 809.43	¹ H NMR (300 MHz, DMSO-d ₆) δ 7.73 (d, J = 7.9 Hz, 1H), 7.62 (s, 1H), 7.53-7.41 (m, 2H), 6.73 (s, 2H), 5.10 (dd, J = 13.2, 5.1 Hz, 1H), 4.50-4.29 (m, 2H), 4.24 (d, J = 10.0 Hz, 4H), 3.87 (s, 6H), 3.67-3.54 (m, 5H), 3.44-3.28 (m, 6H), 3.19-2.86 (m, 8H), 2.79-2.72 (m, 2H), 2.67-2.52 (m, 8H), 2.47-2.31 (m, 2H), 2.06-1.95 (m, 4H).
D73	LCMS (ESI) m/z: [M + H] ⁺ = 808.43	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.1 (br s, 1H), 8.26 (s, 2H, FA), 7.64 (s, 1H), 7.55 (d, J = 8.9 Hz, 3H), 6.61 (s, 3H), 5.11 (dd, J = 13.2, 5.1 Hz, 1H), 4.42 (d, J = 17.1 Hz, 1H), 4.28 (d, J = 17.2 Hz, 1H), 4.22 (s, 2H), 3.80 (s, 6H), 3.62 (s, 2H), 3.50 (s, 3H), 3.40 (t, J = 5.9 Hz, 2H), 3.05 (d, J = 10.7 Hz, 2H), 2.90 (d, J = 12.0 Hz, 3H), 2.71-2.54 (m, 5H), 2.42 (s, 4H), 2.46-2.35 (m, 3H), 2.16 (q, J = 12.1 Hz, 4H), 2.05-1.93 (m, 1H), 1.82-1.58 (m, 6H), 1.41 (s, 2H), 1.32-1.12 (m, 3H).
D74	LCMS (ESI) m/z: [M + H] ⁺ = 701.50	¹ H NMR (300 MHz, DMSO-d ₆) δ 8.15 (s, 1H, FA), 7.64 (s, 1H), 6.68 (s, 2H), 6.64-6.57 (m, 1H), 4.22 (s, 2H), 3.95 (s, 2H), 3.84 (s, 6H), 3.51 (s, 4H), 3.44-3.39 (m, 5H), 3.19-3.08 (m, 3H), 2.86 (s, 4H), 2.70-2.57 (m, 7H), 2.46-2.41 (m, 2H), 2.10 (s, 3H), 1.74 (s, 7H), 1.67-1.55 (m, 6H), 1.45-1.26 (m, 5H).
D75	LCMS (ESI) m/z: [M + H] ⁺ = 859.50	¹ H NMR (400 MHz, Methanol-d ₄) δ 8.22 (s, 1H, FA), 7.58 (s, 1H), 7.38 (d, J = 6.1 Hz, 1H), 6.73 (s, 2H), 5.12 (dd, J = 12.7, 5.4 Hz, 1H), 4.36 (s, 4H), 3.95 (s, 6H), 3.64 (s, 3H), 3.55 (t, J = 5.6 Hz, 4H), 3.50 (t, J = 4.7 Hz, 4H), 3.18-3.06 (m, 2H), 2.98 (s, 4H), 2.90-2.69 (m, 8H), 2.64 (t, J = 5.7 Hz, 2H), 2.19-2.11 (m, 1H), 2.02 (d, J = 13.8 Hz, 2H), 1.80-1.46 (m, 5H).
D76	LCMS (ESI) m/z: [M + H] ⁺ = 820.55	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.07 (s, 1H), 8.19 (s, 2H, FA), 8.09 (s, 1H), 7.82 (s, 1H), 7.63 (s, 1H), 7.42 (d, J = 8.7 Hz, 2H), 6.93 (d, J = 8.7 Hz, 2H), 6.58 (s, 3H), 5.36 (dd, J = 11.9, 5.1 Hz, 1H), 4.21 (s, 2H), 3.78 (s, 6H), 3.52 (s, 2H), 3.50 (s, 3H), 3.42-3.39 (m, 6H), 3.14-3.11 (m, 4H), 2.86-2.79 (m, 3H), 2.60 (d, J = 4.3 Hz, 4H), 2.49-2.44 (m, 5H), 2.29-2.22 (m, 1H), 2.08-2.00 (m, 2H), 1.65-1.58 (m, 2H), 1.40-1.34 (m, 2H), 1.29-1.22 (m, 1H), 1.18-1.08 (m, 2H).
D77	LCMS (ESI) m/z: [M + H] ⁺ = 811.55	¹ H NMR (300 MHz, Methanol-d ₄) δ 8.41 (s, 1H, FA), 7.72 (d, J = 7.8 Hz, 1H), 7.58 (s, 1H), 6.73 (s, 3H), 6.17 (s, 1H), 5.27 (dd, J = 12.3, 5.3 Hz, 1H), 4.36 (s, 4H), 3.95 (s, 6H), 3.64 (s, 3H), 3.60-3.49 (m, 4H), 3.43 (s, 4H), 3.13-3.04 (m, 1H), 2.92-2.76 (m, 9H), 2.72-2.54 (m, 6H), 2.36-2.24 (m, 1H), 2.08-1.95 (m, 2H), 1.78-1.44 (m, 5H).
D78	LCMS (ESI) m/z: [M + H] ⁺ = 858.50	¹ H NMR (400 MHz, Methanol-d ₄) δ 8.50 (s, 2H, FA), 7.58 (s, 1H), 7.51 (d, J = 9.6 Hz, 1H), 6.73 (s, 2H), 5.12 (dd, J = 12.8, 5.4 Hz, 1H), 4.36 (s, 4H), 3.96 (s, 6H), 3.64 (s, 4H), 3.55 (t, J = 5.6 Hz, 3H), 3.44 (s, 4H), 3.14-3.07 (m, 1H), 2.97-2.83 (m, 2H), 2.78 (s, 4H), 2.76-2.69 (m, 5H), 2.66-2.57 (m, 4H), 2.17-2.11 (m, 1H), 2.06-1.98 (m, 2H), 1.75-1.49 (m, 5H).

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Compound No.	LCMS	¹ H NMR
D79	793.47	
D80	826.37	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.05 (s, 1H), 8.13 (s, 1H), 7.66 (d, J = 8.5 Hz, 1H), 7.61 (s, 1H), 7.31 (d, J = 2.2 Hz, 1H), 7.23 (dd, J = 8.7, 2.3 Hz, 1H), 6.61 (s, 1H), 4.19 (s, 2H), 3.79 (s, 6H), 3.76 (s, 2H), 3.48 (s, 3H), 3.30 (s, 6H), 2.99 (t, J = 10.0 Hz, 2H), 2.57 (d, J = 4.1 Hz, 2H), 2.36-2.28 (m, 2H), 1.98 (d, J = 12.8 Hz, 1H), 1.37 (s, 3H), 1.80-1.61 (m, 2H), 1.32-1.18 (m, 2H).
D81	LCMS (ESI) m/z: [M + H] ⁺ = 849.25	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.08 (s, 1H), 8.15 (s, 2H, FA), 7.69 (d, J = 8.5 Hz, 1H), 7.63 (s, 1H), 7.34 (s, 1H), 7.26 (d, J = 8.8 Hz, 1H), 6.75 (s, 1H), 6.64 (s, 2H), 5.07 (dd, J = 12.9, 5.3 Hz, 1H), 4.20 (s, 2H), 3.82 (s, 10H), 3.50 (s, 5H), 3.45-3.41 (m, 8H), 3.05-2.87 (m, 4H), 2.63-2.58 (m, 3H), 2.41-2.26 (m, 3H), 2.06-1.95 (m, 1H), 1.79-1.66 (m, 2H), 1.45-1.36 (m, 3H), 1.33-1.21 (m, 2H), 0.57-0.51 (m, 2H), 0.45-0.39 (m, 2H).
D82	LCMS (ESI) m/z: [M + H] ⁺ = 939.	¹ H NMR (300 MHz, DMSO-d ₆) δ 7.70 (d, J = 8.5 Hz, 1H), 7.64 (s, 1H), 7.35 (d, J = 2.2 Hz, 1H), 7.27 (d, J = 8.9 Hz, 1H), 6.68-6.56 (m, 3H), 5.72-5.59 (m, 2H), 5.27 (dd, J = 13.0, 5.3 Hz, 1H), 4.77 (p, J = 6.2 Hz, 1H), 4.22 (s, 2H), 3.82 (s, 8H), 3.50 (s, 3H), 3.47-3.38 (m, 8H), 3.16-2.96 (m, 4H), 2.90-2.66 (m, 2H), 2.65-2.53 (m, 7H), 2.40-2.26 (m, 3H), 2.14-2.03 (m, 1H), 1.72 (d, J = 12.3 Hz, 2H), 1.47-1.35 (m, 3H), 1.31-1.19 (m, 8H).
D83	LCMS (ESI) m/z: [M + H] ⁺ = 827.45	¹ H NMR (300 MHz, DMSO-d ₆) δ 10.98 (s, 1H), 8.16 (s, 2H, FA), 7.64 (s, 1H), 7.44 (d, J = 11.8 Hz, 1H), 7.29 (d, J = 8.0 Hz, 1H), 6.65-6.52 (m, 3H), 5.09 (dd, J = 13.4, 5.0 Hz, 1H), 4.41-4.19 (m, 5H), 3.80 (s, 7H), 3.65 (s, 4H), 3.50 (s, 5H), 3.41 (s, 2H), 3.04 (s, 4H), 2.98-2.87 (m, 3H), 2.61-2.58 (m, 4H), 2.44-2.30 (m, 4H), 2.07-1.93 (m, 1H), 1.72-1.59 (m, 2H), 1.47-1.10 (m, 6H).
D84	LCMS (ESI) m/z: [M + H] ⁺ = 827.35	¹ H NMR (300 MHz, DMSO-d ₆) δ 10.98 (s, 1H), 8.16 (s, 2H, FA), 7.64 (s, 1H), 7.43 (d, J = 11.6 Hz, 1H), 7.23 (d, J = 7.5 Hz, 1H), 6.66-6.56 (m, 3H), 5.08 (dd, J = 13.2, 5.1 Hz, 1H), 4.40-4.25 (m, 2H), 4.22 (s, 2H), 3.88-3.74 (m, 9H), 3.50 (s, 4H), 3.45-3.38 (m, 5H), 3.11 (s, 4H), 3.06-2.85 (m, 4H), 2.65-2.57 (m, 5H), 2.41-2.32 (m, 4H), 2.04-1.93 (m, 1H), 1.71 (d, J = 12.4 Hz, 2H), 1.44-1.24 (m, 5H).
D85	LCMS (ESI) m/z: [M + H] ⁺ = 809.80	¹ H NMR (400 MHz, DMSO-d ₆) δ 10.95 (s, 1H), 8.18 (s, 2H, FA), 7.64 (s, 1H), 7.55-7.48 (m, 1H), 7.06 (d, J = 7.9 Hz, 2H), 6.65-6.56 (m, 3H), 5.05 (dd, J = 13.3, 5.1 Hz, 1H), 4.33 (d, J = 16.9 Hz, 1H), 4.26-4.17 (m, 3H), 3.81 (s, 6H), 3.71 (s, 2H), 3.50 (s, 3H), 3.41 (t, J = 5.6 Hz, 3H), 3.30-3.24 (m, 5H), 3.02-2.85 (m, 4H), 2.63-2.52 (m, 6H), 2.43-2.23 (m, 6H), 2.00-1.92 (m, 1H), 1.68 (d, J = 12.5 Hz, 2H), 1.44-1.30 (m, 3H), 1.29-1.16 (m, 2H).
D86	LCMS (ESI) m/z: [M + H] ⁺ = 840.40	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.10 (s, 1H), 8.16 (s, 1H, FA), 7.64 (s, 1H), 7.24 (s, 1H), 7.05 (d, J = 13.2 Hz, 1H), 6.66 (s, 2H), 6.60 (d, J = 4.6 Hz, 1H), 5.07 (dd, J = 12.8, 5.4 Hz, 1H), 4.22 (s, 2H), 3.87 (s, 2H), 3.83 (s, 6H), 3.53-3.44 (m, 8H), 3.43-3.38 (m, 3H), 3.09 (d, J = 11.4 Hz, 2H), 2.94-2.83 (m, 1H), 2.63-2.54 (m, 6H), 2.49-2.41 (m, 5H), 2.37-2.30 (m, 2H), 2.06-1.97 (m, 1H), 1.73 (d, J = 12.6 Hz, 2H), 1.47-1.36 (m, 3H), 1.35-1.22 (m, 2H).
D87	LCMS (ESI) m/z: [M + H] ⁺ = 840.95	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.12 (s, 1H), 8.16 (s, 1H, FA), 7.67-7.61 (m, 2H), 7.37 (t, J = 7.7 Hz, 1H), 6.66-6.55 (m, 3H), 5.10 (dd, J = 12.8, 5.4 Hz, 1H), 4.22 (s, 2H), 3.81 (s, 6H), 3.74 (s, 2H), 3.50 (s, 4H), 3.41 (t, J = 5.5 Hz, 5H), 3.22 (t, J = 4.8 Hz, 5H), 2.99 (d, J = 11.2 Hz, 2H), 2.94-2.78 (m, 2H), 2.64-2.56 (m, 5H), 2.40-2.31 (m, 4H), 2.07-1.97 (m, 1H), 1.69 (d, J = 12.5 Hz, 2H), 1.38 (s, 3H), 1.29-1.16 (m, 2H).
D88	LCMS (ESI) m/z: [M + H] ⁺ = 841.55	¹ H NMR (300 MHz, DMSO-d ₆ , D ₂ O) δ 7.80 (d, J = 11.1 Hz, 1H), 8.14(s, 0H, FA), 7.64 (s, 1H), 7.56 (s, 1H), 6.73 (s, 2H), 5.12 (dd, J = 12.8, 5.4 Hz, 1H), 4.31-4.14 (m, 4H), 3.87 (s, 6H), 3.51 (s, 3H), 3.48-3.39 (m, 6H), 3.31-3.06 (m, 4H), 3.06-2.80 (m, 4H), 2.66-2.56 (m, 5H), 2.56-2.53 (m, 4H), 2.10-1.99 (m, 1H), 1.91-1.82 (m, 2H), 1.81-1.15 (m, 6H).
D89	LCMS (ESI) m/z: [M + H] ⁺ = 782.65	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.02 (s, 1H), 8.27 (s, 1H, FA), 7.67-7.55 (m, 1H), 7.28 (t, J = 7.7 Hz, 1H), 6.92 (d, J = 7.4 Hz, 1H), 6.73 (d, J = 8.1 Hz, 1H), 6.67-6.62 (m, 1H), 6.59 (s, 2H), 5.56 (t, J = 5.0 Hz, 1H), 5.11 (dd, J = 13.2, 5.1 Hz, 1H), 4.27-4.06 (m, 5H), 3.83-3.73 (m, 7H), 3.65 (s, 2H), 3.50 (s, 3H), 3.40 (t, J = 5.4 Hz, 2H), 3.28 (t, J = 7.1 Hz, 3H), 3.14-3.00 (m, 4H), 2.97-2.86 (m, 1H), 2.67-2.53 (m, 2H), 2.34-2.25 (m, 1H), 2.09-1.98 (m, 1H), 1.98-1.84 (m, 2H), 1.63-1.50 (m, 2H), 1.43-1.21 (m, 10H).
D90	LCMS (ESI) m/z: [M + H] ⁺ = 755.65	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.12 (s, 1H), 8.18(s, 0H, FA), 7.83 (d, J = 8.3 Hz, 1H), 7.64 (s, 1H), 7.43 (d, J = 2.2 Hz, 1H), 7.35 (dd, J = 8.3, 2.3 Hz, 1H), 6.69 (t, J = 5.4 Hz, 1H), 6.64 (s, 2H), 5.12 (dd, J = 12.9, 5.3 Hz, 1H), 4.19 (dd, J = 13.5, 7.1 Hz, 4H), 3.91 (s, 2H), 3.82 (s, 6H), 3.65-3.55 (m, 4H), 3.50 (s, 3H), 3.41 (t, J = 5.3 Hz, 4H), 3.13-3.04 (m, 2H), 2.96-2.83 (m, 1H), 2.66-2.55 (m, 2H), 2.15-2.00 (m, 3H), 1.85-1.68 (m, 2H), 1.57-1.37 (m, 4H).

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Compound No.	LCMS	¹ H NMR
D91	LCMS (ESI) m/z: [M + H] ⁺ = 811.70	¹ H NMR (300 MHz, Methanol-d ₄) δ 8.54 (br s, 1H, FA), 7.72 (d, J = 8.4 Hz, 1H), 7.61 (s, 1H), 7.41 (d, J = 2.1 Hz, 1H), 7.28 (d, J = 8.7 Hz, 1H), 6.98-6.87 (m, 2H), 5.10 (dd, J = 12.3, 5.3 Hz, 1H), 4.39-4.33 (m, 4H), 3.99 (s, 3H), 3.64 (s, 4H), 3.55 (s, 8H), 3.09 (t, J = 12.1 Hz, 2H), 2.92-2.80 (m, 5H), 2.80-2.71 (m, 5H), 2.64 (d, J = 5.9 Hz, 4H), 2.14 (s, 1H), 2.00 (s, 1H), 1.78-1.45 (m, 5H).
D92	LCMS (ESI) m/z: [M + H] ⁺ = 794.80.	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.09 (s, 1H), 8.16 (s, 2H, FA), 8.08 (d, J = 1.7 Hz, 1H), 7.73-7.64 (m, 2H), 7.42 (d, J = 1.8 Hz, 1H), 7.34 (d, J = 2.2 Hz, 1H), 7.26 (dd, J = 8.7, 2.3 Hz, 1H), 6.61 (q, J = 4.4 Hz, 1H), 5.08 (dd, J = 12.8, 5.4 Hz, 1H), 4.22 (s, 2H), 3.85 (s, 3H), 3.69 (s, 2H), 3.50 (s, 3H), 3.46-3.40 (m, 8H), 2.98-2.83 (m, 3H), 2.64-2.55 (m, 5H), 2.45-2.25 (m, 5H), 2.17 (t, J = 11.4 Hz, 2H), 2.06-1.98 (m, 1H), 1.65 (d, J = 11.8 Hz, 2H), 1.44-1.09 (m, 6H)
D93	848.51	
D94	863.63	
D95	863.56	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.03 (s, 1H), 8.14 (s, 1H), 7.62 (d, J = 8.6 Hz, 2H), 7.27 (d, J = 2.3 Hz, 1H), 7.20 (dd, J = 8.6, 2.4 Hz, 1H), 6.56 (s, 3H), 5.03 (dd, J = 13.0, 5.4 Hz, 1H), 4.19 (s, 2H), 4.10-3.91 (m, 3H), 3.76 (s, 6H), 3.51 (s, 2H), 3.47 (s, 3H), 3.37 (d, J = 5.8 Hz, 1H), 3.15 (d, J = 2.4 Hz, 2H), 2.92 (t, J = 12.1 Hz, 2H), 2.89-2.80 (m, 1H), 2.65 (q, J = 1.9 Hz, 1H), 2.57 (d, J = 4.3 Hz, 3H), 2.34-2.23 (m, 4H), 1.99 (d, J = 14.8 Hz, 3H), 1.77-1.67 (m, 2H), 1.59 (s, 5H), 1.28 (d, J = 7.0 Hz, 1H), 1.24-1.08 (m, 1H).
D96	835.56	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.03 (s, 1H), 8.16 (s, 1H), 7.70-7.49 (m, 2H), 7.28 (d, J = 2.3 Hz, 1H), 7.20 (dd, J = 8.7, 2.4 Hz, 1H), 6.55 (s, 2H), 5.04 (dd, J = 12.8, 5.4 Hz, 1H), 4.19 (s, 2H), 4.01 (d, J = 13.0 Hz, 2H), 3.75 (s, 5H), 3.73-3.59 (m, 2H), 3.47 (s, 3H), 3.38 (t, J = 5.5 Hz, 2H), 3.15 (s, 2H), 2.99-2.88 (m, 2H), 2.88-2.79 (m, 1H), 2.57 (d, J = 4.3 Hz, 3H), 2.43-2.12 (m, 3H), 2.03-1.93 (m, 1H), 1.74 (d, J = 12.8 Hz, 2H), 1.62 (s, 1H), 1.43 (t, J = 7.1 Hz, 1H), 1.24-1.12 (m, 2H), 0.43 (q, J = 4.8 Hz, 1H), 0.22-0.03 (m, 1H).
D97	890.39	
D98	865.52	
D99	837.59	
D100	837.59	
D101	850.44	
D102	851.59	
D103	867.5	
D104	873.43	
D105	831.57	
D106	LCMS (ESI) m/z: [M + H] ⁺ = 878.55	¹ H NMR (400 MHz, Methanol-d ₄) δ 7.56 (s, 1H), 6.77-6.64 (m, 3H), 6.57 (s, 1H), 5.04 (dd, J = 13.2, 5.4 Hz, 1H), 4.48-4.28 (m, 6H), 4.17-3.98 (m, 2H), 3.98-3.86 (m, 10H), 3.76-3.45 (m, 10H), 3.28-3.20 (m, 3H), 3.16-3.01 (m, 2H), 2.94-2.81 (m, 1H), 2.81-2.71 (m, 4H), 2.66-2.56 (m, 2H), 2.43 (qd, J = 13.0, 4.7 Hz, 1H), 2.19-2.07 (m, 1H), 2.06-1.87 (m, 3H), 1.86-1.69 (m, 3H), 1.67-1.48 (m, 2H).
D107	LCMS (ESI) m/z: [M + H] ⁺ = 835.45.	¹ H NMR (300 MHz, Methanol-d ₄) δ 8.45 (br s, 1H, FA), 7.64 (d, J = 8.3 Hz, 1H), 7.58 (s, 1H), 6.82 (d, J = 2.0 Hz, 1H), 6.73 (s, 2H), 6.66 (dd, J = 8.3, 2.1 Hz, 1H), 5.07 (dd, J = 12.3, 5.4 Hz, 1H), 4.46 (s, 2H), 4.36 (s, 2H), 4.21-4.08 (m, 3H), 3.99-3.85 (m, 7H), 3.78 (s, 4H), 3.64 (s, 3H), 3.55 (t, J = 5.4 Hz, 2H), 2.97 (s, 2H), 2.91-2.68 (m, 6H), 2.66-2.58 (m, 4H), 2.51-2.41 (m, 2H), 2.19-2.01 (m, 1H), 1.97-1.75 (m, 4H), 0.98 (d, J = 5.0 Hz, 3H).
D109	LCMS (ESI) m/z: [M + H] ⁺ = 821.50.	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.07 (s, 1H), 8.19 (s, 1H, FA), 7.64 (d, J = 8.3 Hz, 2H), 6.77 (d, J = 2.1 Hz, 1H), 6.65 (dd, J = 8.4, 2.1 Hz, 1H), 6.62-6.57 (m, 3H), 5.05 (dd, J = 12.9, 5.4 Hz, 1H), 4.22 (s, 2H), 3.80 (s, 6H), 3.73 (s, 4H), 3.70 (s, 2H), 3.50 (s, 4H), 3.48-3.45 (m, 6H), 3.09-3.01 (m, 3H), 2.92-2.87 (m, 1H), 2.60 (d, J = 4.3 Hz, 4H), 2.45-2.40 (m, 2H), 2.28 (s, 4H), 2.05-1.97 (m, 1H), 1.78-1.69 (m, 4H).
D110	LCMS (ESI) m/z: [M + H] ⁺ = 794.30	
D111	LCMS (ESI) m/z: [M + H] ⁺ = 643.25.	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.06 (s, 1H), 7.60 (s, 1H), 7.55 (d, J = 8.4 Hz, 1H), 7.22-7.14 (m, 2H), 6.98 (dd, J = 8.5, 2.1 Hz, 1H), 6.64 (s, 2H), 6.58 (d, J = 4.6 Hz, 1H), 5.02 (dd, J = 13.0, 5.3 Hz, 1H), 4.32 (d, J = 5.3 Hz, 2H), 4.21 (s, 2H), 3.86 (s, 6H), 3.49 (s, 3H), 3.41 (d, J = 5.5 Hz, 3H), 2.95-2.81 (m, 1H), 2.59 (d, J = 4.3 Hz, 3H), 2.54 (s, 3H), 1.99 (d, J = 12.5 Hz, 1H).
D112	LCMS (ESI) m/z: [M + H] ⁺ = 657.10.	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.06 (s, 1H), 7.63 (t, J = 4.3 Hz, 2H), 7.31 (d, J = 2.4 Hz, 1H), 7.11-7.01 (m, 1H), 6.64 (s, 2H), 6.58 (d, J = 4.5 Hz, 1H), 5.04 (dd, J = 12.9, 5.3 Hz, 1H), 4.64 (s, 2H), 4.20 (s, 2H), 3.85 (s, 6H), 3.48 (s, 3H), 3.09 (s, 3H), 2.86 (d, J = 12.5 Hz, 1H), 2.68 (p, J = 1.8 Hz, 3H), 2.59 (s, 3H), 2.58 (s, 6H), 2.00 (d, J = 12.5 Hz, 1H).

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Compound No.	LCMS	¹ H NMR
D113	LCMS (ESI) m/z: [M + H] ⁺ = 822.70	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.12 (s, 1H), 8.18 (s, 1H), 7.86 (d, J = 7.7 Hz, 1H), 7.81-7.74 (m, 2H), 7.64 (s, 1H), 6.62 (s, 2H), 6.60-6.57 (m, 1H), 5.14 (dd, J = 12.9, 5.4 Hz, 1H), 4.22 (s, 2H), 3.80 (s, 6H), 3.67 (s, 2H), 3.50 (s, 4H), 3.41 (t, J = 5.6 Hz, 4H), 3.03 (d, J = 10.9 Hz, 2H), 2.97-2.88 (m, 3H), 2.77 (t, J = 12.2 Hz, 1H), 2.63-2.58 (m, 4H), 2.43-2.36 (m, 2H), 2.30-2.21 (m, 2H), 2.12-2.02 (m, 3H), 1.86-1.79 (m, 2H), 1.78-1.63 (m, 4H), 1.43-1.28 (m, 3H), 1.25-1.14 (m, 2H).
D114	LCMS (ESI) m/z: [M + H] ⁺ = 892.45	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.12 (s, 1H), 9.24 (s, 1H, TFA), 8.32 (s, 1H, TFA), 7.86 (d, J = 8.2 Hz, 1H), 7.64 (s, 1H), 7.35-7.26 (m, 2H), 6.75 (s, 2H), 6.62 (d, J = 4.6 Hz, 1H), 5.13 (dd, J = 12.8, 5.4 Hz, 1H), 5.09-5.00 (m, 1H), 4.27-4.10 (m, 4H), 3.89 (d, J = 2.4 Hz, 6H), 3.52 (s, 3H), 3.43-3.41 (m, 4H), 3.24-3.16 (m, 1H), 3.08-2.81 (m, 5H), 2.75-2.69 (m, 1H), 2.66-2.52 (m, 8H), 2.43-2.39 (m, 1H), 2.17-1.96 (m, 6H), 1.94-1.81 (m, 4H), 1.73-1.61 (m, 2H), 1.24 (s, 6H).
D115	LCMS (ESI) m/z: [M + H] ⁺ = 809.35	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.08 (s, 1H), 8.18 (s, 1H, FA), 7.68 (d, J = 8.5 Hz, 1H), 7.63 (s, 1H), 7.34 (d, J = 2.2 Hz, 1H), 7.26 (dd, J = 8.8, 2.3 Hz, 1H), 6.64 (s, 2H), 6.59 (q, J = 4.4 Hz, 1H), 5.07 (dd, J = 12.9, 5.4 Hz, 1H), 4.21 (s, 2H), 3.90 (s, 2H), 3.82 (s, 6H), 3.50 (s, 3H), 3.44-3.39 (m, 8H), 3.37-3.28 (m, 5H), 3.08 (t, J = 8.9 Hz, 2H), 2.94-2.80 (m, 3H), 2.62-2.56 (m, 4H), 2.30 (t, J = 7.2 Hz, 2H), 2.21-2.13 (m, 1H), 2.06-1.94 (m, 2H), 1.61-1.49 (m, 2H), 1.48-1.39 (m, 1H).
D116	LCMS (ESI) m/z: [M + H] ⁺ = 699.40	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.12 (s, 1H), 8.19 (s, 2H, FA), 7.86-7.80 (m, 1H), 7.64 (s, 1H), 7.30-7.23 (m, 2H), 6.60 (s, 3H), 5.12 (dd, J = 12.8, 5.4 Hz, 1H), 4.94 (t, J = 5.5 Hz, 1H), 4.21 (s, 2H), 3.78 (s, 6H), 3.72-3.65 (m, 5H), 3.50 (s, 3H), 3.40 (s, 3H), 3.13-3.08 (m, 2H), 2.91-2.86 (m, 1H), 2.63-2.57 (m, 5H), 2.09-2.00 (m, 1H).
D117	LCMS (ESI) m/z: [M + H] ⁺ = 823.55	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.15 (s, 1H), 9.47 (s, 1H, TFA), 7.93 (d, J = 7.7 Hz, 1H), 7.87 (s, 1H), 7.83-7.77 (m, 1H), 7.64 (s, 1H), 6.74 (s, 2H), 6.63 (s, 1H), 5.16 (dd, J = 12.8, 5.4 Hz, 1H), 4.24 (d, J = 7.7 Hz, 4H), 3.88 (s, 6H), 3.62 (d, J = 10.7 Hz, 2H), 3.52 (s, 3H), 3.44-3.31 (m, 6H), 3.21-3.00 (m, 7H), 2.98-2.72 (m, 4H), 2.66-2.58 (m, 4H), 2.50-2.42 (m, 4H), 2.15-1.93 (m, 5H).
D118	LCMS (ESI) m/z: [M + H] ⁺ = 849.50	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.09 (s, 1H), 8.18 (s, 2H, FA), 7.69 (d, J = 8.5 Hz, 1H), 7.34 (d, J = 2.8 Hz, 2H), 7.26 (dd, J = 8.7, 2.2 Hz, 1H), 6.68-6.56 (m, 3H), 5.08 (dd, J = 12.8, 5.3 Hz, 1H), 4.23 (s, 2H), 3.82 (s, 9H), 3.49-3.30 (m, 10H), 3.07 (d, J = 11.1 Hz, 3H), 2.97-2.73 (m, 3H), 2.63-2.55 (m, 6H), 2.40-2.30 (m, 2H), 2.08-1.97 (m, 1H), 1.73 (d, J = 12.2 Hz, 2H), 1.51-1.18 (m, 5H), 1.00 (s, 2H), 0.95-0.86 (m, 2H).
D119	LCMS (ESI) m/z: [M + H] ⁺ = 766.25	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.06 (s, 1H), 8.95 (s, 1H), 7.73-7.61 (m, 2H), 6.94 (d, J = 2.2 Hz, 1H), 6.86 (dd, J = 8.6, 2.1 Hz, 1H), 6.74 (s, 2H), 6.70 (d, J = 7.2 Hz, 1H), 5.06 (dd, J = 12.8, 5.4 Hz, 1H), 4.28 (s, 2H), 4.22 (s, 2H), 3.88 (s, 7H), 3.59 (t, J = 9.6 Hz, 1H), 3.55-3.48 (m, 5H), 3.45 (t, J = 5.5 Hz, 2H), 3.28-3.23 (m, 1H), 2.97-2.82 (m, 2H), 2.75 (s, 6H), 2.71-2.65 (m, 1H), 2.62-2.53 (m, 3H), 2.13-1.94 (m, 3H), 1.71-1.59 (m, 1H), 1.55-1.41 (m, 1H).
D120	LCMS (ESI) m/z: [M + H] ⁺ = 810.20	
D121	LCMS (ESI) m/z: [M + H] ⁺ = 810.45	
D122	LCMS (ESI) m/z: [M + H] ⁺ = 780.35	
D123	LCMS (ESI) m/z: [M + H] ⁺ = 780.25	
D124	LCMS (ESI) m/z: [M + H] ⁺ = 849.30	¹ H NMR (300 MHz, Methanol-d ₄) δ 7.65 (d, J = 8.3 Hz, 1H), 7.57 (s, 1H), 6.83 (d, J = 2.0 Hz, 1H), 6.73 (s, 2H), 6.66 (dd, J = 8.4, 2.1 Hz, 1H), 5.07 (dd, J = 12.2, 5.5 Hz, 1H), 4.38 (d, J = 15.8 Hz, 4H), 4.02 (d, J = 9.2 Hz, 4H), 3.97 (s, 6H), 3.79 (s, 3H), 3.67-3.63 (m, 4H), 3.55 (t, J = 5.5 Hz, 2H), 2.92-2.69 (m, 7H), 2.67-2.54 (m, 6H), 2.16-2.08 (m, 1H), 1.89 (s, 4H), 1.15 (s, 6H).
D125		¹ H NMR (300 MHz, DMSO-d ₆ , D ₂ O) δ 7.91-7.80 (m, 1H), 7.59 (s, 1H), 7.36-7.25 (m, 2H), 6.70 (s, 2H), 5.11 (dd, J = 12.8, 5.4 Hz, 1H), 4.75 (t, J = 6.8 Hz, 1H), 4.30-4.09 (m, 4H), 3.85 (s, 6H), 3.61-3.60 (m, 5H), 3.47-3.35 (m, 5H), 3.19-2.77 (m, 8H), 2.66-2.57 (m, 5H), 2.12-1.74 (m, 5H), 1.55-1.34 (m, 2H), 1.32-1.16 (m, 6H), 1.05 (t, J = 7.0 Hz, 1H).
D126	LCMS (ESI) m/z: [M + H] ⁺ = 818.45	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.14 (s, 1H), 8.25 (s, 1H, FA), 7.91 (d, J = 8.1 Hz, 1H), 7.88-7.80 (m, 2H), 7.61 (s, 1H), 6.57 (s, 3H), 5.17 (dd, J = 12.9, 5.4 Hz, 1H), 4.21 (s, 2H), 3.77 (s, 6H), 3.62 (s, 3H),

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Compound No.	LCMS	¹ H NMR
D127	LCMS (ESI) m/z: [M + H] ⁺ = 821.50.	3.49 (s, 4H), 3.42-3.39 (d, J = 5.5 Hz, 6H), 3.01 (s, 3H), 2.93-2.70 (m, 2H), 2.59 (d, J = 4.1 Hz, 5H), 2.14-2.00 (m, 1H), 1.64 (s, 4H), 1.41 (s, 6H). ¹ H NMR (400 MHz, Methanol-d ₄) δ 8.49 (s, 2H), 7.72 (d, J = 8.4 Hz, 1H), 7.58 (s, 1H), 7.39 (d, J = 2.4 Hz, 1H), 7.27 (dd, J = 8.5, 2.4 Hz, 1H), 6.73 (s, 2H), 5.61-5.54 (m, 1H), 5.09 (dd, J = 12.5, 5.4 Hz, 1H), 4.36 (s, 4H), 3.95 (s, 6H), 3.64 (s, 4H), 3.56-3.49 (m, 6H), 3.28-3.22 (m, 2H), 3.18-3.13 (m, 2H), 2.91-2.83 (m, 1H), 2.78 (s, 4H), 2.75-2.40 (m, 12H), 2.16-2.08 (m, 1H).
D128	LCMS (ESI) m/z: [M + H] ⁺ = 878.44.	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.10 (s, 1H), 8.16 (s, 1H, FA), 7.83 (d, J = 8.1 Hz, 1H), 7.63 (s, 1H), 7.31-7.25 (m, 2H), 6.63 (s, 2H), 6.58 (q, J = 4.4 Hz, 1H), 5.12 (dd, J = 12.9, 5.4 Hz, 1H), 4.98 (p, J = 6.7 Hz, 1H), 4.22 (s, 2H), 3.81 (s, 6H), 3.74 (s, 2H), 3.50 (s, 3H), 3.41 (t, J = 5.6 Hz, 7H), 3.01 (d, J = 10.0 Hz, 2H), 2.94-2.86 (m, 1H), 2.62-2.57 (m, 4H), 2.46-2.38 (m, 3H), 2.31-2.13 (m, 4H), 2.09-2.00 (m, 1H), 1.96-1.87 (m, 1H), 1.84-1.75 (m, 2H), 1.68-1.47 (m, 5H), 1.41-1.29 (m, 1H), 1.25-1.08 (m, 2H), 0.84 (d, J = 6.4 Hz, 3H).
D129	LCMS (ESI) m/z: [M + H] ⁺ = 821.35.	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.09 (s, 1H), 8.27 (s, 1H, FA), 7.69 (d, J = 8.5 Hz, 1H), 7.64 (s, 1H), 7.34 (d, J = 2.3 Hz, 1H), 7.26 (dd, J = 8.7, 2.2 Hz, 1H), 6.69-6.58 (m, 3H), 5.08 (dd, J = 12.9, 5.4 Hz, 1H), 4.22 (s, 2H), 3.96-3.87 (m, 2H), 3.83 (s, 8H), 3.50 (s, 3H), 3.46-3.39 (m, 8H), 3.14-2.98 (m, 2H), 2.94-2.74 (m, 3H), 2.60 (d, J = 4.2 Hz, 3H), 2.57-2.54 (m, 2H), 2.46-2.41 (m, 2H), 2.38-2.29 (m, 2H), 2.20 (s, 1H), 2.06-1.98 (m, 1H), 1.85-1.78 (m, 1H), 1.76-1.68 (m, 1H), 1.57-1.48 (m, 1H), 1.43-1.32 (m, 1H).
D130	LCMS (ESI) m/z: [M + H] ⁺ = 823.25	¹ H NMR (300 MHz, Methanol-d ₄) δ 7.63 (d, J = 8.5 Hz, 1H), 7.43 (d, J = 2.6 Hz, 1H), 7.28 (dd, J = 5.9, 2.1 Hz, 2H), 7.20 (d, J = 8.4 Hz, 1H), 6.56 (s, 2H), 5.09 (dd, J = 12.3, 5.4 Hz, 1H), 4.68-4.57 (m, 1H), 4.41-4.14 (m, 2H), 3.88 (d, J = 1.1 Hz, 6H), 3.72-3.60 (m, 3H), 3.59-3.49 (m, 4H), 3.30-3.18 (m, 4H), 3.06-2.86 (m, 4H), 2.90-2.76 (m, 5H), 2.78-2.36 (m, 10H), 2.30-2.11 (m, 2H), 2.07-1.76 (m, 3H), 1.44-1.28 (m, 1H).
D131	712.49	
D132	767.52	
D133	746.3	
D134	774.3	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.10 (s, 1H), 8.13 (s, 1H), 7.95-7.84 (m, 2H), 7.83 (s, 2H), 7.60 (s, 1H), 6.56 (s, 3H), 5.73 (s, 2H), 5.13 (dd, J = 12.7, 5.4 Hz, 1H), 4.19 (s, 2H), 3.47 (s, 4H), 3.38 (t, J = 5.6 Hz, 2H), 3.30 (d, J = 8.7 Hz, 1H), 3.28 (s, 4H), 2.87 (td, J = 16.8, 15.2, 5.3 Hz, 1H), 2.57 (d, J = 4.2 Hz, 4H), 2.51 (s, 1H), 2.18 (t, J = 10.0 Hz, 2H), 2.05 (dd, J = 10.4, 4.7 Hz, 1H), 1.94-1.84 (m, 2H), 1.54 (d, J = 5.9 Hz, 4H).
D135	762.4	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.10 (s, 1H), 8.12 (s, 1H), 7.95-7.84 (m, 1H), 7.82 (dd, J = 7.4, 3.8 Hz, 2H), 7.60 (d, J = 7.0 Hz, 1H), 6.68 (s, 1H), 6.61 (d, J = 1.8 Hz, 2H), 6.56 (d, J = 4.3 Hz, 1H), 5.73 (s, 1H), 5.13 (dd, J = 12.8, 5.3 Hz, 1H), 4.19 (s, 2H), 3.86-3.78 (m, 7H), 3.48 (t, J = 3.1 Hz, 3H), 3.39 (t, J = 5.6 Hz, 2H), 3.24 (dd, J = 16.5, 8.3 Hz, 1H), 2.93-2.80 (m, 3H), 2.73 (s, 1H), 2.57 (dd, J = 4.2, 1.5 Hz, 4H), 2.38 (dt, J = 26.0, 9.7 Hz, 2H), 2.19 (dt, J = 20.1, 9.5 Hz, 2H), 2.08-2.00 (m, 1H), 1.95 (dd, J = 13.3, 7.0 Hz, 2H).
D136	LCMS (ESI) m/z: [M + H] ⁺ = 849.60.	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.08 (s, 1H), 8.21 (s, 2H, FA), 7.64 (t, J = 4.1 Hz, 2H), 6.78 (d, J = 2.1 Hz, 1H), 6.68-6.57 (m, 4H), 5.05 (dd, J = 12.9, 5.4 Hz, 1H), 4.22 (s, 2H), 3.81 (s, 6H), 3.77-3.68 (m, 6H), 3.50 (s, 3H), 3.41 (t, J = 5.6 Hz, 2H), 2.98 (s, 2H), 2.94-2.83 (m, 1H), 2.63-2.52 (m, 7H), 2.47-2.19 (m, 6H), 2.10 (d, J = 7.0 Hz, 2H), 2.04-1.95 (m, 1H), 1.80-1.72 (m, 4H), 1.68 (d, J = 12.7 Hz, 2H), 1.61-1.50 (m, 1H), 1.23-1.10 (m, 2H).
D137	LCMS (ESI) m/z: [M + H] ⁺ = 832.40.	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.15 (s, 1H), 8.21 (s, 1H, FA), 7.91 (dd, J = 7.6, 0.9 Hz, 1H), 7.88-7.81 (m, 2H), 7.63 (s, 1H), 6.66-6.51 (m, 3H), 5.17 (dd, J = 12.8, 5.4 Hz, 1H), 4.21 (s, 2H), 3.77 (s, 6H), 3.40 (t, J = 5.5 Hz, 4H), 2.94-2.85 (m, 1H), 2.77 (t, J = 6.8 Hz, 2H), 2.63-2.54 (m, 7H), 2.42 (s, 4H), 2.12-2.00 (m, 1H), 1.62-1.44 (m, 6H), 1.41 (s, 6H).
D138	LCMS (ESI) m/z: [M + H] ⁺ = 835.41.	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.09 (s, 1H), 8.25 (s, 1H, FA), 7.69 (d, J = 8.6 Hz, 1H), 7.64 (s, 1H), 7.35 (d, J = 2.3 Hz, 1H), 7.27 (dd, J = 8.7, 2.3 Hz, 1H), 6.65 (s, 2H), 6.61 (q, J = 4.4 Hz, 1H), 5.08 (dd, J = 12.9, 5.3 Hz, 1H), 4.22 (s, 2H), 4.00-3.92 (m, 1H), 3.92-3.78 (m, 8H), 3.50 (s, 5H), 3.47-3.38 (m, 9H), 3.13-3.05 (m, 1H), 2.94-2.82 (m, 2H), 2.63-2.55 (m, 5H), 2.44-2.24 (m, 4H), 2.07-1.99 (m, 1H), 1.95-1.72 (m, 3H), 1.66-1.49 (m, 3H), 1.41-1.30 (m, 1H).
D139	LCMS (ESI) m/z: [M + H] ⁺ = 824.50.	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.08 (s, 1H), 8.18 (s, 2H, FA), 7.71-7.60 (m, 2H), 7.34 (d, J = 2.1 Hz, 1H), 7.29-7.22 (m, 1H), 6.62-6.55 (m, 3H), 5.07 (dd, J = 12.6, 5.3 Hz, 1H), 4.22 (s, 2H), 3.78 (s, 6H), 3.56-3.49 (m, 6H), 3.45-3.38 (m, 8H), 2.93-2.82 (m, 1H), 2.66-2.53 (m, 8H), 2.47-2.31 (m, 12H), 2.06-1.98 (m, 1H).

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Compound No.	LCMS	¹ H NMR
D140	LCMS (ESI) m/z: [M + H] ⁺ = 908.55	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.08 (s, 1H), 8.15 (s, 1H, FA), 7.72-7.62 (m, 2H), 7.34 (d, J = 2.2 Hz, 1H), 7.26 (d, J = 8.8 Hz, 1H), 6.67-6.56 (m, 3H), 5.07 (dd, J = 12.7, 5.3 Hz, 1H), 4.22 (s, 2H), 3.84-3.73 (m, 8H), 3.60 (s, 2H), 3.50 (s, 3H), 3.46-3.38 (m, 9H), 2.95-2.83 (m, 1H), 2.76-2.65 (m, 3H), 2.64-2.53 (m, 9H), 2.40-2.23 (m, 6H), 2.17 (s, 2H), 2.08-1.98 (m, 1H), 1.91-1.80 (m, 2H), 1.67-1.51 (m, 4H).
D141	LCMS (ESI) m/z: [M + H] ⁺ = 851.51	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.04 (s, 1H), 8.18 (s, 2H), 7.74-7.47 (m, 2H), 7.26 (d, J = 2.2 Hz, 1H), 7.19 (dd, J = 8.7, 2.3 Hz, 1H), 6.54 (s, 3H), 5.03 (dd, J = 12.9, 5.4 Hz, 1H), 4.19 (s, 2H), 3.99 (d, J = 13.0 Hz, 2H), 3.81 (s, 2H), 3.74 (s, 6H), 3.46 (s, 3H), 3.37 (d, J = 5.7 Hz, 2H), 2.95-2.85 (m, 3H), 2.57 (d, J = 4.3 Hz, 3H), 2.14 (t, J = 7.5 Hz, 2H), 2.03-1.95 (m, 1H), 1.75-1.64 (m, 4H), 1.07 (d, J = 5.7 Hz, 7H), 1.00 (d, J = 6.6 Hz, 2H).
D142	LCMS (ESI) m/z: [M + H] ⁺ = 707.34	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.10 (s, 1H), 8.14 (s, 1H), 7.91-7.81 (m, 2H), 7.60 (s, 1H), 6.56 (d, J = 3.9 Hz, 3H), 5.13 (dd, J = 12.7, 5.4 Hz, 1H), 4.19 (s, 2H), 3.77 (s, 5H), 3.55 (s, 2H), 3.48 (d, J = 10.1 Hz, 5H), 3.37 (t, J = 5.7 Hz, 3H), 3.19-3.14 (m, 2H), 2.87 (ddd, J = 16.7, 13.6, 5.4 Hz, 1H), 2.67-2.59 (m, 1H), 2.57 (d, J = 4.2 Hz, 3H), 2.11-1.98 (m, 1H).
D143	LCMS (ESI) m/z: [M + H] ⁺ = 735.38	
D144	LCMS (ESI) m/z: [M + H] ⁺ = 821.5	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.08 (s, 1H), 8.22 (s, 1H, FA), 7.70-7.59 (m, 2H), 6.91 (d, J = 2.1 Hz, 1H), 6.81 (dd, J = 8.6, 2.2 Hz, 1H), 6.63-6.53 (m, 3H), 5.06 (dd, J = 12.6, 5.4 Hz, 1H), 4.21 (s, 2H), 4.00 (q, J = 8.5 Hz, 2H), 3.88-3.81 (m, 2H), 3.78 (s, 6H), 3.68-3.57 (m, 3H), 3.54-3.45 (m, 8H), 3.25-3.16 (m, 4H), 3.10 (q, J = 7.3 Hz, 2H), 2.95-2.81 (m, 1H), 2.62-2.55 (m, 6H), 2.26-2.17 (m, 2H), 2.07-1.96 (m, 1H).
D145	LCMS (ESI) m/z: [M + H] ⁺ = 851.65	¹ H NMR (300 MHz, Methanol-d ₄) δ 7.82-7.76 (m, 1H), 7.58 (s, 1H), 7.51 (d, J = 2.3 Hz, 1H), 7.38 (dd, J = 8.5, 2.2 Hz, 1H), 6.73 (s, 2H), 5.11 (dd, J = 12.4, 5.4 Hz, 1H), 4.37 (d, J = 4.6 Hz, 4H), 4.25 (d, J = 13.6 Hz, 2H), 3.96 (d, J = 3.5 Hz, 6H), 3.75-3.60 (m, 6H), 3.60-3.52 (m, 3H), 3.51-3.41 (m, 2H), 3.24-3.07 (m, 4H), 2.92-2.70 (m, 6H), 2.63 (s, 2H), 2.18-1.85 (m, 4H), 1.84-1.58 (m, 4H), 1.57-1.48 (m, 6H).
D146	LCMS (ESI) m/z: [M + H] ⁺ = 807.55	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.08 (s, 1H), 8.17 (s, 1H, FA), 7.68 (d, J = 8.5 Hz, 1H), 7.63 (s, 1H), 7.34 (d, J = 2.2 Hz, 1H), 7.25 (dd, J = 8.7, 2.3 Hz, 1H), 6.66-6.56 (m, 3H), 5.07 (dd, J = 12.9, 5.4 Hz, 1H), 4.22 (s, 2H), 3.84 (s, 2H), 3.82 (s, 6H), 3.59 (s, 2H), 3.50 (s, 4H), 3.48 (s, 2H), 3.43-3.39 (m, 8H), 2.93-2.84 (m, 1H), 2.62-2.58 (m, 4H), 2.58-2.55 (m, 1H), 2.35 (t, J = 4.9 Hz, 4H), 2.27-2.19 (m, 2H), 2.06-1.99 (m, 1H), 1.95 (t, J = 10.1 Hz, 2H).
D147	LCMS (ESI) m/z: [M + H] ⁺ = 827.35	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.09 (s, 1H), 8.15 (s, 1H, FA), 7.68 (d, J = 8.4 Hz, 1H), 7.64 (s, 1H), 7.34 (s, 1H), 7.25 (dd, J = 8.9, 2.3 Hz, 1H), 6.59 (s, 3H), 5.07 (dd, J = 12.8, 5.4 Hz, 1H), 4.22 (s, 2H), 3.79 (s, 6H), 3.57 (s, 2H), 3.50 (s, 3H), 3.46-3.37 (m, 10H), 2.94-2.82 (m, 1H), 2.65-2.55 (m, 11H), 2.38-2.30 (m, 2H), 2.06-1.98 (m, 1H), 1.81 (t, J = 12.2 Hz, 2H), 1.74-1.54 (m, 2H).
D148	LCMS (ESI) m/z: [M + H] ⁺ = 809.35	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.08 (s, 1H), 8.18 (s, 1H, FA), 7.68 (d, J = 8.5 Hz, 1H), 7.63 (s, 1H), 7.34 (d, J = 2.2 Hz, 1H), 7.26 (dd, J = 8.8, 2.3 Hz, 1H), 6.64 (s, 2H), 6.59 (q, J = 4.4 Hz, 1H), 5.07 (dd, J = 12.9, 5.4 Hz, 1H), 4.21 (s, 2H), 3.90 (s, 2H), 3.82 (s, 6H), 3.50 (s, 3H), 3.44-3.39 (m, 8H), 3.37-3.28 (m, 5H), 3.08 (t, J = 8.9 Hz, 2H), 2.94-2.80 (m, 3H), 2.62-2.56 (m, 4H), 2.30 (t, J = 7.2 Hz, 2H), 2.21-2.13 (m, 1H), 2.06-1.94 (m, 2H), 1.61-1.49 (m, 2H), 1.48-1.39 (m, 1H).
D149	LCMS (ESI) m/z: [M + H] ⁺ = 824.75	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.12 (s, 1H), 8.23 (s, 2H, FA), 7.83 (dd, J = 8.2, 2.8 Hz, 1H), 7.63 (s, 1H), 7.34-7.23 (m, 2H), 6.66-6.57 (m, 3H), 5.12 (dd, J = 12.9, 5.4 Hz, 1H), 4.98-4.90 (m, 1H), 4.69 (p, J = 7.1 Hz, 1H), 4.22 (s, 2H), 3.80 (d, J = 4.7 Hz, 6H), 3.71 (s, 2H), 3.50 (s, 3H), 3.40 (t, J = 5.5 Hz, 2H), 3.01-2.92 (m, 2H), 2.92-2.84 (m, 1H), 2.74-2.68 (m, 1H), 2.65-2.57 (m, 4H), 2.56-2.52 (m, 2H), 2.42-2.25 (m, 3H), 2.22-2.14 (m, 1H), 2.10-1.98 (m, 6H), 1.93-1.57 (m, 4H), 1.49 (s, 1H), 1.25-1.07 (m, 2H).
D150	LCMS (ESI) m/z: [M + H] ⁺ = 769.45	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.09 (s, 1H), 8.21 (s, 2H, FA), 7.68 (d, J = 8.5 Hz, 1H), 7.64 (s, 1H), 7.34 (d, J = 2.2 Hz, 1H), 7.26 (dd, J = 8.7, 2.3 Hz, 1H), 6.66 (t, J = 5.5 Hz, 1H), 6.60 (s, 2H), 5.08 (dd, J = 12.9, 5.4 Hz, 1H), 4.23 (s, 2H), 3.79 (s, 6H), 3.54 (s, 2H), 3.50 (s, 3H), 3.46-3.39 (m, 7H), 3.22 (q, J = 6.5 Hz, 3H), 2.93-2.83 (m, 1H), 2.64-2.53 (m, 6H), 2.43 (t, J = 7.1 Hz, 2H), 2.21 (s, 6H), 2.06-1.98 (m, 1H).
D151	LCMS (ESI) m/z: [M + H] ⁺ = 839.55	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.08 (s, 1H), 8.15 (s, 2H), 7.68 (d, J = 8.5 Hz, 1H), 7.63 (s, 1H), 7.34 (d, J = 2.2 Hz, 1H), 7.25 (dd, J = 8.6, 2.2 Hz, 1H), 6.59 (s, 3H), 5.08 (dd, J = 12.7, 5.3 Hz, 1H), 4.21 (s, 2H),

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Compound No.	LCMS	¹ H NMR
D152	LCMS (ESI) m/z: [M + H] ⁺ = 825.70.	3.79 (s, 6H), 3.76-3.67 (m, 1H), 3.52 (s, 2H), 3.49 (s, 5H), 3.48-3.35 (m, 10H), 2.98-2.80 (m, 1H), 2.75-2.55 (m, 7H), 2.50-2.43 (m, 2H), 2.36-2.26 (m, 2H), 2.19-1.97 (m, 2H), 1.85 (t, J = 10.7 Hz, 1H), 1.61-1.32 (m, 4H). ¹ H NMR (400 MHz, DMSO-d ₆) δ 11.08 (s, 1H), 8.14 (s, 1H, FA), 7.68 (d, J = 8.5 Hz, 1H), 7.63 (s, 1H), 7.35 (s, 1H), 7.26 (dd, J = 8.7, 2.3 Hz, 1H), 6.62 (s, 2H), 6.61-6.57 (m, 1H), 5.08 (dd, J = 12.9, 5.3 Hz, 1H), 4.21 (s, 2H), 3.81 (s, 6H), 3.78-3.73 (m, 1H), 3.65 (s, 2H), 3.50 (s, 4H), 3.47-3.42 (m, 6H), 3.42-3.37 (m, 5H), 2.95-2.82 (m, 2H), 2.74-2.69 (m, 1H), 2.60 (d, J = 4.3 Hz, 5H), 2.58-2.54 (m, 4H), 2.31-2.21 (m, 1H), 2.12-1.97 (m, 2H), 1.68-1.51 (m, 2H).
D153	LCMS (ESI) m/z: [M + H] ⁺ = 835.60.	¹ H NMR (400 MHz, Methanol-d ₄) δ 8.49 (s, 2H, FA), 7.74 (dd, J = 16.5, 8.4 Hz, 1H), 7.57 (s, 1H), 7.27 (s, 1H), 7.13 (d, J = 8.8 Hz, 1H), 6.70 (s, 2H), 5.11 (dd, J = 12.7, 5.4 Hz, 1H), 4.34 (d, J = 11.5 Hz, 4H), 4.05-3.88 (m, 8H), 3.78 (d, J = 12.0 Hz, 2H), 3.71-3.61 (m, 5H), 3.57-3.40 (m, 5H), 3.12-2.99 (m, 2H), 2.89-2.68 (m, 9H), 2.62 (s, 3H), 2.18-2.09 (m, 1H), 1.93 (d, J = 13.8 Hz, 2H), 1.84-1.78 (m, 1H), 1.73-1.63 (m, 1H), 1.52 (s, 3H), 1.39-1.27 (m, 1H).
D154	LCMS (ESI) m/z: [M + H] ⁺ = 859.75.	¹ H NMR (300 MHz, Methanol-d ₄) δ 8.51 (s, 2H), 7.70 (d, J = 8.5 Hz, 1H), 7.57 (s, 1H), 7.37 (d, J = 2.2 Hz, 1H), 7.25 (dd, J = 8.6, 2.3 Hz, 1H), 6.72 (s, 2H), 5.09 (dd, J = 12.3, 5.4 Hz, 1H), 4.37 (d, J = 9.4 Hz, 4H), 3.95 (s, 6H), 3.70-3.62 (m, 4H), 3.60-3.45 (m, 7H), 3.18 (t, J = 12.6 Hz, 2H), 2.95-2.69 (m, 12H), 2.68-2.59 (m, 2H), 2.58-2.46 (m, 1H), 2.19-2.07 (m, 3H), 2.03-1.86 (m, 2H).
D155	LCMS (ESI) m/z: [M + H] ⁺ = 859.50.	¹ H NMR (300 MHz, Methanol-d ₄) δ 7.79 (d, J = 8.4 Hz, 1H), 7.59 (s, 1H), 7.50 (d, J = 2.1 Hz, 1H), 7.41-7.34 (m, 1H), 6.75 (s, 2H), 5.12 (dd, J = 12.4, 5.4 Hz, 1H), 4.50 (s, 2H), 4.36 (s, 2H), 3.96 (s, 6H), 3.94-3.74 (m, 3H), 3.70-3.62 (m, 5H), 3.61-3.39 (m, 8H), 3.39-3.35 (m, 2H), 3.23-3.19 (m, 1H), 2.87-2.71 (m, 6H), 2.68-2.61 (m, 2H), 2.40-2.21 (m, 3H), 2.18-2.09 (m, 1H), 1.96-1.80 (m, 2H).
D156	737.88	
D157	723.39	
D158	738.37	
D159	751.46	
D161	751.39	NA
D162	751.32	NA
D163	765.39	NA
D164	766.4	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.04 (s, 1H), 8.13 (s, 1H), 7.65-7.57 (m, 2H), 6.75 (d, J = 2.1 Hz, 1H), 6.61 (dd, J = 8.4, 2.1 Hz, 1H), 6.56 (s, 2H), 6.54 (d, J = 4.5 Hz, 1H), 5.03 (dd, J = 12.9, 5.4 Hz, 1H), 4.19 (s, 2H), 4.05 (t, J = 7.7 Hz, 2H), 3.80 (dd, J = 8.9, 4.9 Hz, 2H), 3.76 (s, 5H), 3.54 (s, 2H), 3.47 (s, 3H), 3.37 (s, 1H), 3.25 (q, J = 5.9 Hz, 1H), 3.15 (s, 1H), 2.86 (ddd, J = 17.6, 13.9, 5.4 Hz, 1H), 2.61-2.50 (m, 4H), 2.47-2.42 (m, 4H), 2.34-2.29 (m, 4H), 1.99 (dp, J = 12.2, 4.6, 4.0 Hz, 1H).
D165	737.4	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.03 (s, 1H), 8.15 (s, 1H), 7.61 (d, J = 8.8 Hz, 2H), 6.76 (d, J = 2.1 Hz, 1H), 6.62 (dd, J = 8.4, 2.1 Hz, 1H), 6.57 (s, 2H), 6.54 (dd, J = 5.8, 3.3 Hz, 1H), 5.03 (dd, J = 12.9, 5.4 Hz, 1H), 4.19 (s, 2H), 3.90 (q, J = 8.4 Hz, 4H), 3.77 (s, 6H), 3.47 (s, 3H), 3.38 (t, J = 5.5 Hz, 2H), 3.15 (s, 1H), 3.10 (s, 0H), 2.86 (ddd, J = 17.4, 13.9, 5.5 Hz, 1H), 2.77 (s, 2H), 2.61-2.48 (m, 6H), 2.05-1.93 (m, 3H).
D166	737.4	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.04 (s, 1H), 8.13 (s, 1H), 7.65-7.58 (m, 2H), 6.80 (d, J = 2.0 Hz, 1H), 6.72 (dd, J = 8.3, 2.0 Hz, 1H), 6.56 (d, J = 12.0 Hz, 3H), 5.03 (dd, J = 12.9, 5.4 Hz, 1H), 4.28 (dt, J = 8.3, 4.2 Hz, 1H), 4.19 (s, 2H), 3.77 (s, 6H), 3.78-3.66 (m, 2H), 3.47 (s, 3H), 3.38 (t, J = 5.6 Hz, 2H), 3.15 (s, 1H), 2.96-2.79 (m, 2H), 2.72 (s, 1H), 2.57 (d, J = 4.3 Hz, 4H), 2.51 (s, 1H), 2.45 (s, 0H), 2.03-1.85 (m, 2H).
D167	LCMS (ESI) m/z: [M + H] ⁺ = 807.	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.08 (s, 1H), 8.24 (s, 2H, FA), 7.69-7.59 (m, 2H), 6.89 (d, J = 2.1 Hz, 1H), 6.80 (dd, J = 8.6, 2.2 Hz, 1H), 6.60 (s, 3H), 5.05 (dd, J = 12.5, 5.4 Hz, 1H), 4.21 (s, 2H), 3.79 (s, 6H), 3.70-3.63 (m, 4H), 3.54-3.47 (m, 7H), 3.43-3.37 (m, 6H), 3.18-3.10 (m, 4H), 3.04 (s, 2H), 2.91-2.82 (m, 1H), 2.63-2.56 (m, 5H), 2.39-2.31 (m, 1H), 2.15 (t, J = 6.8 Hz, 2H), 2.06-1.95 (m, 1H).
D168	LCMS (ESI) m/z: [M + H] ⁺ = 981.35.	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.09 (s, 1H), 7.69 (d, J = 8.5 Hz, 1H), 7.64 (s, 1H), 7.35 (d, J = 2.3 Hz, 1H), 7.26 (dd, J = 8.7, 2.3 Hz, 1H), 6.69 (s, 2H), 6.60 (q, J = 4.4 Hz, 1H), 5.08 (dd, J = 12.9, 5.4 Hz, 1H), 4.22 (s, 2H), 4.14 (s, 2H), 4.00-3.91 (m, 2H), 3.86 (s, 6H), 3.70-3.56 (m, 2H), 3.50 (s, 3H), 3.45-3.39 (m, 6H), 2.94-2.83 (m, 2H), 2.63-2.56 (m, 6H), 2.50-2.42 (m, 7H), 2.06-1.97 (m, 1H).
D169	LCMS (ESI) m/z: [M + H] ⁺ = 795.35.	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.08 (s, 1H), 8.21 (s, 1H, FA), 7.68 (d, J = 8.5 Hz, 1H), 7.63 (s, 1H), 7.34 (d, J = 2.3 Hz, 1H), 7.26 (dd, J = 8.7, 2.3 Hz, 1H), 6.64-6.54 (m, 3H), 5.07 (dd, J = 12.9, 5.4 Hz, 1H), 4.22 (s, 2H), 3.80 (s, 6H), 3.73 (s, 2H), 3.50 (s, 4H), 3.45-3.37 (m, 8H), 2.95-2.79 (m, 2H), 2.66-2.56 (m, 6H), 2.49-2.44 (m, 3H),

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Compound No.	LCMS	¹ H NMR
D170	LCMS (ESI) m/z: [M + H] ⁺ = 795.35	2.40-2.32 (m, 3H), 2.31-2.26 (m, 2H), 2.07-1.96 (m, 1H), 1.91-1.81 (m, 1H), 1.46-1.34 (m, 1H). ¹ H NMR (400 MHz, DMSO-d ₆) δ 11.08 (s, 1H), 8.19 (s, 1H, FA), 7.68-7.61 (m, 2H), 7.32 (d, J = 2.3 Hz, 1H), 7.24 (dd, J = 8.7, 2.3 Hz, 1H), 6.58 (s, 3H), 5.07 (dd, J = 12.9, 5.4 Hz, 1H), 4.21 (s, 2H), 4.04 (d, J = 12.7 Hz, 2H), 3.77 (s, 6H), 3.51 (d, J = 11.7 Hz, 6H), 3.40 (t, J = 5.6 Hz, 4H), 2.98-2.86 (m, 3H), 2.63-2.55 (m, 5H), 2.48-2.37 (m, 8H), 2.06-1.97 (m, 1H), 1.83 (d, J = 12.1 Hz, 2H), 1.43 (q, J = 11.7, 11.0 Hz, 2H).
D171	LCMS (ESI) m/z: [M + H] ⁺ = 823.55	¹ H NMR (300 MHz, Methanol-d ₄) δ 7.73 (d, J = 8.2 Hz, 1H), 7.57 (s, 1H), 6.96 (d, J = 2.1 Hz, 1H), 6.81 (dd, J = 8.2, 2.1 Hz, 1H), 6.73 (d, J = 4.1 Hz, 2H), 5.09 (dd, J = 12.3, 5.4 Hz, 1H), 4.50-4.20 (m, 9H), 3.96 (d, J = 4.8 Hz, 6H), 3.64 (s, 3H), 3.62-3.52 (m, 4H), 3.25-3.02 (m, 4H), 2.96-2.82 (m, 4H), 2.81-2.69 (m, 5H), 2.67-2.60 (m, 2H), 2.18-1.86 (m, 4H), 1.80-1.52 (m, 4H).
D172	LCMS (ESI) m/z: [M + H] ⁺ = 835.80	¹ H NMR (300 MHz, Methanol-d ₄) δ 8.45 (s, 2H, FA), 7.68 (d, J = 8.4 Hz, 1H), 7.58 (s, 1H), 7.09 (d, J = 2.1 Hz, 1H), 6.94 (dd, J = 8.5, 2.2 Hz, 1H), 6.71 (s, 2H), 5.08 (dd, J = 12.4, 5.4 Hz, 1H), 4.74 (s, 1H), 4.35 (s, 4H), 4.14 (s, 1H), 3.93 (s, 6H), 3.71-3.60 (m, 5H), 3.58-3.44 (m, 4H), 3.24 (d, J = 10.4 Hz, 1H), 3.16-3.01 (m, 3H), 2.97-2.70 (m, 8H), 2.63 (t, J = 5.7 Hz, 2H), 2.28-2.07 (m, 3H), 1.96 (d, J = 13.4 Hz, 2H), 1.75-1.49 (m, 5H).
D173	LCMS (ESI) m/z: [M + H] ⁺ = 851.40	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.08 (s, 1H), 8.20 (s, 2H, FA), 7.69-7.61 (m, 2H), 7.27 (d, J = 2.2 Hz, 1H), 7.21 (dd, J = 8.8, 2.4 Hz, 1H), 6.69-6.54 (m, 3H), 5.07 (dd, J = 12.9, 5.4 Hz, 1H), 4.22 (s, 2H), 4.03-3.93 (m, 3H), 3.79 (s, 7H), 3.63 (s, 3H), 3.50 (s, 3H), 3.40 (t, J = 5.5 Hz, 2H), 2.99-2.84 (m, 5H), 2.59 (d, J = 4.3 Hz, 3H), 2.58-2.53 (m, 4H), 2.25 (s, 3H), 2.23-2.15 (m, 2H), 2.06-1.97 (m, 1H), 1.88-1.71 (m, 2H), 1.63 (d, J = 12.5 Hz, 2H), 1.56-1.43 (m, 2H), 1.31 (s, 3H), 1.21-1.07 (m, 2H).
D174	LCMS (ESI) m/z: [M + H] ⁺ = 849.35	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.09 (s, 1H), 8.18 (s, 1H, FA), 7.68 (d, J = 8.5 Hz, 1H), 7.62 (s, 1H), 7.34 (d, J = 2.2 Hz, 1H), 7.25 (dd, J = 8.7, 2.0 Hz, 1H), 6.59 (s, 3H), 5.07 (dd, J = 12.8, 5.4 Hz, 1H), 4.21 (s, 2H), 3.78 (s, 6H), 3.60-3.37 (m, 15H), 2.96-2.80 (m, 4H), 2.63-2.53 (m, 6H), 2.20 (s, 2H), 2.13-1.95 (m, 3H), 1.60-1.50 (m, 2H), 1.45-1.31 (m, 2H), 1.28-1.13 (m, 1H), 0.39 (s, 2H), 0.17 (s, 2H).
D175	LCMS (ESI) m/z: [M + H] ⁺ = 922.35	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.08 (s, 1H), 8.19 (s, 1H, FA), 7.70-7.66 (m, 1H), 7.63 (d, J = 2.7 Hz, 1H), 7.35 (s, 1H), 7.26 (t, J = 7.9 Hz, 1H), 6.62-6.56 (m, 3H), 5.07 (dd, J = 13.0, 5.4 Hz, 1H), 4.21 (s, 2H), 3.77 (d, J = 2.7 Hz, 6H), 3.61 (s, 2H), 3.57-3.52 (m, 4H), 3.50-3.47 (m, 4H), 3.44-3.40 (m, 7H), 3.34-3.30 (m, 3H), 2.94-2.84 (m, 1H), 2.63-2.58 (m, 6H), 2.57-2.54 (m, 5H), 2.48-2.41 (m, 3H), 2.41-2.23 (m, 2H), 2.06-1.97 (m, 1H), 1.72-1.58 (m, 2H), 1.56-1.41 (m, 2H).
D176	792.29	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.10 (d, J = 15.8 Hz, 1H), 8.14 (s, 2H), 7.78 (d, J = 16.8 Hz, 2H), 7.60 (s, 1H), 6.57 (d, J = 15.8 Hz, 3H), 5.20-4.89 (m, 2H), 4.19 (s, 2H), 3.96 (s, 3H), 3.79 (d, J = 3.5 Hz, 7H), 3.48 (s, 3H), 2.57 (d, J = 4.2 Hz, 3H), 1.88-1.64 (m, 1H), 1.45-1.24 (m, 2H).
D177	764.92	
D178	794.46	
D179	LCMS (ESI) m/z: [M + H] ⁺ = 780.46	
D180	LCMS (ESI) m/z: [M + H] ⁺ = 766.39	
D181	LCMS (ESI) m/z: [M + H] ⁺ = 820.5	
D182	LCMS (ESI) m/z: [M + H] ⁺ = 792.91	
D183	LCMS (ESI) m/z: [M + H] ⁺ = 767.92	
D184	LCMS (ESI) m/z: [M + H] ⁺ = 781.45	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.06 (s, 1H), 8.20 (s, 2H, FA), 7.79-7.50 (m, 2H), 6.76 (d, J = 2.0 Hz, 1H), 6.67 (d, J = 7.8 Hz, 1H), 6.62 (s, 2H), 5.05 (dd, J = 12.6, 5.4 Hz, 1H), 4.88 (t, J = 5.4 Hz, 1H), 4.24 (s, 2H), 4.21-4.10 (m, 1H), 4.00 (d, J = 9.7 Hz, 1H), 3.80 (s, 6H), 3.75-3.59 (m, 4H), 3.51 (s, 4H), 3.46-3.37 (m, 3H), 3.20 (d, J = 6.2 Hz, 2H), 3.08 (t, J = 8.0 Hz, 1H), 2.98-2.80 (m, 1H), 2.71-2.59 (m, 3H), 2.58-2.54 (m, 2H), 2.29 (s, 6H), 2.08-1.91 (m, 2H), 1.76 (d, J = 12.4 Hz, 1H).

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Compound No.	LCMS	¹ H NMR
D185	LCMS (ESI) m/z: [M + H] ⁺ = 795.25	
D186	LCMS (ESI) m/z: [M + H] ⁺ = 867.6	
D187	LCMS (ESI) m/z: [M + H] ⁺ = 795.55	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.06 (s, 1H), 8.22 (s, 3H, FA), 7.66-7.58 (m, 2H), 6.83 (d, J = 2.0 Hz, 1H), 6.74 (dd, J = 8.3, 2.1 Hz, 1H), 6.64-6.56 (m, 3H), 5.05 (dd, J = 12.8, 5.4 Hz, 1H), 4.40-4.31 (m, 1H), 4.22 (s, 2H), 3.80 (s, 8H), 3.62 (s, 2H), 3.50 (s, 3H), 3.41 (t, J = 5.5 Hz, 2H), 3.20 (d, J = 6.7 Hz, 2H), 2.96-2.77 (m, 3H), 2.63-2.53 (m, 6H), 2.44 (t, J = 6.5 Hz, 3H), 2.31-2.25 (m, 6H), 2.07-1.90 (m, 3H).
D188	LCMS (ESI) m/z: [M + H] ⁺ = 809.65	
D189	LCMS (ESI) m/z: [M + H] ⁺ = 795.6	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.06 (s, 1H), 8.23 (s, 3H, FA), 7.76-7.55 (m, 2H), 6.79 (d, J = 2.1 Hz, 1H), 6.71-6.60 (m, 2H), 6.60 (s, 2H), 5.06 (dd, J = 12.7, 5.3 Hz, 1H), 4.23 (s, 2H), 4.06-3.87 (m, 4H), 3.79 (s, 6H), 3.51 (s, 5H), 3.47-3.37 (m, 2H), 3.24-3.13 (m, 2H), 2.98-2.75 (m, 4H), 2.68-2.51 (m, 7H), 2.19 (s, 6H), 2.04 (q, J = 12.6, 9.6 Hz, 3H).
D190	LCMS (ESI) m/z: [M + H] ⁺ = 809.65	
D191	LCMS (ESI) m/z: [M + H] ⁺ = 809.45	
D192	LCMS (ESI) m/z: [M + H] ⁺ = 781.7	
D193	LCMS (ESI) m/z: [M + H] ⁺ = 936.55	
D194	LCMS (ESI) m/z: [M + H] ⁺ = 824.35	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.10 (s, 1H), 8.19 (s, 2H, FA), 7.83 (d, J = 8.3 Hz, 1H), 7.64 (s, 1H), 7.42 (d, J = 2.2 Hz, 1H), 7.34 (dd, J = 8.3, 2.3 Hz, 1H), 6.68 (t, J = 5.7 Hz, 1H), 6.61 (s, 2H), 5.12 (dd, J = 12.9, 5.4 Hz, 1H), 4.25-4.13 (m, 4H), 3.80 (s, 6H), 3.63 (s, 3H), 3.50 (s, 4H), 3.40 (t, J = 5.5 Hz, 2H), 3.30 (s, 2H), 3.24 (s, 2H), 3.03 (t, J = 6.2 Hz, 2H), 2.96-2.81 (m, 1H), 2.66-2.55 (m, 4H), 2.28 (s, 7H), 2.15-2.03 (m, 3H), 1.86-1.71 (m, 4H)
D195	LCMS (ESI) m/z: [M + H] ⁺ = 837.65	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.07 (s, 1H), 8.15 (s, 1H), 7.67 (d, J = 8.5 Hz, 1H), 7.63 (s, 1H), 7.32 (d, J = 2.2 Hz, 1H), 7.25 (dd, J = 8.6, 2.3 Hz, 1H), 6.69 (s, 2H), 6.59 (q, J = 4.4 Hz, 1H), 5.06 (dd, J = 12.8, 5.4 Hz, 1H), 4.22 (s, 2H), 4.16-4.00 (m, 4H), 3.85 (s, 6H), 3.54-3.48 (m, 7H), 3.03-2.89 (m, 4H), 2.86-2.67 (m, 4H), 2.63-2.57 (m, 4H), 2.31-2.25 (m, 2H), 2.21 (s, 3H), 2.10-1.95 (m, 2H), 1.85-1.71 (m, 4H), 1.69-1.59 (m, 1H), 1.54-1.40 (m, 2H), 1.37-1.23 (m, 2H).
D196	LCMS (ESI) m/z: [M + H] ⁺ = 849.60	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.07 (s, 1H), 8.18 (s, 2H, FA), 7.69-7.61 (m, 2H), 6.95 (d, J = 2.1 Hz, 1H), 6.86 (dd, J = 8.6, 2.2 Hz, 1H), 6.63-6.55 (m, 3H), 5.06 (dd, J = 12.6, 5.4 Hz, 1H), 4.21 (s, 2H), 3.79 (s, 6H), 3.70-3.61 (m, 4H), 3.50 (s, 3H), 3.40 (t, J = 5.6 Hz, 2H), 3.30 (dd, J = 11.0, 2.6 Hz, 2H), 3.00-2.91 (m, 4H), 2.91-2.76 (m, 2H), 2.65-2.52 (m, 10H), 2.44-2.38 (m, 2H), 2.32-2.23 (m, 2H), 2.04-1.94 (m, 1H), 1.63 (d, J = 12.2 Hz, 2H), 1.38-1.12 (m, 5H).
D197	LCMS (ESI) m/z: [M + H] ⁺ = 837.42	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.06 (s, 1H), 8.19 (s, 2H, FA), 7.68-7.61 (m, 2H), 6.95 (d, J = 2.1 Hz, 1H), 6.87-6.81 (m, 1H), 6.59 (s, 3H), 5.06 (dd, J = 12.4, 5.4 Hz, 1H), 4.22 (s, 2H), 3.78 (s, 6H), 3.66-3.53 (m, 5H), 3.50 (s, 4H), 3.41 (s, 2H), 3.21-3.10 (m, 4H), 2.90-2.80 (m, 3H), 2.60 (d, J = 4.2 Hz, 3H), 2.43-2.39 (m, 2H), 2.18 (s, 3H), 2.13-1.98 (m, 4H), 1.90-1.79 (m, 1H), 1.62 (d, J = 11.9 Hz, 2H), 1.38-1.04 (m, 6H).
D198	LCMS (ESI) m/z: [M + H] ⁺ = 837.65	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.07 (s, 1H), 8.20 (s, 2H, FA), 7.66-7.61 (m, 2H), 7.11 (d, J = 2.3 Hz, 1H), 7.03 (dd, J = 8.8, 2.4 Hz, 1H), 6.63-6.57 (m, 3H), 5.06 (dd, J = 12.9, 5.4 Hz, 1H), 4.22 (s, 2H), 3.79 (s, 6H), 3.69-3.59 (m, 7H), 3.50 (s, 3H), 3.40 (t, J = 5.5 Hz, 2H), 2.99-2.80 (m, 4H), 2.73 (t, J = 5.1 Hz, 2H), 2.62-2.54 (m, 6H), 2.44 (t, J = 7.2 Hz, 2H), 2.19 (t, J = 11.0 Hz, 2H), 2.06-1.95 (m, 1H), 1.85 (p, J = 6.2, 5.6 Hz, 2H), 1.59 (d, J = 11.9 Hz, 2H), 1.38-1.01 (m, 6H).
D199	LCMS (ESI) m/z: [M + H] ⁺ = 835.60	¹ H NMR (300 MHz, Methanol-d ₄) δ 8.54 (s, 1H), 7.66 (d, J = 8.3 Hz, 1H), 7.58 (s, 1H), 6.84 (d, J = 2.1 Hz, 1H), 6.74-6.66 (m, 3H), 5.07 (dd, J = 12.3, 5.4 Hz, 1H), 4.36 (s, 2H), 4.33 (s, 2H), 4.17 (s, 4H), 3.95 (s, 6H), 3.71-3.62 (m, 4H), 3.60-3.52 (m, 6H), 3.51-3.42 (m, 2H),

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Compound No.	LCMS	¹ H NMR
D200	LCMS (ESI) m/z: [M + H] ⁺ = 851.95	3.12-2.99 (m, 2H), 2.88-2.71 (m, 6H), 2.68-2.59 (m, 4H), 2.17-2.07 (m, 1H), 1.95 (d, J = 13.5 Hz, 2H), 1.73-1.53 (m, 2H), 1.46-1.36 (m, 2H). ¹ H NMR (300 MHz, DMSO-d ₆) δ 11.06 (s, 1H), 8.16 (s, 1H, FA), 7.69-7.60 (m, 2H), 7.32 (d, J = 2.2 Hz, 1H), 7.25 (dd, J = 8.7, 2.3 Hz, 1H), 6.64-6.54 (m, 3H), 5.07 (dd, J = 12.7, 5.4 Hz, 1H), 4.22 (s, 2H), 4.10 (d, J = 13.0 Hz, 2H), 3.80 (s, 6H), 3.69 (s, 2H), 3.50 (s, 3H), 3.44-3.37 (m, 5H), 3.00-2.87 (m, 5H), 2.84-2.66 (m, 2H), 2.63-2.56 (m, 4H), 2.47-2.44 (m, 1H), 2.30-2.17 (m, 5H), 2.06-1.98 (m, 1H), 1.79 (d, J = 12.3 Hz, 2H), 1.65 (d, J = 12.3 Hz, 2H), 1.55-1.41 (m, 2H), 1.39-1.16 (m, 5H).
D201	LCMS (ESI) m/z: [M + H] ⁺ = 795.75	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.07 (s, 1H), 8.19 (s, 1H, FA), 7.67 (d, J = 8.5 Hz, 1H), 7.63 (s, 1H), 7.33 (d, J = 2.1 Hz, 1H), 7.24 (dd, J = 8.7, 2.3 Hz, 1H), 6.62-6.54 (m, 3H), 5.07 (dd, J = 12.7, 5.3 Hz, 1H), 4.22 (s, 2H), 3.79 (s, 6H), 3.53-3.49 (m, 6H), 3.43-3.39 (m, 8H), 2.93-2.82 (m, 3H), 2.60 (d, J = 4.3 Hz, 8H), 2.25-2.16 (m, 1H), 2.10-1.98 (m, 3H), 1.71 (d, J = 11.9 Hz, 2H), 1.47-1.31 (m, 2H).
D202	LCMS (ESI) m/z: [M + H] ⁺ = 894.55	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.08 (s, 1H), 8.14 (s, 1H, FA), 7.70 (d, J = 8.5 Hz, 1H), 7.64 (s, 1H), 7.35 (d, J = 2.3 Hz, 1H), 7.27 (dd, J = 8.8, 2.2 Hz, 1H), 6.72 (s, 2H), 6.60 (q, J = 4.2 Hz, 1H), 5.08 (dd, J = 12.7, 5.4 Hz, 1H), 4.28-4.11 (m, 4H), 3.87 (s, 6H), 3.64 (s, 2H), 3.51 (s, 3H), 3.49-3.40 (m, 9H), 3.15-3.00 (m, 4H), 2.97-2.82 (m, 2H), 2.65-2.54 (m, 10H), 2.47-2.28 (m, 6H), 2.12-1.96 (m, 3H), 1.85-1.54 (m, 2H).
D203	LCMS (ESI) m/z: [M + H] ⁺ = 868.50	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.11 (s, 1H), 8.17 (s, 1H, FA), 7.84 (dd, J = 8.4, 4.2 Hz, 1H), 7.63 (d, J = 3.5 Hz, 1H), 7.43 (s, 1H), 7.36 (d, J = 8.4 Hz, 1H), 6.65-6.53 (m, 3H), 5.12 (dd, J = 12.9, 5.4 Hz, 1H), 4.25-4.15 (m, 4H), 3.79 (d, J = 3.3 Hz, 6H), 3.69-3.54 (m, 5H), 3.50 (s, 3H), 3.45-3.29 (m, 8H), 2.97-2.83 (m, 1H), 2.65-2.54 (m, 7H), 2.47-2.32 (m, 3H), 2.07-1.93 (m, 3H), 1.75-1.60 (m, 2H), 1.59-1.44 (m, 2H).
D204	853.65	
D205	LCMS (ESI) m/z: [M + H] ⁺ = 837.75	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.08 (s, 1H), 8.19 (s, 2H, FA), 7.68-7.62 (m, 2H), 7.31 (d, J = 2.2 Hz, 1H), 7.23 (dd, J = 8.7, 2.3 Hz, 1H), 6.66 (t, J = 5.1 Hz, 1H), 6.60 (s, 2H), 5.07 (dd, J = 12.7, 5.4 Hz, 1H), 4.22 (s, 2H), 4.12-4.01 (m, 2H), 3.79 (s, 6H), 3.59 (s, 2H), 3.50 (s, 3H), 3.41 (t, J = 5.4 Hz, 4H), 3.22-3.13 (m, 4H), 2.95-2.81 (m, 4H), 2.65-2.55 (m, 3H), 2.25 (s, 8H), 2.05-1.96 (m, 1H), 1.89-1.67 (m, 4H), 1.53-1.36 (m, 2H), 1.27-1.12 (m, 2H).
D206	LCMS (ESI) m/z: [M + H] ⁺ = 867.65	
D207	LCMS (ESI) m/z: [M + H] ⁺ = 809.7	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.06 (s, 1H), 8.24 (s, 2H, FA), 7.64 (d, J = 5.9 Hz, 2H), 6.96-6.41 (m, 5H), 5.20-4.86 (m, 1H), 4.46-4.12 (m, 2H), 3.96 (t, J = 6.6 Hz, 6H), 3.79 (s, 6H), 3.65-3.30 (m, 9H), 3.11 (d, J = 6.9 Hz, 2H), 2.77 (s, 4H), 2.23 (s, 6H), 2.15-1.92 (m, 3H), 1.75-1.55 (m, 2H), 1.25-0.97 (m, 3H).
D208	LCMS (ESI) m/z: [M + H] ⁺ = 823.6	
D209	LCMS (ESI) m/z: [M + H] ⁺ = 823.45	
D210	LCMS (ESI) m/z: [M + H] ⁺ = 809.45	
D211	LCMS (ESI) m/z: [M + H] ⁺ = 795.55	
D212	LCMS (ESI) m/z: [M + H] ⁺ = 809.35	
D213	LCMS (ESI) m/z: [M + H] ⁺ = 840.4	
D214	LCMS (ESI) m/z: [M + H] ⁺ = 920.4	
D215	LCMS (ESI) m/z: [M + H] ⁺ = 824.4	
D216	LCMS (ESI) m/z: [M + H] ⁺ = 920.4	

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Compound No.	LCMS	¹ H NMR
D217	LCMS (ESI) m/z: [M + H] ⁺ = 950.5	
D218	LCMS (ESI) m/z: [M + H] ⁺ = 934.55	
D219	LCMS (ESI) m/z: [M + H] ⁺ = 906.5	
D220	LCMS (ESI) m/z: [M + H] ⁺ = 920.5	¹ H NMR (300 MHz, Methanol-d ₄) δ 8.47 (s, 2H, FA), 7.71-7.57 (m, 2H), 7.36 (s, 1H), 7.24 (d, J = 8.6 Hz, 1H), 6.71 (s, 2H), 5.07 (dd, J = 12.3, 5.4 Hz, 1H), 4.38 (s, 4H), 4.10 (t, J = 9.0 Hz, 1H), 3.95 (s, 6H), 3.63 (s, 4H), 3.51 (s, 5H), 3.45-3.39 (m, 1H), 3.23-3.04 (m, 4H), 3.01-2.81 (m, 8H), 2.80-2.60 (m, 8H), 2.58-2.50 (m, 2H), 2.36-2.22 (m, 1H), 2.20-2.03 (m, 3H), 2.00-1.76 (m, 6H), 1.75-1.58 (m, 3H).
D221	LCMS (ESI) m/z: [M + H] ⁺ = 906.5	
D222	NA	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.12 (s, 1H), 8.15 (s, 2H, FA), 7.84 (d, J = 8.3 Hz, 1H), 7.64 (s, 1H), 7.43 (d, J = 2.3 Hz, 1H), 7.35 (dd, J = 8.4, 2.3 Hz, 1H), 6.67-6.56 (m, 3H), 5.12 (dd, J = 12.9, 5.3 Hz, 1H), 4.39-4.35 (m, 1H), 4.25-4.17 (m, 4H), 3.81 (s, 6H), 3.59 (t, J = 4.2 Hz, 2H), 3.50 (s, 4H), 3.41-3.38 (m, 3H), 2.94-2.85 (m, 1H), 2.67-2.57 (m, 7H), 2.37-2.25 (m, 5H), 2.18 (s, 2H), 2.10-2.01 (m, 1H), 1.90-1.73 (m, 4H), 1.63-1.51 (m, 4H), 1.46-1.39 (m, 2H).
D223	LCMS (ESI) m/z: [M + H] ⁺ = 881.4	
D224	LCMS (ESI) m/z: [M + H] ⁺ = 824.4	
D225	LCMS (ESI) m/z: [M + H] ⁺ = 824.4	
D226	LCMS (ESI) m/z: [M + H] ⁺ = 906.45	
D227	LCMS (ESI) m/z: [M + H] ⁺ = 851.6	¹ H NMR (300 MHz, Methanol-d ₄) δ 8.55 (s, 2H, FA), 7.66 (d, J = 8.5 Hz, 1H), 7.61 (s, 1H), 7.33 (d, J = 2.1 Hz, 1H), 7.22 (dd, J = 8.7, 2.2 Hz, 1H), 6.72 (s, 2H), 5.08 (dd, J = 12.3, 5.4 Hz, 1H), 4.44-4.34 (m, 4H), 4.13-4.00 (m, 2H), 3.95 (s, 6H), 3.64 (s, 3H), 3.57 (t, J = 5.4 Hz, 3H), 3.47-3.34 (m, 4H), 3.22 (t, J = 7.0 Hz, 2H), 3.13-2.94 (m, 3H), 2.88 (s, 7H), 2.78-2.62 (m, 4H), 2.24 (s, 2H), 2.16-2.07 (m, 1H), 2.03-1.77 (m, 5H), 1.62 (s, 1H), 1.46-1.29 (m, 2H).
D228	LCMS (ESI) m/z: [M + H] ⁺ = 810.7	
D229	LCMS (ESI) m/z: [M + H] ⁺ = 881.6	
D230	LCMS (ESI) m/z: [M + H] ⁺ = 892.5	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.08 (s, 1H), 8.17 (s, 2H, FA), 7.69 (d, J = 8.5 Hz, 1H), 7.65 (s, 1H), 7.35 (s, 1H), 7.31-7.23 (m, 1H), 6.82 (d, J = 7.4 Hz, 1H), 6.63 (s, 2H), 5.08 (dd, J = 12.8, 5.2 Hz, 1H), 4.22 (s, 2H), 4.03 (q, J = 7.9 Hz, 1H), 3.81 (s, 6H), 3.77-3.70 (m, 4H), 3.61 (s, 2H), 3.50 (s, 3H), 3.47-3.38 (m, 10H), 2.94-2.76 (m, 3H), 2.60-2.54 (m, 2H), 2.45-2.29 (m, 12H), 2.15 (t, J = 10.2 Hz, 2H), 2.08-1.97 (m, 1H), 1.51-1.33 (m, 4H).
D231	LCMS (ESI) m/z: [M + H] ⁺ = 809.45	
D232	LCMS (ESI) m/z: [M + H] ⁺ = 920.45	
D233	LCMS (ESI) m/z: [M + H] ⁺ = 868.45	
D234	LCMS (ESI) m/z: [M + H] ⁺ = 823.45	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.07 (s, 1H), 8.18 (s, 1H, FA), 7.67-7.59 (m, 2H), 6.93-6.85 (m, 2H), 6.77-6.71 (m, 1H), 6.60 (s, 2H), 5.04 (dd, J = 12.8, 5.4 Hz, 1H), 4.23 (s, 2H), 4.14-4.04 (m, 1H), 3.79 (s, 6H), 3.59 (s, 2H), 3.53-3.47 (m, 5H), 3.45-3.41 (m, 3H), 3.41 (s, 2H), 3.13-3.04 (m, 3H), 2.94-2.83 (m, 1H), 2.74-2.69 (m, 1H), 2.62-2.57 (m, 1H), 2.47-2.34 (m, 3H), 2.25 (s, 6H), 2.15 (t, J = 10.9 Hz, 1H), 2.05-1.89 (m, 4H), 1.79-1.68 (m, 2H), 1.65-1.56 (m, 2H).

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Compound No.	LCMS	¹ H NMR
D235	LCMS (ESI) m/z: [M + H] ⁺ = 852.45	
D236	LCMS (ESI) m/z: [M + H] ⁺ = 837.45	
D237	LCMS (ESI) m/z: [M + H] ⁺ = 838.4	
D238	LCMS (ESI) m/z: [M + H] ⁺ = 823.45	
D239	m/z: [M + H] ⁺ = 824.85	
D240	LCMS (ESI) m/z: [M + H] ⁺ = 920.5	
D241	LCMS (ESI) m/z: [M + H] ⁺ = 795.4	
D242	LCMS (ESI) m/z: [M + H] ⁺ = 838.45	
D243	LCMS (ESI) m/z: [M + H] ⁺ = 892.45	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.08 (s, 1H), 8.27 (s, 2H, FA), 7.68 (d, J = 8.5 Hz, 1H), 7.64 (s, 1H), 7.34 (s, 1H), 7.26 (d, J = 8.7 Hz, 1H), 6.77 (s, 1H), 6.58 (s, 2H), 5.08 (dd, J = 12.8, 5.5 Hz, 1H), 4.28 (d, J = 5.9 Hz, 2H), 4.02-3.94 (m, 1H), 3.78 (s, 6H), 3.61-3.50 (m, 4H), 3.50-3.42 (m, 12H), 2.95-2.85 (m, 2H), 2.79-2.67 (m, 3H), 2.64-2.56 (m, 3H), 2.40-2.25 (m, 4H), 2.14 (s, 6H), 2.05-1.81 (m, 3H), 1.76-1.67 (m, 1H), 1.63-1.47 (m, 5H).
D244	LCMS (ESI) m/z: [M + H] ⁺ = 824.45	
D245	LCMS (ESI) m/z: [M + H] ⁺ = 906.45	
D246	LCMS (ESI) m/z: [M + H] ⁺ = 823.45	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.08 (s, 1H), 9.37 (s, 1H, TFA), 9.04 (s, 1H), 7.69 (t, J = 4.1 Hz, 2H), 6.93 (s, 1H, TFA), 6.75 (d, J = 13.2 Hz, 3H), 6.65 (d, J = 8.3 Hz, 1H), 5.06 (dd, J = 12.8, 5.3 Hz, 1H), 4.31-4.09 (m, 4H), 3.89 (d, J = 6.6 Hz, 8H), 3.83 (s, 2H), 3.52 (s, 3H), 3.45 (s, 4H), 3.25-2.79 (m, 8H), 2.76 (d, J = 4.8 Hz, 6H), 2.66-2.60 (m, 3H), 2.16 (d, J = 13.7 Hz, 2H), 1.95 (dd, J = 34.7, 19.8 Hz, 5H).
D247	LCMS (ESI) m/z: [M + H] ⁺ = 892.6	
D248	LCMS (ESI) m/z: [M + H] ⁺ = 810.4	
D249	LCMS (ESI) m/z: [M + H] ⁺ = 796.35	
D250	LCMS (ESI) m/z: [M + H] ⁺ = 838.35	¹ H NMR (400 MHz, DMSO-d ₆) δ 8.42 (s, 2H, FA), 7.90 (d, J = 8.2 Hz, 1H), 7.67 (s, 1H), 7.50 (d, J = 2.2 Hz, 1H), 7.39 (dd, J = 8.3, 2.3 Hz, 1H), 6.78 (s, 2H), 6.59 (d, J = 7.1 Hz, 1H), 5.13 (dd, J = 12.9, 5.4 Hz, 1H), 4.48 (s, 2H), 4.31 (t, J = 5.7 Hz, 2H), 4.25 (s, 2H), 4.06 (q, J = 7.5 Hz, 1H), 3.88 (s, 6H), 3.52 (s, 6H), 3.46-3.39 (m, 2H), 3.11-3.05 (m, 2H), 3.01 (s, 6H), 2.93 (s, 2H), 2.92-2.80 (m, 1H), 2.63 (s, 1H), 2.61-2.53 (m, 2H), 2.35 (s, 2H), 2.10-2.04 (m, 1H), 1.94-1.83 (m, 2H), 1.70 (t, J = 6.8 Hz, 3H), 1.60-1.44 (m, 3H).
D251	LCMS (ESI) m/z: [M + H] ⁺ = 878.85	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.12 (s, 1H), 8.17 (s, 1H, FA), 7.83 (d, J = 8.2 Hz, 1H), 7.63 (s, 1H), 7.33-7.25 (m, 2H), 6.65-6.57 (m, 3H), 5.12 (dd, J = 12.8, 5.4 Hz, 1H), 4.98 (q, J = 6.8 Hz, 1H), 4.22 (s, 2H), 3.79 (s, 6H), 3.63 (s, 2H), 3.50 (s, 3H), 3.41 (t, J = 5.5 Hz, 3H), 2.95-2.84 (m, 3H), 2.64-2.56 (m, 6H), 2.48-2.34 (m, 8H), 2.26-2.15 (m, 2H), 2.09-2.00 (m, 1H), 1.87-1.78 (m, 2H), 1.70-1.55 (m, 6H), 1.42-1.32 (m, 2H), 1.31-1.10 (m, 3H).

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Compound No.	LCMS	¹ H NMR
D252	LCMS (ESI) m/z:	
D253	[M + H] ⁺ = 838.35 LCMS (ESI) m/z:	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.12 (s, 1H), 8.17 (s, 1H, FA), 7.83 (d, J = 8.1 Hz, 1H), 7.64 (s, 1H), 7.35-7.23 (m, 2H), 6.68-6.55 (m, 3H), 5.12 (dd, J = 12.9, 5.4 Hz, 1H), 5.04-4.93 (m, 1H), 4.22 (s, 2H), 3.81 (s, 6H), 3.73 (s, 2H), 3.50 (s, 3H), 3.43-3.41 (m, 3H), 3.01-2.93 (m, 2H), 2.92-2.77 (m, 2H), 2.64-2.56 (m, 6H), 2.45-2.26 (m, 7H), 2.15-1.99 (m, 3H), 1.86-1.76 (m, 2H), 1.71-1.53 (m, 7H), 1.26-1.08 (m, 2H).
D254	[M + H] ⁺ = 878.65 LCMS (ESI) m/z:	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.12 (s, 1H), 8.21 (s, 1H, FA), 7.84 (d, J = 8.3 Hz, 1H), 7.63 (s, 1H), 7.34-7.26 (m, 2H), 6.65-6.56 (m, 3H), 5.12 (dd, J = 12.8, 5.4 Hz, 1H), 5.03 (t, J = 6.9 Hz, 1H), 4.22 (s, 2H), 3.78 (s, 6H), 3.53-3.49 (m, 6H), 3.43-3.39 (m, 6H), 2.96-2.78 (m, 4H), 2.64-2.56 (m, 5H), 2.47-2.44 (m, 1H), 2.15-2.00 (m, 4H), 1.92-1.82 (m, 2H), 1.67-1.43 (m, 9H).
D255	LCMS (ESI) m/z:	¹ H-NMR (400 MHz, DMSO-d ₆) δ 11.09 (s, 1H), 8.15 (s, 1H, FA), 7.69 (d, J = 8.5 Hz, 1H), 7.64 (s, 1H), 7.35 (d, J = 2.2 Hz, 1H), 7.27 (dd, J = 8.7, 2.3 Hz, 1H), 6.69 (s, 2H), 6.61 (q, J = 4.3 Hz, 1H), 5.08 (dd, J = 12.9, 5.4 Hz, 1H), 4.22 (s, 2H), 4.04 (s, 2H), 3.86 (s, 6H), 3.51 (s, 3H), 3.48-3.38 (m, 7H), 3.10-2.83 (m, 5H), 2.73 (t, J = 7.6 Hz, 1H), 2.64-2.53 (m, 6H), 2.39 (s, 4H), 2.08-1.95 (m, 3H), 1.76 (s, 2H), 1.73-1.57 (m, 4H).
D256	LCMS (ESI) m/z:	¹ H NMR (300 MHz, DMSO) δ 11.08 (s, 1H), 8.20 (d, FA, 2H), 7.73-7.61 (m, 2H), 7.39-7.18 (m, 2H), 6.60 (d, 3H), 5.07 (dd, 1H), 4.22 (s, 2H), 3.80 (s, 7H), 3.63 (d, 3H), 3.50 (s, 3H), 3.42 (d, 6H), 2.98-2.80 (m, 3H), 2.60 (d, 4H), 2.51-2.39 (m, 5H), 2.22-2.12 (m, 4H), 2.16-1.95 (m, 1H), 1.69 (d, 2H), 1.60-1.45 (m, 1H), 1.22-1.05 (m, 2H);
D257	LCMS (ESI) m/z:	¹ H NMR (400 MHz, Methanol-d ₄) δ 8.49 (s, 3H, FA), 7.81 (d, J = 8.4 Hz, 1H), 7.56-7.48 (m, 2H), 7.39 (d, J = 8.6 Hz, 1H), 6.64 (s, 2H),
D258	[M + H] ⁺ = 809.50 LCMS (ESI) m/z:	5.12 (dd, J = 12.6, 5.4 Hz, 1H), 4.35 (s, 2H), 4.10-4.02 (m, 3H), 3.90 (s, 6H), 3.89-3.85 (m, 3H), 3.84-3.62 (m, 9H), 3.55-3.48 (m, 3H), 3.08-3.01 (m, 1H), 2.98-2.83 (m, 2H), 2.78 (s, 3H), 2.76-2.70 (m, 2H), 2.64-2.57 (m, 2H), 2.44 (s, 1H), 2.17-1.98 (m, 4H), 1.39-1.27 (m, 2H).
D259	LCMS (ESI) m/z: [M + H] ⁺ = 854.4	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.11 (s, 1H), 8.37 (s, 1H, FA), 7.79 (d, J = 8.3 Hz, 1H), 7.56 (d, J = 30.9 Hz, 2H), 7.47-7.31 (m, 1H), 6.61 (d, J = 5.4 Hz, 3H), 5.10 (dd, J = 13.0, 5.4 Hz, 1H), 4.21 (s, 2H), 3.91 (d, J = 8.9 Hz, 1H), 3.81 (s, 10H), 3.76-3.57 (m, 12H), 3.40 (t, J = 5.6 Hz, 4H), 3.26 (t, J = 10.7 Hz, 2H), 2.98-2.82 (m, 1H), 2.71 (s, 1H), 2.66-2.54 (m, 5H), 2.28-2.20 (m, 1H), 2.09-1.98 (m, 1H), 1.84-1.79 (s, 1H).
D260	LCMS (ESI) m/z: [M + H] ⁺ = 795.4	¹ H NMR (300 MHz, Methanol-d ₄) δ 8.48 (s, 1H, FA), 7.84 (d, J = 8.3 Hz, 1H), 7.60 (s, 1H), 7.46 (s, 1H), 7.40-7.34 (m, 1H), 6.72 (s, 2H), 5.13 (dd, J = 12.4, 5.4 Hz, 1H), 4.44-4.35 (m, 6H), 4.19-4.09 (m, 1H), 3.95 (s, 6H), 3.69-3.48 (m, 8H), 3.43-3.38 (m, 1H), 3.18-3.11 (m, 2H), 2.92-2.72 (m, 13H), 2.65 (s, 2H), 2.21-2.10 (m, 1H), 1.98 (dd, J = 12.8, 7.2 Hz, 1H), 1.82-1.71 (m, 4H), 1.58-1.47 (m, 1H).
D261	LCMS (ESI) m/z: [M + H] ⁺ = 809.45	
D262	LCMS (ESI) m/z: [M + H] ⁺ = 824.4	¹ H NMR (400 MHz, DMSO-d ₆) δ 8.43 (s, 2H, FA), 7.90 (d, J = 8.4 Hz, 1H), 7.68 (s, 1H), 7.50 (d, J = 2.2 Hz, 1H), 7.39 (dd, J = 8.4, 2.3 Hz, 1H), 6.88 (d, J = 6.7 Hz, 1H), 6.77 (s, 2H), 5.14 (dd, J = 12.8, 5.4 Hz, 1H), 4.48 (s, 2H), 4.31 (t, J = 5.2 Hz, 4H), 4.25 (s, 2H), 4.13 (dq, J = 16.8, 8.2 Hz, 1H), 3.88 (s, 6H), 3.52 (s, 6H), 3.42 (t, J = 4.7 Hz, 2H), 3.01 (s, 6H), 2.96-2.84 (m, 4H), 2.63 (d, J = 3.8 Hz, 1H), 2.61-2.53 (m, 3H), 2.40-2.33 (m, 1H), 2.27-1.96 (m, 6H), 1.87-1.82 (m, 1H), 1.81-1.74 (m, 1H).
D263	LCMS (ESI) m/z: [M + H] ⁺ = 810.45	
D264	LCMS (ESI) m/z: [M + H] ⁺ = 796.7	
D265	LCMS (ESI) m/z: [M + H] ⁺ = 838.45	¹ H NMR (400 MHz, DMSO-d ₆ with a drop of D ₂ O) δ 8.40 (s, 2H, FA), 7.88 (d, J = 8.3 Hz, 1H), 7.66 (s, 1H), 7.48 (d, J = 2.3 Hz, 1H), 7.39 (dd, J = 8.4, 2.3 Hz, 1H), 6.74 (s, 2H), 5.12 (dd, J = 12.8, 5.4 Hz, 1H),

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Compound No.	LCMS	¹ H NMR
		4.47 (s, 2H), 4.30 (t, J = 6.1 Hz, 2H), 4.23 (s, 2H), 3.86 (s, 6H), 3.55-3.48 (m, 5H), 3.41 (s, 2H), 3.25 (s, 1H), 3.07-2.97 (m, 9H), 2.96-2.81 (m, 3H), 2.75-2.70 (m, 1H), 2.66-2.55 (m, 5H), 2.34 (s, 1H), 2.24-2.14 (m, 1H), 2.09-1.94 (m, 3H), 1.52-1.42 (m, 2H), 1.03-0.97 (m, 1H).
D266	LCMS (ESI) m/z: [M + H] ⁺ = 824.4	
D267	LCMS (ESI) m/z: [M + H] ⁺ = 838.4	
D268	LCMS (ESI) m/z: [M + H] ⁺ = 810.7	
D269	LCMS (ESI) m/z: [M + H] ⁺ = 824.45	¹ H NMR (400 MHz, Methanol-d ₄) δ 8.51 (s, 2H, FA), 7.81 (d, J = 8.2 Hz, 1H), 7.60 (s, 1H), 7.46 (s, 1H), 7.37 (d, J = 8.2 Hz, 1H), 6.71 (s, 2H), 5.12 (dd, J = 12.4, 5.4 Hz, 1H), 4.49 (s, 2H), 4.37 (d, J = 2.8 Hz, 4H), 3.95 (s, 6H), 3.63 (s, 3H), 3.58-3.52 (m, 2H), 3.37 (s, 2H), 3.27 (d, J = 7.3 Hz, 2H), 3.15 (s, 4H), 2.88 (s, 7H), 2.80-2.58 (m, 4H), 2.17-2.07 (m, 1H), 1.97-1.55 (m, 4H), 1.18-1.06 (m, 1H), 0.70-0.60 (m, 1H), 0.37 (t, J = 5.1 Hz, 1H).
D270	LCMS (ESI) m/z: [M + H] ⁺ = 810.4	
D271	LCMS (ESI) m/z: [M + H] ⁺ = 810.45	
D272	LCMS (ESI) m/z: [M + H] ⁺ = 824.4	
D273	LCMS (ESI) m/z: [M + H] ⁺ = 810.4	
D274	LCMS (ESI) m/z: [M + H] ⁺ = 796.35	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.12 (s, 1H), 8.18 (s, 2H, FA), 7.84 (d, J = 8.3 Hz, 1H), 7.64 (s, 1H), 7.46 (dd, J = 4.0, 2.2 Hz, 1H), 7.37 (dt, J = 8.3, 2.3 Hz, 1H), 6.63 (s, 2H), 6.58 (d, J = 4.9 Hz, 1H), 5.12 (dd, J = 12.9, 5.4 Hz, 1H), 4.26 (d, J = 11.7 Hz, 4H), 3.89 (s, 1H), 3.80 (d, J = 1.9 Hz, 6H), 3.74-3.54 (m, 4H), 3.50 (s, 3H), 3.44 (d, J = 5.7 Hz, 2H), 3.18-3.10 (m, 2H), 3.03-2.98 (m, 1H), 2.92-2.84 (m, 3H), 2.74-2.69 (m, 1H), 2.32 (s, 5H), 2.22 (d, J = 13.6 Hz, 2H), 2.09-2.04 (s, 2H), 1.80 (t, J = 11.3 Hz, 1H), 1.74-1.61 (m, 2H), 1.46 (d, J = 10.2 Hz, 1H). LCMS (ESI) m/z: [M + H] ⁺ = 796.35
D275	LCMS (ESI) m/z: [M + H] ⁺ = 864.45	¹ H NMR (400 MHz, Methanol-d ₄) δ 8.56 (br s, 1H, FA), 7.82 (d, J = 8.3 Hz, 1H), 7.58 (s, 1H), 7.30 (s, 1H), 7.25 (dd, J = 8.2, 2.2 Hz, 1H), 6.72 (s, 2H), 5.12 (dd, J = 12.7, 5.4 Hz, 1H), 4.97 (t, J = 6.6 Hz, 1H), 4.47 (s, 2H), 4.36 (s, 2H), 4.25 (t, J = 9.5 Hz, 2H), 4.01-3.90 (m, 8H), 3.64 (s, 3H), 3.60-3.49 (m, 5H), 3.47-3.42 (m, 1H), 3.23-3.17 (m, 1H), 2.95-2.82 (m, 3H), 2.80-2.67 (m, 5H), 2.64-2.54 (m, 4H), 2.18-2.10 (m, 1H), 2.03 (dd, J = 12.7, 6.2 Hz, 2H), 1.80-1.60 (m, 4H).
D276	LCMS (ESI) m/z: [M + H] ⁺ = 852.39	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.12 (s, 1H), 7.84 (d, J = 8.3 Hz, 1H), 7.70 (t, J = 5.8 Hz, 1H), 7.62 (s, 1H), 7.44 (d, J = 2.3 Hz, 1H), 7.36 (dd, J = 8.3, 2.3 Hz, 1H), 6.62-6.57 (m, 1H), 6.56 (s, 2H), 5.12 (dd, J = 12.9, 5.4 Hz, 1H), 4.19 (dd, J = 13.0, 6.6 Hz, 4H), 3.77 (s, 6H), 3.50 (s, 3H), 3.44 (s, 2H), 3.40 (t, J = 5.5 Hz, 2H), 3.40-3.35 (m, 2H), 3.07 (q, J = 6.5 Hz, 2H), 2.95-2.84 (m, 1H), 2.63-2.54 (m, 5H), 2.09-1.97 (m, 4H), 1.90 (s, 6H), 1.81-1.71 (m, 2H), 1.54-1.35 (m, 4H).
D277	LCMS (ESI) m/z: [M + H] ⁺ = 820.55	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.12 (s, 1H), 8.23 (s, 1H, FA), 7.84 (d, J = 8.3 Hz, 1H), 7.61 (s, 1H), 7.43 (d, J = 2.2 Hz, 1H), 7.35 (dd, J = 8.3, 2.3 Hz, 1H), 7.25 (t, J = 5.8 Hz, 1H), 6.64-6.54 (m, 3H), 5.13 (dd, J = 12.9, 5.4 Hz, 1H), 4.25-4.15 (m, 4H), 4.02-3.85 (m, 2H), 3.77 (s, 6H), 3.57 (s, 2H), 3.49 (s, 3H), 3.43-3.38 (m, 6H), 3.11 (q, J = 6.5 Hz, 2H), 2.96-2.82 (m, 1H), 2.66-2.52 (m, 6H), 2.10-1.99 (m, 1H), 1.97-1.85 (m, 2H).
D278	LCMS (ESI) m/z: [M + H] ⁺ = 834.15	¹ H NMR (400 MHz, Methanol-d ₄) δ 8.48 (s, 1H, FA), 7.80 (d, J = 8.3 Hz, 1H), 7.55 (s, 1H), 7.40 (d, J = 2.3 Hz, 1H), 7.32 (dd, J = 8.4, 2.3 Hz, 1H), 6.63 (s, 2H), 5.11 (dd, J = 12.5, 5.4 Hz, 1H), 4.35 (s, 2H), 4.20 (t, J = 6.2 Hz, 2H), 4.16-4.06 (m, 1H), 4.02 (s, 2H), 3.88 (s, 6H), 3.88-3.83 (m, 4H), 3.63 (s, 3H), 3.53 (t, J = 5.6 Hz, 2H), 3.14 (t, J = 6.9 Hz, 2H), 2.95-2.83 (m, 1H), 2.80-2.67 (m, 5H), 2.65-2.60 (m, 2H), 2.18-2.09 (m, 1H), 1.97-1.86 (m, 2H), 1.80-1.67 (m, 2H).
D279	LCMS (ESI) m/z: [M + H] ⁺ = 806.45	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.12 (s, 1H), 7.86 (d, J = 8.3 Hz, 1H), 7.61 (s, 1H), 7.51 (s, 1H), 7.46 (d, J = 2.3 Hz, 1H), 7.37 (dd, J = 8.3, 2.3 Hz, 1H), 6.62-6.53 (m, 3H), 5.13 (dd, J = 12.9, 5.3 Hz, 1H), 4.24-4.16 (m, 4H), 4.04 (s, 1H), 3.78 (s, 6H), 3.59 (s, 2H), 3.49 (s,

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Compound No.	LCMS	¹ H NMR
D280	LCMS (ESI) m/z: [M + H] ⁺ = 820.60.	3H), 3.47-3.36 (m, 7H), 3.31 (s, 1H), 2.95-2.83 (m, 1H), 2.64-2.53 (m, 7H), 2.10-2.02 (m, 1H). ¹ H NMR (300 MHz, DMSO-d ₆) δ 11.12 (s, 1H), 7.83 (dd, J = 8.5, 7.2 Hz, 1H), 7.62 (s, 1H), 7.53 (d, J = 8.5 Hz, 1H), 7.47 (d, J = 7.2 Hz, 1H), 6.75-6.58 (m, 3H), 5.09 (dd, J = 12.8, 5.4 Hz, 1H), 4.28-4.19 (m, 4H), 3.96-3.61 (m, 8H), 3.50 (s, 3H), 3.40 (t, J = 5.5 Hz, 2H), 3.38 (s, 6H), 3.24-3.14 (m, 2H), 2.97-2.82 (m, 1H), 2.64-2.54 (m, 6H), 2.09-1.88 (m, 3H).
D281	LCMS (ESI) m/z: [M + H] ⁺ = 834.10	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.12 (s, 1H), 10.08 (s, 1H, TFA), 7.83 (dd, J = 8.5, 7.2 Hz, 1H), 7.69 (s, 1H), 7.64 (s, 1H), 7.53 (d, J = 8.5 Hz, 1H), 7.47 (d, J = 7.3 Hz, 1H), 6.71 (s, 2H), 6.61 (q, J = 4.3 Hz, 1H), 5.09 (dd, J = 12.7, 5.5 Hz, 1H), 4.66-4.29 (m, 5H), 4.26-4.14 (m, 5H), 3.86 (s, 6H), 3.51 (s, 3H), 3.44-3.39 (m, 4H), 3.11-3.01 (m, 2H), 2.94-2.81 (m, 1H), 2.64-2.53 (m, 6H), 2.07-1.98 (m, 1H), 1.85-1.76 (m, 2H), 1.70-1.60 (m, 2H).
D282	LCMS (ESI) m/z: [M + H] ⁺ = 806.60.	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.12 (s, 1H), 7.84 (t, J = 7.8 Hz, 1H), 7.62 (s, 1H), 7.55 (d, J = 8.6 Hz, 1H), 7.50 (d, J = 7.2 Hz, 1H), 6.69-6.53 (m, 3H), 5.11 (dd, J = 12.8, 5.4 Hz, 1H), 4.49-4.00 (m, 7H), 3.79 (s, 7H), 3.56-3.47 (m, 4H), 3.44-3.37 (m, 5H), 2.94-2.83 (m, 1H), 2.59 (d, J = 4.2 Hz, 4H), 2.57-2.53 (m, 4H), 2.07-1.98 (m, 1H).
D283	LCMS (ESI) m/z: [M + H] ⁺ = 852.39.	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.11 (s, 1H), 7.82 (dd, J = 8.5, 7.3 Hz, 1H), 7.69 (t, J = 5.8 Hz, 1H), 7.62 (s, 1H), 7.53 (d, J = 8.6 Hz, 1H), 7.45 (d, J = 7.2 Hz, 1H), 6.59 (d, J = 4.4 Hz, 1H), 6.56 (s, 2H), 5.08 (dd, J = 12.9, 5.4 Hz, 1H), 4.24-4.17 (m, 4H), 3.77 (s, 6H), 3.50 (s, 3H), 3.44 (s, 2H), 3.40 (t, J = 5.6 Hz, 2H), 3.37 (s, 2H), 3.11-3.03 (m, 2H), 2.93-2.81 (m, 1H), 2.61-2.55 (m, 5H), 2.07-1.98 (m, 4H), 1.90 (s, 6H), 1.81-1.73 (m, 2H), 1.54-1.38 (m, 4H).
D284	LCMS (ESI) m/z: [M + H] ⁺ = 813.45	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.09 (s, 1H), 8.15 (s, 1H, FA), 7.67 (d, J = 8.5 Hz, 1H), 7.63 (s, 1H), 7.33 (d, J = 2.3 Hz, 1H), 7.25 (dd, J = 8.7, 2.3 Hz, 1H), 6.61 (d, J = 6.9 Hz, 3H), 5.07 (dd, J = 12.9, 5.4 Hz, 1H), 4.21 (s, 2H), 3.84-3.77 (m, 7H), 3.71 (s, 2H), 3.64-3.53 (m, 5H), 3.52-3.46 (m, 4H), 3.43-3.38 (m, 7H), 2.95-2.83 (m, 1H), 2.72 (s, 2H), 2.60-2.53 (m, 9H), 2.28 (s, 3H), 2.06-1.97 (m, 1H).
D285	LCMS (ESI) m/z: [M - H] ⁺ = 774.37.	¹ H NMR (300 MHz, Methanol-d ₄) δ 8.56 (br s, 1.7H, FA), 7.55 (s, 1H), 7.24 (t, J = 7.7 Hz, 1H), 7.09 (d, J = 7.6 Hz, 1H), 6.91 (d, J = 7.9 Hz, 1H), 6.71 (s, 2H), 5.19 (dd, J = 13.4, 5.1 Hz, 1H), 4.39-4.27 (m, 5H), 4.09-3.98 (m, 2H), 3.94 (s, 6H), 3.67-3.50 (m, 14H), 3.40 (t, J = 5.4 Hz, 2H), 3.04-2.91 (m, 2H), 2.85 (s, 6H), 2.61 (s, 2H), 2.53-2.39 (m, 1H), 2.24-2.12 (m, 1H).
D286	LCMS (ESI) m/z: [M + H] ⁺ = 866.30.	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.12 (s, 1H), 8.18 (s, 1.0H, FA), 7.83 (d, J = 8.3 Hz, 1H), 7.68-7.62 (m, 2H), 7.43 (d, J = 2.3 Hz, 1H), 7.35 (dd, J = 8.3, 2.3 Hz, 1H), 6.62-6.56 (m, 3H), 5.12 (dd, J = 12.9, 5.4 Hz, 1H), 4.25-4.13 (m, 4H), 3.78 (s, 6H), 3.54 (s, 2H), 3.50 (s, 3H), 3.42-3.38 (m, 3H), 3.05 (q, J = 6.6 Hz, 2H), 2.94-2.84 (m, 1H), 2.64-2.53 (m, 6H), 2.48 (s, 2H), 2.11 (s, 3H), 2.08-2.01 (m, 1H), 1.85 (s, 6H), 1.80-1.71 (m, 2H), 1.51-1.35 (m, 4H).
D287	LCMS (ESI) m/z: [M + H] ⁺ = 874.35.	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.08 (s, 1H), 8.16 (s, 0.7H, FA), 7.68 (d, J = 8.5 Hz, 1H), 7.61 (s, 1H), 7.35 (d, J = 2.3 Hz, 1H), 7.26 (dd, J = 8.6, 2.3 Hz, 1H), 7.08 (t, J = 5.8 Hz, 1H), 6.58 (s, 3H), 5.07 (dd, J = 12.9, 5.3 Hz, 1H), 4.21 (s, 2H), 4.07-3.95 (m, 1H), 3.78 (s, 6H), 3.58 (s, 2H), 3.49 (s, 3H), 3.47-3.34 (m, 14H), 3.08 (q, J = 6.5 Hz, 2H), 2.94-2.82 (m, 1H), 2.59 (d, J = 4.2 Hz, 5H), 2.54 (s, 2H), 2.42 (t, J = 6.7 Hz, 2H), 2.07-1.99 (m, 1H).
D288	LCMS (ESI) m/z: [M + H] ⁺ = 866.25.	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.14 (s, 1H), 8.19 (s, 0.9H, FA), 7.81 (dd, J = 8.6, 7.3 Hz, 1H), 7.68-7.61 (m, 2H), 7.52 (d, J = 8.6 Hz, 1H), 7.45 (d, J = 7.2 Hz, 1H), 6.64-6.54 (m, 3H), 5.08 (dd, J = 12.9, 5.4 Hz, 1H), 4.26-4.13 (m, 4H), 3.78 (s, 6H), 3.55-3.47 (m, 6H), 3.40 (t, J = 5.5 Hz, 2H), 3.08-3.01 (m, 2H), 2.93-2.82 (m, 1H), 2.63-2.52 (m, 6H), 2.47 (s, 2H), 2.10 (s, 3H), 2.06-1.99 (m, 1H), 1.84 (s, 6H), 1.81-1.73 (m, 2H), 1.51-1.39 (m, 4H).
D289	LCMS (ESI) m/z: [M + H] ⁺ = 854.70.	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.12 (s, 1H), 8.19 (s, 0.9H, FA), 7.83 (d, J = 8.3 Hz, 1H), 7.63 (s, 1H), 7.44 (d, J = 2.3 Hz, 1H), 7.36 (d, J = 8.4 Hz, 1H), 6.58 (s, 3H), 5.12 (dd, J = 12.9, 5.4 Hz, 1H), 4.33-4.26 (m, 2H), 4.21 (s, 2H), 3.97-3.89 (m, 1H), 3.76 (s, 8H), 3.70-3.67 (m, 2H), 3.59 (s, 3H), 3.49 (s, 4H), 2.95-2.76 (m, 5H), 2.63-2.52 (m, 10H), 2.29-2.19 (m, 1H), 2.18-2.10 (m, 1H), 2.09-1.99 (m, 1H), 1.52-1.28 (m, 2H).
D290	LCMS (ESI) m/z: [M + H] ⁺ = 815.50.	¹ H NMR (400 MHz, DMSO-d ₆ , D ₂ O) δ 8.53 (t, J = 5.8 Hz, TFA salt), 7.84 (d, J = 8.3 Hz, 1H), 7.66 (s, 1H), 7.46 (d, J = 2.3 Hz, 1H), 7.40 (dd, J = 8.3, 2.3 Hz, 1H), 6.68 (s, 2H), 5.11 (dd, J = 12.9, 5.4 Hz, 1H), 4.77 (s, 2H), 4.22 (d, J = 4.7 Hz, 4H), 3.86 (s, 6H), 3.58-3.52 (m, 2H), 3.50 (s, 3H), 3.44-3.14 (m, 8H), 2.94-2.83 (m, 4H), 2.75 (s, 6H), 2.65-2.53 (m, 4H), 2.08-1.96 (m, 1H).
D291	LCMS (ESI) m/z:	¹ H-NMR (400 MHz, DMSO-d ₆) δ 11.01 (s, 1H), 8.31 (t, J = 5.3 Hz, 1H), 8.22 (s, 0.8H, FA), 7.64 (s, 1H), 7.48 (t, J = 8.1 Hz, 1H), 6.68-

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Compound No.	LCMS	¹ H NMR
		[M + H] ⁺ = 797.55. 6.59 (m, 4H), 6.49 (d, J = 8.3 Hz, 1H), 5.18 (dd, J = 11.5, 5.6 Hz, 1H), 4.22 (s, 2H), 3.79 (s, 6H), 3.62 (s, 2H), 3.50 (s, 3H), 3.40 (t, J = 5.5 Hz, 2H), 3.14 (q, J = 6.6 Hz, 2H), 3.03 (q, J = 6.5 Hz, 2H), 2.90-2.76 (m, 1H), 2.70-2.58 (m, 2H), 2.58-2.52 (m, 5H), 2.27 (s, 6H), 2.20-2.08 (m, 1H), 1.65-1.54 (m, 2H), 1.44-1.38 (m, 2H), 1.37-1.18 (m, 8H).
D292	LCMS (ESI) m/z: [M + H] ⁺ = 801.50.	¹ H-NMR (400 MHz, Methanol-d ₄) δ 8.55 (brs, 0.9H, FA), 7.53 (s, 1H), 7.43 (t, J = 8.1 Hz, 1H), 6.70-6.63 (m, 3H), 6.51 (d, J = 8.3 Hz, 1H), 5.20-5.13 (m, 1H), 4.41-4.25 (m, 4H), 3.94 (s, 6H), 3.82-3.77 (m, 2H), 3.68 (s, 4H), 3.62-3.57 (m, 5H), 3.56-3.35 (m, 6H), 2.93-2.79 (m, 8H), 2.78-2.69 (m, 1H), 2.62 (s, 3H), 2.58-2.50 (m, 2H), 2.23-2.14 (m, 1H).
D293	764.3	¹ H NMR (400 MHz, DMSO-d ₆) δ 7.72 (s, 1H), 7.41 (d, J = 8.6 Hz, 1H), 6.75-6.62 (m, 4H), 5.06 (dd, J = 13.2, 5.0 Hz, 1H), 4.35 (t, J = 18.7 Hz, 4H), 4.27-4.16 (m, 3H), 4.11-3.98 (m, 3H), 3.88 (s, 6H), 3.75-3.62 (m, 5H), 3.53 (s, 3H), 3.31-3.12 (m, 3H), 3.05-2.82 (m, 8H), 2.81-2.73 (m, 1H), 2.70-2.56 (m, 3H), 2.42-2.32 (m, 1H), 2.18-2.05 (m, 2H), 2.04-1.84 (m, 3H).
D294	866.45	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.13 (s, 1H), 8.20 (s, 0.9H, FA), 7.81 (dd, J = 8.5, 7.2 Hz, 1H), 7.64 (t, J = 5.8 Hz, 1H), 7.58-7.49 (m, 2H), 7.44 (d, J = 7.2 Hz, 1H), 6.58 (s, 2H), 6.54 (q, J = 4.3 Hz, 1H), 5.08 (dd, J = 12.9, 5.4 Hz, 1H), 4.26 (s, 2H), 4.20 (t, J = 6.4 Hz, 2H), 3.79 (s, 6H), 3.56-3.49 (m, 4H), 3.47 (s, 3H), 3.05 (q, J = 6.4 Hz, 2H), 2.88 (ddd, J = 17.3, 14.1, 5.4 Hz, 1H), 2.63-2.55 (m, 1H), 2.55-2.51 (m, 4H), 2.50-2.46 (m, 4H), 2.11 (s, 3H), 2.07-1.99 (m, 1H), 1.85 (s, 6H), 1.77 (t, J = 6.9 Hz, 2H), 1.50-1.39 (m, 4H).
D295	854.7	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.12 (s, 1H), 8.15 (s, 0.7H, FA), 7.83 (d, J = 8.3 Hz, 1H), 7.56 (s, 1H), 7.45 (d, J = 2.2 Hz, 1H), 7.36 (dd, J = 8.3, 2.3 Hz, 1H), 6.61 (s, 2H), 6.55 (q, J = 4.4 Hz, 1H), 5.12 (dd, J = 12.9, 5.4 Hz, 1H), 4.31-4.27 (m, 2H), 4.25 (s, 2H), 3.94-3.88 (m, 1H), 3.79 (s, 8H), 3.75 (s, 2H), 3.72-3.68 (m, 3H), 3.47 (s, 4H), 3.10-3.01 (m, 2H), 2.99-2.81 (m, 3H), 2.69-2.52 (m, 10H), 2.41-2.32 (m, 1H), 2.31-2.21 (m, 1H), 2.08-1.97 (m, 1H), 1.64-1.39 (m, 2H).
D296	813.45	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.08 (s, 1H), 8.16 (s, 1.1H, FA), 7.67 (d, J = 8.5 Hz, 1H), 7.56 (s, 1H), 7.34 (d, J = 2.3 Hz, 1H), 7.25 (dd, J = 8.7, 2.3 Hz, 1H), 6.63 (s, 2H), 6.54 (d, J = 4.5 Hz, 1H), 5.07 (dd, J = 12.9, 5.4 Hz, 1H), 4.27 (s, 2H), 3.82 (s, 6H), 3.73 (s, 2H), 3.63-3.55 (m, 4H), 3.52 (t, J = 5.9 Hz, 2H), 3.46 (s, 3H), 3.45-3.38 (m, 8H), 2.94-2.84 (m, 1H), 2.75 (s, 2H), 2.64-2.54 (m, 9H), 2.35-2.25 (m, 3H), 2.08-1.97 (m, 1H).
D297	802.55	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.10 (s, 1H), 8.19 (s, 0.8H, FA), 7.82-7.72 (m, 2H), 7.54 (s, 1H), 7.48 (d, J = 8.6 Hz, 1H), 7.43 (d, J = 7.2 Hz, 1H), 6.57 (s, 2H), 6.56-6.51 (m, 1H), 5.08 (dd, J = 12.9, 5.4 Hz, 1H), 4.37-4.29 (m, 2H), 4.25 (s, 2H), 3.85-3.77 (m, 8H), 3.59 (t, J = 5.7 Hz, 2H), 3.54-3.50 (m, 4H), 3.46 (s, 3H), 3.34 (q, J = 5.9 Hz, 4H), 2.94 (s, 2H), 2.91-2.83 (m, 1H), 2.62-2.52 (m, 5H), 2.10 (s, 3H), 2.05-1.96 (m, 1H).
D298	827.5	¹ H-NMR (400 MHz, Methanol-d ₄) δ 8.47 (brs, 1.4H, FA), 7.69 (d, J = 8.5 Hz, 1H), 7.50 (s, 1H), 7.34 (d, J = 2.3 Hz, 1H), 7.24 (dd, J = 8.6, 2.3 Hz, 1H), 6.77 (s, 2H), 5.08 (dd, J = 12.4, 5.5 Hz, 1H), 4.50 (s, 2H), 4.09 (s, 2H), 3.97 (s, 6H), 3.92-3.87 (m, 2H), 3.76 (t, J = 5.2 Hz, 2H), 3.61 (s, 3H), 3.51-3.45 (m, 8H), 2.90-2.86 (m, 4H), 2.83 (s, 6H), 2.79-2.71 (m, 10H), 2.17-2.07 (m, 1H).
D299	809.94	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.04 (s, 1H), 8.13 (s, 2H), 7.87 (s, 1H), 7.65 (d, J = 8.5 Hz, 1H), 7.31 (d, J = 2.3 Hz, 1H), 7.23 (dd, J = 8.7, 2.3 Hz, 1H), 6.68 (s, 2H), 6.33 (d, J = 4.5 Hz, 1H), 5.05 (dd, J = 12.9, 5.4 Hz, 1H), 3.81 (s, 5H), 3.65 (d, J = 3.1 Hz, 2H), 3.52 (s, 3H), 3.39 (d, J = 5.4 Hz, 4H), 3.15 (s, 2H), 2.97-2.78 (m, 3H), 2.58 (d, J = 4.3 Hz, 3H), 2.00 (dd, J = 9.2, 4.2 Hz, 1H), 1.64 (d, J = 12.6 Hz, 2H), 1.35 (s, 1H).
D300	738.82	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.03 (s, 1H), 8.14 (s, 1H), 7.85 (d, J = 11.1 Hz, 1H), 7.60 (d, J = 8.3 Hz, 1H), 6.70-6.56 (m, 3H), 5.02 (dd, J = 12.9, 5.4 Hz, 1H), 4.68-4.54 (m, 2H), 4.38 (s, 2H), 3.80 (s, 6H), 3.73 (d, J = 19.6 Hz, 4H), 3.52 (s, 3H), 3.47 (s, 2H), 2.93-2.74 (m, 1H), 2.58 (dd, J = 4.4, 2.9 Hz, 4H), 2.37 (s, 4H), 2.05-1.91 (m, 1H), 1.69 (s, 4H).
D301	749.15	¹ H NMR (400 MHz, Methanol-d ₄) δ 7.41 (d, J = 7.8 Hz, 2H), 6.88 (d, J = 2.2 Hz, 1H), 6.80 (dd, J = 8.2, 2.2 Hz, 1H), 6.70 (s, 2H), 5.14 (dd, J = 13.3, 5.2 Hz, 1H), 4.51 (s, 2H), 4.47-4.20 (m, 4H), 4.15 (s, 2H), 3.95 (s, 6H), 3.77 (s, 4H), 3.61 (s, 3H), 3.55-3.35 (m, 5H), 3.15 (s, 1H), 3.29-3.01 (m, 1H), 2.98-2.85 (m, 1H), 2.80 (d, J = 17.6 Hz, 1H), 2.60 (t, J = 6.4 Hz, 2H), 2.51-2.47 (m, 3H), 2.25-2.04 (m, 5H), 1.85-1.79 (m, 2H), 1.72-1.67 (m, 2H).
D302	737.4	¹ H NMR (400 MHz, Methanol-d ₄) δ 8.33 (s, 2H, FA), 7.91 (s, 1H), 7.39 (d, J = 8.2 Hz, 1H), 6.85 (d, J = 2.2 Hz, 1H), 6.81-6.74 (m, 3H),

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Compound No.	LCMS	¹ H NMR
DD1	LCMS (ESI) m/z: [M + H] ⁺ = 878.5	5.23 (t, J = 3.4 Hz, 2H), 5.14 (dd, J = 13.3, 5.2 Hz, 1H), 5.07 (t, J = 3.4 Hz, 2H), 4.45 (s, 2H), 4.42 (d, J = 16.5 Hz, 1H), 4.36 (d, J = 16.5 Hz, 1H), 4.22 (t, J = 9.4 Hz, 2H), 3.98 (s, 6H), 3.95 (d, J = 10.0 Hz, 2H), 3.71 (s, 3H), 3.68 (s, 4H), 3.20-3.10 (m, 1H), 2.98-2.86 (m, 1H), 2.84-2.69 (m, 3H), 2.61-2.42 (m, 5H), 2.23-2.13 (m, 1H), 1.94-1.90 (m, 4H). ¹ H NMR (400 MHz, Methanol-d ₄) δ 8.56 (s, 1H, FA), 7.55 (s, 1H), 7.49 (td, J = 8.6, 5.9 Hz, 1H), 7.28-7.20 (m, 1H), 7.19-7.12 (m, 1H), 6.63 (s, 2H), 4.36 (s, 2H), 3.96-3.82 (m, 8H), 3.64 (s, 3H), 3.59-3.49 (m, 4H), 3.31-3.24 (m, 2H), 2.94-2.74 (m, 8H), 2.69-2.45 (m, 7H), 1.79 (d, J = 10.9 Hz, 2H), 1.62-1.46 (m, 5H).

Example 67—Preparation of Compounds D303-D375

In analogy to the procedures described in the examples above, compounds D303-D375 were prepared using the appropriate starting materials ²⁰

Compound No.	LCMS	¹ H NMR
D303	669.2	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.07 (s, 1H), 7.67 (d, J = 8.3 Hz, 1H), 7.60 (s, 1H), 6.82 (d, J = 2.1 Hz, 1H), 6.73-6.66 (m, 1H), 6.63-6.53 (m, 3H), 5.06 (dd, J = 12.9, 5.4 Hz, 1H), 4.44-4.36 (m, 3H), 4.20 (s, 2H), 4.16 (s, 2H), 3.73 (s, 6H), 3.48 (s, 3H), 3.42-3.35 (m, 2H), 2.95-2.81 (m, 1H), 2.63-2.57 (m, 4H), 2.57-2.51 (m, 3H), 2.07-1.96 (m, 1H).
D304	752.45	¹ H NMR (300 MHz, DMSO-d ₆) δ 10.97 (s, 1H), 7.65 (s, 1H), 7.38 (d, J = 8.2 Hz, 1H), 6.69 (d, J = 7.5 Hz, 2H), 6.61 (d, J = 5.1 Hz, 3H), 5.08 (dd, J = 13.2, 5.1 Hz, 1H), 4.39-4.11 (m, 4H), 3.81 (s, 6H), 3.79-3.53 (m, 4H), 3.54-3.44 (m, 5H), 3.47-3.36 (m, 3H), 3.00-2.80 (m, 1H), 2.72-2.51 (m, 6H), 2.50-2.10 (m, 4H), 2.05-1.61 (m, 4H), 0.95 (d, J = 6.7 Hz, 3H).
D305	823.45	¹ H NMR (400 MHz, DMSO-d ₆) δ 8.14 (s, 1H, FA), 7.62 (s, 1H), 7.42 (d, J = 8.4 Hz, 1H), 7.26 (dd, J = 8.4, 2.4 Hz, 1H), 7.16 (d, J = 2.3 Hz, 1H), 6.67-6.58 (m, 2H), 6.54 (s, 1H), 5.16 (dd, J = 13.5, 5.1 Hz, 1H), 4.37-4.16 (m, 4H), 3.84-3.70 (m, 10H), 3.50 (s, 3H), 3.40 (t, J = 5.7 Hz, 2H), 3.31 (s, 3H), 3.07-2.92 (m, 5H), 2.82-2.66 (m, 8H), 2.63-2.58 (m, 4H), 2.38 (dd, 2H), 2.05-1.93 (m, 1H), 1.75 (d, J = 12.5 Hz, 2H), 1.48 (s, 3H), 1.35-1.12 (m, 3H).
D306	774.4	¹ H NMR (300 MHz, DMSO-d ₆) δ 10.98 (s, 1H), 7.65 (s, 1H), 7.42 (d, J = 8.6 Hz, 1H), 6.75-6.65 (m, 4H), 6.61 (s, 1H), 5.08 (dd, J = 13.3, 5.0 Hz, 1H), 4.38-4.15 (m, 5H), 3.92 (s, 3H), 3.86 (s, 7H), 3.75 (s, 3H), 3.51 (s, 3H), 2.98-2.83 (m, 2H), 2.60 (s, 7H), 2.43-2.34 (m, 3H), 2.25-2.07 (m, 2H), 2.02-1.92 (m, 1H).
D307	693.2	¹ H NMR (300 MHz, DMSO-d ₆) δ 10.97 (s, 1H), 7.65 (s, 1H), 7.36 (d, J = 9.2 Hz, 1H), 7.17-7.07 (m, 3H), 6.65-6.60 (m, 2H), 6.23-5.73 (m, 1H), 5.08 (dd, J = 13.3, 5.1 Hz, 1H), 4.50 (s, 2H), 4.36-4.12 (m, 4H), 3.83 (s, 6H), 3.49 (s, 3H), 3.43 (d, J = 5.0 Hz, 4H), 2.90 (s, 4H), 2.62 (s, 2H), 2.44-2.29 (m, 2H), 1.98 (d, J = 12.6 Hz, 1H).
D308	738.45	¹ H NMR (400 MHz, DMSO-d ₆) δ 10.97 (s, 1H), 8.18 (s, 1H, FA), 7.65 (s, 1H), 7.37 (d, J = 8.1 Hz, 1H), 6.77-6.65 (m, 2H), 6.63-6.53 (m, 3H), 5.08 (m, J = 13.3, 5.1 Hz, 1H), 4.39-4.12 (m, 4H), 3.79 (s, 6H), 3.57 (s, 4H), 3.50 (d, J = 4.1 Hz, 4H), 3.41 (m, J = 4.9 Hz, 4H), 2.91 (m, J = 17.6, 13.6, 5.3 Hz, 1H), 2.59 (d, J = 4.2 Hz, 6H), 2.44-2.33 (m, 4H), 2.03-1.95 (m, 1H), 1.71 (m, J = 5.3 Hz, 4H).
D309	837.6	¹ H NMR (400 MHz, DMSO-d ₆) δ 10.98 (s, 1H), 7.63 (s, 1H), 7.42 (d, J = 8.5 Hz, 1H), 7.26 (dd, J = 8.5, 2.4 Hz, 1H), 7.15 (d, J = 2.3 Hz, 1H), 6.76-6.50 (m, 3H), 5.10 (dd, J = 13.3, 5.1 Hz, 1H), 4.39-4.14 (m, 4H), 3.98-3.66 (m, 8H), 3.50 (s, 3H), 3.45-3.38 (m, 2H), 3.33-3.26 (m, 4H), 3.00-2.82 (m, 2H), 2.81-2.64 (m, 3H), 2.64-2.56 (m, 5H), 2.56-2.52 (m, 2H), 2.49-2.27 (m, 3H), 2.20 (s, 1H), 2.06-1.90 (m, 1H), 1.83-1.67 (m, 2H), 1.62-1.35 (m, 5H), 1.35-0.94 (m, 6H).
D310	835.5	¹ H NMR (400 MHz, Methanol-d ₄) δ 7.68-7.61 (m, 1H), 7.54 (s, 1H), 7.10 (d, J = 8.0 Hz, 2H), 6.61 (s, 2H), 5.12 (dd, J = 13.3, 5.1 Hz, 1H), 4.63 (s, 1H), 4.49-4.38 (m, 2H), 4.36 (s, 2H), 4.22 (s, 1H), 3.97 (d, J = 12.8 Hz, 2H), 3.87 (s, 7H), 3.63 (s, 4H), 3.53 (t, J = 5.6 Hz, 3H), 3.23 (s, 2H), 3.09 (s, 2H), 2.99-2.86 (m, 3H), 2.85-2.71 (m, 5H), 2.62 (d, J = 6.1 Hz, 2H), 2.54-2.41 (m, 1H), 2.22-2.13 (m, 1H), 1.89 (d, J = 12.5 Hz, 2H), 1.79-1.71 (m, 2H), 1.66 (s, 1H), 1.44 (q, J = 11.2 Hz, 2H), 1.09-0.87 (m, 2H), 0.80 (s, 2H).
D311	885.45	¹ H NMR (400 MHz, DMSO-d ₆) δ 10.96 (s, 1H), 8.20 (s, 1H, FA), 7.61 (s, 1H), 7.50 (d, J = 8.5 Hz, 1H), 7.13 (t, J = 5.8 Hz, 1H), 7.08-6.99 (m, 2H),

Compound No.	LCMS	¹ H NMR
D312	885.45	¹H NMR (400 MHz, DMSO-d₆) δ 10.96 (s, 1H), 8.21 (s, 2H, TFA), 7.62 (s, 1H), 7.42 (d, J = 8.4 Hz, 1H), 7.26 (dd, J = 8.6, 2.4 Hz, 1H), 7.18-7.09 (m, 2H), 6.55 (s, 2H), 6.17-5.79 (m, 1H), 5.10 (dd, J = 13.3, 5.1 Hz, 1H), 4.45-4.13 (m, 5H), 3.86-3.66 (m, 10H), 3.55-3.48 (m, 5H), 3.48-3.41 (m, 4H), 3.40-3.36 (m, 1H), 2.98-2.85 (m, 1H), 2.76-2.66 (m, 6H), 2.65-2.55 (m, 3H), 2.45-2.34 (m, 1H), 2.03-1.96 (m, 1H), 1.77 (d, J = 12.3 Hz, 2H), 1.51-1.47 (m, 3H), 1.28 (d, J = 11.7 Hz, 2H), 0.69-0.65 (m, 2H), 0.51 (s, 2H).
D313	887.65	¹H NMR (400 MHz, Methanol-d₄) δ 7.81 (s, 1H), 7.71 (s, 2H), 7.58 (s, 1H), 6.73 (s, 2H), 5.90 (tt, J = 56.7, 4.2 Hz, 1H), 5.18 (dd, J = 13.3, 5.2 Hz, 1H), 4.77-4.45 (m, 3H), 4.41 (s, 2H), 4.30-4.01 (m, 1H), 3.95 (s, 6H), 3.80 (d, J = 12.1 Hz, 2H), 3.65 (s, 3H), 3.62-3.49 (m, 4H), 3.42 (s, 2H), 3.16 (br s, 1H), 3.00-2.76 (m, 3H), 2.75-2.61 (m, 5H), 2.61-2.42 (m, 2H), 2.26-2.16 (m, 1H), 2.08 (d, J = 13.6 Hz, 2H), 1.82 (s, 1H), 1.67 (s, 10H).
D314	835.5	¹H NMR (400 MHz, Methanol-d₄) δ 7.54 (s, 1H), 7.46 (d, J = 8.4 Hz, 1H), 7.39-7.30 (m, 2H), 6.61 (s, 2H), 5.15 (dd, J = 13.3, 5.2 Hz, 1H), 4.63 (s, 1H), 4.50-4.35 (m, 2H), 4.36 (s, 2H), 4.23 (s, 1H), 4.02-3.85 (m, 7H), 3.81 (d, J = 12.3 Hz, 2H), 3.66-3.61 (m, 4H), 3.56-3.51 (m, 3H), 3.34-3.20 (m, 2H), 3.09 (s, 2H), 2.96-2.77 (m, 4H), 2.80-2.76 (m, 4H), 2.61 (s, 2H), 2.58-2.45 (m, 1H), 2.24-2.15 (m, 1H), 1.92 (d, J = 12.4 Hz, 2H), 1.77 (s, 2H), 1.61 (s, 1H), 1.50 (q, J = 10.8 Hz, 2H), 1.10 (s, 1H), 0.94-0.73 (m, 3H).
D315	871.6	¹H NMR (300 MHz, DMSO-d₆) δ 10.98 (s, 1H), 8.18 (s, 1H, FA), 7.64 (s, 1H), 7.41 (d, J = 8.8 Hz, 1H), 6.76-6.68 (m, 2H), 6.63-6.58 (m, 3H), 5.10 (dd, 1H), 4.39-4.13 (m, 4H), 3.89 (d, J = 7.9 Hz, 2H), 3.79 (s, 6H), 3.69 (d, J = 8.0 Hz, 2H), 3.60 (s, 2H), 3.50 (s, 3H), 3.41 (s, 2H), 3.40 (s, 2H), 2.98-2.82 (m, 3H), 2.59 (d, J = 4.2 Hz, 5H), 2.44-2.34 (m, 4H), 2.18 (d, J = 7.2 Hz, 4H), 2.00 (s, 3H), 1.63 (d, J = 12.6 Hz, 2H), 1.50 (s, 1H), 1.17-1.02 (m, 2H).
D316	862.5	¹H NMR (400 MHz, DMSO-d₆) δ 10.98 (s, 1H), 7.63 (s, 1H), 7.42 (d, J = 8.4 Hz, 1H), 7.26 (dd, J = 8.5, 2.4 Hz, 1H), 7.15 (dd, J = 9.2, 4.1 Hz, 2H), 6.64 (s, 2H), 5.99 (t, J = 4.2 Hz, 1H), 5.09 (dd, J = 13.3, 5.1 Hz, 1H), 4.43-4.10 (m, 5H), 3.78 (d, J = 27.4 Hz, 9H), 3.61 (s, 1H), 3.11 (s, 6H), 2.92 (s, 2H), 2.98-2.85 (m, 4H), 2.72 (d, J = 11.9 Hz, 2H), 2.62 (s, 3H), 2.55 (s, 3H), 2.46-2.30 (m, 3H), 2.06-1.89 (m, 1H), 1.82-1.68 (m, 2H), 1.53 (d, J = 40.7 Hz, 3H), 1.35-1.20 (m, 2H).
D317	862.35	¹H NMR (400 MHz, DMSO-d₆) δ 10.95 (s, 1H), 8.16 (s, 1H, FA), 7.65 (s, 1H), 7.49 (d, J = 8.6 Hz, 1H), 7.13 (t, J = 5.7 Hz, 1H), 7.04 (s, 2H), 6.59 (s, 2H), 6.15-5.81 (m, 1H), 5.09-4.99 (m, 1H), 4.37-4.12 (m, 4H), 3.85 (d, J = 12.6 Hz, 2H), 3.78 (s, 6H), 3.53 (s, 2H), 3.49-3.39 (m, 6H), 2.95-2.74 (m, 1H), 2.71-2.63 (m, 2H), 2.63-2.57 (m, 1H), 2.57-2.54 (m, 3H), 2.47-2.40 (m, 3H), 2.40-2.37 (m, 1H), 2.37-2.28 (m, 4H), 2.03-1.88 (m, 1H), 1.73 (d, J = 13.8 Hz, 2H), 1.50 (s, 1H), 1.42-1.31 (m, 2H), 1.27-1.12 (m, 2H).
D318	840.55	¹H NMR (300 MHz, DMSO-d₆) δ 10.96 (s, 1H), 8.22 (s, 1H, FA), 7.63 (s, 1H), 7.50 (d, J = 8.8 Hz, 1H), 7.12-6.99 (m, 2H), 6.65-6.58 (m, 1H), 6.56 (s, 2H), 5.05 (dd, J = 13.2, 5.1 Hz, 1H), 4.38-4.12 (m, 4H), 3.92-3.81 (m, 4H), 3.78 (s, 6H), 3.49-3.34 (m, 4H), 3.00-2.70 (m, 5H), 2.65-2.54 (m, 4H), 2.41-2.15 (m, 5H), 2.14-1.91 (m, 3H), 1.80-1.67 (m, 2H), 1.60-1.45 (m, 1H), 1.43-1.30 (m, 2H), 1.28-1.17 (m, 2H), 1.16-1.06 (m, 6H).
D319	840.55	¹H NMR (300 MHz, Methanol-d₄) δ 7.80-7.53 (m, 4H), 6.75 (s, 2H), 5.18 (dd, J = 13.3, 5.1 Hz, 1H), 4.62-4.42 (m, 4H), 4.36 (s, 2H), 3.98 (s, 6H), 3.93-3.72 (m, 5H), 3.61-3.50 (m, 3H), 3.29-3.21 (m, 3H), 3.02-2.81 (m, 4H), 2.78 (s, 4H), 2.69-2.60 (m, 2H), 2.57-2.44 (m, 1H), 2.30-2.15 (m, 1H), 2.12-1.97 (m, 2H), 1.91-2.69 (m, 4H), 1.66-1.51 (m, 7H).
D320	887.45	¹H NMR (400 MHz, Methanol-d₄) δ 7.68 (d, J = 9.2 Hz, 1H), 7.58 (s, 1H), 7.20-7.17 (m, 2H), 6.73 (s, 2H), 5.90 (tt, J = 56.6, 4.2 Hz, 1H), 5.13 (dd, J = 13.3, 5.1 Hz, 1H), 4.62 (s, 1H), 4.53-4.33 (m, 4H), 4.13 (s, 1H), 3.95 (s, 7H), 3.92 (s, 1H), 3.64 (s, 3H), 3.60-3.54 (m, 3H), 3.53-3.49 (m, 1H), 3.47 (s, 2H), 3.15 (s, 1H), 3.05-2.93 (m, 3H), 2.96-2.86 (m, 1H), 2.85-2.75 (m, 4H), 2.64 (t, J = 5.2 Hz, 3H), 2.48 (qd, J = 13.2, 4.7 Hz, 1H), 2.23-2.12 (m, 1H), 1.90 (d, J = 12.6 Hz, 2H), 1.68 (s, 7H), 1.66-1.59 (m, 2H), 1.51-1.37 (m, 2H).
D321	849.55	¹H NMR (400 MHz, DMSO-d₆) δ 10.96 (s, 1H), 7.62 (s, 1H), 7.54 (d, J = 8.3 Hz, 1H), 7.11 (s, 2H), 6.72 (s, 2H), 5.04 (dd, J = 13.2, 5.1 Hz, 1H), 4.58 (s, 1H), 4.38-4.14 (m, 4H), 3.97 (s, 3H), 3.89 (s, 7H), 3.62-3.57 (m, 4H), 3.51 (s, 3H), 3.41 (t, J = 5.6 Hz, 2H), 3.34-3.15 (m, 4H), 2.97-2.78 (m, 3H), 2.67-2.55 (m, 6H), 2.56-2.50 (m, 2H), 2.45-2.29 (m, 1H), 2.22 (s, 1H), 2.03-1.81 (m, 3H), 1.82-1.68 (m, 4H), 1.64-1.55 (m, 1H), 1.31 (q, J = 10.7 Hz, 2H).

Compound No.	LCMS	¹ H NMR
D322	899.5	¹ H NMR (300 MHz, Methanol-d ₄) δ 7.69 (d, J = 9.1 Hz, 1H), 7.59 (s, 1H), 7.20 (d, J = 7.1 Hz, 2H), 6.74 (s, 2H), 6.16-5.65 (m, 1H), 5.12 (dd, J = 13.2, 5.1 Hz, 1H), 4.58-4.26 (m, 6H), 4.06-3.86 (m, 9H), 3.76-3.65 (m, 4H), 3.67-3.46 (m, 7H), 3.40 (s, 2H), 3.24 (s, 2H), 3.03 (t, J = 12.2 Hz, 2H), 2.94-2.78 (m, 2H), 2.78-2.69 (m, 2H), 2.67-2.61 (m, 2H), 2.57-2.34 (m, 2H), 2.22-2.02 (m, 3H), 1.94 (d, J = 12.5 Hz, 2H), 1.74 (d, J = 33.3 Hz, 3H), 1.61-1.38 (m, 2H).
D323	840.5	¹ H NMR (300 MHz, Methanol-d ₄) δ 8.49 (s, 2H, FA), 7.58 (s, 1H), 7.45 (d, J = 8.2 Hz, 1H), 7.37-7.30 (m, 2H), 6.73 (s, 2H), 5.15 (dd, J = 13.3, 5.1 Hz, 1H), 4.54-4.46 (m, 1H), 4.43-4.29 (m, 5H), 3.96 (s, 6H), 3.84-3.68 (m, 4H), 3.64 (s, 3H), 3.55 (t, J = 5.3 Hz, 2H), 3.37 (s, 1H), 2.96-2.72 (m, 6H), 2.64 (s, 3H), 2.58-2.44 (m, 3H), 2.24-2.12 (m, 1H), 1.95-1.83 (m, 2H), 1.62-1.42 (m, 9H), 1.42-1.22 (m, 2H).
D324	840.5	¹ H NMR (400 MHz, Methanol-d ₄) δ 7.68 (d, J = 9.3 Hz, 1H), 7.58 (s, 1H), 7.20-7.17 (m, 2H), 6.74 (s, 2H), 5.13 (dd, J = 13.3, 5.1 Hz, 1H), 4.59 (d, J = 13.5 Hz, 1H), 4.50-4.41 (m, 3H), 4.36 (s, 2H), 4.04-3.88 (m, 10H), 3.64 (s, 4H), 3.55 (t, J = 5.6 Hz, 2H), 3.37 (s, 1H), 3.24-3.13 (m, 1H), 3.08-2.84 (m, 6H), 2.83-2.75 (m, 1H), 2.68-2.59 (m, 2H), 2.52-2.41 (m, 1H), 2.23-2.12 (m, 1H), 1.91 (d, J = 12.9 Hz, 2H), 1.77-1.67 (m, 3H), 1.61 (d, J = 6.6 Hz, 6H), 1.52-1.40 (m, 2H).
D325	837.4	¹ H NMR (400 MHz, Methanol-d ₄) δ 8.57 (s, 1H, FA), 7.63 (d, J = 8.6 Hz, 1H), 7.57 (s, 1H), 7.11-7.05 (m, 2H), 6.72 (s, 2H), 5.11 (dd, J = 13.3, 5.1 Hz, 1H), 4.94-4.89 (m, 1H), 4.46 (s, 1H), 4.46-4.37 (m, 2H), 4.36 (s, 3H), 3.97-3.91 (m, 8H), 3.73-3.66 (m, 1H), 3.64 (s, 3H), 3.55 (t, J = 5.6 Hz, 2H), 3.00-2.89 (m, 1H), 2.89-2.80 (m, 3H), 2.78 (s, 5H), 2.63 (t, J = 5.6 Hz, 3H), 2.60-2.39 (m, 4H), 2.21-2.10 (m, 1H), 1.92-1.83 (m, 2H), 1.65 (s, 1H), 1.58-1.48 (m, 8H), 1.44-1.29 (m, 3H).
D326	837.45	¹ H NMR (400 MHz, Methanol-d ₄) δ 8.57 (s, 1H, FA), 7.58 (s, 1H), 7.45 (d, J = 8.4 Hz, 1H), 7.39-7.26 (m, 2H), 6.73 (s, 2H), 5.15 (dd, J = 13.3, 5.1 Hz, 1H), 4.49-4.34 (m, 3H), 4.36 (s, 3H), 3.96 (s, 6H), 3.77 (d, J = 12.3 Hz, 2H), 3.74-3.66 (m, 1H), 3.64 (s, 3H), 3.55 (t, J = 5.6 Hz, 2H), 3.01-2.86 (m, 2H), 2.86-2.76 (m, 6H), 2.75 (s, 1H), 2.64 (t, J = 5.7 Hz, 3H), 2.60-2.42 (m, 4H), 2.26-2.13 (m, 1H), 1.89 (s, 2H), 1.52 (d, J = 6.6 Hz, 9H), 1.48-1.35 (m, 3H).
D327	807.55	¹ H NMR (300 MHz, DMSO-d ₆) δ 10.99 (s, 1H), 8.19 (s, 1H), 7.61 (s, 1H), 7.41 (d, J = 8.4 Hz, 1H), 7.25 (d, J = 8.1 Hz, 1H), 7.15 (s, 1H), 6.76 (s, 2H), 6.60 (d, J = 4.2 Hz, 1H), 5.10 (dd, J = 13.2, 4.8 Hz, 1H), 4.33 (d, J = 17.1 Hz, 1H), 4.25-4.14 (m, 3H), 3.90-3.70 (m, 7H), 3.49 (s, 3H), 3.39 (s, 4H), 2.99-2.84 (m, 2H), 2.78-2.65 (m, 4H), 2.65-2.55 (m, 4H), 2.44-2.30 (m, 10H), 2.04-1.93 (m, 1H), 1.75 (d, J = 11.7 Hz, 2H), 1.55-1.35 (m, 3H), 1.35-1.12 (m, 5H).
D328	876.3	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.09 (s, 1H), 7.65 (t, J = 4.3 Hz, 2H), 7.32-7.09 (m, 3H), 6.58 (s, 2H), 5.99 (tt, J = 56.7, 4.3 Hz, 1H), 5.07 (dd, J = 12.8, 5.4 Hz, 1H), 4.26 (s, 2H), 4.03 (d, J = 12.7 Hz, 2H), 3.77 (s, 6H), 3.51 (s, 2H), 3.46 (m, 2H), 3.44 (m, 2H), 3.43 (m, 2H), 2.91 (q, J = 14.2, 13.3 Hz, 4H), 2.61 (d, J = 3.6 Hz, 1H), 2.35 (d, J = 30.3 Hz, 10H), 2.01 (d, J = 11.0 Hz, 1H), 1.74 (d, J = 12.6 Hz, 2H), 1.57 (s, 1H), 1.35 (d, J = 7.3 Hz, 2H), 1.19 (dd, J = 20.1, 9.1 Hz, 2H).
D329	853.4	¹ H NMR (400 MHz, DMSO-d ₆) δ 10.96 (s, 1H), 7.63 (s, 1H), 7.52 (d, J = 8.5 Hz, 1H), 7.10-7.04 (m, 2H), 6.70 (s, 2H), 6.62 (s, 1H), 5.05 (dd, J = 13.3, 5.1 Hz, 1H), 4.32 (d, J = 16.9 Hz, 1H), 4.25-4.15 (m, 4H), 3.85-3.83 (m, 2H), 3.81-3.73 (m, 8H), 3.72-3.58 (m, 4H), 3.51 (s, 4H), 3.42 (s, 6H), 3.27-2.99 (m, 5H), 2.98-2.84 (m, 4H), 2.64-2.56 (m, 4H), 2.45-2.28 (m, 1H), 2.02-1.89 (m, 1H), 1.75 (d, J = 12.4 Hz, 2H), 1.65-1.48 (m, 3H), 1.33-1.17 (m, 2H).
D330	845.4	¹ H NMR (300 MHz, DMSO-d ₆) δ 10.96 (s, 1H), 8.24 (s, 2H, FA), 7.64 (s, 1H), 7.52 (d, J = 8.4 Hz, 1H), 7.17-7.06 (m, 2H), 6.59 (s, 3H), 5.05 (dd, J = 13.2, 5.0 Hz, 1H), 4.37-4.29 (m, 1H), 4.25-4.19 (m, 3H), 4.11-4.07 (m, 1H), 3.90-3.86 (m, 2H), 3.78 (s, 6H), 3.54-3.47 (m, 5H), 3.40 (t, J = 5.6 Hz, 3H), 3.24 (dd, J = 30.5, 13.2 Hz, 2H), 3.05-2.84 (m, 2H), 2.64-2.57 (m, 4H), 2.47-2.26 (m, 10H), 2.15-2.06 (m, 1H), 2.01-1.81 (m, 3H), 1.53-1.40 (m, 1H), 1.37-1.18 (m, 1H).
D331	655.4	¹ H NMR (400 MHz, DMSO-d ₆) δ 10.94 (s, 1H), 7.60 (s, 1H), 7.51 (d, J = 8.3 Hz, 1H), 6.64-6.49 (m, 5H), 5.04 (dd, J = 13.3, 5.1 Hz, 1H), 4.36-4.28 (m, 4H), 4.24-4.15 (m, 3H), 4.03-3.97 (m, 2H), 3.73 (s, 6H), 3.48 (s, 3H), 3.38 (t, J = 5.5 Hz, 2H), 2.97-2.83 (m, 1H), 2.63-2.53 (m, 6H), 2.42-2.27 (m, 1H), 1.99-1.92 (m, 1H).
D332	849.5	¹ H NMR (400 MHz, DMSO-d ₆) δ 10.99 (s, 1H), 7.62 (s, 1H), 7.49 (d, J = 8.4 Hz, 1H), 7.38 (dd, J = 8.5, 2.3 Hz, 1H), 7.32 (d, J = 2.3 Hz, 1H), 6.73 (s, 2H), 5.09 (dd, J = 13.3, 5.1 Hz, 1H), 4.65-4.43 (m, 1H), 4.44-4.15 (m, 4H), 3.89 (s, 6H), 3.81-3.76 (m, 4H), 3.57-3.31 (m, 4H), 3.41-3.30 (m, 3H), 3.24-3.06 (m, 5H), 2.94-2.84 (m, 4H), 2.75-2.63 (m, 5H), 2.61-2.57 (m, 3H), 2.56-2.52 (m, 1H), 2.14-2.05 (m, 1H), 2.04-1.96 (m, 1H), 1.96-1.79 (m, 4H), 1.74-1.69 (m, 2H), 1.60-1.56 (m, 1H), 1.46-1.36 (m, 2H).

Compound No.	LCMS	¹ H NMR
D333	899.5	¹ H NMR (300 MHz, Methanol-d ₄) δ 7.72-7.56 (m, 4H), 6.74 (s, 2H), 6.16-5.60 (m, 1H), 5.24-5.08 (m, 1H), 4.62-4.22 (m, 6H), 3.97 (s, 6H), 3.80 (d, J = 12.0 Hz, 2H), 3.70-3.46 (m, 8H), 3.44-3.33 (m, 4H), 3.29-3.16 (m, 4H), 3.06 (s, 2H), 2.96-2.68 (m, 2H), 2.68-2.59 (m, 4H), 2.57-2.46 (m, 1H), 2.41 (s, 1H), 2.12 (dt, J = 45.4, 6.8 Hz, 5H), 1.83-1.47 (m, 5H).
D334	875.4	¹ H NMR (300 MHz, Methanol-d ₄) δ 7.59 (s, 1H), 7.39 (d, J = 8.2 Hz, 1H), 7.04-6.67 (m, 4H), 5.14 (dd, J = 13.1, 5.1 Hz, 1H), 4.66-4.48 (m, 2H), 4.45-4.31 (m, 4H), 4.31-4.12 (m, 4H), 3.97 (s, 6H), 3.93-3.89 (m, 2H), 3.87-3.57 (m, 9H), 3.55 (t, J = 5.6 Hz, 3H), 2.98-2.80 (m, 3H), 2.78 (s, 3H), 2.76-2.66 (m, 2H), 2.66-2.60 (m, 2H), 2.57-2.43 (m, 1H), 2.26-2.13 (m, 1H), 2.11-1.98 (m, 1H), 1.93-1.70 (m, 3H).
D335	868.55	¹ H NMR (300 MHz, DMSO-d ₆) δ 10.98 (s, 1H), 8.14 (s, 0.4H, FA), 7.65 (s, 1H), 7.39 (d, J = 8.3 Hz, 1H), 7.33-7.25 (m, 1H), 7.20 (d, J = 8.0 Hz, 1H), 6.73-6.64 (m, 2H), 6.57 (s, 2H), 5.39 (d, J = 8.0 Hz, 1H), 5.08 (dd, J = 13.3, 5.1 Hz, 1H), 4.61 (s, 2H), 4.36-4.16 (m, 2H), 3.80 (s, 6H), 3.72-3.57 (m, 6H), 3.48 (s, 3H), 3.14-2.81 (m, 5H), 2.70-2.56 (m, 5H), 2.49-2.30 (m, 4H), 2.27-2.05 (m, 3H), 2.04-1.58 (m, 6H), 1.51-1.32 (m, 1H).
D336	885.4	¹ H NMR (400 MHz, MeOD) δ 8.47 (s, 2FA, 2H), 7.59 (s, 1H), 7.40 (d, J = 8.2 Hz, 1H), 6.86 (d, J = 2.2 Hz, 1H), 6.78 (dd, J = 8.2, 2.2 Hz, 1H), 6.73 (s, 2H), 6.07-5.71 (m, 1H), 5.14 (dd, J = 13.3, 5.1 Hz, 1H), 4.47-4.32 (m, 6H), 3.95 (s, 6H), 3.67 (d, J = 18.9 Hz, 7H), 3.61-3.48 (m, 6H), 3.15-3.10 (m, 2H), 2.98-2.57 (m, 8H), 2.56-2.42 (m, 3H), 2.21-2.14 (m, 1H), 2.09-2.0 (m, 3H), 1.99-1.93 (m, 5H), 1.58-1.53 (m, 2H).
D337	837.5	¹ H NMR (300 MHz, DMSO-d ₆) δ 7.65 (d, J = 8.5 Hz, 1H), 7.29-7.16 (m, 2H), 6.50 (s, 2H), 4.99 (dd, J = 12.7, 5.5 Hz, 1H), 4.19 (d, J = 16.7 Hz, 4H), 3.96 (s, 3H), 3.77 (s, 6H), 3.50 (s, 4H), 3.34 (s, 8H), 3.10 (s, 2H), 2.94 (t, J = 12.3 Hz, 2H), 2.84-2.72 (m, 1H), 2.63 (d, J = 3.4 Hz, 1H), 2.55 (s, 3H), 2.4-2.5 (m, 1H), 2.18 (s, 2H), 2.10 (s, 3H), 2.06-1.96 (m, 1H), 1.73 (d, J = 12.4 Hz, 2H), 1.57 (s, 3H), 1.17 (d, J = 12.4 Hz, 2H).
D338	900.45	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.13 (s, 1H), 8.03 (t, 1H), 7.86-7.71 (m, 1H), 7.57 (s, 1H), 7.35-7.21 (m, 2H), 6.73-6.59 (m, 3H), 5.12 (dd, J = 12.8, 5.4 Hz, 1H), 4.98 (t, J = 6.8 Hz, 1H), 4.25 (s, 2H), 3.79 (s, 6H), 3.59 (s, 3H), 3.42 (t, J = 5.5 Hz, 3H), 2.99-2.81 (m, 3H), 2.66-2.57 (m, 5H), 2.46-2.37 (m, 2H), 2.37-1.98 (m, 9H), 1.80 (dd, J = 12.3, 6.3 Hz, 2H), 1.70-1.59 (m, 4H), 1.56 (t, J = 5.1 Hz, 2H), 1.48 (s, 1H), 1.17-0.98 (m, 2H).
D339	871.5	¹ H NMR (400 MHz, Methanol-d ₄) δ 8.53 (s, 1H, FA), 7.65-7.54 (m, 2H), 6.70 (s, 2H), 6.61-6.51 (m, 2H), 5.09 (dd, J = 13.2, 5.1 Hz, 1H), 4.61 (s, 2H), 4.47-4.27 (m, 6H), 4.05 (d, J = 7.9 Hz, 2H), 3.93 (s, 6H), 3.74 (d, J = 8.0 Hz, 2H), 3.63 (s, 3H), 3.53 (t, J = 5.6 Hz, 4H), 3.17-3.01 (m, 2H), 2.95-2.85 (m, 1H), 2.82-2.73 (m, 4H), 2.72-2.58 (m, 3H), 2.57-2.39 (m, 3H), 2.37-2.27 (m, 2H), 2.18-2.07 (m, 3H), 2.05-1.97 (m, 2H), 1.96-1.83 (m, 1H), 1.51-1.40 (m, 1H).
D340	853.4	¹ H NMR (400 MHz, DMSO-d ₆) δ 10.99 (s, 1H), 7.63 (s, 1H), 7.44 (d, J = 8.5 Hz, 1H), 7.33-7.26 (m, 1H), 7.20 (s, 1H), 6.70 (s, 2H), 6.62 (s, 1H), 5.10 (dd, J = 13.3, 5.1 Hz, 1H), 4.34 (d, J = 16.8 Hz, 1H), 4.25-4.17 (m, 4H), 3.81-3.77 (m, 8H), 3.76-3.72 (m, 6H), 3.51 (s, 3H), 3.40 (s, 6H), 3.24-3.00 (m, 4H), 2.98-2.85 (m, 3H), 2.76 (t, J = 11.9 Hz, 3H), 2.65-2.52 (m, 4H), 2.43-2.34 (m, 1H), 2.04-1.94 (m, 1 H), 1.78 (d, J = 12.3 Hz, 2H), 1.66-1.45 (m, 3H), 1.39-1.23 (m, 2H).
D341	812.45	¹ H NMR (300 MHz, MeOD) δ 8.52 (s, FA, 1H), 7.57 (s, 1H), 7.45 (d, J = 8.3 Hz, 1H), 7.38-7.27 (m, 2H), 6.68 (s, 2H), 5.15 (dd, J = 13.2, 5.1 Hz, 1H), 4.51-4.33 (m, 4H), 4.11 (s, 2H), 3.91 (s, 6H), 3.77 (d, J = 12.2 Hz, 2H), 3.64 (s, 3H), 3.54 (t, J = 5.7, 5.7 Hz, 2H), 3.15-3.03 (m, 4H), 3.00-2.85 (m, 4H), 2.85-2.68 (m, 6H), 2.66-2.61 (m, 2H), 2.59-2.42 (m, 1H), 2.23-2.13 (m, 1H), 1.87 (d, J = 12.3 Hz, 2H), 1.64-1.55 (m, 3H), 1.51-1.38 (m, 2H).
D342	850.35	¹ H NMR (400 MHz, Methanol-d ₄) δ 8.53 (s, 2H, FA), 8.10-7.81 (m, 3H), 7.59 (s, 1H), 6.73 (s, 2H), 5.16 (dd, J = 12.6, 5.4 Hz, 1H), 4.37 (d, J = 3.4 Hz, 4H), 4.26-4.09 (m, 1 H), 3.95 (s, 6H), 3.80-3.65 (m, 4H), 3.59-3.36 (m, 6H), 3.11 (s, 3H), 2.90 (ddd, J = 17.6, 14.3, 5.1 Hz, 1H), 2.83-2.69 (m, 5H), 2.64 (t, J = 5.5 Hz, 2H), 2.30 (t, J = 13.5 Hz, 1H), 2.21-2.09 (m, 2H), 2.04 (d, J = 12.7 Hz, 3H), 1.96 (s, 1H), 1.78 (s, 3H), 1.65 (s, 2H), 1.58-1.49 (m, 3H), 1.45 (d, J = 6.2 Hz, 3H).
D343	932.4	¹ H NMR (400 MHz, DMSO-d ₆) δ 11.12 (s, 1H), 8.19 (s, FA, 1H), 7.82 (d, J = 8.2 Hz, 1H), 7.62 (s, 1H), 7.32-7.24 (m, 2H), 6.64-6.55 (m, 3H), 5.12 (dd, J = 12.8, 5.4 Hz, 1H), 5.01-4.94 (m, 1H), 4.21 (s, 2H), 3.86-3.71 (m, 8H), 3.54-3.47 (m, 5H), 3.05-2.90 (m, 2H), 2.90-2.76 (m, 5H), 2.73-2.68 (m, 1H), 2.64-2.55 (m, 5H), 2.45-2.40 (m, 4H), 2.07-1.96 (m, 3H), 1.92-1.75 (m, 3H), 1.64-1.45 (m, 6H), 1.32-1.11 (m, 2H).
D344	859.45	¹ H NMR (400 MHz, Methanol-d ₄) δ 7.69 (d, J = 2.1 Hz, 1H), 7.66-7.57 (m, 3H), 6.74 (s, 2H), 6.13-5.65 (m, 1H), 5.22-5.13 (m, 1H), 4.62-4.37 (m, 6H), 3.95 (s, 6H), 3.88-3.74 (m, 3H), 3.65 (s, 3H), 3.60-3.55 (m, 7H), 3.53-3.49 (m, 1H), 3.44 (s, 3H), 3.27-3.20 (m, 2H), 3.19-3.09

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Compound No.	LCMS	¹ H NMR
		(m, 2H), 2.99-2.85 (m, 1H), 2.81-2.76 (m, 1H), 2.66 (t, J = 5.6 Hz, 2H), 2.59-2.44 (m, 1H), 2.25-2.15 (m, 1H), 2.03 (d, J = 13.2 Hz, 2H), 1.77-1.73 (m, 3H), 1.68-1.58 (m, 2H).
D345	918.45	¹ H NMR (400 MHz, Methanol-d ₄) δ 7.58 (s, 1H), 7.48 (d, J = 8.3 Hz, 1H), 7.20 (d, J = 2.5 Hz, 1H), 7.15 (dd, J = 8.3, 2.4 Hz, 1H), 6.74 (d, J = 4.1 Hz, 2H), 5.16 (dd, J = 13.3, 5.1 Hz, 1H), 4.79 (t, J = 6.7 Hz, 1H), 4.50-4.42 (m, 2H), 4.36 (s, 4H), 3.96 (s, 6H), 3.69-3.59 (m, 5H), 3.55 (t, J = 5.6 Hz, 2H), 3.38-3.35 (m, 1H), 3.20-3.08 (m, 2H), 3.00-2.84 (m, 4H), 2.84-2.81 (m, 1H), 2.78 (s, 3H), 2.68-2.58 (m, 3H), 2.57-2.45 (m, 4H), 2.37 (d, J = 14.8 Hz, 1H), 2.24-2.01 (m, 4H), 1.96-1.87 (m, 2H), 1.74-1.52 (m, 6H).
D346	845.4	¹ H NMR (300 MHz, DMSO-d ₆) δ 10.99 (s, 1H), 8.36 (s, 3H, FA), 7.64 (s, 1H), 7.44 (d, J = 8.3 Hz, 1H), 7.30 (d, J = 8.4 Hz, 1H), 7.20 (s, 1H), 6.58 (s, 3H), 5.10 (dd, J = 13.4, 4.9 Hz, 1H), 4.40-4.31 (m, 1H), 4.25-4.16 (m, 3H), 4.02-3.89 (m, 3H), 3.78 (s, 6H), 3.52-3.49 (m, 5H), 3.42-3.37 (m, 3H), 3.23-3.14 (m, 1H), 3.14-3.05 (m, 1H), 2.97-2.84 (m, 2H), 2.62-2.56 (m, 4H), 2.44-2.28 (m, 10H), 2.05-1.94 (m, 2H), 1.92-1.72 (m, 2H), 1.57-1.44 (m, 1H), 1.36-1.22 (m, 1H).
D347	882.4	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.14 (s, 1H), 9.21 (d, J = 73.7 Hz, TFA 2H), 7.86 (d, J = 8.3 Hz, 1H), 7.77 (s, 1H), 7.34 (d, J = 2.3 Hz, 1H), 7.30 (d, J = 8.3 Hz, 1H), 6.77 (s, 2H), 6.66 (d, J = 4.5 Hz, 1H), 6.13 (s, 1H), 5.96 (s, 1H), 5.13 (dd, J = 12.9, 5.3 Hz, 1H), 5.05-4.97 (m, 1H), 4.23 (d, J = 14.7 Hz, 4H), 3.88 (s, 6H), 3.18 (s, 2H), 3.07-2.78 (m, 7H), 2.60 (d, J = 4.1 Hz, 8H), 2.18-1.69 (m, 11H), 1.63-1.39 (m, 2H).
D348	737.4	¹ H NMR (300 MHz, DMSO-d ₆) δ 7.61-7.55 (m, 2H), 7.51 (dd, J = 8.4, 2.2 Hz, 1H), 6.80 (s, 2H), 5.17 (dd, J = 13.3, 5.2 Hz, 1H), 4.54-4.43 (m, 2H), 4.40 (s, 2H), 3.97 (s, 6H), 3.79 (d, J = 12.1 Hz, 2H), 3.67 (s, 3H), 3.48-3.42 (m, 5H), 3.32-3.24 (m, 3H), 3.15-3.01 (m, 6H), 2.92-2.84 (m, 3H), 2.84-2.74 (m, 1H), 2.60-2.42 (m, 1H), 2.24-2.07 (m, 3H), 2.03-1.94 (m, 2H), 1.75-1.67 (m, 3H), 1.60-1.50 (m, 2H).
D349	796.25	¹ H NMR (400 MHz, DMSO-d ₆) δ 10.98 (s, 1H), 10.30-9.27 (m, 2H, TFA), 7.63 (d, J = 7.4 Hz, 1H), 7.41 (d, J = 8.8 Hz, 1H), 6.80-6.67 (m, 4H), 5.07 (dd, J = 13.2, 5.1 Hz, 1H), 4.43-4.26 (m, 6H), 4.24-4.18 (m, 2H), 4.10-3.94 (m, 3H), 3.89 (d, J = 3.2 Hz, 6H), 3.76-3.70 (m, 2H), 3.69-3.59 (m, 5H), 3.51 (s, 3H), 3.27-3.08 (m, 2H), 3.05-2.84 (m, 3H), 2.70-2.56 (m, 4H), 2.45-2.36 (m, 1H), 2.17-2.06 (m, 2H), 2.03-1.88 (m, 3H).
D350	736.25	¹ H NMR (300 MHz, DMSO-d ₆) δ 7.68 (s, 1H), 7.41 (d, J = 8.8 Hz, 1H), 6.79-6.66 (m, 4H), 5.04 (dd, J = 13.2, 5.1 Hz, 1H), 4.34 (dd, J = 17.0, 5.4 Hz, 3H), 4.25-4.14 (m, 3H), 4.04-3.92 (m, 2H), 3.89 (s, 6H), 3.71 (s, 2H), 3.57-3.49 (m, 3H), 3.42 (s, 5H), 3.36 (s, 3H), 3.26-3.10 (m, 1H), 3.05-2.92 (m, 3H), 2.82 (q, J = 11.6, 9.6 Hz, 2H), 2.60 (d, J = 16.7 Hz, 1H), 2.36 (dt, J = 13.5, 6.6 Hz, 1H), 2.11 (d, J = 13.7 Hz, 2H), 1.93 (td, J = 17.8, 16.9, 9.8 Hz, 3H).
D351	695.25	¹ H NMR (400 MHz, Methanol-d ₄) δ 8.56 (s, 1H, FA), 7.66 (d, J = 8.2 Hz, 1H), 6.88 (s, 1H), 6.86 (d, J = 2.1 Hz, 1H), 6.69 (dd, J = 8.4, 2.1 Hz, 1H), 6.65 (s, 2H), 5.08 (dd, J = 12.5, 5.4 Hz, 1H), 4.10-3.98 (m, 2H), 3.98-3.88 (m, 6H), 3.89-3.80 (m, 4H), 3.62 (s, 3H), 3.40 (t, J = 5.6 Hz, 2H), 3.05-2.81 (m, 4H), 2.81-2.68 (m, 3H), 2.56 (t, J = 6.2 Hz, 2H), 2.19-1.97 (m, 5H), 1.85 (q, J = 5.9 Hz, 2H).
D352	750.3	¹ H NMR (400 MHz, DMSO-d ₆) δ 10.97 (s, 1H), 8.21 (s, 2H, FA), 7.47 (s, 1H), 7.37 (d, J = 8.0 Hz, 1H), 6.68 (d, J = 10.2 Hz, 2H), 6.63 (s, 2H), 5.08 (dd, J = 13.3, 5.1 Hz, 1H), 4.38-4.14 (m, 2H), 3.81 (s, 7H), 3.57 (s, 6H), 3.40 (d, J = 16.8 Hz, 4H), 3.29 (dd, J = 6.9, 4.7 Hz, 1H), 3.15 (s, 3H), 2.98-2.84 (m, 1H), 2.77-2.68 (m, 2H), 2.59 (d, J = 16.1 Hz, 2H), 2.42 (d, J = 6.8 Hz, 2H), 2.39 (s, 3H), 2.28 (s, 4H), 1.98 (d, J = 12.0 Hz, 1H), 1.72 (t, J = 5.3 Hz, 4H).
D353	763.6	¹ H NMR (300 MHz, MeOD) δ 7.71 (s, 1H), 7.42 (d, J = 8.2 Hz, 1H), 6.88 (d, J = 2.2 Hz, 1H), 6.80 (d, J = 3.3 Hz, 3H), 5.14 (dd, J = 13.2, 5.1 Hz, 1H), 4.53-4.31 (m, 4H), 3.98 (d, J = 3.5 Hz, 6H), 3.78 (d, J = 23.5 Hz, 4H), 3.70-3.58 (m, 7H), 3.31-3.25 (m, 1H), 3.24-2.96 (m, 7H), 2.96-2.74 (m, 4H), 2.59-2.42 (m, 1H), 2.34-2.00 (m, 10H), 1.78-1.55 (m, 2H).
D354	807.45	¹ H NMR (400 MHz, DMSO-d ₆) δ 10.98 (s, 1H), 8.87 (s, 1H), 8.23 (s, 2H, FA), 7.64-7.47 (m, 2H), 7.07-6.92 (m, 2H), 6.59 (s, 2H), 5.07 (dd, J = 13.4, 5.1 Hz, 1H), 4.86-4.78 (m, 1H), 4.38 (d, J = 17.2 Hz, 1H), 4.25 (d, J = 17.3 Hz, 1H), 3.79 (s, 6H), 3.55 (s, 3H), 3.53 (s, 2H), 2.93-2.77 (m, 5H), 2.64-2.59 (m, 1H), 2.45-2.36 (m, 5H), 2.29-2.17 (m, 4H), 2.08-1.99 (m, 5H), 1.80-1.75 (m, 2H), 1.61-1.51 (m, 6H), 1.45-1.41 (m, 1H), 1.07-0.99 (m, 2H).
D355	868.55	¹ H NMR (300 MHz, DMSO-d ₆) δ 10.98 (s, 1H), 8.14 (s, 0.4H, FA), 7.65 (s, 1H), 7.39 (d, J = 8.3 Hz, 1H), 7.33-7.25 (m, 1H), 7.20 (d, J = 8.0 Hz, 1H), 6.73-6.64 (m, 2H), 6.57 (s, 2H), 5.39 (d, J = 8.0 Hz, 1H), 5.08 (dd, J = 13.3, 5.1 Hz, 1H), 4.61 (s, 2H), 4.36-4.16 (m, 2H), 3.80 (s, 6H), 3.72-3.57 (m, 6H), 3.48 (s, 3H), 3.14-2.81 (m, 5H), 2.70-2.56 (m, 5H), 2.49-2.30 (m, 4H), 2.27-2.05 (m, 3H), 2.04-1.58 (m, 6H), 1.51-1.32 (m, 1H).

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Compound No.	LCMS	¹ H NMR
D356	666.4	¹ H NMR (300 MHz, DMSO-d ₆) δ 10.98 (s, 1H), 8.20 (s, 1H, FA), 7.76 (s, 1H), 7.37 (d, J = 8.1 Hz, 1H), 6.73-6.62 (m, 4H), 5.08 (dd, J = 13.2, 5.0 Hz, 1H), 4.38-4.12 (m, 2H), 3.82 (s, 6H), 3.56 (s, 3H), 3.53-3.47 (m, 6H), 3.02-2.82 (m, 3H), 2.72 (t, J = 7.5 Hz, 2H), 2.65-2.56 (m, 1H), 2.45-2.27 (m, 5H), 2.05-1.94 (m, 3H), 1.81-1.63 (m, 4H).
D357	668.25	¹ H NMR (300 MHz, DMSO-d ₆) δ 8.24 (s, 1H, FA), 7.98 (s, 1H), 7.37 (d, J = 8.1 Hz, 1H), 6.73-6.62 (m, 4H), 5.20 (s, 2H), 5.06 (dd, J = 13.2, 5.1 Hz, 1H), 4.93 (s, 2H), 4.38-4.12 (m, 2H), 3.82 (s, 6H), 3.60-3.51 (m, 9H), 2.98-2.80 (m, 1H), 2.65-2.53 (m, 3H), 2.42-2.26 (m, 3H), 2.03-1.93 (m, 1H), 1.79-1.69 (m, 4H).
D358	682.25	¹ H NMR (300 MHz, DMSO-d ₆) δ 8.22 (s, 1H, FA), 7.98 (s, 1H), 7.62 (d, J = 8.2 Hz, 1H), 6.76 (d, J = 2.0 Hz, 1H), 6.64 (s, 3H), 5.20 (s, 2H), 5.04 (dd, J = 12.8, 5.3 Hz, 1H), 4.93 (s, 2H), 3.81 (s, 6H), 3.72 (s, 4H), 3.56 (s, 3H), 3.53 (s, 2H), 2.90-2.78 (m, 1H), 2.64-2.52 (m, 2H), 2.47-2.35 (m, 4H), 2.05-1.95 (m, 1H), 1.76-1.70 (m, 4H).
D359	668.25	¹ H NMR (300 MHz, DMSO-d ₆) δ 8.22 (s, 1H, FA), 7.98 (s, 1H), 7.48 (d, J = 8.2 Hz, 1H), 6.65 (s, 2H), 6.54-6.43 (m, 2H), 5.20 (s, 2H), 5.01 (dd, J = 13.3, 5.0 Hz, 1H), 4.92 (s, 2H), 4.36-4.02 (m, 2H), 3.82 (s, 6H), 3.61 (s, 9H), 2.88 (t, J = 14.3 Hz, 1H), 2.68-2.50 (m, 5H), 2.40-2.29 (m, 1H), 2.00-1.91 (m, 1H), 1.79-1.73 (m, 4H).
D360	681.4	¹ H NMR (400 MHz, DMSO-d ₆) δ 10.95 (s, 1H), 8.14 (s, 0.2H, FA), 7.50 (d, J = 8.2 Hz, 1H), 6.90 (s, 1H), 6.64 (s, 2H), 6.57-6.45 (m, 2H), 5.49 (s, 1H), 5.04 (dd, J = 13.3, 5.1 Hz, 1H), 4.35-4.16 (m, 2H), 4.10-3.78 (m, 8H), 3.69 (s, 4H), 3.48 (s, 3H), 3.26 (s, 3H), 3.01-2.79 (m, 3H), 2.65-2.55 (m, 2H), 2.48-2.43 (m, 2H), 2.41-2.35 (m, 1H), 2.05-1.77 (m, 5H), 1.69 (s, 2H).
D361	681.35	¹ H NMR (400 MHz, DMSO-d ₆) δ 10.98 (s, 1H), 8.15 (s, 0.2H, FA), 7.39 (d, J = 8.1 Hz, 1H), 6.90 (s, 1H), 6.69 (d, J = 7.6 Hz, 2H), 6.66-6.52 (m, 2H), 5.47 (d, J = 2.8 Hz, 1H), 5.08 (dd, J = 13.3, 5.1 Hz, 1H), 4.35-4.16 (m, 2H), 3.91-3.74 (m, 8H), 3.62 (s, 4H), 3.48 (s, 3H), 3.26 (s, 4H), 2.96-2.87 (m, 1H), 2.85-2.70 (m, 2H), 2.64-2.55 (m, 1H), 2.50-2.45 (m, 2H), 2.43-2.33 (m, 1H), 2.03-1.94 (m, 1H), 1.84 (s, 4H), 1.70 (s, 2H).
D362	627.2	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.12 (s, 1H), 7.88-7.80 (m, 1H), 7.76 (s, 1H), 7.27 (d, J = 7.4 Hz, 2H), 6.67 (s, 2H), 5.12 (dd, J = 12.8, 5.3 Hz, 1H), 4.97 (s, 1H), 3.81 (s, 9H), 3.51 (s, 3H), 3.28-3.12 (m, 3H), 3.01-2.80 (m, 3H), 2.70 (d, J = 7.6 Hz, 2H), 2.60 (d, J = 13.7 Hz, 2H), 2.00 (q, J = 7.5 Hz, 3H).
D363	680.2	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.08 (s, 1H), 8.18 (s, 1H, FA), 7.76 (s, 1H), 7.63 (d, J = 8.3 Hz, 1H), 6.78 (d, J = 2.1 Hz, 1H), 6.69-6.60 (m, 3H), 5.05 (dd, J = 12.8, 5.3 Hz, 1H), 3.81 (s, 6H), 3.73 (s, 3H), 3.51 (s, 5H), 2.96 (t, J = 7.5 Hz, 3H), 2.90-2.80 (m, 1H), 2.78-2.55 (m, 4H), 2.41 (s, 4H), 1.99 (t, J = 7.7 Hz, 3H), 1.72 (s, 4H).
D364	626.25	¹ H NMR (300 MHz, DMSO-d ₆) δ 10.98 (s, 1H), 7.77 (s, 1H), 7.42 (d, J = 8.4 Hz, 1H), 7.29-7.20 (m, 1H), 7.13 (d, J = 2.3 Hz, 1H), 6.67 (s, 2H), 5.09 (dd, J = 13.2, 5.1 Hz, 1H), 4.39-4.13 (m, 2H), 3.83 (s, 6H), 3.60 (s, 2H), 3.51 (s, 3H), 3.16 (s, 4H), 3.01-2.83 (m, 3H), 2.70 (d, J = 7.4 Hz, 2H), 2.58 (s, 5H), 2.41-2.30 (m, 1H), 1.98 (d, J = 8.4 Hz, 3H).
D365	621.35	¹ H NMR (400 MHz, Methanol-d ₄) δ 8.53 (s, 1H, FA), 7.83 (s, 1H), 7.76-7.66 (m, 2H), 7.60 (d, J = 7.9 Hz, 1H), 6.79 (s, 2H), 5.17 (dd, J = 13.3, 5.2 Hz, 1H), 4.60-4.45 (m, 2H), 4.41-4.27 (m, 4H), 4.10 (q, J = 7.6 Hz, 2H), 3.97 (s, 6H), 3.85 (q, J = 8.2 Hz, 1H), 3.66 (s, 3H), 3.06-2.75 (m, 6H), 2.51 (qd, J = 13.2, 4.7 Hz, 1H), 2.24-2.18 (m, 1H), 2.15-2.02 (m, 2H).
D366	640.3	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.09 (s, 1H), 7.76 (s, 1H), 7.67 (d, J = 8.4 Hz, 1H), 7.32 (s, 1H), 7.23 (d, J = 8.6 Hz, 1H), 6.68 (s, 2H), 5.07 (dd, J = 13.1, 5.3 Hz, 1H), 3.83 (s, 6H), 3.67-3.57 (m, 2H), 3.51 (s, 3H), 3.40 (s, 5H), 2.96 (t, J = 7.3 Hz, 2H), 2.86 (d, J = 14.5 Hz, 1H), 2.70 (d, J = 7.5 Hz, 2H), 2.63-2.55 (m, 5H), 1.98 (q, J = 8.7, 7.2 Hz, 3H).
D367	623.15	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.02 (s, 1H), 8.14 (s, 0.4H, FA), 8.01 (s, 1H), 7.74-7.58 (m, 3H), 6.69 (s, 2H), 5.20 (t, J = 3.5 Hz, 2H), 5.11 (dd, J = 13.2, 5.1 Hz, 1H), 4.92 (t, J = 3.4 Hz, 2H), 4.57-4.30 (m, 2H), 4.02-3.89 (m, 4H), 3.86 (s, 6H), 3.67-3.53 (m, 6H), 3.01-2.83 (m, 1H), 2.67-2.55 (m, 1H), 2.49-2.30 (m, 1H), 2.08-1.96 (m, 1H).
D368	623.2	¹ H NMR (300 MHz, DMSO-d ₆ with a drop of D ₂ O) δ 8.17 (s, 0.4H, FA), 7.96 (s, 1H), 7.70 (dd, J = 7.9, 0.7 Hz, 1H), 7.64 (t, J = 1.1 Hz, 1H), 7.52 (dd, J = 7.9, 1.4 Hz, 1H), 6.64 (s, 2H), 5.17 (d, J = 3.5 Hz, 2H), 5.08 (dd, J = 13.2, 5.1 Hz, 1H), 4.92 (d, J = 3.3 Hz, 2H), 4.50-4.19 (m, 2H), 3.82 (s, 6H), 3.71-3.61 (m, 4H), 3.55 (s, 3H), 3.40 (q, J = 7.2 Hz, 1H), 3.29 (t, J = 7.0 Hz, 2H), 2.98-2.80 (m, 1H), 2.67-2.55 (m, 1H), 2.48-2.27 (m, 1H), 2.06-1.96 (m, 1H).
D369	621.3	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.02 (s, 1H), 10.1 (d, 2H, TFA), 7.77 (dd, J = 11.4, 7.3 Hz, 2H), 7.69 (s, 1H), 7.58 (d, J = 7.9 Hz, 1H), 6.78 (d, J = 3.3 Hz, 2H), 5.13 (dd, J = 13.2, 5.1 Hz, 1H), 4.53-4.08 (m, 9H), 3.90 (s, 6H), 3.51 (s, 3H), 2.93 (q, J = 9.5 Hz, 3H), 2.77-2.56 (m, 3H), 2.40-2.24 (m, 1H), 1.99 (q, J = 7.5 Hz, 3H).
D370	651.3	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.01 (s, 1H), 8.15 (1.0 FA, s, 1H), 7.99 (s, 1H), 7.73-7.54 (m, 3H), 6.67 (s, 2H), 5.32-5.05 (m, 3H), 5.00-4.86

Compound No.	LCMS	¹ H NMR
D371	649.35	(m, 2H), 4.58-4.27 (m, 2H), 3.84 (s, 6H), 3.67 (s, 2H), 3.56 (s, 3H), 3.01-2.79 (m, 3H), 2.75-2.55 (m, 2H), 2.47-2.20 (m, 3H), 2.11-1.95 (m, 1H), 1.95-1.80 (m, 2H), 1.76-1.53 (m, 2H). ¹ H NMR (300 MHz, DMSO-d ₆) δ 11.01 (s, 1H), 8.19 (1.0 FA, s, 1H), 7.75 (s, 1H), 7.68-7.55 (m, 3H), 6.66 (s, 2H), 5.11 (dd, J = 13.2, 5.1 Hz, 1H), 4.59-4.25 (m, 2H), 3.81 (s, 6H), 3.54 (s, 2H), 3.51 (s, 3H), 3.01-2.87 (m, 3H), 2.82-2.67 (m, 4H), 2.67-2.54 (m, 2H), 2.48-2.34 (m, 1H), 2.34-2.17 (m, 2H), 2.08-1.91 (m, 3H), 1.91-1.77 (m, 2H), 1.69-1.50 (m, 2H).
D372	765.6	¹ H NMR (300 MHz, DMSO-d ₆) δ 10.97 (s, 1H), 8.17 (s, FA, 1H), 7.99 (s, 1H), 7.37 (d, J = 8.2 Hz, 1H), 6.67 (s, 4H), 5.21 (s, 2H), 5.08 (dd, J = 13.2, 5.0 Hz, 1H), 4.94 (s, 2H), 4.35-4.14 (m, 2H), 3.83 (s, 6H), 3.67 (s, 2H), 3.57 (d, J = 3.8 Hz, 7H), 3.05-2.79 (m, 4H), 2.64-2.55 (m, 1H), 2.42-2.17 (m, 6H), 2.17-2.06 (m, 2H), 2.04-1.91 (m, 1H), 1.84-1.58 (m, 6H), 1.58-1.47 (m, 1H), 1.28-0.95 (m, 2H).
D373	642.2	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.09 (s, 1H), 8.14 (s, FA, 0.2H), 8.00 (s, 1H), 7.68 (d, J = 8.5 Hz, 1H), 7.38-7.17 (m, 2H), 6.68 (s, 2H), 5.21 (s, 2H), 5.07 (dd, J = 12.7, 5.4 Hz, 1H), 4.93 (s, 2H), 3.85 (s, 6H), 3.56 (s, 5H), 3.50-3.38 (m, 4H), 2.97-2.80 (m, 1H), 2.69-2.54 (m, 4H), 2.50-2.40 (m, 2H), 2.11-1.95 (m, 1H).
D374	528.1	¹ H NMR (300 MHz, DMSO-d ₆) δ 11.00 (s, 1H), 7.82 (s, 1H), 7.67-7.42 (m, 3H), 6.78 (s, 2H), 5.13 (dd, 1H), 4.53-4.29 (m, 2H), 3.73 (s, 6H), 3.53 (s, 3H), 3.08-2.99 (m, 2H), 2.69 (d, 4H), 2.28 (s, 1H), 2.13-1.96 (m, 3H).
D375	612.25	¹ H NMR (400 MHz, DMSO-d ₆) δ 10.28 (s, 1H), 8.22 (s, 2H, FA), 7.74 (s, 1H), 7.14 (t, J = 8.0 Hz, 1H), 6.64 (s, 2H), 6.58 (d, J = 7.9 Hz, 1H), 6.35 (s, 1H), 6.28 (d, J = 8.4 Hz, 1H), 3.80 (s, 6H), 3.72 (t, J = 6.7 Hz, 2H), 3.50 (d, J = 3.4 Hz, 6H), 2.96 (t, J = 7.5 Hz, 2H), 2.81-2.66 (m, 5H), 2.66-2.59 (m, 1H), 2.40 (s, 4H), 2.03-1.92 (m, 3H), 1.69 (t, J = 5.5 Hz, 4H).

Example 68—BRD9 bromodomain TR-FRET Competition Binding Assay

This example demonstrates the ability of the compounds of the disclosure to biochemically inhibit BRD9 bromodomain in a competition binding assay.

Procedure: His-Flag-BRD9 (P133-K239; Swiss Prot Q9H8M2; SEQ ID NO:1 mgsshhhhhenlyfq/gdykdddkgslevlfqg/PAENEST-PIQQLLEHFLRQLQRKDPHGFFAFPVTDALAPGYSMII KHPMDFGTMKDKIVANEYKSVTEFKADFKLMCD-NAMTYNRPDTVYYKLAKKILHAGFKMMSK) was cloned, expressed, purified, and then treated with TEV protease. Cleaved His tag was removed by purification. The binding of a biotinylated small molecule ligand of BRD9 was assessed via the LANCE® TR-FRET platform (PerkinElmer), and the compounds were assayed for inhibitory activity against this interaction.

Results: A mixture of biotinylated-ligand and SureLight™ Allophycocyanin-Streptavidin (APC-SA, PerkinElmer AD0201) in 50 mM HEPES (pH 7.4), 50 mM NaCl, 1 mM TCEP (pH 7), 0.01% (v/v) Tween-20, 0.01% (w/v) bovine serum albumin was added to a white 384-well PerkinElmer Proxiplate Plus plate. DMSO or 3-fold serially diluted compounds were then added to the Proxiplate followed by addition of Flag-BRD9. After a 10-minute incubation at room temperature, Eu—W1024 anti-FLAG (PerkinElmer, AD0273) was added. The final reaction mixture that contained 3.75 nM biotinylated ligand, 3 nM Flag-BRD9, 7.5 nM SureLight™ Allophycocyanin-Streptavidin, and 0.2 nM Eu—W1024 anti-FLAG was incubated at room temperature for 90 minutes.

The plates were then read on a PerkinElmer Envision plate reader to determine the ratio of emission at 665 nm over 615 nm. Data was normalized to a DMSO control (100%) and a no protein control (0%) and then fit to a four parameter, non-linear curve fit to calculate an IC₅₀ (μM) as

shown in Table 5. As shown by the results in Table 5, a number of compounds of the present disclosure exhibit an IC₅₀ value of <1 μM for BRD9 binding, indicating their affinity for targeting BRD9.

TABLE 5

Bromodomain TR-FRET Binding	
Compound No.	Bromodomain TR-FRET BRD9 IC ₅₀ (nM)
B1	NT
B2	+
B3	+
B4	+
B5	++++
B6	++
B7	++++
B8	+++
B9	++++
B10	+
B11	+
B12	+
B13	+++
B14	+
B15	+
B16	+
B17	+
B18	+++
B19	+++
B20	+++
B21	++
D1	NT
D2	NT
D3	NT
D4	NT
D5	NT
D6	NT
D7	NT
D8	NT
D9	NT
D10	NT
D11	NT

643

TABLE 5-continued

Bromodomain TR-FRET Binding	
Compound No.	Bromodomain TR-FRET BRD9 IC ₅₀ (nM)
D12	NT
D13	NT
D14	+++
D15	+++
D16	NT
D17	+++
D18	+++
D19	+
D20	+
D21	+++
D22	++++
D23	++
D24	++
D25	NT
D26	++++
D27	++
D28	++++
D29	++++
D30	++
D31	++++
D32	++
D33	+++
D34	+++
D35	+++
D36	+++
D37	++++
D38	++++

“+” indicates inhibitory effect of ≥ 1000 nM;“++” indicates inhibitory effect of ≥ 100 nM;“+++” indicates inhibitory effect of ≥ 10 nM;“++++” indicates inhibitory effect of < 10 nM;

“NT” indicates not tested

Example 69—SYO1 BRD9 NanoLuc Degradation Assay

This example demonstrates the ability of the compounds of the disclosure to degrade a Nanoluciferase-BRD9 fusion protein in a cell-based degradation assay.

Procedure: A stable SYO-1 cell line expressing 3xFLAG-NLuc-BRD9 was generated. On day 0 cells were seeded in 30 μ L media into each well of 384-well cell culture plates. The seeding density was 8000 cells/well. On day 1, cells were treated with 30 nL DMSO or 30 nL of 3-fold serially DMSO-diluted compounds (10 points in duplicates with 1 μ M as final top dose). Subsequently plates were incubated for 6 hours in a standard tissue culture incubator and equilibrated at room temperature for 15 minutes. Nanoluciferase activity was measured by adding 15 μ L of freshly prepared Nano-Glo Luciferase Assay Reagent (Promega N1130), shaking the plates for 10 minutes and reading the bioluminescence using an EnVision reader.

Results: The Inhibition % was calculated using the following formula: % Inhibition = $100 \times (\text{Lum}_{\text{HC}} - \text{Lum}_{\text{Sample}}) / (\text{Lum}_{\text{HC}} - \text{Lum}_{\text{LC}})$. DMSO treated cells are employed as High Control (HC) and 1 μ M of a known BRD9 degrader standard treated cells are employed as Low Control (LC). The data was fit to a four parameter, non-linear curve fit to calculate IC₅₀ (μ M) values as shown in Table 6A, Table 6B, and Table 6C. As shown by the results in Table 6A, Table 6B, and Table 6C, a number of compounds of the present disclosure exhibit an IC₅₀ value of < 1 μ M for the degradation of BRD9, indicating their use as compounds for reducing the levels and/or activity of BRD9 and their potential for treating BRD9-related disorders.

644

TABLE 6A

SYO1 BRD9-NanoLuc Degradation	
Compound No.	SYO1 BRD9-NanoLuc degradation IC ₅₀ (nM)
D1	+++
D2	+++
D3	++
D4	+++
D5	+
D6	++
D7	+++
D8	++
D9	++++
D10	++
D11	+++
D12	+++
D13	+
D14	++
D15	++
D16	+++
D17	+++
D18	+++
D19	+
D20	+
D21	+++
D22	++++
D23	+
D24	+
D25	++++
D26	NT
D27	NT
D28	++++
D29	++++
D30	+
D31	+++
D32	++
D33	+
D34	NT
D35	NT
D36	NT
D37	+++
D38	+++

“+” indicates inhibitory effect of ≥ 1000 nM;“++” indicates inhibitory effect of ≥ 100 nM;“+++” indicates inhibitory effect of ≥ 10 nM;“++++” indicates inhibitory effect of < 10 nM;

“NT” indicates not tested

TABLE 6B

SYO1 BRD9-NanoLuc Degradation	
Compound No.	SYO1 BRD9-NanoLuc degradation IC ₅₀ (nM)
B22	+
B23	+
B24	NT
D39	++++
D40	++++
D41	++++
D42	++++
D43	++++
D44	++++
D45	++++
D46	++++
D47	++++
D48	++++
D49	++++
D50	+++
D51	++++
D52	++++
D53	+++
D54	+
D55	++++
D56	++++
D57	+++

645

TABLE 6B-continued

SYO1 BRD9-NanoLuc Degradation		
Compound No.	SYO1 BRD9-NanoLuc degradation IC ₅₀ (nM)	
D58	++++	5
D59	+++	
D60	++	
D61	++	
D62	+	
D63	++++	10
D64	++++	
D65	++	
D66	++++	
D67	++	
D68	++++	15
D69	++++	
D70	+	
D71	++++	
D72	++++	
D73	++++	20
D74	+	
D75	+++	
D76	++++	
D77	+++	
D78	++++	25
D79	+	
D80	++++	
D81	++++	
D82	++++	
D83	++++	30
D84	++++	
D85	++++	
D86	++++	
D87	++++	
D88	++++	35
D89	++++	
D90	+++	
D91	++++	
D92	+++	
D93	++++	40
D94	NT	
D95	++++	
D96	++++	
D97	++++	
D98	++++	45
D99	++++	
D100	++++	
D101	++++	
D102	++++	
D103	++++	50
D104	++++	
D105	++++	
D106	++++	
D107	++++	
D109	++++	55
D110	++++	
D111	+++	
D112	++++	
D113	++++	
D114	++++	60
D115	++++	
D116	++++	
D117	++++	
D118	++++	
D119	++++	65
D120	++	
D121	++++	
D122	+++	
D123	++++	
D124	++++	
D125	++++	
D126	++	
D127	++++	
D128	++++	
D129	+++	
D130	++++	
D131	++++	
D132	++++	
D133	++++	

646

TABLE 6B-continued

SYO1 BRD9-NanoLuc Degradation		
Compound No.	SYO1 BRD9-NanoLuc degradation IC ₅₀ (nM)	
D134	++++	
D135	++++	
D136	++++	
D137	+++	
D138	++++	
D139	++++	
D140	++++	
D141	++++	
D142	++++	
D143	++++	
D144	+++	
D145	++++	
D146	++++	
D147	++++	
D148	+++	
D149	++++	
D150	+++	
D151	++++	
D152	++++	
D153	+++	
D154	++++	
D155	++++	
D156	++++	
D157	++++	
D158	++++	
D159	++++	
D161	++++	
D162	++++	
D163	++++	
D164	++++	
D165	++++	
D166	++++	
D167	+++	
D168	+++	
D169	++++	
D170	++++	
D171	++++	
D172	++++	
D173	++++	
D174	++++	
D175	++	
D176	+++	
D177	+++	
D178	++++	
D179	++++	
D180	++++	
D181	++++	
D182	++++	
D183	++++	
D184	+++	
D185	++	
D186	+++	
D187	+++	
D188	++	
D189	++	
D190	++	
D191	++	
D192	++	
D193	+	
D194	++	
D195	++++	
D196	++++	
D197	++++	
D198	++++	
D199	++++	
D200	++++	
D201	++++	
D202	+++	
D203	+++	
D204	++	
D205	+++	
D206	++	
D207	+++	
D208	+++	
D209	+++	

647

TABLE 6B-continued

SYO1 BRD9-NanoLuc Degradation		
Compound No.	SYO1 BRD9-NanoLuc degradation IC ₅₀ (nM)	
D210	+++	5
D211	++	
D212	+++	
D213	++	
D214	++	
D215	++	10
D216	+	
D217	+	
D218	+++	
D219	++	
D220	++	15
D221	++	
D222	++++	
D223	+++	
D224	++++	
D225	+++	20
D226	+++	
D227	++++	
D228	+++	
D229	+++	
D230	+	25
D231	++++	
D232	+++	
D233	++	
D234	+++	
D235	++	30
D236	+++	
D237	+	
D238	+++	
D239	+++	
D240	++	35
D241	++	
D242	++	
D243	+	
D244	++	
D245	++	40
D246	++++	
D247	++	
D248	+	
D249	+	
D250	++	45
D251	++++	
D252	++	
D253	++++	
D254	++++	
D255	++++	50
D256	++++	
D257	+	
D258	++	
D259	+++	
D260	+++	55
D261	+++	
D262	++	
D263	++	
D264	+	
D265	++	60
D266	+++	
D267	+++	
D268	+++	
D269	+++	
D270	++	65
D271	+++	
D272	++	
D273	++	
D274	++	
D275	+++	
D276	+++	
D277	++	
D278	+++	
D279	++	
D280	++	
D281	+++	
D282	++	
D283	++++	
D284	+++	

648

TABLE 6B-continued

SYO1 BRD9-NanoLuc Degradation		
Compound No.	SYO1 BRD9-NanoLuc degradation IC ₅₀ (nM)	
D285	NT	
D286	++	
D287	++	
D288	++	
D289	+++	
D290	+	
D291	+++	
D292	++	
D293	++	
D294	+	
D295	++	
D296	+	
D297	+	
D298	+	
D299	++++	
D300	++++	
D301	++++	
D302	++++	
DD1	++++	

“+” indicates inhibitory effect of ≥ 1000 nM;
“++” indicates inhibitory effect of ≥ 100 nM;
“+++” indicates inhibitory effect of ≥ 10 nM;
“++++” indicates inhibitory effect of < 10 nM;
“NT” indicates not tested

TABLE 6C

SYO1 BRD9-NanoLuc Degradation		
Compound No.	SYO1 BRD9-NanoLuc degradation IC ₅₀ (nM)	
D303	++++	35
D304	++++	
D305	+	
D306	++++	
D307	++++	
D308	++++	40
D309	++++	
D310	++++	
D311	++++	
D312	++++	
D313	+	45
D314	++++	
D315	++++	
D316	+	
D317	++++	
D318	++++	50
D319	++++	
D320	++++	
D321	++++	
D322	++++	
D323	++++	55
D324	++++	
D325	++++	
D326	++++	
D327	++++	
D328	+	60
D329	++++	
D330	++++	
D331	+++	
D332	++++	
D333	++++	65
D334	++++	
D335	++++	
D336	++++	
D337	+	
D338	++++	
D339	++++	
D340	++++	
D341	++++	
D342	++++	

649

TABLE 6C-continued

SYO1 BRD9-NanoLuc Degradation	
Compound No.	SYO1 BRD9-NanoLuc degradation IC ₅₀ (nM)
D343	+
D344	++++
D345	++++
D346	++++
D347	++++
D348	++++
D349	+++
D350	++++
D351	+++
D352	+++
D353	++++
D354	++++
D355	++++
D356	++++
D357	++++
D358	++++
D359	++++
D360	++++
D361	++++
D362	++++
D363	++++
D364	++++
D365	++++
D366	++++
D367	++++
D368	++++
D369	++++
D370	++++
D371	++++
D372	++++
D373	++++

650

TABLE 6C-continued

SYO1 BRD9-NanoLuc Degradation	
Compound No.	SYO1 BRD9-NanoLuc degradation IC ₅₀ (nM)
D374	++++
D375	+

“+” indicates inhibitory effect of ≥ 1000 nM;
 “++” indicates inhibitory effect of ≥ 100 nM;
 “+++” indicates inhibitory effect of ≥ 10 nM;
 “++++” indicates inhibitory effect of < 10 nM;
 “NT” indicates not tested

Other Embodiments

All publications, patents, and patent applications mentioned in this specification are incorporated herein by reference in their entirety to the same extent as if each individual publication, patent, or patent application was specifically and individually indicated to be incorporated by reference in its entirety. Where a term in the present application is found to be defined differently in a document incorporated herein by reference, the definition provided herein is to serve as the definition for the term.

While the invention has been described in connection with specific embodiments thereof, it will be understood that invention is capable of further modifications and this application is intended to cover any variations, uses, or adaptations of the invention following, in general, the principles of the invention and including such departures from the present disclosure that come within known or customary practice within the art to which the invention pertains and may be applied to the essential features hereinbefore set forth, and follows in the scope of the claims.

Other embodiments are in the claims.

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Gln Gly Pro Ala Glu Asn Glu Ser Thr Pro Ile Gln Gln Leu Leu Glu
35           40           45

His Phe Leu Arg Gln Leu Gln Arg Lys Asp Pro His Gly Phe Phe Ala
50           55           60

Phe Pro Val Thr Asp Ala Ile Ala Pro Gly Tyr Ser Met Ile Ile Lys
65           70           75           80

His Pro Met Asp Phe Gly Thr Met Lys Asp Lys Ile Val Ala Asn Glu
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Asn Ala Met Thr Tyr Asn Arg Pro Asp Thr Val Tyr Tyr Lys Leu Ala
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ataaaacatc ccatggattt 20

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ccggttcctc ccaggcggca 20

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<210> SEQ ID NO 313
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<210> SEQ ID NO 314
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<210> SEQ ID NO 317
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<210> SEQ ID NO 318
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<210> SEQ ID NO 321
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<210> SEQ ID NO 322
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<400> SEQUENCE: 473

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<223> OTHER INFORMATION: Synthetic Construct

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<400> SEQUENCE: 486

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<212> TYPE: DNA
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<211> LENGTH: 20

<212> TYPE: DNA

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<210> SEQ ID NO 497

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<212> TYPE: DNA

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<212> TYPE: DNA
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<210> SEQ ID NO 502
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<212> TYPE: DNA
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<400> SEQUENCE: 502

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<210> SEQ ID NO 505
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<210> SEQ ID NO 506
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gtatggtgag tagtcgcttg 20

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<210> SEQ ID NO 509
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<210> SEQ ID NO 510
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<210> SEQ ID NO 511
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<210> SEQ ID NO 512
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<210> SEQ ID NO 513
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<220> FEATURE:
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<210> SEQ ID NO 515
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<210> SEQ ID NO 516
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<210> SEQ ID NO 517
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<223> OTHER INFORMATION: Synthetic Construct

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cgaagataac gcggatacct

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<210> SEQ ID NO 658

<211> LENGTH: 20

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gcagtagcgc ttcaggccca 20

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ggaagaatgc agagtataag 20

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actggccagg aacgatgcga 20

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gcagctccaa gatcttccca

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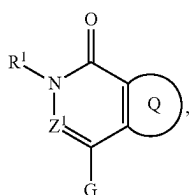
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ggagcaggac aaggtcgggg

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The invention claimed is:

1. A compound having the structure of Formula I:

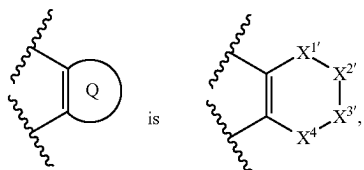


wherein

R¹ is H, optionally substituted C₁-C₆ alkyl, optionally substituted C₂-C₆ alkenyl, optionally substituted C₁-C₆ heteroalkyl, or optionally substituted C₃-C₁₀ carbocyclyl;

Z¹ is CR² or N;

R² is H, halogen, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted C₃-C₁₀ carbocyclyl, optionally substituted C₂-C₉ heterocyclyl, optionally substituted C₆-C₁₀ aryl, or optionally substituted C₂-C₉ heteroaryl;



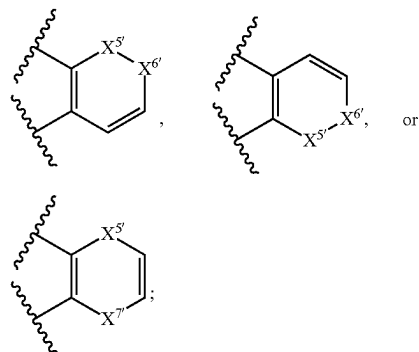
Formula I

30

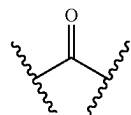
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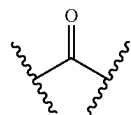
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X¹ is a bond, O, NR³ᵃ,

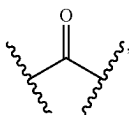


or CR⁴ᵃR⁵ᵃ,
 X² is O, NR³ᵇ,

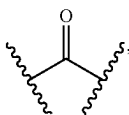


831

or CR^{4b}R^{5b};
X³ is O, NR^{3c};



or CR^{4c}R^{5c};
X⁴ is a bond, O, NR^{3d};



or CR^{4d}R^{5d};
X⁵ is O or NR^{3e} and X⁶ is CR^{4f}R^{5f}, or X⁵ is CR^{4e}R^{5e} and X⁶ is O or NR^{3f};

X⁷ is O, NR^{3g}, or CR^{4g}R^{5g};

X⁸ is O, NR^{3h}, or CR^{4h}R^{5h};

each of R^{3a}, R^{3b}, R^{3c}, and R^{3d} is, independently, H, halogen, hydroxyl, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted C₃-C₁₀ carbocyclyl, optionally substituted C₂-C₉ heterocyclyl, optionally substituted C₆-C₁₀ aryl, optionally substituted C₂-C₉ heteroaryl, optionally substituted C₂-C₆ alkenyl, optionally substituted C₂-C₆ heteroalkenyl, optionally substituted C₁-C₆ acyl, thiol, optionally substituted sulfone, optionally substituted sulfonamide, or optionally substituted amino, or R^{3a} and R^{4b}, R^{4a} and R^{3b}, R^{4b} and R^{4a}, R^{3b} and R^{4c}, R^{4b} and R^{4c}, R^{3c} and R^{4b}, R^{3c} and R^{4d}, R^{4c} and R^{4d}, and/or R^{3d} and R^{4c}, together with the atoms to which each is attached, combine to form optionally substituted C₂-C₉ heterocyclyl;

each of R^{4a}, R^{4b}, R^{4c}, and R^{4d} is, independently, H, halogen, hydroxyl, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted C₃-C₁₀ carbocyclyl, optionally substituted C₂-C₉ heterocyclyl, optionally substituted C₆-C₁₀ aryl, optionally substituted C₂-C₉ heteroaryl, optionally substituted C₂-C₆ alkenyl, optionally substituted C₂-C₆ heteroalkenyl, optionally substituted C₁-C₆ acyl, thiol, optionally substituted sulfone, or optionally substituted amino, or R^{3a} and R^{4b}, R^{4a} and R^{3b}, R^{4b} and R^{4a}, R^{3b} and R^{4c}, R^{4b} and R^{4c}, R^{3c} and R^{4b}, R^{3c} and R^{4d}, R^{4c} and R^{4d}, and/or R^{3d} and R^{4c}, together with the atoms to which each is attached, combine to form optionally substituted C₂-C₉ heterocyclyl;

each of R^{5a}, R^{5b}, R^{5c}, and R^{5d} is, independently, H, halogen, hydroxyl, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted C₃-C₁₀ carbocyclyl, optionally substituted C₂-C₉ heterocyclyl, optionally substituted C₆-C₁₀ aryl, optionally substituted C₂-C₉ heteroaryl, optionally substituted C₂-C₆ alkenyl, optionally substituted C₂-C₆ heteroalkenyl, hydroxyl, thiol, or optionally substituted amino;

each of R^{3e}, R^{3f}, R^{3g}, and R^{3h} is, independently, H, halogen, hydroxyl, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted C₃-C₁₀ carbocyclyl, optionally substituted

832

C₂-C₉ heterocyclyl, optionally substituted C₆-C₁₀ aryl, optionally substituted C₂-C₉ heteroaryl, optionally substituted C₂-C₆ alkenyl, optionally substituted C₂-C₆ heteroalkenyl, optionally substituted C₁-C₆ acyl, thiol, optionally substituted sulfone, optionally substituted sulfonamide, or optionally substituted amino, or R^{3e} and R^{4f} or R^{4e} and R^{3f}, together with the atoms to which each is attached, combine to form optionally substituted heterocyclyl;

each of R^{4e}, R^{4f}, R^{4g}, and R^{4h} is, independently, H, halogen, hydroxyl, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted C₃-C₁₀ carbocyclyl, optionally substituted C₂-C₉ heterocyclyl, optionally substituted C₆-C₁₀ aryl, optionally substituted C₂-C₉ heteroaryl, optionally substituted C₂-C₆ alkenyl, optionally substituted C₂-C₆ heteroalkenyl, optionally substituted C₁-C₆ acyl, thiol, optionally substituted sulfone, optionally substituted sulfonamide, or optionally substituted amino, or R^{3e} and R^{4f} or R^{4e} and R^{3f}, together with the atoms to which each is attached, combine to form optionally substituted heterocyclyl;

each of R^{5e}, R^{5f}, R^{5g}, and R^{5h} is, independently, H, halogen, hydroxyl, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted C₃-C₁₀ carbocyclyl, optionally substituted C₂-C₉ heterocyclyl, optionally substituted C₆-C₁₀ aryl, optionally substituted C₂-C₉ heteroaryl, optionally substituted C₂-C₆ alkenyl, optionally substituted C₂-C₆ heteroalkenyl, hydroxyl, thiol, or optionally substituted amino; and

G is optionally substituted C₆-C₁₀ aryl, optionally substituted C₃-C₁₀ carbocyclyl, optionally substituted C₂-C₉ heteroaryl, or C₂-C₉ heterocyclyl,

or a pharmaceutically acceptable salt thereof.

2. A compound having the structure of Formula II:

A-L-B

Formula II,

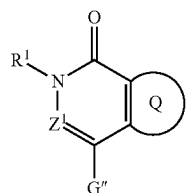
where

B is a degradation moiety,

L is a linker, and

A has the structure of Formula III:

Formula III



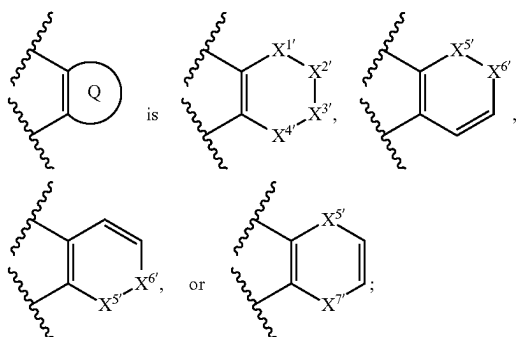
where

R¹ is H, optionally substituted C₁-C₆ alkyl, optionally substituted C₂-C₆ alkenyl, optionally substituted C₁-C₆ heteroalkyl, or optionally substituted C₃-C₁₀ carbocyclyl;

Z¹ is CR² or N;

R² is H, halogen, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted C₃-C₁₀ carbocyclyl, optionally substituted C₂-C₉ heterocyclyl, optionally substituted C₆-C₁₀ aryl, or optionally substituted C₂-C₉ heteroaryl;

833



X^{1i} is a bond, O, NR^{3ai} , or $CR^{4ai}R^{5ai}$,
 X^{2i} is O, NR^{3bi} , or $CR^{4bi}R^{5bi}$,
 X^{3i} is O, NR^{3ci} , or $CR^{4ci}R^{5ci}$,
 X^{4i} is a bond, O, NR^{3di} , or $CR^{4di}R^{5di}$,
 X^{5i} is O, NR^{3ei} , or $CR^{4ei}R^{5ei}$,
 X^{6i} is O, NR^{3fi} , or $CR^{4fi}R^{5fi}$,
 X^{7i} is O, NR^{3gi} , or $CR^{4gi}R^{5gi}$,
 each of R^{3ai} , R^{3bi} , R^{3ci} , and R^{3di} is, independently, H,



halogen, hydroxyl, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, optionally substituted C_1 - C_6 acyl, thiol, optionally substituted sulfone, optionally substituted sulfonamide, or optionally substituted amino, or R^{3ai} and R^{4bi} , R^{4ai} and R^{3bi} , R^{4bi} and R^{4ai} , R^{3bi} and R^{4ci} , R^{4bi} and R^{4ci} , R^{3ci} and R^{4bi} , R^{3ci} and R^{4di} , R^{4ci} and R^{4di} , and/or R^{3di} and R^{4ci} , together with the atoms to which each is attached, combine to form optionally substituted C_2 - C_9 heterocyclyl;

R^{3i} is absent, optionally substituted C_1 - C_6 alkylene, optionally substituted C_1 - C_6 heteroalkylene, optionally substituted C_3 - C_{10} carbocyclylene, optionally substituted C_2 - C_9 heterocyclylene, optionally substituted C_6 - C_{10} arylene, optionally substituted C_2 - C_9 heteroarylene, optionally substituted C_2 - C_6 alkenylene, optionally substituted C_2 - C_6 heteroalkenylene, optionally substituted sulfone, optionally substituted sulfonamide, or optionally substituted amino;

each of R^{4ai} , R^{4bi} , R^{4ci} , and R^{4di} is, independently, H, halogen, hydroxyl, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, thiol, optionally substituted sulfone, or optionally substituted amino, or R^{3ai} and R^{4bi} , R^{4ai} and R^{3bi} , R^{4bi} and R^{4ai} , R^{3bi} and R^{4ci} , R^{4bi} and R^{4ci} , R^{3ci} and R^{4bi} , R^{3ci} and R^{4di} , R^{4ci} and R^{4di} , and/or R^{3di} and R^{4ci} , together with the atoms to which each is attached, combine to form optionally substituted C_2 - C_9 heterocyclyl;

834

each of R^{5ai} , R^{5bi} , R^{5ci} , and R^{5di} is, independently, H, halogen, hydroxyl, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, hydroxyl, thiol, or optionally substituted amino;

each of R^{3ei} , R^{3fi} , and R^{3gi} is, independently, H,



halogen, hydroxyl, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, optionally substituted C_1 - C_6 acyl, thiol, optionally substituted sulfone, optionally substituted sulfonamide, or optionally substituted amino, or R^{3e} and R^{4f} or R^{4e} and R^{3f} , together with the atoms to which each is attached, combine to form optionally substituted heterocyclyl;

each of R^{4ei} , R^{4fi} , and R^{4gi} is, independently, H, halogen, hydroxyl, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, optionally substituted C_1 - C_6 acyl, thiol, optionally substituted sulfone, optionally substituted sulfonamide, or optionally substituted amino, or R^{3ei} and R^{4fi} or R^{4ei} and R^{3fi} , together with the atoms to which each is attached, combine to form optionally substituted heterocyclyl;

each of R^{5ei} , R^{5fi} , and R^{5gi} is, independently, H, halogen, hydroxyl, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, hydroxyl, thiol, or optionally substituted amino;

G^i is



optionally substituted C_6 - C_{10} aryl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heteroaryl, or C_2 - C_9 heterocyclyl;

G^i is optionally substituted C_3 - C_{10} carbocyclylene, C_2 - C_9 heterocyclylene, optionally substituted C_6 - C_{10} arylene, or optionally substituted C_2 - C_9 heteroarylene; and

835

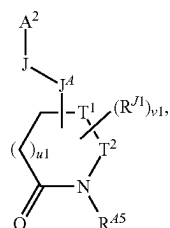
A¹ is a bond between A and the linker,
where one of R^{3at}, R^{3bt}, R^{3ct}, R^{3dt}, R^{3et}, R^{3ft}, and R^{3gt} is



or G^u is



wherein B has the structure of Formula Y:



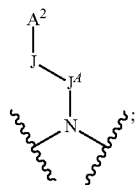
where

A² is a bond between the degradation moiety and the linker;

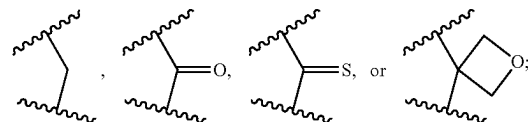
v1 is 0, 1, 2, 3, 4, or 5;

u1 is 1, 2, or 3;

T¹ is a bond or



T² is



R^{5A} is H, optionally substituted C₁-C₆ alkyl, or optionally substituted C₁-C₆ heteroalkyl;

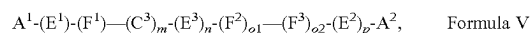
each R⁷¹ is, independently, halogen, optionally substituted C₁-C₆ alkyl, or optionally substituted C₁-C₆ heteroalkyl;

J⁴ is absent, O, optionally substituted amino, optionally substituted C₁-C₆ alkyl, or optionally substituted C₁-C₆ heteroalkyl; and

J is absent, optionally substituted C₃-C₁₀ carbocyclylene, optionally substituted C₆-C₁₀ arylene, optionally substituted C₂-C₉ heterocyclylene, or optionally substituted C₂-C₉ heteroarylene;

836

wherein L has the structure of Formula V:



wherein

A¹ is a bond between the linker and A;

A² is a bond between B and the linker;

each of m, n, o1, o2, and p is, independently, 0 or 1;

each of E¹ and E² is, independently, O, S, NR^N, optionally substituted C₁₋₁₀ alkylene, optionally substituted C₂₋₁₀ alkenylene, optionally substituted C₂₋₁₀ alkynylene, optionally substituted C₂₋₁₀ polyethylene glycol, or optionally substituted C₁₋₁₀ heteroalkylene;

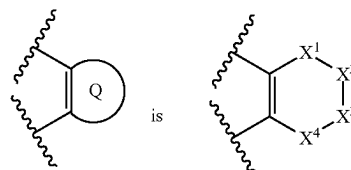
E³ is optionally substituted C₁-C₆ alkylene, optionally substituted C₁-C₆ heteroalkylene, O, S, or NR^N;

each R^N is, independently, H, optionally substituted C₁₋₄ alkyl, optionally substituted C₂₋₄ alkenyl, optionally substituted C₂₋₄ alkynyl, optionally substituted C₂₋₆ heterocyclyl, optionally substituted C₆₋₁₂ aryl, or optionally substituted C₁₋₇ heteroalkyl;

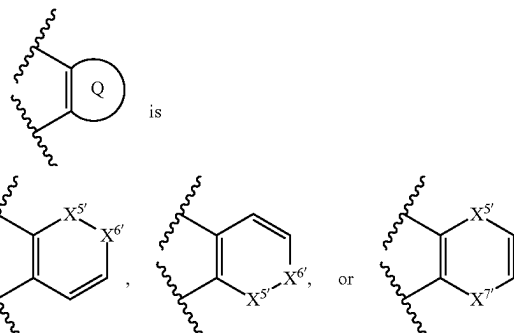
C³ is carbonyl, thiocarbonyl, sulphonyl, or phosphoryl; and

each of F¹, F², and F³ is, independently, optionally substituted C₃-C₁₀ carbocyclylene, optionally substituted C₂₋₁₀ heterocyclylene, optionally substituted C₆-C₁₀ arylene, or optionally substituted C₂-C₉ heteroarylene; or a pharmaceutically acceptable salt thereof.

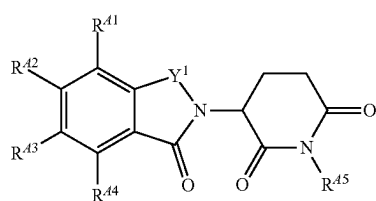
3. The compound of claim 2, wherein



4. The compound of claim 3, wherein



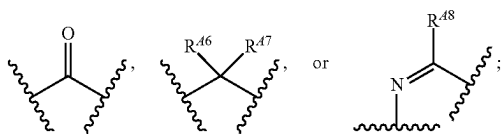
5. The compound of claim 2, wherein the structure of Formula Y has the structure of Formula A:



Formula A

837

wherein in
Y¹ is



R⁴⁵ is H, optionally substituted C₁-C₆ alkyl, or optionally substituted C₁-C₆ heteroalkyl;

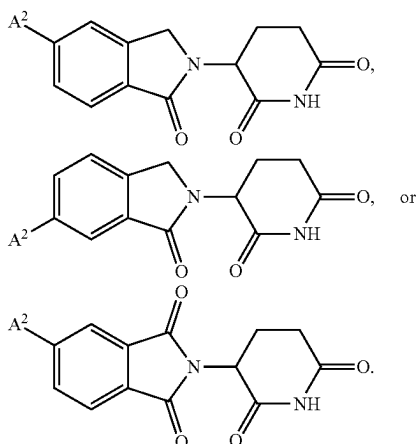
R⁴⁶ is H or optionally substituted C₁-C₆ alkyl; and R⁴⁷ is H or optionally substituted C₁-C₆ alkyl; or R⁴⁶ and R⁴⁷, together with the carbon atom to which each is bound, combine to form optionally substituted C₃-C₆ carbocyclyl or optionally substituted C₂-C₅ heterocyclyl; or R⁴⁶ and R⁴⁷, together with the carbon atom to which each is bound, combine to form optionally substituted C₃-C₆ carbocyclyl or optionally substituted C₂-C₅ heterocyclyl;

R⁴⁸ is H, optionally substituted C₁-C₆ alkyl, or optionally substituted C₁-C₆ heteroalkyl;

each of R⁴¹, R⁴², R⁴³, and R⁴⁴ is, independently, H, A², halogen, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted C₃-C₁₀ carbocyclyl, optionally substituted C₂-C₉ heterocyclyl, optionally substituted C₆-C₁₀ aryl, optionally substituted C₂-C₉ heteroaryl, optionally substituted C₂-C₆ alkenyl, optionally substituted C₂-C₆ heteroalkenyl, optionally substituted —O—C₃-C₆ carbocyclyl, hydroxyl, thiol, or optionally substituted amino; or R⁴¹ and R⁴², R⁴² and R⁴³, and/or R⁴³ and R⁴⁴, together with the carbon atoms to which each is attached, combine to form (N); and

(N) is optionally substituted C₆-C₁₀ aryl, optionally substituted C₃-C₁₀ carbocyclyl, optionally substituted C₂-C₉ heteroaryl, or C₂-C₉ heterocyclyl, any of which is optionally substituted with A², wherein one of R⁴¹, R⁴², R⁴³, and R⁴⁴ is A², or (N) is substituted with A², or a pharmaceutically acceptable salt thereof.

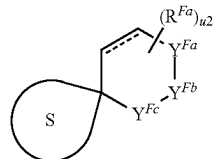
6. The compound of claim 5, wherein the structure of Formula A is



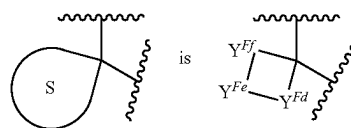
838

7. The compound of claim 2, wherein the degradation moiety comprises the structure of Formula FA:

Formula FA



where



or a bicyclic moiety which is substituted with A² and substituted with one or more groups independently selected from H, R^{FF1}, and oxo;

---- is a single bond or a double bond;

u2 is 0, 1, 2, or 3;

A² is a bond between the degrader and the linker;

Y^{Fa} is CR^{Fb}R^{Fc}, C=O, C=S, C=CH₂, SO₂, S(O), P(O)Oalkyl, P(O)NHalkyl, P(O)N(alkyl)₂, P(O)alkyl, P(O)OH, P(O)NH₂;

Y^{Fb} is NH, NR^{FF1}, CH₂, CHR^{FF1}, C(R^{FF1})₂, O, or S;

Y^{Fc} is CR^{Fd}R^{Fe}, C=O, C=S, C=CH₂, SO₂, S(O), P(O)Oalkyl, P(O)NHalkyl, P(O)N(alkyl)₂, P(O)alkyl, P(O)OH, P(O)NH₂;

each of R^{Fb}, R^{Fc}, R^{Fd}, and R^{Fe} is, independently, H, alkyl, aliphatic, heteroaliphatic, aryl, heteroaryl, carbocyclyl, hydroxyl, alkoxy, amino, —NHalkyl, or —Nalkyl₂;

or R^{Fb} and R^{Fc}, together with the carbon atom to which each is attached, combine to form a 3-, 4-, 5-, or 6-membered spirocarbocyclylene, or a 4-, 5-, or 6-membered spiroheterocyclylene comprising 1 or 2 heteroatoms selected from N and O;

or R^{Fd} and R^{Fe}, together with the carbon atom to which each is attached, combine to form a 3-, 4-, 5-, or 6-membered spirocarbocyclylene, or a 4-, 5-, or 6-membered spiroheterocyclylene comprising 1 or 2 heteroatoms selected from N and O; and

or R^{Fd} and R^{Fb}, together with the carbon atoms to which each is attached, combine to form a 1, 2, 3, or 4 carbon bridged ring;

each of Y^{Fd} and Y^{Fe} is, independently, CH₂, CHR^{FF2}, C(R^{FF2})₂, C(O), N, NH, NR^{FF3}, O, S, or S(O);

Y^{Fe} is a bond or a divalent moiety attached to Y^{Fd} and Y^{Ff} that contains 1 to 5 contiguous carbon atoms that form a 3 to 8-membered ring,

wherein 1, 2, or 3 carbon atoms can be replaced with a nitrogen, oxygen, or sulfur atom;

wherein one of the ring atoms is substituted with A² and the others are substituted with one or more groups independently selected from H and R^{FF1}; and

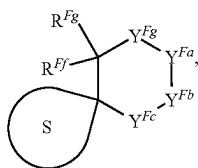
wherein the contiguous atoms of Y^{Fe} can be attached through a single or double bond;

each R^{FF1} is, independently, H, alkyl, alkenyl, alkynyl, aliphatic, heteroaliphatic, carbocyclyl, halogen, hydroxyl, amino, cyano, alkoxy, aryl, heteroaryl, heterocyclyl, alkylamino, alkylhydroxyl, or haloalkyl;

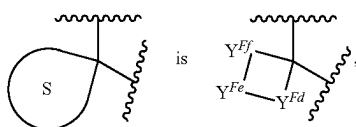
each R^{FF2} is, independently, alkyl, alkene, alkyne, halogen, hydroxyl, alkoxy, azide, amino, —C(O)H, —C(O)OH, —C(O)(aliphatic, including alkyl), —C(O)O(ali-

839

phatic, including alkyl), —NH(aliphatic, including alkyl), —N(aliphatic including alkyl)(aliphatic including alkyl), —NHSO₂alkyl, —N(alkyl)SO₂alkyl, —NHSO₂aryl, —N(alkyl)SO₂aryl, —NHSO₂alkenyl, —N(alkyl)SO₂alkenyl, —NHSO₂alkynyl, —N(alkyl)SO₂alkynyl, aliphatic, heteroaliphatic, aryl, heteroaryl, heterocyclic, carbocyclic, cyano, nitro, nitroso, —SH, —Salkyl, or haloalkyl; and
 R^{FF3} is alkyl, alkenyl, alkynyl, —C(O)H, —C(O)OH, —C(O)alkyl, or —C(O)Oalkyl,
 wherein if Y^{Fd} or Y^{Ff} is substituted with A^2 , then Y^{Fe} is a bond, or a pharmaceutically acceptable salt thereof.
 8. The compound of claim 2, wherein the degradation moiety comprises the structure of Formula FB:



where



or a bicyclic moiety which is substituted with A^2 and substituted with one or more groups independently selected from H, R^{FF1} , and oxo;

A^2 is a bond between the degrader and the linker;

Y^{Fa} is $CR^{Fb}R^{Fc}$, C=O, C=S, C=CH₂, SO₂, S(O), P(O)Oalkyl, P(O)NHalkyl, P(O)N(alkyl)₂, P(O)alkyl, P(O)OH, P(O)NH₂;

each of Y^{Fb} and Y^{Fg} is, independently, NH, NR^{FF1} , CH₂, CHR^{FF1} , $C(R^{FF1})_2$, O, or S;

Y^{Fc} is $CR^{Fd}R^{Fe}$, C=O, C=S, C=CH₂, SO₂, S(O), P(O)Oalkyl, P(O)NHalkyl, P(O)N(alkyl)₂, P(O)alkyl, P(O)OH, P(O)NH₂;

each of R^{Fb} , R^{Fc} , R^{Fd} , R^{Fe} , R^{Ff} , and R^{Fg} is, independently, H, alkyl, aliphatic, heteroaliphatic, aryl, heteroaryl, carbocyclyl, hydroxyl, alkoxy, amino, —NHalkyl, or —Nalkyl₂;

or R^{Fb} and R^{Fc} , together with the carbon atom to which each is attached, combine to form a 3-, 4-, 5-, or 6-membered spirocarbocyclene, or a 4-, 5-, or 6-membered spiroheterocyclene comprising 1 or 2 heteroatoms selected from N and O;

or R^{Fd} and R^{Fe} , together with the carbon atom to which each is attached, combine to form a 3-, 4-, 5-, or 6-membered spirocarbocyclene, or a 4-, 5-, or 6-membered spiroheterocyclene comprising 1 or 2 heteroatoms selected from N and O;

or R^{Ff} and R^{Fg} , together with the carbon atom to which each is attached, combine to form a 3-, 4-, 5-, or 6-membered spirocarbocyclene, or a 4-, 5-, or 6-membered spiroheterocyclene comprising 1 or 2 heteroatoms selected from N and O;

or R^{Fd} and R^{Fb} , together with the carbon atoms to which each is attached, combine to form a 1, 2, 3, or 4 carbon bridged ring;

840

or R^{Fd} and R^{Ff} , together with the carbon atoms to which each is attached, combine to form a 1, 2, 3, or 4 carbon bridged ring;

or R^{Fb} and R^{Fg} , together with the carbon atoms to which each is attached, combine to form a 1, 2, 3, or 4 carbon bridged ring;

each of Y^{Fd} and Y^{Ff} is, independently, CH₂, CHR^{FF2} , $C(R^{FF2})_2$, C(O), N, NH, NR^{FF3} , O, S, or S(O);

Y^{Fe} is a bond or a divalent moiety attached to Y^{Fd} and Y^{Ff} that contains 1 to 5 contiguous carbon atoms that form a 3 to 8-membered ring,

wherein 1, 2, or 3 carbon atoms can be replaced with a nitrogen, oxygen, or sulfur atom;

wherein one of the ring atoms is substituted with A^2 and the others are substituted with one or more groups independently selected from H and R^{FF1} ; and

wherein the contiguous atoms of Y^{Fe} can be attached through a single or double bond;

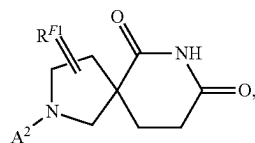
each R^{FF1} is, independently, H, alkyl, alkenyl, alkynyl, aliphatic, heteroaliphatic, carbocyclyl, halogen, hydroxyl, amino, cyano, alkoxy, aryl, heteroaryl, heterocyclyl, alkylamino, alkylhydroxyl, or haloalkyl;

each R^{FF2} is, independently, alkyl, alkene, alkyne, halogen, hydroxyl, alkoxy, azide, amino, —C(O)H, —C(O)OH, —C(O)(aliphatic, including alkyl), —C(O)O(aliphatic, including alkyl), —NH(aliphatic, including alkyl), —N(aliphatic including alkyl)(aliphatic including alkyl), —NHSO₂alkyl, —N(alkyl)SO₂alkyl, —NHSO₂aryl, —N(alkyl)SO₂aryl, —NHSO₂alkenyl, —N(alkyl)SO₂alkenyl, —NHSO₂alkynyl, —N(alkyl)SO₂alkynyl, aliphatic, heteroaliphatic, aryl, heteroaryl, heterocyclic, carbocyclic, cyano, nitro, nitroso, —SH, —Salkyl, or haloalkyl; and

R^{FF3} is alkyl, alkenyl, alkynyl, —C(O)H, —C(O)OH, —C(O)alkyl, or —C(O)Oalkyl,

wherein if Y^{Fd} or Y^{Ff} is substituted with A^2 , then Y^{Fe} is a bond, or a pharmaceutically acceptable salt thereof.

9. The compound of claim 2, wherein the degradation moiety comprises the structure of Formula F1:



Formula F1

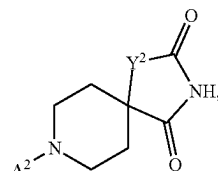
wherein

A^2 is a bond between the degrader and the linker; and

R^{F1} is absent or O,

or a pharmaceutically acceptable salt thereof.

10. The compound of claim 2, wherein the degradation moiety comprises the structure of Formula F2:



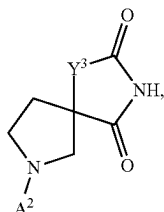
Formula F2

841

wherein

A² is a bond between the degrader and the linker; and
R^{F1} is absent or O,
or a pharmaceutically acceptable salt thereof.

11. The compound of claim 2, wherein the degradation moiety comprises the structure of Formula G:

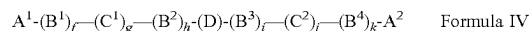


Formula G

wherein

A² is a bond between the degrader and the linker; and
R^{F1} is absent or O,
or a pharmaceutically acceptable salt thereof.

12. The compound of claim 2, wherein the linker has the structure of Formula IV:



842

wherein

A¹ is a bond between the linker and A;

A² is a bond between B and the linker;

each of B¹, B², B³, and B⁴ is, independently, optionally substituted C₁-C₂ alkyl, optionally substituted C₁-C₃ heteroalkyl, O, S, S(O)₂, or NR^N;

each R^N is, independently, H, optionally substituted C₁₋₄ alkyl, optionally substituted C₂₋₄ alkenyl, optionally substituted C₂₋₄ alkynyl, optionally substituted C₂₋₆ heterocyclyl, optionally substituted C₆₋₁₂ aryl, or optionally substituted C₁₋₇ heteroalkyl;

each of C¹ and C² is, independently, carbonyl, thiocarbonyl, sulphonyl, or phosphoryl;

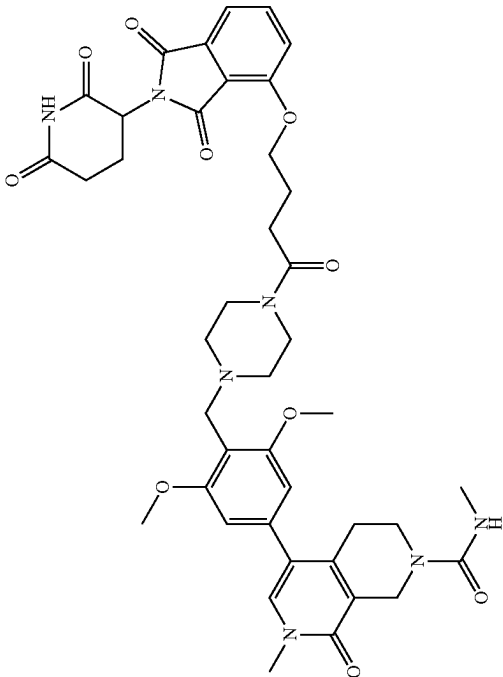
each of f, g, h, i, j, and k is, independently, 0 or 1; and

D is optionally substituted C₁₋₁₀ alkyl, optionally substituted C₂₋₁₀ alkenyl, optionally substituted C₂₋₁₀ alkynyl, optionally substituted C₂₋₆ heterocyclyl, optionally substituted C₆₋₁₂ aryl, optionally substituted C₂-C₁₀ polyethylene glycol, or optionally substituted C₁₋₁₀ heteroalkyl, or a chemical bond linking A¹-(B¹)_f-(C¹)_g-(B²)_h to -(B³)_i-(C²)_j-(B⁴)_k-A².

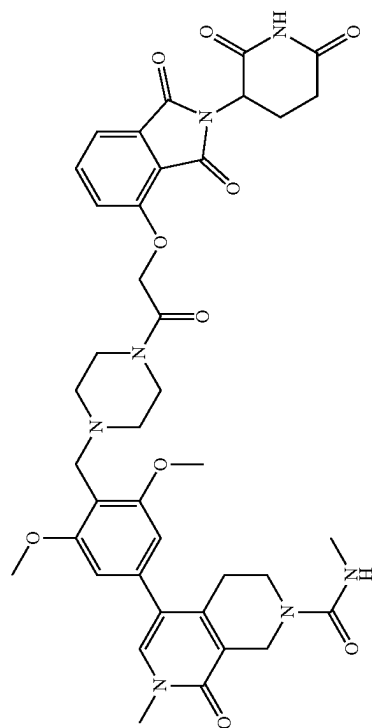
13. The compound of claim 2, wherein the compound has the structure of A any one of compounds D1-D31, D32-D211, and D212-D343, or a pharmaceutically acceptable salt thereof

843

844

Compound No.	Structure
D1	

D2



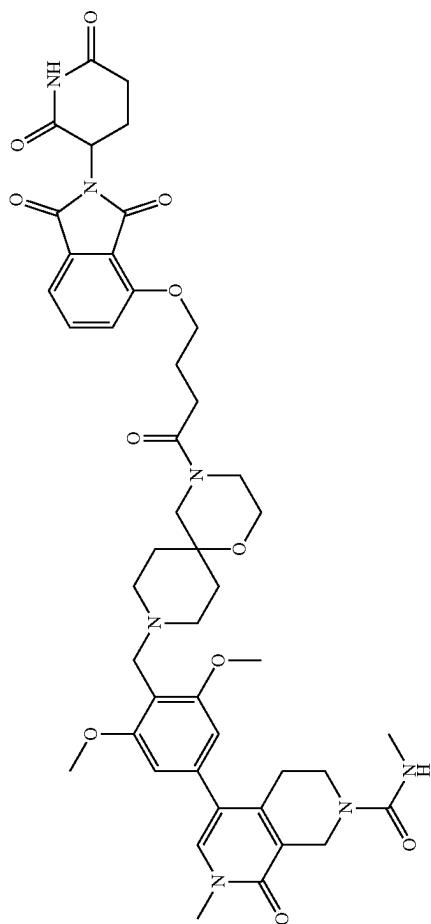
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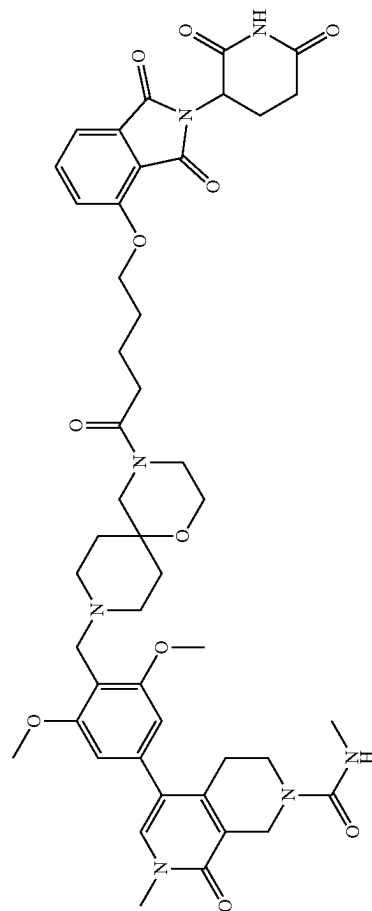
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Compound
No. Structure

D3



D4



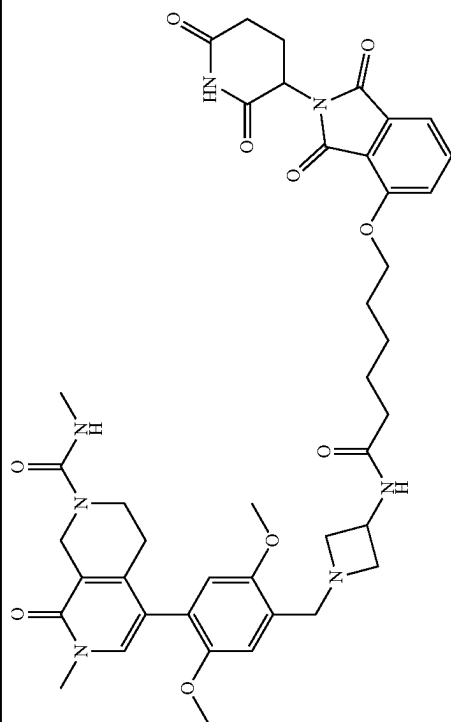
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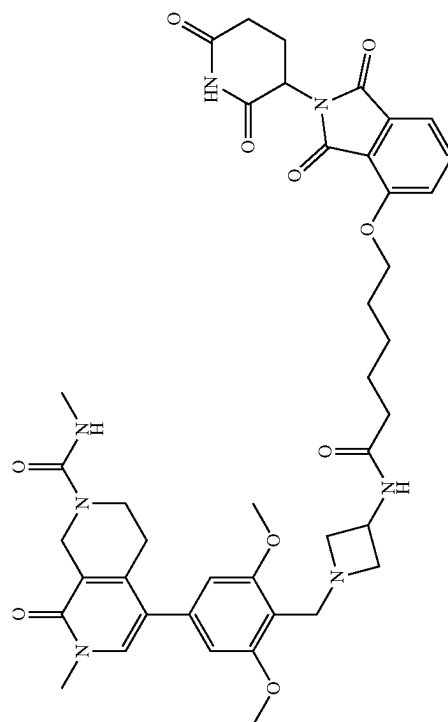
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Compound No.	Structure
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D15



D16



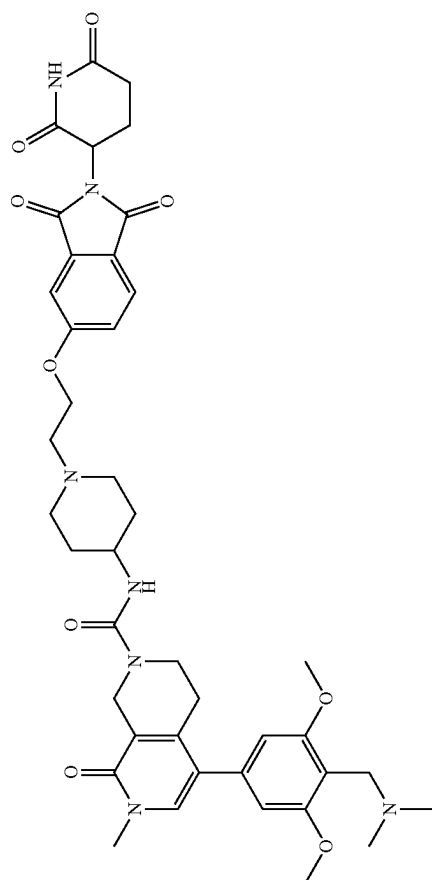
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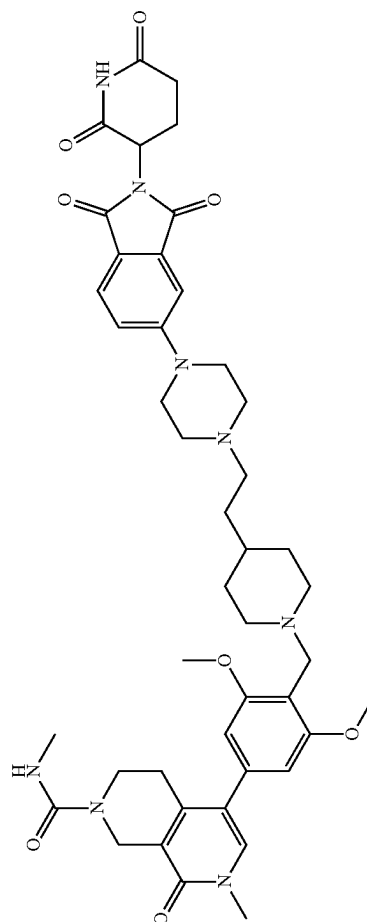
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Compound No.	Structure
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D26



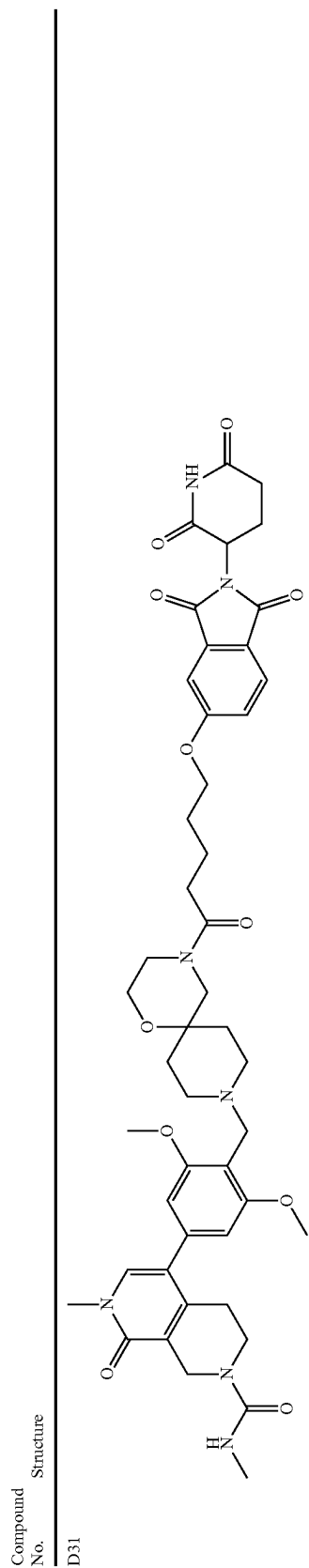
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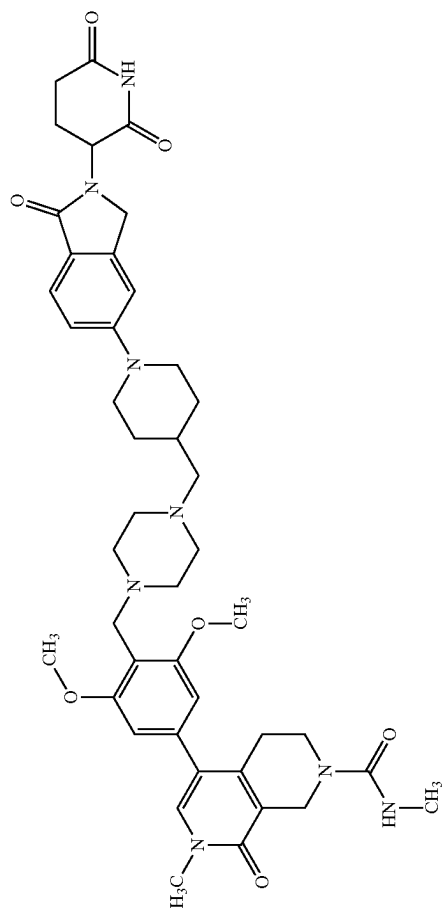
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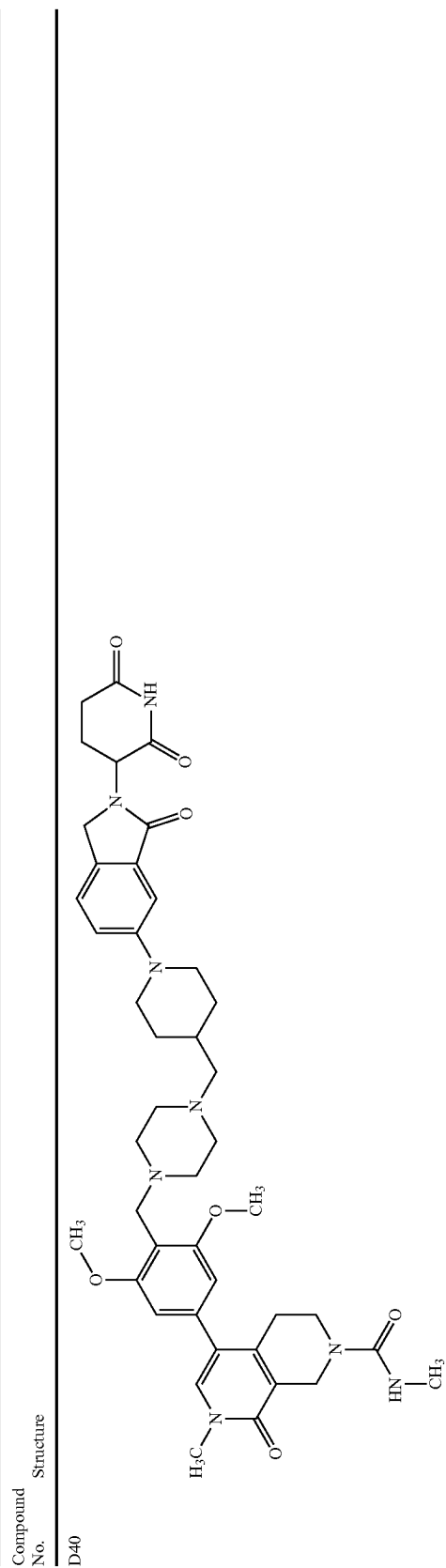
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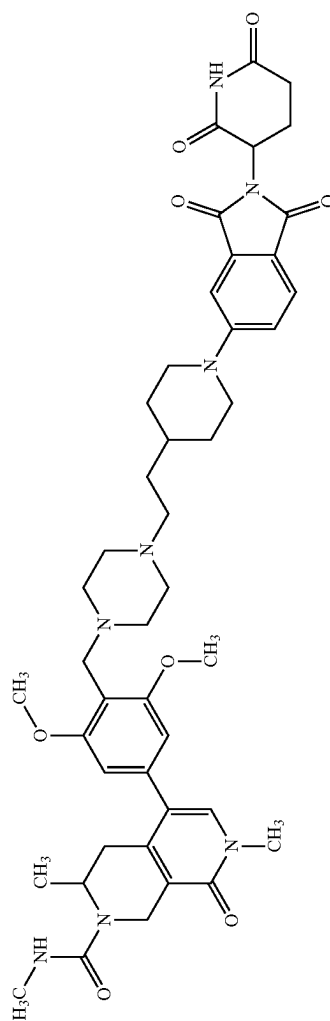
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854

-continued



D41



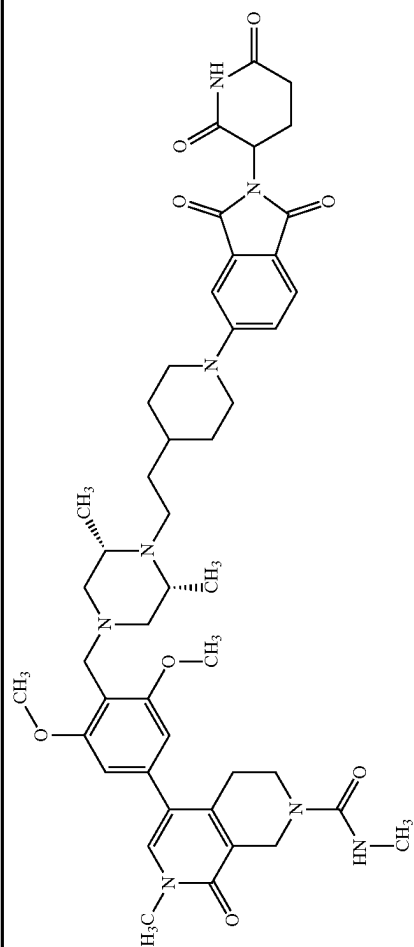
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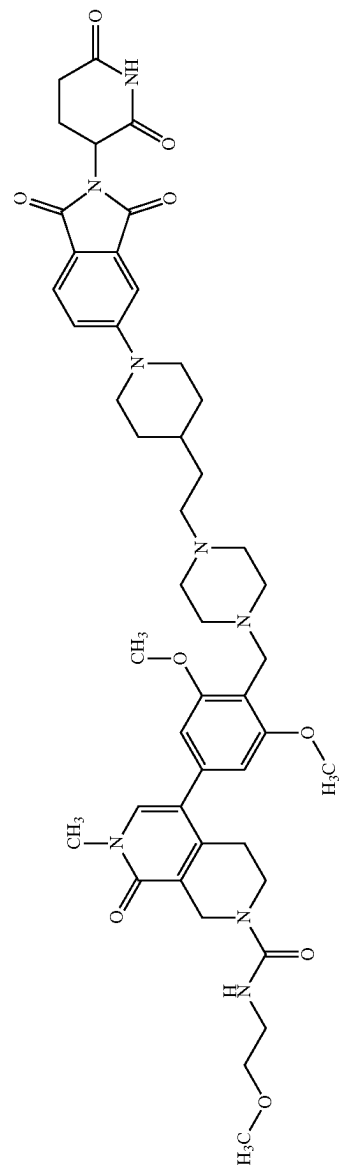
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Compound No.	Structure
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D42



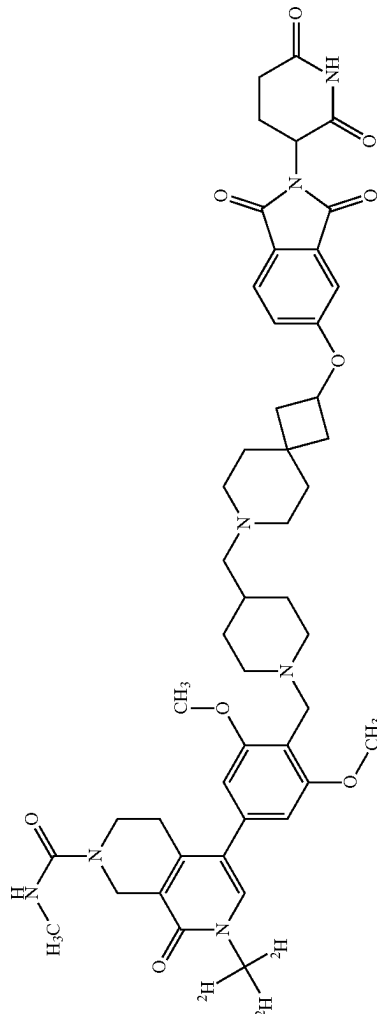
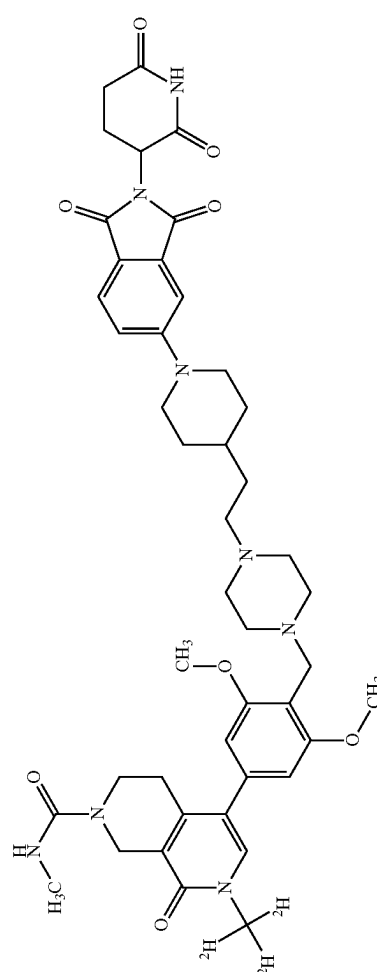
D43



857

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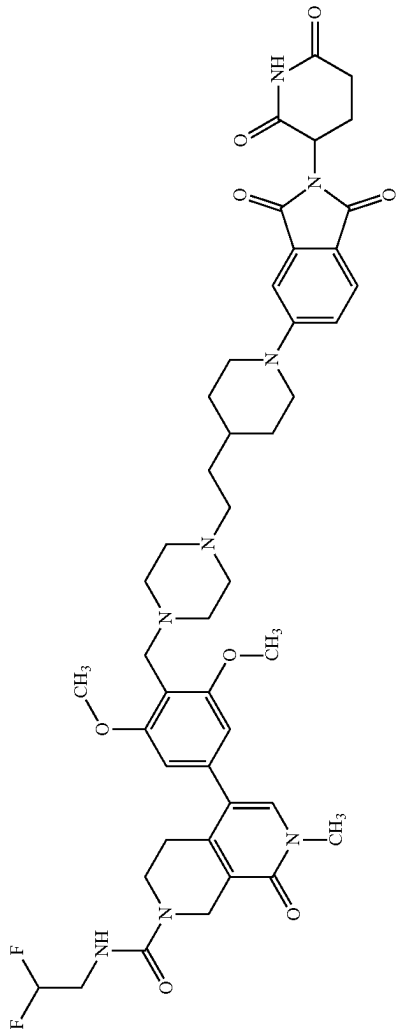
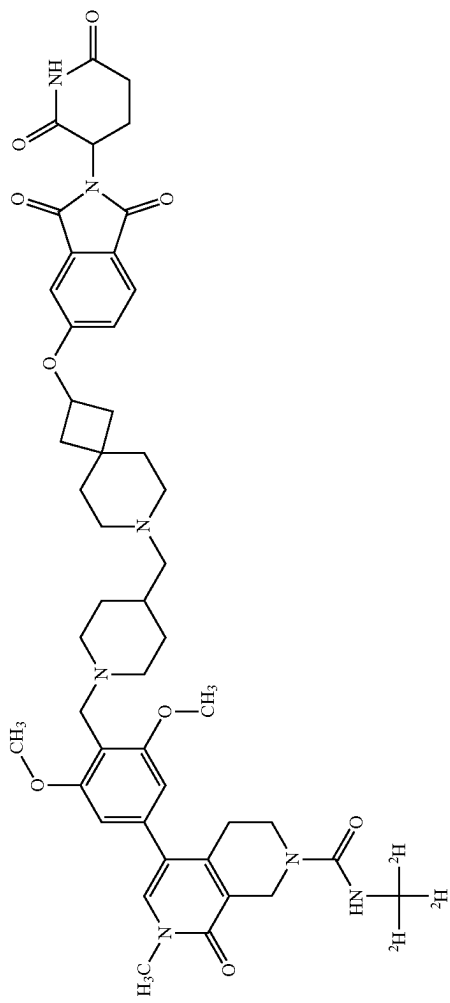
-continued

Compound No.	Structure
D44	 Chemical structure of compound D44: A pyridine ring substituted with a methylamino group (H₃C-NH-C=O), a 2,2,2-trideuterioethyl group, a 4-methoxyphenyl group, and a 4-methoxy-2-((4-methoxyphenyl)methyl)piperidin-1-yl group. The piperidine ring is further substituted with a 4-methoxyphenyl group and a 4-methoxy-2-((4-methoxyphenyl)methyl)piperidin-1-yl group. The piperidine ring is also substituted with a 4-methoxyphenyl group and a 4-methoxy-2-((4-methoxyphenyl)methyl)piperidin-1-yl group.
D45	 Chemical structure of compound D45: A pyridine ring substituted with a methylamino group (H₃C-NH-C=O), a 2,2,2-trideuterioethyl group, a 4-methoxyphenyl group, and a 4-methoxy-2-((4-methoxyphenyl)methyl)piperidin-1-yl group. The piperidine ring is further substituted with a 4-methoxyphenyl group and a 4-methoxy-2-((4-methoxyphenyl)methyl)piperidin-1-yl group. The piperidine ring is also substituted with a 4-methoxyphenyl group and a 4-methoxy-2-((4-methoxyphenyl)methyl)piperidin-1-yl group.

859

860

-continued

Compound No.	Structure
D46	 Chemical structure of compound D46: A complex molecule featuring a central pyridine ring substituted with a 2-fluoroethyl group, a methyl ester, and a 4-methyl-1H-pyridin-2-yl group. This central ring is connected via a methylene bridge to a piperazine ring, which is further linked to a 1,2,3,4-tetrahydropyridine ring. The tetrahydropyridine ring is substituted with a methyl group and a 1,2,3,4-tetrahydropyridin-5-yl group. The tetrahydropyridine ring is also connected via a methylene bridge to a piperazine ring, which is further linked to a 1,2,3,4-tetrahydropyridine ring. The tetrahydropyridine ring is substituted with a methyl group and a 1,2,3,4-tetrahydropyridin-5-yl group. The tetrahydropyridine ring is also connected via a methylene bridge to a piperazine ring, which is further linked to a 1,2,3,4-tetrahydropyridine ring. The tetrahydropyridine ring is substituted with a methyl group and a 1,2,3,4-tetrahydropyridin-5-yl group.
D47	 Chemical structure of compound D47: A complex molecule featuring a central pyridine ring substituted with a 2-methoxyethyl group, a methyl ester, and a 4-methyl-1H-pyridin-2-yl group. This central ring is connected via a methylene bridge to a piperazine ring, which is further linked to a 1,2,3,4-tetrahydropyridine ring. The tetrahydropyridine ring is substituted with a methyl group and a 1,2,3,4-tetrahydropyridin-5-yl group. The tetrahydropyridine ring is also connected via a methylene bridge to a piperazine ring, which is further linked to a 1,2,3,4-tetrahydropyridine ring. The tetrahydropyridine ring is substituted with a methyl group and a 1,2,3,4-tetrahydropyridin-5-yl group. The tetrahydropyridine ring is also connected via a methylene bridge to a piperazine ring, which is further linked to a 1,2,3,4-tetrahydropyridine ring. The tetrahydropyridine ring is substituted with a methyl group and a 1,2,3,4-tetrahydropyridin-5-yl group.

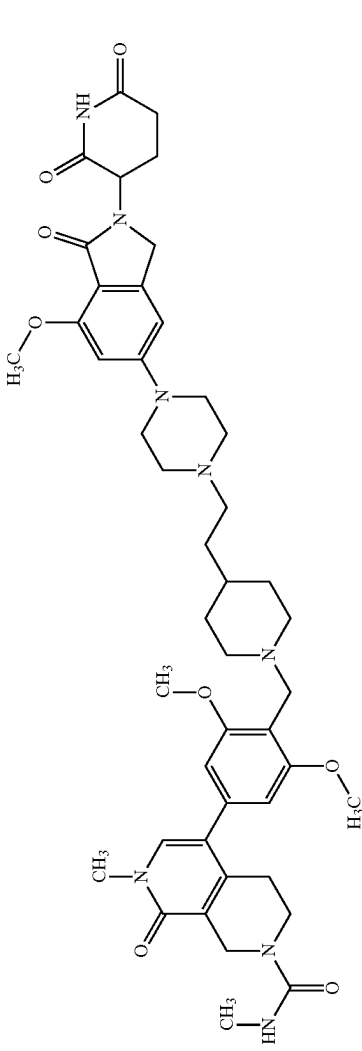
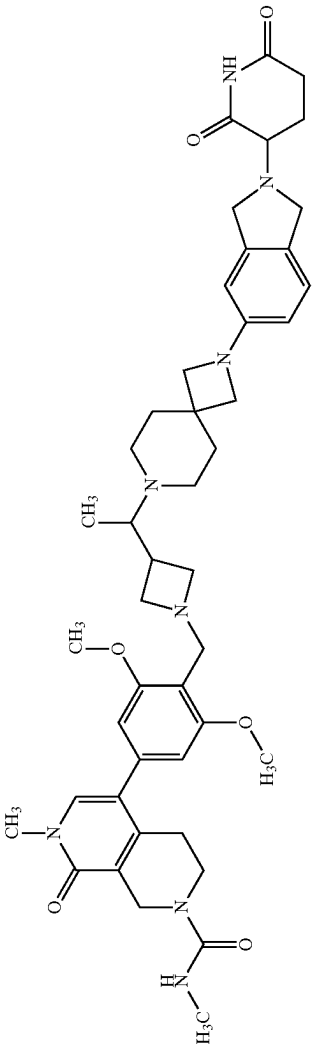
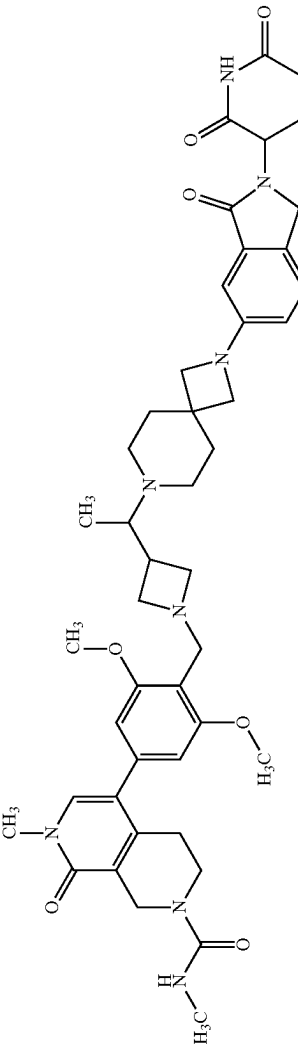
Compound No.	Structure
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CN(C)C(=O)N1CCc2c(cnc2C1=O)c3ccc(cc3N4CCCCC4CCN5CCN(C5Cc6cc(OC)c7ccc(cc67)N8CCCCN8)cc8)C(=O)N9C(=O)CCCC9=OCN(C)C(=O)c1ccc2c(c1)cnc2C3=CC=C(C=C3)C(OC)C(C4=CC=CC=C4C5=CC=CC=C5N(CCCN6CCN(CCCN7CCN(C8=CC=CC=C8C(=O)NC8=O)N7)C6)N5)N4

863

864

-continued

Compound No.	Structure
D50	
D51	
D52	

-continued

Compound No.	Structure
D53	
D54	
D55	

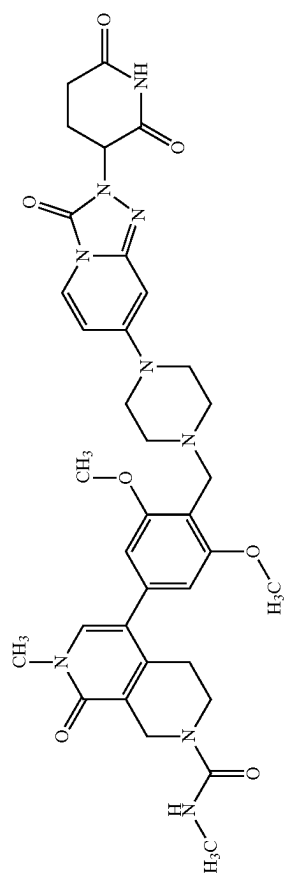
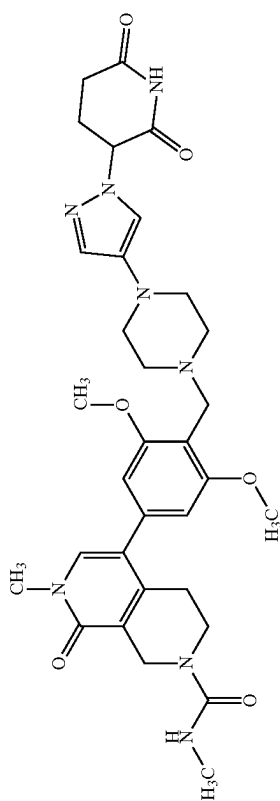
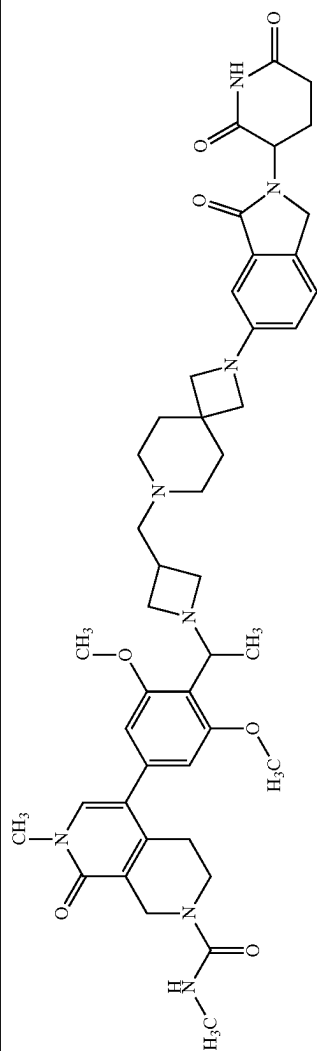
867

868

-continued

Compound No.	Structure
D56	
D57	
D58	

Compound No.	Structure
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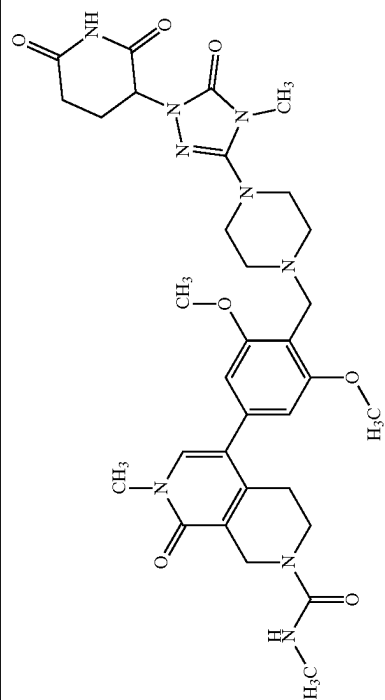
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872

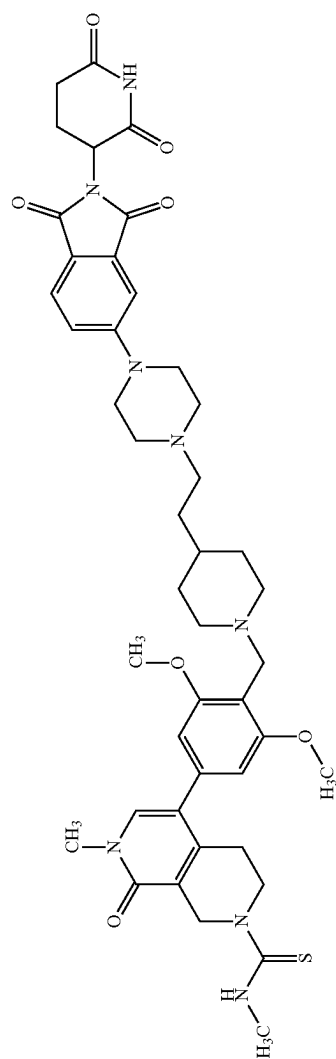
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Compound No.	Structure
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D62



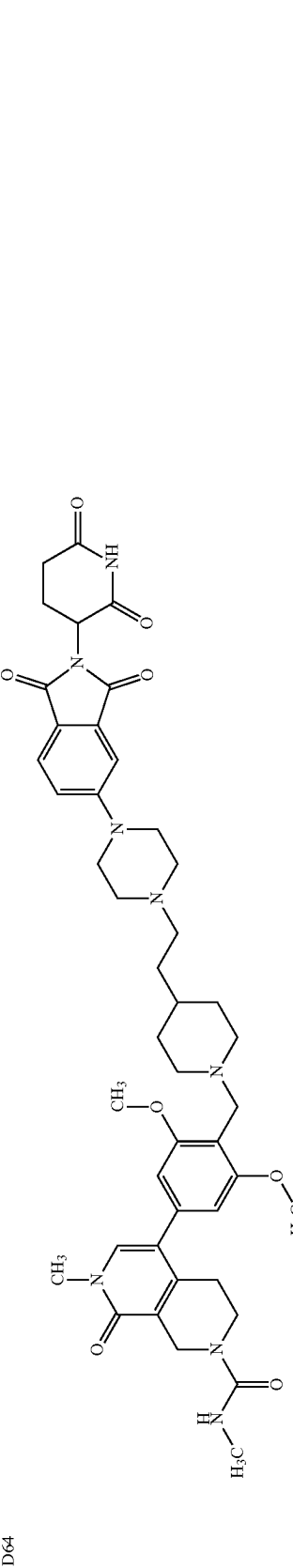
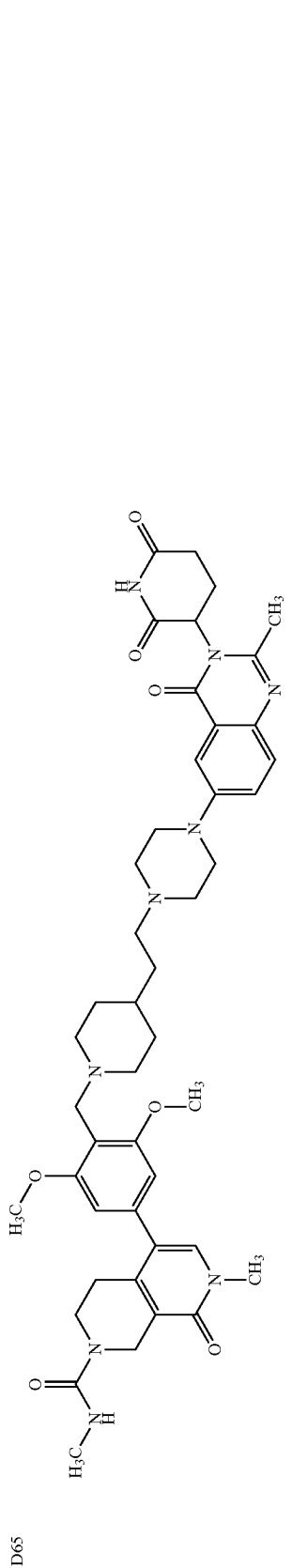
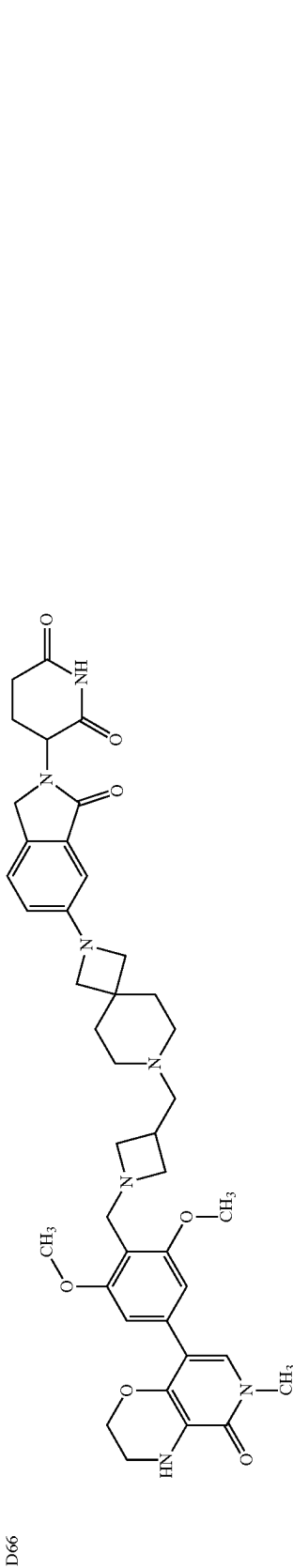
D63



873

874

-continued

Compound No.	Structure
D64	
D65	
D66	

-continued

Compound No.	Structure
D67	
D68	
D69	

Compound No.	Structure
D70	
D71	
D72	

879

880

-continued

Compound No.	Structure
D73	
D75	

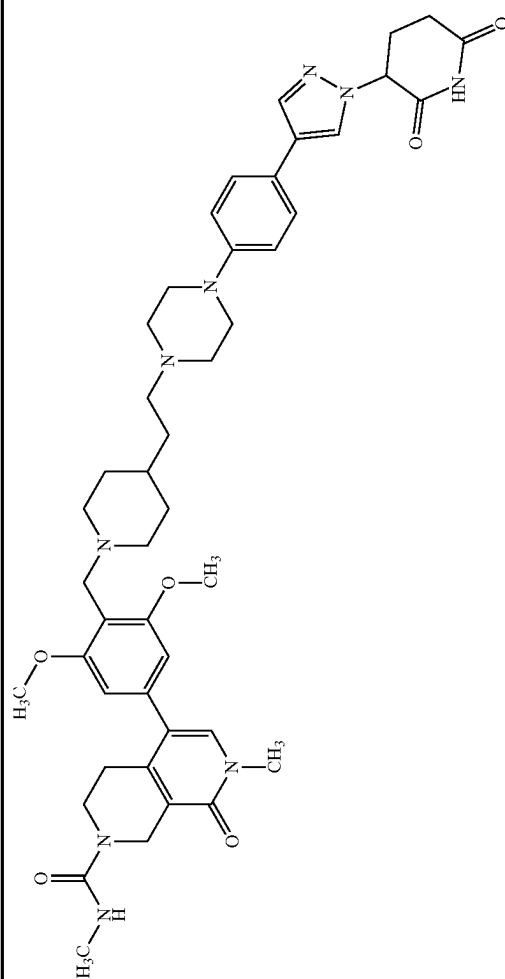
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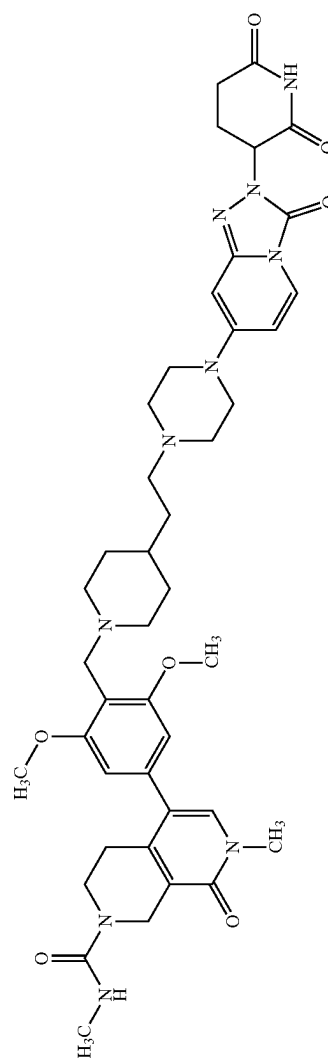
-continued

Compound No.	Structure
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D76



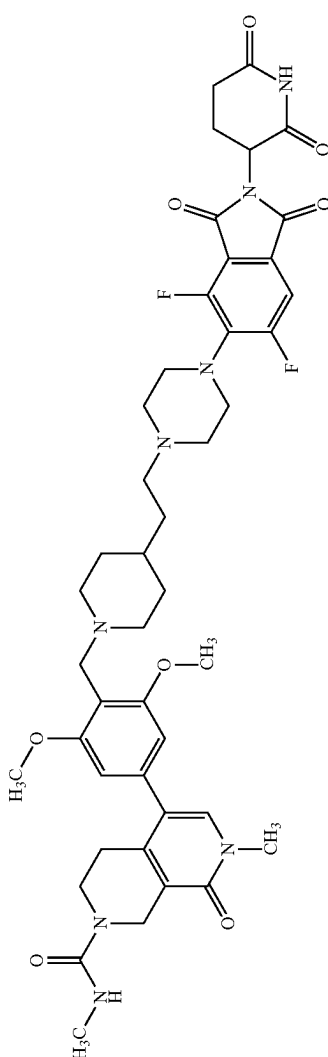
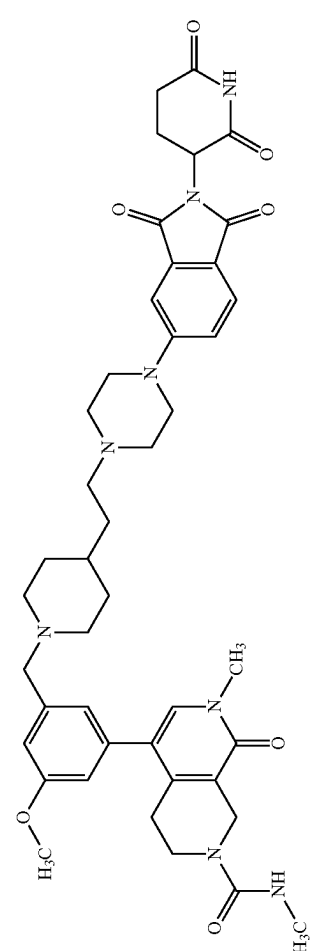
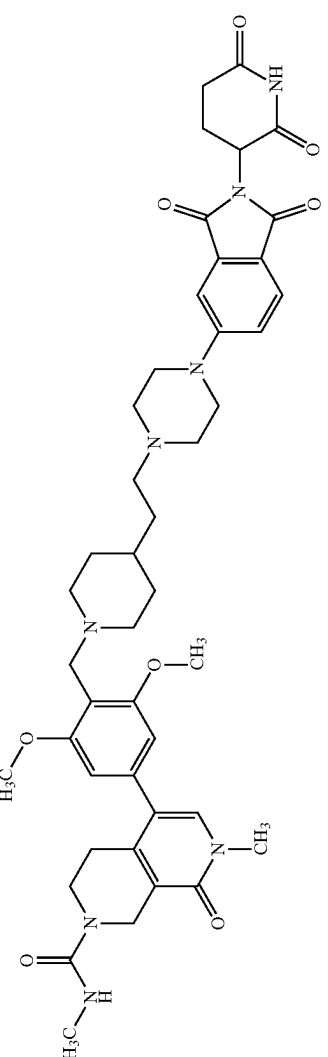
D77



883

884

-continued

Compound No.	Structure
D78	
D79	
D81	

Compound No.	Structure
D82	
D83	

887

888

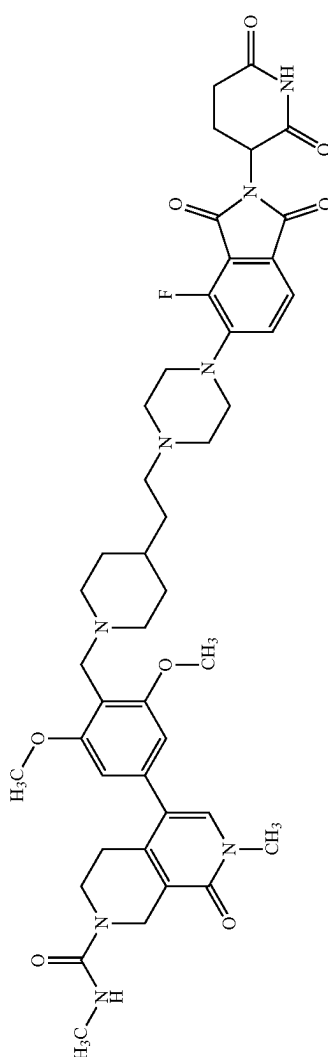
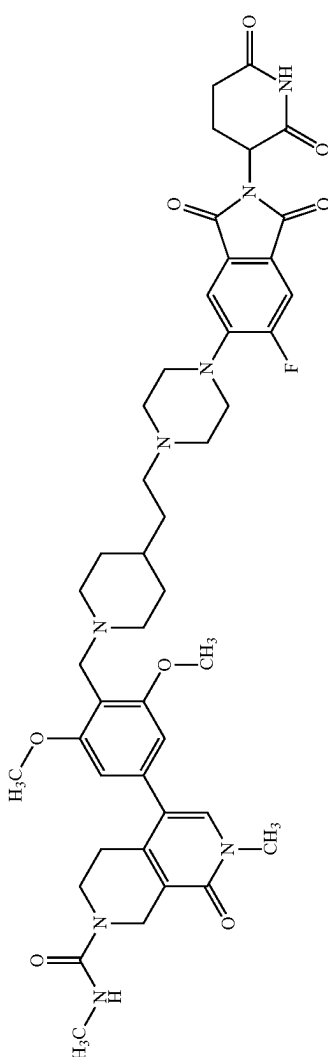
-continued

Compound No.	Structure
D84	 <chem>CNC(=O)c1ccc2c(c1)ncn2C3=CC=C(C=C3)C(OC)C4=CC=C(C=C4)CN(CCCN5CCN(C5c6ccc7c(c6)c(=O)[nH]c7=O)c8ccccc8)CCN9CCN(C9)c10ccccc10</chem>
D85	 <chem>CNC(=O)c1ccc2c(c1)ncn2C3=CC=C(C=C3)C(OC)C4=CC=C(C=C4)CN(CCCN5CCN(C5c6ccccc6)CCN7CCN(C7)c8ccccc8)CCN9CCN(C9)c10ccccc10</chem>
D86	 <chem>CNC(=O)c1ccc2c(c1)c(F)c3c2c(=O)[nH]c3=OCC4=CC=C(C=C4)CN(CCCN5CCN(C5c6ccc7c(c6)c(=O)[nH]c7=O)c8ccccc8)CCN9CCN(C9)c10ccccc10</chem>

889

890

-continued

Compound No.	Structure
D87	 Chemical structure of compound D87: A complex molecule featuring a central pyridine ring substituted with a methyl group at the 2-position and a methylamino group at the 3-position. This pyridine is linked via its 4-position to a benzene ring, which also bears a methoxy group and a methylamino group. The benzene ring is further connected to a piperidine ring, which is linked to a 4-fluoro-1H-indolizino[1,2-b]pyridine system. This system is substituted with a methylamino group and a methylamino group, and is further linked to a piperidine ring, which is connected to a 4-fluoro-1H-indolizino[1,2-b]pyridine system. The structure is highly symmetrical and complex, with multiple rings and functional groups.
D88	 Chemical structure of compound D88: A complex molecule featuring a central pyridine ring substituted with a methyl group at the 2-position and a methylamino group at the 3-position. This pyridine is linked via its 4-position to a benzene ring, which also bears a methoxy group and a methylamino group. The benzene ring is further connected to a piperidine ring, which is linked to a 4-fluoro-1H-indolizino[1,2-b]pyridine system. This system is substituted with a methylamino group and a methylamino group, and is further linked to a piperidine ring, which is connected to a 4-fluoro-1H-indolizino[1,2-b]pyridine system. The structure is highly symmetrical and complex, with multiple rings and functional groups.

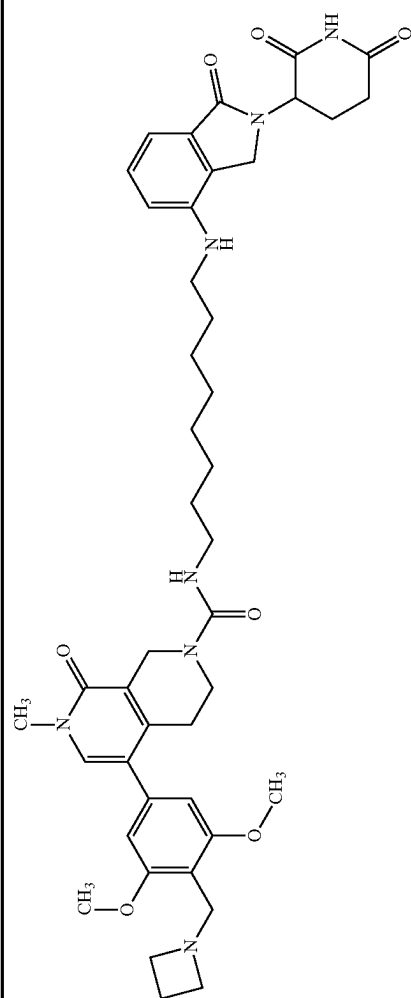
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892

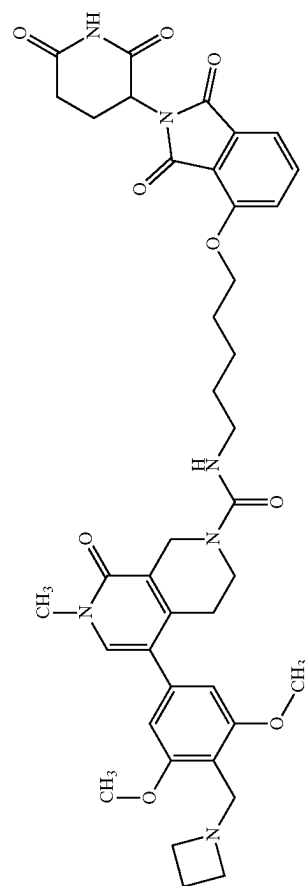
-continued

Compound No.	Structure
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D89



D90



-continued

Compound No.	Structure
D91	
D92	
D93	

895

896

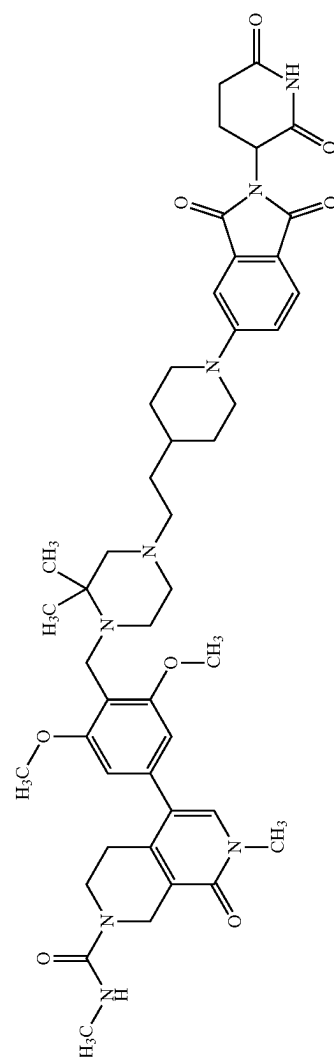
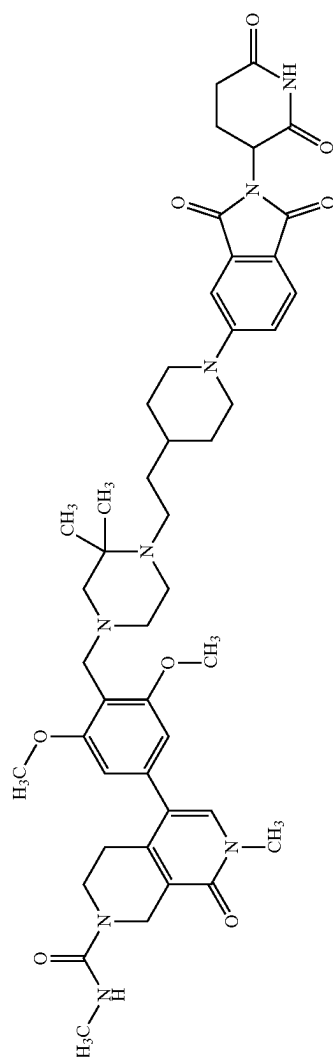
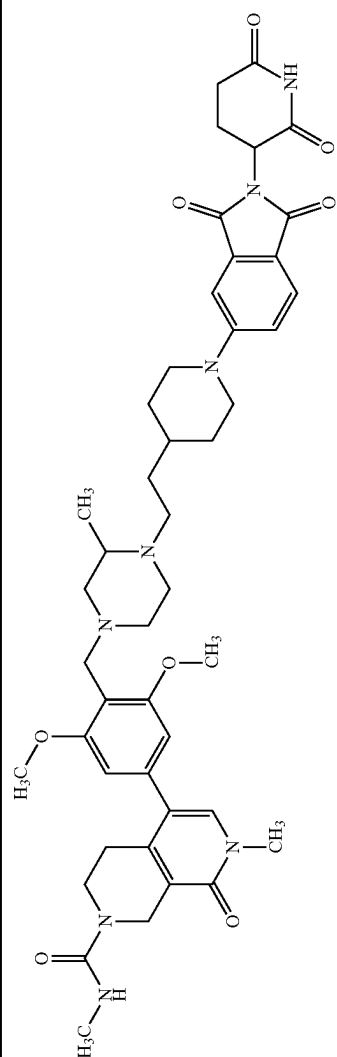
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Compound No.	Structure
D94	
D95	
D96	

-continued

Compound No.	Structure
D97	
D98	
D99	

Compound No.	Structure
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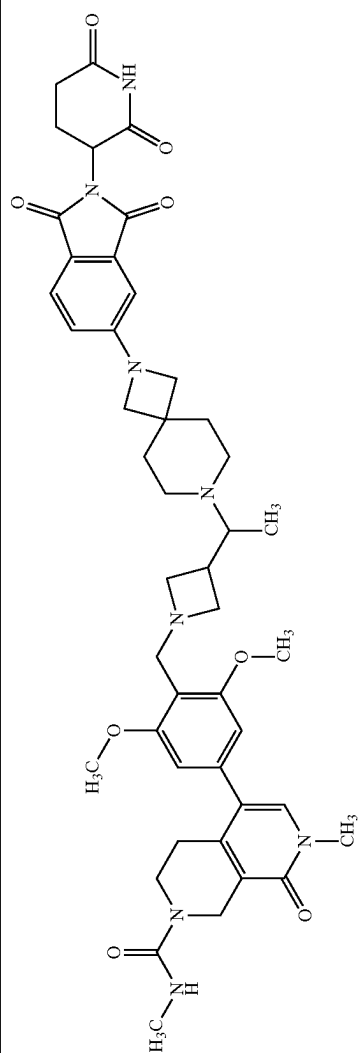
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902

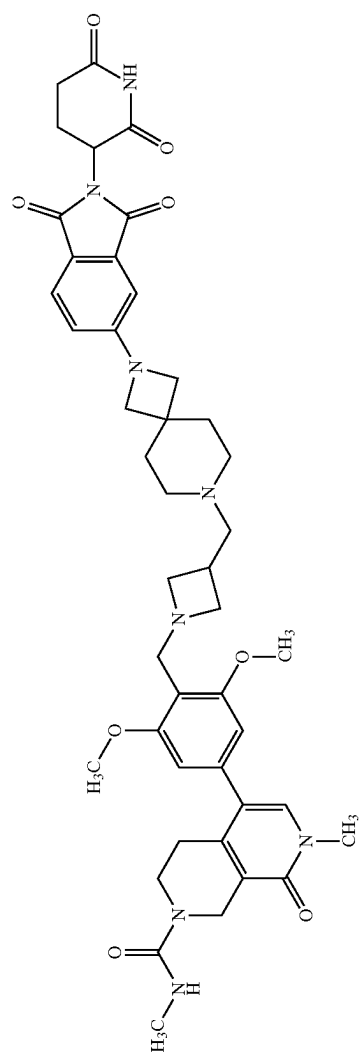
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Compound No.	Structure
D103	
D104	
D106	

Compound No.	Structure
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D109



-continued

[illegible]

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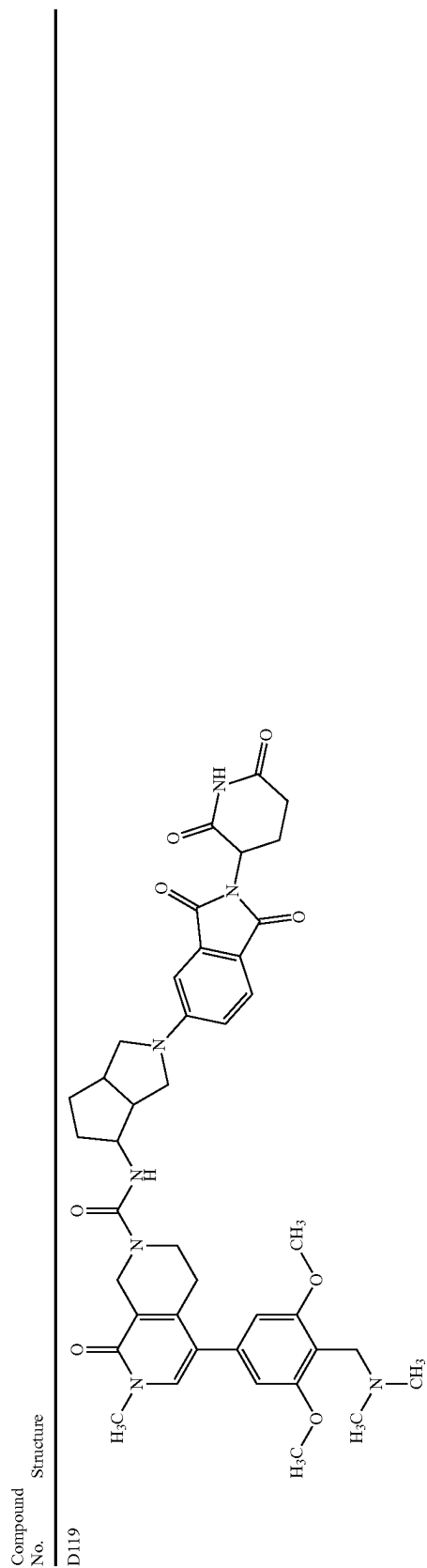
Compound No.	Structure
D114	
D115	
D116	

[illegible]

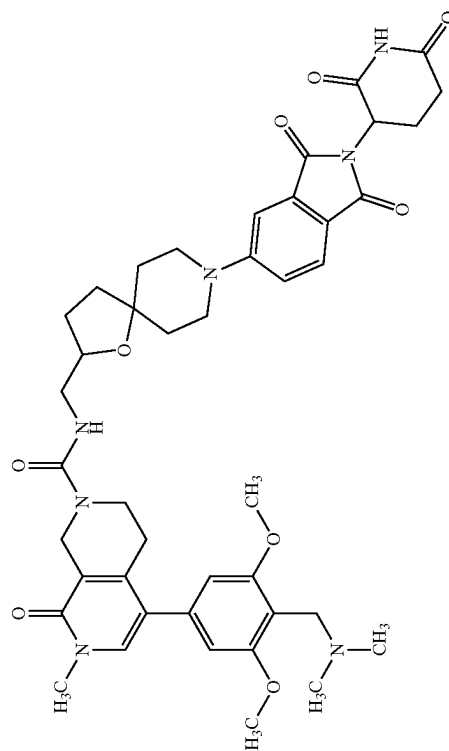
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912

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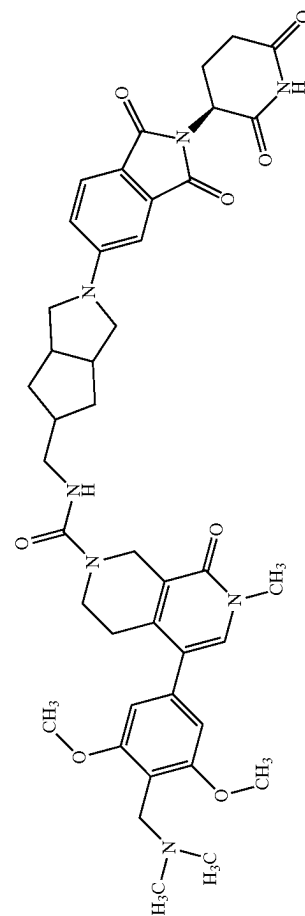
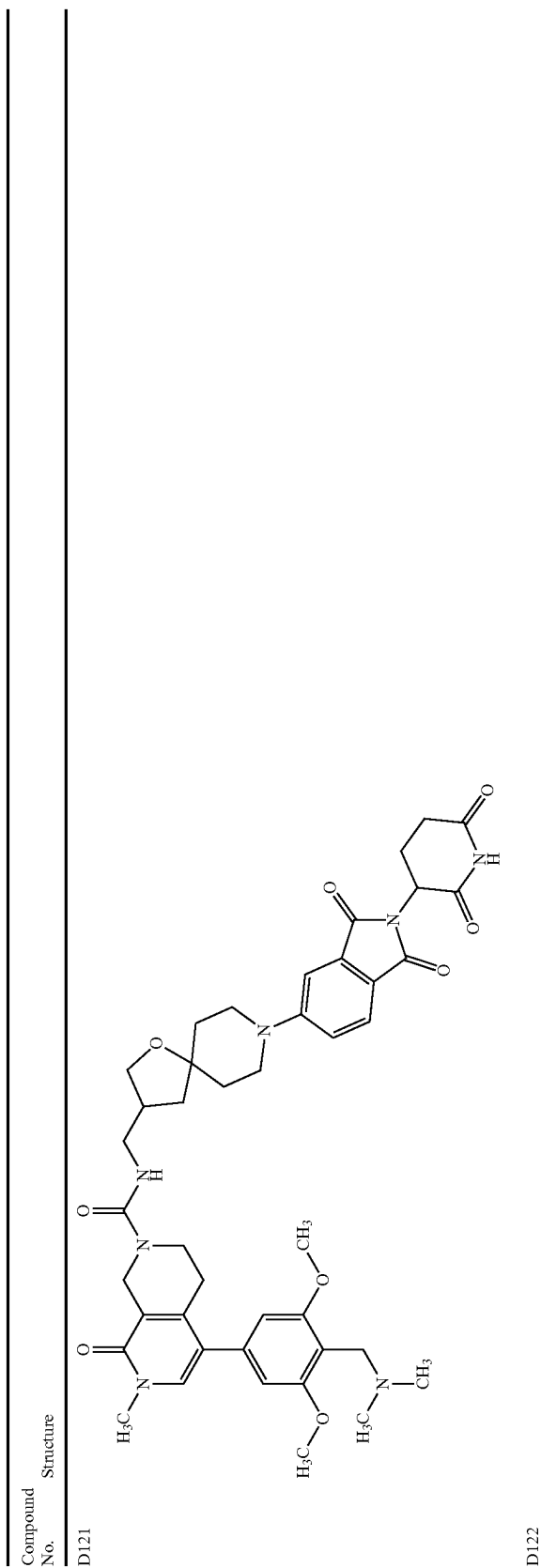
D120



913

914

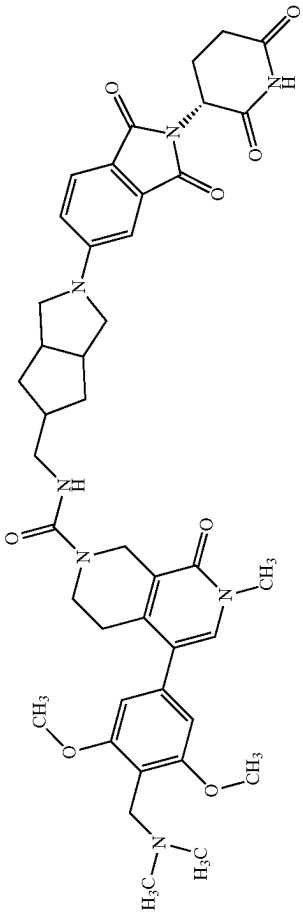
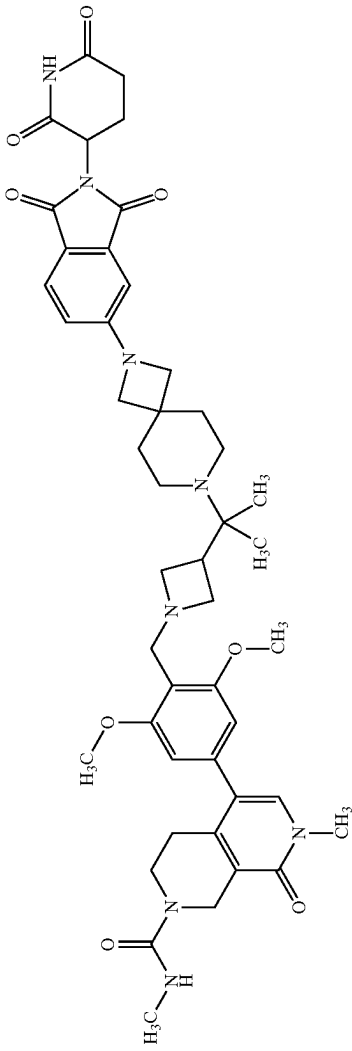
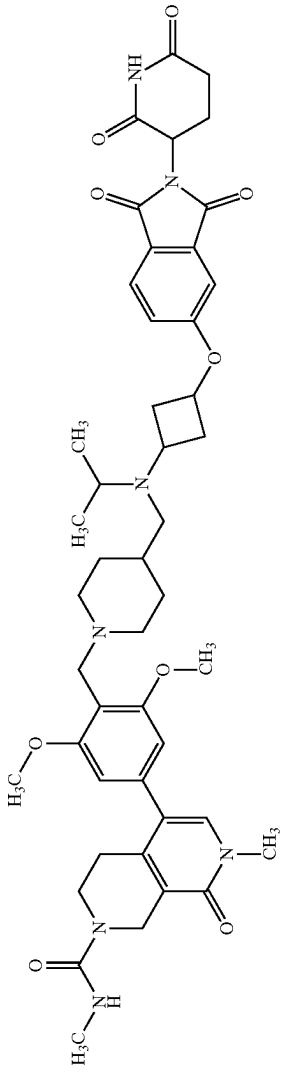
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915

916

-continued

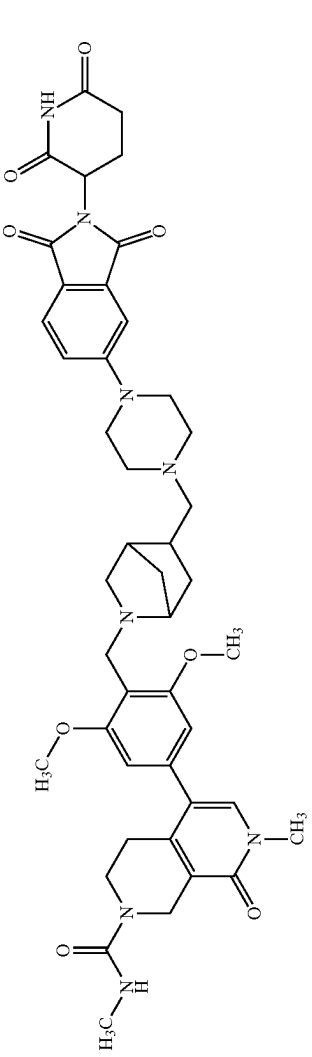
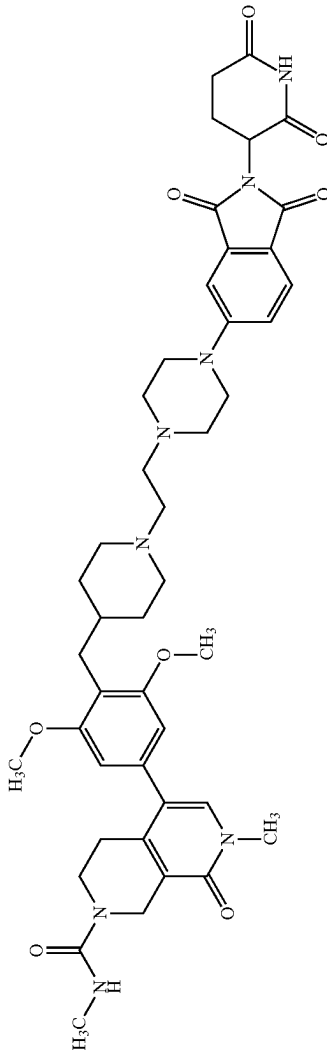
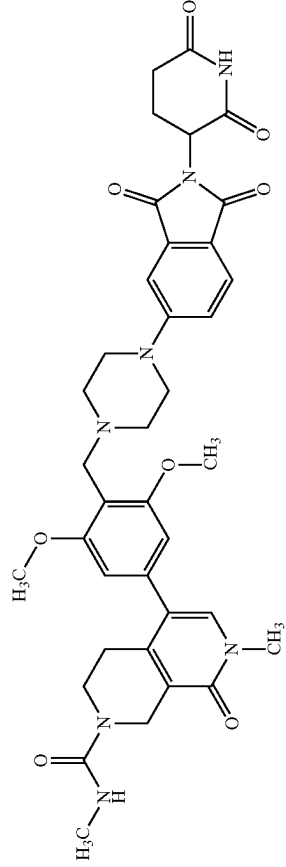
Compound No.	Structure
D123	
D124	
D125	

Compound No.	Structure
D126	
D127	
D128	

919

920

-continued

Compound No.	Structure
D129	
D130	
D131	

Compound No.	Structure
D132	
D133	
D134	

923

924

-continued

Compound No.	Structure
D135	
D136	
D137	

925

926

-continued

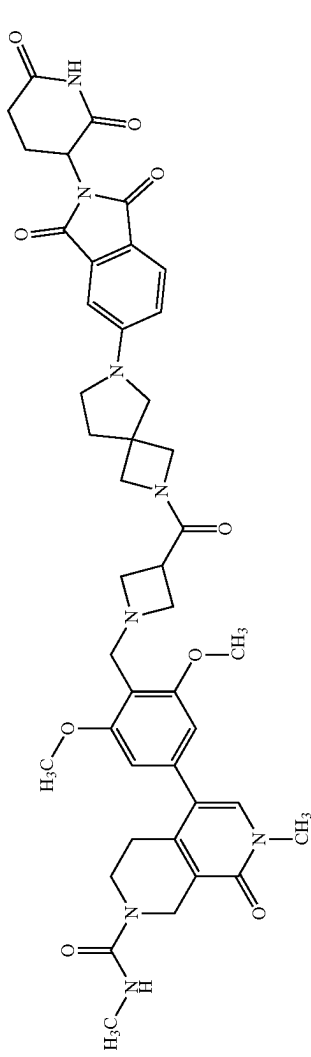
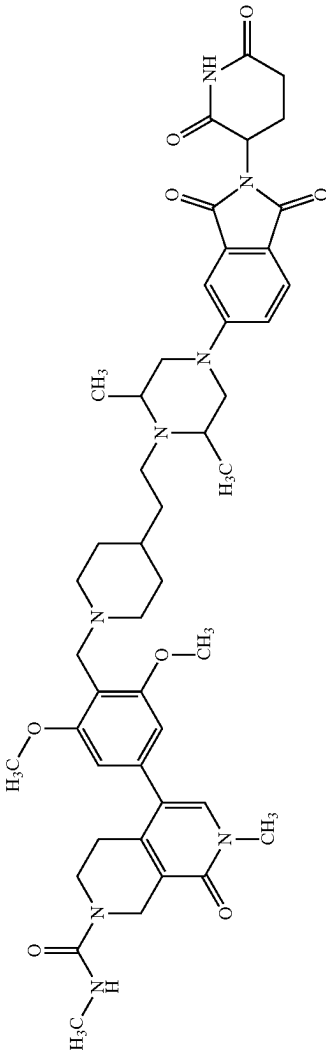
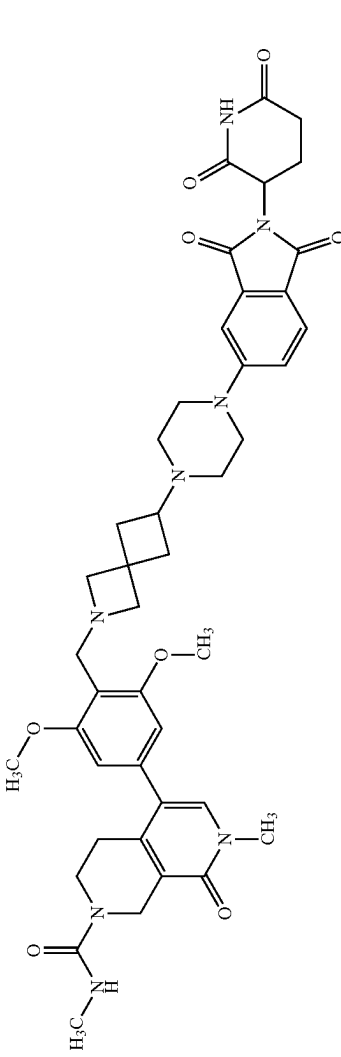
Compound No.	Structure
D138	
D139	
D140	

[illegible]

929

930

-continued

Compound No.	Structure
D144	
D145	
D146	

-continued

Compound No.	Structure
D147	
D148	
D149	

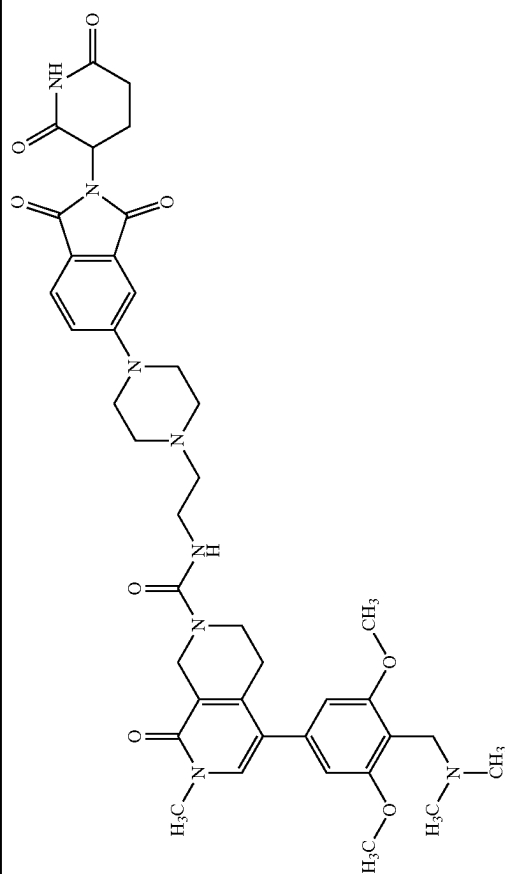
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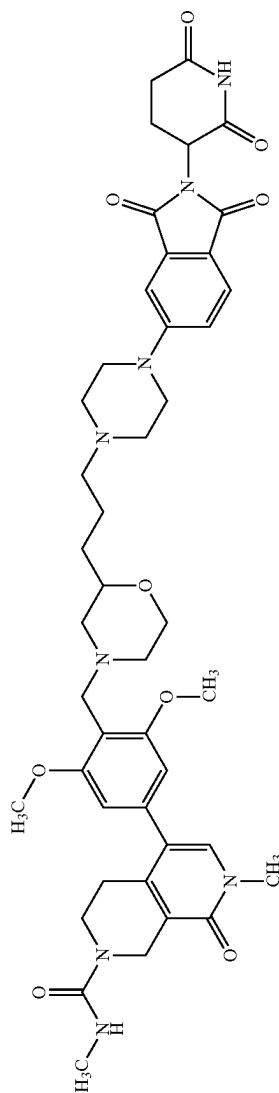
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Compound No.	Structure
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D150



D151



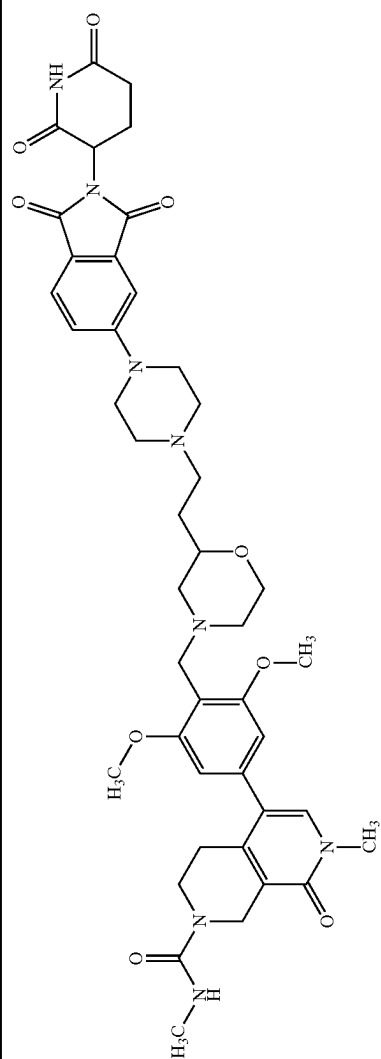
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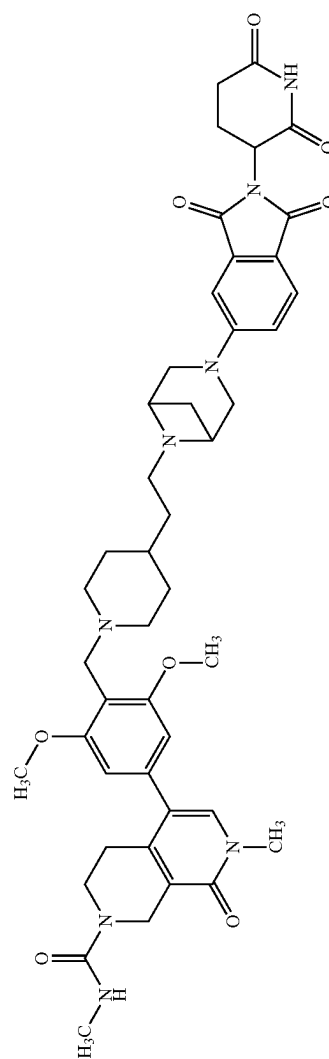
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Compound No.	Structure
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D152



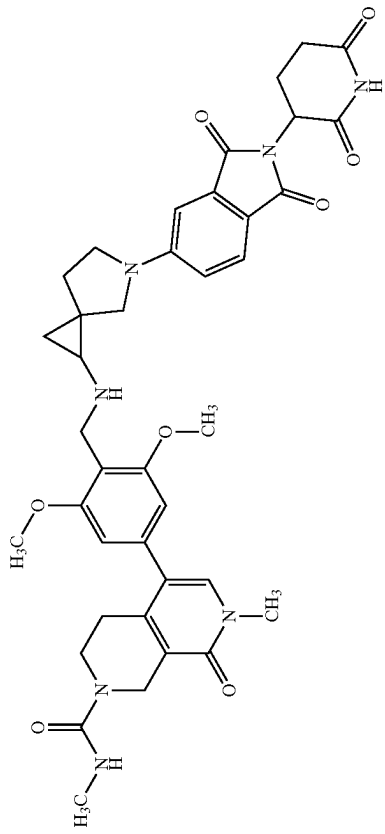
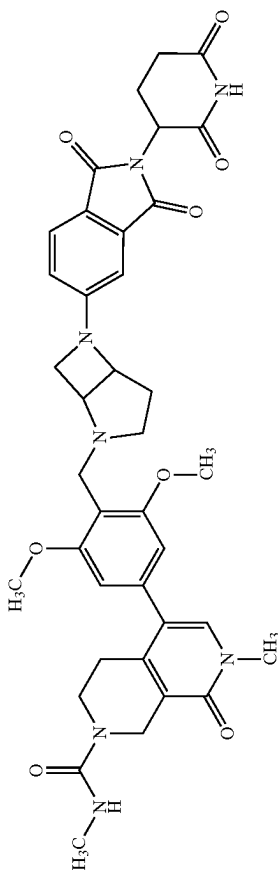
D153



939

940

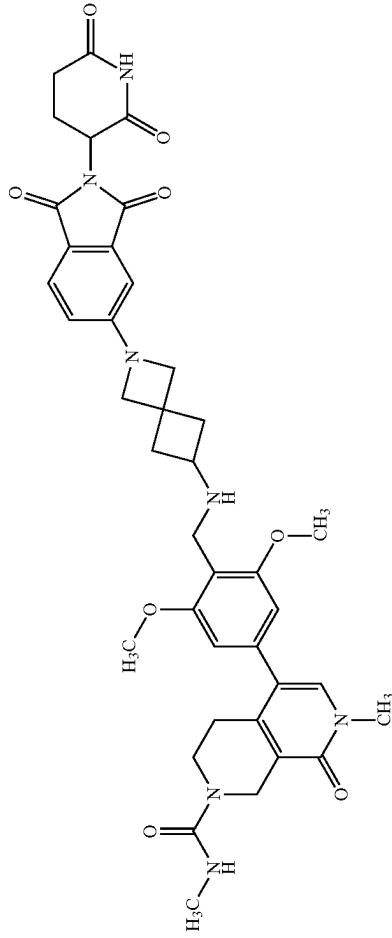
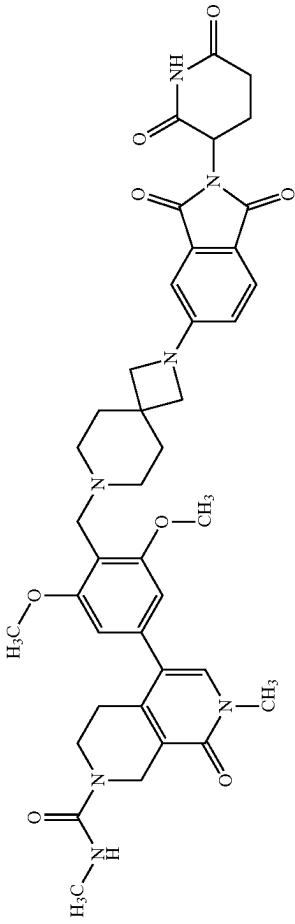
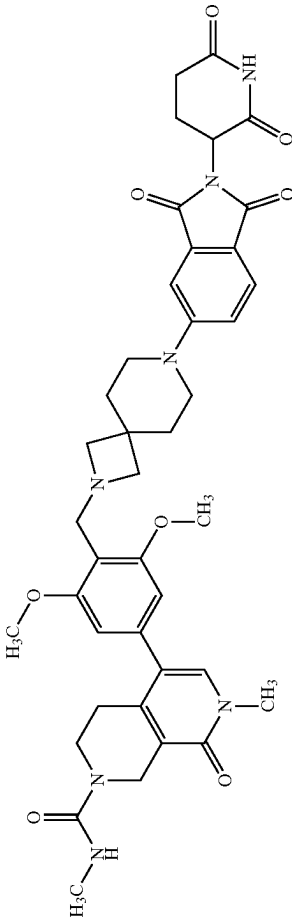
-continued

Compound No.	Structure
D156	 Chemical structure of compound D156: A complex molecule featuring a central pyridine ring substituted with a methyl group (CH₃) and a methylamino group (NHCH₃). This central ring is connected to a benzene ring, which is further substituted with a methoxy group (OCH₃) and a methoxymethyl group (CH₂OCH₃). The methoxymethyl group is linked to a bicyclic amine system (a cyclopropane fused to a pyrrolidine ring). This bicyclic system is connected to a benzimidazole ring system, which is substituted with a methyl group (CH₃) and a methylamino group (NHCH₃).
D157	 Chemical structure of compound D157: A complex molecule featuring a central pyridine ring substituted with a methyl group (CH₃) and a methylamino group (NHCH₃). This central ring is connected to a benzene ring, which is further substituted with a methoxy group (OCH₃) and a methoxymethyl group (CH₂OCH₃). The methoxymethyl group is linked to a bicyclic amine system (a cyclopropane fused to a pyrrolidine ring). This bicyclic system is connected to a benzimidazole ring system, which is substituted with a methyl group (CH₃) and a methylamino group (NHCH₃).

941

942

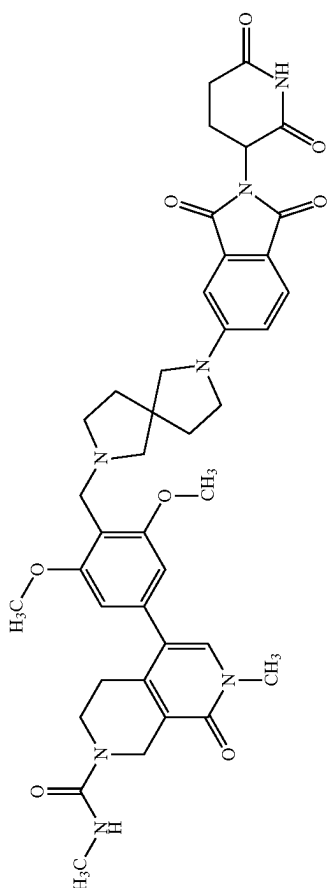
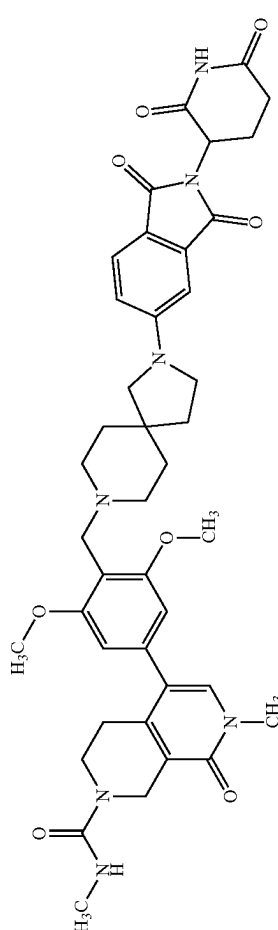
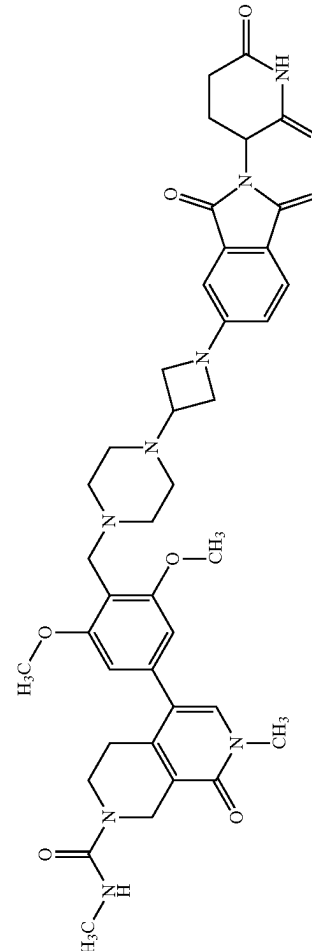
-continued

Compound No.	Structure
D158	 Chemical structure of compound D158: A 6-methoxy-2-methyl-1,2,3,4-tetrahydropyridine-5-carboxamide derivative. The pyridine ring is substituted with a methyl group at position 2, a methoxy group at position 6, and a carboxamide group at position 5. The pyridine ring is connected via a methylene bridge to a piperidine ring, which is further connected via a methylene bridge to a benzimidazole ring system. The benzimidazole ring is substituted with a methoxy group at position 2 and a carboxamide group at position 3. The benzimidazole ring is connected via a methylene bridge to a piperidine ring, which is further connected via a methylene bridge to a benzimidazole ring system. The benzimidazole ring is substituted with a methoxy group at position 2 and a carboxamide group at position 3.
D159	 Chemical structure of compound D159: A 6-methoxy-2-methyl-1,2,3,4-tetrahydropyridine-5-carboxamide derivative. The pyridine ring is substituted with a methyl group at position 2, a methoxy group at position 6, and a carboxamide group at position 5. The pyridine ring is connected via a methylene bridge to a piperidine ring, which is further connected via a methylene bridge to a benzimidazole ring system. The benzimidazole ring is substituted with a methoxy group at position 2 and a carboxamide group at position 3. The benzimidazole ring is connected via a methylene bridge to a piperidine ring, which is further connected via a methylene bridge to a benzimidazole ring system. The benzimidazole ring is substituted with a methoxy group at position 2 and a carboxamide group at position 3.
D161	 Chemical structure of compound D161: A 6-methoxy-2-methyl-1,2,3,4-tetrahydropyridine-5-carboxamide derivative. The pyridine ring is substituted with a methyl group at position 2, a methoxy group at position 6, and a carboxamide group at position 5. The pyridine ring is connected via a methylene bridge to a piperidine ring, which is further connected via a methylene bridge to a benzimidazole ring system. The benzimidazole ring is substituted with a methoxy group at position 2 and a carboxamide group at position 3. The benzimidazole ring is connected via a methylene bridge to a piperidine ring, which is further connected via a methylene bridge to a benzimidazole ring system. The benzimidazole ring is substituted with a methoxy group at position 2 and a carboxamide group at position 3.

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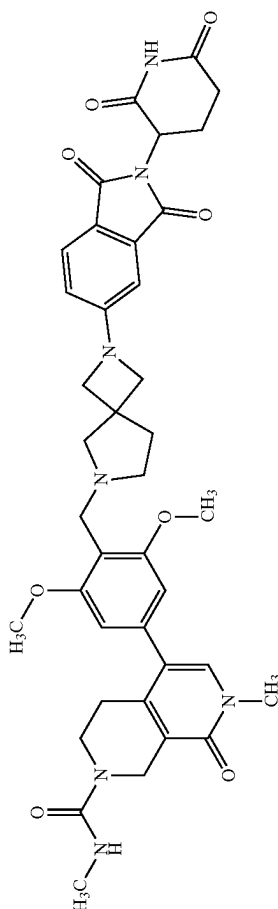
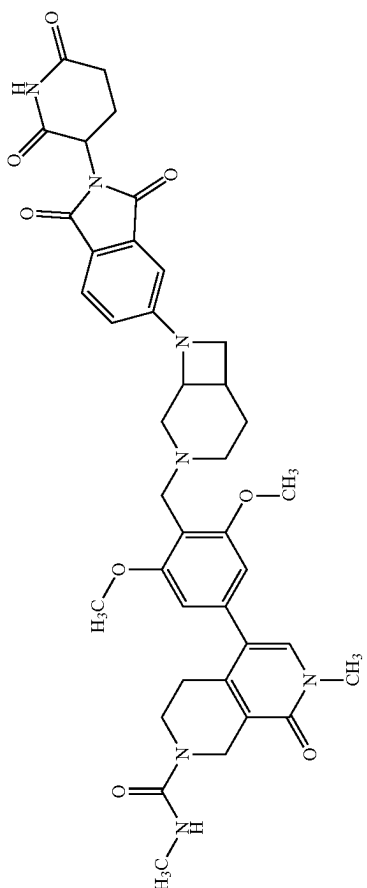
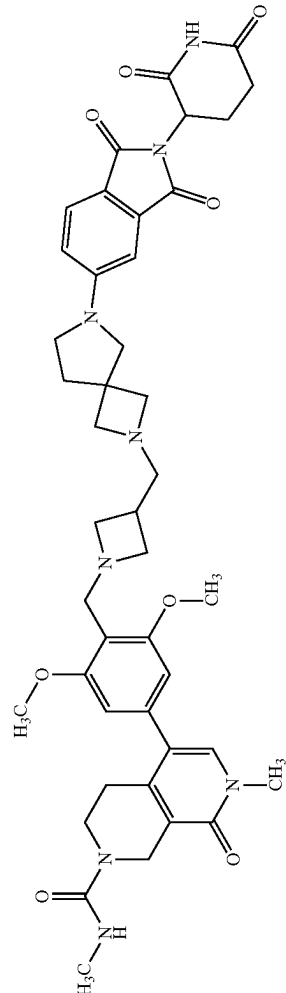
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Compound No.	Structure
D162	
D163	
D164	

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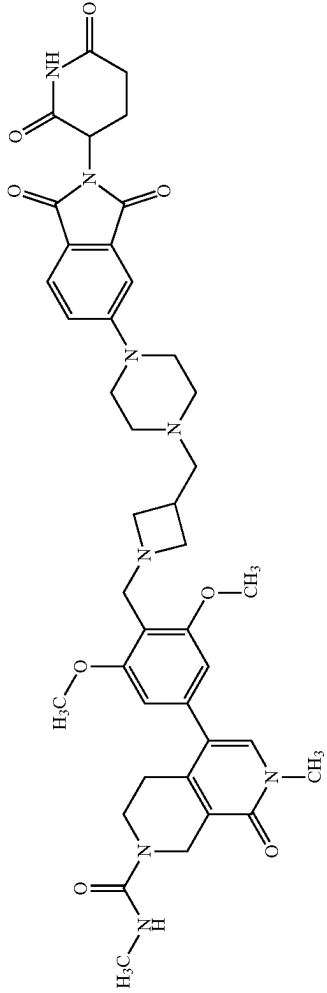
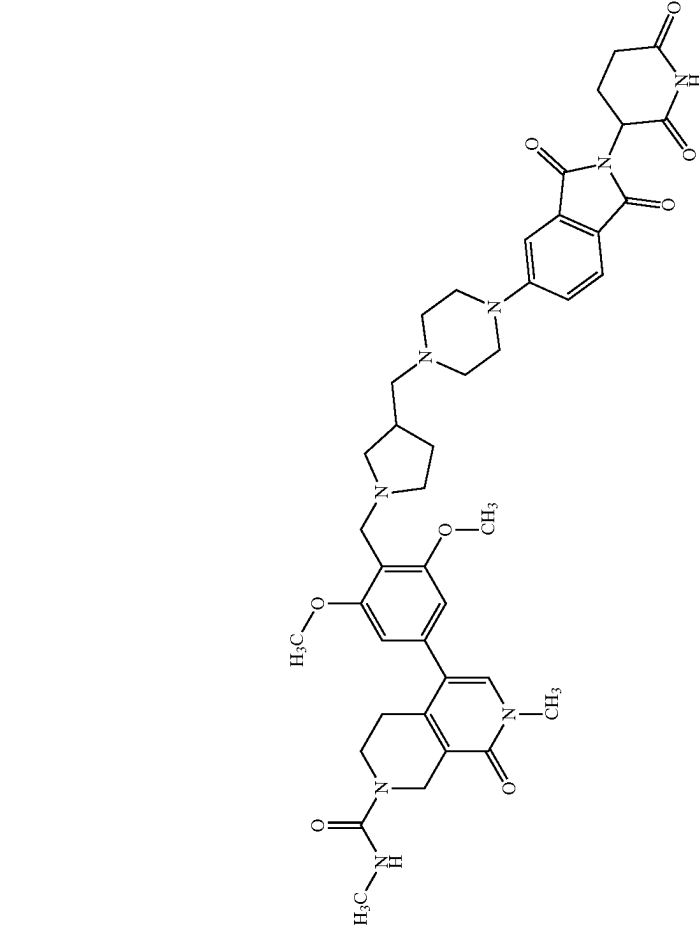
-continued

Compound No.	Structure
D165	
D166	
D167	

947

948

-continued

Compound No.	Structure
D168	 Chemical structure of compound D168: A complex molecule featuring a central benzene ring substituted with a methoxy group (H₃C-O-), a methyl 2-(4-methoxyphenyl)-6-methyl-1,2,3,4-tetrahydropyridine-5-carboxylate group, and a 1-(4-methoxyphenyl)-4-methyl-1H-pyrazole-5-carboxylate group. The pyrazole ring is further substituted with a 1-(4-methoxyphenyl)-4-methyl-1H-pyrazole-5-carboxylate group.
D169	 Chemical structure of compound D169: A complex molecule featuring a central benzene ring substituted with a methoxy group (H₃C-O-), a methyl 2-(4-methoxyphenyl)-6-methyl-1,2,3,4-tetrahydropyridine-5-carboxylate group, and a 1-(4-methoxyphenyl)-4-methyl-1H-pyrazole-5-carboxylate group. The pyrazole ring is further substituted with a 1-(4-methoxyphenyl)-4-methyl-1H-pyrazole-5-carboxylate group.

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-continued

Compound No.	Structure
D170	
D171	
D172	

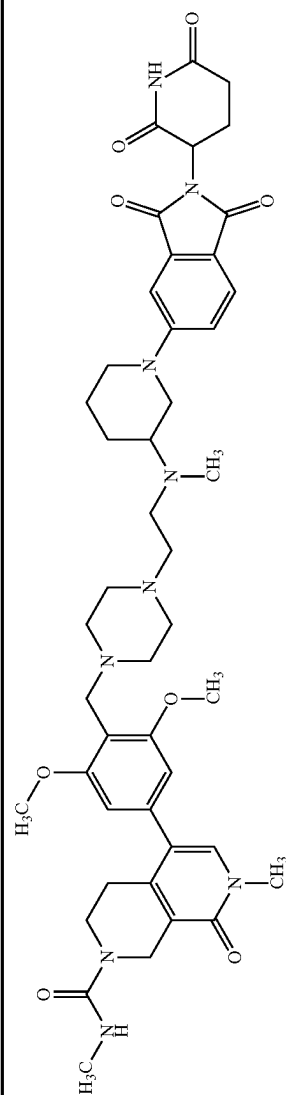
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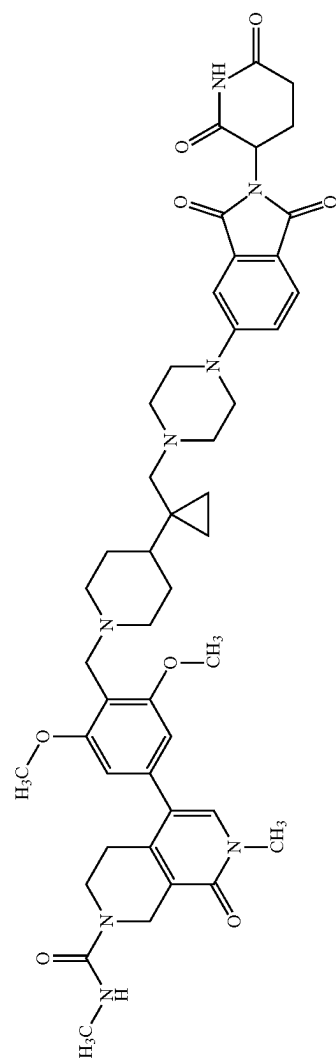
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Compound No.	Structure
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D173



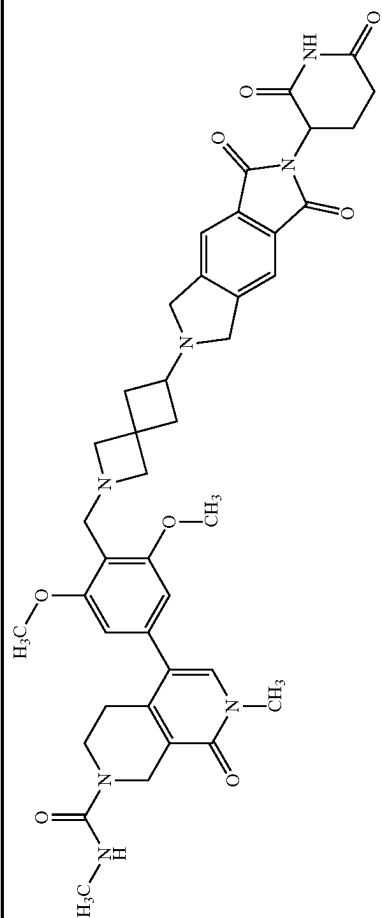
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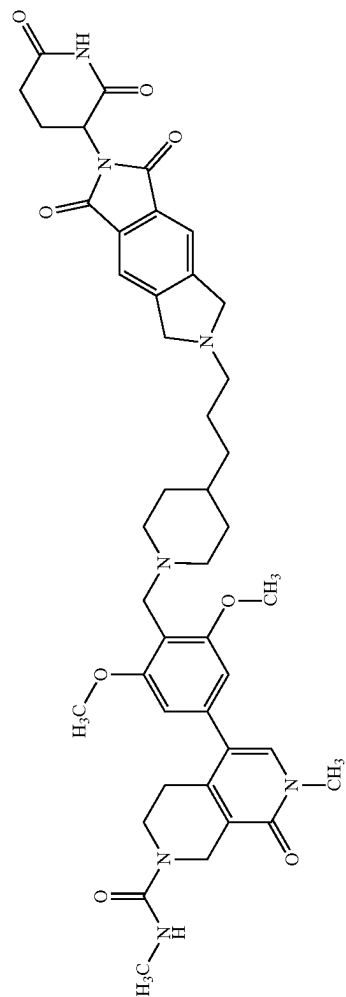
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[illegible]

Compound No.	Structure
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D178



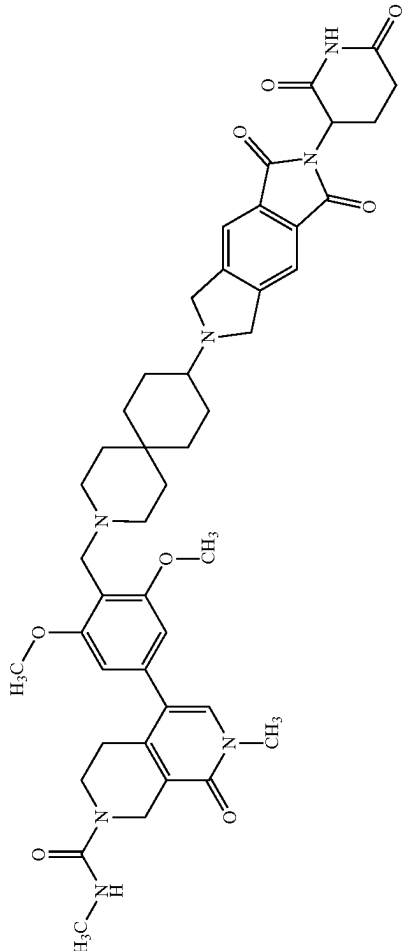
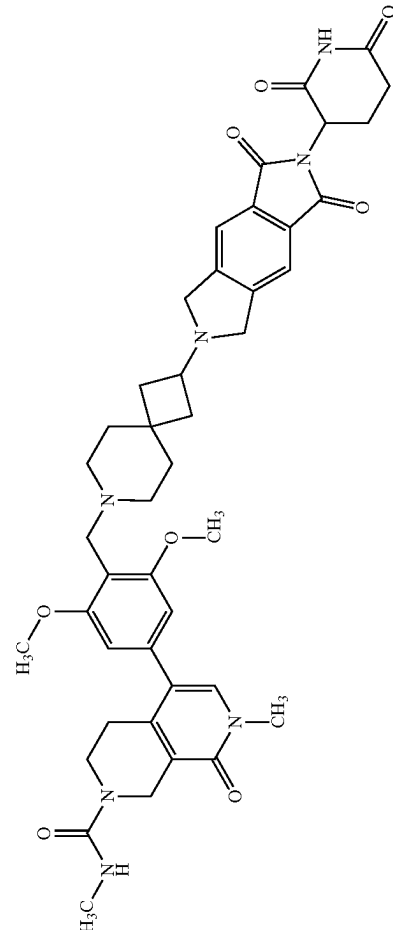
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Compound No.	Structure
D179	
D180	

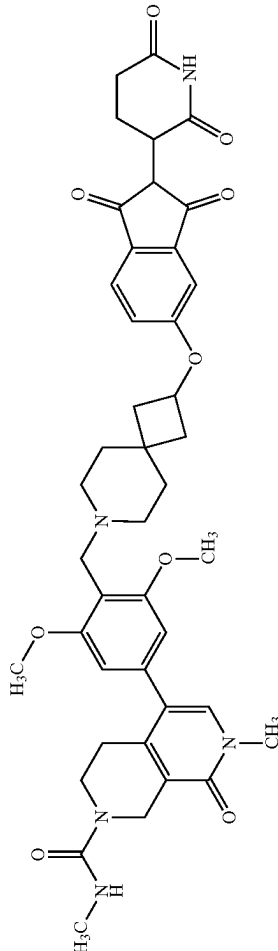
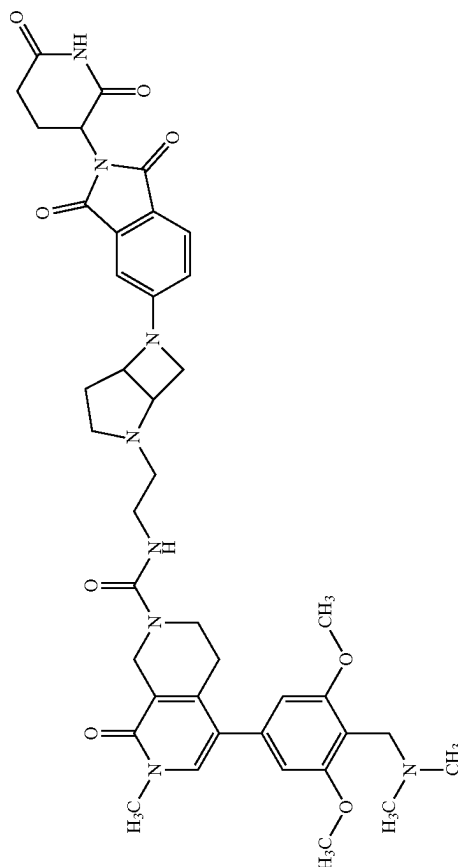
959

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-continued

Compound No.	Structure
D181	 Chemical structure of compound D181: A complex molecule featuring a central pyridine ring substituted with a methyl group (CH₃) and a methylamino group (NHCH₃). This pyridine ring is connected to a benzene ring, which is further substituted with a methoxy group (H₃C-O) and a methoxy group (O-CH₃). The benzene ring is linked to a piperidine ring, which is connected to a cyclohexane ring. The cyclohexane ring is further connected to a piperidine ring, which is linked to a benzimidazole system. The benzimidazole system is substituted with a methyl group (CH₃) and a methylamino group (NHCH₃).
D182	 Chemical structure of compound D182: A complex molecule featuring a central pyridine ring substituted with a methyl group (CH₃) and a methylamino group (NHCH₃). This pyridine ring is connected to a benzene ring, which is further substituted with a methoxy group (H₃C-O) and a methoxy group (O-CH₃). The benzene ring is linked to a piperidine ring, which is connected to a cyclobutane ring. The cyclobutane ring is further connected to a piperidine ring, which is linked to a benzimidazole system. The benzimidazole system is substituted with a methyl group (CH₃) and a methylamino group (NHCH₃).

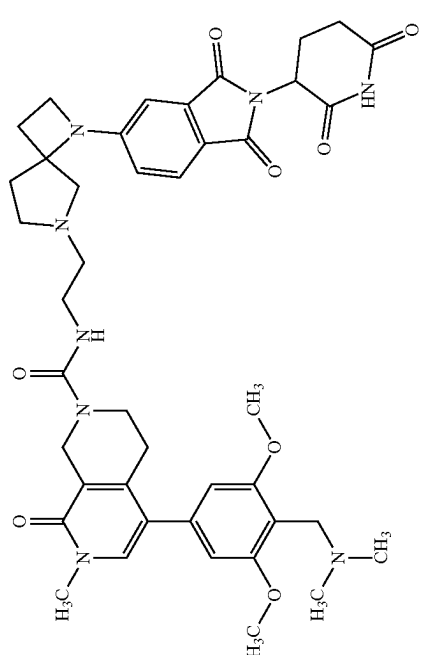
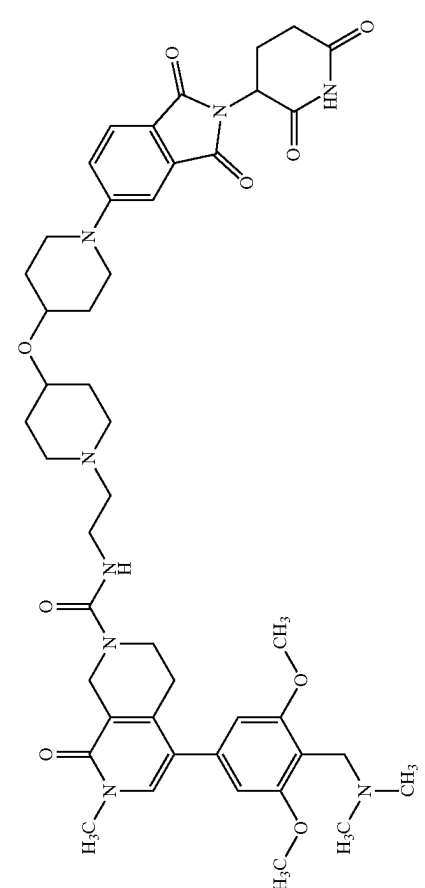
-continued

Compound No.	Structure
D183	 Chemical structure of compound D183: A complex molecule featuring a central pyridine ring substituted with a methylamino group, a methoxy group, and a 4-methoxyphenyl group. This central ring is connected via a methylene bridge to a piperidine ring, which is further linked to a cyclobutane ring. The cyclobutane ring is connected to a 4-methoxyphenyl group, which is in turn linked to a 1,2,3,4-tetrahydronaphthalene-1-carboxamide moiety. The tetrahydronaphthalene moiety is substituted with a methylamino group and a methoxy group. The entire structure is terminated by a 2,6-pyridinedicarboxamide group.
D184	 Chemical structure of compound D184: A complex molecule featuring a central pyridine ring substituted with a methylamino group, a methoxy group, and a 4-methoxyphenyl group. This central ring is connected via a methylene bridge to a piperidine ring, which is further linked to a cyclobutane ring. The cyclobutane ring is connected to a 4-methoxyphenyl group, which is in turn linked to a 1,2,3,4-tetrahydronaphthalene-1-carboxamide moiety. The tetrahydronaphthalene moiety is substituted with a methylamino group and a methoxy group. The entire structure is terminated by a 2,6-pyridinedicarboxamide group.

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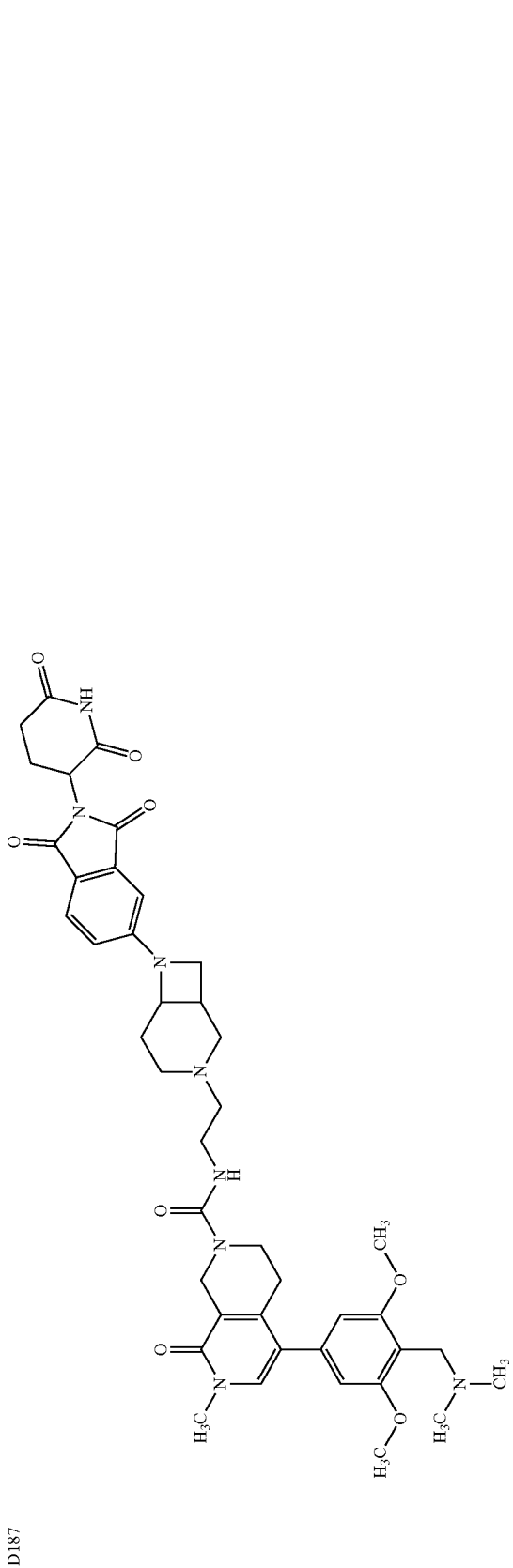
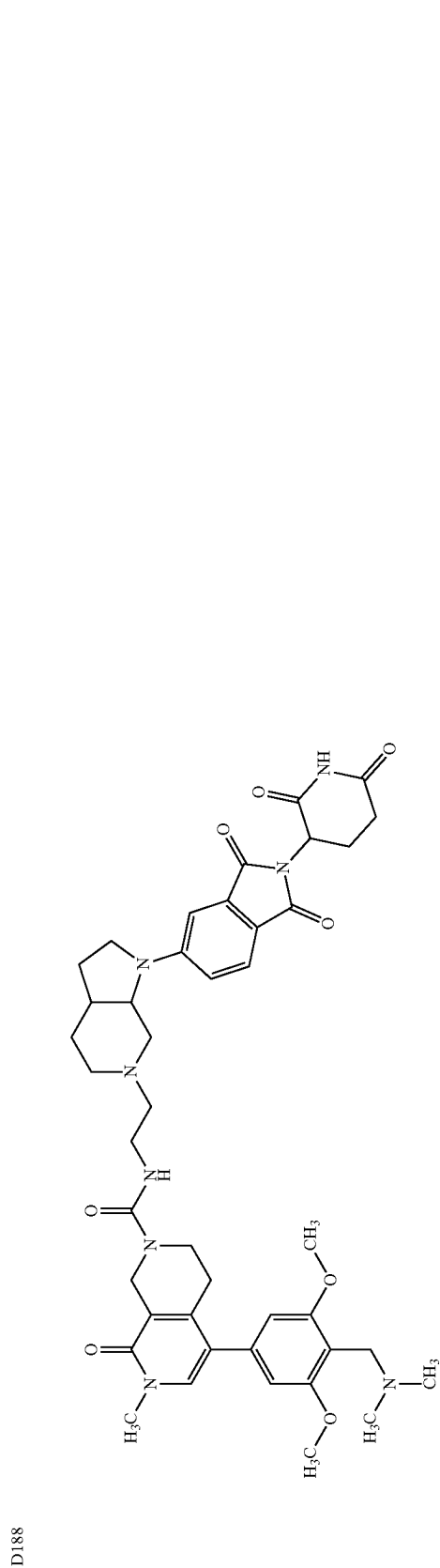
-continued

Compound No.	Structure
D185	 Chemical structure of compound D185: A 4-methoxy-2-(dimethylaminomethyl)phenyl group is connected at the para position to a 6-methoxy-2-methyl-1,2,3,4-tetrahydropyrimidin-5(1H)-one. This is further connected via an amide linkage to a 2-(2,2,5-trimethyl-1,3-dioxane-5-carbonyl)pyrrolidine. The pyrrolidine ring is substituted with a 1-(2,6-dimethyl-4-(2-oxo-2,3,4,5-tetrahydro-1H-benzothiazol-5-yl)ethyl)pyrrolidin-2-yl group.
D186	 Chemical structure of compound D186: A 4-methoxy-2-(dimethylaminomethyl)phenyl group is connected at the para position to a 6-methoxy-2-methyl-1,2,3,4-tetrahydropyrimidin-5(1H)-one. This is further connected via an amide linkage to a 2-(2,2,5-trimethyl-1,3-dioxane-5-carbonyl)pyrrolidine. The pyrrolidine ring is substituted with a 1-(2,6-dimethyl-4-(2-oxo-2,3,4,5-tetrahydro-1H-benzothiazol-5-yl)ethyl)pyrrolidin-2-yl group.

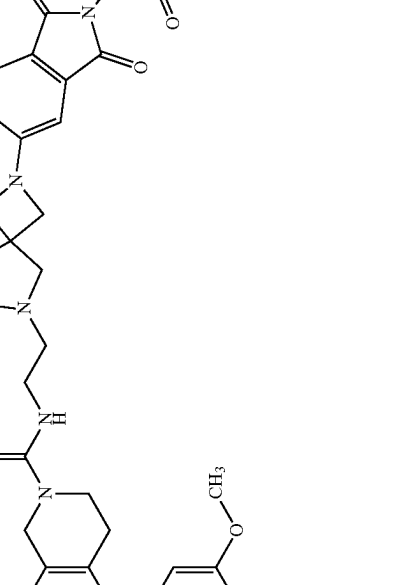
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-continued

Compound No.	Structure
D187	 Chemical structure of compound D187: A complex molecule featuring a central benzene ring substituted with a methoxy group (-OCH₃) and a dimethylaminomethyl group (-CH₂N(CH₃)₂). This central ring is connected to a 6-methyl-1,2,3,4-tetrahydropyridine-2-carbonyl group. The carbonyl group is further linked via an amide bond to a 4-(4,5,6,7-tetrahydro-1H-indolizin-2-yl)butan-1-amine derivative, which is terminated by a 2,6-pyridinedione moiety.
D188	 Chemical structure of compound D188: A complex molecule featuring a central benzene ring substituted with a methoxy group (-OCH₃) and a dimethylaminomethyl group (-CH₂N(CH₃)₂). This central ring is connected to a 6-methyl-1,2,3,4-tetrahydropyridine-2-carbonyl group. The carbonyl group is further linked via an amide bond to a 4-(4,5,6,7-tetrahydro-1H-indolizin-2-yl)butan-1-amine derivative, which is terminated by a 2,6-pyridinedione moiety.

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Compound No.	Structure
D189	
D190	

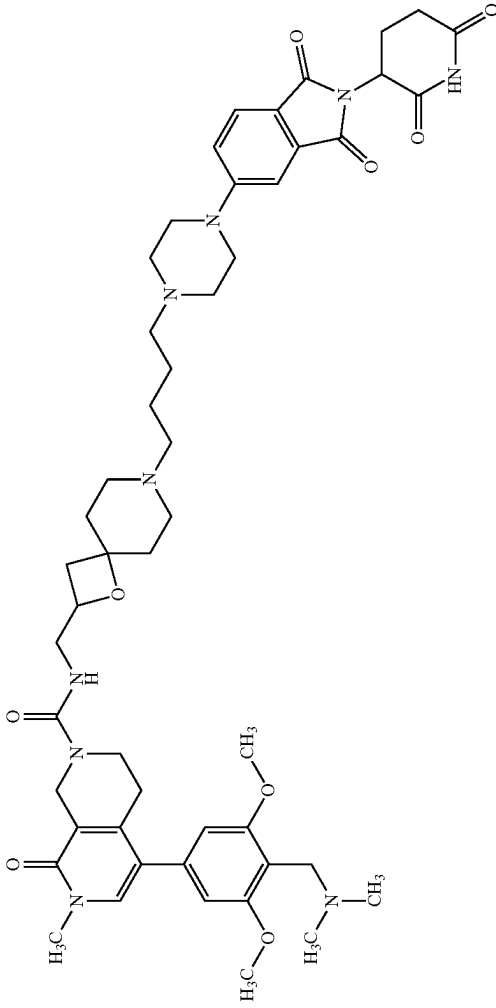
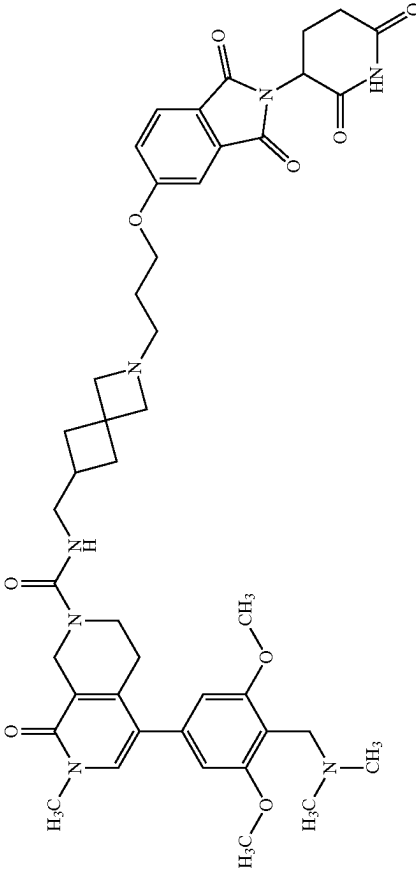
-continued

Compound No.	Structure
D191	
D192	

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-continued

Compound No.	Structure
D193	 Chemical structure of compound D193: A 6-methoxy-2-(dimethylaminomethyl)-3,4-dihydroquinoline-2-carboxamide derivative. The quinoline ring is substituted with a methoxy group at position 6 and a dimethylaminomethyl group at position 3. The amide group is connected via a 1,3-dioxolane ring to a piperazine ring, which is further connected via a 1,4-bis(piperazin-2-yl)butane linker to a benzimidazole-2-carboxamide derivative. The benzimidazole ring is substituted with a methoxy group at position 6 and a dimethylaminomethyl group at position 3.
D194	 Chemical structure of compound D194: A 6-methoxy-2-(dimethylaminomethyl)-3,4-dihydroquinoline-2-carboxamide derivative. The quinoline ring is substituted with a methoxy group at position 6 and a dimethylaminomethyl group at position 3. The amide group is connected via a 1,3-dioxolane ring to a piperazine ring, which is further connected via a 1,4-bis(piperazin-2-yl)butane linker to a benzimidazole-2-carboxamide derivative. The benzimidazole ring is substituted with a methoxy group at position 6 and a dimethylaminomethyl group at position 3.

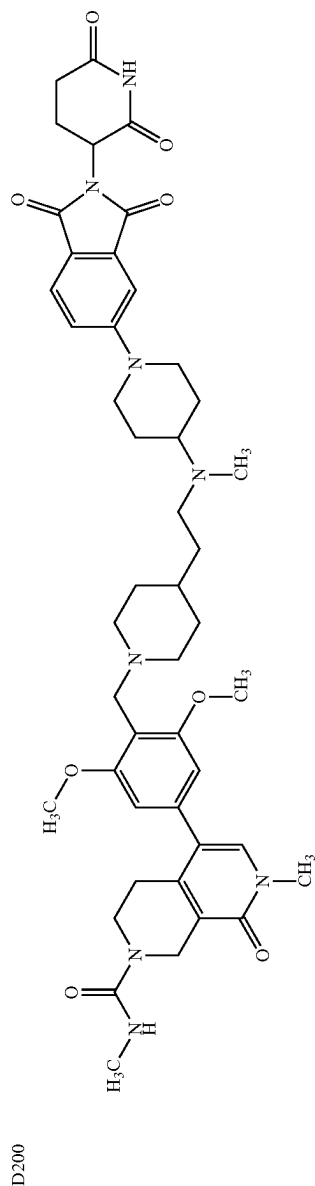
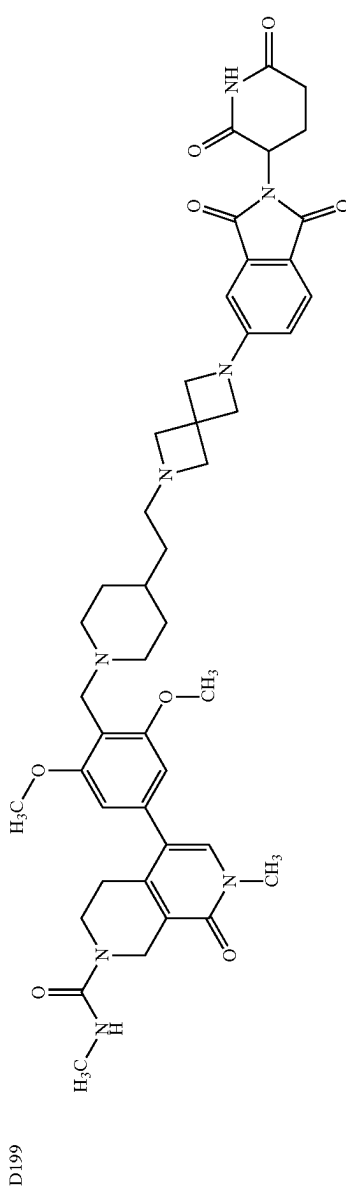
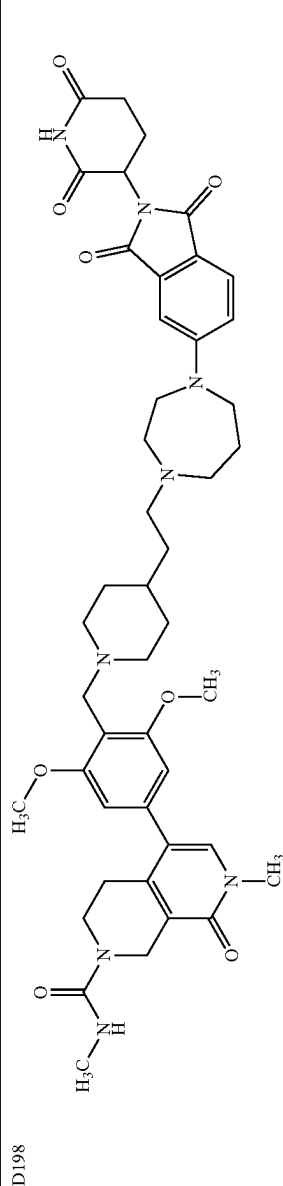
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Compound No.	Structure
D195	
D196	
D197	

Compound No.	Structure
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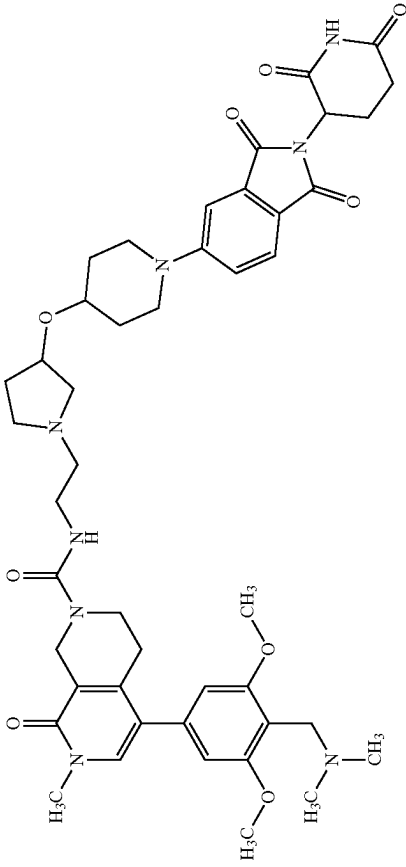
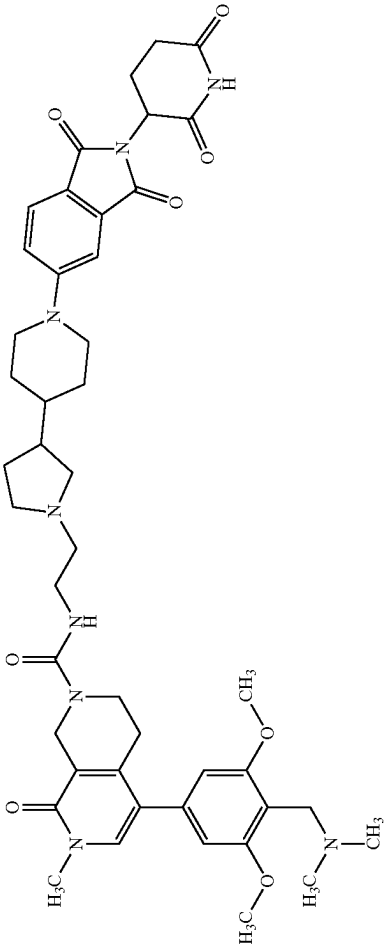
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Compound No.	Structure
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D203	<chem>CN(C)C(=O)N1CCc2c(c1)nc(C)nc2C3=CC=C(C=C3C4=CC=CC=C4N5C(=O)C(=O)N5C6=CC=CC=C6N7C(=O)C(=O)N7)C8CCN(C8)CC9CCCCC9C10CCCCC10C11OC11</chem>

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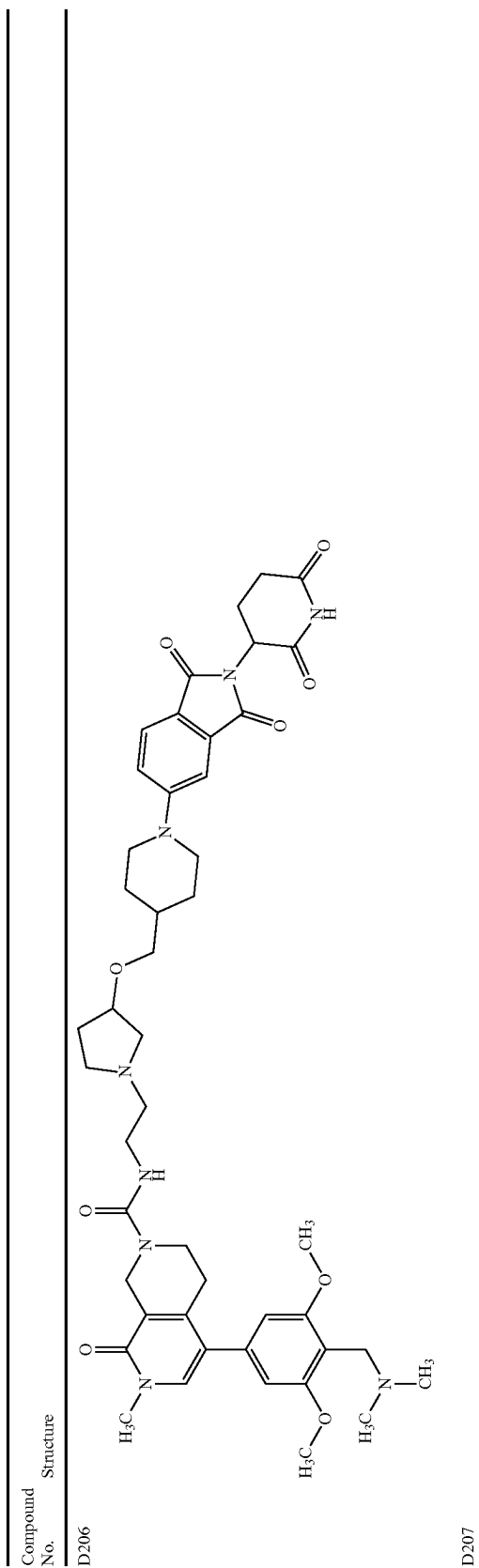
-continued

Compound No.	Structure
D204	 Chemical structure of compound D204: A 6-methoxy-2-(dimethylamino)benzoic acid derivative. The carboxylic acid group is converted to an amide, linked to a 4-(dimethylamino)pyrrolidine-1-yl group. This is further linked via an ether to a 4-(1-methyl-2-oxo-2,3-dihydro-1H-benzimidazol-5-yl)pyrrolidine-1-yl group.
D205	 Chemical structure of compound D205: A 6-methoxy-2-(dimethylamino)benzoic acid derivative. The carboxylic acid group is converted to an amide, linked to a 4-(dimethylamino)pyrrolidine-1-yl group. This is further linked via an ether to a 4-(1-methyl-2-oxo-2,3-dihydro-1H-benzimidazol-5-yl)pyrrolidine-1-yl group.

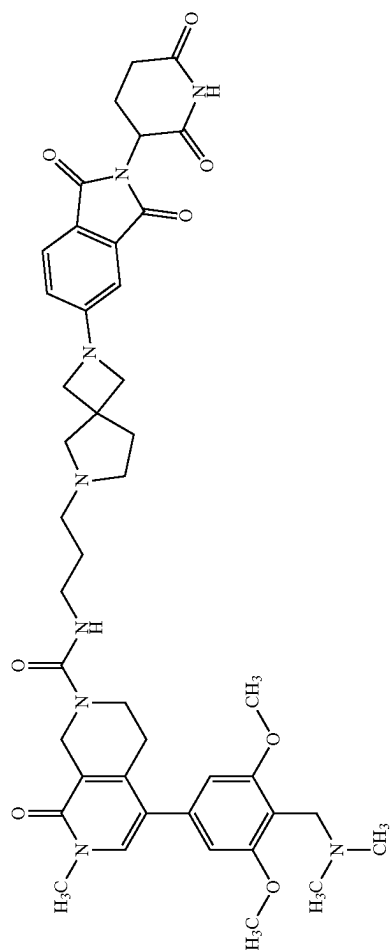
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-continued



D207



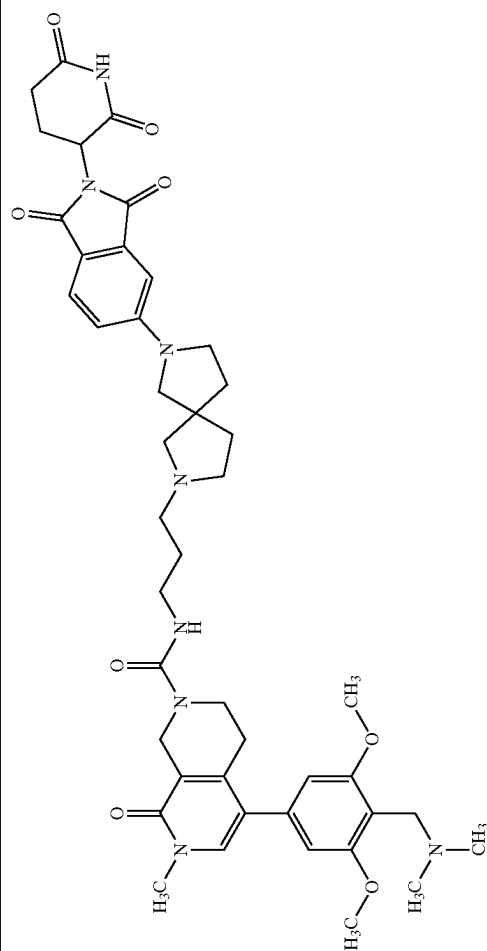
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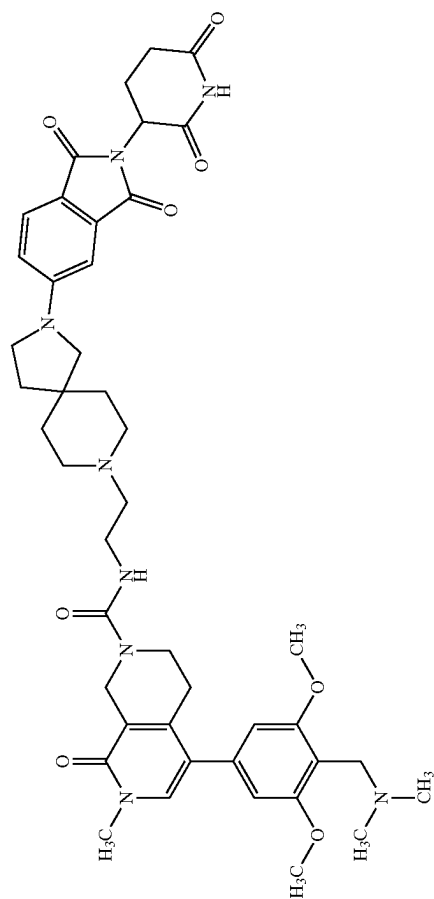
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Compound No.	Structure
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D208



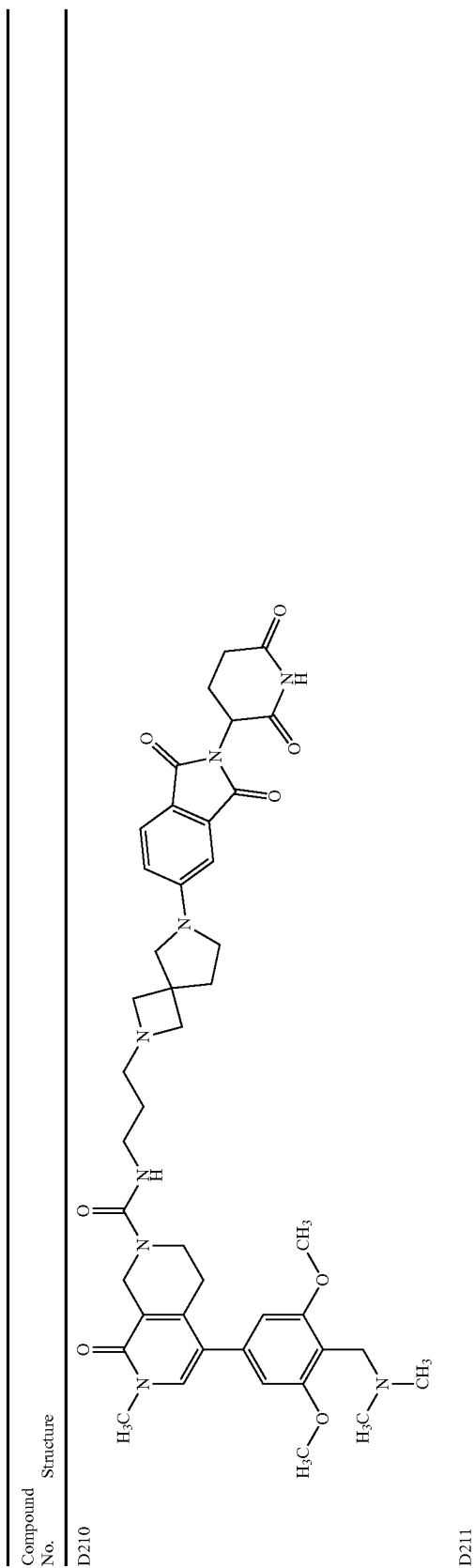
D209



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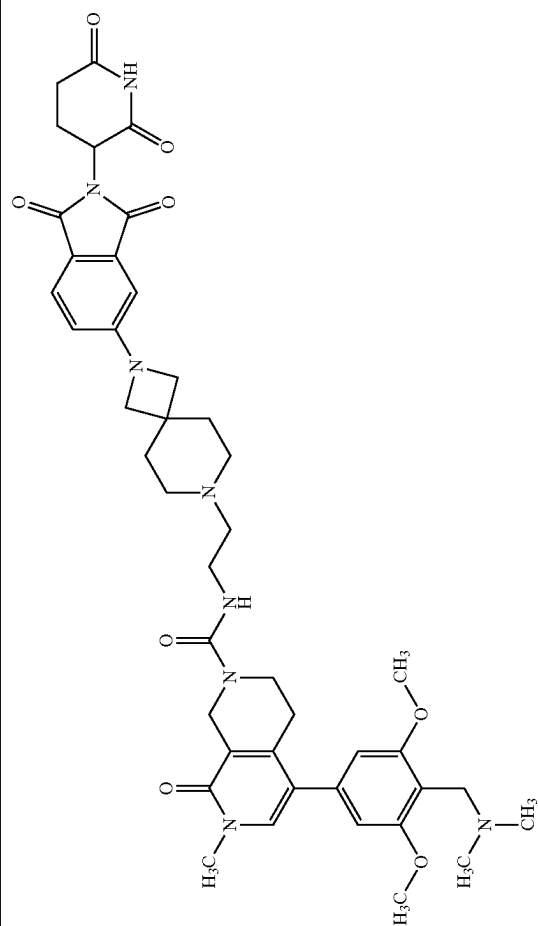
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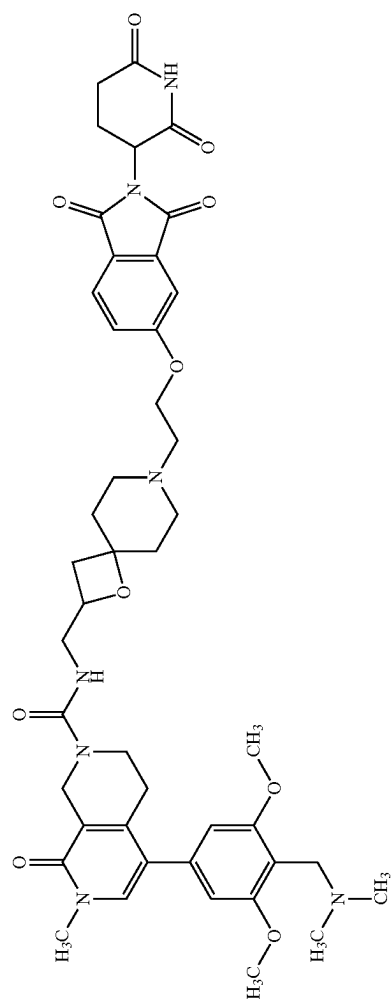
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Compound No.	Structure
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D212



D213



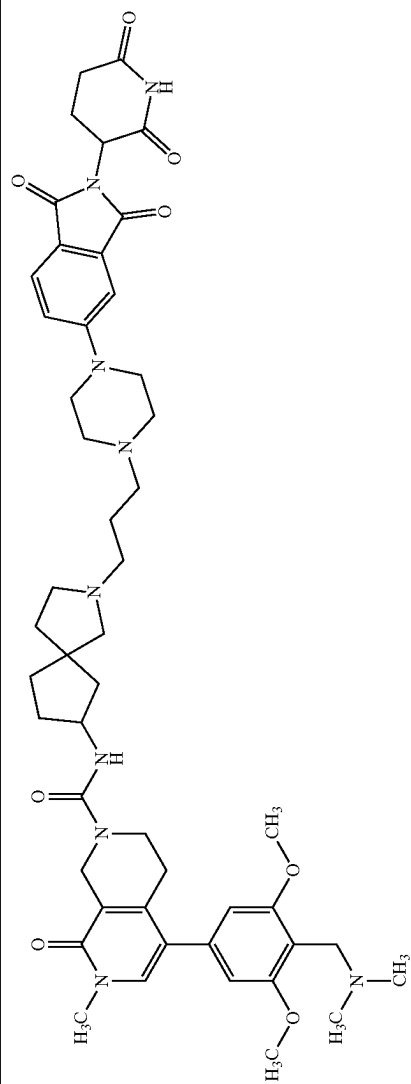
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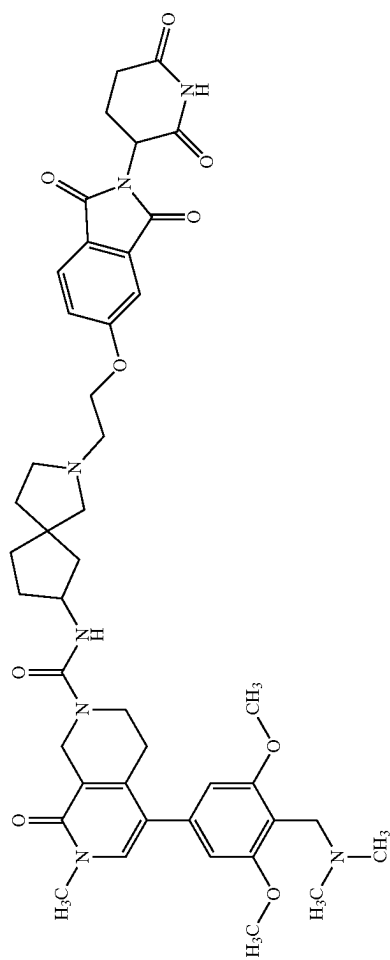
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Compound No.	Structure
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D214



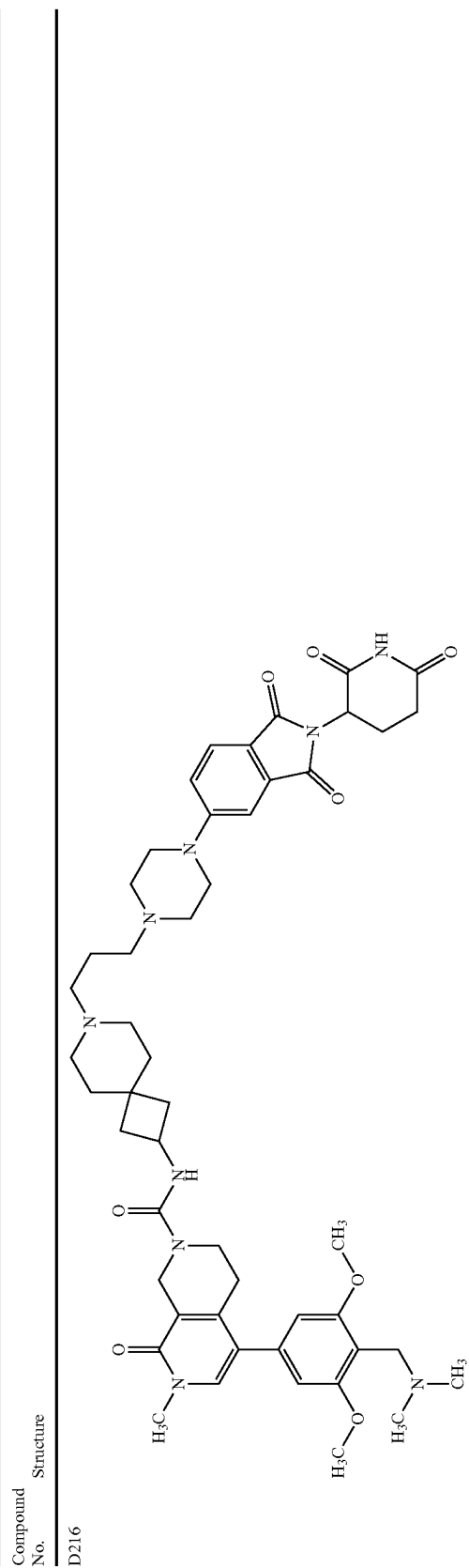
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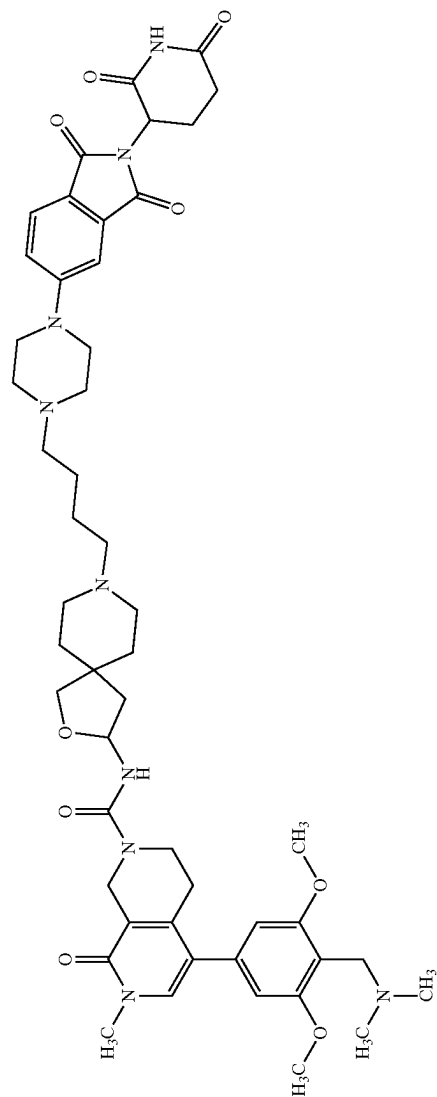
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992

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D217



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Compound No.	Structure
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D219	

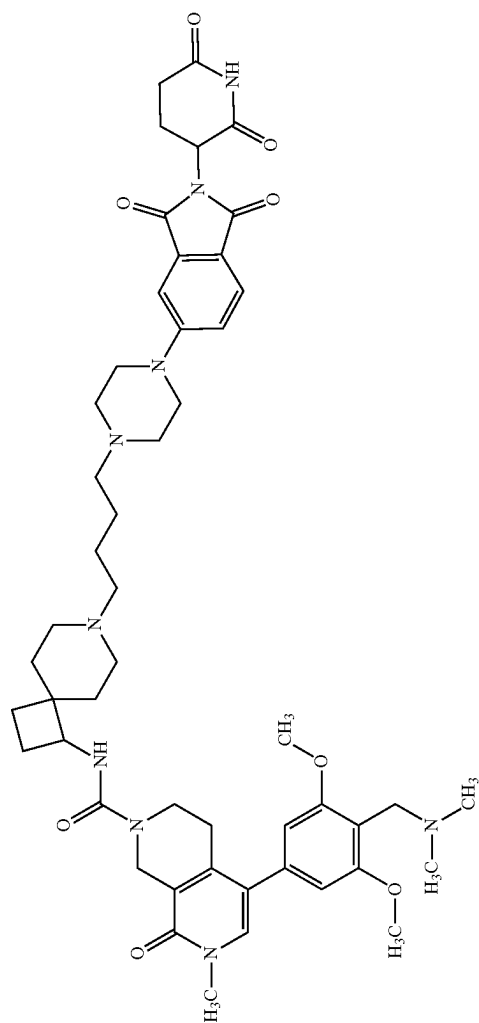
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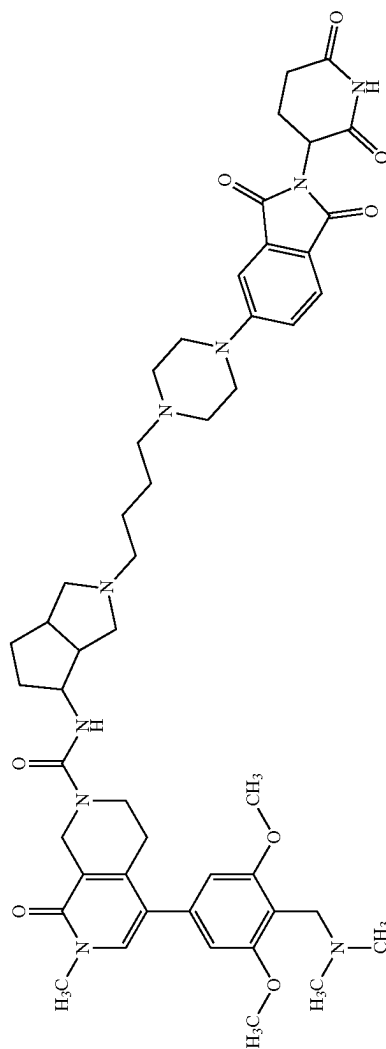
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Compound No.	Structure
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D220



D221



Compound No.	Structure
D222	
D223	

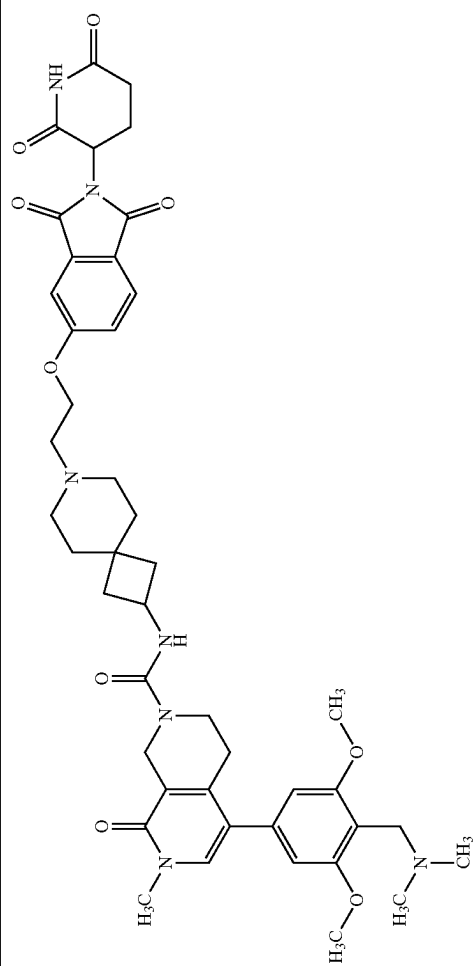
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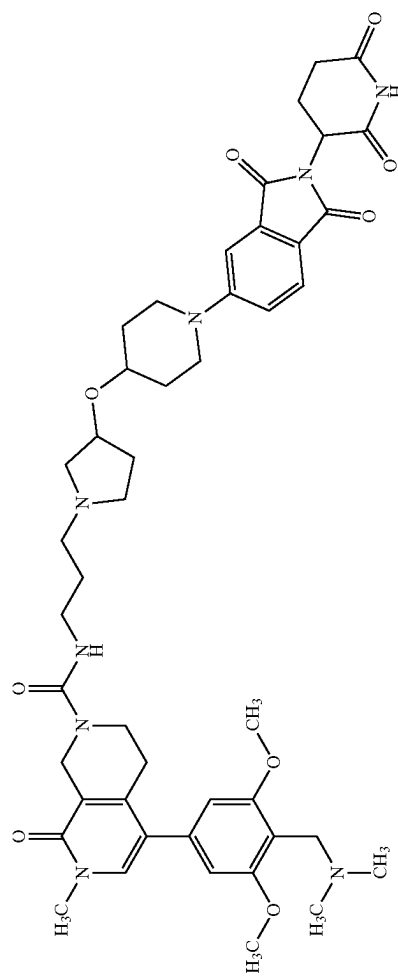
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Compound No.	Structure
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D224



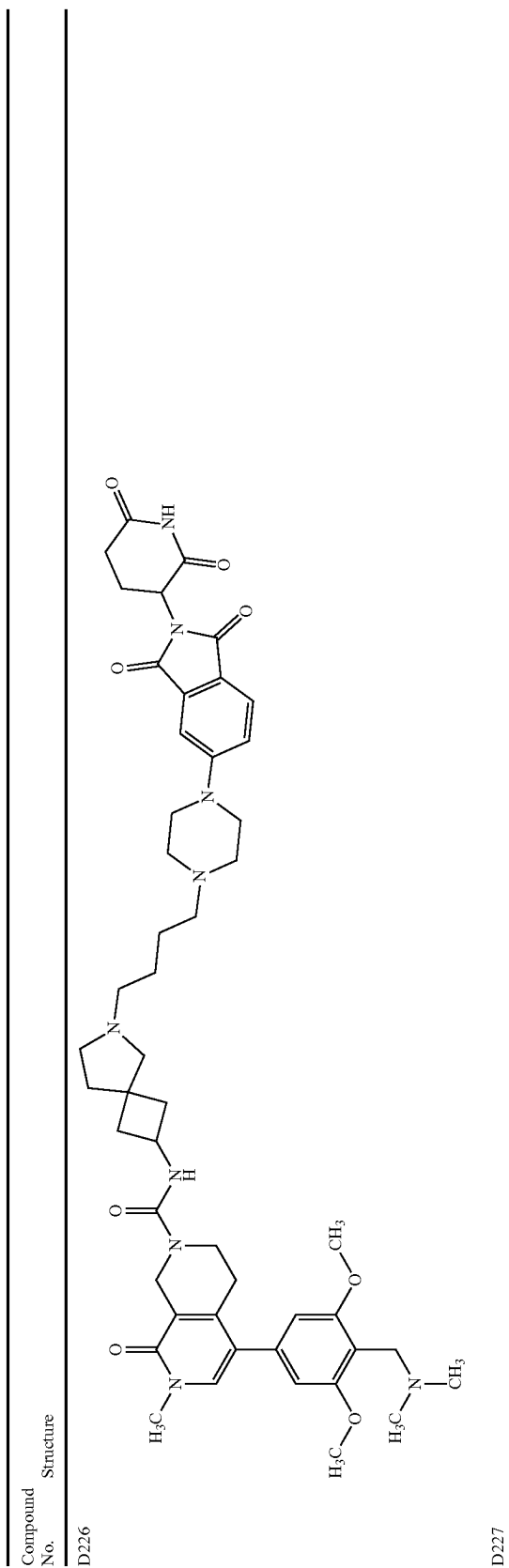
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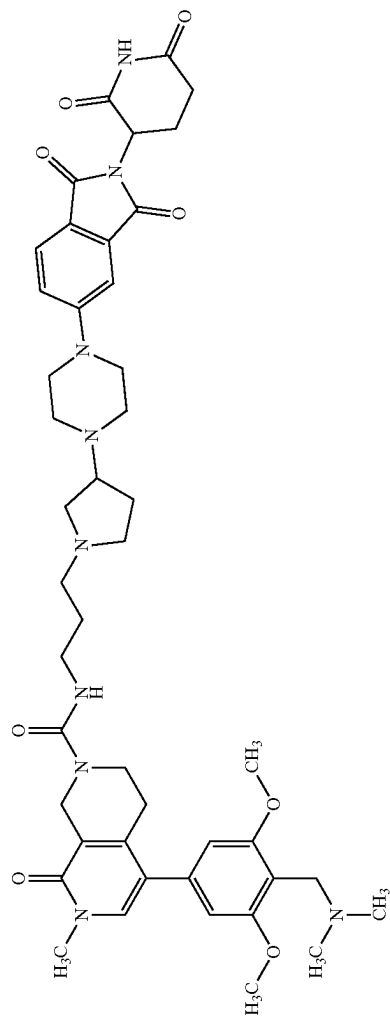
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1002

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D227



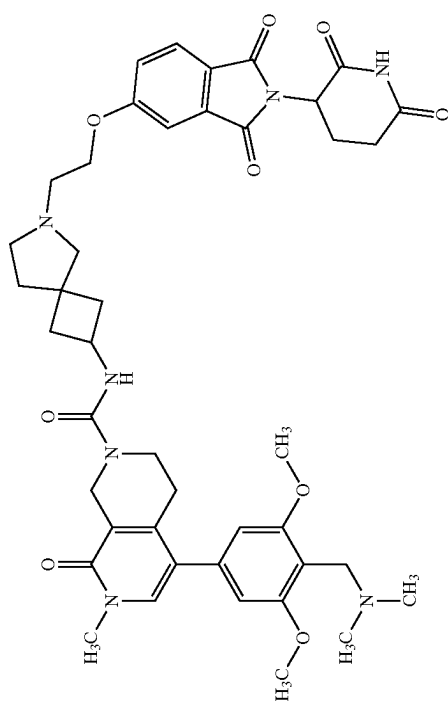
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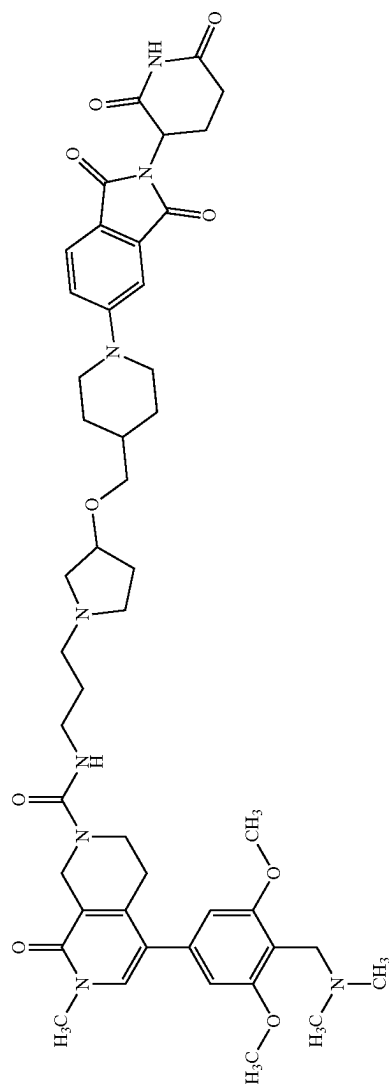
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Compound No.	Structure
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D228



D229



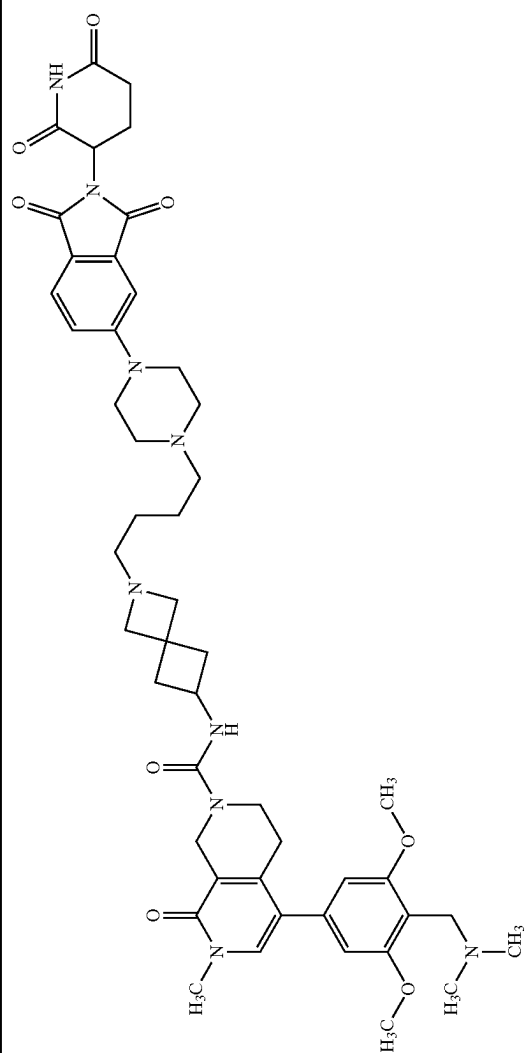
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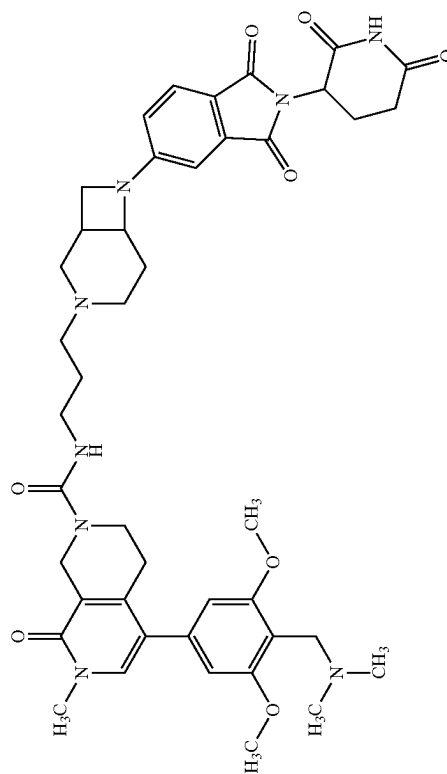
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Compound No.	Structure
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D230



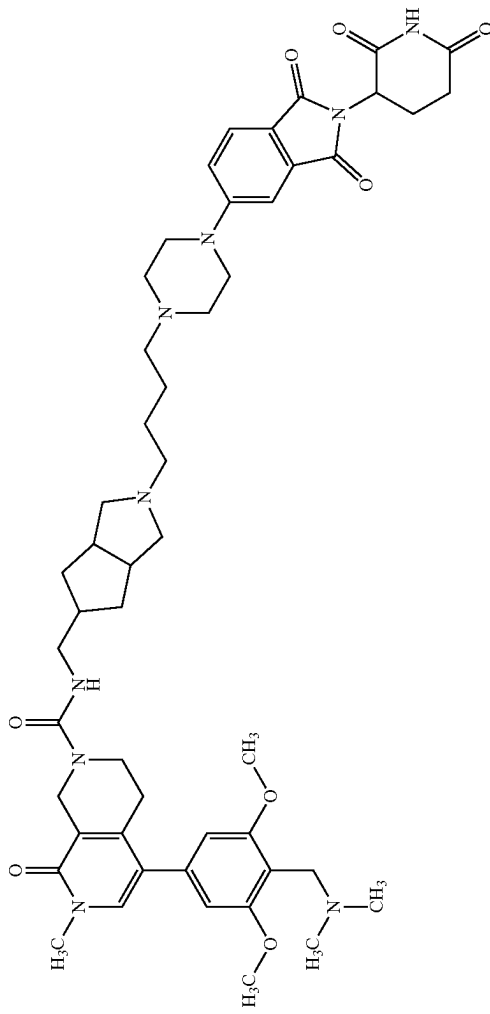
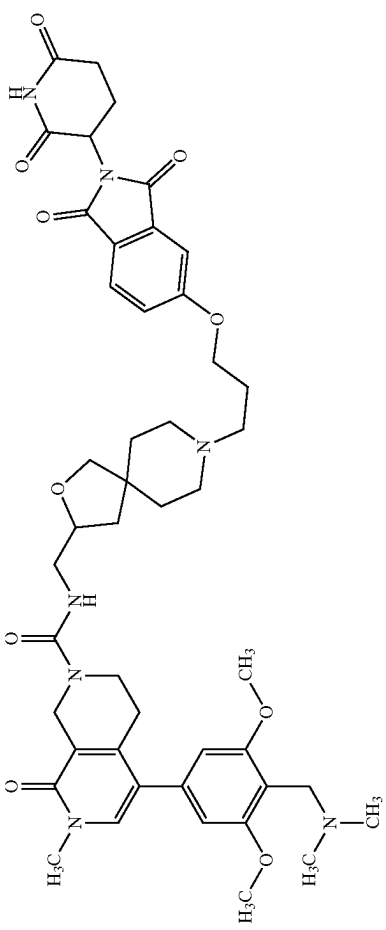
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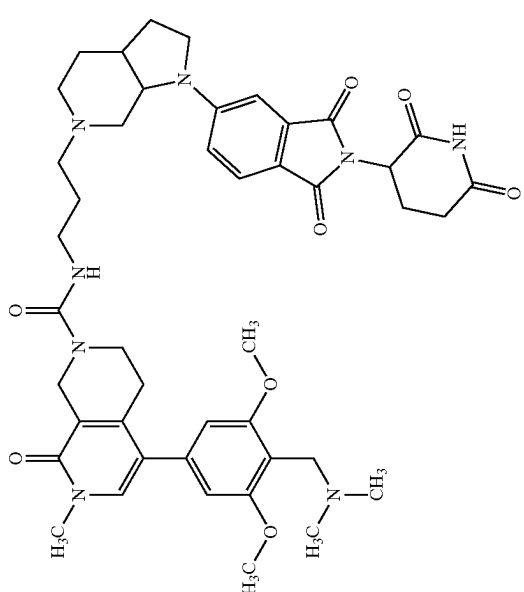
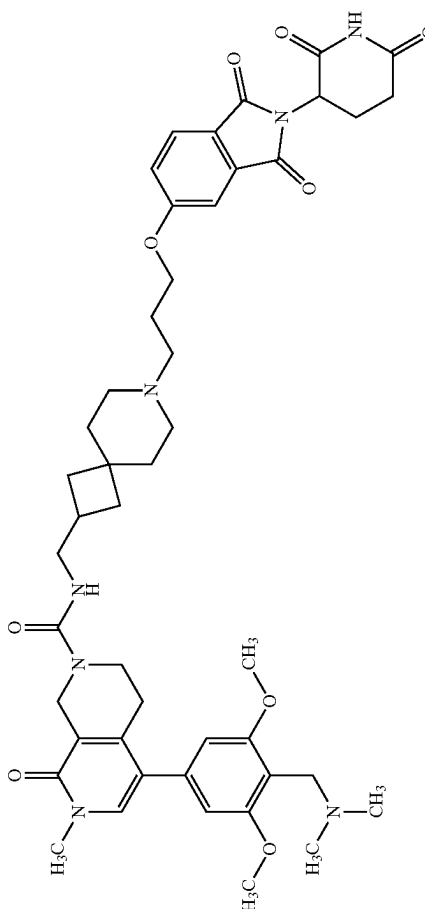
-continued

Compound No.	Structure
D232	 Chemical structure of compound D232: A complex molecule featuring a central benzene ring substituted with a methoxy group (-OCH₃) and a dimethylaminomethyl group (-CH₂N(CH₃)₂). This central ring is connected to a pyridine ring, which is further substituted with a methoxy group and a dimethylaminomethyl group. The pyridine ring is linked via an amide bond to a bicyclic system (indane), which is connected to a long chain containing a piperazine ring and a phthalimide moiety.
D233	 Chemical structure of compound D233: A complex molecule featuring a central benzene ring substituted with a methoxy group (-OCH₃) and a dimethylaminomethyl group (-CH₂N(CH₃)₂). This central ring is connected to a pyridine ring, which is further substituted with a methoxy group and a dimethylaminomethyl group. The pyridine ring is linked via an amide bond to a bicyclic system (indane), which is connected to a long chain containing a piperazine ring and a phthalimide moiety.

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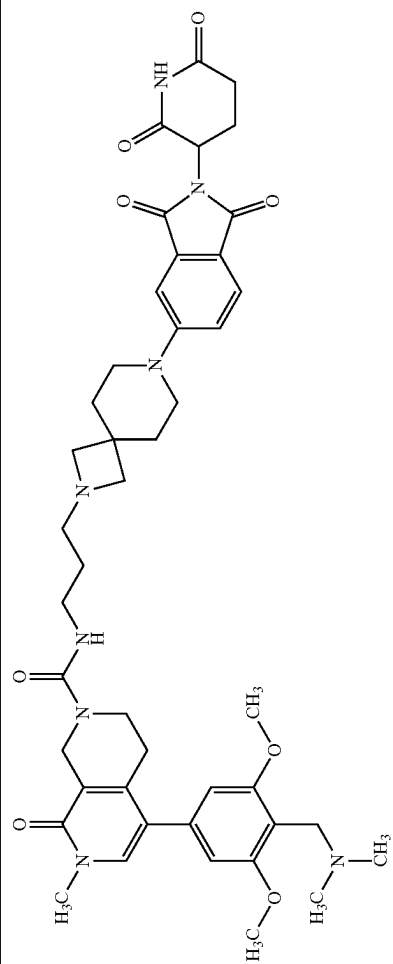
-continued

Compound No.	Structure
D234	 Chemical structure of compound D234: A 6-methoxy-2-(dimethylamino)phenyl group is connected at the 4-position to a 6-methoxy-2-(dimethylamino)pyridine-3-carbonyl group. The carbonyl group is further connected via an amide linkage to a 3,4-dihydro-1H-pyrido[1,2-a]pyrimidin-2-yl group. This pyrimidine ring is connected at its 4-position to a 1,2,3,4-tetrahydro-1H-pyridine-2-yl group. The pyridine ring is connected at its 3-position to a 1,2,3,4-tetrahydro-1H-pyridine-2-yl group. The pyridine ring is connected at its 3-position to a 1,2,3,4-tetrahydro-1H-pyridine-2-yl group. The pyridine ring is connected at its 3-position to a 1,2,3,4-tetrahydro-1H-pyridine-2-yl group.
D235	 Chemical structure of compound D235: A 6-methoxy-2-(dimethylamino)phenyl group is connected at the 4-position to a 6-methoxy-2-(dimethylamino)pyridine-3-carbonyl group. The carbonyl group is further connected via an amide linkage to a 3,4-dihydro-1H-pyrido[1,2-a]pyrimidin-2-yl group. This pyrimidine ring is connected at its 4-position to a 1,2,3,4-tetrahydro-1H-pyridine-2-yl group. The pyridine ring is connected at its 3-position to a 1,2,3,4-tetrahydro-1H-pyridine-2-yl group. The pyridine ring is connected at its 3-position to a 1,2,3,4-tetrahydro-1H-pyridine-2-yl group. The pyridine ring is connected at its 3-position to a 1,2,3,4-tetrahydro-1H-pyridine-2-yl group.

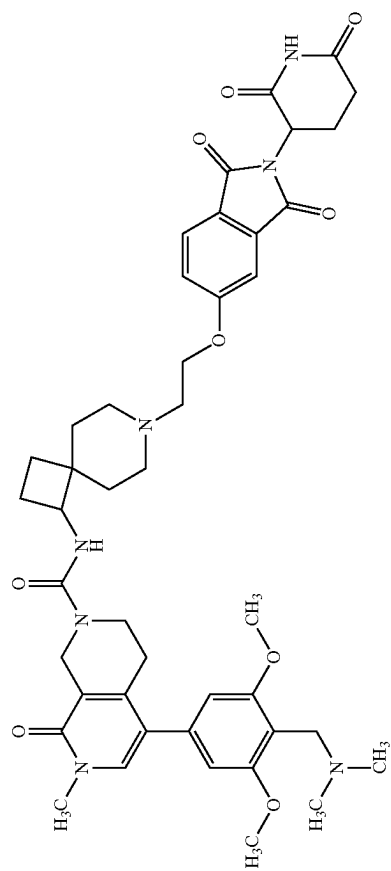
Compound No.	Structure
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Compound No.	Structure
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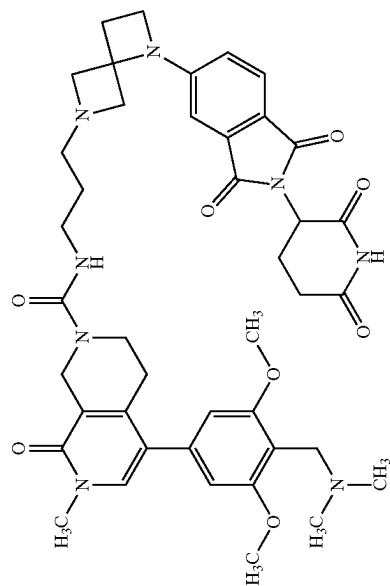
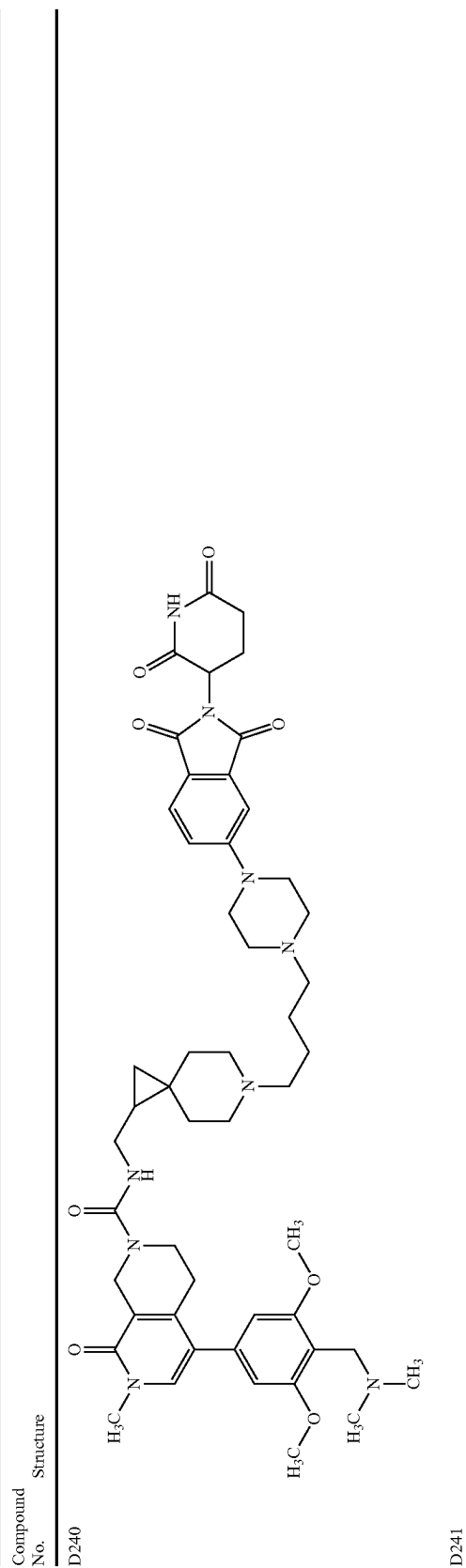
D239



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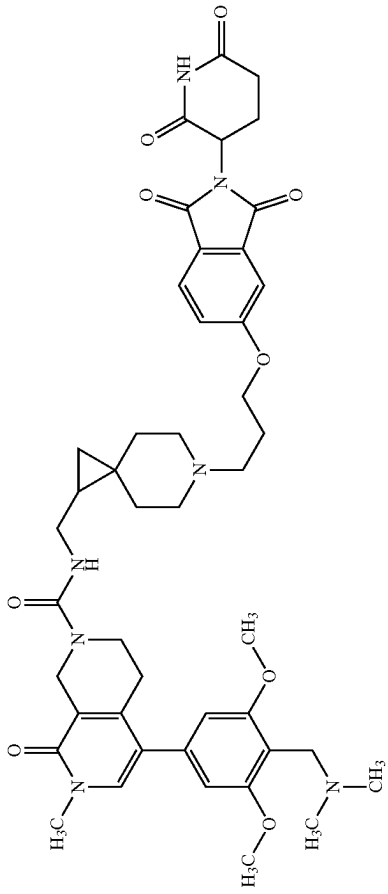
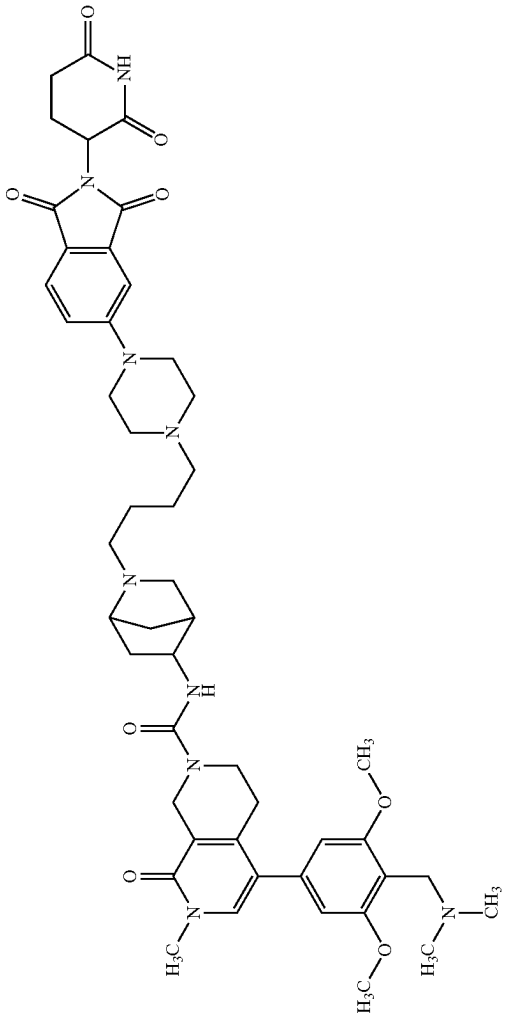
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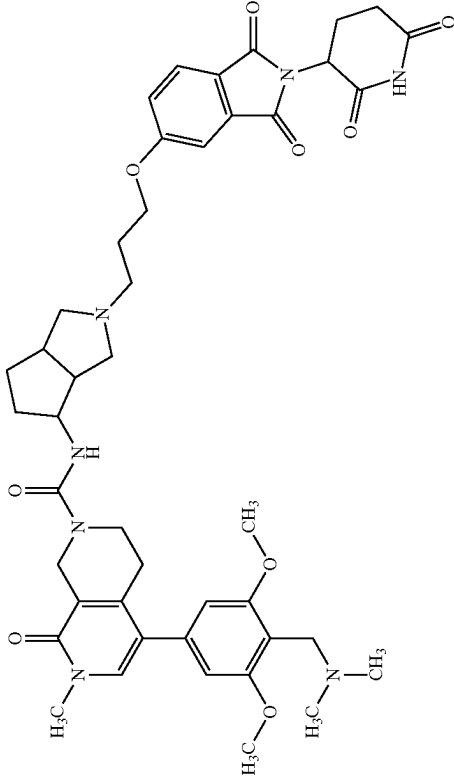
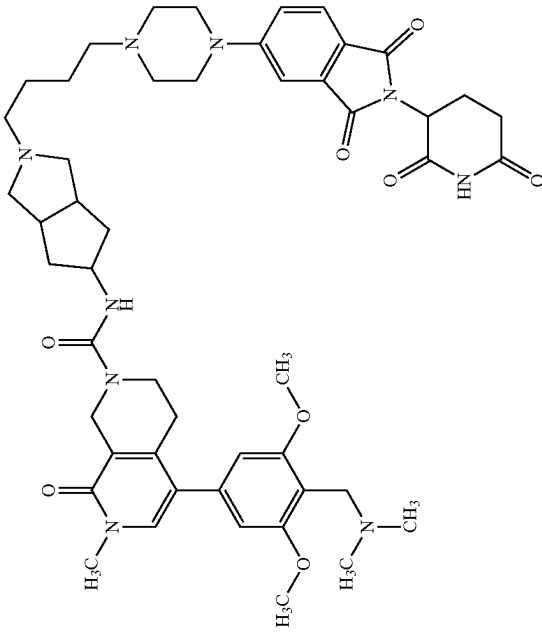
-continued

Compound No.	Structure
D242	 Chemical structure of compound D242: A 6-methoxy-2-(dimethylaminomethyl)benzamide derivative. The benzamide core is substituted with a methoxy group at the 6-position and a dimethylaminomethyl group at the 2-position. The amide nitrogen is connected to a 1,2,3,4-tetrahydropyridine ring. This ring is further connected to a 1,2,3,4-tetrahydropyridine ring, which is connected to a 1,2,3,4-tetrahydropyridine ring. The amide nitrogen is also connected to a 1,2,3,4-tetrahydropyridine ring. The amide nitrogen is also connected to a 1,2,3,4-tetrahydropyridine ring. The amide nitrogen is also connected to a 1,2,3,4-tetrahydropyridine ring.
D243	 Chemical structure of compound D243: A 6-methoxy-2-(dimethylaminomethyl)benzamide derivative. The benzamide core is substituted with a methoxy group at the 6-position and a dimethylaminomethyl group at the 2-position. The amide nitrogen is connected to a 1,2,3,4-tetrahydropyridine ring. This ring is further connected to a 1,2,3,4-tetrahydropyridine ring, which is connected to a 1,2,3,4-tetrahydropyridine ring. The amide nitrogen is also connected to a 1,2,3,4-tetrahydropyridine ring. The amide nitrogen is also connected to a 1,2,3,4-tetrahydropyridine ring. The amide nitrogen is also connected to a 1,2,3,4-tetrahydropyridine ring.

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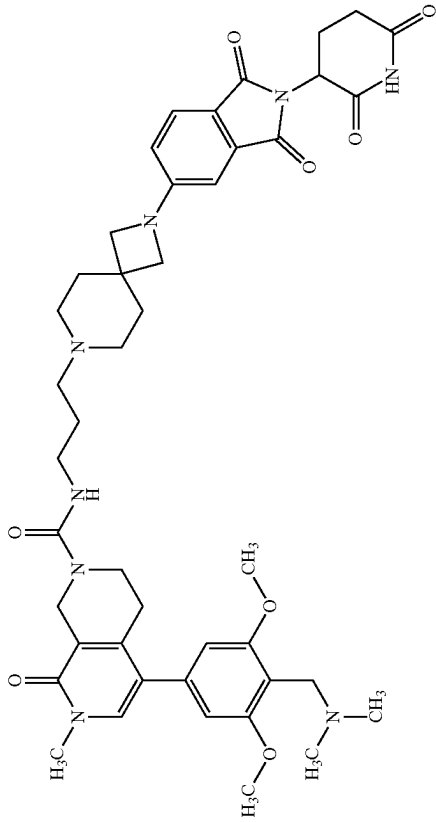
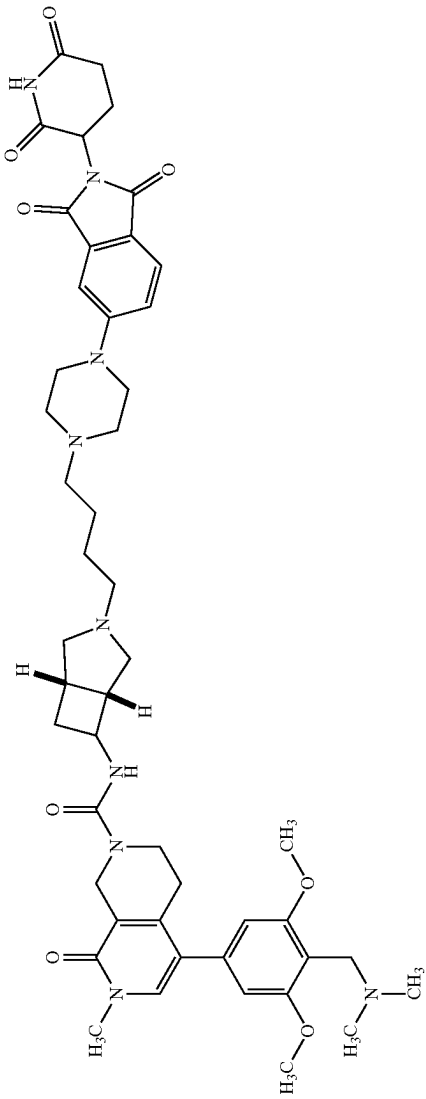
-continued

Compound No.	Structure
D244	 Chemical structure of compound D244: A complex molecule featuring a central benzene ring substituted with a methoxy group (-OCH₃) and a dimethylaminomethyl group (-CH₂N(CH₃)₂). This central ring is connected to a pyridine ring (with a methyl group on the nitrogen) via a methylene bridge. The pyridine ring is further linked to a piperidine ring, which is connected to a bicyclic amine system (a decalin derivative). This system is linked via an amide bond to a long chain containing an ether linkage, which is connected to a benzimidazole ring system. The benzimidazole ring is substituted with a methoxy group and a dimethylaminomethyl group, and is further linked to a piperidine ring, which is connected to a bicyclic amine system (a decalin derivative). The piperidine ring is also linked to a benzimidazole ring system, which is substituted with a methoxy group and a dimethylaminomethyl group, and is further linked to a piperidine ring, which is connected to a bicyclic amine system (a decalin derivative).
D245	 Chemical structure of compound D245: A complex molecule featuring a central benzene ring substituted with a methoxy group (-OCH₃) and a dimethylaminomethyl group (-CH₂N(CH₃)₂). This central ring is connected to a pyridine ring (with a methyl group on the nitrogen) via a methylene bridge. The pyridine ring is further linked to a piperidine ring, which is connected to a bicyclic amine system (a decalin derivative). This system is linked via an amide bond to a long chain containing a piperazine ring, which is connected to a benzimidazole ring system. The benzimidazole ring is substituted with a methoxy group and a dimethylaminomethyl group, and is further linked to a piperidine ring, which is connected to a bicyclic amine system (a decalin derivative). The piperidine ring is also linked to a benzimidazole ring system, which is substituted with a methoxy group and a dimethylaminomethyl group, and is further linked to a piperidine ring, which is connected to a bicyclic amine system (a decalin derivative).

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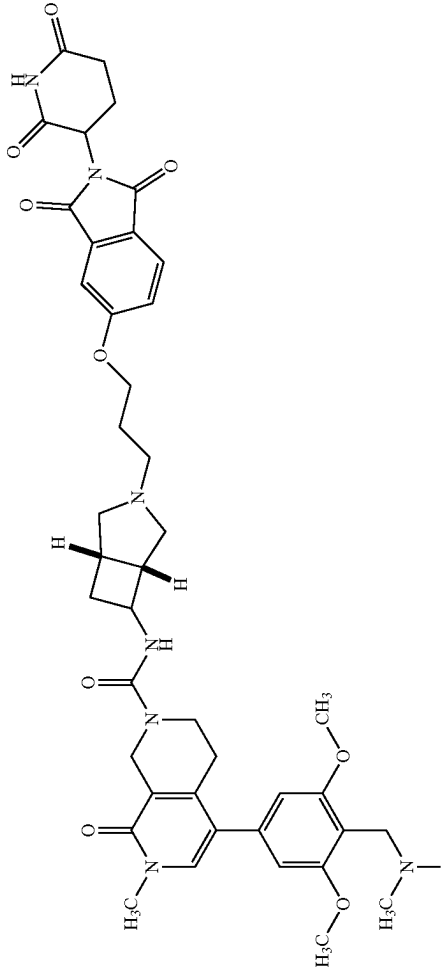
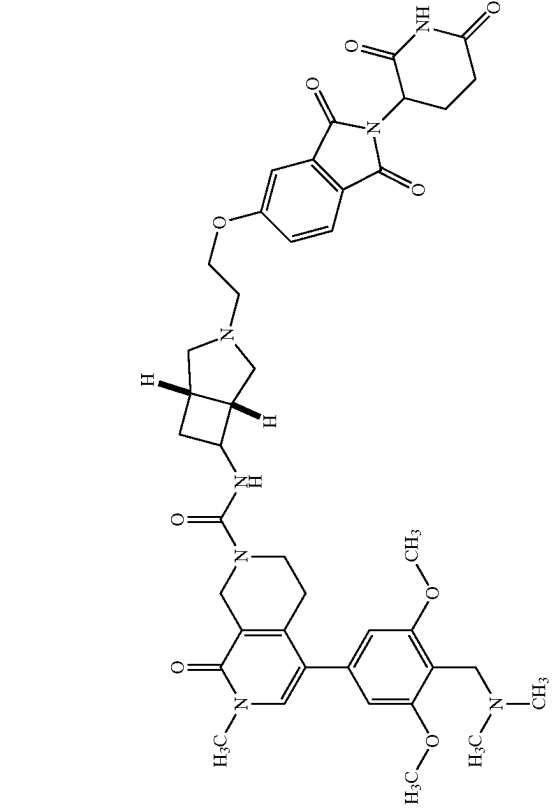
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Compound No.	Structure
D246	 Chemical structure of compound D246: A 6-methoxy-2-(dimethylamino)benzyl group is attached to the 4-position of a 1-methyl-2-(dimethylamino)-6-methoxy-4-((4-(2,6-dimethyl-1,3-dioxan-5-yl)butyl)amino)pyrimidin-5(1H)-one. The pyrimidinone ring is further substituted with a 1-((2,6-dimethyl-1,3-dioxan-5-yl)amino)-2,4-dioxo-1,2,3,4-tetrahydro-1H-benzimidazole-5-yl group at the 2-position.
D247	 Chemical structure of compound D247: A 6-methoxy-2-(dimethylamino)benzyl group is attached to the 4-position of a 1-methyl-2-(dimethylamino)-6-methoxy-4-((4-(2,6-dimethyl-1,3-dioxan-5-yl)butyl)amino)pyrimidin-5(1H)-one. The pyrimidinone ring is further substituted with a 1-((2,6-dimethyl-1,3-dioxan-5-yl)amino)-2,4-dioxo-1,2,3,4-tetrahydro-1H-benzimidazole-5-yl group at the 2-position.

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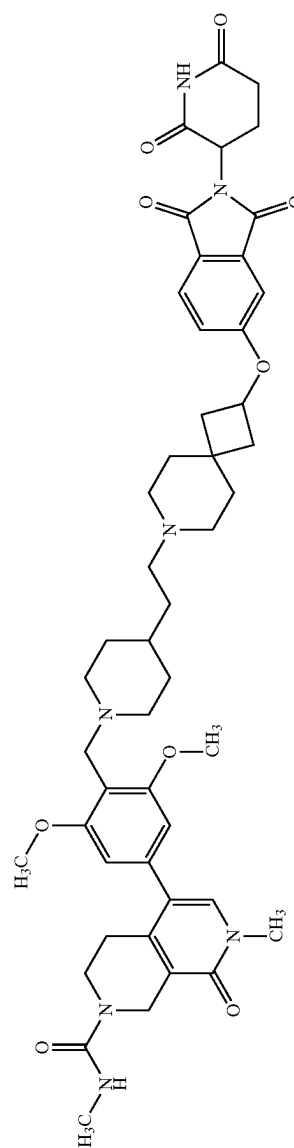
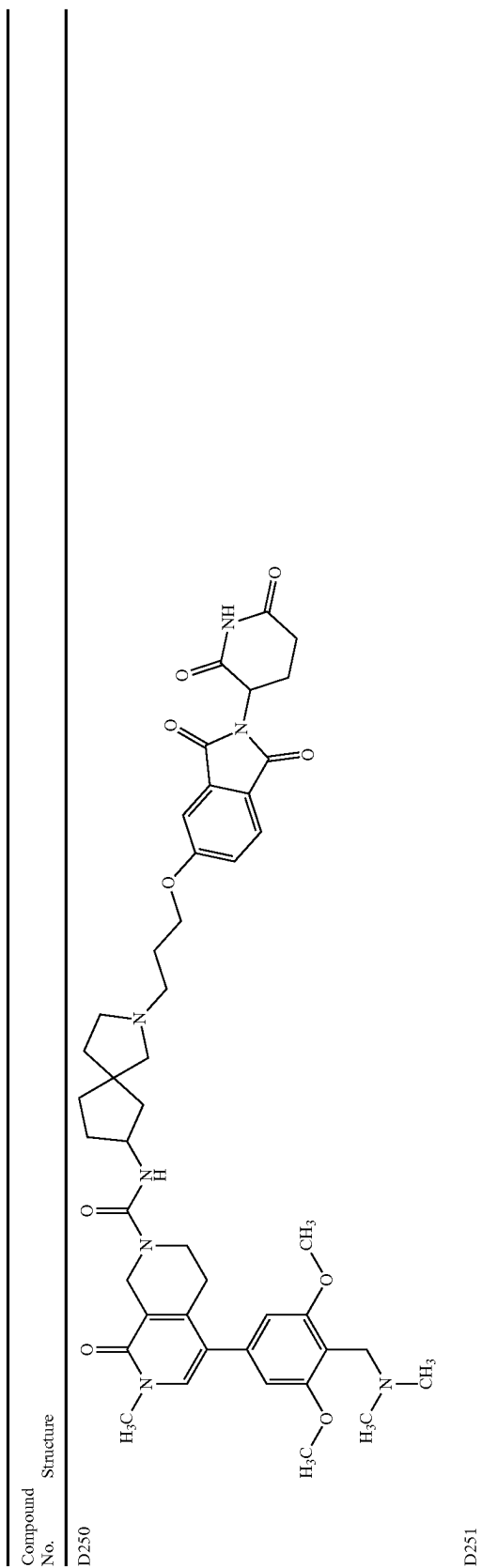
-continued

Compound No.	Structure
D248	 Chemical structure of compound D248: A complex molecule featuring a 6-methyl-2,4-dimethoxyphenyl group connected via a methylene bridge to a dimethylamino group. This is linked to a piperidine ring, which is further connected to a bicyclic system (a cyclobutane fused to a pyrrolidine). The bicyclic system is connected via a propyl chain to a 4-(2,4-dimethoxyphenyl)-2,4-dihydroquinoline-3-carboxamide moiety, which is finally linked to a 2,4-dimethoxyphenyl group connected via a methylene bridge to a dimethylamino group.
D249	 Chemical structure of compound D249: A complex molecule featuring a 6-methyl-2,4-dimethoxyphenyl group connected via a methylene bridge to a dimethylamino group. This is linked to a piperidine ring, which is further connected to a bicyclic system (a cyclobutane fused to a pyrrolidine). The bicyclic system is connected via a propyl chain to a 4-(2,4-dimethoxyphenyl)-2,4-dihydroquinoline-3-carboxamide moiety, which is finally linked to a 2,4-dimethoxyphenyl group connected via a methylene bridge to a dimethylamino group.

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Compound No.	Structure
D252	
D253	
D254	

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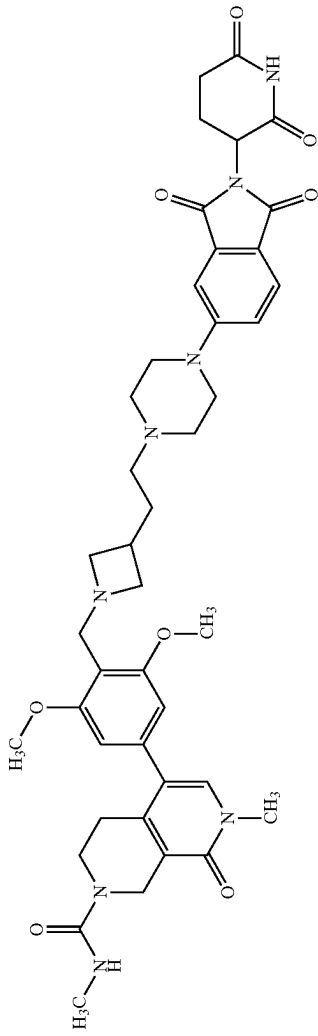
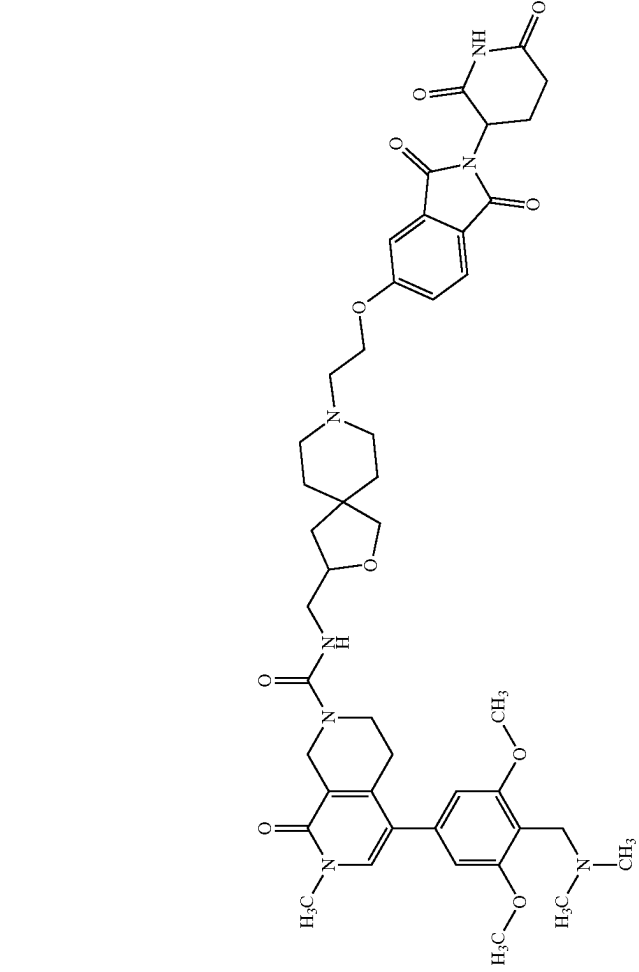
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Compound No.	Structure
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D256	<chem>CN(C)C(=O)N1CCc2c(c1)nc(C)c2C3=CC=C(C=C3)C(OC)C4=CC=C(C=C4)N5CCN(C5)C6=CC=CC=C6N7C(=O)NC(=O)CC7</chem>
D257	<chem>CN(C)C(=O)N1CCc2c(c1)nc(C)c2C3=CC=C(C=C3)C(OC)C4=CC=C(C=C4)N5CCN(C5)C6=CC=CC=C6N7C(=O)NC(=O)CC7</chem>

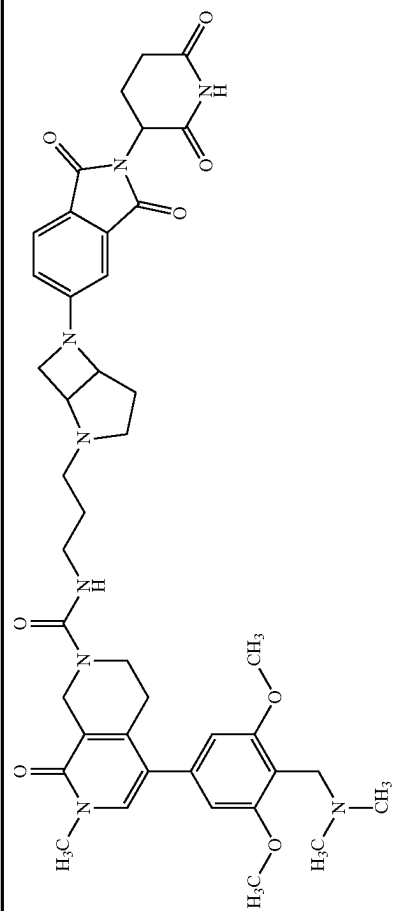
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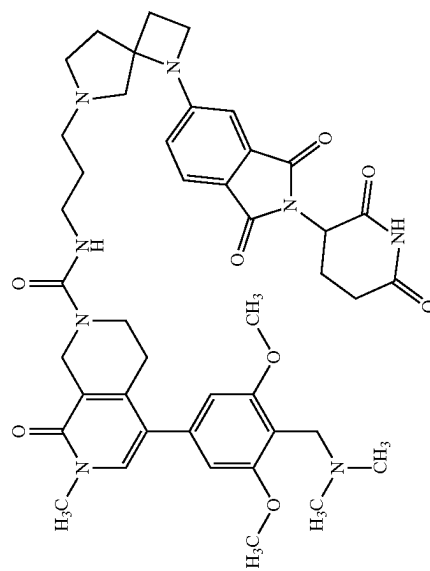
-continued

Compound No.	Structure
D258	
D259	

Compound No.	Structure
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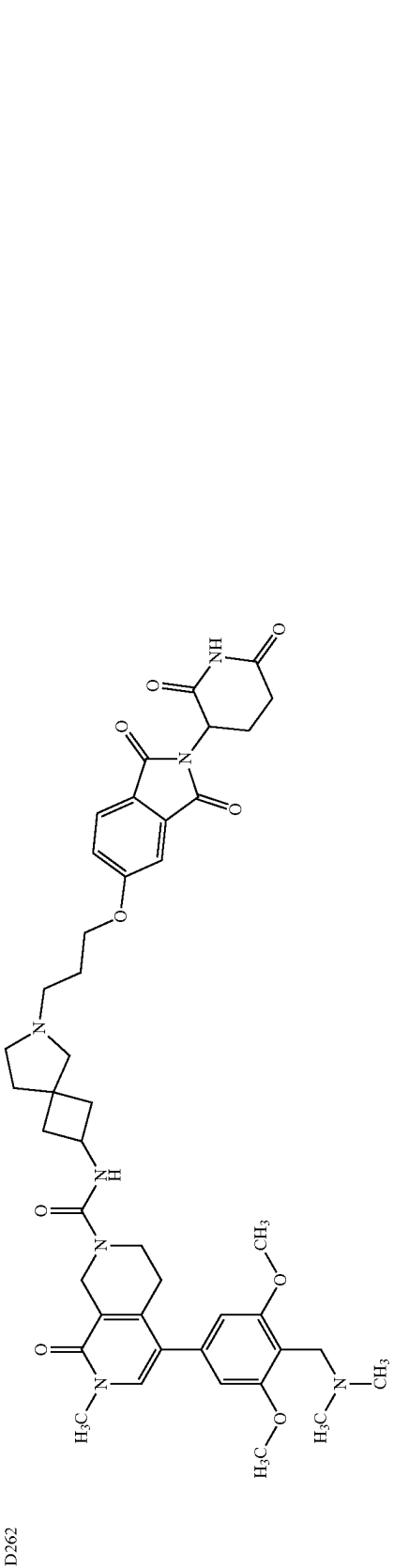
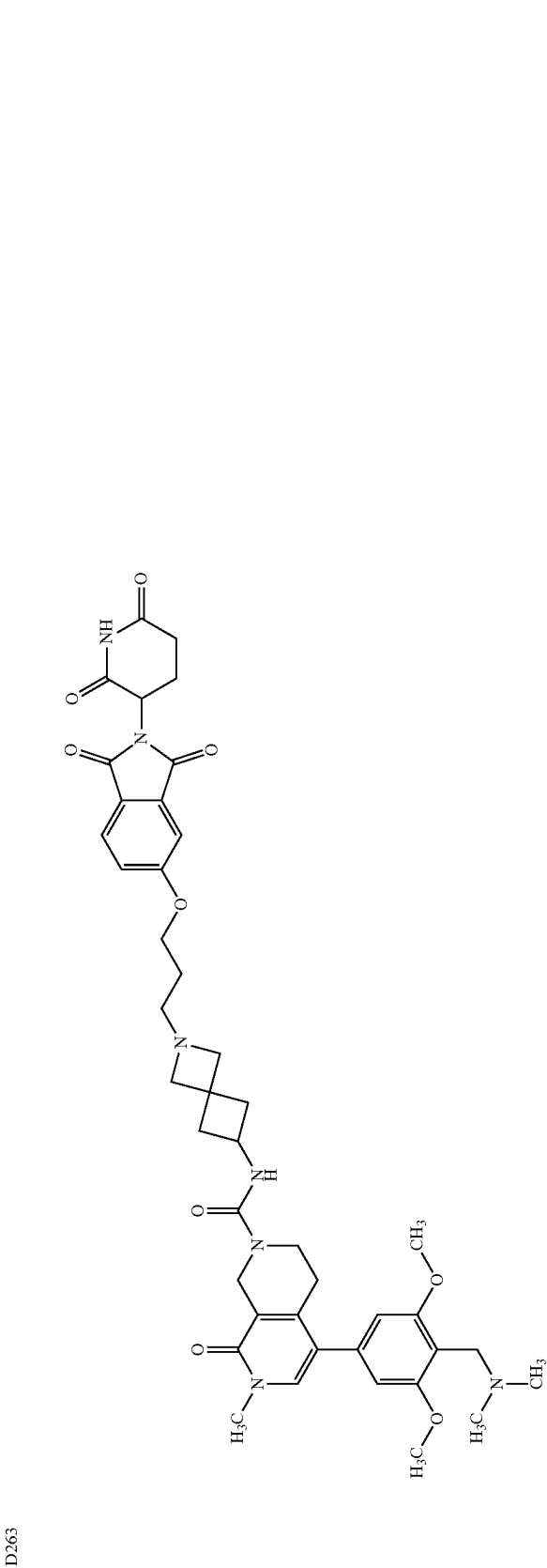
D261



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Compound No.	Structure
D262	 Chemical structure of compound D262: A 6-methoxy-2-(dimethylaminomethyl)benzimidazole core. The benzimidazole ring is substituted at the 6-position with a methoxy group and at the 2-position with a dimethylaminomethyl group. The benzimidazole ring is connected via its 2-position to a 4-(dimethylaminomethyl)benzimidazole ring. This second benzimidazole ring is further substituted at its 4-position with a methoxy group and at its 2-position with a dimethylaminomethyl group. The benzimidazole ring is connected via its 2-position to a 4-(dimethylaminomethyl)benzimidazole ring. This second benzimidazole ring is further substituted at its 4-position with a methoxy group and at its 2-position with a dimethylaminomethyl group. The benzimidazole ring is connected via its 2-position to a 4-(dimethylaminomethyl)benzimidazole ring. This second benzimidazole ring is further substituted at its 4-position with a methoxy group and at its 2-position with a dimethylaminomethyl group.
D263	 Chemical structure of compound D263: A 6-methoxy-2-(dimethylaminomethyl)benzimidazole core. The benzimidazole ring is substituted at the 6-position with a methoxy group and at the 2-position with a dimethylaminomethyl group. The benzimidazole ring is connected via its 2-position to a 4-(dimethylaminomethyl)benzimidazole ring. This second benzimidazole ring is further substituted at its 4-position with a methoxy group and at its 2-position with a dimethylaminomethyl group. The benzimidazole ring is connected via its 2-position to a 4-(dimethylaminomethyl)benzimidazole ring. This second benzimidazole ring is further substituted at its 4-position with a methoxy group and at its 2-position with a dimethylaminomethyl group. The benzimidazole ring is connected via its 2-position to a 4-(dimethylaminomethyl)benzimidazole ring. This second benzimidazole ring is further substituted at its 4-position with a methoxy group and at its 2-position with a dimethylaminomethyl group.

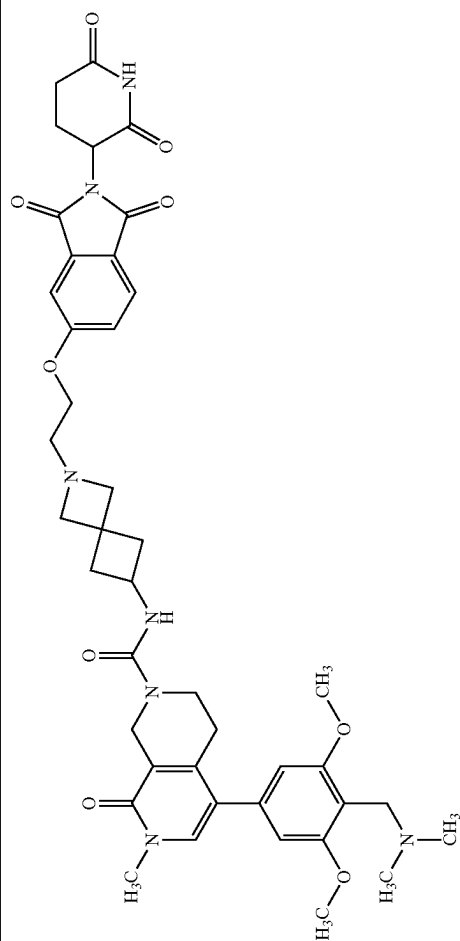
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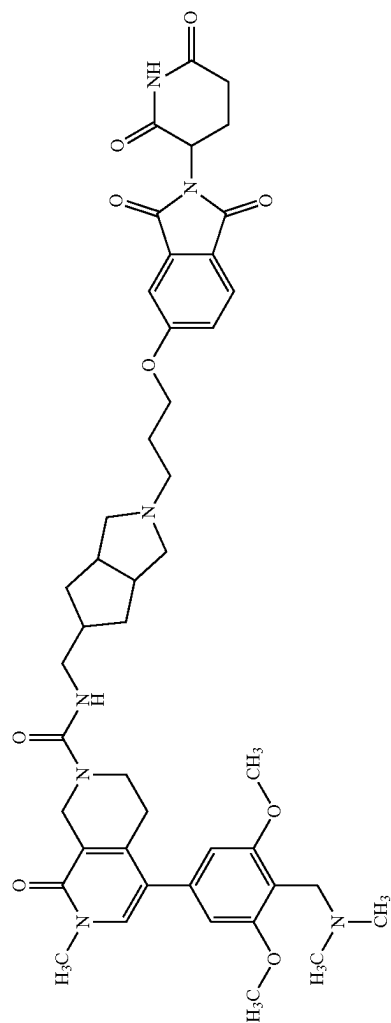
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Compound No.	Structure
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D264



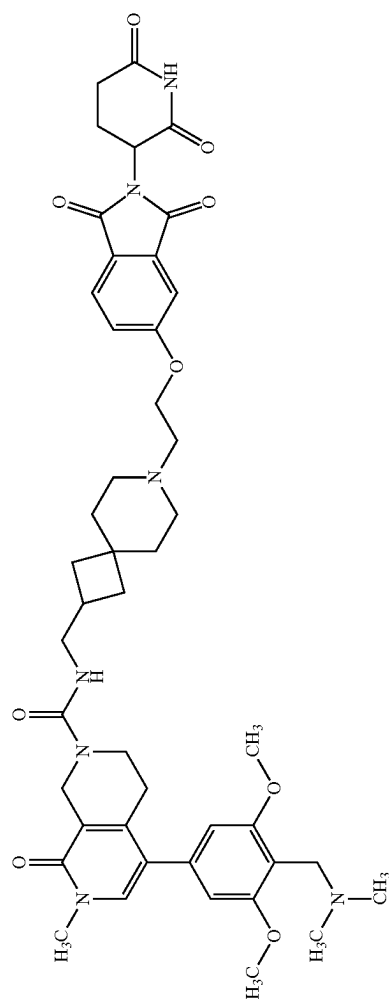
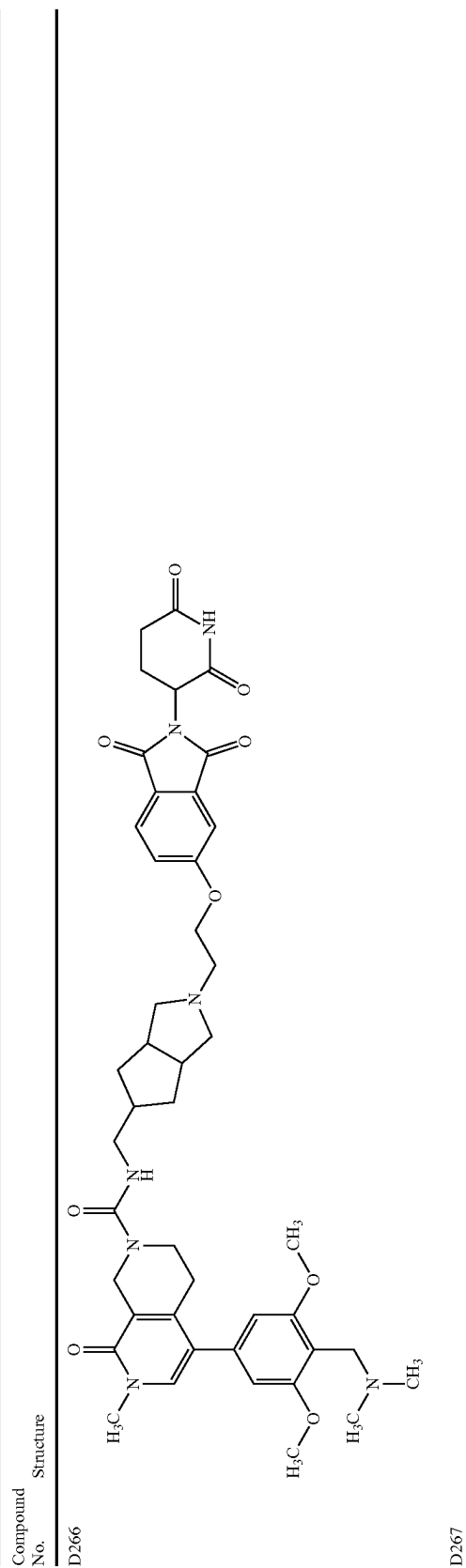
D265



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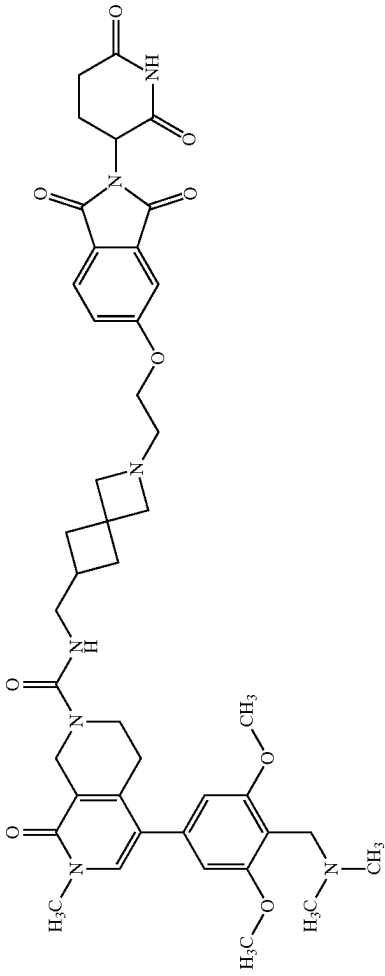
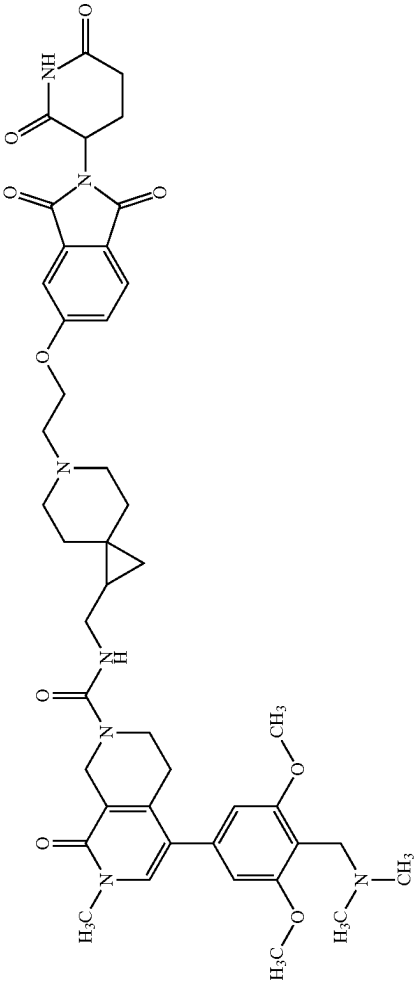
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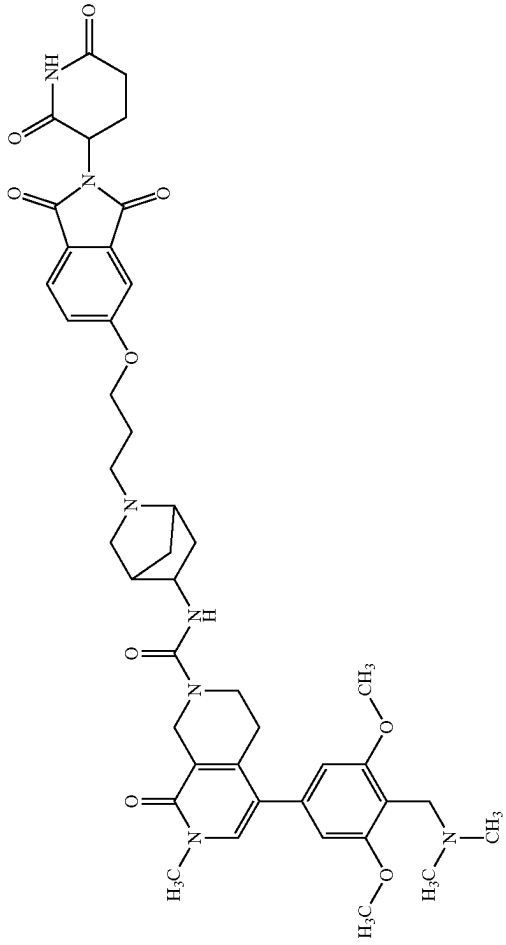
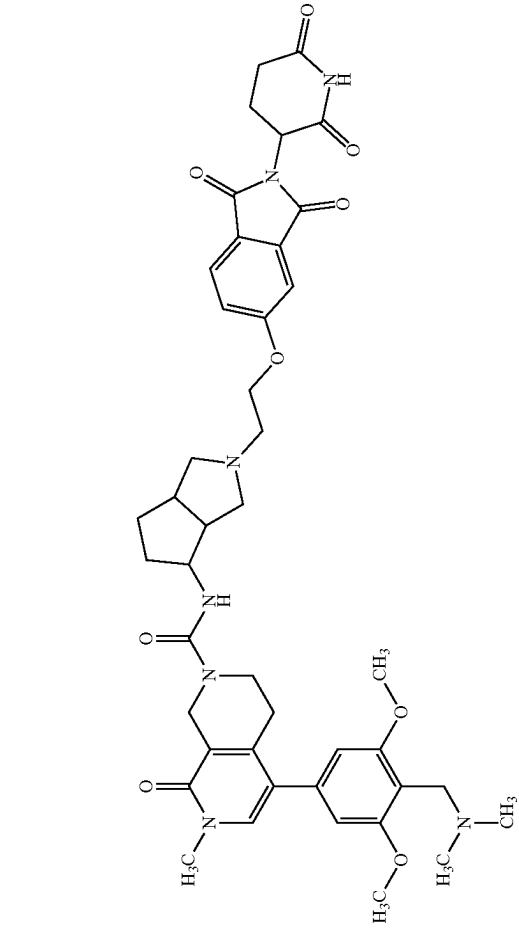
-continued

Compound No.	Structure
D268	 Chemical structure of compound D268: A 6-methoxy-2-(dimethylaminomethyl)phenyl group is attached to the 4-position of a 1-methyl-2,3-dihydroquinolin-4(1H)-one. The nitrogen at position 1 is connected via an amide bond to a 2,2,3,3-tetramethyl-1-azabicyclo[2.2.1]heptane ring. This bicyclic ring is further connected via an ether linkage to a 1,2,3,4-tetrahydroquinolin-5(1H)-one moiety.
D269	 Chemical structure of compound D269: A 6-methoxy-2-(dimethylaminomethyl)phenyl group is attached to the 4-position of a 1-methyl-2,3-dihydroquinolin-4(1H)-one. The nitrogen at position 1 is connected via an amide bond to a 2,2,3,3-tetramethyl-1-azabicyclo[2.2.1]heptane ring. This bicyclic ring is further connected via an ether linkage to a 1,2,3,4-tetrahydroquinolin-5(1H)-one moiety.

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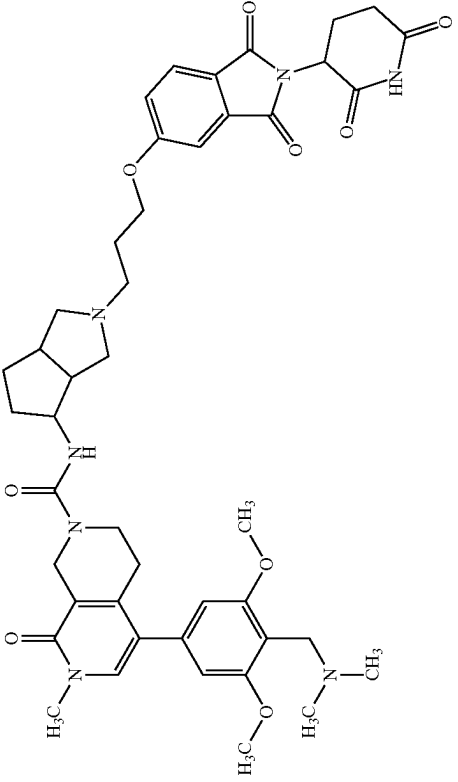
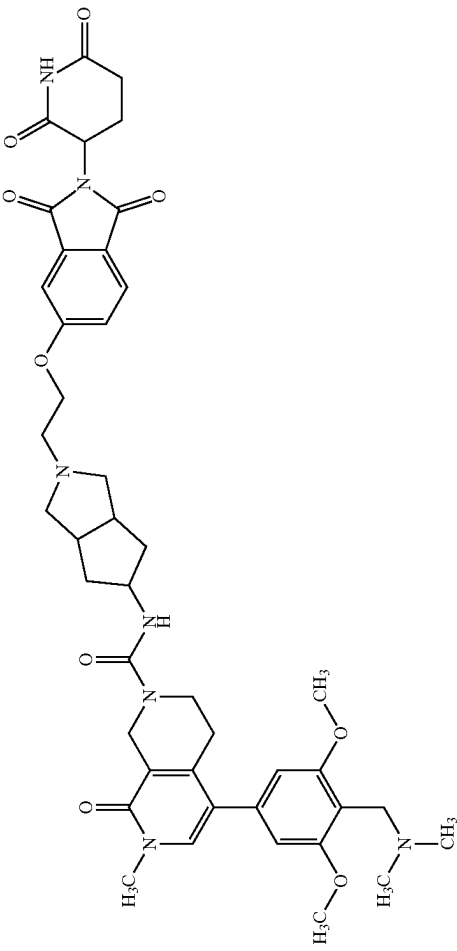
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Compound No.	Structure
D270	 Chemical structure of compound D270: A complex molecule featuring a central benzene ring substituted with a methoxy group (-OCH₃) and a dimethylaminomethyl group (-CH₂N(CH₃)₂). This central ring is connected to a 4-methyl-1,2,3,4-tetrahydropyridine-6-carbonyl group, which is further linked via an amide bond to a bicyclic amine (8-azabicyclo[3.2.1]octane). This bicyclic amine is connected through a 3-ethoxypropyl chain to a 4-ethoxyphenyl group, which is finally linked via an amide bond to a 2-piperidinone ring.
D271	 Chemical structure of compound D271: A complex molecule featuring a central benzene ring substituted with a methoxy group (-OCH₃) and a dimethylaminomethyl group (-CH₂N(CH₃)₂). This central ring is connected to a 4-methyl-1,2,3,4-tetrahydropyridine-6-carbonyl group, which is further linked via an amide bond to a bicyclic amine (8-azabicyclo[3.2.1]octane). This bicyclic amine is connected through a 3-ethoxypropyl chain to a 4-ethoxyphenyl group, which is finally linked via an amide bond to a 2-piperidinone ring.

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Compound No.	Structure
D272	 Chemical structure of compound D272: A 6-methoxy-2-(dimethylaminomethyl)benzimidazole ring is connected at the 4-position to a 2-methoxy-5-(2,2,5-trimethyl-1,3-dioxane-5-carbonyl)benzimidazole ring. The latter is further connected at the 2-position via a 3-oxo-1,2,3,4-tetrahydroquinoline-5-yl group to a 2-oxo-1,2,3,4-tetrahydroquinoline-5-yl group, which is finally connected at the 2-position to a 2-oxo-1,2,3,4-tetrahydroquinoline-5-yl group.
D273	 Chemical structure of compound D273: A 6-methoxy-2-(dimethylaminomethyl)benzimidazole ring is connected at the 4-position to a 2-methoxy-5-(2,2,5-trimethyl-1,3-dioxane-5-carbonyl)benzimidazole ring. The latter is further connected at the 2-position via a 3-oxo-1,2,3,4-tetrahydroquinoline-5-yl group to a 2-oxo-1,2,3,4-tetrahydroquinoline-5-yl group, which is finally connected at the 2-position to a 2-oxo-1,2,3,4-tetrahydroquinoline-5-yl group.

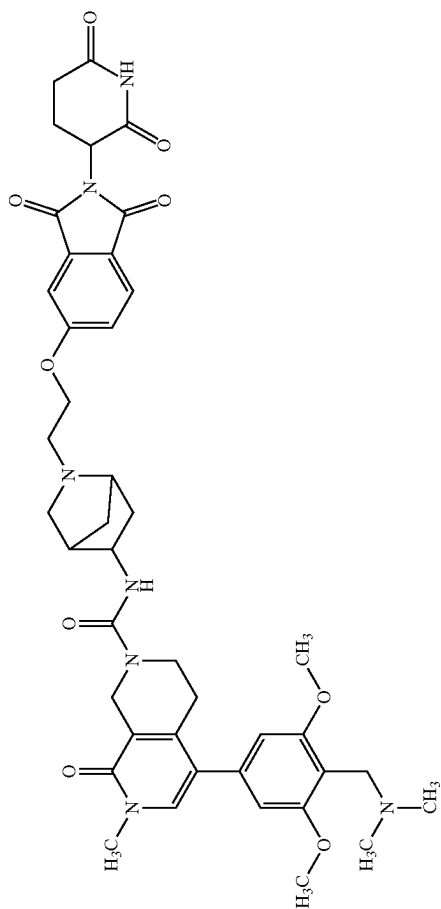
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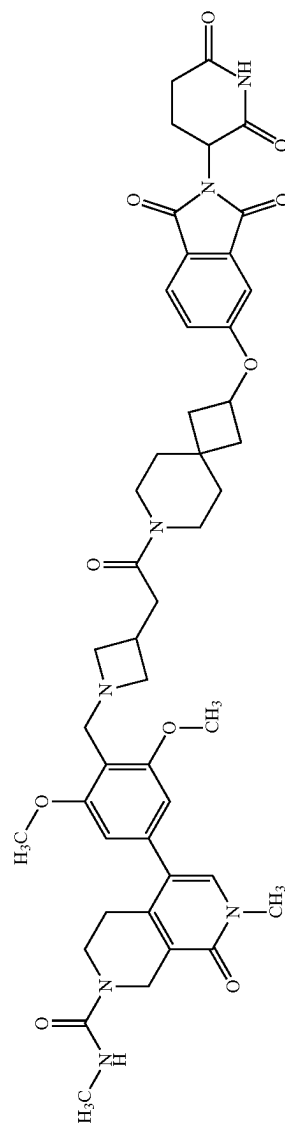
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Compound No.	Structure
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D274



D275



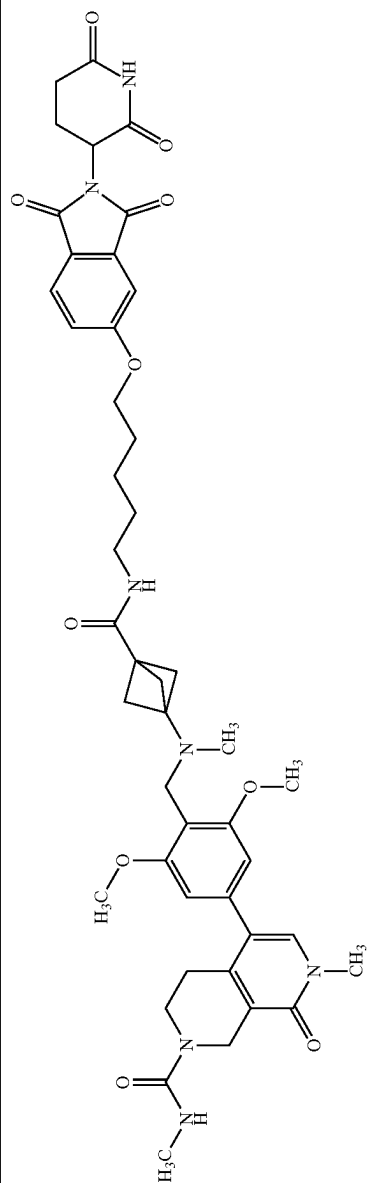
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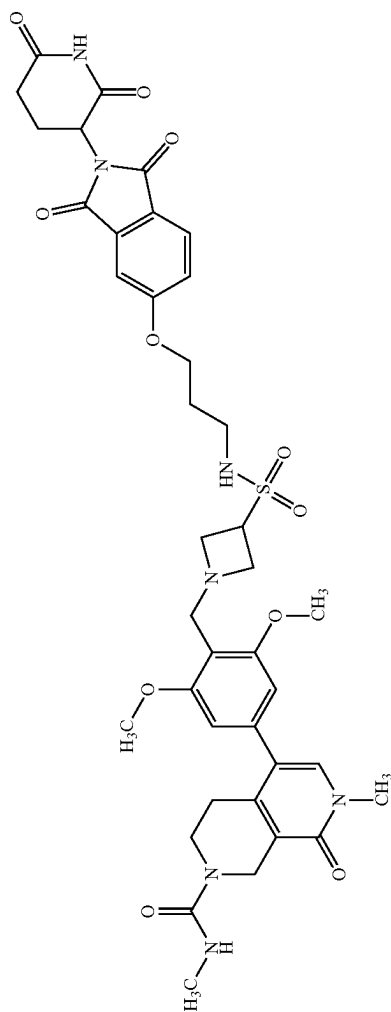
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Compound No.	Structure
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D276



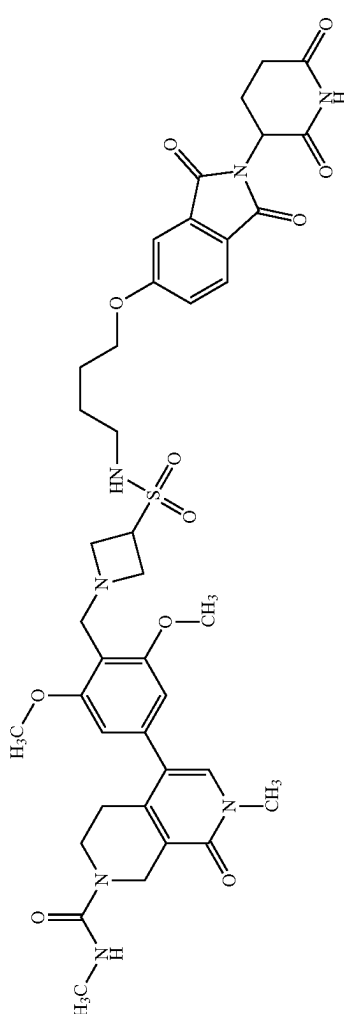
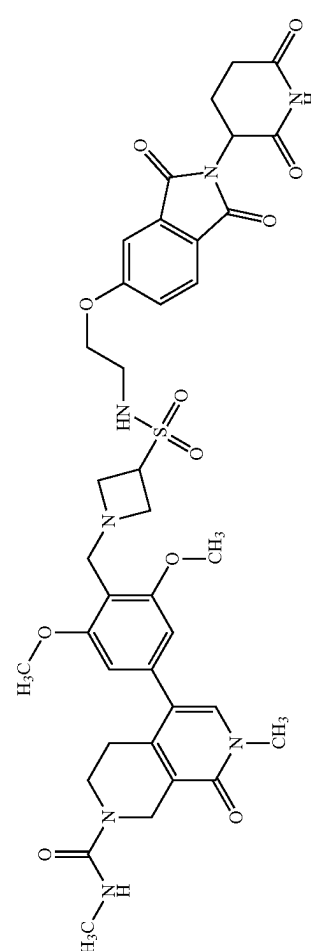
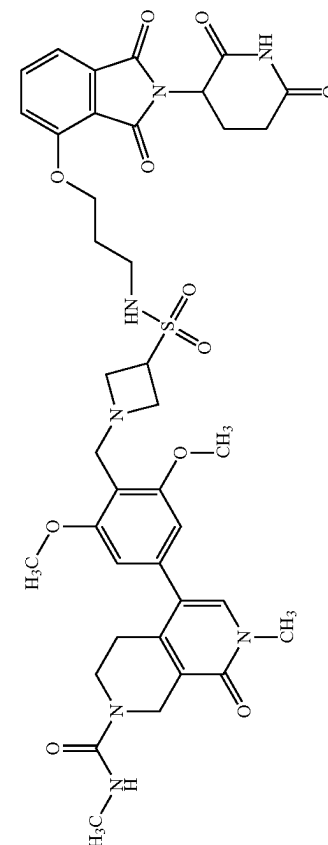
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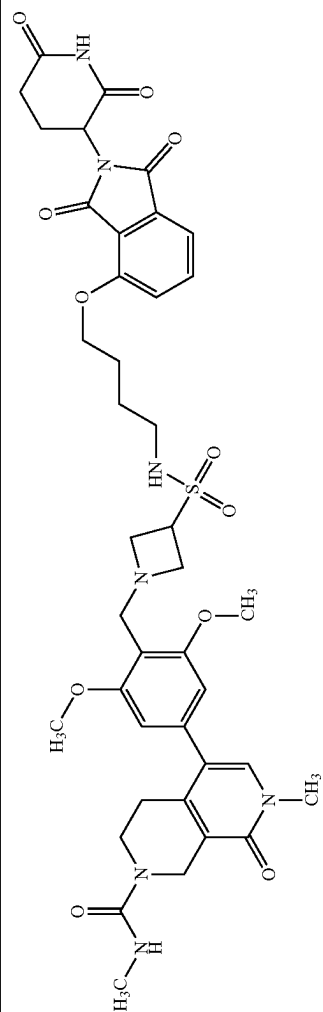
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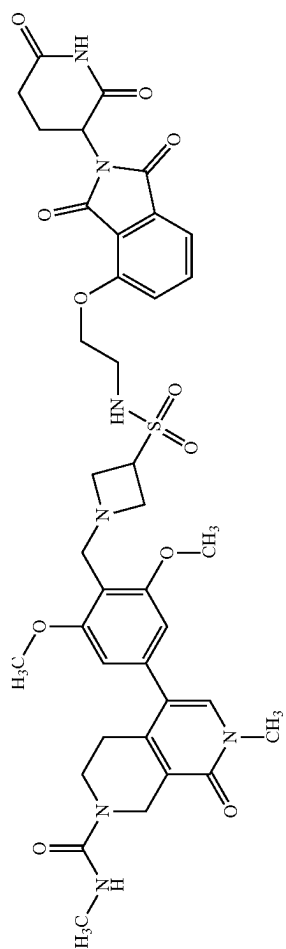
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Compound No.	Structure
D278	 Chemical structure of compound D278: A complex molecule featuring a central pyridine ring substituted with a methyl group (CH₃) and a carbonyl group (C=O). This central ring is connected via a methylene bridge to a benzene ring, which is further substituted with a methoxy group (OCH₃) and a methoxyethyl chain (CH₂CH₂OC(=O)N(CH₃)). The methoxyethyl chain is linked to a pyridine ring, which is substituted with a methyl group (CH₃) and a carbonyl group (C=O). This pyridine ring is connected via a methylene bridge to a benzene ring, which is further substituted with a methoxy group (OCH₃) and a methoxyethyl chain (CH₂CH₂OC(=O)N(CH₃)). The methoxyethyl chain is linked to a pyridine ring, which is substituted with a methyl group (CH₃) and a carbonyl group (C=O). This pyridine ring is connected via a methylene bridge to a benzene ring, which is further substituted with a methoxy group (OCH₃) and a methoxyethyl chain (CH₂CH₂OC(=O)N(CH₃)).
D279	 Chemical structure of compound D279: A complex molecule featuring a central pyridine ring substituted with a methyl group (CH₃) and a carbonyl group (C=O). This central ring is connected via a methylene bridge to a benzene ring, which is further substituted with a methoxy group (OCH₃) and a methoxyethyl chain (CH₂CH₂OC(=O)N(CH₃)). The methoxyethyl chain is linked to a pyridine ring, which is substituted with a methyl group (CH₃) and a carbonyl group (C=O). This pyridine ring is connected via a methylene bridge to a benzene ring, which is further substituted with a methoxy group (OCH₃) and a methoxyethyl chain (CH₂CH₂OC(=O)N(CH₃)). The methoxyethyl chain is linked to a pyridine ring, which is substituted with a methyl group (CH₃) and a carbonyl group (C=O). This pyridine ring is connected via a methylene bridge to a benzene ring, which is further substituted with a methoxy group (OCH₃) and a methoxyethyl chain (CH₂CH₂OC(=O)N(CH₃)).
D280	 Chemical structure of compound D280: A complex molecule featuring a central pyridine ring substituted with a methyl group (CH₃) and a carbonyl group (C=O). This central ring is connected via a methylene bridge to a benzene ring, which is further substituted with a methoxy group (OCH₃) and a methoxyethyl chain (CH₂CH₂OC(=O)N(CH₃)). The methoxyethyl chain is linked to a pyridine ring, which is substituted with a methyl group (CH₃) and a carbonyl group (C=O). This pyridine ring is connected via a methylene bridge to a benzene ring, which is further substituted with a methoxy group (OCH₃) and a methoxyethyl chain (CH₂CH₂OC(=O)N(CH₃)). The methoxyethyl chain is linked to a pyridine ring, which is substituted with a methyl group (CH₃) and a carbonyl group (C=O). This pyridine ring is connected via a methylene bridge to a benzene ring, which is further substituted with a methoxy group (OCH₃) and a methoxyethyl chain (CH₂CH₂OC(=O)N(CH₃)).

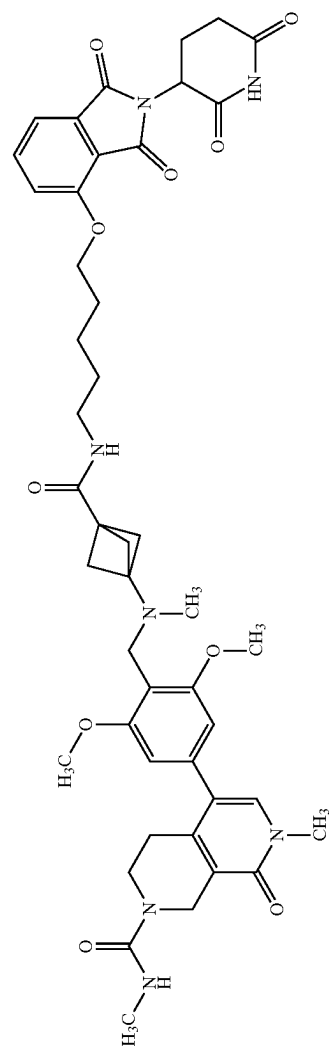
Compound No.	Structure
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D282



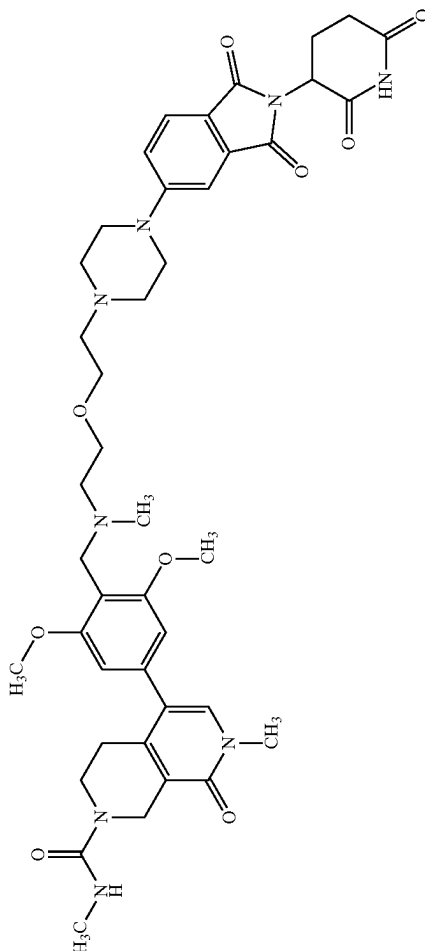
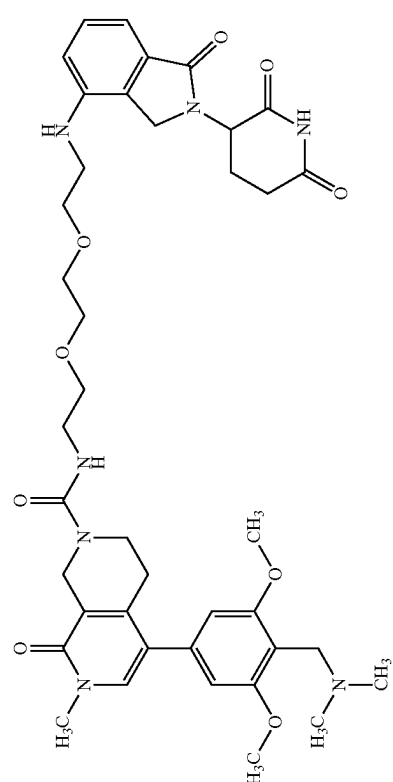
D283



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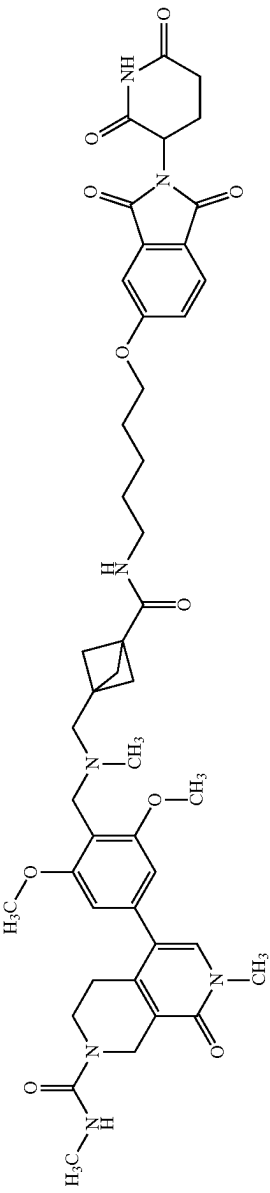
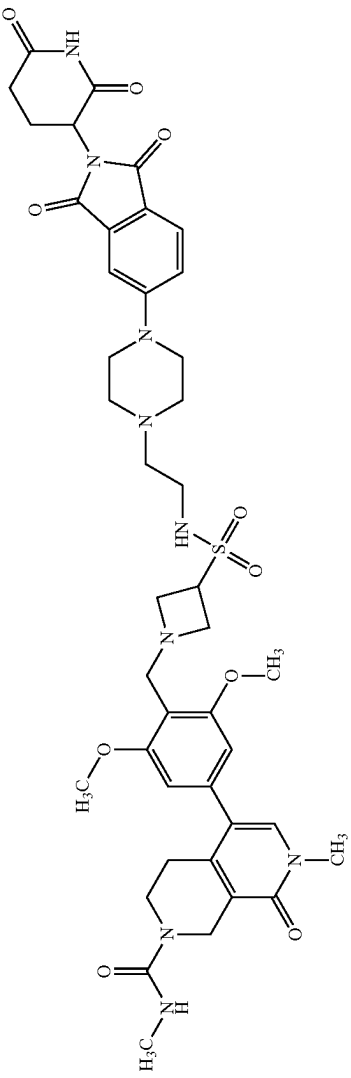
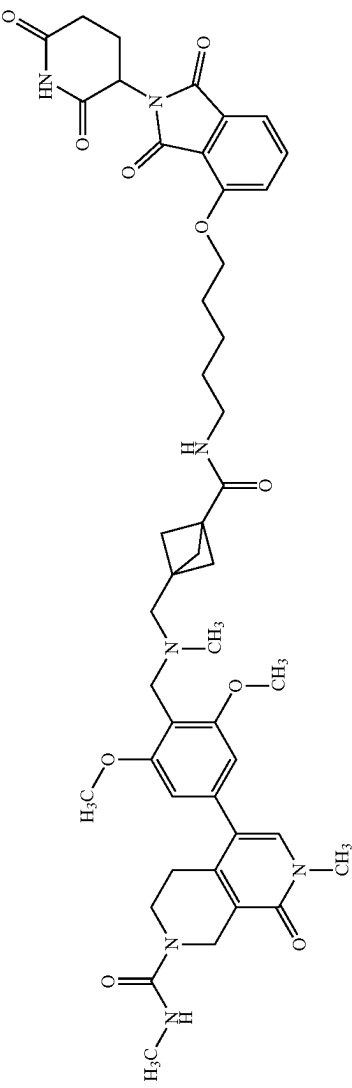
-continued

Compound No.	Structure
D284	 Chemical structure of compound D284: A complex molecule featuring a central benzene ring substituted with a methoxy group (H₃C-O-), a dimethylamino group (-N(CH₃)₂), and a side chain containing a piperidine ring, a carbonyl group, and a methyl group. The side chain is further substituted with a methoxy group and a dimethylamino group. The molecule also includes a benzimidazole ring system and a dimethylamino group.
D285	 Chemical structure of compound D285: A complex molecule featuring a central benzene ring substituted with a methoxy group (H₃C-O-), a dimethylamino group (-N(CH₃)₂), and a side chain containing a piperidine ring, a carbonyl group, and a methyl group. The side chain is further substituted with a methoxy group and a dimethylamino group. The molecule also includes a benzimidazole ring system and a dimethylamino group.

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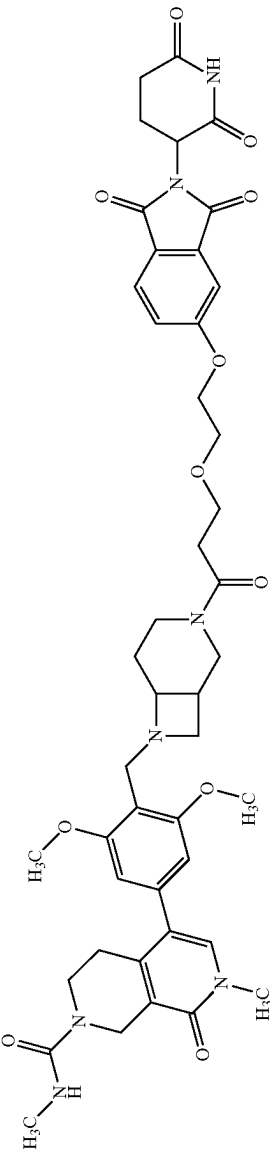
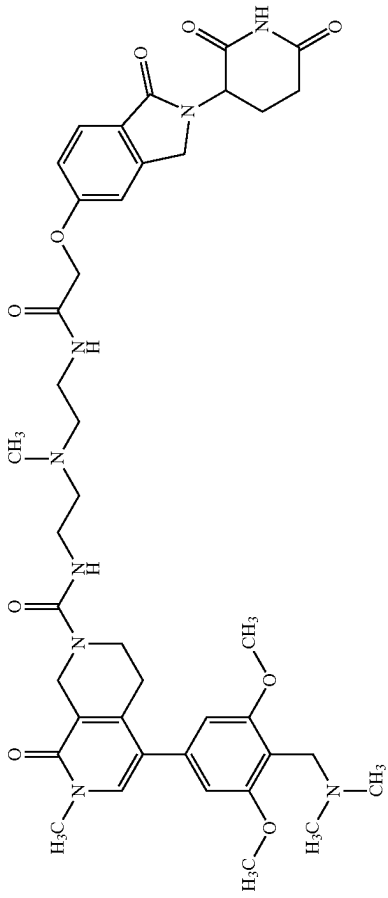
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Compound No.	Structure
D286	
D287	
D288	

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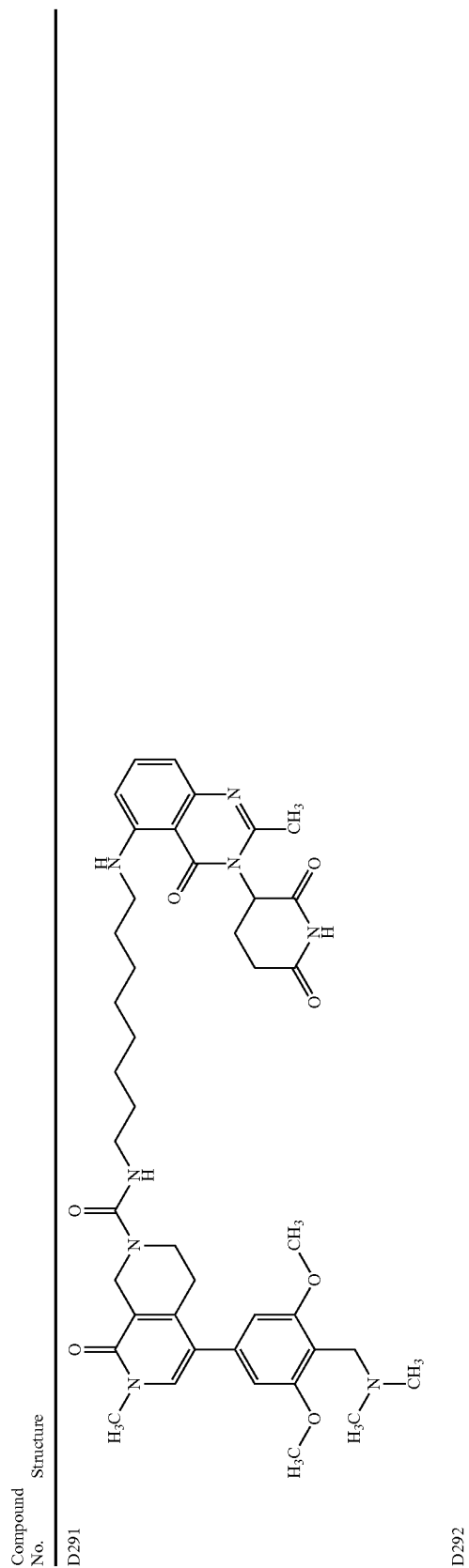
-continued

Compound No.	Structure
D289	 Chemical structure of compound D289: A complex molecule featuring a central benzene ring substituted with a methoxy group and a methyl group. This benzene ring is connected to a pyridine ring, which is further substituted with a methyl group and a methylamino group. The pyridine ring is linked via a methylene group to a piperidine ring. The piperidine ring is connected to a carbonyl group, which is further linked to a chain containing two ether linkages and a terminal amide group. The amide group is connected to a benzimidazole ring system, which is substituted with a methyl group and a methylamino group. The benzimidazole ring is further substituted with a methyl group and a methylamino group.
D290	 Chemical structure of compound D290: A complex molecule featuring a central benzene ring substituted with a methoxy group and a methyl group. This benzene ring is connected to a pyridine ring, which is further substituted with a methyl group and a methylamino group. The pyridine ring is linked via a methylene group to a piperidine ring. The piperidine ring is connected to a carbonyl group, which is further linked to a chain containing two ether linkages and a terminal amide group. The amide group is connected to a benzimidazole ring system, which is substituted with a methyl group and a methylamino group. The benzimidazole ring is further substituted with a methyl group and a methylamino group.

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-continued



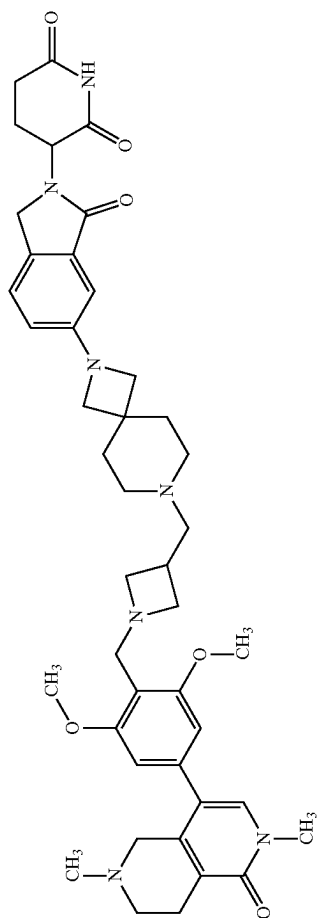
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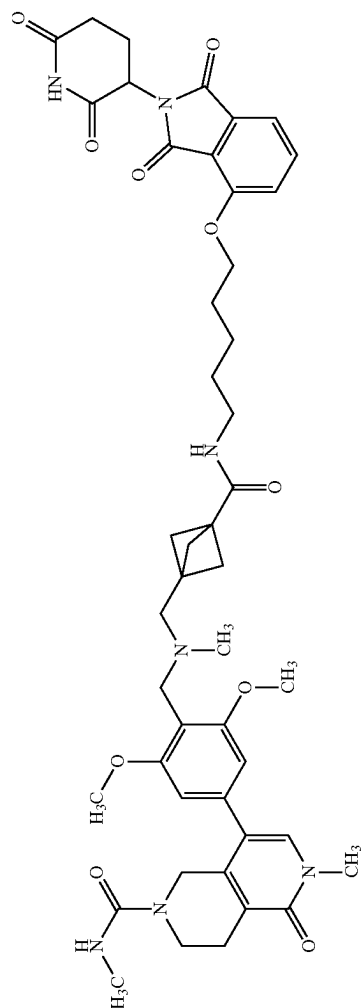
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Compound No.	Structure
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D293



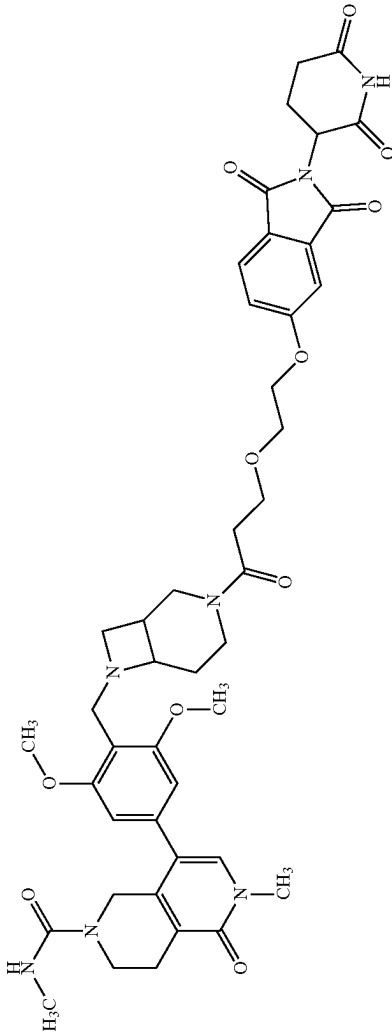
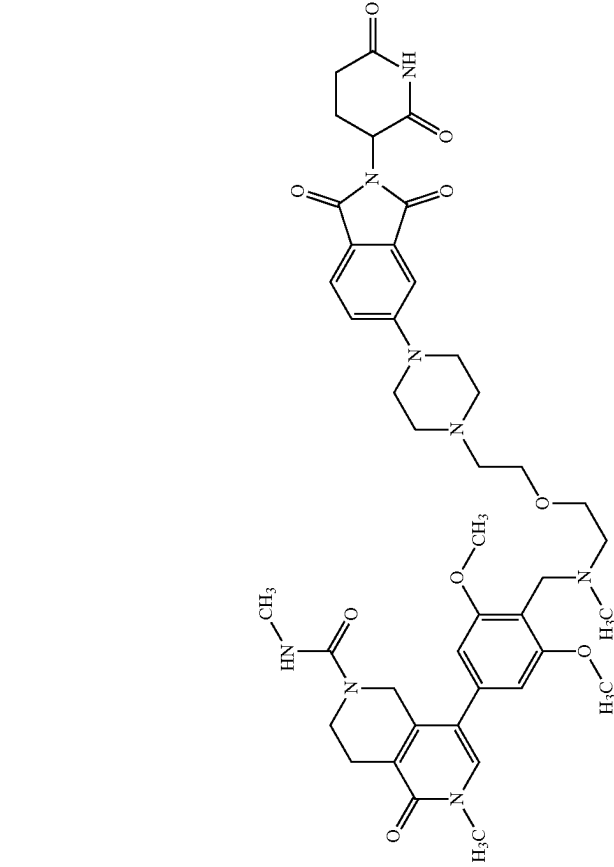
D294



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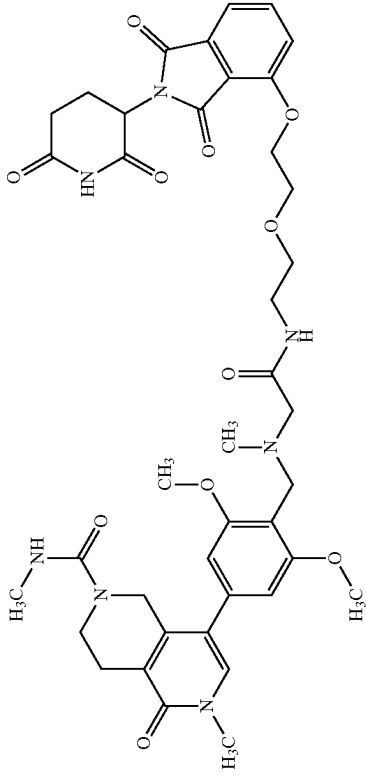
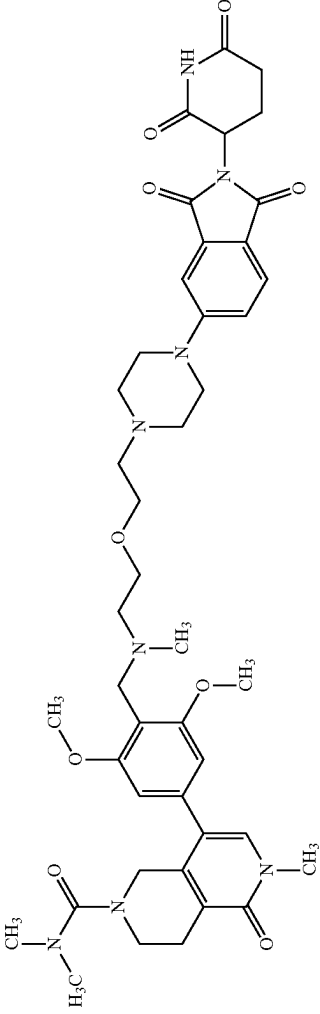
-continued

Compound No.	Structure
D295	
D296	

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-continued

Compound No.	Structure
D297	
D298	

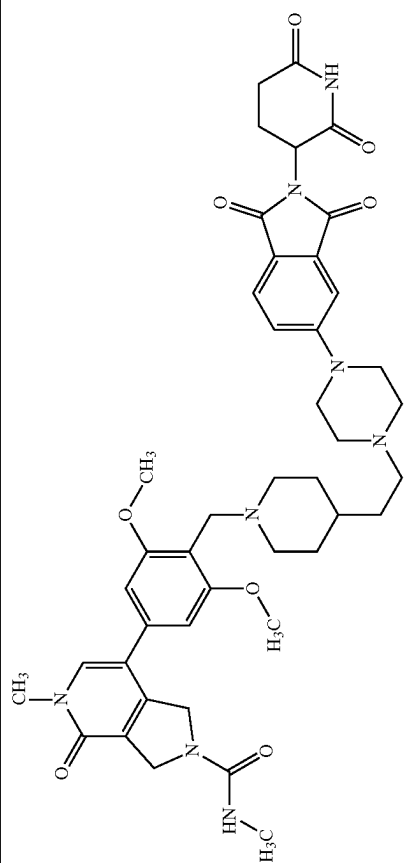
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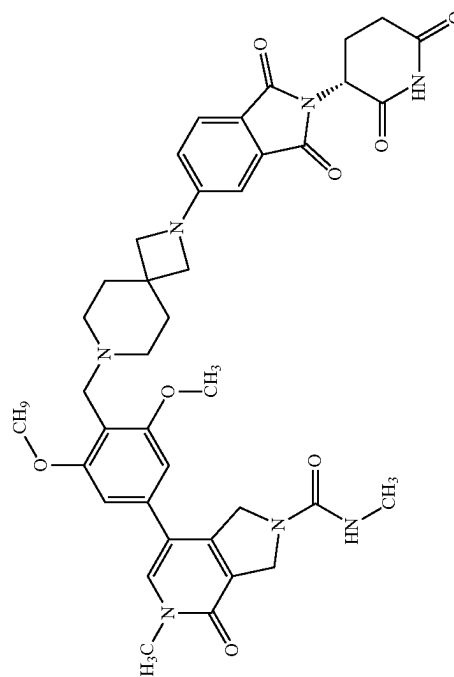
-continued

Compound No.	Structure
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D299



D300



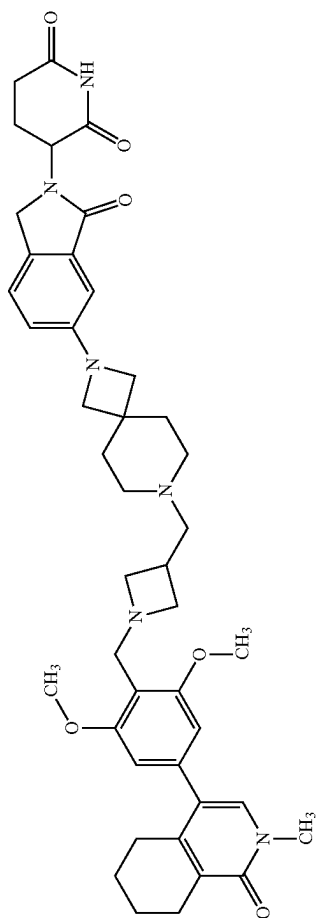
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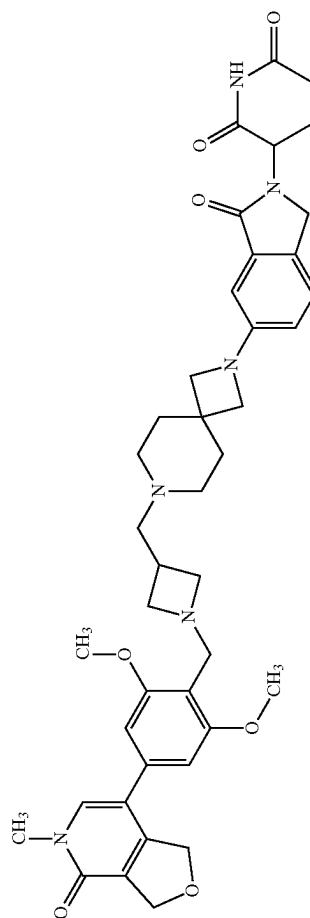
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Compound No.	Structure
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D301



D302



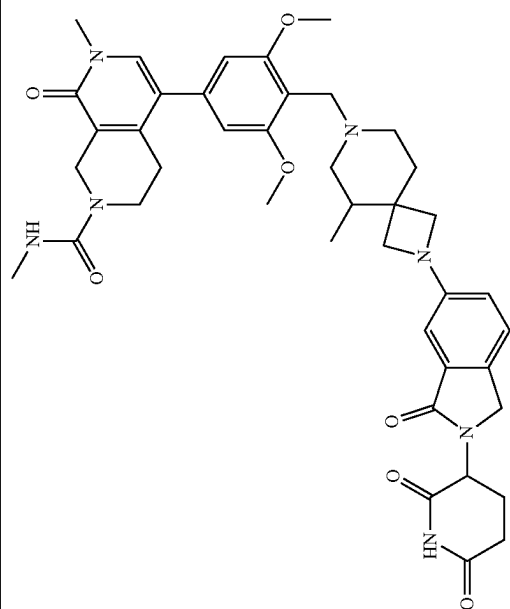
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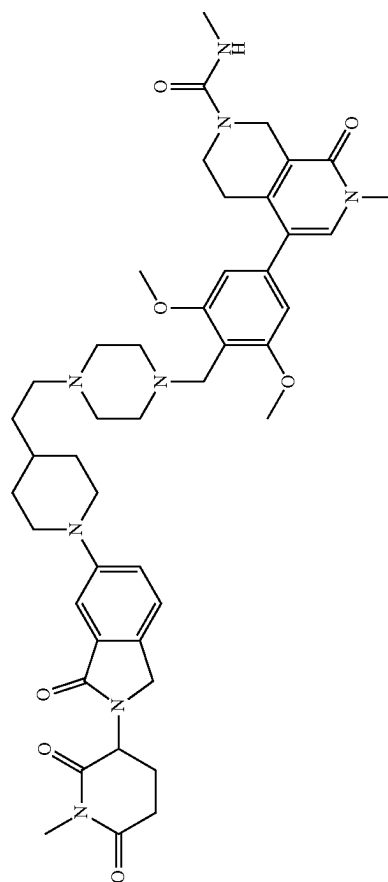
-continued

Compound No.	Structure
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D304



D305



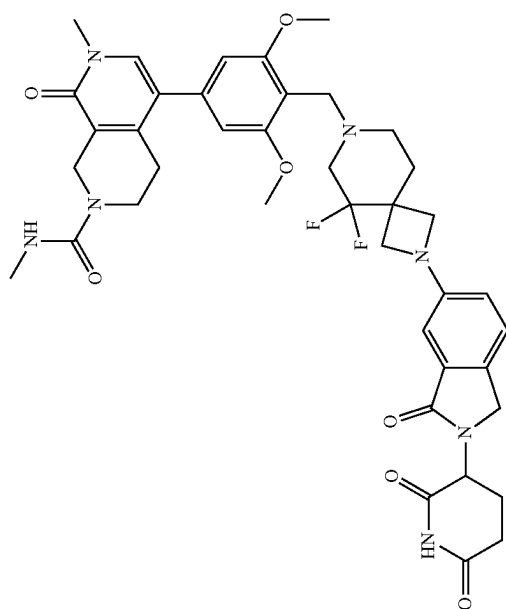
1075

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-continued

Compound No.	Structure
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D306



Compound No.	Structure
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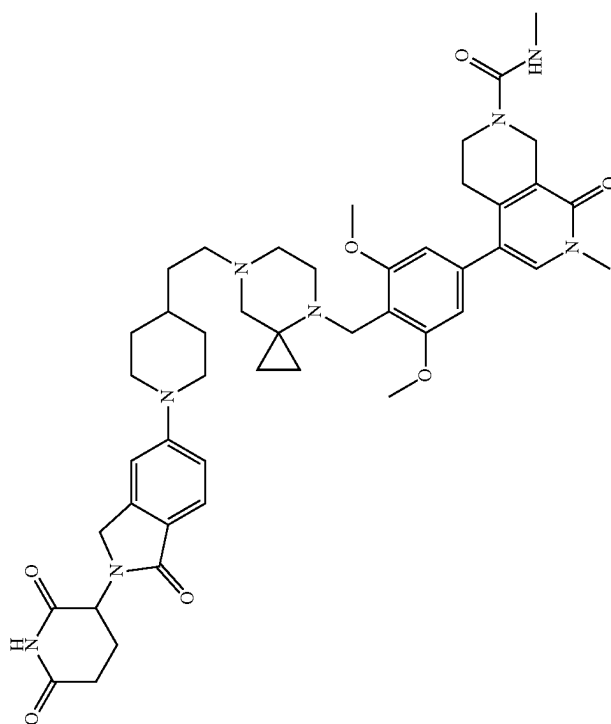
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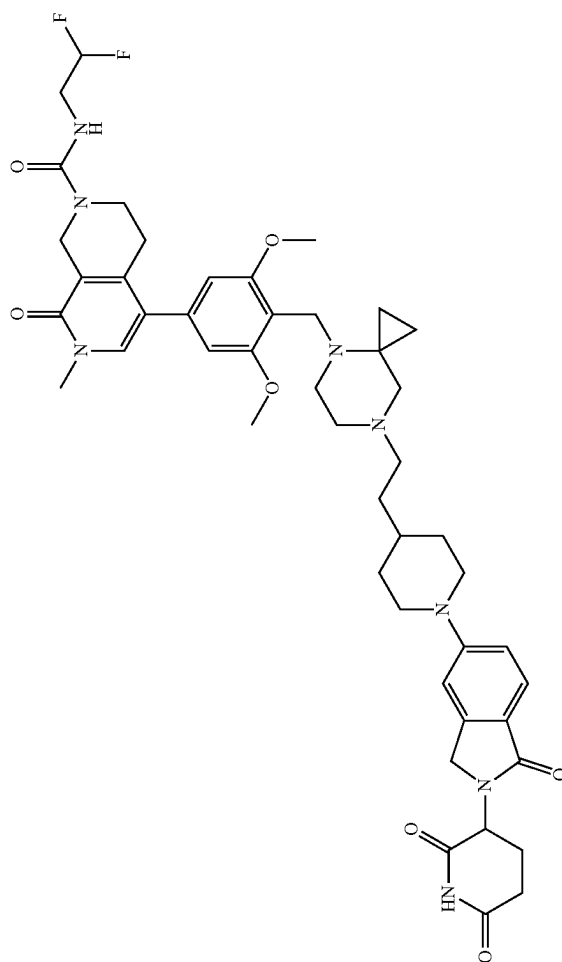
Compound No.	Structure
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D310



-continued

Compound No.	Structure
D311	



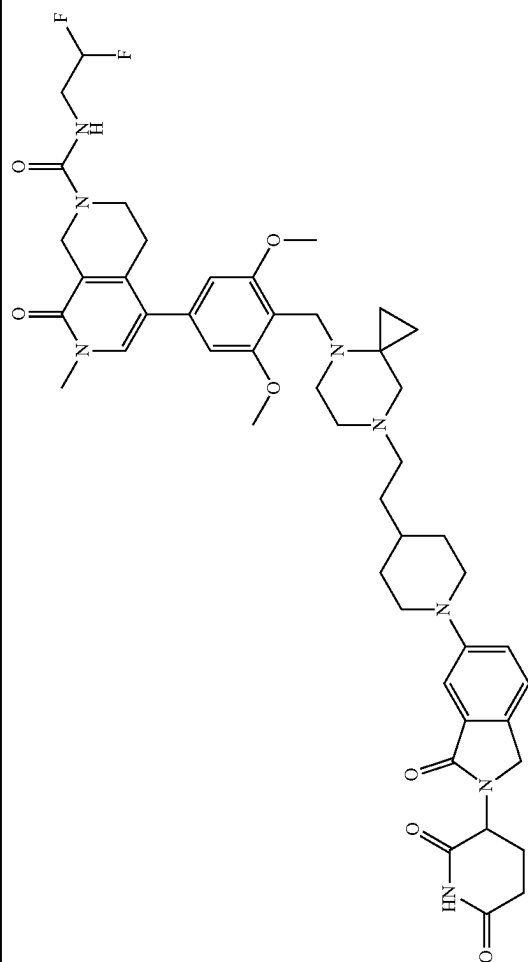
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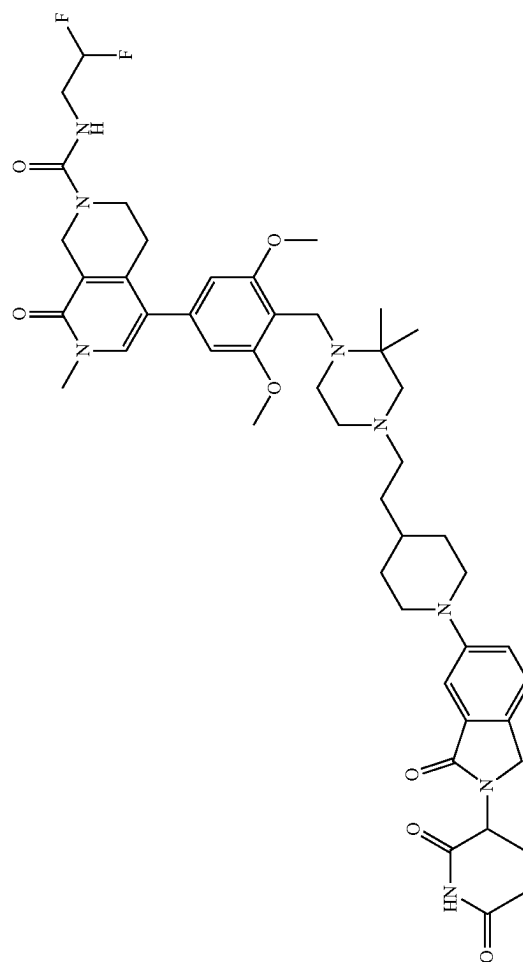
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Compound No.	Structure
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D312



D313



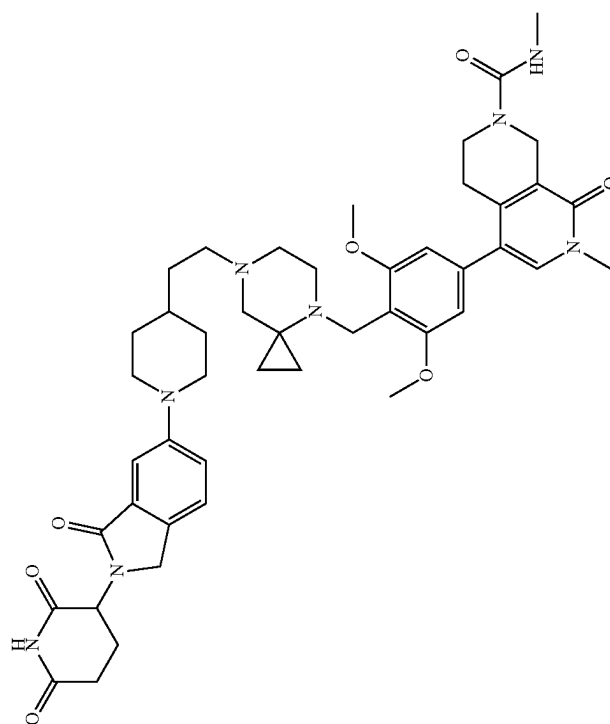
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-continued

Compound No.	Structure
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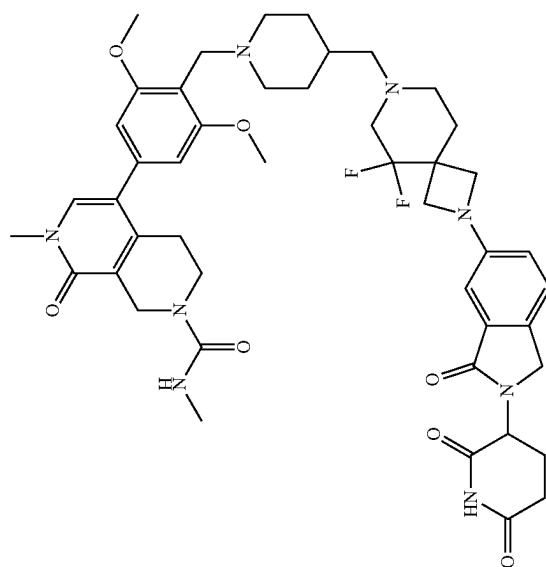
D314



-continued

Compound No.	Structure
1	
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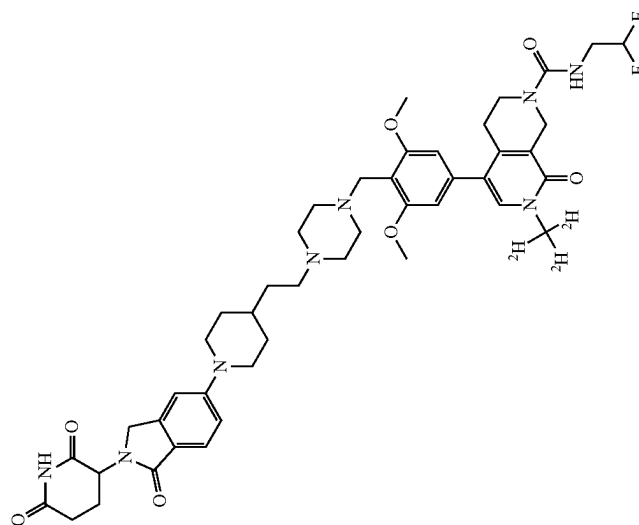
D315



-continued

Compound No.	Structure
1	
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11	
12	
13	
14	
15	

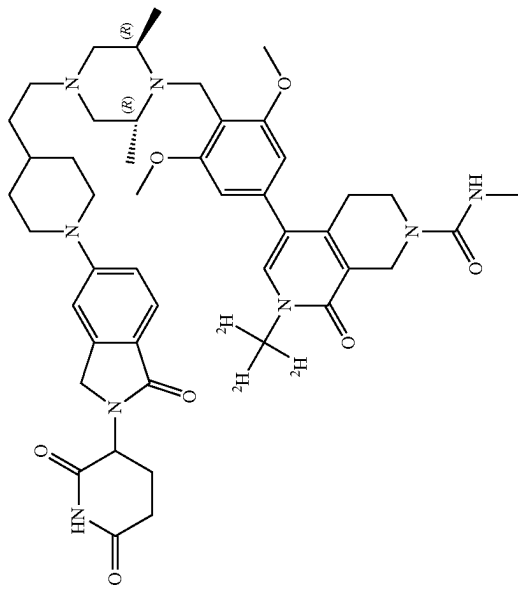
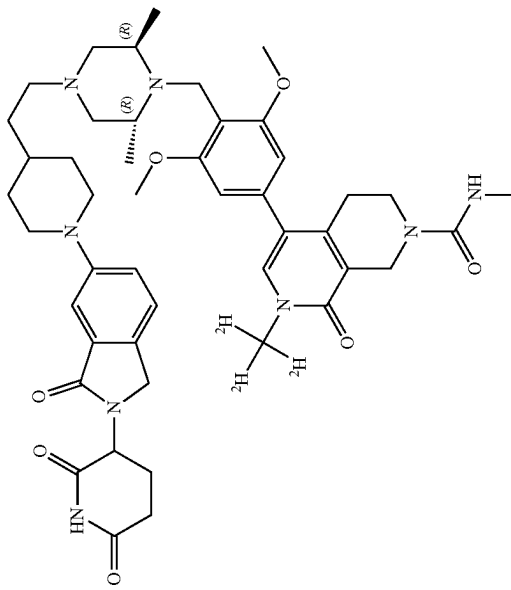
D317



1095

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-continued

Compound No.	Structure
D318	 Chemical structure of compound D318. It features a central pyridine ring substituted with a 4-methoxyphenyl group, a 4-methoxyphenyl group, and a 4-methoxyphenyl group. The pyridine ring is also substituted with a 4-methoxyphenyl group. The structure includes a piperidine ring, a morpholine ring, and a 4-methoxyphenyl group. Stereochemistry is indicated with (R) and (S) labels.
D319	 Chemical structure of compound D319. It is similar to D318 but with a different substitution pattern on the pyridine ring. The structure includes a piperidine ring, a morpholine ring, and a 4-methoxyphenyl group. Stereochemistry is indicated with (R) and (S) labels.

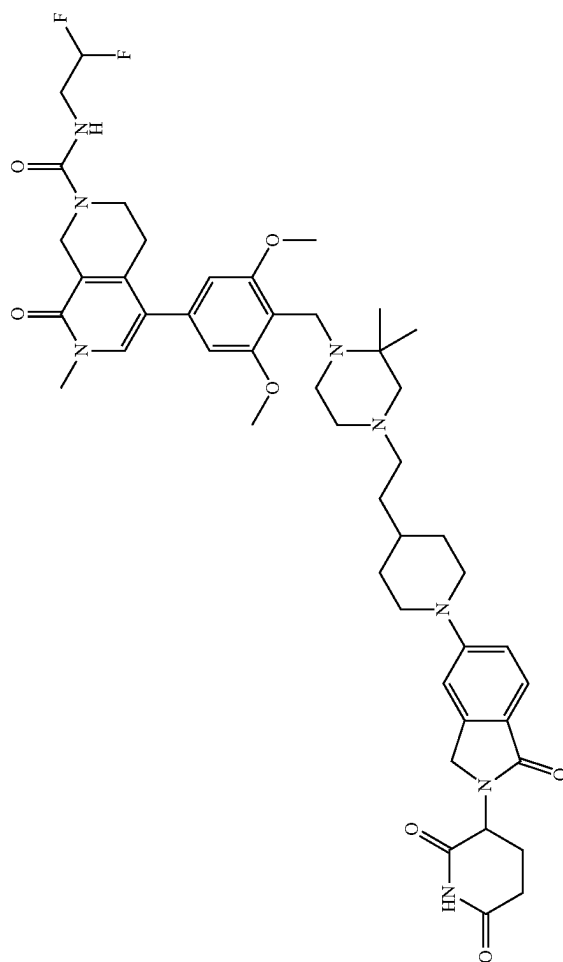
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-continued

Compound No.	Structure
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D320

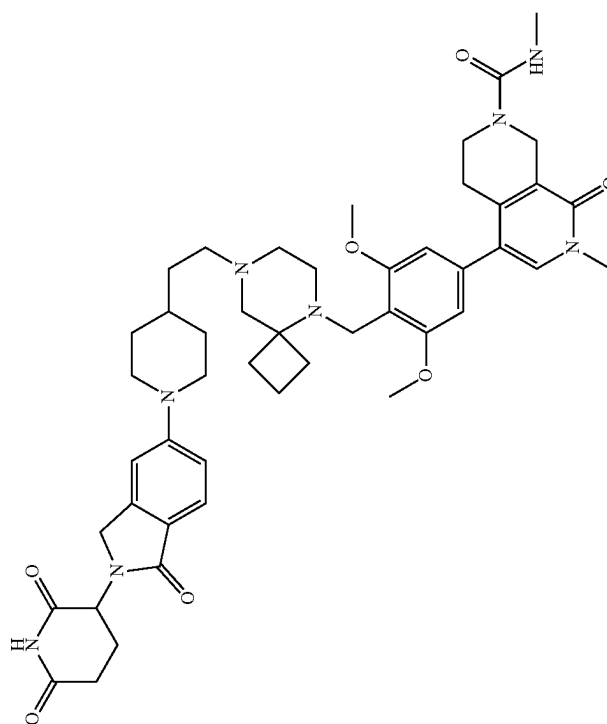


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1100

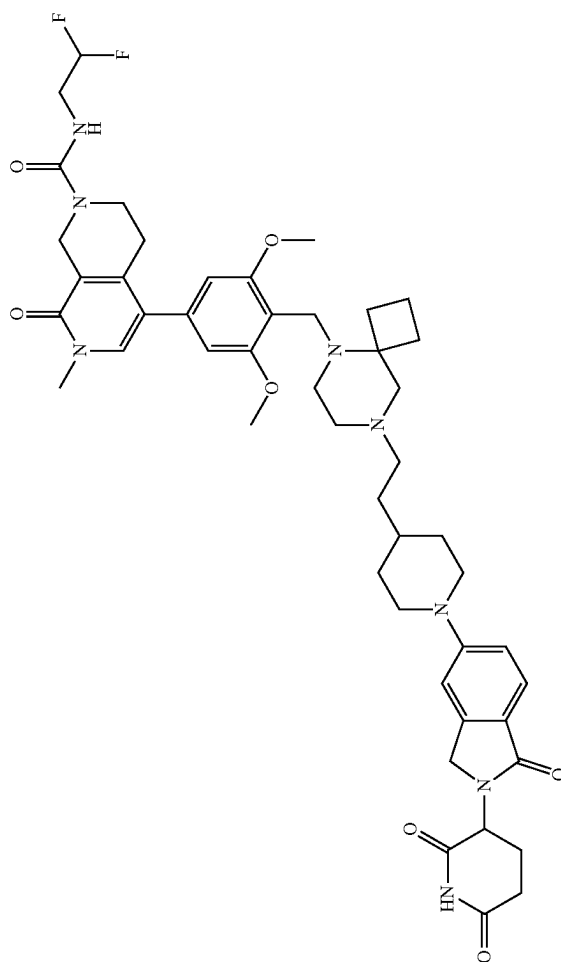
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Compound No.	Structure
D321	



-continued

Compound No.	Structure
D322	



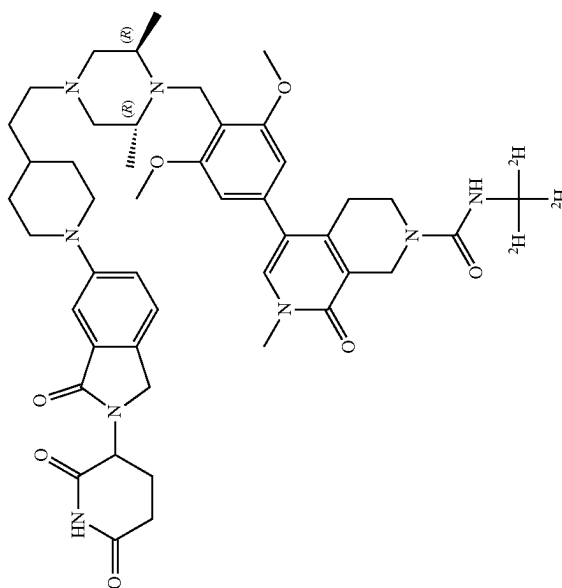
1103

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1104

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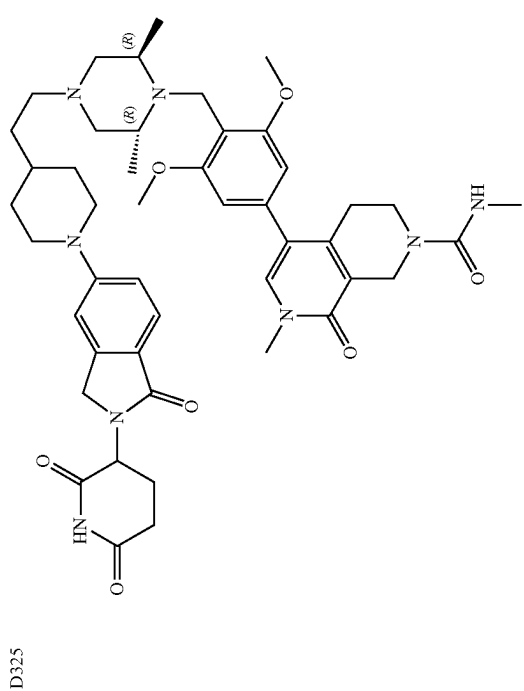
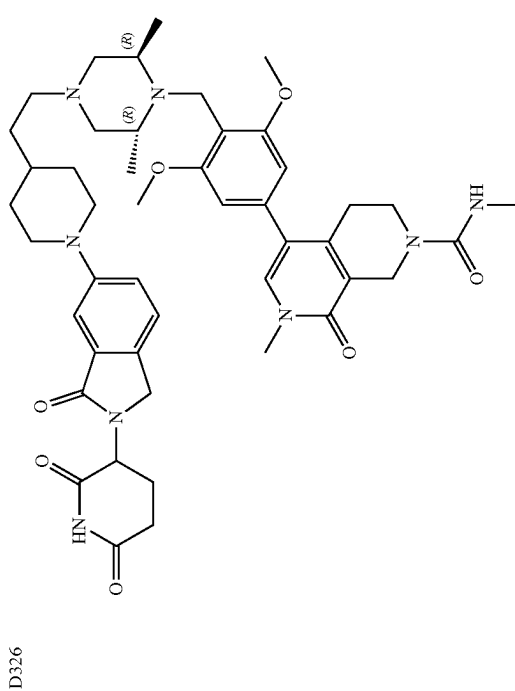
Compound No.	Structure
D323	



1107

1108

-continued

Compound No.	Structure
D325	 Chemical structure of compound D325: A complex molecule featuring a central benzene ring substituted with a methoxy group and a (1R,2R)-2-methyl-2,6,7,8-tetrahydro-1H-pyrimidin-5-yl group. This central ring is also connected to a 1-methyl-2,4-dihydroquinolin-3(1H)-one moiety and a 1-methyl-2,4-dihydroquinolin-3(1H)-one moiety. The structure includes stereochemical indicators (R) and a dashed bond.
D326	 Chemical structure of compound D326: A complex molecule featuring a central benzene ring substituted with a methoxy group and a (1R,2R)-2-methyl-2,6,7,8-tetrahydro-1H-pyrimidin-5-yl group. This central ring is also connected to a 1-methyl-2,4-dihydroquinolin-3(1H)-one moiety and a 1-methyl-2,4-dihydroquinolin-3(1H)-one moiety. The structure includes stereochemical indicators (R) and a dashed bond.

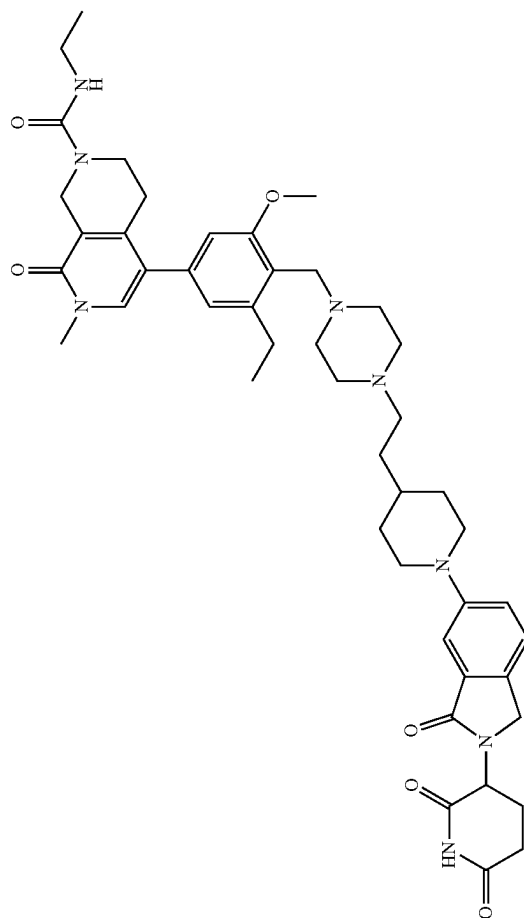
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1110

-continued

Compound No.	Structure
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D327



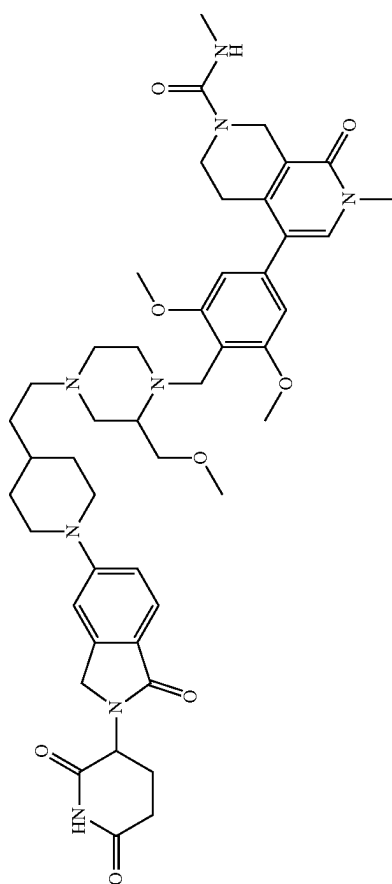
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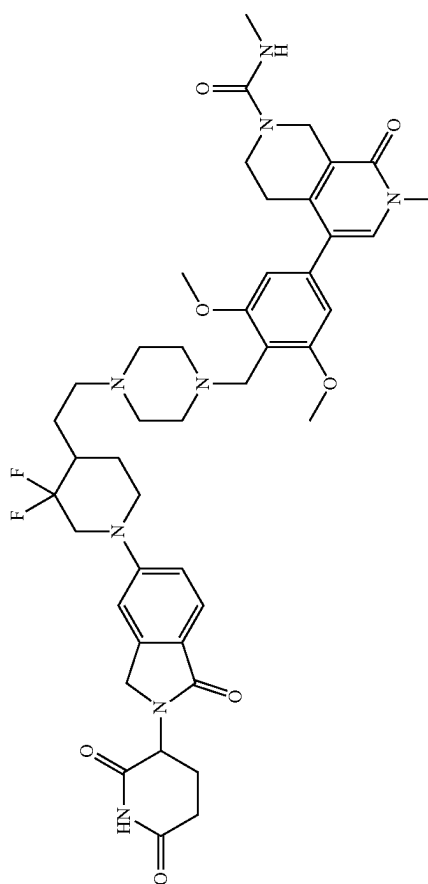
-continued

Compound No.	Structure
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D329



D330



Compound No.	Structure
D333	
D334	

Compound No.	Structure
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CNC(=O)c1ccc2c(c1)c(=O)n(C)c2-c3cc(OC)c(CN4CCN(CC4)C(F)(F)CN5CCN(CC5)C6CCN(C6)C7C=CC8C(=O)NCC8=CC=C7N9C(=O)CCC(=O)N9)c(O)c3

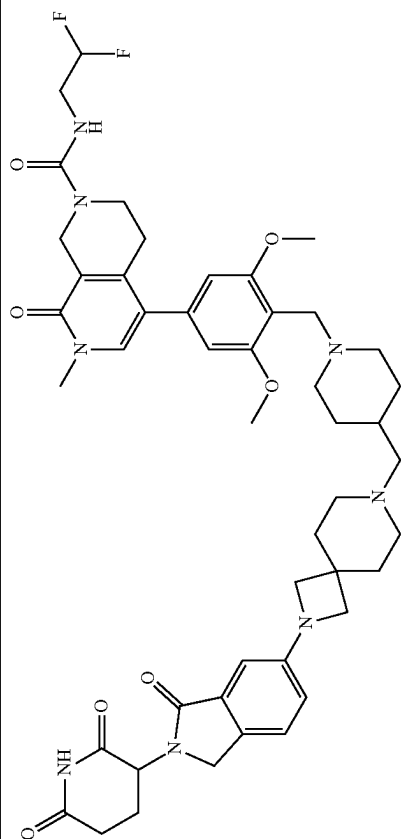
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1122

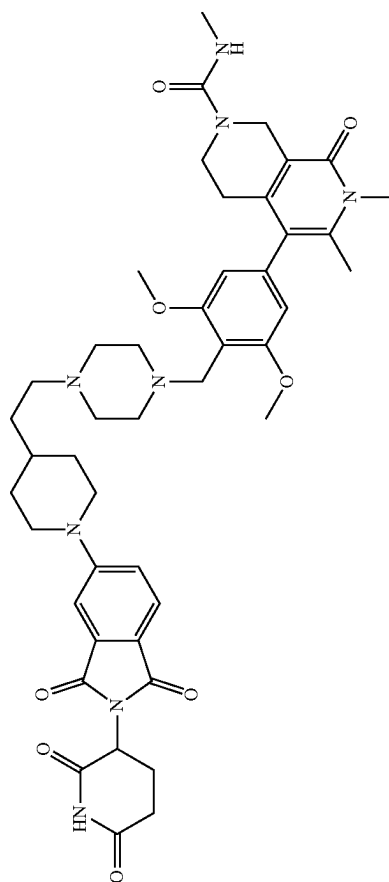
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Compound No.	Structure
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D336



D337



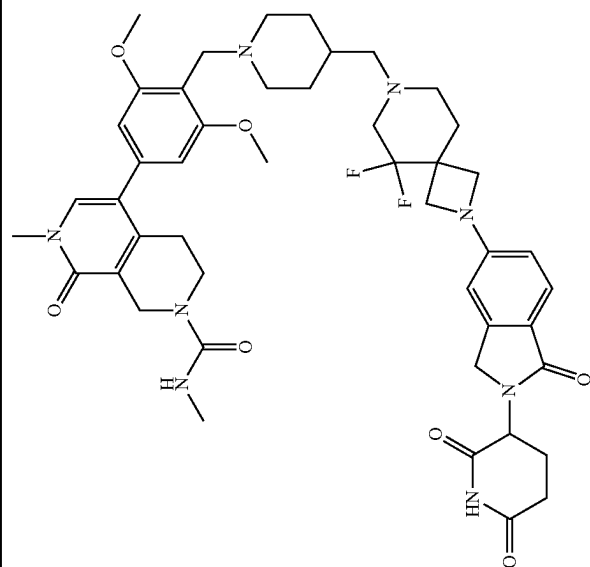
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1126

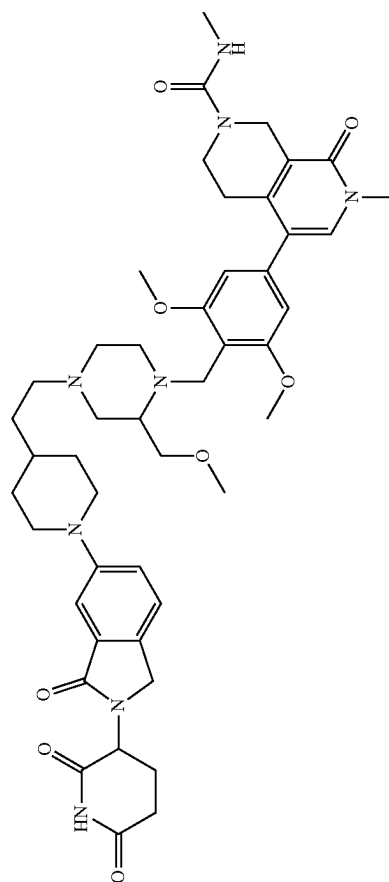
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Compound No.	Structure
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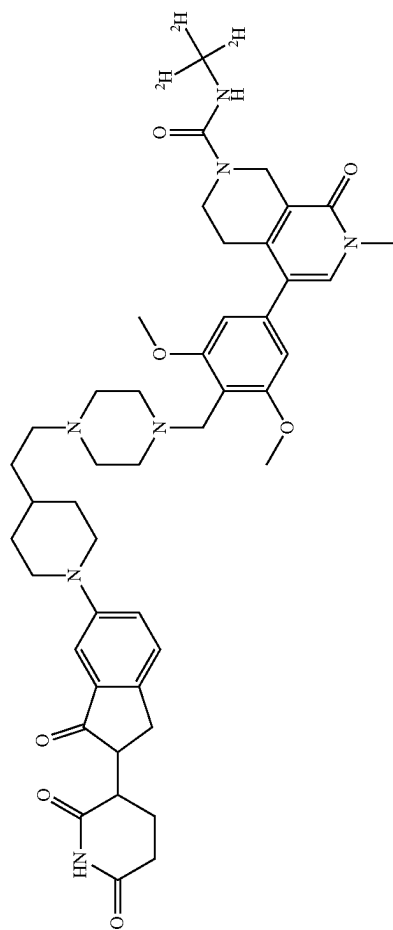
D339



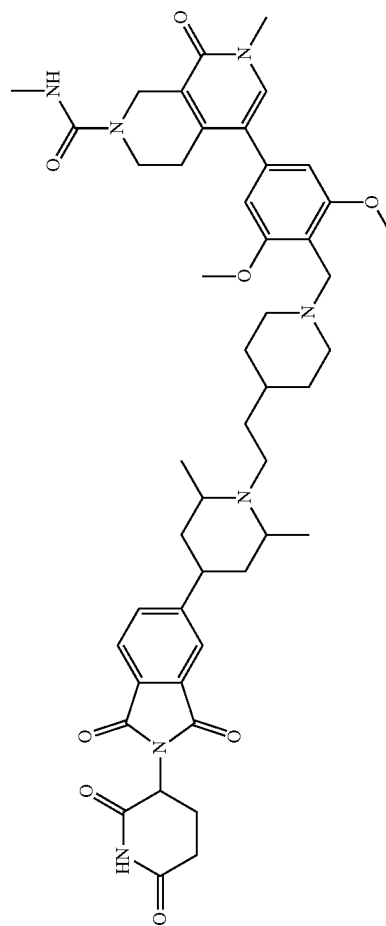
D340



Compound No.	Structure
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D342



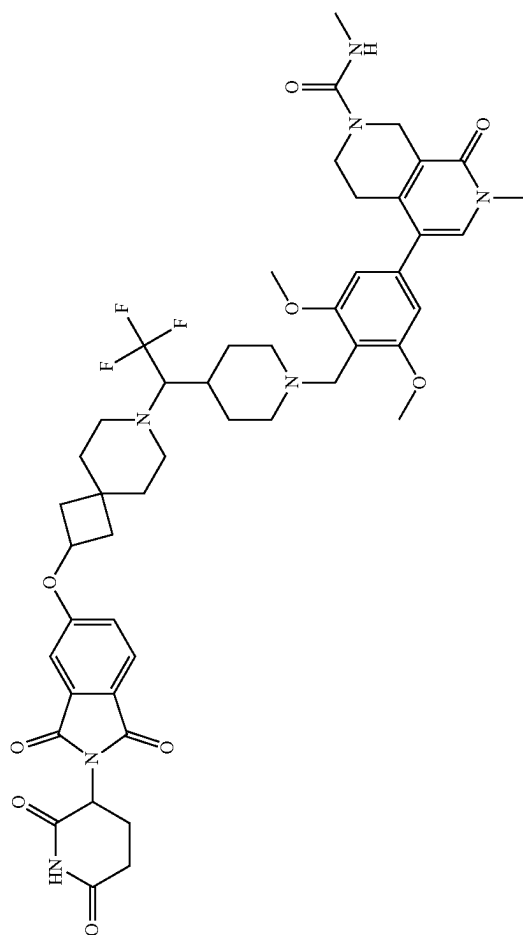
1129

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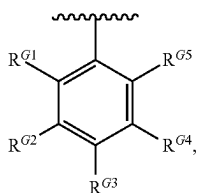
Compound No.	Structure
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D343



1131

14. The compound of claim 1, wherein G" is



wherein

each of R^{G1} , R^{G2} , R^{G3} , R^{G4} , and R^{G5} is, independently, H, halogen, optionally substituted C_1 - C_6 alkyl, optionally substituted C_1 - C_6 heteroalkyl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heterocyclyl, optionally substituted C_6 - C_{10} aryl, optionally substituted C_2 - C_9 heteroaryl, optionally substituted C_2 - C_6 alkenyl, optionally substituted C_2 - C_6 heteroalkenyl, optionally substituted $-O-C_3$ - C_6 carbocyclyl, optionally substituted $-C_1$ - C_3 alkyl- C_3 - C_6 carbocyclyl, optionally substituted $-C_1$ - C_3 alkyl- C_2 - C_5 heterocyclyl, hydroxyl, thiol, or optionally substituted amino; or R^{G1} and R^{G2} , R^{G2} and R^{G3} , R^{G3} and R^{G4} , and/or R^{G4} and R^{G5} , together with the carbon atoms to which each is attached, combine to form optionally substituted C_6 - C_{10} aryl, optionally substituted C_3 - C_{10} carbocyclyl, optionally substituted C_2 - C_9 heteroaryl, or optionally substituted C_2 - C_9 heterocyclyl.

15. The compound of claim 1, wherein the compound has the structure of any one of compounds B1-B6 in the below table, or a pharmaceutically acceptable salt thereof

Compound No.	Structure
B1	
B2	

1132

-continued

Compound No.	Structure
B3	
B4	
B5	
B6	

1133

16. A pharmaceutical composition comprising the compound of claim 2, or a pharmaceutically acceptable salt thereof, and a pharmaceutically acceptable excipient.

17. A method of treating a synovial cancer in a subject having synovial cancer the method including administering 5
to the subject having synovial cancer an effective amount of a compound of claim 2 or a pharmaceutically acceptable salt thereof.

1134

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